

Eleuterio F. Toro

Computational Bodily Fluid Dynamics

Models, Algorithms and Applications



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Eleuterio F. Toro
Laboratory of Applied Mathematics
Department of Civil, Environmental
and Mechanical Engineering
University of Trento
Trento, Italy

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*To my Postgraduate Students and
Postdoctoral Research Fellows.
For the rewarding learning experience
in pursue of knowledge for a better world*

Preface

Human fluid anatomy and physiology, bodily fluid dynamics, mathematical models via differential equations, modern computational algorithms and medical applications. This is the overall subject of this book, which is aimed at students and researchers in biomedical engineering, applied mathematics, computational physics and various branches of medicine. It is a largely inter- and multidisciplinary subject, built upon fundamental works that have been carried out over the centuries by many researchers.

William Harvey (1578–1657) is credited with one of the most fundamental discoveries ever made about the anatomy and physiology of the human body [1, 2]. He understood the circulatory nature of blood flow involving two closely interconnected systems, namely the arterial and the venous systems. As all great discoveries, Harvey's was initially surrounded by bitter controversy and active opposition from the medical establishment. It was not until 20–30 years later that universities around the world came to accept what, 400 years on, is taken for granted. Over the centuries, scientists from a wide array of disciplines have built upon Harvey's legacy, that of a most fascinating world of a unified cardiovascular system, comprising circulating blood in an exceedingly complex network of billions of elastic vessels connected to a pumping heart.

Bodily fluid physiology is currently recognised to go beyond the heart, arterial and venous blood flow, to include other extracellular fluid compartments, namely interstitial fluid in the interstitium and lymph in the lymphatic system. Moreover, local circulations include cerebrospinal fluid in the craniospinal cavity; perilymph and endolymph in the inner ear; and vitreous and aqueous humour in the eye. Furthermore, fluid physiology is not just the sum of several extracellular fluid compartments. A fundamental aspect is the dynamic and interactive nature of the various fluid systems.

Fluids in the central nervous system constitute a representative example. High-pressure arterial blood flows into the cranial cavity in systole, interacts dynamically, in a time span of about one second, with brain tissue, interstitial fluid, cerebrospinal fluid and the venous blood compartment. The transiently increased arterial blood volume is accommodated into the fixed-volume cranial cavity through the

timely response of the brain venous blood that flows towards the heart and the transient displacement of the cerebrospinal fluid into the spinal canal. Importantly, the mechanics of fluid flow, the interactive dynamics of fluid compartments and the overall body fluid homeostasis are strongly modulated through the regulatory function of the nervous and endocrine systems. A holistic approach is therefore essential to the understanding of fluid physiology and pathology involving the delicate interactive nature of the cardiovascular system and various fluid compartments.

For centuries, scientists from mechanics and applied mathematics have attempted to set fluid physiology in a theoretical framework. Almost two centuries after Harvey's discovery, the Swiss mathematician Leonhard Euler (1775) made the first documented attempt to construct a mathematical model to represent the flow of blood in elastic vessels. He did so in terms of a proposed first-order system of two partial differential equations [3, 4]. Then, however, solving the governing equations in order to provide insight into the physics of blood flow proved too difficult, even for Leonhard Euler. He stated “... *since there appears to be no way in which this can be completely resolved and the investigation can be considered to transcend human powers, the work will end here*”.

The two equations that Euler stated in his manuscript, are often called *the Euler equations* in the haemodynamics literature. For a precise definition of the Euler equation see the classical books of Landau and Lifshitz [5]; Courant and Friedrichs [6]; and Liepmann and Roshko [7]. Nowadays, particularly in the computational fluid dynamics community, the name *Euler equations* refers to a system that describes the dynamics of compressible media, comprising a continuity equation, momentum equation(s) and the energy equation, supplemented with an equation of state to account for the thermodynamics of the particular medium under consideration [8–10].

In this book, we shall refer to the original two partial differential equations proposed by Euler for the dynamics of blood flow, as well as extensions and variations of them, as the *blood flow equations*, or the *equations of haemodynamics*. Solving the blood flow equations remained a challenge until the twentieth century. Simpler mathematical models, based on ordinary differential equations, were proposed by Otto Frank (1899) [11]. Frank's original models, called today compartmental models, or zero-dimensional models, constitute an important class of theoretical tools available for simulating fluid physiology.

From Harvey's discovery of the circulatory nature of blood as a single arterial-venous system, there was a long journey to the point of formulating and solving the equations for blood flow in anatomically and physiologically realistic blood-vessel network configurations. This was left to scientific advances produced almost four centuries later. Technological achievements of the twentieth century produced imaging modalities (X-ray, ultrasound and magnetic resonance) that today allow the *reconstruction* of the hidden flow domain, or parts of it, from imaging data, thus aiding the determination of the computational geometry, or computational domain. Moreover, imaging techniques can also provide measured flow data that is essential for validating computational predictions. The twentieth century also witnessed two more relevant scientific and technological advances that underpin the field of

computational science today. First, the design of numerical algorithms to compute approximate solutions to partial differential equations, such as those proposed by Euler, for example.

The fundamental contributions of mathematicians such as Sergei Godunov [12], Peter Lax [13] and James Glimm [14], to give just three names, created the foundations for the development of modern numerical algorithms for solving non-linear systems of evolutionary partial differential equations [8–10], including the blood flow equations. But of course, numerical algorithms by themselves would be useless without the existence of fast computing machines, computers, also a product of the twentieth century. Often, and erroneously in the literature, the credit for the wonders of computational science are ascribed solely to computing machines.

Today, we find ourselves at the forefront of integrating the groundbreaking advancements in fluid physiology, mathematics, computational algorithms, and various other technologies that have evolved over an extensive period of time. There exist advanced mathematical techniques to analyse the equations for fluid flow, sophisticated computational algorithms to solve the equations, powerful computing machines to carry out the calculations and refined imaging techniques to visualise cardiovascular anatomy and measure quantities of fundamental interest, such as flow and concentrations of chemicals.

This book attempts to put together mathematical models for all major bodily fluid compartments, detailed analysis of the governing differential equations and advanced computational algorithms to solve them accurately and efficiently, along with the use of modern imaging and experimental techniques. The full package points to the distinct possibility of simulating the time-dependent dynamics of all major interacting bodily fluid compartments in the entire human body. Our underlying, holistic approach aims at continuously increasing the complexity of models to account for more detailed anatomy and physiology as it appears reasonable, with the purpose of understanding the full system as resulting from the interaction of many pieces put together, as in nature.

We believe the end product constitutes a useful framework for improving our understanding of bodily fluid physiology and for undertaking studies of major classes of human diseases, such as arterial hypertension and neurological disorders, to give just two examples. Only a better understanding of diseases will lead to the design of more effective therapeutical strategies.

This Book: Its Genesis, Its Target

This book originates from essentially two distinct but complementary areas of academic activity. The first source is mathematical modelling of fluid dynamics, the development of computational algorithms to solve evolutionary partial differential equations and application of models and methods to simulate fluid flow phenomena in industrial, environmental and aerospace contexts. Activities in this area have involved research; publication of research articles and books; and university postgraduate

teaching to master and Ph.D. students of science and engineering in several universities, in several countries, for more than four decades. Examples include postgraduate courses delivered in Cranfield University, Manchester Metropolitan University and Cambridge University, UK. Relevant to the contents of this book was a module on *Numerical Methods for Environmental Problems* for master students and a *Winter School for Numerical Methods for Hyperbolic Equations* for Ph.D. students and researchers, in the Faculty of Engineering of the University of Trento, Italy.

The second component of this book, broadly described as bodily fluid physiology and related diseases, emerged as a new research topic in my group, for Master and Ph.D. students in 2009. These teaching and research activities then led to the creation in 2012 of the specialism *Modelling and Simulation of Biomedical Problems* in the master course of the Mathematics Department, University of Trento. One of the several modules of the full course was *Computational Haemodynamics*. This course was designed and taught by the author for several years; the lecture notes for that course form the core material of this book. In addition, much of the contents and applications have emerged from research performed with my master and Ph.D. students and from fruitful collaboration with many academics in several countries, including biologists, physiologists, medical doctors, engineers, mathematicians, physicists and magnetic resonance imaging specialists. Such collaboration activities attest to the fact that the subject of this book is highly interdisciplinary, as is medicine nowadays.

This book focuses on mathematical models for the integrated fluid dynamics of bodily fluids in humans, including arterial blood, venous blood, interstitial fluid and cerebrospinal fluid. The emphasis is on integrating all major fluid compartments throughout the human body, aiming to replicate their real functional and anatomical connectivity. Many of the models and methods presented could be extended to include other systems, such as the lymphatic and urinary systems. However, this general approach comes at the cost of certain simplifications. The mathematical models are algebraic (non-differential), zero-dimensional systems consisting of ordinary differential equations in time and one-dimensional, cross-sectional averaged systems of non-linear partial differential equations.

While the simplicity of these models may initially seem to limit the book's scope, they retain the key features of bodily fluid dynamics, particularly fluid flow in compliant (non-rigid) conduits. Moreover, resulting mathematical models can predict physiological behaviour at affordable computational cost. In particular, it has been demonstrated that global mathematical models for the entire human circulation and its interaction with cerebrospinal fluid and brain dynamics can be very useful in predicting physio-pathological phenomena in the craniospinal cavity [15–19].

This book may be used directly for teaching computational methods for bodily fluid dynamics to master and Ph.D. students, or advanced level undergraduate students, in academic departments, such as engineering, mathematics, physics, computer science and even biology and medicine. The book may also prove useful to researchers in universities and research laboratories in both the fields of computational biofluid dynamics and numerical analysis of partial differential equations.

Contents of This Book

This book consists of 21 chapters divided into four parts, as detailed below.

Part I is concerned with all major human bodily fluid systems, their anatomy and physiology, some basic concepts in fluid mechanics and some notions on mathematical modelling. This part is divided into five chapters, as follows:

1. The Cardiovascular System.
2. Notions of Fluid Mechanics and Transport.
3. Interstitial Fluid and Lymph.
4. Cerebrospinal Fluid.
5. Fluids in the Eye and Inner Ear.

This part is meant for scientists wishing to get involved in the subject of computational bodily fluid dynamics, for example engineers, physicists, applied mathematicians, numerical analysts and computer scientists.

Part II is concerned with the governing partial differential equations of bodily fluid mechanics, their mathematical analysis and construction of exact solutions. This part contains six chapters:

6. The Equations of Haemodynamics.
7. Hyperbolic Equations I: The Linear Case.
8. Hyperbolic Equations II: The Non-linear Case.
9. The Riemann Problem for Blood Flow.
10. Generalised Mathematical Models.
11. Mathematical Models for Spinal CSF Dynamics.

Part III is concerned with computational algorithms for solving the differential equations that model fluid flow. This part contains six chapters:

12. Basic Notions on Numerical Methods.
13. Conservative Methods.
14. Methods for Non-conservative Systems.
15. Non-linear Spatial Reconstruction Methods.
16. High-Order Methods for Scalar Equations.
17. High-Order Methods for Non-linear Systems.

Part IV is concerned with comprehensive reviews of two major fields of applications, simulations by means of a global closed loop mathematical model, ending with some concluding remarks and perspectives:

18. Neurological Disorders and Neurofluids.
19. Arterial Hypertension and Bodily Fluids.
20. Simulating Intracranial and Essential Hypertension.
21. Concluding Remarks and Perspectives.

How Could This Book Be Used

The contents of this book could be used in several ways. One possibility is to use each separate part as a single set of lectures for a specific university course at the masters or advanced undergraduate levels.

For example, all five chapters of Part I could form the bases for a course on human fluid physiology for medical students, biomedical engineers, physicists, applied mathematicians, numerical analysts and computer scientists.

All six chapters of Part II could be integrated into a set of lectures for applied mathematicians, engineers of various disciplinary sectors and physicists. The core subject would be partial differential equations for fluid flow, centred on bodily fluids as the motivating application.

All six chapters of Part III could form the material for a course on advanced computational algorithms for solving partial differential equations that model fluid flow. The course would be suitable for numerical analysts, applied mathematicians, biomedical engineers, physicists and computer scientists.

The three chapters of Part IV could be integrated with other courses on advanced computational bodily fluid dynamics, or as a supplement to any of the above parts.

A more comprehensive use of this book would be for a course on computational haemodynamics, or bodily fluid dynamics, as done in the Mathematics Department of the University of Trento. In fact there would be enough material for two courses or for a single course with appropriate selection of contents.

Several chapters of the book end with a list of practical exercises and suggested miniprojects that can be used as supplementary course material in a teaching setting. The contents of this book as a whole could also be of use to medical researchers with a mathematics, engineering or physics background. Much of the material is advanced. Some current research issues are also addressed. All contents are backed up by extensive and up-to-date lists of references.

The mathematical models and the numerical methodologies studied are transferable to several scientific and technological areas that involve fluid dynamics, evolutionary partial differential equations and numerical methods for solving such PDEs. This book will contribute to education and training of young scientists

across applied mathematics, physics, engineering (e.g. mechanical, environmental, industrial, bio-medical, aeronautical), computer science and medicine.

Trento, Italy

Eleuterio F. Toro
 OBE, Ph.D., Dr. h.c. (USACH), Dr.
 h.c. (UFRO), Dr. h.c. (UCT)
 Professor Emeritus, University
 of Trento, Italy
 Academic Visitor, Cavendish
 Laboratory
 Department of Physics, University
 of Cambridge, UK
 Life Fellow, Clare Hall, University
 of Cambridge, UK
eleuterio.toro@unitn.it
<https://www.eleuteriotoro.com>

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Part I
Elements of Fluid Physiology

Chapter 1

The Cardiovascular System



Abstract This chapter, the first of the book, focuses on the human cardiovascular system, described as a rapid transport network through which blood carries red blood cells, white blood cells, platelets, oxygen, nutrients, carbon dioxide, waste products, hormones, heat, antibodies, and many other substances. The three key components of the cardiovascular system—the heart, blood, and blood vessels—are examined in detail. A dedicated section covers blood, its components, and its functions. Following this, the structure of the human heart is explored, along with an in-depth study of the cardiac cycle. The final section focuses on blood vessels, beginning with an overview of the circulatory routes. Each type of vessel is then examined in detail, starting with arteries, followed by veins, and concluding with capillaries in the microvasculature. Throughout, the chapter introduces the fundamental concepts of human cardiovascular anatomy and physiology while also considering the broader context of fluid physiology and mathematical modelling. The chapter concludes with a summary and recommendations for further reading.

1.1 Overview of the Human Cardiovascular System

The living human body is a complex mechanical, electrical and chemical system involving the dynamic interaction of fluids, gases and solids, all aimed at supporting life. This chapter is concerned with the human cardiovascular system, as depicted in Fig. 1.1. It is composed of the heart, the blood and two distinct blood vessel networks formed by arteries and veins, which are connected by the microvasculature. In place of **cardiovascular system**, one also speaks of **circulatory system**, although sometimes the literature excludes the blood from the definition of cardiovascular system. Here we adopt the more inclusive definition and use cardiovascular system as equivalent to circulatory system. As we shall see shortly, the circulatory system is further divided into the **systemic circulation** and the **pulmonary circulation**.

A major function of the cardiovascular system is the pumping of oxygenated blood by the heart, the system's pump, into the diverging arterial network that branches out into smaller and smaller vessels, ending up in the bed of capillaries of tissues and organs, where exchange of materials between the flowing blood and

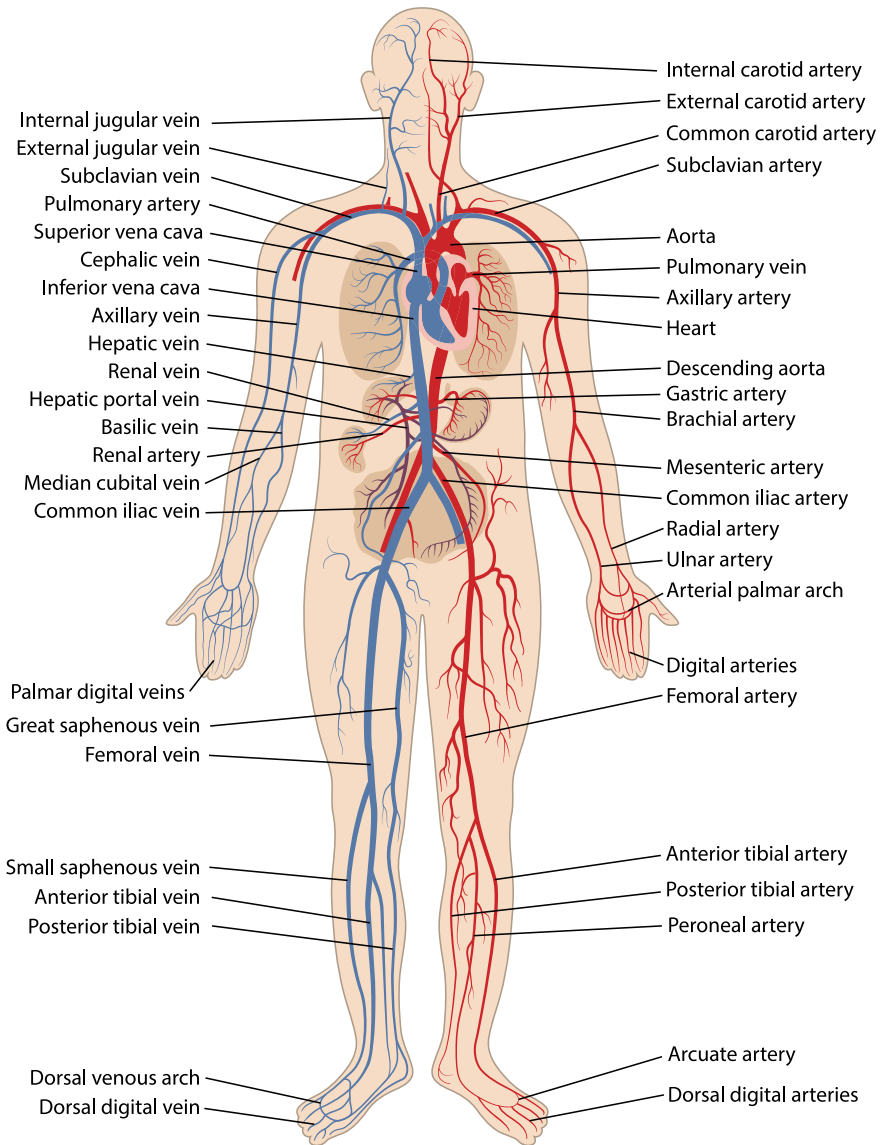


Fig. 1.1 The human cardiovascular system is composed of the heart, the blood and a network of blood vessels comprising arteries, veins and the microvasculature (not shown). The full circulatory system includes the systemic circulation and the pulmonary circulation. Conventionally, arterial blood is represented in red and venous blood in blue. Only major vessels are shown (from Shutterstock, with permission)

the surrounding interstitium occurs. The exchange includes a **delivery** function and a **removal, or clearance**, function. Materials delivered to tissues include oxygen, glucose, amino acids, fatty acids, vitamins, drugs and water. Materials removed contain metabolic waste products, including urea and creatinine, through the **urinary system**. The **clearance** function of the circulation is executed by the return of the oxygen-depleted blood, through a convergent network of venous vessels, back to the heart. Figure 1.3 depicts the human heart. The functions so far described concern the **systemic circulation** and involve the left ventricle of the heart, the contractions of which pump out oxygenated arterial blood, and the right atrium of the heart that receives the oxygen-depleted venous blood from the full body. Figure 1.8 depicts the circulation routes of blood.

In order to replenish the oxygen content of blood, another circulation system is necessary, namely the **pulmonary circulation**. The right ventricle of the heart pumps the oxygen-depleted venous blood returned into the right atrium of the heart, to the lungs, where exchange of gases takes place: carbon dioxide is discharged and oxygen is loaded into the blood, to be transported back to the heart left atrium, in readiness for pumping it out again by the left ventricle and thus repeat the cycle. The function of transport of oxygen and carbon dioxide involves the pulmonary circulation and the **respiratory system**, while the blood transport of nutrients to tissues and organs involves the **digestive system**. A major part of nutrients come from the small intestine, where nutrients are absorbed by capillaries into the blood stream and transported back to the heart via the venous system, and then into the arterial system that will deliver such nutrients at the level of the microvasculature.

Additional functions of the cardiovascular system include transport and delivery of hormones from the place of production by glands, to a target organ in need of regulation. Hormones are chemical messengers dissolved in blood plasma or bound to blood plasma proteins. Thermoregulation is another important function of the CVS, in concert with other systems. The CVS, through flowing blood, transports heat from deep organs to the skin surface, thus regulating heat loss from the skin to the environment. Furthermore, reproduction ensures the continuation of the species. This would not be feasible without the pressure-assisted erection function bestowed by the cardiovascular system and its interaction with various control and bodily-fluid systems.

There is a fluid system, the **lymphatic system**, that complements the transport function of the systemic and pulmonary circulations. This system consists of a fluid called **lymph** and a network of lymphatic vessels, nodes and organs. The lymphatic system collects excess **interstitial fluid** (ISF) from the interstitium, transports it along the lymphatic network, through lymph nodes and organs, to finally discharge it into the venous system. The lymphatic system also carries nutrients from the small intestine back to the venous system. All contents of the lymphatic network end up in the venous system and hence into the circulatory system. Analogous fluid dynamics and exchange functions are present in the central nervous system (CNS), composed of the brain, the spinal cord and neural parts of the ocular and auditory systems. Cerebrospinal fluid (CSF) plays a distinctive role in the CNS through a function that

is analogous to that of the lymphatic system in the periphery. CNS fluid systems are currently the subject of intense research activity and much remains to be understood.

In summary, the human cardiovascular system is primarily a fast transport system that works in conjunction with other fluid transport systems in the body. The dynamics of the flowing blood and exchange of materials between the intravascular space and the interstitium is closely regulated by the central nervous system and the endocrine system.

In the following sections we describe in more detail the various components and functions of the cardiovascular system, starting with the blood.

1.2 Blood

Blood is a multi-material mixture, often defined as a two-phase suspension of **formed elements** (45%) in an aqueous solution called **plasma** (55%). Blood as a whole constitutes about 20% of the total extracellular fluid volume of the body, amounting to about 8% of the total body mass. Next, we describe blood plasma and formed elements separately.

1.2.1 Blood Plasma and Solutes

Plasma is the aqueous part of blood. It is composed of 91.5% water and 7% dissolved substances, which include various types of solutes. **Metabolism solutes** include nutrients (glucose, lipids, amino acids, vitamins), hormones, respiratory gases (oxygen, carbon dioxide) and waste products resulting from metabolic processes. A second group of plasma solutes includes **salts and ions**. Plasma is a dilute salt solution and in water, salts dissociate into their component ions. The main plasma ions are: sodium (Na^+), chloride (Cl^-), bicarbonate (HCO_3^-), calcium (Ca^{2+}), magnesium (Mg^{2+}), zinc (Zn^{2+}) and potassium (K^+). In the human body these ions are called **electrolytes** and serve various functions related to their electric charge, concentration gradient, or play a role as signalling molecules. A physiological role includes the functioning of muscles and nerves and formation of bone. Electrolytes also play a role in osmosis, a mechanism that affects **osmotic pressure** and thus the movement of water in the body. From the point of view of haemodynamics and the movement of fluids and solutes in the body, the role of electrolytes is important, as we shall see.

Proteins in blood plasma are another group of solutes; they include serum albumin, antibodies, fibrinogen and prothrombin. Serum albumin (or just blood albumin), produced in the liver, is the most abundant protein and has an effect on **osmosis**. Antibodies play a role in immunity, while fibrinogen and prothrombin are key players in blood coagulation. In fact, anomalous amounts of fibrinogen in the blood plasma may be associated with an increased risk of thrombosis. The dissolved substances in plasma, as a whole, play a role in the rheology of blood as determinants of blood

plasma viscosity and thus flow resistance. Blood plasma alone is more viscous than water. Whole blood, that is plasma plus formed cellular elements (red blood cells, white blood cells and platelets) is even more viscous than plasma alone, as we shall see in Chap. 2. Blood density is $\rho = 1060 \text{ kg/m}^3$ and blood temperature is 38°C , this is about 1°C higher than oral or rectal body temperature.

Next we study the formed, or cellular, elements of blood.

1.2.2 Cellular Elements of Blood

Suspended in plasma, the aqueous part of blood, are the so-called **formed elements**, or **cellular elements**, which constitute about 45% of the whole blood volume. See Fig. 1.2. Formed elements include **red blood cells** (RBCs), or **erythrocytes**, which occupy 99% of the suspension volume, with the remaining 1% being occupied by **white blood cells** (WBCs), or **leukocytes**, and fragments of cells called **platelets**. Table 1.2 shows numbers and main functions of formed elements.

Blood cells are formed in the **red bone marrow**, from the last 3 months before birth and for the rest of life; they enter the bloodstream through enlarged and leaky capillaries that surround red bone marrow cells and fibres. We next study each type of the formed elements in more detail.

Red Blood Cells or Erythrocytes

Red blood cells have the shape of biconcave discs; they have a diameter of 7 to $8 \mu\text{m}$ and a large surface area. See Fig. 1.2. A healthy adult male has about 4.5 to 6.5 million RBCs per microlitre (μL) of blood, while a healthy adult female has 3.9 to 5.6 million RBCs per microlitre; see Table 1.1. Just to give an idea of the meaning of

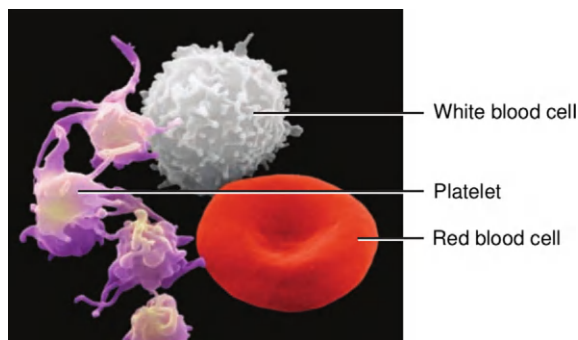


Fig. 1.2 Formed elements of blood: red blood cells, white blood cells and platelets. Their dimensions in terms of diameter are: RBCs: 7 to $8 \mu\text{m}$; WBCs: 10 to $14 \mu\text{m}$ (large ones) and 6 to $9 \mu\text{m}$ (small ones); platelets: 2 to $4 \mu\text{m}$. Recall that $1 \mu\text{m} = 10^{-6} \text{ m}$ (from Tortora and Derrickson [1], with permission)

Table 1.1 Formed elements of blood: red blood cells, white blood cells and platelets. Second column shows numbers and third column shows main functions. Recall that $1\mu\text{L} = 10^{-6}\text{L}$

Name	Numbers in millions per μL	Functions
Red blood cells	3.9–5.6 females; 4.5–6.5 males	Transport of O_2 and CO_2
White blood cells	0.005 to 0.01	Combat pathogens
Platelets	0.15 to 0.4	Platelet plug, blood clotting

such numbers, one drop of blood is about $50\mu\text{L}$. A word of caution: these numbers tend to vary a lot in the literature.

RBCs travel through the circulation and squeeze through the narrow capillaries of diameter $5\text{--}10\mu\text{m}$, in single-file flow, so-called **bolus flow**. As they pass through capillaries, RBCs deform without rupturing but suffer damage, which determines their short life of about 4 months. Note that mature erythrocytes have no nuclei; hence they cannot manufacture proteins and thus cannot repair themselves. New mature RBCs enter the circulation from red bone marrow at the rate of about 2 million per second; at this pace the rate of production balances the equally high rate of RBCs destruction, keeping numbers of RBCs steady. Ruptured RBCs are removed from the circulation through destruction by macrophages in the spleen, liver and red bone marrow. The breakdown products are recycled through a complex chain of events that includes formation of new RBCs in the red bone marrow.

Red blood cells perform the fundamental role of transporting respiratory gases; they deliver oxygen to tissues and collect carbon dioxide. Almost the entire interior of a red blood cell is packed with molecules called **haemoglobin**, which in turn are loaded with oxygen. One RBC contains 280 million molecules of haemoglobin. The large surface area of a RBC optimises diffusion of gas molecules into and out of it. Oxygen is picked up from the lungs and, as blood flows through tissue capillaries, it is released from haemoglobin; after some chemical reactions, they first diffuse into the interstitial fluid across the capillary wall and then into the cells. In this manner, about 25% of arterial oxygen is released into tissues. Haemoglobin molecules also pick up about 23% of carbon dioxide, as blood flows through tissues, which is then released from the haemoglobin into the lungs. Red blood cells and their contained haemoglobin molecules have a significant influence on blood flow and blood pressure, in the context of haemodynamics, as we shall see in successive chapters. Table 1.1 shows that RBCs outnumber the other types of formed elements.

The percentage of volume occupied by erythrocytes in blood is called **haematocrit** and is an important quantity, both physiologically and clinically. In females the haematocrit ranges from 35.5% to 44.9%; in men it ranges from 38.3% to 48.6%. Under pathological conditions these percentages may vary significantly from their physiological values. The concentration of RBCs dominates the effect of formed elements on the **rheological properties of blood**. That is, haematocrit affects the viscosity of blood and therefore the resistance to blood flow in the cardiovascular system. **Haemorheology** is the discipline that deals with the study of blood flow and

the deformation behaviour of its formed elements; this subject is both of scientific and clinical interest. See Chap. 2 for further details. As already stated, haemoglobin has a role in the **regulation** of blood flow and blood pressure. This effect on blood flow and pressure comes in through **nitric oxide** (NO), which is a gaseous hormone produced by the endothelial cells in the blood vessel wall. NO binds to the travelling haemoglobin in the blood and, under particular circumstances, is released locally causing **vasodilation**, increasing blood flow and enhancing oxygen delivery to cells. Details on regulation mechanisms are discussed in Chap. 2.

White Blood Cells (WBCs) or Leukocytes

The fundamental role of white blood cells of all types is the defence of the body, by combating invading microorganisms and other foreign substances that have entered the body, by means of phagocytosis and activation of the immune system response. WBCs have a nuclei but, unlike RBCs, they do not have haemoglobin. WBCs enter the blood stream from red bone marrow; WBCs occupy less than 1 % of the total blood volume and are outnumbered by RBCs by a factor of about 700; see Table 1.1. WBCs are divided into two main types: **agranular** cells and **granular** cells; see Table 1.2.

Agranular cells include:

- **Lymphocytes**. Lymphocytes are further divided into **T-lymphocytes** (or T cells), **B-lymphocytes** (or B cells) and **natural killer cells** (or NK cells).
- **Monocytes**. Monocytes are macrophages in blood and are further divided into **fixed macrophages** and **wandering macrophages**. The former reside in a specific tissue, for example in the lungs or the spleen.

Granular white blood cells are circulating cells and divide into the following three groups:

- **Neutrophils** constitute a large portion, between 60 % and 70% of all white blood cells in humans. They migrate to sites of injury, stick to the blood vessel wall and

Table 1.2 White blood cells (WBCs), or leukocytes, divide into two main types: agranular cells and granular cells. Percentages are given in third column, while dimensions are given in the last column. Recall that 99% of the volume of all formed elements is occupied by red blood cells

Name	Type	Percentage of total WBCs (%)	Diameter (μm)
Lymphocyte (T, B, NK)	Agranular	20–25	6–14
Monocyte	Agranular	3–8	12–20
Neutrophil	Granular	60–70	10–12
Eosinophil	Granular	2–4	10–12
Basophil	Granular	0.5–1	8–10

push their way through to the infected area, where they ingest, or phagocytize, microorganisms, foreign particles and dead matter. Via **chemotaxis** neutrophils are attracted by different chemicals released by microbes and inflamed tissues.

- **Eosinophils** occupy between 2% and 4% of all white blood cells. They leave the blood stream through the capillary wall and enter the tissue fluid in the interstitium. They are believed to release enzymes, such as histaminase, that combat the effects of histamine and other substances involved in inflammation during allergic reactions. Eosinophils also phagocytize antigen-antibody complexes and are effective against certain parasitic worms. A high eosinophil count often indicates an allergic condition or a parasitic infection.
- **Basophils** occupy between 0.5% and 1% of all white blood cells. They enhance the body's response to injury or infection by rupturing their granules and releasing chemicals. They are attributed a role in causing allergic responses.

Emigration is the process that distinguishes WBCs from RBCs. Under physiological conditions red blood cells remain in the blood within the circulatory system, while **most white blood cells leave the bloodstream** and gather at sites of pathogen invasion or inflammation. As a matter of fact, only 2% of the total **lymphocyte population** remains in the circulating blood at any time. Lymphocytes (T cells, B cells, NK cells) continually recirculate, from blood to interstitial spaces of tissues, to lymphatic fluid and back to blood via the lymphatic system that drains into the venous system. Most of the lymphocyte population resides in the **lymphatic system**, the skin and lungs. Lymphocytes are said to be the **major soldiers in the immune system battles**.

Granular WBCs and monocytes leave the bloodstream to fight injury or infection and never return to it. Monocytes are transported from the blood into the tissues, where they enlarge and differentiate into macrophages. White blood cells are distinguished for involving not just the cardiovascular system but also the interstitial fluid compartment, the lymphatic transport network and the immune system.

Platelets or Thrombocytes

The fundamental function of platelets is **haemostasis**, that is a defence mechanism related to prevention and control of bleeding. Platelets are blood cells, or cell fragments, formed in the red bone marrow, from where they enter the blood circulation. They are described as irregularly shaped fragments, with many vesicles, no nucleus and incapable of cell division. See Fig. 1.2. Each microlitre of blood contains between 0.15 to 0.4 million platelets and are more abundant than white blood cells, see Table 1.1. Their size is smaller than that of other formed elements, with a diameter of 2-4 μm . Compare with the size of white blood cells in Table 1.2. The life span of platelets is between 5 and 9 days, after which they are removed from the blood and destroyed by macrophages in the spleen and liver. The concentration of platelets is higher in the spleen than in the peripheral blood.

As already indicated, platelets play an important role in **coagulation**, or **blood clotting**. Damage to blood vessels or tissues initiates bleeding, which in turn initiates

coagulation by aggregation of platelets at the damaged site, temporarily stopping blood loss. The substance **thromboplastin** is released by platelets at injured tissue. Thromboplastin, in combination with other blood components form the enzyme **prothrombin activator**. This in turn converts **prothrombin** to **thrombin**. At this point, the soluble protein **fibrinogen** enters the scene. Thrombin converts fibrinogen to **fibrin**. This is an insoluble thread-like protein which, along with the trapped red blood cells, form the clot, stopping blood loss.

1.2.3 Functions of Flowing Blood

Blood in a living healthy human flows continuously in the systemic and pulmonary circulations, which are functionally linked to other bodily fluid systems, such as the interstitial fluid that bathes body cells and is constantly renewed by the blood stream. The circulatory system is also linked to the lymphatic system, which collects excessive ISF from tissues and transports it back to the blood circulation in the venous district. In the central nervous system (CNS), blood is also linked to the cerebrospinal fluid (CSF), which surrounds the brain parenchyma and the spinal cord. Flowing blood is a **fast transport system**. Flowing blood is also a **regulatory system** that contributes to maintain body fluid haemostasis and body cells water content through its osmotic pressure and **Starling forces**. The venous system combines the functions of transport, regulation and protection. It collects and transports back to heart CO_2 , metabolic wastes and toxic materials; it holds more than 2/3 of the total blood, playing a major blood volume regulatory function; it serves as sink for ISF, lymph and CSF. The clearance, or removal, function of the venous system in the CNS has become a major focus of attention in recent years. Finally, it is worth pointing out that there are numerous diseases that are linked to blood disorders, regarding both blood composition and blood dynamics.

In the next section we study another of the three components of the cardiovascular system, the heart.

1.3 The Human Heart

The heart is one of the three components of the cardiovascular system, the other two being the blood, studied in Sect. 1.2, and the blood vessels, to be studied in Sect. 1.4. The heart is an electrically activated **pump** that triggers into action all three components of the cardiovascular system by pumping blood to the circulatory vessel network. The human heart is an organ of about 12 cm in length by 9 cm in width and 6 cm in thickness, see Fig. 1.3; it has an average mass in adults of 250 g in females and 300 g in males. Each minute, the heart pumps out about 5 litres of blood through the body. During a lifetime, the heart pumps out about 200 million litres of blood, through the execution of three billion (3×10^9) contractions.

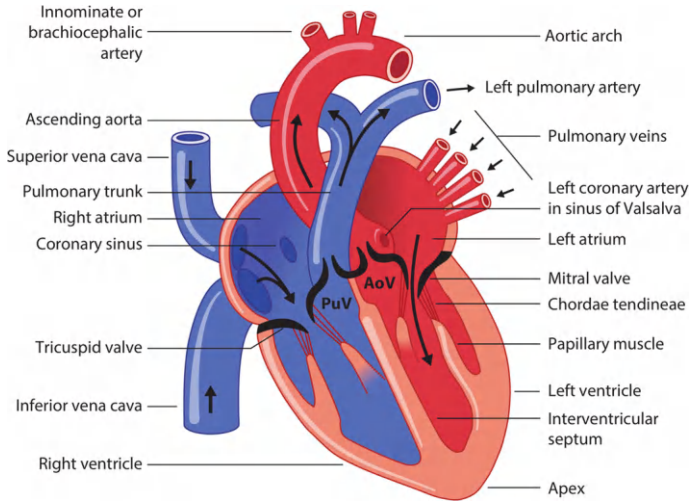


Fig. 1.3 The human heart. The right ventricle (RV) and left ventricle (LV) pump out simultaneously the same volume of blood. RV pumps used-up blood to the pulmonary circulation via the pulmonary arteries. LV pumps oxygenated blood to the arterial system of the systemic circulation via the aorta. Oxygen-depleted blood from the body returns to the right atrium (RA) via the superior and inferior venae cavae. Oxygen-rich blood from the lungs is returned to the left atrium (LA) via the pulmonary veins. The opening of the tricuspid and mitral valves allow the passage of blood from the atria (right and left) into their respective ventricles. The aortic valve (AoV) controls the passage of oxygen-rich arterial blood from the left ventricle to the aorta. The pulmonary valve (PuV) controls the passage of oxygen-depleted venous blood from the right ventricle to the pulmonary trunk. Note the four chambers, the four apertures and the four valves (from Herring and Paterson [2], with permission)

In this section we first introduce the main features of heart structures, such as chambers, apertures and valves. Then we describe the dynamics of blood flow in the heart and its connection to the systemic and pulmonary circulations, via the third component of the cardiovascular system, namely the blood vessels.

1.3.1 Gross Structure of the Heart

Here we give a succinct overview of the main structures of the heart, which include four chambers, four apertures and four valves, as detailed below.

Four Chambers

The **right atrium** (RA) and its corresponding **right ventricle** (RV) are two of the four heart chambers that constitute the right side of the heart; see Fig. 1.3. The **left atrium** (LA) and the **left ventricle** (LV) are the other two heart chambers, which constitute the left side of the heart. The left-side chambers contain oxygenated arterial blood,

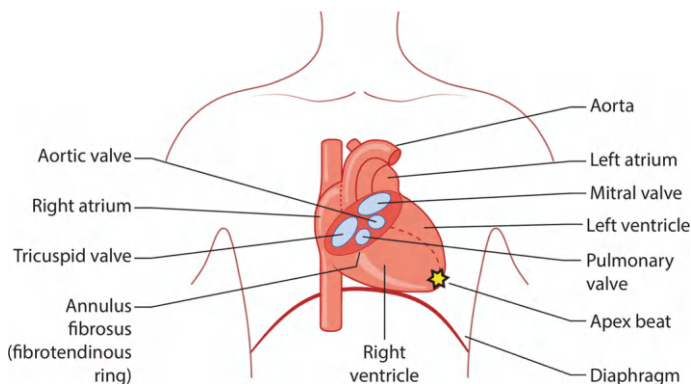


Fig. 1.4 Four heart apertures and four corresponding heart valves connecting four chambers. Note relative size of inlet and outlet valves; inlet valves have a larger cross-sectional area than the outlet valves (from Herring and Paterson [2], with permission)

as illustrated by the red colour in the figure, while the right-side chambers contain venous, oxygen-depleted blood, depicted in blue. The top chambers, LA and RA, are contractile reservoirs that receive blood. The right atrium receives venous blood from the two major veins, the superior vena cava and the inferior vena cava. The left atrium receives blood from the **superior pulmonary veins**, but note that such blood is actually oxygen-rich (arterial) blood from the lungs. The bottom chambers, LV and RV, are muscular pumps that deliver blood to the arterial network, through contraction. The right ventricle pumps oxygen-depleted blood to the pulmonary arteries of the **pulmonary circulation**, while the left ventricle pumps oxygen-rich blood to the arterial network of the **systemic circulation**. The four cardiac chambers are built upon a ring called the **annulus fibrosus**; see Fig. 1.4.

Four Apertures in the Ventricles

The four apertures of the heart are defined by the **annulus fibrosus**; see Fig. 1.4. Two apertures connect ventricles to atria, on the left and right sides, while the other two connect ventricles to the circulation; see Fig. 1.3. The right ventricle aperture connects the right ventricle to the pulmonary arteries and the vessel network of the pulmonary circulation. The fourth aperture connects the left ventricle to the aorta and thus to the vessel network of the systemic circulation. Each of the four apertures is associated with a valve, as discussed below.

Four Valves

Each of the four apertures referred to above is controlled by a valve. The four valves are grouped in an oblique plane, as depicted in Fig. 1.4. Note their relative size. The **atrioventricular valves**, or A-V valves, connect atria to ventricles; these are

the **tricuspid valve** and the **mitral valve**. The tricuspid valve controls the aperture between the right atrium and the right ventricle; it is an **inlet valve**, as it controls the inlet of venous blood from the right atrium into the right ventricle. The mitral valve controls the aperture between the left atrium and the left ventricle; it is also an inlet valve, as it controls the inlet of arterial blood from the left atrium into the left ventricle. The other two valves are the **pulmonary valve** (PuV) and the **aortic valve** (AoV). The pulmonary valve controls the aperture between the right ventricle and the pulmonary arteries via the pulmonary trunk; it is an **outlet valve** and controls the passage of blood from the right ventricle out to the pulmonary circulation. The aortic valve controls the aperture between the left ventricle and the aorta; it is also an outlet valve and controls the passage of arterial blood from the left ventricle out to the systemic circulation via the aorta. The outlet valves are also called **semi-lunar valves**, due to their structure.

Having introduced the main anatomical structures of the heart we next describe the blood flow through all four chambers, apertures and valves during the time taken by one cardiac cycle.

1.3.2 The Cardiac Cycle

The heart possesses a system whereby rhythmical electrical impulses are generated, which in turn cause rhythmical contractions of the heart muscle. Such impulses are generated in the **sinus node**, located in the wall of the right atrium. The impulses are then propagated through the whole heart in a short time, first through the atria and then through the ventricles, with a broadly-varying time delay of about 0.1 s. The cyclically generated electrical impulses give rise to a cycle of rhythmical contractions of the atria and the ventricles of the heart; this is known as the **cardiac cycle**. The contractions are associated with blood movement in the heart chambers and the blood vessels of the pulmonary and systemic circulations. The cardiac cycle is also defined as *the cardiac events that take place in the time period between the beginning of one heart beat to the beginning of the next*.

The **heart rate** (HR) is defined as the **number of heart beats** per minute, that is

$$HR = \text{heart rate} = \text{number of beats/min} . \quad (1.1)$$

One may measure the heart rate directly by taking the **pulse**. The heart beat sends a pressure wave through the arterial system; this produces a noticeable pulse that is felt, for example, on the palm side of the wrist, just below the base of the thumb. The reciprocal of the heart rate gives the **duration of the cardiac cycle** (DCC), that is

$$DCC = \text{duration of cardiac cycle} = 1 \text{ min/number of beats} . \quad (1.2)$$

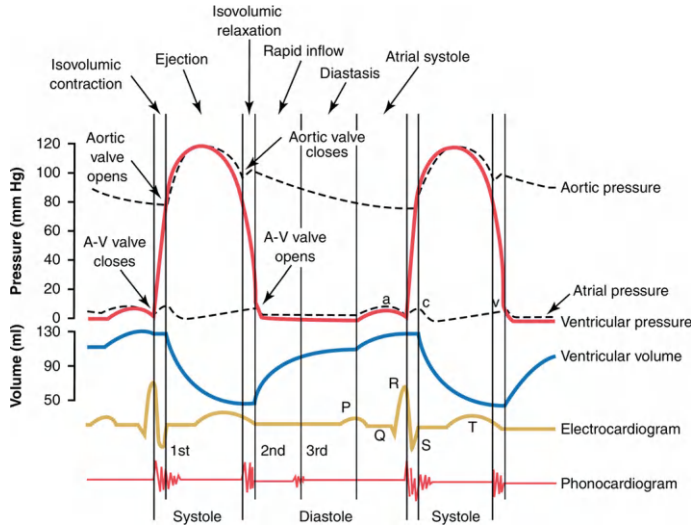


Fig. 1.5 Illustration of the cardiac cycle for the left heart. The analysis of the cardiac cycle may conveniently be started from the beginning of the diastolic phase and its associated continuous increase in ventricular volume, initiated by the opening of the atrioventricular valves and identified by the second heart sound in the phonocardiogram. The cardiac cycle is completed by the systolic phase, characterised by a rapid increase in ventricular pressure and a rapid decrease in ventricular volume (from Hall [3], with permission)

Example. For $HR = 72$ beats/min, $DCC = 1$ min/72. That is,

$$DCC = (1/72) \text{ min} = 0.01388 \text{ min} = 0.01388 \times 60 \text{ s} = 0.8333 \text{ s} .$$

The heart rate, and hence the duration of the cardiac cycle, varies from person to person; it also depends strongly on whether the body is at rest or under exercise conditions.

We next give a detailed description of the cardiac cycle for the left heart. Figure 1.5 depicts cardiac events associated with the left heart within the time period defined by the cardiac cycle. There are three pressure traces of special interest, namely the left ventricular pressure, the left atrial pressure and the aortic pressure. Also of interest is the left ventricular volume variation with time. The bottom traces show the electrocardiogram and the phonocardiogram. The cardiac cycle exhibits two distinct periods, namely **diastole** and **systole**, as seen at the bottom of the figure.

Diastole: Filling of the Ventricles

The analysis of the cardiac cycle may conveniently begin by identifying the start of the diastole in Fig. 1.5, when the atrioventricular (A-V) valves (tricuspid and mitral) open, allowing blood from the atria to flow into the relaxing ventricles. This happens

immediately after the second sound shown in the phonocardiogram. The relatively long diastolic period lasts for about 2/3 of the cardiac cycle duration and ends when the atrioventricular valves close, just before the sharp rise in ventricular pressure.

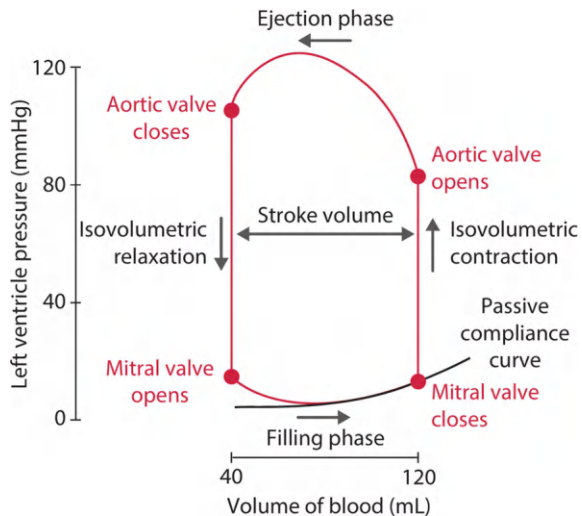
There are many cardiac parameters that change during the cardiac cycle, but the ventricular volume curve of Fig. 1.5 is most representative. The blood volume rises from its minimum value called **end-systolic volume**, or ESV , to its maximum value, called **end-diastolic volume**, or EDV . Normally ESV varies between 40 ml and 50 ml, while EDV varies between 110 ml and 120 ml. Another important quantity associated with the cardiac cycle is the **stroke volume** (SV), which is defined as the volume of blood pumped out of the left ventricle during a systolic contraction. In fact SV can be calculated in terms of left ventricular end-diastolic volume and left ventricular end-systolic volume, namely

$$SV = EDV - ESV. \quad (1.3)$$

Figure 1.6 illustrates the concept of stroke volume. The **increase in ventricular diastolic volume** takes place in three stages, all three of similar durations, as discussed below.

1. **Rapid filling of the ventricle.** The ventricle fills very rapidly after the sudden opening of the mitral valve. Blood flows from the left atrium into the left ventricle. See also the bottom side of Fig. 1.6, labelled filling phase.
2. **Slow filling of the ventricle.** A small amount of blood is added from the atrium, which continuously receives blood (arterial) from the pulmonary veins.
3. **Atrial contraction (atrial systole).** An additional 20 % of blood is added from the atrial contraction, which is clearly identified by the **a-wave** in the atrial pressure trace of Fig. 1.5.

Fig. 1.6 Pressure-volume loop of the cardiac cycle for the left ventricle of the human heart under rest, upright position. Compare with Fig. 1.5. Each of the four corners of the loop is characterised by the opening and closure of the mitral and aortic valves (circles). Note the stroke volume, the difference between end-diastolic blood volume and end-systolic blood volume (from Herring and Paterson [2], with permission)



During the diastole the ventricle pressure is at its lowest level and remains slightly below the corresponding atrial pressure, which is what permits the continuous passage of blood from the atria to the ventricles, in the ventricular filling phase; see traces in Fig. 1.5. Both atrial and ventricular pressures remain virtually constant during diastole, except for the small rise signaled by the **a-wave** resulting from atrial contraction (atrial systole). The aortic pressure decreases continuously throughout the diastolic phase but remains well above both the atrial and ventricular pressures.

Two useful concepts related to heart pressure are **preload** and **afterload**. **Preload** is entirely equivalent to left ventricle **end-diastolic pressure**, or EDV, and is associated to the stretch of the ventricle at the end of diastole, in readiness for the strong contraction and the corresponding large pressure increase in systole. **Afterload** is defined as the pressure that the left ventricle must overcome in order to open the aortic valve and push the blood into the systemic circulation. For obvious reasons, afterload is also known as **systemic vascular resistance**.

Systole: Emptying of the Ventricles

The remaining, second period of the cardiac cycle corresponds to the **systolic phase**, which takes about 1/3 of the cardiac cycle duration. Systole starts from the closing of the A-V valves, which gives rise to the first heart sound, as seen in the phonocardiogram in Fig. 1.5. Systole is characterised by a sudden ventricular pressure increase ended by a sudden fall, along with a large and continuous decrease in ventricular volume, as seen in Fig. 1.5. In analogy with the diastolic phase, the systolic phase is also divided into three distinct stages, namely

1. **Isovolumetric contraction.** As the left ventricular contraction begins, the ventricular pressure rises abruptly, causing the mitral valve to close, while the aortic valve remains closed. As seen Fig. 1.5, despite the sudden increase in ventricular pressure, ventricular volume remains constant and hence the name isovolumetric contraction. See also the right hand side of Fig. 1.6. Note that ventricular pressure is now above atrial pressure.
2. **Ejection of blood.** This phase of systole begins with the opening of the aortic valve caused by the ventricular pressure increase, normally above 80 mmHg. The period of ejection is characterised by a sudden decrease of ventricular volume caused by ventricular emptying into the aorta. Most of the ventricular ejection, about 70 %, takes place in the first third of the systolic phase. For a brief period the ventricular pressure rises above the aortic pressure. See also the top side of Fig. 1.6.
3. **Isovolumetric relaxation.** The end of systole is marked by ventricular relaxation, whereby ventricular pressure decreases very rapidly, thus reversing the pressure gradient across the aortic valve and causing their sudden closure. Then, for a short time after the valve closure, the ventricle continues to relax very rapidly, without changing the ventricular volume. Hence the name isovolumetric relaxation. See also the left hand side of Fig. 1.6. A characteristic feature of this phase is the kink

in the aortic pressure trace, called **dichrotic wave** or **dichrotic notch**. This is classically attributed to the closure of the aortic valve, which in turn is produced by the reversing of the pressure gradient across the valve.

Electrocardiography Waves (ECG) and Sound Waves

Cardiac electrical events during the cardiac cycle are associated with ECG waves, while heart valve closure is associated with heart sound waves. See electrocardiogram and phonocardiogram traces at the bottom of Fig. 1.5. The **P-wave** is generated by electrical activation of the atria, marking the start of atrial contraction. This wave is followed by the **QRS-wave** system, which is produced by the electrical activation of the ventricles, setting the immediate start of ventricular contraction. The **T-wave** is produced by the electrical recharging of the ventricles, marking the onset of diastole. The opening and closure of valves cause sound waves. The **first heart sound wave** is associated with the onset of ventricular contraction, at the time of closure of the atrio-ventricular valves. The **second heart sound wave** is associated with the onset of diastole, at the time of closure of the semi-lunar valves, and follows the **T-wave**.

The Pressure-Volume Curve

Another representation of the cardiac cycle is through a plot of ventricular blood pressure against ventricular blood volume, as shown in the approximately rectangular region of Fig. 1.6 for the left ventricle. The event begins at the bottom left corner, signalling the opening of the mitral valve and the start of the ventricular filling phase, which is completed by the mitral valve closure. The bottom right-hand side corner identifies the closure of the mitral valve. There follows the period of isovolumetric contraction, in which the ventricular pressure rises very steeply, while ventricular volume remains constant; compare with Fig. 1.5. This stage is completed by the opening of the aortic valve, identified by the top right-hand side corner, after which the ejection phase begins. During the ejection phase the ventricular pressure varies continuously. It first increases, reaches a maximum and then decreases until the top left-hand side corner is reached, when the aortic valve closes. At this point the isovolumetric relaxation period starts, which is completed at the left-hand side bottom corner, when the mitral valve opens. The whole process is then repeated. Note the representation of **stroke volume** in Fig. 1.6; see also Eq. (1.6).

So far we have studied the cardiac cycle in terms of the left heart, that is the left atrium, left ventricle, mitral valve and aortic valve. We now consider the cardiac cycle for the right heart.

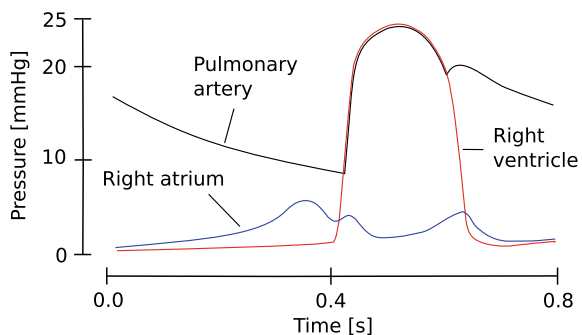
1.3.3 The Right Heart and Vein Waves

The cardiac cycle for the right heart (right atrium, right ventricle, tricuspid valve and pulmonary valve) is illustrated in Fig. 1.7, which displays traces for right atrial pressure, right ventricular pressure and pulmonary artery pressure. There is a strong analogy, in the shape of traces, between the left heart and right heart, as seen by comparing Figs. 1.5 and 1.7. But there is a major difference in the range of pressures. Pressures in the left heart are much higher than those in the right heart, in fact by a factor of 5 or 6. The right heart is connected to the low-pressure, low resistance venous blood of the systemic circulation and the pulmonary circulation. The left heart is connected to the high-pressure, high resistance arterial blood vessel network of the systemic circulation. A note of caution on Fig. 1.7. During the first part of systole, **the right ventricular pressure actually rises above that of the pulmonary artery**, while in plots shown in the literature, this is not always obvious. In our Fig. 1.7 these two pressures are distinct, though virtually indistinguishable. See left ventricular pressure in Fig. 1.5.

As already pointed out, the right heart is linked to the low-pressure venous system, in particular to the great venae cavae and beyond. Jugular veins are normally in open communication with the right atrium, see Fig. 1.13. Therefore, flow and pressure in the internal and external jugular veins are directly affected by the dynamics of the right heart cardiac cycle, generating the well-recognised **jugular venous pulse**. This is of clinical significance in assessing the ensuing **central venous pressure** cycle. The atrial pressure cycle has a direct upstream effect on flow and pressure in the venae cavae and hence in the jugular veins, observed in the form of waves. It is worth noting however, that most individuals possess valves in their internal jugular veins [4], which may affect the pressure and flow communication between jugular veins and the right atrium.

The next section deals with the third component of the cardiovascular system, namely the blood vessels that circulate the blood in the body.

Fig. 1.7 Cardiac cycle pressure traces for right heart. Compare with Fig. 1.5 for the left heart. Note that pressures for left heart are much higher than for the left heart, by a factor of 5 or 6



1.4 Blood Vessels

We start with a brief overview of the full circulatory system, the vessel networks and the direction of the circulating blood.

1.4.1 *Routes of the Circulating Blood*

The blood vessels of the cardiovascular system constitute a system of interconnected networks of compliant tubes that, in a **to and fro** fashion, (a) transport blood between heart and tissues and (b) between heart and lungs. The first is a process performed by the **systemic circulation**, while the second is performed by the **pulmonary circulation**, as defined earlier. Figure 1.8 depicts the complete human circulatory system in which the blood flow direction is indicated by arrows. Long black arrows refer to the systemic circulation originating at the left ventricle and ending at the right atrium, while shorter black arrows refer to the pulmonary circulation originating at the right ventricle and ending at the left atrium. In most of the circulation, blood outflows from the heart to the arterial system, large arteries, smaller arteries, arterioles, capillaries. Then blood performs a **U-turn** [1] at the capillary bed and returns to the heart following post-capillary veins, venules, veins, venae cavae and the heart. But there are special cases in which blood passes from one capillary bed to another capillary bed through a vein. This is called **portal circulation** and the hepatic portal circulation is a prominent example, as shown by red arrows in Fig. 1.8.

In the **systemic circulation**, arterial blood (red) is pumped by the heart left ventricle at high pressure into the aorta, which branches into an upper network that perfuses the neck and head, and a lower network that perfuses the rest of the body, arriving to the lower limbs. The districts supplied with arterial blood are shown in red and lie on the right hand side of Fig. 1.8. Systemic arterial vessels form a divergent network of gradually smaller vessels until they reach the capillaries; these are the vessels of the smallest size, where the exchange of materials take place. From then on arterial blood becomes venous blood (blue) and the venous converging vessels network is formed; this carries oxygen-depleted blood and wastes back to the inferior and superior venae cavae, and back to the right atrium of the heart. There are **local systemic circulations** that supply blood to, and collect venous blood from, major organs such as the heart itself (coronary circulation), the brain, the liver, the spleen, the kidneys and the stomach. Note the hepatic portal circulation in Fig. 1.8.

The **pulmonary circulation** starts from the heart right ventricle that pumps venous blood just returned to the right atrium. This venous blood is pumped to the lungs via the pulmonary trunk and the pulmonary arteries. See Figs. 1.3 and 1.8. Note that the **pulmonary arteries** carry venous (blue) blood. This arterial network branches out into increasingly smaller pulmonary blood vessels, reaching the pulmonary capillaries. Here the venous blood unloads carbon dioxide and uploads oxygen. The newly oxygenated (red) blood is returned to the heart left atrium via the **pulmonary veins**.

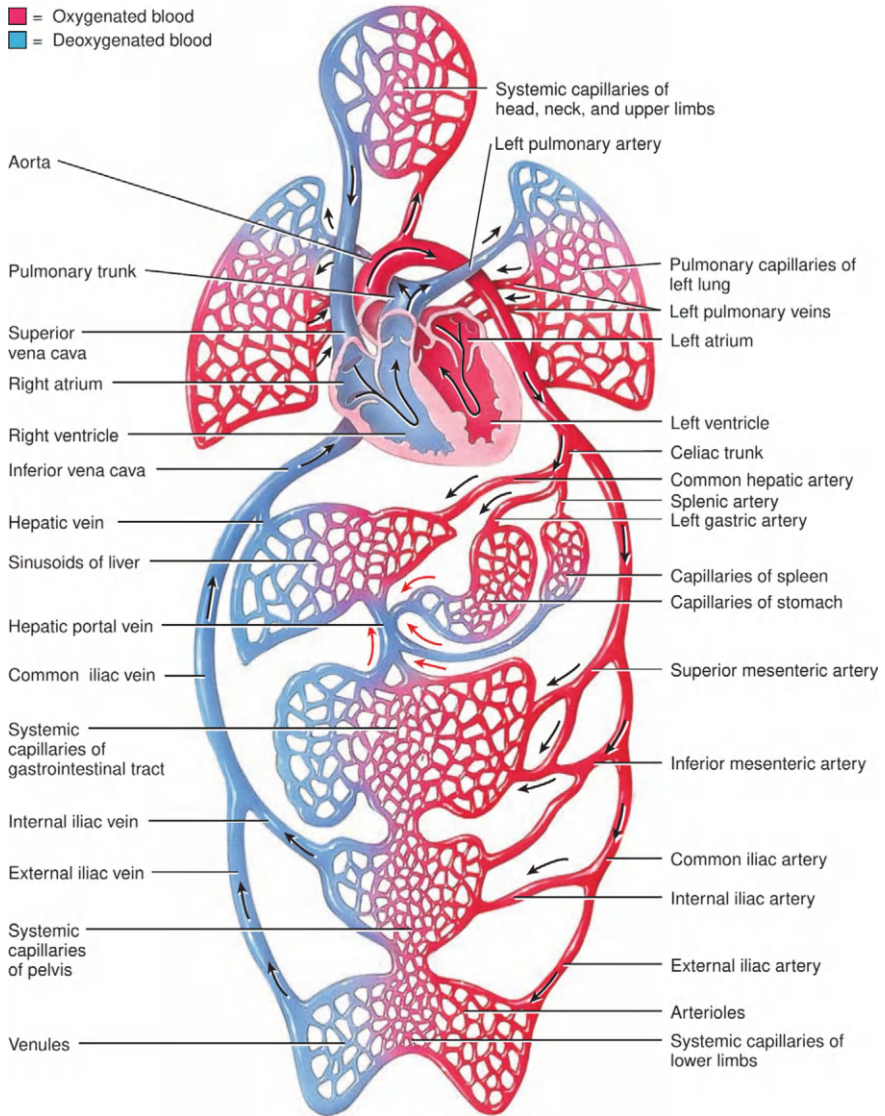


Fig. 1.8 Sketch of blood circulation routes and direction. Long black arrows indicate the systemic circulation, short black arrows the pulmonary circulation and red arrows refer to the hepatic portal circulation (from Tortora and Derrickson [1], with permission)

Table 1.3 Pressure units conversion table

Convert from	Pa (N/m^2)	Bar	Atmosphere	mmHg	mmH ₂ O
1 Pa (N/m^2)	1	10^{-5}	9.87×10^{-6}	0.0075	0.1

Table 1.4 Blood vessel measures for the human systemic circulation network. Lengths are given in cm; area in cm^2 ; velocity in cm/s and pressure in mmHg (data from A. Sequeira in Farina et al. [7]). For comparison see Burton [5] and Formaggia et al. [6]

Vessel	Number	Diameter	Wall Thick.	Cr. Sec. Area	Mean vel.	Mean pres.
Aorta	1	3	2×10^{-1}	7	12	93.75
Arteries	8×10^3	10^{-1}	10^{-1}	8×10^{-3}	45	90.0
Arterioles	10^7	5×10^{-3}	2×10^{-3}	2×10^{-5}	5	52.5
Capillaries	10^{10}	8×10^{-4}	10^{-4}	5×10^{-7}	0.1	22.5
Venules	4×10^7	10^{-2}	2×10^{-4}	7.9×10^{-5}	2	11.25
Veins	8×10^3	1.8×10^{-1}	5×10^{-2}	10^{-1}	10	7.5
Vena cava	2	3	15×10^{-2}	6	14	3.75

Note that the pulmonary veins carry arterial (red), oxygen-rich blood into the right atrium in readiness for a next cycle through a contraction of the left ventricle of the heart. See Figs. 1.3 and 1.8.

As already pointed out, we can distinguish three types of blood vessels: arterial vessels, venous vessels and capillaries. Figure 1.10 shows the network of main arteries, Fig. 1.13 shows the network of main veins and Fig. 1.15 illustrates the capillaries. Table 1.4 shows vessel measures, including number of vessels, diameter, wall thickness, cross-sectional area, mean velocity and mean pressure. The reader is warned that most quantities shown in Table 1.4 vary significantly in the literature; see Burton [5] and Formaggia et al. [6] (Tables 1.1 and 1.2) for comparison. For the readers convenience, Table 1.3 recalls conversion of pressure units. Figure 1.9 shows blood pressure distribution in the systemic and the pulmonary circulations, covering all blood vessel districts. For the reader’s convenience we provide pressure units conversion in Table 1.3.

1.4.2 Arteries

Blood vessels of the arterial network of the **systemic circulation** carry oxygen and nutrients to the body tissues in the blood pumped by the left ventricle of the heart; the arterial network starts from the aorta, from where it branches continuously into smaller arteries, then arterioles and finally capillaries. See Fig. 1.10 for the network of main human arteries and Fig. 1.9 for the pressure distribution as blood travels from the aorta all the way to the venae cavae. In the **pulmonary circulation**, the arterial

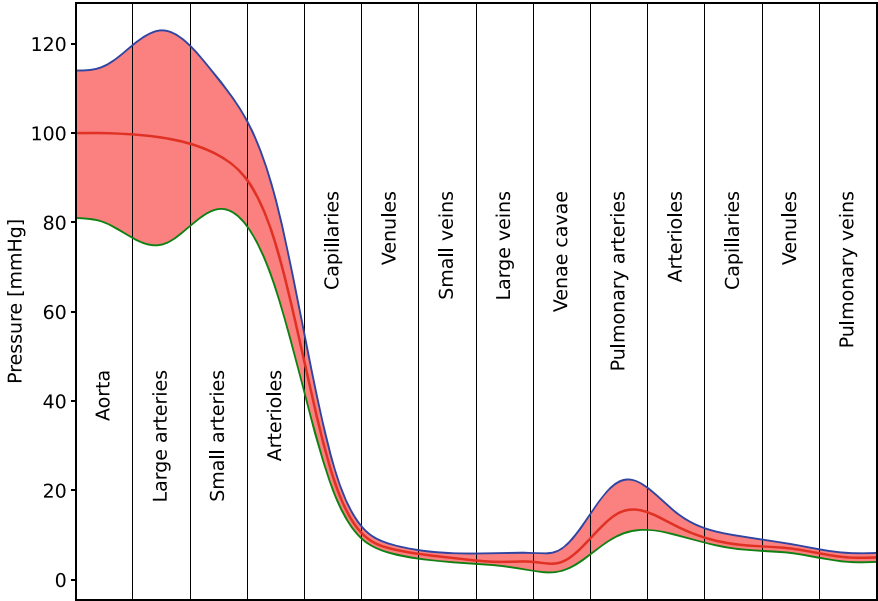


Fig. 1.9 Illustration of blood pressure distribution in the different kinds of blood vessels of the circulatory system, for a subject in the supine position. The red band accounts for the variability of pressure values, emphasising the oscillatory character of these. The thick red line represents the mean value of pressure (data from Hall [3])

network starts in the pulmonary trunk and transports venous, oxygen-depleted blood to the lungs. See Fig. 1.9 for the pressure distribution as blood travels from the pulmonary trunk all the way to the pulmonary veins.

Table 1.4 shows a few measures for the arterial and venous networks. The first four rows, from the aorta to capillaries, show data for the arterial network, while rows 5, 6 and 7 give data for the venous vessels. Column 1 identifies the types of blood vessels and column 2 shows the numbers of vessels; they go from the single aorta to 8000 arteries. The numbers then grow as we move to the 10 million arterioles and then to 10 billion capillaries; Tortora and Derrickson [1] speak of 20 billion capillaries, rather than 10. Column 3 shows the corresponding diameters of the lumen of the vessels; they vary from centimetres to microns. The huge variations in diameter are also seen for wall thickness, shown in column 4. Column 5 shows cross-sectional areas; column 6 shows the mean velocities, while the last column shows the mean pressures observed in the various districts of the arterial network. The mean pressure decreases monotonically along the network. For the arterial network pressure varies from 94 mmHg in the aorta to 22.5 mmHg in capillaries. See Fig. 1.9 for comparison.

The left side of Fig. 1.9 illustrates the variation of pressure along the arterial network in the systemic circulation, up to the level of capillaries. Of course, the pressure continues to decrease monotonically along the venous network, to be discussed shortly. Due to cardiac pulsations one observes an oscillatory behaviour of pressure

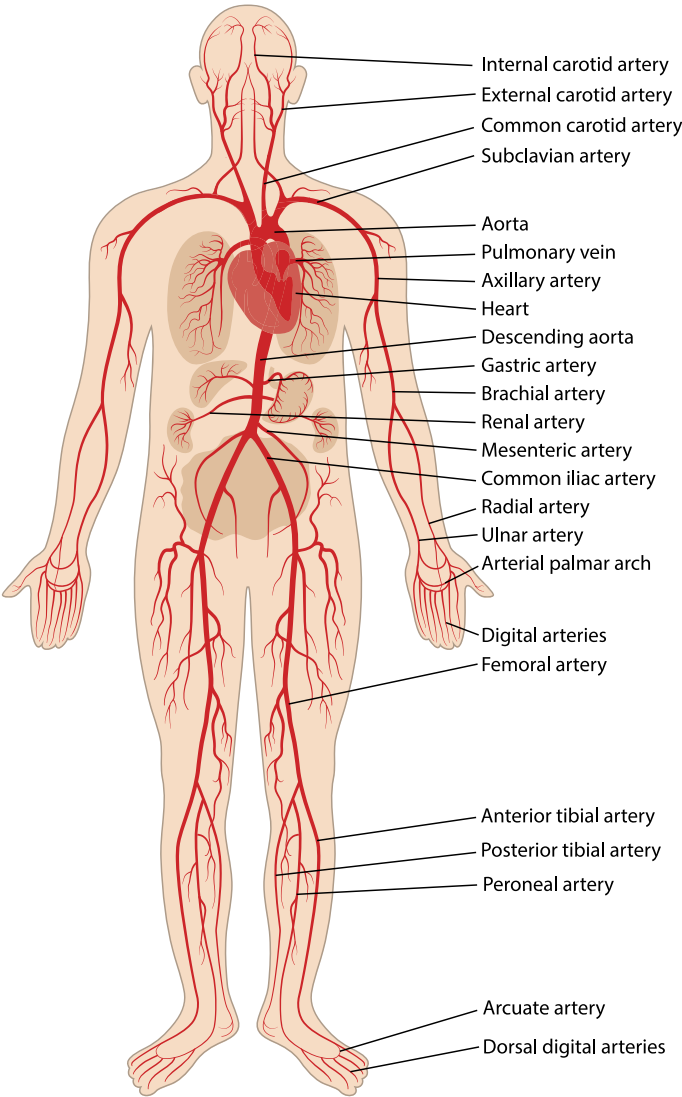


Fig. 1.10 Major arteries of the human cardiovascular system. See Table 1.4 for various measures for diameter, wall thickness, cross-sectional area, mean velocity and mean pressure. See also Fig. 1.9 for the pressure distribution (from Shutterstock, with permission)

on the left side of Fig. 1.9. The oscillations are contained within the red broad band and vary between about 80 mmHg in diastole and about 120 mmHg in systole. These values vary amongst individuals and conditions under which pressure is measured. The right side of Fig. 1.9 shows the pressure distribution along the network of the pulmonary circulation. Pressures are much smaller than in the systemic circulation and also exhibit an oscillatory behaviour in the larger arterial vessels. Again, pressure decreases towards the venous part of the pulmonary circulation.

Arteries are compliant tubes that withstand the highest pressures in the vascular system under the pumping action of the heart. Their diameters vary in the form of **expansions** and **contractions**, as volumes of blood are pumped by the ventricles towards the systemic and pulmonary circulation networks. The leading pressure wave stretches the diameter of the artery. Generally, one speaks of **vasodilation** and **vasoconstriction** to refer to the increase and decrease in vessel diameter, respectively.

Figure 1.11, top frame (a), shows an instant at which the left ventricle contracts and pushes blood into the aorta, the wall of which is stretched as the force of the high pressure blood passes through. The bottom frame (b) of Fig. 1.11 shows an instant at which the **arterial wall recoils**, as the left ventricle relaxes in diastole, releasing its stored energy, which keeps pushing the blood towards the capillaries, ensuring smooth supply of blood during diastole. As a consequence of this pumping function at high pressure, the structure of the wall of the arteries exhibits special characteristics. They are muscular elastic tubes, as discussed next.

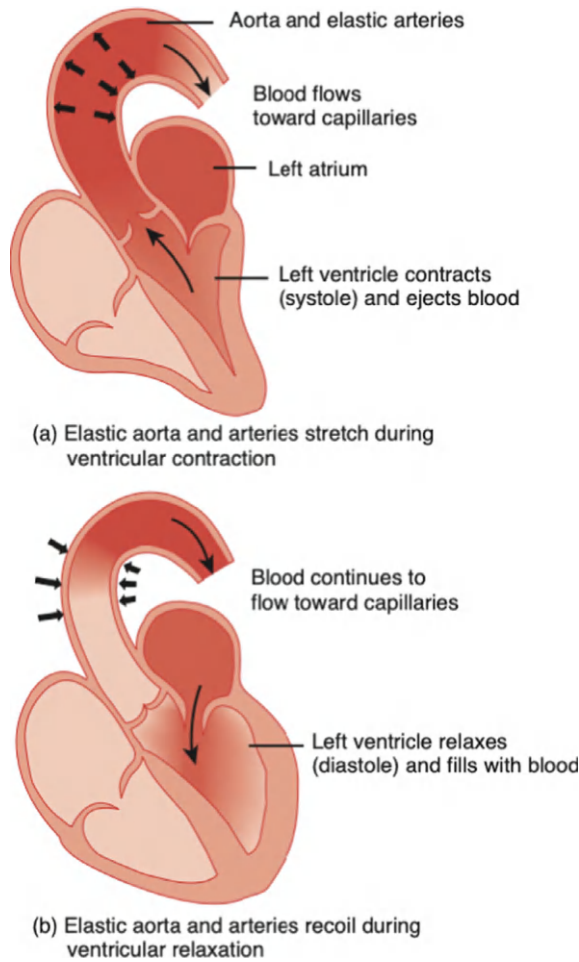
The Wall of Arterial Vessels

Figure 1.12 shows the structure of the wall of a small artery. This structure is common to all types of arteries, not capillaries, and includes three layers. Starting from the interior of the vessel, these are: the **tunica intima** (inner most), the **tunica media** (middle) and the **tunica adventitia** (outer most) [2, 8]. The tunica intima, the most inner layer, contains a sheet of endothelial cells resting on a layer of connective tissue. The tunica media is mainly made up of smooth muscle cells, embedded in a matrix of elastin and collagen fibres. The tunica media is bounded by two sheets of elastin, the internal elastic lamina and the external elastic lamina. The tunica adventitia is a sheath of connective tissue, separated from the intima by the external elastic membrane. The outer border of the adventitia is not clearly defined.

The inner layer of endothelial cells is the main barrier to extravasation of blood plasma and solutes to the interstitium. As a matter of fact, there is another interior layer that coats the endothelium on the lumen side; this is called the **glycocalyx**, which is ignored in most presentations of the cardiovascular system. The **glycocalyx** plays an important role as a sieving barrier for macromolecules. As stated in [9, 10], accounting for the glycocalyx is essential in mathematical models for the transport of materials across vessel walls, between the vessel lumen and the interstitium.

The arterial network is completed with the network of capillaries, whose end function is to deliver oxygen and nutrients to adjacent tissues and to reabsorb carbon dioxide and wastes from the interstitium, on the venular side of capillaries.

Fig. 1.11 Stretching (top) and recoiling (bottom) of aorta wall, as blood from the contracting left ventricle is pumped towards capillaries. The stretching stores energy in the elastic arterial wall that is then released during the recoiling phase, as the left ventricle relaxes in diastole (from Tortora and Derrickson [1], with permission)



1.4.3 Veins

Figure 1.13 shows the network of major venous vessels of the human circulatory system. The venous network constitutes the other major network of blood vessels of the cardiovascular system, which is continuous with the arterial system, shown in Fig. 1.10, through the capillaries, shown in Fig. 1.15. The venous system begins from the venular end of capillaries in the capillary bed, followed by post-capillary venules, and then by a convergent venous network of increasingly larger diameters, until the superior and inferior venae cavae are reached, which are the largest veins in the venous network.

Table 1.4 shows a few characteristics of the venous vessels, starting on the row below the capillaries. Column 2 shows the number of different types of venous

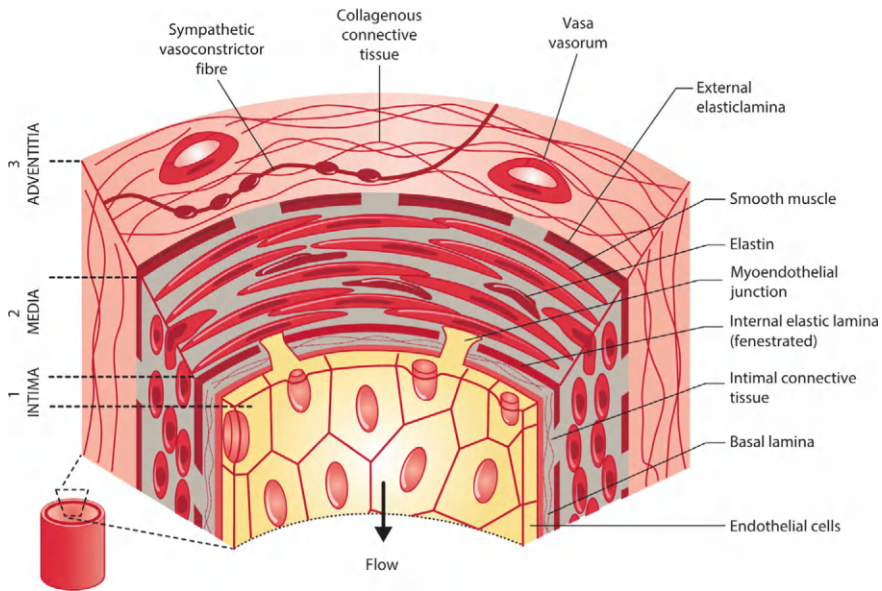


Fig. 1.12 The artery wall includes three layers. From the interior of the vessel these are: the **tunica intima**, the **tunica media** and the **tunica adventitia** (from Herring and Paterson [2], with permission)

vessels. The number of venules is comparable to the number of arterioles; veins are comparable in number to small arteries. There are two large veins, the inferior vena cava and the superior vena cava; see Fig. 1.13. Concerning diameter, venules are comparable to arterioles, whereas veins can have a much larger diameter than small arteries. The venae cavae are also larger in diameter than the aorta. Regarding wall thickness of veins; this increases monotonically as one moves from venules to the venae cavae. Venule wall thickness is much smaller than that of arterioles. The remaining venous segments are again thinner than the corresponding arterial vessels. In general, veins are much more **compliant than arteries**, they deform more easily than arteries. Veins even collapse, under physiological conditions. Their larger diameter and cross-sectional area is consistent with the larger volume of blood stored in veins.

As anticipated by Fig. 1.18, veins hold about 2/3, or more, of the total blood volume, for which the venous network constitutes a blood reservoir for the body [11, 12]. Table 1.4 also shows mean pressure along the full cardiovascular network. The pressure in the venous vessels is much smaller than that in arteries and decreases monotonically from venules to the venae cavae. This fact is confirmed in Fig. 1.9, which shows the mean pressure distribution along the full cardiovascular network, for the systemic circulation on the left and for the pulmonary circulation on the right. It is seen that pressure in veins is, in general, much smaller than in arteries, in both circulations.

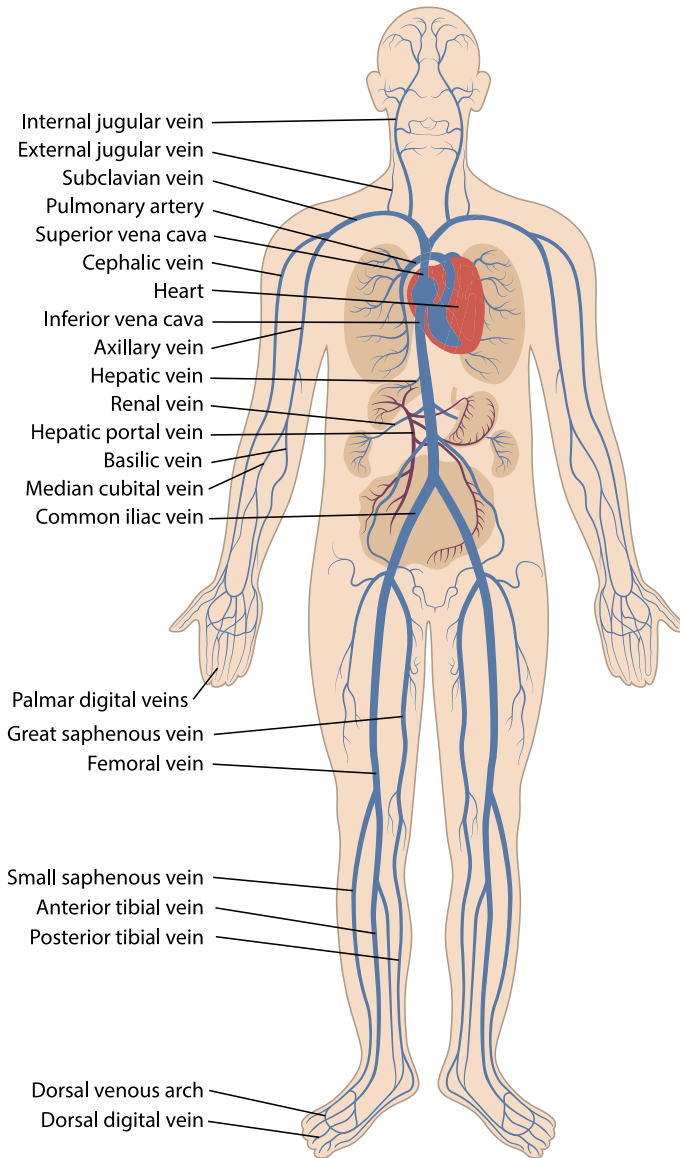


Fig. 1.13 Major veins of the human cardiovascular system; compare with the arterial system depicted in Fig. 1.10; see also Table 1.4 for measures for venous vessels. The venous system is continuous with the arterial system at the capillary bed; see Fig. 1.15 (from Shutterstock, with permission)

In the systemic circulation, the main function of the venous system is to return the oxygen-depleted blood from capillaries to the heart right atrium. From the material exchange process at the level of capillaries, the venous system reabsorbs solutes across the wall of the venular end of capillaries, including carbon dioxide and other metabolic wastes. Once in the venous blood, these materials are transported back to the heart, after which they enter the **pulmonary circulation** for unloading carbon dioxide and picking up oxygen. Note the reverse role of arteries and veins here, as already pointed out previously. It is the **pulmonary artery** that carries the **venous blood** ejected by the right ventricle and it is the **pulmonary veins** that transport **oxygenated** arterial blood back to the left atrium. See Fig. 1.3.

The Wall of Venous Vessels

As already stated, the wall of veins is thinner and much more compliant than that of arteries; this is a consequence of the wall structure. The wall of veins has the same three layers of the wall of arteries: tunica intima (interna), tunica media and tunica externa. See Fig. 1.14. The venous wall lacks, compared to that of arteries, the internal and external elastic lamina. See also Fig. 1.12. A notable difference between arteries and veins is that veins have valves, as shown in Fig. 1.14. Venous valves are prominent in the legs. Their function is to ensure unidirectional blood flow, from tissue to heart, by preventing the return of blood flow. Contrary to what some

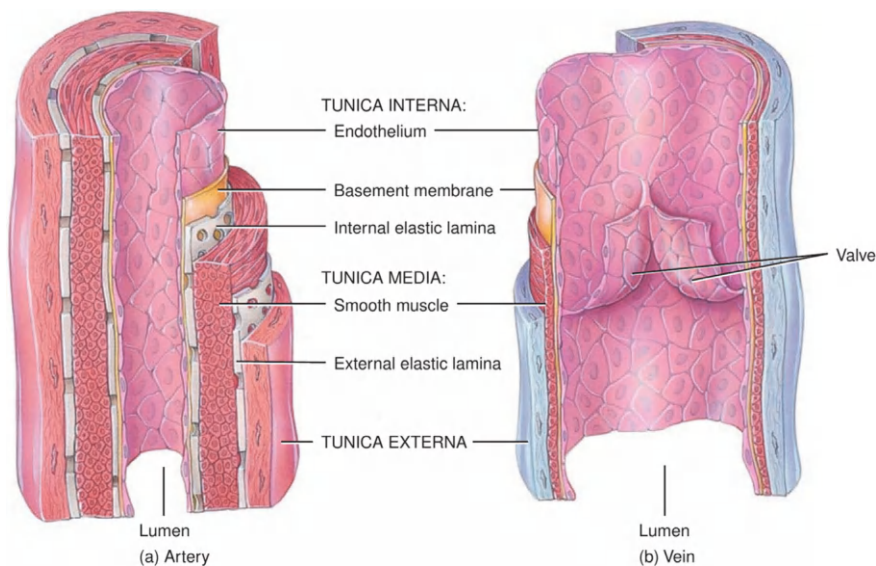


Fig. 1.14 Vein wall structure (right) compared to artery wall structure (left). Artery and vein walls have the same layers, except for the internal and external elastic laminae present in arteries and absent in veins. Additionally, some veins exhibit valves to prevent retrograde flow (from Tortora and Derrickson [1], with permission)

textbooks state, venous valves are also present in the internal and external jugular veins, and in the azygos vein. Their presence has long been established from cadaver dissection studies. However, sound knowledge on their prevalence, morphology, competence and their role in cerebral venous return is largely incomplete; see review [4] and references therein. Further details on veins are found in Chaps. 18 and 19.

Remarks on the Venous System

Veins in general have received much less attention than their counterparts, arteries, both at the physiological and clinical levels. However, during the last decade or so, it has become apparent that the venous system requires more attention, particularly in the context of cranio-spinal haemodynamics and neurological diseases. See [11–17] and references therein. As already stated, the arterial system delivers oxygen and nutrients, essential to life. Equally important is the **clearance of metabolic waste**, efficiently. This has become more evident in the studies of neurological diseases, in which the **clearance of neurotoxic materials** is increasingly becoming the focus of promising research. For the venous system we recommend the papers by Pang [11, 12] and Schaller [18]. For venous flow in the head and neck we recommend the work of Doepp et al. [19] and that of Wilson [20].

Moreover, and this is relevant to the themes of this book, in the mathematical modelling of the cardiovascular system, veins have been largely neglected. Often, in fact, the cardiovascular system tends to be understood as the arterial system alone. In the last few years it has become apparent that the mathematical modelling of blood flow in veins is more challenging than that in arteries. As already stated, veins are highly deformable (highly non-linear) and collapse under physiological conditions. This mechanical behaviour poses challenges to the mathematical modelling and to the design of suitable computational algorithms [4, 13, 17, 21]. See also Chaps. 18 and 19.

Venous Volume and Role in Arterial Hypertension

As already pointed out, figures given in the literature for the blood volume held by the venous system are highly variable. The figure of 2/3 of the total blood volume is the most common. However there are authors that propose figures between 70% and 85%; see London et al. [22, 23] and Safar et al. [24]. Importantly, in Chap. 19 we unveil the important role of the venous system in arterial hypertension.

1.4.4 Capillaries and the Microcirculation

Tortora and Derrickson [1] define the **microcirculation** as the flow of blood from a metarteriole of the body, through capillaries and into postcapillary venules. See Fig. 1.15. The main purpose of the cardiovascular system is realised in the microcirculation, where transport across the vessel wall takes place, that is **capillary**

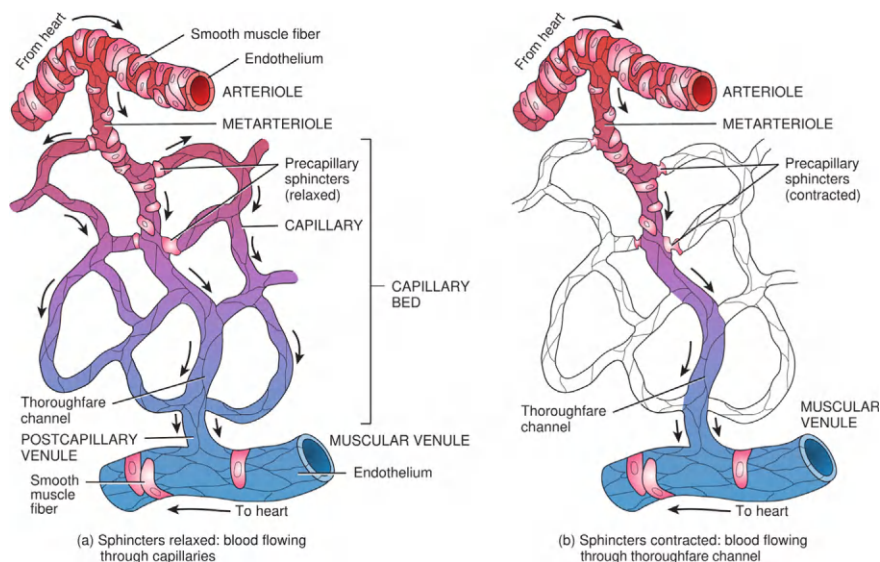


Fig. 1.15 The microvasculature. Capillaries are placed between arterioles, on the arterial side, and venules, on the venous side. Capillaries are the centre of the blood microcirculation, they join the arterial and venous networks of Figs. 1.13 and 1.10. Precapillary sphincters determine the flow in the capillary bed. Compare the left frame (precapillary sphincters relaxed) to the right frame (precapillary sphincters contracted) (from Tortora and Derrickson [1], with permission)

exchange of many materials between the blood stream and the interstitium. In particular, oxygen and nutrients are delivered to tissues and carbon dioxide and wastes are removed from the interstitium and transported back to the heart, partly by the venous system and partly by the lymphatic system; see Chap. 3.

The Capillary Bed

Figure 1.15 shows the capillary bed that connects the arterial to the venous system. A capillary bed consists of a network of capillaries containing between 10 and 100 capillaries. Starting from the arterial side (top) and finishing on the venous side (bottom), the microvessels include: arterioles, metarterioles, capillaries, postcapillary venules and venules. Arterioles are known as **resistance vessels**, due to their **resistance to flow** caused by their small radii and by contraction of the smooth muscle of their wall, that is **vasoconstriction**. **Vasodilation** is the opposite, that is relaxation of the wall smooth muscle and enlargement of the cross-sectional area. Vasodilation and vasoconstriction also have an effect on blood pressure; pressure will rise with vasoconstriction and will decrease with vasodilation. **Precapillary sphincters** control through which capillaries blood will flow. Vasodilation and vasoconstriction are a response to regulatory mechanisms operated by the nervous system and the endocrine system. Regulation mechanisms are discussed in Chap. 2.

Capillaries are the most numerous blood vessels in the cardiovascular system, of the order of 10 billion. Capillaries are the smallest blood vessels in the cardiovascular system, they have the smallest vessel diameter as well as the thinnest vessel wall. See Table 1.4. Nonetheless, capillaries have collectively, the largest cross-sectional area of the entire circulatory system. All of these features facilitate the fulfilment of the main mission of capillaries, that is **capillary exchange**. Figure 1.9 shows mean pressures in both systemic and pulmonary circulations. The pressure in capillaries is the lowest.

Blood flow in capillaries is characterised as **bolus flow**, in the sense that red blood cells travel in single-file fashion by squeezing and deforming their shape to accommodate to the capillary radio; see Fig. 1.17. Recall that the size of RBCs is comparable to the diameter of capillaries. Moreover, flow in capillaries is intermittent and controlled by precapillary sphincters; see Fig. 1.15. Therefore, flow in some capillaries may cease altogether for seconds and even for minutes. Given the large number of capillaries one can speak of **averaged capillary blood velocity**. As shown in Table 1.4, blood velocity (averaged) in capillaries is the lowest in the full circulatory system, of the order of 1 mm/s. Such low blood velocities facilitates material exchange.

The Wall of Capillaries

Capillaries are exchange vessels and thus the characteristics of their wall structure will reflect the material exchange demands of the tissue they serve. Figure 1.16 shows three types of capillaries, namely continuous, fenestrated and sinusoid. Their chief difference lies in their permeability to substances in the lumen and the interstitium. In general, the wall of capillaries consists of a single layer of endothelial cells, surrounded externally by a thin **basement membrane**, as depicted in Fig. 1.16 for each of the three types of capillaries. Compare with the wall structure of arteries shown in Fig. 1.12 and veins shown in Fig. 1.14. Table 1.4 gives some gross measures on capillaries, such as diameter, wall thickness, cross-section area, mean velocity and mean pressure. Regarding vessel dimensions see also [5] and [6].

Endothelial cells are separated by **junctions** or **intercellular clefts**, which constitute small passageways between the interior of the capillary and the interstitium, with a spacing between 6 and 7 nm. The clefts, even if narrow, constitute an important factor in the transport by diffusion of water molecules and small solutes, across the wall. The characteristics of the clefts between endothelial cells in the capillary wall may vary significantly from organ to organ. In the brain, for example, such **intercellular clefts** are called **tight junctions**. These, as their name suggests, are extremely selective in controlling the passage of materials across the wall. These tight junctions are a key component of the **blood-brain barrier**, or BBB, which keeps brain tissue separated from blood, by selectively impeding influx of most blood-borne substances. Very different from the brain capillaries are those in the liver, in which clefts between endothelial cells in the wall are widely open, permitting the passage of almost all dissolved plasma substances. Endothelial cells themselves may possess features that facilitate, or otherwise, the exchange of materials between

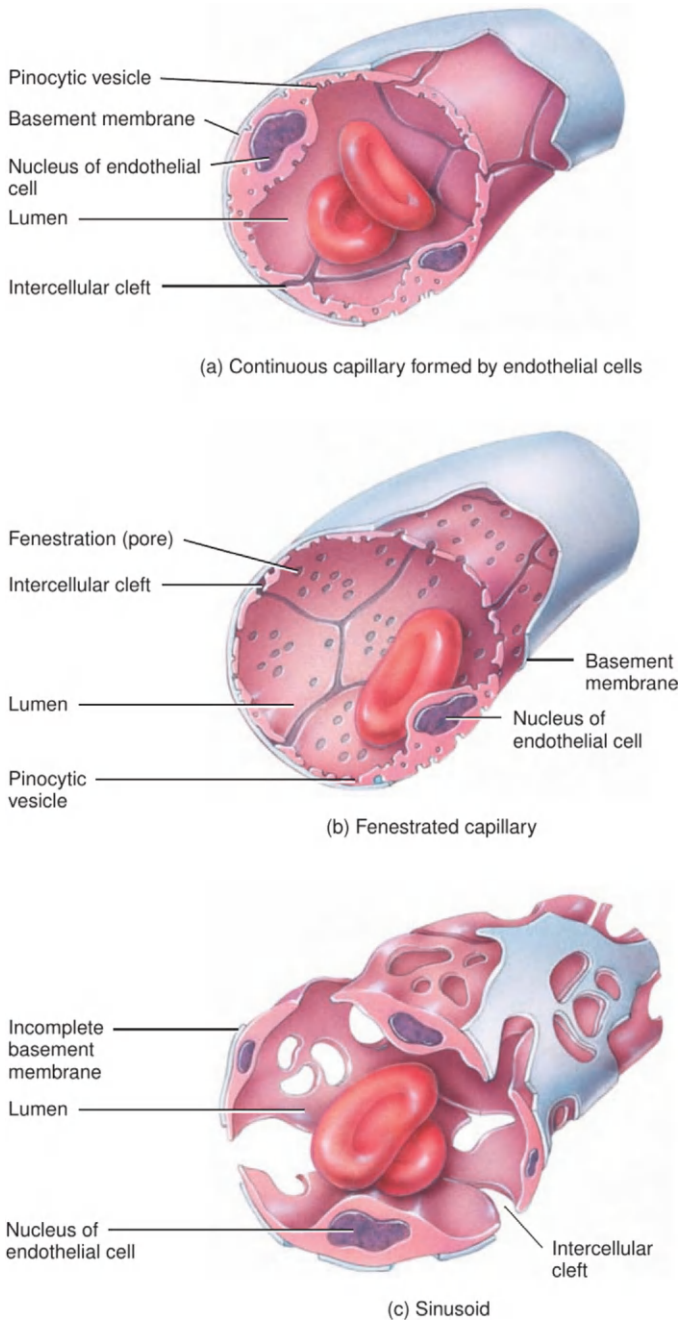


Fig. 1.16 Capillary wall for three types of capillaries: continuous, fenestrated and sinusoid. The wall is composed of a single layer of endothelial cells and the basement membrane. The types of capillaries are distinguished by the degree of fenestration of the endothelial cells, the size of the intercellular clefts and the structure of the basement membrane (from Tortora and Derrickson [1], with permission)

blood and interstitial fluid in the interstitium. In continuous capillaries, endothelial cells have no pores or gaps, other than the clefts. The endothelial cells of fenestrated capillaries have many fenestrations, or pores, of size between 70 and 100 nm. Sinusoids have endothelial cells with large fenestrations and large intercellular clefts. Moreover, sinusoids have also incomplete or absent basement membrane. Continuous capillaries are found in the brain and lungs. Fenestrated capillaries are found in the kidneys, in the villi of the small intestine and in the ciliary processes of the eye. Sinusoids are found in the liver and the spleen.

Capillaries, the Microcirculation and Other Fluid Systems

The capillary bed, in many ways, represents the fulfilment of the mission of the complete cardiovascular system. Two main functions stand out, namely

- Unloading of oxygen from red blood cells and nutrients; this is accomplished by diffusive transport across the capillary wall into the surrounding interstitial space and the cells.
- Loading of carbon dioxide and wastes from the interstitial space; this is accomplished by reabsorption across the wall of the venular end of capillaries, back into the vessel lumen.

Figure 1.17 illustrates these functions of capillaries, which are facilitated by the structure of the vessel wall just described. The exchange of materials across the vessel wall is governed by the **Starling forces**, and is determined by (a) the **hydrostatic pressure** difference between the fluids either side of the wall and (b) the **osmotic pressure** difference between the solutes either side of the wall. A mathematical description of the **Starling principle** is given in Chap. 2.

The microvessels just discussed are crucial to the microcirculation of blood, obviously, but their function must be complemented by other bodily fluid systems, namely

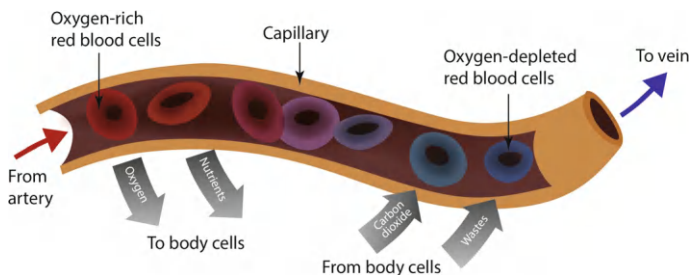


Fig. 1.17 Capillary blood flow and capillary exchange of materials between the vessel lumen and the interstitial fluid between body cells. Red blood cells are transported from the arterial side towards the venous side; RBCs unload oxygen and nutrients that will diffuse across the capillary wall towards the interstitial fluid surrounding body cells; RBCs upload carbon dioxide and wastes that have diffused from the interstitium across the vessel wall to be transported back to heart by the venous side of the blood stream (Courtesy of Violeta Da Rold)

the **interstitial fluid** system in the interstitium and the **lymphatic system**. The capillary exchange involves reabsorption of fluids and solutes from the interstitium at the venous end of capillaries. The excess ISF in the interstitium, not reabsorbed into the vessel lumen, is collected by the lymphatics system that carries these fluids to the venous system and hence to the circulatory system. The lymphatic system is described in Chap. 3, along with other major bodily fluid systems that are interconnected to the cardiovascular system.

1.4.5 Classes of Vessels and Distribution of Blood

All blood vessels in the cardiovascular system transport blood in the body. But beyond this common function, blood vessels are characterised by performing additional functions and, consequently, different types of vessels have different characteristics; these are summarised below, for the systemic circulation:

- **Arteries.** Arteries in the systemic circulation carry blood pumped by the left ventricle at high pressure and high velocity to the body tissues. As seen in Fig. 1.9 mean pressure in the aorta and large arteries, for example, are of the order of 100 mmHg. Consequently, arteries have strong muscular walls. See Table 1.4.
- **Arterioles.** Arterioles are the last segments of the arterial network before the capillaries. Arterioles are blood flow control vessels, they control the amount of blood released into the capillaries. See Fig. 1.15. The strong muscular wall of arterioles contracts and relaxes, thus tremendously altering the blood flow through capillaries and thus to tissues. In other words, arterioles control the **resistance to blood flow** in the body, for which they are also called **resistance vessels**.
- **Capillaries.** Capillaries are the most numerous of blood vessels, the function of which is to carry out the exchange of materials between the blood inside the vessel and the surrounding interstitial fluid, see Fig. 1.17. To facilitate this most important function the vessel wall of the capillaries is very thin and largely porous, although the porosity depends strongly on the particular organ of interest. See Figs. 1.16 and 1.17.
- **Venules.** Venules are the first vessel segments of the venous network, after the capillaries. They collect venous blood from capillaries and branch into a convergent network of venous vessels. Part of the reabsorption of materials from the interstitium into the lumen takes place across the venule wall.
- **Veins.** Veins transport venous blood from venules back to the right atrium. As seen in Table 1.4 and Fig. 1.9, mean pressure and mean velocity in veins is very low. Veins contain most of the blood in the cardiovascular system, as illustrated in Fig. 1.18. For this reason, the venous system is effectively a **blood reservoir**.

Figure 1.18 displays the distribution of blood in the cardiovascular system. Systemic veins and venules contain most of the blood, about $2/3$, compared to the 13% of systemic arteries and arterioles. Systemic capillaries account for 7%. Vessels in the

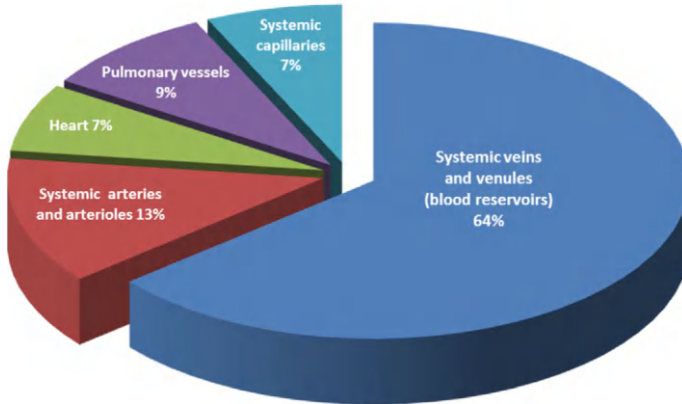


Fig. 1.18 Distribution of blood in the cardiovascular system. The venous system holds about $2/3$ of the total blood volume in the body (Courtesy of Dr B. Toro. Data from Tortora and Derrickson [1])

pulmonary circulation (arteries, veins and capillaries) contain 7 %, while the heart vessels (arteries, veins and capillaries) contain 7% of the total. All these figures vary in the literature.

1.5 Summary and Further Reading

In this chapter we have presented a brief introduction to the human cardiovascular system, consisting of the heart, the blood and the complex network of blood vessels comprising a divergent network of arteries joined by the microvasculature to a convergent network of veins. A separate section is devoted to each component, in the wider scenario of human fluid physiology and keeping in mind the essential role of the cardiovascular system as a fast transport system. This chapter is the first of five introductory chapters aimed at providing background knowledge necessary for the rest of the book. The physiological aspects of this chapter are linked to the physics and mathematics of fluid dynamics introduced in Chap. 2. Much of the material of this Chapter was gathered from classical textbooks, e.g. Hall and Guyton [3], Levick [8], Tortora and Derrickson [1], Mohrman and Heller [25], Silverthorn [26] and Klabunde [27].

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Chapter 2

Notions of Fluid Mechanics and Transport



Abstract In this introductory chapter, we begin by revisiting key concepts from classical fluid mechanics for incompressible fluids, with a particular focus on fluid compartments in the human body, especially blood. We start by covering basic fluid dynamics principles, methods of study, and illustrative mathematical models for blood flow, expressed in terms of differential equations. The cardiovascular system is viewed as a fluid transport system, where both advection and diffusion transport mechanisms are present. Next, we review the clinical significance of arterial blood pressure, providing several examples. This is followed by a discussion of simple algebraic laws of flow, including Bernoulli's, Darcy's, and Poiseuille's laws, as well as Starling's principle of fluid exchange in the microvasculature. The elementary biofluid dynamics section continues with an exploration of blood rheology, viscosity, and their role in hemodynamics, with particular emphasis on the microvasculature. The final section addresses the regulation mechanisms of the cardiovascular system, within the context of bodily fluid dynamics, transport, and tissue perfusion. This chapter on fundamental fluid dynamics concepts serves as a bridge between Chap. 1, which focuses on the human cardiovascular system, and the subsequent chapters on bodily fluids. The chapter concludes with a brief summary and suggestions for further reading.

2.1 Fluid Mechanics Concepts

Continuum fluid mechanics is a very broad field of study, with a large range of applications in science and technology, including physiology and medicine. One distinguishes between compressible fluids and incompressible fluids, the latter being more relevant to the themes of this book. The distinction between compressible and incompressible fluids rests upon physical grounds. However, it is noted that the associated methodologies tend to be quite different, especially from the point of view of mathematical models, mathematical techniques and computational algorithms. There are numerous textbooks on continuum fluid mechanics. For incompressible fluids we particularly recommend the classical book of Sir Horace Lamb [1], that of

Curle and Davies [2] and the comprehensive book of Panton [3]. For compressible fluids we recommend the books by Liepmann and Roshko [4], Landau and Lifshitz [5] and Anderson [6]. The present book is concerned with theoretical fluid mechanics of human bodily fluids. In particular we are interested in the flow of blood in the full cardiovascular system, as depicted in Fig. 2.1. Relevant textbooks on bodily fluids include Fung [7, 8], Pedley [9], Westerhof et al. [10] and the recent book by Grotberg [11], to name but a few.

2.1.1 Examples, Physical Principles and Methods

Perhaps the most familiar example of fluid mechanics to the general reader is the flow of water in a channel or river. The associated scientific discipline called hydraulics is one of the oldest branches of fluid mechanics, well known to the Arabs and Romans, being today an important area of environmental and geophysical fluid dynamics. Less obvious to the general reader will be the fluid dynamics associated to the flight of a bird, a passenger aircraft at cruising speed of Mach number $M_a = 0.8$, or the re-entry of a space vehicle into the earth atmosphere at hypersonic speed, $M_a = 25$. Even less obvious will be the fluid dynamics during the respiration process, or the flow of blood from the left ventricle of heart all the way to the microvasculature and back to the right atrium; see Fig. 2.1. Fluid mechanics today embraces a very broad range of scientific and technological disciplines, covering environmental sciences, multiple areas of industrial applications, aerospace sciences, and more recently, biological fluids and medicine.

An essential property of all fluids is that they flow; that is they deform under the action of a shearing force. Water flows naturally under the action of gravitational forces. The periodic contractions of the wall of the left ventricle of the heart and the associated wall displacement exert a force on the fluid (arterial blood) causing deformation of the body of fluid inside the ventricle and thus flow of blood through the blood vessel network of the cardiovascular system. The deformation of the vessel solid wall caused by the haemodynamic stresses are more evident on the arterial side of the circulation via the wall stretch in systole and wall recoil in diastole. We note in passing that the phenomenon of vessel wall deformation pertains to the field of solid mechanics. Body forces, such as gravitational force, play a major role in modulating the dynamics of bodily fluids in general. Postural changes of the body will cause displacement of fluid compartment boundaries generating fluid deformation and therefore flow. Veins, for example, are easily compressed by surrounding tissue and organs, causing flow-generating forces. The human body spends about two thirds of its daily life under continuous postural changes that modulate fluid dynamics processes. More dramatically, trauma involved in a traffic accident, or other violent event, will activate strong external forces that will therefore act on all bodily fluid compartments, with tragic consequences. Traumatic brain injury is currently an important branch of medicine that addresses related issues. Most of the human body materials are fluids in the form of liquids and gases, whose composition

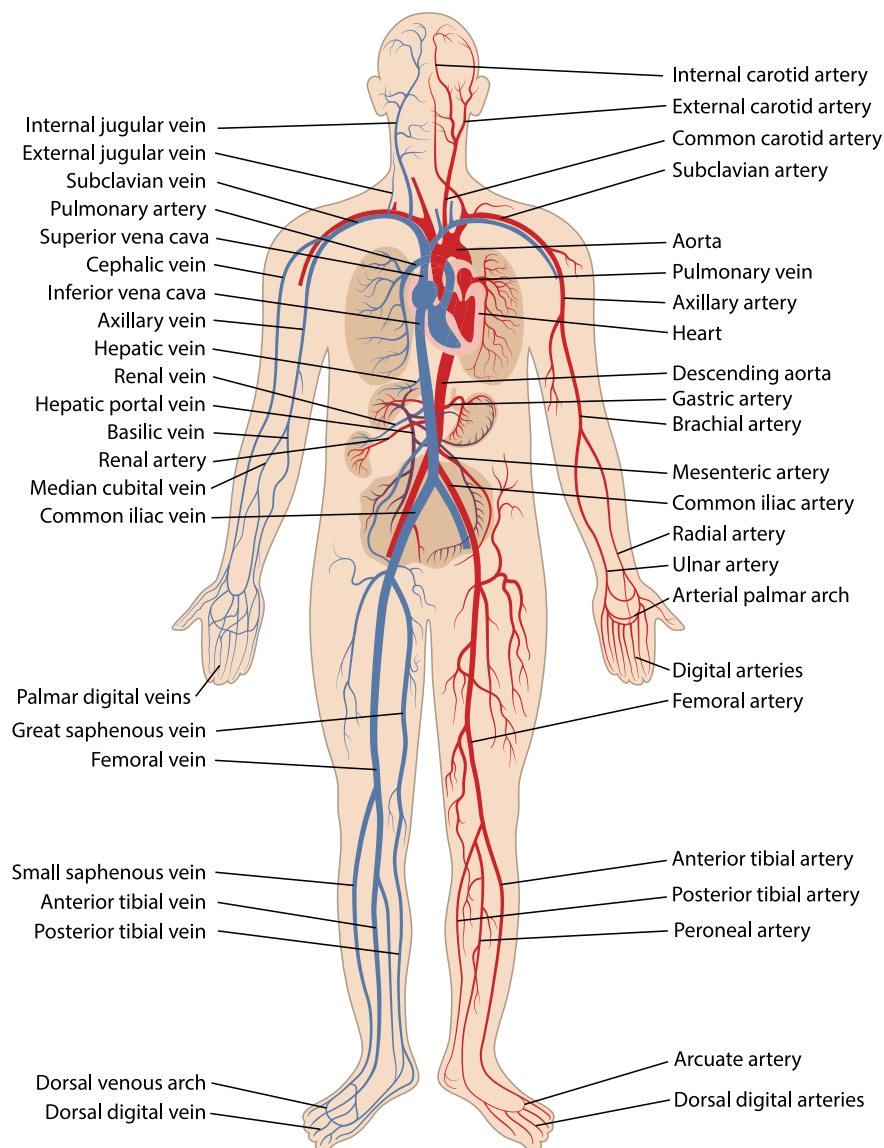


Fig. 2.1 The human cardiovascular system is composed of the heart, blood and a network of blood vessels comprising arteries, veins and the microvasculature (not shown). The full circulatory system includes the systemic circulation and the pulmonary circulation. Conventionally, arterial blood is represented in red and venous blood in blue. Only major vessels are shown (from Shutterstock, with permission)

and dynamics functionally unify the entire body. Classical study methodologies in fluid dynamics include experimental and theoretical approaches.

The experimental method is the oldest systematic method of study in the physical sciences. Modern experimental methods make use of large and sophisticated laboratories supplemented with visualization techniques and data acquisition methodologies. The wind tunnel and the shock tube have been essential experimental tools for studying the physics of compressible fluids [12]. The shock tube problem studied in shock tubes has its hydraulics analogue, the dam-break problem, studied in hydraulics laboratories [13]. In biological flows, glass tubes have been used profusely to study the behaviour and the properties of blood. Experimental methods have been, are and will always be, a fundamental tool of study in science and engineering. Data acquired from experiments is also of fundamental importance for other methodologies, as we shall see.

Theoretical fluid dynamics is a study methodology that rests upon three physical laws, namely (i) conservation of mass, (ii) balance of momentum and (iii) conservation of energy. These principles may be expressed through mathematical statements in the form of time-dependent partial differential equations (PDEs), or integral equations. The derivation of the equations of fluid dynamics from first principles is found in any classical book of fluid dynamics; see for example [1, 2] and [3]. In fully specifying the mathematical problem representing the physical fluid dynamics problem, one often requires additional laws, typically from thermodynamics, as well as experimentally determined parameters, such as viscosity. Moreover, one must specify the **spatial domain of the equations**. For flow of water in a single channel of regular shape this may be a trivial task, but not so for an aircraft in flight with flexing wings in flutter; the task will be exceedingly challenging for the case of flow of gases in the lungs during respiration. See Chaps. 6 and 10 of this book.

Having precisely specified the spatial domain one then requires the prescription of **boundary conditions**; this is a very challenging physical problem, to start with. Much physical knowledge of the problem is required to express mathematically what the boundaries are doing to the fluid and how the fluid is responding. As most realistic problems are time dependent, one also requires to specify **initial conditions**. In the end one has to solve an initial-boundary value problem (IBVP). But before attempting to look for solutions to the problem, strictly speaking, one should determine the **well-posedness** of the IBVP. That is, whether the IBVP has a solution, and if so, whether the solution is unique. In the case of multiple solutions one should determine which of the solutions represents the physical problem under consideration; this task usually requires the judicious application of entropy conditions. Well-posedness is a very challenging theoretical task and is often bypassed when concerned with complex applications. But it is not enough to rigorously show that a solution exists. The next problem is to find actual solutions to the equations. The full equations are often coupled non-linear systems with stiff or non-stiff source terms and depend on time and three spatial coordinates, that is four independent variables. Not mentioned so far is the problem associated with **parameters** in the equations, as distinct from the **unknowns** of the equations, the **dependent variables**. In practice such parameters, e.g. coefficient of viscosity or Young's modulus of elasticity, are measured or

estimated. Such measurements or estimates may be subject to large errors, adding considerable **uncertainty** to the calculated results for the unknowns. For physiological flows this is an issue to be well aware of. Then, obtaining solutions to the full equations for the **dependent variables** represents yet another major challenge. Under strong simplifications one may obtain analytical solutions in closed form.

Analytical methods, coupled to experimental methods, have for a long time been an essential tool in theoretical fluid dynamics. Major advances in theoretical fluid dynamics are due to the 19th century hydrodynamicists e.g. [1, 14]. Indeed, an analytical closed-form solution allows us to relate the interdependence of the various parameters of the problem and the unknowns, such as pressure and velocity. However, the necessary simplifications to the equations, in order to obtain analytical solutions, are often too strong, with much of the physics of the problem being lost. Alternative, or supplementary, methods are needed.

Computational methods represent an alternative to analytical methods. A major advantage of computational methods is that one may retain the complexity of the full equations and account quite satisfactorily for the complexity of the domain and its boundaries. A major challenge here is the design of reliable numerical algorithms, to solve the equations on a computer. Strictly speaking, this task poses also some theoretical challenges to the numerical analyst, to make sure that the numerical method is stable, convergent and of adequate accuracy. As for well-posedness for the IBVP, this theoretical challenge is often bypassed when solving very complex computational problems, resorting instead to the analysis of methods for simplified model equations that mimic some of the features of the original problem. Modern computational methods rely strongly on analytical methods and experimental methods, the use of which is mandatory for partially verifying the validity of computed solutions, even if for simplified situations. Advanced high performance computing techniques and symbolic manipulators are also essential supplementary tools in the practice of computational simulations.

The degree of maturity reached in the development of mathematical models, the advancements in the design computational algorithms to solve the equations and the ever increasing computing power available has facilitated the emergence of **Commercial Packages** to solve practical industrial problems. These tools, essentially black boxes, have their strengths and limitations. One of their strengths is generality, they can tackle many classes of problems. Their main limitation is accuracy. Overall, for industrial applications, commercial packages are very useful if used judiciously.

Regarding background on computational methods we recommend, for example, the books of Hirsch [15, 16], Godlewski and Raviart [17], LeVeque [18], Quarteroni and Valli [19] and Toro [12, 13]. For computational methods for physiological flows we strongly recommend the book edited by Formaggia, Quarteroni and Veneziani [20]; the books by Tu, Inthavong and Wong [21]; Vassilevsky, Olshanskii, Simakov, Kolobov and Danilov [22] and the very recent textbook of Quarteroni, Dedé, Manzoni and Vergara [23].

2.1.2 *Distinctive Features of Bodily Fluid Mechanics*

The study of the mechanics of bodily fluids is much more challenging than in many other fields of science and technology. To begin with, experimental methods for physiological flows are quite recent, as compared to industrial and aerospace disciplines, for example. The masses of experimental data available in aerospace disciplines have contributed significantly, not only to garner a basic understanding of the physical phenomena of interest, but also in the development and application of other study methodologies. This is the case, for example, in the formulation of mathematical models, the measurement or estimation of unknown parameters in the models and in developing validation and verification procedures for analytical and computational methods.

In order to appreciate what is distinctive in bodily fluid mechanics, let us compare the problem of flow of water in a channel, or the flow of gas in a cylindrical shock tube, with the problem of flow of blood in the human body, as represented in Fig. 2.1. To start with, the bodily fluid system cannot be seen, touched, measured or observed using traditional techniques long available for use in conventional fluid mechanics. It is appropriate to recognise, however, that cadaver studies, practised for millennia, have made significant contributions to the fields of anatomy and physiology. This is particularly so for elucidating important anatomical aspects of the cardiovascular system, despite the limitation of missing distinctive aspects of living bodies, of course. A major breakthrough in the last few decades, is the emergency of imaging technologies, such as magnetic resonance imaging (MRI). Such techniques, even with their own limitations, permit the visualisation of the vascular anatomy and also the measurement of important physical quantities such as flow and transport, but not pressure, unfortunately.

In vitro experimental research in physiological flows can provide valuable physical information, but suffers from serious limitations. Experimental conditions are highly artificial, the system under study is isolated, domains are generally rigid and highly regular, flow conditions are often assumed to be steady (time independent). Then, even if results are reliable, it is difficult to relate such findings to the true physiology of living human subjects. Nonetheless, in vitro experimental research can be useful for evaluating parameters that may be used for closure of mathematical models. Studies of blood rheology, for example, cannot do without in vitro experimental research in glass tubes. In vivo experiments, usually performed in animals and to some extent in living humans, are indeed more realistic than in vitro experiments. While in vivo experiments have proved quite effective in research aimed at pharmaceutical production, for instance, they are very limited for physiological/pathological fluid dynamics, which is the main theme of interest here. Still, their contribution is essential to the advancement of knowledge.

Mathematical modelling, coupled with advanced computational algorithms and powerful computing machines, while a success story for industrial fluid dynamics, for instance, is still at an early stage in bodily fluid dynamics. To illustrate the challenges posed let us consider the fluid dynamics problem of blood flow in the cardiovascular

system alone, comprising the systemic circulation and the pulmonary circulation, see Fig. 2.1. That is, for simplicity, we are excluding many other physiological aspects that are closely coupled to the fluid dynamics of blood, such as the transport and exchange of materials between blood and tissues; this would bring in the role of other fluid materials and systems such as interstitial fluid, cerebrospinal fluid, the lymphatic system, the urinary system, the digestive system, for example. Assuming we have a mathematical model for the fluid dynamics of blood, the next problem to face is the definition of the **domain**. Needless to say, we cannot account for the billions of blood vessels and vessel junctions in the human body, first because at present we cannot determine their precise geometry, that is lengths, diameters and connectivity. Current, MRI techniques cannot image very small vessels. But even if we could determine the computational geometry from refined MRI data, the resulting computational problem would be intractable, even with very approximate mathematical models. Assuming we choose to represent only major vessels for the arterial and venous systems, we then need an approximate method for representing the small vessels and the microvasculature that serve to connect the arterial and venous systems.

A distinctive feature of blood flow is that vessels are compliant, elastic, conduits. Mathematically, we have a fluid flow problem in which the flow domain has moving boundaries, that is the spatial position of the boundary is not known *a priori*, as would be the case for fluid flow in a rigid tube. In other words, we need to formulate equations and solution procedures to describe the fluid flow problem inside the vessel, the internal flow distribution, coupled to the associated deformation of the vessel wall, the structural problem. Generally this problem is described as a **fluid-structure interaction** (FSI) problem [20]. The vessel wall deforms in response to the passage of blood, associated pressure waves, as well as in response to control mechanisms that regulate the calibre of blood vessels.

The fluid flow problem is affected by several factors, including the geometrical aspects of the cardiovascular system; blood properties such as viscosity and its two-phase character; blood vessel wall mechanical properties; respiration; regulatory mechanisms and spontaneous contractions in veins, as in the lymphatic system. In addition, different flow patterns are present under physiological conditions, laminar flow, turbulent flow and bolus flow in capillaries. Assuming we have somehow approximated the fluid dynamics problem in the cardiovascular system, the next step could be that of accounting for the final mission of the cardiovascular system, which is realised through delivery and removal of materials at the capillary level, essentially by means of slow, diffusive transport across the vessel wall. In this manner, the cardiovascular system is connected to the interstitium and the interstitial fluid compartment. Blood is not just a moving fluid per se; it is a transporter fluid flowing between the heart and the body. Ultimately, blood must transport oxygen and nutrients to tissues according to metabolic needs and transport back to heart, and then to lungs, waste products of metabolism. These processes activate the full circulatory system, as well as control mechanisms involving two major additional systems: the nervous system and the endocrine system.

The selection of suitable computational algorithms to solve the associated time-dependent partial differential equations in domains with moving boundaries is another task to be faced. In spite of tremendous advances in the numerical analysis of evolutionary PDEs in the last few decades, the choice of appropriate numerical methods is by no means trivial. While for industrial problems there have been significant advances in the development of industrial commercial packages, for bodily fluid dynamics problems, as described here, this is far from being a realistic choice, at least for the time being. We therefore encourage the reader to be rigorous in the choice of methods, which essentially means that simulation codes will not be general black-box packages, but will very much be research codes in constant development and scrutiny.

To give a flavour of the issues discussed so far, in the next section we present, in an informal manner, some mathematical models for blood flow in a segment of a single blood vessel.

2.1.3 Blood Flow in Single Vessel: Mathematical Model

As already stated, theoretical fluid mechanics is based on mathematical models, often expressed in terms of time-dependent partial differential equations or PDEs. As a way of introducing basic and relevant concepts that are central to this textbook, we now present a simple mathematical model for blood flow in a segment of a blood vessel, as depicted in Fig. 2.2. The fluid dynamics of blood inside the vessel interacting with the displacement of the compliant vessel wall can be represented by the following non-linear, first-order system of PDEs

$$\left. \begin{aligned} \partial_t A + \partial_x(uA) &= s_1 , \\ \partial_t(uA) + \partial_x(\alpha Au^2) + \frac{A}{\rho} \partial_x p &= s_2 . \end{aligned} \right\} \quad (2.1)$$

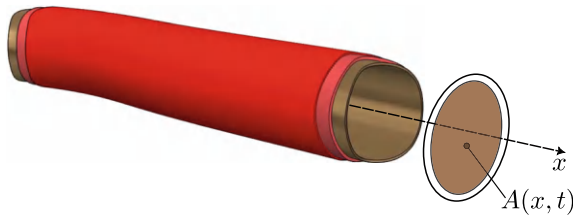


Fig. 2.2 Sketch of a blood vessel segment represented as a one-dimensional structure oriented in the axial direction x . At each position x one can define the cross-sectional area $A(x, t)$ that is used to define cross-sectional area averaged one-dimensional quantities (courtesy of Professor A. Saviglia, University of Trento, Italy)

Here the Coriolis coefficient α and the source terms are

$$\alpha(\beta) = \frac{\beta + 2}{\beta + 1}, \quad s_1 = 0, \quad s_2 = -2(\beta + 2)\pi \frac{\mu}{\rho} u, \quad \beta > 0. \quad (2.2)$$

The first equation in (2.1), called the **continuity equation**, expresses the physical principle of conservation of mass; the second equation expresses **balance of momentum**. The conserved quantities are cross-sectional area $A(x, t)$ and flow $q(x, t) = A(x, t)u(x, t)$. The quantities s_1 and s_2 , defined in (2.2), are generally called **source terms**. A rigorous derivation of the non-linear blood flow model (2.1)–(2.2) from the Navier-Stokes equations is carried out in Chap. 6.

It is important to clearly distinguish between unknowns and parameters, in system (2.1)–(2.2), as detailed below.

Unknowns and tube law. System (2.1)–(2.2) contains two PDEs and the following three unknowns, namely

- $A(x, t)$: cross-sectional area of the vessel at position x and time t ;
- $u(x, t)$: cross-sectional area averaged velocity of blood at a vessel cross section;
- $p(x, t)$: cross-sectional area averaged pressure of blood inside the vessel.

System (2.1)–(2.2) has more unknowns than equations, so in order to close the system an extra equation is required, called **closure condition**. In haemodynamics such closure condition is called **tube law**; it is an algebraic expression that relates the pressure $p(x, t)$ in the lumen to the mechanics of the vessel wall displacement via the cross-sectional area $A(x, t)$. In other words, in the model (2.1)–(2.2) **the vessel wall is not rigid**. The tube law couples the elastic properties of the compliant vessel wall to the fluid dynamics inside the vessel and accounts for an essential feature of blood flow, that is the displacement of the vessel wall. The choice of tube law depends on the mechanical properties of the vessel wall and therefore one must distinguish between arteries and veins, for example. As an example, here we adopt a simple tube law for arteries only, namely

$$p(x, t) = p_e(x, t) + \mathbb{P}(A; K), \quad \mathbb{P}(A; K) = K(x) \left(\sqrt{\frac{A}{A_0}} - 1 \right), \quad (2.3)$$

with

$$K(x) = \frac{\sqrt{\pi}}{(1 - \nu^2)} \frac{E(x)h_0(x)}{\sqrt{A_0(x)}}. \quad (2.4)$$

The **transmural pressure** $\mathbb{P}(A; K)$ is defined as the pressure difference

$$\mathbb{P}(A; K) := p(x, t) - p_e(x, t). \quad (2.5)$$

In the arguments of the transmural pressure, note the distinction between unknowns and parameters; A is an unknown of the equations, while K is a parameter.

Parameters of the equations. The parameters of the Eqs. (2.1)–(2.3) are

- ρ : density of blood ρ assumed to be a known constant, usually set to $\rho = 1060 \text{ kg/m}^3$;
- μ : blood viscosity, derived from the rheological properties of blood and assumed a known constant here;
- α : Coriolis coefficient, linked to the choice of velocity profile for one-dimensional representation of variables; see Chap. 6 for details.
- $A_0(x)$: cross-sectional area of the vessel at **equilibrium**, that is when the **transmural pressure** vanishes and $u = 0$;
- $h_0(x)$: vessel wall thickness;
- $E(x)$: Young's modulus of elasticity;
- ν : Poisson ratio, taken as $\nu = 1/2$ for incompressible media;
- $p_e(x, t)$: external pressure, assumed for the moment to be a prescribed function.

Note that in general the geometric and mechanical parameters $A_0(x)$, $h_0(x)$ and $E(x)$ vary with distance along the length of the vessel and are thus functions of x ; $p_e(x, t)$ may vary with both space and time. The determination of parameters is a major issue in computational haemodynamics and hence a source of uncertainties. Just for a moment, think of the task of determining the single function $h_0(x)$, or reconstructed versions of it, from measured data, for all vessels of the cardiovascular system!

The system of Eqs. (2.1)–(2.3) is not completely amenable to analytical treatment, apart from some special situations, not even for a single vessel segment, as we shall see in Chap. 9, for example. To appreciate the challenge think, for example, of extending the model to the full cardiovascular system composed of billions of vessels connected through virtually innumerable vessel junctions. One requires sophisticated numerical methods to solve the equations and appropriate models to account for vessel junctions.

Linearised Approximation

In attempting to apply analytical techniques to fully resolve the mathematical problem posed by system (2.1)–(2.3), it is an instructive exercise to try to simplify the system, by looking, for example, at linearised versions of the equations. There are many ways of doing this. In what follows we shall first assume that all parameters are constant and known in advance. Then we simplify the meaning of the unknowns as follows. We consider a small perturbation $a(x, t)$ of the equilibrium cross-sectional area A_0 , assumed constant, and a small velocity perturbation $v(x, t)$ of a stationary flow of speed $u_0 = 0$. That is

$$A(x, t) = A_0 + a(x, t) , \quad u(x, t) = 0 + v(x, t) . \quad (2.6)$$

Then, assume the perturbations $a(x, t)$ to be small compared to A_0 and $v(x, t)$ to be small. Assume in addition that derivatives of the perturbations are also small and hence products of these small quantities can be neglected. By substituting $A(x, t)$ and $u(x, t)$ from Eq. (2.6) into Eq. (2.1) and neglecting second-order terms we obtain a system of **linearised blood flow equations** for the new unknowns $a(x, t)$ and $v(x, t)$, namely

$$\left. \begin{aligned} \partial_t a + A_0 \partial_x v &= 0 , \\ \partial_t v + \frac{c^2}{A_0} \partial_x a &= 0 . \end{aligned} \right\} \quad (2.7)$$

Here

$$c = \sqrt{\frac{K \sqrt{A_0}}{2\rho}} \quad (2.8)$$

is a **wave speed**, a constant here; c is the speed of vanishingly small-amplitude waves, analogous to the speed of sound in a compressible gas. As will be seen in detail in Chap. 7 the first-order linear system (2.7) is amenable to analytical treatment. As a matter of fact, it can also be expressed as the classical second-order linear wave equation, in either of the unknowns, namely

$$\partial_t^{(2)} a(x, t) = c^2 \partial_x^{(2)} a(x, t) \quad \text{or} \quad \partial_t^{(2)} v(x, t) = c^2 \partial_x^{(2)} v(x, t) . \quad (2.9)$$

For mathematical techniques to study linear systems see Chap. 7 and for non-linear systems see Chap. 8. Apart from its mathematical beauty, Eqs. (2.7) are not very useful in practice even though they have the merit of retaining variations in space and time and the key feature of compliance of the blood vessel wall through the variable $a(x, t)$.

Another approach to approximate the equations is to somehow give up variations in space, or account for them in a very approximate manner. This is what **zero-dimensional models do**, as seen in Chap. 6.

2.2 Transport in the Cardiovascular System

As already pointed out the cardiovascular system, composed of the pumping heart, the blood and the network of blood vessels, is a **rapid transport** system in which the moving blood inside the vessels serves as the transporter for cellular elements (red, white, platelets), oxygen, nutrients, metabolic wastes, and a variety of substances that are relevant to regulation mechanisms, as we shall see in Sect. 2.5. To accomplish its mission, the cardiovascular system works in conjunction with other structures and mechanisms that support **slow transport** by diffusion to deliver oxygen and

nutrients to tissues and to collect carbon dioxide and wastes from these. Therefore, transport of materials to and from the body tissues is thus accomplished through a fast mode of transport, called **advective transport**, and a slow mode of transport, called **diffusion transport**. In what follows we introduce the reader to these two main modes of transport.

2.2.1 Advective Transport

Transport in the cardiovascular system, as depicted in Fig. 2.1, is predominantly advective in character, also called **convective transport** or **bulk flow**. Blood constituents, such as cellular elements, oxygen, nutrients, metabolic waste products and heat, are transported by a moving medium, the blood as a whole, whose movement is originated from the energy of the pumping heart. Blood is therefore a carrier medium moving at high velocity with the mission of transporting vital components for long distances in short times. For example, oxygen from the lungs is transported to the smallest blood vessels in the extremities in about 30 s; transport by diffusion would take five years [24]. Essentially, blood components suspended in plasma will travel with the velocity u of whole blood. Mathematically, assuming a certain blood chemical species with concentration $\eta(x, t)$, the advective transport of such species with u could be represented by the following partial differential equation (PDE)

$$\partial_t \eta(x, t) + u \partial_x \eta(x, t) = 0. \quad (2.10)$$

Here x represents distance (position) along a blood vessel and t is time. As usual, the symbols $\partial_t \eta(x, t)$ and $\partial_x \eta(x, t)$ represent partial derivatives of $\eta(x, t)$ in time and space, respectively. Equation (2.10) simply says that the concentration of the chemical species represented by $\eta(x, t)$ travels with the velocity u of the flowing blood. Of course, a major problem is to know the velocity of blood, which will depend on space x and time t . If u is constant, the solution of the initial value problem for (2.10) is immediate. For the more general case we may add equation (2.10) to system (2.1) for blood flow in a vessel. The resulting 3×3 system will couple blood flow to transport of a chemical species with concentration $\eta(x, t)$. In fact one may add more equations to system (2.1) to account for the transport of additional chemical species. Moreover, Eq. (2.10) may include additional terms, such as an algebraic source term to account for chemical reactions and a second-order differential term to account for dissipative effects. The resulting model would be an advection-diffusion-reaction equation. In subsequent chapters of this book we shall study models and methods to deal with the transport of species in the blood stream.

2.2.2 Transport by Diffusion

Diffusion transport is due entirely to the random fluctuations of particles suspended in a medium due to temperature, called **Brownian motion**. There is no other extra source of energy in this case, for which diffusion transport is considered a passive mode of transport. The key to diffusion transport is the concentration gradient (spatial) of the substance of interest. As in the previous section, assume a chemical species with concentration $\eta(x, t)$. The concentration gradient is denoted as $\nabla\eta = \partial_x\eta(x, t)$. If $\nabla\eta$ is uniform, then there will be no **net movement of particles** in any given direction. In other words, if the concentration is spatially constant, there will be no diffusive transport. Diffusion transport results from spatial variations in concentration and transport takes place **down the concentration gradient**. That is, net movement of particles in a given direction will result from a zone of large concentration towards a zone of low concentration. One of the first theoretical works on diffusion transport is attributed to the famous physicist Albert Einstein who, amongst other things, identified the limitations of diffusion, the so-called **Einstein's law of diffusion**. This law states that the time t taken for randomly jumping particles to move a distance x in a specified direction is proportional to the square of x , written as $t \propto x^2$. After identifying the proportionality parameter Einstein's law of diffusion becomes

$$t = \frac{1}{2D}x^2, \quad (2.11)$$

where D is the solute diffusion coefficient. Diffusion transport turns out to be very slow over long distances. Table 2.1 shows some examples. The first column shows the district of interest. The third column shows the time taken for a glucose molecule to diffuse a distance x , shown in the second column, in a specified direction. The last two rows make the point that diffusion transport is wholly inadequate for those two districts.

The density η (number of particles per unit volume) of Brownian particles at position x and time t satisfies the classical diffusion equation

$$\partial_t\eta(x, t) = \alpha\partial_x^{(2)}\eta(x, t). \quad (2.12)$$

Table 2.1 Illustration of Einstein's law of diffusion in various districts of the human body shown in the first column. The third column shows the time taken for a glucose molecule to diffuse a distance x , shown in the second column, in a specified direction (data from Levick [24])

Example in vivo	Distance x	Time t
Neurovascular gap	0.1 μm	0.000005 s
Capillary wall	1.0 μm	0.0005 s
Capillary to cell	10.0 μm	0.05 s
Skin, artery wall	1.0 mm	9.26 min
Left ventricle	1.0 cm	15.4 h

Here $\partial_x^{(2)}\eta(x, t)$ is the second-order partial derivative in space of the function $\eta(x, t)$ and α is the **diffusion coefficient**. Equation (2.12) is also known as the **heat equation** and appears in various fields of application.

For long distances, advection is much faster than diffusion; advective transport does not rely on **passive random Brownian motion** of particles, but on a moving carrier fluid (the transporter) that carries the particles with it. This is of course independent of the fact that Brownian motion continues to act if concentration gradients exist. Physiologically, the mission of the cardiovascular system is accomplished through combining advection and diffusion transport.

Consider, for example, the transport of oxygen, starting from gas in the respiratory system. The distance between the lungs and the blood stream is very short and thus transport is accomplished by diffusion. Then, fast advective transport by the cardiovascular system is activated to carry the oxygen to the capillary bed. Then across the capillary wall oxygen is transported by diffusion into the interstitium and from here to the cells. Finally, some of the interstitial fluid, including solutes and metabolic waste, is reabsorbed from the interstitium through diffusion, into the blood stream, from where advective transport will carry the molecules fast, back to the heart.

There are districts of the human body in which the mode of transport of solutes, notably in the brain interstitium, is still unknown. Currently, there is active research on neurodegenerative diseases, such as Alzheimer's disease, aimed at elucidating which is the dominant mode of transport responsible for the clearing of neurotoxic materials from the brain tissue, such as amyloid- β and tau [25–27].

A more realistic model for transport that would replace (2.10) and (2.12) is the advection-diffusion reaction equation

$$\partial_t \eta(x, t) + u \partial_x \eta(x, t) = \alpha \partial_x^{(2)} \eta(x, t) + s(\eta) , \quad (2.13)$$

where $s(\eta)$ is a source term that would account for chemical reactions of the relevant species. More advanced descriptions of transport phenomena and mathematical modelling are found in the book of Formaggia, Quarteroni and Veneziani [20]. See also D'Angelo [28]; Tavelli, Dumbser and Casulli [29] and Possenti et al. [30].

We next introduce some very simple, basic laws of fluid mechanics applied to some physical aspects of blood flow and pressure.

2.3 Simple Laws for Flow and Pressure

We introduce the reader to some very simple but useful laws, profusely applied by physicians and cardiovascular physiologists [24, 31], to blood flow q and pressure p . Mathematically, these laws will be expressed purely in terms algebraic expressions, rather than differential equations, thus radically simplifying the mathematical aspects. Their derivation, however, comes from both experimental and theoretical fluid mechanics, applied to bodily fluid dynamics [7, 9, 10, 32, 33]. Mathematical

manipulation of differential equations, under suitable assumptions, lead to the simple algebraic laws we are concerned with in this section. But before embarking on simple laws of flow, we first revise some very basic concepts associated to blood pressure.

2.3.1 Mean Arterial Pressure

Pressure is a fundamental haemodynamical quantity of clinical value in most branches of medicine. Figure 2.3 depicts the blood pressure curve in the aorta as function of time, within one cardiac cycle; three reference values of pressure are depicted, namely systolic pressure, diastolic pressure and mean pressure. Usually, blood-pressure values are given relative to atmospheric pressure, which is 760 mmHg.

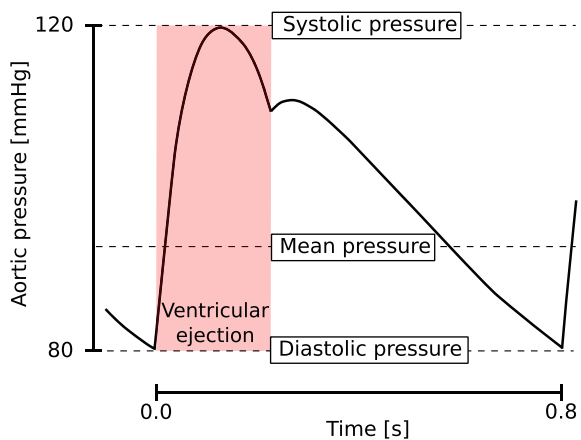
Estimates and Examples

The mean arterial pressure P_a (or *MAP*) at rest may be estimated as

$$P_a = P_d + \frac{1}{3} PP \quad \text{with } PP = P_s - P_d : \text{pulse pressure} . \quad (2.14)$$

Here P_s is **systolic pressure**; P_d is **diastolic pressure** and PP is **pulse pressure**, the difference between systolic and diastolic pressures. Both P_s and P_d can be measured under rest conditions. Typical values are $P_s = 120$ mmHg and $P_d = 80$ mmHg. Conventionally this is written as 120/80 mmHg. One can calculate the pulse pressure as $PP = 40$ mmHg and estimate the mean arterial pressure P_a according to (2.14). See Fig. 2.3.

Fig. 2.3 Blood pressure in the aorta as function of time, within one cardiac cycle. Three reference values of pressure are shown, namely systolic pressure, diastolic pressure and mean pressure



Strictly speaking, for an assumed pressure function $p(t)$, as the curve in Fig. 2.3, the mean arterial pressure P_a is given by the integral average

$$P_a = \frac{1}{DCC} \int_0^{DCC} p(t) dt , \quad (2.15)$$

where DCC is the duration of the cardiac cycle. Depending on the form of the function $p(t)$ one could evaluate (2.15) analytically, or numerically. Thus (2.14) is simply an estimate for (2.15). On the other hand, the pressure function $p(t)$ in (2.15) is not always available.

Under rest conditions estimate (2.14) seems to be quite accurate, but is not so under exercise conditions. Instead, under exercise conditions, the modified empirical formula is used

$$P_a = P_d + 0.01(P_s - P_d) \exp(4.14 - 40.74/HR) , \quad (2.16)$$

where HR is heart rate, the number of heart beats, or heart contractions, per minute; see Moran et al. [34] for details.

Example 1: arterial blood pressure under rest conditions. $P_s = 123$ mmHg; $P_d = 70$ mmHg and $HR = 60$. On applying formulae (2.14) and (2.16), we obtain, respectively

$$P_a^{rest} = 87.7 , \quad P_a^{exercise} \approx 86.9 . \quad (2.17)$$

For this example, formulae for rest and exercise conditions give similar results.

Example 2: blood pressure under exercise conditions. $P_s = 140$ mmHg; $P_d = 75$ mmHg and $HR = 92$. From formulae (2.14) and (2.16) we obtain

$$P_a^{rest} = 96.7 , \quad P_a^{exercise} \approx 101.2 . \quad (2.18)$$

Compared to (2.17), results for this second example are quite different.

Arterial Hypertension

Arterial hypertension is a major subject of clinical and research interest that involves pressure values. Following Levick [24] *arterial hypertension might be defined, simplistically, as a chronic, usually progressive, raised arterial blood pressure*. Arterial hypertension affects more than one billion people worldwide, a quarter of the world population. Hypertension is diagnosed if repeated measurements of resting brachial artery pressure exceed 140/90 mmHg in a patient under 50 years of age, or 160/95 mmHg in older patients. The precise causes of arterial hypertension are still unknown. Predisposing factors include high dietary salt intake, obesity and excessive alcohol consumption. Chapter 19 of this book examines arterial hypertension in some detail.

In this textbook we shall study mathematical methods to theoretically quantify haemodynamical quantities, such as pressure, velocity and flow in the full circulatory system. Next we introduce Bernoulli's law.

2.3.2 Bernoulli's Law

Physically, flow occurs due to a difference of total energy, or **total energy gradient**, as it is commonly called. The simplest case is that of a pressure gradient. There are three modes of mechanical energy that affect flow, namely

$$\left. \begin{aligned} \text{Pressure energy} &= PE, \\ \text{Potential energy of blood relative to heart} &= \rho ghV, \\ \text{Kinetic energy} &= \frac{1}{2}\rho u^2 V. \end{aligned} \right\} \quad (2.19)$$

Here P is pressure; V is volume of blood; ρ is blood density; $g = 9.81 \text{ m/s}^2$ is acceleration due to gravity; h is height relative to heart position and u is blood velocity. The **total mechanical energy per unit volume of blood** is defined by Bernoulli's law [24] as

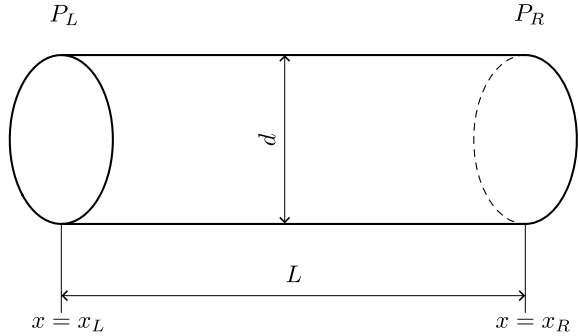
$$E_M = P + \rho gh + \frac{1}{2}\rho u^2. \quad (2.20)$$

Flow takes place down a total energy gradient, not just pressure gradient. In fact flow may occur against an **adverse pressure gradient**, as long as the total energy gradient allows it. That is, flow may occur from a point of low pressure to a point of higher pressure. One example in haemodynamics is furnished by blood flow in a stenosed vessel. Another example of flow against an adverse pressure gradient is provided by lymph flow in the lymphatic system.

2.3.3 Darcy's Law of Flow

Darcy's law is usually associated with the broad field of flow in porous media. The law originates from the work of the civil engineer Henry Darcy, who was concerned with flow of water through beds of gravel, sand or soil [10, 24, 33]. We shall apply the law to bodily fluids in humans. To start with, let us consider the very simple case of blood flow along an idealised segment of a blood vessel, as depicted in Fig. 2.4. Let us assume that the vessel is a straight, rigid tube of constant cross-sectional area, diameter d and length L and that the flow of a liquid (blood plasma or water) through it is steady, non-pulsatile and driven by a constant **pressure drop**, or **pressure head**

Fig. 2.4 Sketch of steady, non-pulsatile flow of water or plasma along a long, rigid tube of constant diameter d , driven by a constant pressure drop $\Delta P = P_L - P_R$



$$\Delta P = P_L - P_R , \quad (2.21)$$

where P_L and P_R are prescribed constant pressures, upstream (left) and downstream (right) of the tube, respectively. Often, ΔP is called **pressure gradient**. This is mathematically incorrect, as the gradient would be $\Delta P/L$, where L denotes the distance between the points associated with the two pressure values. Under the assumed conditions one may directly apply the well-known Darcy law of flow [24] and write

$$Q = L_p \Delta P = L_p (P_L - P_R) , \quad (2.22)$$

where flow Q is the volume of blood transferred per unit time, thus a rate of transport, and L_p is the constant of proportionality called the **hydraulic conductance** of the tube. L_p is a measure of **ease of flow**, and depends on tube diameter d . The wider the tube, the greater the hydraulic conductance and the bigger the flow generated by a given pressure drop ΔP . Thus a large artery or vein, has a bigger conductance and carries a bigger flow than a narrow arteriole, say. Introducing **resistance** R as the reciprocal of hydraulic conductance L_p , we write

$$R = \frac{1}{L_p} . \quad (2.23)$$

R represents **difficulty of flow**, opposition to flow, resistance to flow. Flow Q can now be expressed as

$$Q = \frac{(P_L - P_R)}{R} = \frac{\Delta P}{R} \quad (2.24)$$

and hence resistance R may be written as

$$R = \frac{\Delta P}{Q} . \quad (2.25)$$

Darcy's law is also known as Ohm's law, or as Poiseuille's law, or as the Hagen-Poiseuille law [10]. The law relates three variables: flow Q , pressure difference ΔP and resistance R . Given any two of them, through experimental measurements for instance, the third variable can be calculated from (2.24). Below we give an example in which Darcy's law (2.24) can be applied, even if very approximately.

Blood Flow in the Full Systemic Circulation

Pressure in the great veins is close to atmospheric pressure. There is thus a large pressure difference driving blood from arteries to veins. In reality, arterial pressure is pulsatile, because the heart ejects blood intermittently (systole), with rests in-between (diastole). See Fig. 2.3. During diastole the systemic arterial pressure decays from about 120 mmHg to about 80 mmHg, while pulmonary artery pressure decays from 25 mmHg to 10 mmHg.

Despite its simplicity, **Darcy's law of flow** (2.24) may be applied, in a very approximate manner, to the full systemic circulation, where now $\Delta P = P_L - P_R$ is the pressure drop across the entire systemic circulation, R is the **total peripheral resistance** TPR , and Q is the blood flow through the vasculature, which is equivalent to the **cardiac output** (CO). With specific notation Darcy's law now reads

$$CO = \frac{P_a - P_{cv}}{TPR}, \quad (2.26)$$

where P_a is mean arterial pressure, also denoted as MAP , and P_{cv} is **central venous pressure**, also denoted as $CV P$ (close to atmospheric pressure). Therefore, mean arterial pressure becomes

$$P_a = TPR \times CO + P_{cv}. \quad (2.27)$$

As pressure is measured relative to atmospheric pressure we may take $P_{cv} \approx 0$ and hence

$$P_a \approx TPR \times CO. \quad (2.28)$$

We conclude that the mean pressure may be estimated if one knows the total peripheral resistance and the cardiac output. Alternatively, on rearranging (2.28), one may estimate the total peripheral resistance TPR , if mean pressure P_a and cardiac output CO are known.

Darcy's law, of flow or the Hagen-Poiseuille law, can be used to construct realistic computational models consisting of very large blood vessel networks. In this approach, the law is applied to each vessel segment.

Cerebral Perfusion Pressure

We may apply Darcy's law of flow to cerebral haemodynamics, to estimate **cerebral blood flow** CBF to the brain and **cerebral perfusion**, as follows

$$CBF = CPP/CVR . \quad (2.29)$$

Here CPP is **cerebral perfusion pressure** and CVR is **cerebral vascular resistance**, with

$$CPP = P_a - \max \{ICP, JVP\} . \quad (2.30)$$

Here ICP is **intracranial pressure** and JVP is **jugular venous pressure**. Low mean-arterial pressure P_a or high intracranial pressure ICP or high jugular venous pressure JVP , compromise cerebral perfusion. ICP is the pressure in the brain parenchyma and is estimated from the pressure in the cerebrospinal fluid (CSF) in the cranial cavity. Clinically, ICP is estimated invasively from lumbar puncture (spinal tap) to measure CSF pressure in the spinal subarachnoid space. Note the profound implications of the very simple formulae (2.29), (2.30) for cerebral blood flow and cerebral perfusion pressure. CBF depends on cerebral vascular resistance CVR , arterial haemodynamics via P_a , venous haemodynamics via JVP and cerebrospinal fluid dynamics via ICP .

2.3.4 Poiseuille's Law

Poiseuille's flow, or Hagen-Poiseuille flow [10, 33], is defined as the flow in a cylindrical tube, such as that of Fig. 2.4, satisfying the following conditions:

1. The length L of the tube is large,
2. The tube is rigid and of constant cross-sectional area, of diameter d ,
3. The pressure drop $\Delta P = P_L - P_R$ is measured in the full length L ,
4. The fluid is viscous, with coefficient of viscosity μ ,
5. The flow is steady (time-independent) and laminar (non-turbulent).

Then, according to Poiseuille's law of flow, the flow Q is given as

$$Q = \Delta P \frac{\pi d^4}{128 L \mu} . \quad (2.31)$$

Comparing to Darcy's law of flow (2.24), Poiseuille's law gives **laminar resistance** as

$$R = \frac{128 L \mu}{\pi d^4} . \quad (2.32)$$

Hence now, given the pressure head ΔP , flow Q is known. Or, given flow Q , the pressure head ΔP is determined.

The resistance to flow R depends on geometric factors L and d , as well as the rheological factor μ , the viscosity of blood. Resistance to flow increases as the length L increases; the resistance also increases if viscosity μ increases. Resistance is inversely proportional to the diameter d of the tube **to the fourth power**. To appreciate the effect of cross-sectional area reduction through reduction of d , assume the diameter d is reduced by a factor of 4, then the resistance increases by a factor of 256. Narrow vessels in the microvasculature will have a large resistance to flow.

2.3.5 Capillary Exchange Between Blood and ISF

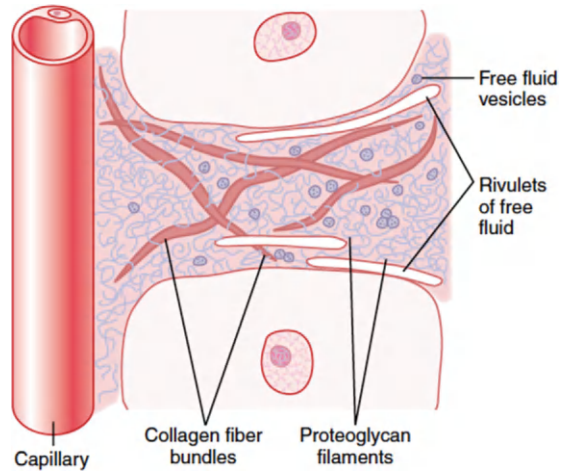
The most important mission of the cardiovascular system is realised in the microvasculature, through material exchange at the capillary wall, between the blood plasma inside the capillary lumen and the interstitial fluid in the surrounding interstitium. In this section we discuss, qualitatively, the main physiological factors that affect such material exchange. With this background, the following Sect. 2.3.6 will approach the subject more formally, in terms of the well-known Starling equation for fluid filtration.

Blood Flow and Capillary Exchange

As already described in Chap. 1, the circulatory system carries blood around the body, and along with its cellular formed elements and solutes in the plasma. The travelling of blood and its contents take the following route: oxygen-rich blood departs from the left ventricle of the heart, it continues into the arterial system, reaching the microvasculature (arterioles, capillaries, venules). Here, after unloading oxygen and nutrients, and uploading carbon dioxide and wastes, blood makes a **U turn**, by reversing direction towards larger veins, the venae cavae and the right heart atrium. Oxygen-depleted blood is first drained into the right ventricle, from where it is pumped to the lungs, and back to the left atrium of the heart. Then, the advective cycle of the circulatory system is repeated.

Here we are interested in the final mission of the systemic circulation, which is to deliver oxygen and nutrients to tissues and to remove carbon dioxide and metabolic wastes at the capillary level, just before the **U turn**. This event takes place through **material exchange across the wall of the capillary**. This material exchange is between two material compartments, namely (a) the blood plasma in the vessel lumen and (b) the interstitial fluid between the cells in the interstitium. Such compartments are separated by the capillary wall, which acts as a semi-permeable, or **selectively permeable**, membrane. Figure 2.5 depicts a capillary and the interstitium next to it, composed of cells and ISF between the cells. Recall that ISF is a solution of chemical composition similar to that of plasma, composed of water and solutes in various concentrations; see Chap. 3.

Fig. 2.5 Interstitium next to the capillary wall. The interstitium includes cells and interstitial fluid between the cells. The interstitial fluid is a solution of chemical composition similar to that of blood plasma. See Chap. 3 for details. (Figure from Hall [35], with permission)



Modes of Transport Across Membranes

Hall [35] discusses modes of transport, specifically across the cell membrane, which separates two compartments: the intracellular fluid and the extracellular fluid, which in fact is the interstitial fluid ISF. The specific structure of the cell membrane consists of a lipid bilayer and large numbers of protein molecules in the lipid. Proteins contribute to the transport of substances across the membrane, including channel proteins. One mode of transport is by **simple diffusion**, which is characterised by direct diffusion through the lipid substance of the bilayer and through carrier proteins in the lipid bilayer. Recall that the energy that causes diffusion is the energy of the normal random molecular motion of matter. **Facilitated diffusion** is a form of diffusion that makes use of carrier proteins, but still relying on random molecular motion. In **active transport** carrier proteins cause the substances to move against energy gradients. In this case, extra energy is needed, in addition to kinetic energy from the natural random movement of molecules.

In this chapter we are specifically interested in transport of materials, particularly water molecules and solutes, across the capillary wall, which is composed by a layer of endothelial cells and the basement membrane. See Chap. 1. Gases, such as oxygen and carbon dioxide, move freely by diffusion across the endothelium. An important contribution to capillary exchange comes from **bulk flow**, understood as mass movement of fluid driven by hydrostatic and osmotic pressure gradients. A form of diffusion transport, called osmosis, also plays an important role.

Osmosis Notions

Osmosis concerns the movement of water molecules by diffusion, from a zone of large concentration of water molecules to another of lower concentration or water molecules. Water, a solvent for all living matter [31], moves freely in and out of almost every cell in the body. Such dynamics is facilitated by (i) water-filled ion channels and (ii) special water channels created by the protein **aquaporin**. A key element of osmosis is the presence of **selectively permeable** membranes, or barriers. That is, membranes are permeable to water molecules but not, or less, permeable, to solutes. Classically, the concept of osmosis [31, 35, 36] is introduced via an experimental arrangement, such as that illustrated in Fig. 2.6.

A U-shape tube is divided into two chambers left (L) and right (R) by a selectively permeable membrane. Initially, both chambers hold the same volume of materials. In the present example, the left chamber is filled with pure water, which can freely flow into the right chamber through the membrane. The right chamber holds a solution, in which the solute molecules cannot cross the membrane into the left chamber. Departing from these initial condition, water molecules from the more **diluted** left chamber, move by diffusion to the more **concentrated** solution in the right chamber, across the membrane. This process causes a volume increase in the right chamber; part of the water volume initially in the left chamber has moved into the right chamber, while the solute molecules in the right chamber have remained in the right chamber. Figure 2.6 shows the classical case in which **equilibrium** has been reached.

What is shown in the example of Fig. 2.6 is a the special case, in which the left chamber contains no solutes. In general, the left chamber could also contain solutes

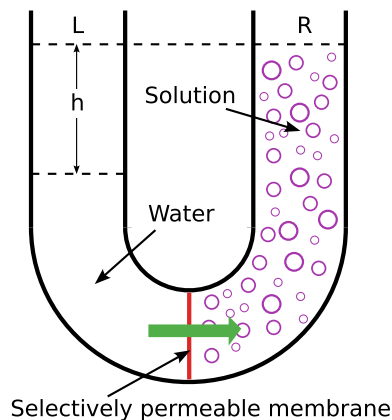


Fig. 2.6 Illustration of osmosis. Two sections L and R of a U-shaped tube are separated by a selectively permeable membrane, initially holding the same volume of materials. The left chamber section L is filled with pure water, which can freely flow into the right chamber through the membrane. The right chamber section R holds a solution, in which the solute molecules cannot cross the membrane into the left chamber

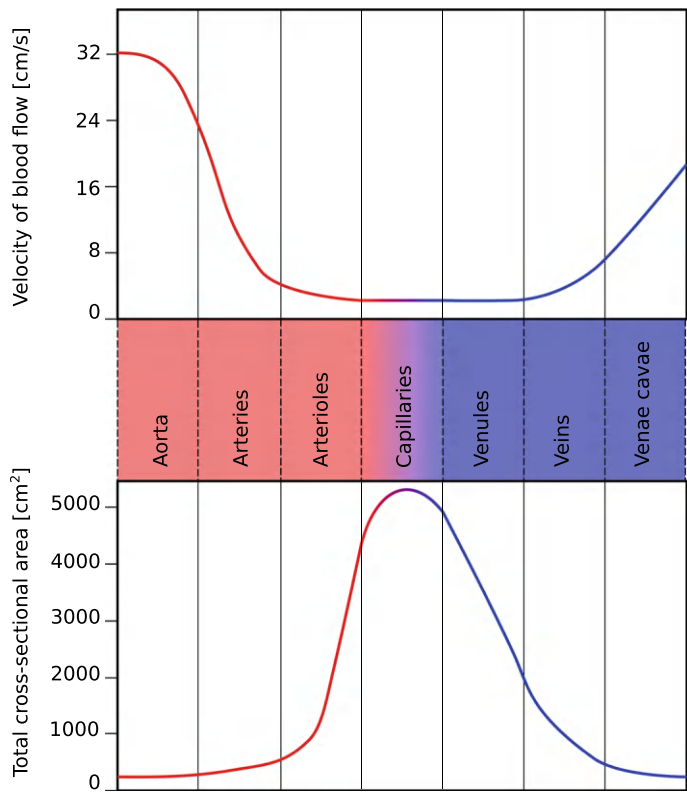
at a given concentration, in general, different from that of the solute in the right chamber, in any case creating a concentration gradient, which is a key element in the definition of osmosis. Osmosis is defined in [31], as **the movement of water across a membrane in response to a solute concentration gradient**. **Osmotic pressure** is defined as the pressure needed to oppose osmosis. In Sect. 2.3.6 we shall apply some of these concepts to establish the Starling equation for plasma filtration across the capillary wall.

Velocity, Surface Area and Capillary Wall

The material exchange at the capillary is strongly dependent on three main factors, namely the velocity of blood through the capillary, the capillary-wall surface area available for exchange, and the structure of the capillary wall. We succinctly review the role of blood velocity, capillary surface area and structure of the capillary wall, in determining the efficiency of the capillary exchange.

1. **Blood velocity at capillaries is slow.** When blood reaches the capillary bed, velocity reduces to its minimum value in the circulatory system, of the order of 1 mm/s. Figure 2.7, top frame (a), shows the average velocity of blood through the main compartments of the systemic circulation. Velocity at the capillary is the lowest. Lower velocities mean longer times for the exchange process to take place.
2. **Surface area in the capillary bed is large.** Even though the surface area of the capillary wall of each capillary is very small, due to the small diameter of the capillary vessel, of the order of 0.01 mm, the total surface area of the capillary bed as a whole is huge, due to the huge number of capillary vessels, of the order of 10 billion, covering a total length of over 80000 km. Figure 2.7, bottom frame (b), shows the total cross-sectional area of blood vessels through the systemic circulation. The maximum value is reached in the capillaries, which is consistent with the minimal velocity, as seen in frame (a). The resulting capillary surface area in humans is of the order of 6300 m^2 [31].
3. **The capillary wall.** The efficiency of the capillary exchange depends also on the structure of the wall of the capillary, that is its permeability. As described in Chap. 1, the structure of the capillary wall depends on the particular function of that capillary, which depends largely on its location in the body. The wall of capillaries consists of a single layer of endothelial cells, surrounded externally by a thin **basement membrane**. There are three types of capillaries, namely continuous, fenestrated and sinusoid. Their chief difference lies in their permeability to substances in the lumen and the interstitium. Endothelial cells themselves may possess features that facilitate, or otherwise, the material exchange process. In continuous capillaries, endothelial cells have no pores or gaps, other than the clefts. The endothelial cells of fenestrated capillaries have many fenestrations, or pores, of size between 70 and 100 nanometres. Sinusoids have endothelial cells with large fenestrations and large intercellular clefts. Moreover, sinusoids also have incomplete or even absent basement membrane. Continuous capillaries are found

(a) Velocity of blood flow in the circulatory system



(b) Total cross-sectional area of blood vessels

Fig. 2.7 Blood average velocity, top frame, and vessel total cross-sectional area, bottom frame, in various compartments of the systemic circulation

in the brain and lungs. Fenestrated capillaries are found kidneys, in the villi of the small intestine and in the ciliary processes of the eye. Sinusoids are found in the liver and the spleen.

In summary, slow blood velocity through the capillary means longer times for material exchange. Increased surface area in capillary bed means the material exchange is greatly enhanced. The specific location of the capillary will determine its function, its wall structure and thus its permeability.

In the next section we apply some of these concepts to establish the Starling equation for fluid exchange across the capillary wall.

2.3.6 Starling's Principle of Fluid Exchange

The Starling principle of fluid exchange is concerned with a quantitative description of exchange of materials across the capillary wall. The exchange is between two compartments, the blood plasma in the capillary lumen and the interstitial fluid in the interstitium. Such compartments are separated by the capillary wall, which acts as a **selectively permeable membrane**, as depicted in Fig. 2.8.

The structural elements of the capillary depicted in Fig. 2.8, frame (b), include: the vessel plasma (top); the glycocalyx attached to the endothelium and partly blocking the cleft between endothelial cells; the layer of endothelial cells and the interstitial fluid in the interstitium (bottom). The forces that determine the plasma ultrafiltration across the vessel wall result from four pressure values, two hydrostatic pressures P_c and P_i , and two osmotic pressures π_p and π_i . The two hydrostatic pressures P_c and P_i denote, respectively, the capillary blood pressure in the lumen and the interstitial fluid pressure in the interstitium. The pressure difference

$$\Delta P = P_c - P_i \quad (2.33)$$

drives plasma filtration across the capillary wall, as in Darcy's law (2.24). The osmotic pressures π_p and π_i denote, respectively, blood plasma osmotic pressure due mainly to plasma proteins, and ISF osmotic pressure in the interstitium, due to solutes. The osmotic pressure difference

$$\Delta \pi = \pi_p - \pi_i \quad (2.34)$$

opposes plasma transport from the capillary across the wall and into the interstitium. The pressure head $\Delta P = P_c - P_i$ is called the **hydraulic push**, while $\Delta \pi = \pi_p - \pi_i$ is called the **osmotic suction** [24]. Figure 2.8, top frame (a), depicts the effect of the various pressures via four arrows, including some typical numbers for such pressures.

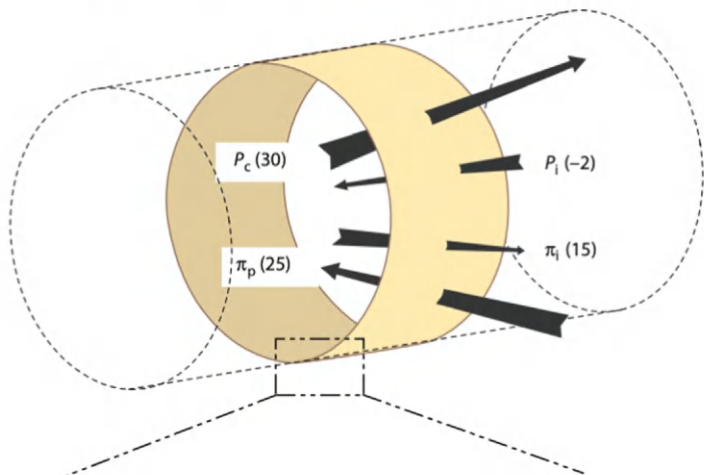
The combined force resulting from the two pressure heads (2.33) and (2.34) determines the **volume filtered per unit time** J_v across the wall, which is proportional to the difference of such pressure jumps, that is

$$J_v \propto (\Delta P - \Delta \pi) = (P_c - P_i) - (\pi_p - \pi_i) . \quad (2.35)$$

Net filtration results from the **hydraulic push** ΔP minus the **osmotic suction** $\Delta \pi$.

The proportionality constant in (2.35) depends on the **surface area** S of the capillary wall and on the **hydraulic conductance** L_p of the wall. Moreover, as the capillary wall does not constitute a **perfect semi-permeable membrane**, some of the plasma proteins will cross the membrane (the vessel wall) and therefore the plasma solutes will not fully exert their colloid osmotic pressure. To account for the **imperfection** of the vessel wall as a membrane, one introduces the **reflection coefficient**

(a) Traditional Starling principle: $\text{filtration force} = (P_c - P_i) - \sigma(\pi_p - \pi_i)$



(b) Revised Starling principle: $\text{filtration force} = (P_c - P_i) - \sigma(\pi_p - \pi_g)$

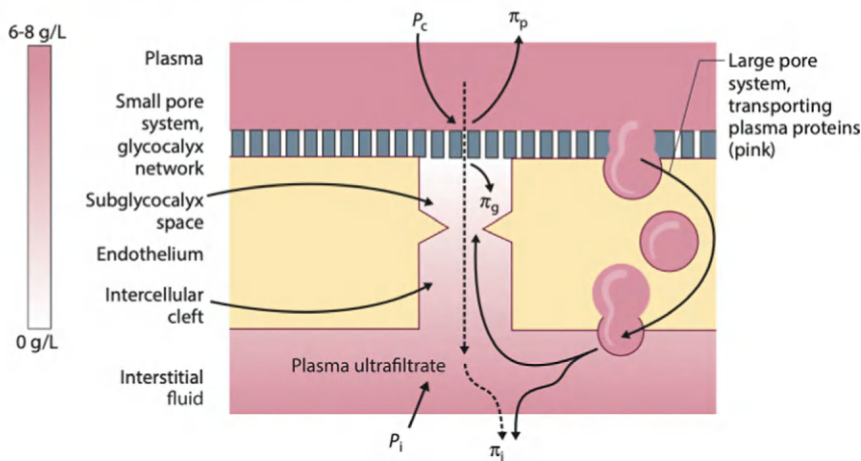


Fig. 2.8 Capillary exchange between plasma and the interstitial fluid across the capillary wall. Frame **b** depicts the main structures: the vessel plasma (top); the glycocalyx attached to the endothelium and partly blocking the cleft between endothelial cells; the layer of endothelial cells and the interstitial fluid in the interstitium (bottom). Frame **a** depicts the interior and exterior of a capillary, along with arrows to represent the two hydraulic pressures P_c and P_i , and the two osmotic pressures π_p and π_i . Net filtration results from the **hydraulic push** ΔP minus the **osmotic suction** $\Delta\pi$. In the revised Starling's equation π_g is substituted for π_i (from Herring and Paterson [37], with permission)

$$\sigma = \frac{\Delta\pi_{effec}}{\Delta\pi_{ideal}}, \quad (2.36)$$

where $\Delta\pi_{ideal}$ accounts for a **perfect** semi-permeable membrane from which all plasma proteins **reflect**; that is they do not cross it. $\Delta\pi_{effec}$ accounts for the fact that the vessel wall may be an imperfect membrane, where some proteins pass through and some reflect. For example $\sigma = 0.8$ means that 80% of the plasma osmotic pressure is exerted at the capillary wall. By introducing the proportionality constant $L_p S$ and the reflection coefficient σ we obtain the **traditional Starling equation for fluid filtration**

$$J_v = L_p S [(P_c - P_i) - \sigma(\pi_p - \pi_i)]. \quad (2.37)$$

Here again, L_p is the hydraulic conductance of the wall and S is the surface area of the wall. See Fig. 2.8, top frame (a).

There is a **revised Starling principle**, in which the pressure π_i is replaced by the more accurate value π_g , the osmotic pressure in the sub-glycocalyx space. That is

$$J_v = L_p S [(P_c - P_i) - \sigma(\pi_p - \pi_g)]. \quad (2.38)$$

See Fig. 2.8, bottom frame (b).

The Starling equation for fluid filtration has been given entirely in terms of an algebraic relation. See the analogy with Darcy's and Poiseuille's laws of flow (2.24) and 2.31. Further details are found in [24, 36] and [31]. For more elaborate, differential mathematical models for fluid and solute filtration across the capillary wall see, for example, [38, 39] and references therein.

2.3.7 Capillary Reabsorption at the Venous End

The literature is not unanimous when describing the process of capillary exchange. For example, Silverthorn [31] and Levick [24] assign different roles to the venous end of capillaries, when it comes to reabsorption of fluid from the interstitium. It is a established fact that the capillary is not homogeneous along its length; one can distinguish an arterial segment from the arterial side and a venous segment continuous with the venous side. According to Silverthorn [31], **most capillaries show transition from net filtration at the arterial end to net absorption at the venous end**, noting the exceptions offered by capillaries in the kidney and the intestine. As an example, Silverthorn considers the case in which the hydrostatic pressure in the plasma is $P_c = 32$ mmHg at the arterial end and $P_c = 15$ mmHg at the venous end of the capillary. Assuming also that osmotic pressure in the plasma, due to blood proteins, is constant along the length of the capillary, and given by $\pi_c = 25$ mmHg and that the osmotic pressure in the interstitium is $\pi_i = 0$ mmHg along the length of the capillary. Then, assuming a linear variation of the hydrostatic pressure, along the

full length of the capillary, an intersection point $\hat{x}$ exists at which $P_c(\hat{x}) = \pi_c(\hat{x})$. Downstream of this point there must be net absorption from the interstitium into the capillary.

Levick [24] argues that the state of reabsorption into capillaries is exceptional, being present only transiently. On the other hand, the recent work of Hu et al. [40], based on a Magnetic Resonance Imaging study of cerebral haemodynamics, reports a major role of cerebral venous reabsorption, directly from perivenous spaces. In any case of course, the lymphatic system will reabsorb the excess of ISF in the interstitium and take it back to the venous system and thus to the circulation. See Chap. 3.

The next section gives an introduction to blood rheology, with the aim of determining values for the blood viscosity.

2.4 Blood Rheology

This section is concerned with the rheology of blood, **haemorheology**, that is the deformation behaviour of blood under dynamic conditions. Deformation of a material is understood as relative displacement of material points within the body of material. In general, the response of materials to applied loads is described by **constitutive equations**. These are tensor equations relating strain tensors to stress tensors, the so-called stress-strain relationships. For relevant background the reader is referred to Fung [8] and Humphrey and Delange [33], for example. The presentation of blood rheology here, will be elementary.

Resistance to blood flow is strongly influenced by the rheological properties of blood itself, ultimately expressed through the viscosity. In order to understand such properties it is necessary to recall that, physically, blood is a **two-phase suspension** of formed elements in an aqueous solution called plasma [41]. As seen in Chap. 1, blood plasma is mainly composed of water but also includes organic molecules, proteins and salts. Formed elements include red blood cells (RBCs), white blood cells (WBCs) and platelets. The viscosity of blood results from three main factors: (i) the viscosity of plasma, (ii) the rheological behaviour of formed elements and (iii) the volume fraction of formed elements. The latter contribution is dominated by the **haematocrit** He , that is the volume fraction of RBCs. Typically $He \approx 45\%$, also written as $He \approx 0.45$. The mechanical properties of RBCs are an important factor, which is expressed in their deformability and aggregability. The viscosity of blood will also depend on vessel diameter, shear rate and temperature. In fact blood viscosity decreases as temperature increases, which is true for all liquids. For gases, the opposite is true; their viscosity increases as the temperature increases.

2.4.1 Viscosity. Newtonian and Non-Newtonian Fluids

As hinted by the simple Poiseuille law (2.31)–(2.32), blood viscosity is **resistance to flow**, and is therefore an important parameter affecting blood fluidity and, ultimately, tissue perfusion. Classically, as suggested by Newton [8], viscosity can be introduced through the simplified arrangement of Fig. 2.9, which depicts a layer of fluid between two parallel solid plates, with constant velocity gradient $\frac{du}{dn}$. A shearing force F is applied to the top plate, which is in contact with the top fluid layer, with a contact area A . The resulting **shear stress** $S = F/A$, which is a friction force per unit area, displaces the top fluid layer in contact with the plate at the velocity u_{top} of the top plate. The bottom fluid layer remains attached to the stationary plate, that is $u_{bot} = 0$. The internal fluid distortion is measured through the velocity difference Δu between two layers divided by the distance Δd between them, that is $\Delta u/\Delta d$. In fact this is related to the **shear strain** τ , or **shear rate**, which is defined as

$$\tau = \frac{du}{dn} . \quad (2.39)$$

Here τ accounts for the fluid distortion induced by the **shear stress**. The shear rate is the derivative of u in the normal (perpendicular) direction of flow. Shear rate is therefore the change in velocity per unit distance in the normal direction to flow. For the special arrangement of Fig. 2.9 we have $\tau = u_{top}/d$. Viscosity is defined as

$$\mu = \frac{\text{shear stress}}{\text{shear rate}} = \frac{S}{\tau} = S / \frac{du}{dn} , \quad (2.40)$$

which gives the classical stress-strain relationship

$$S = \mu \frac{du}{dn} . \quad (2.41)$$

Here μ is called the **coefficient of viscosity**, or simply viscosity. Again with reference to Fig. 2.9, viscosity represents the friction between two moving planes of fluid, or as Levick puts it [24], *viscosity is the shear stress needed to generate one unit of shear rate*. The viscosity μ as defined is often called **dynamic viscosity**, while $\nu = \mu/\rho$ is called **kinematic viscosity**, where ρ is blood density.

A note regarding dimensions of viscosity. First, the dimensions of shear stress: [shear stress] = [force/area] = Pa (pascal), that is N/m^2 . The dimensions of shear rate: [shear rate] = [velocity/length] = $[\frac{1}{s}]$. Therefore the dimensions of viscosity in the SI system are

$$[\mu] = [Ns/m^2] = [Pa.s] . \quad (2.42)$$

In practice, other units for viscosity are also used. In the system of units *centimetre-gram-second*, the unit of force is the *dyne* and the unit of viscosity is the *poise* (P),

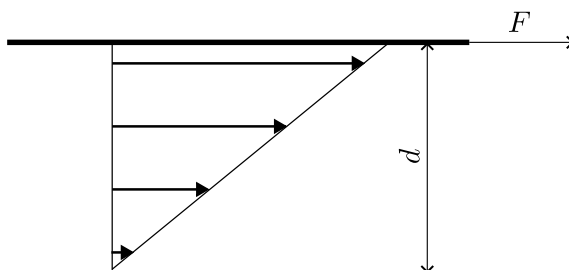


Fig. 2.9 Fluid flow between two solid parallel plates with constant velocity gradient du/dn . A shearing force F is applied to the top plate, while the bottom plate remains stationary. Successive fluid layers are displaced at varying velocity; the top layer moves with top plate, while the bottom layer remains stationary

Table 2.2 Viscosity in $[cP]$ units for various physiological fluids at the fixed temperature of 37°C . Note the huge influence of formed elements on the viscosity of whole blood. Most given values change with gender and physiological conditions, as well as temperature

Fluid	Viscosity (dynamic) $[cP]$
Water	0.6913
Blood plasma	1.1 to 1.3
Whole blood	3 to 4
Cerebrospinal fluid (CSF)	0.7 to 1.0
Interstitial fluid (ISF)	1.5
Lymph (thoracic duct)	1.23
Urine	0.8 to 1.0
Saliva	1.05 to 1.3

with $1 Pa.s = 10P$. One often uses **one millipascal per second**, ($0.001 Pa.s$) as the unit for viscosity. Then

$$1mPa.s = 0.001Pa.s = 0.01P = 1cP \text{ (one centipoise)}. \quad (2.43)$$

In Table 2.2 we compare the viscosity of various physiological fluids at the fixed temperature of 37°C . Blood plasma, the aqueous part of blood, is mainly composed of water, but blood plasma viscosity is larger than that of water, almost twice as large. The third row shows the viscosity of whole blood, putting in evidence the huge contribution of formed elements to blood viscosity. For comparison, Table 2.2 also shows the viscosity of other physiological fluids of interest, namely cerebrospinal fluid, interstitial fluid, lymph, urine and saliva.

A **Newtonian fluid** is defined as a fluid of **constant** viscosity, while the viscosity of a **non-Newtonian fluid** is **variable**, it depends on the shear rate. Viscosity may decrease with shear rate, which characterises **shear-thinning** fluids. Instead, in **shear-thickening** fluids, the viscosity increases as the shear rate is increased.

2.4.2 Viscosity of Blood Plasma

Blood plasma, the aqueous part of blood, is the suspending phase for cellular elements (RBCs, WBCs and platelets); see Chap. 1. Blood plasma is composed of 91.5% water and 7% of dissolved substances. It is the 7% portion of dissolved substances that increases the viscosity of blood plasma with respect to that of pure water, as illustrated in Table 2.2. We note also that blood plasma behaves as a Newtonian fluid, its viscosity is independent of shear rate. However, its value depends quite strongly on concentration of dissolved substances, which vary significantly under physiological and pathological states. The dissolved substances in blood plasma include three types of solutes. The first group is composed of **metabolism solutes**, which in turn include nutrients (glucose, lipids, amino acids, vitamins), hormones, respiratory gases (oxygen, carbon dioxide) and wastes. These substances are all related to metabolism, either needed for or produced by. The second group of plasma solutes includes **salts and ions**.

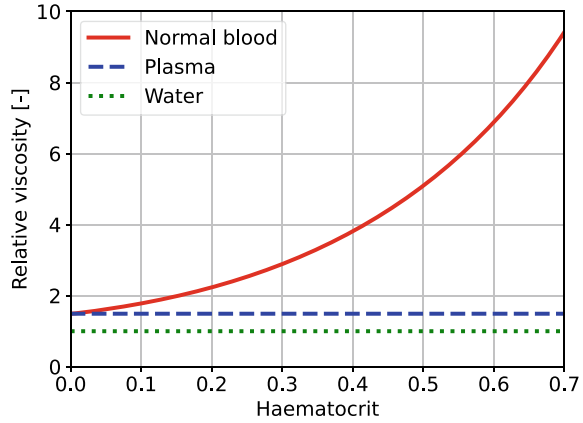
Plasma is a dilute salt solution, and in water salts dissociate into their component ions. In the human body plasma ions are called **electrolytes** and serve various functions, including a role in osmosis, a mechanism that affects **osmotic pressure** and thus the movement of water in the body. See Sect. 2.3.6 on the Starling principle for water and solute transport across the wall of capillaries. The third group of dissolved substances in blood plasma are **proteins**, which include serum albumin, antibodies, fibrinogen and prothrombin. Serum albumin has an effect in **osmosis**; antibodies play a role in immunity, while fibrinogen and prothrombin are key players in blood coagulation. In fact, anomalous amounts of fibrinogen in the blood plasma may be associated with an increased risk of thrombosis. Blood, plasma and formed elements, is denser and more viscous than water, with density $\rho = 1060 \text{ kg/m}^3$ at temperature of 38°C , about 1°C higher than oral or rectal body temperature.

The protein content of plasma is a major determinant of its viscosity, which is more evident in disease processes. According to [41], plasma viscosity can increase from its reference range of $1.1\text{--}1.29 \text{ cP}$ to $5.0\text{--}6.0 \text{ cP}$, under abnormally high protein levels (hyperproteinemia). In particular, **fibrinogen** is identified for its contribution to the increase of plasma viscosity. Hydration affects blood plasma rheology, generally influencing concentration of dissolved substances and thus plasma viscosity [42].

2.4.3 Haematocrit and Viscosity

Haematocrit He is the volume fraction of red blood cells in blood. It is expressed either as a percentage or as a fraction in decimal form. Under physiological conditions He varies between 35.5% and 44.9% in adult females and between 38.3% and 48.6% in adult males. He is high in humans populating geographical zones at high altitude. He is also high in younger people; in newborns He varies between 55.0% and 68%.

Fig. 2.10 Relative viscosity of normal blood as function of haematocrit He , compared to viscosity of blood plasma and viscosity of water. At the reference value of $He = 0.45$ blood is about 4 to 5 times more viscous than water (data from Hall [35])



Haematocrit has a profound effect on blood viscosity. As seen in Table 2.2 whole blood viscosity is higher than plasma viscosity. The excess value is due to the presence of cellular elements, RBCs, WBCs and platelets. These formed elements contribute to the disturbance of the flow streamlines and therefore to the increased resistance to flow and thus to the increase of whole blood viscosity. Red blood cells are the main contributors, by their shear volume fraction; they occupy 99% of all formed elements. Blood viscosity increases strongly with haematocrit. Figure 2.10 shows blood plasma viscosity and whole-blood viscosity as a function of haematocrit, both normalised with respect to the reference viscosity of water. Obviously, the viscosity of plasma and water are constant with respect to He , but whole-blood viscosity increases sharply with He . At the normal reference value $He = 0.45$, blood is about 4 to 5 times more viscous than water.

In addition to concentration of RBCs, that is He , to blood viscosity, there three additional contributions of RBCs which emanate from (i) their deformability, (ii) their aggregability and (iii) their sedimentation, as will be explained shortly.

Simple Formulae to Describe Viscosity

Many attempts have been made to produce mathematical formulae to represent the distribution of blood viscosity as a function of various parameters. For theoretical approaches see, for example, Popel and Johnson [43] and Farina et al. [44, 45]. For approaches employing fitting of experimental data see Pries et al. [46] and [47], for example. A very simple formula for blood viscosity that accounts for plasma viscosity and haematocrit He is the so-called generalised Einstein's formula for the viscosity of a two-phase liquid/solid suspension [48], namely

$$\mu = \mu_0 \times (1 + c_0 He) . \quad (2.44)$$

Here μ_0 represents the viscosity of the aqueous part of the two-phase mixture, H_e is the volume fraction of the solid phase and c_0 is a coefficient. For blood μ_0 is the viscosity of plasma, H_e is the **haematocrit**, typically $H_e = 0.45$, and $c_0 = 2.5$. See Westerhof et al. [10]. In fact, for $\mu_0 = 1.3$ and $\mu_{water} = 0.69$, formula (2.44) gives $\mu = 4.0$. See Fig. 2.10.

Next we discuss the behaviour of red blood cells and its effect on whole-blood viscosity.

RBC Deformability

Physiologically, RBCs are highly deformable and under high shear rates tend to orient themselves with the streamlines so as to reduce resistance and thus reduce viscosity. The deformability of RBCs depends fundamentally on their cellular and mechanical properties, as well as shear rates. It is known that at low shear rates $\mu_{rigRBC} < \mu$ and at high shear rates $\mu_{rigRBC} > \mu$, where μ is blood viscosity and μ_{rigRBC} is the viscosity for rigid (non-deformable) RBCs.

RBC Aggregability and Sedimentation

Red blood cells have a tendency to aggregate, that is to stick together. RBC aggregates resemble stacks of coins and are called **rouleaux**. **RBC aggregability** depends on several factors, but the following two are prominent: the surface properties of RBCs and the composition of plasma proteins. Large plasma proteins promote aggregability of RBCs. Fibrinogen, for example, in large concentrations will have a two-fold effect on blood viscosity. First, it will directly increase plasma viscosity and then, by promoting RBC aggregation will in turn increase blood viscosity. On the other hand, there are also dis-aggregability factors; RBC aggregates tend to be dispersed by sufficiently large fluid forces. Aggregability is highly sensitive to shear rate at the low end of shear rates but not so much at the higher end.

Blood exhibits shear thinning behaviour, μ decreases as the shear rate increases. At present, there is no complete understanding of the precise mechanisms at work in explaining the balance between aggregating and dis-aggregating forces acting on RBCs and thus on blood viscosity [41]. **RBC sedimentation** is yet another feature of the behaviour of red blood cells that affects RBC distribution across the vessel and resistance to flow. In response to gravitational forces RBCs tend to accumulate on the lower side of the vessel. Naturally, the degree of such effect will depend on the orientation of the particular vessel with respect to the gravitational field. In the human body, especially under constant postural changes, all possible vessel orientations with respect to gravity are possible.

2.4.4 Vessel Diameter: The Fahraeus-Lindqvist Effect

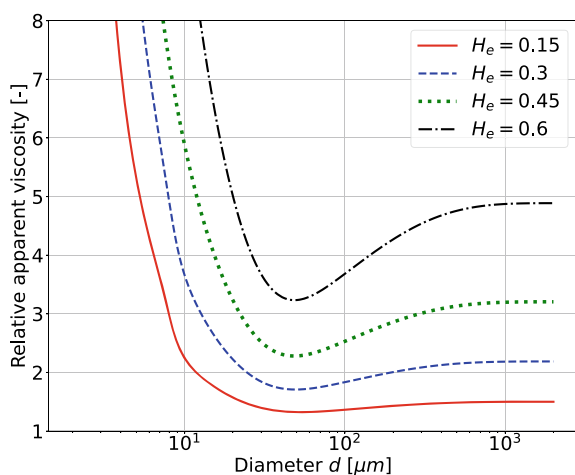
The **geometrical factor** in the form of the **vessel diameter** has a profound effect on the rheological behaviour of blood as a whole. Recall that the vessel diameter in the human vasculature changes by four orders of magnitude, from 3 microns to 3 cm. White blood cells and platelets have little effect on blood viscosity, which is partly explained by their very low volume fractions, as compared to RBCs. However, in very narrow vessels the effect of WBCs on viscosity may be larger than that of RBCs. This is particularly the case in disease states, when WBCs are activated and are more rigid [41].

Here we examine the effect of vessel diameter on blood viscosity; we also discuss the so-called the Fahraeus-Lindqvist effect and study empirical formulae for blood viscosity as a function of vessel diameter and haematocrit.

Variation of Viscosity with Vessel Diameter

It was first reported by Fahraeus and Lindqvist, in their seminal paper [49], that blood viscosity decreases sharply as the vessel diameter decreases. This phenomenon has since been called the **Fahraeus-Lindqvist effect** and is highly relevant to haemodynamics in the microvasculature. The Fahraeus-Lindqvist effect has been the subject of intensive study for almost one century, using *in vitro* and *in vivo* experimental methods, as well as theoretical methods based on fluid dynamics principles; see for example [41, 43–47] to name but a few. Figure 2.11 shows relative blood viscosity functions of diameter, for fixed values of He , constructed by fitting curves to a large set of *in vivo* experimental data [47]. Curves for four fixed values of haematocrit He are shown, namely 0.15, 0.3, 0.45, 0.6. It is seen that for a curve $\mu = \mu(d)$ corresponding to a fixed haematocrit He , it is possible to identify two critical values

Fig. 2.11 *In vivo* relative viscosity of blood as a function of vessel diameter for four fixed values of haematocrit He : 0.15, 0.3, 0.45, 0.6. Curves are obtained by fitting experimental data (data from Pries, Neuhaus and Gaetgens [47])



for the diameter d_{min} and d_{max} , with corresponding values μ_{min} and μ_{max} for the relative viscosity, resulting in three intervals for the diameter. For the precise interpretation of the values He in Fig. 2.11 see the original work of Pries, Neuhaus and Gaetgens [47].

With reference to Fig. 2.11 we observe the following, as the diameter d **decreases**:

1. $I_R = [d_{max}, \infty)$: Constant viscosity, $\mu = \mu_{max} = \text{constant} \forall d \in I_R$.
2. $I_M = [d_{min}, d_{max}]$: Decreasing viscosity, as d decreases, $\mu(d) < \mu_{max} \forall d \in I_M$.
3. $I_L = (0, d_{min}]$: Increasing viscosity, as d decreases, $\mu(d) > \mu_{min} \forall d \in I_L$.

Note that the values d_{min} and d_{max} are not very sensitive to the chosen value of the haematocrit He . However, μ_{min} and μ_{max} are highly sensitive to He . Clearly, as the haematocrit increases, the viscosity increases, as already seen in Fig. 2.10. For any fixed value of He , the relative viscosity tends to a constant value in $I_R = [d_{max}, \infty)$, and is thus independent of diameter, as diameter increases. That is to say, for large-diameter vessels blood viscosity is independent of diameter. As diameter decreases, a sudden decrease in viscosity is observed in $I_M = [d_{min}, d_{max}]$, reaching a minimum value of viscosity μ_{min} . Further decrease in diameter shows a sharp increase in viscosity in $I_L = (0, d_{min}]$. The minimum value for viscosity is close to that of plasma alone, for small values of the haematocrit. It is worth mentioning that viscosity curves exist for in vitro experimental data [46]; such curves display a qualitative behaviour analogous to the in vivo data curves of Fig. 2.11, but shifted to the left. The main difference is in d_{min} , which for the in vitro experimental data is visibly smaller than for the in vivo data.

Biophysics of the Fahraeus-Lindqvist Effect

The Fahraeus-Lindqvist phenomenon is intimately connected to the blood flow patterns in blood vessels of sufficiently small diameter, where the dimensions of formed elements, such as red blood cells, become comparable with the vessel diameter. Red blood cells tend to accumulate in the centre of the vessels, leaving a lubricating plasma layer close to the wall, essentially free from RBCs. This means that at any fixed position along the vessel length the haematocrit distribution is a function of radius, that is $He = He(r)$. The formed central core of RBCs of large He moves faster than the fluid particles in the plasma viscous layer close to the wall. Recall that the vessel wall endothelium is coated by the glycocalyx, which contributes to slowing down the flow close to the wall. Some aspects of the Fahraeus-Lindqvist effect are discussed by Farina et al. [44, 45] in the framework of theoretical fluid dynamics of the phenomenon. They also propose mathematical formulae for the viscosity as a function of vessel diameter. The up-to-date review of Secomb [50] is highly recommended. See also the textbooks by Roselli and Diller [51] and that by Fasano and Sequeira [52].

Computational models for haemodynamical simulations need to input reliable values of parameters in the models, such as viscosity. At present, it seems as if the fitted viscosity curves of Pries and collaborators based on in vivo data [46, 47]

offer a realistic answer to the problem. See [53] for an application of these viscosity functions in the framework of a global, closed-loop model for the human circulation.

In the next section we discuss regulation mechanisms of the human body in the framework of bodily fluid dynamics and the essential vascular requirement of tissue perfusion.

2.5 Haemodynamics and Regulation Mechanisms

The circulatory system, comprising the systemic and pulmonary circulations, constitute a major system of bodily fluids in humans. The dynamics of blood in the full circulatory system results from the automatic contraction of the heart ventricles and the elastic response of blood vessels, accounting for the purely mechanical aspects. But there are other factors at play, regulation mechanisms, that contribute with the bio part to the fluid mechanics in living organisms. These regulation mechanisms are part of general homeostatic mechanisms operated by the nervous system and the endocrine system; in particular, they act on blood vessels and the heart so as to modulate the blood fluid dynamics to maintain adequate oxygen and nutrient supply to all tissues at all times.

2.5.1 Blood Flow, Pressure and Vascular Resistance

According to the simple laws of Darcy and Poiseuille, blood flow Q depends on pressure head ΔP and resistance R , namely

$$Q = \frac{\Delta P}{R}, \quad \text{with} \quad R = \frac{128L\mu}{\pi d^4}. \quad (2.45)$$

Recall that blood flow Q is the volume of blood that flows through any tissue per unit time, measured in mL/min . Other relevant quantities include cardiac output CO , which is the volume of blood that circulates through the systemic circulation per minute. CO is identical to the volume of blood through the pulmonary circulation; it is the total blood flow and, as seen earlier $CO = HR \times SV$. The cardiac output CO will be distributed to all body tissues perfused by the blood vessel network, for which the very simple law (2.45) proposed for a single vessel segment, turns out to be useful in more general situations. The larger the pressure difference ΔP , the larger the blood flow Q ; the larger the resistance R the smaller the flow Q . Pressure difference and resistance vary throughout the circulatory system. Essentially, the blood flow to a specific tissue depends on pressure difference and resistance, which in turn however, are functions of other parameters, as we shall see.

Blood pressure varies significantly along the vessel network, from the aorta to the vena cavae. It exhibits an oscillatory component from the aorta to the arterioles, with

systolic and diastolic peaks. The mean pressure MAP in the systemic circulation decreases essentially monotonically with distance from the left ventricle, from just over 80 mmHg to essentially zero when entering the right ventricle. By rearranging Eq. (2.28) with appropriate notation we may write

$$MAP \approx CO \times SVR, \quad (2.46)$$

where SVR is the systemic vascular resistance. Then, for a fixed resistance, mean arterial pressure MAP increases with cardiac output CO , that is with heart rate HR or stroke volume SV or both.

We recall that vascular resistance is **difficulty of flow**, that is, opposition to flow. As illustrated by Eq. (2.45), resistance depends on length L of the blood vessel, blood viscosity μ and the diameter d of the vessel. Of these factors, the vessel diameter d is a prominent determinant of vascular resistance, hence of blood flow and pressure. For vessels of small diameter d , such as arterioles, capillaries and venules, resistance according to Eq. (2.46) will be large; this is particularly evident through d to the 4th power. This will have the effect of decreasing blood flow and increasing mean arterial pressure. In the complete vascular system, small vessels contribute the most to the **systemic vascular resistance** SVR , or **total peripheral resistance** (TPR).

Venous return (VR) is yet another parameter affecting CO . Venous return is the volume of blood collected by the venous network and flowing back to the right atrium. Such return is in part due to the residual pressure, that is the pressure difference between the venules and the right atrium, which is about 16 mmHg. Other contributions to venous return come from (i) skeletal muscle pumps and (ii) the respiratory pump. Both kinds of pumps make use of opening and closing of vein valves, coupled to compression and decompression of veins. The most obvious example of skeletal muscle pumps is the contraction of the muscles in the legs, which ensures venous blood return to the heart against the force of gravity. The respiratory pump makes use of compression and decompression of veins in the thoracic and abdominal cavities. Figure 2.12 describes the dependence factors of mean arterial pressure MAP ; the left side shows factors that affect cardiac output CO , while the right side displays factors affecting systemic vascular resistance SVR . The final dependence of mean arterial pressure, is the product $MAP = CO \times R$.

Regulation mechanisms are aimed at modulating the haemodynamics driven by mechanical determinants. Such regulation takes various forms and acts on various parameters, notably the vessel radius d in vasoconstriction and vasodilation, as we shall see.

2.5.2 Autoregulation: The Cardiovascular Centre

Regulation of the full body homeostasis is carried out by the nervous system, the endocrine system, and sometimes by a combination of both. The nervous systems send impulses that travel along axons and neurons. The endocrine system releases

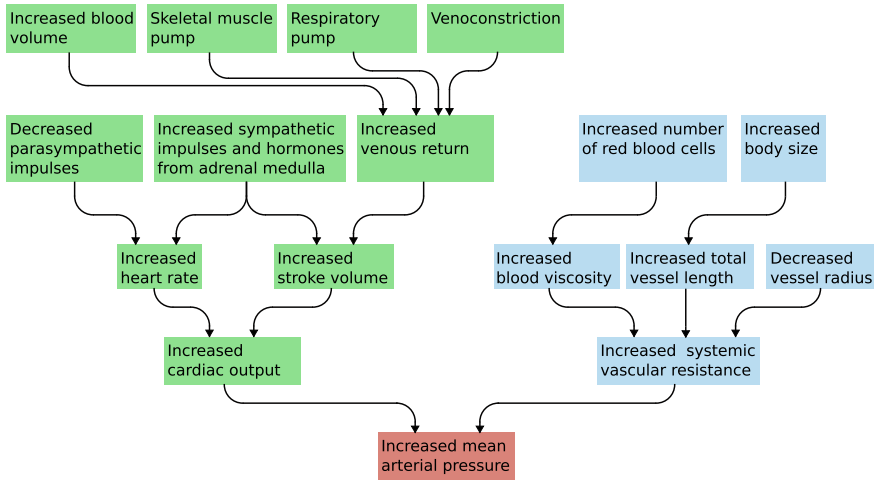


Fig. 2.12 Dependence factors of cardiac output CO (left) and systemic vascular resistance SVR (right), giving the final dependence of mean arterial pressure, as the product $MAP = CO \times R$ (data from Tortora and Derrickson [36])

hormones, mostly into the interstitial fluid (ISF) system, which then enter the blood stream across vessel wall; once in the blood stream such hormones travel to target cells. Note the two forms of transport of hormones: transport from ISF to blood across the vessel wall by diffusion and then fast transport in the blood stream. The brain stem houses the **cardiovascular centre** (CVC) in the medulla oblongata, depicted in Fig. 2.13. The top right-hand side rectangle shows the inputs to the cardiovascular centre, while the bottom right-hand side rectangle depicts the response in the form of output to the heart. Inputs derive from two sources, namely **higher brain centres** and **sensory receptors**. Ensuing output effectors from the cardiovascular centre to the heart are propagated along sympathetic and parasympathetic nerves of the Autonomic Nervous System.

The input source constituted by **higher brain centres**, includes the cerebral cortex, the limbic system and the hypothalamus. We note that the limbic system, formed by brain structures located on both sides of the thalamus, is associated to emotions, behaviour, motivation, long-term memory and olfaction. That is to say, emotional states may elicit responses that affect the physics of blood flow dynamics. The second class of inputs to the cardiovascular system include proprioceptors, chemoreceptors and baroreceptors. Proprioceptors monitor the position of limbs and muscles during body activity and send nerve impulses to the cardiovascular centre, which constitute major stimuli for a fast rise of the heart rate at the start of a physical activity. Chemoreceptors monitor chemical changes in the blood, such as concentrations of chemicals. Baroreceptors monitor changes in blood pressure in major arteries and veins through sensing the vessel stretching produced by the pressure of the flowing

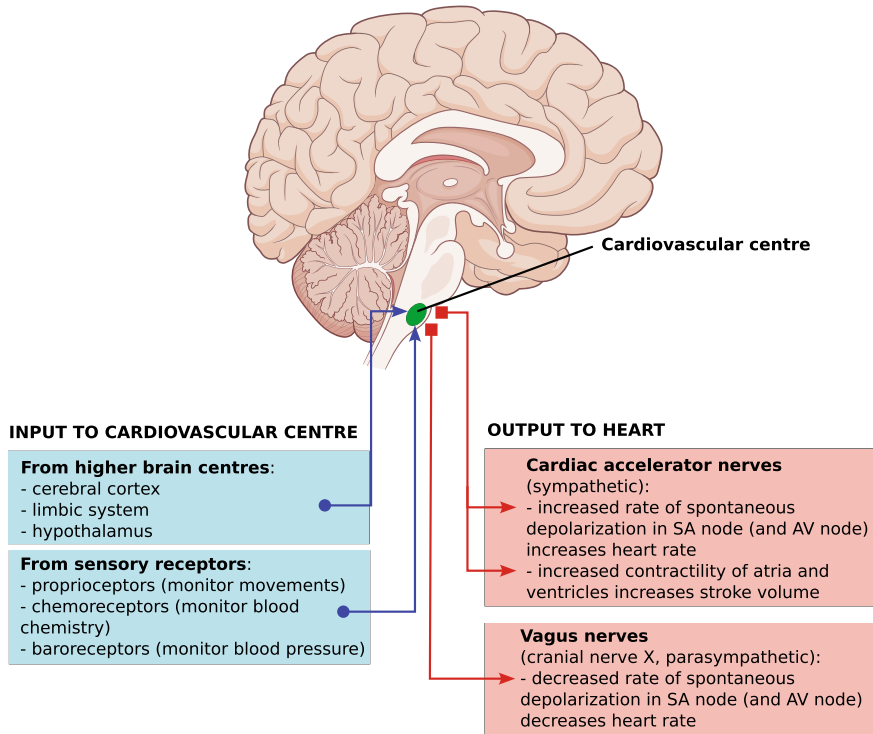


Fig. 2.13 The cardiovascular centre, located in the medulla oblongata. Input from higher brain centres, proprioceptors, baroreceptors and chemoreceptors. Outputs through the sympathetic and parasympathetic divisions of the autonomic nervous system. Regulated functions include heart rate, stroke volume, blood vessel diameter and systemic vascular resistance (data from Tortora and Derrickson [36])

blood. Outputs to effectors include **sympathetic neurons** from the medulla oblongata and **parasympathetic neurons**. See Fig. 2.13.

Regulatory roles of the cardiovascular centre include: regulation of the heart rate and stroke volume; activation of neural systems that regulate blood pressure and blood flow to specific tissues, and activation of a collection of neurons that form the **vasomotor centre**. The vasomotor centre controls the diameter of blood vessels through vasodilation (augmentation of diameter) and vasoconstriction (reduction of diameter). The cardiovascular centre sends impulses via vasomotor nerves to the innervated wall of arterioles throughout the body, thus determining the resting level of the systemic vascular resistance.

2.5.3 *Autoregulation of Pressure*

Autoregulation of a tissue is the ability to automatically adjust its blood flow to match its metabolic needs. Autoregulation is particularly important for certain tissues, such as the heart, skeletal muscle and the brain, in which even regional variations of blood distribution may cater for particular needs, according to function. In each capillary bed, local changes can regulate vasomotion and thus blood flow and pressure. There are two kinds of stimuli that cause autoregulation of blood flow and blood pressure, namely physical changes and chemical changes.

Physical changes include temperature and myogenic response. Increase in temperature causes vasodilation and thus increase of blood flow, while decrease in temperature causes vasoconstriction and hence reduction of blood flow. **Myogenic response** refers to the reaction of the smooth muscle of the vessel to stretching caused by local pressure. Smooth muscle contracts more forcefully when it is stretched and relaxes when stretching lessens.

Chemical changes relate to the concentration of a variety of chemicals released into the blood stream. Such chemicals alter the vessel diameter through vasodilation and vasoconstriction. For example, nitric oxide (NO) is an important vasodilator; NO is released by endothelial cells of the vessel wall. Other types of cells that release chemicals to alter the vessel diameter include white blood cells, platelets and macrophages. Other chemicals released by cells into the blood stream include potassium ions, hydrogen ions and lactic acid.

This final section of this chapter is meant to give the reader an elementary presentation on regulation mechanisms and how these affect the physics of the fluid dynamics of blood. For further reading on standard material on the subject we recommend the following textbooks: Levick [24], Hall and Guyton [35], and Tortora and Derrickson [36]. Advanced studies are found in research papers in the literature. As one example of research material we mention the work of Payne [54], who proposed mathematical models to predict the regulation response of the cerebral vasculature to changes in arterial blood pressure, arterial concentration of carbon dioxide and neural stimulation. See also the works of Ursino and Gianessi [55] and Toro et al. [53] on mathematical models for regulation mechanisms in cerebral haemodynamics.

2.6 Summary and Further Reading

In this chapter we have presented a succinct introduction to several aspects of the fluid mechanics of bodily fluids, with particular emphasis on blood flow in the cardiovascular system. Note that each section of the chapter would deserve an entire book by itself. Appropriate references are cited for further study. Ideally, the reader of this book should have some background knowledge on theoretical fluid mechanics, to start with. To cover this aspect we strongly recommend the book of Curle and Davies [2]; see also the classic book of Lamb [1]. The reader should also have

some specialised knowledge of bio-fluid dynamics, for which the books by Fung [7], Pedley [9], Humphrey and Delange [33] and Grotberg [11] are recommended. More specifically, for haemodynamics we suggest studying the book of Westerhof, Stergiopulos and Noble [10], in conjunction with this chapter and possibly with the rest of this book. For physiological aspects we suggest the books of Hall and Guyton [35]; Levick [24]; Mohrman and Heller [56]; Silverthorn [31] and the comprehensive textbook of Tortora and Derrickson [36].

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Chapter 3

Interstitial Fluid and Lymph



Abstract The fluid portion of an adult human's total body mass is 60% in males and 55% in females. This fluid is distributed into two major compartments: intracellular fluid (2/3) and extracellular fluid (1/3). In this chapter, we primarily focus on extracellular fluids, which include blood plasma, interstitial fluid, and lymph in the lymphatic system. We begin with an overview of all major fluid compartments, distinguishing between intracellular fluids, extracellular fluids, and the subdivisions of the latter. Next, we examine the interstitial fluid (ISF) and its direct connection to the vascular and lymphatic systems. A section is then dedicated entirely to the lymphatic system and its anatomical connections with other systems. The chapter concludes with a summary and suggestions for further reading.

3.1 Body Fluid Compartments and Material Exchange

The purpose here is first to identify the two major fluid compartments in the human body, extracellular fluids (ECF) and intracellular fluid (ICF), and then discuss material exchange amongst the various fluid compartments. We discuss the following: material exchange between blood and interstitial fluid (ISF) in the interstitium; the collecting of excess ISF and proteins by the lymphatic system to be returned to the circulation; and material exchange between ISF and intracellular fluid.

3.1.1 Intracellular and Extracellular Fluids

The percentage of fluid of the total body mass of an adult human is about 60% in males and about 55% in females. Here, **fluids** are understood as body water and their dissolved substances, or solutes [1]. Figure 3.1 shows the distribution of body fluids, first into two major compartments, namely intracellular fluid and extracellular fluids, the first occupying 2/3 of the total and the later the remaining 1/3. Intracellular fluid is the fluid within the cell, while extracellular fluids are the remaining fluids, in turn constituted by the interstitial fluid, occupying 80%, and blood plasma, occupying

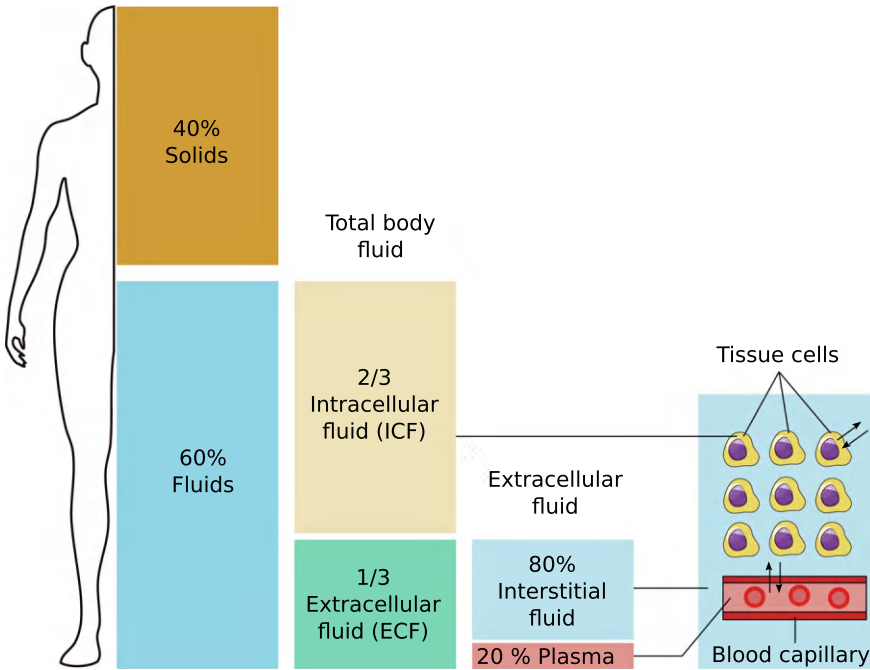


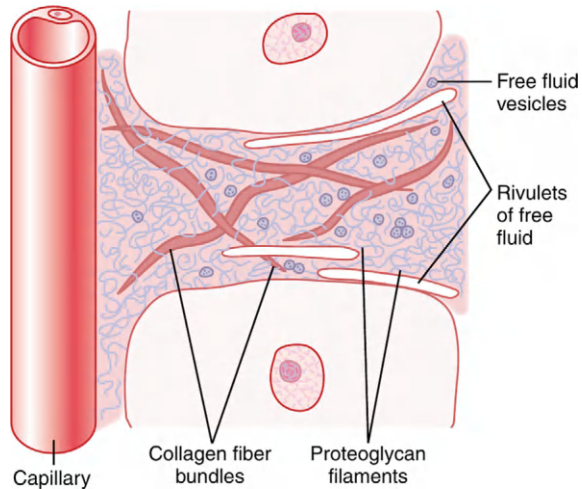
Fig. 3.1 Fluids in the human body. The percentage of fluid of the total body mass of an adult human is about 60% in males and about 55% in females. Note the exchange of materials between blood and interstitial fluid and between interstitial fluid and intracellular fluid. Not shown is the collecting of excess interstitial fluid and proteins by the lymphatic system that returns lymph to the venous system and thus to the circulation

the remaining 20%. ISF is associated with other well-defined fluid compartments, as we shall see, the lymph in the lymphatic system and cerebrospinal fluid in the central nervous system (CNS). In this book we are exclusively concerned with the dynamics of extracellular fluids.

3.1.2 The Interstitium and Interstitial Fluid

The **interstitium** is formed by the spaces between the cells of the body and occupies about 17% of the total body volume. The fluid in the interstitium is called **interstitial fluid**. Interstitial fluid bathes the cells and originates from filtration and diffusion of water and solutes from the blood plasma at the capillary wall. Figure 3.2 depicts the interstitium next to a capillary, along with its structure and components. Not every material in the interstitium is fluid. There are two solid structures, namely (a) collagen fibre bundles and (b) proteoglycan filaments. Most of the interstitial fluid is entrapped in the proteoglycan filament structures, while some small portion of *free*

Fig. 3.2 Interstitial fluid in the interstitium next to a capillary segment. Solid structures in the interstitium include **a** collagen fiber bundles and **b** proteoglycan filaments, giving ISF gel-like characteristics (from Hall [2], with permission)



fluid is formed by the free-fluid vesicles and the so called *rivulets* of free fluid. Under physiological conditions the free-fluid portion is very small, of the order of 1%. However, under the pathological condition of tissue oedema, the free-fluid portion may well reach 50% or more.

The combined presence of solid structures and the entrapped fluid and solutes in the proteoglycan filaments gives the whole fluid/solid mixture the physical characteristics of a **gel**. Such characteristics of ISF determine its transport properties. ISF mainly diffuses through the gel, molecule by molecule, noting that diffusion through the gel is almost the same as diffusion through free fluid [2]. As seen in Chap. 2, diffusion is a very efficient transport mode for short distances, such as that between a capillary and a neighbouring tissue cell. For the transport of oxygen, carbon dioxide, electrolytes, small molecular weight nutrients and cellular excreta, such diffusion-dominated transport mode is sufficient. Interstitial fluid originates primarily from filtration of blood plasma from the circulation at the capillary wall. ISF fills in the narrow spaces between the cells, as suggested by Fig. 3.1.

As already discussed in Chaps. 1 and 2, the cardiovascular system exchanges materials with the interstitium at the capillary level, see Fig. 3.2. Typically, oxygen and nutrients are transported across the capillary wall and carbon dioxide and wastes are reabsorbed into the vessel lumen, from the interstitium. Figure 3.1, right-hand side rectangle, illustrates the exchange of materials between the systemic capillary lumen and the interstitial fluid surrounding the cells in the interstitium, and exchange between ISF and intracellular fluid in the cells.

Types of Capillaries and Modes of Transport

Capillaries are of different types, depending on the district of the circulatory system of the human body. Their type will determine the mode of transport of materials

across the vessel wall. **Continuous capillaries** have continuous endothelial cell membranes; the gaps between endothelial cells, the intercellular clefts, are apertures that allow the passage of materials across the vessel wall. Typically, continuous capillaries are found in the brain and lungs. **Fenestrated capillaries** have fenestrated or porous endothelial cell membranes and are typically found in the kidneys and villi in the small intestine. **Sinusoids** are large capillaries with large fenestrations and large intercellular clefts; they are typically found in the liver and the spleen. A more detailed discussion on types of capillaries is found in Chap. 1.

Obviously, the wall transport properties and modes of transport depend on the particular type of capillary. As seen in Chap. 2, simple diffusion is one mode of transport of materials between the capillary lumen and the interstitial fluid compartment. Substances move down their concentration gradients, from regions of large concentration to regions of low concentration. This description applies to oxygen, carbon dioxide, amino acids and hormones. Diffusion of some substances may occur through fenestrations, intercellular clefts and also through the endothelial cell itself, by penetrating the plasma membrane of endothelial cells.

Clearly, the difference in the types of capillaries will have an impact on the transport of materials across the vessel wall. Some materials can also cross the capillary wall by another form of transport called **transcytosis**, whereby some substances in the blood plasma are trapped within a pinocytotic **vesicle**, which then enters the endothelial cell, moves across it and exits and releases the substance on the other side of the wall. **Filtration** and **reabsorption** are transport modes resulting from pressure gradients. These bring in the concept of **bulk flow**, understood as the process in which a large number of particles in a fluid move together in the same direction. This is also known as **advective transport**, sometimes also known as **convective transport**. As seen in Chap. 2, advective transport is much faster than diffusion transport, for sufficiently large spatial scales.

3.1.3 Plasma-ISF Exchange and Lymph Absorption

Figure 3.3 depicts in more detail the process of fluid movement involving three extracellular fluid compartments: blood plasma in a systemic capillary lumen; interstitial fluid in the interstitial space and lymphatic fluid, or lymph, in the lymphatic system.

Let us consider the fluid filtration at the capillary wall between the capillary lumen and the interstitial space, that is between blood plasma and ISF. The section of blood vessel shown in Fig. 3.3 presents two distinct portions, the arterial end (red) of the capillary on the left-hand side receiving oxygenated blood from arterioles, and the venous end (blue) absorbing oxygen-depleted blood into the venous system, to be transported to venules and back to heart. Note the fall in blood (hydrostatic) pressure between the the arterial end, $BHP = 35 \text{ mmHg}$ and the venous end of the capillary, $BHP = 16 \text{ mmHg}$. Here we adopt the notation of [1]. The pressure difference $\Delta BHP = (35 - 16) \text{ mmHg} = 19 \text{ mmHg}$ ensures blood flow from the arterial end towards the venous end.

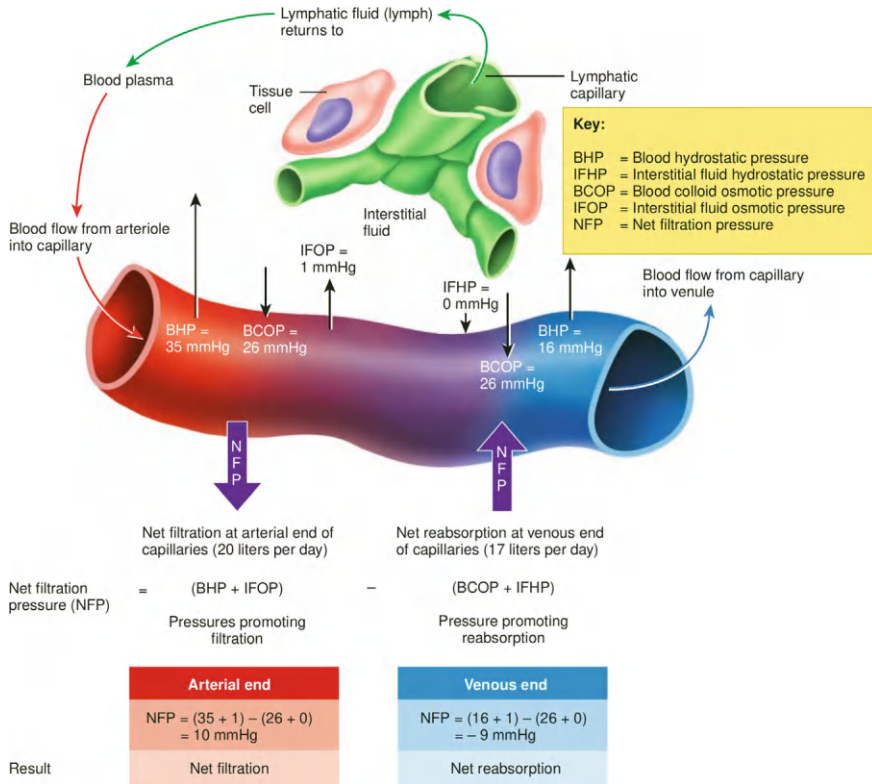


Fig. 3.3 Blood capillary exchange according to the Starling forces and absorption of interstitial fluid into the lymphatic system at lymphatic capillaries (from Tortora and Derrickson [1], with permission)

As already discussed in Chap. 2, the exchange of materials across the capillary vessel wall depends on two pressure differences. First, the hydrostatic pressure difference between the plasma hydrostatic pressure $P_c = BHP$ inside the capillary and the ISF hydrostatic pressure $P_i = IFHP$ in the interstitial space. Then the osmotic pressure difference between the plasma osmotic pressure $\pi_c = BCOP$ inside the capillary and the ISF osmotic pressure $\pi_i = IFOP$ in the interstitial space. As seen in Chap. 2, the Starling principle gives an expression for the net filtration of fluid across the vessel wall involving the two pressure differences. The Starling equation for fluid filtration is

$$J_v = L_p S [(P_c - P_i) - \sigma(\pi_c - \pi_i)] . \quad (3.1)$$

Net filtration results from the **hydraulic push** ΔP minus the **osmotic suction** $\Delta\pi$. Here σ is the reflection coefficient and accounts for the **imperfection** of the vessel wall as a membrane. Further details are found in [3] and [1]. See also

the classical works of Katchasky [4], Kedem and Katchasky [5] and Rippe and Haraldsson [6]. In the example illustrated in Fig. 3.3, the reflection coefficient has been assumed to be $\sigma = 1$. Moreover, the ISF hydrostatic pressure has been assumed to be $IFHP = 0$ mmHg and the ISF osmotic pressure has been taken as $IFOP = 1$ mmHg. Moreover, the pressures values have been re-arranged from Eq. (3.1). The pressure at the arterial end is $NFP = 10$ mmHg, resulting in filtration, and $NFP = -9$ mmHg at the venous end, resulting in reabsorption.

Remarks:

1. According to Levick [3], most tissues are normally in a steady state of filtration. That is, normally most tissues **do not continuously reabsorb fluid** at the venous end of the capillary, as stated earlier. But Levick's assertion is suitably modified when accounting for different kinds of tissue and transient changes of Starling forces.
2. Overall, filtration and reabsorption do not balance each other. It is estimated that about 15% of filtered blood plasma out to the ISF is not reabsorbed back into the capillary lumen.
3. As seen in Fig. 3.3, top part, the interstitial fluid ISF is connected to the lymphatic system. In fact, the excess of ISF fluid, not reabsorbed into the vascular system, enters the lymphatic system and is transported back to the venous system and thus to the circulation. See Sect. 3.2 for more details.

3.1.4 The Cell, Its Structure and Functions

The cell is the smallest living unit in the human body. Cell biology is a huge scientific subject by itself and much of the background is found in numerous human physiology books; see for example those by Tortora and Derrickson [1] and Silverthorn [7]. The cell represents the second level of structural organization in the body, after the chemical level comprising atoms and molecules. Higher levels, in succession, include tissues, organs, systems and the full body, formed by all its parts functioning together. The human body contains about 200 types of specialized cells, responsible for a multitude of functions. Central to the functions of a cell are huge numbers of chemical reactions aimed at supporting life processes. Figure 3.4 demonstrates the main components of a cell.

Structures of the Cell

Essentially, the cell structures of a typical cell include the cell membrane, the cytoplasm and the nucleus, as depicted in Fig. 3.4. In what follows we briefly discuss these cell components.

The **cell membrane**, also called **plasma membrane**, is the outer surface of the cell that physically separates its inner contents from the external environment, the interstitium, providing also structural support to the cell in order to maintain its shape.

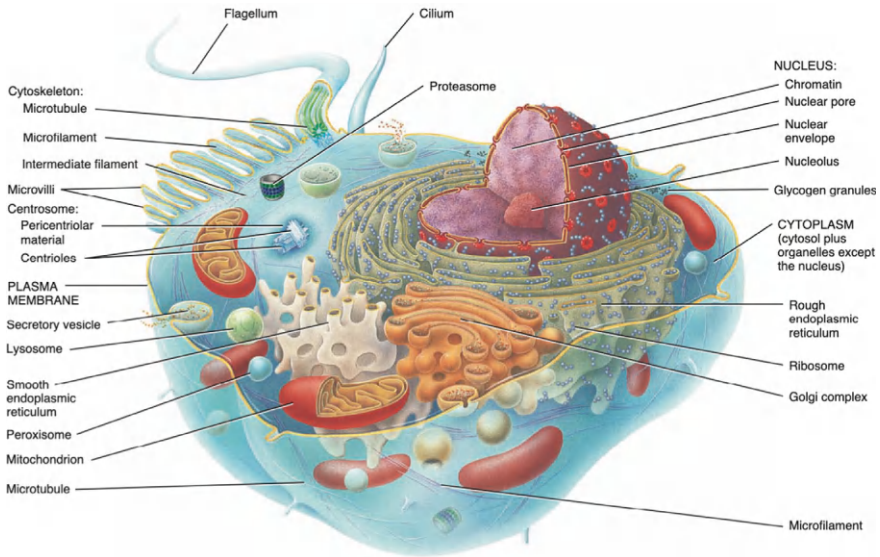


Fig. 3.4 Structures of a typical cell: cell membrane (or plasma membrane), cytoplasm and nucleus (from Tortora and Derrickson [1], with permission)

The cell membrane is akin to the wall of the capillary that separates blood plasma in the circulatory system from the interstitial space. The cell membrane constitutes a selectively permeable barrier that regulates the flow of materials into and out of the cell. In particular, the cell membrane allows the transport of ions and nutrients from the interstitium to the interior of the cell. Vice-versa, the cell membrane allows the transport of cell excreta, that is cellular wastes, from the interior of the cell to the exterior, the interstitium. The cell membrane itself contains proteins that enable communication between the cell and its environment, as well as between cells. See Fig. 3.4. Also, cell membrane proteins provide structural support to the cell and create specialised junctions between cells. Cells are not isolated objects floating in interstitial fluid. Cells form tissue, in which cells are held together by support structures.

The **cytoplasm** of a cell is formed by all the contents in the interior of the cell between the plasma membrane, and the nucleus. Such space in turn includes **organelles** and the fluid inside the cell, the intracellular fluid (ICF), also called **cytosol**. The **cytosol** is composed of water, dissolved substances and suspended proteins, such as enzymes and other molecules. There are several types of organelles, each creating sub-compartments and performing specific functions. Examples of organelles shown in Fig. 3.4 include Golgi complex, cytoskeleton and the specially important **mitochondria**.

The **nucleus**, a large organelle, contains DNA and is the cell's control centre; it is enclosed by the **nuclear envelope**, which separates the nucleus from the cytoplasmic space. The nuclear envelope is a double-membrane structure pierced by

nuclear pores, which participate in the transport of materials between the nucleus and surrounding cytosol in the cytoplasmic space of the cell. The nucleus contains randomly scattered granular material, or **chromatin**, and some **nucleoli**, usually between one and four. Chromatin are composed of DNA and proteins, while nucleoli contain genes and proteins.

Exchange of Materials Across the Cell Membrane

As already stated, the very final mission of the cardiovascular system is to supply oxygen and nutrients to the cell across the cell membrane, and to remove excreta from cell metabolism, e.g. carbon dioxide. The multitude of functions of the cell rest on a very complex architecture of the cell membrane. For a century or so, scientists have been postulating a double-layer structure model for the cell membrane consisting of proteins, lipids and usually a small amount of carbohydrate. For example, the cell membrane of a red blood cell contains 49% protein, 43% lipid and 8% carbohydrate. The currently adopted model for the cell membrane is referred to as the **fluid mosaic model** [7].

To conclude this brief discussion on the cell, we refer the reader to Fig. 3.5, which shows the chemical concentrations in the three body fluid compartments, namely the intracellular fluid, the interstitial fluid and blood plasma. The graphic bars show that in general body fluid compartments are in chemical disequilibrium. The chemical concentrations of ISF and blood plasma are essentially identical, apart from protein concentration which is virtually zero in the interstitial fluid. Recall that at the capillary wall, there is virtually no transport of proteins from the blood plasma into the interstitial fluid.

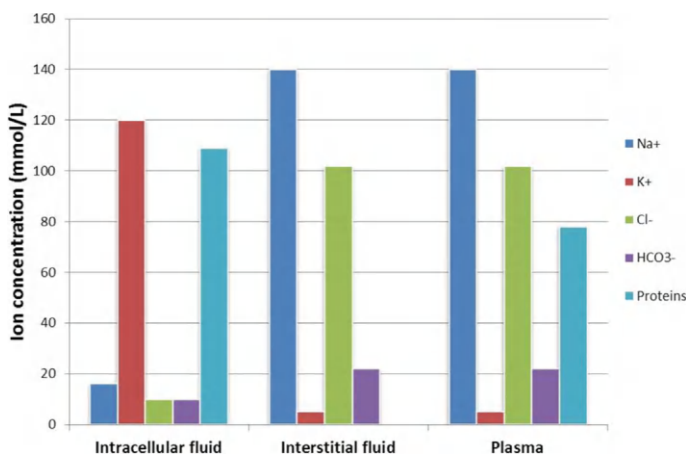


Fig. 3.5 Chemical concentrations in three body fluid compartments: plasma, ISF and ICF (Courtesy of Dr. Brigitte Toro, data from Silverthorn [7])

3.2 The Lymphatic System

Figure 3.6 depicts the human lymphatic system, which is a proper vascular system akin to the cardiovascular system, though not circulatory. This section is devoted to a succinct introduction to the lymphatic system, emphasising its fluid dynamics and transport features, as well as its anatomical and functional connections to the interstitial fluid and blood compartments. Appropriate references are given for further study.

The lymphatic system has analogous components to (i) blood, (ii) blood vessel network and (iii) the pumping heart. In the lymphatic system the fluid is called **lymph**; the vessel network is a complex network of lymphatic vessels, coupled to lymphoid organs; and the pumping mechanism relies primarily on pressure gradients and spontaneous contractions of the lymphatic vessel wall, coupled to a system of valves. Gaspare Aselli is credited [8] for having first recognised lymph and lymphatic vessels in 1622, while William Hunter, more than a century later, is credited for having provided a more complete view of the lymphatic system [3]. The lymphatic system collects excess interstitial fluid from the interstitium, becoming lymph, and transports it back to the circulation, carrying with it proteins, waste products, immune cells, viruses, bacteria and even cancer cells. The three functions of the lymphatic system are:

1. Body-fluid balance preservation through anatomical and functional connections of three major fluid compartments: blood, interstitial fluid and lymph;
2. Nutritional function through connection between the lymphatic system and the digestive system and
3. Immune function through connection between the lymphatic system and the immune system.

The lymphatic system is present in all tissues of the body, but is absent in **avascular tissues**, such as cartilage, the epidermis and the cornea of the eye; it is also absent in bones, superficial portions of the skin, portions of the spleen and red bone marrow. Interestingly, also the brain parenchyma lacks a lymphatic system, which has been the source of much controversy over the years. Virtually all classical books on brain anatomy and physiology state that lymphatic vessels are absent in the brain. Very recently, however, lymphatic vessels in the brain meninges were discovered, independently, by Louveau et al. [9] and Aspelund et al. [10]. For further discussion on this point see Chap. 4 and references therein.

3.2.1 The Lymphatic System and Bodily Fluids: Overview

The lymphatic system is a one-way fluid transport system that collects excess interstitial fluid from the interstitium and transports it to the circulation. Figures 3.3 and 3.7 depict the connection between various body fluids: blood in the arterial end of

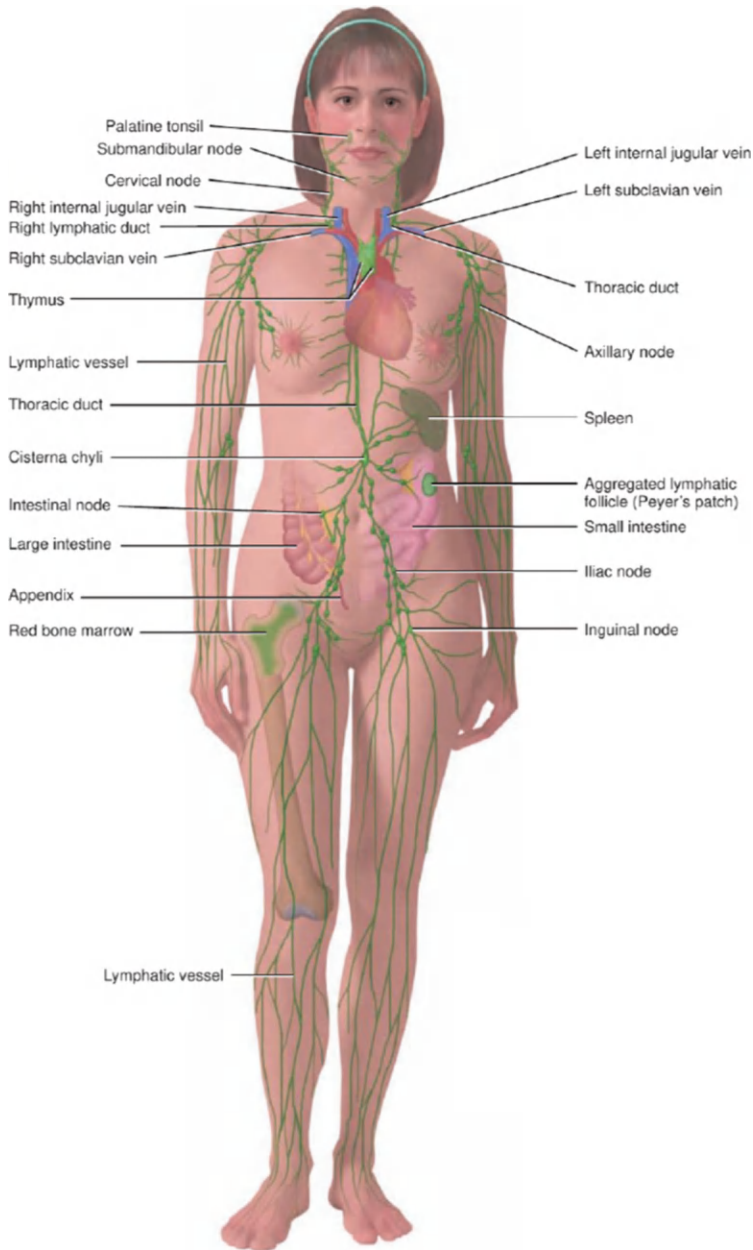
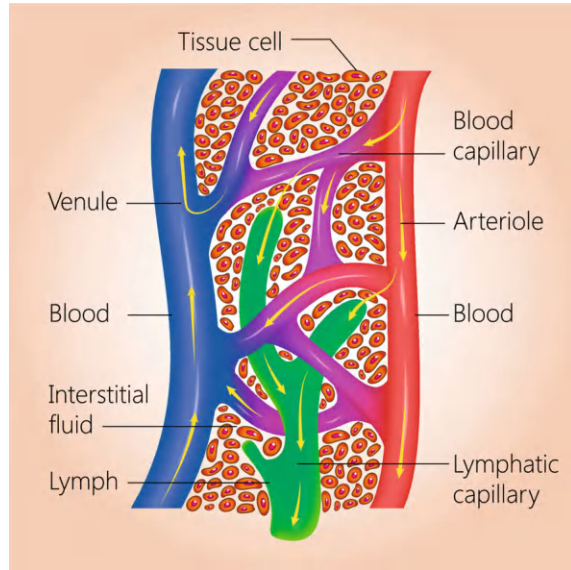


Fig. 3.6 The lymphatic system in the human body. Excess interstitial fluid from the interstitium enters the lymphatic system network at lymphatic capillaries; the fluid, then called **lymph**, drains into the venous system at: **a** the end of the **thoracic duct**, near the junction between the left internal jugular vein and the left subclavian vein; and **b** the end of the **right lymphatic duct**, near the junction between the right internal jugular vein and the right subclavian vein. Lymph is also absorbed directly into the venous system of lymphatic nodes (from Tortora and Derrickson [1], with permission)

Fig. 3.7 Body fluids: blood at the arterial end of capillary; interstitial fluid between cells; blood at the venous end of capillary and lymph in lymphatic vessel. Lymph is transported to the venous system and thus to the circulation (from Shutterstock, with permission)

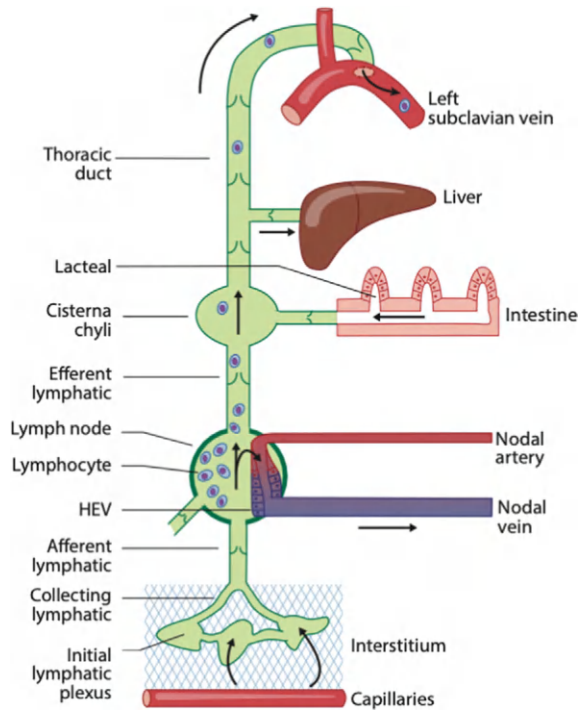


the capillary; interstitial fluid between cells in the interstitium; blood in the venous end of the capillary; and lymph in the lymphatic vessel. As already discussed in Sect. 3.1.3, at the arterial end of the capillary there is net filtration of plasma into the interstitium and at the venous end of the capillary there is net absorption of IFS into the blood vessels. Levick [3] disputes the net-absorption function of venular capillaries; see Sect. 3.1.3. Starling forces govern this process of filtration and absorption; see Fig. 3.3. The excess IFS not absorbed into the venous end of the capillary, which is estimated to be 10% in [7] and 17% in [2], enters the lymphatic system at the level of the terminal capillary of the lymphatic network. Figure 3.8 shows the principal components and routes for lymph formation, flow and drainage. Lymph is formed in the interstitium, flows along the lymphatic network, passing through nodes and draining into the neck veins. Part of the lymph is absorbed in the node, directly into the circulation.

Lymph is transported back to the circulation, through a very complex network of lymph vessels and nodes, as shown in Figs. 3.6 and 3.8. Three main structures of the lymphatic system stand out. First, the complex network of lymphatic vessels, which includes different types of vessels. Then there are numerous lymphatic nodes, between 600 and 700 in humans, according to [8]. Examples of nodes include the cervical lymph nodes, the lumbar lymph nodes and the pelvic lymph nodes. Figure 3.6 depicts all major components of the lymphatic system, from lymphatic capillaries all the way to the neck veins.

As we are mainly interested in the fluid dynamics and transport aspects of the lymphatic system, we have so far placed the emphasis on the network of lymphatic vessels and nodes. However, the lymphatic system also includes **lymphoid organs**,

Fig. 3.8 The lymphatic system. Sketch of principal components and routes for lymph flow. Lymph is formed in the interstitium, flows along the lymphatic network, passing through lymph nodes and draining into the neck veins. Lymph is also formed in the intestine. Part of the lymph is absorbed in the High Endothelial Venules (HEV) of the node and returned directly to the circulation. The HEV also facilitate the traffic of immune cells between the nodal lymph and the nodal vasculature (from Herring and Paterson [11], with permission)



which concern the immune function of the lymphatic system. Lymphoid organs are divided into **primary lymphoid organs** and **secondary lymphoid organs**. The primary lymphoid organs include the bone marrow and the thymus, where immune cells are created. Secondary lymphoid organs is where immune response takes place and include lymph nodes, the spleen, tonsils and specialised mucosa tissue. Of these, the lymph node plays in addition a fundamental role in the fluid mechanics and transport of lymph and will therefore be considered in more detail in Sect. 3.2.4.

Lymph Flow

The fluid dynamics of the lymphatic system of Fig. 3.6 is summarised in Fig. 3.8, which describes the lymph flow, starting from ISF in the interstitium, into the lymphatic system and draining into venous system, either at the lymphatic node or in the left and right subclavian veins in the neck. More precisely, the following steps are observed:

1. Interstitial fluid in the interstitium enters the lymphatic system at the level of the lymphatic capillaries as depicted, for example, in Fig. 3.9, top frame. Lymph also enters the lymphatic system in the intestine to fulfil the nutritional function, see Fig. 3.9, bottom frame; lymphatic capillaries in the intestine are called **lacteals** and the lymph is called **chyle**.

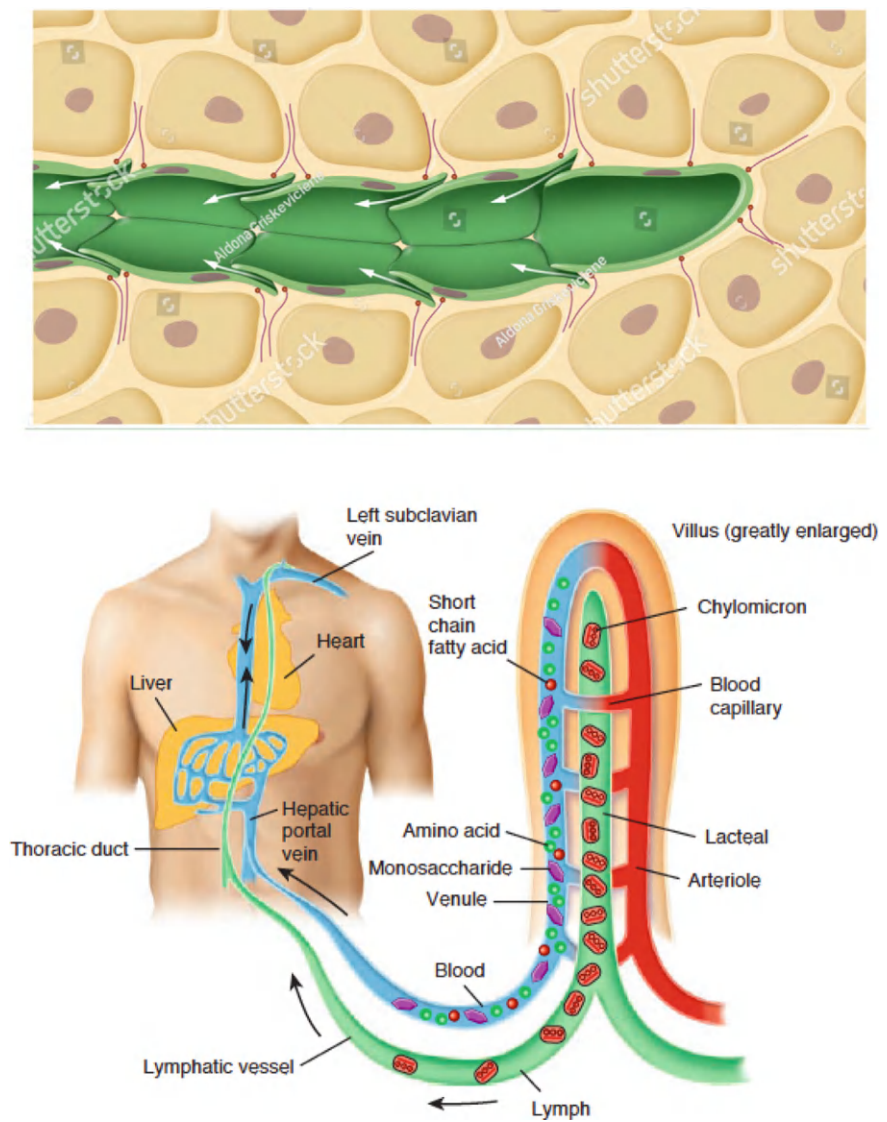


Fig. 3.9 Lymphatic capillaries. Top frame shows interstitial fluid entering the lymphatic system at the lymphatic capillary (from Shutterstock, with permission). Bottom frame: lymph (chyle) transporting nutrients from intestine lymphatic capillaries (lacteals) (from Tortora and Derrickson [1], with permission)

2. Once inside the lymphatic capillaries the fluid is called lymph, which then flows along a series of conduits, including **collecting lymphatics**, which are equipped with lymphatic valves to ensure one-way flow, as in veins in the circulatory system. See Fig. 3.11.
3. Lymph flow continues along lymphatic vessels to enter **lymph nodes** through **afferent** lymphatic vessels penetrating the node. Lymph nodes are very complex three-dimensional compliant structures. See Figs. 3.8 and 3.12.
4. Lymph flow comes out of the node through **efferent** lymphatic vessels. Part of the fluid is absorbed into the venous system inside the node, to be returned directly to the circulation. This takes place in the so-called **HEV**, High Endothelial Venule.
5. Much of the immune function of the lymphatic system is realised inside the lymph node, in conjunction with other lymphoid organs.
6. Post-nodal lymph continues its course towards two main emptying sites into the venous circulation: the end of the **right lymphatic duct** and the end of the **thoracic duct**. Figure 3.13 depicts the gross asymmetry in the lymph emptying process.

Following Tretbar et al. [8], the lymphatic transport system is divided into three categories, namely (a) the superficial system that drains the skin and subcutaneous tissues; (b) the deeper system that drains muscles, joints, synovial sheaths and bones, and (c) the visceral system that drains the small intestine, spleen, liver, thymus and lungs.

3.2.2 *Lymphatic Capillaries and Lymph Formation*

As already pointed out, the portion of interstitial fluid in the intercellular space that is not absorbed into the venous end of blood capillaries, enters the lymphatic capillaries and thus into the lymphatic system and ultimately into the circulation. The ISF portion that enters the lymphatics at lymphatic capillaries, according to [7], is of the order of 10% and according to [2], of the order of 17% of the total ISF. Lymph is formed at the rate of about 3 litres per day. The quantity of lymph formation varies significantly in the literature and is based on thoracic duct discharge data.

Lymphatic Capillaries

Lymphatic capillaries, also called **initial lymphatics** or **terminal lymphatics**, constitute the start of the lymphatic vessel network, as depicted in Figs. 3.6, 3.7 and 3.8. Figure 3.9, top frame, shows interstitial fluid entering the lymphatic capillary in the interstitium; the bottom frame shows lymph, called **chyle** here, transported from specialised lymphatic capillaries, called **lacteals**; these carry dietary lipids, ultimately into the blood circulation, to fulfil the nutrition function of the lymphatic system. See also Fig. 3.8. Two thirds of all lymph in humans is formed in the liver and intestines. Lymphatic capillaries are tiny vessels, though of larger diameter than

blood capillaries, of about 10 to 50 microns. Their wall is constituted by a single layer of endothelial cells, with incomplete basement membrane. The endothelial cells are separated by clefts (gaps) of about 14 nm; the wall thickness ranges from 50 to 100 nm.

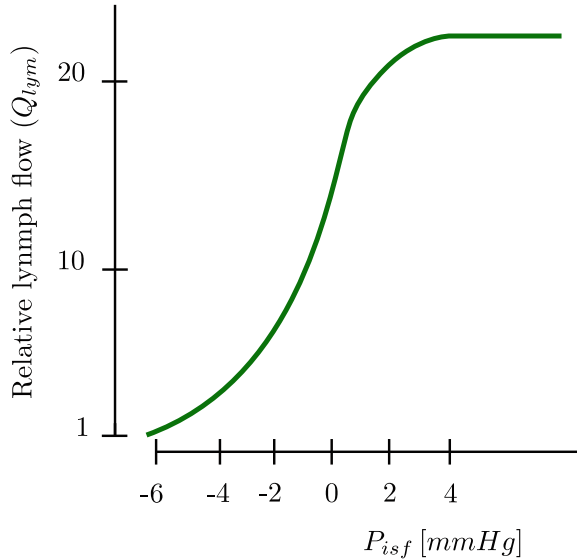
The unique structure of lymphatic capillaries allows ISF from the interstitium to enter the capillary vessel, but not to return to the interstitium. The one layer of overlapping endothelial cells is attached to surrounding tissues by means of anchoring filaments, which contain elastic fibres. ISF enters the lymphatics by means of pressure differences, as discussed below. Lymphatic capillaries are the beginning of the lymphatic system network, as seen in Figs. 3.6, 3.8 and 3.9. The remaining downstream elements of the network will be discussed in subsequent sections.

Lymph Formation and Composition

Lymph composition [8] is essentially that of ISF in the interstitium and includes: 96% water, proteins, lipids, carbohydrates, enzymes, glucose, urea, hormones, dissolved gases (oxygen, carbon dioxide), cells (lymphocytes and macrophages), toxins, bacteria, viruses, cellular debris and other body wastes. Colloids are also present in lymph and include: sodium, potassium, magnesium, chloride, zinc, copper, calcium and phosphorous. The concentrations of colloids in lymph are about the same as those in blood plasma. Large particulate matter that cannot be absorbed directly into the blood across the venous end of the blood capillary enters the lymphatic system. Recall that small amounts of proteins from blood enter continuously the interstitium and minute amounts, if any, return to blood via absorption at the venous end of the blood capillary. Hence, the vital function of returning proteins to blood is performed by the lymphatic system.

The flow of lymph into lymph capillaries depends on the relative values fluid pressures of ISF and lymph, as the respective osmotic pressures are essentially the same. See Sect. 3.1.3 on Starling forces. Let us denote by P_{isf} the fluid pressure in the interstitium and by P_{lym} the fluid pressure of lymph inside the lymphatic capillary. If $P_{isf} > P_{lym}$ then ISF flows into the lymphatic capillary. If $P_{isf} < P_{lym}$ then the higher pressure inside the lymphatic capillary closes the endothelial gap, a virtual valve formed by overlapping endothelial cells, so that lymph cannot escape back to the interstitium. The pressure P_{isf} in turn depends on (a) blood capillary pressure, (b) decreased plasma osmotic pressure, (c) increased ISF osmotic pressure and (d) increased blood capillary permeability. Then, ISF pressure P_{isf} determines the rate of lymph formation Q_{lym} , as illustrated in Fig. 3.10 for the case of a dog's leg [2]. As P_{isf} increases Q_{lym} increases, but up to a point when P_{isf} becomes just above atmospheric pressure (zero relative pressure). After that, Q_{lym} becomes constant. Higher values of P_{isf} compress the external surfaces of lymph capillaries, impeding lymph flow into the lymph capillary. Note the range of negative ISF pressure, which in fact contributes to keeping tissues together [2]. Several factors affect lymph flow. These include surrounding skeletal muscle, movement of the body, pulsation of nearby arteries and compression of tissues externally. Under rest conditions the flow of ISF into lymphatics is essentially zero [2].

Fig. 3.10 Relative lymph flow into the lymphatic capillary as a function of ISF pressure P_{isf} in the interstitium. As P_{isf} increases Q_{lym} increases up to a constant value, when P_{isf} becomes just above atmospheric pressure (data from Hall [2])



Lymphatic Capillary Pump

The lymphatic capillary performs a pumping action on the lymph down the lymphatic network. As already explained, once the pressure P_{isf} inside the capillary is sufficiently high, the gap between the overlapping endothelial cells closes and the pressure P_{isf} promotes flow down the lymphatic network. In addition to this pumping action of the capillary, there is another pumping contribution resulting from **contractions of the capillary wall**. The walls of the capillary contain contractile filaments. Hall [2] suggests that at least part of lymph pumping results from rhythmical contraction of the lymphatic capillary wall. On the other hand, Levick [3] (page 207) argues that, from the fact that the wall of lymphatic capillaries lacks smooth muscle in most tissues, pumping in lymphatic capillaries can only be **extrinsic**, external to the lymph capillary. **Intrinsic** pumping derives from spontaneous contractions of the wall of lymphangions. Intrinsic pumping in larger lymphatic vessels down the network is a major determinant of unidirectional lymphatic flow, in combination with mechanisms that include the action of lymphatic valves. This point is addressed next.

3.2.3 Collecting Lymphatics and Lymph Propulsion

As depicted in Figs. 3.6, 3.8 and 3.9 the lymphatic system consists of a network of lymphatic vessels, nodes and lymphoid organs, starting from lymphatic capillaries and ending with lymphatic ducts that empty into the neck veins. The gradually increasing number of components of the chain, after the capillary, include

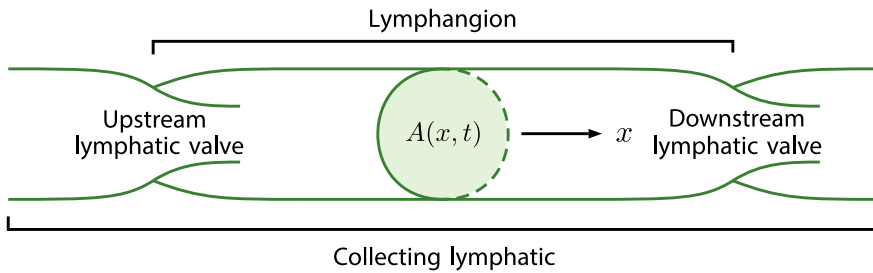


Fig. 3.11 Collecting lymphatic, lymphangion and valves (Courtesy of Dr. C. Contarino, data from Agarwal, Contarino and Toro [12]). The cross-sectional area $A(x, t)$ changes with position x and time t during lymphangion contraction

pre-collectors and **collecting lymphatics**. Collecting lymphatics, as sketched in Fig. 3.11, are a central component of the system, consisting of two valves enclosing the **lymphangion**. The lymphangion is a short, contractile lymphatic vessel, often described as a **mini-heart**. The spontaneous contractions of the lymphangion's wall produces a lymph pumping action along the network. This pumping action is required, as pressure in the lymphatic capillaries is about 0, while the pressure in the neck veins downstream is of the order of 3 mmHg.

The wall of collecting lymphatic vessels possesses abundant encircling smooth muscle, which effects spontaneous rhythmic contractions, giving rise to **intrinsic** propulsion. The frequency of these contractions is of the order of 8 to 15 cycles per minute. The rhythmic contraction of the lymphangion, in combination with the coordinated role of valves, is entirely analogous to the cardiac cycle event [3]. The following stages are observed:

- Diastolic filling phase: proximal (downstream) valve is closed and distal (upstream) valve is open.
- Isovolumetric contraction: both valves are closed.
- Ejection phase: proximal valve is open and distal valve is closed.
- Isovolumetric relaxation: both valves are closed.

Lymph pressures may reach 40 mmHg to 50 mmHg during the contraction cycle. In the ejection phase, the ejection fraction is about 25%. In this way the lymphatic system takes care of volume homeostasis by preventing excess of ISF in the interstitium.

It is accepted that lymph propulsion involves several mechano-biological, auto-regulatory systems to provide optimal flow in various scenarios. In particular, frequency of lymphatic contractions is found to depend on the circumferential stretch of the vessel wall and on the wall shear stress. See Kunert et al. [13], Gashev [14], Roose et al. [15] and Margaritis and Black [16], for example. The contraction mechanisms of lymphangions and lymph propulsion are major subjects of current research interest. See, for example, the recent communications on both experimental and mathematical modelling by Moore and Bertram [17]; Contarino and Toro [12] and Jafarnejad et al. [18], to name just a few.

3.2.4 The Lymph Node

This succinct introduction to the lymph node is mainly focused on its role in the fluid dynamics and transport of lymph in the lymphatic system, its connection to blood and body fluid homeostasis. We start from a description of the main physical structures of the node; there follows a discussion on the flow of lymph inside the node and the exchange of materials between lymph and blood. We end up with some remarks on the immune function of the lymph node.

Structures of the Node

Lymph nodes are **lymphoid organs** of the lymphatic system; they are connected through the network of lymphatic vessels and are scattered around the body, usually in groups, as illustrated in Fig. 3.6. Prominent lymph node groups are cervical lymph nodes (deep and superficial), axillary lymph nodes, lumbar lymph nodes, pelvic lymph nodes and inguinal lymph nodes. The human body is reported to have between 600 to 700 lymph nodes [8], of variable dimensions, with lengths between 1 mm and 25 mm. As depicted in Fig. 3.12 the lymph node exhibits a complex kidney-shaped three-dimensional structure, some details of which we discuss next.

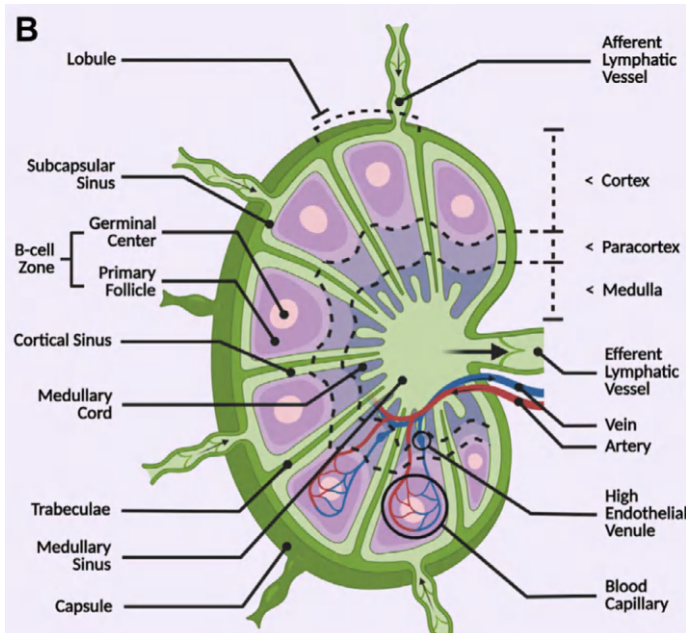


Fig. 3.12 Lymph node. Simplified sketch showing the main components, such as node capsule, afferent and efferent lymphatic vessel, various districts of the node parenchyma (cortex, paracortex, medulla) and node vasculature (arteries and veins) (from Shou et al. [19], with permission)

The **capsule** is the cover of the lymph node, which is constituted by a dense membrane of connective tissue that protrudes well into the node interior through finger-like structures called **trabeculae**. The capsule, along with trabecular reticular fibres and fibroblast, constitute the **stroma** of the lymph node, which provides structural support to the node. The trabeculae divide the parenchyma of the lymph node, circumferentially, into various compartments called **lobules**. Radially, the parenchyma is divided into **cortex**, **paracortex** and **medulla**. Some authors use, respectively, the terms outer cortex and inner cortex, for cortex and paracortex [1].

Lymph enters the node through **afferent** lymph vessels, one corresponding to each lobule; see Fig. 3.12. Lymph leaves the node through **efferent lymphatic vessels** at the **hilum**, not shown in Fig. 3.12, to continue its course along the lymphatics network. Usually, the number of afferent vessels is larger than that of efferent vessels. Both afferent and efferent lymphatic vessels are equipped with valves to ensure one-way flow. The node is prominently vascularised by arteries entering the node and veins exiting the node, also at the hilum. As will be discussed shortly, material exchange occurs between the lymph node and the blood in the interior of the node, resulting in net lymph absorption into the venous system, locally.

Lymph Flow and Material Exchange in the Node

Lymph entering the node through afferent vessels follows a series of tortuous conduits within the node before exiting through one or more efferent lymphatic vessels at the **hilum**. The lymph flow paths, coloured in light green in Fig. 3.12, are called **sinuses** and are traversed by reticular fibres which are likely to offer considerable resistance to flow. The sinuses contain incoming lymph, which transports lymphocytes and macrophages, amongst many other materials. The first channel is the **subcapsular sinus**, just beneath the capsule. From here, according to [19], 90% of lymph flows through trabecular sinuses around the lobule, reaching medullary sinuses, from which lymph drains into efferent lymph vessels to exit the node at the hilum. The remaining 10% of lymph percolates into the cortex and then into the paracortex. This portion of lymph is divided into two parts. One part drains into medullary sinuses to then exit the node through efferent vessels, while the other part is absorbed into paracortical blood vessels, returning some of the lymph directly to the blood circulation, right inside the node. Caution is required with figures regarding the amounts of lymph that drain into the venous system directly at the nodes and the rest of lymph that carries on flowing in the lymphatic network, eventually draining into the neck veins. This **fluid exchange** function of the node across the interface between the blood in the node and the lymph in the node, the **blood-lymph barrier**, has been known for some time and is realised through the so-called **High Endothelial Venules** (HEV). See Sect. 3.1.3 for details on the analogous case of plasma-ISF exchange at the blood capillary.

In 1982 Adair et al. [20] reported comparisons of prenodal and postnodal lymph flows and protein concentrations, in experiments performed on a dog popliteal lymph node. They concluded that lymph nodes modify lymph protein concentration and colloid osmotic pressure, apart from the case in which these are already at equilibrium

values. In other words, post-nodal lymph is not representative of interstitial fluid. More precisely, they found that virtually all of the change in the protein concentration in lymph inside the node resulted from the transport of protein-free fluid either into or out of the lymph in the node, across the blood-lymph barrier. The direction of lymph flow, in or out, across the blood-lymph barrier, depends on what is required to establish equilibrium of Starling forces; see Sect. 3.1.3.

In a subsequent study, Adair and Guyton [21] investigated the effect of increased venous pressure P_{ven} in the node on the efferent lymph flow Q_{ef} and efferent lymph protein concentration C_{ef} . They found that as P_{ven} increased efferent lymph flow Q_{ef} invariably increased and C_{ef} decreased. Their study shows that the lymph node performs as a fluid exchange chamber. As expected, such exchange across the blood-lymph barrier responds to the Starling forces, in which both node venous pressure and the lymph colloid osmotic pressure play a role. Importantly, venous pressure and the role of the venous system in general tend to be ignored in the literature; but see Chaps. 18 and 19 of this book.

There is agreement in the literature as to the absorption of lymph into the venous system of the node; see [20] and [21], for example. What is a matter of controversy is the actual amount of fluid absorbed. Following Shou et al. [19], this amount does not exceed 10%, while according to Tretbar et al. [8] this quantity would almost be 50%. Levick [3] also discusses this issue and by judging from his Fig. 11.17, the amount of lymph absorbed into the vasculature in the node would be close to 50%, which would agree with Tretbar et al. [8]. This means that estimates for daily lymph formation based on post-nodal lymph emptying in neck veins is only a part of the formed lymph in the body.

Immune Cells: Proliferation and Trafficking

Daily, many foreign substances may enter the body through three main portals of entry: the skin, the intestine tract and the lungs. Additional substances may be produced internally, such as cancer cells, that may also appear as foreign substances. It is the task of the immune system to defend the body from ensuing diseases. Recall from Chap. 1 that lymphocytes, the *major soldiers of the immune system battles*, include three types of cells: T cells, B cells and NK (natural killer) cells. Only 2% of lymphocytes remain in the blood circulation at any time, while the rest circulates from blood to ISF, to lymph, to lymph nodes and back to blood.

Much of the immune function of the lymphatic system is realised in the lymphatic node. The cortex of the node, see Fig. 3.12, houses lymphatic nodules, or follicles, which in turn house dendritic cells, macrophages and dense aggregates of B cells, proliferated in the node. One may distinguish between primary and secondary follicles. The paracortex of the node houses T cells and dendritic cells that enter the node from other tissues. Proliferation of T cells takes place in the paracortex; T cells then leave the node to migrate to tissues with antigenic activity. The **medulla** of the node contains macrophages, antibody-producing plasma cells and B cells that have migrated from the cortex into the medulla.

Activated lymphocytes in the node are released to the blood stream in lymph leaving the node through efferent lymphatic vessels; hence efferent lymph has a higher white blood cell count. B and T cells also migrate directly to blood within the node through specialised post-capillary venules, the **High Endothelial Venules (HEV)**. These venules are characterised by having a wall consisting of a single layer of **thick** endothelial cells, which possess surface receptors specific for B and T lymphocytes. These thick endothelial cells facilitate the movement of lymphocytes between the lymph in the node and the nodal blood circulation. Therefore, lymphocytes can leave the node either through lymph efferent vessels or directly through the nodal vasculature. In addition, HEV offer a route for **lymph absorption** directly into the nodal venous blood circulation, as alluded to previously.

3.2.5 *Lymph is Drained into Neck Veins: Gross Asymmetry*

The lymphatic network eventually drains its lymph content into the venous system in two places, as depicted in Fig. 3.13. The first site is the end of the **thoracic duct**, located near the junction between the left internal jugular vein and the left subclavian vein. The second site of lymph discharge is the end of the **right lymphatic duct**, placed between the right internal jugular vein and the right subclavian vein.

As demonstrated in Fig. 3.13 the draining of lymph into the venous system is grossly asymmetric. Lymph from the right top quarter of the body drains into the right lymphatic duct, while lymph from the remaining three quarters of the body, the lower half and the top left half, drains into the thoracic duct. In a resting human, according to Hall [2], 100 mL per hour of lymph drain into the thoracic duct, while 20 mL per day discharges through other channels. Assuming that these are the only two lymph draining sites, one can therefore estimate the total lymph flow as being 2 to 3 litres per day [2]. See previous remarks on daily lymph formation estimated from post-nodal lymph flow.

The overall fluid balance preservation function of the lymphatic system is seen through the following numbers. According to Levick, see Fig. 11.7 of [3], 20 litres per day of interstitial fluid are generated by filtration at blood capillaries. Of this amount, 8 litres enter the lymphatic capillaries to travel down to the afferent lymphatic vessels draining into lymph nodes. At lymph nodes 4 litres of lymph are absorbed into the venous system in the lymph vasculature, that is, half of the lymph is returned to the venous circulation at the lymph nodes. The remaining 4 litres of lymph exit the node through the efferent lymphatic vessels, flow through the lymphatic network, eventually draining into the neck veins. As already pointed out, there is disagreement amongst various authors as to the precise amount of lymph absorbed inside lymph nodes and hence the amount of formed lymph, as measured at the neck veins.

There are many aspects of the anatomy and physiology of the lymphatic system which remain uncertain; some of these are currently the subject of intensive research activity. Interestingly, in some ways the lymphatic system is analogous to the venous system in that it receives interstitial fluid at the lymphatic capillary, that then becomes

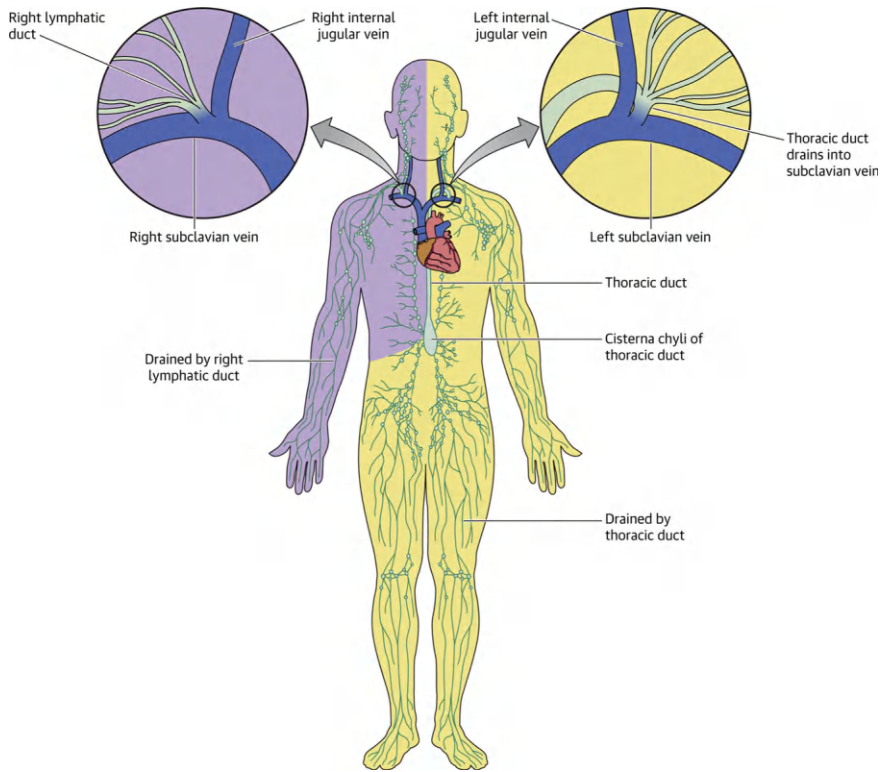


Fig. 3.13 Discharge of lymph into the venous system at two sites: the end of the right lymphatic duct and the end of the thoracic duct; see circled areas. Note the very evident asymmetry in the lymph draining process (from Fudim et al. [22], with permission)

lymph; this is transported back to the circulation, via the venous system. The venous system reabsorbs fluid from the interstitium at the venular end of the blood capillary and returns it to the circulation. Both the venous and lymphatic systems are intimately linked and play a major role in the **clearing of metabolic wastes** from the body. The clearing, or removal, function of these systems is not always sufficiently emphasised. A theme of current interest these days is the recent discovery of lymphatic vessels in the brain meninges [9, 10]. Such lymphatic transport system connects cerebrospinal fluid in the central nervous system with the peripheral lymphatic system in the cervical region. For further discussion on this point see Chap. 4 and references therein. Collectively, all of these strongly interconnected extracellular fluids in the central nervous system were recently termed **Neurofluids** in [23], as the acronym for a European research proposal. The term neurofluids has subsequently been adopted in [24–27], amongst others.

This section on the human lymphatic system has been introductory in character. The literature on the lymphatic system is vast. The fundamentals are found

in most textbooks on human anatomy and physiology [1–3, 7, 8]. There are many research publications and comprehensive reviews available, amongst which we mention Swartz [28]; Gashev [14]; Margaris and Black [16]; Roose and Swartz [15]; Kunert et al. [13]; Moore and Bertram [17]; Contarino and Toro [12]; Jafarnejad et al. [18]; Shou et al. [19], to name just a few.

3.3 Conclusions and Further Reading

In this chapter we have studied two extracellular fluids, interstitial fluid and lymphatic fluid. Attention has been drawn to material exchange processes between compartments and the unifying role of the venous system, noting that all fluids considered drain, ultimately, into the venous system. The contents of this chapter must be seen in the more general context of extracellular fluid compartments, the cardiovascular system studied in Chap. 1 and the cerebrospinal fluid system to be studied in Chap. 4. Advanced topics of current research interest have been given specialised references in each separate section. For basic physiological aspects of bodily fluids we suggest the following classical books: Hall [2]; Tortora and Derrickson [1]; Silverthorn [7]; Levick [3] and Herring and Paterson [11].

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Chapter 4

Cerebrospinal Fluid



Abstract This chapter focuses on the cerebrospinal fluid (CSF) compartment within the central nervous system (CNS). It builds upon the discussions of body fluid compartments in the preceding chapters. CSF is traditionally described as a clear, colorless liquid that surrounds the brain and spinal cord parenchyma, providing mechanical and chemical protection to these delicate structures, among other functions. Three major aspects of CSF stand out in the scientific literature: CSF generation, CSF motion (or circulation), and CSF absorption. All three areas of CSF physiology present knowledge gaps, which contribute to ongoing controversy. One of the few certainties regarding CSF is that most neurological disorders are linked to CSF pathophysiology. CSF delivers nutrients to the CNS and plays a crucial role in clearing toxic metabolites from the brain and spinal cord into the venous system, either directly or indirectly via the lymphatic system. Abnormalities in CSF physiology can lead to the accumulation of toxic substances in neural tissues, such as amyloid- β ($A\beta$) and tau, which are central to the pathogenesis of Alzheimer's disease. In this chapter, we first identify the CSF spaces in the CNS and describe the formation and composition of CSF. We then explore CSF dynamics and its functions in CNS physiology. Following that, we discuss two major recent discoveries relevant to CSF physiology: the meningeal lymphatic system and the glymphatic system. The next two sections specifically address CSF in the spinal canal and associated pathologies. The final section integrates all major CNS extracellular fluid compartments and their interactive dynamics, using an MRI study of a patient with idiopathic intracranial hypertension (IIH) and a healthy control for comparison. The chapter concludes with a summary and suggestions for further reading.

4.1 The Cerebrospinal Fluid System

Cerebrospinal fluid (CSF) constitutes a distinct extracellular fluid system that is relevant to the anatomy and physiology of the central nervous system (CNS); that is, the brain, the spinal cord, the neuronal parts of the ocular, and of the auditory and equilibrium systems. Classically, CSF is described in textbooks as a clear, colourless

liquid that surrounds the brain parenchyma and spinal cord, providing mechanical and chemical protection to the CNS delicate structures.

4.1.1 CSF Spaces of the Central Nervous System

Figure 4.1 depicts CSF filling craniospinal spaces (blue). The left frame depicts a global view of CSF spaces, including the cranial subarachnoid space (CSAS) surrounding the brain parenchyma, the spinal subarachnoid space (SSAS) around the spinal cord and the central canal inside the spinal cord. The right frame shows CSF in the cerebral subarachnoid space around the brain parenchyma in more detail, including the four cerebral ventricles (left lateral, right lateral, the third and the fourth), and the aqueduct of Sylvius, which connects the third and fourth ventricles. The CSF spaces are located between the arachnoid membrane and the pia mater. The craniospinal cavity in humans (called the cerebral cavity in [1]) houses the brain and the spinal cord; it has a total volume of 1.6 L to 1.7 L. The volume of CSF is about 150 mL, of which 25 mL fill the four ventricles and 125 mL fill the subarachnoid

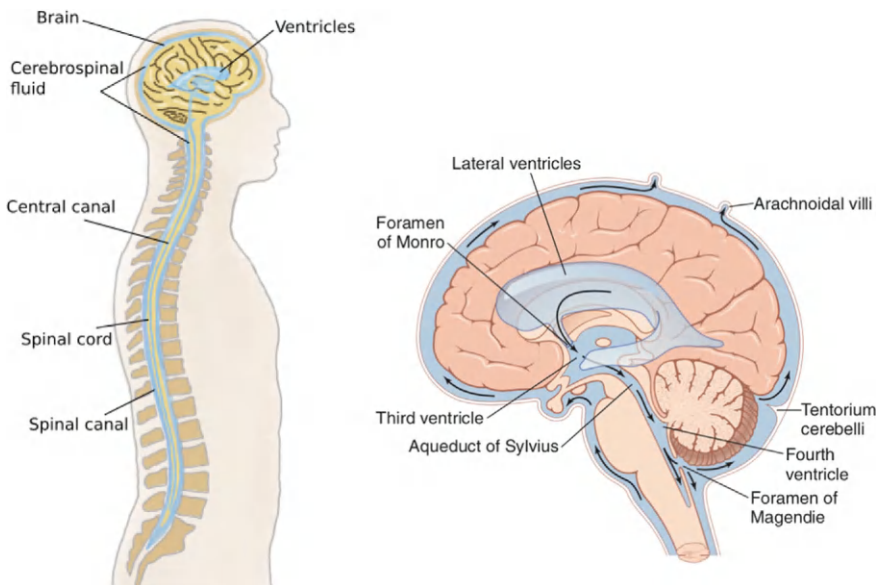


Fig. 4.1 Cerebrospinal fluid spaces in craniospinal cavities shown in blue. Left frame: global view of CSF spaces, including the cerebral subarachnoid space (CSAS), spinal subarachnoid space (SSAS) around the spinal cord and the central canal inside the spinal cord (courtesy of Dr. Mariia Dvoriashyna, University of Oxford, UK). Right frame: cerebral subarachnoid space (CSAS) around the brain parenchyma; the four cerebral ventricles (left lateral, right lateral, the third and the fourth) and the aqueduct of Sylvius connecting the third and fourth ventricles (from Hall [1], with permission)

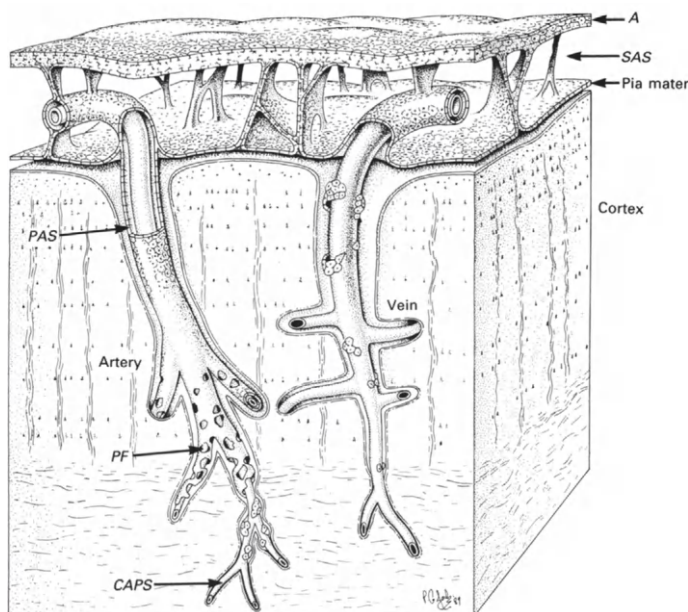


Fig. 4.2 Blood vessels in subarachnoid space penetrating the brain parenchyma and forming perivascular spaces filled with CSF (from Zhang, Inman and Weller [4], with permission)

spaces, cerebral and spinal [2]. These figures vary a lot in the literature and may, in any case, underestimate the true quantities [3]. Additional CSF-filled spaces in the CNS are found around blood vessels penetrating the brain parenchyma, as depicted in Fig. 4.2. The CSAS, labelled just SAS in the figure, lies between the arachnoid (A) and the pia mater. Large arteries and veins in CSAS penetrate the brain parenchyma and carry with them a layer of cells derived from the pia mater, thus forming a **perivascular space** between the pia-mater derived cells and the abluminal side of the vessel wall.

The left hand side of the figure shows the perivascular space around an artery, labelled PAS. The perivascular spaces follow arteries and veins as far as arterioles and venules, respectively. At this stage we note that in some publications, authors use the term **paravascular** (along side), instead of **perivascular** (around). Also, the definition of perivascular spaces and their boundaries changes significantly in the literature. Compare [5] with [6], for example. Two important additional CSF spaces are the **optic nerve sheath** around the **optic nerve** in the ocular system and the **cochlear aqueduct** in the auditory system. The latter communicates the CSF system to the perilymphatic fluid compartment in the inner ear. Chapter 5 provides more details about these CSF spaces, not always suitably identified in the CSF literature.

More generally, the brain parenchyma, the spinal cord and their communication channels with external compartments, are completely surrounded by CSF; see Hladky and Barrand [7]. This includes, for example, the perineural spaces of all pairs of

cranial nerves (twelve in all) and the vasculature. Fahmy et al. [8] conclude, from a recent MRI study, that *all CNS neuro- and vascular-communication channels are surrounded with CSF*. This means that (i) the perineural spaces of all nerves of the cranial and spinal nerves are filled with CSF, and (ii) the perivascular spaces of MRI-visible vasculature are also filled with CSF. Clearly identifying all CSF spaces in the CNS is most important when discussing CSF generation, CSF flow, CSF absorption and CSF functions, in both physiology and pathophysiology.

4.1.2 Formation and Composition of CSF

Classically, CSF is said to be secreted in the **choroid plexus** of the four cerebral ventricles, from the lumen of the blood capillary across the layer formed by the capillary wall and a layer of ependymal cells around the capillary's exterior surface; this is the **blood–cerebrospinal fluid barrier**. See Fig. 4.3. The choroid plexus is described as a growth of blood vessels, covered by a thin layer of **ependymal cells**, that projects into the temporal horn of each lateral ventricle, into the posterior portion of the third ventricle, and on the roof of the fourth ventricle [1]. Figure 4.3, left frame,

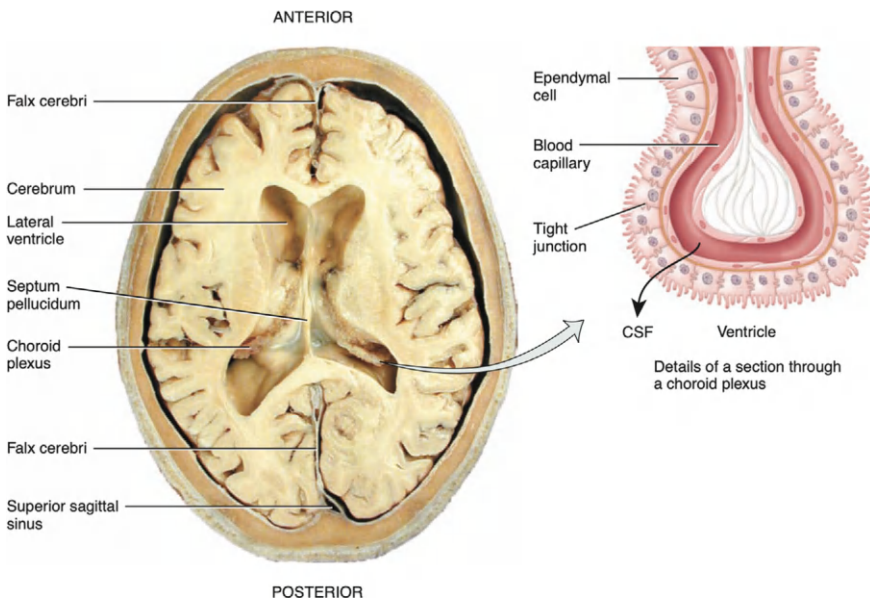


Fig. 4.3 Cerebrospinal fluid formation. Left frame: choroid plexus on the roof of the four cerebral ventricles. Right frame: blood capillary and layer of ependymal cells around the blood capillary wall. CSF secreted into the ventricle lumen originates from arterial blood in the capillary. Ependymal cells with tight junctions form the **blood–cerebrospinal fluid barrier** (from Tortora and Derrickson [9], with permission)

exhibits the location of the choroid plexus for each ventricle. The right frame shows details of a cross section through the choroid plexus. CSF is generated at the rate of about 500 mL per day. As the total volume of CSF is estimated as 150 mL, the volume of CSF generated in a single day is 3 to 4 times the entire volume of CSF in the CNS. That is, the CSF volume renews itself about 3 to 4 times per day. All cerebral ventricles contribute to the generation of CSF, but the lateral ventricles are prominent, generating about 2/3 of the total CSF formation.

This classical description of CSF formation is disputed by many authors. Apart from the choroid plexus of ventricles, there are other sources of CSF production. The ependymal surface of all four cerebral ventricles is a source of small amounts of CSF secretion [1]. As stated in [10], this layer plays a role in cerebral fluid balance, and in secretion. Additional small amounts of CSF are also generated by arachnoid membranes and by the brain parenchyma itself, along perivascular spaces surrounding blood vessels that penetrate the brain, see Fig. 4.2. See the early works of Rennels et al. [11] and Cserr et al. [12]. CSF production depends on many factors. Also, the CSF generation varies strongly with time within the day. As in all tissues in the body, interstitial fluid (ISF) in the CNS is formed at the blood capillary wall in the brain and spinal cord parenchyma. In the CNS, the blood-ISF interface is called the **blood-brain barrier** (BBB). The endothelial cells forming the wall of brain capillaries are equipped with **tight junctions**, which make the BBB a very selective barrier between brain tissue and blood. Nonetheless, interstitial fluid is formed, which then drains into the brain interstitium. CSF and ISF exchange in the brain parenchyma at the perivascular spaces.

Table 4.1 shows the chemical composition of cerebrospinal fluid, as compared with blood plasma. See [7, 13] for reviews.

From Table 4.1 we observe the following:

1. Sodium ion concentration of CSF is approximately equal to that of plasma.
2. Chloride ion concentration of CSF is about 15% greater than that in plasma.
3. Potassium ion concentration of CSF is approximately 40% less than that of plasma;
4. Glucose concentration of CSF is 30% to 40% less than that of plasma.
5. pH of CSF is virtually identical to that of blood plasma.
6. Overall, as compared to blood plasma, CSF is lower in protein and K^+ , and higher in Cl^- and Mg^{2+} [5].

To be noted, CSF has a 99% water content, as compared to blood plasma that has 92% water content. Due to its optimal chemical composition, CSF provides chemical protection and the right environment for neural signalling in the CNS. Ionic composition is crucial to these functions, as slight alterations may cause serious disruption of action potentials and postsynaptic potentials. It is noted in addition that the osmotic pressure of CSF is approximately equal to that of plasma.

Table 4.1 Composition of CSF compared to that of blood plasma (data from Hladky and Barrand 2014 [13])

Component	CSF	Plasma
	<i>Concentration/mg (100 ml)⁻¹</i>	
Protein	16 to 38	6300 to 8500
Sugar	45 to 80	80 to 120
Amino acids	1.5 to 3	4.5 to 9
Creatinine	0.5 to 2.2	0.7 to 2
Uric acid	0.4 to 2.8	2.9 to 6.9
Urea	5 to 39	22 to 42
Colesterol	—	100 to 150
Lactic acid	8 to 25	10 to 32
Phosphate (inorganic)	3.4	4.7
	<i>Concentration/mmol kg⁻¹</i>	
Na ⁺	147	150
K ⁺	2.86	4.63
Ca ²⁺	1.14	2.35
Mg ²⁺	1.1	0.8
Cl ⁻	113	99
HCO ₃ ⁻	23.3	26.8
pH	7.3	7.4

4.2 CSF Dynamics and Functions

CSF functions are strongly dependent on CSF fluid dynamics and the multiple anatomical connections between the CSF spaces and other body fluid compartments. The transport role of CSF is highlighted. CSF is a two-way carrier fluid. First, it transports oxygen, glucose and other substances from blood to neurons and glial cells. Second, it removes metabolic wastes from the tissues back to the blood in multiple ways.

4.2.1 Flow and Drainage of CSF into the Venous System

The basic process of CSF generation and drainage into the circulation determines the physiological environment in the craniospinal compartment. Intracranial pressure (ICP), for example, is one of the many parameters that affect the physiology and pathology of the CNS. ICP depends in part on a delicate balance between CSF

production and absorption. Our understanding of these processes is currently undergoing substantial revision.

CSF Motion: Conventional Description

We start from the classical theory of CSF motion, as described in textbooks e.g. [1]. CSF is mainly formed in the lateral ventricles. See Fig. 4.1, right-hand side frame. From here CSF flows to the third ventricle through two narrow openings called **interventricular foramina**, also called **foramina of Monro**. From the third ventricle, after small amounts of CSF are added, flow of CSF continues to the fourth ventricle through the **aqueduct of Sylvius**, also called **cerebral aqueduct**. From the fourth ventricle, CSF exits through three openings on the roof of the ventricle. One such opening is called **foramen of Magendie**, or **median aperture**, through which CSF leaves the fourth ventricle and enters the **cisterna magna**, and hence the cerebral subarachnoid space CSAS. The cisterna magna is the CSF space behind the medulla oblongata, just below the cerebellum.

The other two openings, one either side, called **foramina of Luschka**, or **lateral apertures**, allow the efflux of CSF from the fourth ventricle into the subarachnoid space. The cisterna magna is continuous with the cerebral subarachnoid that surrounds the cerebrum and the spinal subarachnoid space that surrounds the spinal cord. The **central canal** configuration is highly variable amongst individuals and varies with age in each subject. If any, a very small amount of CSF flows in the central canal, for which it is usually neglected in most descriptions of spinal CSF. However, the central canal is relevant when it comes to the study of spinal canal CSF related pathologies; see Sects. 4.4.2 and 4.5. For more details see Hall [1], for example.

As to CSF drainage into the circulation. Conventionally, CSF is considered to be mainly absorbed directly into the venous system through the **arachnoid villi**, which are finger-like structures that project into the **superior sagittal sinus** and other minor venous sinuses. See black arrows in Fig. 4.1, right-hand side frame. Conglomerates of these villi form macroscopic structures called **arachnoidal granulations** that protrude into the sinuses. The endothelial cells covering the villi have sufficiently large vesicular passages through the cells that facilitate flow of CSF, dissolved protein molecules and also larger particles, such as red and white blood cells [1].

An additional well-established CSF drainage path is offered by perineural sheaths around the olfactory nerves from the brain to the mucous lining of the nose, exiting the skull through a perforation of the **cribriform plate** called **foramen magnum**. Along this route CSF reaches the conventional (peripheral) lymphatic system in the nasal submucosa [14]. Additionally, CSF also drains into the peripheral lymphatics along perineural sheaths around spinal nerves [2]. In the spinal subarachnoid space, the part of CSF absorbed by the epidural venous plexus and spinal nerve sheaths enters the peripheral lymphatic system, while the remaining CSF flows rostrally towards the cranial subarachnoid space during the cardiac cycle, as depicted in Fig. 4.5.

In summary, generated CSF, and partly ISF, are reabsorbed into the venous system through the following routes:

1. Direct drainage into the venous system through arachnoidal granulations.
2. Drainage into the peripheral lymphatic system through perineural spaces around the olfactory nerves in the brain and spinal nerves.

The second CSF drainage pathway is in effect an indirect route into the venous system and thus into the circulation. This is because the lymphatic system drains part of the lymph directly into the venous system at the lymph nodes and the rest into the neck veins, as described in Chap. 3.

CSF Drainage: New Developments

Our description of CSF dynamics so far has followed the classical theory of CSF found in most textbooks, such as [1, 9, 14, 15]. In recent years, however, the classical view of both CSF generation and reabsorption has been strongly questioned, in good part as the result of intensive research activities, largely motivated by a broad range of neurodegenerative and related diseases. See for example [6, 7, 16, 17], and the many references therein. The breadth of the research involving the CSF system has increased in recent years, partly owed to the so-called **glymphatic system** [5, 18–20], which proposes an explanation as to how CSF and ISF accomplish a lymphatic function in the brain parenchyma. See Sect. 4.3 for more details on meningeal lymphatics and the glymphatic system.

Recent developments include a newly established CSF drainage route via the meningeal lymphatics [21, 22], whereby CSF reaches the deep cervical lymph nodes of the peripheral lymphatic system. Another interesting new development on CSF drainage routes, with potentially broad implications, is reported by Hu and collaborators [23]. From a refined MRI study they conclude that there is a direct, substantial cerebral vascular contribution to the clearance of CNS substances. This clearance contribution is effected via direct CSF reabsorption across the blood-brain barrier into the venous system in the brain parenchyma. In other words, in addition to the direct CSF drainage into the venous system through arachnoidal granulations, there is another direct CSF drainage into the venous system inside the brain parenchyma. This new CSF drainage directly into the venous system offers a fast route for clearance of CNS substances, as compared to the indirect routes via the lymphatic system.

In summary, generated CSF, and partly ISF, are reabsorbed into the venous system in at least four ways, two of them already contained in the classical CSF theory, namely

1. direct drainage into the venous sinuses through arachnoidal granulations,
2. indirect drainage via the peripheral lymphatic system through perineural spaces around the olfactory nerves in the brain and spinal nerves. To these, two more drainage pathways may be added
3. indirect drainage to the venous system via the newly discovered meningeal lymphatics, and

4. direct drainage into the brain parenchyma venous system.

In any case, CSF ends up in the venous system and thus in the circulation. The CSF theory is still in active development. For some of the latest works and critical reviews of the glymphatic system see, for example, [3, 6, 7, 13, 16, 24–29], and the many references therein.

4.2.2 Functions of CSF

Cerebrospinal fluid performs multiple functions, amongst which we emphasise the mechanical and chemical protection to provide a stable environment for accurate neuronal signalling, and the lymphatic function to eliminate neural-tissue excess fluid and waste products into the venous system.

Mechanical and Chemical Protection

As illustrated in Fig. 4.1, CSF surrounds brain and spinal cord. CSF bathes all major solid structures of the central nervous system thus providing mechanical protection to the delicate neural tissue, acting as a shock-absorbing medium. CSF, brain and spinal cord have comparable specific gravity and thus under **physiological conditions** brain and spinal cord float in CSF, preventing contact of delicate brain and spinal cord tissue with hard surrounding structures. This protection function, however may fail, for example, due to a sudden blow to the head. The protection may also fail without actual physical contact with the head, but due to rapid acceleration or rapid deceleration, as in vehicular accidents and in the so-called **shaken baby syndrome**. Central nervous system fluids are tightly regulated in order to provide an optimal chemical environment for accurate neuronal signalling. For example, an optimal environment requires low concentrations of K^+ , Ca^{2+} and protein, in order to optimise signalling and minimise neurotoxicity. See Table 4.1. Even slight changes in the ionic composition of CSF within the brain can seriously disrupt production of action potentials and postsynaptic potentials [9]. Moreover, neuronal and astroglial waste must be promptly removed.

Lymphatic Function of CSF

As is well known, the brain has a very high metabolic rate. Accounting for only 2% of the total body mass, the brain accounts for 15% of the total body metabolism under resting conditions. This figure can increase by as much as 150% under exercise conditions [1]. It is therefore puzzling [19] that the parenchyma of the brain, and spinal cord, does not have a lymphatic system, as elsewhere in the body. The high metabolic rate of neurons requires a means for rapid elimination of their metabolic by-products. As seen in Chap. 3 the lymphatic system elsewhere in the body is characterised by a network of lymphatic vessels, nodes and lymphoid organs, the functions of which

include removal of excess interstitial fluid and clearance of metabolic wastes and toxic materials. The absence of a lymphatic system in the brain parenchyma continues to be true, in spite of the recent discovery of a meningeal lymphatic system [21, 22].

As a matter of fact, the CSF system, in combination with ISF, performs the function of a lymphatic system, though the precise mechanisms remain elusive. As depicted in Fig. 4.2, arteries and veins penetrate the surface of the brain, carrying with them a layer of pia mater and thus forming a perivascular space between the pia mater and the vessel wall. The perivascular spaces follow arteries and veins as far as arterioles and venules [1], thus forming a vast fluid network that emulates the lymphatic system present in the rest of the body; see Chap. 3. As elsewhere in the body, fluid and solutes are transported across the BBB to form ISF that fills the interstitial space. The excess of the combined materials CSF and ISF is drained along perivascular spaces into the subarachnoid spaces and then reabsorbed into the venous system and other conduits, as described in Sect. 4.2. The transport function of perivascular spaces also serves to eliminate other particulate debris out of the brain, for instance dead white blood cells and other infectious matter produced by infections. Effectively, CSF allows exchange of nutrients and waste products between the blood and nervous tissue.

The lymphatic function of the CSF system has long been recognised, for example in the pioneering work of Cserr and collaborators [12]. Currently, the precise lymphatic function of the CSF system is the subject of much research, as well as controversy, as discussed in Sect. 4.3.

4.3 Lymphatics and Glymphatics

Here we briefly discuss two topics of interest in CNS fluid physiology, the first concerns the discovery of a meningeal lymphatic system in 2015 and the second regards the so-called glymphatic system put forward in 2012.

4.3.1 *The Discovery of Meningeal Lymphatics*

For centuries it has been stated in the literature that the central nervous system lacks a lymphatic system, as present in other parts of the body. As already mentioned, this fact was partly disproved in 2015, see [21, 22]. Both groups of researchers discovered in mice, a functional lymphatic network in the meningeal layers, not in the brain parenchyma. They noted that cerebrospinal fluid is drained into the meningeal lymphatic vessels, which in turn communicate directly with deep cervical lymph nodes in the periphery. According to the authors in [21] *the newly discovered meningeal lymphatic vessels are a novel path for cerebrospinal fluid drainage and represent a more conventional path for immune cells to egress the central nervous system*. Further studies have confirmed that meningeal lymphatic vessels are also

present in humans. See Visanji, Lang and Muñoz [30], Louveau et al. [31] and the review by Hershenhouse et al. [32].

4.3.2 The Glymphatic System

Overview

During the last decade there has been intensive research activity aimed at explaining the lymphatic function in the brain parenchyma, in the absence of a conventional lymphatic network in this district. Nedergaard and collaborators published a series of papers reporting studies in live mice, starting from Iliff et al. [18]. Their studies revealed rapid exchange of CSF and ISF in the brain in a brain-wide transport system, as represented in Fig. 4.4, which identifies the salient three elements of the proposed glymphatic system, namely

1. a **paraarterial** CSF influx route,
2. a **paravenous** ISF clearance route, and
3. an intracellular transastrocytic path that couples the two extracellular perivascular routes 1. and 2.

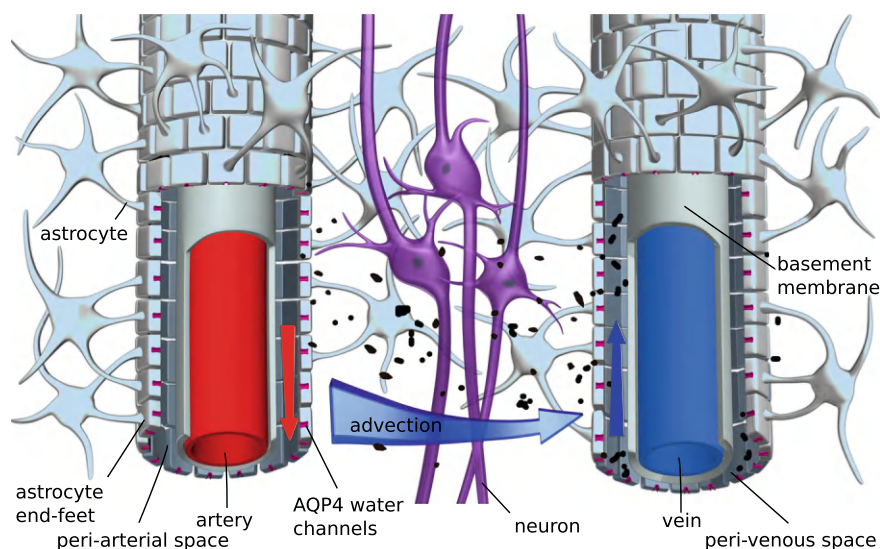


Fig. 4.4 Representation of main components of the glymphatic system. Cerebrospinal fluid flows along arterial perivascular space, shown on left-hand side, enters the interstitial fluid space, facilitated by aquaporin-4 water channels. The fluid flows advectively through interstitial spaces transporting solutes along, reaching the venous perivascular space, shown on the right-hand side, and leaving the brain (courtesy of Professor Helene Benveniste, Yale Medical School, USA)

For an introduction to the glymphatic system we recommend the concise presentation of Nedergaard [19] and the extended work of Jessen et al. [5], the recent work of Wardlaw et al. [33] and the review of Benveniste et al. [20].

It is useful to recall that the central nervous system vasculature possesses some unique features. In particular, arterioles, capillaries and venules in the brain parenchyma are surrounded by astrocytic vascular endfeet, which create the outer wall of the perivascular space surrounding the vasculature. Therefore, the perivascular space is bounded internally by the abluminal surface of the blood vessel and externally by the **astrocyte end feet** attached to the vessel [5, 19]. Deeper into the brain parenchyma, just before the capillary level, the perivascular space disappears and becomes the basal lamina. This structure is replicated in reverse in the venous system, starting from post-capillary venules that initiate a CSF-filled perivascular space [5].

Aquaporin-4 (AQP4) is the name given to water channels on the vascular endfeet of astrocytes, see Fig. 4.4. Once CSF penetrates the arterial perivascular space, AQP4 facilitates flow out of the periarterial space and into the interstitium of the brain parenchyma. Here, CSF exchanges with ISF produced by the brain parenchyma microvasculature, as is usual in all tissues. The resulting fluid in the interstitium, let us still call it ISF, moves **advectively** across the interstitium, carrying material with it towards the venous perivascular space. According to the theory, as first formulated, ISF exits the brain through the perivenous route, reaching the lymphatic vessels in the neck and finally into the circulation at the thoracic duct (left) and at the right lymphatic duct (right); see Chap. 3.

Glymphatics Related Works

The glymphatic theory has some common elements with early works on CNS fluids. In particular, the component (1) of the glymphatic system, regarding CSF flow into the brain parenchyma along perivascular spaces, may be seen as a refinement of the early work of Rennels and collaborators [11]. Their experimental work on cats and dogs revealed fast CSF inflow along perivascular spaces around large blood vessels penetrating the brain parenchyma. They also suggested that the observed advective flow may be facilitated by pulsations of the cerebral arteries. Element (2) of the glymphatic theory regarding the ISF clearance route, connects well to the early work of Cserr [34], who reported results from experimental work on animals.

According to Cserr, cerebral interstitial fluid, formed as elsewhere in the body at the blood capillary wall, drains from the cerebral interstitium into cerebrospinal fluid in the subarachnoid space. From here there are two possible routes, namely (a) the fluid can be absorbed as CSF into the dural sinuses, e.g. the superior sagittal sinus, or (b) be drained via the olfactory nerve sheath and nasal submucosa into the deep cervical (peripheral) lymphatic system. There is also a portion of ISF that is reabsorbed into blood at the venous end of the capillary, as elsewhere in the body, before reaching the subarachnoid space. See also Cserr et al. [12]. On these bases it may be argued that the glymphatic theory is an extension and unification of these

earlier works, with the addition of element (3) that explains the connection between (1) and (2) via the role of aquaporin-4.

Substantial related work has been reported over the years by Weller and colleagues. See for example Weller et al. [35] and Carare et al. [36]. In their proposed model named IPAD, for **Intramural Peri-Arterial Drainage**, ISF and solutes moves in the opposite direction to that of arterial blood, instead of the same direction, as proposed by the glymphatic theory.

Current Issues

Since the glymphatic theory was first formulated, a number of controversial issues have been raised. One such issue regards the fluid dynamics forces involved, for example to explain the movement of periarterial CSF towards perivenous CSF. The authors of the theory suggest, as Rennels and collaborators did [11], that arterial pulsations would provide the forces. Another issue is the advective character of ISF in the brain interstitium [5]. There are many publications on this theme, some supporting the advection flow hypothesis, some arguing that the flow is diffusion dominated, others that the flow is of the advection-diffusion type [37]. The role of aquaporin-4, element (3) of the theory, has also been brought into question. See Smith et al. [26], for example. As already mentioned in Sect. 4.2 Hu and collaborators [23] have added another aspect to the glymphatic theory, namely the question of venous absorption of solutes from the perivenous spaces in the brain via the BBB. The study suggests that transport from the perivenous space to the vein lumen is significant. These finding may have implications for elucidating the waste-clearance function of the glymphatic system; see also [38]. These and other aspects of the glymphatic theory have been comprehensively reviewed in many recent publications. Here we mention the works of Abbott et al. [6]; Benveniste et al. [20] and Hladky and Barrand [29].

More often than not, the reference to the central nervous system is centred exclusively on the brain. The spinal district of the CSF is often not mentioned or considered separately. The neural parts of the ocular and auditory systems are hardly mentioned as being part of the CNS. The spinal CSF system will be treated in detail in Sect. 4.4, while fluids in the neural parts of the ocular and inner-ear systems will be examined in Chap. 5.

4.4 CSF in the Spinal Canal

The cerebrospinal fluid compartment was introduced in Sect. 4.1, including CSF formation, chemical composition, reabsorption, interconnections, dynamics and main functions. The focus there was on cranial CSF. This section is exclusively concerned with CSF in the spinal subarachnoid space SSAS. We first address spinal CSF motion and then review major pathologies associated with spinal CSF dynamics, such as spina bifida, Chiari malformations and syringomyelia.

4.4.1 Spinal Canal and CSF Motion

This section is exclusively focused on cerebrospinal fluid in the spinal canal. Figure 4.5, right-hand side frame, shows a schematic view of the full CSF spaces, including the cranial cavity and the spinal canal. Red arrows depict the circulatory CSF flow. The left bottom frame shows a schematic view of a cross section of the spinal canal containing the spinal cord and CSF in the SSAS between the dura mater and the pia mater. Here we consider the spatial domain between the head and the sacrum composed of the spinal cord, which is continuous with the brain, enclosed by the CSF-filled spinal subarachnoid space.

The spinal cord is encased by the pia mater, while the SSAS is bounded externally by the spinal meninges, composed of the arachnoid and the dura matter, and internally by the pia mater, which separates CSF from cord tissue. Note that the spinal subarachnoid space also includes multiple nerve roots across it, as illustrated

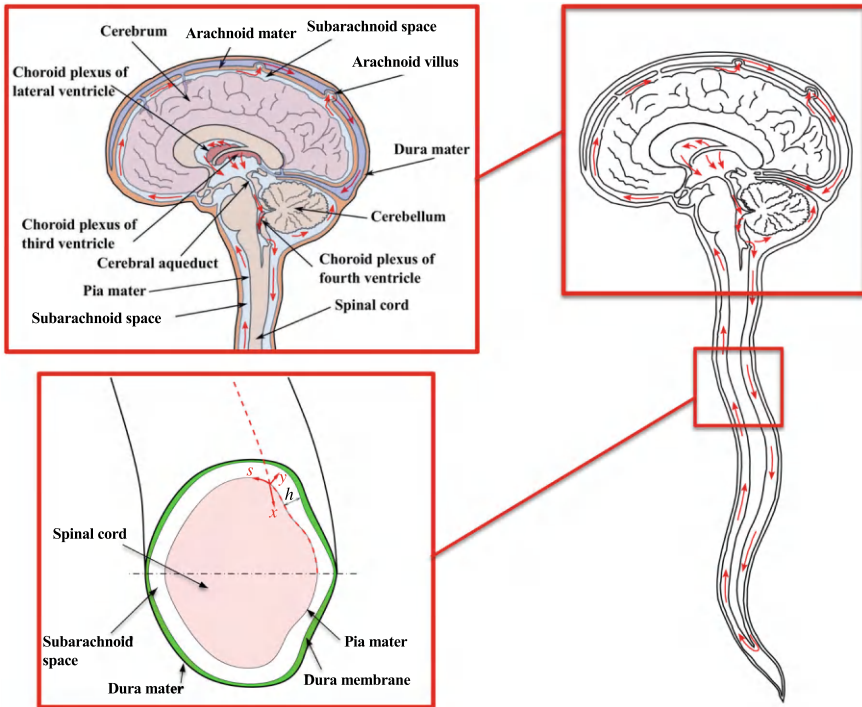


Fig. 4.5 Cerebrospinal fluid in craniospinal spaces: cranial cavity (CSAS) and spinal cavity, or spinal canal (SSAS). Top left frame: schematic overview of CSF production, circulation and absorption. Right frame: schematic view of the full CSF spaces including cranial cavity and spinal canal. Left bottom frame: schematic view of a cross section of the spinal canal containing the spinal cord and CSF between the dura mater and the pia mater (Courtesy of Professor Carlos Martínez, modified from Sánchez et al. [27])

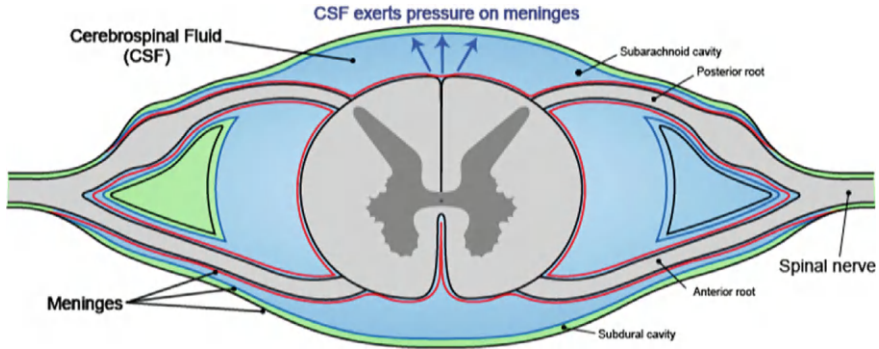


Fig. 4.6 Cross-section of spinal cord and surrounding cerebrospinal fluid in spinal subarachnoid space (SSAS). The spinal cord is surrounded by the pia matter, while the SSAS is surrounded by the meninges. The tiny black dot in the interior of the grey mater of the spinal cord represents the central canal. Note the presence of spinal nerve roots traversing the spinal canal (from <https://en.m.wikipedia.org/wiki/File:Gray770-en.svg>)

in Fig. 4.6. Nerve roots have a significant influence on the to-and-fro motion of CSF in the spinal CSF space. In the interior of the spinal cord is the **central canal**, a very small and not always continuous conduit; see Fig. 4.1, left hand side.

The length of the spinal canal is typically between $L = 0.6\text{ m}$ and $L = 0.8\text{ m}$. The cross-sectional area varies significantly along the length of the spinal canal; according to [39], the hydraulic diameter varies between $d = 5\text{ mm}$ and $d = 15\text{ mm}$. The estimated volume of CSF in the spinal canal in an adult human varies between $V = 40\text{ mL}$ and $V = 60\text{ mL}$ [2, 27, 40].

Within each cardiac cycle, there is pulsatile flow of cerebrospinal fluid from the cranial cavity CSAS into the spinal canal SSAS, and back. See Fig. 4.5. The origin of this CSF motion is the increased volume of high-pressure arterial blood pumped into the cranial cavity during systole. By virtue of the Monro-Kellie doctrine [41] there corresponds an efflux of other intracranial fluid contents, namely venous blood towards the jugular and vertebral veins and ultimately to the heart, and CSF towards the spinal subarachnoid space. This part is similar to the dynamical interaction in CSAS involving arterial blood, CSF, brain parenchyma and venous blood. Intermittent perturbations may result from transients due to cough, sneeze, change of posture, through the vertebral venous system.

The highly harmonious time-dependent interacting dynamics of several material compartments during the cardiac cycle means that, from the clinical and mathematical modelling points of view, it is essential to recognize this **holistic** setting in order to remain close to the physiology of the phenomena of interest. The seminal, early works of Williams [42, 43] are highly instructive in this respect. Particularly interesting is his *Demonstration Analogue for Ventricular and Intraspinal Dynamics (DAVID)*. This physical model for CSF motion includes the contributions from arterial and venous vasculature, inside and outside the cranial cavity, the heart, the abdominal venous system, the epidural vein, and even the presence of a meningocele. The more

recent in vitro model of Martin and Loth [44] is also very useful in understanding the physical problem of interest.

Other modern works dealing with CSF motion interacting with arterial blood, CSF and venous blood include those of Bateman et al. [45–49], Stoodley et al. [50], Hemley et al. [51], Liu [52] and Liu et al. [53]. An insightful description of the dynamical interaction of CSF with other fluid compartments in the craniospinal space is found in the works of Beggs [54, 55]. See also Sect. 4.6.

4.4.2 *The Central Canal*

The central canal resides in the interior of the gray matter of the spinal cord and spans from the fourth ventricle all the way to the **filum terminale**, a narrow prolongation of the spinal cord [56]. Figure 4.6 represents this minute feature as a single dot in the centre. See also Fig. 4.1, left-hand side. The central canal in humans may be filled with CSF, but it usually obliterates with age and therefore does not play a significant role in the CSF motion in the spinal canal, as it does in other species [57–59]. However, the presence of the central canal is important when trying to unravel the origin and development of spinal-cord related pathologies, such as syringomyelia [59]. Moreover, the central canal may act as a sink for potentially toxic substances drained away from the area of spinal cord injury [59, 60]. Further discussions on the central canal in the context of the spinal canal and the spinal cord are found in Sect. 4.5 and references therein, to be discussed next.

4.5 Pathologies Associated with Spinal CSF Dynamics

There are several diseases that are specifically linked to the spinal cord, the causes of which are not always well understood. Here we restrict our attention to some spinal cord pathologies that may have a fluid mechanics component and that could potentially be studied by the models and methods presented in this book. See Chap. 11.

4.5.1 *Spina Bifida*

Spina bifida is a congenital malformation resulting from defective development of the **neural tube** [61]. This is the structure in an embryo that becomes the brain, spinal cord and the surrounding tissues and fluids. Three types of spina bifida are recognised, all three exhibiting clefts of the vertebral column along with associated signs of spinal canal contents escaping through gaps in the vertebrae. Figure 4.7 illustrates one type of spina bifida.

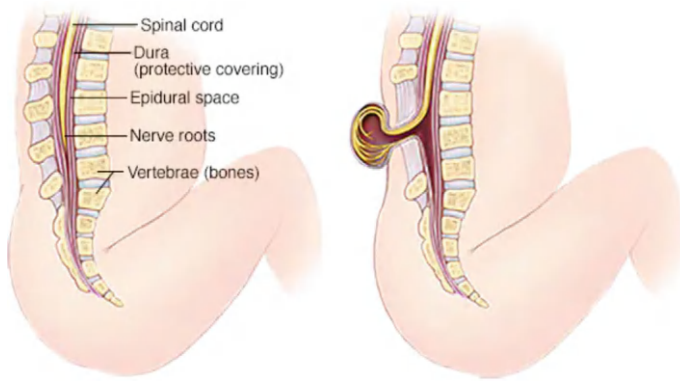


Fig. 4.7 Illustration of spina bifida. Left frame depicts normal case. Right frame depicts spinal cord with myelomeningocele type of spina bifida (or open spina bifida) (from Mayo Clinic)

The least serious case is called **spina bifida occulta**, which is characterised by small gaps between vertebrae. In this instance the vertebrae fail to completely enclose the spinal cord, which still is normal in form and is covered by the skin of the back [61]. In most cases spina bifida occulta goes undetected.

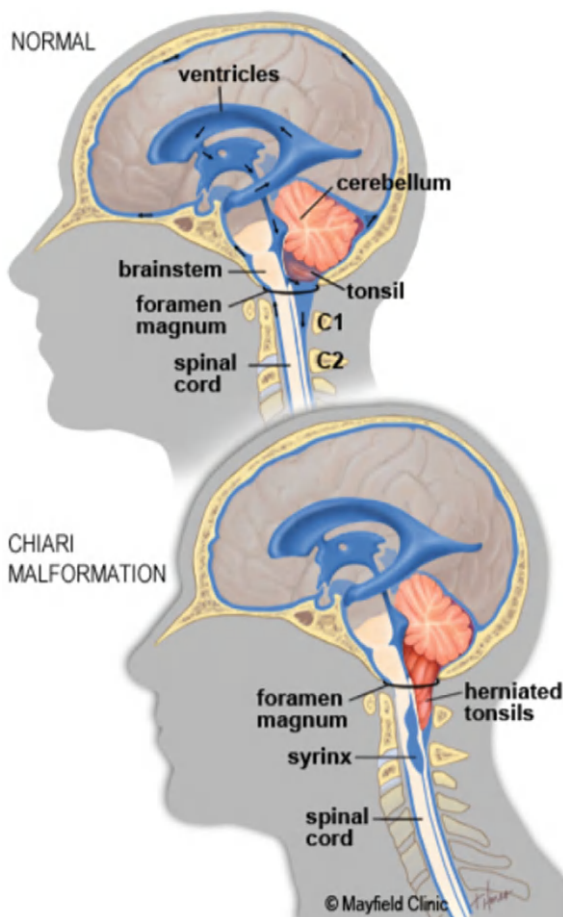
The rarest case is called **meningocele**, in which a sack of fluid, or pouch, bulges through an opening of the vertebrae. Nerves are not affected, and the spinal cord is not in the sack. The consequences are not as serious as in the third type called **myelomeningocele**, illustrated in Fig. 4.7, right-hand side. This is the most severe type, also called **open spina bifida**. In this case, meninges protrude through the vertebral defect, forming a fluid-filled pouch, similar to that of meningocele, but this time the pouch includes elements of the spinal cord or nerve roots. See Hassan et al. [62] for an up to date review of spina bifida and the many references therein.

Needless to say, the dynamics of CSF flow in the spinal canal is grossly disrupted by the particular geometric configuration that characterises spina bifida. The consequences of disrupted CSF dynamics are multiple. Spina bifida is often associated with another condition called syringomyelia, to be discussed in Sect. 4.5.3.

4.5.2 Chiari Malformations

Chiari malformations are abnormalities involving various craniospinal structures, such as the fourth ventricle, the foramen magnum, the cerebellum, the brain stem, the cranio-cervical junction and the spinal canal. Figure 4.8 illustrates the normal anatomy on top compared with the anomalous case of Chiari malformation on the bottom. In these deformities the cerebellum herniates through the skull and penetrates downwards into the spinal canal. One consequence is compression of the brain stem, cranial nerves and the cervical spinal cord. Moreover, the resulting blockage of the

Fig. 4.8 Chiari malformation. Top frame: normal anatomy of cerebellum. Bottom frame: illustration of Chiari I malformation; cerebellar tonsils herniate across the skull, through the foramen magnum, into the spinal canal, interfering with normal CSF flow and possibly causing accumulation of fluid, called cyst or syrinx, inside the spinal cord (from Mayfield Chiari Centre, with permission)



spinal subarachnoid space interferes with the normal fluid dynamics of CSF up and down the SSAS during the cardiac cycle. The anomalous CSF fluid dynamics may include build up of CSF in the cranial SAS, resulting in raised intracranial pressure. Often, Chiari malformations are associated with build up of fluid inside the spinal cord in the form of **cyst** or **syrinx**, a condition called syringomyelia; see Sect. 4.5.3.

Chiari malformations are considered to be congenital conditions, though acquired cases have been reported. Symptoms are varied and will depend on the particular type of Chiari malformation, but typically will include headache, neck pain, dizziness, sensory changes, vertigo, limb weakness, fatigue and hearing loss [63–66].

Chiari malformations were first described and characterised by Dr. Hans Chiari in 1891, who identified four types, which we briefly describe below.

Chiari I

This is the mildest case and is characterised by displacement of cerebellar tonsils more than 5 mm, caudally through the foramen magnum. In this type, the brain stem and the fourth ventricle maintain a degree of normality, though the latter may be smaller than normal and mildly distorted. Three subtypes of Chiari I are identified in the literature, two of these are defined as congenital, while the third is defined as acquired, usually involving deformities of the foramen magnum. It is reported that 20% to 70% of Chiari I cases are associated with syringomyelia/hydromyelia; the percentage depends on the degree of disruption of the fluid dynamics of CSF between the spine and cranium [64]. Chiari I is most commonly diagnosed in adolescents and adults. In many cases of Chiari I, symptoms may be absent, but if they occur, any of those listed above may be included. Additional features of Chiari I include bone deformation at the upper level of the spine, scoliosis and spina bifida occulta; see Sect. 4.5.1.

Chiari II

This form of Chiari malformation is congenital and affects infants. Characteristic features of Chiari II include downward displacement of medulla oblongata, fourth ventricle, and cerebellum; and elongation of fourth ventricle and pons. Chiari II occurs almost exclusively in subjects with myelomeningocele, a form of spina bifida; see Sect. 4.5.1. Symptoms of Chiari II are mainly a consequence of myelomeningocele, grossly disturbed CSF dynamics and hydrocephalus. Such specific symptoms include breathing anomalies, involuntary eye movements and loss of arm strength. In the literature, this type of Chiari malformation is called **Arnold-Chiari malformation**.

Chiari III

This type of Chiari malformation affects infants: it is rare, life threatening and characterised by severe herniation of the cerebellum and brain stem through a defect in the back of the head or neck of the infant, forming a fluid-filled sack. Chiari III is associated with high mortality rates and severe neurological deficits in patients that survive.

Chiari IV

This is the most rare and most severe type of Chiari malformation. It affects infants and is often fatal. Chiari IV is characterised by anomalous development of the cerebellum; malformations of the brain and brain stem may also be present.

The pathophysiology of Chiari malformations is currently not well understood. Further details on the physiology and clinical aspects are found in [63–66], for example. As anticipated, Chiari malformations are strongly connected to other pathologies, such as spina bifida and syringomyelia, to be discussed in Sect. 4.5.3. Of the several aspects involved in Chiari malformations, the fluid dynamics of CSF between

the cerebral SAS and the spinal SAS stands out as prominent. Altered CSF dynamics in the spinal canal will affect the entire craniospinal fluid dynamics, to start with.

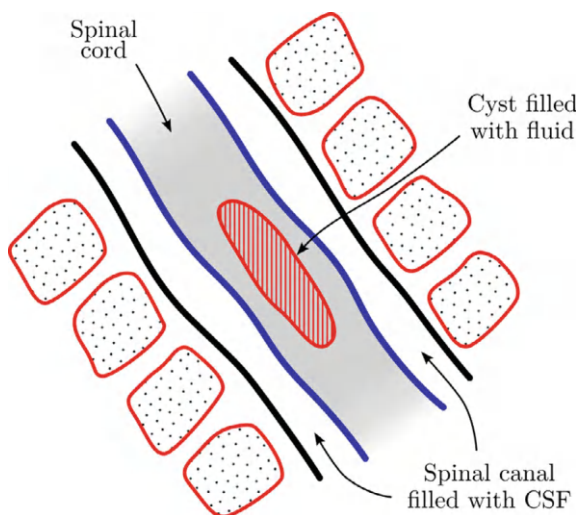
In the next section we deal with a related spinal anomaly called syringomyelia.

4.5.3 Syringomyelia

Syringomyelia is defined as a chronically progressive illness characterised by the presence of longitudinal fluid-filled cavities called cyst, or syrinx, in the interior the spinal cord; see Fig. 4.9. The majority of syrinxes are associated with Chiari malformations [67]. Syringomyelia is a neuropathology that may interfere with nerve signal transmission, the symptoms of which will include chronic pain, sensory deficits and, ultimately, paralysis.

Historically, several variations of what is understood today as syringomyelia, have been considered [68–70]. For example, the case in which the fluid inside the cysts is CSF, or some other fluid indistinguishable from CSF, has been termed **communicating syringomyelia** [71]. Available evidence suggests that the origin of CSF inside the cyst is due to CSF travelling from the fourth ventricle down the central canal. This type has also been called **hydromyelia** in the literature [68]. According to [71] there are other ways in which fluid within a syrinx can be originated. These include liquefaction of haematomata, after traumatic paraplegia, and transudation of fluid from intrinsic spinal tumours. In other words, there are several possible causes for the fluid-filled cavities to form over time, but the precise mechanisms by which the phenomenon takes place are poorly understood. Chiari malformations, which are

Fig. 4.9 Syringomyelia. A cyst, or syrinx, located in the interior of the spinal cord. The fluid inside the cyst could be CSF, or some other CSF-like fluid. The spinal cord is surrounded by the spinal canal, filled with CSF (courtesy of Violeta Da Rold)



characterised by the protrusion of brain tissue into the spinal canal, may give rise to syringomyelia.

For a discussion on the mechanisms of syringomyelia, its origin, the causes of progression and the nature of separate fluid compartments, see the classical works of Williams [71, 72] and the most recent review of Klekamp [59]. Much attention has been paid to the possible origin of syringomyelia, how it is formed, what is the nature of the fluids inside the cysts and their possible connections. The review of Klekamp [59] comprehensively analyses all theories on syringomyelia available till 2002.

There is consensus in the scientific community that mechanical forces, fluid dynamics and wave propagation play an important role in the study of syringomyelia [42, 43, 69–71, 73–76]. The fluid inside the fluid-filled cavity will be subject to periodic forces exerted by pressure waves in the cerebrospinal fluid in the spinal canal, resulting from the periodic cardiac-cycle pulsations. The to and fro motion of CSF in the spinal canal may thus increase the volume of the cyst and damage the cord tissue, potentially leading to disabilities [71].

Klekamp [59], in addition to presenting a historical review of the various theories of syringomyelia, proposes his own theory, the so-called **oedema hypothesis**, which is summarised in his rather broad definition of syringomyelia, namely: *syringomyelia is understood as a state of chronic interstitial edema of the spinal cord due to accumulation of extracellular fluid*. In view of the latest developments on brain fluid dynamics, the theory proposed by Klekamp is appealing.

Our interpretation of this theory implies that the formation of cysts, predominantly in the central canal or in the spinal cord extracellular space and characterised by extracellular fluid accumulation, is due to failure of ISF/CSF clearance mechanisms. Klekamp's proposed mechanisms are all fluid dynamics based and include obstruction of CSF flow, altered blood flow in the cord, increased viscosity of ISF, obstruction of perivascular spaces and altered venous flow in the cord. Martinez-Arizala et al. [77] demonstrated experimentally that dorsal spinal venous occlusion in the rat produces syringomyelia. In vivo studies of the mechanical aspects of syringomyelia are difficult, for obvious reasons. Instead, experimental studies based on in vitro models are more easily carried out; see for example the early works of Williams [42, 43], Martinez-Arizala et al. [77] and the more recent works of Martin and collaborators [44, 73].

Mathematical models for studying syringomyelia have also been communicated by several research groups. Examples include Martin et al. [78], Bertram [67], Martin et al. [73], Elliott et al. [75], Cirovic et al. [69] and Cirovic and Rusbridge [70]. Mathematical modelling is a useful approach for studying the fluid dynamics of bodily fluids, CSF in particular. In Chap. 11 we present, in detail, a practical mathematical model to simulate the motion of CSF in the spinal canal, coupled to the dynamics of the deforming spinal cord. Moreover, such model can be coupled to all major fluid compartments in the human body to provide realistic simulations. See [79, 80] and [81] for further details; see also Chap. 20 of this book.

4.6 Neurofluids Interacting Dynamics. Case Study

In this final section of this chapter we put together major CNS extracellular fluid compartments through their dynamical interaction in the time frame of the cardiac cycle. We do so by means of an MRI study of one Idiopathic Intracranial Hypertension (IIH) patient and one healthy control. The material compartments of interest here include the arterial system, the venous system and cerebrospinal fluid in the full craniospinal cavity. Collectively, all major interconnected extracellular fluids in the CNS were recently termed **Neurofluids** in [82], as the acronym for a European research proposal. The term neurofluids has subsequently been adopted in [83–86], amongst others. In order to illustrate the close interaction of neurofluids we have chosen the MRI case study reported in [87].

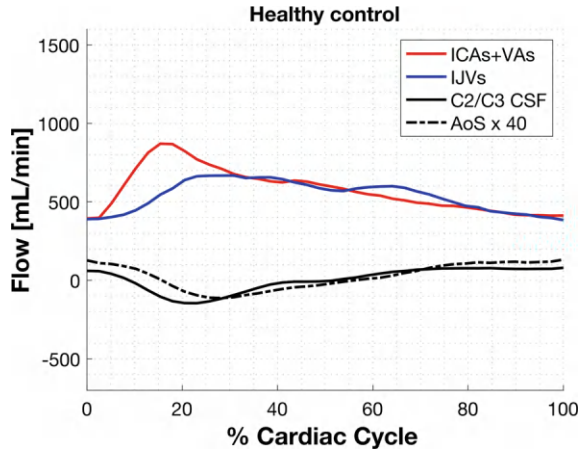
4.6.1 Craniospinal CSF Dynamics

The cerebrospinal fluid dynamics in the full craniospinal space is a very complex event that depends primarily on the cardiac cycle and involves intracranial and extracranial components [41, 87]. The time-dependent events involve a dynamic interaction of various material compartments in the CNS. Following the high-pressure arterial inflow during systole, there is outflow of venous blood, displacement of cerebrospinal fluid into the spinal canal, in a to and fro fashion. There is also a dynamic response from brain and spinal parenchyma tissue, as well as interstitial fluid. According to the Monro-Kellie hypothesis the intracranial volume in the rigid skull encasement is kept constant during the cardiac cycle. Therefore, the increased arterial volume during systole must necessarily cause the displacement of other fluids, such as CSF and venous blood, occupying part of the intracranial space.

Craniospinal Dynamics in a Healthy Subject

In order to gain an understanding of the phenomenon it is helpful to consider Fig. 4.10, which depicts the flow dynamics of the real case of a healthy 26 year old subject [87]. The figure shows MRI measured flow q against time t , for one cardiac cycle. The MRI measured quantities are for arterial blood flow (ICAs+VAs), venous blood flow in the internal jugular veins (IJVs); CSF flow at the C2/C3 level (C2/C3 CSF) and CSF flow in the aqueduct of Sylvius, multiplied by a factor of 40 (AoSx40). Here ICAs means internal carotid arteries; VAs means vertebral arteries and IJVs means left and right internal jugular veins. It is seen that the arterial blood systolic peak (ICAs+VAs) occurs roughly at a time that marks the 20% of the cardiac cycle, after which flow q then gradually decreases for the rest of the cardiac cycle. In response to the increased arterial blood volume, venous blood exits the cranium, which is seen in the trace of venous flow in the internal jugular veins (IJVs), also displaying a maximum value, shortly after the maximum

Fig. 4.10 Healthy control. CSF and blood flow over a cardiac cycle in 26 year healthy mail subject. Arterial blood in internal carotid arteries and vertebral arteries (ICAs+VAs); venous blood in internal jugular veins (IJVs); CSF at C2/C3 level (C2/C3 CSF) and CSF in the aqueduct of Sylvius (AoSx40) (Courtesy of Dr. Christian Contarino; data from Agarwal, Contarino and Toro [87])



of the arterial blood trace. The time difference between these two maxima is called **arterio-venous delay**. Also, in response to the arterial pulse, cerebrospinal fluid is displaced, exhibiting corresponding extrema and defining corresponding time delays.

Disturbed Craniospinal Dynamics in IIH Patient

Here we consider the case of a patient diagnosed with Idiopathic Intracranial Hypertension (IIH), a condition of unknown etiology frequently associated with dural sinus stenosis [87]. The patient considered was a 34 year old obese female, who reported multiple episodes of neck pain, throbbing headache, and progressive vision loss. Magnetic resonance venography revealed the presence of stenoses in the transverse sinuses. This intracranial anatomical anomaly of the venous system hampers the outflow of venous blood from the cranial cavity and alters completely the intracranial interactive dynamics involving CSF, arterial blood, venous blood and brain parenchyma.

Figure 4.11 shows a comparison of MRI measured flow q against time t , for one cardiac cycle, between a healthy control, shown in bottom-right frame, and the IIH patient, shown in frames (a) to (e). Full details are found in [87]. Frame (a) shows MRI measurements before treatment; the bottom-right frame shows key to the traces for arterial blood flow (ICAs+VAs), venous blood flow in the internal jugular veins (IJVs); CSF flow at the C2/C3 level (C2/C3 CSF) and CSF flow in the aqueduct of Sylvius, multiplied by a factor of 40 (AoSx40). Compare frame (a) to the healthy-control, bottom right. The main differences are seen in the venous flow and the CSF flow in the aqueduct of Sylvius. The venous outflow curve of the patient is well below that of the healthy control, even though the arterial inflow is larger in the patient. This means that venous flow occurs not through the main outflow routes given by the internal jugular veins, but through collaterals. Then, there is a large arterio-venous delay, seen by comparing the systolic peaks for arterial and venous flow. The curves

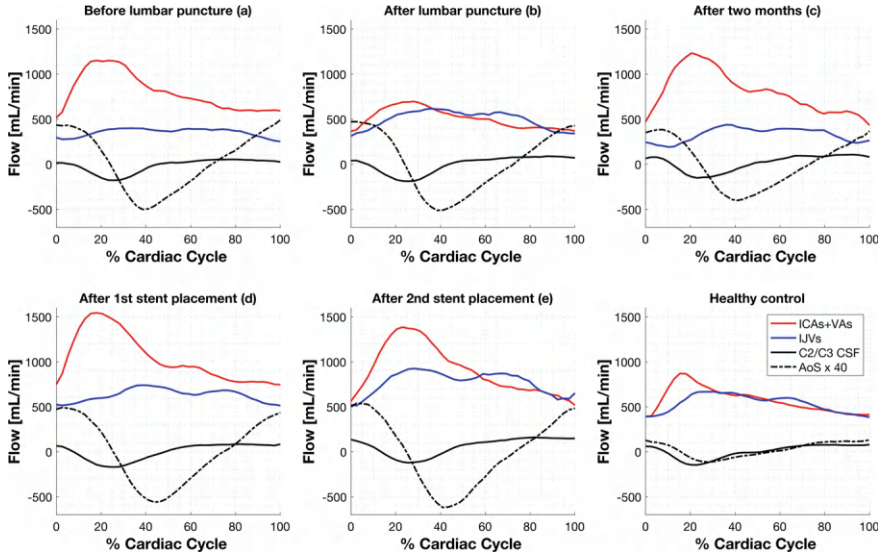


Fig. 4.11 IIH patient compared with healthy control. CSF and blood flow over a cardiac cycle at different time points and treatments on IIH patient, frames **a** to **e**, and in a healthy subject (Courtesy of Dr. Christian Contarino; data from Agarwal, Contarino and Toro [87])

for CSF flow through the aqueduct of Sylvius reveal accentuated pulsatility in the patient, suggesting high intracranial pressure (ICP). Unfortunately, pressure cannot be measured by MRI.

Frames (b) to (e) in Fig. 4.11 correspond to flow quantities measured after various treatments were tried, all described in [87]. These included lumbar puncture (LP) to remove CSF, compare (a) with (b); pharmaceutical medical therapy, compare (a) with (c); and venous stent placements deployed at two separate times, see (d) and (e). Compare all IIH patient cases (a)-(e) with healthy control, last frame. Detailed discussion is found in [87].

In summary, the dynamics of CSF in the craniospinal cavity is a complex, time-dependent event that is intimately linked to the dynamics of arterial blood inflow, venous blood outflow and brain dynamics (tissue and ISF). The flow alterations observed in the IIH patient are quite obvious in the venous blood traces. Surprisingly, the treatment also affected the arterial blood traces quite significantly; see [87] and [47] for further discussion on this point.

4.6.2 Brain Fluids and CSF Pressure

As already discussed, the dynamic interaction of various compartments during the cardiac cycle involves high pressure input of arterial blood during systole and exiting

of CSF from the cranial cavity into the spinal compartment and exiting of venous blood from the cranium directly into the vertebral and internal jugular veins, and back to the heart. As seen in Fig. 4.11, perturbing any of these fluid compartments alters the entire brain dynamics. In the study of Sect. 4.6.1, the perturbation results from anomalous cerebral venous outflow, due to intracranial venous stenoses. Theoretically, one would expect the perturbed cerebral venous outflow to cause an increased intracranial venous pressure and increased intracranial pressure ICP (increased CSF pressure). Potentially, such intracranial hypertension would affect perfusion and reduce the clearance of metabolic wastes. Unfortunately, pressure cannot be measured through magnetic resonance imaging.

There are various invasive methods to measure the intracranial pressure, that is the CSF pressure. One such methods used by clinicians is the **lumbar puncture** method. Pressure of CSF is a quantity of primary importance, both in physiology and pathology. In a clinical setting, CSF pressure p tends to be regarded as a constant in a subject in the supine position (lying horizontally), even though p changes with time within the cardiac cycle. A value of $p = 10$ mmHg is regarded as normal in a healthy subject, but individual variations can be significant, with $5 \text{ mmHg} \leq p \leq 15 \text{ mmHg}$.

In the event of an increase in pressure in the superior sagittal sinus, such as in venous sinus stenosis, venous drainage from cortical and bridging veins would be hampered resulting in two potential outcomes: (a) increased storage of venous blood volume in the cortical veins and, (b) increased collateral flow that would redirect the volume of venous blood through a longer, less efficient path within the cranial cavity, resulting in a longer cerebral circulation time [88]. In [87], it is hypothesised that this might result in a *permanent* increased volume of venous blood in the cranial cavity, and that such extra *stored* blood in the cortical veins be **tumoral**; see [87] for discussion.

Sophisticated global mathematical models for the human circulation are, in principle, capable of predicting the intracranial pressure under the conditions described in this section. In Müller and Toro [79] and in Toro et al. [80] it is shown that both intracranial and extracranial venous outflow strictures perturb the entire craniospinal fluid dynamics, including an increase in intracranial pressure. The level of pressure increase depends on the level of cerebral venous outflow collateralization of each individual.

4.7 Conclusions and Further Reading

The cerebrospinal fluid system has been presented, devoting separate sections to CSF in the cranial cavity, CSF dynamics in the spinal canal and CSF-dynamics related pathologies in the spine. The presentation has kept a unifying view of all major extracellular fluid compartments of the central nervous system. The arterial and venous compartments were introduced in Chap. 1, interstitial fluid and the lymphatic system were presented in Chap. 3; the present Chap. 4 added the cerebrospinal fluid system. Bodily fluids of the eye and the inner ear will be introduced in Chap. 5.

Attention is consistently drawn to the unifying role of the venous system, noting that all fluids considered drain ultimately into the venous system.

The presentation has been introductory in nature. Still, we have interspersed some aspects that are the subject of current research. The reader is made aware of the fact that we have not included other major systems, such as the digestive system, the urinary system and the respiratory system. Neither have we included the nervous system, the endocrine system, along with their major regulatory function in overall body fluid homeostasis. Preference has been given to systems that might be modelled by the methods of this book.

References for advanced topics of current research interest have been given in each separate section. For basic physiological aspects of bodily fluids we suggest the following classical books: Hall [1]; Tortora and Derrickson [9]; Silverthorn [15]; Levick [14] and Herring and Paterson [89].

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Chapter 5

Fluids in the Eye and Inner Ear



Abstract In this chapter, we review the major extracellular fluid compartments in the eye and inner ear of humans. We begin with the extracellular fluids that characterize the ocular system: the aqueous humor in the anterior cavity of the eye, the vitreous humor in the vitreous chamber, and cerebrospinal fluid in the optic nerve sheath. Intraocular pressure is defined, and Goldmann's equation for intraocular pressure is discussed. Briefly, we also touch on two topics of current interest: the lymphatic system and the newly proposed glymphatic system in the human eye. Next, we turn to the fluid compartments of the inner ear, starting with a basic introduction to human ear anatomy. Two extracellular fluid systems—endolymph and perilymph—are introduced, as these compartments are integral to the inner ear's fluid physiology. Anatomical and functional communications between these fluid compartments are examined. After describing hearing function under physiological conditions, we discuss Meniere's disease, a well-known inner ear disorder. Relevant to Meniere's disease are the classical animal experiments by Friis and Qvortrup, which explore the role of disturbed venous fluid dynamics in understanding the disease. We also discuss the role of venous occlusions in inner ear pathologies. The chapter concludes with a summary and suggestions for further reading.

5.1 Fluids in the Ocular System

Vision is essential to human survival. The eyes detect visible light, the part of the electromagnetic spectrum that exhibits colours, in which wavelengths range between $400\ \mu\text{m}$ and $700\ \mu\text{m}$. Objects absorb or reflect visible light, depending on the wavelength. The vision system involves the physics of vision, neurological aspects and fluid mechanics of several interconnected fluid compartments. In this section we address specifically the fluid dynamics aspects of all major extracellular fluid compartments that characterise the vision system, that is cerebrospinal fluid, vitreous body in the vitreous chamber, and aqueous humour in the anterior and posterior chambers. We emphasise the anatomical and functional interconnections amongst such fluid systems, viewed in a holistic fluid mechanics framework. For a classical description of the vision system we recommend the textbooks of Tortora and

Derrickson [2], Hall [5] and Silverthorn [58]. This section also contains brief reviews of recent research advances in ocular fluid dynamics, including the lymphatic and the newly discovered glymphatic systems.

5.1.1 Anatomy of the Human Vision System

Here we briefly review all major structures of the human vision system, as depicted in Fig. 5.1, which include the eyeball and associated internal and external accessory structures. Accessory structures include the eyebrows, the upper and lower eyelashes, the eyelids, the lacrimal apparatus, and extrinsic eye muscles. Salient posterior features include the neural parts that connect the eye to the central nervous system, namely the retina, the optic disc, the lamina cribrosa (not shown in Fig. 5.1) and the

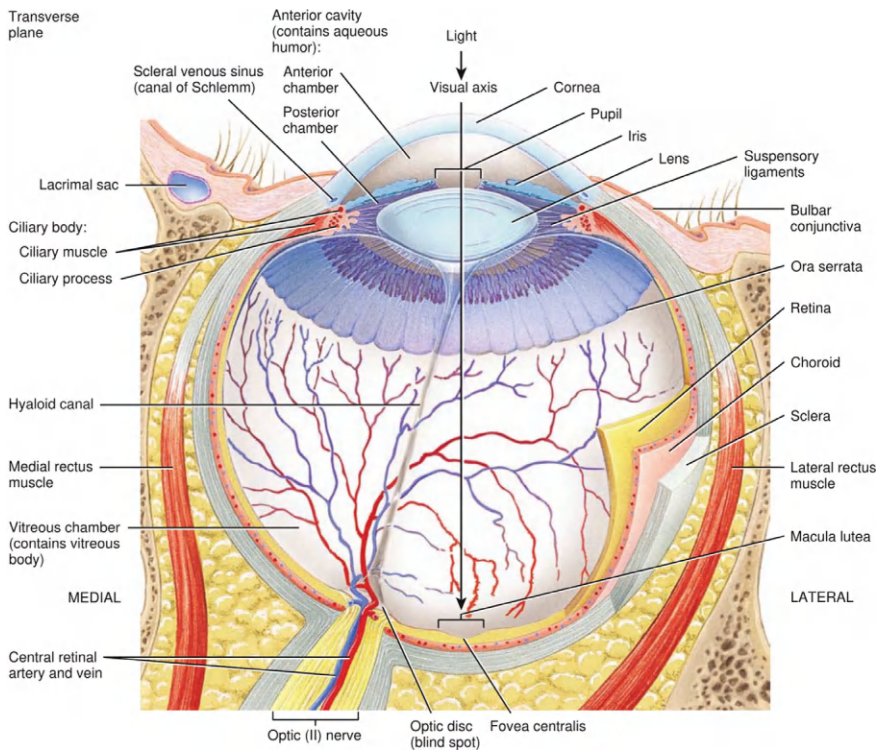


Fig. 5.1 Anatomical description of the human eye. Salient structures include the wall of the eyeball, which features three layers: the fibrous tunic, the vascular tunic and the retina; three fluid chambers: the anterior chamber that contains aqueous humour, the posterior chamber that also contains aqueous humor and the vitreous chamber that contains vitreous humour (from Tortora and Derrickson [2], with permission)

optic nerve. Anterior structures of the eye include the cornea, the iris, the pupil and the lens. The blood vasculature is most evident in the choroid and retina, and in the optic nerve, featuring the central retinal artery and the central retinal vein. A note of caution. The network of arteries and veins depicted in Fig. 5.1 are meant to represent the vasculature located in the interior layers of the wall of the eyeball, not within the vitreous chamber, as it may appear.

The eyeball of a human adult has a diameter of about $d = 2.5$ cm, giving a volume of $V = 8.18 \text{ cm}^3 = 8.18 \mu\text{L}$ and a surface area $S = 19.63 \text{ cm}^2$. The anterior $1/6$ of the surface area is exposed to the external environment, that is about 3.27 cm^2 ; the rest is fitted into the orbit. The wall of the eyeball, as seen in Fig. 5.1, consists of three layers: (1) the fibrous tunic, (2) the vascular tunic, and (3) the retina. These three layers are clearly distinguishable in most of the eyeball boundary, except in the anterior part of the eye. The interior of the eyeball features three major chambers: (1) **the anterior chamber** that contains **aqueous humour**, (2) **the posterior chamber** that contains **aqueous humour** and (3) **the vitreous chamber** that contains the **vitreous body**, or **vitreous humour** or just **vitreous**.

Notwithstanding quoted numbers for the volumes of the various fluid compartments vary in the literature, here we give the following. The average volume of the anterior chamber (aqueous humour) is $220 \mu\text{L}$, of the posterior chamber (aqueous humour) is $60 \mu\text{L}$ and of the vitreous cavity (vitreous humour) is $5 \mu\text{L}$ to $6 \mu\text{L}$. The vitreous cavity volume is often given as being more than $2/3$ of the volume of the eye; 80% of the total volume is also quoted. The total volume of the average adult eye is approximately $6.5 \mu\text{L}$ to $7.0 \mu\text{L}$, which is less than the figure of $8.18 \mu\text{L}$ given above.

In what follows we describe these major eye components in more detail, noting again that the emphasis is on major extra-cellular fluid compartments.

5.1.2 *The Wall of the Eyeball*

We describe in some detail the three layers of the eyeball wall. **The fibrous tunic** constitutes the surface layer of the eyeball, that is the outer layer of the eyeball wall. It is divided into two parts. The anterior part constituted by the cornea and the posterior part constituted by the sclera. The cornea is a curved, transparent solid structure that lacks blood vessels (avascular). Oxygen from the air and from the eyelid vasculature dissolves in tears and is transmitted to the cornea via the tear film. The cornea derives its macromolecules and nutrients from the aqueous humour. The sclera, the white part of the eye, covers the entire surface of the eyeball, except the cornea. The layer formed by the sclera is constituted by dense connective tissue, mostly made of collagen fibres and fibroblast.

The vascular tunic, or uvea is the middle layer of the eyeball wall. It consists of three parts: the choroid, situated in the posterior part of the wall, constitutes most of the surface area; the ciliary body is situated near the anterior part; the iris in the anterior part of the eye completes the vascular tunic layer.

The choroid is a highly vascularised part of the middle layer of the eyeball wall, providing a route for the supply of nutrients to the retina and a means for draining used up venous blood and metabolic wastes. The ciliary body contains the ciliary muscle and the ciliary processes, which are protrusions of folds in the interior surface of the ciliary body. The ciliary processes are highly vascularised and their blood capillaries perform a most important function: they secrete aqueous humour, a fluid that fills the anterior and posterior compartments shown in Fig. 5.1. See also Fig. 5.3.

The retina is the vascularised inner layer of the eyeball wall, which lines the posterior three quarters of the eyeball. In turn the retina is composed of two sub-layers, namely a pigmented layer and the neural (or sensory) layer, see Fig. 5.1. As noted in [2] the surface of the retina is the only place in the body in which arterial and venous blood vessels can be viewed directly, by means of the **ophthalmoscope**. Pathological changes in such blood vessels can be observed, such as those resulting from hypertension, diabetes mellitus, as well as age-related macular disease. In the next section we give more details about the retina.

5.1.3 Neural Parts of the Eye

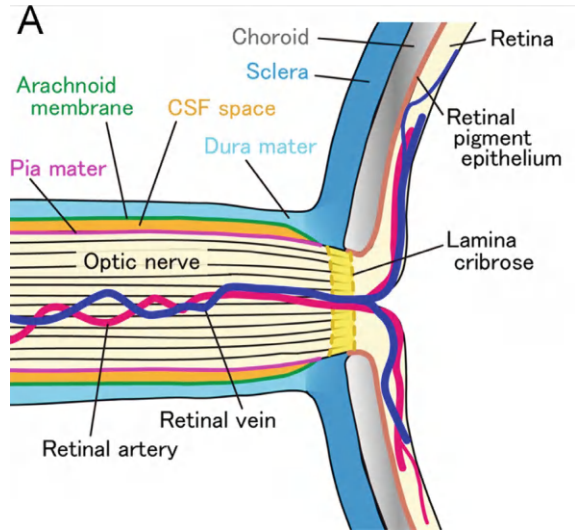
Here we give more details on the neural, or sensory, parts of the eye, including the retina, the lamina cribrosa and the optic nerve.

The retina is an outgrowth of the brain and constitutes the start of the visual pathway; see Fig. 5.1. It processes visual data before sending neural impulses to the brain through the axons of the optic nerve. Retinal neurons form three distinct layers: the photoreceptor layer, the bipolar cell layer, and the ganglion cell layer. Photoreceptors start the process of converting light rays into nerve impulses that are sent to the brain. There are two types of photoreceptors, called **rods** and **cones**. The human retina has 6 million cones and 120 million rods [2]. Visual information flows from photoreceptors to bipolar cells and from here to ganglion cells. As noted in [2] the axons of ganglion cells extend posteriorly to the optic disc and exit the eyeball along the optic nerve. See Fig. 5.2.

The optic disc is defined as the position of exit of axons leaving the eyeball. The optic disc contains no rods and no cones, and therefore an image that strikes cannot be seen. The optic disc is the **blind spot** of the eye, see Fig. 5.1.

The lamina cribrosa is defined in [3] as a fenestrated, three-dimensional network of load-bearing trabeculae, that provides structural and nutrient support to the retinal ganglion cell axons as they leave the eye on their path towards the brain; many of the trabeculae contain capillaries. See Fig. 5.2. Biomechanically, the lamina cribrosa is regarded as a weak spot, as its thickness is only one-third of that of the sclera at the scleral canal. This eye structure is responsible for providing structural support to the optic nerve head and allowing an open pathway for the the axons to exit the eyeball. The lamina cribrosa plays a major role in glaucoma. According to Wang et al. [4] the lamina cribrosa supports a pressure difference between the eye and the brain, with

Fig. 5.2 The optic nerve sheath around the optic nerve includes a space filled with CSF. The central retinal artery and central retinal vein penetrate the optic nerve and continue towards the eye up to a few millimeters before the optic nerve reaches the lamina cribrosa (from Nakashima et al. [1], with permission)



intraocular pressure (IOP) usually greater than intracranial pressure (ICP). Typical average values are $IOP = 15 \text{ mmHg}$ and $ICP = 11 \text{ mmHg}$.

The optic nerve, also called **cranial nerve II**, is the neural structure that connects the retina to the visual cortex. It carries sensory nerve impulses from the retina to the visual centres of the brain. See Fig. 5.1. The optic nerve begins at the optic disc, located at the back of the eye. See Fig. 5.2. After emerging from the back of the eye, the optic nerves from both eyes cross over, forming an X-type structure called **optic chiasm**. From here nerve fibres travel in optic tracts to various zones of the brain.

5.1.4 CSF in the Optic Nerve Sheath

Anatomically, the dura of the brain extends as a sheath around the optic nerve and then connects with the sclera of the eye. As depicted in Fig. 5.2, the optic nerve sheath defines a space that is continuous with the entire CSF system. Moreover, the retinal artery and retinal vein pierce the optic nerve sheath a few millimetres behind the eye and then pass along with the optic nerve fibres into the eye itself. In passing through the optic nerve sheath, the internal pressure in these blood vessels is directly affected by the CSF pressure in the optic nerve sheath that acts as external pressure on these compliant vessels. Furthermore, the interactive dynamics of all fluid compartments within the craniospinal space will directly affect the CSF pressure and volume in the optic nerve sheath, which in turn will impact on the ocular fluid dynamics.

According to Hall [5], there are three ways in which elevated CSF pressure in the craniospinal cavity can affect the eye:

1. It drives fluid first into the optic nerve sheath and then along the spaces between the optic nerve fibres to the interior of the eyeball.
2. The resulting high pressure hampers outward fluid flow in the optic nerve, causing accumulation of excess fluid in the **optic disc** at the centre of the retina.
3. The CSF pressure in the optic nerve sheath will hamper venous outflow in the retinal vein; this will increase the retinal capillary pressure throughout the eye, which will induce retinal oedema and reduce eye perfusion.

Oedema can affect the optic disc in a more obvious manner than in other parts of the retina. This is due to its tissue being much more distensible than those of the remainder of the retina. The optic disc then swells into the eye cavity, a phenomenon called **papilledema**, which can be visualised in eye pathologies linked to various conditions, characterised by intracranial hypertension, caused, for example, by traumatic lesions, tumors and venous stenoses. The underlying condition may be idiopathic and one then speaks of Idiopathic Intracranial Hypertension (IIH). Stenoses in transverse venous sinuses are often associated with IIH. These venous outflow strictures cause elevated CSF pressure, distended optic nerve sheath, followed by a series of clinical consequences, including visual loss [1, 6–8].

In the next two sections we describe two major eye fluid systems. We start with the aqueous humour in the anterior and posterior chambers.

5.1.5 Aqueous Humour

Aqueous humour is a fluid specific to the visual system. In this section we study various, basic aspects, such as its definition, functions, generation, absorption and fluid dynamics. The material is based on classical text books as well as advanced research works in the open literature.

Aqueous Humour Definition

Aqueous humour is contained in two compartments or chambers, the **anterior chamber** and the **posterior chamber**, situated either side of the **iris**. See Figs. 5.1 and 5.3. Aqueous humour is an optically clear fluid generated from blood capillaries in the **ciliary body**, a highly vascularised district that is effectively a continuation of the wall of the eyeball. The composition of aqueous humour is similar to that of blood plasma, from which it is generated by filtration, secretion and diffusion; it is also similar to that of cerebrospinal fluid. Water in aqueous humour constitutes about 99.9% of the total content; the rest includes organic and inorganic ions, carbohydrates, glutathione, urea, amino acids and proteins, oxygen and carbon dioxide. Aqueous humour is said to be analogous to a blood surrogate for the eye **avascular structures**, such as the lens and the cornea. As blood and CSF, aqueous humour

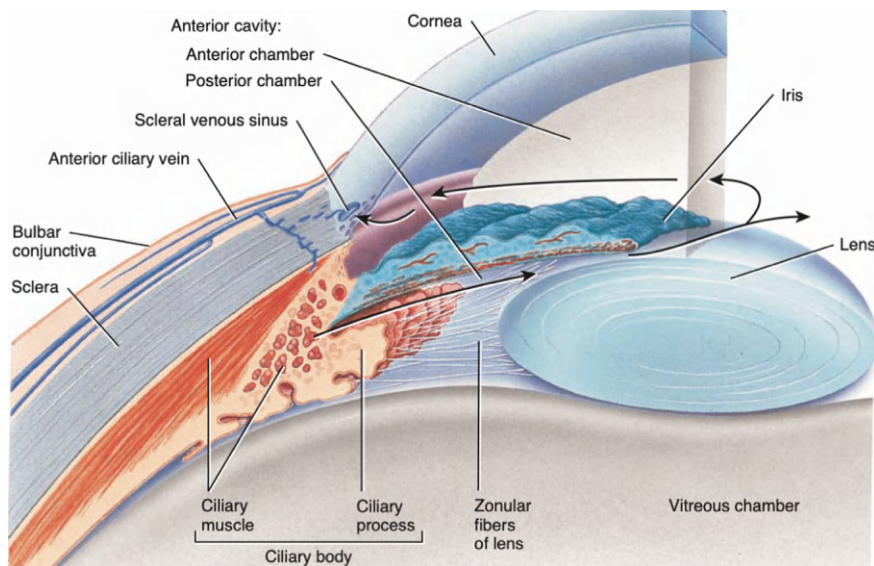


Fig. 5.3 The lens separates two major cavities in the interior of the eyeball: the anterior cavity and the vitreous chamber. The anterior cavity is formed by the anterior and posterior chambers, both containing aqueous humour. The vitreous cavity contains vitreous humour (from Tortora and Derrickson [2], with permission)

fulfils both a nutritional function in supplying nutrients to the avascular tissues of the lens and the cornea, as well as a removal function to clear metabolic wastes.

Aqueous Humour Generation and Absorption

Classically [2, 5], the dynamics of aqueous humour generation and absorption is described as follows. Aqueous humour is secreted by the **ciliary processes** in the **ciliary body** into the posterior chamber; then aqueous humour flows around the lens into the anterior chamber. See arrows in Fig. 5.3. From here aqueous humour is traditionally described as being absorbed into the venous system via two routes:

1. The **trabecular meshwork (TM)**, also called **conventional pathway**.
2. The **uveoscleral pathway**, also called **non-conventional pathway**.

A third outflow route, the **subconjunctival outflow route**, is also recognised in the literature [9]. This third route will be discussed in more detail in Sect. 5.1.8. See Smith et al. [10] for new developments on aqueous humour outflow routes.

The generation of aqueous humour in the ciliary body follows a **circadian rhythm**, that is a 24-hour cycle. Generation of aqueous humour is higher in the morning than at night. The mechanisms that explain the circadian rhythm are not well understood [11]. In healthy adults the generation rate of aqueous humour is of

the order of $2.4 \pm 0.6 \mu\text{L}/\text{min}$. This figure corresponds to daytime measurements in human adults aged 20 to 83 years [11].

From the trabecular meshwork, which is a porous structure, aqueous humour flows into the lumen of the **canal of Schlemm**, also called **scleral venous sinus**. This is an opening located at the junction of the sclera with the cornea. From the canal of Schlemm aqueous humour flows into collector channels, into aqueous veins, and finally into episcleral veins [9]. The pressure of aqueous humour in the episcleral veins is of the order of 8 mmHg to 10 mmHg. According to Lee et al. [9] the conventional pathway accounts for about 90% of the total aqueous humour outflow, under physiological conditions. The **non-conventional pathway**, or uveoscleral pathway, is composed of the uveal meshwork and the anterior face of the ciliary muscle. Aqueous humour enters the connective tissue between the muscle bundles, through the suprachoroidal space, and out through the sclera [11]. See Figs. 5.1 and 5.3. Most resistance to aqueous humour flow is found in the trabecular meshwork, of the order of 75%, while the uveoscleral pathway contributes with the remaining 25% of resistance. These figures vary significantly in the literature [9].

The overall process of generation and absorption of aqueous humour is analogous to that of cerebrospinal fluid in the cranial cavity; see Chap. 4 for details. Aqueous humour is generated from blood capillaries and is eventually reabsorbed into the venous system. The balance of generation and absorption of aqueous humour, and its dynamics in the anterior and posterior chambers, determines the **intraocular pressure** (IOP). Analogously, the balance of generation and absorption of cerebrospinal fluid, and its dynamics in the cranio-spinal cavity, determines the **intracranial pressure** (ICP). The intraocular pressure is a quantity of major physiological and clinical interest; normal values of IOP are between 12 mmHg and 20 mmHg, with an average of about 15 mmHg. IOP is also called eye pressure; elevated IOP is referred to as ocular hypertension. There is consensus that IOP is mainly determined by the aqueous humour, in the anterior and posterior chambers, but a role is also recognized to the vitreous humour in the vitreous chamber [2]. See also Sect. 5.1.6.

Recent research on aqueous humour generation and absorption includes the work of Dvoriashyna and collaborators [12], who constructed a mathematical model that predicts the production rate of aqueous humor by the ciliary processes as well as its composition in the posterior chamber. Also worth-mentioning is recent research in which the existence of an eye lymphatic system is acknowledged, which constitutes another means of draining aqueous humour, eventually to the venous system; see Sect. 5.1.8 for details.

Aqueous Humour Dynamics

Motion of aqueous humour in the anterior and posterior chambers fulfils a most important physiological role, for which three mechanisms are recognised in the literature, namely

1. Secretion of aqueous humour in the ciliary body and absorption through various mechanisms referred to above, naturally create pressure gradients that induce motion of aqueous humour.
2. Temperature gradients between the cornea and the iris generate aqueous humour motion.
3. Eye movement, of various kinds, also induce aqueous humour motion.

There are various reasons for studying the fluid dynamics of aqueous humour. First of all, aqueous humour motion is crucial for the understanding of eye fluid physiology, generally. Clinically, aqueous humour motion is of interest to understand motion and deposition of particulate matter, in the context of eye drug administration. Also of clinical interest is the estimation of the magnitude of shear stresses caused by fluid motion, acting on various eye structures, such as endothelial cells, for example. The intraocular pressure, a physical parameter of primary importance, emerges from the complete dynamics of aqueous humour, coupled to its generation and absorption.

There are many, both experimental and theoretical, studies in the literature concerned with aqueous humour motion. Experimental works include the early communication of Holm [13], for example, which was concerned with pupillary aqueous humour flow in living humans. Through a photogrammetric procedure they succeeded in estimating the volume of pupillary flow in the living eye. The more recent work of Takahashi and Atsumi [14] reported on the development of an automatic analyser for eye movement, applicable to specific subjects. Their study has become a reference communication concerning eye movement, in which they identify various types of movements and provide measurement of relevant parameters that have proved useful for subsequent research.

Mathematical modelling, coupled to analytical and semi-analytical techniques, have also been applied to the study of aqueous humour motion. This approach relies on strong simplifications to the fluid physiology so as to derive tractable mathematical models that may yield analytical, or semi-analytical, solutions to the problem. This is a powerful approach for identifying relevant parameters of the problem and their explicit interdependence, as well as aiding our understanding of the overall mechanisms at play.

Canning, Dewynne and Fitt [15] studied the flow of aqueous humour in the anterior chamber resulting from temperature differences between the front and back of the anterior chamber. In constructing their model they assumed the flow to have small Reynolds numbers, to be viscous dominated and driven by temperature gradients. Their model is capable of predicting various pathophysiological features. Fitt and González [16] considered more broadly the fluid mechanics of aqueous humour in the anterior chamber. They deployed an asymptotic analysis approach to study the motion of aqueous humour in the anterior chamber of the eye resulting from five types of mechanisms, including flow caused by rapid eye movement during sleep.

Repetto and collaborators have communicated a series of papers concerned with various aspects of ocular fluid dynamics [17–22]. The last paper is specifically concerned with aqueous humour fluid dynamics, the rest being concerned with the vitreous humour; see Sect. 5.1.6. Dvoriashyna and collaborators have also communicated various works concerned with mathematical modelling of ocular fluid dynamics [12, 23–25]. In [23], for example, the authors study the fluid dynamics of aqueous humour in the posterior chamber of the human eye, subject to perturbations of the normal flow from the posterior to the anterior chamber. The perturbations considered were (i) blockage of the pupil and (ii) iridotomy, that is a hole through the iris to facilitate flow from the posterior to the anterior chamber. In [24] the authors study the flow induced by eye rotations in the anterior chamber of the human eye. By applying lubrication theory and under sufficiently strong assumptions, they obtain analytical solutions. For more realistic cases they deploy numerical methods to solve the associated system of ordinary differential equations.

Mathematical modelling, in conjunction with computational techniques, has also been applied to the simulation of aqueous humour flow. Abouali and collaborators [26] investigated saccadic eye movement effects on the fluid dynamics of aqueous humour in the anterior chamber. They solved the time-dependent, three-dimensional continuity and Navier-Stokes equations using a commercial CFD package. As a validation exercise they first compared numerical solutions against existing analytical solutions for simplified situations. They also compared computed solutions against the experimental data of Repetto et al. [17] for the case of vitreous humour in [27]. The authors used the computational tool to study in detail the fluid dynamics of aqueous humour resulting from saccadic eye movement under physically realistic conditions. For details on their findings see [26]. Modarreszadeh and collaborators [28] applied the same computational model of Abouali and collaborators [26] to study the fluid dynamics of aqueous humour in the anterior chamber due to rapid eye movement. Repetto et al. [22] deployed a mathematical model for aqueous humour flow in the anterior chamber, in the presence of intraocular lenses. The governing equations were solved with the open source software *OpenFOAM*. From the model results they concluded that the alterations of the aqueous humour dynamics in the anterior chamber, resulting from lens implants, were unlikely to cause medical complications to the patient.

An overview of aqueous humour in the anterior and posterior chambers has been presented. We next study the vitreous humour in the vitreous chamber.

5.1.6 Vitreous Humour

This section is devoted to the largest material compartment of the eye, namely the vitreous chamber and its content, the vitreous humour, or vitreous body, or simply vitreous. The human vitreous is described as a transparent aqueous solution which fills the vitreous chamber of the eye, located between the lens and the retina; see Fig. 5.1.

Composition of Vitreous

The vitreous humour occupies about 80% of the total volume of the eyeball and comprises 99% water, but has a gel-like structure at birth due to a network of collagen fibrils and large molecules of hyaluronic acid. The vitreous humour characteristics vary from individual to individual. They also vary with age; aging causes the vitreous humour to undergo a process of progressive liquefaction, so that at the age of 80 to 90 years, more than half of the vitreous humour is liquid. Hall [5] describes vitreous humour as *a gelatinous mass held together by a fine fibrillar network composed primarily of greatly elongated proteoglycan molecules*. Table 5.1 shows concentrations of hyaluronic acid and collagen in vitreous humour of human, porcine, bovine and ovine. The complexity of the vitreous body compartment is revealed in the preface to Sebag's book [29] written by CL Schepens: *The vitreous body long has been the most mysterious of all ocular structures, owing perhaps to its seeming simplicity*. Clinical aspects of the vitreous are discussed in the recent review of Asencio-Duran et al. [30], and references therein.

The characterisation of the vitreous humour from the mechanical and chemical points of view, is important for several reasons. Examples include:

1. Understanding the physiology of vision, generally.
2. Gaining insight into vision pathologies linked to vitreous properties, such as liquefaction with age.
3. Studying injection and transport processes associated with intravitreal drug delivery, of much clinical importance.
4. Informing the design of vitreous substitutes.
5. Aiding the construction of mathematical models based on assumptions close to reality.

For the biochemistry of the vitreous humour we refer the reader to the recent communication of Aretz et al. [32] and references therein. They conducted a comprehensive study of the **proteome** of the vitreous body; the authors describe their work as *the most comprehensive human vitreous proteome coverage and list reported to date*. Recall that the proteome is the set of proteins present in an organism, under specific conditions. The authors identified 1111 unique proteins in the human vitreous body proteome, distributed into many groups with distinct biological functions. They

Table 5.1 Concentration of hyaluronic acid and collagen in vitreous humour of human, porcine, bovine and ovine (data from Shafae et al. [31])

Vitreous humour	Hyaluronic acid ($\mu\text{g/mL}$)	Collagen ($\mu\text{g/mL}$)
Human	140–340	280–1360
Porcine	160	150
Bovine	50–570	684
Ovine	100–1070	384

concluded: *this highly complex mixture strikingly differs from the human plasma proteome. Thus human vitreous fluid seems to be a unique body fluid.* See [31] for the biochemical composition of vitreous in humans, porcine, bovine, and ovine. See [33] for the correlation between age and biochemical compositions of the human vitreous.

Rheological Properties of Vitreous

To discuss mechanical properties of the vitreous humour, one begins by adopting an appropriate definition for it. A reasonable question to ask is this: is vitreous humour a fluid or is it a solid? In many publications vitreous humour is defined as a fluid, perhaps in reference to its 99% water content. However, a more appropriate definition is **hydrogel**. Generally, a hydrogel is defined as a **biphasic material** comprising a mixture of a porous permeable solid phase and at least 10% of a fluid phase, mainly composed of water [34]. As already stated the water content of vitreous humour is reported as 99%, while the solid phase is composed of collagen fibrils (type II, IX, V, IX, and IV) interspersed with hyaluronic acid. Table 5.1 gives concentrations of hyaluronic acid and collagen in vitreous humour of human, porcine, bovine and ovine. The rheological properties of the vitreous humour have proved hard to determine reliably. Here we shall briefly review some published works on the subject. For useful background on biomechanics, particularly of viscoelastic materials, see the classical book of Fung [35] and the more recent book of Humphrey and Delange [36]. For general background see the introductory Sect. 11.4 of Chap. 11 of [36].

The overall resistance to deformation of the hydrogel is called the **complex modulus** and is denoted as G^* . Contributions to G^* come from the elastic solid phase G' and a viscous contribution G'' from the liquid phase. G' is called the **elastic modulus** or **storage modulus**, while G'' is called **viscous modulus** or **loss modulus**. Thus, the overall rheological properties of vitreous will be determined by the mechanical properties of the collagen and hyaluronan as well as their concentrations, see Table 5.1.

Several techniques for quantifying the mechanical properties of the mammalian vitreous have been reported, including acoustic techniques, magnetic resonance imaging techniques [37] and shear rheometry, which has become popular in recent years [38]. One of the first studies of the human vitreous humour rheology is attributed to Lee et al. [39], who reported rheological properties as function of location within the eye, making it evident that such properties are not homogenous in space. Recall also that they are not homogenous in time either, as these properties change strongly with age, vitreous liquefaction figuring prominently.

Shafaie et al. [31] investigated the rheological properties of vitreous samples from humans and mammals (ovine, porcine and bovine), and carried out detailed comparison of the resulting measurements. In addition, the authors also put forward a *first generation vitreous mimic*, with a composition of 4.5 mg/L hyaluronic acid and complex viscosity of $0.3 \pm 0.01 Pa$; they found it to be a better mimic of the human vitreous than the mammalian samples evaluated. As to rheological properties of vitreous samples of human, bovine and porcine, their results include values for the storage modulus G' . Their reported measures are $1.4 \pm 0.95 Pa$, $1.7 \pm 0.31 Pa$,

$1.4 \pm 0.14 Pa$, respectively. For the ovine vitreous, they reported the value $G' = 4.2 \pm 0.62 Pa$, which is significantly higher than the other measures.

The authors concluded that the rheological properties of human vitreous are comparable to those of porcine and bovine, while ovine vitreous has greater elasticity and viscosity than human vitreous. Shafaie et al. [31] also evaluated the diffusion properties of the vitreous samples used in their study. They found that, even though the rheological properties of vitreous samples considered are similar, diffusion properties are not. The diffusion of fluorescein through these vitreous samples showed substantial differences in measures of steady-state flux and diffusion coefficient. Diffusion properties of vitreous are important from the clinical point of view. Ocular diseases associated with the posterior segment of the eye motivates the development of drugs to be delivered by intravitreal injections, directly into the vitreous body, rather than via the systemic route. Important aspects are drug penetration and residence time, which are strongly dependent on the material properties of vitreous.

Tram and Swindle-Reilly [38] carried out a comprehensive study of the rheology of the human vitreous humour. The main novelty of their research was the evaluation of the influence of age on the rheological properties. Their study demonstrated that, with age (i) the vitreous gel becomes stiffer and (ii) the vitreous liquid becomes less elastic. Liquefaction of vitreous with increasing age is a well documented fact, whereby the volume of liquid vitreous is known to increase. The hallmark of liquefaction is phase separation, resulting in pockets of liquid inside the vitreous. The authors note that quantitative studies of age-related vitreous liquefaction remain elusive, particularly regarding its effect on mechanical properties. By considering two groups, older than 65 years (old humans) and younger than 65 years (young humans), amongst other results, they reported the following values: $G_{old}^* = 1.11 \pm 0.64 Pa$, $G_{young}^* = 0.53 \pm 0.39 Pa$ and $G_{porcine}^* = 0.24 \pm 0.16 Pa$. That is $G_{old}^* > G_{young}^* > G_{porcine}^*$.

Table 5.2 shows a comparison of results for storage and loss moduli for human and porcine vitreous, from the works of Shafeie et al. [31] and Tram et al. [38], in which both groups use the same methodology, namely shear rheometry. To be noted, Tram et al. [38] used 23 human samples and 15 porcine samples, while Shafein et al. [31] used 3 pairs of samples for both human and porcine vitreous. For more results and discussion on age-related changes in the human vitreous see the original paper of Tram et al. [38] and references therein. For latest results on human vitreous see Luo et al. [40].

Table 5.2 Comparison of results for storage and loss moduli for human and porcine vitreous (data from Tram et al. [38] and Shafaie et al. [31])

Species	G' [38]	G' [31]	G'' [38]	G'' [31]
Human	$6.5 \pm 3.0 Pa$	$1.4 \pm 0.95 Pa$	$0.96 \pm 0.47 Pa$	$0.7 \pm 0.37 Pa$
Porcine	$5.0 \pm 0.58 Pa$	$1.4 \pm 0.14 Pa$	$0.65 \pm 0.22 Pa$	$0.4 \pm 0.14 Pa$

Exchange in the Vitreous Chamber

Regarding the dynamics of the vitreous body, Smith et al. [10] state that it is generally accepted that for all practical purposes the *fluid* content of the vitreous chamber is stagnant and that the **aqueous humour** occupying the anterior and posterior chambers exits the eye only via anterior pathways, as already described in Sect. 5.1.5. However, this view has been challenged by Aretz et al. [32], amongst others, who state that there is exchange of materials between the vitreous chamber and surrounding structures. In particular, they state that **soluble proteins** in the vitreous humour are thought to originate from blood plasma by filtration through fenestrated capillaries of the ciliary body stroma via the iris root.

Furthermore, Aretz et al. [32] state that *there is a continuous exchange between the aqueous humor and the vitreous humor located in the posterior part of the eye behind the lens*. Moreover, proteins from vitreous chamber tissues, including the **retinal pigment epithelium** (RPE), may impact on the pattern of soluble proteins present in the vitreous. Recall that the RPE lies between the Bruch membrane (next to the choroid) and the retina, as depicted in Fig. 5.4. The RPE is a fundamental component of the retina that plays essential roles in visual functions [41].

The recent work of Smith et al. [10] also challenges the view that the fluid content of the vitreous chamber is stagnant. In particular, they also claim that there is strong evidence suggesting there is a substantial fluid flow across the retinal pigment epithelium. They identify strong and independent lines of evidence supporting fluid flow across the RPE. They conclude by stating that *on balance, it appears the current evidence favours the view that there is normally a significant aqueous flow across the RPE in vivo*.

Boulton and Barker [41] state that the RPE functions both as a selective barrier to and a vegetative regulator of the overlying **photoreceptor layer**, thereby playing a key role in its maintenance. Moreover, consistently with Smith et al. [10], they state that *the RPE regulates the transport of nutrients and waste products to and from the retina*. The RPE also contributes to maintain retinal homeostasis through the release of diffusible factors. Hall [5] also suggests the existence of material exchange by filtration and diffusion between the vitreous chamber and the vasculature in the retina.

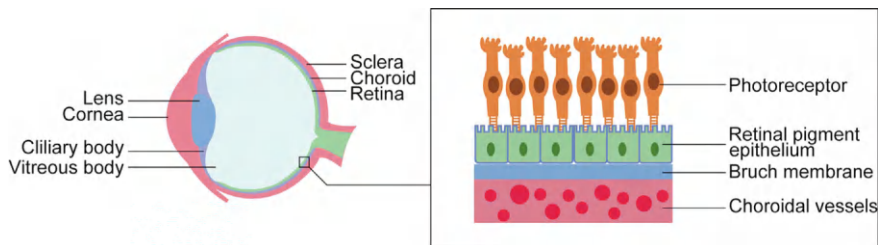


Fig. 5.4 Retinal Pigment Epithelium (RPE) between the Bruch membrane (next to the choroid) and the retina (from Sun et al. [42])

In line with [5, 10, 32] and others, Dvoriashyna and collaborators [25] have recently communicated a novel mathematical model to simulate fluid and solute transport across the RPE. The model confirms the expectation that the flow across the RPE is driven by Starling forces determined by osmotic gradients across cell junctions. See Chap. 2 for background on Starling forces. Amongst other outcomes, the model of [25] gives estimates on how water transport is affected by inhibition of membrane ion channels and carbonic anhydrase. The authors note that carbonic anhydrase inhibitors are used clinically to prevent fluid accumulation in the sub-retinal space, which lies between the neural layer of the retina and the RPE.

Recently, Dorsam et al. [43] have communicated results from numerical simulations of aqueous humour flow in the anterior and posterior chambers, as well as the vitreous chamber. An underlying assumption was that aqueous humour also leaves the eye through the vitreous chamber and the retina; see Sect. 5.1.5. As a mathematical model for the dynamics of aqueous humour the authors adopted the Stokes and Darcy partial differential equations. They also considered the transport of injected material via an additional advection-diffusion-reaction equation for the concentration. To solve the equations the authors used the finite element method built in the software **deal.II**. This contribution is a significant step forward in approaching the physiology of eye fluid compartments. Of course there are many potential improvements that could be undertaken.

Eye fluid compartments are not strictly rigid, they are not isolated amongst themselves or from the rest of the body; the vasculature plays a mayor role in the dynamics of all fluid compartments. Notably, there is exchange of materials between fluid compartments, which must be taken into account in order to better understand salient aspects of the fluid physiology, especially under pathological conditions. As it is usual in mathematical models for bodily fluids one faces the problem of having to choose values for a variety of parameters. In the study of Dorsam et al. [43], important parameters include the porosity in the trabecular meshwork, the rate of aqueous humour production and the pressure of the episcleral veins. By appropriate choice of parameters they predicted values for intraocular pressure and other quantities of physiological relevance.

Vitreous Humour Motion Induced by Eye Movement

Motion of the vitreous humour may be induced by various types of eye movements. **Saccade** is just one out of four well-documented types, defined collectively as rapid eye movements that may range from reading to gazing around a room. These eye movements are characterized by an extremely high angular acceleration and are known to induce shear stress at the wall of the vitreous chamber and the retina. The shear stresses induced by vitreous motion are thought to play an important role in the pathogenesis of retinal detachment and associated eye pathologies. There are many studies on this topic in the literature, using experimental methods, theoretical studies through simplified mathematical models and theoretical simulations involving more complex mathematical models requiring sophisticated numerical approaches.

Repetto and collaborators[17] conducted an experimental study of the vitreous motion induced by saccadic eye movements. In order to emulate the vitreous chamber and the dynamics of its content, the authors constructed a physical model consisting of a cylinder in perspex material filled with glycerol. The underlying assumptions were that the vitreous chamber was a rigid cavity and that the vitreous humour was a viscous homogeneous fluid (glycerol). They visualised the three-dimensional motion of the glycerol and compared measurements with calculations from a simplified mathematical model, obtaining satisfactory agreement. Despite the strong assumptions made, relevant physical information was extracted from the experiments. For example, the maximum shear stress at the wall significantly increased with the fluid viscosity. This finding has clinical implications, in the context of stresses on the retina and potential occurrence of retinal detachment.

Later, Repetto and collaborators [18] communicated a mathematical model for vitreous motion induced by saccadic eye rotations. They assumed the vitreous humour to be a Newtonian fluid. Their work focused on the effect of the shape of the domain on the flow field. Despite the strong simplifications underlying the mathematical model, their theoretical predictions were shown to be in good qualitative agreement with published experimental data. A finding of relevance for the problem of intravitreal drug injections is that vitreous motion facilitates drug diffusion through the eye ball, as compared to the stationary-fluid case. Through numerical simulations, Repetto et al. [19] studied the effect of traction, induced by saccadic eye movement, exerted by detached vitreous on the retina. As is well known, such traction may result in tearing of the retina and even detachment. They found that for various configurations considered, eye rotations can generate large traction on the retina.

Bonfiglio et al. [20] carried out an experimental study on vitreous motion induced by eye rotation. They found that these induce vitreous motion and stresses on the retina, and within the vitreous body. They also noted that vitreous viscoelasticity has a strong influence on motion, which has implications for the designing of vitreous substitute materials possessing the required properties.

Abouali et al. [27] carried out simulations of vitreous body dynamics, based on numerical solutions of the time-dependent, three-dimensional Navier-Stokes equations. They studied the fluid dynamics of vitreous body in the vitreous chamber caused by saccadic eye movement, following vitreous liquefaction. In particular, they studied the effect of shear stress on possible retinal detachment. The authors also developed an analytic approximation for the maximum shear stress acting on the wall of the spherical cavity subjected to sinusoidal rotations. This solution puts in evidence the parameters that affect maximum shear stress, such as density of fluid, kinematic viscosity and radius of the sphere simulating the eyeball. This analytical solution is also useful for validating full-fledged numerical simulations.

A clinical note: posterior vitreous detachment. With increasing age, collagen fragments aggregate into thicker fibres, causing weakening of vitreoretinal adhesion at their interface. These combined processes eventually result in posterior vitreous detachment (PVD) in approximately 50% of individuals older than 50 years of age.

See Kakehashi et al. [44] for a comprehensive study on variations in PVD and a clinical classification system for PDV.

5.1.7 Goldmann's Equation for Intraocular Pressure

Intraocular pressure keeps the eyeball inflated, ensuring an adequate shape that fulfils optical requirements. Under physiological conditions, IOP is of the order of 15 mmHg. Glaucoma is a family of vision-threatening optic neuropathies strongly related to elevated IOP, with subsequent injury to the optic nerve [11, 45]. Glaucoma causes irreversible but potentially preventable vision loss. Conventionally, elevation of IOP is a central tenet of glaucoma pathology and treatment [11]. In the ophthalmology literature, occasionally one finds reference to the Goldmann equation for intraocular pressure [9]. Often however, the quantities involved are not clearly defined, particularly their physical meaning and dimensions. For a review of the Goldman equation and extensions see the paper of Brubaker [46].

As a matter of fact, the Goldmann equation is equivalent to Darcy's law of flow applied to two connected compartments. See Chap. 2 of this book. Let us define the first compartment as the union of the anterior and posterior aqueous humour chambers, with aqueous humour pressure $P_L = IOP$, and the second compartment as the episcleral vein downstream, with pressure $P_R = P_{vein}$. Define the flow rate of aqueous humour (or just flow) between these two compartments as Q . Then, Darcy's law of flow reads

$$Q = \frac{P_L - P_R}{R} = \frac{IOP - P_{vein}}{R}, \quad (5.1)$$

where R is the total resistance to flow between the two compartments, which includes resistance from both the trabecular and uveoscleral pathways. Instead of *resistance to flow* R one can introduce the *ease of flow* [47] as the reciprocal of resistance $1/R$. But here we shall keep R in the equations. From (5.1) it follows that IOP is given as

$$IOP = Q \times R + P_{vein}. \quad (5.2)$$

This very simple estimate tells us that IOP will increase (i) if flow Q increases, (ii) if resistance R to outflow increases, and (iii) if episcleral vein pressure P_{vein} increases. Of these three factors, the last one is rarely appreciated, even though it is reasonable to consider that cerebral venous hypertension, secondary to downstream venous outflow strictures, could have an effect on IOP [8]. Extensions to (5.2) are found in [46]. See also [8], where analogous extensions were proposed for intracranial pressure in the context of Idiopathic Intracranial Hypertension, noting that intraocular pressure is in many ways analogous to the intracranial pressure (ICP) in the craniospinal cavity [48, 49].

5.1.8 *Lymphatics and Glymphatics in the Eye*

Major functions of the lymphatic system include immune surveillance, absorption of lipids, tumour progression and metastasis, removal of excess interstitial fluid from the interstitium, clearance of wastes and transport of fluid and solutes to the venous system, either at the level of lymph nodes or further downstream into the neck veins at the thoracic duct and the right lymphatic duct. See Chap. 3.

Traditionally, the eye has been considered to be devoid of a lymphatic network to collect excess ISF [2]. Recent works, however, provide evidence for the presence of lymphatic vessels in the visual system and for the potential contribution of eye lymphatics to drainage of ocular fluids and clearance of metabolic wastes [4, 9, 50, 51]. Furthermore, in analogy with the brain parenchyma, it is proposed that the eye also possesses a **glymphatic system** that relies on networks of perivascular channels to drain excess fluid from tissue, eventually, to the venous system [4, 45].

The fluid transport function of both the lymphatics and the proposed glymphatics may well contribute to the drainage of aqueous humour and thus to the regulation of intraocular pressure, usually regarded as a risk factor for damage to the optic nerve and the onset of glaucoma. Furthermore, both the lymphatics and the proposed glymphatics are concerned with the removal of toxic waste products that may be responsible for eye retinal neurodegenerative processes. It is reasonable to argue that such function may be as important as elevated IOP. The recent demonstration of the existence of the lymphatic and glymphatic fluid systems is likely to change our view of ocular fluid dynamics in health and disease. In what follows we give a brief account of both the lymphatic and glymphatic systems.

Eye Lymphatics

For general background on the human lymphatic system, the reader is referred to Chap. 3. Eye lymphatics in the conjunctiva have, for a long time, been known to exist [9], though such knowledge has not yet been transferred to eye physiology textbooks. Moreover, dura lymphatics have also been recently discovered in the optic nerve [4]; this eye dural lymphatic system is entirely analogous to the brain dural lymphatic system discovered in 2015 [52, 53].

As pointed out earlier, two major drainage pathways are well characterized, namely the conventional trabecular meshwork/Schlemm's canal pathway, and the non-conventional uveoscleral route through the supraciliary and the suprachoroidal spaces. Via these pathways, aqueous humour reaches the blood venous circulation in either episcleral veins or choroidal vessels, respectively. The uveoscleral route allows aqueous humour flow along the supraciliary and suprachoroidal spaces. From here, aqueous humour flows to choroidal vessels, eventually reaching the venous blood circulation.

Figure 5.5 depicts conventional routes for aqueous humour dynamics. It also shows additional routes for aqueous humour outflow, namely the subconjunctival space and the **lymphatic vessels of the conjunctiva**, which then drain aqueous

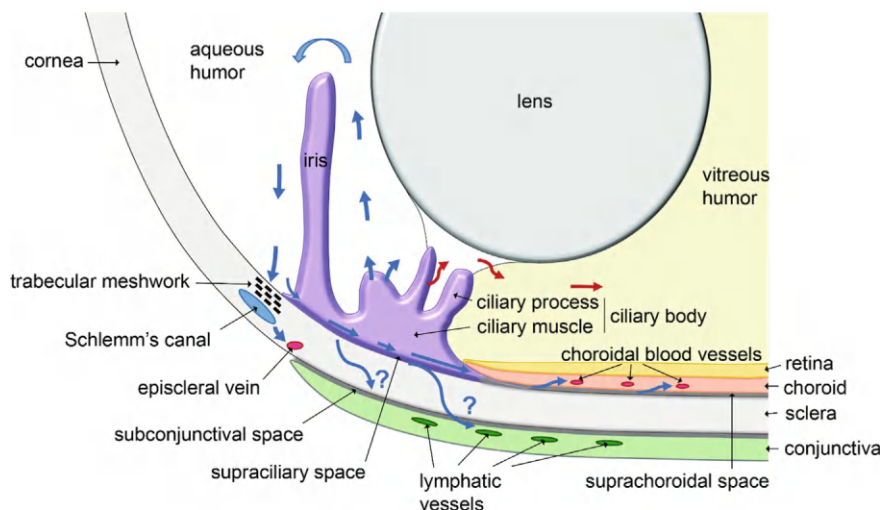


Fig. 5.5 Revisiting the routes for aqueous humour flow in the anterior chamber. Blue arrows: outflow; red arrows: inflow. In addition to the well established aqueous humour pathways, new routes are suggested, the subconjunctival space and the lymphatic vessels of the conjunctiva (from Subileau and Vittet [45], with permission)

humour to the peripheral lymphatics, namely the neck lymph nodes. Potentially, the extended network for aqueous humour outflow will have an effect on intraocular pressure, which is considered to be a major risk factor for the progression of glaucoma. Moreover, the additional routes for aqueous humour drainage [9] contribute to the elimination of metabolic wastes from the retina [4]. For a recent review of these issues see Subileau and Vittet [45] and references therein.

Eye Glymphatics

Very recently, the existence of an **eye glymphatic system** in rodents has been demonstrated [4, 45]. The eye glymphatic system parallels the brain glymphatics described in Chap. 3, which is characterised in part by flow in periarterial and perivenous spaces. Figure 5.6 depicts the glymphatic system in the eye, taken from Subileau and Vittet [45]. The figure shows the periarterial spaces around a retinal artery, shown in red arrows. The fluid in the periarterial spaces is cerebrospinal fluid [4] that flows in the direction of arterial blood [54] towards the eye. Similarly, the perivenous spaces, as exemplified by the space around the retinal vein shown by blue arrows in Fig. 5.6, transport ISF/CSF and solutes from the retina towards peripheral lymphatics. The perivenous spaces provide a metabolic waste clearance route for the highly active retinal neurons. As in the brain, such clearance pathway depends on glial water channel aquaporin-4. In [4] it is argued that the flow is driven by a pressure difference across the **lamina cribrosa**, which normally supports an ocular/intracranial pressure difference. Under physiological conditions intraocular pressure is higher

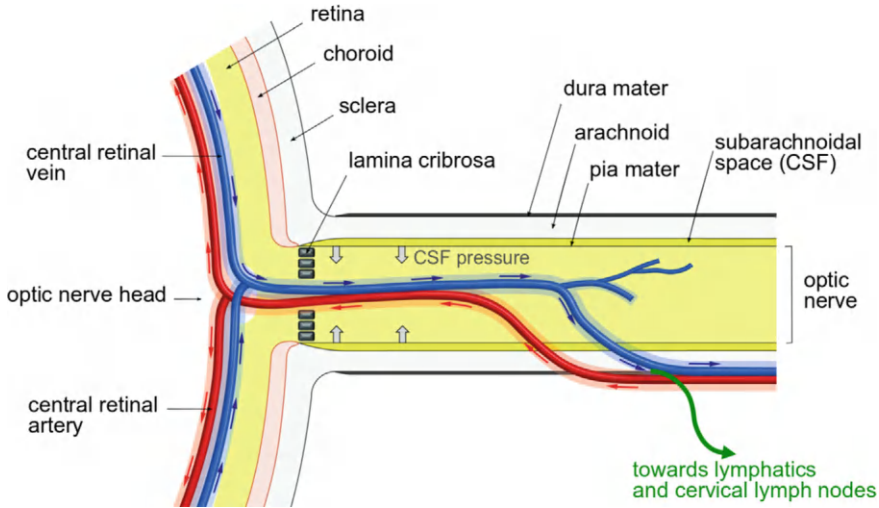


Fig. 5.6 The eye glymphatic system: optic nerve parietal and perivenous spaces (Courtesy of Professor D. Vittet and Professor M. Subileau. Data from [45])

than intracranial pressure. A distinguishing function of the glymphatic system is the elimination of metabolic waste products from the eye retinal layers. Such wastes are transported further downstream from the perivenous spaces to peripheral lymphatics, cervical lymph nodes and eventually the venous system.

Elimination of retina waste products via the glymphatic system depends on (a) transport of CSF from the subarachnoid space into the eye along periarterial spaces, (b) transport of fluid and solutes from the eye along perivenous spaces, (c) drainage of these materials into the peripheral lymphatics, and eventually (d) into the venous system. Also, the balance between generation and absorption of aqueous humour and the maintenance of intraocular pressure are strongly dependent on transport of aqueous humour along these identified pathways. In all of these interconnections, the venous system plays a mayor role.

Summary on Ocular Fluids

After a succinct description of the principal aspects of the eye anatomy following classical textbooks we have concentrated more specifically on the fluid dynamics aspects of all major extracellular fluid compartments. Of these, we have given prominence to the vitreous humour, the aqueous humour and the cerebrospinal fluid in the optic nerve sheath, as well as their roles in the eye fluid dynamics physiology and pathology, such as glaucoma. Recent relevant works include Salerni et al. [55], Dvoriashyna et al. [25], Levin and Cohen [56], and Luo et al. [40]. We have not explicitly included the vascular system. Blood, arterial and venous, plays a major role in the fluid physiology of the eye as a whole, as all fluid compartments are

interconnected. Up-to-date research works providing a comprehensive view of the fluid mechanics of the eye are found in the book edited by Guidoboni, Harris and Sacco [57] and references therein, which is strongly recommended for further study. Recent advances in ocular fluid dynamics, such as the lymphatics and glymphatics, have been mentioned and briefly described with appropriate up-to-date references for further study. See [4] and [45], and references therein.

5.2 Fluids in the Inner Ear

This section focuses on the anatomical structures and major fluid compartments involved in the two functions of the ear: **hearing** and **equilibrium**. The inner ear houses and protects the neurological structures employed for the hearing function, performed by the cochlear apparatus, and the equilibrium function, performed by the vestibular apparatus. Background material is found in most classical books in human physiology; see for example Silverthorn [58], Tortora and Derrickson [2] and Hall [5]. Here we are also interested in research-related concepts centred on the interacting fluid dynamics of various fluid compartments in the inner ear, with the emphasis on the fluids that characterise the auditory system, namely perilymph and endolymph.

5.2.1 Ear Anatomical Structures

Here we give a very brief description of the ear main anatomical structures. Full details are found in most classical books on human anatomy [2, 5, 58]. Figure 5.7 depicts the structures of the three anatomical regions of the human ear, namely the outer ear, the middle ear and the inner ear. The **outer ear** includes the auricle, the external auditory canal and the eardrum, also called tympanic membrane. The **middle ear** is the air-filled cavity lying between the eardrum and a thin bony partition that contains two small openings, namely the oval window and the round window.

The **round window** is covered by a membrane, and in analogy with the eardrum, is called the secondary tympanic membrane. The **oval window** is attached to the base of the stapes, the third of the middle ear **auditory ossicles**. These are three small bone structures that, according to their shapes, are called the malleus (hammer), the incus (anvil) and the stapes (stirrup). These are the smallest bones in the body. The **Eustachian tube**, also called **auditory tube**, connects the middle ear with the superior portion of the throat, called nasopharynx; it plays a role in equalizing the pressure in the middle ear with that in the atmosphere. This function is most evident in daily life in the so-called *popping of ears*, as pressures equalize.

The **inner ear**, or **labyrinth**, is formed by a complex series of canals, as shown in Fig. 5.7. The inner ear is the district we are mostly interested in, whose main structures are depicted in Figs. 5.7 and 5.8. One distinguishes three main interconnected structures, namely the **semi-circular canals**, the **vestibule** and the **cochlea**.

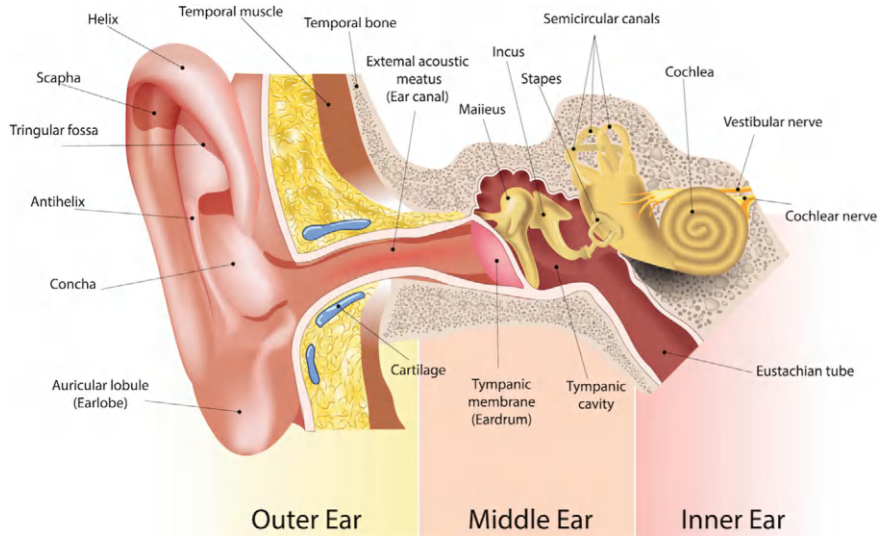


Fig. 5.7 Human ear main anatomical structures found in three districts: the outer ear, the middle ear and the inner ear (from Shutterstock, with permission)

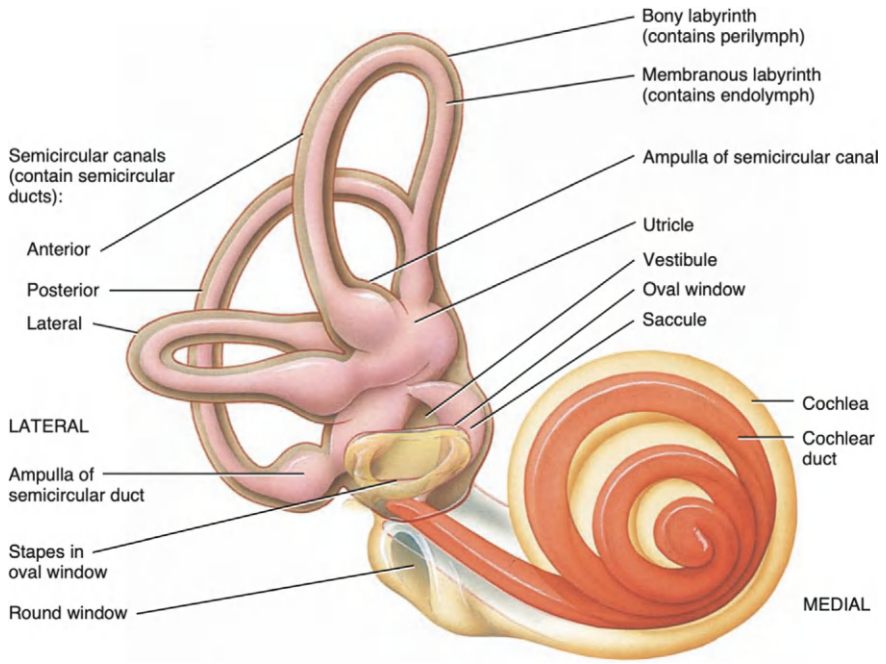


Fig. 5.8 Inner ear main components. The semi-circular canals and the vestibule are responsible for the equilibrium function; the cochlea is responsible for the hearing function. Note the presence of the fluids perilymph and endolymph. See also Fig. 5.9 (from Tortora and Derrickson [2], with permission)

The first two, that is the **semi-circular canals** and the **vestibule**, contain receptors to perform the **equilibrium** function, while the cochlea contains receptors to perform the **hearing** function. Then one distinguishes an outer bony labyrinth, whose external boundary is bone, and an enclosed inner membranous labyrinth. A very simplified geometrical analogue of this arrangement is a set of three coiled, side-by-side, tubes, as we discuss next.

5.2.2 The Cochlea: Perilymph and Endolymph

Here we discuss the perilymphatic and endolymphatic fluid compartments as in the cochlea; see the right hand side of Fig. 5.8. A representative description of the geometry of these fluid compartments is given in Fig. 5.9, which depicts a cross-section through one of the turns of the cochlea, which may be described as a system of coiled, side by side, fluid-filled tubes.

These three tubes are named scala vestibuli (top), containing perilymph; scala media (middle) containing endolymph; and scala tympani (bottom) containing perilymph. The scala vestibuli is separated from the scala media by the **Reissner membrane**, or **vestibular membrane**. The Reissner membrane is very thin and, as such, it is said to have a negligible effect on wave propagation in the fluids. However, the

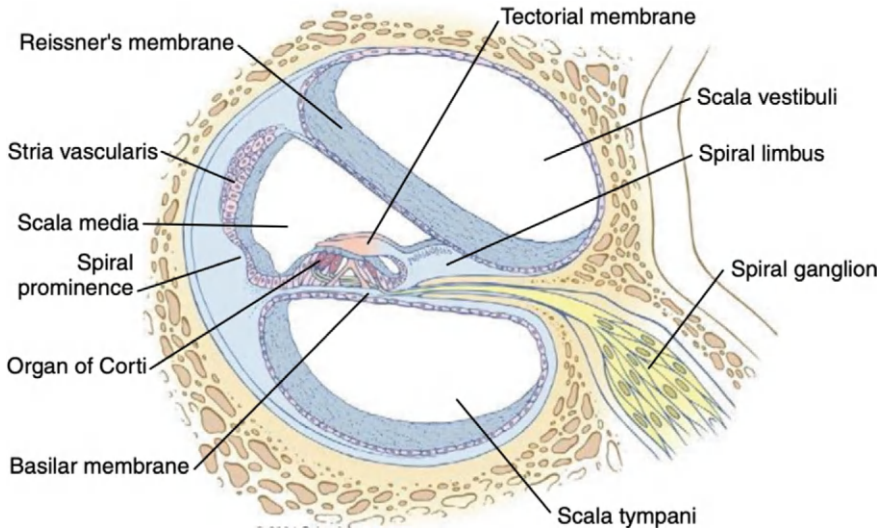


Fig. 5.9 Cross section of the cochlea depicting three fluid channels labelled: scala vestibuli (containing perilymph), scala media (containing endolymph) and scala tympani (containing perilymph). Endolymph is separated from perilymph by two membranes: the Reissner membrane and the basilar membrane (from Hall [5], with permission)

Table 5.3 Chemical composition of perilymph and endolymph (data from [59])

Component	Perilymph	Endolymph
	<i>Concentration/mg (100 ml)⁻¹</i>	
Protein	200 to 400	126
	<i>Concentration/mmol kg⁻¹</i>	
<i>Na⁺</i>	140	5
<i>K⁺</i>	10	144

Reissner membrane is important in keeping separate two kinds of fluids of different chemical composition, perilymph in the scala vestibuli and endolymph in the scala media. The scala tympani is separated from the scala media by the **basilar membrane**. On the surface of the basilar membrane lies the **organ of Corti**, which contains the **hair cells** that generate nerve impulses in response to sound vibrations. Also shown in Fig. 5.9 are spiral ganglion neural cells, which are ultimately responsible for sending signals, via the cochlear nerve, to the central nervous system.

The chemical composition of perilymph and endolymph is usually measured in terms of concentration of proteins and the electrolytes sodium Na^+ and potassium K^+ . Table 5.3 shows chemical concentrations for perilymph and endolymph. The most striking difference is in the electrolytes, as the concentrations of Na^+ and K^+ are virtually inverted. As noted by Schuknecht [59], anatomical and physiological studies show that the vestibular and auditory nerve fibers of the inner ear are bathed in perilymph. Moreover, perilymph has electrolyte levels similar to CSF, thus being suitable for the bathing of sensory and neural structures and for neurotransmission. On the other hand, the high potassium level of endolymph is toxic to sensory and neural structures and therefore blocks excitation and transmission. It is suggested that sudden hearing loss and acute vertiginous episodes may be linked to **potassium intoxication** arising from the rupture of the membranous labyrinth [59].

5.2.3 Anatomical Communications of Fluid Compartments

There are anatomical communication pathways between the perilymphatic and the endolymphatic fluid compartments in the inner ear and other CNS fluid compartments. Figure 5.10 depicts documented pathways amongst endolymphatic spaces, starting from the connection between the entire endolymphatic space to the **endolymphatic sac** via the **endolymphatic duct** [60]. Then, the cochlear duct is connected to the saccule via the so-called ductus reuniens. The saccular duct then connects the saccule to other compartments, such as the utricle via the utricular duct and the sinus. In addition, a most important connection, not shown in Fig. 5.10, is that between the perilymphatic space and the cerebrospinal fluid in the subarachnoid space, through the cochlear aqueduct. This connection is consistent with the fact that

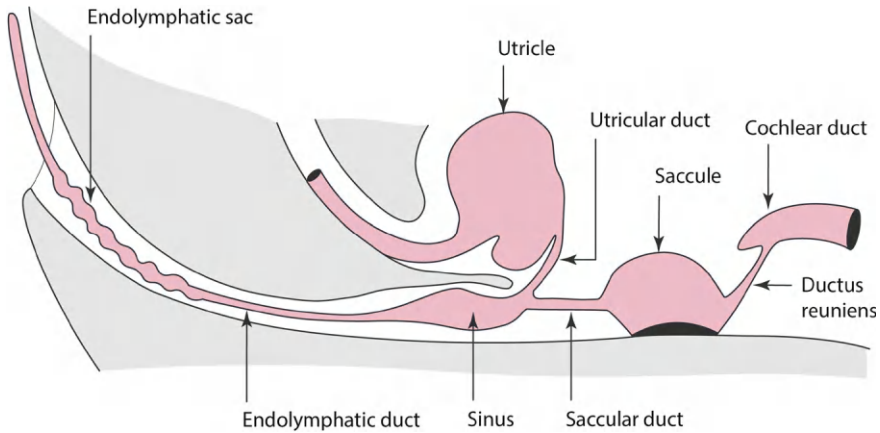


Fig. 5.10 Pathways amongst endolymphatic spaces, emphasising the direction of flow of endolymph towards the endolymphatic sac. Not shown is the connection between the perilymphatic space and the cerebrospinal fluid in the cerebral subarachnoid space via the cochlear aqueduct (Courtesy of Violeta Da Rold; data from Schuknecht [60])

the chemical composition of perilymph is similar to that of CSF, as seen in Table 5.3. Further details are provided by Schuknecht and Rütger [60] and references therein. As we shall see later in Sect. 5.2.5, these anatomical connections amongst fluid compartment are most important in, potentially, providing a better understanding of ear pathologies, such as Ménière’s disease. Pathological, congenital malformation of these connections lead to disruption of the fluid dynamics and associated transport processes in the inner ear.

5.2.4 The Hearing Function

As already stated, the ear is a sensory organ that is specialized for performing two distinct functions, namely **hearing and equilibrium**. The vestibular complex of the inner ear is the primary sensor for equilibrium, while the remainder of the ear is employed for hearing. Here we focus on some aspects of the hearing function; details about the equilibrium function are found, for example, in [58] and [2].

Figure 5.11 depicts the main inner ear components involved in the hearing function. It also shows the sequence of multiple transduction steps to convert sound waves, ultimately, into coded information about sound through the cochlear branch of the vestibulocochlear nerve (cranial nerve VIII) and the brain; these are:

1. Sound waves are directed into the external auditory canal towards the tympanic membrane (or eardrum).
2. Sound waves strike the tympanic membrane causing it to vibrate.

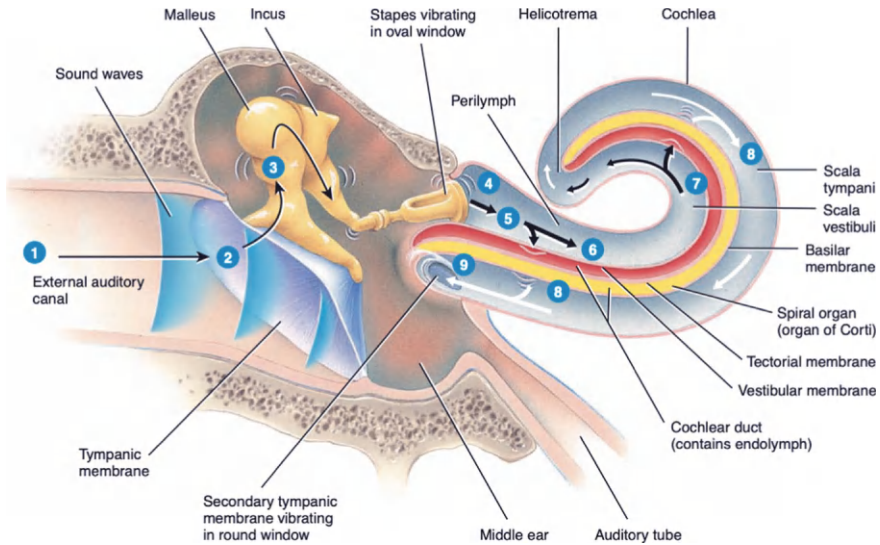


Fig. 5.11 Inner ear main components involved in the hearing function, which begins with sound waves being directed into the external auditory canal towards the eardrum, and ending with the production of receptor potentials that ultimately lead to the generation of nerve impulses (from Tortora and Derrickson [2], with permission)

3. The mechanical vibrations of the tympanic membrane are transmitted to the three bones (malleus, incus and stapes) in the middle ear, causing them in turn to vibrate.
4. As the stapes is attached to the oval window membrane, its vibrations cause motion of the perilymphatic fluid of the cochlea, as liquid waves.
5. The perilymph waves cause movement of the compliant membranes separating perilymph in the vestibular and basal ducts from endolymph in the cochlear duct. Such membrane movements bend hair cells and open ion channels creating electrical signals that alter neurotransmitter release.
6. Pressure waves are transmitted from the scala vestibuli to the scala tympani. As shown by number 9 in the figure, the waves eventually reach the round window, which then bulges out into the middle ear.
7. The propagating pressure waves in the scala vestibuli deform the flexible walls of both the scala vestibuli and the scala tympani. The pressure waves are transmitted into the endolymphatic fluid contained in the cochlear duct.
8. The pressure waves in the endolymph cause the basilar membrane to vibrate. The vibrations move the hair cells of the spiral organ against the tectorial membrane, leading to bending of the hair cell stereocilia, producing receptor potentials that ultimately lead to the generation of nerve impulses.

There are many additional aspects to the hearing function of the ear, such as the physical aspects of frequency, amplitude and duration of sound waves. Details on these, as well as neural aspects of hearing, are found for example in [2, 5] and [58].

5.2.5 *Ménière's Disease*

Here we consider a particular inner ear pathology, namely Ménière's disease (MD), a condition first described by the French physician Prosper Ménière in 1861. Ménière's disease is a pathology of the inner ear characterized by intermittent vertigo episodes, tinnitus, aural pressure and temporal sensorineural hearing loss. The fluid dynamics of the inner ear is central to the understanding of various mechanisms associated with several hearing disorders. Potentially therefore, such disorders involve the dynamics of the arterial and venous blood systems, as well as that of the perilymph and endolymph fluid compartments. Here we briefly review the potential role of the endolymphatic sac and the inner-ear blood circulation in MD.

The Endolymphatic Sac

The normal onset of MD is unilateral, but often extends to the other ear, with a registered 40% of bilateralization cases observed in the first five years, as reported by Bruno et al. [61]. MD is often described as a multifactorial pathology including genetic factors, autoimmune mechanisms, environmental conditions, infection and even psychological aspects. Though the aetiology of MD remains uncertain, a fluid dynamics component of MD is currently recognised. For example, disorders in the dynamics of the endolymphatic system have for a long time been associated with MD.

The volume and pressure of endolymph in the endolymphatic sac is controlled by the balance between production and absorption of endolymph. This phenomenon is analogous to that of CSF in the brain; see Chap. 4. It is also analogous to that of aqueous humour in the eye; see Sect. 5.1.5 of this chapter. The volume of endolymph in the endolymphatic sac may increase to the point of triggering **endolymphatic hydrops**, a condition associated with MD. The dynamics of endolymph is dominated by the existence of five well-documented pathways for endolymph to flow to the endolymphatic sac, namely the endolymphatic duct, the endolymphatic sinuses, the utricular ducts, the saccular ducts, and the ductus reuniens [60]; see Fig. 5.10. MD patients are known to suffer from a range of anomalies in any of these channels, as documented in the early work of Schuknecht and Rütger [60].

Reported anomalies include blockages in the aforementioned endolymph pathways. External shunt operation has been applied for many years in order to arrest the progression of endolymphatic hydrops. The classical work of Guild [62] on the dynamics of endolymph is recommended, both for its historical as well as scientific value. Needless to say that the endolymphatic sac and associated structures, such as the endolymphatic duct, possess vasculature and therefore, corresponding circulation-related anomalies are then to be expected.

The Inner-Ear Blood Circulation

Anomalous inner fluid dynamics includes also the blood circulation, ranging from compromised supply of nutrients through the arterial system, to inefficient elimination of proteins and metabolic waste through the venous system. Blood supply to the inner ear is provided by the labyrinthine artery, which usually branches from the anterior inferior cerebellar artery (AICA) or, more rarely, directly from the basilar artery. The labyrinthine artery is actually a set of arterioles, whose small diameter contributes to attenuate intense blood flow and to guarantee a smooth blood supply to the inner ear [63].

Drainage of inner ear venous blood is instead accomplished essentially by two veins, namely the vein of the cochlear aqueduct and the labyrinthine vein. The former usually empties into the superior bulb of the internal jugular vein, while the latter drains into the sigmoid sinus, into the superior petrosal sinus or into the transverse sinus.

The pioneering early works of Gussen [64] drew attention to the role of the blood circulation in the inner ear to provide a plausible explanation of the generation of endolymphatic hydrops observed in MD patients. More particularly, Gussen emphasised the role of the venous system in the overall inner ear circulation, through the identification of two relevant aspects. First the accentuated individual, congenital anatomical variability of the inner ear venous system, and second, the fluid dynamics aspects of venous blood.

The concept of **venous insufficiency** was introduced to mean compromised venous drainage from the inner ear district and associated venous hypertension. Gussen [64] identifies fluid transport mechanisms, increased venous pressure and insufficient venous drainage of the vestibular organs as crucial mechanisms in the pathophysiology of Ménière's disease.

5.2.6 The Animal Experiment of Friis and Qvortrup

Friis and Qvortrup [65] conducted an experimental study that is relevant to our understanding of the role of disturbed venous blood flow in MD. The experiment was performed in the rat using in vivo fluorescence microscopy. The authors claim that their animal study is also relevant to humans, given that the venous vasculature in the rat is similar to that of humans. Consider frame (a) of Fig. 5.12. Arterial blood is supplied to the cochlea (C) and the vestibular apparatus (V) by the basilar artery (BA). See also the anterior vestibular artery (AVA) and the posterior vestibular artery (PVA). The posterior meningeal artery (PMA) supplies arterial blood to the microcirculation of the endolymphatic sac (ES). Venous drainage from the cochlea, the vestibular apparatus and the endolymphatic sac is effected by the vein of the vestibular aqueduct (VVA), which drains venous blood into the sigmoid sinus (SS). The experiment of Friis and Qvortrup is depicted in frame (b) of Fig. 5.12. The authors, artificially caused an obstruction in the distal part of the **vein of the vestibular aqueduct (VVA)**, at a

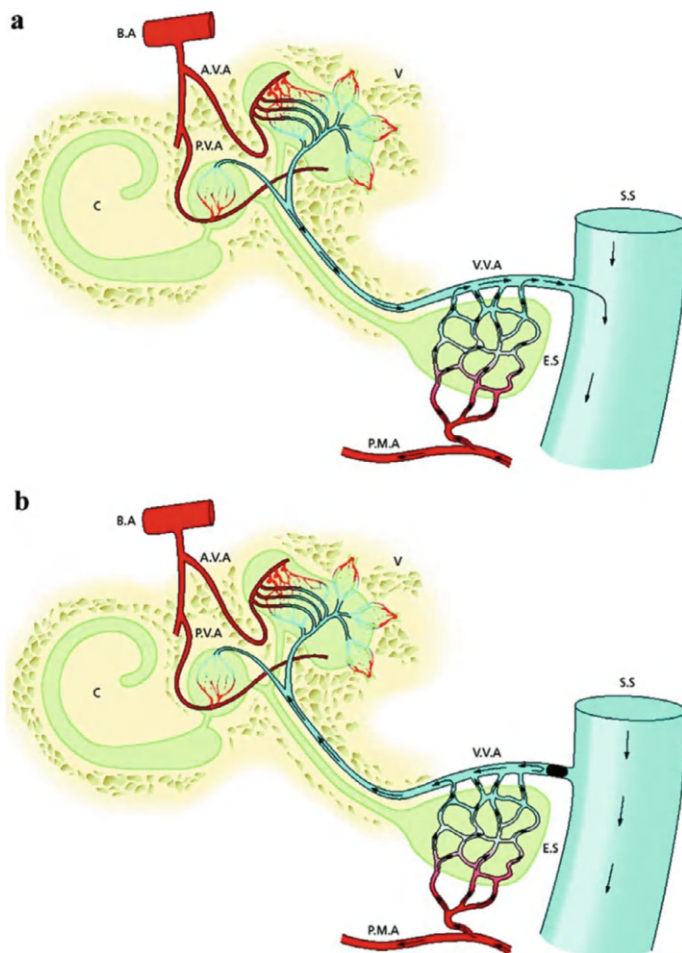


Fig. 5.12 Schematic representation of the animal experiment of Friis and Qvortrup [65]. The top frame **a** depicts the normal situation: arterial blood is supplied to the cochlea (C), the vestibular apparatus (V) and the endolymphatic sac (ES), while venous blood is drained by the vein of the vestibular aqueduct (VVA), which transports venous blood into the sigmoid sinus (SS), and subsequently to the heart. The bottom frame **b** depicts the experiment and its consequences. The obstruction of the VVA placed just before the SS causes sustained reverse venous blood flow, towards the inner ear and the endolymphatic sac, as well as increased venous pressure (from Friis and Qvortrup [65], with permission)

position just before the VVA drains into the sigmoid sinus; see black blot. Through the experiment they demonstrated that such obstruction causes sustained reversed venous blood flow in the VVA towards the inner ear, as illustrated by the black arrows. The sustained reversed flow continued until the animal was killed.

Based on their experimental study Friis and Qvortrup [65] note that the resulting anomalous venous blood flow causes increased venous pressure and potentially could trigger multiple mechanisms, including altered venous reabsorption, disturbed endolymph homeostasis and deranged electrolyte composition in the inner ear fluids. Furthermore, the authors observe that occlusion of the VVA in humans has been reported in connection with MD. Such occlusions may derive from thrombi formation. A thrombus's configuration may change in time, leading to temporary or chronic blood flow occlusion, and consequently to fluctuating symptoms, as observed in MD. The animal experiment of Friis and Qvortrup [65] has strengthened the fluid dynamics aspects of MD advanced by Gussen [64]. It also strengthens the role of more recent findings concerned with pathologies secondary to venous occlusions more generally, as discussed below.

5.2.7 Venous Occlusions: Discussion

The notion of reversed venous flow caused by downstream obstructions has been central to the CCSVI (Chronic Cerebrospinal Venous Insufficiency) theory put forward by Zamboni and collaborators, in the context of neurological conditions, multiple sclerosis in particular [66]. The CCSVI theory is based on the widely confirmed observation that congenital anatomical venous malformations in the head and neck veins cause downstream obstructions to venous outflow from the central nervous system. The CCSVI theory, as an explanation for multiple sclerosis has been the subject of much controversy. Nonetheless, the CCSVI notion has reinvigorated research on the much neglected venous system and potentially associated diseases. Zhou et al. [67] carried out a clinical study of intracranial hypertension induced by internal jugular vein stenoses, in the context of Idiopathic Intracranial Hypertension (IIH). The relevant point of this study was the experimental confirmation that distal cerebral venous outflow strictures cause intracranial hypertension. The symptoms that characterised the cohort of patients considered in [67] are reported to have virtually disappeared after treating internal jugular vein stenosis by stenting. In a related, more recent study by Primiani et al. [68], essentially, the CCSVI theory was applied to explain a condition described as **cognitive fatigue** and **brain fog**. Moreover, the condition is reported as having been resolved through surgical intervention via stenting, which eliminated venous outflow congestion and venous hypertension.

Returning to inner ear anomalies, such as MD, the CCSVI theory has recently been invoked to establish a strong association between venous outflow anomalies in the neck veins, as in CCSVI, and MD. See for example the clinical studies of Alpini et al. [69, 70] and Bruno et al. [61, 71]. The fluid dynamics aspects of disturbed venous flow in MD have also drawn the attention of mathematical modellers.

Using the global, closed loop mathematical model of Müller and Toro [72], a theoretical study of the effect of venous outflow anomalies on inner-ear circulation in the context of MD was carried out by Toro et al. [73]. The computational study revealed that venous outflow strictures cause (i) intracranial venous hypertension (elevated venous pressure in the cranial cavity) and (ii) increased intracranial pressure (elevated ICP). In the updated study of Toro et al. [49] it was confirmed that venous obstructions, both intracranial and extracranial, cause intracranial venous hypertension and intracranial hypertension. They also established that the level of hypertension depends crucially on the level of collateralisation in the venous network of the specific subject under study. See Chaps. 18 and 20 for related results and discussions.

Summary on Inner Ear Fluids

We have presented a brief overview of the inner ear, its anatomical structures and fluid compartments. The emphasis has been on perilymphatic and endolymphatic fluid compartments, which characterise the inner-ear fluid physiology and potentially associated pathologies. In addition to the fundamentals, found in most classical books in human anatomy and physiology [2, 5, 58], we have chosen Ménière's disease to illustrate the relevance of inner-ear fluid compartments, including the role of the vasculature. To conclude we note the relevance of animal experiments, recent human clinical studies and computational simulations in advancing our understanding of the role of disturbed fluid dynamics in inner-ear related pathologies.

5.3 Conclusions and Further Reading

In this chapter we have reviewed major extracellular fluid compartments in the eye and the inner ear, in humans. Concerning ocular fluids we have given a succinct account of the principal aspects of the eye anatomy following classical textbooks, focusing more specifically on the aqueous humour, vitreous humour and the cerebrospinal fluid in the optic nerve sheath, as well as their roles in the eye fluid dynamics physiology and pathology, such as glaucoma. Recent advances in ocular fluid dynamics, such as the lymphatics and glymphatics, have been mentioned and briefly described with appropriate up-to-date references for further study [4] and [45]. As to inner ear fluids, we have given emphasis to perilymphatic and endolymphatic fluid compartments, which characterise the inner-ear fluid physiology and potentially associated pathologies. Ménière's disease has been chosen to illustrate the relevance of inner-ear fluid compartments, including the role of the vasculature. We have also described the animal experiments of Friis and Qvortrup [65] and discussed the potential role of venous occlusions in various pathologies, including inner-ear related diseases, such as MD.

For research related issues we have given appropriate specialised references in Sect. 5.1 for ocular fluids and in Sect. 5.2 for inner-ear fluids. For basic anatomical and physiological aspects of the eye and the ear, and associated bodily fluids, we suggest the following classical books: Hall [5]; Tortora and Derrickson [2] and Silverthorn [58].

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Part II
Equations, Analysis and Exact Solutions

Chapter 6

The Equations of Haemodynamics



Abstract In this chapter, we derive the governing equations for the dynamics of blood flow in compliant vessels. We begin by briefly reviewing the Navier-Stokes equations for compressible materials, augmented by a closure condition through an equation of state. These equations are particularly relevant to specialised medical applications, such as Traumatic Brain Injury and Extracorporeal Shock Wave Lithotripsy. The Navier-Stokes equations are then specialised for incompressible media, such as liquids at low pressure and velocity. Formulations for solving these equations are reviewed, including the artificial compressibility method and the stream function-vorticity approach. To prepare for the derivation of blood flow equations in compliant conduits, we define suitable control volumes and recall the Reynolds transport theorem. We then proceed with a detailed derivation of the one-dimensional averaged equations for the dynamics of blood flow in compliant arteries and veins. This results in a system of two first-order, time-dependent, nonlinear partial differential equations. Since these two equations contain three unknowns, rendering the system undetermined, a closure condition is added by incorporating the mechanics of the compliant vessel wall via a tube law (or pressure law). A simple example of a determined system is provided, using a basic tube law valid only for blood flow in compliant arteries. References to later chapters on generalised models, analysis, and exact solutions are included. Finally, from the established one-dimensional equations, we derive zero-dimensional models, or compartmental models, which consist of systems of ordinary differential equations in time. The chapter concludes with a summary, conclusions, and suggested references for further study.

6.1 The Navier-Stokes Equations: Brief Review

We start from the general Navier-Stokes (NS) partial differential equations (PDEs) for compressible materials and then specialise to the incompressible case. The choice between the compressible and incompressible NS equations depends on the particular fluid regime of interest. There are many situations of medical relevance in which the compressible NS equations must be adopted as the fluid dynamics model. A very prominent example is the subject of **Traumatic Brain Injury** (TBI), see Laccarino

et al. [1]. TBI is understood as an alteration in brain function resulting from blunt or penetrating force to the head, as in traffic accidents and war, for example. The velocity and pressure fields involved in causing TBI require the adoption of compressible fluid dynamics models. Another example characterised by compressible flow regimes is **Extracorporeal Shock Wave Lithotripsy** (ESWL). This is a non-invasive treatment to break up kidney stones. For a review on ESWL see Lee et al. [2]. Related mathematical modelling and computational aspects are found in [3, 4], and references therein. For general applications of shock waves in medicine see Takayama and Ohtani [5] and Loske [6].

The incompressible NS equations offer an adequate model for modelling physical situations involving low speed and pressure. Such conditions hold under physiological or near-physiological conditions, for which the incompressible NS equations are widely used as the starting point for constructing models for haemodynamics. In this chapter we shall be concerned with sub-models derived from the incompressible NS equations.

6.1.1 Equations from Conservation Principles

Here we first state the general physical laws of conservation of mass, momentum and total energy, in both differential and integral forms.

Differential form of the Equations

In differential conservation law form the physical laws of conservation of mass, momentum and total energy read

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{U}) = 0, \quad (6.1)$$

$$\partial_t (\rho \mathbf{U}) + \nabla \cdot [\rho \mathbf{U} \otimes \mathbf{U} + p \mathbf{I} - \Pi] = \rho \mathbf{B}, \quad (6.2)$$

$$\partial_t E + \nabla \cdot [(E + p) \mathbf{U} - \mathbf{U} \cdot \Pi + \mathbf{Q}] = \rho (\mathbf{U} \cdot \mathbf{B}). \quad (6.3)$$

Here the spatial domain is $\Omega \subset \mathbb{R}^3$ and $t \in \mathbb{R}$ is time. The vector of conserved quantities is $\mathbf{Q} = [\rho, \rho \mathbf{U}, E]^T$, where ρ is density, or mass density, $\mathbf{U} = [u, v, w]^T$ is the velocity vector and $E = \rho (\frac{1}{2} \mathbf{U} \cdot \mathbf{U} + e)$ is total energy. An **equation of state** will give the specific internal energy e in terms of two other variables, usually chosen amongst density, pressure and entropy. In the momentum Eq. (6.2) p is pressure, $\mathbf{I}$ is the unit tensor and Π is the viscous stress tensor, yet to be specified. The right hand sides of (6.2) and (6.3) include source terms due to body forces $\mathbf{B} = [b_1, b_2, b_3]^T$, gravitational forces for example. The tensor product $\mathbf{U} \otimes \mathbf{U}$ and the unit tensor $\mathbf{I}$ are given as

$$\mathbf{U} \otimes \mathbf{U} = \begin{pmatrix} u^2 & uv & uw \\ vu & v^2 & vw \\ wu & wv & w^2 \end{pmatrix}, \quad \mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (6.4)$$

Integral Form of the Equations

A more general formulation of the governing Eqs. (6.1)–(6.3) is the integral form. As a matter of fact the differential form may be derived from the integral form under appropriate smoothness assumptions, when viscous terms are absent. Consider a three-dimensional control volume V bounded by S , with outward unit normal vector $\mathbf{n}$. Then, the integral form of Eqs. (6.1)–(6.3) reads

$$\frac{d}{dt} \int \int \int_V \rho \, dV = - \int \int_S \mathbf{n} \cdot (\rho \mathbf{U}) \, dS, \quad (6.5)$$

$$\left. \begin{aligned} \frac{d}{dt} \int \int \int_V (\rho \mathbf{U}) \, dV &= - \int \int_S [\mathbf{U}(\mathbf{n} \cdot \rho \mathbf{U}) + p \mathbf{n} - \mathbf{n} \cdot \Pi] \, dS \\ &\quad + \int \int \int_V \rho \mathbf{B} \, dV, \end{aligned} \right\} \quad (6.6)$$

$$\left. \begin{aligned} \frac{d}{dt} \int \int \int_V E \, dV &= - \int \int_S [\mathbf{n} \cdot (E \mathbf{U} + p \mathbf{U} + \mathbf{Q}) - \mathbf{U} \cdot (\mathbf{n} \cdot \Pi)] \, dS \\ &\quad + \int \int \int_V \rho (\mathbf{U} \cdot \mathbf{B}) \, dV. \end{aligned} \right\} \quad (6.7)$$

For details see Chap. 1 of [7].

6.1.2 Viscous Stresses

The stresses in a fluid, given by a tensor $\mathbf{S}$, are due to the effects of the thermodynamic pressure p and the viscous stresses Π . The stress tensor can be written as

$$\mathbf{S} = -p\mathbf{I} + \Pi, \quad (6.8)$$

where $p\mathbf{I}$ is the spherically symmetric tensor due to p , $\mathbf{I}$ is the unit tensor as in (6.4) and Π is the viscous stress tensor. It is desirable to express $\mathbf{S}$ in terms of flow variables already defined. For the pressure contribution this is achieved by defining p in terms of other thermodynamic variables via an **equation of state**. Recall that equations of state are approximate statements about the nature of a material. In defining the viscous stress contribution Π one may resort to the **Newtonian approximation**, whereby Π is related to the derivatives of the velocity field $\mathbf{U} = [u, v, w]^T$ via the **deformation tensor**

$$\mathbf{D} = \begin{bmatrix} u_x & \frac{1}{2}(v_x + u_y) & \frac{1}{2}(w_x + u_z) \\ \frac{1}{2}(u_y + v_x) & v_y & \frac{1}{2}(w_y + v_z) \\ \frac{1}{2}(u_z + w_x) & \frac{1}{2}(v_z + w_y) & w_z \end{bmatrix}. \quad (6.9)$$

The Newtonian assumption is an idealisation in which the relationship between Π and $\mathbf{D}$ is linear and homogeneous, that is Π will vanish only if $\mathbf{D}$ vanishes, and the medium is isotropic with respect to this relation. An isotropic medium is that in which there are no preferred directions. By denoting the stress tensor as

$$\Pi = \begin{bmatrix} \tau^{xx} & \tau^{xy} & \tau^{xz} \\ \tau^{yx} & \tau^{yy} & \tau^{yz} \\ \tau^{zx} & \tau^{zy} & \tau^{zz} \end{bmatrix}, \quad (6.10)$$

the Newtonian approximation becomes

$$\Pi = 2\mu\mathbf{D} + (\mu_b - \frac{2}{3}\mu)(\text{div}\mathbf{U})\mathbf{I}. \quad (6.11)$$

Or in full

$$\left. \begin{aligned} \tau^{xx} &= \frac{4}{3}\mu u_x - \frac{2}{3}\mu(v_y + w_z) + \mu_b \text{div}\mathbf{U}, \\ \tau^{yy} &= \frac{4}{3}\mu v_y - \frac{2}{3}\mu(w_z + u_x) + \mu_b \text{div}\mathbf{U}, \\ \tau^{zz} &= \frac{4}{3}\mu w_z - \frac{2}{3}\mu(u_x + v_y) + \mu_b \text{div}\mathbf{U}, \\ \tau^{xy} &= \tau^{yx} = \mu(u_y + v_x), \\ \tau^{yz} &= \tau^{zy} = \mu(v_z + w_y), \\ \tau^{zx} &= \tau^{xz} = \mu(w_x + u_z). \end{aligned} \right\} \quad (6.12)$$

In the Newtonian relationship (6.11) there are two scalar quantities that are still undetermined; these are the **coefficient of shear viscosity** μ and the **coefficient of bulk viscosity** μ_b . Approximate expressions for these quantities are obtained from experimentation, as well as from molecular theory. In particular, for monatomic gases the molecular theory based on the hard sphere assumption gives $\mu_b = 0$, which is found to agree well with experiment. For polyatomic gases $\mu_b \neq 0$; appropriate values for μ_b are to be obtained experimentally.

Concerning the coefficient of shear viscosity μ , it is observed that, as long as the temperature T is not too high, μ depends strongly on temperature and only slightly on pressure. Again, molecular theory and experimentation suggest that μ be proportional to T^n . In fact, in the molecular theory $n = \frac{1}{2}$. A relatively accurate relation between μ and T is the Sutherland formula

$$\mu = C_1 \left[1 + \frac{C_2}{T} \right]^{-1} \sqrt{T}, \quad (6.13)$$

where C_1 and C_2 are two experimentally adjustable constants. For T measured in kelvin (K), μ has the units of $\text{kg m}^{-1}\text{s}^{-1}$. For the case of air one has

$$C_1 = 1.46 \times 10^{-6}, \quad C_2 = 112 \text{ K}. \quad (6.14)$$

Sutherland's formula (6.13) describes the dependence of η on T rather well for a wide range of temperatures, provided no dissociation or ionisation take place. These phenomena occur at very high temperatures, where the dependence of η on pressure p , in addition to temperature T , cannot be neglected. Useful background on high temperature gas dynamics is found in the books by Anderson [8] and by Clarke and McChesney [9].

Remark. In mainstream haemodynamics, variations of blood viscosity with temperature, enter the subject of blood rheology; see Chap. 2.

Equation of State (EOS)

A closure condition that completes the number of required equations in order to have a determined problem is required. Such closure condition is called **equation of state**. For a general compressible material the caloric EOS is a functional relationship involving three quantities, usually pressure p , density ρ and specific internal energy e . One may also use the specific volume $v = 1/\rho$ instead of density ρ , or entropy s instead of e . The derived expression for the **speed of sound** a will depend on the choice of dependent and independent variables. Two possible choices and their respective expressions for the sound speed a are:

$$\left. \begin{aligned} p &= p(\rho, e), \quad a = \sqrt{\frac{p}{\rho^2} p_e + p_\rho}, \\ e &= e(\rho, p), \quad a = \sqrt{\frac{p}{\rho^2 e_p} - \frac{e_\rho}{e_p}}, \end{aligned} \right\} \quad (6.15)$$

where subscripts denote partial derivatives. For example, $p_e = \frac{\partial p}{\partial e}$. The equation of state of a material reflects its thermodynamical properties. The equations that govern the dynamics of different materials could be the same but the difference amongst the various media is accounted for by the equations of state. For compressible liquids there is a wide range of equations of state that can be used. For details, see for example, Ivings et al. [10] and references therein.

It is worth remarking that algebraic expressions for the EOS in (6.15) are not always available. For complex materials scientists may have to resort to experiments in order to construct a pressure function in terms of other variables. This situation in compressible fluid dynamics is entirely analogous to the problem of closing the

PDEs system for blood flow with a **tube law**, the equivalent of the equation of state. See Chap. 10.

Heat Conduction

Influx of energy contributes to the rate of change of total energy E . We denote by $\mathbf{Q} = [q_1, q_2, q_3]^T$ the energy flux vector, which results from (i) heat flow due to temperature gradients, (ii) diffusion processes in gas mixtures and (iii) radiation. Let us just consider the effect (i) above, where $\mathbf{Q}$ is the heat flux vector caused by temperature gradients. In a similar manner to that in which viscous stresses were related to gradients of the velocity vector $\mathbf{V}$, one can relate $\mathbf{Q}$ to gradients of temperature T via Fourier's heat conduction law, namely

$$\mathbf{Q} = -\kappa \nabla T . \quad (6.16)$$

This expression for $\mathbf{Q}$ is then inserted into the energy Eqs. (6.3) or (6.7).

Up to this point we have assumed equations for a general compressible material. For further details see Chap. 1 of [7].

6.2 The Incompressible NS Equations

Here we specialise the equations to the case of an incompressible Newtonian fluid, with constant density ρ , constant viscosity μ and no heat conduction. In this case, only the continuity and momentum equations apply, which in differential form read

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{U}) = 0 \quad (6.17)$$

and

$$\partial_t (\rho \mathbf{U}) + \nabla \cdot [\rho \mathbf{U} \otimes \mathbf{U} + p \mathbf{I} - \Pi] = \rho \mathbf{B} . \quad (6.18)$$

As the bulk viscosity $\mu_b = 0$, see [11], the components of the viscous stress tensor simplify to

$$\left. \begin{aligned} \tau^{xx} &= \frac{4}{3}\mu u_x - \frac{2}{3}\mu(v_y + w_z) , \\ \tau^{yy} &= \frac{4}{3}\mu v_y - \frac{2}{3}\mu(w_z + u_x) , \\ \tau^{zz} &= \frac{4}{3}\mu w_z - \frac{2}{3}\mu(u_x + v_y) , \\ \tau^{xy} &= \tau^{yx} = \mu(u_y + v_x) , \\ \tau^{yz} &= \tau^{zy} = \mu(v_z + w_y) , \\ \tau^{zx} &= \tau^{xz} = \mu(w_x + u_z) . \end{aligned} \right\} \quad (6.19)$$

Then, the three-dimensional, time-dependent Navier-Stokes equations for an incompressible non-heat conducting fluid can be written as

$$\left. \begin{aligned} \partial_x u + \partial_y v + \partial_z w &= 0, \\ \partial_t u + u \partial_x u + v \partial_y u + w \partial_z u + \frac{1}{\rho} \partial_x p &= g_{(x)} + \frac{\mu}{\rho} (\partial_x^{(2)} u + \partial_y^{(2)} u + \partial_z^{(2)} u), \\ \partial_t v + u \partial_x v + v \partial_y v + w \partial_z v + \frac{1}{\rho} \partial_y p &= g_{(y)} + \frac{\mu}{\rho} (\partial_x^{(2)} v + \partial_y^{(2)} v + \partial_z^{(2)} v), \\ \partial_t w + u \partial_x w + v \partial_y w + w \partial_z w + \frac{1}{\rho} \partial_z p &= g_{(z)} + \frac{\mu}{\rho} (\partial_x^{(2)} w + \partial_y^{(2)} w + \partial_z^{(2)} w). \end{aligned} \right\} \quad (6.20)$$

Here $[x, y, z] \in \Omega \subset \mathbb{R}^3$ and $t \in \mathbb{R}$ is time. The four unknowns in the equations are the three components of the velocity vector $\mathbf{U}(x, y, z, t) = [u(x, y, z, t), v(x, y, z, t), w(x, y, z, t)]^T$ and the pressure $p(x, y, z, t)$. The spatial domain Ω is fixed and assumed to be known. To define the initial-boundary value problem for (6.20), initial and boundary conditions are to be specified. The **parameters** of equations (6.20) include density ρ , assumed to be constant; viscosity μ , assumed to be constant and $\mathbf{G} = [g_{(x)}, g_{(y)}, g_{(z)}]^T$, assumed up to here to be the gravitational field.

By introducing the **kinematic viscosity** via the ratio

$$\nu = \frac{\mu}{\rho}, \quad (6.21)$$

equation (6.20) can be written more succinctly as

$$\left. \begin{aligned} \nabla \cdot \mathbf{U} &= 0 && : \text{continuity}, \\ \partial_t \mathbf{U} + (\mathbf{U} \cdot \nabla) \mathbf{U} + \frac{1}{\rho} \nabla p &= \mathbf{G} + \nu \nabla^2 \mathbf{U} && : \text{momentum}. \end{aligned} \right\} \quad (6.22)$$

Here $\text{div} \mathbf{U} \equiv \nabla \cdot \mathbf{U}$ is the **divergence operator** applied to the vector $\mathbf{U}$, ∇p is the **gradient operator** applied to the scalar p and ∇^2 is the **Laplacian operator** applied to the vector $\mathbf{U}$.

Remark: Strictly speaking the Navier-Stokes equations are just the three equations for the three velocity components in the momentum balance law in (6.22). The first equation in (6.22) is called the equation of continuity, or the equation for conservation of mass.

6.2.1 Stream Function-Vorticity Formulation

In order to compute solutions to the incompressible NS Eqs. (6.20), or (6.22), it is useful to formulate them appropriately. Here we present two formulations, in which we ignore the gravitation field $\mathbf{G}$ in (6.22). For ease of presentation we restrict attention to the two-dimensional case, in Cartesian coordinates. That is

$$\partial_x u + \partial_y v = 0 , \quad (6.23)$$

$$\partial_t u + u \partial_x u + v \partial_y u + \frac{1}{\rho} \partial_x p = \nu [\partial_x^2 u + \partial_y^2 u] , \quad (6.24)$$

$$\partial_t v + u \partial_x v + v \partial_y v + \frac{1}{\rho} \partial_y p = \nu [\partial_x^2 v + \partial_y^2 v] . \quad (6.25)$$

We have a set of three Eqs. (6.23)–(6.25) for the three unknowns u , v , p . In principle, given a domain Ω along with initial and boundary conditions, one should be able to solve this problem. Suitable reformulations of the equations may facilitate the computational task.

The stream function-vorticity formulation is attractive for the 2D case, but not so much for the three-dimensional case. In two space dimensions one first considers the **magnitude of the vorticity vector**

$$\zeta = \partial_x v - \partial_y u . \quad (6.26)$$

Introducing a **stream function** ψ we have

$$u = \partial_y \psi , \quad v = -\partial_x \psi . \quad (6.27)$$

By combining Eqs. (6.24) and (6.25), so as to eliminate pressure p , and using (6.26) we obtain

$$\partial_t \zeta + u \partial_x \zeta + v \partial_y \zeta = \nu [\partial_x^2 \zeta + \partial_y^2 \zeta] . \quad (6.28)$$

This is called the **vorticity transport equation**, a second-order advection-diffusion PDE, of **parabolic** type. A relevant observation, from the numerical point of view, is that the advection terms of the left hand side of Eq. (6.28) can be written in conservative form and hence we have

$$\partial_t \zeta + \partial_x (u \zeta) + \partial_y (v \zeta) = \nu [\partial_x^2 \zeta + \partial_y^2 \zeta] . \quad (6.29)$$

This follows from the fact that $\partial_x u + \partial_y v = 0$, which was also used to obtain (6.28) from (6.24)–(6.25). In order to solve (6.29) one requires the solution for the stream function ψ , which in turn is related to the vorticity ζ via

$$\partial_x^2 \psi + \partial_y^2 \psi = -\zeta . \quad (6.30)$$

This is called the **Poisson equation**, which is a PDE of **elliptic type**. There are numerical schemes available to solve (6.29)–(6.30) using the *apparent* decoupling of the parabolic-elliptic problem.

6.2.2 Artificial Compressibility Formulation

The artificial compressibility formulation was put forward by Chorin [12], originally for the steady state case. Let us consider the two-dimensional NS Eqs. (6.23)–(6.25), written in non-dimensional form, namely

$$\partial_x u + \partial_y v = 0 , \quad (6.31)$$

$$\partial_t u + u \partial_x u + v \partial_y u + \partial_x p = \alpha [\partial_x^2 u + \partial_y^2 u] , \quad (6.32)$$

$$\partial_t v + u \partial_x v + v \partial_y v + \partial_y p = \alpha [\partial_x^2 v + \partial_y^2 v] . \quad (6.33)$$

The following non-dimensionalisation has been used

$$\left. \begin{aligned} u &\leftarrow u/V_\infty , \quad v \leftarrow v/V_\infty , \quad p \leftarrow \frac{p}{\rho_\infty V_\infty^2} , \\ x &\leftarrow x/L , \quad y \leftarrow y/L , \quad t \leftarrow t V_\infty / L , \\ \alpha &= 1/R_{eL} , \quad R_{eL} = \frac{V_\infty L}{\nu_\infty} . \end{aligned} \right\} \quad (6.34)$$

Here R_{eL} is a dimensionless quantity called the Reynolds number.

A crucial step in the artificial compressibility formulation is the introduction of a constant, non-zero *parameter* c^2 , where c represents a finite **wave propagation speed**. A second step in the formulation is the introduction of an **artificial compressibility** term $\partial_t p$, where t now denotes an artificial time. Then multiplying (6.31) by c^2 and adding the term $\partial_t p$ to the first equation one has

$$\partial_t p + \partial_x (uc^2) + \partial_y (vc^2) = 0 . \quad (6.35)$$

By using Eq. (6.31) the advective terms in (6.32)–(6.33) can be written in **conservative form**, so that the modified system becomes

$$\left. \begin{aligned} \partial_t p + \partial_x (uc^2) + \partial_y (vc^2) &= 0 , \\ \partial_t u + \partial_x (u^2 + p) + \partial_y (uv) &= \alpha [\partial_x^2 u + \partial_y^2 u] , \\ \partial_t v + \partial_x (uv) + \partial_y (v^2 + p) &= \alpha [\partial_x^2 v + \partial_y^2 v] . \end{aligned} \right\} \quad (6.36)$$

Equations (6.36) can be written in compact form as

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) + \partial_y \mathbf{G}(\mathbf{Q}) = \mathbf{D} , \quad (6.37)$$

where

$$\left. \begin{aligned} \mathbf{Q} &= \begin{bmatrix} p \\ u \\ v \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} c^2 u \\ u^2 + p \\ uv \end{bmatrix}, \\ \mathbf{G} &= \begin{bmatrix} c^2 v \\ uv \\ v^2 + p \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \\ \alpha (\partial_x^2 u + \partial_y^2 u) \\ \alpha (\partial_x^2 v + \partial_y^2 v) \end{bmatrix} \end{aligned} \right\} \quad (6.38)$$

The vector $\mathbf{D}$ is a function of second-order spatial derivatives. Equations (6.37)–(6.38) are called the **artificial compressibility equations**, which are equivalent to the original equations in the **steady state** limit.

Formulations and Solution Methods

The left-hand side of (6.37)–(6.38) forms a non-linear hyperbolic system. A solution method for the corresponding Riemann problem was presented by Elsworth and Toro [13]; see also [14]. Once the solution of the Riemann problem is available one can implement a Godunov-type method to approximate the advection part, that is left-hand side of (6.37)–(6.38). For the diffusion terms on the right-hand side of (6.37)–(6.38) one can apply conventional finite difference approximations, for example. However, the resulting method is subject to a very stringent Courant stability condition, see Chap. 12. An attractive practical option is the **time advection-diffusion splitting** approach, potentially yielding a second order accurate method. For details see Brown [15], Toro and Brown [16], and Siviglia and Toro [17]. Improvements can result from suitable approximations to the solution of the Riemann problem [13]. Advanced, related methods have been presented by Shapiro and Drikakis [18] and by Bassi et al. [19, 20]. A yet unexplored option is the **hyperbolisation approach**, whereby, at the continuous level, the full parabolic Eqs. (6.37)–(6.38) may be approximated by a hyperbolic system with stiff source terms. For details on this approach see [21–23].

So far the discussion on the artificial compressibility approach has focused on the steady incompressible NS equations. It is possible to extend the approach to time dependent problems using the **dual time technique**. For details see, for instance, Domini et al. [24] and references therein. The relevant concepts regarding the artificial compressibility approach, illustrated via the two-dimensional equations, extend directly to the three dimensional incompressible Navier-Stokes equations.

Fluid/Structure Interaction

So far we have assumed that the spatial domain Ω for the NS equations is known a priori. In haemodynamics, the core subject of this book, this is not so, as blood vessels are **compliant** conduits, giving rise to **moving-boundary problems**. In practice one may assume prescribed **rigid** blood vessels, solve the NS equations for the internal blood flow distributions, and regard the solution as an approximation to the problem. Rigid models are used profusely in computational bodily fluid dynamics. A more

realistic approach is to acknowledge that the full problem is of the **fluid/structure interaction** (FSI) type. The domain itself, the structure, is an unknown of the problem, as is the internal fluid flow distribution, the fluid problem. Strictly speaking these two coupled problems must be solved in a coupled manner. For details see Formaggia et al. [25] and references therein.

For general background on the Navier-Stokes equations see, for example, Chorin and Marsden [26]; Panton [27]; Toro [7] and White [28].

In the following section we construct simplified FSI one-dimensional models that account for boundary compliance, that is, the interaction between the internal fluid flow distribution and the position of the moving boundary, the compliant vessel wall.

6.3 Reynolds' Transport Theorem

Throughout this book we adopt a simplified mathematical model for describing flow of blood in compliant vessels. The governing equations will be time dependent, quasi-one-dimensional, non linear and endowed with tube laws that will account for the mechanics of compliant arteries and veins, the latter being much more deformable. Our models will retain a fundamental feature of haemodynamics, namely the compliant character of blood vessels. In other words, the moving-boundary character of the full problem is retained. The resulting model is able to reproduce the propagation of pressure and flow waves in arteries and veins with a satisfactory level of accuracy, as established, for example, in [25, 29–36].

There are various ways of deriving the sought one-dimensional averaged equations in the literature; see Formaggia et al. [25] and references therein. Here we reproduce the approach proposed by Hughes and Lubliner [37] and Hughes [38]. Our presentation follows that of Chap. 10 in Formaggia et al. [25].

6.3.1 Defining a Compliant Blood Vessel Segment

Consider a generic compliant blood vessel in three space dimensions, with axial coordinate x , between $x = x_L$ and $x = x_R$, defining a control volume $V(t)$ with boundary $S(t)$, as shown in Fig. 6.1. The following notation is assumed:

- $s(x, y, z, t)$ is a generic cross section at x ;
- s_L is left cross section at $x = x_L$, fixed, normal to direction x ;
- s_R is right cross section at $x = x_R$, fixed, normal to direction x .

The boundary $S(t)$ of $V(t)$ is decomposed as

$$S(t) = S_w \cup \{s_L, s_R\}, \quad (6.39)$$

where $S_w = S_w(x, y, z, t)$ is the moving vessel wall.

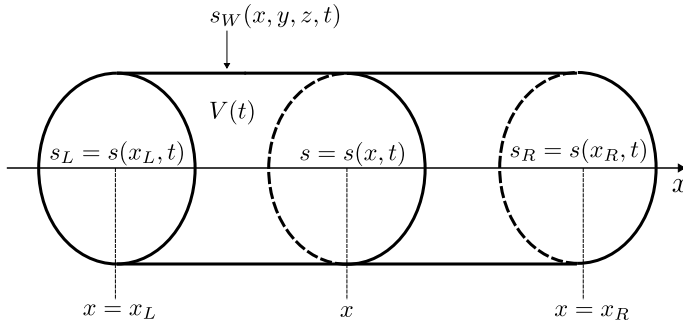


Fig. 6.1 Sketch of section of compliant blood vessel in three dimensions, between $x = x_L$ and $x = x_R$ ($x_L < x_R$), defining control volume $V(t)$ with boundary $S(t) = S_w \cup \{s_L, s_R\}$

6.3.2 Application of the Reynolds Transport Theorem

The Reynolds transport theorem is a multidimensional version of the Leibniz integral rule. For our problem Reynolds' theorem states

$$\frac{d}{dt} \int_{V(t)} \mathcal{F}(\mathbf{x}, t) dV = \int_{V(t)} \partial_t \mathcal{F}(\mathbf{x}, t) dV + \int_{S_w} \mathcal{F} \mathbf{U}_w \cdot \mathbf{n} d\sigma . \quad (6.40)$$

Here $\mathbf{x} = [x, y, z] \in \Omega \subset \mathbb{R}^3$; $\mathcal{F}(\mathbf{x}, t)$ is a continuous function; $\mathbf{n}$ is the outward unit normal vector to $S(t)$; $\mathbf{U}_w$ is the velocity of the vessel wall S_w . At the end positions $x = x_L$ and $x = x_R$, $\mathbf{U}_w \cdot \mathbf{n} = 0$. Let us assume the general case, in which there may be fluid filtration across the vessel wall (a permeable wall). Then the *relative velocity* is

$$\mathbf{U}_r = \mathbf{U}_w - \mathbf{U} . \quad (6.41)$$

It is convenient to define the cross-sectional average $\bar{a}$ of a generic continuous function $a(\mathbf{x}, t)$ at section s of area $A(x, t)$ as follows

$$\bar{a} = \frac{1}{A} \int_s a(\mathbf{x}, t) d\sigma \quad (6.42)$$

and note that

$$A(x, t) = \int_s d\sigma . \quad (6.43)$$

Obviously, the volume integral may be written thus

$$\int_{V(t)} a(\mathbf{x}, t) dV = \int_{x_L}^{x_R} \left[\int_s a(\mathbf{x}, t) d\sigma \right] dx , \quad (6.44)$$

which by virtue of (6.42) becomes

$$\int_{V(t)} a(\mathbf{x}, t) dV = \int_{x_L}^{x_R} A \bar{a}(\mathbf{x}, t) dx . \quad (6.45)$$

From now on we assume $a = \mathcal{F}$. Because x_L and x_R are independent of time we have

$$\frac{d}{dt} \int_{V(t)} \mathcal{F}(\mathbf{x}, t) dV = \frac{d}{dt} \int_{x_L}^{x_R} A \bar{\mathcal{F}}(\mathbf{x}, t) dx = \int_{x_L}^{x_R} \partial_t (A \bar{\mathcal{F}}(\mathbf{x}, t)) dx . \quad (6.46)$$

The second term on the right-hand side of (6.40), in view of (6.41), becomes

$$\int_{S_w} \mathcal{F} \mathbf{U}_w \cdot \mathbf{n} d\sigma = \int_{S_w} \mathcal{F} \mathbf{U}_r \cdot \mathbf{n} d\sigma + \int_{S_w} \mathcal{F} \mathbf{U} \cdot \mathbf{n} d\sigma . \quad (6.47)$$

For an impermeable wall, the first term on the right-hand side of (6.47) is zero. In view of (6.39), the second term on right-hand side of (6.47) becomes

$$\int_{S_w} \mathcal{F} \mathbf{U} \cdot \mathbf{n} d\sigma = \int_{S(t)} \mathcal{F} \mathbf{U} \cdot \mathbf{n} d\sigma - \int_{S_L} \mathcal{F} \mathbf{U} \cdot \mathbf{n} d\sigma - \int_{S_R} \mathcal{F} \mathbf{U} \cdot \mathbf{n} d\sigma . \quad (6.48)$$

Since u is the x component of the velocity vector $\mathbf{U}$, then (6.48) becomes

$$\int_{S_w} \mathcal{F} \mathbf{U} \cdot \mathbf{n} d\sigma = \int_{S(t)} \mathcal{F} \mathbf{U} \cdot \mathbf{n} d\sigma + \int_{S_L} \mathcal{F} u d\sigma - \int_{S_R} \mathcal{F} u d\sigma . \quad (6.49)$$

In view of Gauss' theorem, the first term on the right-hand side of (6.49) becomes

$$\int_{S(t)} \mathcal{F} \mathbf{U} \cdot \mathbf{n} d\sigma = \int_{V(t)} \nabla \cdot (\mathcal{F} \mathbf{U}) dV \quad (6.50)$$

and thus (6.49) becomes

$$\int_{S_w} \mathcal{F} \mathbf{U} \cdot \mathbf{n} d\sigma = \int_{V(t)} \nabla \cdot (\mathcal{F} \mathbf{U}) dV + \int_{S_L} \mathcal{F} u d\sigma - \int_{S_R} \mathcal{F} u d\sigma . \quad (6.51)$$

From (6.42) we may write the last two terms in (6.51) as

$$\int_{S_L} \mathcal{F} u d\sigma - \int_{S_R} \mathcal{F} u d\sigma = - \int_{x_L}^{x_R} \frac{\partial}{\partial x} (A \bar{\mathcal{F}} u) dx \quad (6.52)$$

and thus (6.51) becomes

$$\left. \begin{aligned} \int_{S_w} \mathcal{F} \mathbf{U}_w \cdot \mathbf{n} d\sigma &= \int_{S_w} \mathcal{F} \mathbf{U}_r \cdot \mathbf{n} d\sigma + \int_{V(t)} \nabla \cdot (\mathcal{F} \mathbf{U}) dV \\ &\quad - \int_{x_L}^{x_R} \frac{\partial}{\partial x} (A \overline{\mathcal{F} u}) dx . \end{aligned} \right\} \quad (6.53)$$

By inserting (6.46) and (6.53) into (6.40) we obtain

$$\left. \begin{aligned} \int_{x_L}^{x_R} \frac{\partial}{\partial t} (A \overline{\mathcal{F}}) dx &= \int_{V(t)} \partial_t \mathcal{F} dV + \int_{V(t)} \nabla \cdot (\mathcal{F} \mathbf{U}) dV \\ &\quad + \int_{S_w} \mathcal{F} \mathbf{U}_r \cdot \mathbf{n} d\sigma - \int_{x_L}^{x_R} \frac{\partial}{\partial x} (A \overline{\mathcal{F} u}) dx . \end{aligned} \right\} \quad (6.54)$$

Now we apply (6.44) with $a = \mathcal{F}$ to the first three terms in (6.54), to obtain

$$\left. \begin{aligned} \int_{x_L}^{x_R} \frac{\partial}{\partial t} (A \overline{\mathcal{F}}) dx &= \int_{x_L}^{x_R} \left(\int_s \frac{\partial}{\partial t} \mathcal{F} d\sigma \right) dx + \int_{x_L}^{x_R} \left(\int_s \nabla \cdot (\mathcal{F} \mathbf{U}) d\sigma \right) dx \\ &\quad + \int_{x_L}^{x_R} \left(\int_{\partial s} \mathcal{F} \mathbf{U}_r \cdot \mathbf{n} d\xi \right) dx - \int_{x_L}^{x_R} \frac{\partial}{\partial x} (A \overline{\mathcal{F} u}) dx . \end{aligned} \right\} \quad (6.55)$$

After rearranging, (6.55) becomes

$$\frac{\partial}{\partial t} (A \overline{\mathcal{F}}) + \frac{\partial}{\partial x} (A \overline{\mathcal{F} u}) = \int_s \left(\frac{\partial}{\partial t} \mathcal{F} + \nabla \cdot (\mathcal{F} \mathbf{U}) \right) d\sigma + \int_{\partial s} \mathcal{F} \mathbf{U}_r \cdot \mathbf{n} d\xi . \quad (6.56)$$

Remarks: The function $\mathcal{F}$ in (6.56) is generic. Appropriate choices for $\mathcal{F}$ will yield the sought governing blood flow equations. Note also that for an impermeable wall the last term on the right hand side of (6.56) vanishes.

6.4 One-Dimensional Blood Flow Equations

In this section we first derive a one-dimensional model for blood flow in a single compliant blood vessel and then consider a simplified version of the governing partial differential equations, coupled to a very simple tube law valid for arteries, as distinct from veins.

6.4.1 Derivation of the Equations

Here we use Reynolds's transport theorem (6.40), as dealt with in the previous section, to derive the equations for conservation of mass and the balance of momentum.

Equation for Conservation of Mass

To derive the equation of continuity, or equation for conservation of mass, we use the area-averaged version (6.56) of Reynolds' transport theorem, with $\mathcal{F} = 1$. By recalling Eq. (6.22), for an incompressible fluid one has

$$\nabla \cdot \mathbf{U} = 0, \quad (6.57)$$

equation (6.56) gives the sought **equation of continuity** or equation of **conservation mass**

$$\frac{\partial}{\partial t} A + \frac{\partial}{\partial x} (A\bar{u}) = \int_{\partial s} \mathbf{U}_r \cdot \mathbf{n} d\xi. \quad (6.58)$$

Here $A(x, t)$ is the unknown cross-sectional area and $\bar{u}(x, t)$ is the unknown cross-sectional averaged velocity. The right hand side of (6.58) allows for vessel wall filtration; in the case of an impermeable wall, Eq. (6.58) becomes

$$\frac{\partial}{\partial t} A + \frac{\partial}{\partial x} (A\bar{u}) = 0. \quad (6.59)$$

This is a homogenous equation, that is, there is no source term.

Equation for the Balance of Momentum

To begin with, we set $\mathcal{F} = u$ in Eq. (6.56). In addition we note that

$$\nabla \cdot (u\mathbf{U}) = u\nabla \cdot \mathbf{U} + \mathbf{U} \cdot \nabla u. \quad (6.60)$$

Then, by using the total derivative

$$\frac{Du}{Dt} = \frac{\partial}{\partial t} u + \mathbf{U} \cdot \nabla u \quad (6.61)$$

in Eq. (6.56) and noting (6.57) we obtain

$$\frac{\partial}{\partial t} (A\bar{u}) + \frac{\partial}{\partial x} (A\bar{u}^2) = \int_s \frac{Du}{Dt} d\sigma + \int_{\partial s} u\mathbf{U}_r \cdot \mathbf{n} d\xi. \quad (6.62)$$

Now we invoke the equation for momentum balance

$$\int_{V(t)} \frac{D(\rho \mathbf{U})}{Dt} dV = \int_{V(t)} \rho \mathbf{B} dV + \int_{S(t)} \mathbf{T} n d\sigma, \quad (6.63)$$

where ρ is density (assumed constant here), $\mathbf{B}$ is a general **body force** vector, that may include the gravitational field, and $\mathbf{T}$ is the **Cauchy stress tensor**. By applying the divergence theorem to the last term on the right-hand side of (6.63) we obtain

$$\int_{V(t)} \frac{D\mathbf{U}}{Dt} dV = \int_{V(t)} \mathbf{B} dV + \frac{1}{\rho} \int_{V(t)} \nabla \cdot \mathbf{T} dV. \quad (6.64)$$

We now invoke the **constitutive law** for a fluid

$$\mathbf{T} = -p\mathbf{I} + \mathbf{D}, \quad (6.65)$$

where p is the internal fluid pressure, $\mathbf{I}$ is the identity tensor and $\mathbf{D}$ is the **tensor of deviatoric stresses**. Equation (6.64) can now be written thus

$$\int_{x_L}^{x_R} \left(\int_s \frac{D\mathbf{U}}{Dt} dS \right) dx = \int_{x_L}^{x_R} \left(\int_s \left[\mathbf{B} + \frac{1}{\rho} (-\nabla p + \nabla \cdot \mathbf{D}) \right] d\sigma \right) dx. \quad (6.66)$$

For the x -component Eq. (6.66) becomes

$$\int_s \frac{Du}{Dt} d\sigma = \int_s \left[b + \frac{1}{\rho} \left(-\frac{\partial}{\partial x} p + d \right) \right] d\sigma, \quad (6.67)$$

with the obvious notation for b and d . Returning to (6.62) we have

$$\frac{\partial}{\partial t} (A\bar{u}) + \frac{\partial}{\partial x} (A\bar{u}^2) = \int_s \left[b + \frac{1}{\rho} \left(-\frac{\partial}{\partial x} p + d \right) \right] d\sigma + \int_{\partial s} u \mathbf{U}_r \cdot \mathbf{n} d\xi. \quad (6.68)$$

In terms of cross-sectional averages we have

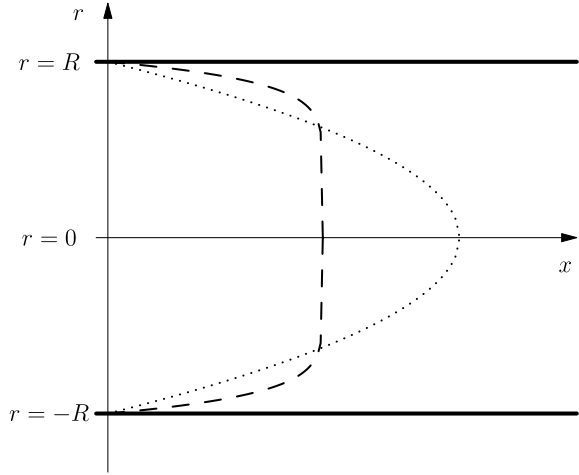
$$\frac{\partial}{\partial t} (A\bar{u}) + \frac{\partial}{\partial x} (A\bar{u}^2) = \frac{A}{\rho} [\rho \bar{b} - \frac{\partial}{\partial x} \bar{p} + \bar{d}] + \int_{\partial s} u \mathbf{U}_r \cdot \mathbf{n} d\xi. \quad (6.69)$$

Before arriving at the final form of the blood flow equations, some parameters must be defined.

6.4.2 Coriolis Coefficient and Viscous Resistance

There are some parameters that remain to be determined in the momentum Eq. (6.69). One of them is the **Coriolis coefficient** (also called momentum correction coefficient)

Fig. 6.2 Prescribed velocity profiles (6.70) depending on the parameter β . For $\beta = 2$ the parabolic velocity profile is obtained (dotted line). The blunt velocity profile (dashed line) is obtained from $\beta = 9$. Vessel wall depicted by thick continuous lines



due to the presence of $\overline{u^2}$ instead of $\overline{u}^2$ in the second term of the left hand side of (6.69). Another parameter is the **viscous resistance**, due to $\overline{d}$ on the right hand side of (6.69). For both cases one needs to specify a cross-section **velocity profile** for the axial component u of the velocity vector. Smith et al. [39] proposed the velocity profile

$$b(\beta) = \frac{\beta + 2}{\beta} \left[1 - \left(\frac{r}{R} \right)^\beta \right] \overline{u}, \quad (6.70)$$

where $\beta > 0$ is a parameter, R is the radius of the vessel and $r \in [0, R]$ is radial distance. Figure 6.2 depicts examples of velocity profiles for two choices of the parameter β .

To resolve the problem due to $\overline{u^2}$ one introduces the so-called **Coriolis coefficient** α via

$$\overline{u^2} = \alpha(\overline{u})^2 = \frac{1}{A} \int_s u^2 d\sigma. \quad (6.71)$$

Then, use of (6.70) into (6.71) gives

$$\alpha(\beta) = \frac{\beta + 2}{\beta + 1}. \quad (6.72)$$

We now turn to the **viscous forces** represented by $\overline{d}$ in (6.69). Following Formaggia et al. [25] we assume the linear relationship

$$\frac{A}{\rho} \overline{d} = -R_{vis} \overline{u}, \quad (6.73)$$

where $R_{vis} > 0$ represents **viscous resistance** per unit length of tube. Then the momentum Eq. (6.69) becomes

$$\frac{\partial}{\partial t}(A\bar{u}) + \frac{\partial}{\partial x}(\alpha A(\bar{u})^2) + \frac{A}{\rho} \frac{\partial}{\partial x} \bar{p} = A\bar{b} - R_{vis}\bar{u} + \int_{\partial s} u \mathbf{U}_r \cdot \mathbf{n} d\xi, \quad (6.74)$$

where the viscous resistance coefficient is expressed in terms of the parameter β , that is

$$R_{vis} = 2(\beta + 2)\pi \frac{\mu}{\rho}. \quad (6.75)$$

The sought blood flow governing equations are (6.59) and (6.74).

Remarks on the velocity profile.

1. The velocity profile (6.70) is a prescribed function that depends on the parameter $\beta > 0$ [39].
2. Velocity profiles, depending on β , reproduce a flat velocity profile, at one extreme, to a parabolic velocity profile, passing through blunted velocity profiles.
3. Figure 6.2 shows two examples of velocity profiles, namely a parabolic velocity profile (dotted line) obtained from $\beta = 2$ and a blunt velocity profile (dashed line) obtained from $\beta = 9$, giving the often used viscous resistance coefficient $R_{vis} = 22\pi \frac{\mu}{\rho}$. Smith et al. [39] used this particular choice in their studies.
4. The resulting Coriolis coefficient $\alpha(\beta)$ in (6.72 varies in the interval $[1, 2)$. It is a monotone decreasing and concave up function; $\alpha(\beta)$ tends to 1 as β tends to ∞ .

6.4.3 Resulting Differential Equations

Here we give the final form of the governing Eqs. (6.59) and (6.74) resulting from assuming an *impermeable wall*, zero **body forces** and parameters derived from the choice of velocity (6.70). For notational convenience we drop bars from the unknowns. Then the continuity and momentum Eqs. (6.59) and (6.74) become

$$\left. \begin{aligned} \partial_t A + \partial_x(uA) &= s_1, \\ \partial_t(uA) + \partial_x(\alpha Au^2) + \frac{A}{\rho} \partial_x p &= s_2. \end{aligned} \right\} \quad (6.76)$$

Here the Coriolis coefficient α and the source terms in (6.76) are

$$\alpha(\beta) = \frac{\beta + 2}{\beta + 1}, \quad s_1 = 0, \quad s_2 = -2(\beta + 2)\pi \frac{\mu}{\rho} u, \quad \beta > 0. \quad (6.77)$$

The two first-order non-linear partial differential Eq. (6.76) govern the fluid dynamics of blood inside the vessel interacting with the displacement of the compliant vessel

wall, once a pressure law, or tube law, is prescribed for the pressure p . The first equation, called the **continuity equation**, expresses the physical principle of conservation of mass; the second equation expresses the law of **balance of momentum**. The quantities s_1 and s_2 are generally called **source terms**.

It is important to clearly distinguish between unknowns and parameters, in system (6.76), as detailed below. System (6.76) contains two PDEs and the following three unknowns, namely

- $A(x, t)$: cross-sectional area of the blood vessel at position x and time t ;
- $u(x, t)$: cross-sectional averaged velocity of blood at a vessel cross section;
- $p(x, t)$: cross-sectional averaged pressure of blood inside the blood vessel.

Remarks: conserved variables and conservative form. The physically **conserved variables** in Eq. (6.76) are cross-sectional area $A(x, t)$ and flow $q(x, t) = A(x, t)u(x, t)$. We note that, mathematically, the continuity equation in (6.76) is written in **conservative form**, or **divergence form**, that is in the form

$$\partial_t q + \partial_x f(q) = 0, \quad (6.78)$$

where $q(x, t)$ is a generic function called the **conserved variable** and $f(q)$ is the **flux function**, or **physical flux**. The momentum balance equation in (6.76), due to the third term on the left hand side, is written in non-conservative form. Whether the PDEs can be expressed in conservative form or not, has profound mathematical and numerical implications. See Chaps. 8 and 14 for further details on these mathematical concepts.

6.4.4 Tube Law: A Simple Closure Condition

System (6.76) has more unknowns than equations, so in order to close the system an extra equation is required, called **closure condition**. In haemodynamics such closure condition is called **tube law**; it is an algebraic expression that relates the pressure $p(x, t)$ in the lumen to the mechanics of the vessel wall displacement via the cross-sectional area $A(x, t)$. In other words, in the model (6.76) **the vessel wall is not rigid**. The tube law couples the elastic properties of the compliant vessel wall to the fluid dynamics inside the vessel and accounts for an essential feature of blood flow, that is the compliance of the vessel wall. The choice of tube law depends on the mechanical properties of the vessel wall and therefore one must distinguish between arteries and veins, for example. Here, as an example, we adopt a simple tube law that is valid for arteries, namely

$$p(x, t) = p_e(x, t) + \mathbb{P}(A; K), \quad \mathbb{P}(A; K) = K(x) \left(\sqrt{\frac{A}{A_0}} - 1 \right), \quad (6.79)$$

Here $p_e(x, t)$ is the **external pressure**, $\mathbb{P}(A; K)$ is the **transmural pressure** and $K(x)$ is the **coefficient of elasticity**, defined as

$$K(x) = \frac{\sqrt{\pi}}{(1 - \nu^2)} \frac{E(x)h_0(x)}{\sqrt{A_0(x)}} . \quad (6.80)$$

The **transmural pressure** $\mathbb{P}(A; K)$ is defined as the pressure difference between the internal and the external pressures, that is

$$\mathbb{P}(A; K) := p(x, t) - p_e(x, t) . \quad (6.81)$$

In what follows we define the parameters of the equations emerging from the tube law.

6.4.5 Parameters of the Equations

The notation in the transmural pressure $\mathbb{P}(A; K)$ distinguishes the argument A , an unknown, from the argument K , a parameter. The parameters in the equations (6.76) for the dynamics, along with the tube law (6.79) are:

1. ρ : density of blood is assumed to be a known constant, usually set to $\rho = 1060 \text{ kg/m}^3$;
2. μ : blood viscosity, derived from rheological properties of blood, and assumed a known constant here;
3. α : Coriolis coefficient, linked to the choice of velocity profile for one-dimensional representation of variables;
4. $A_0(x)$: cross-sectional area of the vessel at **equilibrium**, that is when the transmural pressure $\mathbb{P}(A; K)$ and velocity u vanish;
5. $h_0(x)$: vessel wall thickness;
6. $E(x)$: Young's modulus of elasticity;
7. ν : Poisson ratio, taken as $\nu = 1/2$ for incompressible media;
8. $p_e(x, t)$: external pressure, assumed for the moment to be a prescribed function of distance x and time t .

In general, the geometric and mechanical parameters $A_0(x)$, $h_0(x)$ and $E(x)$ in (6.80) vary with distance x along the length of the vessel and are thus functions of x . The determination of parameters is a major issue in computational bodily fluid dynamics, and hence a major source of uncertainty. It is worth noting, that despite its simplicity, system (6.76), along with the simple tube law (6.79)–(6.80), are not completely amenable to analytical treatment, apart from some special situations, as we shall see in Chap. 9, for example.

In the next section we construct still simpler models, consisting of systems of ordinary differential equations.

6.5 Zero-Dimensional Models for Blood Flow

Here, the simple governing Eqs. (6.76), (6.77), (6.79) are further simplified. This is carried out by first introducing the concept of **compartment**, which refers to a particular **district** of the body, or a single blood vessel, or even a segment of a vessel. In fact the full body may be **modelled** (represented) by a finite set of compartments, connected in an appropriate manner. The emerging models from this approach are called **compartmental models**, or **zero-dimensional models** (0D), or **lumped-parameter models**, or simply **lumped models**.

Within each compartment, haemodynamics variables are **constant in space** (spatial homogeneity) and vary only in time. In a 0D model one studies the behaviour of a compartment in relation to other compartments. Zero-dimensional models may be constructed directly from one-dimensional models, such as system (6.76), (6.77), (6.79), by defining spatial integral averages in the chosen compartment. In this manner the one-dimensional time dependent representation via PDEs becomes a system of ordinary differential equations (ODEs) in time, for quantities that are spatially averaged, constant, in the compartment.

Zero-dimensional models in cardiovascular physiology can be traced back to the pioneering work of Otto Frank [40]. See also [25, 41–46], and the many references therein.

6.5.1 Spatial Integral Averaging

Let us consider a segment $[x_1, x_2]$ of a blood vessel, of length $\Delta s = x_2 - x_1$. Let us then define integral averages in $[x_1, x_2]$ for flow $Q(x, t) = A(x, t)u(x, t)$, pressure $p(x, t)$ and cross-sectional area $A(x, t)$, as follows:

$$\left. \begin{aligned} \hat{Q}(t) &= \frac{1}{\Delta s} \int_{x_1}^{x_2} Q(x, t) dx , \\ \hat{p}(t) &= \frac{1}{\Delta s} \int_{x_1}^{x_2} p(x, t) dx , \\ \hat{A}(t) &= \frac{1}{\Delta s} \int_{x_1}^{x_2} A(x, t) dx . \end{aligned} \right\} \quad (6.82)$$

Next we simplify the PDE system (6.76) by neglecting the inertial term $\partial_x (\alpha A u^2)$ in the momentum balance law, to obtain

$$\left. \begin{aligned} \partial_t A + \partial_x (uA) &= s_1 = 0 , \\ \partial_t (uA) + \frac{A}{\rho} \partial_x p &= s_2 . \end{aligned} \right\} \quad (6.83)$$

It is desirable to understand the meaning of the simplified one-dimensional system of PDEs (6.83) obtained by neglecting the inertial term $\alpha A u^2$ in the momentum equation in (6.76). Ghitti et al. [46] carried out a dimensional analysis and compared the size of the inertial term with the pressure term, in the full momentum equation in (6.76). They found that in fact the inertial term is significantly smaller than the pressure term if the non-dimensional number

$$Fr = |u|/c \quad (6.84)$$

is small. Here c is the wave speed emerging from the eigenstructure analysis of the full Eqs. (6.76). Fr is called here **Froude number**, in analogy to incompressible, free-surface shallow fluids [47, 48]. This number is also called **speed index** in the literature [49]. For compressible materials this number is called **Mach number**.

Ghitti et al. [46] tested the condition $Fr \ll 1$ on data from two well-known arterial vessel networks and concluded that Fr is effectively small for a useful range of flow conditions in blood flow dynamics. For flow conditions outside this range one must return to the full equations (6.76). See the recent works [45] and [46], in which the strong assumptions alluded to above to construct 0D models, are considerably relaxed.

Remark: pressure system. The simplified PDE system (6.83) is identical to the so-called **pressure system** in the Toro-Vázquez (TV) splitting method [50], when applied to the full blood flow equations [51]. See Chap. 13 for details on the physical and mathematical features of the pressure system.

In order to construct 0D models from the **pressure system** (6.83), additional simplifying assumptions are made, namely

1. Variations of the elasticity coefficient $K(x)$ in the tube law (6.79) are small and $K(x)$ can thus be replaced by an averaged, constant K .
2. Variations of cross-sectional area $A(x, t)$ around the equilibrium cross-sectional area $A_0(x)$ are small; $A_0(x)$ is assumed to be independent of x , constant and still denoted by A_0 .

6.5.2 Zero-Dimensional Equations

Now we proceed to integrate in space, in the interval $[x_1, x_2]$, the continuity equation, the first equation in (6.83). We obtain an ordinary differential equation in time for the averaged cross-sectional area $\hat{A}(t)$, namely

$$\Delta s \frac{d}{dt} \hat{A}(t) + Q(x_2, t) - Q(x_1, t) = 0. \quad (6.85)$$

By integrating in space in the interval $[x_1, x_2]$ the momentum equation in (6.83), term by term, we obtain

$$\left. \begin{aligned} \frac{1}{\Delta s} \int_{x_1}^{x_2} \partial_t Q(x, t) dx &= \frac{d}{dt} \hat{Q}(t) , \\ \frac{1}{\Delta s} \int_{x_1}^{x_2} \frac{A_0}{\rho} \partial_x p(x, t) dx &= \frac{A_0}{\rho \Delta s} [p(x_2, t) - p(x_1, t)] , \\ \frac{1}{\Delta s} \int_{x_1}^{x_2} s_2(x, t) dx &= -\Omega \frac{1}{\Delta s} \int_{x_1}^{x_2} Q(x, t) dx = \Omega \hat{Q}(t) , \end{aligned} \right\} \quad (6.86)$$

with

$$\Omega = \frac{8\mu\pi}{\rho A_0} . \quad (6.87)$$

Thus the averaged momentum equation becomes the ODE

$$\left(\frac{\rho \Delta s}{A_0} \right) \frac{d}{dt} \hat{Q}(t) + \Omega \left(\frac{\rho \Delta s}{A_0} \right) \hat{Q}(t) + p(x_2, t) - p(x_1, t) = 0 . \quad (6.88)$$

At this stage one could adopt system (6.85), (6.88) as the sought governing ODE system, in terms of the unknowns $\hat{A}(t)$ and $\hat{Q}(t)$, with the pressure terms $p(x_1, t)$ and $p(x_2, t)$ evaluated through direct application of the tube law (6.79). This is the approach taken in [46]. Conventionally, however, the equations are formulated in terms of $\hat{p}(t)$ and $\hat{Q}(t)$, as follows. Assuming the tube law (6.79) for arteries, differentiating with respect to time and linearising by replacing A by the constant A_0 we obtain

$$\partial_t p = \partial_t p_e + \frac{K}{2A_0} \partial_t A . \quad (6.89)$$

Integration of (6.89) with respect to x in interval $[x_1, x_2]$ gives

$$\left. \begin{aligned} \partial_t \left(\frac{1}{\Delta s} \int_{x_1}^{x_2} p(x, t) dx \right) &= \partial_t \left(\frac{1}{\Delta s} \int_{x_1}^{x_2} p_e(x, t) dx \right) \\ &+ \frac{K}{2A_0} \partial_t \left(\frac{1}{\Delta s} \int_{x_1}^{x_2} A(x, t) dx \right) . \end{aligned} \right\} \quad (6.90)$$

Further manipulations of (6.90) yield a relationship between $\hat{A}(t)$ and the averaged pressure $\hat{p}(t)$, including $\hat{p}_e(t)$, namely

$$\frac{d}{dt} \hat{A}(t) = \frac{2A_0}{K} \frac{d}{dt} \hat{p}(t) - \frac{2A_0}{K} \frac{d}{dt} \hat{p}_e(t) . \quad (6.91)$$

Substitution of (6.91) into (6.85) yields the continuity equation in the form

$$\frac{2A_0 \Delta s}{K} \frac{d}{dt} \hat{p}(t) + Q(x_2, t) - Q(x_1, t) = \frac{2A_0 \Delta s}{K} \frac{d}{dt} \hat{p}_e(t) . \quad (6.92)$$

The resulting zero-dimensional equations are obtained by collecting the Eqs. (6.92) and (6.88), namely

$$\left. \begin{aligned} C \frac{d}{dt} \hat{p}(t) + Q(x_2, t) - Q(x_1, t) &= C \frac{d}{dt} \hat{p}_e(t) , \\ L \frac{d}{dt} \hat{Q}(t) + R \hat{Q}(t) + p(x_2, t) - p(x_1, t) &= 0 . \end{aligned} \right\} \quad (6.93)$$

Here, the coefficients C , L and R are defined as

$$\left. \begin{aligned} C &= \frac{2A_0 \Delta s}{K} : \text{Capacitance} , \\ L &= \frac{\rho \Delta s}{A_0} : \text{Inductance} , \\ R &= \frac{\rho \Delta s \Omega}{A_0} : \text{Resistance} . \end{aligned} \right\} \quad (6.94)$$

The system of ordinary differential Eq. (6.93) governs the time evolution of the spatially averaged pressure $\hat{p}(t)$ and flow $\hat{Q}(t)$ within the vessel segment $[x_1, x_2]$. Obviously $\hat{p}(t)$ and $\hat{Q}(t)$ are constant in space within $[x_1, x_2]$. The function values $Q(x_1, t)$, $Q(x_2, t)$, $p(x_1, t)$, $p(x_2, t)$ permit the connection with neighbouring compartments. For details see [25] and references therein.

6.5.3 The Electric Analogy

The terminology introduced for the coefficients C , L and R in (6.94) resembles that of **electric circuits** in electrical engineering. Here capacitance means a mass storage term due to compliance; inductance is associated with the inertial term in the momentum equation and resistance is due to viscosity and radii of vessels. The electrical analogy is no coincidence, as in the theory of electric circuits one has laws and equations that are entirely analogous to those for blood flow. The simplest electric circuit is a series electric circuit, consisting of an **electromotive source**, e.g. a battery, and a **resistor**, an electric light bulb, for example. Then, according to Ohm's law, one may write the voltage drop ΔE_R across the resistor as

$$\Delta E_R = RI \longleftrightarrow I = \frac{\Delta E_R}{R} , \quad (6.95)$$

where I is the electric current flowing through the resistor and R is resistance. We remark that this equation is entirely analogous to Darcy's law of flow, introduced in Chap. 2 applied to blood flow in a vessel, namely $Q = \Delta P/R$, where Q is blood flow, ΔP is pressure drop and R is vascular resistance. See (6.95).

More complex electric circuits introduce **inductors**, that oppose changes in current, and **capacitors**, that store energy. Analogous laws to (6.95) can be written for the voltage drop ΔE_L across the inductor and for the voltage drop ΔE_C across the capacitor, namely

$$\Delta E_L = L \frac{d}{dt} I, \quad \Delta E_C = \frac{1}{C} q. \quad (6.96)$$

Here q denotes charge. Then Kirchhoff's second law for the flow of current is invoked, namely: *in a closed circuit, the impressed voltage $E(t)$ is equal to the sum of the voltage drops ΔE_* in the rest of the circuit*. For example, for a simple RL-circuit (resistance-inductor circuit), by virtue of (6.95) and (6.96) one derives a linear ordinary differential equation

$$L \frac{d}{dt} I + RI = E(t), \quad (6.97)$$

where $E(t)$ is a prescribed electromotive source.

Some closing remarks concerning the derivation of the zero-dimensional model (6.93) are in order. First we note that important contributions to computational haemodynamics have come from researchers with a background in electrical engineering, particularly via 0D models. Then we note that the inertial term αAu^2 in the momentum equation has been neglected, resulting in the pressure system of PDEs (6.83). The dimensional analysis carried out in [46] states that for small Froude numbers Fr , this is justified. Interestingly, neglecting αAu^2 implies that the effect of the Coriolis coefficient α disappears altogether from Eq. (6.76). Compare the pressure system (6.83) with full system (6.76).

Historically, important works on 0D models include Frank [40], Guyton et al. [41], Westerhof et al. [42], Ursino et al. [43, 44], to name but a few. See also [25, 45] and [46], and references therein.

6.6 Summary and Concluding Remarks

In this Chapter we have first established the foundations for deriving the governing equations for the dynamics of blood flow in compliant conduits e.g. arteries or veins. We began with the compressible Navier-Stokes equations of fluid flow dynamics, supplemented by appropriate thermodynamic statements that provide a closure condition via an equation of state. Reference is made to medical applications based on compressible fluid dynamics models, including Traumatic Brain Injury and Extracorporeal Shock Wave Lithotripsy, a non-invasive treatment for breaking up kidney stones. Next, we reviewed the incompressible Navier-Stokes equations, which are valid for most bodily fluid dynamics applications. To facilitate the computation of solutions to these equations, two reformulations were presented: the Stream Function-Vorticity and the Artificial Compressibility formulations. In preparation

for deriving practical models relevant to this book, Reynolds' transport theorem is also stated.

We then provided a detailed derivation of the one-dimensional averaged equations for the dynamics of blood flow in compliant arteries and veins. These equations form a system of two first-order, time-dependent, nonlinear partial differential equations. The system contains three unknowns, making it undetermined. Similarly to compressible fluid dynamics, in which an equation of state incorporating the thermodynamics is needed to close the system, here we incorporate the mechanics of the compliant blood vessel wall through a tube law (or pressure law). An example of a determined system was provided, where the closure condition is a simple tube law. This law, presented in algebraic form, incorporates purely elastic effects and is strictly valid only for compliant arteries, not veins. Finally, from the established 1D equations, we derive 0D models (compartmental models), consisting of systems of ordinary differential equations in time.

This is the first chapter in Part II of this book, titled **Equations, Analysis, and Exact Solutions**. The following two Chaps. 7 and 8, provide theoretical tools for the mathematical analysis of the equations presented here. Chapter 9 contains a detailed exact solver for the Riemann problem for these equations, applicable to both arteries and veins. Chapter 10 presents generalised models for blood flow and their mathematical analysis.

Useful references for this chapter include the book edited by Formaggia, Quarteroni, and Veneziani [25], written by renowned specialists; all of its chapters are highly relevant to the overall theme of this Chapter and are thus strongly recommended. For the general equations of fluid dynamics we recommend Chorin and Marsden [26], Panton [27], and White [28]. Toro [7], Chap. 1, offers a concise presentation of the fluid dynamics equations for both the compressible and incompressible cases. For the derivation of the one-dimensional averaged equations for the dynamics of blood flow in compliant arteries and veins, we recommend Chap. 10 of [25] as well as the classical works of Hughes and Lubliner [37] and Hughes [38]. Regarding 0D models, reading the work of Frank [40] is essential from a historical perspective, as are the papers by Guyton et al. [41], Westerhof et al. [42], and Ursino et al. [43, 44], to name just a few. For recent works on compartmental models, we recommend [45] and [46], along with the references therein.

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Chapter 7

Hyperbolic Equations I: The Linear Case



Abstract This chapter focuses on the basic mathematical theory of linear constant-coefficient hyperbolic partial differential equations, whose solutions depend on both space and time. We begin by introducing the initial-value problem for the scalar linear advection equation with a constant characteristic speed, then define the geometric concept of characteristic curves and derive the exact solution. Next, we examine the solution to the Riemann problem as a special case. The effect of boundaries and the need for boundary conditions are then discussed, leading to the definition of the initial-boundary value problem, with relevant examples provided. The linear advection equation is then generalised to include source terms, variable characteristic speeds, and multiple spatial dimensions. We proceed by introducing general linear hyperbolic systems, providing definitions for eigenvalues, eigenvectors, and hyperbolicity. Characteristic variables are also introduced, and the general initial-value problem is solved analytically. The special case of the Riemann problem is revisited and solved exactly. A case study is included, consisting of a linear system of two equations derived from the linearisation of the nonlinear blood flow equations for arteries. We derive the eigenstructure and solve the general initial-value problem. Again, the Riemann problem is defined and solved, and a specific problem is solved in detail. Finally, a relaxation technique is introduced, whereby a parabolic advection-diffusion-reaction equation is approximated by a hyperbolic 2×2 system of equations with stiff source terms. Exact solutions to a special initial-value problem are explicitly found. A carefully chosen set of exercises is provided at the end of the chapter.

7.1 Scalar Equations

Many problems in science and engineering are governed by partial differential equations (PDEs) of the advection-diffusion-reaction type. Here we study a simplified PDE that emulates wave propagation phenomena and is useful for introducing basic theoretical notions for studying hyperbolic PDEs.

7.1.1 Basic Notions

In the scalar case (a single equation), we start from the advection-diffusion-reaction type PDE

$$\partial_t q(x, t) + \partial_x f(q(x, t)) = s(x, t, q(x, t)) + \partial_x (\alpha(x, t, q) \partial_x q(x, t)) , \quad (7.1)$$

where

1. $q(x, t)$ is the *unknown* of the equation, called the *dependent variable*; $q(x, t)$ is a function of two *independent variables* x , that represents spatial variations, and t , usually associated with time variations; for convenience we shall often omit the arguments of functions;
2. $f(q)$ is a prescribed function of q called the *flux*, or the *physical flux function*;
3. $s(x, t, q)$ is a prescribed function, called the *source function* or the *source term*, or *reaction term*;
4. $\alpha(x, t, q)$ is a prescribed function, called the *diffusion coefficient*;
5. Partial derivatives with respect to a generic variable s will be denoted by $\partial_s^{(k)} q(x, t)$, where k is the order.

The left-hand side $\partial_t q(x, t) + \partial_x f(q(x, t))$ involves first-order partial derivatives and is called the *advection part* and is associated with *hyperbolic equations*. The term involving the second-order spatial partial derivative $\partial_x (\alpha(x, t, q) \partial_x q(x, t))$ is called the *diffusion term* and gives the full equation its *parabolic* character. Equations of type (7.1) are called *advection-diffusion-reaction* equations and are of parabolic type. Wave propagation and transport phenomena can be described by equations of type (7.1).

The simplest example of (7.1) is the linear advection-diffusion-reaction equation

$$\partial_t q(x, t) + \lambda \partial_x q(x, t) = \beta q(x, t) + \alpha \partial_x^{(2)} q(x, t) . \quad (7.2)$$

Here the flux function $f(q)$ and the source term $s(q)$ are both linear functions of q , namely

$$f(q) = \lambda q , \quad s(q) = \beta q , \quad (7.3)$$

with λ , β and α prescribed constants. We call (7.2) the *model advection-diffusion-reaction equation*. There are cases of special interest. For example, in the absence of diffusion we just have advection and reaction terms, the *hyperbolic part*. In the absence of the flux derivative term and the source term we have the well-known diffusion equation (parabolic), namely

$$\partial_t q(x, t) = \alpha \partial_x^{(2)} q(x, t) . \quad (7.4)$$

In the absence of all spatial derivatives equation (7.1) becomes a first-order ordinary differential equation (ODE)

$$\frac{d}{dt}q(t) \equiv q'(t) = s(t, q). \quad (7.5)$$

In this chapter we shall primarily be concerned with linear advection-reaction equations.

7.1.2 The Linear Advection Equation

We first study the simplest case of the PDE (7.2), the linear advection equation in an infinite spatial domain, along with an initial condition (IC) at the initial time $t = 0$, namely

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \lambda \partial_x q = 0, \quad -\infty < x < \infty, t > 0, \\ \text{IC: } q(x, 0) = h(x), \quad -\infty < x < \infty. \end{array} \right\} \quad (7.6)$$

Here λ is the characteristic speed and the initial condition $h(x)$ is a prescribed function of distance x . The spatial domain is infinite; there are no spatial boundaries and therefore (7.6) defines a pure *initial-value problem*, or IVP.

Characteristic Curves and the Solution

Characteristic curves, or simply *characteristics*, constitute an important notion in the study of hyperbolic equations. Characteristics are functions $x(t)$ in the x - t half-plane satisfying the following IVP for an ODE

$$\left. \begin{array}{l} \text{ODE: } x'(t) = \lambda, \\ \text{IC: } x(0) = x_0, \end{array} \right\} \quad (7.7)$$

where x_0 is the initial condition at time $t = 0$. The function

$$x = x_0 + \lambda t \quad (7.8)$$

is the exact solution of IVP (7.7). This can be verified by direct differentiation of (7.8) with respect to time t , which reproduces the ODE, and evaluation of the solution (7.8) at time $t = 0$, gives the initial condition x_0 in (7.7).

Figure 7.1 illustrates a characteristic curve $x(t)$ in (7.8) departing from the point x_0 , for the case of positive characteristic speed λ , in the t - x plane. The point x_0 is called the *foot of the characteristic*. Characteristics for the linear advection equation are straight lines.

In practice it is more common to represent characteristics in the x - t plane, as illustrated in Fig. 7.2 for a family of characteristics. The inclination, or slope, of the

Fig. 7.1 Characteristic curve $x(t) = x_0 + \lambda t$ in the t - x plane given by (7.8), departing from the initial point x_0 , for the case of positive characteristic speed λ

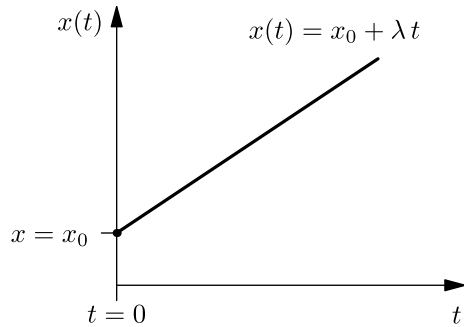
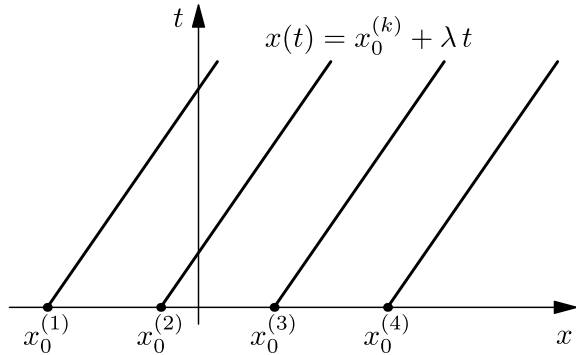


Fig. 7.2 Family of characteristic curves $x(t)$ in the x - t plane, for the case of positive characteristic speed λ , for four different initial conditions. Compare with Fig. 7.1



characteristics depends on the characteristic speed λ ; in fact the slope is $\frac{1}{\lambda}$. In the linear case, characteristic curves are all parallel, which is a distinguishing feature of linear hyperbolic equations with constant coefficients. The importance of characteristics is twofold, as we now explain. Considering the time-rate of change (or total derivative) of $q(x(t), t)$ along the characteristic curve (7.8) and noting from (7.7) that $x'(t) = \lambda$ we obtain

$$\frac{dq}{dt} = \partial_t q + \partial_x q \frac{dx}{dt} = \partial_t q + \lambda \partial_x q = 0. \quad (7.9)$$

That is, the time-rate of change dq/dt along the characteristic curve $x(t) = x_0 + \lambda t$ is zero, or in other words, $q(x, t)$ is *constant along the characteristic curve*. These observations allow us to find the analytical expression for the solution of the IVP (7.6). The value of $q(x, t)$ on the characteristic curve passing through a generic point (x, t) is equal to the value of q at x_0 , the foot of the characteristic. That is

$$q(x, t) = q(x_0 + \lambda t, t) = q(x_0, 0) = h(x_0). \quad (7.10)$$

But from (7.8) $x_0 = x - \lambda t$; therefore the solution of IVP (7.6) is

$$q(x, t) = h(x - \lambda t). \quad (7.11)$$

The exact solution is simply the initial condition h in (7.6) evaluated at the position $x - \lambda t$. Formula (7.11), or (7.10), says that the solution at the point (x, t) is found by evaluating the initial condition at the foot of the characteristic of slope λ passing through the point (x, t) . Figure 7.3 illustrates three possible cases, depending on the sign of the characteristic speed λ .

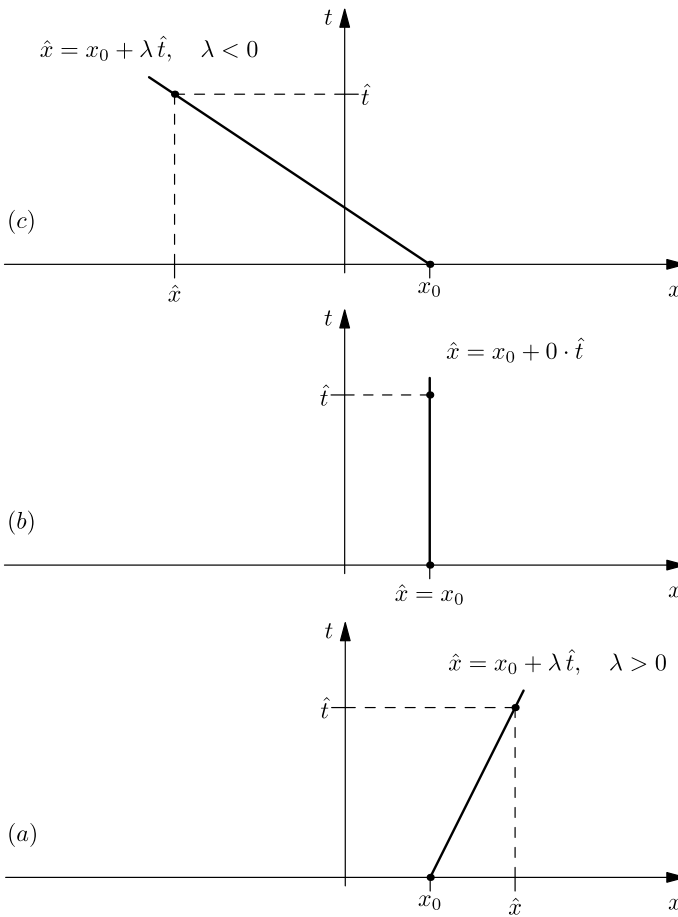


Fig. 7.3 The solution at a point $(\hat{x}, \hat{t})$ is found by tracing the characteristic that passes through the point $(\hat{x}, \hat{t})$ back to its foot x_0 . Frame **a** positive speed λ ; frame **b** zero speed λ (infinite slope $\frac{1}{\lambda}$); frame **c** negative speed λ

Exercise 7.1 Assume that the initial function $h(x)$ in the IVP (7.6) is differentiable. Verify that the function $q(x, t)$ in (7.11) is the exact solution of the initial-value problem (7.6).

Solution First we calculate the partial derivatives $\partial_t q$ and $\partial_x q$ of the proposed solution (7.11). We have

$$\left. \begin{aligned} \partial_x q(x, t) &= \partial_x h(x - \lambda t) = h'(x - \lambda t), \\ \partial_t q(x, t) &= \partial_t h(x - \lambda t) = -\lambda h'(x - \lambda t). \end{aligned} \right\} \quad (7.12)$$

Substituting these partial derivatives into the PDE in (7.6) we obtain the identity

$$\partial_t q + \lambda \partial_x q = -\lambda h' + \lambda h' = 0, \quad (7.13)$$

and thus the function $q(x, t) = h(x - \lambda t)$ is the solution of the PDE in (7.6) for any differentiable function $h(x)$. The initial condition is also satisfied.

Remark. Note that in the geometric construction of the solution (7.11) to IVP (7.6) it was not necessary to assume regularity of the function $h(x)$. In fact $h(x)$ can even be discontinuous and its *discontinuities* (singularities) will simply be propagated along characteristics.

7.1.3 Initial-Value Problem: An Example

Here we study in detail the following IVP

$$\left. \begin{aligned} \text{PDE: } \partial_t q + \lambda \partial_x q &= 0, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } q(x, 0) &= h(x) = \begin{cases} 0 & \text{if } x < -1, \\ 1 - x^2 & \text{if } -1 \leq x \leq 1, \\ 0 & \text{if } x > 1. \end{cases} \end{aligned} \right\} \quad (7.14)$$

Direct application of formula (7.11) gives the solution of (7.14) as

$$\left. \begin{aligned} q(x, t) &= h(x - \lambda t) = \begin{cases} 0 & \text{if } x - \lambda t < -1, \\ 1 - (x - \lambda t)^2 & \text{if } -1 \leq x - \lambda t \leq 1, \\ 0 & \text{if } x - \lambda t > 1, \end{cases} \\ \text{or} \\ q(x, t) &= h(x - \lambda t) = \begin{cases} 0 & \text{if } x < -1 + \lambda t, \\ 1 - (x - \lambda t)^2 & \text{if } -1 + \lambda t \leq x \leq 1 + \lambda t, \\ 0 & \text{if } x > 1 + \lambda t. \end{cases} \end{aligned} \right\} \quad (7.15)$$

Table 7.1 Solution values $q(x, 1)$ for IVP (7.14) for $\lambda = 1$ at time $t = 1$, as function of distance x , according to (7.16)

x	$q(x, 1)$	Condition on x
-2	0	$x < 0$
-1	0	$x < 0$
$-\frac{1}{2}$	0	$x < 0$
0	$1 - (0 - 1)^2 = 0$	$0 \leq x \leq 2$
$\frac{1}{2}$	$1 - (\frac{1}{2} - 1)^2 = \frac{3}{4}$	$0 \leq x \leq 2$
1	$1 - (1 - 1)^2 = 1$	$0 \leq x \leq 2$
$\frac{3}{2}$	$1 - (\frac{3}{2} - 1)^2 = \frac{3}{4}$	$0 \leq x \leq 2$
2	$1 - (2 - 1)^2 = 0$	$0 \leq x \leq 2$
3	0	$x > 2$

The characteristic speed λ is a **parameter** of the problem and must be specified in the expression (7.15) for the solution. For convenience here we choose $\lambda = 1$. Then the solution depends on space and time. For a specified output, say $t = 1$, the solution becomes a function of space x alone and from (7.15), we have

$$q(x, t) = h(x - \lambda t) = \begin{cases} 0 & \text{if } x < 0, \\ 1 - (x - 1)^2 & \text{if } 0 \leq x \leq 2, \\ 0 & \text{if } x > 2. \end{cases} \quad (7.16)$$

Table 7.1 gives tabulated values $q(x, 1)$ as function of x , according to (7.16).

Figure 7.4 illustrates the complete solution process. We start from the initial condition of the problem displayed in the bottom frame (a). Then we construct the picture of characteristics, shown in x - t space in frame (b); finally in frame (c) we display a plot of solution profiles $q(x, t_k)$ at different times t_k . The middle frame (b) also shows the complete domain of the problem divided into three regions by the characteristic curves departing from the points $x = -1$ and $x = 1$. Any curve departing from a point $x_0 < -1$ carries the constant value $q(x_0, 0) = h(x_0) = 0$. Thus the solution in the entire region $\mathcal{R}_0$ is $q(x, t) = 0$. Similarly, any characteristic curve departing from a point $x_0 > 1$ carries the constant value $q(x_0, 0) = h(x_0) = 0$, that is the solution in the entire region $\mathcal{R}_2$ is $q(x, t) = 0$. Characteristics departing from a point x_0 , with $-1 < x_0 < 1$, carry the non-zero value $q(x_0, 0) = h(x_0)$, that is the solution in the entire region $\mathcal{R}_1$ is the initial parabola translated in time. The characteristic departing from $x_0 = 0$ carries the value $q(0, 0) = h(0) = 1$. At any fixed time $t = \hat{t}$, profiles $q(x, \hat{t})$ as functions of distance x can be drawn by constructing a table of $q(x, \hat{t})$ against x , as in Table 7.1. See also frame (c) of Fig. 7.4.

Next we consider a very special initial value problem, called the Riemann problem.

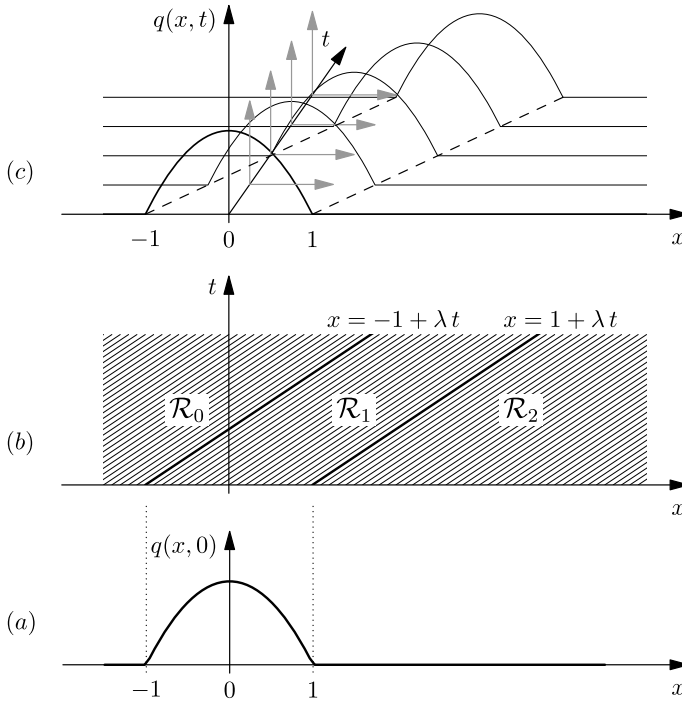


Fig. 7.4 Solution of initial value problem (7.14). Frame **a** displays the initial condition $q(x, 0)$; frame **b** displays picture of characteristics in x - t space and frame **c** shows solution profiles $q(x, t_k)$ at different times t_k

7.1.4 The Riemann Problem

The Riemann problem for the linear advection equation is the special IVP

$$\begin{aligned} \text{PDE: } & \partial_t q + \lambda \partial_x q = 0, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } & q(x, 0) = h(x) = \begin{cases} q_L \text{ (constant)} & \text{if } x < 0, \\ q_R \text{ (constant)} & \text{if } x > 0, \end{cases} \end{aligned} \quad (7.17)$$

where q_L (left of 0) and q_R (right of 0) are two specified constants. From (7.11) it is obvious that the solution is

$$q(x, t) = h(x - \lambda t) = \begin{cases} q_L & \text{if } x - \lambda t < 0, \\ q_R & \text{if } x - \lambda t > 0, \end{cases} \quad (7.18)$$

or, in terms of the characteristic $x = \lambda t$ emanating from the origin

$$q(x, t) = \begin{cases} q_L & \text{if } \frac{x}{t} < \lambda, \\ q_R & \text{if } \frac{x}{t} > \lambda. \end{cases} \quad (7.19)$$

Figure 7.5 illustrates the solution of the Riemann problem for the case in which the characteristic speed λ is positive. Frame (a) shows the piece-wise constant initial condition. Frame (b) shows the picture of characteristics in x - t space; the characteristic curve $x = 0 + \lambda t$ departing from $x = 0$ divides the x - t half-plane into two regions $\mathcal{R}_0$ and $\mathcal{R}_1$, with $q(x, t) = q_L$ in $\mathcal{R}_0$ and $q(x, t) = q_R$ in $\mathcal{R}_1$. Note that any characteristic departing from a point $x_0 < 0$ carries the constant value $q(x, t) = q_L$ and any characteristic departing from a point $x_0 > 0$ carries the constant value $q(x, t) = q_R$. The characteristic departing from the point $x_0 = 0$ carries the singularity, with no value of $q(x, t)$ being assigned. Finally frame (c) in Fig. 7.5 shows solution profiles at various times t_k .

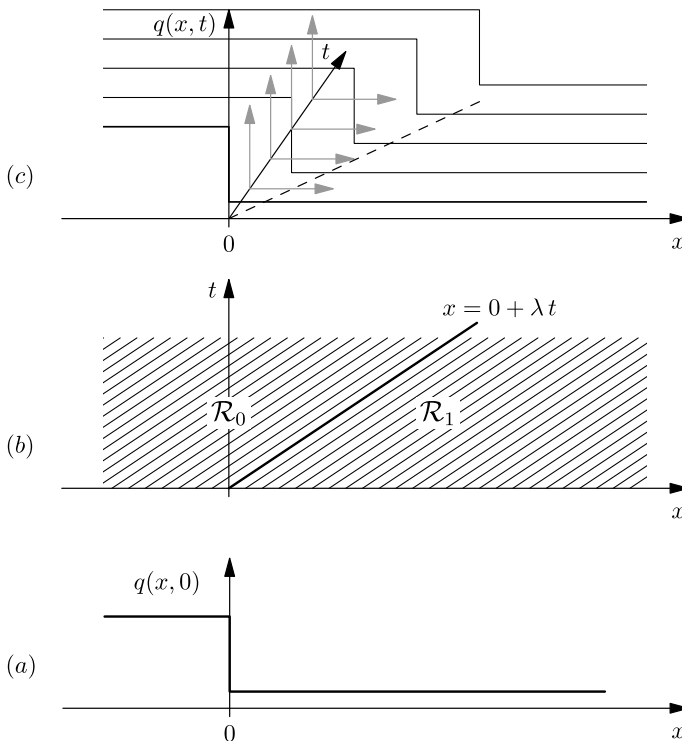


Fig. 7.5 Solution of Riemann problem (7.17). Frame **a** displays the piece-wise constant initial condition $q(x, 0)$; frame **b** displays picture of characteristics in x - t space for $\lambda > 0$ and frame **c** shows solution profiles $q(x, t_k)$ at different times t_k

Remark. Note that the solution (7.19) of the Riemann problem (7.17) is a function of the ratio x/t , for which it is called a *similarity solution* and is often written as $q(x/t)$.

Next we consider an initial value problem in a finite domain, for which the presence of boundaries must be accounted for.

7.1.5 Effect of Boundaries

So far we have only considered the pure initial-value problem for the simplest scalar hyperbolic equation, namely the linear advection equation. As the boundaries have been placed at infinity there has been no need to consider *boundary conditions*. However in most practical problems the spatial domain is finite, boundaries are present and one needs to consider the *initial-boundary value problem*, or IBVP. In order to illustrate the situation let us consider the IBVP for the one-dimensional linear advection equation in a finite spatial domain, namely

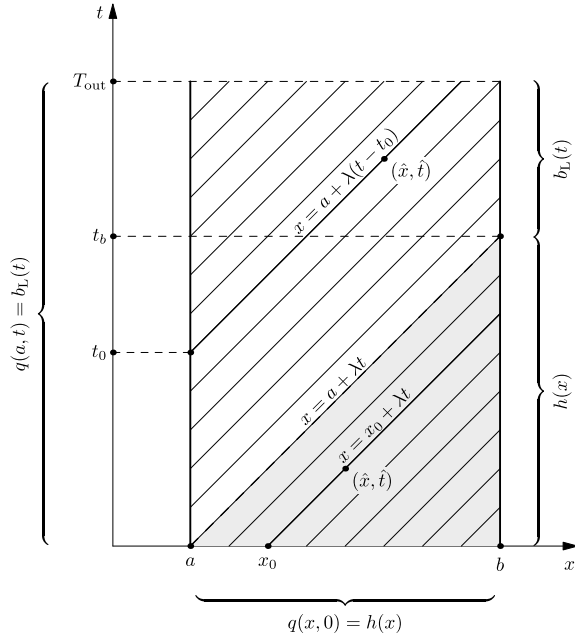
$$\left. \begin{array}{lll} \text{PDE: } \partial_t q + \lambda \partial_x q = 0, & x \in (a, b), & t > 0, \\ \text{IC: } q(x, 0) = h(x), & x \in (a, b), & t = 0, \\ \text{BCs: } q(a, t) = b_L(t), & q(b, t) = b_R(t), & t \geq 0. \end{array} \right\} \quad (7.20)$$

In this type of problems, in addition to the initial condition $h(x)$, the solution also depends on the boundary conditions $b_L(t)$ at $x = a$, $b_R(t)$ at $x = b$, two *prescribed* functions of time. Let us consider the case in which $\lambda > 0$. Figure 7.6 illustrates the situation of interest. The domain is $[a, b] \times [0, T_{out}]$, where T_{out} is a specified final, or output time. We note however that for hyperbolic problems the domain is *open* in the time direction, which means that $T_{out} > 0$ can be chosen arbitrarily. Clearly, for any point $(\hat{x}, \hat{t})$ in the space-time domain $[a, b] \times [0, \infty)$ there are two situations.

1. The point $(\hat{x}, \hat{t})$ lies to the right of the characteristic $x = a + \lambda t$, that is $(\hat{x} - a)/\hat{t} \geq \lambda$. Then the corresponding solution depends only on the initial condition $h(x)$ at the point $x_0 = \hat{x} - \lambda \hat{t} \geq a$.
2. The point $(\hat{x}, \hat{t})$ is such that $(\hat{x} - a)/\hat{t} < \lambda$. Then the corresponding solution depends only on the boundary condition $b_L(t)$ on the left. In this case it is necessary to find the time t_0 at which the characteristic through $(\hat{x}, \hat{t})$ intersects the vertical line $x = a$. The reader is encouraged to verify that $t_0 = \hat{t} + (a - \hat{x})/\lambda$.

We emphasize that in the case of positive characteristic speed, as the one considered here, the boundary condition $b_R(t)$ at $x = b$ is *redundant*; in this case the solution $q(b, t)$ on the right boundary will be determined either from the initial condition in the interior of the domain $[a, b]$ if $t \leq t_b$, or from the boundary condition $b_L(t)$ at $x = a$ if $t > t_b$. Here $t_b = (b - a)/\lambda$ is the time at which the characteristic departing from $x = a$ intersects the vertical line $x = b$.

Fig. 7.6 Picture of characteristics for the initial-boundary value problem (7.20) in a finite spatial domain $[a, b]$ illustrating the effect of a left boundary condition for the case of positive characteristic speed λ



Then the general solution of the IBVP for $\lambda > 0$ is given as

$$q(x, t) = \begin{cases} h(x - \lambda t) & \text{if } x - \lambda t > a, \\ b_L\left(t + \frac{a - x}{\lambda}\right) & \text{if } x - \lambda t \leq a. \end{cases} \quad (7.21)$$

7.1.6 Generalizations of the Linear Advection Equation

The PDE in (7.6) is the simplest hyperbolic PDE. Here we mention a number of possible generalizations.

Linear Advection-Reaction Equation

We consider the linear advection-reaction equation and the associated initial-value problem

$$\left. \begin{array}{ll} \text{PDE: } \partial_t q + \lambda \partial_x q = \beta q, & -\infty < x < \infty, \quad t > 0, \\ \text{IC: } q(x, 0) = h(x), & -\infty < x < \infty, \end{array} \right\} \quad (7.22)$$

where the initial condition $h(x)$ is a prescribed function of distance, λ is a prescribed constant characteristic speed and $\beta \leq 0$ is a prescribed *reaction rate* constant. It is easily verified that the exact solution of the IVP (7.22) is

$$q(x, t) = h(x - \lambda t)e^{\beta t}. \quad (7.23)$$

Note that when $\beta = 0$, solution (7.23) reproduces solution (7.11) for the simple *homogeneous* (no source term) linear advection equation. We remark that for stability of the PDE one requires the condition $\beta \leq 0$. The function (7.23) represents a wave traveling at speed λ , attenuated in time by the action of the factor $e^{\beta t}$. For $\beta > 0$ the solution would increase in time without bound.

Note that the time-rate of change of q along a characteristic $x = x_0 + \lambda t$ is

$$\frac{d}{dt}q(x_0 + \lambda t, t) = \partial_t q(x, t) + \lambda \partial_x q(x, t) = \beta q(x, t) \neq 0. \quad (7.24)$$

In other words, in the presence of a non-vanishing source term $s(x, t, q(x, t))$ the solution $q(x, t)$ is no longer constant along characteristics, unlike the homogeneous case.

Linear Advection Equation with Variable Coefficient

We now consider the linear advection equation with a characteristic speed that varies in space and time, namely $\lambda \equiv \lambda(x, t)$. The associated general IVP is

$$\left. \begin{array}{ll} \text{PDE: } \partial_t q + \lambda(x, t) \partial_x q = 0, & -\infty < x < \infty, \quad t > 0, \\ \text{IC: } q(x, 0) = h(x), & -\infty < x < \infty. \end{array} \right\} \quad (7.26)$$

The corresponding characteristic curves satisfy the IVP

$$\left. \begin{array}{ll} \text{ODE: } x'(t) = \lambda(x, t), & t > 0, \\ \text{IC: } x(0) = x_0. \end{array} \right\} \quad (7.28)$$

Characteristic curves are no longer straight lines and therefore no longer parallel to each other, as illustrated in the following example.

Exercise 7.2 Solve the IVP

$$\left. \begin{array}{ll} \text{PDE: } \partial_t q + Cx \partial_x q = 0, & -\infty < x < \infty, \quad t > 0, \\ \text{IC: } q(x, 0) = h(x) = \begin{cases} 1 & \text{if } |x| \leq 1, \\ 0 & \text{if } |x| > 1, \end{cases} \end{array} \right\} \quad (7.29)$$

where the characteristic speed is $\lambda(x, t) = Cx$, with $C \leq 0$ a constant.

Solution To solve the problem we first study the associated characteristic curves. These are solutions of the IVP

$$x' = Cx, \quad x(0) = x_0, \quad (7.30)$$

The exact solution is

$$x(t) = x_0 e^{Ct}. \quad (7.31)$$

The characteristic curves (7.31) tend to zero as time t tends to infinity, and are no longer straight lines, as for the case of constant λ . However, we still have the property that the solution is constant along characteristics, which allows us to easily find the solution. From (7.31) we have $x_0 = x e^{-Ct}$ and therefore

$$q(x, t) = h(x_0) = h(x e^{-Ct}) = \begin{cases} 1 & \text{if } |x e^{-Ct}| \leq 1, \\ 0 & \text{if } |x e^{-Ct}| > 1. \end{cases} \quad (7.33)$$

Figure 7.7 illustrates the complete solution for IVP (7.29), starting from the initial condition of the problem shown in frame (a). Frame (b) shows the picture of characteristics in x - t space and frame (c) shows plots of solution profiles $q(x, t_k)$ at different times t_k .

Note that the middle frame (b) also shows that the complete domain of the problem is divided into three disjoint regions $\mathcal{R}_0$, $\mathcal{R}_1$ and $\mathcal{R}_2$ by the two characteristic curves departing from the points $x = -1$ and $x = 1$. Any curve starting at a point $x_0 < -1$ in region $\mathcal{R}_0$ carries the constant value $q(x_0, 0) = h(x_0) = 0$. Thus the solution in the entire region $\mathcal{R}_0$ is $q(x, t) = 0$. Similarly, any characteristic curve departing from a point $x_0 > 1$ in region $\mathcal{R}_2$ carries the constant value $q(x_0, 0) = h(x_0) = 0$, that is the solution in the entire region $\mathcal{R}_2$ is $q(x, t) = 0$. Characteristics departing from a point x_0 , with $-1 < x_0 < 1$ in region $\mathcal{R}_1$ carry non-zero values of $q(x, t)$ in the region $\mathcal{R}_1$, which contains the initial square wave transported in time. Compare Fig. 7.7 with Figs. 7.4 and 7.5.

Linear Advection-Reaction in Multiple Space Dimensions

The generalization of the linear advection-reaction equation in (7.22) to multiple space dimensions reads

$$\partial_t q + \lambda_1 \partial_x q + \lambda_2 \partial_y q + \lambda_3 \partial_z q = \beta q \quad (x, y, z) \in \mathbb{R}^3, \quad t > 0, \quad (7.34)$$

where the unknown is a function $q(x, y, z, t)$, with x , y and z denoting the Cartesian coordinate directions. Here λ_1 , λ_2 and λ_3 are the components of the characteristic speed in the x , y and z directions, respectively. The initial-value problem for (7.34) with initial condition

$$q(x, y, z, 0) = h(x, y, z) \quad (7.36)$$

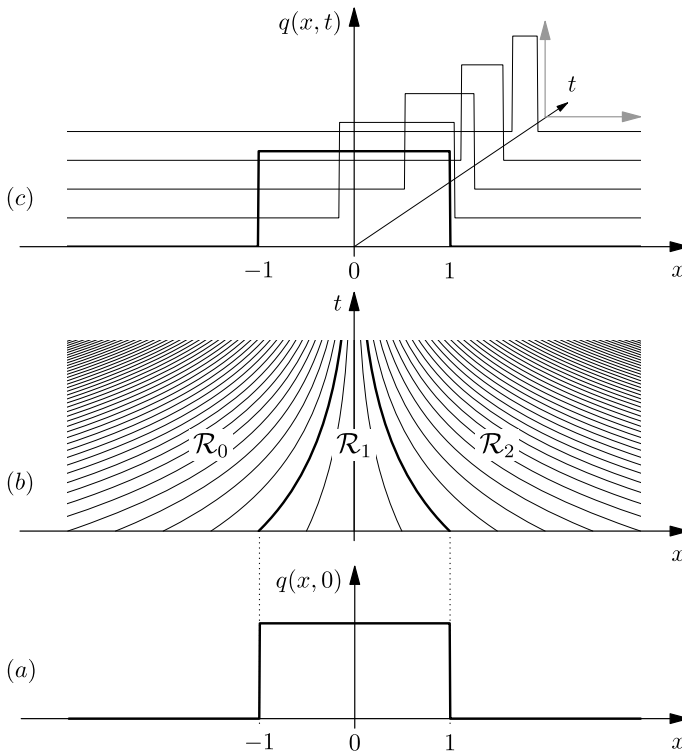


Fig. 7.7 Solution of IVP (7.29) to the linear advection equation with variable characteristic speed $\lambda(x, t) = Cx$, with $C \leq 0$. Frame **a** shows the initial condition; frame **b** shows the picture of characteristics and frame **c** shows solution profiles at various times t_k

has solution

$$q(x, y, z, t) = h(x - \lambda_1 t, y - \lambda_2 t, z - \lambda_3 t) e^{\beta t}, \quad (7.38)$$

as the reader can easily verify.

Summary

We have first studied the mathematical aspects of the linear advection equation with constant characteristic speed λ , emphasizing the physical meaning of the equation in terms of wave propagation phenomena. We introduced the important concept of *characteristic curve* and constructed the exact solution of the general initial-value problem, and of the Riemann problem in particular. We also studied the initial-boundary value problem and discussed the important issue of boundary conditions. Then we considered various extensions. The first extension was to add a source term, the resulting equation is then *inhomogeneous*. We also considered the case of variable characteristic speed $\lambda(x, t)$, the equation still being linear but with variable

coefficient. We also considered the inhomogeneous linear advection equation with constant coefficients in three space dimensions.

It is expected that the reader will reinforce these theoretical concepts by solving the proposed exercises at the end of the chapter. Equipped with this background on scalar linear equations the reader will be in an adequate position to study systems of linear equations, which is the subject of the next section.

7.2 Systems of Linear Hyperbolic Equations

Here we study some basic mathematical aspects of linear systems of hyperbolic equations with constant coefficients in one space dimension. Then, in order to illustrate the theory we study in complete detail a simple example with physical meaning, namely the linearised blood flow equations. The chapter is concluded with a list of exercises. The reader is advised to first read in detail the previous section on the linear scalar case before proceeding with the rest of the chapter.

7.2.1 Basic Theoretical Notions and Definitions

We first present the form of the equations and give some relevant definitions. Consider systems of m one-dimensional time-dependent first-order equations with source terms

$$\partial_t \mathbf{Q}(x, t) + \mathbf{A} \partial_x \mathbf{Q}(x, t) = \mathbf{S}(\mathbf{Q}(x, t)), \quad (7.39)$$

with

$$\mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_m \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & & a_{2m} \\ \vdots & & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mm} \end{bmatrix}, \quad \mathbf{S}(\mathbf{Q}) = \begin{bmatrix} s_1(\mathbf{Q}) \\ s_2(\mathbf{Q}) \\ \vdots \\ s_m(\mathbf{Q}) \end{bmatrix}. \quad (7.40)$$

Here $\mathbf{Q}$ is the vector of unknowns, $\mathbf{A}$ is the coefficient matrix of the system, assumed constant, and $\mathbf{S}(\mathbf{Q})$ is the vector of source terms. Note that (7.39) is the direct extension of the linear advection equation with source term

$$\partial_t q(x, t) + \lambda \partial_x q(x, t) = \beta q(x, t) \quad (7.41)$$

studied in the previous section, where the coefficient matrix $\mathbf{A}$ is analogous to the characteristic speed λ . However in the system case the characteristic speeds will be given by the *eigenvalues* of the matrix $\mathbf{A}$, leading to the definition of hyperbolicity, as we now see.

Eigenvalues

The eigenvalues of system (7.39) are the eigenvalues of its coefficient matrix $\mathbf{A}$, and these in turn are the roots of the *characteristic polynomial*

$$P(\lambda) \triangleq \det(\mathbf{A} - \lambda \mathbf{I}) = 0, \quad (7.42)$$

where $\mathbf{I}$ is the $m \times m$ identity matrix and λ is a parameter.

Remark: The polynomial equation $P(\lambda) = 0$ may have real roots, complex roots, distinct roots or repeated roots. The roots (eigenvalues) are denoted as λ_i , for $i = 1, 2, \dots, m$. Moreover, finding the eigenvalues may turn out to be a very challenging task.

Associated with the eigenvalues one has left and right eigenvectors, which we now define.

Right Eigenvectors

The right eigenvector $\mathbf{R}_i$ of the coefficient matrix $\mathbf{A}$ corresponding to the eigenvalue λ_i is the column vector

$$\mathbf{R}_i = [r_{1i} \ r_{2i} \ \cdots \ r_{ii} \ \cdots \ r_{mi}]^T, \quad \text{such that} \quad \mathbf{A}\mathbf{R}_i = \lambda_i \mathbf{R}_i. \quad (7.43)$$

This says that the product of the matrix $\mathbf{A}$ by the vector $\mathbf{R}_i$ equals the product of the scalar λ_i by the vector $\mathbf{R}_i$.

Left Eigenvectors

The left eigenvector $\mathbf{L}_i$ of the coefficient matrix $\mathbf{A}$, corresponding to the eigenvalue λ_i is the row vector

$$\mathbf{L}_i = [l_{i1} \ l_{i2} \ \cdots \ l_{ii} \ \cdots \ l_{im}], \quad \text{such that} \quad \mathbf{L}_i \mathbf{A} = \lambda_i \mathbf{L}_i. \quad (7.44)$$

We are now in a position to define hyperbolic systems and related systems.

Hyperbolic System

A system of the form (7.39) is said to be *hyperbolic* if the coefficient matrix $\mathbf{A}$ has m real eigenvalues and a corresponding complete set of m linearly independent eigenvectors. That is, each set of left or right eigenvectors

$$\left. \begin{array}{l} \mathbf{L}_1, \mathbf{L}_2, \dots, \mathbf{L}_i, \dots, \mathbf{L}_{m-1}, \mathbf{L}_m, \\ \mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_i, \dots, \mathbf{R}_{m-1}, \mathbf{R}_m \end{array} \right\} \quad (7.45)$$

is linearly independent and generates $\mathbb{R}^m$. Conventionally, the eigenvalues of a hyperbolic system, being real, are written in increasing order as

$$\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_i \leq \dots \leq \lambda_{m-1} \leq \lambda_m. \quad (7.46)$$

We remark that the eigenvalues are not required to be distinct to have a hyperbolic system; what is important is that there is a complete set of linearly independent eigenvectors, corresponding to the real eigenvalues.

Strictly Hyperbolic System

A hyperbolic system of the form (7.39) is said to be *strictly hyperbolic* if all the eigenvalues of the system are distinct. Note that a strictly hyperbolic system is obviously hyperbolic, as distinct eigenvalues will ensure a complete set of linearly independent eigenvectors.

Non-strictly Hyperbolic System

A system of the form (7.39) is said to be *non-strictly hyperbolic* if all the eigenvalues of the system are real, but not distinct, and a complete set of corresponding linearly independent eigenvectors exists. In other words, some of the real roots of the characteristic polynomial are repeated, but what is important is that a complete set of linearly independent eigenvectors exists. This situation is quite common in practice.

Weakly Hyperbolic System

A system of the form (7.39) is said to be *weakly hyperbolic* if all eigenvalues are real but no complete set of linearly independent eigenvectors exists.

Elliptic System

A system of the form (7.39) is said to be *elliptic* if all its eigenvalues are complex. Conventionally, elliptic systems of partial differential equations are more commonly associated with second-order equations.

Mixed Elliptic/Hyperbolic System

A system of the form (7.39) is said to be *mixed elliptic/hyperbolic* if some of its eigenvalues are real and the remaining ones are complex. Systems of this type are present in the area of mathematical modelling of compressible multiphase flows, but also in biological fluid dynamics [1]; see Chap. 11. Systems of this kind are very challenging, both mathematically and numerically; physically, they are the subject of scrutiny.

Orthonormality of Eigenvectors

Two eigenvectors $\mathbf{L}_i$ and $\mathbf{R}_j$ are said to be *orthonormal* if

$$\mathbf{L}_i \cdot \mathbf{R}_j = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \quad (7.47)$$

Next we study a simplified, artificial system of PDEs, which is suitable for exercising the theoretical concepts introduced so far.

7.2.2 Example: An Abstract Model System

Consider the following homogeneous linear system with constant coefficients

$$\partial_t \mathbf{Q} + \mathbf{A} \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}), \quad (7.48)$$

with

$$\mathbf{Q} = \begin{bmatrix} q_1(x, t) \\ q_2(x, t) \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 0 & a \\ b & 0 \end{bmatrix}, \quad \mathbf{S}(\mathbf{Q}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad (7.49)$$

where a and b are real numbers. We first find the eigenvalues by finding the roots of the characteristic polynomial in (7.42). We have

$$P(\lambda) \doteq \det(\mathbf{A} - \lambda \mathbf{I}) = \det \left(\begin{bmatrix} 0 & a \\ b & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0. \quad (7.50)$$

Two solutions are found, namely

$$\lambda_1 = -\sqrt{ab}, \quad \lambda_2 = \sqrt{ab}. \quad (7.51)$$

Several possibilities emerge:

1. If a and b are two non-zero real numbers of the same sign, then the eigenvalues are real and distinct and the system is *strictly hyperbolic*.
2. If $a = 0$ or $b = 0$ then the eigenvalues are real but not distinct, $\lambda_1 = \lambda_2 = 0$. The system is hyperbolic but *not strictly hyperbolic*.
3. If a and b are real, distinct from zero and of opposite sign, then the eigenvalues are complex and the system is *elliptic*.

Let us assume the first case above, that is the system is strictly hyperbolic. Let us now find the two linearly independent eigenvectors corresponding to the eigenvalues λ_1 and λ_2 . To find the right eigenvectors we proceed as follows. Assume a right eigenvector $[r_1, r_2]^T$ corresponding to an eigenvalue λ so that

$$\begin{bmatrix} 0 & a \\ b & 0 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} \lambda r_1 \\ \lambda r_2 \end{bmatrix}, \quad (7.52)$$

from which we obtain

$$a r_2 = \lambda r_1, \quad b r_1 = \lambda r_2. \quad (7.53)$$

These two equations are not independent and so effectively we have an overdetermined system, that is a single equation for two unknowns r_1 and r_2 . This leads to a *one-parameter family of solutions*. To begin with, we choose to use the first equation, which gives

$$r_2 = \frac{\lambda}{a} r_1. \quad (7.54)$$

To find the two right eigenvectors corresponding to the two eigenvalues we use (7.54) for each eigenvalue in turn. Since in each case we have two unknowns r_1 and r_2 we need to make a choice for one of them, r_1 say. For $\lambda = \lambda_1 = -\sqrt{ab}$ and setting $r_1 = \beta_1$, a parameter, we obtain the two components of the first eigenvector $\mathbf{R}_1$ as

$$r_1 = \beta_1, \quad r_2 = -\frac{\sqrt{ab}}{a} \beta_1 = -\sqrt{\frac{b}{a}} \beta_1. \quad (7.55)$$

This is the sought one-parameter family of solutions.

For $\lambda = \lambda_2 = \sqrt{ab}$ and setting $r_1 = \beta_2$ we obtain the two components of the second eigenvector $\mathbf{R}_2$ as

$$r_1 = \beta_2, \quad r_2 = \frac{\sqrt{ab}}{a} \beta_2 = \sqrt{\frac{b}{a}} \beta_2. \quad (7.56)$$

Then the two right eigenvectors are

$$\mathbf{R}_1 = \beta_1 \begin{bmatrix} 1 \\ -\sqrt{b/a} \end{bmatrix}, \quad \mathbf{R}_2 = \beta_2 \begin{bmatrix} 1 \\ \sqrt{b/a} \end{bmatrix}. \quad (7.57)$$

The quantities β_1 and β_2 are arbitrary parameters and are assumed to be real numbers. They determine the length of the vectors and are called *scaling factors*.

The vectors (7.57) are linearly independent; system (7.47) is hyperbolic, provided a and b are two non-zero real numbers of equal sign. The system is also strictly hyperbolic, as $\lambda_1 \neq \lambda_2$.

Exercise 7.3 Find the left eigenvectors corresponding to the two real eigenvalues (7.51), stating the assumptions made.

Solution *Left to the reader as an exercise.*

7.2.3 Diagonalization and Characteristic Variables

Consider now the matrix $\mathbf{R} = [\mathbf{R}_1, \dots, \mathbf{R}_i, \dots, \mathbf{R}_m]$ whose columns are the right eigenvectors and the diagonal matrix Λ , whose diagonal entries are the eigenvalues of the system, namely

$$\mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1m} \\ r_{21} & r_{22} & & r_{2m} \\ \vdots & & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mm} \end{bmatrix}, \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_m \end{bmatrix}. \quad (7.58)$$

Next we recall the following property of matrices arising from hyperbolic systems.

Proposition 1 *If $\mathbf{A}$ is the coefficient matrix of a hyperbolic system (7.39) then the following identities hold*

$$\mathbf{A} = \mathbf{R}\Lambda\mathbf{R}^{-1} \quad \text{or} \quad \Lambda = \mathbf{R}^{-1}\mathbf{A}\mathbf{R}, \quad (7.59)$$

where $\mathbf{R}^{-1}$ denotes the inverse of the matrix $\mathbf{R}$. In this case the coefficient matrix $\mathbf{A}$ is said to be diagonalisable. The system (7.39) is said to be diagonalisable if the coefficient matrix $\mathbf{A}$ is diagonalisable.

Proof *(Left to the reader as an exercise)* ■

The existence of the inverse matrix $\mathbf{R}^{-1}$ makes it possible to define new variables $\mathbf{C} = [c_1, c_2, \dots, c_m]^T$ via the transformation

$$\mathbf{C} = \mathbf{R}^{-1}\mathbf{Q} \quad \Longleftrightarrow \quad \mathbf{Q} = \mathbf{R}\mathbf{C}, \quad (7.60)$$

so that the linear system (7.39), when expressed in terms of $\mathbf{C}$, becomes *completely decoupled*, in a sense to be defined. The new variables $\mathbf{C}$ are called *characteristic variables*.

Next we write the governing PDEs (7.39) in terms of the characteristic variables $\mathbf{C}$, for which we need the partial derivatives in (7.39). Since $\mathbf{A}$ is constant, $\mathbf{R}$ is also constant and therefore from (7.60) these derivatives are calculated as follows

$$\partial_t \mathbf{Q} = \mathbf{R} \partial_t \mathbf{C}, \quad \partial_x \mathbf{Q} = \mathbf{R} \partial_x \mathbf{C}. \quad (7.61)$$

Direct substitution of these expressions into equation (7.39) gives

$$\mathbf{R} \partial_t \mathbf{C} + \mathbf{A} \mathbf{R} \partial_x \mathbf{C} = \mathbf{S}. \quad (7.62)$$

Multiplication of this equation from the left by $\mathbf{R}^{-1}$ and use of (7.59) gives

$$\partial_t \mathbf{C} + \Lambda \partial_x \mathbf{C} = \hat{\mathbf{S}}, \quad \hat{\mathbf{S}} = \mathbf{R}^{-1} \mathbf{S}. \quad (7.63)$$

This is called the *canonical form* or *characteristic form* of system (7.39). The simple nature of this system is more apparent when written in full. Assuming $\hat{\mathbf{S}} = \mathbf{0}$ we have

$$\partial_t \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix} + \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_m \end{bmatrix} \partial_x \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \quad (7.64)$$

Clearly, each equation i -th of this system is of the form

$$\partial_t c_i + \lambda_i \partial_x c_i = 0, \quad i = 1, \dots, m \quad (7.65)$$

and involves the *single unknown* $c_i(x, t)$, being completely decoupled from the remaining equations and variables. The full system is therefore *decoupled*.

Equation (7.65) is identical to the linear advection equation (7.41); now the characteristic speed is λ_i . At any chosen point $(\hat{x}, \hat{t})$ in the x - t half-plane there are m characteristic curves $x_i(t)$ passing through $(\hat{x}, \hat{t})$ and satisfying the m ODEs

$$x'_i(t) = \lambda_i, \quad \text{for } i = 1, \dots, m. \quad (7.66)$$

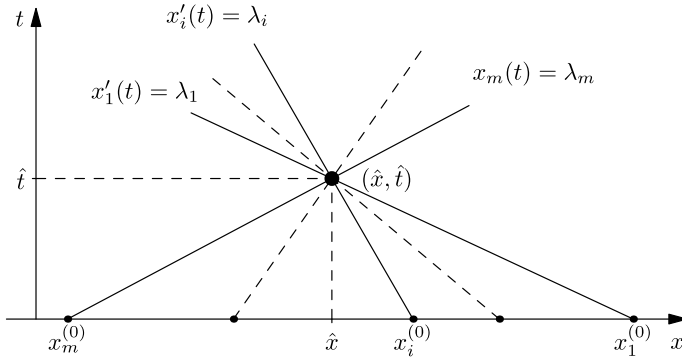


Fig. 7.8 The solution at a point $(\hat{x}, \hat{t})$ depends on the initial condition at the foot $x_i^{(0)}$ of each characteristic $x_i(t) = x_i^{(0)} + \lambda_i t$

See Fig. 7.8. Each characteristic curve $x_i(t) = x_i^{(0)} + \lambda_i t$ intersects the x -axis at the initial point $x_i^{(0)}$, which is the *foot of the characteristic* passing through the point $(\hat{x}, \hat{t})$. The point $x_i^{(0)}$ is given as

$$x_i^{(0)} = \hat{x} - \lambda_i \hat{t}, \quad \text{for } i = 1, 2, \dots, m. \quad (7.67)$$

Since each equation (7.65) is just a linear advection equation, from the previous Sect. 7.1 we know that the solution at $(\hat{x}, \hat{t})$ is given by

$$c_i(\hat{x}, \hat{t}) = c_i^{(0)}(x_i^{(0)}) = c_i^{(0)}(\hat{x} - \lambda_i \hat{t}), \quad \text{for } i = 1, 2, \dots, m, \quad (7.68)$$

where $c_i^{(0)}(x)$ is the initial condition, at the initial time. The initial conditions for the characteristic variables are obtained from the transformation (7.60) applied to the initial condition $\mathbf{Q}(x, 0)$.

Given the assumed order (7.46) of the distinct eigenvalues the following inequalities are satisfied.

$$x_m^{(0)} < x_{m-1}^{(0)} < \dots < x_2^{(0)} < x_1^{(0)}. \quad (7.69)$$

Domain of Dependence

The interval $[x_m^{(0)}, x_1^{(0)}]$ is called the *domain of dependence* of the point $(\hat{x}, \hat{t})$. See Fig. 7.8. The solution at $(\hat{x}, \hat{t})$ depends exclusively on initial data at points within the interval $[x_m^{(0)}, x_1^{(0)}]$. This is a distinguishing feature of hyperbolic equations. The initial data outside the domain of dependence can be changed in any manner we wish but this will not affect the solution at the point $(\hat{x}, \hat{t})$.

7.2.4 The General Initial-Value Problem

We now study the general IVP for the general linear homogeneous (no source terms) hyperbolic system

$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \mathbf{A} \partial_x \mathbf{Q} = \mathbf{0}, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } \quad \mathbf{Q}(x, 0) = \mathbf{Q}^{(0)}(x), \end{array} \right\} \quad (7.70)$$

where the vector $\mathbf{Q}^{(0)}(x)$ of initial conditions is denoted as

$$\mathbf{Q}^{(0)}(x) = \begin{bmatrix} q_1^{(0)}(x) \\ q_2^{(0)}(x) \\ \vdots \\ q_m^{(0)}(x) \end{bmatrix}. \quad (7.71)$$

We find the general solution of the IVP (7.70) by first solving the corresponding homogeneous IVP for the canonical system (7.63), or (7.64), in terms of the characteristic variables $\mathbf{C}$ and appropriate initial condition. We solve

$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{C} + \Lambda \partial_x \mathbf{C} = \mathbf{0}, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } \quad \mathbf{C}(x, 0) = \mathbf{C}^{(0)}(x), \end{array} \right\} \quad (7.72)$$

with

$$\mathbf{C}^{(0)}(x) = [c_1^{(0)}(x), \dots, c_2^{(0)}(x), \dots, c_m^{(0)}(x)]^T = \mathbf{R}^{-1} \mathbf{Q}^{(0)}(x). \quad (7.73)$$

The solution of IVP (7.72) is direct. By considering each unknown $c_i(x, t)$ in (7.65) and its corresponding initial data $c_i^{(0)}(x)$ we write its solution immediately, as in (7.68), namely

$$c_i(x, t) = c_i^{(0)}(x - \lambda_i t), \quad \text{for } i = 1, 2, \dots, m. \quad (7.74)$$

See Fig. 7.8. The solution of the general IVP in terms of the original variables $\mathbf{Q}$ is now obtained by transforming back according to (7.60), namely $\mathbf{Q} = \mathbf{R}\mathbf{C}$, which written in full gives

$$\left. \begin{array}{l} q_1 = c_1(x, t)r_{11} + c_2(x, t)r_{12} + \dots + c_m(x, t)r_{1m}, \\ q_2 = c_1(x, t)r_{21} + c_2(x, t)r_{22} + \dots + c_m(x, t)r_{2m}, \\ \vdots \\ q_m = c_1(x, t)r_{m1} + c_2(x, t)r_{m2} + \dots + c_m(x, t)r_{mm}, \end{array} \right\} \quad (7.75)$$

or

$$\begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_m \end{bmatrix} = c_1(x, t) \begin{bmatrix} r_{11} \\ r_{21} \\ \vdots \\ r_{m1} \end{bmatrix} + c_2(x, t) \begin{bmatrix} r_{12} \\ r_{22} \\ \vdots \\ r_{m2} \end{bmatrix} + \dots + c_m(x, t) \begin{bmatrix} r_{1m} \\ r_{2m} \\ \vdots \\ r_{mm} \end{bmatrix}, \quad (7.76)$$

or more succinctly

$$\mathbf{Q}(x, t) = \sum_{i=1}^m c_i(x, t) \mathbf{R}_i. \quad (7.77)$$

This means that the function $c_i(x, t)$ is the coefficient of $\mathbf{R}_i$ in an *eigenvector expansion* of the solution vector $\mathbf{Q}(x, t)$. But according to (7.74), $c_i(x, t) = c_i^{(0)}(x - \lambda_i t)$ and hence

$$\mathbf{Q}(x, t) = \sum_{i=1}^m c_i^{(0)}(x - \lambda_i t) \mathbf{R}_i. \quad (7.78)$$

Thus, given a point (x, t) in the x - t plane, the solution $\mathbf{Q}(x, t)$ at this point depends only on the initial data at the m points $x_0^{(i)} = x - \lambda_i t$, the *feet of the characteristics*. These are the intersections of the characteristic lines associated with the speeds λ_i with the x -axis, as depicted in Fig. 7.8. The solution (7.78) for $\mathbf{Q}$ can be seen as the superposition of m waves, each of which is advected independently without change in shape. The i -th wave has shape $c_i^{(0)}(x) \mathbf{R}_i$ and propagates with speed λ_i .

7.2.5 The Riemann Problem

We now study the Riemann problem for the homogeneous version of the hyperbolic, constant coefficient hyperbolic system (7.70), namely the IVP

$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \mathbf{A} \partial_x \mathbf{Q} = \mathbf{0}, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } \quad \mathbf{Q}(x, 0) = \mathbf{Q}^{(0)}(x) = \begin{cases} \mathbf{Q}_L & \text{if } x < 0, \\ \mathbf{Q}_R & \text{if } x > 0. \end{cases} \end{array} \right\} \quad (7.79)$$

The initial data consists of two constant vectors $\mathbf{Q}_L$ and $\mathbf{Q}_R$, separated by a discontinuity at $x = 0$. This is a special case of IVP (7.70). We assume that the system is *strictly hyperbolic* and we order the real and distinct eigenvalues as in (7.46). The structure of the solution of the Riemann problem (7.79) is depicted in Fig. 7.9, in the x - t plane. The solution consists of a fan of m waves emanating from the origin, one

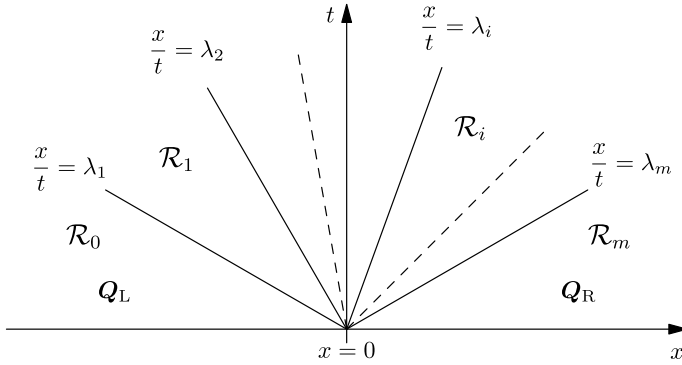


Fig. 7.9 Structure of the solution of the Riemann problem. There are m waves that divide the half x - t plane into $m + 1$ regions (wedges) $\mathcal{R}_i$, of constant $\mathbf{Q}$, with $i = 0, 1, \dots, m$

wave for each eigenvalue λ_i . The speed of the wave i is the eigenvalue λ_i . These m waves divide the x - t half plane into $m + 1$ regions

$$\mathcal{R}_i = \left\{ (x, t) / -\infty < x < \infty; t \geq 0; \lambda_i < \frac{x}{t} < \lambda_{i+1} \right\}, \quad i = 1, \dots, m-1. \quad (7.80)$$

Region $\mathcal{R}_0$ lies to the left of the 1-wave and corresponds to the initial value $\mathbf{Q}_L$; region $\mathcal{R}_m$ lies to the right of the m -wave and corresponds to the initial value $\mathbf{Q}_R$. Solving the Riemann problem means finding constant values for $\mathbf{Q}$ in regions $\mathcal{R}_i$ for $i = 1, \dots, m-1$. The problem is solved as follows. The form of the solution is that given by (7.78) for the general IVP. For the specific case of the Riemann problem (7.79) we need to find the coefficients, that is the characteristic variables $c_i(x, t)$. The initial conditions $\mathbf{Q}_L, \mathbf{Q}_R$ must also have the form (7.78). We define the vectors

$$\left. \begin{aligned} \mathbf{Q}_L &= [q_{1L}, \dots, q_{iL}, \dots, q_{mL}]^T, & \mathbf{Q}_R &= [q_{1R}, \dots, q_{iR}, \dots, q_{mR}]^T, \\ \mathbf{C}_L &= [c_{1L}, \dots, c_{iL}, \dots, c_{mL}]^T, & \mathbf{C}_R &= [c_{1R}, \dots, c_{iR}, \dots, c_{mR}]^T, \\ \mathbf{C}_L &= \mathbf{R}^{-1} \mathbf{Q}_L, & \mathbf{C}_R &= \mathbf{R}^{-1} \mathbf{Q}_R, \end{aligned} \right\} \quad (7.81)$$

where $\mathbf{C}_L, \mathbf{C}_R$ are the characteristic variables. Then the sought coefficients are found by using the specific initial conditions of the Riemann problem, which must satisfy

$$\mathbf{Q}_L = \sum_{i=1}^m c_{iL} \mathbf{R}_i = \mathbf{R} \mathbf{C}_L, \quad \mathbf{Q}_R = \sum_{i=1}^m c_{iR} \mathbf{R}_i = \mathbf{R} \mathbf{C}_R. \quad (7.82)$$

Equations (7.82) define two systems of linear equations for the coefficients $\mathbf{C}_L$ and $\mathbf{C}_R$, namely

$$\left. \begin{aligned} \begin{bmatrix} r_{11} & r_{21} & \cdots & r_{1m} \\ r_{21} & r_{22} & \cdots & r_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mm} \end{bmatrix} \begin{bmatrix} c_{1L} \\ c_{2L} \\ \vdots \\ c_{mL} \end{bmatrix} &= \begin{bmatrix} q_{1L} \\ q_{2L} \\ \vdots \\ q_{mL} \end{bmatrix} \\ \begin{bmatrix} r_{11} & r_{21} & \cdots & r_{1m} \\ r_{21} & r_{22} & \cdots & r_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mm} \end{bmatrix} \begin{bmatrix} c_{1R} \\ c_{2R} \\ \vdots \\ c_{mR} \end{bmatrix} &= \begin{bmatrix} q_{1R} \\ q_{2R} \\ \vdots \\ q_{mR} \end{bmatrix} \end{aligned} \right\} \quad (7.83)$$

In terms of the characteristic variables we have m scalar Riemann problems for the PDEs

$$\left. \begin{aligned} \text{PDE: } \partial_t c_i + \lambda_i \partial_x c_i &= 0, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } c_i^{(0)}(x) &= \begin{cases} c_{iL} & \text{if } x < 0, \\ c_{iR} & \text{if } x > 0, \end{cases} \end{aligned} \right\} \quad (7.84)$$

for $i = 1, \dots, m$. From the previous results, see equation (7.74), we know that the solutions of these scalar Riemann problems are given by

$$c_i(x, t) = c_i^{(0)}(x - \lambda_i t) = \begin{cases} c_{iL} & \text{if } x - \lambda_i t < 0 \Leftrightarrow x/t \leq \lambda_i, \\ c_{iR} & \text{if } x - \lambda_i t > 0 \Leftrightarrow x/t > \lambda_i. \end{cases} \quad (7.85)$$

For a given point (x, t) there is an integer $I(x, t)$, that depends on (x, t) , and an associated eigenvalue λ_I such that (x, t) belongs to region $\mathcal{R}_I$, that is

$$\lambda_I < x/t < \lambda_{I+1}. \quad (7.86)$$

See Fig. 7.10. Then we have

$$\left. \begin{aligned} x - \lambda_i t > 0 & \quad \text{for } i = 1, 2, \dots, I & \rightarrow \text{coefficients } c_{iR}, \\ x - \lambda_i t < 0 & \quad \text{for } i = I + 1, I + 2, \dots, m & \rightarrow \text{coefficients } c_{iL}. \end{aligned} \right\} \quad (7.87)$$

Note that the integer $I = I(x, t)$ is the maximum value of the sub-index i for which $x - \lambda_i t > 0$. We can then write the final solution to the Riemann problem (7.79) in terms of the original variables as

$$\mathbf{Q}(x, t) = \sum_{i=1}^I c_{iR} \mathbf{R}_i + \sum_{i=I+1}^m c_{iL} \mathbf{R}_i. \quad (7.88)$$

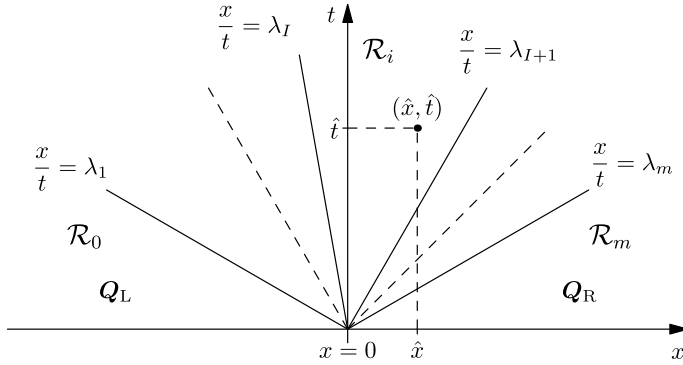


Fig. 7.10 The solution of the Riemann problem at a point $(\hat{x}, \hat{t})$ depends on the associated index $I = I(\hat{x}, \hat{t})$

We next show that it is possible to obtain simpler expressions for the solution of the Riemann problem involving a single linear algebraic system for the coefficients.

Proposition 2 *The solution of the Riemann problem may be expressed as*

$$\mathbf{Q}(x, t) = \mathbf{Q}_L + \sum_{i=1}^I \delta_i \mathbf{R}_i, \quad \text{or} \quad \mathbf{Q}(x, t) = \mathbf{Q}_R - \sum_{i=I+1}^m \delta_i \mathbf{R}_i, \quad (7.89)$$

where $\Delta \mathbf{C} = [\delta_1, \dots, \delta_i, \dots, \delta_m]^T$ and $\mathbf{R} \Delta \mathbf{C} = \Delta \mathbf{Q} = \mathbf{Q}_R - \mathbf{Q}_L$. The coefficients δ_i are the jumps in the individual characteristic variables and are found by solving the linear system

$$\sum_{i=1}^m \delta_i \mathbf{R}_i = \Delta \mathbf{Q}. \quad (7.90)$$

Proof (Left to the reader) ■

Remarks: The definition of the number I in (7.89) remains as for (7.88). Note from (7.89) that the solution can be built from the data state $\mathbf{Q}_L$ by adding jumps $\delta_i \mathbf{R}_i$ across all waves to the left of the point $(\hat{x}, \hat{t})$. Alternatively, one starts from the data state $\mathbf{Q}_R$ and subtracts the corresponding wave jumps. See Fig. 7.10.

In the next section we introduce an example to which the studied theory on linear hyperbolic systems can be applied directly.

7.3 Linearised Blood Flow Equations

In this section we carry out a detailed mathematical analysis of a linear system of hyperbolic equations that models blood flow in arterial vessels. We first derive the equations, starting from a non-linear model.

7.3.1 Derivation of a Linear Blood Flow Model

Consider the non-linear equations for blood flow

$$\left. \begin{aligned} \partial_t A + \partial_x(uA) &= 0, \\ \partial_t(uA) + \partial_x(Au^2) + \frac{A}{\rho} \partial_x p &= 0. \end{aligned} \right\} \quad (7.91)$$

These equations are a special case of the blood flow equations, informally introduced in Chap. 2 and rigorously derived in Chap. 6. The unknowns of system (7.91) are:

- $A(x, t)$: cross-sectional area of the vessel at position x and time t ;
- $u(x, t)$: cross-sectional area averaged velocity of blood at a vessel cross section;
- $p(x, t)$: cross-sectional area averaged pressure of blood inside the vessel.

The density of blood ρ is a **parameter** of the problem and is assumed to be a known constant; typically $\rho = 1050 \text{ kg/m}^3$.

There are two PDEs in (7.91) and three unknowns, namely $A(x, t)$, $u(x, t)$ and $p(x, t)$. In order to close the system, an extra *closure condition* is required, called *tube law*; this relates $p(x, t)$ to the vessel wall displacement via the cross-sectional area $A(x, t)$, thus coupling elastic properties of the compliant vessel wall to the fluid dynamics inside the vessel. Here we adopt a simple tube law for arteries (as distinct from veins)

$$p = p_e(x, t) + \mathbb{P}(A; K), \quad \mathbb{P}(A; K) = K(x) \left(\sqrt{\frac{A}{A_0}} - 1 \right), \quad (7.92)$$

with

$$K(x) = \frac{\sqrt{\pi}}{(1 - \nu^2)} \frac{E(x)h_0(x)}{\sqrt{A_0(x)}}. \quad (7.93)$$

The **parameters** that emerge from the tube law are

- $A_0(x)$: cross-sectional area of the vessel at *equilibrium*, that is when the *transmural pressure* $\mathbb{P} := p(x, t) - p_e(x, t) = 0$ and $u = 0$;
- $h_0(x)$: vessel wall thickness;
- $E(x)$: Young's modulus;

- ν : Poisson ratio, taken as $\nu = 1/2$ for incompressible media;
- $p_e(x, t)$: external pressure, a prescribed function.

Note that in general the parameters $A_0(x)$, $h_0(x)$ and $E(x)$ vary with distance along the length of the vessel and are thus functions of x ; $p_e(x, t)$ may vary with both space and time. In what follows we shall assume that all such parameters are constant and known in advance.

We now recall from Chap. 2 a linearised version of Eq. (7.91) by considering a small perturbation $a(x, t)$ of the equilibrium cross-sectional area A_0 and a small velocity perturbation $v(x, t)$ of a stationary flow with $u(x, t) = 0$, namely

$$A(x, t) = A_0 + a(x, t) , \quad u(x, t) = 0 + v(x, t) . \quad (7.94)$$

Then, assume $a(x, t)$ to be small compared to A_0 and $v(x, t)$ to be small. Assume in addition that derivatives of the perturbations are also small and hence products of these small quantities can be neglected. Substituting $A(x, t)$ and $u(x, t)$ from Eq. (7.94) into Eq. (7.91) and neglecting second order terms we obtain a system of *linearised blood flow equations* for the new unknowns $a(x, t)$ and $v(x, t)$, namely

$$\partial_t a + A_0 \partial_x v = 0 , \quad \partial_t v + \frac{c^2}{A_0} \partial_x a = 0 , \quad (7.95)$$

where

$$c = \sqrt{\frac{K \sqrt{A_0}}{2\rho}} \quad (7.96)$$

is called *wavespeed*, a constant here. This is the speed of vanishingly small-amplitude waves.

7.3.2 Eigenstructure of the Linear System

In matrix form Eqs. (7.95) read

$$\partial_t \mathbf{Q} + \mathbf{A} \partial_x \mathbf{Q} = \mathbf{0} , \quad (7.97)$$

where the vector of unknowns $\mathbf{Q}(x, t)$ and the coefficient matrix $\mathbf{A}$ are respectively given as

$$\mathbf{Q} = \begin{bmatrix} a \\ v \end{bmatrix} , \quad \mathbf{A} = \begin{bmatrix} 0 & A_0 \\ \frac{c^2}{A_0} & 0 \end{bmatrix} . \quad (7.98)$$

We note in passing that equations (7.97)–(7.98) reproduce the well known linear second-order wave equation, for both $a(x, t)$ and $v(x, t)$, namely

$$\partial_t^{(2)} a = c^2 \partial_x^{(2)} a \quad \text{and} \quad \partial_t^{(2)} v = c^2 \partial_x^{(2)} v . \quad (7.99)$$

The reader is encouraged to verify this. We remark that the second-order linear wave equation (7.99), either for a or for v , is a popular second-order hyperbolic model for wave propagation phenomena.

Eigenvalues

First we compute the eigenvalues of the system given by the eigenvalues of the matrix $\mathbf{A}$ in (7.98). These are the roots of the characteristic polynomial, as in (7.42), and are given as

$$\lambda_1 = -c , \quad \lambda_2 = c . \quad (7.100)$$

We next find the corresponding eigenvectors.

Eigenvectors

A left eigenvector $\mathbf{L}_1 = [l_1, l_2]$ of $\mathbf{A}$ corresponding to the eigenvalue λ_1 is found from the equation

$$[l_1, l_2] \begin{bmatrix} 0 & A_0 \\ \frac{c^2}{A_0} & 0 \end{bmatrix} = [\lambda_1 l_1, \lambda_1 l_2] , \quad (7.101)$$

which leads to the algebraic equations

$$l_2 \frac{c^2}{A_0} = \lambda_1 l_1 , \quad l_1 A_0 = \lambda_1 l_2 . \quad (7.102)$$

These two algebraic equations are not independent (verify) and thus we effectively have a single equation, which has a one-parameter family of solutions. By assigning an arbitrary parameter value α_1 to l_1 in the second equation we obtain

$$l_1 = \alpha_1 , \quad l_2 = -\alpha_1 \frac{A_0}{c} . \quad (7.103)$$

An analogously procedure applied to the eigenvector λ_2 gives

$$l_1 = \alpha_2 , \quad l_2 = \alpha_2 \frac{A_0}{c} . \quad (7.104)$$

Collecting results we have

$$\mathbf{L}_1 = \alpha_1 \left[1, -\frac{A_0}{c} \right] , \quad \mathbf{L}_2 = \alpha_2 \left[1, \frac{A_0}{c} \right] . \quad (7.105)$$

Analogous calculations for the set of right eigenvectors lead to

$$\mathbf{R}_1 = \beta_1 \begin{bmatrix} 1 \\ -\frac{c}{A_0} \end{bmatrix}, \quad \mathbf{R}_2 = \beta_2 \begin{bmatrix} 1 \\ \frac{c}{A_0} \end{bmatrix}. \quad (7.106)$$

Here $\alpha_1, \alpha_2, \beta_1$ and β_2 are four arbitrary *scaling parameters*, the values of which are usually chosen on the grounds of convenience. For example $\alpha_1 = \alpha_2 = \beta_1 = \beta_2 = 1$. If we want the left and right eigenvectors to be *orthonormal*, then the scaling parameters must be chosen to satisfy

$$\alpha_1 \beta_1 = 1/2, \quad \alpha_2 \beta_2 = 1/2. \quad (7.107)$$

Prescribing $\alpha_1 = \alpha_2 = 1$ gives $\beta_1 = \beta_2 = \frac{1}{2}$. Then the matrices $\mathbf{L}$ and $\mathbf{R}$ formed by the left and right eigenvectors are given by

$$\mathbf{L} = \begin{bmatrix} 1 & -\frac{A_0}{c} \\ 1 & \frac{A_0}{c} \end{bmatrix}, \quad \mathbf{R} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -\frac{c}{A_0} & \frac{c}{A_0} \end{bmatrix}. \quad (7.108)$$

It is easy to show that

$$\mathbf{R}^{-1} = \mathbf{L}, \quad (7.109)$$

where $\mathbf{R}^{-1}$ denotes the inverse matrix of $\mathbf{R}$ (verify).

7.3.3 Equations in Characteristic Variables

First recall that the characteristic variables are defined as $\mathbf{C} = [c_1, c_2]^T = \mathbf{R}^{-1}\mathbf{Q}$. But in view of the relationship (7.109) it is preferable to use

$$\mathbf{C} = \mathbf{L}\mathbf{Q}, \quad (7.110)$$

as it is more direct. For our system, written in full, we have

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{A_0}{c} \\ 1 & \frac{A_0}{c} \end{bmatrix} \begin{bmatrix} a \\ v \end{bmatrix}, \quad (7.111)$$

which gives the characteristic variables as

$$c_1 = a - \frac{A_0}{c}v, \quad c_2 = a + \frac{A_0}{c}v. \quad (7.112)$$

Now, the governing equations can be expressed in terms of the new variables $\mathbf{C}$, the characteristic variables. From (7.110), as $\mathbf{L}$ is constant, we obtain

$$\partial_t \mathbf{Q} = \mathbf{L}^{-1} \partial_t \mathbf{C}, \quad \partial_x \mathbf{Q} = \mathbf{L}^{-1} \partial_x \mathbf{C}. \quad (7.113)$$

Then (7.97) becomes

$$\mathbf{L}^{-1} \partial_t \mathbf{C} + \mathbf{A} \mathbf{L}^{-1} \partial_x \mathbf{C} = \mathbf{0}. \quad (7.114)$$

Multiplying (7.114) from the left by $\mathbf{L}$ gives

$$\partial_t \mathbf{C} + (\mathbf{L} \mathbf{A} \mathbf{L}^{-1}) \partial_x \mathbf{C} = \mathbf{0}. \quad (7.115)$$

It is easily verified that

$$\mathbf{L} \mathbf{A} \mathbf{L}^{-1} = \Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} -c & 0 \\ 0 & c \end{bmatrix} \quad (7.116)$$

and thus the equations in characteristic variables read

$$\partial_t \mathbf{C} + \Lambda \partial_x \mathbf{C} = \mathbf{0}. \quad (7.117)$$

The equations have become completely decoupled, as is more obviously seen when written in full

$$\left. \begin{aligned} \partial_t c_1 - c \partial_x c_1 &= 0, \\ \partial_t c_2 + c \partial_x c_2 &= 0. \end{aligned} \right\} \quad (7.118)$$

Each equation, separately, is a linear advection equation, whose solution is immediate, for given initial conditions.

7.3.4 The General Initial-Value Problem

We want to solve the initial-value problem (IVP)

$$\left. \begin{aligned} \text{PDEs: } \partial_t \mathbf{Q} + \mathbf{A} \partial_x \mathbf{Q} &= \mathbf{0}, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } \mathbf{Q}(x, 0) &= \mathbf{Q}^{(0)}(x). \end{aligned} \right\} \quad (7.119)$$

Here, the initial condition $\mathbf{Q}^{(0)}(x)$ at time $t = 0$ is an arbitrary function of x alone. This IVP is most easily solved in terms of the characteristic variables $\mathbf{C}$ by replacing the IVP (7.119) by the equivalent IVP

$$\left. \begin{aligned} \text{PDEs: } \partial_t \mathbf{C} + \Lambda \partial_x \mathbf{C} &= \mathbf{0}, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } \mathbf{C}(x, 0) &= \mathbf{C}^{(0)}(x) = \mathbf{L} \mathbf{Q}^{(0)}(x). \end{aligned} \right\} \quad (7.120)$$

From (7.112) the initial conditions in (7.120) become

$$\left. \begin{aligned} c_1^{(0)}(x) &= a^{(0)}(x) - \frac{A_0}{c} v^{(0)}(x), \\ c_2^{(0)}(x) &= a^{(0)}(x) + \frac{A_0}{c} v^{(0)}(x). \end{aligned} \right\} \quad (7.121)$$

Hence the solution of (7.120) in terms of characteristic variables is

$$\left. \begin{aligned} c_1(x, t) &= c_1^{(0)}(x - \lambda_1 t) = a^{(0)}(x + ct) - \frac{A_0}{c} v^{(0)}(x + ct), \\ c_2(x, t) &= c_2^{(0)}(x - \lambda_2 t) = a^{(0)}(x - ct) + \frac{A_0}{c} v^{(0)}(x - ct). \end{aligned} \right\} \quad (7.122)$$

To recover the solution of IVP (7.119) in terms of the original variables one uses the relationship $\mathbf{Q} = \mathbf{RC}$, which gives

$$\left. \begin{aligned} a(x, t) &= \frac{1}{2} [a^{(0)}(x + ct) + a^{(0)}(x - ct) + \frac{A_0}{c} (-v^{(0)}(x + ct) + v^{(0)}(x - ct))] , \\ v(x, t) &= \frac{1}{2} [v^{(0)}(x + ct) + v^{(0)}(x - ct) + \frac{A_0}{c} (-a^{(0)}(x + ct) + a^{(0)}(x - ct))] . \end{aligned} \right\} \quad (7.123)$$

This is the general solution of IVP (7.119) for arbitrary initial conditions $\mathbf{Q}^{(0)}(x)$. We next study the solution of a special initial-value problem, with a special type of initial conditions.

7.3.5 The Riemann Problem

The Riemann problem for system (7.97)–(7.98) is the special initial-value problem

$$\left. \begin{aligned} \text{PDEs: } \partial_t \mathbf{Q} + \mathbf{A} \partial_x \mathbf{Q} &= \mathbf{0}, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } \mathbf{Q}(x, 0) &= \mathbf{Q}^{(0)}(x) = \begin{cases} \mathbf{Q}_L & \text{if } x < 0, \\ \mathbf{Q}_R & \text{if } x > 0. \end{cases} \end{aligned} \right\} \quad (7.124)$$

$\mathbf{Q}_L$ and $\mathbf{Q}_R$ are two constant vectors. Figure 7.11 illustrates the structure of the solution. The problem is to find the solution $\mathbf{Q}^*$ in the intermediate region $\mathcal{R}_1$, called the *Star Region*. From (7.89) the solution at any point (x, t) can be written as

$$\mathbf{Q}(x, t) = \mathbf{Q}_L + \sum_{i=1}^I \delta_i \mathbf{R}_i, \quad (7.125)$$

where the positive integer $I(x, t)$ is such that

$$\lambda_I < \frac{x}{t} < \lambda_{I+1} \quad (7.126)$$

and the coefficients δ_i are the solution of the following $m \times m$ linear algebraic system

$$\sum_{i=1}^m \delta_i \mathbf{R}_i = \mathbf{Q}_R - \mathbf{Q}_L = \Delta \mathbf{Q}. \quad (7.127)$$

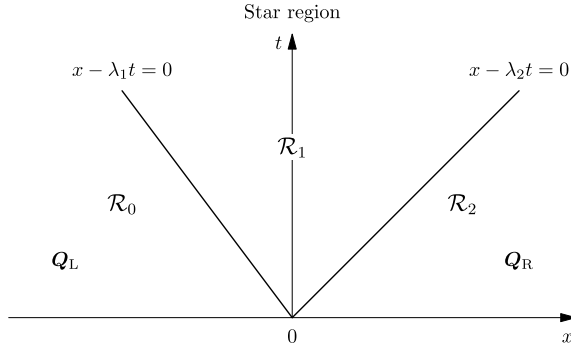


Fig. 7.11 Structure of the solution of the Riemann problem for the linearised blood flow equations. The two characteristic lines associated with the eigenvalues $\lambda_1 = -c$ and $\lambda_2 = c$ divide the half x - t plane into three constant wedges. The solution is unknown in the $\mathcal{R}_1$ region, the *Star Region*

Here $m = 2$ is the number of equations in system (7.124). For our particular equations the linear algebraic system is

$$\delta_1 \begin{bmatrix} -\frac{1}{2} \frac{c}{A_0} \\ \frac{1}{2} \frac{c}{A_0} \end{bmatrix} + \delta_2 \begin{bmatrix} \frac{1}{2} \frac{c}{A_0} \\ \frac{1}{2} \frac{c}{A_0} \end{bmatrix} = \begin{bmatrix} \Delta a \\ \Delta v \end{bmatrix} = \begin{bmatrix} a_R - a_L \\ v_R - v_L \end{bmatrix}. \quad (7.128)$$

This gives

$$\delta_1 + \delta_2 = 2\Delta a, \quad -\delta_1 + \delta_2 = 2\frac{A_0}{c} \Delta v. \quad (7.129)$$

The solution is

$$\delta_1 = \Delta a - \frac{A_0}{c} \Delta v, \quad \delta_2 = \Delta a + \frac{A_0}{c} \Delta v. \quad (7.130)$$

Here we are interested in the solution $\mathbf{Q}^* = [a^*, v^*]^T$ in the *star region* $\mathcal{R}_1$, which is given by the points (x, t) such that

$$\lambda_1 = -c < \frac{x}{t} < \lambda_2 = c, \quad (7.131)$$

with $I = 1$ in (7.125). See also (7.87). Then the solution is given by

$$\mathbf{Q}^* = \mathbf{Q}_L + \delta_1 \mathbf{R}_1, \quad (7.132)$$

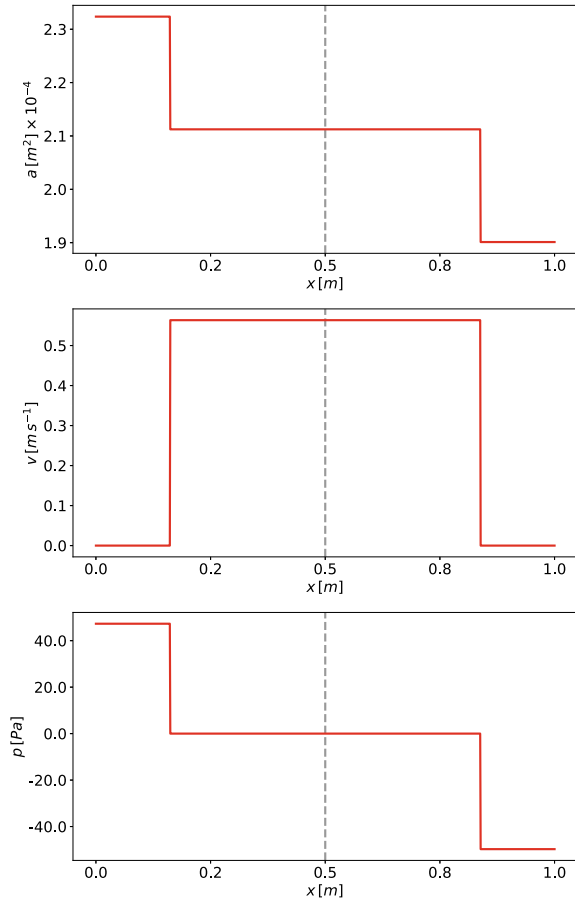
which yields

$$\left. \begin{aligned} a^* &= \frac{1}{2}(a_L + a_R) - \frac{1}{2} \frac{A_0}{c} (v_R - v_L), \\ v^* &= \frac{1}{2}(v_L + v_R) - \frac{1}{2} \frac{c}{A_0} (a_R - a_L). \end{aligned} \right\} \quad (7.133)$$

Numerical Example of a Riemann Problem

Here we consider a specific example of a Riemann problem for the linearised blood flow equations. We take the spatial domain $[0, 1]$ and an initial discontinuity placed at $x_0 = 0.5 \text{ m}$. Values of parameters are: $A_0 = 2.1124 \times 10^{-4} \text{ m}^2$; $h_0 = 8.2 \times 10^{-4} \text{ m}$; $\nu = 0.5$; $\rho = 1050 \text{ kg/m}^3$; $E = 5 \times 10^5 \text{ Pa}$. Initial velocities are $v_L = 0 \text{ m/s}$ and $v_R = 0 \text{ m/s}$. Area perturbation data are $a_L = 1.1 \times A_0 \text{ m}^2$ and $a_R = 0.9 \times A_0 \text{ m}^2$. The solution consists of two discontinuities travelling left and right, as depicted in Fig. 7.12 at the output time $T_{out} = 0.06 \text{ s}$.

Fig. 7.12 Solution of Riemann for the linearised blood flow equations at the output time $T_{out} = 0.06 \text{ s}$. Quantities shown: are area $a(x, T_{out})$, velocity $v(x, T_{out})$ and pressure $p(x, T_{out})$. Dotted vertical lines show position of initial discontinuity



7.4 Hyperbolic Systems from Parabolic Equations

In this section we introduce a procedure to approximate *advection-diffusion-reaction* partial differential equations by **hyperbolic systems** with stiff source terms, following the Cattaneo relaxation approach [2]. The methodology has been pioneered by Nishikawa [3] for the steady state case and has more recently been extended by Toro and Montecinos [4, 5] to the time-dependent case. The approach is applicable to the blood flow equations with a viscoelastic tube law leading to a parabolic system. See Montecinos, Müller and Toro [6] and Chap. 10.

Here we apply the approach to a linear parabolic equation, resulting in a system of linear hyperbolic equations, which are subsequently studied following the methods of this chapter.

7.4.1 Equations, Formulations and Analysis

Consider the time-dependent, advection-diffusion-reaction equation, with a stiff or non-stiff reaction term as follows

$$\partial_t q_1(x, t) + \hat{\lambda} \partial_x q_1(x, t) = \alpha \partial_x^2 q_1(x, t) + \beta q_1(x, t) . \quad (7.134)$$

The unknown is the function $q_1(x, t)$, $\hat{\lambda}$ is the characteristic speed, $\alpha > 0$ is a diffusion coefficient and $\beta \leq 0$ is a reaction-rate coefficient. Here we allow for stiff reaction, for which $|\beta| \gg 1$, noting however that the formulation works equally well for a non-stiff source term, or no source term.

We introduce a *relaxation time* ϵ satisfying $0 < \epsilon \ll 1$ and an auxiliary function $q_2(x, t)$ such that

$$q_2(x, t) \rightarrow \partial_x q_1(x, t) \quad \text{as} \quad \epsilon \rightarrow 0 . \quad (7.135)$$

Then, based on (7.135), we consider the following additional evolution partial differential equation

$$\partial_t q_2(x, t) = [\partial_x q_1(x, t) - q_2(x, t)] \frac{1}{\epsilon} . \quad (7.136)$$

Equations (7.134) and (7.136) constitute a relaxation system

$$\left. \begin{aligned} \partial_t q_1(x, t) + \hat{\lambda} \partial_x q_1(x, t) - \alpha \partial_x^2 q_1(x, t) &= \beta q_1(x, t) , \\ \partial_t q_2(x, t) - \frac{1}{\epsilon} \partial_x q_1(x, t) &= -q_2(x, t) \frac{1}{\epsilon} . \end{aligned} \right\} \quad (7.137)$$

These equations approximate (7.134) as $\epsilon \rightarrow 0$ and can be written in the form of a system of hyperbolic balance laws with source terms, namely

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}) , \quad (7.138)$$

with

$$\mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} , \quad \mathbf{F} = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} \hat{\lambda} q_1 - \alpha q_2 \\ -q_1/\epsilon \end{bmatrix} , \quad \mathbf{S} = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} \beta q_1 \\ -q_2/\epsilon \end{bmatrix} . \quad (7.139)$$

Note that irrespectively of the nature of the source term $s(q_1)$ in the original Eq. (7.134), the relaxation system (7.138)–(7.139) is stiff due to the new source term $-q_2/\epsilon$, which we now prove to be hyperbolic.

Proposition 3 *The relaxation system (7.138)–(7.139) is strictly hyperbolic for all non-zero values of the relaxation parameter ϵ .*

Proof Written in quasilinear form, system (7.138)–(7.139) reads

$$\partial_t \mathbf{Q} + \mathbf{A} \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}) , \quad (7.140)$$

in which the Jacobian matrix is

$$\mathbf{A} = \frac{\partial \mathbf{F}}{\partial \mathbf{Q}} = \begin{bmatrix} \partial f_1 / \partial q_1 & \partial f_1 / \partial q_2 \\ \partial f_2 / \partial q_1 & \partial f_2 / \partial q_2 \end{bmatrix} = \begin{bmatrix} \hat{\lambda} & -\alpha \\ -\frac{1}{\epsilon} & 0 \end{bmatrix} . \quad (7.141)$$

The eigenvalues of $\mathbf{A}$ are the roots of the characteristic polynomial $|\mathbf{A} - \lambda \mathbf{I}| = 0$, where $\mathbf{I}$ is the identity matrix and λ is a parameter. They are easily found to be

$$\lambda_1 = \frac{1}{2} \hat{\lambda} - \sqrt{\left(\frac{1}{2} \hat{\lambda}\right)^2 + \frac{\alpha}{\epsilon}} , \quad \lambda_2 = \frac{1}{2} \hat{\lambda} + \sqrt{\left(\frac{1}{2} \hat{\lambda}\right)^2 + \frac{\alpha}{\epsilon}} , \quad (7.142)$$

which are real and distinct and thus the system is strictly hyperbolic. Note that the associated wave speeds satisfy the pattern $\lambda_1 < \hat{\lambda} < \lambda_2$. The right eigenvectors, for appropriate scalings, are

$$\mathbf{R}_1 = \begin{bmatrix} \epsilon \lambda_1 \\ -1 \end{bmatrix} , \quad \mathbf{R}_2 = \begin{bmatrix} \epsilon \lambda_2 \\ -1 \end{bmatrix} , \quad (7.143)$$

which for $\lambda_1 \neq \lambda_2$ are linearly independent. The relaxation system (7.138)–(7.139) is thus strictly hyperbolic, as claimed. $\blacksquare$

7.4.2 Exact Solution

Next we find a special exact solution to the relaxation system (7.138)–(7.139).

Proposition 4 *For all values of ϵ and β satisfying*

$$\beta = -1/\epsilon, \quad (7.144)$$

the initial value problem for system (7.138)–(7.139) with initial conditions

$$\mathbf{Q}^{(0)}(x) = \mathbf{Q}(x, 0) = \begin{bmatrix} q_1^{(0)}(x) \\ q_2^{(0)}(x) \end{bmatrix} \quad (7.145)$$

has exact solution given by

$$q_1(x, t) = \left. \begin{aligned} & \frac{e^{-\frac{1}{\epsilon}t}}{(\lambda_2 - \lambda_1)} \lambda_1 [-q_1^{(0)}(x - \lambda_1 t) - \epsilon \lambda_2 q_2^{(0)}(x - \lambda_1 t)] \\ & + \frac{e^{-\frac{1}{\epsilon}t}}{(\lambda_2 - \lambda_1)} \lambda_2 [q_1^{(0)}(x - \lambda_2 t) + \epsilon \lambda_1 q_2^{(0)}(x - \lambda_2 t)] , \end{aligned} \right\} \quad (7.146)$$

and

$$q_2(x, t) = \left. \begin{aligned} & -\frac{e^{-\frac{1}{\epsilon}t}}{\epsilon(\lambda_2 - \lambda_1)} [q_1^{(0)}(x - \lambda_1 t) + \epsilon \lambda_2 q_2^{(0)}(x - \lambda_1 t)] \\ & + \frac{e^{-\frac{1}{\epsilon}t}}{\epsilon(\lambda_2 - \lambda_1)} [q_1^{(0)}(x - \lambda_2 t) + \epsilon \lambda_1 q_2^{(0)}(x - \lambda_2 t)] . \end{aligned} \right\} \quad (7.147)$$

Proof The matrix of right eigenvectors is

$$\mathbf{R} = \begin{bmatrix} \epsilon \lambda_1 & \epsilon \lambda_2 \\ -1 & -1 \end{bmatrix} \quad (7.148)$$

and the characteristic variables are

$$\mathbf{C} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \mathbf{R}^{-1} \mathbf{Q} . \quad (7.149)$$

We can express system (7.138)–(7.139) in characteristic variables as

$$\partial_t \mathbf{C} + \Lambda \partial_x \mathbf{C} = \hat{\mathbf{S}} , \quad (7.150)$$

with diagonal coefficient matrix

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \quad (7.151)$$

and transformed source term as

$$\hat{\mathbf{S}} = \begin{bmatrix} \hat{s}_1 \\ \hat{s}_2 \end{bmatrix} = \mathbf{R}^{-1} \mathbf{S} , \quad (7.152)$$

with

$$\hat{s}_1 = -\frac{1}{\epsilon} c_1 , \quad \hat{s}_2 = -\frac{1}{\epsilon} c_1 . \quad (7.153)$$

Thus system (7.138)–(7.139) becomes decoupled and the exact solutions for $c_1(x, t)$ and $c_2(x, t)$ can be computed as

$$\left. \begin{aligned} c_1(x, t) &= c_1^{(0)}(x - \lambda_1 t) e^{\frac{1}{\epsilon} t} , \\ c_2(x, t) &= c_2^{(0)}(x - \lambda_2 t) e^{-\frac{1}{\epsilon} t} . \end{aligned} \right\} \quad (7.154)$$

Transforming back to the original variables we obtain the solution for the initial value problem for (7.138)–(7.139) with general initial condition (7.145) under the restriction (7.144); it is given by (7.146)–(7.147), as claimed. ■

The exact solution to the relaxation system just constructed can be useful for assessing the performance of numerical methods intended for such type of systems.

7.5 Concluding Remarks and Further Reading

We have studied some basic mathematical notions on linear hyperbolic partial differential equations and have illustrated the theory through its application to some examples solved in detail. In particular, we have applied the theory to the study of a linear system that models blood flow in arteries. Moreover, as a novel example, we introduced a relaxation procedure whereby advection-diffusion equations (parabolic equations) may be approximated by hyperbolic equations with stiff source terms. The provided background on linear hyperbolic equations will be useful for studying simplified models of practical interest and, more importantly, for studying non-linear hyperbolic equations. The reader is encouraged to solve the exercises listed at the end of the chapter.

For further reading on the subject of this chapter we strongly recommend the following books: Zachmanoglou and Thoe [7]; Logan [8] and Jeffrey [9].

7.6 Exercises

7.1 Consider the following Riemann problem for the homogeneous linear advection equation with characteristic speed $\lambda = 1$.

$$\begin{aligned} \text{PDE: } & \partial_t q + \lambda \partial_x q = 0, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } & q(x, 0) = h(x) = \begin{cases} q_L = 1 & \text{if } x < 0, \\ q_R = 0 & \text{if } x > 0. \end{cases} \end{aligned} \quad (7.155)$$

Perform the following tasks:

1. Draw the initial profile $h(x)$ at time $t = 0$.
2. Draw a picture of characteristics in the x - t half-plane, with curves emanating from the following points: $x_0^{(1)} = -2$, $x_0^{(2)} = -1$, $x_0^{(3)} = 0$, $x_0^{(4)} = 1$ and $x_0^{(5)} = 2$.
3. For each characteristic k emanating from the point $x_0^{(k)}$, identify the value $h(x_0^{(k)})$ of the initial condition.
4. Using the results above construct the complete solution to the Riemann problem (7.155) in the x - t half-plane.
5. Using the above construction draw the solution profiles $q(x, t_k)$ corresponding to the constant times $t_1 = 1$ and $t_2 = 2$.
6. Find the value of the solution along the t -axis for times $t_1 = 1$, $t_2 = 2$ and $t_3 = 3$. Is the solution of the Riemann problem constant along the t -axis, for all times?
7. Repeat all the above exercises for the Riemann problem (7.155) with characteristic speed $\lambda = -1$.

7.2 Consider the following initial value problem for the homogeneous linear advection equation with characteristic speed $\lambda = 1$ and initial condition

$$\text{IC: } q(x, 0) = h(x) = \begin{cases} 0 & \text{if } x < -1, \\ 1 - x^2 & \text{if } -1 \leq x \leq 1, \\ 0 & \text{if } x > 1. \end{cases} \quad (7.156)$$

Perform the following tasks:

1. Draw the initial profile $h(x)$.
2. Draw a picture of characteristics in the x - t half plane, with curves emanating from the following points: $x_0^{(1)} = -2$, $x_0^{(2)} = -1$, $x_0^{(3)} = 0$, $x_0^{(4)} = 1$ and $x_0^{(5)} = 2$.
3. For each characteristic k emanating from the point $x_0^{(k)}$, identify the value $h(x_0^{(k)})$ of the initial condition.
4. Using the results above construct the complete solution to the initial-value problem (7.156) in the x - t half-plane.
5. Using the above construction draw the solution profiles $q(x, t_k)$ corresponding to the constant times $t_1 = 1$, $t_2 = 2$ and $t_3 = 3$.

6. Find the value of the solution along the t -axis for times $t_1 = 1$, $t_2 = 2$ and $t_3 = 3$.
7. Is the solution of the initial-value problem constant along the t -axis, for all times?
8. Is the integral in space of the solution profiles $q(x, t_k)$ preserved?
9. Repeat all the above exercises for the Riemann problem (7.155) with characteristic speed $\lambda = -1$.

7.3 Consider the initial value problem for the linear advection-reaction equation

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \lambda \partial_x q = \beta q, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } \quad q(x, 0) = h(x), \quad -\infty < x < \infty, \end{array} \right\} \quad (7.157)$$

with λ and β two real constants. Assume that the general initial condition $q(x, 0) = h(x)$ is differentiable. Show that the exact solution of (7.157) is

$$q(x, t) = h(x - \lambda t) e^{\beta t}. \quad (7.158)$$

7.4 Consider the following Riemann problem for the inhomogeneous linear advection equation

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \lambda \partial_x q = \beta q, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } \quad q(x, 0) = h(x) = \begin{cases} q_L = 1 & \text{if } x < 0, \\ q_R = 0 & \text{if } x > 0, \end{cases} \end{array} \right\} \quad (7.159)$$

with characteristic speed $\lambda = 1$ and reaction-rate coefficient $\beta = -1$.

Perform the following tasks:

1. Draw the initial condition $h(x)$.
2. Draw a picture of characteristics in the x - t half plane, with curves emanating from the following points: $x_0^{(1)} = -2$, $x_0^{(2)} = -1$, $x_0^{(3)} = 0$, $x_0^{(4)} = 1$, $x_0^{(5)} = 2$.
3. For each characteristic k emanating from the point $x_0^{(k)}$, identify the value $h(x_0^{(k)})$ of the initial condition at the foot of the characteristic $x_0^{(k)}$.
4. Does the value $h(x_0^{(k)})$ at the foot of the characteristic remain constant in time along the characteristic? If it varies in time, how does it vary? Hint: look at the function (7.158).
5. Using the results above illustrate the complete solution to the Riemann problem (7.159) in the x - t half-plane.
6. Draw solution profiles $q(x, t_k)$ corresponding to the constant times $t_1 = 1$, $t_2 = 2$ and $t_3 = 3$.
7. Is the solution of the Riemann problem constant along the t axis?

8. Calculate the following time-integral average of the solution right at the interface $x = 0$ (along the t -axis)

$$\hat{q} = \frac{1}{T} \int_0^T h(-\lambda t) e^{\beta t} dt, \quad (7.160)$$

where T is a given positive time.

7.5 Study the IBVP (7.159) for the case $\lambda < 0$ and discuss in full detail the solution procedure, illustrating graphically the geometric situation, as done for the case $\lambda > 0$.

7.6 Study the solution of the IBVP (7.159) in the domain $[a, b]$, with $\lambda = 1$, initial condition $h(x) = 0$ and left boundary condition

$$b_L(t) = t/\sqrt{t^2 + \epsilon}, \quad t \in [0, \infty). \quad (7.161)$$

Impose a condition on the right boundary $x = b$, such that the right-travelling wave goes past $x = b$ undisturbed. Take $a = 0$, $b = 1$, $\epsilon = 0.1$ and plot the exact solution at times $T_{out} = \frac{1}{2}$, $T_{out} = 1$ and $T_{out} = \frac{3}{2}$.

7.7 Consider the abstract in-homogeneous linear system

$$\partial_t \mathbf{Q} + \mathbf{A} \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}), \quad (7.162)$$

where $\mathbf{Q}(x, t)$ is the vector of unknowns, $\mathbf{A}$ is a constant coefficient matrix and $\mathbf{S}(\mathbf{Q})$ is the source term vector, all given as

$$\mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 0 & a \\ a & 0 \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix}, \quad a > 0. \quad (7.163)$$

Perform the following tasks:

1. Show that the eigenvalues of the system are $\lambda_1 = -a$ and $\lambda_2 = a$.
2. Show that the corresponding left eigenvectors are

$$\mathbf{L}_1 = \alpha_1 [1, -1], \quad \mathbf{L}_2 = \alpha_2 [1, 1]. \quad (7.164)$$

3. Show that the corresponding right eigenvectors are

$$\mathbf{R}_1 = \beta_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \mathbf{R}_2 = \beta_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad (7.165)$$

4. Show that imposing orthonormality of left and right eigenvectors leads to the condition $\alpha_1 \beta_1 = 1/2$ and $\alpha_2 \beta_2 = 1/2$.

5. Choose $\alpha_1 = \alpha_2 = 1$ and $\beta_1 = \beta_2 = 1/2$. Show that the inverse matrix of $\mathbf{R}$ formed by the columns of right eigenvectors is

$$\mathbf{R}^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}. \quad (7.166)$$

Note that $\mathbf{L} = \mathbf{R}^{-1}$, where $\mathbf{L}$ is the matrix whose rows are the left eigenvectors of $\mathbf{A}$.

7.8 For system (7.162)–(7.163) perform the following tasks:

1. Show that the vector of characteristic variables is

$$\mathbf{C} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} q_1 - q_2 \\ q_1 + q_2 \end{bmatrix}. \quad (7.167)$$

2. Show that decoupled equations with source terms in characteristic variables read

$$\left. \begin{aligned} \partial_t c_1 - a \partial_x c_1 &= \frac{1}{2}(s_1 - s_2), \\ \partial_t c_2 + a \partial_x c_2 &= \frac{1}{2}(s_1 + s_2). \end{aligned} \right\} \quad (7.168)$$

3. Show that if the original source term in (7.162)–(7.163) is chosen as

$$\mathbf{S} = \frac{1}{2} \begin{bmatrix} (\gamma_1 + \gamma_2)q_1 + (\gamma_2 - \gamma_1)q_2 \\ (\gamma_2 - \gamma_1)q_1 + (\gamma_1 + \gamma_2)q_2 \end{bmatrix}, \quad (7.169)$$

where γ_1 and γ_2 are any two arbitrary real numbers, then the decoupled inhomogeneous system reads

$$\left. \begin{aligned} \partial_t c_1 - a \partial_x c_1 &= \gamma_1 c_1, \\ \partial_t c_2 + a \partial_x c_2 &= \gamma_2 c_2. \end{aligned} \right\} \quad (7.170)$$

7.9 For system (7.162)–(7.163) perform the following tasks:

1. Show that the exact solution of the IVP for the inhomogeneous system (7.170) is

$$\left. \begin{aligned} c_1(x, t) &= c_1^{(0)}(x + at)e^{\gamma_1 t}, \\ c_2(x, t) &= c_2^{(0)}(x - at)e^{\gamma_2 t}, \end{aligned} \right\} \quad (7.171)$$

where $c_1^{(0)}(x)$ and $c_2^{(0)}(x)$ are the initial conditions for the characteristic variables $c_1(x)$ and $c_2(x)$ respectively. Recall that the initial conditions are given as $\mathbf{C}^{(0)} = \mathbf{R}^{-1}\mathbf{Q}^{(0)}$, where $\mathbf{Q}^{(0)}$ is the initial condition for (7.162)–(7.163) in terms of the original variables.

- Find the final expression for the corresponding solution in terms of the original variables $\mathbf{Q}$.

7.10 Consider the Riemann problem for the homogeneous version of system (7.162)–(7.163)

$$\begin{aligned} \text{PDEs: } \partial_t \mathbf{Q} + \mathbf{A} \partial_x \mathbf{Q} &= \mathbf{0}, \\ \text{ICs: } \mathbf{Q}(x, 0) &= \mathbf{Q}^{(0)}(x) = \begin{cases} \mathbf{Q}_L & \text{if } x < 0, \\ \mathbf{Q}_R & \text{if } x > 0, \end{cases} \end{aligned} \quad (7.172)$$

where

$$\mathbf{Q}_L = \begin{bmatrix} q_{1L} \\ q_{2L} \end{bmatrix}, \quad \mathbf{Q}_R = \begin{bmatrix} q_{1R} \\ q_{2R} \end{bmatrix} \quad (7.173)$$

are two constant vectors. Perform the following tasks

- Find the general solution of the Riemann problem (7.172)–(7.173).
- Sketch the structure of the solution of the Riemann problem in the x - t plane and identify regions of constant values for the solution.
- Choose the following special values: $a = 1$, $q_{1L} = 1$, $q_{2L} = 2$, $q_{1R} = 1/2$, $q_{2R} = 1/4$. Plot the solution profiles $q_1(x, t_k)$ and $q_2(x, t_k)$ at time $T_{out} = 1$ in the spatial domain $[-2, 2]$.

7.11 Consider the diffusion equation (parabolic)

$$\partial_t q_1(x, t) = \alpha \partial_x^2 q_1(x, t), \quad (7.174)$$

with $\alpha > 0$ the diffusion coefficient, a prescribed constant. Perform the following tasks:

- Utilize the relaxation procedure to express Eq. (7.174) as a system of the form (7.138)–(7.139). Recall that the relaxation parameter ϵ satisfies the condition $0 < \epsilon < 1$.
- Find the eigenstructure of the system (eigenvalues and eigenvectors)
- Is the system hyperbolic?
- Solve exactly the Riemann problem with data $\mathbf{Q}_L$ and $\mathbf{Q}_R$ for the homogeneous case.
- Display the structure of the solution of the Riemann problem in the x - t plane.
- Select specific data for the parameters $\alpha > 0$ and ϵ , and for the initial conditions in the Riemann problem and plot solution profiles at a specified time. See Fig. 7.12, for example.

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Chapter 8

Hyperbolic Equations II: The Non-Linear Case



Abstract This chapter focuses on the mathematical theory of nonlinear hyperbolic equations, in which discontinuities can form over time from smooth initial conditions. We begin with scalar nonlinear equations, studying rarefaction and shock waves. Emphasis is placed on shock formation from continuous initial data, which renders invalid the differential form of the equation, as derivatives may no longer be defined. Generalised solutions, which accommodate both continuous and discontinuous solutions, arise from introducing the integral form of the conservation laws. These solutions now satisfy the differential form in smooth regions and the Rankine-Hugoniot condition at discontinuities. However, the enlarged set of solutions also includes physically non-admissible, spurious solutions. Examples of non-uniqueness are provided, and a selection criterion—the Lax entropy condition—is introduced to eliminate unwanted solutions. The Riemann problem for the Burgers equation is then introduced and solved. Systems of hyperbolic equations are discussed, along with definitions and examples. Eigenvalues, eigenvectors, and the notion of hyperbolicity are introduced, as well as characteristic fields and their properties. The integral form for systems of hyperbolic equations is presented, along with the corresponding Rankine-Hugoniot conditions. Nonlinear blood flow equations are introduced as a relevant example of a system of hyperbolic equations. The mathematical concepts studied are applied to analyse a simplified version of the blood flow equations, assuming the Coriolis coefficient to be $\alpha = 1$. These equations are also expressed in terms of mathematical conservation laws, but using physical variables rather than conserved variables. A cautionary note is provided to ensure that the reader correctly interprets such equations. The chapter concludes with a summary and recommendations for further study, along with a list of references. A selection of exercises is included at the end of the chapter.

8.1 Scalar Non-Linear Equations

We start this chapter by studying single (scalar) non-linear first-order equations. For useful background the reader is advised to consult Chap. 7 on the linear case.

8.1.1 Definitions and Examples

Consider the single first-order partial differential equation for the unknown function $q(x, t)$

$$\partial_t q + \partial_x f(q) = 0. \quad (8.1)$$

This equation is called a *conservation law*, in which q is the *conserved variable* and $f(q)$ is the *flux function*, a prescribed function of the unknown q ; $f(q)$ is also called the *physical flux*. If $q(x, t)$ and $f(q)$ are sufficiently smooth then all derivatives in (8.1) are defined and the equation is said to be written in *differential, conservative form*. By virtue of the chain rule we may express (8.1) in the alternative form (non-conservative), called the *quasi-linear form*, as follows

$$\partial_t q + \lambda(q) \partial_x q = 0, \quad \frac{d}{dq} f(q) = f'(q) := \lambda(q). \quad (8.2)$$

Here $\lambda(q)$ is called *characteristic speed*, which in general, is a function of the unknown q . Note the analogy between the non-linear equation in (8.2) and the linear advection equation studied in Chap. 7, in which the characteristic speed λ is either a constant or a function of the independent variables x and t . In (8.2) the characteristic speed depends on the solution q itself.

Scalar conservation laws may be classified by the properties of the characteristic speed $\lambda(q)$, which depend on the flux function $f(q)$. An important property of $\lambda(q)$ is *monotonicity*, or otherwise, giving rise to essentially three cases:

1. **Convex flux:** $\lambda(q)$ is a monotone *increasing* function of q , that is

$$\frac{d}{dq} \lambda(q) = \lambda'(q) = f''(q) > 0, \quad \forall q. \quad (8.3)$$

2. **Concave flux:** $\lambda(q)$ is a monotone *decreasing* function of q , that is

$$\frac{d}{dq} \lambda(q) = \lambda'(q) = f''(q) < 0, \quad \forall q. \quad (8.4)$$

3. **Non-convex, non-concave flux:** $\lambda(q)$ vanishes for some q , that is

$$\frac{d}{dq}\lambda(q) = \lambda'(q) = f''(q) = 0, \quad \text{for some } q. \quad (8.5)$$

We now give some examples.

The inviscid Burgers equation. The equation, the flux and its derivatives are

$$\left. \begin{aligned} \partial_t q + \partial_x f(q) &= 0 \\ f(q) &= \frac{1}{2}q^2, \quad \lambda(q) = f'(q) = q, \quad \lambda'(q) = f''(q) = 1 > 0. \end{aligned} \right\} \quad (8.6)$$

The flux is convex and in this special case the characteristic speed $\lambda(q)$ is identical to the conserved variable q . Consequently, Burgers equation can be written as

$$\partial_t q + q \partial_x q = 0. \quad (8.7)$$

As a result of the monotone increasing behaviour of the characteristic speed, larger values of the conserved variable q will propagate faster than smaller values of q , leading to wave distortion and eventually to shock formation.

A traffic flow equation. A simple model for traffic flow is a conservation law of the form (8.1) with flux given as follows

$$f(q) = u_{\max} \left(1 - \frac{q}{q_{\max}} \right) q. \quad (8.8)$$

In this equation $u_{\max}$ and $q_{\max}$ are two prescribed, positive constant parameters of the equation, with $0 < q \leq q_{\max}$. The associated flux derivatives are

$$\left. \begin{aligned} \lambda(q) = f'(q) &= u_{\max} \left(1 - \frac{2q}{q_{\max}} \right), \\ \lambda'(q) = f''(q) &= -\frac{2u_{\max}}{q_{\max}} < 0. \end{aligned} \right\} \quad (8.9)$$

The flux is concave. In this case larger values of the conserved variable q will propagate more slowly than smaller values of q , the opposite to the Burgers Eq. (8.7).

Cubic flux equation. Another example of a conservation law (8.1) is furnished by choosing the cubic flux function

$$f(q) = q^3 \quad (8.10)$$

and associated flux derivatives

$$\begin{aligned} \lambda(q) = f'(q) = 3q^2, \\ \lambda'(q) = f''(q) = 6q \end{aligned} \left\{ \begin{array}{ll} < 0 & \text{if } q < 0, \\ = 0 & \text{if } q = 0, \\ > 0 & \text{if } q > 0. \end{array} \right\} \quad (8.11)$$

This equation is non-convex/non-concave for values of q in the set of real numbers and $\lambda'(q) = 0$ for $q = 0$. Solutions of this type of equations exhibit a peculiar behaviour and will not be considered further.

8.1.2 Solution Along Characteristics

Some of the theory studied for the linear advection equation in Chap. 7 is also valid for the non-linear case. Let us then consider the initial-value problem (IVP) for the non-linear scalar conservation law

$$\begin{aligned} \text{PDE: } \partial_t q + \partial_x f(q) &= 0, \\ \text{IC: } q(x, 0) &= h(x). \end{aligned} \quad (8.12)$$

Consider now characteristic curves $(x(t), t)$ satisfying the IVP

$$\begin{aligned} \text{ODE: } x'(t) &= \lambda(q(x(t), t)), \\ \text{IC: } x(0) &= x_0. \end{aligned} \quad (8.13)$$

Then, by regarding both q and x to be functions of t we find the **total derivative** of q along a curve $(x(t), t)$

$$\begin{aligned} \frac{d}{dt} q(x(t), t) &= \partial_t q(x(t), t) + \partial_x q(x(t), t) x'(t), \\ &= \partial_t q(x(t), t) + \lambda(q) \partial_x q(x(t), t) = 0. \end{aligned} \quad (8.14)$$

This says that $q(x, t)$ is constant along the characteristic curve satisfying the IVP (8.13) and therefore $\lambda(q)$ is also constant along the characteristic. Hence characteristic curves are straight lines, just as for the linear advection case. The value of $q(x, t)$ along each curve is the value of $q(x, t)$ at the initial point $x(0) = x_0$, the foot of the characteristic. We therefore may write the solution at any point (x, t) on the characteristic as

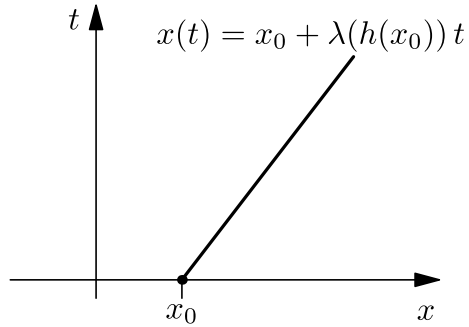
$$q(x(t), t) = h(x_0). \quad (8.15)$$

The characteristic speed $\lambda(q)$ may then be evaluated at x_0 so that the solution curves of IVP (8.13) are

$$x(t) = x_0 + \lambda(h(x_0))t. \quad (8.16)$$

Figure 8.1 depicts the situation.

Fig. 8.1 Characteristic curve $x(t) = x_0 + \lambda(h(x_0))t$ emanating from the point x_0 , called the foot of the characteristic



Equations (8.15) and (8.16) may be regarded as the analytical solution of IVP (8.12). Given a point (x, t) in the x - t half-plane the solution at (x, t) is found by evaluating the initial condition $h(x)$ at the foot x_0 of the characteristic. The initial point x_0 depends on the chosen point (x, t) in the x - t plane and from (8.16) we have

$$x_0 = x - \lambda(h(x_0))t, \quad (8.17)$$

which is a nonlinear equation, the complexity of which depends on the prescribed function $h(x)$, the initial condition of IVP (8.12). The solution given by (8.15)–(8.16) is *implicit*, which is more apparent if we substitute x_0 from (8.17) into (8.15) to obtain

$$q(x, t) = h(x_0(x, t)) = h(x - \lambda(h(x_0(x, t)))t). \quad (8.18)$$

This solution is identical in form, but only in form, to the solution of the linear advection equation studied in Chap. 7.

We now verify that (8.18) is solution of (8.12), as claimed. From (8.18) and (8.17) we obtain the t and x partial derivatives of $q(x, t)$, that is

$$\left. \begin{aligned} \partial_t q(x, t) &= \partial_t h(x_0(x, t)) = -h'(x_0) \partial_t x_0(x, t), \\ \partial_x q(x, t) &= \partial_x h(x_0(x, t)) = h'(x_0) \partial_x x_0(x, t). \end{aligned} \right\} \quad (8.19)$$

Then we find the t and x partial derivatives of $x_0(x, t)$ from (8.17). It is easy to verify that

$$\partial_t x_0 = -\frac{\lambda(h(x_0))}{1 + \lambda'(h(x_0))h'(x_0)t}, \quad \partial_x x_0 = \frac{1}{1 + \lambda'(h(x_0))h'(x_0)t}. \quad (8.20)$$

By substituting the partial derivatives (8.20) into the expressions for $\partial_t q$ and $\partial_x q$ in (8.19), and these in turn into the PDE in (8.12) we verify that the function (8.18) is in fact a solution of the non-linear equation in (8.12).

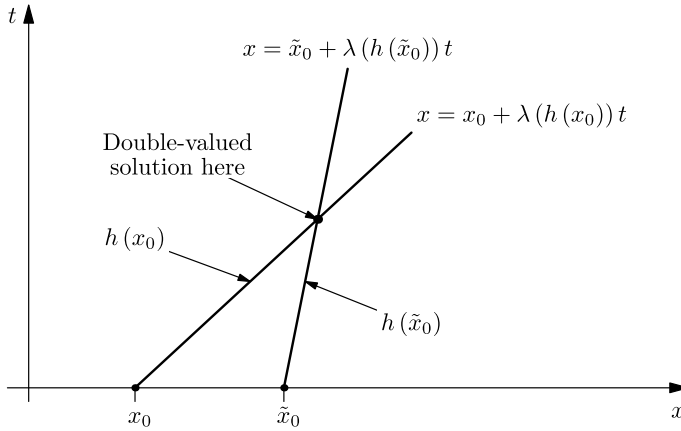


Fig. 8.2 Crossing of characteristics. Characteristics emanating from x_0 and $\tilde{x}_0$ carry different initial values $h(x_0)$ and $h(\tilde{x}_0)$ leading to multi-valued solutions at the point of intersection

Remarks. Solutions to non-linear equations may also be obtained by following characteristics, as done for the linear case in Chap. 7. However there are substantial differences to the linear case, as we now point out.

1. In the non-linear case the solution is *implicit*, see (8.18). Computing the actual solution in general requires the use of an iterative numerical procedure.
2. More importantly, the method of solution following characteristics breaks down when two characteristic curves intersect, as these carry different constant values from the initial condition, leading to non-admissible multi-valued solutions. See Fig. 8.2.
3. Note also that the denominators in (8.20) may be zero and thus the partial derivatives may be undefined. Hence the differential form of the equation is rendered invalid. Alternative, non-differential, formulations of the equations are then necessary.
4. The vanishing of the denominator in (8.20) actually gives the time $t = t_b$ at which shock formation occurs.

8.1.3 Smooth Solutions: Rarefaction Waves

Here we study a class of smooth solutions called rarefaction waves, in which the non-linear phenomenon of wave distortion is evident. We restrict ourselves to convex equations, such as the Burgers equation. We study two types of rarefactions.

Case 1: Rarefaction Wave From Smooth Data

We consider the following IVP with smooth initial condition

$$\left. \begin{aligned} \partial_t q + \partial_x f(q) &= 0, \quad f''(q) > 0, \\ q(x, 0) &= h(x) = \begin{cases} q_L & \text{if } x < x_L, \\ q_L + \frac{x - x_L}{x_R - x_L}(q_R - q_L) & \text{if } x_L \leq x \leq x_R, \\ q_R & \text{if } x > x_R. \end{cases} \end{aligned} \right\} \quad (8.21)$$

Figure 8.3 depicts the situation; the bottom frame (a) shows the piece-wise linear initial condition; the middle frame (b) shows the corresponding picture of characteristics in the x - t plane and the top frame (c) shows solution profiles at various times. In frame (b) the two main characteristic curves are identified, namely

$$\left. \begin{aligned} \text{Tail of the rarefaction: } &x = x_L + \lambda(q_L)t, \\ \text{Head of the rarefaction: } &x = x_R + \lambda(q_R)t. \end{aligned} \right\} \quad (8.22)$$

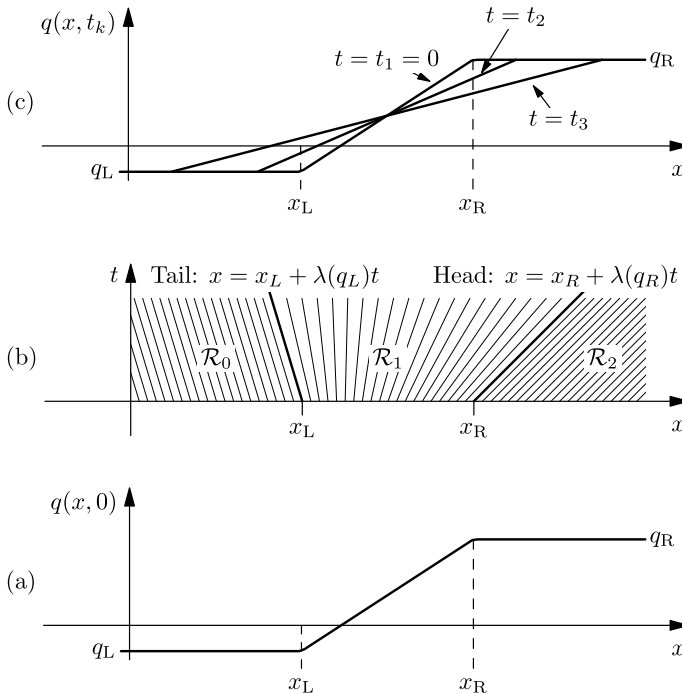


Fig. 8.3 Rarefaction solution from smooth initial condition. Bottom frame **a**: initial condition $q(x, 0)$ at time $t = 0$. Middle frame **b**: picture of characteristics in x - t space divided into three regions $\mathcal{R}_0$, $\mathcal{R}_1$, $\mathcal{R}_2$. Top frame **c**: solution profiles at various times

The *head* and *tail* of the rarefaction divide the x - t half-plane into three regions. A region $\mathcal{R}_0$ to the left of the tail, a region $\mathcal{R}_2$ to the right of the head and a middle region $\mathcal{R}_1$ between the tail and head.

Clearly, the solution in region $\mathcal{R}_0$ is the initial data q_L and in region $\mathcal{R}_2$ is the initial data q_R . The solution in region $\mathcal{R}_1$ is found by tracing characteristics back to the initial condition. For a given point (x, t) in region $\mathcal{R}_1$ we have

$$x_L + \lambda(q_L)t < x < x_R + \lambda(q_R)t \quad (8.23)$$

and the foot $x_0(x, t)$ of the characteristic curve passing through (x, t) is found to be

$$x_0(x, t) = \frac{x(x_R - x_L) + t(x_L\lambda(q_R) - x_R\lambda(q_L))}{x_R - x_L + t(\lambda(q_R) - \lambda(q_L))}. \quad (8.24)$$

Thus, at a given time t the solution as a function of x is given as follows:

$$q(x, t) = \begin{cases} q_L & \text{if } x < x_L + \lambda(q_L)t, \\ q_R & \text{if } x > x_R + \lambda(q_R)t, \\ h(x_0(x, t)) & \text{otherwise.} \end{cases} \quad (8.25)$$

Case 2: Rarefaction Wave From a Riemann Problem

We now consider an IVP for a non-linear conservation law with convex flux function and with discontinuous initial condition, namely

$$\begin{aligned} \partial_t q + \partial_x f(q) &= 0, & f''(q) &> 0, \\ q(x, 0) &= h(x) = \begin{cases} q_L & \text{if } x < 0, \\ q_R & \text{if } x > 0. \end{cases} \end{aligned} \quad (8.26)$$

This type of IVP is called **Riemann problem**; its initial condition may be seen as the limiting case of IVP (8.21). Here we assume $q_L \leq q_R$ and we look for *similarity* solutions of the form $q(x, t) = g(x/t)$, where $g(\psi)$ is a differentiable function of a single variable $\psi = x/t$. Then

$$\partial_t q = -\frac{x}{t^2} g' \left(\frac{x}{t} \right), \quad \partial_x q = \frac{1}{t} g' \left(\frac{x}{t} \right). \quad (8.27)$$

Substitution of these partial derivatives into the PDE $\partial_t q + \lambda(q)\partial_x q$ gives

$$g' \left(\frac{x}{t} \right) \frac{1}{t} \left[\lambda \left(g \left(\frac{x}{t} \right) \right) - \frac{x}{t} \right] = 0, \quad (8.28)$$

leading to

$$\lambda\left(g\left(\frac{x}{t}\right)\right) = \frac{x}{t}. \quad (8.29)$$

Consequently, the complete solution of the Riemann problems (8.26) is

$$q(x, t) = \begin{cases} q_L & \text{if } x/t < \lambda(q_L), \\ \lambda(x/t) & \text{if } \lambda(q_L) \leq x/t \leq \lambda(q_R), \\ q_R & \text{if } x/t > \lambda(q_R), \end{cases} \quad (x, t) \in \mathcal{R}_0, \mathcal{R}_1, \mathcal{R}_2. \quad (8.30)$$

Figure 8.4 depicts the situation; the bottom frame (a) shows the piece-wise constant (discontinuous) initial condition; the middle frame (b) shows the corresponding picture of characteristics in the x - t plane and the top frame (c) shows solution profiles at various times. In frame (b) we depict the characteristics associated with the head and the tail, which divide the entire x - t half-plane into three regions: $\mathcal{R}_0$, $\mathcal{R}_1$ and $\mathcal{R}_2$. Rarefactions of this type are called *centred rarefactions*, as all the characteristic curves that carry a change emerge from a single point, in this case the origin $x = 0$.

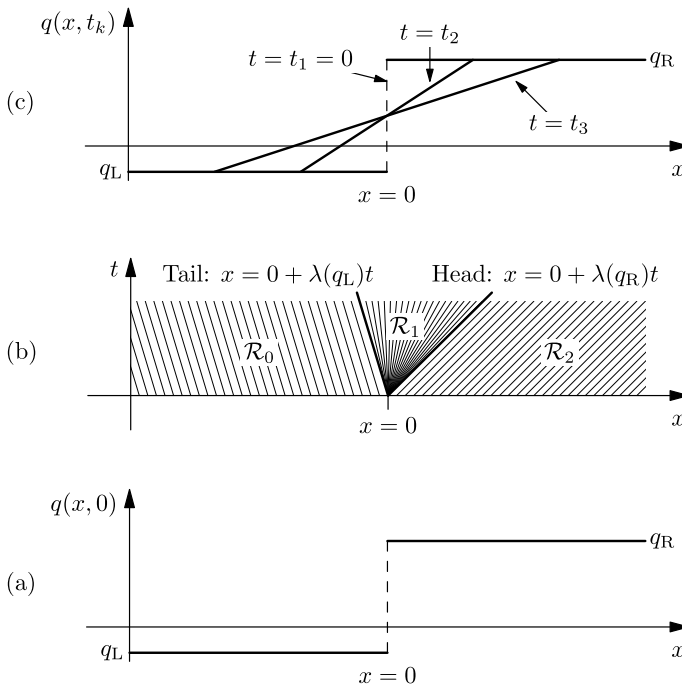


Fig. 8.4 Rarefaction solution from a Riemann problem. Bottom frame **a**: initial condition $q(x, 0)$ at time $t = 0$ consisting of two constant states separated (discontinuously) at $x = 0$. Middle frame **b**: picture of characteristics dividing the x - t space into three regions $\mathcal{R}_0$, $\mathcal{R}_1$ and $\mathcal{R}_2$. Top frame **c**: solution profiles at various times

Burgers equation: If the flux function in (8.21) is taken as $f(q) = \frac{1}{2}q^2$, then the resulting PDE is Burgers' equation, for which characteristic speed is identical to the conserved variable, that is $\lambda(q) = q$. Therefore, for Burgers equation the solution inside the rarefaction is simply $q(x, t) = x/t$.

8.1.4 Discontinuous Solutions: Formation of Shocks

For linear problems, see Chap. 7, discontinuities are present in the solution if present at the initial time and they propagate along characteristics. For non-linear equations the situation is different. First, discontinuities may arise from smooth initial conditions, in finite time. Then, discontinuities will not in general propagate along characteristics. While solutions remain smooth (no crossing of characteristics) one may still use the implicit formula (8.18) to find the solution, using an iterative method. In order to study a major feature of non-linear equations, formation of shocks, we first consider the following example.

Non-linear Problem Solved by the Method of Characteristics

An illustration of shock formation is given through the following initial-value problem for the Burgers equation

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \partial_x f(q) = 0, \quad f(q) = \frac{1}{2}q^2, \\ \text{IC: } q(x, 0) = h(x) = \begin{cases} 1 & \text{if } x < 0, \\ 1 - x & \text{if } 0 \leq x \leq 1, \\ 0 & \text{if } x > 1. \end{cases} \end{array} \right\} \quad (8.31)$$

Figure 8.5 shows the initial condition at time $t = 0$ in frame (a), and the corresponding picture of characteristics in frame (b). We observe that characteristics emanating from the interval $[0, 1]$ cross at time $t = t_b = 1$. For times t , with $t < t_b$ we can use the method of characteristics to find the solution using the implicit formula (8.18). Given a time t_* and a distance x_* , the task is to find the solution $q(x_*, t_*)$. Using (8.18) and remembering that for Burgers' equation $\lambda(q) = q$ we write $q(x_*, t_*) = h(x_* - q(x_*, t_*)t_*)$. After defining $q_* = q(x_*, t_*)$ the above algebraic equation becomes $q_* = h(x_* - q_*t_*)$. If $0 \leq x_* - q_*t_* \leq 1$, then from (8.31) $q_* = h(x_* - q_*t_*) = 1 - (x_* - q_*t_*)$, whose solution for q_* is

$$q_* = \frac{1 - x_*}{1 - t_*}. \quad (8.32)$$

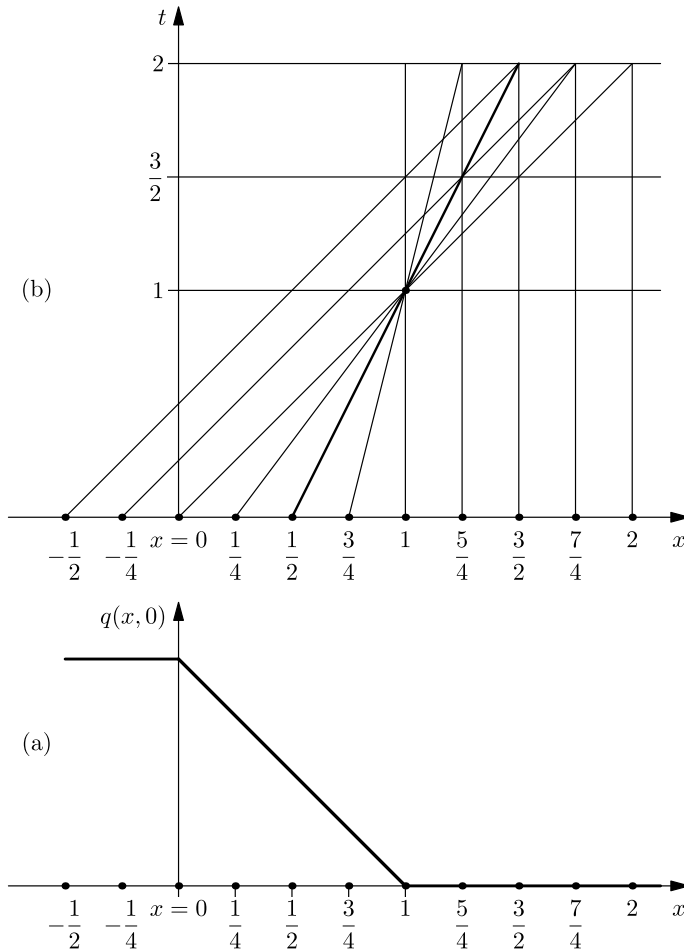


Fig. 8.5 Initial value problem with smooth, piece-wise linear initial condition for the Burgers equation, IVP (8.31). Frame **a**: initial condition at time $t = 0$; frame **b**: corresponding picture of characteristics

Note that, trivially, if $x_* - q_* t_* < 0$ then the solution is $q(x_*, t_*) = 1$ and if $x_* - q_* t_* > 1$ the solution is $q(x_*, t_*) = 0$. Thus the general analytical solution at a point (x, t) for $t < 1$ is given as

$$q(x, t) = \begin{cases} 1 & \text{if } x < t, \\ \frac{1-x}{1-t} & \text{if } t \leq x \leq 1, \\ 0 & \text{if } x > 1. \end{cases} \quad (8.33)$$

At time $t = 1$ characteristics cross, a shock wave forms and the method of characteristics is no longer applicable to obtain admissible solutions.

Multi-Valued Solutions

We shall attempt to evolve the solution for times greater than unity, still using the method of characteristics. In Fig. 8.6 we show five solution profiles at five different times: $t = 0$ (initial condition), $t = 1/2$, $t = 1$, $t = 3/2$ and $t = 2$. These profiles have been obtained by evolving the five points on the initial profile denoted as P_1 , P_2 , P_3 , P_4 , P_5 . Each of the points P_k has an associated characteristic speed given by the initial condition at the appropriate position, recalling again that $\lambda(q) = q$. In Table 8.1, second column, we give these speeds. For example, P_1 moves with speed 1 and P_5 moves with speed zero (stationary). The remaining columns of Table 8.1 give the position of the various points at various times. The reader is encouraged to verify these numbers.

Figure 8.6 shows the corresponding solution profiles at times $t = 0$ (initial condition), $t = 1/2$, $t = 1$, $t = 3/2$ and $t = 2$. Note the steepening of the wave at time

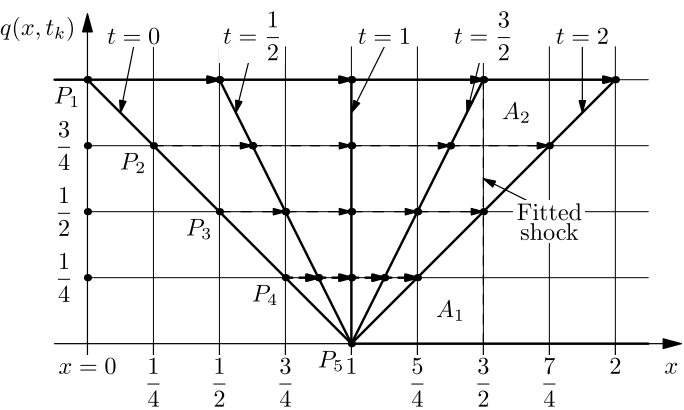


Fig. 8.6 Analytical solution of the initial-value problem (8.31) using the method of characteristics at five different times, namely: $t = 0$ (initial condition), $t = 1/2$, $t = 1$, $t = 3/2$ and $t = 2$. See Table 8.1

Table 8.1 Time variation of the spatial position of the five points on the initial profile at time $t = 0$, Fig. 8.5a. See also Fig. 8.6, where solution profiles are shown

Point	Speed	Time $t = 0$	Time $t = 1/2$	Time $t = 1$	Time $t = 3/2$	Time $t = 2$
P_1	1.0	0	$1/2$	1	$3/2$	2
P_2	$3/4$	$1/4$	$5/8$	1	$11/8$	$7/4$
P_3	$1/2$	$1/2$	$3/4$	1	$5/4$	$3/2$
P_4	$1/4$	$3/4$	$7/8$	1	$9/8$	$5/4$
P_5	0	1	1	1	1	1

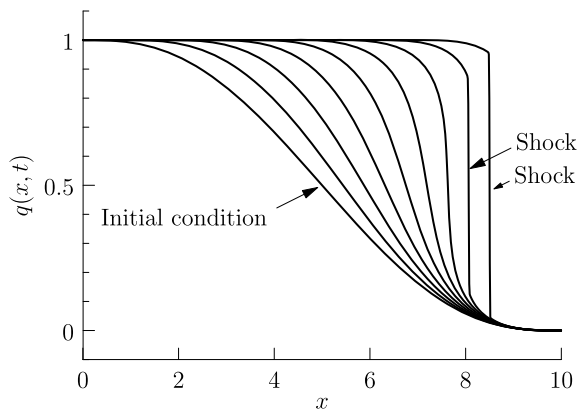
$t = 1/2$, with respect to the initial profile. At time $t = 1$ a shock forms, which is defined by the vertical segment accumulating all the initial values $0 \leq q(x, 0) \leq 1$. The convention is to regard this multivalued segment as a singularity (the shock), with no value assigned to the discontinuity. At $t = 3/2$ we note that the solution profile has folded over becoming triple-valued in the interval $1 < x < 3/2$. For example at $x = 5/4$ the solution takes on the three values $q(5/4, 3/2) = 0$, $q(5/4, 3/2) = 1/2$ and $q(5/4, 3/2) = 1$. At time $t = 2$ the triple-valued interval has become larger, namely $x \in [1, 2]$.

In the classical theory of hyperbolic equations these multivalued solutions are not admitted as valid solutions. Instead, as soon as the wave becomes multivalued, a shock wave is fitted. At time $t = 1$ there is no ambiguity. However, at larger times there appears to be ambiguity as to where the shock is to be placed. To resolve this difficulty consider the multivalued profile at $t = 2$ in Fig. 8.6 and the two areas A_1 and A_2 associated with the interval $[1, 2]$ in which the curve is tripled-valued. Imposing the condition $A_1 = A_2$ gives a shock solution with the discontinuity positioned at $x = 3/2$. This is called the *equal area rule*. See [1] for details. Fitting the shock in the correct location is related to the property of *conservation* of hyperbolic conservation laws. This is discussed in more detail later in this chapter.

A Numerical Example of Shock Formation

To further illustrate the phenomenon of shock formation we consider an initial-value problem for the Burgers equation with continuous initial condition $q(x, 0) = h(x)$ given by a fifth degree polynomial. Figure 8.7 shows a sequence of computed solutions $q(x, t_k)$, for various time values. Note how the smooth initial profile $q(x, 0) = h(x)$ at time $t = 0$ steepens as time evolves, distorting it to the extreme of forming a discontinuity. This is radically different to the linear case, as seen in Chap. 7. The numerical method used here automatically fits a shock in the correct position. This property of (some) numerical methods is called **conservation** and will be studied further in Chap. 13.

Fig. 8.7 Shock wave formation from smooth initial condition at time $t = 0$. The shown profiles at various times were computed with the Godunov first-order method [2] on a fine mesh



The phenomenon of shock formation calls for extension of the definition of solution to non-linear equations. To this end the equations are reformulated in terms of integral relations that no longer require continuity of the solution.

8.1.5 Integral Forms of the Equation

Consider the general in-homogeneous case in which the governing equation includes a source term. The differential conservative form reads

$$\partial_t q(x, t) + \partial_x f(q(x, t)) = s(q(x, t)). \quad (8.34)$$

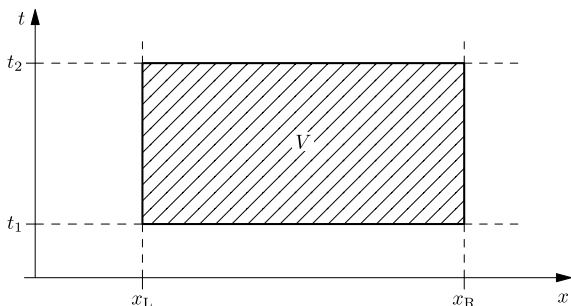
Equation (8.34) is called a **balance law** and when the source term $s(q(x, t)) = 0$ is called a **conservation law**. We shall study integral forms of (8.34) in order to include discontinuous solutions. We shall also derive a condition to be satisfied at discontinuities, precisely where the differential form breaks down. This is done by considering a general *control volume* V in the x - t plane

$$V = [x_L, x_R] \times [t_1, t_2], \quad (8.35)$$

as depicted in Fig. 8.8. To obtain a first version of the integral form of Eq. (8.34) one starts by integrating (8.34) in space and time in the control volume V given in (8.35), namely

$$\int_{x_L}^{x_R} \int_{t_1}^{t_2} [\partial_t q(x, t) + \partial_x f(q(x, t))] dt dx = \int_{x_L}^{x_R} \int_{t_1}^{t_2} s(q(x, t)) dt dx. \quad (8.36)$$

Fig. 8.8 Control volume $V = [x_L, x_R] \times [t_1, t_2]$ in x - t space



On rearranging the space and time integrals we obtain

$$\begin{aligned} \int_{x_L}^{x_R} \left[\int_{t_1}^{t_2} \partial_t q(x, t) dt \right] dx = & - \int_{t_1}^{t_2} \left[\int_{x_L}^{x_R} \partial_x f(q(x, t)) dx \right] dt \\ & + \int_{t_1}^{t_2} \left[\int_{x_L}^{x_R} s(q(x, t)) dx \right] dt . \end{aligned} \quad (8.37)$$

By performing the space and time integrations above, the **integral form of the conservation law** is derived and reads

$$\begin{aligned} \int_{x_L}^{x_R} q(x, t_2) dx = & \int_{x_L}^{x_R} q(x, t_1) dx - \left[\int_{t_1}^{t_2} f(q(x_R, t)) dt - \int_{t_1}^{t_2} f(q(x_L, t)) dt \right] \\ & + \int_{x_L}^{x_R} \int_{t_1}^{t_2} s(q(x, t)) dt dx . \end{aligned} \quad (8.38)$$

In the absence of the source term this equation states that *the total amount of $q(x, t)$ in the interval $[x_L, x_R]$ at time $t = t_2$ is equal to the total amount of $q(x, t)$ in the interval $[x_L, x_R]$ at time $t = t_1$ plus a difference of time integrals of the fluxes at the extreme points of the interval.* In the presence of a source term this statement is appropriately modified to account for the volume integral of the source term on the right-hand side of (8.38).

For a number of reasons it is also convenient to obtain an *averaged version* of the integral form (8.38) by dividing through by Δx and, trivially, introducing $\Delta t / \Delta t$ in the time integrals. We obtain

$$\begin{aligned} \frac{1}{\Delta x} \int_{x_L}^{x_R} q(x, t_2) dx = & \frac{1}{\Delta x} \int_{x_L}^{x_R} q(x, t_1) dx \\ & - \frac{\Delta t}{\Delta x} \left[\frac{1}{\Delta t} \int_{t_1}^{t_2} f(q(x_R, t)) dt - \frac{1}{\Delta t} \int_{t_1}^{t_2} f(q(x_L, t)) dt \right] \\ & + \frac{\Delta t}{\Delta x \Delta t} \int_{x_L}^{x_R} \int_{t_1}^{t_2} s(q(x, t)) dx dt . \end{aligned} \quad (8.39)$$

This averaged formulation of the integral form is used directly to construct numerical methods of the **finite volume** type, to approximate hyperbolic equations numerically. See Chap. 13.

An alternative version of the integral form of equation (8.34) is

$$\frac{d}{dt} \int_{x_L}^{x_R} q(x, t) dx = f(q(x_L, t)) - f(q(x_R, t)) + \int_{x_L}^{x_R} s(q(x, t)) dx. \quad (8.40)$$

This relation says that the time-rate of change of the total amount of $q(x, t)$ in the interval $[x_L, x_R]$ is purely due to the difference of fluxes at the extreme points of the interval $[x_L, x_R]$ plus the spatial integral of the source. Note that by integrating Eq. (8.40) in time reproduces again equation (8.38). For most of this chapter we shall use the homogeneous version of (8.38), or (8.40) (no source term), that is

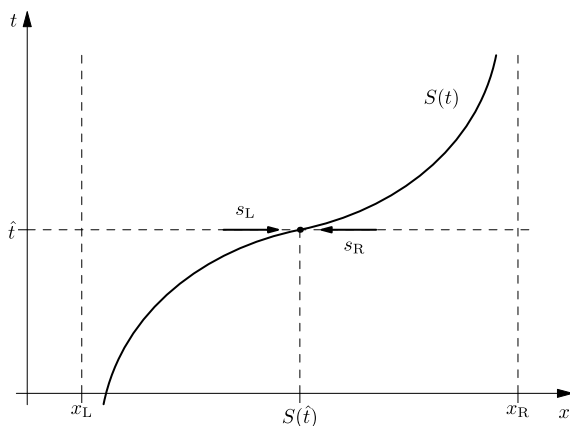
$$\int_{x_L}^{x_R} q(x, t_2) dx = \int_{x_L}^{x_R} q(x, t_1) dx + \int_{t_1}^{t_2} f(q(x_L, t)) dt - \int_{t_1}^{t_2} f(q(x_R, t)) dt. \quad (8.41)$$

Generalized or weak solution. A generalized, or weak solution of the balance law (8.34) is a function $q(x, t)$ that satisfies the integral form (8.38), or (8.40), of the balance law.

8.1.6 Rankine-Hugoniot Jump Condition

Consider a solution $q(x, t)$ such that $q(x, t)$ and $f(q)$, and their derivatives are continuous everywhere except on a line $S = S(t)$ on the x - t plane across which $q(x, t)$ has a jump discontinuity. See Fig. 8.9.

Fig. 8.9 Shock path discontinuity $S = S(t)$ in x - t plane and associated shock speed s . Across the shock path the solution is discontinuous and obeys the Rankine-Hugoniot jump condition. Away from the shock the solution is smooth and obeys the differential equations. Throughout the domain the solution obeys the integral form of the equations



Beware of the use of s for the source term (as $s(q)$) and for the shock speed. Select now two fixed points x_L and x_R on the x -axis such that $x_L < S(t) < x_R$. Enforcing the (homogeneous) conservation law in integral form (8.40) on $[x_L, x_R]$ leads to

$$f(q(x_L, t)) - f(q(x_R, t)) = \frac{d}{dt} \int_{x_L}^{S(t)} q(x, t) dx + \frac{d}{dt} \int_{S(t)}^{x_R} q(x, t) dx. \quad (8.42)$$

We now use the Leibniz formula

$$\frac{d}{d\alpha} \int_{\xi_1(\alpha)}^{\xi_2(\alpha)} g(\xi, \alpha) d\xi = \int_{\xi_1(\alpha)}^{\xi_2(\alpha)} \frac{\partial}{\partial \alpha} g(\xi, \alpha) d\xi + g(\xi_2, \alpha) \frac{\partial \xi_2}{\partial \alpha} - g(\xi_1, \alpha) \frac{\partial \xi_1}{\partial \alpha} \quad (8.43)$$

to obtain

$$\begin{aligned} f(q(x_L, t)) - f(q(x_R, t)) &= [q(s_L, t) - q(s_R, t)] s \\ &\quad + \int_{x_L}^{S(t)} \partial_t q(x, t) dx + \int_{S(t)}^{x_R} \partial_t q(x, t) dx, \end{aligned} \quad (8.44)$$

where $q(s_L, t)$ is the limit of $q(S(t), t)$ as x tends to $S(t)$ from the left, $q(s_R, t)$ is the limit of $q(S(t), t)$ as x tends to $S(t)$ from the right and

$$s = \frac{d}{dt} S(t) \quad (8.45)$$

is the **speed of the discontinuity**. As the partial derivative $\partial_t q(x, t)$ is bounded the relevant integrals in (8.44) vanish identically as $S(t)$ is approached from left and right and we obtain

$$f(q(s_L, t)) - f(q(s_R, t)) = [q(s_L, t) - q(s_R, t)] s. \quad (8.46)$$

Defining the jumps

$$\Delta f = f(q(s_R, t)) - f(q(s_L, t)); \quad \Delta q = q(s_R, t) - q(s_L, t), \quad (8.47)$$

we can relate the speed s of the discontinuity to the jumps as

$$\Delta f = s \Delta q. \quad (8.48)$$

This relation at the shock is called the *Rankine-Hugoniot condition*. For the scalar case considered here one can solve for the speed s as

$$s = \frac{\Delta f}{\Delta q}. \quad (8.49)$$

Therefore, in order to admit discontinuous solutions we may formulate the problem in terms of PDEs, which are valid in smooth parts of the solution, and the Rankine-Hugoniot condition across discontinuities.

Example: Burgers' equation. Assume a shock wave of speed s with states q_L and q_R either side of the discontinuity, modelled by the inviscid Burgers equation in (8.6). The Rankine-Hugoniot condition gives

$$f(q_R) - f(q_L) = \frac{1}{2}q_R^2 - \frac{1}{2}q_L^2 = s(q_R - q_L), \quad (8.50)$$

from which the shock speed is given by

$$s = \frac{1}{2}(q_L + q_R). \quad (8.51)$$

We remark that this is a very special case, in which the shock speed is a simple arithmetic average of the characteristic speeds either side of the shock.

8.1.7 Non-Uniqueness and the Entropy Condition

Having extended the notion of solution via the integral formulation to include solutions containing discontinuities, we end up with too many solutions, as we now demonstrate via the following example.

$$\begin{aligned} \text{PDE: } \partial_t q + \partial_x f(q) &= 0, & f(q) &= \frac{1}{2}q^2, \\ \text{IC: } q(x, 0) = h(x) &= \begin{cases} 0 & \text{if } x \leq 0, \\ 1 & \text{if } x > 0. \end{cases} \end{aligned} \quad (8.52)$$

From the result of equation (8.30) we can calculate the solution following characteristics, to obtain the smooth solution

$$q(x, t) = \begin{cases} 0 & \text{if } x/t < 0, \\ x/t & \text{if } 0 \leq x/t \leq 1, \\ 1 & \text{if } x/t > 1. \end{cases} \quad (8.53)$$

Figure 8.10 illustrates the solution and the corresponding picture of characteristics.

In addition to solution (8.53) we can find another, discontinuous, solution to the same problem, given as

$$q(x, t) = \begin{cases} 0 & \text{if } x/t \leq s = 1/2, \\ 1 & \text{if } x/t > s = 1/2. \end{cases} \quad (8.54)$$

Figure 8.11 depicts the discontinuous solution (8.54) and the corresponding picture of characteristics. To be a solution this function must satisfy the integral form of the conservation law

$$\int_{x_L}^{x_R} q(x, t_2) dx = \int_{x_L}^{x_R} q(x, t_1) dx + \int_{t_1}^{t_2} [f(q(x_L, t)) - f(q(x_R, t))] dt. \quad (8.55)$$

This can be verified as follows. Consider a general finite volume $V = [x_L, x_R] \times [t_1, t_2]$ as shown in Fig. 8.12. We calculate the various terms of the integral form as follows

$$\left. \begin{aligned} \int_{x_L}^{x_R} q(x, t_2) dx &= \int_{x_L}^{x_1} q(x, t_2) dx + \int_{x_1}^{x_R} q(x, t_2) dx = 1 \times (x_R - x_2), \\ \int_{x_L}^{x_R} q(x, t_1) dx &= \int_{x_L}^{x_1} q(x, t_1) dx + \int_{x_1}^{x_R} q(x, t_1) dx = 1 \times (x_R - x_1), \\ \int_{t_1}^{t_2} f(q(x_L, t)) dt - \int_{t_1}^{t_2} f(q(x_R, t)) dt &= -\frac{1}{2}(t_2 - t_1). \end{aligned} \right\} \quad (8.56)$$

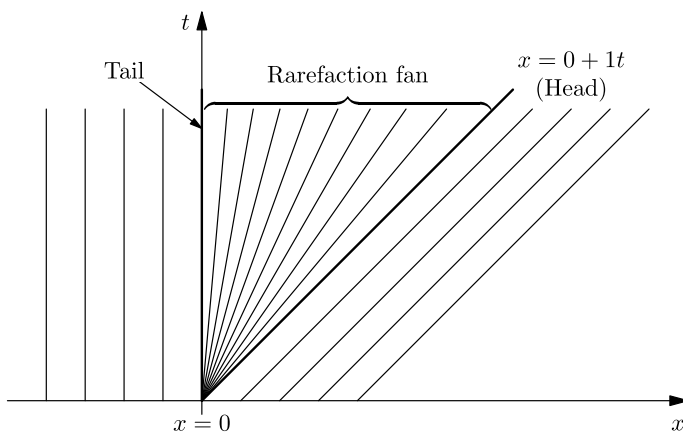


Fig. 8.10 Rarefaction (smooth) solution to Riemann problem (8.52) for the Burgers equation

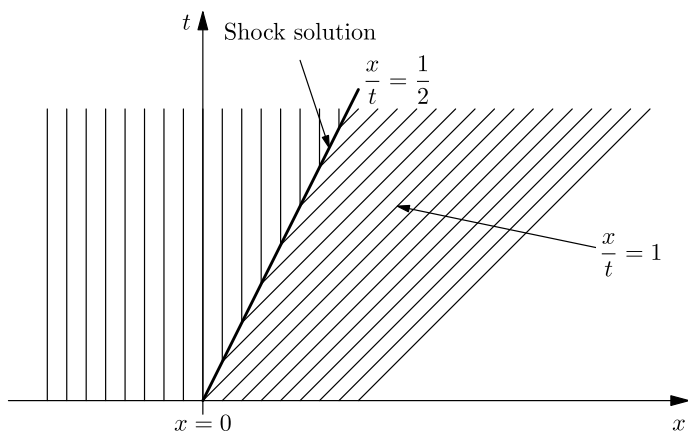


Fig. 8.11 Shock (discontinuous) solution (8.54) to problem (8.52)

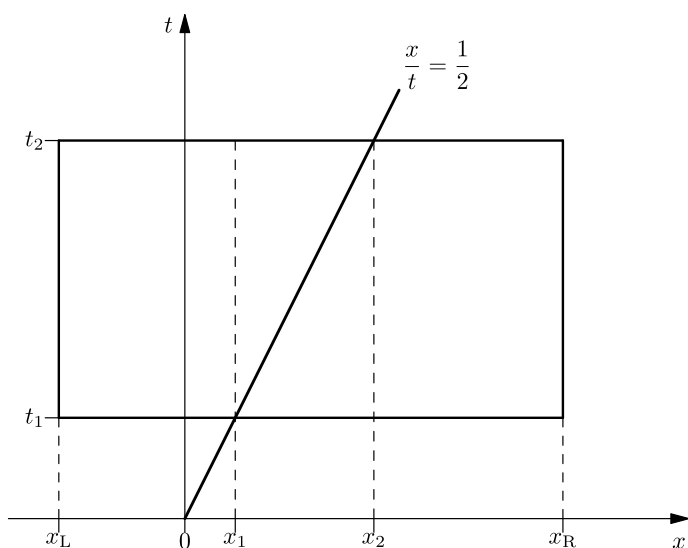


Fig. 8.12 Control volume on which to verify the integral form of the conservation law

Enforcing the conservation law (8.55) requires the condition

$$\frac{\Delta x}{\Delta t} = \frac{1}{2}, \quad (8.57)$$

which is the *speed of the shock*. Therefore the proposed solution (8.54) satisfies the integral form of the conservation law and is therefore a generalized (or weak) solution of the Burgers equation in (8.52).

The Lax Entropy Condition

There are now solutions that are spurious, that is non admissible. The question is how to distinguish admissible from non-admissible solutions. This requires the adoption of selection criterion, as explained below.

We consider a discontinuous solution to be admissible if arisen from *compression*. The *compressibility* condition is ensured by imposing the *Lax entropy condition*. Given a shock solution of speed s , the Lax entropy condition for a scalar equation reads

$$\lambda(q_L) > s > \lambda(q_R), \quad (8.58)$$

where $\lambda(q_L)$ and $\lambda(q_R)$ are characteristic speeds computed to the left and to the right of the shock, respectively.

The entropy condition means that characteristics on the left and on the right of the shock *run into the shock*. That is, the shock is *compressed* by the characteristics either side. See Fig. 8.13. Thus the proposed shock solution (8.54), illustrated in Fig. 8.11, is not an admissible solution, which is called **rarefaction shock**, or **entropy-violating shock**. The admissible solution is the (smooth) rarefaction solution (8.53) illustrated in Fig. 8.10.

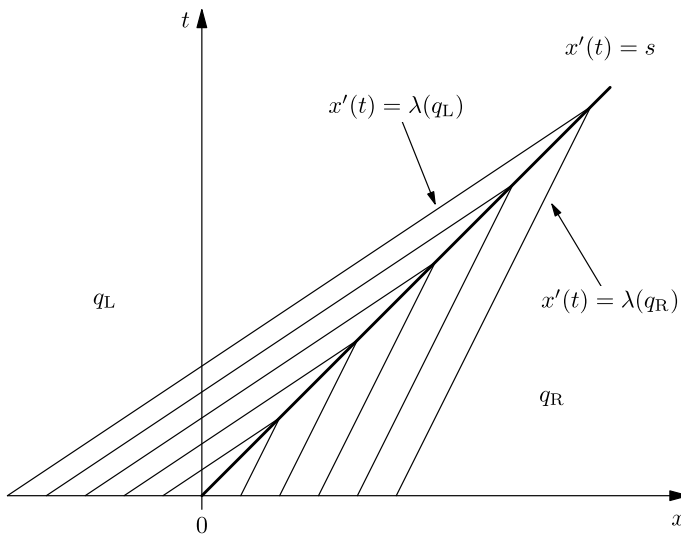


Fig. 8.13 Picture of characteristics for an entropy-satisfying shock. Characteristics from either side of the shock run into the shock

8.1.8 The Riemann Problem for Burgers' Equation

We end up this section on scalar non-linear conservation laws by solving the Riemann problem for the inviscid Burgers equation

$$\begin{aligned} \text{PDE: } \partial_t q + \partial_x f(q) &= 0, & f(q) &= \frac{1}{2}q^2, \\ \text{IC: } q(x, 0) &= \begin{cases} q_L & \text{if } x < 0, \\ q_R & \text{if } x > 0. \end{cases} \end{aligned} \quad (8.59)$$

From the previous discussion the exact solution is a *single wave* emanating from the origin. In view of the entropy condition this wave is either a shock, when $q_L > q_R$, or a rarefaction, when $q_L \leq q_R$. The complete solution is

$$q(x, t) = \begin{cases} \begin{cases} q_L & \text{if } x - st < 0 \\ q_R & \text{if } x - st > 0 \end{cases} & \text{if } q_L > q_R \text{ (shock),} \\ \text{where } s = \frac{1}{2}(q_L + q_R) \\ \begin{cases} q_L & \text{if } \frac{x}{t} < q_L \\ \frac{x}{t} & \text{if } q_L < \frac{x}{t} < q_R \\ q_R & \text{if } \frac{x}{t} > q_R \end{cases} & \text{if } q_L \leq q_R \text{ (rarefaction).} \end{cases} \quad (8.60)$$

Figure 8.14 illustrates the solution (8.60) of the Riemann problem for Burgers' Eq. (8.59). The bottom frame illustrates the shock case, while the top frame illustrates the rarefaction case. It is worth remarking that the inclination of the characteristic lines and the shock path depend on the values for q_L and q_R . For example, the shock path would be inclined towards the left if $s = \frac{1}{2}(q_L + q_R) < 0$.

The reader will find instructive exercises for scalar non-linear conservation laws, at the end of the chapter. In the next section we study systems of hyperbolic conservation laws.

8.2 Systems of Non-Linear Hyperbolic Equations

Here we study some very basic properties of systems of non-linear hyperbolic equations. We restrict the attention to notions that are directly useful to the study of the one-dimensional blood flow equations, of special relevance to this book. Before proceeding, the reader is encouraged to revise Chap. 7, especially the section concerned with *systems of linear* hyperbolic equations.

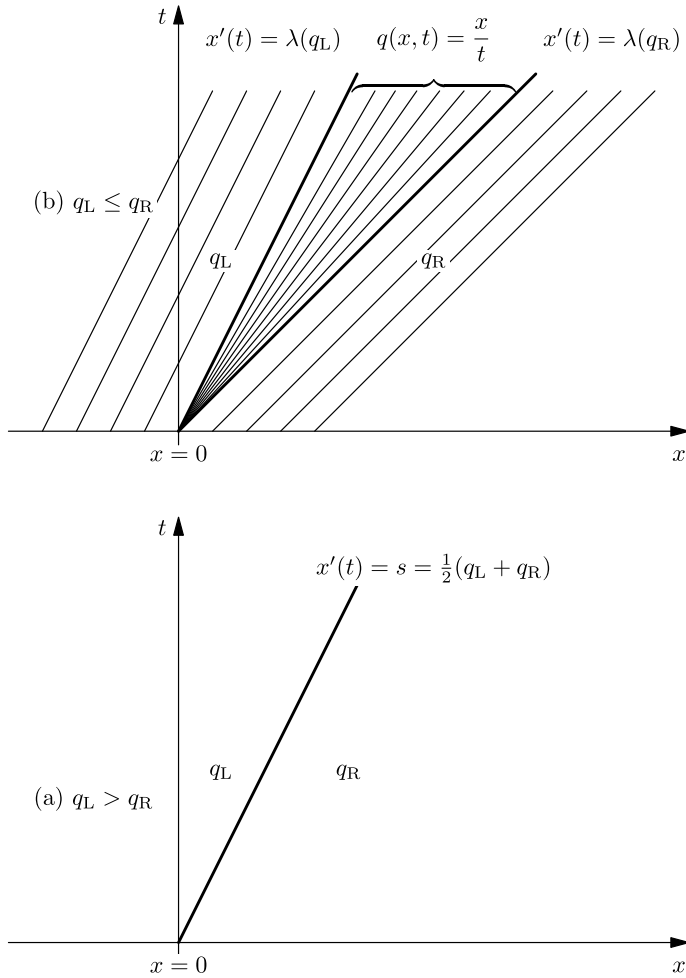


Fig. 8.14 The Riemann problem for the Burgers equation. Frame **a** illustrates the case in which the solution is a shock wave, when the initial data satisfies $q_L > q_R$. Frame **b** shows the case in which the solution is a rarefaction wave, when the initial data satisfies $q_L \leq q_R$.

8.2.1 Systems in Conservative Form

A general system of m first-order partial differential equations in one spatial dimension reads

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}), \quad (8.61)$$

where

$$\mathbf{Q}(x, t) = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_m \end{bmatrix}, \quad \mathbf{F}(\mathbf{Q}) = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_m \end{bmatrix}, \quad \mathbf{S}(\mathbf{Q}) = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_m \end{bmatrix}. \quad (8.62)$$

The independent variables are x and t , while $\mathbf{Q}(x, t)$ is the vector of dependent variables, called *conserved variables*, and are the unknowns of the problem; $\mathbf{F}(\mathbf{Q})$ is the flux vector; $\mathbf{S}(\mathbf{Q})$ is the vector of source terms. Flux and source term vectors are prescribed vector-valued functions of the unknowns. The source term vector $\mathbf{S}(\mathbf{Q})$ does not contain differential terms of the unknowns. Each component k of the vectors $\mathbf{F}(\mathbf{Q})$ and $\mathbf{S}(\mathbf{Q})$ in (8.62) is a function of the components $q_i = q_i(x, t)$ of the vector of unknowns $\mathbf{Q}(x, t)$, that is

$$\left. \begin{aligned} q_i &= q_i(x, t), \\ f_i &= f_i(q_1(x, t), \dots, q_m(x, t)), \\ s_i &= s_i(x, t, q_1(x, t), \dots, q_m(x, t)). \end{aligned} \right\} \quad (8.63)$$

The model linear advection-reaction equation

$$\partial_t q(x, t) + \lambda \partial_x q(x, t) = \beta q(x, t), \quad \lambda : \text{constant}, \quad \beta : \text{constant} \quad (8.64)$$

studied in Chap. 7 can be written in the form (8.61), where the flux is $f(q) = \lambda q$, a linear function of the unknown function $q(x, t)$; the source term is $s(q) = \beta q(x, t)$, also a linear function of the unknown function $q(x, t)$.

A system of equations of the form (8.61) is called a system of *balance laws*. When the source term vector is identically zero we call (8.61) a *system of conservation laws* and the equations are called homogeneous; otherwise we speak of inhomogeneous equations. In (8.61) the equations are written in *differential conservative form*. This form is only valid for sufficiently smooth solutions, that is when the partial derivatives are actually defined. For solutions with discontinuities one resorts to a generalised form, the *integral form* of the conservation laws, as seen earlier in this chapter for the scalar case.

8.2.2 Systems in Quasi-Linear Form: The Jacobian Matrix

In order to study the mathematical character of equations (8.61) it is convenient to first express the system in quasi-linear form as follows

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}). \quad (8.65)$$

$\mathbf{A}(\mathbf{Q})$ is called the *Jacobian matrix* of $\mathbf{F}(\mathbf{Q})$ and is yet to be defined. We then consider just the *principal part* of the equations, that is the left-hand side of (8.65) containing

the differential terms. The source terms do not contain differential terms of the unknowns and hence do not affect the mathematical character of the equations.

For illustrative purposes we first consider the special case of a homogeneous system of two equations, namely

$$\left. \begin{aligned} \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) &= \mathbf{0}, \\ \mathbf{Q} &= \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}, \quad \mathbf{F}(\mathbf{Q}) = \begin{bmatrix} f_1(q_1, q_2) \\ f_2(q_1, q_2) \end{bmatrix}. \end{aligned} \right\} \quad (8.66)$$

By writing each equation separately and applying the chain rule to the flux-term components we have

$$\left. \begin{aligned} \partial_t q_1 + \frac{\partial}{\partial q_1} f_1(q_1, q_2) \partial_x q_1 + \frac{\partial}{\partial q_2} f_1(q_1, q_2) \partial_x q_2 &= 0, \\ \partial_t q_2 + \frac{\partial}{\partial q_1} f_2(q_1, q_2) \partial_x q_1 + \frac{\partial}{\partial q_2} f_2(q_1, q_2) \partial_x q_2 &= 0. \end{aligned} \right\} \quad (8.67)$$

In matrix form system (8.67) reads

$$\partial_t \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \begin{bmatrix} \frac{\partial f_1}{\partial q_1} & \frac{\partial f_1}{\partial q_2} \\ \frac{\partial f_2}{\partial q_1} & \frac{\partial f_2}{\partial q_2} \end{bmatrix} \partial_x \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad (8.68)$$

or

$$\left. \begin{aligned} \partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} &= \mathbf{0}, \\ \mathbf{A}(\mathbf{Q}) &= \frac{\partial \mathbf{F}}{\partial \mathbf{Q}} = \begin{bmatrix} \frac{\partial f_1}{\partial q_1} & \frac{\partial f_1}{\partial q_2} \\ \frac{\partial f_2}{\partial q_1} & \frac{\partial f_2}{\partial q_2} \end{bmatrix}, \end{aligned} \right\} \quad (8.69)$$

where $\mathbf{A}(\mathbf{Q})$ is the *Jacobian matrix*. For an $m \times m$ one-dimensional non-linear system of the form (8.61), it is trivial to show that the quasi-linear form is as (8.69) with the Jacobian matrix given as

$$\mathbf{A}(\mathbf{Q}) := \frac{\partial \mathbf{F}}{\partial \mathbf{Q}} = \begin{bmatrix} \frac{\partial f_1}{\partial q_1} & \frac{\partial f_1}{\partial q_2} & \cdots & \frac{\partial f_1}{\partial q_m} \\ \frac{\partial f_2}{\partial q_1} & \frac{\partial f_2}{\partial q_2} & \cdots & \frac{\partial f_2}{\partial q_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial q_1} & \frac{\partial f_m}{\partial q_2} & \cdots & \frac{\partial f_m}{\partial q_m} \end{bmatrix}. \quad (8.70)$$

As done for linear systems in Chap. 7, we are now in a position to study the eigenstructure of non-linear first-order systems.

8.2.3 Eigenstructure: Eigenvalues and Eigenvectors

The concepts of eigenstructure and hyperbolicity for a non-linear system (8.65) are the same as for linear systems studied in Chap. 7, which the reader is encouraged to revise before proceeding with this section. Recall that for the non-linear case the coefficient matrix $\mathbf{A}(\mathbf{Q})$ in (8.70) is a function of the vector of unknown conserved variables $\mathbf{Q}$. Therefore the eigenvalues and eigenvectors are all functions of the vector of conserved variables $\mathbf{Q}$.

The eigenvalues of system (8.65) are the eigenvalues of the Jacobian matrix $\mathbf{A}(\mathbf{Q})$, which in turn are the roots of the characteristic polynomial

$$P(\lambda) = |\mathbf{A}(\mathbf{Q}) - \lambda \mathbf{I}| = 0, \quad (8.71)$$

where $\mathbf{I}$ is the identity matrix and λ is a parameter. The roots (eigenvalues) are denoted as $\lambda_1(\mathbf{Q}), \lambda_2(\mathbf{Q}), \dots, \lambda_m(\mathbf{Q})$. The corresponding left and right eigenvectors, assumed to exist, are denoted as

$$\left. \begin{array}{l} \mathbf{L}_1(\mathbf{Q}) ; \quad \mathbf{L}_2(\mathbf{Q}) ; \quad \dots ; \quad \mathbf{L}_m(\mathbf{Q}), \\ \mathbf{R}_1(\mathbf{Q}) ; \quad \mathbf{R}_2(\mathbf{Q}) ; \quad \dots ; \quad \mathbf{R}_m(\mathbf{Q}). \end{array} \right\} \quad (8.72)$$

If all eigenvalues are real, then we assume them to be ordered in increasing order as follows

$$\lambda_1(\mathbf{Q}) \leq \lambda_2(\mathbf{Q}) \leq \dots \leq \lambda_m(\mathbf{Q}). \quad (8.73)$$

We next recall the definition of hyperbolicity, as already given for linear systems, namely

Hyperbolic system. A one-dimensional $m \times m$ first order system of the form (8.65) is *hyperbolic* if the Jacobian matrix $\mathbf{A}(\mathbf{Q})$ as in (8.70) has m real eigenvalues $\lambda_1(\mathbf{Q}), \lambda_2(\mathbf{Q}), \dots, \lambda_m(\mathbf{Q})$ and a corresponding set of m linearly independent eigenvectors $\mathbf{R}_1(\mathbf{Q}), \mathbf{R}_2(\mathbf{Q}), \dots, \mathbf{R}_m(\mathbf{Q})$.

Strictly/non-strictly hyperbolic system. A hyperbolic system is said to be *strictly hyperbolic* if all eigenvalues are distinct. Otherwise the system is said to be *non-strictly hyperbolic*.

Remarks:

1. Strictly hyperbolic system. Note that if the system has m real and distinct eigenvalues, then the system can be classified automatically as hyperbolic (strictly) without the need to find a corresponding complete set of linearly independent eigenvectors.
2. Weakly hyperbolic system; elliptic system and mixed elliptic/hyperbolic system. These definitions are as in Chap. 7.
3. Multidimensional first-order systems. All the above concepts can be extended to three-dimensional systems. For details see [3, 4] and [5], for example.

8.2.4 Characteristic Fields

Consider a system of m hyperbolic equations of the form (8.65) with real eigenvalues $\lambda_i(\mathbf{Q})$ and corresponding right eigenvectors $\mathbf{R}_i(\mathbf{Q})$. The *characteristic speed* $\lambda_i(\mathbf{Q})$ defines a *characteristic field*, which is called the λ_i -field, or the $\mathbf{R}_i$ -field, or simply the i -field.

Before giving some relevant definitions, we recall that the *gradient* of an eigenvalue $\lambda_i(\mathbf{Q})$ is given by the vector

$$\nabla \lambda_i(\mathbf{Q}) = \left[\frac{\partial \lambda_i}{\partial q_1}, \frac{\partial \lambda_i}{\partial q_2}, \dots, \frac{\partial \lambda_i}{\partial q_m} \right]. \quad (8.74)$$

We are now in a position to define two classes of characteristic fields.

Linearly degenerate field. A $\lambda_i(\mathbf{Q})$ -characteristic field is said to be *linearly degenerate* if

$$\nabla \lambda_i(\mathbf{Q}) \cdot \mathbf{R}_i(\mathbf{Q}) = 0, \quad \forall \mathbf{Q} \in \mathbb{R}^m, \quad (8.75)$$

where $\mathbb{R}^m$ is the set of real-valued vectors of m components.

Genuinely non-linear field. A $\lambda_i(\mathbf{Q})$ -characteristic field is said to be *genuinely non-linear* if

$$\nabla \lambda_i(\mathbf{Q}) \cdot \mathbf{R}_i(\mathbf{Q}) \neq 0, \quad \forall \mathbf{Q} \in \mathbb{R}^m. \quad (8.76)$$

Remarks:

1. *Phase space.* The *phase space* is the space of vectors $\mathbf{Q} = [q_1, \dots, q_m]^T$ in $\mathbb{R}^m$. This is also called *state space*. For the special case of a 2×2 system we speak of the *phase plane* q_1 - q_2 .
2. *Linear system.* Note that for a linear hyperbolic system with constant coefficients the eigenvalues λ_i are all constant and therefore $\nabla \lambda_i(\mathbf{Q}) = \mathbf{0}$. Hence all characteristic fields of a linear hyperbolic system with constant coefficients are *linearly degenerate*.

3. *Characteristic fields and waves.* Isolated waves and their features are associated to specific characteristic fields and their nature. For example, *shock waves* and *rarefaction waves* are associated to genuinely non-linear fields. Linearly degenerate fields are associated with *contact discontinuities*.
4. A **contact discontinuity** associated with a λ_i characteristic field is a discontinuous wave for which the λ_i characteristic curves either side are parallel to the wave. Recall that for a rarefaction, the λ_i characteristic curves either side of the wave, diverge; for an admissible shock, such curves converge.

8.2.5 Generalised Solutions: The Integral Form

In order to cater for discontinuous solutions one needs to adopt the integral form of the conservation laws. One form is a generalisation of that for the scalar case and is obtained by integration of (8.61) in space and time in the control volume V given by (8.35). The result is the integral form of the conservation laws written as

$$\left. \begin{aligned} \int_{x_L}^{x_R} \mathbf{Q}(x, t_2) dx &= \int_{x_L}^{x_R} \mathbf{Q}(x, t_1) dx \\ &+ \int_{t_1}^{t_2} \mathbf{F}(\mathbf{Q}(x_L, t)) dt - \int_{t_1}^{t_2} \mathbf{F}(\mathbf{Q}(x_R, t)) dt \\ &+ \int_{t_1}^{t_2} \int_{x_L}^{x_R} \mathbf{S}(\mathbf{Q}(x, t)) dx dt . \end{aligned} \right\} \quad (8.77)$$

An alternative integral form is

$$\frac{d}{dt} \int_{x_L}^{x_R} \mathbf{Q}(x, t) dx = \mathbf{F}(\mathbf{Q}(x_L, t)) - \mathbf{F}(\mathbf{Q}(x_R, t)) + \int_{x_L}^{x_R} \mathbf{S}(\mathbf{Q}(x, t)) dx . \quad (8.78)$$

These are generalizations of the integral form (8.38) for the scalar case.

Rankine-Hugoniot Conditions

Across a discontinuity, the conservative form of the equations produces the Rankine-Hugoniot jump conditions across the discontinuous wave, which is purely an algebraic relation involving the wave speed and jumps in flux and conserved variables across the wave, that is

$$\Delta \mathbf{F} = S \Delta \mathbf{Q} . \quad (8.79)$$

Here S is the speed of the discontinuous wave, while the jumps in flux and conserved variables across the wave are defined as

$$\Delta \mathbf{F} = \mathbf{F}(\mathbf{Q}_R) - \mathbf{F}(\mathbf{Q}_L); \quad \Delta \mathbf{Q} = \mathbf{Q}_R - \mathbf{Q}_L. \quad (8.80)$$

Here $\mathbf{Q}_L$ and $\mathbf{Q}_R$ are the limiting states from the left and right of the discontinuity. Note that now, unlike the scalar case, it is not possible to solve directly for the speed S in terms of the jump vectors $\Delta \mathbf{F}$ and $\Delta \mathbf{Q}$.

Next we apply some of the above theoretical concepts to study the equations that govern flow of blood.

8.3 The Equations for Blood Flow

The governing equations for blood flow in a blood vessel have already been introduced in Chap. 6. Here we simply state the equations and then proceed to their mathematical analysis applying the theoretical concepts introduced in this chapter.

8.3.1 *Recalling the Equations and the Tube Law*

The equations arise from imposing conservation of mass and balance of momentum and consist of the following non-linear, first-order system of partial differential equations

$$\left. \begin{aligned} \partial_t A + \partial_x (Au) &= s_1, \\ \partial_t (Au) + \partial_x (\alpha Au^2) + \frac{A}{\rho} \partial_x p &= s_2. \end{aligned} \right\} \quad (8.81)$$

The source terms s_1 , s_2 and the Coriolis coefficient α in (8.81) are

$$s_1 = 0, \quad s_2 = -2(\beta + 2)\pi \frac{\mu}{\rho} u, \quad \alpha(\beta) = \frac{\beta + 2}{\beta + 1}, \quad \beta > 0. \quad (8.82)$$

These equations were derived in Chap. 6, in which

- $A(x, t)$: cross-sectional area of the vessel at position x and time t ;
- $u(x, t)$: cross-sectional averaged velocity of blood at a vessel cross section;
- $p(x, t)$: cross-sectional averaged pressure of blood inside the vessel;
- $q(x, t) = A(x, t)u(x, t)$: flow (or flow rate);
- ρ : blood density, constant;
- α : Coriolis coefficient;
- μ : kinematic viscosity.

The unknowns of the problem are cross-sectional area $A(x, t)$, velocity $u(x, t)$ (or flow $q(x, t)$) and pressure $p(x, t)$. As there are three unknowns and only two equations, one requires a **closure condition**, called **tube law** in haemodynamics. Usually, one expresses algebraically one of the unknowns in terms of the remaining ones. In haemodynamics such algebraic expression relates the mechanics of the vessel wall to the internal flow distribution in which the pressure is written as a function of cross-sectional area, namely

$$p := p(A) . \quad (8.83)$$

In Chap. 10 we study general tube laws for arteries and veins.

Next we determine the complete eigenstructure of the system without assuming any specific form of the tube law, but considering separately the cases $\alpha = 1$ and $\alpha \neq 1$ for the Coriolis coefficient. The conclusions will be valid for both arteries and veins.

8.3.2 Eigenstructure and Wave Speed for $\alpha = 1$

Here the tube law is assumed in general form as in (8.83) but the Coriolis coefficient in (8.81) is taken to be $\alpha = 1$ and $\mu = 0$. Equation (8.81) may then be written as

$$\left. \begin{aligned} \partial_t A + \partial_x (Au) &= 0 , \\ \partial_t (Au) + \left(\frac{A}{\rho} \frac{\partial p}{\partial A} - u^2 \right) \partial_x A + 2u \partial_x (Au) &= 0 . \end{aligned} \right\} \quad (8.84)$$

The expression for the momentum equation above is derived through algebraic manipulations of the momentum equation in (8.81) involving the following: (i) expanding the spatial derivative of the second term, (ii) expressing the pressure gradient as $\partial_x p = \frac{\partial p}{\partial A} \partial_x A$, and (iii) adding and subtracting $u^2 \partial_x A$. In matrix form Eqs. (8.84) read

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}) , \quad (8.85)$$

with

$$\mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \equiv \begin{bmatrix} A \\ Au \end{bmatrix} , \quad \mathbf{A}(\mathbf{Q}) = \begin{bmatrix} 0 & 1 \\ \frac{A}{\rho} \frac{\partial p}{\partial A} - u^2 & 2u \end{bmatrix} . \quad (8.86)$$

Equations (8.85)–(8.86) are written in terms of the physically conserved quantities A and $q \equiv Au$. We note however that the equations are not written in conservation-law form. From the point of view of numerical approximation, the non-conservative Eqs. (8.85)–(8.86) can be solved straightforwardly using a path-conservative method, for example. See Chap. 14.

The eigenvalues of system (8.85)–(8.86) are easily found to be

$$\lambda_1 = u - c , \quad \lambda_2 = u + c , \quad (8.87)$$

where the **wave speed** emerges naturally as

$$c = \sqrt{\frac{A}{\rho} \frac{\partial p}{\partial A}}. \quad (8.88)$$

Straightforward calculations for the corresponding left eigenvectors give

$$\mathbf{L}_1 = \alpha_1 \left[1, -\frac{1}{u+c} \right], \quad \mathbf{L}_2 = \alpha_2 \left[1, -\frac{1}{u-c} \right], \quad (8.89)$$

with α_1 and α_2 two arbitrary scaling factors. Analogous calculations for the right eigenvectors give

$$\mathbf{R}_1 = \beta_1 \begin{bmatrix} 1 \\ u-c \end{bmatrix}, \quad \mathbf{R}_2 = \beta_2 \begin{bmatrix} 1 \\ u+c \end{bmatrix}, \quad (8.90)$$

where β_1 and β_2 are arbitrary scaling factors. Values for the scaling factors are usually chosen on convenience grounds. For example, if we impose orthonormality of left and right eigenvectors, then the coefficients must satisfy the following relations

$$\alpha_1 \beta_1 = \frac{(u+c)}{2c}, \quad \alpha_2 \beta_2 = -\frac{(u-c)}{2c}. \quad (8.91)$$

Prescribing $\beta_1 = \beta_2 = \frac{1}{2c}$ gives $\alpha_1 = (u+c)$ and $\alpha_2 = -(u-c)$. Then the corresponding matrices $\mathbf{L}$ and $\mathbf{R}$ formed by the left and right eigenvectors are given by

$$\mathbf{L} = \begin{bmatrix} (u+c) & -1 \\ -(u-c) & 1 \end{bmatrix}, \quad \mathbf{R} = \frac{1}{2c} \begin{bmatrix} 1 & 1 \\ (u-c) & (u+c) \end{bmatrix}. \quad (8.92)$$

It is then easy to show that

$$\mathbf{R}^{-1} = \mathbf{L}, \quad (8.93)$$

where the matrix $\mathbf{R}^{-1}$ denotes the inverse matrix of $\mathbf{R}$ (verify).

The Equations in Terms of Primitive Variables

Equation (8.81) may also be expressed in terms of the **primitive, or physical variables** A and u , again in non conservation-law form. By expanding derivatives in the momentum equation and using the continuity equation we obtain

$$\left. \begin{aligned} \partial_t A + u \partial_x A + A \partial_x u &= 0, \\ \partial_t u + u \partial_x u + \frac{1}{\rho} \frac{\partial p}{\partial A} \partial_x A &= 0. \end{aligned} \right\} \quad (8.94)$$

In matrix form we may write the equations as

$$\partial_t \mathbf{V} + \mathbf{M}(\mathbf{V}) \partial_x \mathbf{V} = \mathbf{0}, \quad (8.95)$$

with

$$\mathbf{V} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \equiv \begin{bmatrix} A \\ u \end{bmatrix}, \quad \mathbf{M}(\mathbf{V}) = \begin{bmatrix} u & A \\ \frac{1}{\rho} \frac{\partial p}{\partial A} & u \end{bmatrix}. \quad (8.96)$$

The eigenvalues of system (8.95)–(8.96) are given by (8.87); these are identical to those for the formulation of the equations in terms of conserved variables. The eigenvectors, however, are different. The left eigenvectors are

$$\mathbf{L}_1 = \alpha_1 [1, -A/c], \quad \mathbf{L}_2 = \alpha_2 [1, A/c], \quad (8.97)$$

with α_1 and α_2 two arbitrary scaling factors. Analogous calculations for the set of right eigenvectors give

$$\mathbf{R}_1 = \beta_1 \begin{bmatrix} 1 \\ -c/A \end{bmatrix}, \quad \mathbf{R}_2 = \beta_2 \begin{bmatrix} 1 \\ c/A \end{bmatrix}, \quad (8.98)$$

where β_1 and β_2 are arbitrary scaling factors.

We have performed the complete analysis of the eigenstructure of the system of blood flow equations for the special case in which Coriolis coefficient is $\alpha = 1$ and a generic tube law of the form $p = p(A)$. The eigenstructure and wave speed for general α will be studied in detail in Chap. 10.

In what follows we assume a specific tube law, namely a tube law for blood flow in arterial vessels.

8.4 Simplified Equations for Blood Flow in Arteries

Here we introduce a specific tube law for arteries, given as

$$p(x, t) := p(A) = p_e(x, t) + K(x) \left[\left(\frac{A(x, t)}{A_0(x)} \right)^{1/2} - 1 \right], \quad (8.99)$$

with

$$K(x) = \frac{\sqrt{\pi}}{(1 - \nu^2)} \frac{E(x) h_0(x)}{\sqrt{A_0(x)}}, \quad (8.100)$$

where

- $p_e(x, t)$: external pressure at position x and time t ;
- $K(x)$: coefficient of elasticity;

- $E(x)$: Young's modulus of elasticity;
- $A_0(x)$: equilibrium cross-sectional area;
- $h_0(x)$: vessel wall thickness;
- ν : Poisson's ratio, taken as $\nu = 1/2$ for incompressible media.

8.4.1 Conservative Formulation in Conserved Variables

The physically conserved quantities in Eq. (8.81) are A and $q \equiv Au$. As already noted the equations are not written in conservation-law form, due to the term $\frac{A}{\rho} \partial_x p$ in the momentum equation. In order to easily express the equations in conservation-law form in what follows we make the following assumptions

1. The parameters h_0 , A_0 , E in the tube law (8.99)–(8.100) for arteries are assumed to be constant;
2. The Coriolis coefficient is constant and assumed $\alpha = 1$;
3. The Poisson ratio is constant and set to $\nu = 1/2$.

As a consequence, the coefficient of elasticity K is also constant. Some algebraic manipulations based on these assumptions applied to the term $\frac{A}{\rho} \partial_x p$ in the momentum equation give

$$\frac{A}{\rho} \partial_x p = \gamma \partial_x A^{3/2}, \quad \gamma = \frac{K}{3\rho\sqrt{A_0}} = \frac{4\sqrt{\pi} E h_0}{9 \rho A_0}. \quad (8.101)$$

Then the full system (8.81) can be written in conservation-law form as

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}), \quad (8.102)$$

where

$$\left. \begin{aligned} \mathbf{Q} &= \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \equiv \begin{bmatrix} A \\ Au \end{bmatrix}, \quad \mathbf{F}(\mathbf{Q}) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \equiv \begin{bmatrix} Au \\ Au^2 + \gamma A^{3/2} \end{bmatrix}, \\ \mathbf{S}(\mathbf{Q}) &= \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \end{aligned} \right\} \quad (8.103)$$

Here $\mathbf{Q}$ is the vector of conserved variables, $\mathbf{F}(\mathbf{Q})$ is the vector of physical fluxes and $\mathbf{S}(\mathbf{Q})$ is the vector of source terms. Conservative methods can now be used to solve these equations; see Chap. 13.

8.4.2 Conservative Formulation in Physical Variables

Returning to Eq. (8.94) and noting that for smooth solutions

$$u \partial_x u = \partial_x \left(\frac{1}{2} u^2 \right) \quad (8.104)$$

one can write the momentum in (8.94) in *conservation-law form*. In fact, by noting the definition of wave speed (8.88), the full momentum equation can be written in conservation-law form. Hence the full system can be written in conservation-law form

$$\partial_t \mathbf{V} + \partial_x \mathbf{G}(\mathbf{V}) = \mathbf{S}(\mathbf{V}) , \quad (8.105)$$

where

$$\mathbf{V} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \equiv \begin{bmatrix} A \\ u \end{bmatrix} , \quad \mathbf{G}(\mathbf{V}) = \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} \equiv \begin{bmatrix} Au \\ \frac{1}{2} u^2 + 3\gamma A^{1/2} \end{bmatrix} , \quad (8.106)$$

and source term

$$\mathbf{S}(\mathbf{V}) = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ 0 \end{bmatrix} . \quad (8.107)$$

Now the variable $v_2 = u$ is a *conserved variable*, but only in a *mathematical sense*, not in a physical sense. Therefore the conservative formulation (8.105)–(8.106) is conservative purely in a mathematical sense, not in a physical sense.

For smooth solutions all formulations are equivalent, but not so for generalised solutions admitting shock waves (elastic jumps). In fact it can be demonstrated that shock waves from the mathematical conservative formulation (8.105)–(8.106) are different from the physical conservative formulation (8.102)–(8.103). They have different wave strengths and different speeds. Great care is needed when adopting formulations of the governing equations for real practical purposes. An analogous situation arises from the shallow water equations; see [6] for a full discussion on this issue.

In Chap. 9, the blood flow equations, for both arteries and veins, will be studied in much more detail.

8.5 Concluding Remarks and Further Reading

We have studied the main mathematical features of systems of non-linear hyperbolic equations. For the scalar case the study has been reasonably detailed. For the system case we have limited ourselves to some of the main issues, such as the integral form of

the equations, the Rankine-Hugoniot jump conditions at shocks and various possible formulations of the equations. Exercises are given at the end of the chapter.

For further reading on the subject we strongly recommend the following books: Zachmanoglou and Thoe [7]; Logan [8]; Jeffrey [9]; Godlewski and Raviart [3]; Smoller [10] and Sharma [11].

8.6 Exercises

8.1 Consider the following initial-value problem for the Burgers equation

$$\begin{aligned} \text{PDE: } \partial_t q + \partial_x f(q) &= 0, & f(q) &= \frac{1}{2}q^2, \\ \text{IC: } q(x, 0) = h(x) &= \begin{cases} -1 & \text{if } x < 0, \\ 1 & \text{if } x > 0. \end{cases} \end{aligned} \quad (8.108)$$

1. Draw the picture of characteristics associated with the initial condition in the x - t plane.
2. Find the rarefaction solution $q(x, t)$ and draw the corresponding picture of characteristics in the x - t plane.
3. Verify that the above solution satisfies the integral form of the conservation law for an arbitrary quadrilateral control volume.
4. Draw solution profiles at times $t_0 = 0$, $t_1 = 1$ and $t_2 = 2$.
5. Show that the function

$$q(x, t) = \begin{cases} -1 & \text{if } x < 0, \\ 1 & \text{if } x > 0 \end{cases} \quad (8.109)$$

is a solution (shock) of the integral form of the conservation law for Burgers' equation.

6. Find the shock speed of the above solution for an arbitrary quadrilateral control volume.
7. By drawing the picture characteristics in the x - t plane show that the above solution (8.109) is entropy violating.

8.2 Study all the possible wave configurations emerging from the solution of the Riemann problem for Burgers' equation and provide specific examples for each case. *Hint:* In addition to the fact that the solution could be either a shock wave or a rarefaction wave, the wave could travel to the left, to the right or be stationary (in the case of a shock) or transcritical or transonic (in the case of a rarefaction).

8.3 Consider the Riemann problem for the Burgers equation

$$\begin{aligned} \text{PDE: } \partial_t q + \partial_x f(q) &= 0, & f(q) &= \frac{1}{2}q^2, \\ \text{IC: } q(x, 0) = h(x) &= \begin{cases} \hat{q} & \text{if } x < 0, \\ -\hat{q} & \text{if } x > 0, \end{cases} \end{aligned} \quad (8.110)$$

where $\hat{q}$ is a positive constant.

1. Show that the function

$$q(x, t) = \begin{cases} \hat{q} & \text{if } x < 0, \\ -\hat{q} & \text{if } x > 0 \end{cases} \quad (8.111)$$

is a *weak solution* of the problem. Hint: select an arbitrary control volume in the x - t plane and show that the integral form of the conservation law is satisfied.

2. By applying the Rankine-Hugoniot condition verify that the shock solution (8.111) has speed zero (stationary shock).
3. Plot the solution (8.111) in the x - t plane, including the shock path and the picture of characteristics on either side of the discontinuity.
4. Verify that the solution (8.111) satisfies the Lax entropy condition.

8.4 Consider the following initial-value problem for the Burgers equation

$$\begin{aligned} \text{PDE: } \partial_t q + \partial_x f(q) &= 0, & f(q) &= \frac{1}{2}q^2, \\ \text{IC: } q(x, 0) = h(x) &= \begin{cases} 0 & \text{if } x < 0, \\ 1 & \text{if } 0 \leq x \leq 1/2, \\ 0 & \text{if } x > 1/2. \end{cases} \end{aligned} \quad (8.112)$$

1. Draw the picture of characteristics and the structure of the solution in the x - t plane. Hint: note that the solution initially contains two distinct waves and there is a *critical time* $\hat{t}$ at which the left wave catches up with the right wave.
2. Calculate $\hat{t}$ above.
3. Plot the solution profile at times $t_1 = \hat{t}/2$ and at $t_2 = \hat{t}$.
4. What happens after $\hat{t}$? Represent the wave structure and the picture of characteristics in the x - t plane up to times much longer than $\hat{t}$.

8.5 Consider the following initial-value problem for the traffic flow equation

$$\begin{aligned} \text{PDE: } \partial_t q + \partial_x f(q) &= 0, & f(q) &= u_{\max}(1 - q/q_{\max})q, \\ \text{IC: } q(x, 0) = h(x) &= \begin{cases} q_L & \text{if } x < 0, \\ q_R & \text{if } x > 0, \end{cases} \end{aligned} \quad (8.113)$$

where $q_{\max}$ and $u_{\max}$ are positive constants.

1. Find the characteristic speed $\lambda(q)$.
2. Assume the initial data q_L, q_R to be connected by a shock wave. Apply the Rankine-Hugoniot condition and find an expression for the shock speed.
3. Find conditions on the initial data q_L, q_R for the shock to satisfy the Lax entropy condition. Compare your conclusions with those for the Burgers equation case.
4. What are the conditions on the initial data q_L, q_R so as to generate a rarefaction (smooth) solution.

8.6 Consider the inhomogeneous equation

$$\partial_t q(x, t) + \partial_x f(q(x, t)) = s(q(x, t)). \quad (8.114)$$

By imitating the procedure to obtain the Rankine-Hugoniot condition for the homogeneous case ($s = 0$) attempt to obtain the Rankine-Hugoniot condition across a shock wave for the in-homogeneous case that includes a source term $s(q) \neq 0$.

8.7 Starting from the integral form of (8.114)

$$\frac{d}{dt} \int_{x_L}^{x_R} q(x, t) dx = f(q(x_L, t)) - f(q(x_R, t)) + \int_{x_L}^{x_R} s(q(x, t)) dx, \quad (8.115)$$

and using the fundamental theorem of calculus show that a *smooth* weak solution also satisfies the differential form (8.114) of the balance law.

8.8 Consider the conservative form of the Eqs. (8.102)–(8.103).

1. Find the eigenvalues.
2. Find the left eigenvectors.
3. Find the right eigenvectors
4. Is the system hyperbolic?
5. Determine the nature of the two characteristic fields.

8.9 Consider the equations of haemodynamics augmented with an equation for a passive scalar $\eta(x, t)$, namely

$$\left. \begin{aligned} \partial_t A + \partial_x (Au) &= s_1 := 0, \\ \partial_t (Au) + \partial_x (\alpha Au^2) + \frac{A}{\rho} \partial_x p &= s_2 := 0, \\ \partial_t (A\eta) + \partial_x (Au\eta) &= s_3 := 0. \end{aligned} \right\} \quad (8.116)$$

Assume $\alpha = 1$ and a tube law of the form $p = p(A)$.

1. Express the equations in matrix form in terms of the conserved variables.
2. Find the eigenvalues.
3. Find the left eigenvectors.
4. Find the right eigenvectors.
5. Under what conditions is the system hyperbolic?
6. Determine the nature of the three characteristic fields.

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Chapter 9

The Riemann Problem for Blood Flow



Abstract In this chapter we derive the exact solution to the Riemann problem for the non-linear system of blood flow equations, under the assumptions that the Coriolis coefficient is $\alpha = 1$, and with constant geometric and mechanical parameters. This solution serves as a reference for assessing numerical methods, applying boundary conditions, and designing Godunov-type numerical schemes. Additionally, it is valuable for devising and evaluating approximate Riemann solvers used in such numerical methods. Importantly, the exact solution to these non-linear problems enhances our understanding of the governing equations as a mathematical model for blood flow in compliant vessels. To proceed, one must also incorporate the tube law for the mechanics of the vessel wall, which distinguishes arteries from veins. First, we solve the Riemann problem for arteries, where the two blood flow equations are supplemented by a third equation for a passive scalar, representing the transport of chemical species. The main properties of the equations are reviewed, and rarefaction wave solutions are analysed using generalised Riemann invariants. An all-rarefaction exact solution is derived. Shock waves are introduced via the Rankine-Hugoniot conditions. The complete set of equations, encompassing rarefactions, shocks, and contacts, is presented, including a single non-linear algebraic equation for the cross-sectional area. It is shown that this equation has a unique solution within the physically valid range of positive cross-sectional areas. A numerical solution and a sampling procedure are implemented to obtain the full solution to the Riemann problem. Next, we solve the Riemann problem for veins. This problem is more complex than that for arteries, and all steps are rigorously justified. The overall methodology follows the same approach used for arteries, and sample solutions are provided. The chapter concludes with a summary and suggestions for further reading. A set of mini-projects is proposed for exercises based on the content of this chapter and related topics in the book.

9.1 The Riemann Problem for Blood Flow in Arteries

In this section we solve exactly the Riemann problem for blood flow in arteries with constant material properties. The case for veins is studied in Sect. 9.5. For background see Chap. 6 for the equations of haemodynamics and Chap. 8 for their mathematical properties.

9.1.1 Recalling the Equations and the Riemann Problem

Here, in addition to the classical two equations for the laws of conservation of mass and balance of momentum, we add an *advection equation for a passive scalar* $\eta(x, t)$, advected with the fluid speed $u(x, t)$. The motivation for adding the tracer equation is twofold. First, transport of solutes is an essential feature of the circulation; second, from the mathematical and numerical points of view, equations of tracer-advection type pose their own challenges.

The Equations in Conservation-law Form

Departing from Chaps. 6 and 8, and adding the passive scalar η our full system, written in conservation-law form, reads

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}), \quad (9.1)$$

where

$$\left. \begin{aligned} \mathbf{Q} &= \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \equiv \begin{bmatrix} A \\ Au \\ A\eta \end{bmatrix}, & \mathbf{F}(\mathbf{Q}) &= \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \equiv \begin{bmatrix} Au \\ Au^2 + \gamma A^{3/2} \\ A\eta u \end{bmatrix}, \\ \mathbf{S}(\mathbf{Q}) &= \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \end{aligned} \right\} \quad (9.2)$$

Here we have taken the Coriolis coefficient to be $\alpha = 1$ and the source terms to be zero, consistent with $\mu = 0$. To close the system we have assumed a tube law of the form $p := p(A)$. Specifically, for arterial vessels we have adopted here

$$p(x, t) := p(A) = p_e(x, t) + K(x) \left[\left(\frac{A(x, t)}{A_0(x)} \right)^{1/2} - 1 \right], \quad (9.3)$$

with

$$K(x) = \frac{\sqrt{\pi}}{(1 - \nu^2)} \frac{E(x)h_0(x)}{\sqrt{A_0(x)}}. \quad (9.4)$$

For constant K and Poisson's ratio set to $\mu = 1/2$, the constant coefficient γ in (9.2) is given as

$$\gamma = \frac{K}{3\rho\sqrt{A_0}} = \frac{4\sqrt{\pi}}{9} \frac{Eh_0}{\rho A_0}. \quad (9.5)$$

The Jacobian matrix of system (9.1)–(9.2) is given as

$$\mathbf{A}(\mathbf{Q}) = \frac{\partial \mathbf{F}}{\partial \mathbf{Q}} = \begin{bmatrix} 0 & 1 & 0 \\ c^2 - u^2 & 2u & 0 \\ -u\eta & \eta & u \end{bmatrix}. \quad (9.6)$$

The eigenvalues of $\mathbf{A}(\mathbf{Q})$, written in increasing order, are

$$\lambda_1 = u - c, \quad \lambda_2 = u, \quad \lambda_3 = u + c, \quad (9.7)$$

where c is the **wave speed** given as

$$c = \sqrt{\frac{A}{\rho} \frac{\partial p}{\partial A}} = \sqrt{\frac{3}{2} \gamma \sqrt{A}}. \quad (9.8)$$

The corresponding right eigenvectors are

$$\mathbf{R}_1 = \beta_1 \begin{bmatrix} 1 \\ u - c \\ \eta \end{bmatrix}, \quad \mathbf{R}_2 = \beta_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{R}_3 = \beta_3 \begin{bmatrix} 1 \\ u + c \\ \eta \end{bmatrix}, \quad (9.9)$$

where β_1 , β_2 and β_3 are arbitrary scaling factors. Unless otherwise stated, we shall assume the scaling parameters to be chosen as $\beta_1 = \beta_2 = \beta_3 = 1$.

The Riemann Problem

The Riemann problem for system (9.1) is the initial-value problem

$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0}, \quad -\infty < x < \infty, \quad t > 0, \\ \text{ICs: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{Q}_L : \text{constant} & \text{if } x < 0, \\ \mathbf{Q}_R : \text{constant} & \text{if } x > 0. \end{cases} \end{array} \right\} \quad (9.10)$$

The initial conditions are given by the two constant states $\mathbf{Q}_L$ and $\mathbf{Q}_R$. Useful background on the Riemann problem for linear systems is found in Chap. 7.

Figure 9.1 depicts the structure of the exact solution of the Riemann problem (9.10) for the homogeneous blood flow Eqs. (9.1). There are three wave families associated to the three real eigenvalues $\lambda_1 = u - c$, $\lambda_2 = u$ and $\lambda_3 = u + c$, but note that, unlike the linear case, the speeds of the waves are not always given by the eigenvalues. The middle wave is always a contact discontinuity, while the left and right waves are non-linear and can be either shocks (elastic jumps) or rarefactions.

It is worth noting that all four wave configurations shown in Fig. 9.2 are subcritical, but may display a supercritical pattern, that is, all waves may be placed on one side of the t -axis. The three wave families depicted in Fig. 9.1 separate four constant states,

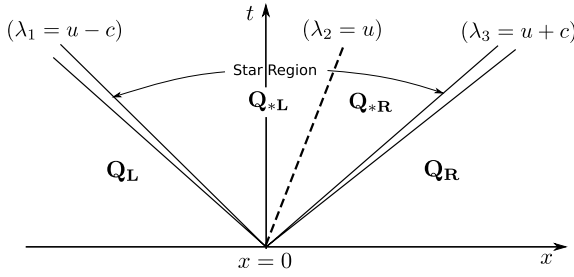


Fig. 9.1 Structure of the solution of the Riemann problem (9.10) in x - t half-plane. There are three wave families associated with the three eigenvalues separating four constant states Q_L , Q_{*L} , Q_{*R} and Q_R . The unknown states to be found are Q_{*L} and Q_{*R} that comprise the *Star Region*

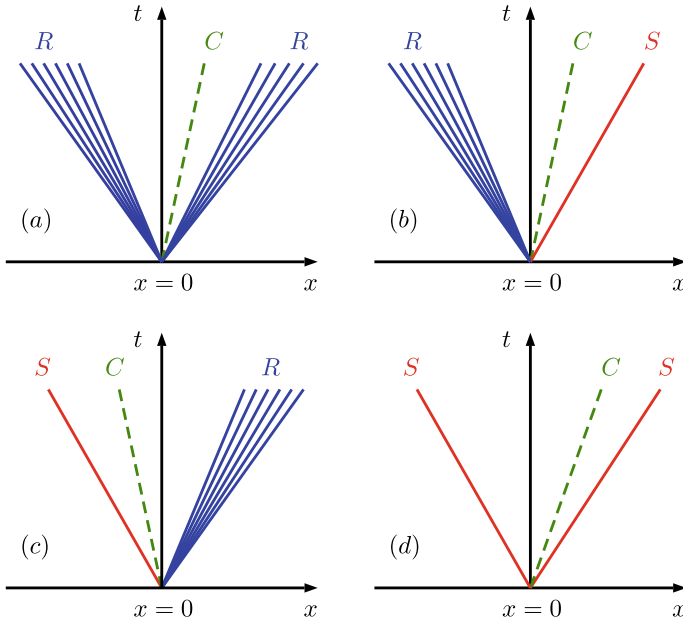


Fig. 9.2 All four possible wave patterns arising from the solution of the Riemann problem (9.10) for arteries. Here R: rarefaction; C: contact and S: shock

namely Q_L , Q_{*L} , Q_{*R} and Q_R . The unknown states to be found are Q_{*L} and Q_{*R} , which comprise the *Star Region*, while Q_L and Q_R are known data states.

Figure 9.2 depicts all four possible wave patterns arising from the solution of the Riemann problem.

- Frame (a) depicts the case of left rarefaction/contact/right rarefaction;
- Frame (b) depicts the pattern left rarefaction/contact/right shock;
- Frame (c) corresponds to left shock/contact/right rarefaction and
- Frame (d) which depicts the pattern left shock/contact/right shock.

In order to find the solution of the Riemann problem we first introduce some basic notions on the characteristic fields present in the solution of the Riemann problem.

9.1.2 Characteristic Fields and Simple Wave Relations

Here we recall the character of the wave families present in the solution of the Riemann problem, as depicted in Figs. 9.1 and 9.2. We also introduce some mathematical concepts that allow a straightforward classification of the waves.

Nature of Characteristic Fields

The character of the waves can be determined by using the eigenvalues and eigenvectors. For example, for the middle wave with eigenvalue $\lambda_2 = u$ it is easily verified that the associated characteristic field is *linearly degenerate*, that is

$$\nabla \lambda_2(\mathbf{Q}) \cdot \mathbf{R}_2(\mathbf{Q}) = 0 \quad \forall \mathbf{Q} \in \mathbb{R}^3. \quad (9.11)$$

The linear middle wave is called a *contact discontinuity*.

It is also straightforward to show that the characteristic fields corresponding to $\lambda_1 = u - c$ and $\lambda_3 = u + c$ are genuinely non-linear, that is

$$\nabla \lambda_K(\mathbf{Q}) \cdot \mathbf{R}_K(\mathbf{Q}) \neq 0 \quad \forall \mathbf{Q} \in \mathbb{R}^3 \quad \text{for } K = 1 \text{ and } K = 3. \quad (9.12)$$

The waves associated with these genuinely non-linear characteristic fields are either shocks or rarefactions. See Chap. 7.

Generalised Riemann Invariants and Wave Jumps

It is convenient to determine relationships between the various quantities involved *across* the various wave families in physical space x - t , see Fig. 9.1. These can be provided by the *Generalised Riemann Invariants* [1, 2]. Consider a general hyperbolic system of m equations and the wave associated with the $\lambda_j(\mathbf{Q})$ -characteristic field with the corresponding right eigenvector

$$\mathbf{R}_j(\mathbf{Q}) = [r_{1j}, r_{2j}, \dots, r_{mj}]^T. \quad (9.13)$$

The generalised Riemann invariants apply *across* the wave structure of the j -th wave and lead to $m - 1$ ordinary differential equations in phase space:

$$\frac{dq_1}{r_{1j}} = \frac{dq_2}{r_{2j}} = \frac{dq_3}{r_{3j}} = \dots = \frac{dq_m}{r_{mj}}. \quad (9.14)$$

Equation (9.14) relates ratios of dq_i to the respective component r_{ij} of the right eigenvector $\mathbf{R}_j$. It is emphasised that the generalised Riemann invariants are to be interpreted as meaning

$$dq_i \propto r_{ij}. \quad (9.15)$$

For example, if $r_{ij} = 0$ then $dq_i = 0$ and therefore q_i does not change across the wave. As a matter of fact, by inspecting the components of the eigenvectors (9.9) of the system, one can anticipate whether or not some of the components of the vector of unknowns change across the waves in the Riemann problem.

Consider an eigenvector analysis of jumps in conserved variables across the middle wave associated with the eigenvalue $\lambda_2 = u$ and eigenvector $\mathbf{R}_2 = [0, 0, 1]^T$. Since the first two components of $\mathbf{R}_2$ are zero, this means that the jumps across the middle wave of the first two components of the vector $\mathbf{Q} = [q_1, q_2, q_3]^T \equiv [A, Au, A\eta]^T$ are zero, or that the respective quantities are constant. That is, $dA = 0$, which implies that A is constant across the contact discontinuity; in other words $A_{*L} = A_{*R} = A_*$, see Fig. 9.3. Also, $d(Au) = Adu + u dA = 0$. But $dA = 0$ and therefore $du = 0$, which means that u is constant across the contact discontinuity and therefore we can write $u_{*L} = u_{*R} = u_*$, see Fig. 9.3. Collecting results, across the contact discontinuity we have

$$A_{*L} = A_{*R} = A_*, \quad u_{*L} = u_{*R} = u_*. \quad (9.16)$$

For the left and right non-linear waves, the jumps in variables across these waves are less obvious and their determination requires a different approach. However, we may anticipate that the concentration η is constant across both non-linear waves and jumps across the contact discontinuity are discontinuous. We may write

$$\eta_{*L} = \eta_L \text{ across the left wave, } \eta_{*R} = \eta_R \text{ across the right wave.} \quad (9.17)$$

Figure 9.3 depicts the situation.

9.1.3 Solution Strategy

Next we outline the solution strategy for solving the Riemann problem (9.10), in which we use the above concepts. There are essentially three tasks to be performed:

1. **The Star Problem.** The first task is to find two unknowns in the *Star Region*, namely $A_* = A_{*L} = A_{*R}$ and $u_* = u_{*L} = u_{*R}$, see Figs. 9.1 and 9.3. To this end we connect the unknown state $\mathbf{Q}_{*L}$ to the known state $\mathbf{Q}_L$ via appropriate wave relations embodied in a function f_L across the left wave associated with the eigenvalue $\lambda_1 = u - c$. See Fig. 9.3. Similarly, the unknown state $\mathbf{Q}_{*R}$ will be connected to the data $\mathbf{Q}_R$ via wave relations across the right wave associated with the eigenvalue $\lambda_3 = u + c$, embodied in a function f_R . See Fig. 9.3.

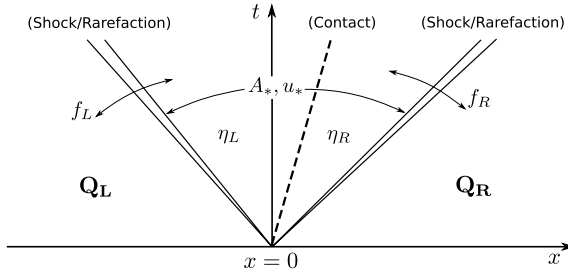


Fig. 9.3 Revised structure of the solution of the Riemann problem for the blood flow equations. Cross-sectional area A and velocity u in the *Star Region* are constant across the contact discontinuity, while η remains constant across the non-linear waves but jumps discontinuously across the contact discontinuity. The functions f_L and f_R represent wave jumps, yet to be determined, across the non-linear waves; they connect the unknowns to the initial data

The sought functions f_L and f_R will result from the application of *generalised Riemann invariants* across rarefaction waves and *Rankine-Hugoniot conditions* across shock waves (elastic jumps).

2. **Solution inside rarefactions waves.** Once the *Star Problem* has been solved, the next task consists of finding the solution inside rarefactions waves; these are smooth waves of finite width. The task is performed by using Riemann invariants.
3. **Sampling the solution.** The final problem is to sample the complete solution, that is, find $\mathbf{Q}(x, t)$, $\forall x \in \mathbb{R}$, $\forall t > 0$.

As shown in Fig. 9.2 there are four possible wave patterns. The middle wave is always a contact discontinuity, while the outer waves can be either rarefactions or shocks, and this is not known in advance. Determining the wave type associated to the genuinely non-linear fields is part of the solution procedure. To this end we study separately the cases of rarefactions and shocks, starting with rarefaction waves.

9.2 Rarefaction Waves

First we study the wave configuration in which both non-linear waves are rarefaction waves for which the application of generalised Riemann invariants determines the wave jumps across these waves.

9.2.1 The Left Wave is an Isolated Rarefaction Wave

Assume that the left wave of the solution of the Riemann problem is an isolated rarefaction wave connecting the left data state $\mathbf{Q}_L$ to the unknown state $\mathbf{Q}_{*L}$, the left

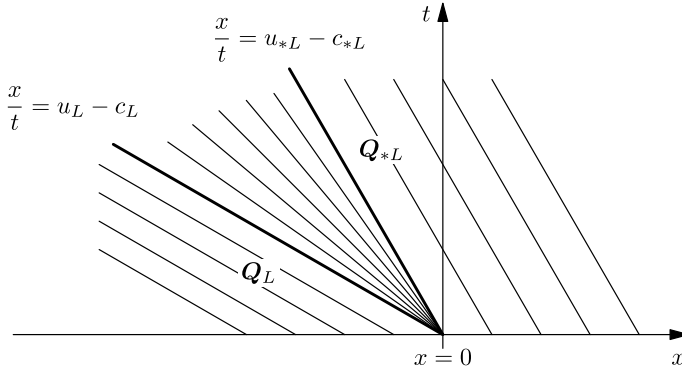


Fig. 9.4 Isolated left rarefaction wave associated with eigenvalue $\lambda_1 = u - c$, connecting the unknown constant state $\mathbf{Q}_{*L}$ to the constant data state $\mathbf{Q}_L$. The smooth wave is bounded by the *head*, given by the characteristic $x/t = u_L - c_L$, and the *tail*, given by the characteristic $x/t = u_{*L} - c_{*L}$

part of the *Star Region*. See Fig. 9.4. Consider the characteristic field associated with the eigenvalue $\lambda_1 = u - c$ and its eigenvector $\mathbf{R}_1 = [1, u - c, \eta]^T$. Application of (9.14) gives the following relations across the $\lambda_1(\mathbf{Q})$ wave

$$\frac{dA}{1} = \frac{d(Au)}{u - c} = \frac{d(A\eta)}{\eta}. \quad (9.18)$$

Then, equating the first and second ratios yields

$$du + \frac{c}{A} dA = 0. \quad (9.19)$$

Inserting the wave speed

$$c = \sqrt{\frac{3}{2} \gamma \sqrt{A}} \quad (9.20)$$

into (9.19) and integrating in phase space gives

$$u + 4c = \text{constant} \rightarrow u_* + 4c_* = u_L + 4c_L, \quad (9.21)$$

as anticipated in (9.17). This means that the quantity $u + 4c$ remains constant as we cross the wave structure associated with $\lambda_1 = u - c$. Moreover, using equality of the second and third ratios in (9.18) implies

$$\eta = \text{constant} \rightarrow \eta_{*L} = \eta_L. \quad (9.22)$$

That is, the concentration $\eta(x, t)$ is unaffected by the wave associated with $\lambda_1(\mathbf{Q})$. Using (9.21) to connect $u_* = u_{*L}$ to u_L , see Fig. 9.3, we obtain a relation between u_{*L} and u_L via a function f_L , namely

$$u_{*L} = u_L - f_L; \quad f_L = 4(c_{*L} - c_L). \quad (9.23)$$

The function $u_{*L} = u_L - f_L$ defines a *wave curve* across a left rarefaction, called **rarefaction curve**.

Solution Inside a Left Rarefaction Wave

Consider a left rarefaction wave associated with the eigenvalue $\lambda_1 = u - c$, as depicted in Fig. 9.5, and a point $\hat{P} = (\hat{x}, \hat{t})$ inside the wave. Consider a characteristic line through $\hat{P} = (\hat{x}, \hat{t})$ and the origin $(0, 0)$, of slope (known)

$$\hat{u} - \hat{c} = \frac{\hat{x}}{\hat{t}}. \quad (9.24)$$

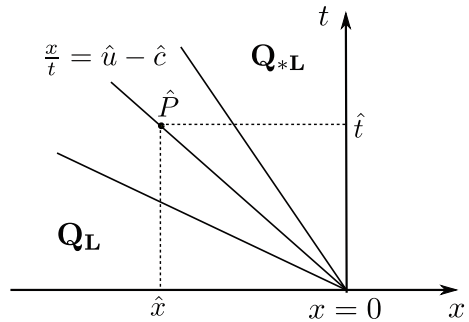
The unknowns of the problem are $\hat{u} = u(\hat{x}, \hat{t})$, $\hat{c} = c(\hat{x}, \hat{t})$ and $\hat{\eta}(\hat{x}, \hat{t})$. As η does not change across the wave, $\hat{\eta}(\hat{x}, \hat{t}) = \eta_L$. To find the rest of the solution we apply the left Riemann invariant (9.21) to connect the point $\hat{P}$ to the left initial condition to obtain

$$\hat{u} + 4\hat{c} = u_L + 4c_L. \quad (9.25)$$

Equations (9.24) and (9.25) are two equations for two unknowns, whose solution is immediate, so that the complete solution is

$$\hat{u} = \frac{1}{5} \left(u_L + 4c_L + 4\frac{\hat{x}}{\hat{t}} \right), \quad \hat{c} = \frac{1}{5} \left(u_L + 4c_L - \frac{\hat{x}}{\hat{t}} \right), \quad \hat{\eta} = \eta_L. \quad (9.26)$$

Fig. 9.5 Point $\hat{P} = (\hat{x}, \hat{t})$ inside left rarefaction wave. Solution at point $\hat{P}$ is sought



9.2.2 The Right Wave is an Isolated Rarefaction Wave

Assume a right rarefaction wave, as depicted in Fig. 9.6, connecting the constant states $\mathbf{Q}_{*R}$ (behind the wave, left) and $\mathbf{Q}_R$ (right). As done for the left rarefaction wave we make use of generalised Riemann invariants (9.14). For the right wave field $\lambda_3(\mathbf{Q}) = u + c$ the Riemann invariants are

$$u - 4c = \text{constant} \rightarrow u_* - 4c_* = u_R - 4c_R \quad (9.27)$$

and

$$\eta = \text{constant} \rightarrow \eta_{*R} = \eta_R, \quad (9.28)$$

with η generally changing discontinuously across the contact wave. So the concentration only changes across the contact wave and remains constant across rarefaction waves. We use (9.27) to connect the u_R to $u_* = u_{*R}$ behind the right rarefaction, from which we obtain a relation between u_{*R} and u_R via a function f_R , namely

$$u_{*R} = u_R + f_R; \quad f_R = 4(c_{*R} - c_R). \quad (9.29)$$

For the right rarefaction the **rarefaction curve** is the function $u_{*R} = u_R + f_R$.

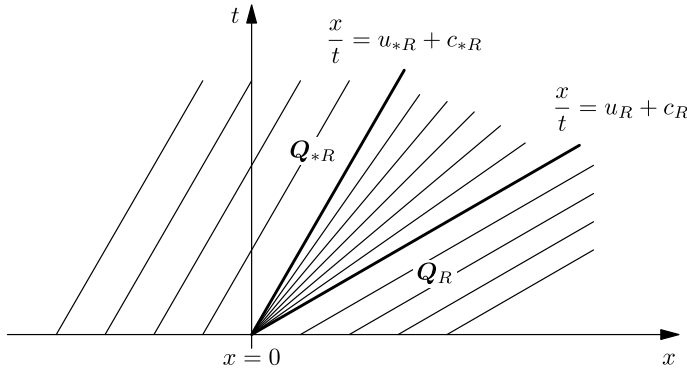


Fig. 9.6 Isolated right rarefaction wave associated with eigenvalue $\lambda_3 = u + c$, connecting the unknown constant state $\mathbf{Q}_{*R}$ to the constant data state $\mathbf{Q}_R$. The smooth wave is bounded by the *head*, given by the characteristic $x/t = u_R + c_R$, and the *tail*, given by the characteristic $x/t = u_{*R} + c_{*R}$

Solution Inside a Right Rarefaction Wave

For a right rarefaction the procedure to find the solution inside the wave is entirely analogous to the left rarefaction case; the result is

$$\hat{u} = \frac{1}{5} \left(u_R - 4c_R + 4\frac{\hat{x}}{\hat{t}} \right), \quad \hat{c} = \frac{1}{5} \left(-u_R + 4c_R + \frac{\hat{x}}{\hat{t}} \right), \quad \hat{\eta} = \eta_R. \quad (9.30)$$

The reader is encouraged to verify result (9.30).

9.2.3 All-Rarefaction Solution

Here we consider the special wave pattern depicted in frame (a) of Fig. 9.2, that is the pattern: left rarefaction/contact/right rarefaction. As seen earlier, for the left rarefaction we have

$$u_{*L} = u_L - f_L; \quad f_L = 4(c_{*L} - c_L). \quad (9.31)$$

For the right rarefaction we have

$$u_{*R} = u_R + f_R; \quad f_R = 4(c_{*R} - c_R). \quad (9.32)$$

But recall that from (9.16)

$$u_{*L} = u_{*R} = u_* \text{ and } A_{*L} = A_{*R} = A_*. \quad (9.33)$$

From the second relation above it follows that

$$c_{*L} = c_{*R} = c_*. \quad (9.34)$$

Use of Eqs. (9.33)–(9.34) into (9.31)–(9.32) leads to

$$\left. \begin{aligned} u_* &= u_L - f_L; \quad f_L = 4(c_* - c_L), \\ u_* &= u_R + f_R; \quad f_R = 4(c_* - c_R). \end{aligned} \right\} \quad (9.35)$$

Subtracting the first equation from the second equation in (9.35) gives

$$f(A) \equiv f_L + f_R + u_R - u_L = 4(c_* - c_L) + 4(c_* - c_R) + u_R - u_L = 0. \quad (9.36)$$

That is

$$4(c_* - c_L) + 4(c_* - c_R) + u_R - u_L = 0. \quad (9.37)$$

This is a linear algebraic equation for the single unknown c_* , the wave speed in the *Star Region*. Its exact closed-form solution is immediate, namely

$$c_* = \frac{1}{2}(c_L + c_R) - \frac{1}{8}(u_R - u_L). \quad (9.38)$$

The exact solution for particle velocity u_* is obtained from (9.35) as

$$u_* = \frac{1}{2}(u_L + u_R) - 2(c_R - c_L). \quad (9.39)$$

Using c_* in the expression for the wave speed (9.20) we obtain the exact solution for the cross-sectional area in the *Star Region*, that is

$$A_* = \left(\frac{2c_*^2}{3\gamma} \right)^2. \quad (9.40)$$

The above solution for A_* , u_* and c_* is exact when both non-linear waves are rarefaction waves. As we shall see later, it turns out that solution (9.39)–(9.40) is a good approximation for other wave patterns. For example for the case of (a) left rarefaction/contact/right shock; (b) left shock/contact/right rarefaction and even for the case (c) left shock/contact/right shock. See Fig. 9.2.

Composite Wave Curves

The wave curves for the two rarefactions associated with f_L and f_R give rise to a more general **composite wave curve** obtained by subtracting the first equation from the second equation in (9.35), namely

$$f(A) \equiv f_L(A; \mathbf{Q}_L) + f_R(A; \mathbf{Q}_R) + u_R - u_L, \quad (9.41)$$

noting that both f_L and f_R are actually functions of A_* and that $f(A_*) = 0$. In the general case the functions f_L and f_R will account for different types of waves, not just rarefactions, as we shall see in the following section. By analysing the function $f(A)$ it can be shown that the solution for A_* exists and is unique, within a physically admissible range for the parameters of the problem.

In the next section we study the case in which the non-linear waves are shock waves. The main task is to determine the form of the functions f_L and f_R for the case of shock waves.

9.3 Shock Waves or Elastic Jumps

As illustrated in Fig. 9.2 the outer non-linear waves may also be shock waves. Here we study such waves assuming in each case an isolated shock wave. That is, we assume the special case in which the solution of the Riemann problem contains a single non-trivial wave consisting of a shock wave. We start from the left non-linear wave in Fig. 9.3.

9.3.1 The Left Wave is an Isolated Shock Wave

Consider a left-facing shock wave of speed S_L associated with the eigenvalue $\lambda_1 = u - c$, as depicted in Fig. 9.7. The Rankine-Hugoniot conditions are applicable to this type of discontinuous waves, see Chap. 8 for background. Next, in order to facilitate the analysis, we define a transformation of frame of reference, namely

$$\hat{u}_{*L} = u_{*L} - S_L ; \quad \hat{u}_L = u_L - S_L, \quad (9.42)$$

in which the shock wave has speed $\hat{S}_L = 0$ (stationary shock). The vector of conserved variables and fluxes are denoted as $\hat{\mathbf{Q}}$ and $\hat{\mathbf{F}}$ respectively. The Rankine-Hugoniot conditions applied to the stationary shock read

$$\Delta \hat{\mathbf{F}} = \hat{S}_L \Delta \hat{\mathbf{Q}} = 0.$$

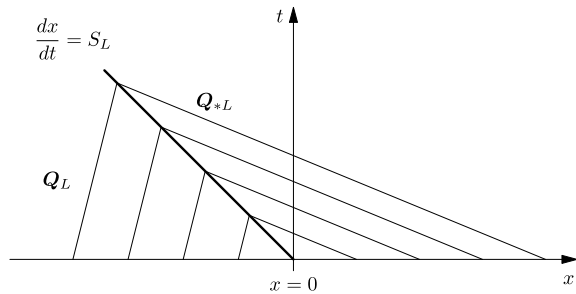
That is

$$\left. \begin{aligned} A_{*L} \hat{u}_{*L} &= A_L \hat{u}_L, \\ A_{*L} \hat{u}_{*L}^2 + \gamma A_{*L}^{3/2} &= A_L \hat{u}_L^2 + \gamma A_L^{3/2}, \\ A_{*L} \hat{u}_{*L} \eta_{*L} &= A_L \hat{u}_L \eta_L. \end{aligned} \right\} \quad (9.43)$$

The first of Eq. (9.43) says that the mass flux is constant across the shock wave, that is

$$M_L \equiv A_{*L} \hat{u}_{*L} = A_L \hat{u}_L, \quad (9.44)$$

Fig. 9.7 Isolated left-facing shock wave of speed S_L associated with eigenvalue $\lambda_1 = u - c$, connecting $\mathbf{Q}_L$ (ahead) and $\mathbf{Q}_{*L}$ (behind)



in which the parameter M_L denotes the constant. Substitution of (9.44) into the third of Eqs. (9.43) gives

$$\eta_{*L} = \eta_L, \quad (9.45)$$

which means that the concentration $\eta(x, t)$ is constant across the shock wave. Recall that the same conclusion was drawn previously for the case of a rarefaction wave. From the second equation in (9.43) we may write

$$(A_{*L}\hat{u}_{*L})\hat{u}_{*L} + \gamma A_{*L}^{3/2} = (A_L\hat{u}_L)\hat{u}_L + \gamma A_L^{3/2}, \quad (9.46)$$

which, after using (9.44) yields

$$M_L = -\gamma \left(\frac{A_{*L}^{3/2} - A_L^{3/2}}{\hat{u}_{*L} - \hat{u}_L} \right). \quad (9.47)$$

From (9.44) we also have

$$\hat{u}_{*L} = \frac{M_L}{A_{*L}}, \quad \hat{u}_L = \frac{M_L}{A_L}, \quad (9.48)$$

which if substituted into (9.47) yield

$$M_L = \sqrt{\frac{\gamma A_L A_{*L} (A_{*L}^{3/2} - A_L^{3/2})}{A_{*L} - A_L}}. \quad (9.49)$$

The choice of sign above is assisted by invoking the Lax entropy condition; see Chap. 8. From (9.42)

$$\hat{u}_{*L} - \hat{u}_L = u_{*L} - u_L, \quad (9.50)$$

from which (9.47) gives

$$u_{*L} = u_L - \frac{\gamma (A_{*L}^{3/2} - A_L^{3/2})}{M_L}. \quad (9.51)$$

This expression relates u_{*L} to u_L via a function f_L , namely

$$u_{*L} = u_L - f_L; \quad f_L = \sqrt{\frac{\gamma (A_{*L} - A_L) (A_{*L}^{3/2} - A_L^{3/2})}{A_L A_{*L}}}. \quad (9.52)$$

This equation defines the **shock wave curve** involving the function f_L depicted in Fig. 9.3 for the case in which the left non-linear wave is a shock wave. Compare with f_L in Eq. (9.23) for the case of a left rarefaction. See Fig. 9.3. An expression

for the shock speed can be obtained as follows. From (9.42) $S_L = u_L - \hat{u}_L$, which, after using (9.48) gives the shock speed as

$$S_L = u_L - M_L/A_L. \quad (9.53)$$

Next we consider the case in which the right non-linear wave is a shock wave.

9.3.2 The Right Wave is an Isolated Shock Wave

Consider a right-facing shock of speed S_R associated with the eigenvalue $\lambda_3 = u + c$, as depicted in Fig. 9.8.

Proceeding as for the left-facing shock wave case, we next define the transformation

$$\hat{u}_{*R} = u_{*R} - S_R; \quad \hat{u}_R = u_R - S_R. \quad (9.54)$$

Then, the Rankine-Hugoniot conditions applied to the corresponding stationary shock give

$$\left. \begin{aligned} A_{*R} \hat{u}_{*R} &= A_R \hat{u}_R, \\ A_{*R} \hat{u}_{*R}^2 + \gamma A_{*R}^{3/2} &= A_R \hat{u}_R^2 + \gamma A_R^{3/2}, \\ A_{*R} \hat{u}_{*R} \eta_{*R} &= A_R \hat{u}_R \eta_R. \end{aligned} \right\} \quad (9.55)$$

The first of Eqs. (9.55) says that the mass flux is constant across the shock wave, that is

$$M_R \equiv -A_{*R} \hat{u}_{*R} = -A_R \hat{u}_R. \quad (9.56)$$

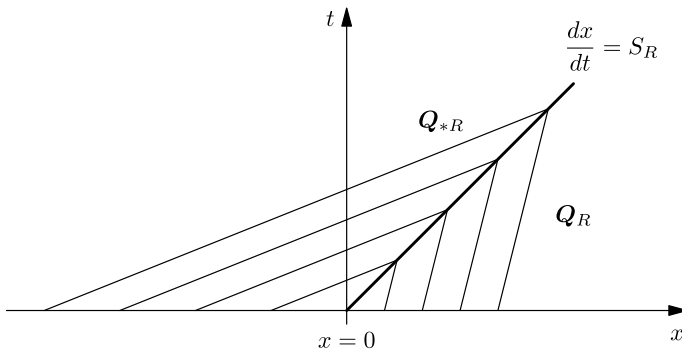


Fig. 9.8 Isolated right-facing shock wave of speed S_R associated with eigenvalue $\lambda_3 = u + c$, connecting constant states Q_R (ahead) and Q_{*R} (behind)

Also, from the first and the third equations we conclude that

$$\eta_{*R} = \eta_R, \quad (9.57)$$

that is, the concentration $\eta(x, t)$ is unaffected by the shock wave, just as for the rarefaction case. From the second equation in (9.55)

$$(A_{*R}\hat{u}_{*R})\hat{u}_{*R} + \gamma A_{*R}^{3/2} = (A_R\hat{u}_R)\hat{u}_R + \gamma A_R^{3/2}, \quad (9.58)$$

which, after using (9.56) yields

$$M_R = \gamma \left(\frac{A_{*R}^{3/2} - A_R^{3/2}}{\hat{u}_{*R} - \hat{u}_R} \right). \quad (9.59)$$

From (9.56)

$$\hat{u}_{*R} = -\frac{M_R}{A_{*R}}, \quad \hat{u}_R = -\frac{M_R}{A_R}, \quad (9.60)$$

which, if substituted into (9.59) give

$$M_R = \sqrt{\frac{\gamma A_R A_{*R} (A_{*R}^{3/2} - A_R^{3/2})}{A_{*R} - A_R}}. \quad (9.61)$$

From (9.54)

$$\hat{u}_{*R} - \hat{u}_R = u_{*R} - u_R, \quad (9.62)$$

from which (9.59) becomes

$$u_{*R} = u_R + \frac{\gamma(A_{*R}^{3/2} - A_R^{3/2})}{M_R}, \quad (9.63)$$

which leads to a relation between u_{*R} and u_R via a function f_R , namely

$$u_{*R} = u_R + f_R; \quad f_R = \sqrt{\frac{\gamma(A_{*R} - A_R)(A_{*R}^{3/2} - A_R^{3/2})}{A_R A_{*R}}}. \quad (9.64)$$

This is the **shock wave curve** for a right shock associated with the eigenvalue $\lambda_3 = u + c$. The sought function f_R has now been determined. Compare with f_R in Eq. (9.29) for the case of a right rarefaction. See Fig. 9.3. An expression for the speed of the right shock is obtained from (9.54) $S_R = u_R - \hat{u}_R$, which after using (9.60) gives

$$S_R = u_R + M_R/A_R. \quad (9.65)$$

We have completed the study of all possible waves in the solution of the Riemann problem for blood flow in arteries. See Fig. 9.2. Compared to the *all-rarefaction case* it would be interesting to consider the *all-shock case*. Unfortunately, for this wave pattern it is not possible to obtain closed-form exact solutions. We recall that the middle linear wave remains a contact discontinuity, across which only the passive scalar changes, discontinuously. As a matter of fact, the solution for the concentration $\eta(x, t)$ is virtually decoupled from the rest of the solution of the Riemann problem and can simply be added once the advection speed u_* has been determined.

Next we collect all the equations for the general case in which the non-linear waves could be either shocks or rarefactions.

9.4 Assembling the Equations and the Solution

So far we have developed all the wave relations needed to find A_* and u_* in the *Star Region*, noting that the solution for the concentration $\eta(x, t)$ is decoupled from the remaining variables and may thus temporarily be neglected. Recall that we first connected the single unknown A_* to the data state $\mathbf{Q}_L$ and to the data state $\mathbf{Q}_R$ via two wave curves determined by two functions

$$f_L = f_L(A_*; A_L); \quad f_R = f_R(A_*; A_R), \quad (9.66)$$

as illustrated in Fig. 9.3. These functions depend on A_* , the *wave type* (shock or rarefaction) and the initial conditions for the cross-sectional area A_L and A_R . We are now in the position to state theorem 9.1 below.

Theorem 9.1 *The solution A_* in the Star Region for the Riemann problem (9.10) for arteries is the root of the non-linear algebraic equation*

$$f(A) \equiv f_L(A; A_L) + f_R(A; A_R) + \Delta u = 0, \quad \Delta u \equiv u_R - u_L, \quad (9.67)$$

where for the left wave

$$f_L = \begin{cases} 4(c - c_L) & \text{if } A \leq A_L \text{ (rarefaction),} \\ \sqrt{\frac{\gamma(A - A_L)(A^{3/2} - A_L^{3/2})}{A_L A}} & \text{if } A > A_L \text{ (shock)} \end{cases} \quad (9.68)$$

and for the right wave

$$f_R = \begin{cases} 4(c - c_R) & \text{if } A \leq A_R \text{ (rarefaction),} \\ \sqrt{\frac{\gamma(A - A_R)(A^{3/2} - A_R^{3/2})}{A_R A}} & \text{if } A > A_R \text{ (shock).} \end{cases} \quad (9.69)$$

Once the root A_* of (9.67) is known, the solution for the velocity u_* follows as

$$u_* = \frac{1}{2}(u_L + u_R) + \frac{1}{2}[f_R(A_*, A_R) - f_L(A_*, A_L)]. \quad (9.70)$$

Proof We present a sketch of the proof. First recall that the particle velocity u_* and the cross-sectional area A_* are constant in the *Star Region*. Then, the function f_L is used to relate u_* to the left initial condition $\mathbf{Q}_L$ across the left wave. In case the left wave is a rarefaction we use (9.23) and if the left wave is a shock we use the relation (9.52). Analogously, the function f_R is used to relate u_* to the right initial condition $\mathbf{Q}_R$ across the right wave. If the right wave is a rarefaction we use relation (9.29) and if it is a shock we use (9.64). As u_* is constant in the Star Region, we can eliminate u_* resulting in Eq. (9.67), as claimed.

Then the particle velocity could be written in terms of the function f_L , for both the rarefaction and shock cases; see (9.23) and (9.52). So we could compute u_* directly from f_L once A_* is known. Alternatively, we could compute u_* directly from f_R using (9.29) or (9.64). Solution (9.70) results from the arithmetic mean of the two possible solutions. This concludes the outline of the proof. ■

9.4.1 Remarks on Wave Curves

The solution of the Riemann problem may be approached in several ways. One possibility is to interpret the derived four wave curves by plotting their locus in the pressure-velocity plane, as presented by Courant and Friedrichs for the Euler equations, as early as 1948 [3]. See also the discussion in [4]. The sought solution results from the intersection of the wave curves in the pressure-velocity plane. This approach is useful in providing insight into the features of the solution.

Alternatively, one can proceed as suggested by Toro [5] for the Euler equations. This approach results in a single non-linear equation $f(A) = 0$ to be solved, providing a very efficient method; see Eq. (9.67). The sought solution A_* is the intersection of the function $f(A) = 0$ with the A -axis, as shown in Fig. 9.9. First note that the function $f(A)$ is monotone increasing and concave down. For the physically admissible $A \geq 0$, the sought solution for the cross-sectional area exists and is unique. See [6] for detailed analysis of $f(A)$ wave curves.

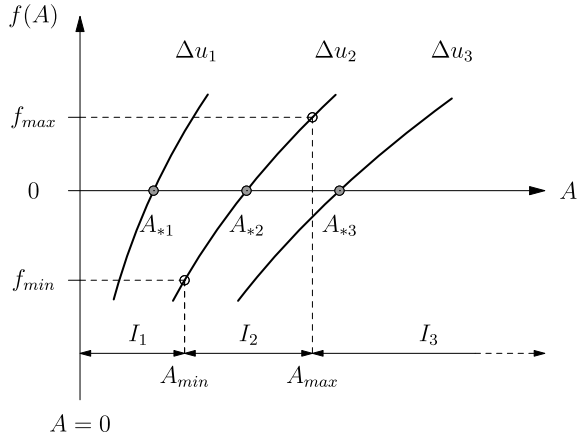
As seen in Fig. 9.9 there are three possibilities, for given initial data $\mathbf{W}_L = [A_L, u_L]$ and $\mathbf{W}_R = [A_R, u_R]$, namely

1. **Rarefaction/shock case.** The velocity difference $\Delta u = u_R - u_L$ in (9.67) is such that the zero of the function $f(A)$ satisfies

$$A_{min} := \min\{A_L, A_R\} \leq A_* = A_{*2} \leq A_{max} := \max\{A_L, A_R\}. \quad (9.71)$$

The solution lies in interval $I_2 = [A_{min}, A_{max}]$, as illustrated in Fig. 9.9 for $\Delta u = \Delta u_2$. Here there are two possible wave patterns: shock/rarefaction or rarefaction/shock. See Fig. 9.2.

Fig. 9.9 Behaviour of the function $f(A)$ and the solution A_* . The curve $f(A)$ is monotone increasing and concave down and the solution for A_* exists and is unique in the physically admissible range of non-negative cross-sectional area. There are three cases, which depend on the size of the velocity difference $\Delta u = u_R - u_L$



2. **Rarefaction/rarefaction case.** The velocity difference Δu is such that the zero of the function $f(A)$ satisfies

$$A_* = A_{*1} \leq A_{min} \leq A_{max}. \quad (9.72)$$

The solution lies in interval $I_1 = [0, A_{min}]$, as illustrated in Fig. 9.9 for $\Delta u = \Delta u_1$.

3. **Shock/shock case.** The velocity difference Δu is such that the zero of the function $f(A)$ satisfies

$$A_* = A_{*3} \geq A_{max} \geq A_{min}. \quad (9.73)$$

The solution lies in interval $I_3 = [A_{max}, \infty)$, as illustrated in Fig. 9.9 for $\Delta u = \Delta u_3$.

9.4.2 Numerical Solution in the Star Region

Here we compute the solution for the cross-sectional area A_* and the particle velocity u_* in the *Star Region*. The solution for $\eta(x, t)$ follows trivially, as we shall see. To find A_* we solve numerically the algebraic non-linear Eq. (9.67), for which several iterative methods are available in the literature. Here we use the Newton-Raphson method

$$A^{(k+1)} = A^{(k)} - \frac{f(A^{(k)})}{f'(A^{(k)})}, \quad \text{for } k = 0, 1, \dots, K. \quad (9.74)$$

The iterative process (9.74) is stopped whenever the relative change ΔA in A is smaller than a prescribed, small positive tolerance $TOL > 0$, that is when

$$\Delta A \equiv \frac{|A^{(k+1)} - A^{(k)}|}{(A^{(k+1)} + A^{(k)})/2} \leq TOL. \quad (9.75)$$

Typically, one takes $TOL = 10^{-6}$. Knowing A_* , the solution for u_* follows directly from (9.70).

Guess value for A_* . The Newton-Raphson method (9.74) requires a guess value $A^{(0)}$ to start the iteration. Care is required in providing a reliable initial guess to avoid negative iterates and thus failure of the iteration process. Here it is suggested to use the *all-rarefaction* solution (9.38), (9.40) for the initial guess, repeated here

$$A^{(0)} = \left(\frac{2c_*^2}{3\gamma} \right)^2, \quad c_* = \frac{1}{2}(c_L + c_R) - \frac{1}{8}(u_R - u_L). \quad (9.76)$$

Recall that this has been obtained by assuming *a priori* that both non-linear waves are rarefaction waves, which in fact is the exact solution if effectively the actual solution contains two rarefaction waves.

Existence and uniqueness. By analysing the function $f(A)$ it can be shown, that the solution for A_* exists and is unique, within a physically admissible range for the parameters of the problem.

In the next section we describe a procedure to evaluate the solution across the complete wave structure and to determine the solution profile $\mathbf{Q}(x, \hat{t})$ for any chosen time $t = \hat{t}$.

9.4.3 Solution Sampling Procedure

So far we know the solution in the *Star Region* for A_* and u_* , and the solution inside rarefactions, if present in the solution. Now, given any given point (x, t) in the x - t half plane, $-\infty < x < \infty$ and $t \geq 0$, we want to compute the solution $\mathbf{Q}(x, t)$; see Fig. 9.10. We call this task the *sampling procedure*. To facilitate the sampling we first introduce the *contact discontinuity* path

$$x/t = u_* \quad (9.77)$$

and divide the half-plane into two disjoint subregions, namely

$$\mathcal{R}_L = \left\{ (x, t) / \frac{x}{t} < u_* \right\}, \quad \mathcal{R}_R = \left\{ (x, t) / \frac{x}{t} > u_* \right\}. \quad (9.78)$$

Next, for a specified time $t = \hat{t}$ we determine a spatial sampling interval $[x_L, x_R]$ such that

$$x_L < \hat{t} \hat{S}_L - \epsilon, \quad x_R > \hat{t} \hat{S}_R + \epsilon, \quad (9.79)$$

where $\hat{S}_L$ and $\hat{S}_R$ are the smallest and largest signal speeds in the solution of the Riemann, and $\epsilon > 0$ is a small, conveniently chosen parameter with dimensions of length. To sample the solution we introduce the **velocity** sampling parameter

$$\xi = x/\hat{t}. \quad (9.80)$$

Then, for the given time $\hat{t}$, the solution $\mathbf{Q}(x, \hat{t})$, called the profile at time $\hat{t}$, is only a function of $x \in [x_L, x_R]$ containing the full wave system, as depicted in Fig. 9.10. To perform the sampling we use the physical variables $\mathbf{W} = [A, u, \eta]^T$, rather than the conserved variables, and make use of ξ to locate the sampling point and assign the corresponding solution $\mathbf{W}(\xi)$. There are two cases:

1. **The sampling point lies to the left of the contact.** The solution $\mathbf{W}(\xi)$ for the chosen $(x, \hat{t})$ belongs to region $\mathcal{R}_L$ and depends on the type of wave present in that region. There are two possibilities:
 - **Left shock.** If the left wave is a shock of speed S_L , then $\mathcal{R}_L$ is again subdivided into two subregions and the solution is

$$\mathbf{W}(\xi) \equiv \begin{cases} \mathbf{W}_* = [A_*, u_*, \eta_L]^T & \text{if } S_L \leq \xi \leq u_*, \\ \mathbf{W}_L = [A_L, u_L, \eta_L]^T & \text{if } \xi < S_L, \end{cases} \quad (9.81)$$

where the shock speed S_L is given by (9.53).

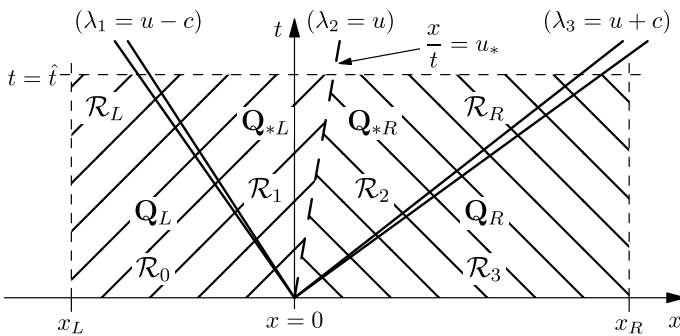


Fig. 9.10 Sampling the solution profile $\mathbf{Q}(x, \hat{t})$ across the complete wave structure, at time $t = \hat{t}$. To ease the task the domain is divided into two subregions by the path of the contact discontinuity $x/t = u_*$. The sampling interval $[x_L, x_R]$ is chosen in accordance with $\hat{t}$ so as to include the full wave system in the sampling

- **Left rarefaction.** If the left wave is a rarefaction, then $\mathcal{R}_L$ is subdivided into three subregions and the solution is

$$\mathbf{W}(\xi) = \begin{cases} \mathbf{W}_L = [A_L, u_L, \eta_L]^T & \text{if } \xi < u_L - c_L, \\ \mathbf{W}_{Lfan} = [\hat{A}_L, \hat{u}_L, \eta_L]^T & \text{if } u_L - c_L \leq \xi \leq u_* - c_*, \\ \mathbf{W}_* = [A_*, u_*, \eta_L]^T & \text{if } u_* - c_* \leq \xi \leq u_*, \end{cases} \quad (9.82)$$

where $\hat{u}_L$ and $\hat{c}_L$ denote the particle velocity and the wave speed inside the left rarefaction, given by (9.26); $\hat{A}_L$ can be derived from $\hat{c}_L$ using the definition of wave speed (9.8).

2. **The sampling point lies to the right of the contact.** The solution $\mathbf{W}(\xi)$ for given $(x, \hat{t})$ belongs to region $\mathcal{R}_R$ and depends on the type of the right wave present. Again there are two possibilities.

- **Right shock.** If the right wave is a shock wave of speed S_R , then $\mathcal{R}_R$ is subdivided into two subregions and the solution is

$$\mathbf{W}(\xi) \equiv \begin{cases} \mathbf{W}_* = [A_*, u_*, \eta_R]^T & \text{if } u_* \leq \xi \leq S_R, \\ \mathbf{W}_R = [A_R, u_R, \eta_R]^T & \text{if } \xi > S_R, \end{cases} \quad (9.83)$$

where the shock speed S_R is given by (9.65).

- **Right rarefaction.** If the right wave is a rarefaction, then $\mathcal{R}_R$ is subdivided into three subregions and the solution is

$$\mathbf{W}(\xi) = \begin{cases} \mathbf{W}_R = [A_R, u_R, \eta_R]^T & \text{if } \xi > u_R + c_R, \\ \mathbf{W}_{Rfan} = [\hat{A}_R, \hat{u}_R, \eta_R]^T & \text{if } u_* + c_* \leq \xi \leq u_R + c_R, \\ \mathbf{W}_* = [A_*, u_*, \eta_R]^T & \text{if } u_* \leq \xi \leq u_* + c_*, \end{cases} \quad (9.84)$$

where $\hat{u}_R$ and $\hat{c}_R$ inside the right rarefaction are given by (9.30); $\hat{A}_R$ can be derived from $\hat{c}_R$ using the definition of wave speed (9.8).

The Vacuum State

So far in this chapter we have assumed that the vacuum state (no blood) is absent, that is $A(x, t) > 0$. However, as shown by Toro and Sivilgia [7], the Riemann problem for arteries admit the vacuum state, either as an initial condition or as a result of interaction of the data states via the Riemann problem, in which case the **Star Region** is a vacuum state. For arteries the admission of vacuum is controversial, under physiological conditions. For veins, however, it makes more sense to allow for the vacuum state, as veins are collapsible under physiological conditions. This point will be discussed later in Sec. 9.5 about veins.

The complete solution process of the Riemann problem and evaluation of the exact solution has been concluded. In the next section we present results for some specific examples.

9.4.4 Numerical Examples for Arteries

In this section we evaluate the exact solution of four Riemann problems for the equations for blood flow in arteries with constant mechanical and geometrical parameters.

Definition of Test Problems

We have selected a suite of four problems. Parameters of the problems are given in Table 9.1 and initial conditions for the four test problems are given in Table 9.2.

Exact Solutions

Exact solutions for **Test 1** and **Test 2** are shown in Fig. 9.11, in left and right columns respectively. In both cases the exact solution consists of a left rarefaction, a right-travelling contact discontinuity and a right shock. Two differences to note. The contact discontinuity in Test 2 is faster than that of Test 1 and the left rarefaction in Test 2 is transcritical, or transonic, or sonic, or critical.

Exact solutions for **Test 3** and **Test 4** are shown in Fig. 9.12, in left and right columns respectively. The solution of Test 3 consists of two rarefactions and a stationary contact discontinuity; the solution of Test 4 consists of two shocks and a stationary contact discontinuity.

Table 9.1 Parameters for Riemann problems in Table 9.2

Domain length	x_0	h_0	A_0	E	μ	ρ
0.5	0.25	5.0×10^{-4}	3.14×10^{-4}	3.0×10^5	0.5	1000

Table 9.2 Four Riemann problems for the augmented equations for blood flow in arteries. The last column describes the wave pattern that emerges from the exact solution of the Riemann problem, where R: rarefaction; C: contact and S: shock

Test	A_L	u_L	ψ_L	A_R	u_R	ψ_R	T_{out}	Wave pattern
1	3.5×10^{-4}	0.0	1.0	3.0×10^{-4}	0.0	0.0	0.05	RCS
2	10.0×10^{-4}	0.0	1.0	1.0×10^{-4}	0.0	0.0	0.04	R(sonic)CS
3	3.14×10^{-4}	-0.5	1.0	3.14×10^{-4}	0.5	0.0	0.05	RCR
4	3.14×10^{-4}	0.5	1.0	3.14×10^{-4}	-0.5	0.0	0.05	SCS

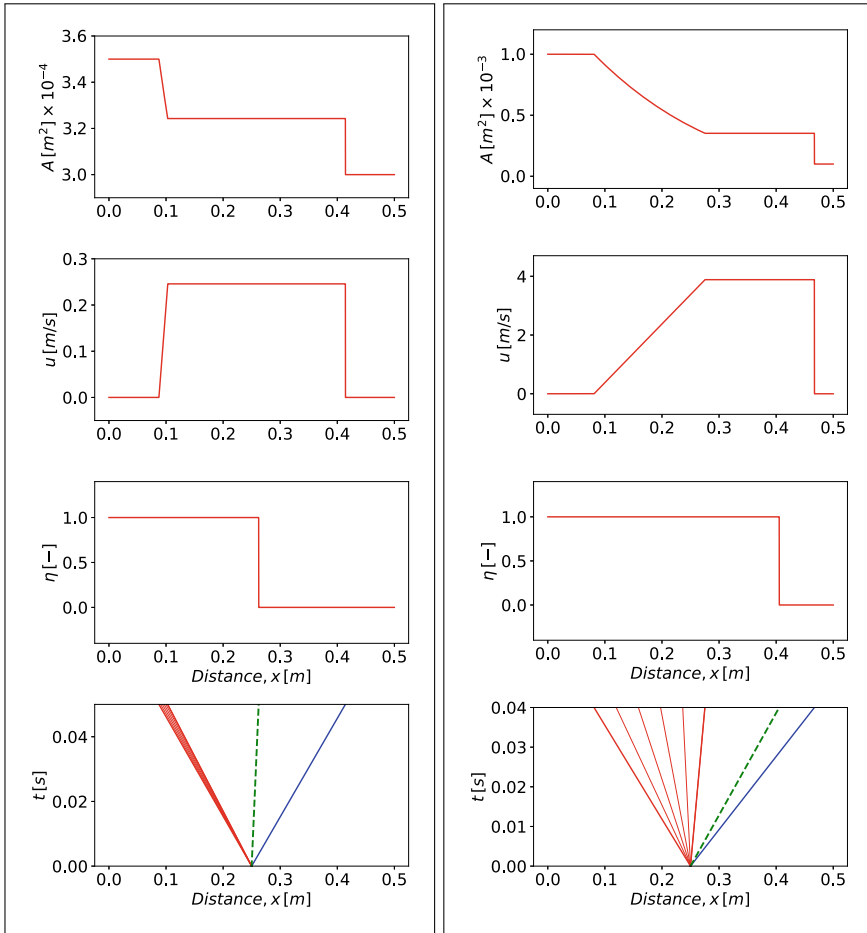


Fig. 9.11 Exact solution of Riemann problems **Test 1** (left column) and **Test 2** (right column) at times $T_{out} = 0.05$ s and $T_{out} = 0.04$ s, respectively. For each test we show, from top to bottom, cross-sectional area A , velocity u , passive scalar η and the corresponding wave pattern in the x - t plane. The solution for Test 1 consists of a left rarefaction, a right shock and a right-moving contact. The full wave pattern is sub-critical. The solution for Test 2 exhibits an analogous wave pattern, but the left rarefaction is **transcritical**, or **sonic**

9.5 The Riemann Problem for Blood Flow in Veins

In this section we solve exactly the Riemann problem for blood flow in veins. As a matter of fact the equations for the dynamics of blood flow are identical to those for blood flow in arteries. The difference between arteries and veins stems from the tube law, which accounts for the mechanical behaviour of the vessel wall. We first recall the equations and a generalised elastic tube law that includes both arteries and veins.

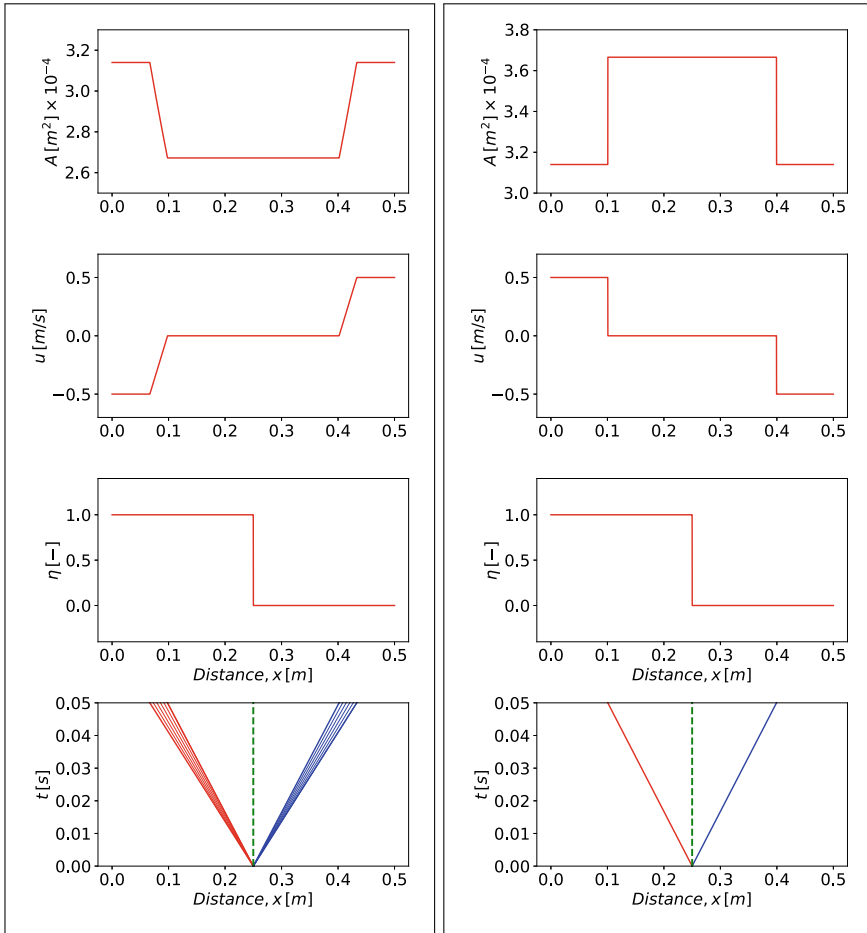


Fig. 9.12 Exact solution of Riemann problems **Test 3** (left column) and **Test 4** (right column) at time $T_{out} = 0.05$ s, for both tests. For each test we show, from top to bottom, cross-sectional area A , velocity u , passive scalar η and the corresponding wave pattern in the x - t plane. The solution for Test 3 consists of two rarefaction waves and a stationary contact. The full wave pattern is sub-critical. The solution for Test 4 consists of two shock waves and a stationary contact discontinuity. The flow pattern is sub-critical

9.5.1 Equations and Tube Law

The mechanical behaviour of the vessel wall of veins is substantially different to that in arteries. Veins collapse under physiological conditions. A familiar example is furnished by the internal jugular veins in humans. In most healthy individuals, in the sitting position, such veins are collapsed; in this case venous blood from the brain is re-directed to the vertebral veins and other collaterals.

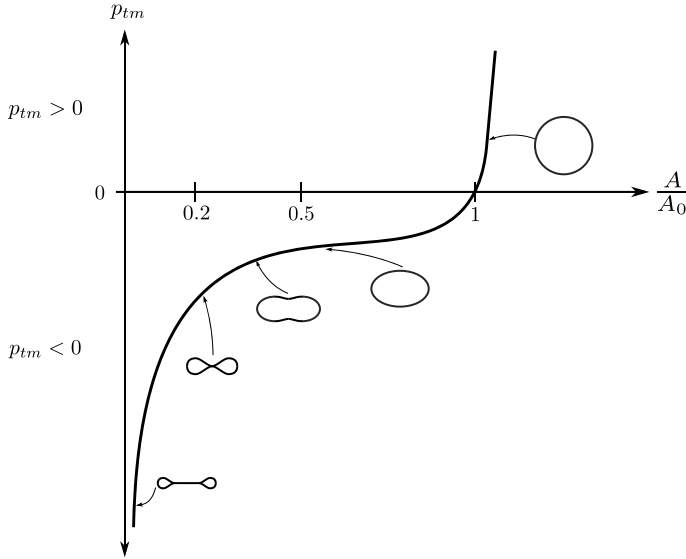


Fig. 9.13 Tube law for veins. Variation of transmural pressure $p = p - p_e$ with normalized cross-sectional area for veins, along with the corresponding changes in shape of the cross-sectional area for veins, as the transmural pressure varies. Adapted from Shapiro [8]

The high deformability of veins represents a challenge to mathematical models and numerical methods, much more so than for arteries. Figure 9.13 is a sketch of the tube law for veins, depicting the transmural pressure as a function of normalised cross-sectional area. Note the characteristic sigmoidal behaviour of the pressure function and the extreme changes in vein cross-sectional area, from a circular shape at high transmural pressures to a virtually collapsed state at low (large negative) transmural pressures. Also, the highly non-linear behaviour reveals how very small changes in transmural pressure cause large variations in cross-sectional area, consistent with the highly deformable behaviour of veins. In order to allow for both arteries and veins in our blood flow models we generalise the *elastic tube law* (9.3) as follows

$$p(x, t) = p_e(x, t) + K(x)\phi(A(x, t); A_0(x)), \quad (9.85)$$

with

$$\phi(A(x, t); A_0(x)) = \left[\left(\frac{A(x, t)}{A_0(x)} \right)^m - \left(\frac{A(x, t)}{A_0(x)} \right)^n \right]. \quad (9.86)$$

The distinction between arteries and veins comes through the *coefficient of elasticity* $K(x)$ and the exponents m and n , namely

$$K(x) = \begin{cases} K_a(x) = \frac{E(x)}{1-\nu^2} \frac{h_0(x)}{r_0(x)}, & m = \frac{1}{2}, n = 0 \quad (\text{arteries}), \\ K_v(x) = \frac{E(x)}{12(1-\nu^2)} \left(\frac{h_0(x)}{r_0(x)} \right)^3, & m \approx 10, n = -\frac{3}{2} \quad (\text{veins}). \end{cases} \quad (9.87)$$

Here $p_e(x, t)$ is the external pressure, $A_0(x)$ is vessel cross-sectional area at equilibrium, ν , $h_0(x)$, $r_0(x)$, $E(x)$ are respectively the Poisson ratio (set to $\nu = 1/2$), wall-thickness at equilibrium, cross-sectional radius at equilibrium and Young's modulus of elasticity. Generally, the parameters $m \geq 0$ and $n \leq 0$ are real numbers, which for hyperbolicity must satisfy some constraints, as described in [9]. See also Chap. 10.

Remark: In the definition (9.87) for the coefficient of elasticity $K(x)$ we have used the vessel radius, instead of the area as done for arteries in Eq. (9.4). The arterial case in Eq. (9.4) is reproduced by $K_a(x)$ in (9.87) for $m = 1/2$ and $n = 0$.

Regarding the differential equations for the dynamics of blood flow we restrict our attention here to the classical 2×2 system for mass and momentum, neglecting the presence of a passive scalar. There is no fundamental reason for this choice. Regarding geometrical and mechanical parameters along the length of the vessel, as done for the arterial case we assume them to be constant. The general tube law (9.85)–(9.87), which accommodates both arteries and veins, is built into the momentum equation. The resulting system, written in conservation-law form, reads

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}), \quad (9.88)$$

where

$$\left. \begin{aligned} \mathbf{Q} &= \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \equiv \begin{bmatrix} A \\ Au \end{bmatrix} \\ \mathbf{F}(\mathbf{Q}) &= \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \equiv \begin{bmatrix} Au \\ Au^2 + \frac{K}{\rho} A_0 \left\{ \frac{m}{m+1} \left(\frac{A}{A_0} \right)^{m+1} - \frac{n}{n+1} \left(\frac{A}{A_0} \right)^{n+1} \right\} \end{bmatrix} \\ \mathbf{S}(\mathbf{Q}) &= \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \end{aligned} \right\} \quad (9.89)$$

We have assumed $\alpha = 1$ for the Coriolis coefficient and zero source terms.

The eigenstructure of the system is as for the arterial case, which we repeat here. The eigenvalues and eigenvectors are

$$\lambda_1 = u - c, \quad \lambda_2 = u + c, \quad (9.90)$$

with corresponding right eigenvectors

$$\mathbf{R}_1 = \beta_1 \begin{bmatrix} 1 \\ u - c \end{bmatrix}, \quad \mathbf{R}_2 = \beta_2 \begin{bmatrix} 1 \\ u + c \end{bmatrix}, \quad (9.91)$$

where β_1 and β_2 are arbitrary scaling factors, which unless otherwise stated, will be chosen as $\beta_1 = \beta_2 = 1$. Now the **wave speed** c for veins is

$$c = \sqrt{\frac{K_v}{\rho} \left[m \left(\frac{A}{A_0} \right)^m - n \left(\frac{A}{A_0} \right)^n \right]}. \quad (9.92)$$

For convenience c may be written as

$$c = \sqrt{(m+1) \gamma_1 A^m + (n+1) \gamma_2 A^n}, \quad (9.93)$$

where

$$\gamma_1 = \frac{m K_v}{(m+1) \rho A_0^m} > 0 \quad \text{and} \quad \gamma_2 = -\frac{n K_v}{(n+1) \rho A_0^n} < 0. \quad (9.94)$$

Sufficient conditions for the wave speed to be real, and thus for hyperbolicity, are $m \geq 0$ and $n \leq 0$.

9.5.2 Solution of the Riemann Problem for Veins

The Riemann problem is the special Cauchy problem

$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0}, \quad -\infty < x < \infty, \quad t > 0, \\ \text{ICs: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{Q}_L : \text{constant} & \text{if } x < 0, \\ \mathbf{Q}_R : \text{constant} & \text{if } x > 0. \end{cases} \end{array} \right\} \quad (9.95)$$

Here the initial conditions are given by the two constant states $\mathbf{Q}_L$ and $\mathbf{Q}_R$. Useful background on the Riemann problem for linear systems is found in Chap. 7. Details on the Riemann problem for blood flow in arteries are given in Sec. 9.1. Figure 9.14 depicts the structure of the solution of the Riemann problem (9.95) in the x - t plane. As for the arterial case, the main step in determining the exact solution consists of computing the **Star State** $\mathbf{Q}_*$, that is the cross sectional area A_* and the particle velocity u_* in the **Star Region**. See Theorem 9.2 below.

Theorem 9.2 *The solution A_* in the Star Region for the Riemann problem (9.95) for veins is the root of the non-linear algebraic equation*

$$f(A) \equiv f_L(A; A_L) + f_R(A; A_R) + \Delta u = 0, \quad \Delta u = u_R - u_L, \quad (9.96)$$

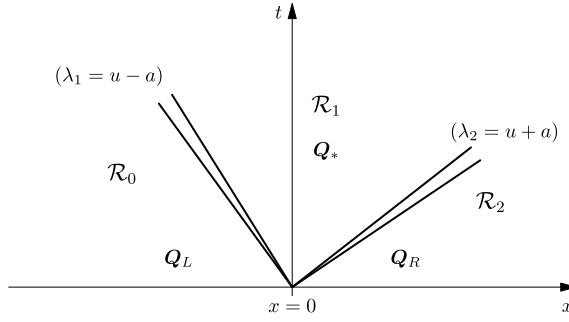


Fig. 9.14 Structure of the solution of the Riemann problem (9.95) for veins in x - t plane with initial conditions $\mathbf{Q}_L$ and $\mathbf{Q}_R$. There are three constant regions separated by two non-linear wave families. As we have chosen to exclude the equation for a passive scalar η , there is no contact discontinuity present. The unknowns of the problem are the vector $\mathbf{Q}_*$ in the **Star Region** and the wave types, which can be either shocks or rarefactions

where the functions f_K , for $K = L, R$, are

$$f_K = \begin{cases} \int_{A_K}^A \frac{c(\tau)}{\tau} d\tau & \text{if } A \leq A_K, \\ \sqrt{\frac{A - A_K}{A A_K} [\gamma_1 (A^{m+1} - A_K^{m+1}) + \gamma_2 (A^{n+1} - A_K^{n+1})]} & \text{if } A > A_K. \end{cases} \quad (9.97)$$

The solution for the particle velocity u_* in the **Star Region** follows as

$$u_* = \frac{1}{2} (u_L + u_R) + \frac{1}{2} (f_R(A_*; A_R) - f_L(A_*; A_L)). \quad (9.98)$$

Proof The proof is similar to that for the Riemann problem for arteries and involves the application of Rankine-Hugoniot jump conditions across shock waves and generalised Riemann invariants across rarefaction waves. Full details are found in [10].

■

Numerical Solution in the Star Region

To solve the non-linear Eq. (9.96) one can use, for example, a Newton-Raphson or a Regula-Falsi method. Care is required in choosing the *initial guess* for a Newton-Raphson method. For arteries, if one assumes that both non-linear waves are rarefaction waves, then both the velocity u_* and the cross-sectional area A_* can be explicitly calculated; see Sect. 9.2.3. The resulting A_* can be used as an initial guess for the complete non-linear problem. However for veins the situation is more complicated and it seems as if at present there is no *all-rarefaction counterpart*. A mean value of the data, if used with caution, is a practical option for the initial guess in a Newton-Raphson iteration procedure.

We note that for veins, assuming that the two non-linear waves are rarefaction waves, the following expression for u_* can be formally written

$$u_* = \frac{1}{2} (u_L + u_R) - \frac{1}{2} \int_{A_L}^{A_R} \frac{c(\tau)}{\tau} d\tau. \quad (9.99)$$

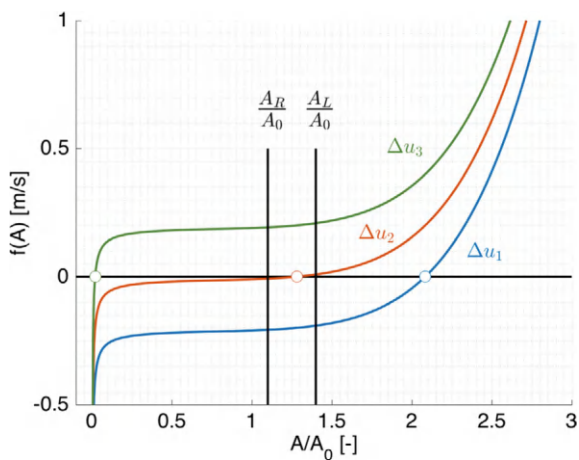
But unlike the arterial case, full determination of u_* for veins requires numerical computation of the integral on the right hand side of (9.99). This expression could be useful if instead of the non-linear Eq. (9.96) for A one could formulate an equation in terms of u_* , then (9.99) could be used as a reliable guess value for the iteration. Once the solution in the *Star Region* has been calculated, the complete solution can be obtained through a sampling procedure, as done for the arterial case in Sec. 9.4.3.

In the next section we discuss some distinguishing properties of the solution of the Riemann problem for veins. The discussion is based on the work of Spiller et al. [10], where full details are found.

9.5.3 Distinguishing Properties of the Solution

Here we study some distinguishing properties of the solution of the Riemann problem for veins. We start by analysing the behaviour of the function $f(A)$ in (9.96). Figure 9.15 depicts $f(A)$ for fixed data values A_L, A_R for cross-sectional area and for three representative values of the parameter $\Delta u = u_R - u_L$. The cross-sectional area shown in the horizontal axis is dimensionless, as is normalised with respect to the equilibrium cross-sectional area A_0 ; the function $f(A)$ in the vertical axis has dimensions of velocity. For the plots of Fig. 9.15 we have defined

Fig. 9.15 Behaviour of function $f(A)$ in (9.96) for two prescribed normalised cross-sectional areas, with $A_R < A_L$, and three prescribed values of the velocity difference $\Delta u = u_R - u_L$. Here $A_L = 1.4A_0$, $A_R = 1.1A_0$, $\Delta u_1 = -0.2$ [m/s], $\Delta u_2 = 0$ [m/s] and $\Delta u_3 = 0.2$ [m/s]. The root, shown by white a circle, decreases as Δu increases



$$A_m = \min \left\{ \frac{A_L}{A_0}, \frac{A_R}{A_0} \right\}, \quad A_M = \max \left\{ \frac{A_L}{A_0}, \frac{A_R}{A_0} \right\}, \quad (9.100)$$

with $A_m = \frac{A_R}{A_0}$ and $A_M = \frac{A_L}{A_0}$. Each of the three values of $\Delta u = u_R - u_L$ represents a distinct class of solutions, as detailed below:

1. **Two-shock solution:** curve $f(A)$ for Δu_1 . The corresponding solution A_{*1} in the *Star Region* is obviously the intersection of the curve $f(A)$ with the horizontal axis and is marked with a white circle. Note that

$$A_{*1} > A_M > A_m.$$

2. **Rarefaction/shock solution:** curve $f(A)$ for Δu_2 . The solution A_{*2} satisfies

$$A_M > A_{*2} > A_m.$$

3. **Two-rarefaction solution:** curve $f(A)$ for Δu_3 : The solution A_{*3} satisfies

$$A_M > A_m > A_{*3}.$$

Note that $\Delta u_1 < \Delta u_2 < \Delta u_3$ implies the corresponding roots satisfy $A_{*1} > A_{*2} > A_{*3}$. Furthermore, the root A_* tends to zero as Δu tends to infinity, as will be proved.

Next we prove some general properties of the function $f(A)$. For convenience, let us first rewrite f_K as follows

$$f_K = \begin{cases} \int_{A_K}^A \frac{c(\tau)}{\tau} d\tau & \text{if } A \leq A_K, \\ \sqrt{\frac{A-A_K}{AA_K}} g(A) & \text{if } A > A_K, \end{cases} \quad (9.101)$$

where

$$g(A) = \gamma_1(A^{m+1} - A_K^{m+1}) + \gamma_2(A^{n+1} - A_K^{n+1}). \quad (9.102)$$

Lemma 1 *The function $g(A)$ in (9.102) satisfies the following properties*

1. $g'(A) = c^2(A) > 0$ for $A \in (0, +\infty)$,
2. $g \in C^1(0, +\infty)$,
3. $g(A_K) = 0$,
4. $g(A) > 0$ for $A > A_K$,
5. $\lim_{A \rightarrow A_K} \frac{g(A)}{A - A_K} = g'(A_K) = c^2(A_K)$.

Proof The derivative of g is

$$g'(A) = (m+1)\gamma_1 A^m + (n+1)\gamma_2 A^n, \quad (9.103)$$

which is positive, as all terms are positive. Moreover, by comparing (9.103) to (9.93) we see that $g'(A) = c^2$, as claimed. The second point above follows from the fact that g and its derivative g' are continuous. The third point follows from direct evaluation of the function (9.102). The fourth point follows from the first and third points. The proof for the last point follows from direct evaluation of the following

$$\lim_{A \rightarrow A_K} \frac{g(A) - g(A_K)}{A - A_K} = g'(A_K) = c^2(A_K). \quad (9.104)$$

■

As a consequence of the Lemma just proved, we can also state that the function $f(A)$ in (9.96) takes on real values for $A \in (0, +\infty)$.

Lemma 2 *The function $f(A)$ in (9.96) satisfies the following properties:*

1. $f \in C^1(0, +\infty)$,
2. f is strictly increasing,
3. $\lim_{A \rightarrow +\infty} f(A) = +\infty$.

Proof It is sufficient to prove the claimed properties for the functions f_K in (9.101). Concerning point 1 above we note that the functions f_K are continuous at $A = A_K$ and continuous in their entire domain. Moreover, direct evaluation of the derivative of f_K gives

$$f'_K(A) = \begin{cases} \frac{c(A)}{A} & \text{if } A \leq A_K, \\ \frac{1}{2\sqrt{AA_K}} \sqrt{\frac{A - A_K}{g(A)}} \left[\frac{A_K}{A} \frac{g(A)}{A - A_K} + g'(A) \right] & \text{if } A > A_K. \end{cases} \quad (9.105)$$

It is obvious that the functions above are continuous and continuity also holds at the connecting points $A = A_K$; we have

$$\lim_{A \rightarrow A_K^+} f'_K(A) = \frac{\sqrt{g'(A_K)}}{A_K} = \frac{c(A_K)}{A_K}, \quad (9.106)$$

from which it follows that $f \in C^1$. Regarding point 2, note that the functions f'_K are always positive; this is obvious for the rarefaction branch, while for the shock branch this follows from Lemma 1. The third point results from a direct (omitted) calculation. Hence the proof of the lemma has been completed. ■

Lemma 3 *If $n < 0$, then the following integral diverges*

$$\int_0^A \frac{c(\tau)}{\tau} d\tau = \lim_{\hat{A} \rightarrow 0^+} \int_{\hat{A}}^A \frac{c(\tau)}{\tau} d\tau = +\infty. \quad (9.107)$$

Proof After inserting the wave speed (9.93) into the integral above we have

$$\left. \begin{aligned} & \int_0^A \frac{\sqrt{(m+1)\gamma_1\tau^m + (n+1)\gamma_2\tau^n}}{\tau} d\tau \\ & \lim_{\hat{A} \rightarrow 0^+} \int_{\hat{A}}^A \sqrt{(m+1)\gamma_1\tau^{m-2} + (n+1)\gamma_2\tau^{n-2}} d\tau \geq \\ & \lim_{\hat{A} \rightarrow 0^+} \int_{\hat{A}}^A \sqrt{(n+1)\gamma_2\tau^{n-2}} d\tau. \end{aligned} \right\} = \quad (9.108)$$

The inequality follows from the positivity of all quantities involved and for $\hat{A}$ small enough so that $\hat{A} < A$. Computing the right-hand side of inequality (9.108) one obtains

$$\lim_{\hat{A} \rightarrow 0^+} \int_{\hat{A}}^A \sqrt{(n+1)\gamma_2\tau^{n-2}} d\tau = \frac{2}{n} \sqrt{(n+1)\gamma_2} \lim_{\hat{A} \rightarrow 0^+} \left(A^{\frac{n}{2}} - \hat{A}^{\frac{n}{2}} \right) = +\infty. \quad (9.109)$$

In view of the theorem of divergence of improper integrals, since the integral (9.109) diverges, then the left-hand side of inequality (9.108) diverges, and the lemma is thus proved. ■

Lemma 4 *If $n < 0$ then*

$$\lim_{A \rightarrow 0^+} f(A) = -\infty. \quad (9.110)$$

Proof Assuming $A_L \neq 0$ and $A_R \neq 0$, for A sufficiently small we have $A < A_L$ and $A < A_R$. It follows that f is made up of two rarefaction branches and therefore

$$f(A) = \int_{A_R}^A \frac{c(\tau)}{\tau} d\tau + \int_{A_L}^A \frac{c(\tau)}{\tau} d\tau + u_R - u_L. \quad (9.111)$$

Computing $f(0)$ one obtains

$$f(0) = \int_{A_R}^0 \frac{c(\tau)}{\tau} d\tau + \int_{A_L}^0 \frac{c(\tau)}{\tau} d\tau + u_R - u_L. \quad (9.112)$$

In view of Lemma 3, the statement has been proved. ■

Theorem 9.3 *If $n < 0$ then the non-linear equation $f(A) = 0$ in (9.96) has a solution $A > 0$ for $\Delta u \in (-\infty, +\infty)$ and the solution is unique.*

Proof From Lemma 4, Lemma 2 and Bolzano's theorem [11], the claim follows. ■

Theorem 9.4 *Let $A_* \equiv A_*(\Delta u)$ be the solution of the non-linear equation $f(A) = 0$ in (9.96). Then*

$$\lim_{\Delta u \rightarrow +\infty} A_*(\Delta u) = 0. \quad (9.113)$$

Proof Let us define h as

$$h(A) \equiv f_L(A; A_L) + f_R(A; A_R), \quad (9.114)$$

where f_L and f_R are defined in (9.97). The function h does not depend on Δu . Therefore, from (9.96)

$$f(A) = h(A) + \Delta u. \quad (9.115)$$

Since A_* is the solution of $f(A) = 0$, then

$$h(A_*) = -\Delta u. \quad (9.116)$$

From Lemmas 2 and 4 it follows that $h \in C^1(\mathbb{R}^+)$, h is strictly increasing and the image of h is $\mathbb{R}$. Therefore, the inverse h^{-1} of h is well defined from $\mathbb{R}$ to $\mathbb{R}^+$, and hence

$$A_* = h^{-1}(-\Delta u). \quad (9.117)$$

From Lemma 4, it can be deduced that the function h satisfies

$$\lim_{A \rightarrow 0^+} h(A) = -\infty \quad (9.118)$$

and the inverse h^{-1} satisfies

$$0 = \lim_{\Delta u \rightarrow -\infty} h^{-1}(\Delta u) = \lim_{\Delta u \rightarrow +\infty} h^{-1}(-\Delta u). \quad (9.119)$$

Inserting (9.117) into (9.119) we obtain

$$\lim_{\Delta u \rightarrow +\infty} A_*(\Delta u) = 0 \quad (9.120)$$

and the theorem is thus proved. ■

9.6 Representative Vein Riemann Problems

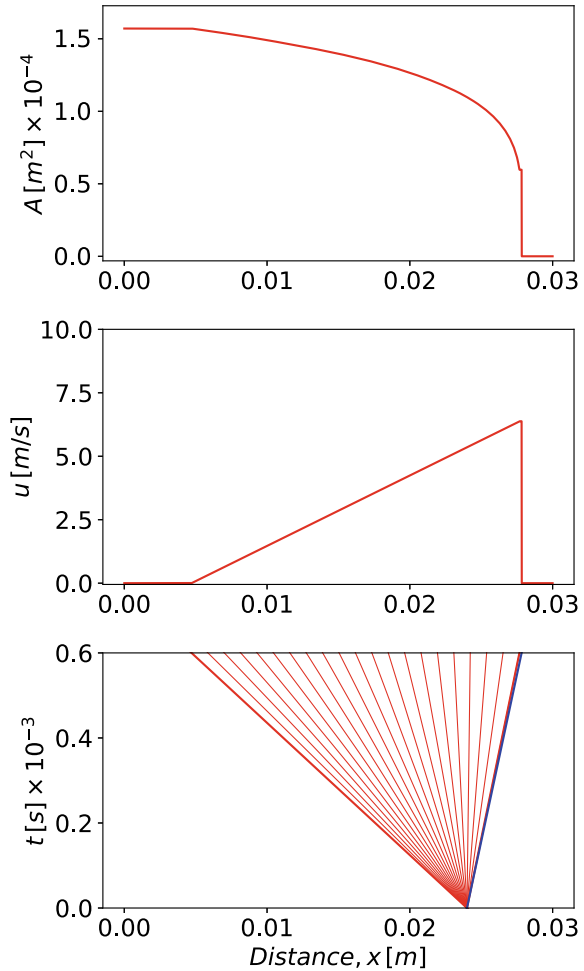
Here we show results for some selected examples and discuss the distinctive features of veins, as compared to arteries. We solve two types of Riemann problems that are illustrative of veins, as distinct from arteries. To this end we take data from [12] for the *right emissary vein*, located in the head, which has length $L = 3 \cdot 10^{-2}$ [m], equilibrium radius $r_0 = 10^{-3}$ [m], wall thickness $h_0 = r_0 10^{-2} = 10^{-5}$ [m] and equilibrium cross-sectional area $A_0 = \pi r_0^2$ [m²]. The density of blood is taken as $\rho = 1050$ [kg/m³] and the Young's modulus as $E = 10^5$ [Pa].

9.6.1 Near Vacuum Data State $A_R = A_\epsilon$

Here we consider the Riemann problem defined by the following initial conditions: $A_L = 5A_0$ [m^2], $u_L = 0$ [m/s], $A_R = A_\epsilon = 10^{-9}$ [m^2], $u_R = 0$ [m/s]. The position of the initial discontinuity is $x_0 = 0.8L$ [m] and output time is $T_{out} = 6 \cdot 10^{-4}$ [s]. The solution of this Riemann problem consists of a left transcritical rarefaction wave and a right shock wave, as depicted in Fig. 9.16.

The top frame depicts the normalised cross-sectional area, the middle frame shows the blood velocity and the bottom frame displays the x - t diagram of the resulting wave system. The solution for the cross-sectional area is highly non-linear, especially in the vicinity of $x \approx 0.9L$, near the tail of the rarefaction. The constant **Star Region**

Fig. 9.16 Case 1: The Riemann problem for a nearly collapsed right state. The initial condition is: $A_L = 5A_0$ [m^2], $u_L = 0$ [m/s], $A_R = A_\epsilon = 10^{-9}$ [m^2], $u_R = 0$ [m/s], $x_0 = 0.8L$ [m] and output time is $t = 6 \cdot 10^{-4}$ [s]. The solution consists of a left transcritical rarefaction and a right shock. The top frame depicts the normalised cross-sectional area, the middle frame the blood velocity and the bottom frame displays the x - t diagram of the resulting wave system. The **Star Region** that separates the rarefaction wave from the shock, is very narrow and hardly visible, that is, the tail of the rarefaction almost coincides with the shock path



that separates the rarefaction wave from the shock, is very narrow and hardly visible; the tail of the rarefaction almost coincides with the shock path.

9.6.2 Near Vacuum Star State $A_* = A_\epsilon$

Here we consider a test problem whose solution consists of two very strong rarefaction waves travelling in opposite directions. We assume the following initial conditions: $A_L = 2.14A_0$ [m^2], $u_L = -0.3$ [m/s], $A_R = 2.14A_0$ [m^2], $u_R = 0.3$ [m/s], $x_0 = 0.5L$ [m]. The solution of this Riemann problem is depicted in Fig. 9.17 at

Fig. 9.17 Case 2: nearly collapsed Star State. Initial condition is:

$$A_L = 2.14A_0 \text{ [m}^2\text{]},$$

$$u_L = -0.3 \text{ [m/s]},$$

$$A_R = 2.14A_0 \text{ [m}^2\text{]},$$

$$u_R = 0.3 \text{ [m/s]},$$

$$x_0 = 0.5L \text{ [m]} \text{ and output}$$

$$\text{time } t = 0.015 \text{ [s].}$$

The top frame depicts the normalised

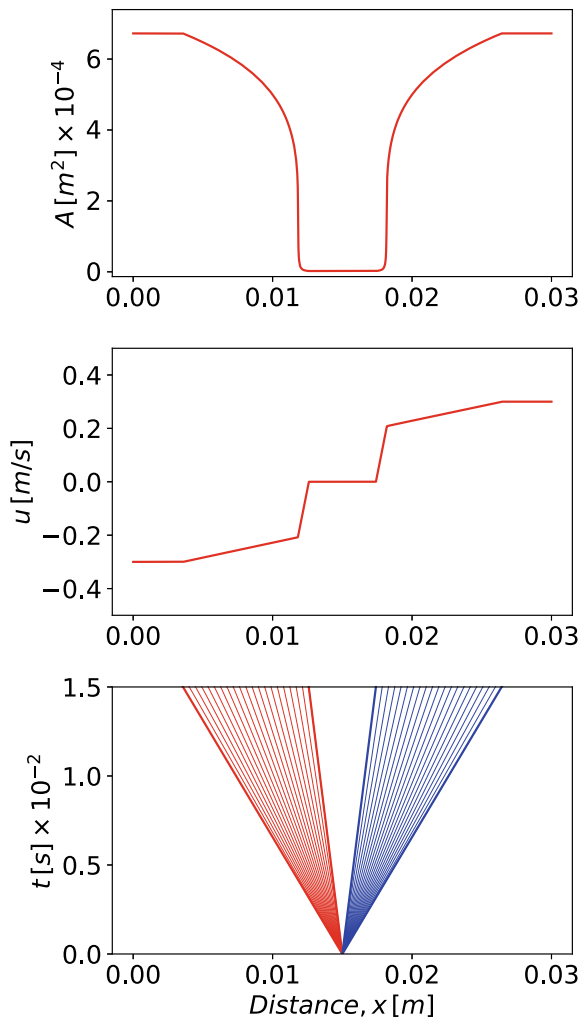
cross-sectional area, the middle

frame the blood

velocity and the bottom

frame shows the x - t diagram

of the resulting wave system



the output time $T_{out} = 0.015$ [s]. The rarefaction waves carry a strong reduction in cross-sectional area and connect the left and right data states to a (nearly) collapsed state $A_* = A_\epsilon > 0$. Note the very strong variation of the solution, especially visible in the cross-sectional area A , near the tail of the rarefactions, close to the **Star Region**.

9.6.3 Remarks on Vein Collapse

Vein collapse in the Riemann problem from non-collapsed initial states through sufficiently strong rarefaction waves in venous vessels does not happen for finite velocity difference Δu in (9.96), as rigorously proved above. This situation is distinct from other systems of PDEs, such as the equations of gas dynamics [2] and the shallow water equations [13]. Moreover, truly collapsed initial states are not admissible in veins, but are open to arbitrary choice through specified states A_ϵ with $0 < A_\epsilon \ll 1$. In other words, the **vacuum state** is not permitted for veins. This is in contrast to arteries, which do admit truly collapsed states either as initial conditions or as emerging from the solution of the equations, as shown by Toro and Siviglia [14]. The situation is somehow unexpected. Physiologically, the more collapsible blood vessels are veins, as compared to arteries. However, their roles regarding vacuum seem to have been inverted through the adopted mathematical models. This situation is the direct consequence of the tube law for veins, as the governing partial differential equations for the dynamics of blood flow are the same for arteries and veins. For further details on vein collapse see Spiller et al. [10].

9.7 Concluding Remarks and Further Reading

In this chapter we have solved exactly the Riemann problem for blood flow assuming constant geometrical and mechanical parameters for the vessels and Coriolis coefficient set to unity. We first studied the arterial case in detail, followed by the venous case, treated more succinctly. The exact solution can be used as a reference solution for assessing numerical methods, for applying boundary conditions and for designing Godunov-type numerical methods. Knowing the exact solution of the Riemann problem is also helpful in devising and assessing approximate Riemann solvers for use in Godunov-type numerical methods. Last but not least, the exact solutions to these non-linear problems are valuable for gaining a better understanding of the governing equations as a mathematical model for blood flow in compliant vessels.

For further reading on the subject we strongly recommend the following articles Toro and Siviglia [14]; Toro and Siviglia [9]; Spiller, Toro, Vázquez-Cendón and Contarino [10].

9.8 Miniprojects

Here we proposed two mini projects which set into action most of the relevant contents of the previous chapters, this one in particular. These mini projects can be tackled by the reader or can be assigned to students.

9.1 Equations in primitive variables but in conservation form. Consider the blood flow equations for arteries in terms of **primitive variables**, written in conservation-law form

$$\partial_t \mathbf{V} + \partial_x \mathbf{G}(\mathbf{V}) = \mathbf{S}(\mathbf{V}), \quad (9.121)$$

where

$$\mathbf{V} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \equiv \begin{bmatrix} A \\ u \end{bmatrix}, \quad \mathbf{G}(\mathbf{V}) = \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} \equiv \begin{bmatrix} Au \\ \frac{1}{2}u^2 + 3\gamma A^{1/2} \end{bmatrix} \quad (9.122)$$

and zero source term

$$\mathbf{S}(\mathbf{V}) = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (9.123)$$

1. Find the eigenvalues and left and right eigenvectors.
2. Determine the nature of the characteristic fields.
3. Solve the Riemann problem exactly.
4. Implement the solver using your favorite programming language.
5. Solve all the Riemann problem examples in Sect. 9.4.4 and compare the solutions computed from (9.121)–(9.123) with those obtained from the conservative-variable formulation shown in Figs. 9.11 and 9.12.
6. Comment on the comparisons, paying particular attention to differentiating smooth from discontinuous solutions.
7. Pay special attention to wave speeds.

9.2 Riemann problems solved with a tube law for veins. Solve all the Riemann problem examples in Sect. 9.4.4 with the equations in terms of conserved variables but using the tube law for veins. Compare your results with those shown in Figs. 9.11 and 9.12. Comment on the results.

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Chapter 10

Generalised Blood Flow Models and Analysis



Abstract Averaged one-dimensional mathematical models for blood flow in compliant vessels require equations describing the dynamics of fluid flow in the form of partial differential equations (PDEs), along with a constitutive equation for the vessel wall mechanics—also known as the tube law—to account for the vessel’s compliance. The coupling of these two components results in a simple fluid-structure interaction model for blood flow in compliant (as opposed to rigid) vessels. In this chapter, we study elastic and viscoelastic tube laws, which are required to satisfy two basic criteria: First, they must accurately represent the physiological phenomena of interest. Second, when coupled with the PDEs governing blood fluid dynamics, they must preserve certain mathematical properties. Here we analyse the resulting PDEs for each tube law considered. Mathematically admissible elastic tube laws lead to hyperbolic equations. We establish their hyperbolic character and examine their complete eigenstructure, and the nature of their characteristic fields. Viscoelastic tube laws give rise to parabolic equations. We approximate these parabolic equations by hyperbolic equations with stiff source terms and analyse the resulting hyperbolised PDEs. Tube laws involve several geometric and mechanical parameters, which must typically be determined through experimentation. In this chapter, we examine both elastic and viscoelastic tube laws with constant and variable parameters, analysing the resulting PDEs. A case study of animal and human blood vessels is presented, in which a parameter estimation process is applied, and the resulting tube laws are used to replicate experimental data. It is seen that viscoelastic tube laws more accurately represent the physiology of compliant blood vessels compared to elastic tube laws and are therefore recommended for practical applications. The chapter concludes with a summary, a discussion, and a list of recommended references for further study.

10.1 Fluid Dynamics and Vessel Wall Mechanics

The equations for the fluid dynamics of blood flow in the human cardiovascular system were derived in Chap. 6. These equations, to be determined, must be supplemented with values for the Coriolis coefficient and expressions for the velocity profile. Moreover, the two partial differential equations (PDEs) include three

unknowns, namely cross-sectional area $A(x, t)$, cross-sectional averaged velocity $u(x, t)$ and cross-sectional averaged pressure $p(x, t)$. Therefore, a closure condition is needed in order to have a determined mathematical problem. A common procedure to close a system is to establish a functional relationship between two of the unknowns, adding in this manner the extra missing equation. For the problem under study, the two chosen unknowns to be related are vessel cross-sectional area A and vessel lumen pressure p in the functional form $p = p(A)$, which is known as **tube law**, or simply **pressure law**.

Experimental and theoretical approaches are used to derive tube laws. Physically, the tube law adds an essential component to the mathematical model, namely the mechanics of the compliant vessel wall. Hence, the resulting mathematical model couples the blood fluid dynamics, represented by PDEs, to the vessel wall displacement, represented by the tube law. Effectively, one ends up with a fluid/structure interaction model, in its simplest setting.

In this section, we discuss two relevant issues arising in the construction of mathematical models for the dynamics of blood flow in compliant blood vessels. In Sect. 10.1.1 we recall the governing partial differential equations and identify the presence of unknown parameters and the need to find a closure condition, the tube law. In Sect. 10.1.2 we discuss the salient points regarding tube laws for the mechanics of the vessel wall.

10.1.1 Differential Equations for the Fluid Dynamics

The one-dimensional (1D) averaged equations for the dynamics of blood flow were established in Chap. 6, which arise from the basic physical laws of conservation of mass and balance of momentum. For ease of presentation, the resulting system of non-linear, first-order system of partial differential equations is recalled here as

$$\left. \begin{aligned} \partial_t A + \partial_x(Au) &= s_1, \\ \partial_t(Au) + \partial_x(\alpha Au^2) + \frac{A}{\rho} \partial_x p &= s_2. \end{aligned} \right\} \quad (10.1)$$

The source terms s_1 , s_2 and the Coriolis coefficient α in (10.1) are

$$s_1 = 0, \quad s_2 = -2(\beta + 2)\pi \frac{\mu}{\rho} u, \quad \alpha(\beta) = \frac{\beta + 2}{\beta + 1}, \quad \beta > 0. \quad (10.2)$$

The unknowns and parameters are:

- $A(x, t)$ is cross-sectional area of the vessel at position x and time t (unknown);
- $u(x, t)$ is cross-sectional averaged velocity of blood at a vessel cross section (unknown);
- $p(x, t)$ is cross-sectional averaged pressure of blood inside the vessel (unknown);
- ρ is blood density, constant (parameter);

- α is the Coriolis coefficient, or momentum correction factor (parameter);
- μ is kinematic viscosity (parameter).

The flow, or flow rate, is defined as $q(x, t) = A(x, t)u(x, t)$. As stated, the unknowns of the problem are $A(x, t)$, $u(x, t)$ (or flow $q(x, t)$) and $p(x, t)$. As there are three unknowns and only two equations, one requires a **closure condition**, called **tube law**, or **pressure law**.

10.1.2 Tube Laws for the Mechanics of the Vessel Wall

The mechanics of the compliant vessel wall under the dynamic conditions of pulsatile flow is closely linked to the specific function of the blood vessel, as discussed in the early paper by Burton [1]. Distinction must be made, for example, on the type of vessel, function, location and size. Arteries and veins perform very different functions and have therefore very different mechanical properties. See Chap. 1 on the anatomy and physiology of the blood vessel network of the human cardiovascular system.

The function of the blood vessel is closely linked to its histology, that is the composition of the blood vessel wall, from the intima to the adventitia. Components of the blood vessel wall include elastic fibres, collagen fibrils and smooth muscle cells, the distribution of which within the wall varies according to function and location in the circulation. The histology of the vessel wall is closely linked to mechanical parameters, associated, for example, with elasticity and viscoelasticity.

To construct models for the mechanics of the blood vessel wall there are essentially two main approaches, namely theoretical mechanical models and phenomenological models. Theoretically based approaches include the **Quasilinear Viscoelasticity Theory** of Fung, see Fung [2] and Valdéz-Jasso[3], and the class of **Generalised String Models**, see Veneziani [4], Quarteroni et al. [5] and Formaggia et al. [6]. For background on **phenomenological models**, the interested reader is referred to Langewouters et al. [7, 8].

In this chapter, we shall consider string-type models, which under suitable linearisations [6] yield the following functional expression for the pressure

$$\mathbb{P} = p - p_e = a_0 \left[\sqrt{\frac{A}{A_0}} - 1 \right] + a_1 \partial_t A + a_2 \partial_t^{(2)} A + a_3 \partial_x^{(2)} \left[\sqrt{\frac{A}{A_0}} - 1 \right]. \quad (10.3)$$

Here $\mathbb{P}$ denotes the **transmural pressure**, p is the internal blood pressure, p_e is the external pressure. The right hand side of (10.3) includes an algebraic term and differential terms, whose coefficients a_0, a_1, a_2, a_3 are functions of geometric and mechanical properties of the vessel wall. See also Veneziani [4] and Quarteroni et al. [5]. Each term on the right hand side has a mechanical interpretation. The first term with coefficient a_0 is called **elastic stress**; the second term with coefficient a_1 is called **wall viscosity**; the third term with coefficient a_2 is called **wall inertia** and the last term with coefficient a_3 is called **longitudinal stress**.

Generally, the coefficients a_0, a_1, a_2, a_3 in (10.3) will depend on several parameters $k_1, \dots, k_s$, and one may therefore re-write (10.3) in the form

$$\mathbb{P} = \mathbb{P}(A, \partial_t^{(1)} A, \partial_t^{(2)} A, \partial_x^{(2)} A; k_1, \dots, k_s). \quad (10.4)$$

That is, $\mathbb{P}$ is a function of $A, \partial_t^{(1)} A, \partial_t^{(2)} A, \partial_x^{(2)} A$, with parameters $k_1, \dots, k_s$.

In this Chapter, we shall be concerned exclusively with two types of tube laws: elastic and viscoelastic. Elastic tube laws will only include the first (algebraic) term on the right hand side of (10.3), or modifications of it. Viscoelastic tube laws will include the elastic term, or modifications of it, and the second (differential) term on the right hand side of (10.3).

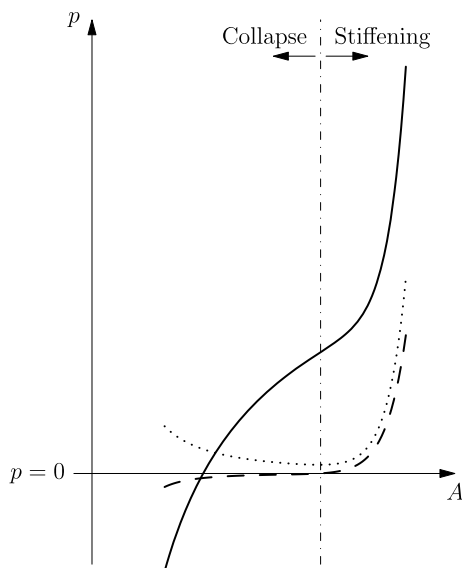
Quantifying the parameters $\mathbf{Z} = \{k_1, \dots, k_s\}$ in (10.4) is a major issue that involves both experimentation and theory, all with the aim of enabling the function (10.4) to accurately reproduce the mechanics of the vessel wall. Determining the optimal set of parameters, called **parametrisation**, requires refined experimentation with both animals and humans.

The set of parameters in the tube law (10.4) may be geometrical or mechanical in character. They may be assumed constant or variable with distance x along the vessel. Anatomically, parameters are actually variable. For example, the vessel wall thickness $h_0(x)$ is naturally space dependent under both normal and pathological conditions. The insertion of grafts or stents makes the x dependency more apparent. In the case of a stent, parameter variations are virtually discontinuous at the interface between healthy tissue and the material of the stent. Therefore, mathematical models must admit variable parameters, with possible discontinuities; this poses challenges both from the mathematical and the computational points of view.

The tube law is utilised to predict mechanical behaviour outside the set of parameters used for parametrisation of the tube law. Bioengineers can then use (10.4) to infer mechanical properties of the blood vessel wall to gain a better understanding of the mechanical properties of the blood vessel. As argued by Valdéz-Jasso et al. [3], a good understanding of the mechanical properties of blood vessels may provide useful insight into the biology of blood vessels under physiological and pathological conditions. Moreover, such understanding may also be useful in tracking the disease progression and, most importantly, aid in the vascular graft design, as well as in its clinical implementation. For this kind of utilization of tube laws, accurate parametrization is crucial, always aiming at the correct description of the vessel wall mechanics.

For the purpose of closing the set of PDEs for the fluid dynamics of blood in the cardiovascular system, one must also consider the mathematical aspects of the tube law. A fundamental requirement of tube laws is that the resulting partial differential equations be well-posed. For first-order PDEs wellposedness is ensured by hyperbolicity [9]. Better still, if the associated characteristic fields are either linearly degenerate or genuinely non linear. See Chap. 8 for background on these theoretical issues. In this section we are particularly interested in adopting tube laws that comply

Fig. 10.1 Representation of a generic tube-law curve $p = p(A)$ by the continuous line, exhibiting desirable features, such as $\partial_A p > 0$ (monotone increasing). The first derivative $\partial_A p$ of $p(A)$ is shown by the dotted line and the second derivative $\partial_A^{(2)} p$ of $p(A)$ is shown by the dashed line. The vertical dashed-dotted line shows the reference area A_0 as the inflexion point of $p(A)$, at which the transmural pressure equals the reference pressure p_0 . For $A < A_0$, $\partial_A^{(2)} p < 0$ (collapse) and for $A > A_0$, $\partial_A^{(2)} p > 0$ (stiffening)



with the alluded mathematical requirements, for which desirable properties of tube laws are illustrated in Fig. 10.1, which depicts the function $p = p(A)$.

The literature on the biomechanics of soft tissue, such as blood vessel walls and associated tube laws, is vast. See Fung [2] for a comprehensive treatment of the subject and various theoretical approaches. An approach to the construction of tube laws for use in 1D models is found in [10], Chap. 3; see also [6]. Additional relevant studies are those of Armentano et al. [11]; Zócalo et al. [12]; Segers et al. [13]; Valdéz-Jasso et al. [3, 14]; Bia et al. [15]; and many more.

Works on the actual coupling of tube laws to the differential equations for the fluid dynamics of blood flow include, for example, Segers et al. [13]; Brook, Falle and Pedley [16]; Olufsen [17]; Liang et al. [18]; Raymond et al. [19]; Alastruey et al. [20]; Müller and Toro [21]; Mynard and Smolich [22]; Toro et al. [23]; Ghigo et al. [24]; Bertaglia et al. [25–27]; Müller et al. [28]; and many more.

In this Chapter, we introduce generalisations of the mathematical models for blood flow presented in previous chapters. We study tube laws that account for major physical properties of the blood vessel wall, such as elasticity and viscoelasticity, allowing for variation of parameters, including discontinuities. On the assumption that such tube laws do account for the mechanics of the blood vessel wall, we focus on the mathematical analysis of the PDEs coupled to the tube laws. Major tasks include finding the eigenstructure of the system of PDEs, that is eigenvalues, eigenvectors, and determining the nature of the associated characteristic fields.

10.2 Elastic Tube Laws

Here we study an elastic tube laws with constant geometric and mechanical parameters, to close the PDE system (10.1)–(10.2). We carry out an analysis of the resulting equations and constraints on the tube-law parameters to preserve desired mathematical properties of the complete model, such as hyperbolicity.

10.2.1 Elastic Tube Law for Arteries

As already stated, the equations for the dynamics of blood flow (10.1)–(10.2) must be augmented by a tube law, or pressure law, in order to close the system, and to account for the compliant vessel wall mechanics. The resulting system accounts for the interaction of the internal flow distribution and the domain boundary motion.

Tube Law and Parameters

The simplest case of an elastic tube law, applicable to arteries only, and directly given for the transmural pressure $\mathbb{P}^E$, is the non-linear algebraic pressure-area relation

$$\left. \begin{aligned} \mathbb{P}^E(A; K) &= p - p_e(x, t) = K(x)\varphi(A; A_0) , \\ \varphi(A; A_0) &= \left[\left(\frac{A}{A_0} \right)^{1/2} - 1 \right] , \\ K(x) &= \frac{\sqrt{\pi}}{(1 - \nu^2)} \frac{E(x)h_0(x)}{\sqrt{A_0(x)}} . \end{aligned} \right\} \quad (10.5)$$

The parameters emerging from the tube law include the following:

1. $p_e(x, t)$ is the external pressure;
2. $A_0(x)$ is the blood vessel cross-sectional area at equilibrium;
3. ν is the Poisson ratio (set to $\nu = 1/2$ for incompressible media);
4. $h_0(x)$ is the vessel wall thickness;
5. $E(x)$ is the Young's modulus of elasticity.

Recall that conditions at equilibrium for $A_0(x)$ and $h_0(x)$ above refer to the situation characterised by $\mathbb{P}^E(A; K) = p - p_e(x, t) = 0$ and $u = 0$.

Tube law (10.5) accounts for just the first term in (10.3), whose mechanical meaning may be interpreted by introducing the strain

$$\varepsilon = \frac{R - R_0}{R_0} = \varphi(A; A_0) = \left[\left(\frac{A}{A_0} \right)^{1/2} - 1 \right] , \quad (10.6)$$

where R and R_0 denote radius and equilibrium radius of the vessel, respectively. Therefore $K(x)\varphi(A; A_0)$ is the **elastic stress**. At this stage we note that this elastic tube law can be generalised by suitable choices for $\varphi(A; A_0)$; see Sect. 10.2.2.

Equations and Their Eigenstructure

Here we carry out a mathematical analysis of the resulting Eqs. (10.1)–(10.2) when coupled to the tube law (10.5). We study the eigenstructure of the system, for which we focus on the principal part of the equations. The Coriolis coefficient α is often assumed to be $\alpha = 1$, but here we take it to be arbitrary.

We start by considering the momentum equation in (10.1). By expanding spatial derivatives in the second term and applying the chain rule to the pressure gradient term we have

$$\partial_t(Au) + \frac{A}{\rho} \frac{\partial p}{\partial A} \partial_x A + \alpha u \partial_x(Au) + (Au) \partial_x(\alpha u) = 0 .$$

By adding and subtracting the term $\alpha u^2 \partial_x A$ and rearranging, the momentum equation becomes

$$\partial_t(Au) + \left(\frac{A}{\rho} \frac{\partial p}{\partial A} - \alpha u^2 \right) \partial_x A + 2\alpha u \partial_x(Au) = 0 . \quad (10.7)$$

System (10.1)–(10.2) is now

$$\left. \begin{aligned} \partial_t A + \partial_x(Au) &= s_1 , \\ \partial_t(Au) + \left(\frac{A}{\rho} \frac{\partial p}{\partial A} - \alpha u^2 \right) \partial_x A + 2\alpha u \partial_x(Au) &= s_2 . \end{aligned} \right\} \quad (10.8)$$

Written in matrix form, system (10.8) reads

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}) , \quad (10.9)$$

with

$$\mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \equiv \begin{bmatrix} A \\ Au \end{bmatrix} , \quad \mathbf{A}(\mathbf{Q}) = \begin{bmatrix} 0 & 1 \\ \frac{A}{\rho} \frac{\partial p}{\partial A} - \alpha u^2 & 2\alpha u \end{bmatrix} , \quad \mathbf{S} = \begin{bmatrix} 0 \\ s_2 \end{bmatrix} . \quad (10.10)$$

It is easily seen that the eigenvalues of the matrix $\mathbf{A}$ in (10.10) are

$$\lambda_1 = \alpha u - c_\alpha , \quad \lambda_2 = \alpha u + c_\alpha . \quad (10.11)$$

The α -**modified wave speed** c_α emerges naturally as

$$c_\alpha = \sqrt{\alpha(\alpha - 1)u^2 + \frac{A}{\rho} \frac{\partial p}{\partial A}} = \sqrt{\alpha(\alpha - 1)u^2 + c^2}, \quad (10.12)$$

where c is the conventional wave speed for $\alpha = 1$, namely

$$c = \sqrt{\frac{A}{\rho} \frac{\partial p}{\partial A}}, \quad (10.13)$$

which is real if the tube law satisfies

$$\frac{\partial p}{\partial A} > 0. \quad (10.14)$$

It is easily verified that the corresponding left eigenvectors are

$$\mathbf{L}_1 = \alpha_1 \left[1, \frac{\alpha u - c_\alpha}{c^2 - \alpha u^2} \right], \quad \mathbf{L}_2 = \alpha_2 \left[1, \frac{\alpha u + c_\alpha}{c^2 - \alpha u^2} \right], \quad (10.15)$$

with α_1 and α_2 , two scaling factors. Analogously, the right eigenvectors are

$$\mathbf{R}_1 = \beta_1 \begin{bmatrix} 1 \\ \alpha u - c_\alpha \end{bmatrix}, \quad \mathbf{R}_2 = \beta_2 \begin{bmatrix} 1 \\ \alpha u + c_\alpha \end{bmatrix}, \quad (10.16)$$

where β_1 and β_2 are arbitrary scaling factors.

Therefore, the complete system of PDEs (10.1)–(10.2) for the dynamics of blood flow, coupled to the mechanics of the compliant vessel wall, expressed by the tube law (10.5), constitutes a hyperbolic system. That is, the eigenvalues are real (and distinct), with an associated complete set of linearly independent eigenvectors. The novelty here is the presence of the Coriolis coefficient α in the equations, which as seen below, has implications for their numerical approximation. See Spilimbergo et al. [29] for further details.

Remarks on the Coriolis coefficient:

1. Hyperbolicity requires

$$\alpha(\alpha - 1)u^2 + c^2 > 0, \quad 1 \leq \alpha \leq 2 \quad (10.17)$$

in (10.12), which is always satisfied if condition (10.14) on the tube law is satisfied.

2. Note that for $\alpha > 1$ there is margin in (10.12) for (10.14) to be violated and still retain hyperbolicity.

3. In general $c_\alpha \geq c$ and $\alpha|u| \geq |u|$. Hence

$$\alpha u + c_\alpha \geq u + c \quad \text{for } u \geq 0 \text{ and } \alpha > 1. \quad (10.18)$$

4. Therefore, the maximum wave speeds emerging from the α -dependent system are larger than those from the conventional case ($\alpha = 1$).
5. Condition (10.18) has implications when enforcing the Courant stability condition in numerical methods for solving Eqs. (10.1)–(10.2) for $\alpha > 1$. See Chap. 12 for background on numerical concepts.

For a comprehensive analysis of the PDEs for general α and the exact solution of the associated Riemann problem, see Spilimbergo et al. [29]. Next we consider tube laws that include venous blood vessels.

10.2.2 Elastic Tube Laws for Arteries and Veins

All too often, one considers the circulation to consist of just arteries. For global models of the circulation one needs tube laws for both arteries and veins [21–23, 28]. As seen in Chap. 9, the mechanics of the wall of veins is much more demanding on the analysis of the equations than that for arteries; see also Chap. 1 on the vein function and the vein wall composition, which ultimately determine the vein's mechanical behaviour.

For the rest of this section, we consider the differential equations (10.1)–(10.2) with $\alpha = 1$, with all geometric and mechanical parameters in the tube law assumed constant.

Tube Law and Parameters

Both experimental and theoretical investigations have led researchers to propose tube laws of the form

$$\left. \begin{aligned} \mathbb{P}^E(A; K) &:= p - p_e = K(x)\varphi(A; A_0), \\ \varphi(A; A_0) &= \left[\left(\frac{A}{A_0} \right)^m - \left(\frac{A}{A_0} \right)^n \right]. \end{aligned} \right\} \quad (10.19)$$

See Flaherty et al. [30], Shapiro [31] and Brook et al. [16]. Tube laws such as (10.19), are meant to include both arteries and veins; this is realised through the **coefficient of elasticity** $K(x)$ and the exponents m and n , namely

$$K(x) = \begin{cases} K_a(x) = \frac{E(x)}{1 - \nu^2} \frac{h_0(x)}{r_0(x)}, & m = \frac{1}{2}, n = 0 \quad (\text{arteries}), \\ K_v(x) = \frac{E(x)}{12(1 - \nu^2)} \left(\frac{h_0(x)}{r_0(x)} \right)^3, & m \approx 10, n = -\frac{3}{2} \quad (\text{veins}). \end{cases} \quad (10.20)$$

Most parameters in (10.19)–(10.20) are as defined as for tube law (10.5). The new parameters are the real numbers m and n , and the vessel radius $r_0(x)$ at equilibrium for veins, instead of the area $A_0(x)$ used for the arterial case. Note that the special values $m = 1/2$ and $n = 0$ in (10.19) reproduce the simple tube law (10.5) for arteries. For veins, Brook et al. [16] proposed the values $m = 10$ and $n = -3/2$ in (10.20).

Equations and Their Eigenstructure

The resulting system of PDEs (10.1)–(10.2), written in conservation-law form, reads

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}), \quad (10.21)$$

where

$$\mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \equiv \begin{bmatrix} A \\ Au \end{bmatrix}, \quad \mathbf{S}(\mathbf{Q}) = \begin{bmatrix} 0 \\ s_2 \end{bmatrix}, \quad \mathbf{F}(\mathbf{Q}) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} Au \\ Au^2 + \frac{KA_0}{\rho} \left\{ \frac{m}{m+1} \left(\frac{A}{A_0} \right)^{m+1} - \frac{n}{n+1} \left(\frac{A}{A_0} \right)^{n+1} \right\} \end{bmatrix}. \quad (10.22)$$

Note again that the Coriolis coefficient has been assumed $\alpha = 1$ here. After calculating the Jacobian matrix $\mathbf{A}(\mathbf{Q})$ from the flux vector $\mathbf{F}(\mathbf{Q})$ above, the eigenvalues are found to be

$$\lambda_1 = u - c, \quad \lambda_2 = u + c, \quad (10.23)$$

with corresponding right eigenvectors

$$\mathbf{R}_1 = \beta_1 \begin{bmatrix} 1 \\ u - c \end{bmatrix}, \quad \mathbf{R}_2 = \beta_2 \begin{bmatrix} 1 \\ u + c \end{bmatrix}. \quad (10.24)$$

Here β_1 and β_2 are again arbitrary scaling factors, which unless otherwise stated, will be chosen as $\beta_1 = \beta_2 = 1$. Simple calculations give the left eigenvectors.

As seen in Chap. 9 the **wave speed** c for veins is

$$c = \sqrt{\frac{K_v}{\rho} \left[m \left(\frac{A}{A_0} \right)^m - n \left(\frac{A}{A_0} \right)^n \right]}. \quad (10.25)$$

For convenience c is written as

$$c = \sqrt{(m+1)\gamma_1 A^m + (n+1)\gamma_2 A^n}, \quad (10.26)$$

where

$$\gamma_1 = \frac{mK_v}{(m+1)\rho A_0^m} > 0 \quad \text{and} \quad \gamma_2 = -\frac{nK_v}{(n+1)\rho A_0^n} < 0. \quad (10.27)$$

Sufficient conditions for the wave speed to be real, and thus for hyperbolicity to hold, are $m \geq 0$ and $n \leq 0$; see also Spiller et al. [32]. Sharper constraints were found by Toro and Siviglia [33], $m \geq 0$ and $n \geq -2$, and by Colombo et al. [34], $m \geq 0$ and $n \in [-2, 0]$.

In Chap. 9 we carried out a full study of Eqs. (10.1)–(10.2) with tube law (10.19)–(10.20) for arteries and veins, for the case of Coriolis coefficient $\alpha = 1$; see also Spiller et al. [32] for a detailed exact solution of the Riemann problem for veins. Spilimbergo et al. [35] studied in full detail the eigenstructure of equations (10.1)–(10.2) with tube law (10.19)–(10.20) for arteries and veins for the case of general Coriolis coefficient α , and also solved the associated Riemann problem exactly.

Brook et al. [16] used an equation of the type (10.19)–(10.20) for modelling the blood flow in the vein of a giraffe. For this purpose they stipulated choices for the coefficient of elasticity $K_v(x)$ and for the exponents m and n , namely $m = 10$ and $n = -3/2$. More recently, Müller and Toro [21] applied a version of (10.19)–(10.20) that accommodates both arteries and veins in their global model for the human circulation. See also the classical works of Pedley [36] and Elad et al. [37]. More recent works on global models include Montecinos et al. [38], Mynard and Smolich [22], Strocchi et al. [39], Toro et al. [23], Müller et al. [28].

So far in this chapter we have considered elastic tube laws but with constant geometric and mechanical parameters. In the next section, we extend the study to variable parameters.

10.3 Elastic Tube Law with One Variable Parameter

In this section we consider variable geometric and mechanical parameters, but limiting ourselves to the case of an elastic tube law for arteries and a single variable parameter; see Toro and Siviglia [40]. This case has been generalised by Toro and Siviglia [33] and by Spilimbergo et al. [41], to include multiple variable parameters. Still more general models involving viscoelastic tube laws and multiple variable parameters are dealt with in Sect. 10.6, following Montecinos, Müller and Toro [38].

10.3.1 Governing Equations

Here we recall the basic haemodynamics Eqs. (10.1)–(10.2), namely

$$\left. \begin{aligned} \partial_t A + \partial_x(Au) &= s_1, \\ \partial_t(Au) + \partial_x(\alpha Au^2) + \frac{A}{\rho} \partial_x p &= s_2, \\ s_1 &= 0, \quad s_2 = -2(\beta + 2)\pi \frac{\mu}{\rho} u, \quad \alpha(\beta) = \frac{\beta + 2}{\beta + 1}, \quad \beta > 0. \end{aligned} \right\} \quad (10.28)$$

We consider the elastic tube law valid for arteries only, as seen in Eq. (10.19). Here, following [42], this tube is expressed with a minor change in notation as

$$\left. \begin{aligned} \mathbb{P}^E(A; K) &= p - p_e(x, t) = \psi(A; K), \\ \psi(A; K) &= K(x) \left[\left(\frac{A}{A_0} \right)^m - 1 \right], \quad m = 1/2, \\ K(x) &= \frac{\sqrt{\pi}}{(1 - \nu^2)} \frac{E(x)h_0(x)}{\sqrt{A_0(x)}}. \end{aligned} \right\} \quad (10.29)$$

Definitions for the unknowns and parameters are given after Eqs. (10.1)–(10.2). The Coriolis coefficient and the external pressure are assumed to be $\alpha = 1$ and $p_e(x, t) = 0$, respectively. Note that the tube law (10.29) is a special case of tube law (10.19)–(10.20), with $m = 1/2$ and $n = 0$. For convenience we shall use m in place of $1/2$ in (10.29), for the rest of this section. Of particular interest here are the geometric parameters $A_0(x)$ and $h_0(x)$ and the mechanical parameter $E(x)$, which in general vary with x . These variable parameters will enter the pressure gradient in the momentum equation in (10.28). In [40], only $K(x)$ was considered as the variable parameter. As a matter of fact, if $A_0(x)$ is assumed constant, then the variation of $K(x)$ includes the combined variations of $h_0(x)$ and $E(x)$, as seen in (10.29).

The variable parameter $K(x)$ enters the pressure gradient in the momentum equation in (10.28). That is

$$\partial_x p = \psi_A \partial_x A + \psi_K \partial_x K, \quad (10.30)$$

with

$$\psi_A = \frac{\partial \psi}{\partial A} = \frac{mK}{A} \left(\frac{A}{A_0} \right)^m, \quad \psi_K = \frac{\partial \psi}{\partial K} = \left(\frac{A}{A_0} \right)^m - 1. \quad (10.31)$$

The complete system of PDEs now reads

$$\left. \begin{aligned} \partial_t A + \partial_x(uA) &= 0, \\ \partial_t(uA) + \partial_x(Au^2) + \frac{A}{\rho} \psi_A \partial_x A &= -\frac{A}{\rho} \psi_K \partial_x K. \end{aligned} \right\} \quad (10.32)$$

What is new in (10.32) is the presence of a new **source term** that includes the spatial gradient $\partial_x K$ of the coefficient of elasticity K . We remark that $K(x)$ here is analogous to the variation of the channel bottom $b(x)$ in shallow water flow models [43, 44]. Source terms of this kind are often called **geometric source terms**, and are notoriously difficult to discretise correctly. See, for example, [44–46] and references therein.

Toro and Siviglia [40] presented an alternative formulation of the model by considering the variable $K(x)$, comprising $h_0(x)$ and $E(x)$, as a new unknown of the problem. This is accomplished by adding the following obvious partial differential equation

$$\partial_t K(x) = 0. \quad (10.33)$$

The enlarged system from (10.32) and (10.33), written in non-conservative form, reads

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}), \quad (10.34)$$

where

$$\left. \begin{aligned} \mathbf{Q} &= \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \equiv \begin{bmatrix} A \\ Au \\ K \end{bmatrix}, \quad \mathbf{A}(\mathbf{Q}) = \begin{bmatrix} 0 & 1 & 0 \\ \frac{A}{\rho} \psi_A - u^2 & 2u & \frac{A}{\rho} \psi_K \\ 0 & 0 & 0 \end{bmatrix}, \\ \mathbf{S}(\mathbf{Q}) &= \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ s_2 \\ 0 \end{bmatrix}. \end{aligned} \right\} \quad (10.35)$$

Toro and Siviglia [42] reported satisfactory results obtained from formulation (10.34)–(10.35), which may even include discontinuous variations of the parameter $K(x)$. However, the application of Godunov-type methods, or any other method, becomes more involved than for the classical conservative formulation of the PDEs. Further details are found in [42]. Nonetheless, the approach has found application in dealing with the case of multiple parameters [21, 23, 33, 38, 41]. A disadvantage of the approach is the enlarged 3×3 system of equations. Moreover, system (10.34)–(10.35) is written in non-conservative form, and it cannot be expressed in conservative form with a corresponding physical flux function. For mathematical issues on conservative systems see Chaps. 8, 13 and for non-conservative systems see Chap. 14.

10.3.2 Eigenstructure and Characteristic Fields

The eigenstructure of the first-order system (10.34)–(10.35) is that of the principal part of the system and is given by the eigenvalues and corresponding eigenvectors.

Proposition 10.1 *The eigenvalues of (10.34)–(10.35) are given by*

$$\lambda_1 = u - c, \quad \lambda_2 = 0, \quad \lambda_3 = u + c, \quad (10.36)$$

where

$$c = \sqrt{\frac{A}{\rho}} \psi_A = \sqrt{\frac{A}{\rho}} \mathbb{P}_A^E \quad (10.37)$$

is the wave speed, as usual. The eigenvalues are real if

$$\mathbb{P}_A^E = \frac{\partial \mathbb{P}^E(A)}{\partial A} > 0. \quad (10.38)$$

See definitions in (10.29) and relation 10.31. See also Fig. 10.1.

Proof By definition the eigenvalues of system (10.34)–(10.35) are the eigenvalues of the matrix $\mathbf{A}$, which in turn are the roots of the characteristic polynomial

$$P(\lambda) = \text{Det}(\mathbf{A} - \lambda \mathbf{I}) = 0, \quad (10.39)$$

where $\mathbf{I}$ is the identity matrix and λ is a parameter. Simple calculations give

$$P(\lambda) = \lambda (\lambda^2 - 2u\lambda + u^2 - c^2) = 0,$$

from which the result (10.36) follows. ■

Proposition 10.2 *The right eigenvectors of $\mathbf{A}$ corresponding to the eigenvalues (10.36) are*

$$\mathbf{R}_1 = \gamma_1 \begin{bmatrix} 1 \\ u - c \\ 0 \end{bmatrix}, \quad \mathbf{R}_2 = \gamma_2 \begin{bmatrix} \frac{c^2}{u^2 - c^2} \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{R}_3 = \gamma_3 \begin{bmatrix} 1 \\ u + c \\ 0 \end{bmatrix}, \quad (10.40)$$

where γ_1 , γ_2 and γ_3 are arbitrary scaling factors.

Proof For an arbitrary right eigenvector $\mathbf{R} = [r_1, r_2, r_3]^T$ we have

$$\mathbf{A}\mathbf{R} = \lambda \mathbf{R}, \quad (10.41)$$

which gives the algebraic system

$$\left. \begin{aligned} r_2 &= \lambda r_1, \\ (u^2 - c^2)r_1 + 2ur_2 + \frac{A}{\rho} \psi_K r_3 &= \lambda r_2, \\ 0 &= \lambda r_3. \end{aligned} \right\} \quad (10.42)$$

By substituting λ in (10.42) by the appropriate eigenvalues in (10.36) in turn we arrive at the sought result. ■

Proposition 10.3 *The λ_1 - and λ_3 -characteristic fields are genuinely non-linear if the tube law exponent m satisfies $m \neq -2$. The λ_2 -characteristic field is linearly degenerate.*

Proof Since $\lambda_2 = 0$ it follows that $\nabla \lambda_2 = \mathbf{0}$ and thus

$$\nabla \lambda_2 \cdot \mathbf{R}_2 = 0. \quad (10.43)$$

Therefore, the λ_2 -characteristic field is linearly degenerate as claimed. For the two non-linear characteristic fields, simple algebraic manipulations give

$$\nabla \lambda_1 \cdot \mathbf{R}_1 = -\frac{m(m+2)K \left(\frac{A}{A_0}\right)^m}{2A\sqrt{m\rho K \left(\frac{A}{A_0}\right)^m}}, \quad \nabla \lambda_3 \cdot \mathbf{R}_3 = \frac{m(m+2)K \left(\frac{A}{A_0}\right)^m}{2A\sqrt{m\rho K \left(\frac{A}{A_0}\right)^m}}. \quad (10.44)$$

Therefore the $\lambda_1(\mathbf{Q})$ - and $\lambda_3(\mathbf{Q})$ -characteristic fields are genuinely non-linear provided $m \neq -2$. ■

10.3.3 The Riemann Problem

The Riemann problem for the homogeneous version of system (10.34)–(10.35) is the special Cauchy problem with piece-wise constant initial condition, namely

$$\left. \begin{aligned} \partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} &= \mathbf{0}, \quad x \in \mathbb{R}, \quad t > 0, \\ \mathbf{Q}(x, 0) &= \begin{cases} \mathbf{Q}_L & \text{if } x < 0, \\ \mathbf{Q}_R & \text{if } x > 0. \end{cases} \end{aligned} \right\} \quad (10.45)$$

In terms of physical variables A , u and K the initial conditions $\mathbf{Q}_L$ and $\mathbf{Q}_R$ above, correspond to $\mathbf{W}_L = [A_L, u_L, K_L]^T$ and $\mathbf{W}_R = [A_R, u_R, K_R]^T$, respectively. For the case of constant parameters, the complete solution to the Riemann problem (10.45) was given in Chap. 9. What is new here is the presence of a third equation representing the variable parameter K , which gives rise to an intermediate characteristic field associated with the intermediate eigenvalue $\lambda_2 = 0$ in (10.36).

The structure of the similarity solution of the problem is depicted in Fig. 10.2 in the x - t half plane. There are three wave families. The left family is associated with the eigenvalue λ_1 , the middle wave family is associated with λ_2 and the right wave family is associated with λ_3 . Waves associated with the **genuinely non-linear** characteristic fields λ_1 and λ_3 are either shocks (discontinuous solutions) or rarefactions (smooth solutions), while the wave associated with the **linearly degenerate** characteristic

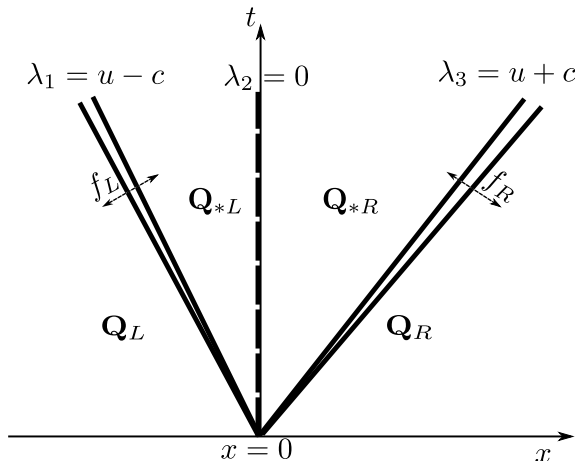


Fig. 10.2 Structure of the solution of the Riemann problem (10.45) for the 3×3 blood flow model of this section. There are three wave families, associated respectively with the eigenvalues $\lambda_1 = u - c$, $\lambda_2 = 0$ and $\lambda_3 = u + c$. The unknowns of the problem lie in the wedge between the outer non-linear waves, called the **Star Region**, which is divided by the middle eigenvalue into Q_{*L} and Q_{*R} . Also shown are the functions f_L and f_R that represent jumps across the non-linear waves connecting data states to unknown states Q_{*L} and Q_{*R} (Courtesy of Prof. A. Siviglia, University of Trento, Italy)

field λ_2 is a contact discontinuity, which represents a jump in material properties represented by K at the interface $x = 0$, with $K = K_L$ if $x < 0$ and $K = K_R$ if $x > 0$.

The entire solution consists of four constant states, namely Q_L (data), Q_{*L} , Q_{*R} and Q_R (data), separated by three waves. The unknown states to be found are Q_{*L} (left of the contact) and Q_{*R} (right of the contact). If any of the λ_1 and λ_3 waves is a rarefaction then there will be a smooth transition between two adjacent constant states. In order to solve exactly the entire initial-value problem one needs to establish appropriate jump conditions across each characteristic field to connect the unknown states Q_{*L} and Q_{*R} to the initial conditions Q_L (left) and Q_R (right), respectively. In terms of physical variables $\mathbf{W}$, the unknowns in the **Star Region** are

$$\mathbf{W}_{*L} = [A_{*L}, u_{*L}, K_{*L}]^T, \quad \mathbf{W}_{*R} = [A_{*R}, u_{*R}, K_{*R}]^T. \quad (10.46)$$

In what follows we establish jump conditions across each characteristic field.

10.3.4 Jumps Across Shocks and Rarefaction Waves

The sought jump conditions are for the non-linear characteristic fields, associated with shocks or rarefactions, and the linearly degenerate characteristic field, associated with a stationary contact discontinuity.

Jumps Across Shocks: Rankine-Hugoniot Conditions

As already noted, the proposed model equations in (10.45) cannot be expressed in conservation-law form, and in principle, the classical Rankine-Hugoniot conditions cannot be applied. For the case of the Riemann problem, however, the vessel properties h_0 and E , and thus K , are constant across the non-linear waves. Hence, across the genuinely non-linear characteristic fields (rarefactions and shocks) it is possible to express the equations in conservation-law form, so that the classical Rankine-Hugoniot conditions apply. Therefore the shock relations obtained in Chap. 9 are still applicable to these two wave families, as shown below.

Proposition 10.4 (Left shock) *If the left λ_1 -wave is a left-facing shock wave of speed S_L then*

$$u_{*L} = u_L - f_L, \quad f_L = \sqrt{\frac{B_L(A_{*L} - A_L)(A_{*L}^{m+1} - A_L^{m+1})}{A_L A_{*L}}} \quad (10.47)$$

and the shock speed is given as

$$S_L = u_L - \frac{M_L}{A_L}, \quad M_L = \sqrt{\frac{B_L A_L A_{*L} (A_{*L}^{m+1} - A_L^{m+1})}{A_{*L} - A_L}}. \quad (10.48)$$

Here

$$B = \frac{mK}{\rho(m+1)A_0^m} : \text{constant}. \quad (10.49)$$

Proof (omitted). For details see [42]. ■

Proposition 10.5 (Right shock) *If the right λ_3 -wave is a right-facing shock wave of speed S_R then*

$$u_{*R} = u_R + f_R, \quad f_R = \sqrt{\frac{B_R(A_{*R} - A_R)(A_{*R}^{m+1} - A_R^{m+1})}{A_R A_{*R}}} \quad (10.50)$$

and the shock speed is given as

$$S_R = u_R + \frac{M_R}{A_R}, \quad M_R = \sqrt{\frac{B_R A_R A_{*R} (A_{*R}^{m+1} - A_R^{m+1})}{A_{*R} - A_R}}. \quad (10.51)$$

Here

$$B = \frac{mK}{\rho(m+1)A_0^m} : \text{constant} . \quad (10.52)$$

Proof (omitted). For details see [42]. ■

Note the meaning of the functions f_L and f_R in Fig. 10.2. They connect known data states to the unknowns of the problem.

Jumps Across Rarefactions

It is possible to establish jump relations across rarefaction waves by means of generalised Riemann invariants; see Chap. 9 for background on Riemann invariants.

Proposition 10.6 (Left rarefaction wave). *Across a left rarefaction wave associated with the characteristic field $\lambda_1 = u - c$ the following relations hold*

$$u_{*L} = u_L - f_L , \quad f_L = \frac{2}{m}(c_{*L} - c_L) . \quad (10.53)$$

Proof (omitted). For details see [42]. ■

Proposition 10.7 (Right rarefaction wave). *Across a right rarefaction wave associated with the characteristic field $\lambda_3 = u + c$ the following relations hold*

$$u_{*R} = u_R + f_R , \quad f_R = \frac{2}{m}(c_{*R} - c_R) . \quad (10.54)$$

Proof (omitted). For details see [42]. ■

10.3.5 Jumps Across the Stationary Contact

We aim at establishing jump conditions across the stationary contact discontinuity associated with the eigenvalue $\lambda_2 = 0$. For variable material properties it is not possible to express the equations in conservation-law form and therefore it is not possible to apply the classical Rankine-Hugoniot jump conditions.

Proposition 10.8 (Stationary contact wave). *Across the contact discontinuity associated with the eigenvalue $\lambda_2 = 0$ the following relations hold*

$$Au = q = \text{constant} , \quad \frac{1}{2}\rho u^2 + \psi = \text{constant} , \quad (10.55)$$

leading to

$$A_{*L}u_{*L} = q_{*L} = A_{*R}u_{*R} = q_{*R} , \quad \frac{1}{2}\rho u_{*L}^2 + \psi_{*L} = \frac{1}{2}\rho u_{*R}^2 + \psi_{*R} . \quad (10.56)$$

Proof To establish jump conditions, and thus prove the above proposition, we adopt two alternative approaches, leading to identical results. In what follows we perform the calculations for each case separately.

Use of Riemann Invariants

We first apply generalised Riemann invariants across the linearly degenerate field. This approach was advocated by Embid and Baer [47] to analyse the Baer-Nunziato equations for two-phase compressible flow, a well-known non-conservative system [48]. From (10.40) we obtain

$$\frac{dA}{\frac{A}{\rho} \frac{\psi_K}{u^2 - c^2}} = \frac{d(Au)}{0} = \frac{dK}{1} . \quad (10.57)$$

The second member of the above equation states immediately that across the contact discontinuity $d(Au) = 0$ and thus $Au = \text{constant}$, proving the first result in (10.55). Relations (10.57) also state that across the contact wave A and K change; see Fig. 10.3. Equating the first and third members in (10.57) gives

$$\frac{A}{\rho} \left(\frac{\psi_K}{u^2 - c^2} \right) dK = dA . \quad (10.58)$$

Using $d(Au) = 0$ just proved and $c^2 = \frac{A}{\rho} \psi_A$ from the definition of wave speed in (10.37) we may write

$$\psi_K dK = -\rho u du - \psi_A dA , \quad (10.59)$$

or

$$\rho u du + \psi_A dA = -\psi_K dK . \quad (10.60)$$

But $\psi = \psi(A; K)$ and thus $d\psi = \psi_A dA + \psi_K dK$. Therefore $\rho u du + d\psi = 0$, which after integration gives the second sought result in (10.55). ■

The Thin-Layer Approach

Now we adopt the thin-layer approach advocated by Schwendeman et al. [49] and used to analyse the Baer-Nunziato equations; see also [50]. It is assumed that the transition layer containing the contact discontinuity is vanishingly thin and that the solution is smooth within the layer. Assuming the layer travels with constant speed S we define the independent variable

$$\xi = x - St , \quad S = \text{constant} , \quad (10.61)$$

which measures distance across the layer. We now study the governing equations locally. For any function $G(x, t)$ we have

$$\frac{\partial G}{\partial x} = \frac{\partial G}{\partial \xi} \frac{\partial \xi}{\partial x}, \quad \frac{\partial G}{\partial t} = \frac{\partial G}{\partial \xi} \frac{\partial \xi}{\partial t}. \quad (10.62)$$

Then the continuity equation in (10.28) gives

$$\frac{\partial A}{\partial \xi} \frac{\partial \xi}{\partial t} + \frac{\partial Au}{\partial \xi} \frac{\partial \xi}{\partial x} = 0, \quad (10.63)$$

or

$$d((u - S)A) = 0. \quad (10.64)$$

Hence with $S = 0$ we obtain $d(Au) = 0$, which is the first sought result in (10.55). Analogous manipulations for the momentum equation give $\frac{1}{2}\rho u^2 + \psi = \text{constant}$, which is the second sought result in (10.55) and the result is thus proved. ■

10.3.6 Complete Solution of the Riemann Problem

All the necessary relations to obtain the solution of the Riemann problem in the **Star Region** have been established. The procedure to find the **complete solution** is embodied in the following proposition.

Proposition 10.9 *The solution of the Riemann problem in the **Star Region** is given by the solution of the following non-linear system*

$$\left. \begin{aligned} f_1(x_1, x_2) &= x_2 - u_L + f_L(x_1) &&= 0, \\ f_2(x_1, x_2, x_3, x_4) &= x_2x_1 - x_4x_3 &&= 0, \\ f_3(x_1, x_2, x_3, x_4) &= \frac{1}{2}\rho(x_2^2 - x_4^2) + K_L \left[\left(\frac{x_1}{A_{0L}} \right)^m - 1 \right] \\ &\quad - K_R \left[\left(\frac{x_3}{A_{0R}} \right)^m - 1 \right] &&= 0, \\ f_4(x_3, x_4) &= x_4 - u_R - f_R(x_3) &&= 0. \end{aligned} \right\} \quad (10.65)$$

The unknowns of the problem are

$$\mathbf{X} = [x_1, x_2, x_3, x_4] \equiv [A_{*L}, u_{*L}, A_{*R}, u_{*R}], \quad (10.66)$$

with

$$f_L(x_1) = \begin{cases} \sqrt{\frac{B_L(x_1 - A_L)(x_1^{m+1} - A_L^{m+1})}{A_L x_1}} & \text{if } A_{*L} > A_L, \\ \frac{2}{m} (D_L x_1^{m/2} - c_L) & \text{if } A_{*L} \leq A_L, \end{cases} \quad (10.67)$$

and

$$f_R(x_3) = \begin{cases} \sqrt{\frac{B_R(x_3 - A_R)(x_3^{m+1} - A_R^{m+1})}{A_R x_3}} & \text{if } A_{*R} > A_R, \\ \frac{2}{m} (D_R x_3^{m/2} - c_R) & \text{if } A_{*R} \leq A_R, \end{cases} \quad (10.68)$$

with constants

$$D_L = \sqrt{\frac{m K_L}{\rho A_{0L}^m}}, \quad D_R = \sqrt{\frac{m K_R}{\rho A_{0R}^m}}. \quad (10.69)$$

The wave speeds c_L and c_R are evaluated on the data according to (10.37). The constants K_L and K_R are evaluated on the data from (10.29) and B is given by (10.49) and (10.52).

Proof The proof involves putting together all relevant results stated previously. Details are found in [42]. ■

Remarks:

- **Complete solution.** The numerical solution of the non-linear system (10.65) gives the four unknowns in the **Star Region**. The rest of the solution follows through a **sampling procedure** by applying the wave relations studied in the previous sections. Part of the sampling process for finding the complete solution involves computing the solution at a point inside a rarefaction wave. Details of the solution sampling procedure are given in Chap. 9.
- **Resonance.** A limitation of our formulation for discontinuous material properties is the potential occurrence of **resonance**. This happens when either of the eigenvalues $\lambda_1 = u - c$ or $\lambda_3 = u + c$ vanish, locally creating critical flow, with a non-linear wave overlapping the stationary wave associated with the eigenvalue $\lambda_2 = 0$. Discussions on theoretical issues regarding resonance are found in [51, 52], and references therein. Another difficulty with the formulation is the loss of uniqueness, as shown by preliminary results obtained by the authors. These are pending issues that require further investigations.

10.3.7 Sample Solutions

We consider two test problems for a straight tube of length 1 m , of constant equilibrium cross sectional area $A_0 = 2.1124 \times 10^{-4}\text{ m}^2$, of wall thickness $h_0 = 8.2 \times 10^{-4}\text{ m}$, Young's modulus $E = 3.0 \times 10^5\text{ N/m}^2$, exponent in tube law (10.29) $m = \frac{1}{2}$ and Poisson ratio $\nu = \frac{1}{2}$. The coefficient K is taken as

$$K_0 = \frac{4\sqrt{\pi}}{3} \frac{Eh_0}{\sqrt{A_0}}. \quad (10.70)$$

Blood density is taken as $\rho = 1050\text{ kg/m}^3$. Initial data for two Riemann problems are given in Table 10.1. Table 10.2 shows values for the exact solution in the **Star Region** for areas A_{*L} , A_{*R} and velocities u_{*L} , u_{*R} . Note that $K_{*L} = K_L$ and $K_{*R} = K_R$. These numbers by themselves can be useful to test numerical methods.

Figure 10.3 shows exact solution profiles for Test 1 at the output time $T_{out} = 0.012\text{ s}$. Quantities shown are: vessel diameter, velocity and Froude number $|u|/c$. Also shown is the resulting wave configuration in the $x-t$ plane. The solution consists of a left facing rarefaction, a stationary contact discontinuity and a right facing shock wave.

Figure 10.4 shows exact solution profiles for Test 2 at the output time $T_{out} = 0.0075\text{ s}$. Quantities shown are: vessel diameter, velocity and Froude number. See also the wave configuration in the $x-t$ plane. The solution consists of a left rarefaction wave, a stationary contact discontinuity and a right rarefaction wave. Note the close proximity of the left rarefaction to the stationary contact, giving rise to a **near resonance situation**.

Table 10.1 Initial conditions for two tests consisting of Riemann problems, for which the exact solution is computed. See values in the **Star Region** given in Table 10.2

Test	$A_L\text{ [m}^2\text{]}$	$u_L\text{ [m/s]}$	K_L	$A_R\text{ [m}^2\text{]}$	$u_R\text{ [m/s]}$	K_R
1	3.000×10^{-4}	-2.6575×10^{-5}	$50 \times K_0$	$3.000E - 04$	6.1230×10^{-6}	K_0
2	3.100×10^{-4}	-3.9967	K_0	3.100×10^{-4}	11.9943	$40 \times K_0$

Table 10.2 Exact solution in the **Star Region** for the two Riemann problems given in Table 10.1

Test	$A_{*L}\text{ [m}^2\text{]}$	$u_{*L}\text{ [m/s]}$	K_{*L}	$A_{*R}\text{ [m}^2\text{]}$	$u_{*R}\text{ [m/s]}$	K_{*R}
1	2.0119×10^{-4}	12.8103	K_L	6.3403×10^{-4}	4.0648	K_R
2	9.3917×10^{-5}	0.9625	K_L	2.0784×10^{-4}	0.4349	K_R

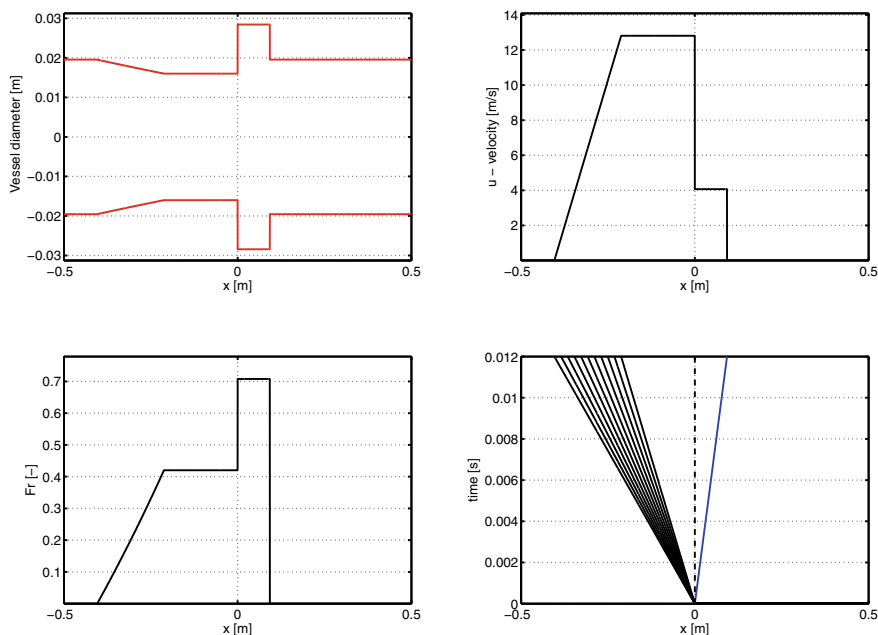


Fig. 10.3 Test 1: Exact solution at the output time $T_{out} = 0.012$ s for initial conditions given in Table 10.1. The solution consists of a left rarefaction wave, a stationary contact discontinuity and a right shock wave (Courtesy of Prof. A. Siviglia, University of Trento, Italy)

10.3.8 Closing Remarks

A simple mathematical model for blood flow in arteries with an elastic tube law and a single discontinuity in the mechanical properties of the arterial vessel wall has been presented. The equations have been analysed and the exact solution to the Riemann problem has been derived. These solutions can be useful in constructing and assessing numerical methods for solving the PDEs resulting from more complex models. The formulation and methodology, presented here in a simplest setting, have become the bases for the construction of more complex models, including elastic tube laws for arteries and veins, and multiple variable geometric and mechanical parameters [33, 41]. The formulation and methodology have also been extended to viscoelastic tube laws for arteries and veins, and multiple variable geometric and mechanical parameters [38]. Such models have been implemented in the Müller-Toro global model for the entire human circulation [21, 23]. The contents of this section, including the physical, physiological and mathematical aspects, have also a pedagogical value in a teaching/research environment for graduate students and researchers.

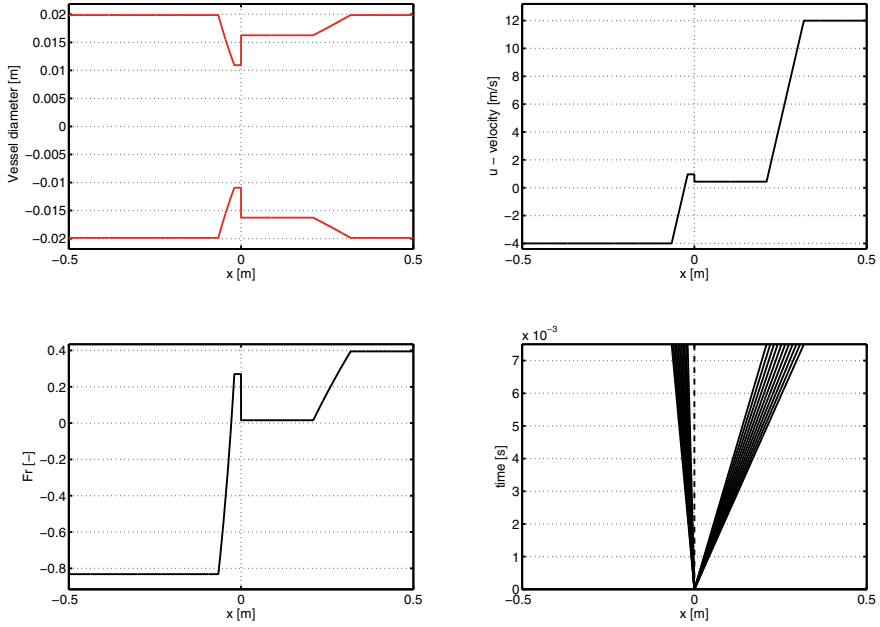


Fig. 10.4 Test 2: Exact solution at the output time $T_{out} = 0.0075$ s for initial conditions given in Table 10.1. The solution consists of a left rarefaction wave, a stationary contact discontinuity and a right rarefaction wave (Courtesy of Prof. A. Sivilgia, University of Trento, Italy)

In readiness for the introduction of viscoelastic tube laws in later sections, which render the equations **parabolic**, in the next section we study a mathematical approach for approximating parabolic equations by hyperbolic equations with stiff source terms.

10.4 Hyperbolisation: Advection-Diffusion-Reaction PDEs

Consider the linear advection-diffusion-reaction (ADR) equation

$$\partial_t q_1(x, t) + \lambda \partial_x q_1(x, t) = \alpha \partial_x^2 q_1(x, t) + \beta q_1(x, t) \quad (10.71)$$

for the unknown function $q_1(x, t)$. Here λ is a wave propagation speed (constant), $\alpha > 0$ is a constant diffusion coefficient and $\beta \leq 0$ is the reaction term coefficient, also a constant. Though not necessary to explain the hyperbolisation, or relaxation, approach to be introduced, we allow for the possibility of a stiff reaction source term $\beta q_1(x, t)$, that is $|\hat{\beta}| \gg 1$.

10.4.1 Hyperbolic System from a Parabolic Equation

By mimicking Cattaneo's law [53], as anticipated in Chap. 7 we consider a **relaxation time** ε , with $0 < \varepsilon \ll 1$, and an auxiliary function $q_2(x, t)$ such that

$$q_2(x, t) \longrightarrow \partial_x q_1(x, t) \text{ as } \varepsilon \longrightarrow 0. \quad (10.72)$$

Consistent with (10.72) we now assume the following evolution partial differential equation for the auxiliary function $q_2(x, t)$

$$\partial_t q_2(x, t) = [\partial_x q_1(x, t) - q_2(x, t)] \frac{1}{\varepsilon}. \quad (10.73)$$

The pair of Eqs. (10.71) and (10.73) give rise to the following system of PDEs

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}), \quad (10.74)$$

with

$$\left. \begin{aligned} \mathbf{Q} &= \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} q_1 \\ \partial_x q_1 \end{bmatrix}, \quad \mathbf{F}(\mathbf{Q}) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} \lambda q_1 - \alpha q_2 \\ -q_1/\varepsilon \end{bmatrix}, \\ \mathbf{S}(\mathbf{Q}) &= \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} \beta q_1 \\ -q_2/\varepsilon \end{bmatrix}. \end{aligned} \right\} \quad (10.75)$$

Analysis of the eigenstructure of system (10.74)–(10.75) gives the following real and distinct eigenvalues

$$\lambda_1 = \frac{1}{2}\lambda - \sqrt{\left(\frac{1}{2}\lambda\right)^2 + \frac{\alpha}{\varepsilon}}, \quad \lambda_2 = \frac{1}{2}\lambda + \sqrt{\left(\frac{1}{2}\lambda\right)^2 + \frac{\alpha}{\varepsilon}} \quad (10.76)$$

and the corresponding linearly independent right eigenvectors, suitably normalised

$$\mathbf{R}_1 = \begin{bmatrix} \varepsilon \lambda_1 \\ -1 \end{bmatrix}, \quad \mathbf{R}_2 = \begin{bmatrix} \varepsilon \lambda_2 \\ 1 \end{bmatrix}. \quad (10.77)$$

Obviously system (10.74)–(10.75) is (strictly) hyperbolic. The infinite wave propagation speed of the parabolic problem has become a pair of finite speeds represented by the eigenvalues λ_1 and λ_2 in (10.76). A choice for the **relaxation parameter** ε must be made in practice, for which the theoretical condition (10.72) is not helpful. A practical selection was put forward by Montecinos et al. [38, 54], where the choice of a finite ε is subject to the size Δx of the mesh and the order of accuracy $\mathcal{O}$ of the numerical method used to approximate (10.74)–(10.75).

The hyperbolisation approach has some drawbacks. First, a single equation became a system of equations, and second, a new, stiff source term appears on the right hand side of (10.74)–(10.75). Therefore, the chosen numerical method must be able to deal with stiff source terms. We remark that even if the original equations do not contain a stiff source term, the hyperbolisation (relaxation) approach gives an augmented system with a stiff source term. For definitions of stiff source terms see Chap. 15 of [55], or Chap. 13 of [44]. Further considerations on these issues are found in [38, 54, 56].

10.4.2 Advection-Diffusion Splitting

From the point of view of computing solutions to the emerging hyperbolic system (10.74)–(10.75) it is attractive to consider applying the TV splitting approach of Toro and Vázquez [42]. Originally, in the TV approach one splits the advection terms from the pressure terms, allowing for the possibility of using an implicit discretisation of the pressure terms, as done in [57].

Here we consider splitting the advection term from the diffusion term, allowing for the possibility of an implicit treatment of the latter as follows. Split the flux vector into an **advection** part and a **diffusion** part in this manner

$$\mathbf{F}(\mathbf{Q}) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \mathbf{F}^{(a)}(\mathbf{Q}) + \mathbf{F}^{(d)}(\mathbf{Q}) = \begin{bmatrix} \lambda q_1 \\ 0 \end{bmatrix} + \begin{bmatrix} -\alpha q_2 \\ -q_1/\varepsilon \end{bmatrix}, \quad (10.78)$$

with the obvious notation for the advection and diffusion fluxes $\mathbf{F}^{(a)}(\mathbf{Q})$ and $\mathbf{F}^{(d)}(\mathbf{Q})$, respectively. Following the TV approach we first form two systems of PDEs, namely

- **Advection system**

$$\left. \begin{aligned} \partial_t \mathbf{Q} + \partial_x \mathbf{F}^{(a)}(\mathbf{Q}) &= \mathbf{0}, \\ \mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} &= \begin{bmatrix} q_1 \\ \partial_x q_1 \end{bmatrix}, \quad \mathbf{F}^{(a)}(\mathbf{Q}) = \begin{bmatrix} \lambda q_1 \\ 0 \end{bmatrix}. \end{aligned} \right\} \quad (10.79)$$

- **Diffusion system**

$$\left. \begin{aligned} \partial_t \mathbf{Q} + \partial_x \mathbf{F}^{(d)}(\mathbf{Q}) &= \mathbf{0}, \\ \mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} &= \begin{bmatrix} q_1 \\ \partial_x q_1 \end{bmatrix}, \quad \mathbf{F}^{(d)}(\mathbf{Q}) = \begin{bmatrix} -\alpha q_2 \\ -q_1/\varepsilon \end{bmatrix}. \end{aligned} \right\} \quad (10.80)$$

Then, for each subsystem one derives corresponding numerical fluxes, whose sum gives the numerical flux for complete, unsplit system (10.74)–(10.75). More details

on the numerical aspects of the TV approach are found in Chaps. 13 and 14. In any case one considers Riemann problems for each separate subsystem, as explained below.

10.4.3 Riemann Problem for the Diffusion System

As the TV splitting depends fundamentally on the diffusion system, we now discuss the associated Riemann problem, that is

$$\left. \begin{aligned} \partial_t \mathbf{Q} + \partial_x \mathbf{F}^{(d)}(\mathbf{Q}) &= \mathbf{0}, \quad x \in \mathbb{R}, \quad t > 0, \\ \mathbf{Q}(x, 0) &= \begin{cases} \mathbf{Q}_L & \text{if } x < 0, \\ \mathbf{Q}_R & \text{if } x > 0. \end{cases} \end{aligned} \right\} \quad (10.81)$$

$\mathbf{Q}$ and $\mathbf{F}^{(d)}(\mathbf{Q})$ are given in (10.80). The real eigenvalues are found to be

$$\lambda_1 = -\sqrt{\frac{\alpha}{\varepsilon}}, \quad \lambda_2 = \sqrt{\frac{\alpha}{\varepsilon}}. \quad (10.82)$$

Note the subsonic character of the system; one always has $\lambda_1 < 0 < \lambda_2$. The corresponding right eigenvectors, suitably normalised, are

$$\mathbf{R}_1 = \begin{bmatrix} \alpha \\ -\lambda_1 \end{bmatrix}, \quad \mathbf{R}_2 = \begin{bmatrix} \alpha \\ -\lambda_2 \end{bmatrix}. \quad (10.83)$$

Using the methods described in Chap. 7, one easily finds the solution in the **Star Region**, namely

$$\left. \begin{aligned} q_1^* &= \frac{\lambda_2 q_1^R - \lambda_1 q_1^L + \alpha(q_2^R - q_2^L)}{\lambda_2 - \lambda_1}, \\ q_2^* &= \frac{\lambda_1 \lambda_2 (q_1^R - q_1^L) + \alpha(\lambda_1 q_2^R - \lambda_2 q_2^L)}{\alpha(\lambda_1 - \lambda_2)}. \end{aligned} \right\} \quad (10.84)$$

Given the subsonic character of the system, no sampling is involved to find the Godunov state needed for flux evaluation in a numerical setting. See Chaps. 12, 14 and 13 for background on numerical aspects.

Having introduced the hyperbolisation approach in a simple setting, we next apply it to the PDEs resulting from adopting a **viscoelastic tube law**. To this end we follow closely the presentation of Colombo et al. [34].

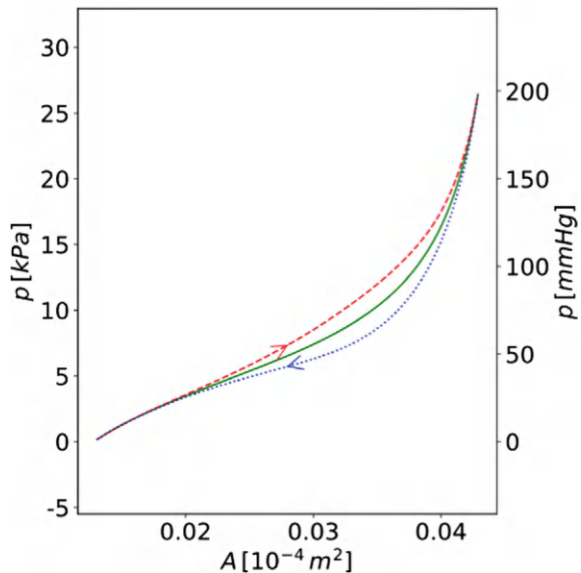
10.5 Viscoelastic Tube Laws and Parameter Estimation

This section improves the representation of the compliant vessel wall mechanics, previously represented by purely elastic tube laws. Here we introduce viscoelastic tube laws, which have proved to model more accurately wave propagation in the cardiovascular system, when coupled to the fluid dynamics of blood flow. For example, numerical simulations from elastic tube laws produce high frequency variations that viscoelastic simulations do not have. However, great care is required when interpreting results that have both physical and numerical components [20, 21, 24, 58]. Importantly, the experimentally observed phenomenon of **hysteresis** is mimicked quite satisfactorily by viscoelastic tube laws, but not at all by elastic tube laws. Figure 10.5 depicts a sketch of the hysteresis loop in the p - A plane, displaying the loading and unloading phases. Also seen is the single straight line from the elastic tube law, joining the extreme points.

For the rest of this section we shall assume that all geometric and mechanical parameters are constant. A major aspect concerns the study of the mathematical properties of the governing equations resulting from the introduction of new parameters via the viscoelastic tube laws. To this end we first approximate the resulting parabolic equations by a hyperbolic system with stiff source terms, following the hyperbolisation procedure introduced in Sect. 10.4 via a model advection-diffusion-reaction PDE.

The benefits of the hyperbolisation approach are two fold. First, to establish well possessedness, it allows a mathematical analysis of the viscoelastic equations using methodology applicable to hyperbolic equations. Second, it facilitates

Fig. 10.5 Hysteresis phenomenon in the pressure-area plane from a viscoelastic tube law during the loading phase (red dashed line) when $\partial_t A > 0$, and the unloading phase (blue dotted line) when $\partial_t A < 0$. The elastic tube law curve, shown by the continuous green line, cannot represent the hysteresis loop



the application of advanced higher order methods of the Godunov type [55]. For steady differential equations, the hyperbolisation approach has been pioneered by Nishikawa and collaborators [59–61]. For time-dependent advection-diffusion-reaction PDEs, the approach has more recently been developed by Toro and collaborators [38, 54, 56, 62]. Very recently the approach has been extended to higher order PDEs by Ketcheson and Biswas [63, 64].

Colombo et al. [34] exploited the hyperbolic approximation and the associated mathematical analysis tools to study the problem of estimation of parameters for viscoelastic tube laws.

10.5.1 Hyperbolisation and Analysis of the Equations

We start by recalling the PDEs for the dynamics of blood flow, namely

$$\left. \begin{aligned} \partial_t A + \partial_x (Au) &= s_1, \\ \partial_t (Au) + \partial_x (\alpha Au^2) + \frac{A}{\rho} \partial_x p &= s_2, \\ s_1 &= 0, \quad s_2 = -2(\beta + 2)\pi \frac{\mu}{\rho} u, \quad \alpha(\beta) = \frac{\beta + 2}{\beta + 1}, \quad \beta > 0. \end{aligned} \right\} \quad (10.85)$$

Viscoelastic Tube Laws

As seen in previous sections, Eq. (10.85) need to be supplemented by a closure condition. Here we assume the following viscoelastic tube law

$$\left. \begin{aligned} \mathbb{P}(A; K) &:= p - p_e = \mathbb{P}^E(A; K) + \mathbb{P}^V(A; K), \\ \mathbb{P}^E(A; K) &= K(x)\varphi(A; A_0) + p_0, \\ \varphi(A; A_0) &= \left[\left(\frac{A}{A_0} \right)^m - \left(\frac{A}{A_0} \right)^n \right], \\ \mathbb{P}^V(A; K) &= C_v \partial_t A, \\ C_v(A; A_0) &= \frac{\Gamma}{A_0 \sqrt{A}}. \end{aligned} \right\} \quad (10.86)$$

Here $\mathbb{P}^E$ denotes the elastic contribution and $\mathbb{P}^V$ denotes the viscoelastic contribution. As for the elastic case studied previously, the distinction between arteries and veins comes through the **coefficient of elasticity** $K(x)$ and the exponents m and n , namely

$$K(x) = \begin{cases} K_a(x) = \frac{E(x)}{1 - \nu^2} \frac{h_0(x)}{r_0(x)}, & m = \frac{1}{2}, n = 0 \quad (\text{arteries}), \\ K_v(x) = \frac{E(x)}{12(1 - \nu^2)} \left(\frac{h_0(x)}{r_0(x)} \right)^3, & m \approx 10, n = -\frac{3}{2} \quad (\text{veins}). \end{cases} \quad (10.87)$$

Most parameters in (10.86)–(10.87) are as defined for (10.19)–(10.20). New parameters include Γ in the viscoelastic part. The elastic term includes the reference pressure p_0 . For an interpretation of p_0 see Fig. 10.1.

Hyperbolising the Parabolic Equations

From the viscoelastic tube law (10.86)–(10.87) we write the transmural pressure in full as

$$\mathbb{P}(A; K) = K(x)\varphi(A; A_0) + p_0 + C_v(A; A_0)\partial_t A, \quad (10.88)$$

where K is chosen from (10.87) for arteries or veins. From the momentum equation in (10.85) we express the time derivative in terms of a spatial derivative, namely

$$\partial_t A = -\partial_x(Au) = -\partial_x q. \quad (10.89)$$

This direct relationship between temporal and spatial derivatives defines a single variable, namely flow q , to which we apply the hyperbolisation (or relaxation) procedure introduced in Sect. 10.4, thus augmenting the system (10.85) by one equation. By mimicking Cattaneo's law [53], we consider a **relaxation time** ε , with $0 < \varepsilon \ll 1$, and define the auxiliary function $\eta(x, t)$ thus

$$\eta(x, t) \longrightarrow \partial_x q(x, t), \quad \text{as } \varepsilon \longrightarrow 0. \quad (10.90)$$

Consistent with (10.90) we now assume the following evolution equation for $\eta(x, t)$

$$\partial_t \eta(x, t) = [\partial_x q(x, t) - \eta(x, t)] \frac{1}{\varepsilon}. \quad (10.91)$$

Replacing q by η , by virtue of (10.90), and use of (10.89) into (10.88) yields

$$\mathbb{P}(A; K) = K(x)\varphi(A; A_0) + p_0 - C_v(A; A_0)\partial_x \eta. \quad (10.92)$$

The pressure gradient term $\frac{A}{\rho} \partial_x p$ (10.85) is calculated using the transmural pressure expression (10.92). The resulting 3×3 system from (10.85) and (10.91), in quasilinear form reads

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q})\partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}), \quad (10.93)$$

with

$$\begin{aligned}
 \mathbf{Q} &= \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \equiv \begin{bmatrix} A \\ Au \\ \eta \end{bmatrix}, \quad \mathbf{A}(\mathbf{Q}) = \begin{bmatrix} \frac{A}{\rho} \frac{\partial p}{\partial A} + \frac{0}{2\rho} - \alpha u^2 & 1 & 0 \\ 0 & 2\alpha u & -\frac{A}{\rho} C_v \\ -\frac{1}{\varepsilon} & 0 & 0 \end{bmatrix}, \\
 \mathbf{S}(\mathbf{Q}) &= \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ s_2 \\ -\frac{\eta}{\varepsilon} \end{bmatrix}.
 \end{aligned} \tag{10.94}$$

Note that the hyperbolised system has a **stiff source term** $-\eta/\varepsilon$, which has arisen from the hyperbolisation procedure; from the computational point of view this source term represents a challenge. In what follows, however, we are concerned with the principal part of PDEs (10.93), as the source terms $\mathbf{S}(\mathbf{Q})$ do not enter the mathematical analysis of the equations. Moreover, for the rest of this section we assume the Coriolis coefficient to be $\alpha = 1$.

Eigenstructure and Hyperbolicity

Here we analyse the mathematical character of the resulting equations, coupled to a viscoelastic tube law, and subsequently hyperbolised. The reader is referred to Chap. 8 for necessary background on the theory of hyperbolic equations.

Proposition 10.10 *For $\alpha = 1$ the eigenvalues of (10.93)–(10.94) are*

$$\lambda_1 = u - c_{vh}, \quad \lambda_2 = u, \quad \lambda_3 = u + c_{vh}, \tag{10.95}$$

with the wave speed c_{vh} for the viscoelastic model given as

$$c_{vh} = \sqrt{\frac{A}{\rho} \frac{\partial p}{\partial A} + \frac{\eta C_v}{2\rho} + \frac{AC_v}{\rho\varepsilon}}. \tag{10.96}$$

Proof (omitted). For details see Colombo et al. [34]. ■

Proposition 10.11 *Hyperbolicity of Eqs. (10.93)–(10.94) requires that the discriminant in (10.93) be strictly positive, that is*

$$\frac{A}{\rho} \frac{\partial p}{\partial A} + \frac{\eta C_v}{2\rho} + \frac{AC_v}{\rho\varepsilon} > 0. \tag{10.97}$$

Proof After some algebraic manipulations the above inequality has been shown to be valid, for sufficiently small ε [34, 38]. Therefore the resulting viscoelastic model (10.93)–(10.94) is actually hyperbolic. ■

Proposition 10.12 *The right eigenvectors for $\mathbf{A}(\mathbf{Q})$ in (10.94), corresponding to the eigenvalues (10.95) are*

$$\mathbf{R}_1 = \begin{bmatrix} \alpha \\ u - c_{vh} \\ -\frac{1}{\varepsilon} \end{bmatrix}, \quad \mathbf{R}_2 = \begin{bmatrix} 1 \\ 0 \\ \frac{(c_*^2 - u^2)\rho}{C_v A} \end{bmatrix}, \quad \mathbf{R}_3 = \begin{bmatrix} \alpha \\ u + c_{vh} \\ -\frac{1}{\varepsilon} \end{bmatrix}, \quad (10.98)$$

where

$$c_* = \sqrt{\frac{A}{\rho} \frac{\partial p}{\partial A} + \frac{\eta C_v}{2\rho}}. \quad (10.99)$$

Proof For details see [38], and [34]. ■

Next we analyse the nature of the characteristic fields

Proposition 10.13 *(i) The $\lambda_2(\mathbf{Q})$ -characteristic field is linearly degenerate. (ii) If system (10.93)–(10.94) is hyperbolic and if $\varepsilon > 0$ is sufficiently small, then the $\lambda_1(\mathbf{Q})$ - and $\lambda_3(\mathbf{Q})$ -characteristic fields are genuinely non-linear.*

Proof Details are omitted. For further information see [38], and [34]. ■

Remarks:

1. As observed by Colombo et al. [34], in determining the nature of the characteristic fields, there are no additional mathematical constraints on the parameters of the viscoelastic tube law, only requiring hyperbolicity, which is always fulfilled if $\varepsilon > 0$ is chosen to be sufficiently small.
2. Mathematically, the viscoelastic tube law, in the setting of hyperbolisation, is much more generous than the purely elastic tube law, as it considerably relaxes the constraints imposed by the latter. For example, the exponents m and n in system (10.93)–(10.94) are constrained by the conditions $m > 0$ and $n \in [-2, 0]$ for the purely elastic tube law. Instead, in the viscoelastic case m and n can be any real numbers.
3. Importantly, the viscoelastic tube law, coupled to the hyperbolisation technique, has enabled us to better understand the mathematical nature of the resulting governing equations.
4. Tube laws must be parametrised in practice. This is a major subject, which is discussed next for the viscoelastic tube law chosen in this section.

10.5.2 Parameter Estimation for Tube Laws

As stated earlier, tube laws are designed to represent the mechanical properties of blood vessel walls via a pressure-area relationship. However, accurate determination of the parameters contained in the tube-law functional form poses challenging tasks. In order to, at least partly, address these challenges, experimental measurements

and rigorous assessment of proposed tube laws are of vital importance. The vessel network of the complete circulation is highly heterogeneous. Therefore, research aimed at designing tube laws must account for this heterogeneity, by recognizing, for example, the marked differences between arteries and veins, vessel size, and pressure ranges. There are many relevant studies available in the open literature, that have addressed some of these issues.

Zócalo et al. [12] experimentally characterised vein mechanical properties in sheep through pressure-diameter measurements; they noted district differences and pressure-range dependence. Bia et al. [15] performed an *in vitro* study of mechanical properties of carotid and aortic arterial segments in sheep and their correlation to the amount of vessel wall constituents, such as collagen, elastin and vascular smooth muscle cells.

Valdéz-Jasso et al. [14] performed an experimental and theoretical study of elastic and viscoelastic properties in the large arteries of 11 sheep. They noted differences along 7 locations studied. Experimental measurements were analysed using a two-parameter elastic model and a four-parameter Kelvin viscoelastic model. They remarked that the inclusion of viscoelastic behaviour is important in order to capture pressure-area dynamics.

Valdéz-Jasso et al. [3] obtained *in vivo* measurements of 10 ovine aortas and 10 human carotid arteries, and *ex vivo* measurements in eleven male merino sheep. They also analysed theoretical linear and non-linear viscoelastic models, developed from the framework of the Quasilinear Viscoelasticity Theory of Fung [2]. Three models were considered, namely a four-parameter (linear) Kelvin viscoelastic model and two five-parameter nonlinear viscoelastic models, an arctangent model and a sigmoid model.

Ghigo et al. [24] used a nonlinear Kelvin–Voigt model for the viscoelasticity of the sheep arterial wall and studied its influence on pulse waves. They confirmed that smaller arteries are stiffer, that is the viscoelastic coefficient increases towards peripheral vessels, to compensate for the increase of the Young's modulus, resulting in a higher damping effect. They incorporated their vessel-wall mechanical model in a one-dimensional fluid dynamics model for blood flow in an arterial network for the sheep, and computed pulse waves in the network.

The rest of this section is based on the recent work of Colombo et al. [34], which was concerned with the assessment of elastic and viscoelastic tube laws, utilising animal and human data. This work focused on the two major requirements that must be fulfilled by the tube law, if it is to be coupled to the fluid dynamics of the circulation. First, there are mathematical constraints imposed by the resulting PDEs, see Sect. 10.5.1; second, physiological accuracy is required, as compared with experimental data.

10.5.3 Experimental Data Acquisition

In the study of Colombo et al. [34], data sets from 12 adult sheep and 15 humans were considered, as detailed below.

- **Ovine arterial vessels:** (1) Ascending aorta (AA); (2) Brachiocephalic trunk (BT); (3) Carotid artery (CA); (4) Proximal descending aorta (PDA); (5) Medial descending aorta (MDA); (6) Distal descending aorta (DDA); (7) Femoral artery (FA); (8) Main pulmonary (MPA), and (9) Left pulmonary artery (LPA).
- **Ovine veins:** (1) Inferior vena cava (IVC); (2) Superior vena cava (SVC); (3) Femoral vein (FV), and (4) Jugular vein (JV).
- **Human vessels:** (1) Femoral artery (FA) and (2) Saphenous vein (SV).

Acquisition of pressure-diameter, or pressure-area data, was carried out in 2 stages. In the first stage, selected vessels from anaesthetised animals were instrumented for measurements, and *in vivo* data was collected. In the second stage, vessel segments from euthanised animals were mounted in custom cannulae within the flow circuit of an *in vitro* system, and *in vitro* data was collected. Then, *in vivo* data was used to verify the quality of the acquired *in vitro* data from the *in vitro* system. Data, in the form time series, were collected for 10 to 20 cardiac cycles, at intervals of 5 ms.

Regarding the pressure ranges used in the *in vitro* tests. For the ovine arteries, physiological pressure ranges were used. For the ovine veins, both low (physiological) and high (arterial pressure range) pressures were used. It is worth noting here that the performance of proposed tube laws depends strongly on the pressure ranges they are expected to operate in. For further details see Colombo et al. [34].

The set parameters to be estimated is denoted by $\mathbf{Z} = \{K, m, n, \Gamma\}$ to accommodate both elastic tube laws, for which $\Gamma = 0$, and viscoelastic tube laws. See Eqs. (10.86)–(10.87). The parameter estimation procedure is also constrained by the mathematical considerations dealt with in Sect. 10.5.1 and specified in Table 10.3. However, it is important to note that such mathematical constraints are only relevant in the case in which the tube law is to be coupled to the equations of fluid dynamics of blood flow.

Table 10.3 Constraints on the set of parameters $\mathbf{Z} = \{K, m, n, \Gamma\}$ in the optimization procedure for the elastic (Elastic(a) and Elastic(b)) and viscoelastic (Viscoelastic(a) and Viscoelastic(b)) tube laws

Case	K	m	n	Γ
Elastic(a)	$K > 0$	$m > 0$	$n \in [-2, 0]$	$\Gamma = 0$
Elastic(b)	$K > 0$	$m = 10.0$	$n = -3/2$	$\Gamma = 0$
Viscoelastic(a)	$K > 0$	$m > 0$	$n \in [-2, 0]$	$\Gamma \geq 0$
Viscoelastic(b)	$K > 0$	$m \in \mathbb{R}$	$n \in \mathbb{R}$	$\Gamma \geq 0$

The parameter estimation procedure is fully described in [34], and therefore we omit its reproduction here.

10.5.4 Sample Results from Parameter Estimation

From the study by Colombo et al. [34], here we consider four cases containing two classes of results: an elastic class and a viscoelastic class, as defined in Table 10.3, where for each case, conditions on the set of parameters $\mathbf{Z} = \{K, m, n, \Gamma\}$ are specified. There are two cases for the elastic tube law in (10.86)–(10.87), and two cases for the viscoelastic tube law in (10.86)–(10.87).

Optimal Parameters: Results and Discussion

From the copious results on optimal parameters reported by Colombo et al. [34] we highlight the following observations: of the four parameters in the set $\mathbf{Z} = \{K, m, n, \Gamma\}$, K and Γ are the ones that permit a more physiological interpretation of their optimal values reported. This observation supports the validity of viscoelastic tube law as a model, as compared with purely elastic tube laws. For instance, regarding the optimal values for the viscosity parameter Γ , it is worth noting that these do not differ significantly between the two viscoelastic cases, and for all types of vessel considered in their study.

On the other hand, the physiological interpretation of the parameters m and n is more challenging. The results from Colombo et al. [34] tell us that the usual ranges recommended in the literature are not valid in general. Moreover, variations can be very large and they not only depend on the vessel type, e.g. arteries versus veins, but also on the pressure ranges considered.

Regarding the optimal values for m and n , we note that for the case Elastic(a) in Table 10.3, these can be very different from the classical values $m = 10$ and $n = -3/2$ of case Elastic(b), for veins. The Elastic(b) case, for the sheep jugular vein, gives $n = -0.4$ and $m = 18.69$. Then, variations of n within the same class of vessels can be large. For instance, for sheep veins at low pressure (physiological), $n = -135.08$ for the inferior vena cava (IVC) and $n = 5.68$ for the jugular vein for the Elastic(b) case.

Variations can be large for the same vessel but for different pressure ranges. For example, for Elastic(a) $n = -1.6$ for the inferior vena cava at high pressure and $n = -0.8$ at low (physiological) pressure. As expected, the variations of m and n in the viscoelastic cases are still larger. Even for the same class of venous vessels variations are large. For example, at low pressure $n = -10.16$ for the sheep inferior vena cava (IVC) and $n = 5.65$ for the sheep jugular vein (JV). Then, variations with pressure range are also significant. For example, for the Viscoelastic(b) case for the inferior vena cava (IVC) at high pressure $n = -49.9$ and for low pressure $n = -10.16$.

From the above, partial observations, one could state that the viscoelastic tube law is to be preferred in applications. Moreover, detailed parametrisation of the tube

law is essential, not only for the local purpose of studying the mechanics of a single vessel, but also for coupling the tube law to the fluid dynamics in mathematical models for the circulation; see [24, 25, 38] and [23], for example.

Pressure-Area Loops and Pressure-Time Histories

Here we present a partial assessment of the performance of the elastic and viscoelastic tube laws with optimised parameters from the parameter estimation procedure, against experimental measurements. We do so in terms of pressure-area hysteresis loops and corresponding pressure vs. time curves, obtained for one cardiac cycle. Figure 10.6 depicts experimental pressure-area hysteresis loops shown by red lines, with measured data from [34]. The full black line depicts the averaged experimental $p(A)$ curve, displaying the ascending and descending phases.

Figures 10.7, 10.8, 10.9, 10.10, 10.11 and 10.12 show comparisons between the experimental curves depicted by full black line and the optimised tube-law cases, for one cardiac cycle; the viscoelastic curves are shown by dashed red lines and the elastic curves by dashed green lines. In each figure the top frame shows pressure-area hysteresis loops and the bottom frame shows pressure vs. time curves. Three sets of results are presented:

1. **Ascending aorta (AA).** Figure 10.7 shows results for the ascending aorta (AA) for the Elastic(a) and Viscoelastic(a) tube-law cases. These results are to be compared with those of Fig. 10.8 for the Elastic(b) and Viscoelastic(b) tube-law cases.

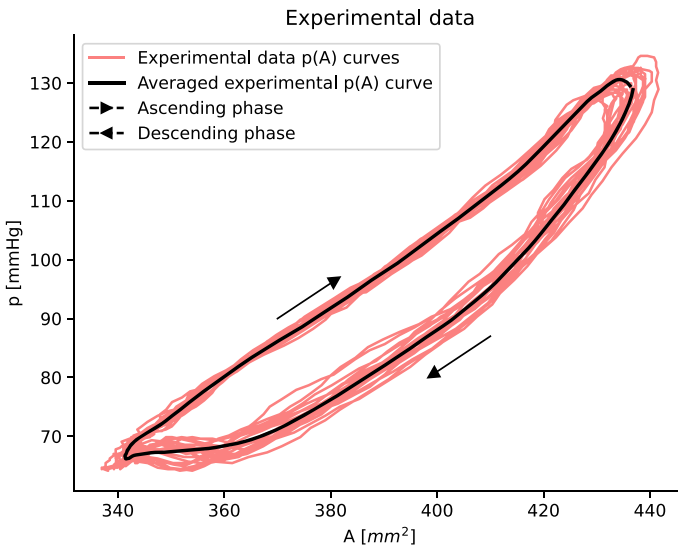


Fig. 10.6 Experimental hysteresis pressure-area loops shown by red lines. Black line depicts averaged $p(A)$ curve, displaying ascending and descending phases (courtesy of Ms. Chiara Colombo, University of Trento, Italy. Data from [34])

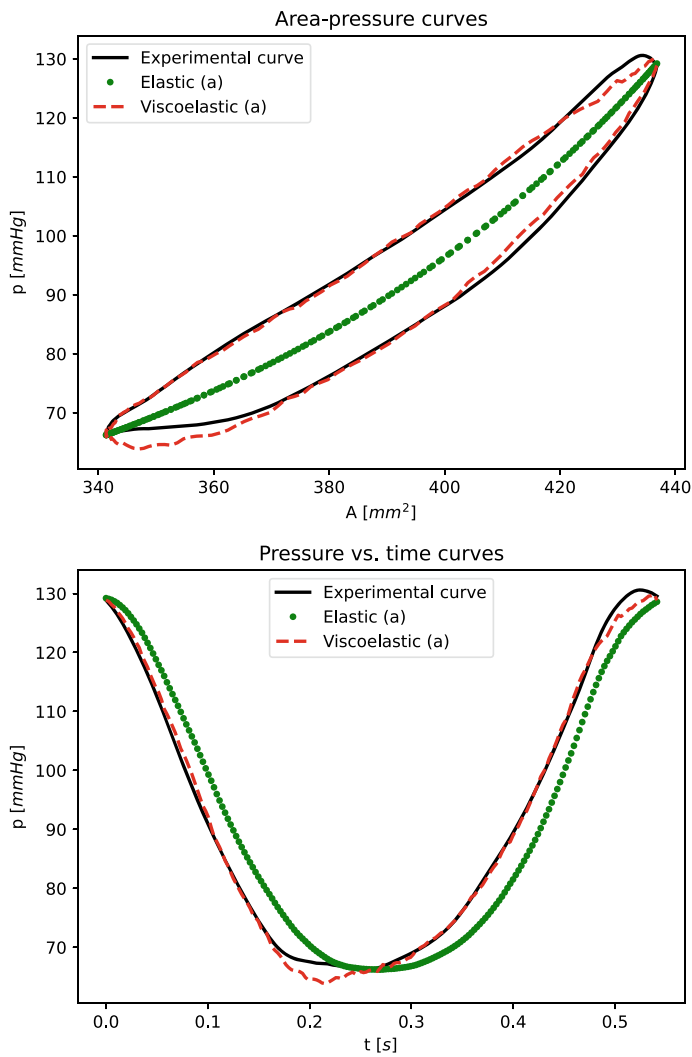


Fig. 10.7 Ascending aorta (AA) tube law results. Curves from elastic and viscoelastic tube laws are compared with experimental averaged curves. Cases shown: Elastic(a) and Viscoelastic(a) (courtesy of Ms. Chiara Colombo, University of Trento, Italy. Data from [34])

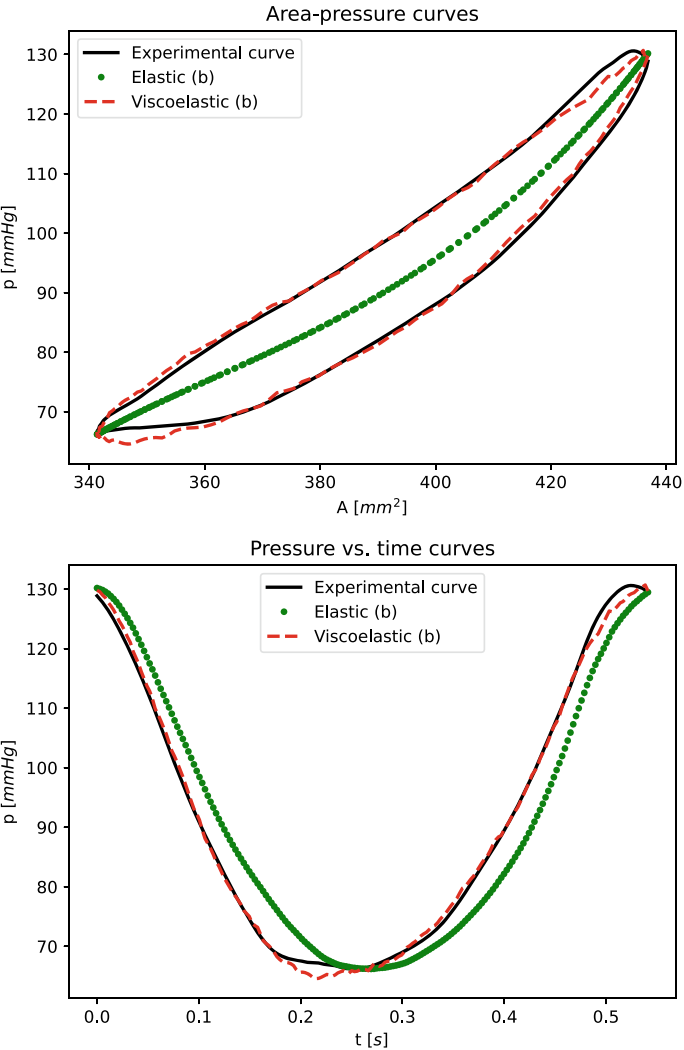


Fig. 10.8 Ascending aorta (AA) tube law results. Curves from elastic and viscoelastic tube laws are compared with experimental averaged curves. Cases shown: Elastic(b) and Viscoelastic(b) (courtesy of Ms. Chiara Colombo, University of Trento, Italy. Data from [34])

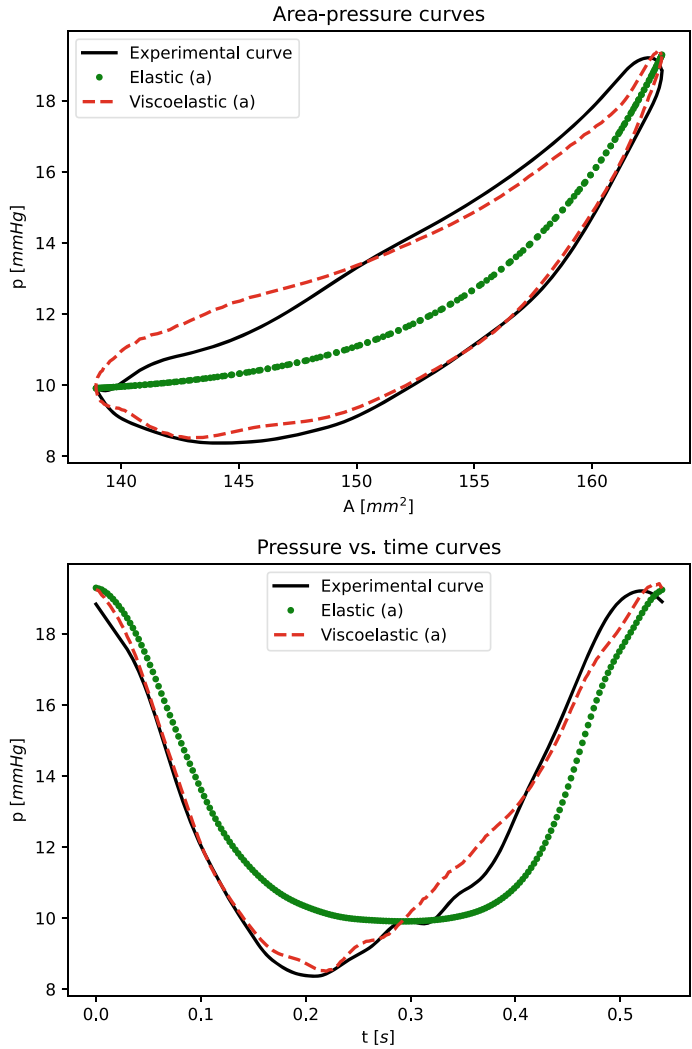


Fig. 10.9 Left pulmonary artery (LPA) tube law results. Curves from elastic and viscoelastic tube laws are compared with experimental averaged curves. Cases shown: Elastic(a) and Viscoelastic(a) (courtesy of Ms. Chiara Colombo, University of Trento, Italy. Data from [34])

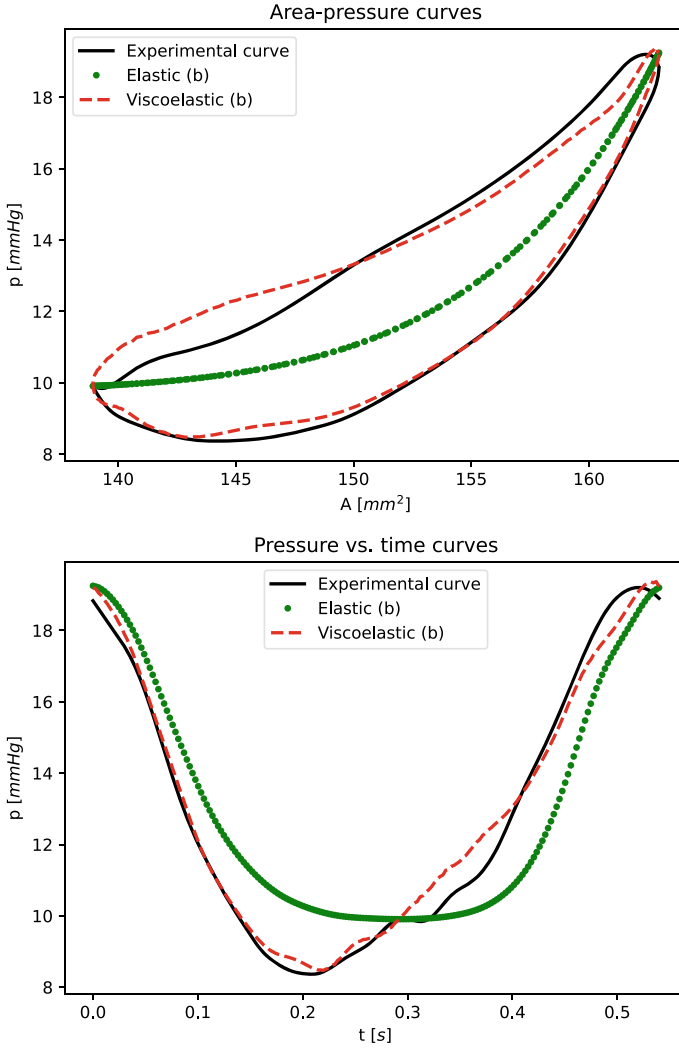


Fig. 10.10 Left pulmonary artery (LPA) tube law results. Curves from elastic and viscoelastic tube laws are compared with experimental averaged curves. Cases shown: Elastic(b) and Viscoelastic(b) (courtesy of Ms. Chiara Colombo, University of Trento, Italy. Data from [34])

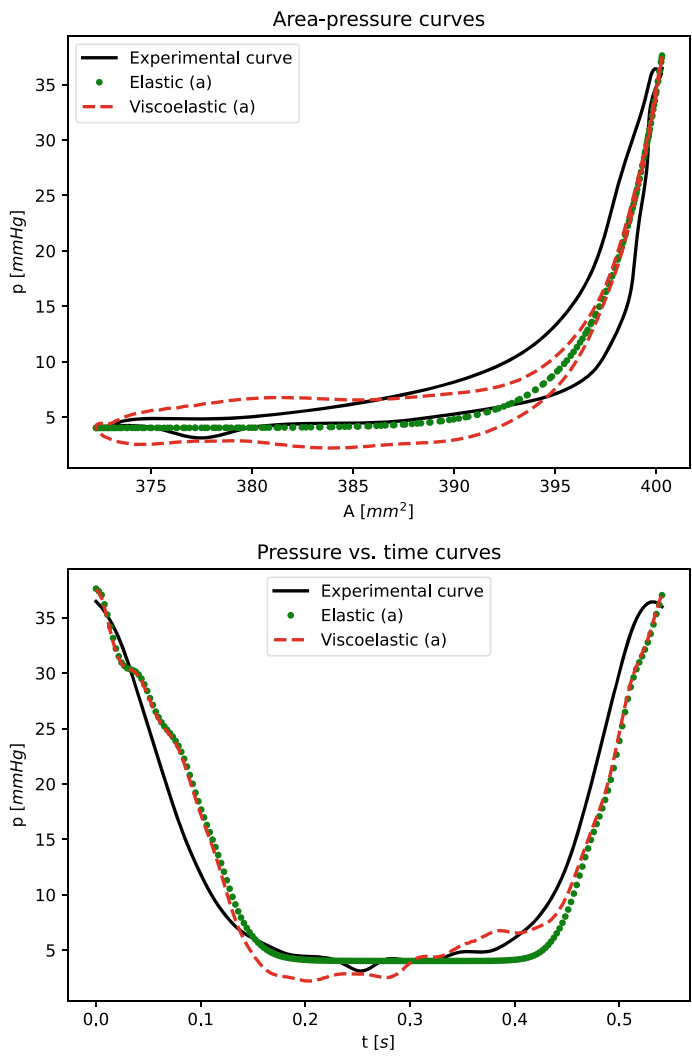


Fig. 10.11 Inferior vena cava (IVC) tube law results. Curves from elastic and viscoelastic tube laws are compared with experimental averaged curves. Cases shown: Elastic(a) and Viscoelastic(a) (courtesy of Ms. Chiara Colombo, University of Trento, Italy. Data from [34])

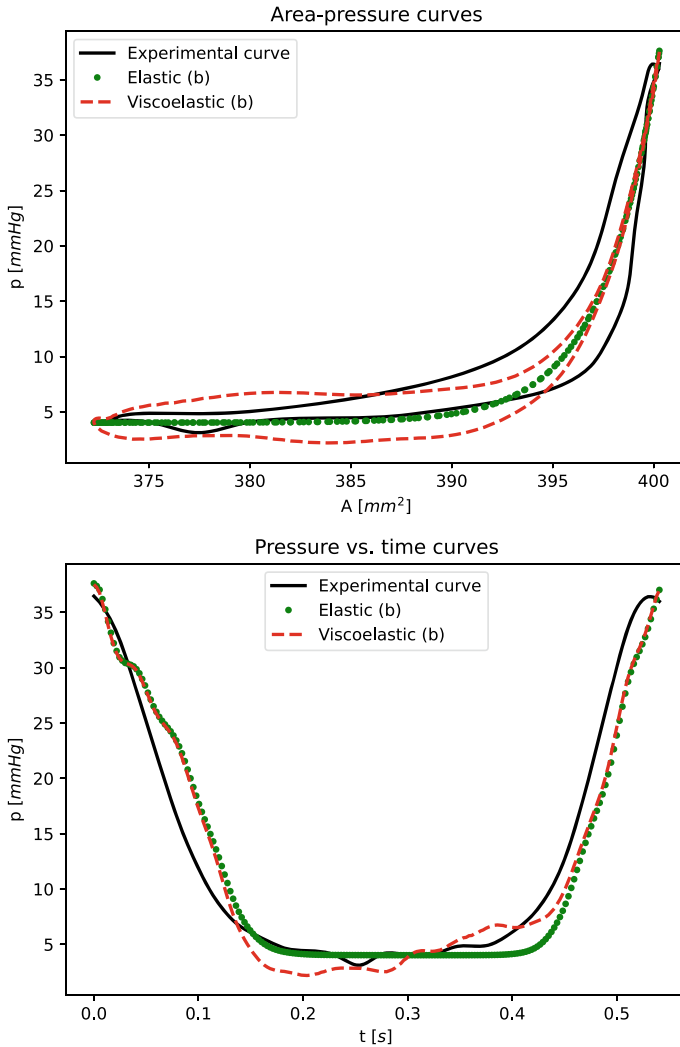


Fig. 10.12 Inferior vena cava (IVC) tube law results. Curves from elastic and viscoelastic tube laws are compared with experimental averaged curves. Cases shown: Elastic(b) and Viscoelastic(b) (courtesy of Ms. Chiara Colombo, University of Trento, Italy. Data from [34])

2. **Left pulmonary artery (LPA).** Figure 10.9 shows results for the left pulmonary artery (LPA) for the Elastic(a) and Viscoelastic(a) tube-law cases. These results are to be compared with those of Fig. 10.10 for the Elastic(b) and Viscoelastic(b) tube-law cases.
3. **Inferior vena cava (IVC).** Figure 10.11 shows results for the inferior vena cava (IVC) for the Elastic(a) and Viscoelastic(a) tube-law cases. These results are to be

compared with those of Fig. 10.12 for the Elastic(b) and Viscoelastic(b) tube-law cases.

Overall, results are very informative. As expected, the performance of the optimised tube laws depends strongly on the vessel type and the pressure range under which the tests were performed. One conclusion is clear: elastic tube laws are unable to represent a distinguishing physiological feature of blood vessels, namely the hysteresis loop. In the elastic case such loop is represented by just a single line in the interior of the loop, connecting its extreme points. Results are particularly satisfactory for the ascending aorta, shown in Figs. 10.7 and 10.8.

Less satisfactory are the results for the left pulmonary artery, shown in Figs. 10.9 and 10.10. The elastic tube law is rather inaccurate for the pressure time histories, in both the elastic and viscoelastic cases.

The results for the inferior vena cava, shown in Figs. 10.11 and 10.12, are satisfactory, noting that veins are notoriously nonlinear in the physiological pressure range. In this test at low (physiological) pressure the strong non-linearity of the tube laws is captured well, by both the elastic and viscoelastic tube laws. In the pressure-area curves in the top frame, one can observe the large area variations with very small pressure changes. And yet the viscoelastic tube law captures well this crucial feature of veins.

10.5.5 *Closing Remarks*

A viscoelastic tube law containing four parameters has been studied in this section. We noted that when the tube law is to be coupled to the PDEs for the fluid dynamics of blood, then the mathematical character of the governing fluid dynamics equations must be analysed and the constraints derived must be imposed in any associated parametrisation procedure. However, if the tube law remains a constitutive equation for the vessel wall mechanics per se, then the mathematical constraints are not applicable. In this Section the mathematical analysis of the equations has been facilitated by approximating the parabolic PDEs resulting from the viscoelastic tube law, by an enlarged hyperbolic system with stiff source terms, following the hyperbolisation methodology introduced in Sect. 10.4.

In this section all geometric and mechanical parameters have been assumed to be constant. In the next section we relax this condition and study viscoelastic tube laws with multiple, variable geometric and mechanical parameters. Analysis of the derived PDEs is performed in order to identify constraints on the tube law for preserving desirable mathematical properties of the governing PDEs.

10.6 Viscoelastic Models with Multiple Parameters

This section contains a generalisation of the models presented in previous sections. Multiple geometric and mechanical parameters are allowed to be variable with distance x , the tube law is viscoelastic and the resulting equations are hyperbolised and analysed. We follow the work of Montecinos, Müller and Toro [54].

10.6.1 Equations, Tube Law and Hyperbolisation

We start from the equations that govern the dynamics of blood flow, coupled to the wall mechanics equation through a viscoelastic tube law. For convenience we recall these equations here.

Equations with Variable Parameters and Viscoelastic Tube Law

The governing PDEs are:

$$\left. \begin{aligned} \partial_t A + \partial_x(Au) &= s_1, \\ \partial_t(Au) + \partial_x(\alpha Au^2) + \frac{A}{\rho} \partial_x p &= s_2, \\ s_1 &= 0, \quad s_2 = -2(\beta + 2)\pi \frac{\mu}{\rho} u, \quad \alpha(\beta) = \frac{\beta + 2}{\beta + 1}, \quad \beta > 0. \end{aligned} \right\} \quad (10.100)$$

The viscoelastic tube law is

$$\left. \begin{aligned} \mathbb{P}(A; K) &:= p - p_e = \mathbb{P}^E(A; K) + \mathbb{P}^V(A; K), \\ \mathbb{P}^E(A; K) &= K(x)\varphi(A; A_0) + p_0, \\ \varphi(A; A_0) &= \left[\left(\frac{A}{A_0} \right)^m - \left(\frac{A}{A_0} \right)^n \right], \\ \mathbb{P}^V(A; K) &= C_v(A; K) \partial_t A, \\ C_v(A; K) &= \frac{\Gamma}{A_0 \sqrt{A}}. \end{aligned} \right\} \quad (10.101)$$

Regarding the elastic case, the distinction between arteries and veins comes through the **coefficient of elasticity** $K(x)$ and the exponents m and n , namely

$$K(x) = \begin{cases} K_a(x) = \frac{E(x)}{1 - \nu^2} \frac{h_0(x)}{r_0(x)}, \quad m = \frac{1}{2}, \quad n = 0 & \text{(arteries),} \\ K_v(x) = \frac{E(x)}{12(1 - \nu^2)} \left(\frac{h_0(x)}{r_0(x)} \right)^3, \quad m \approx 10, \quad n = -\frac{3}{2} & \text{(veins).} \end{cases} \quad (10.102)$$

All parameters for (10.100) have already been defined in previous sections, e.g. Sect. 10.1.1. For the viscoelastic tube law (10.101)–(10.102), all parameters were defined in Sect. 10.5.1.

In this section we consider the following parameters to be variable and functions of distance x only, not of time t :

1. the coefficient of elasticity $K(x)$,
2. the equilibrium cross-sectional area $A_0(x)$ and
3. the external pressure $p_e(x)$.

From the continuity equation in (10.100) we express the time derivative of A in terms of the spatial derivative of q , that is

$$\partial_t A = -\partial_x(Au) = -\partial_x q . \quad (10.103)$$

Then, the pressure gradient $\partial_x p$ in the momentum equation in (10.100) can be calculated from the tube law (10.101), acknowledging (10.103) and the variation of parameters $K(x)$, $A_0(x)$ and $p_e(x)$. The result is

$$\begin{aligned} \partial_x p = [1] \partial_x p_e + [\varphi] \partial_x K + [K(\partial_A \varphi) - (\partial_A C_v)(\partial_x q)] \partial_x A \\ + [K(\partial_{A_0} \varphi) - (\partial_{A_0} C_v)(\partial_x q)] \partial_x A_0 - [C_v] \partial_x^{(2)} q . \end{aligned} \quad (10.104)$$

Substitution of $\partial_x p$ into the momentum equation in (10.100) and adding three trivial equations for the time derivative of the three parameters $K(x)$, $A_0(x)$, $p_e(x)$, we obtain the enlarged system

$$\left. \begin{aligned} \partial_t A + \partial_x q &= 0 , \\ \partial_t q + \partial_x \left(\alpha \frac{q^2}{A} \right) &= - \left[\frac{A}{\rho} \right] \partial_x p_e - \left[\frac{A}{\rho} \varphi \right] \partial_x K \\ &\quad - \frac{A}{\rho} [K(\partial_A \varphi) - (\partial_A C_v)(\partial_x q)] \partial_x A \\ &\quad - \frac{A}{\rho} [K(\partial_{A_0} \varphi) - (\partial_{A_0} C_v)(\partial_x q)] \partial_x A_0 \\ &\quad + \frac{A}{\rho} [C_v] \partial_x^{(2)} q + s_2 , \\ \partial_t K &= 0 , \\ \partial_t A_0 &= 0 , \\ \partial_t p_e &= 0 . \end{aligned} \right\} \quad (10.105)$$

Now we express (10.105) in quasilinear form as

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}) + \mathbf{S}^v(\mathbf{Q}) . \quad (10.106)$$

Here

$$\left. \begin{aligned}
 \mathbf{Q} &= [A, q, K, A_0, p_e]^T, \\
 \mathbf{A} &= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ c^2 - u^2 + \frac{C_v \partial_x q}{2\rho} & 2u & \frac{A}{\rho} \varphi & \frac{A}{A_0} \left(\frac{C_v \partial_x q}{\rho} - c^2 \right) & \frac{A}{\rho} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \\
 \mathbf{S} &= [0, s_2, 0, 0, 0]^T, \\
 \mathbf{S}^v(\mathbf{Q}) &= \left[0, \frac{C_v A}{\rho} \partial_x^{(2)} q, 0, 0, 0 \right]^T.
 \end{aligned} \right\} \quad (10.107)$$

Hyperbolisation of the Parabolic Equations

As in Sects. 10.4 and 10.5 we introduce a new variable η and a relaxation parameter $\varepsilon > 0$ such that

$$\eta \rightarrow \partial_x q, \quad \varepsilon \rightarrow 0. \quad (10.108)$$

In addition, by mimicking Cattaneo's law [53] we introduce an evolution equation for η , namely

$$\partial_t \eta = \frac{1}{\varepsilon} (\partial_x q - \eta). \quad (10.109)$$

With an extra unknown η , system (10.105) is replaced by the following enlarged system

$$\partial_t \mathbf{W} + \mathbf{M}(\mathbf{W}) \partial_x \mathbf{W} = \mathbf{S}(\mathbf{W}), \quad (10.110)$$

where

$$\left. \begin{aligned}
 \mathbf{W} &= [A, q, K, A_0, p_e, \eta]^T, \\
 \mathbf{A} &= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ c^2 - u^2 + \frac{a_\Gamma}{2} & 2u & \frac{A}{\rho} \varphi & \frac{A}{A_0} (a_\Gamma - c^2) & \frac{A}{\rho} & -\frac{A}{\rho} C_v \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{\varepsilon} & 0 & 0 & 0 & 0 \end{bmatrix}, \\
 \mathbf{S} &= [0, s_2, 0, 0, 0, -\frac{1}{\varepsilon} \eta]^T.
 \end{aligned} \right\} \quad (10.111)$$

Here

$$c^2 = \frac{A}{\rho} K \partial_A \varphi, \quad u = \frac{q}{A}, \quad a_\Gamma = \frac{C_v \eta}{\rho}. \quad (10.112)$$

10.6.2 Eigenstructure and Characteristic Fields

Having established the governing equations with a viscoelastic tube law, with three variable parameters and suitably hyperbolised, we now study their eigenstructure and the nature of the associated characteristic fields.

Proposition 10.14 *System (10.110) with coefficient matrix (10.111) is hyperbolic if*

$$\varepsilon^{-1} \geq -\frac{\eta}{2A} - \frac{\rho c^2}{C_v A}, \quad (10.113)$$

with eigenvalues

$$\lambda_1 = u - \tilde{c}, \quad \lambda_2 = \lambda_3 = \lambda_4 = \lambda_5 = 0, \quad \lambda_6 = u + \tilde{c}, \quad (10.114)$$

where

$$\tilde{c} = \sqrt{c^2 + \omega}, \quad \omega = \frac{\varphi A}{\rho \varepsilon} + \frac{a_\Gamma}{2} \quad (10.115)$$

and linearly independent right eigenvectors

$$\begin{aligned} \mathbf{R}_1 &= \begin{bmatrix} 1 \\ u - \tilde{c} \\ 0 \\ 0 \\ 0 \\ -\frac{1}{\varepsilon} \end{bmatrix}, & \mathbf{R}_2 &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{(c^2 + a_\Gamma/2) - u^2}{C_v A} \rho \end{bmatrix}, & \mathbf{R}_3 &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ \frac{\varphi}{C_v} \end{bmatrix}, \\ \mathbf{R}_4 &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ \frac{(a_\Gamma - c^2)}{C_v A_0} \rho \end{bmatrix}, & \mathbf{R}_5 &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ \frac{1}{C_v} \end{bmatrix}, & \mathbf{R}_6 &= \begin{bmatrix} 1 \\ u + \tilde{c} \\ 0 \\ 0 \\ 0 \\ -\frac{1}{\varepsilon} \end{bmatrix}. \end{aligned} \quad (10.116)$$

Proof (omitted). For details see [38]. ■

Proposition 10.15 *The characteristic fields associated to eigenvectors $\mathbf{R}_1$ and $\mathbf{R}_6$ are genuinely non-linear.*

Proof We introduce the notation for eigenvalues $\lambda_1 = \lambda_-$ and $\lambda_6 = \lambda_+$, with corresponding sub-indices for the right eigenvectors. The result follows by noting that

$$\nabla_{\mathbf{W}} \lambda_{\pm} \cdot \mathbf{R}_{\pm} = -\frac{1}{q} \pm \frac{1}{\bar{c}} \left(c \frac{\partial c}{\partial A} + \frac{\partial \omega}{\partial A} \right) - (u \pm \bar{c}) \frac{A}{q^2} \pm \frac{1}{2\bar{c}} \frac{\partial \omega}{\partial \eta} \neq 0 \quad (10.117)$$

and the proof is thus complete. ■

Proposition 10.16 *The characteristic fields associated to the eigenvectors $\mathbf{R}_2$ to $\mathbf{R}_5$ are linearly degenerate.*

Proof The proof follows by noting that for these eigenvectors, the associated eigenvalues are identically zero. Therefore

$$\nabla_{\mathbf{W}} \lambda_k \cdot \mathbf{R}_k = \mathbf{0}, \text{ for } k = 2, 3, 4, 5. \quad (10.118)$$

The proof is complete. ■

10.6.3 Closing Remarks

In this section we have presented a generalised blood flow model endowed with a viscoelastic tube law applicable to arteries and veins, in which three variable geometric and mechanical parameters are allowed to vary with distance x along the vessel. In addition, the resulting parabolic equations due to the viscoelastic component of the tube law are approximated by a hyperbolic system with stiff source terms, following the hyperbolisation procedure introduced in [38, 54, 56]. See also Sects. 10.4 and 10.5. The model of this section extends that of Sect. 10.3, in which the tube law was elastic but one parameter was allowed to vary with x . It also extends that of Sect. 10.5, in which the tube law was viscoelastic, but the geometric and mechanical parameters were constant.

The hyperbolisation procedure leading to hyperbolic systems has two main benefits. First, it allows the mathematical analysis of the equations employing tools from the theory of hyperbolic equations to establish hyperbolicity and thus well-posedness. Second, existing sophisticated numerical methods [44, 55] of the Godunov type and of high order of accuracy can be employed to solve the full haemodynamical problem involving the PDEs for the fluid dynamics and the tube law for the mechanics of the blood vessel wall. To this end, a practical choice of the relaxation time ε is necessary; the theoretical conditions $0 < \varepsilon \ll 1$ and $\varepsilon \rightarrow 0$ are not practical. Montecinos et al. [54] proved a functional form for the optimal choice for ε , which depends on the mesh size Δx , the order of accuracy $\mathcal{O}$ of the numerical scheme and a constant. The model of this section has been applied to study the entire human circulation in a holistic setting [21, 23].

In this section we have assumed the Coriolis coefficient to be $\alpha = 1$. A more general model has been presented by Spilimbergo et al. [29], in which α is allowed to vary in its full range $[1, 2]$.

10.7 Summary and Concluding Remarks

In this chapter we have presented generalised one-dimensional averaged mathematical models for blood flow in compliant vessels. Generalisations involve tube laws to account for the mechanics of the blood vessel wall interacting with the internal blood flow, and the role of geometrical and mechanical parameters characterising the vessel wall tissue.

Tube laws must satisfactorily represent the physiological phenomena of interest and, when coupled to the PDEs for the fluid dynamics of blood, preservation of key mathematical properties of the full mathematical model must be ensured. After a brief introduction to the field of tube laws we have concentrated on elastic and viscoelastic tube laws and analysed the mathematical properties of the PDEs, such as hyperbolicity and nature of characteristic fields. Viscoelastic tube laws give rise to parabolic equations, which have been approximated by hyperbolic equations with stiff source terms following a relaxation approach, and subsequently are analysed.

Highlights of the chapter are found in Sect. 10.3 that presents the solution of the Riemann problem for a model with elastic tube law and a variable mechanical parameter admitting discontinuities; these contents have a methodological value in a teaching/research context. Section 10.5 presents a case study for animal and human blood vessels, in which a parameter estimation process is applied and the resulting tube laws are used to reproduce experimental data. Illustrative results confirm that viscoelastic tube laws represent the physiology of compliant blood vessels more accurately than elastic tube laws, and are therefore recommended for practical use. Finally, Sect. 10.6 presents a complete study of the PDEs for a viscoelastic tube law and multiple variable parameters.

For further study we recommend the classical text book of Fung [2] on the basic science of the mechanics of soft tissue, where the reader can also find details of Fung's **Quasilinear Viscoelasticity Theory** at the bases of the construction of tube laws; see also Valdéz-Jasso et al. [3]. Also recommended is the book edited by Formaggia, Quarteroni and Veneziani [10] and the many references therein. The bases of the class of **Generalised String Models** are found there, which constitute a starting point for the construction of tube laws; see also Veneziani [4], Quarteroni et al. [5] and Formaggia et al. [6]. For the specific topics of this chapter we recommend [33, 42] for the setting of a framework for dealing with variable geometric and mechanical parameters, including material discontinuities. For approximating parabolic equations by hyperbolic systems with stiff source terms we recommend [54, 56] and [38]. For implementation of viscoelastic tube laws with variable parameters for simulating the human circulation we recommend [21–23], and [28].

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Chapter 11

Mathematical Model for Spinal CSF Dynamics



Abstract This chapter presents a co-axial, one-dimensional, time-dependent mathematical model for the dynamics of cerebrospinal fluid (CSF) in the spinal canal, coupled with the dynamics of the deformable spinal cord. The model may prove useful for a range of pathologies associated with disturbed CSF fluid dynamics. Two examples include Chiari malformations and syringomyelia, both of which are linked to anomalous CSF flow in the spinal subarachnoid space. Another relevant condition involves visual impairment due to disturbed CSF dynamics in the optic nerve sheath, coupled with the deformable optic nerve. The model could also be applied to simulate CSF/ISF flow in the perivascular spaces surrounding penetrating arterial and venous vessels in the brain or spinal cord. The one-dimensional, co-axial model presented here consists of four non-linear, time-dependent partial differential equations. Mathematically, these equations have some special characteristics. First, they cannot be expressed in conservation-law form, which creates the challenge of defining discontinuous solutions that obey the classical Rankine-Hugoniot conditions. Second, the non-conservative nature of the equations restricts the range of suitable numerical methods for solving them. Moreover, the equations are of mixed elliptic-hyperbolic type, which could make the problem potentially ill-posed. Fortunately, mathematical analysis of the system reveals a sufficiently large parameter region in which the equations are hyperbolic, ensuring that the system is well-posed. The model has also been coupled with the CSF dynamics in the cranial cavity within the framework of the global, closed-loop mathematical model for the human circulation. Preliminary results are promising. The chapter concludes with a selection of references for further study.

11.1 Modelling CSF in the Spinal Canal

The construction of mathematical models for CSF dynamics, in terms of differential equations, goes back to the pioneering work of Marmarou and collaborators [1], and others, during the 70s. The simplest mathematical models are zero-dimensional (0D) models. These models consist of systems of ordinary differential equations (ODEs) that model variations of the unknowns in time but not in space. For background on

0D models see Chap. 6. For modern examples of the 0D approach applied to CSF dynamics see [2–4], to name but a few. More complete mathematical models include time and space variations and are expressed in terms of systems of partial differential equations (PDEs). Relevant works include [5–29]. In addition to spatial variations, models are also distinguished between rigid models and fluid/structure interaction models. In the first category, the spatial domain is assumed to be fixed in time, that is **rigid**. In the second category the spatial domain is time-dependent and its precise determination is coupled to the flow distribution inside such domain.

The purpose of this chapter is to present a non-rigid, one-dimensional, co-axial time-dependent mathematical model for the dynamics of cerebrospinal fluid in the spinal subarachnoid space SSAS, surrounding the spinal cord. See Chap. 4 for background on CSF and CSF spaces. The derived model will account for the dynamics of CSF in the spinal canal, coupled to the deformation of the spinal cord. Subject to appropriate modifications, the proposed spinal CSF model may also be adapted to model fluid flow in other spaces in the central nervous system that are geometrically analogous to the spinal canal. Other CSF spaces in which the co-axial models presented in this chapter are potentially applicable include: the optic nerve sheath filled with cerebrospinal fluid in, coupled to the dynamics of the enclosed deformable optic nerve; flow in perivascular spaces surrounding penetrating arterial and venous blood vessels in the brain and spinal cord.

Potentially, the models may also be adapted to simulate the dynamics of inner-ear fluids. For each of these fluid domains there is a range of pathologies that may be associated to anomalies in the corresponding CSF fluid dynamics. For example, Chiari malformations and syringomyelia are associated with anomalous CSF flow in the spinal subarachnoid space. Visual impairments are potentially associated with the dynamics of CSF in the optic nerve sheath, coupled to the enclosed deformable optic nerve; a corresponding CSF model may open the way to study a range of pathologies arising in ophthalmology.

The model presented here builds upon the one-dimensional works of Cirovic and collaborators [19, 20, 30], as well as extensions reported by Toro and collaborators [26, 29, 31]. Despite the simplifying assumptions, the model retains two major features: the deformability of the spinal cord is accounted for, and the interaction between the fluid flow of CSF and the coupled response of the deformable spinal cord. In other words, the model retains the fluid/structure interaction feature that distinguishes the fluid flow phenomena of interest here.

In the next section we first adopt a framework to construct one-dimensional mathematical models for the dynamics of CSF in the spinal canal.

11.2 Equations for Flow Inside a Flexible Conduit

Here we consider the basic equations for fluid flow inside a flexible-walled conduit, as depicted in Fig. 11.1. The classical equations for longitudinal waves in such conduit, see Lighthill [32], in terms of physical, or primitive, variables are

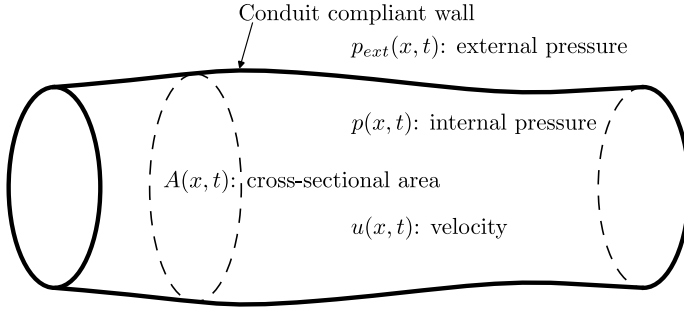


Fig. 11.1 One-dimensional axial model for flow inside a generic compliant conduit. The sought unknowns are cross-sectional area $A(x, t)$, axial velocity $u(x, t)$ and internal pressure $p(x, t)$. A parameter of the problem is the assumed external pressure $p_{ext}(x, t)$

$$\left. \begin{aligned} \partial_t A + \partial_x(Au) &= 0, \\ \partial_t u + u\partial_x u + \frac{1}{\rho}\partial_x \mathcal{P} &= 0. \end{aligned} \right\} \quad (11.1)$$

These equations are statements of conservation of mass and balance of momentum. $A(x, t)$ is the cross-sectional area that depends on space x along the length of the conduit and time t ; $u(x, t)$ is the particle velocity, $\mathcal{P} = p(x, t) - p_{ext}(x, t)$ is the **transmural pressure**, or **excess pressure**; $p(x, t)$ is the internal pressure and $p_{ext}(x, t)$ is the external pressure. The density ρ of the fluid is assumed to be constant. System (11.1) is endowed with the pressure closure condition called **tube law**

$$\mathcal{P} := \mathcal{P}(A) = p(x, t) - p_{ext}(x, t). \quad (11.2)$$

Equations (11.1) are written in terms of the conserved variable A and the non-conserved variable u . The first equation is written in conservation-law form but the second one is written in non-conservative form. Therefore the system is written in non-conservative form. See Chap. 14 for background on non-conservative systems. Introducing $\mathcal{P}$ from (11.2) into the momentum equation in (11.1) gives

$$\partial_t u + u\partial_x u + \frac{1}{\rho} \frac{d\mathcal{P}}{dA} \partial_x A = 0. \quad (11.3)$$

Written in matrix form, system (11.1) becomes

$$\partial_t \mathbf{V} + \mathbf{M}(\mathbf{V})\partial_x \mathbf{V} = \mathbf{0}, \quad (11.4)$$

with

$$\mathbf{V} = \begin{bmatrix} A \\ u \end{bmatrix}, \quad \mathbf{M}(\mathbf{V}) = \begin{bmatrix} u & A \\ \frac{1}{\rho} \frac{d\mathcal{P}}{dA} & u \end{bmatrix}. \quad (11.5)$$

It is easily seen that the eigenvalues of system (11.4)–(11.5) are

$$\lambda_1 = u - c, \quad \lambda_2 = u + c, \quad (11.6)$$

where c is the **wave speed**

$$c = \sqrt{\frac{A}{\rho} \frac{d\mathcal{P}}{dA}}. \quad (11.7)$$

We note that the expression (11.7) for the wave speed comes out naturally from the eigenstructure analysis of the governing equations. The resulting eigenvalues (11.6) are both real and distinct, provided the following condition on the tube law holds

$$\frac{d\mathcal{P}}{dA} > 0. \quad (11.8)$$

That is, $\mathcal{P}(A)$ must be a monotone increasing function of cross-sectional area A . System (11.4) is hyperbolic if condition (11.8) on the tube law is satisfied.

The left eigenvectors corresponding to the eigenvalues λ_1 and λ_2 are respectively given as

$$\mathbf{L}_1(\mathbf{V}) = \alpha_1 [1 - A/c], \quad \mathbf{L}_2(\mathbf{V}) = \alpha_2 [1 + A/c], \quad (11.9)$$

where α_1 and α_2 are arbitrary scaling parameters. Analogously, the right eigenvectors are

$$\mathbf{R}_1(\mathbf{V}) = \beta_1 \begin{bmatrix} 1 \\ -c/A \end{bmatrix}, \quad \mathbf{R}_2(\mathbf{V}) = \beta_2 \begin{bmatrix} 1 \\ c/A \end{bmatrix}, \quad (11.10)$$

with β_1 and β_2 two arbitrary scaling parameters.

In the next section, the generic one-dimensional framework (11.1) is applied to the construction of a co-axial model for CSF in the spinal subarachnoid space.

11.3 The Cirovic-Kim Equations

The model presented here is based on the works of Cirovic and collaborators [19, 20, 30], as well as on extensions, analysis and applications by Toro and collaborators [26, 29, 31]. Following [30], let us consider the configuration of Fig. 11.2 depicting two co-axial compliant tubes. The inner compliant tube represents the deformable spinal cord, with associated unknowns: cross-sectional area $A_c(x, t)$, axial particle velocity $u_c(x, t)$ and internal pressure $p_c(x, t)$. The compliant outer tube for CSF is bounded externally by the dura and the spinal cord; it has associated variables:

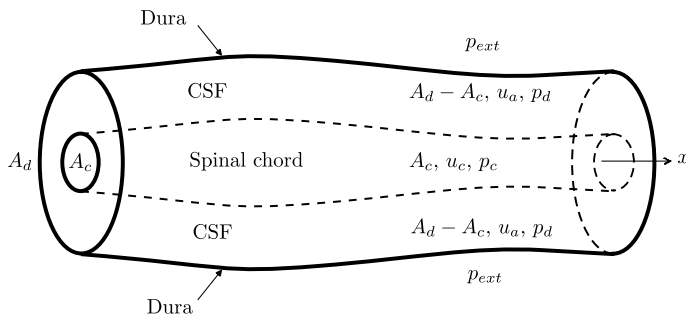


Fig. 11.2 Co-axial tube configuration depicting side view of an outer tube, externally bounded by the dura and an inner tube representing the spinal cord. Here A_d is total cross-sectional area of the outer tube; A_c is cross-sectional area of the deformable spinal cord; u_a is axial velocity of CSF; u_c is axial velocity of cord tissue; p_{ext} is external pressure to dura; p_d is internal pressure of CSF in the spinal canal of cross sectional area $A_d - A_c$; p_c is internal pressure in cord tissue

cross-sectional area $A_d(x, t) - A_c(x, t)$, axial particle velocity $u_a(x, t)$ and pressure $p_d(x, t)$.

On application of conservation of mass and balance of momentum, as in (11.1), in each domain, Cirovic and Kim obtained the following system of first-order partial differential equations

$$\partial_t \mathbf{V} + \mathbf{M}(\mathbf{V}) \partial_x \mathbf{V} = \mathbf{0}. \quad (11.11)$$

Here $\mathbf{V} = [A_c, u_c, A_d, u_a]^T$ is the vector of unknowns and $\mathbf{M}(\mathbf{V})$ is the coefficient matrix

$$\mathbf{M}(\mathbf{V}) = \begin{bmatrix} u_c & A_c & 0 & 0 \\ c_c^2/A_c & u_c & c_d^2/A_d & 0 \\ u_c - u_a & A_c & u_a & A_d - A_c \\ 0 & 0 & c_d^2/A_d & u_a \end{bmatrix}. \quad (11.12)$$

In the equations of Cirovic and Kim (11.11)–(11.12), the symbols c_d and c_c represent the wave speeds of the form (11.7) in the CSF and in the spinal cord respectively, implicitly assuming tube laws of the form (11.2) for these media.

A slight generalization of the equations of Cirovic and Kim (11.11)–(11.12) results by assuming different densities ρ_d and ρ_c for the CSF and the spinal cord respectively. By denoting their ratio as

$$R = \frac{\rho_d}{\rho_c} \quad (11.13)$$

the equations become

$$\left. \begin{aligned} \partial_t A_c + u_c \partial_x A_c + A_c \partial_x u_c &= 0, \\ \partial_t u_c + u_c \partial_x u_c + \frac{c_c^2}{A_c} \partial_x A_c + R \frac{c_d^2}{A_d} \partial_x A_d &= 0, \\ \partial_t A_d + (u_c - u_a) \partial_x A_c + A_c \partial_x u_c + u_a \partial_x A_d + (A_d - A_c) \partial_x u_a &= 0, \\ \partial_t u_a + u_a \partial_x u_a + \frac{c_d^2}{A_d} \partial_x A_d &= 0. \end{aligned} \right\} \quad (11.14)$$

In matrix form equations (11.14) read

$$\partial_t \mathbf{V} + \mathbf{M}(\mathbf{V}) \partial_x \mathbf{V} = \mathbf{0}, \quad (11.15)$$

with

$$\mathbf{V} = \begin{bmatrix} A_c \\ u_c \\ A_d \\ u_a \end{bmatrix}, \quad \mathbf{M}(\mathbf{V}) = \begin{bmatrix} u_c & A_c & 0 & 0 \\ \frac{c_c^2}{A_c} & u_c & \frac{c_d^2}{A_d} & 0 \\ u_c - u_a & A_c & u_a & A_d - A_c \\ 0 & 0 & R \frac{c_d^2}{A_d} & u_a \end{bmatrix}. \quad (11.16)$$

Compare Eqs. (11.15)–(11.16) with (11.11)–(11.12). Both systems are expressed in non-conservative form. Moreover, two of the unknowns are the non-conservative quantities u_c and u_a .

11.4 Equations in Conserved Variables

Here we further modify the equations in two ways: we express the equations in terms of conserved variables and we add source terms to account for viscous losses. The resulting system of equations reads

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}). \quad (11.17)$$

The vector of unknowns are the conserved variables

$$\mathbf{Q} = \begin{bmatrix} A_c \\ A_c u_c \\ A_d - A_c \\ (A_d - A_c) u_a \end{bmatrix} \quad (11.18)$$

and the coefficient matrix $\mathbf{A}(\mathbf{Q})$ is

$$\mathbf{A}(\mathbf{Q}) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ c_c^2 + A_c c_d^2 / A_d - u_c^2 & 2u_c & A_c c_d^2 / A_d & 0 \\ 0 & 0 & 0 & 1 \\ (A_d - A_c) c_d^2 / A_d & 0 & (A_d - A_c) c_d^2 / A_d - u_a^2 & 2u_a \end{bmatrix}. \quad (11.19)$$

The source term vector $\mathbf{S}(\mathbf{Q})$ is given as

$$\mathbf{S}(\mathbf{Q}) = \begin{bmatrix} 0 \\ -8\pi\mu_c \frac{u_c}{\rho} \\ 0 \\ -8\pi\mu_{csf} \frac{u_d}{\rho} \left[\frac{A_d - A_c}{A_d + A_c - (A_d - A_c)/\log(\sqrt{\frac{A_d}{A_c}})} \right] \end{bmatrix}. \quad (11.20)$$

In the source terms (11.20) accounting for viscous losses [26, 31], μ_c is the dynamic viscosity for the cord and μ_{csf} is the dynamic viscosity of CSF in the spinal subarachnoid space.

We note that, in spite of having used the conserved variables to express the governing equations, the resulting system of Eqs. (11.17)–(11.20) cannot be expressed in conservation-law form. This fact has implications when having to choose numerical schemes to compute approximate solutions to the equations, as discussed later in this chapter. For details on non-conservative problems see Chap. 14.

11.5 Closure Conditions: Tube Laws and Wave Speeds

All models (11.1), 11.15 and (11.17) contain more unknowns than equations and therefore closure conditions, called tube laws, are required to have a determined system of equations. Simple, linear tube laws were proposed by Cirovic and Kim [20], in which transmural pressures are directly proportional to the relative change in cross-sectional areas, with respect to equilibrium cross-sectional areas. For the CFS in the spinal SAS, the following tube law is adopted

$$p_d = p_{ext} + L_d \left(\frac{A_d}{A_{0d}} - 1 \right), \quad (11.21)$$

where L_d is the elastance coefficient for the dura and A_{0d} denotes the equilibrium cross-sectional area for the outer tube bounded by the dura. From (11.7), the corresponding wave speed is

$$c_d = \sqrt{T_d A_d}, \quad T_d = \frac{L_d}{\rho A_{0d}}. \quad (11.22)$$

Similarly, for the spinal cord the following tube law is adopted

$$p_c = p_d + L_c \left(\frac{A_c}{A_{0c}} - 1 \right), \quad (11.23)$$

where L_d is the elastance coefficient for the cord and A_{0c} denotes the cord equilibrium cross-sectional area. The corresponding wave speed is

$$c_c = \sqrt{T_c A_c}, \quad T_c = \frac{L_c}{\rho A_{0c}}. \quad (11.24)$$

As stated in [20], the tube law (11.23) is a first approximation to model the spinal cord composed of very soft tissue and extracellular fluids.

A non-linear tube law for the CSF has recently been adopted by Celant et al. [31], namely

$$\tilde{p}_d = p_{0d} e^{\tilde{E}_{ld}(A_d - A_{0d})}. \quad (11.25)$$

The exponential distribution in (11.25) characterises the CSF pressure with respect to the CSF volume. The point (A_{0d}, p_{0d}) represents the reference set point of the curve, which is obtained from a baseline simulation using the linear tube law (11.21). The elastance coefficient $\tilde{E}_{ld}$ in (11.25) is defined such that the elastance at the reference point of the linear tube law (11.21) is equal to the elastance of the non-linear tube law (11.25):

$$\left. \frac{\partial p_d}{\partial A_d} \right|_{A_{0d}} = \left. \frac{\partial \tilde{p}_d}{\partial A_d} \right|_{A_{0d}}. \quad (11.26)$$

That is

$$\tilde{E}_{ld} = \frac{E_{ld}}{A_{0d} p_{0d}}. \quad (11.27)$$

The wave speed $\tilde{c}_d$ of spinal cerebrospinal fluid in the spinal subarachnoid space using the non-linear tube law (11.25) is computed in the usual way; see (11.7).

Model (11.17) for CSF flow in the spinal canal coupled to the dynamics of the spinal cord, along with the non-linear tube law (11.25) was recently implemented in the global model of Müller and Toro [33, 34] to simulate craniospinal fluid dynamics connected to the full human circulation. See [31] for details.

11.6 Mathematical and Numerical Issues

System (11.17) has two properties that make it especial from the mathematical and computational points of view. First, the system cannot be written in conservation-law form, but only in non-conservative form, as in (11.17). Second, system (11.17) is of **mixed elliptic/hyperbolic type**. See Chap. 8 for background on systems of non-linear hyperbolic equations and relevant definitions; Chap. 14 for background on non-conservative systems and Chap. 12 for background on numerical methods. In what follows we address the above mentioned issues in more detail.

11.6.1 Hyperbolicity and Well-Posedness

Here we are concerned with the mathematical character of equations (11.17). Clearly, before deploying numerical methods to solve the equations approximately, a key issue to consider is its **well-posedness**, in the sense of Hadamard [35]. That is,

1. There exists a solution to the equations.
2. The solution is unique.
3. The solution depends continuously on the parameters of the problem.

From the classical paper of Lax [36], it is known that **hyperbolicity** is a necessary condition for well-posedness. Therefore, one needs to determine whether system (11.17) is hyperbolic. As stated in [26], all too often in practice, it is assumed *a priori* that first-order, time-dependent systems of non-linear partial differential equations will be hyperbolic. However, this is not always the case and the present system of equations (11.17) for the dynamics of cerebrospinal fluid flow in the spinal subarachnoid space coupled to the dynamics of the spinal cord, constitutes one example. Preliminary results on the analysis of system (11.17) were reported for the first time by Toro et al. [26], followed by a comprehensive theoretical and numerical study reported in a subsequent paper by Scoz, Bertazzi and Toro [29]. For background on analogous mathematical issues in the field of compressible multiphase flows, see the classical paper of Stewart and Wendroff [37]. See also the work of Romenski, Resnyanski and Toro [38]. In the recent work [29], the authors identified necessary and sufficient conditions for the system (i) to be strictly hyperbolic and (ii) conditions under which the system is mixed elliptic-hyperbolic. Relevant definitions and mathematical background are found in Chap. 8. For full details see [29].

Here we illustrate the results reported in [29] by displaying various regions of the parameter space in Fig. 11.3, along with the mathematical properties of the equations in each of them. The regions of interest lie in the parameter space defined by

$$A_c = A_{0c} , \quad A_{0c} \leq A_d \leq 9A_{0c} , \quad -40 \leq \Delta u = u_c - u_a \leq 40 . \quad (11.28)$$

Green zones in Fig. 11.3 represent the subdomains in which the equations are strictly hyperbolic; red zones represent subdomains in which the equations are of mixed elliptic-hyperbolic type; along black lines the equations can be of hyperbolic type. As in other fields of application, such as compressible multiphase flow [37], the key parameter here is the so called **slip velocity** $\Delta u = u_c - u_a$, that is the difference in velocity between the cord and the surrounding CSF. It is seen in the figure that for a generous physiological range of Δu the equations are hyperbolic and therefore well posed. The equations are also hyperbolic for very large values of $|\Delta u|$, which are probably physiologically unrealistic and therefore of no practical use. Under pathological conditions, however, one could not rule out *a priori*, falling inside the mixed elliptic/hyperbolic region. In this case, solutions are highly unstable, as reported in [29]. We remark that the instability refers to the solutions of the differential equations, not of numerical methods to compute such solutions.

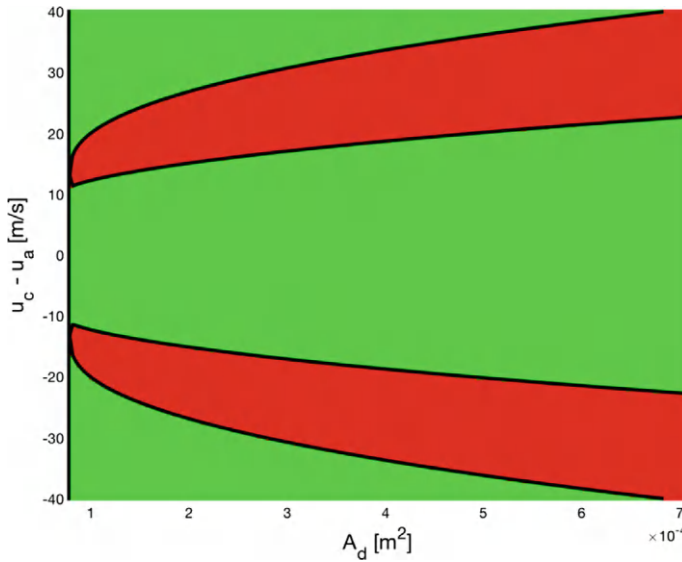


Fig. 11.3 Mathematical character of equations (11.17) with the linear tube laws (11.21) and (11.23). Various regions in the parameters space are shown. Green: the equations are strictly hyperbolic. Red: the equations are of mixed elliptic-hyperbolic type. Black lines: the equations can be of hyperbolic type (courtesy of Ms. Alessia Scoz, University of Trento, Italy. Modified from Scoz et al. [29])

11.6.2 Numerical Issues

Solving equations that may have complex eigenvalues and be of mixed elliptic-hyperbolic type is very challenging from the numerical point of view. First, as the eigenvalues may be complex, the use of popular upwind methods is not possible. Therefore, there is no option but to use **centred methods**, as distinct from **upwind methods**. This is because the computation of eigenvalues is challenging and they may turn out to be complex [26, 29].

Another difficulty is posed by the non-conservative form of the equations (11.17). Naturally, this means that the so popular conservative methods in terms of a numerical flux are not applicable. The reader is advised to consult Chap. 12 on basic numerical concepts, Chap. 13 on conservative methods and Chap. 14 on, precisely, non-conservative systems and numerical methods to solve such equations. The recent work of Celant, Müller and Toro [31] is recommended; it addresses all the issues discussed here, and illustrates the practical value of the CSF model (11.17) to simulate craniospinal fluid dynamics in the realistic setting of the human circulation in the full body; see also [39] and Chap. 20.

11.7 Concluding Remarks and Further Reading

This chapter was concerned with a co-axial, one-dimensional time dependent mathematical model for the dynamics of cerebrospinal fluid in the spinal canal. The model, consisting of four non-linear time-dependent partial differential equations, has two challenging features: it is of mixed-elliptic hyperbolic type and is written in non-conservative form. The co-axial model is potentially applicable to other 1D-type domains in the central nervous system. For example, it could be adapted to simulate the cerebrospinal fluid in the optic nerve sheath, coupled to the dynamics of the enclosed deformable optic nerve. It could also be applied to simulate flow in perivascular spaces surrounding penetrating blood vessels in the brain or in the spinal cord. The model may also be adapted to simulate the dynamics of inner-ear fluids.

For each of the bodily fluid dynamics areas in which the model can be applied, there is a range of pathologies that may be associated to anomalies in the corresponding CSF fluid dynamics. For example, Chiari malformations and syringomyelia are associated with anomalous CSF flow in the spinal subarachnoid space. Visual impairments are potentially associated to the dynamics of CSF in the optic nerve sheath, coupled to the enclosed deformable optic nerve; a corresponding CSF model may open the way to study a range of pathologies arising in ophthalmology.

The following references for further study are recommended: Berkouk et al. [7]; Cirovic and Kim [30]; Toro et al. [26]; and Scoz, Bertazzi and Toro [29].

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Part III
Computational Algorithms for PDES

Chapter 12

Basic Notions on Numerical Methods



Abstract This chapter introduces key concepts in the numerical approximation of first-order partial differential equations of hyperbolic type. We begin with the simplest method, the finite difference method, applied to the basic linear advection equation. We derive several well-known schemes for hyperbolic systems and discuss their essential properties, including Godunov's upwind, Godunov's centred, Lax-Friedrichs, Lax-Wendroff, Fromm, Warming-Beam, and FORCE schemes. Theoretical concepts for analysing numerical methods are introduced, such as local truncation error, order of accuracy, modified equation, linear stability, and monotonicity. We also present a numerical version of the von Neumann method for studying the linear stability of numerical methods, which is applicable to high-order methods in multiple space dimensions. A shortcut for analysing accuracy is introduced, simplifying the evaluation of a numerical method's accuracy, even in multiple space dimensions. Godunov's theorem is stated and proven, demonstrating that methods of high-order accuracy must be nonlinear, even when applied to linear equations. The class of conservative methods is introduced, and properties of the numerical flux are examined. Classical finite difference methods are reformulated in conservative form by identifying the appropriate numerical flux. Boundary conditions, including reflective and transmissive, are discussed. We also cover the calculation of a stable time step for explicit methods, as determined by the Courant stability condition, with an emphasis on providing reliable estimates of the maximal wave speed to ensure stability in nonlinear problems. Numerical results are presented to assess the performance of the methods. Mesh refinement, along with concepts of error, mesh-independent solutions, convergence rates, and efficiency, is discussed and demonstrated through practical examples. Finally, a list of exercises is provided at the end of the chapter.

12.1 Finite Difference Approximation

We consider the following **model** initial boundary value problem (IBVP)

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \lambda \partial_x q = 0, \quad x \in [a, b] \subset \mathbb{R}, \quad t > 0, \\ \text{IC: } q(x, 0) = h(x), \quad x \in [a, b], \quad t = 0, \\ \text{BCs: } q(a, t) = b_L(t), \quad q(b, t) = b_R(t), \quad t \geq 0. \end{array} \right\} \quad (12.1)$$

The governing partial differential equation (PDE) is the linear advection equation (LAE) with constant wave propagation speed λ . The spatial domain is defined by $[a, b] \subset \mathbb{R}$; $h(x)$ is the initial condition (IC) at the initial time $t = 0$, a prescribed function of x ; $b_L(t)$ and $b_R(t)$ are prescribed functions of time and define boundary conditions (BCs) at $x = a$ (left) and at $x = b$ (right), respectively. We note that IBVP (12.1) is technically overprescribed, as only one boundary condition is needed, depending on the sign of λ ; see Chap. 7.

The numerical approximation of IBVP (12.1) involves three stages. One first discretises the space-time domain, then derives approximations to the partial derivatives in the PDE, and finally produces a discrete representation of the PDE on the chosen mesh; this final component is the sought numerical method.

12.1.1 Discretisation of the Space-Time Domain

Here we first perform a **partition of the spatial domain** $[a, b]$ into $M + 2$ equidistant points x_i defined as follows

$$x_i = a + i \Delta x, \quad i = 0, \dots, M + 1, \quad \Delta x = \frac{b - a}{M + 1}. \quad (12.2)$$

M is a prescribed positive integer and represents the number of **interior points**: $x_1, x_2, \dots, x_M$. The two **boundary points** are $x_0 = a$ and $x_{M+1} = b$; Δx is the **mesh size** (or mesh spacing), the distance between two successive points.

Next we perform a **partition of the temporal domain** $[0, T_{out}]$ into a set of time points, or time levels,

$$t_n = n \Delta t, \quad n = 0, \dots, N_{out}, \dots, \quad (12.3)$$

where $t_0 = 0$ is the initial time and $T_{out} = \Delta t N_{out}$ is the chosen **output time**; Δt is the **timestep**. Note that the time domain is infinite, but in practice one chooses to evolve the problem up to a finite, final time T_{out} , with an associated positive integer N_{out} . The discretisation of the space-time domain produces two discretization parameters,

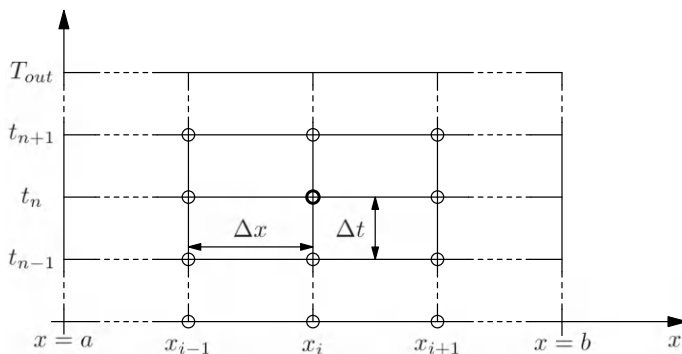


Fig. 12.1 Discretization of the continuous space-time domain $[a, b] \times [0, T_{out}]$ into a mesh consisting of a finite number of points (x_i, t_n) . Here the points are equidistant, both in space and time, but in general such restriction is not necessary

Δx and Δt , assumed constant for the moment. Moreover, we also assume a fixed relationship between Δt and Δx of the form

$$\Delta x = K \Delta t, \quad K > 0 \text{ a constant.} \quad (12.4)$$

Figure 12.1 shows the space-time discretization of the continuous domain $[a, b] \times [0, T_{out}]$. The resulting x - t mesh is determined by the mesh parameters Δx and Δt . The mesh size Δx results from choosing M in (12.2), which means choosing the number of interior points in the partition of $[a, b]$. In general there is complete freedom in choosing the spatial spacing and thus Δx . One could have chosen irregular spacing between the points, giving rise to a set of spacings $\Delta x_{i+\frac{1}{2}} = x_{i+1} - x_i$ for the distance between points x_i and x_{i+1} . For a regular mesh $\Delta x_{i+\frac{1}{2}} := \Delta x$ is constant. The choice of Δt is generally constrained by the chosen spatial discretization and by the wave speeds present in the PDE in Eq. (12.1). In this chapter we deal exclusively with the regular spatial mesh case.

12.1.2 Finite Difference Approximation to Derivatives

The next step is to substitute the partial derivatives $\partial_t q$ and $\partial_x q$ in the PDE in (12.1) by approximations based on the points of the chosen finite difference mesh. To illustrate how this is done we first consider a general smooth function $g(z)$, as depicted in Fig. 12.2. On the assumption that $g(z)$ and all its derivatives $g^{(k)}(z_0)$ are smooth at the point z_0 , we can derive approximations to the derivatives of $g(z)$, as follows. From Taylor's theorem we may find $g(z_0 + \Delta z)$ as

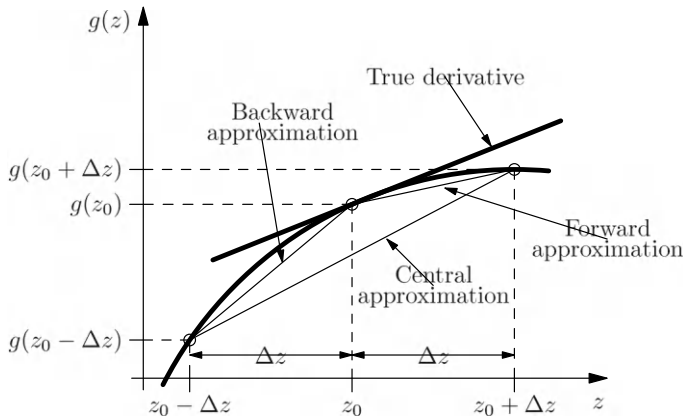


Fig. 12.2 Finite difference approximation to derivatives of the function $g(z)$ at the central point z_0 . Approximations are illustrated by the slopes of straight segments for the backward approximation, the central approximation and the forward approximation. The slopes of these segments are meant to approximate the true derivative $g'(z_0)$

$$g(z_0 + \Delta z) = g(z_0) + \Delta z g'(z_0) + \frac{1}{2} \Delta z^2 g^{(2)}(z_0) + \frac{1}{6} \Delta z^3 g^{(3)}(z_0) + \mathcal{O}(\Delta z^4). \quad (12.5)$$

Here $\mathcal{O}(\Delta z^4)$ is the asymptotic form of the truncating error.

Forward approximation to $g'(z_0)$. First, from (12.5) we write

$$g(z_0 + \Delta z) - g(z_0) = \Delta z g'(z_0) + \frac{1}{2} \Delta z^2 g^{(2)}(z_0) + \frac{1}{6} \Delta z^3 g^{(3)}(z_0) + \mathcal{O}(\Delta z^4),$$

which, on division through by Δz , yields

$$g'(z_0) = \frac{g(z_0 + \Delta z) - g(z_0)}{\Delta z} + \mathcal{O}(\Delta z). \quad (12.6)$$

Here $\mathcal{O}(\Delta z)$ is the **leading term** of the full error, which represents the first term of an infinite power series. Note the form of the error in the approximation (12.6); it is said to be of **first order**, as the leading term of the error $\mathcal{O}(\Delta z)$ involves the **first power** of the spacing Δz . Now, by neglecting the leading term and higher order terms we write the approximation to $g'(z_0)$ as

$$g'(z_0) \approx \frac{g(z_0 + \Delta z) - g(z_0)}{\Delta z}. \quad (12.7)$$

Approximation (12.7) to $g'(z_0)$ is **one-sided**; it is called **forward approximation** to the derivative, as it involves the central point z_0 and its neighbour $z_0 + \Delta z$ on the right hand side.

Backward approximation to $g'(z_0)$. From Taylor's theorem

$$g(z_0 - \Delta z) = g(z_0) - \Delta z g'(z_0) + \frac{1}{2} \Delta z^2 g^{(2)}(z_0) - \frac{1}{6} \Delta z^3 g^{(3)}(z_0) + \mathcal{O}(\Delta z^4) . \quad (12.8)$$

Then, the following first-order, one-sided **backward approximation** results:

$$g'(z_0) = \frac{g(z_0) - g(z_0 - \Delta z)}{\Delta z} + \mathcal{O}(\Delta z) . \quad (12.9)$$

Centred (or central) approximation to $g'(z_0)$. By subtracting (12.8) from (12.5) and rearranging we have

$$g'(z_0) = \frac{g(z_0 + \Delta z) - g(z_0 - \Delta z)}{2\Delta z} + \mathcal{O}(\Delta z^2) . \quad (12.10)$$

This is a second-order approximation, called the **centred or central approximation**, as it uses information from the left and right of the centre point z_0 .

Figure 12.2 illustrates the function $g(z)$, the exact first derivative, and three approximations to it at the central point z_0 : forward, backward and centred.

One-sided second order approximations. Here we find a second-order approximation to the first derivative $g'(z_0)$ using the three equidistant points $z_0 - 2\Delta z$, $z_0 - \Delta z$ and z_0 . Consider Taylor expansions at the two points $z_0 - 2\Delta z$ and $z_0 - \Delta z$ as follows:

$$g(z_0 - 2\Delta z) = g(z_0) - 2\Delta z g'(z_0) + \frac{1}{2} (2\Delta z)^2 g^{(2)}(z_0) - \frac{1}{6} (2\Delta z)^3 g^{(3)}(z_0) + \mathcal{O}(\Delta z^4) \quad (12.11)$$

and

$$g(z_0 - \Delta z) = g(z_0) - \Delta z g'(z_0) + \frac{1}{2} \Delta z^2 g^{(2)}(z_0) - \frac{1}{6} \Delta z^3 g^{(3)}(z_0) + \mathcal{O}(\Delta z^4) . \quad (12.12)$$

The reader may easily verify that suitable use of (12.11) and (12.12) gives

$$g'(z_0) = \frac{3g(z_0) - 4g(z_0 - \Delta z) + g(z_0 - 2\Delta z)}{2\Delta z} + \mathcal{O}(\Delta z^2) . \quad (12.13)$$

This is a second-order, one-sided approximation to the first derivative $g'(z_0)$ at the point z_0 .

Exercise 12.1 Using (12.5) and (12.12) show that

$$g''(z_0) = \frac{g(z_0 + \Delta z) - 2g(z_0) + g(z_0 - \Delta z)}{\Delta z^2} + \mathcal{O}(\Delta z^2) . \quad (12.14)$$

This is second-order, centred approximation to the second derivative $g''(z_0)$.

12.1.3 The Method of Finite Differences

The finite difference method applied to the IBVP (12.1) produces a discrete representation of the PDE on the finite difference mesh given by a finite number of points (x_i, t_n) , as shown in Fig. 12.1. Then, the exact value $q(x_i, t_n)$ at the mesh point (x_i, t_n) is approximated by q_i^n , that is

$$q_i^n \approx q(x_i, t_n) . \quad (12.15)$$

Here the subscript i in q_i^n denotes the spatial position, while the superscript n in q_i^n (not an exponent) denotes the time level n . The next step is to replace the partial derivatives $\partial_t q(x, t)$ and $\partial_x q(x, t)$ in the PDE in (12.1) by finite difference approximations, as seen earlier. To this end we consider the generic point (x_i, t_n) of the mesh. We then write three approximations to the spatial and temporal derivatives in the PDE in (12.1).

The spatial partial derivative $\partial_x q(x, t)$ in (12.1) at the point (x_i, t_n) can be approximated in at least the three following forms:

$$\partial_x q(x_i, t_n) = \begin{cases} \frac{q(x_{i+1}, t_n) - q(x_i, t_n)}{\Delta x} + \mathcal{O}(\Delta x) , & \text{Forward ,} \\ \frac{q(x_i, t_n) - q(x_{i-1}, t_n)}{\Delta x} + \mathcal{O}(\Delta x) , & \text{Backward ,} \\ \frac{q(x_{i+1}, t_n) - q(x_{i-1}, t_n)}{2\Delta x} + \mathcal{O}(\Delta x^2) , & \text{Centred .} \end{cases} \quad (12.16)$$

Similarly, **the temporal partial derivative** $\partial_t q(x, t)$ in (12.1) at the point (x_i, t_n) may be approximated as

$$\partial_t q(x_i, t_n) = \begin{cases} \frac{q(x_i, t_{n+1}) - q(x_i, t_n)}{\Delta t} + \mathcal{O}(\Delta t) , & \text{Forward ,} \\ \frac{q(x_i, t_n) - q(x_i, t_{n-1})}{\Delta t} + \mathcal{O}(\Delta t) , & \text{Backward ,} \\ \frac{q(x_i, t_{n+1}) - q(x_i, t_{n-1})}{2\Delta t} + \mathcal{O}(\Delta t^2) , & \text{Centred .} \end{cases} \quad (12.17)$$

Note that from (12.16) and (12.17), there are nine possible combinations of spatial and temporal approximations, potentially giving rise to nine numerical methods. Not all combinations lead to useful methods. Next we select some of these approximations to construct our first numerical method to solve (12.1).

12.1.4 The Method of Godunov: Finite Difference Version

The first useful numerical method that can be constructed by using the methodology outlined earlier, is the finite difference version of Godunov's method [2]. This is

accomplished by selecting from (12.16) and (12.17), the following finite difference approximations to partial derivatives in the PDE:

$$\left. \begin{aligned} \partial_t q(x_i, t_n) &= \frac{q(x_i, t_{n+1}) - q(x_i, t_n)}{\Delta t} + \mathcal{O}(\Delta t) , \\ \partial_x q(x_i, t_n) &= \begin{cases} \frac{q(x_i, t_n) - q(x_{i-1}, t_n)}{\Delta x} + \mathcal{O}(\Delta x) & \text{if } \lambda > 0 , \\ \frac{q(x_{i+1}, t_n) - q(x_i, t_n)}{\Delta x} + \mathcal{O}(\Delta x) & \text{if } \lambda < 0 . \end{cases} \end{aligned} \right\} \quad (12.18)$$

The time derivative has been approximated by a first-order forward-in-time formula, while the space derivative has been approximated by a first-order, one-sided **upwind** formula, according to the sign of the speed λ of the PDE in (12.1). In this case the information is taken from the side from which the **wind blows**. For this reason this method is known as the **upwind scheme**. For linear equations this scheme was first proposed by Courant, Isaacson and Rees [1]; the scheme is also known as the CIR scheme. Godunov [2] extended the **upwind method** in **conservation form** to solve non-linear systems of **hyperbolic conservation laws**.

Consider the PDE in (12.1) with $\lambda > 0$. Straightforward use of formulae (12.17) to substitute the partial derivatives in the PDE gives the formula

$$\frac{q_i^{n+1} - q_i^n}{\Delta t} + \lambda \left(\frac{q_i^n - q_{i-1}^n}{\Delta x} \right) = 0 . \quad (12.19)$$

Note however, that in arriving at this finite difference equation, we have imposed two types of approximations. First, exact values $q(x_i, t_n)$ have been replaced by approximate discrete values q_i^n . Then, all truncation errors in the finite difference approximations to derivatives have been neglected.

In formula (12.19) we assume that all values at time level n are known. Then by solving for the single unknown q_i^{n+1} at the future time level $n + 1$ we obtain the final form of the **time-marching** numerical scheme as

$$q_i^{n+1} = q_i^n - \frac{\lambda \Delta t}{\Delta x} (q_i^n - q_{i-1}^n) . \quad (12.20)$$

It is convenient to introduce a **dimensionless quantity**, called the **Courant number** (or the Courant-Isaacson-Lewis number, or CFL number)

$$c = \frac{\lambda \Delta t}{\Delta x} = \frac{\lambda}{\Delta x / \Delta t} . \quad (12.21)$$

Note that c is the ratio of two speeds, the speed λ in the PDE in (12.1) and the **mesh speed** $\Delta x / \Delta t$. Then the Godunov method may be written in succinct form as

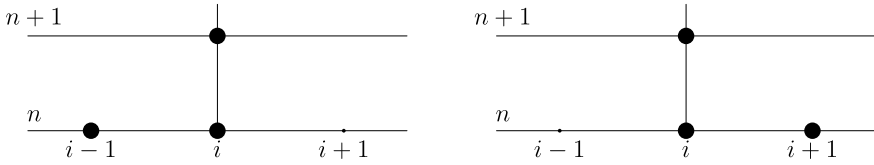


Fig. 12.3 Stencils for Godunov's method for $\lambda > 0$ (left) and for $\lambda < 0$ (right)

$$\left. \begin{aligned} q_i^{n+1} &= q_i^n - c (q_i^n - q_{i-1}^n) \quad \text{if } \lambda > 0, \\ q_i^{n+1} &= q_i^n - c (q_{i+1}^n - q_i^n) \quad \text{if } \lambda < 0. \end{aligned} \right\} \quad (12.22)$$

Figure 12.3 shows the stencils for each of the two cases, for $\lambda > 0$ (left) and for $\lambda < 0$ (right). **The stencil** of a method is the set of points of the mesh that **take part** in the updating formula of the method. In Fig. 12.3 the points of the stencils are the large full circles.

The upwind scheme given by the formulae (12.22) provides an **explicit time-marching** procedure, whereby the unknown value q_i^{n+1} at the point i at the future time level $n + 1$ can be **predicted** by using the known solution at the time level n (the present time), at points i and $i - 1$ when $\lambda > 0$ and points i and $i + 1$ when $\lambda < 0$. Note that the mesh points that lie on the boundaries $x = a$ and $x = b$ must be updated separately, using the boundary conditions of the problem in (12.1). Obviously, at time level 0 one uses the initial condition $h(x)$ to provide the discrete initial values. Here we set $q_i^0 = h(x_i)$, but other methods may require different ways of using $h(x)$ to set the initial conditions.

12.1.5 Other Well-Known Finite Difference Methods

Here we present several well-known finite difference methods.

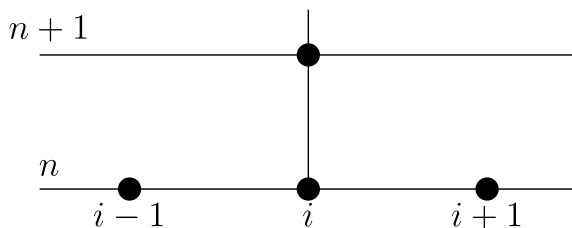
The FTCS Method

The FTCS method (Forward-in-Time Centred-in-Space) results from the following approximations to partial derivatives

$$\left. \begin{aligned} \partial_t q(x_i, t_n) &= \frac{q(x_i, t_{n+1}) - q(x_i, t_n)}{\Delta t} + \mathcal{O}(\Delta t), \\ \partial_x q(x_i, t_n) &= \frac{q(x_{i+1}, t_n) - q(x_{i-1}, t_n)}{2\Delta x} + \mathcal{O}(\Delta x^2). \end{aligned} \right\} \quad (12.23)$$

In fact the name of the scheme derives from the types of finite difference approximations used. In time, one uses a **forward approximation**, while in space one

Fig. 12.4 Stencil for the FTCS method. Note the symmetric distribution of points at the base of the stencil



uses a **centred, or central, approximation**. Substitution of (12.23), after neglecting truncation errors, into the PDE gives

$$\frac{q_i^{n+1} - q_i^n}{\Delta t} + \lambda \left(\frac{q_{i+1}^n - q_{i-1}^n}{2\Delta x} \right) = 0. \quad (12.24)$$

Solving for q_i^{n+1} gives the FTCS numerical scheme

$$q_i^{n+1} = q_i^n - \frac{1}{2}c(q_{i+1}^n - q_{i-1}^n). \quad (12.25)$$

Figure 12.4 shows the stencil of the FTCS scheme, whose symmetric configuration reflects the centred character in space.

As seen in (12.23), the time derivative approximation is first-order accurate and the space derivative approximation is actually second-order accurate. Therefore, naively, one would have expected FTCS to be better than Godunov's method. Unfortunately, FTCS is **unconditionally unstable** and therefore useless. The concept of stability of a numerical method will be introduced later. There are two ways of rescuing FTCS, namely (i) through a simple modification that gives rise to the explicit Lax-Friedrichs scheme and (ii) replacing the explicit time discretization by an implicit time discretization, not pursued here.

The Lax-Friedrichs Method (LF)

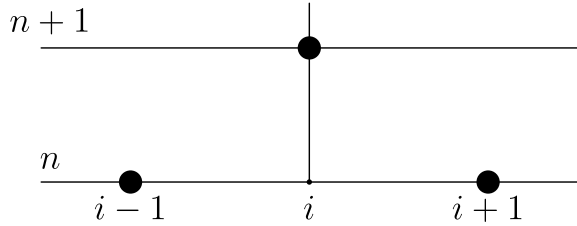
The Lax-Friedrichs scheme results from replacing q_i^n in the approximation to $\partial_t q(x_i, t_n)$ in FTCS, Eq. (12.23), by the mean of its neighbours, namely

$$q_i^n \longrightarrow \frac{1}{2}(q_{i-1}^n + q_{i+1}^n). \quad (12.26)$$

Then, the finite difference approximation of the PDE gives

$$\frac{q_i^{n+1} - \frac{1}{2}(q_{i-1}^n + q_{i+1}^n)}{\Delta t} + \lambda \left(\frac{q_{i+1}^n - q_{i-1}^n}{2\Delta x} \right) = 0, \quad (12.27)$$

Fig. 12.5 Stencil for the Lax-Friedrichs method. Note that the central point q_i^n is missing from the stencil. Compare with Fig. 12.4



yielding the Lax-Friedrichs scheme

$$q_i^{n+1} = \frac{1}{2}(1+c)q_{i-1}^n + \frac{1}{2}(1-c)q_{i+1}^n . \quad (12.28)$$

Figure 12.5 shows the stencil of the LF scheme and note that the centre of the stencil q_i^n is missing. The updated value q_i^{n+1} depends exclusively on the neighbours.

The Lax-Wendroff Method (LW)

The construction of the LW scheme is somewhat different to that of previously presented methods. LW results from the following procedure:

1. The solution at (x_i, t_{n+1}) is expressed as a Taylor series in time

$$q(x_i, t_{n+1}) = q(x_i, t_n) + \Delta t \partial_t q(x_i, t_n) + \frac{1}{2} \Delta t^2 \partial_t^{(2)} q(x_i, t_n) + \mathcal{O}(\Delta t^3) . \quad (12.29)$$

2. The Cauchy-Kovalevskaya procedure is applied to the PDE in (12.1) to replace time derivatives by space derivatives, namely

$$\partial_t q(x, t) = -\lambda \partial_x q(x, t) , \quad \partial_t^{(2)} q(x, t) = \lambda^2 \partial_x^{(2)} q(x, t) . \quad (12.30)$$

3. Using (12.30) into (12.29) one obtains

$$q(x_i, t_{n+1}) = q(x_i, t_n) - \Delta t \lambda \partial_x q(x_i, t_n) + \frac{1}{2} \Delta t^2 \lambda^2 \partial_x^{(2)} q(x_i, t_n) + \mathcal{O}(\Delta t^3) . \quad (12.31)$$

Now the Taylor series expansion contains only space derivatives.

4. Second-order, centred finite difference approximations to the space derivatives in (12.31) read

$$\left. \begin{aligned} \partial_x q(x_i, t_n) &= \frac{q(x_{i+1}, t_n) - q(x_{i-1}, t_n)}{2\Delta x} + \mathcal{O}(\Delta x^2) , \\ \partial_x^{(2)} q(x_i, t_n) &= \frac{q(x_{i+1}, t_n) - 2q(x_i, t_n) + q(x_{i-1}, t_n))}{\Delta x^2} + \mathcal{O}(\Delta x^2) . \end{aligned} \right\} \quad (12.32)$$

5. Use of (12.32) into (12.31) leads to the Lax-Wendroff scheme

$$q_i^{n+1} = \frac{1}{2}c(1+c)q_{i-1}^n + (1-c^2)q_i^n - \frac{1}{2}c(1-c)q_{i+1}^n . \quad (12.33)$$

The stencil for the Lax-Wendroff method is identical to that of the FTCS method, see Fig. 12.4.

Exercise 12.2 Show that for any order k , the Cauchy-Kovalevskaya procedure applied to LAE gives

$$\partial_t^{(k)} q(x, t) = (-\lambda)^k \partial_x^{(k)} q(x, t) . \quad (12.34)$$

Proof Hint: use induction.

A Note on the Cauchy-Kovalevskaya Procedure

The Cauchy-Kovalevskaya (CK) procedure was used to prove existence and uniqueness for the initial-value problem for differential equations, more than 150 years ago, in the classical Cauchy-Kovalevskaya theorem. For detailed background on the Cauchy-Kovalevskaya theorem see for example [3, 4] and [5]. See also discussion in [6]. More recently, in the numerical context, the CK procedure has been used to construct the second-order Lax-Wendroff method, in which first and second-order time derivatives are converted to space derivatives, as seen in (12.30). In more recent developments, the higher-order ADER finite volume methods [7] used the CK procedure for derivatives of any order (12.34), leading to numerical schemes of arbitrary order of accuracy in both space and time. See Chap. 16 for the scalar case and Chap. 17 for non-linear systems.

The FORCE Method

The finite difference version of the FORCE method due to Toro and collaborators [8, 9] results from an arithmetic mean of the Lax-Friedrichs and Lax-Wendroff schemes, namely:

$$\begin{aligned} q_i^{n+1} = & \frac{1}{2} \left[\frac{1}{2}(1+c)q_{i-1}^n + \frac{1}{2}(1-c)q_{i+1}^n \right] \\ & + \frac{1}{2} \left[\frac{1}{2}c(1+c)q_{i-1}^n + (1-c^2)q_i^n - \frac{1}{2}c(1-c)q_{i+1}^n \right] , \end{aligned} \quad (12.35)$$

giving the FORCE scheme as

$$q_i^{n+1} = \frac{1}{4}(1+c)^2 q_{i-1}^n + \frac{1}{2}(1-c^2)q_i^n + \frac{1}{4}(1-c)^2 q_{i+1}^n . \quad (12.36)$$

The stencil for the FORCE method is identical to that of the FTCS method, see Fig. 12.4.

Having constructed several finite difference methods, we next introduce some theoretical concepts that will allow us to analyse and assess the methods.

12.2 Theoretical Notions

Here we introduce some elementary theoretical notions on numerical methods for solving the linear advection equation, the simplest hyperbolic equation. To this end we consider the pure initial value problem (IVP)

$$\left. \begin{array}{l} \text{PDE: } \partial_t q(x, t) + \lambda \partial_x q(x, t) = 0, \quad -\infty < x < \infty, \quad t > 0, \\ \text{IC: } q(x, 0) = h(x), \quad -\infty < x < \infty, \end{array} \right\} \quad (12.37)$$

where λ is constant and $h(x)$ is the initial condition, with $h(x) \rightarrow 0$ as $|x| \rightarrow \infty$.

12.2.1 The General Form of a Scheme

We study finite difference methods to solve (12.37), for which it is convenient to write such methods in the **general form**

$$q_i^{n+1} = H(q_{i-l}^n, q_{i-l+1}^n, \dots, q_i^n, \dots, q_{i+r}^n), \quad (12.38)$$

where H is a real-valued function of $l + r + 1$ real variables and, as before, $q_i^n \approx q(x_i, t_n)$ is a **point-wise value** that approximates the true solution $q(x, t)$ at the mesh point (x_i, t_n) , with $x_i = i \Delta x$, $t_n = n \Delta t$. Consistent with (12.37) we also assume that $q_i^n \rightarrow 0$ as $|i| \rightarrow \infty$.

As an example, we write the Godunov finite difference method (12.22) in general form (12.38). In this case the function H takes the form

$$H = cq_{i-1}^n + (1-c)q_i^n \text{ for } \lambda > 0; \quad H = (1+c)q_i^n - cq_{i+1}^n \text{ for } \lambda < 0. \quad (12.39)$$

As defined in (12.21), $c = \lambda \Delta t / \Delta x$ is the Courant number.

12.2.2 Local Truncation Error and Accuracy

The partial differential equation in (12.37) has an associated (exact) **differential operator**

$$L_e(q) \equiv \partial_t q(x, t) + \lambda \partial_x q(x, t) = 0 \quad (12.40)$$

acting on functions $q(x, t)$, whereas the general scheme (12.38) has an associated (approximate) **numerical operator**

$$L_a(q_i^n) \equiv q_i^{n+1} - H(q_{i-l}^n, q_{i-l+1}^n, \dots, q_i^n, \dots, q_{i+r}^n) = 0 \quad (12.41)$$

acting on discrete point values q_i^n .

Local truncation error (LTE). The local truncation error is defined as

$$\tau_i^n = \frac{1}{\Delta t} L_a(q(x_i, t_n)) . \quad (12.42)$$

That is, the local truncation error is **the numerical operator** applied to the **exact solution** evaluated at the mesh point (x_i, t_n) and **scaled** by Δt . This form of the LTE is also known as the **scaled** LTE and is the form adopted here. Sometimes in the literature, the scaling factor is not included in the definition of LTE.

In order to calculate the LTE of a particular scheme one first assumes the solution to be smooth. Then, application of Taylor's theorem at the **centre of the stencil**, the point (x_i, t_n) , yields

$$\left. \begin{aligned} \tau_i^n &= \partial_t q(i \Delta x, n \Delta t) + \lambda \partial_x q(i \Delta x, n \Delta t) \\ &+ \mathcal{O}(\Delta t^k) + \mathcal{O}(\Delta x^m) . \end{aligned} \right\} \quad (12.43)$$

Here k and m are positive integers. The first line of (12.43) is identically $L_e(q(i \Delta x, n \Delta t)) = 0$. Hence, the local truncation error is what remains, which has the form

$$\tau_i^n = \mathcal{O}(\Delta t^k) + \mathcal{O}(\Delta x^m) . \quad (12.44)$$

Note that the right hand side of (12.44) denotes the **leading term** of τ_i^n , an infinite power series expansion.

Order of accuracy of a scheme. Scheme (12.38), with local truncation error (12.44), is said to be of **order of accuracy k in time and m in space**. Then, noting the fixed relationship $\Delta x = K \Delta t$ introduced in (12.4), the order of accuracy p of scheme (12.38) is defined as

$$p = \min\{k, m\} . \quad (12.45)$$

Consistency of a scheme. The scheme (12.38) is said to be consistent with the partial differential equation in (12.37) if

$$\tau_i^n \rightarrow 0 \text{ as } \Delta x \rightarrow 0 \text{ (or } \Delta t \rightarrow 0) . \quad (12.46)$$

Example: The Local Truncation Error of Godunov's Method

First we write the scheme in general form (12.38), which for $\lambda > 0$ reads

$$q_i^{n+1} = H = c q_{i-1}^n + (1 - c) q_i^n . \quad (12.47)$$

Therefore, from definition (12.42)

$$\tau_i^n = \frac{1}{\Delta t} [q(i\Delta x, (n+1)\Delta t) - (cq((i-1)\Delta x, n\Delta t) + (1-c)q(i\Delta x, n\Delta t))] . \quad (12.48)$$

Taylor expansions about the point (x_i, t_n) are necessary for only two of the three points, namely

$$\left. \begin{aligned} q(i\Delta x, (n+1)\Delta t) &= q(i\Delta x, n\Delta t) + \Delta t \partial_t q(i\Delta x, n\Delta t) \\ &\quad + \frac{1}{2} \Delta t^2 \partial_t^{(2)} q(i\Delta x, n\Delta t) + \mathcal{O}(\Delta t^3) , \\ q((i-1)\Delta x, n\Delta t) &= q(i\Delta x, n\Delta t) - \Delta x \partial_x q(i\Delta x, n\Delta t) \\ &\quad + \frac{1}{2} \Delta x^2 \partial_x^{(2)} q(i\Delta x, n\Delta t) + \mathcal{O}(\Delta x^3) . \end{aligned} \right\} \quad (12.49)$$

Substituting (12.49) into (12.48) and noting

$$c\Delta x/\Delta t = \lambda , \quad c\Delta x^2/\Delta t = \lambda\Delta x$$

we obtain

$$\left. \begin{aligned} \tau_i^n &= \partial_t q(i\Delta x, n\Delta t) + \lambda \partial_x q(i\Delta x, n\Delta t) \\ &\quad + \frac{1}{2} \Delta t \partial_t^{(2)} q(i\Delta x, n\Delta t) - \frac{1}{2} \lambda \Delta x \partial_x^{(2)} q(i\Delta x, n\Delta t) \\ &\quad + \mathcal{O}(\Delta t^2) + \mathcal{O}(\Delta x^2) . \end{aligned} \right\} \quad (12.50)$$

Then, the leading term of the local truncation error is

$$\tau_i^n = \frac{1}{2} \Delta t \partial_t^{(2)} q(i\Delta x, n\Delta t) - \frac{1}{2} \lambda \Delta x \partial_x^{(2)} q(i\Delta x, n\Delta t) = \mathcal{O}(\Delta t) + \mathcal{O}(\Delta x) . \quad (12.51)$$

As

$$\tau_i^n = \mathcal{O}(\Delta t) + \mathcal{O}(\Delta x)$$

the Godunov scheme is **first-order** accurate in both space and time. Furthermore, by converting $\partial_t^{(2)} q(i\Delta x, n\Delta t)$ to $\lambda^2 \partial_x^{(2)} q(i\Delta x, n\Delta t)$ via the Cauchy-Kovalevskaya procedure we obtain

$$\tau_i^n = \frac{1}{2} \lambda \Delta x (c-1) \partial_x^{(2)} q(i\Delta x, n\Delta t) . \quad (12.52)$$

The local truncation error has a **diffusive** form given by a second-order spatial derivative and a coefficient $\frac{1}{2} \lambda \Delta x (c-1)$. The LTE τ_i^n vanishes if $\Delta x = 0$ (not practical), $c = 0$ and $c = 1$. That is, by using CFL number $c = 1$ in the Godunov's method, the local truncation error vanishes and the numerical scheme reproduces the exact values at the mesh points.

12.2.3 The Modified Equation

Generally, a numerical method does not solve the intended PDE but, due to errors inherent in the scheme, it solves more accurately, a different PDE, called the **modified equation**. We introduce the concept of the **modified equation** via the Godunov method for $\lambda > 0$.

In calculating the local truncation error we assume that $q(x, t)$ is a sufficiently smooth solution of the original PDE in (12.37). Suitable application of Taylor's theorem leads to an expression of the type (12.50). As $q(x, t)$ is a solution of the PDE in (12.1) for the first line on the right hand side of (12.50) is zero, obtaining the LTE as in (12.51). Had we also neglected the second line on the right hand side, as if $q(x, t)$ were a solution of the PDE

$$\partial_t q + \lambda \partial_x q + \frac{1}{2} [\Delta t \partial_t^{(2)} q - \lambda \Delta x \partial_x^{(2)} q] = 0 \quad (12.53)$$

at $(i \Delta x, n \Delta t)$, then the LTE would have been of order $\mathcal{O}(\Delta t^2)$, $\mathcal{O}(\Delta x^2)$, that is second-order accurate. Equation (12.53) is the **modified equation** for Godunov's method for $\lambda > 0$ in LAE. This is a partial differential equation that Godunov's method solves to second-order accuracy in both space and time. Note that the modified Eq. (12.53) can be written as

$$\partial_t q + \lambda \partial_x q = \frac{1}{2} [\lambda \Delta x \partial_x^{(2)} q - \Delta t \partial_t^{(2)} q] . \quad (12.54)$$

Assuming the fixed relation (12.4) between the mesh parameters Δx and Δt we use $\mathcal{O}(\Delta t^m)$ also for $\mathcal{O}(\Delta x^m)$. Then, applying the Cauchy-Kovalevskaya procedure to the modified Eq. (12.54) to convert the second-order time derivative to a space derivative, we have

$$\left. \begin{aligned} \partial_t q &= -\lambda \partial_x q + \frac{1}{2} [\lambda \Delta x \partial_x^{(2)} q - \Delta t \partial_t^{(2)} q] = -\lambda \partial_x q + \mathcal{O}(\Delta t) , \\ \partial_x \partial_t q &= -\lambda \partial_x^{(2)} q + \mathcal{O}(\Delta t) , \\ \partial_t^{(2)} q &= -\lambda (\partial_x \partial_t q + \mathcal{O}(\Delta t)) = -\lambda [-\lambda \partial_x^{(2)} q + \mathcal{O}(\Delta t)] + \mathcal{O}(\Delta t) , \\ \partial_t^{(2)} q &= \lambda^2 \partial_x^{(2)} q + \mathcal{O}(\Delta t) . \end{aligned} \right\} \quad (12.55)$$

Then, the final form of the modified equation is

$$\partial_t q + \lambda \partial_x q = \alpha \partial_x^{(2)} q ; \quad \alpha = \frac{1}{2} \lambda \Delta x (1 - c) . \quad (12.56)$$

This is a second-order PDE of the advection-diffusion type (parabolic). The diffusion term $\alpha \partial_x^{(2)} q$ explains why solutions of the inviscid equation in (12.37) behave as solutions of a viscous equation, having the effect of **numerical viscosity** (or artificial),

not of **physical viscosity**. The coefficient α is called the coefficient of **numerical viscosity** (or artificial viscosity).

First-order numerical methods have modified equations of the advection-diffusion type, with **numerical viscosity**. Second-order numerical methods applied to (12.37) have modified equation of the **dispersion type**

$$\partial_t q + \lambda \partial_x q = \gamma \partial_x^{(3)} q, \quad (12.57)$$

with coefficient of **numerical dispersion** γ . The Lax-Wendroff method has modified equation

$$\partial_t q + \lambda \partial_x q = \gamma \partial_x^{(3)} q; \quad \gamma = \frac{1}{6} \lambda \Delta x^2 (c^2 - 1). \quad (12.58)$$

Dispersive errors show up in the form of **phase errors**. The LTE vanishes for the Lax-Wendroff method when $\lambda = 0$, $\Delta x = 0$ (not practical) and $c = 1$.

12.2.4 Linear Stability and the von Neumann Method

Stability is an essential property of a numerical method, noting that there are various interpretations as to what stability means. The adopted meaning depends on what particular features of the numerical solution are regarded as relevant. Classically, **unbounded growth of errors**, as time increases, stands out as a prominent feature. **Spurious oscillations and over/under shoots** also stand out as important features of a numerical solution. Then, there is the distinction between **linear stability** and **non-linear stability** and the various associated methodologies for the corresponding analysis. For background on the general subject of stability of a numerical method we recommend the textbooks [10–13] and [14]. Here we restrict ourselves to a brief introduction to linear stability.

Application of the von Neumann Method

We are concerned with the analysis of **linear stability** of **linear methods** to solve linear equations, typically the linear advection equation. More specifically, we introduce the historical **von Neumann method**, or Fourier series method, to study linear stability. In addition to linearity, one also assumes periodicity of solutions in the finite spatial domain $[a, b]$ and constant mesh spacing Δx . In the von Neumann method the numerical solution is expressed as a finite, truncated Fourier series, in complex exponential form. One then studies the growth in time of a function that coincides with the assumed series at the initial time. Due to the assumed linearity it suffices to study a single mode, which facilitates the algebraic manipulations that, in any case, may become rather challenging.

The von Neumann method first introduces a **trial function**, or **test function**

$$q_i^n = A^n e^{I\theta i} \quad \text{with } -\pi \leq \theta \leq \pi, \quad (12.59)$$

where the amplitude A is a complex number and A^n means A **raised to the power n** . For convenience, and contrary to convention, we use here $I := \sqrt{-1}$ to denote the complex unity, and keep i to identify a mesh spatial position in the numerical method; $\theta = P\Delta x$ is an angle and P is the **wave number** in the x -direction. For stability one requires

$$||A|| \leq 1, \quad (12.60)$$

called the **stability condition**. In this manner, the amplitude raised to the power n in (12.59), where n is the time level, will not increase unboundlessly, as time marching proceeds into the future.

Example 12.3 Linear stability of Godunov's method for $\lambda > 0$. Let us analyse the linear stability of Godunov's method for $\lambda > 0$, that is the method

$$q_i^{n+1} = q_i^n - c(q_i^n - q_{i-1}^n). \quad (12.61)$$

Substituting q_{i-1}^n , q_i^n and q_i^{n+1} in the scheme (12.61) by $A^n e^{I\theta(i-1)}$, $A^n e^{I\theta i}$ and $A^{n+1} e^{I\theta i}$, according to (12.59), gives

$$A^{n+1} e^{I\theta i} = A^n e^{I\theta i} - c[A^n e^{I\theta i} - A^n e^{I\theta(i-1)}]. \quad (12.62)$$

Cancelling $A^n e^{I\theta i}$ we obtain the complex number

$$A = [1 + c(\cos \theta - 1)] + I[-c \sin \theta]. \quad (12.63)$$

Then, the squared modulus of A is

$$||A||^2 = [1 + c(\cos \theta - 1)]^2 + c^2 \sin^2 \theta, \quad (12.64)$$

which after manipulations becomes

$$||A||^2 = 1 - 2c(1 - c)(1 - \cos \theta). \quad (12.65)$$

We impose the stability condition (12.60) in terms of $||A||^2 \leq 1$, which results in the following condition

$$1 - 2c(1 - c)(1 - \cos \theta) \leq 1. \quad (12.66)$$

Manipulation of (12.66) leads to the stability condition

$$0 \leq c \leq 1 . \quad (12.67)$$

It is easy to perform the analysis for the negative wave speed, $\lambda < 0$. The general stability condition that includes both positive and negatives wave speeds is

$$0 \leq |c| \leq 1 . \quad (12.68)$$

In Sect. 12.3.3 we discuss how the stability condition is used to select the time step Δt of the scheme.

Algebraic Shortcut to Linear Stability Analysis

The algebraic manipulations in the analysis of the linear stability of a scheme may become rather involved, particularly for schemes with a wide stencil and of high-order of accuracy. For the special case of a three-point (linear) scheme of the form

$$q_i^{n+1} = b_{-1}q_{i-1}^n + b_0q_i^n + b_1q_{i+1}^n , \quad (12.69)$$

with constant coefficients b_{-1} , b_0 and b_1 , there is a shortcut to the stability analysis, due to Billett [15]. The following two conditions on the coefficients of the scheme must be satisfied

$$(i) \quad b_0(b_{-1} + b_1) \geq 0 , \quad (ii) \quad b_0(b_{-1} + b_1) + 4b_{-1}b_1 \geq 0 . \quad (12.70)$$

In Sect. 12.2.7 we shall study an even simpler criterion for linear stability, based on the viscous form of the scheme.

Exercise 12.4 Apply (12.70) to the Godunov scheme for both $\lambda > 0$ and $\lambda < 0$ to obtain the stability condition (12.68).

Numerical Shortcut to Linear Stability Analysis

The analysis resulting from imposing the stability condition (12.60) may yield very complicated algebraic expressions, making it sometimes impossible to derive clean, simple algebraic conditions, such as (12.68). The complexity of the analysis depends largely on the size of the stencil, which is usually determined by the order of accuracy of the scheme. The analysis of schemes is notoriously hard in two and three space dimensions. However, there is a simple **numerical approach** for obtaining a reliable indication of the stability of a scheme, when the algebra becomes intractable.

For schemes in one space dimension, whose coefficients depend on just one Courant number c , assuming the trial function (12.59) in the specific scheme of interest, one derives an algebraic expression for the amplification factor

$$A = A(c, \theta) , \quad (12.71)$$

where c is the Courant number and θ is the phase angle in the range $-\pi \leq \theta \leq \pi$. For example, for scheme (12.61) one obtains (12.63), which in this case is sufficiently simple to perform all the necessary algebraic manipulations to arrive at an explicit condition on the Courant number.

For complicated expressions, it is suggested to adopt a numerical procedure for computing the modulus $||A||$ of the complex number $A = A(c, \theta)$ and produce a counterplot of $||A||$ in the $c - \theta$ plane, for a large number of values of the point (c, θ) in the rectangle $[0, c_{max}] \times [0, \pi]$. Here c_{max} is an *a priori* estimate of C_{lim} , the sought stability limit. The technique is extremely simple and can be applied to any scheme for the linear advection equation in one and multiple space dimensions. For details see [16] and [17], page 559. See also [18] and [19] for examples of a recent application of the methodology.

12.2.5 Analysis of the Accuracy of a Scheme

In addition to linear stability, the accuracy of a scheme must also be the subject of theoretical analysis, whenever possible. We note in advance, however, that the algebra involved may become rather cumbersome. In Sect. 12.5.2 we shall introduce an empirical method for estimating the accuracy of a scheme. For theoretical studies there is a lemma that simplifies matters considerably, in practice. But before stating such lemma we introduce a definition.

Linear scheme. A scheme of the form

$$q_i^{n+1} = \sum_{k=-l}^{k=r} b_k q_{i+k}^n \quad (12.72)$$

to solve the PDE in (12.37), with constant coefficients b_k , is called a **linear scheme**. For example, Godunov's method for $\lambda > 0$ written as a linear scheme reads

$$q_i^{n+1} = cq_{i-l}^n + (1-c)q_i^n, \quad (12.73)$$

for which the parameters in (12.72) are: $l = 1, r = 0, b_{-1} = c, b_0 = 1 - c$.

Lemma 12.5 Accuracy lemma: A linear scheme of the form (12.72) is said to be p -th order accurate in space and time ($p \geq 0$), in the sense of local truncation error, if and only if

$$\sum_{k=-l}^r k^\eta b_k = (-c)^\eta, \quad \eta = 0, 1, \dots, p. \quad (12.74)$$

Here c is the Courant number. Purely for notational convenience we introduce the notation $0^0 = 1$ (mathematically meaningless).

Proof For the proof see Roe [20] and Toro [6], for example. For an extension of this result to two and three space dimensions see Billett and Toro [16].

Example 12.6 As an example, we apply the accuracy lemma to the Godunov upwind method (12.73) for $\lambda > 0$. Then, we need to verify identity (12.74) for all possible non-negative integer values of η .

$$\text{For } \eta = 0 : \quad (-1)^0 \times c + 0^0 \times (1 - c) = c + 1 - c = 1 = (-c)^0 .$$

This merely says that the sum of the coefficients of the scheme is unity.

$$\text{For } \eta = 1 : \quad (-1)^1 \times c + 0^1 \times (1 - c) = -c = (-c)^1 .$$

Therefore the Godunov scheme is first-order accurate. But just for curiosity, let us try the second-order case:

$$\text{For } \eta = 2 : \quad (-1)^2 \times c + 0^2 \times (1 - c) = c \neq (-c)^2 .$$

Thus the Godunov scheme is **not** of accuracy greater than one, except for $c = 1$.

In the next section we combine accuracy to another important property of numerical methods, namely monotonicity, and study the consequences.

12.2.6 Godunov's Theorem and Monotone Schemes

Here we first introduce the definition of a monotone scheme. This property, valid only for scalar equations, ensures that the numerical solution given by a scheme will not exhibit **new extrema**, and will thus avoid spurious oscillations with computed values lying outside the interval $[q_{min}, q_{max}]$ containing the initial conditions.

Monotone scheme. A numerical scheme of the form

$$q_i^{n+1} = H(q_{i-l}^n, q_{i-l+1}^n, \dots, q_i^n, \dots, q_{i+r}^n) \quad (12.75)$$

is called **monotone** if all partial derivatives of the function H satisfy

$$\frac{\partial}{\partial q_k^n} H(q_{i-l}^n, q_{i-l+1}^n, \dots, q_i^n, \dots, q_{i+r}^n) \geq 0, \quad i - l \leq k \leq i + r. \quad (12.76)$$

An immediate consequence of this definition is that a linear scheme of the form (12.72) is monotone if and only if all its coefficients b_k are non-negative. For example, the Godunov's method (12.73) for $\lambda > 0$ is monotone because the constant coefficients are non-negative, that is $b_{-1} = c \geq 0$ and $b_0 = 1 - c \geq 0$, if the stability condition $c \leq 1$ is satisfied.

Remark: we shall occasionally use the word *monotonicity* of a scheme to mean that the scheme is monotone in the sense of the definition above. For further details on monotone methods and related concepts see Sweby [21] and Chap. 13 of [6].

Exercise 12.7 Consider the Lax-Wendroff method (12.34), written as a linear scheme

$$q_i^{n+1} = b_{-1}q_{i-1}^n + b_0q_i^n + b_1q_{i+1}^n ,$$

with coefficients $b_{-1} = \frac{1}{2}c(1+c)$; $b_0 = (1+c^2)$ and $b_1 = -\frac{1}{2}c(1-c)$. Verify that the method is not monotone and will therefore produce spurious oscillations outside the range $[q_{\min}, q_{\max}]$, containing the initial conditions. Consider both $\lambda > 0$ and $\lambda < 0$ separately.

Remark: There are numerical schemes defined as monotone schemes in the literature, but produce **internal spurious oscillations**, that is, the solution may oscillate but still remaining within the interval $[q_{\min}, q_{\max}]$. The Lax-Friedrichs method is a known example, though not very well-known, in the literature.

Theorem 12.8 (Godunov's Theorem)

There are no monotone, linear schemes (12.72) for the linear advection equation in (12.37) with constant λ , of accuracy two or higher.

Proof It is sufficient to prove that there is no second order, linear, monotone method for LAE in (12.37). To prove this **we proceed by contradiction** and assume there is a second order, linear, monotone method for LAE in (12.37). Then, from the accuracy lemma conditions (12.74) we must have:

$$s_\eta = \sum_{k=-l}^r k^\eta b_k = \begin{cases} s_0 = 1 , & \text{for } \eta = 0 , \\ s_1 = -c , & \text{for } \eta = 1 , \\ s_2 = c^2 , & \text{for } \eta = 2 . \end{cases} \quad (12.77)$$

From (12.77), after some algebraic manipulations, one obtains

$$\left. \begin{aligned} s_2 &= \sum_{k=-l}^r k^2 b_k \\ &= \sum_{k=-l}^r (k+c)^2 b_k - 2c \sum_{k=-l}^r k b_k - c^2 \sum_{k=-l}^r b_k \\ &= \left[\sum_{k=-l}^r (k+c)^2 b_k \right] - 2cs_1 - c^2 s_0 . \end{aligned} \right\} \quad (12.78)$$

Use of (12.77) into (12.78) gives

$$c^2 = \left[\sum_{k=-l}^r (k+c)^2 b_k \right] + c^2 . \quad (12.79)$$

In general, Eq. (12.79) is a contradiction. This is because the first term on the right hand side is non-negative, as from monotonicity $b_k \geq 0$ and the sum of all coefficients must add up to unity; this means that not all coefficients vanish. Then $(k + c)^2 > 0$, with the exception of $c = -k$, which would give a very special, not practical scheme. The contradiction (12.79) allows us to state that the Godunov's theorem has been proved. ■

We next discuss some of the **consequences of Godunov's theorem**: First, it emerges that schemes that are linear and monotone are at most first-order accurate. However, first-order methods are too inaccurate to be of practical use. One must therefore search for other classes of schemes, which is down to finding ways of circumventing Godunov's theorem. The key to this lies on the assumption of linear schemes. Thus a necessary condition for a numerical scheme to be oscillation-free (without new extrema) and of high-order of accuracy (for smooth solutions) is to be **non-linear**. In simple terms: **schemes must be non-linear, even when applied to linear equations**. Since 1959, numerical analysts have made useful advances in understanding the deep consequences of the apparently innocent Godunov's theorem. Much remains to be done to circumvent Godunov's theorem and design good numerical methods for solving problems of practical use. This problem will be taken up in Chap. 16 for the scalar case.

12.2.7 Viscous Form for a Three-Point Scheme

Consider a scalar conservation law

$$\partial_t q + \partial_x f(q) = 0, \quad (12.80)$$

with unknown q and flux function $f(q)$. The simplest example is the linear advection equation in (12.37), for which the flux function is $f(q) = \lambda q$. The viscous form for a three-point scheme can be written as

$$q_i^{n+1} = q_i^n - \frac{1}{2} \frac{\Delta t}{\Delta x} [f(q_{i+1}^n) - f(q_{i-1}^n)] + \frac{1}{2} \left(d_{i+\frac{1}{2}} \Delta q_{i+\frac{1}{2}} - d_{i-\frac{1}{2}} \Delta q_{i-\frac{1}{2}} \right). \quad (12.81)$$

The function $d_{i+\frac{1}{2}}$ is called the **coefficient of numerical viscosity**, in the sense of the viscous form of the scheme, and in general has the form

$$d_{i+\frac{1}{2}} = d_{i+\frac{1}{2}}(q_{i-k+1}^n, q_{i-k+1}^n, \dots, q_i^n, \dots, q_{i+k}^n). \quad (12.82)$$

Here we study the viscous form of a three-point linear scheme for the linear advection equation

$$q_i^{n+1} = b_{-1} q_{i-1}^n + b_0 q_i^n + b_1 q_{i+1}^n. \quad (12.83)$$

The coefficients b_{-1} , b_0 and b_1 are constant. Assume the scheme to be at least first-order. Then from the accuracy lemma, see (12.74), we have

$$b_{-1} + b_0 + b_1 = 1, \quad b_{-1} - b_1 = c. \quad (12.84)$$

System (12.84) gives a one-parameter family of solutions. From the first equation we introduce the parameter d

$$d = b_{-1} + b_1 = 1 - b_0 \quad (12.85)$$

and thus we obtain expressions for the three coefficients in terms of d , that is

$$b_{-1} = \frac{1}{2}(d + c), \quad b_0 = 1 - d, \quad b_1 = \frac{1}{2}(d - c). \quad (12.86)$$

Therefore, in terms of d , scheme (12.83) gives a one-parameter family of schemes

$$q_i^{n+1} = q_i^n - \frac{1}{2}c(q_{i+1}^n - q_{i-1}^n) + \frac{1}{2}d(q_{i+1}^n - 2q_i^n + q_{i-1}^n). \quad (12.87)$$

This is the **viscous form** and d is the **coefficient of numerical viscosity** of the scheme. Specific values of d give specific schemes, as we shall see. The viscous form (12.87) for a general three-point scheme (12.83) is very convenient for analysing some basic properties of the scheme.

Exercise 12.9 Show that the stability condition for a general three-point scheme (12.83), written in viscous form (12.87) is

$$c^2 \leq d \leq 1. \quad (12.88)$$

Exercise 12.10 Show that a truncation error analysis of the scheme (12.83) has a **coefficient of numerical viscosity**, in the sense of local truncation error analysis, given as

$$\alpha_{visc} = \frac{1}{2}\Delta x \lambda \left(\frac{d - c^2}{c} \right). \quad (12.89)$$

Thus, effectively, the coefficient d is a measure of the truncation error of the scheme.

Further use of the viscous form (12.87) reveals that well-known schemes are obtained by an appropriate choice of the parameter d . In particular, we can reproduce the Lax-Friedrichs, FORCE, Godunov upwind, Godunov centred and Lax-Wendroff schemes by appropriately choosing d , namely

$$d = \begin{cases} d_{LF} = 1 & \rightarrow \text{Lax-Friedrichs} , \\ d_{FORCE} = \frac{1}{2}(1 + c^2) & \rightarrow \text{FORCE} , \\ d_{GodU} = |c| & \rightarrow \text{Godunov upwind} , \\ d_{GodC} = 2c^2 & \rightarrow \text{Godunov centred} , \\ d_{LW} = c^2 & \rightarrow \text{Lax-Wendroff} . \end{cases} \quad (12.90)$$

The $c - d$ plot of Fig. 12.6 displays the schemes in (12.90) derived from the viscous form.

Theorem 12.11 *The Godunov upwind scheme for the linear advection equation is the monotone scheme with the smallest local truncation error. That is, the most accurate scheme in the class of monotone methods is Godunov's method.*

Proof Monotonicity requires positivity (non-negativity) of the coefficients b_{-1} , b_0 and b_1 given by (12.86). This leads to the condition $d \geq c$ and the result follows, as α_{visc} takes its minimum value, while maintaining monotonicity, when $d = c$, which gives precisely the Godunov's method. ■

So far we have considered finite difference methods, their derivation and some of their theoretical properties. Next we introduce another class of numerical methods, called conservative methods. As a matter of fact, most finite difference methods can be reformulated as conservative methods, as we shall see.

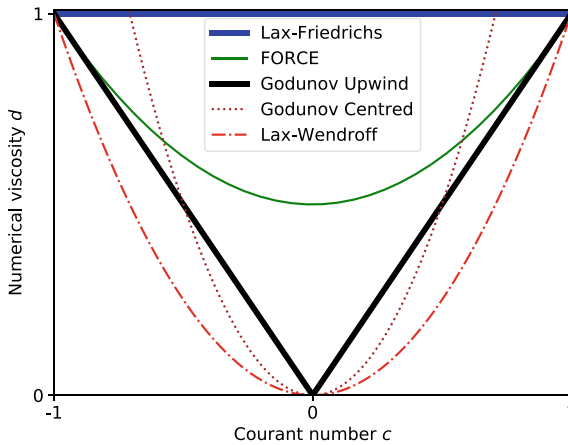


Fig. 12.6 Numerical schemes represented in terms of their coefficient of numerical viscosity $d = d(c)$, as a function of Courant number c , obtained from the viscous form of the schemes. For all linearly stable methods d is bounded below by d_{LW} (Lax-Wendroff) and bounded above by $d_{LF} = 1$ (Lax-Friedrichs). For all monotone schemes the coefficient of numerical viscosity $d = d(c)$ lies inside the triangle formed by the thick lines, with the Lax-Friedrichs scheme above and the Godunov upwind method below. The Lax-Friedrichs scheme is the least accurate of all linearly stable schemes, as $d = 1$ is the maximum value of the coefficient of numerical viscosity. The Lax-Wendroff scheme is the most accurate linearly stable scheme, with second-order viscosity equal to zero. The Lax-Wendroff and Godunov centred schemes are non-monotone. The most accurate scheme in the class of monotone methods is the Godunov upwind method

12.3 Conservative Methods

Consider the initial-boundary value problem for a **conservation law**

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + f(q(x, t)) = 0, \quad x \in [a, b], \quad t > 0, \\ \text{IC: } q(x, 0) = h(x), \quad x \in [a, b], \quad t = 0, \\ \text{BCs: } q(a, t) = b_L(t), \quad q(b, t) = b_R(t), \quad t \geq 0. \end{array} \right\} \quad (12.91)$$

Here $q(x, t)$ is the unknown and $f(q)$ is the **flux function** or **physical flux**. The simplest example is the linear advection equation for which $f(q) = \lambda q$. A well-known non-linear example is the inviscid Burgers equation, for which $f(q) = \frac{1}{2}q^2$. See Chap. 8.

12.3.1 Definitions and Examples

Conservative method. A conservative method to solve (12.91) is a method written in the form

$$q_i^{n+1} = q_i^n - \frac{\Delta t}{\Delta x} (f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}}), \quad (12.92)$$

where $f_{i+\frac{1}{2}}$ is called the **numerical flux**, which is generally expressed as

$$f_{i+\frac{1}{2}} = f_{i+\frac{1}{2}}(q_{i-l}^n, \dots, q_i^n, q_{i+1}^n, \dots, q_{i+r}^n), \quad (12.93)$$

where l and r are two non-negative integers. Figure 12.7 shows a typical mesh configuration for a conservative method. The domain $[a, b]$ is subdivided into a set of M disjoint subintervals $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$, for $i = 1, \dots, M$. These intervals are called **cells**, or **volumes**, or **elements**. The mesh spacing and cell centre are given as

$$\Delta x = \frac{b-a}{M} = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}, \quad x_i = \frac{1}{2}(x_{i-\frac{1}{2}} + x_{i+\frac{1}{2}}). \quad (12.94)$$

The mesh is defined by the position of the cell boundaries $x_{i+\frac{1}{2}}$, for $i = 0, \dots, M$ and times t_n , for $n = 0, \dots, N_{out}$, as defined previously.

Formula (12.92) is an expression for updating the solution in a cell i at time level n , to the new time level $n+1$, using known data at the old time level n . The numerical solution q_i^n in (12.92) in the cell I_i at time n is usually allocated to the centre x_i of the cell I_i . To fully define the scheme, one requires an expression for the numerical flux $f_{i+\frac{1}{2}}$ in (12.92), which is defined at the inter-cell boundary $x_{i+\frac{1}{2}}$. Boundary conditions are applied at the boundaries of the domain $x = a$ and $x = b$, which coincide with the inter-cell boundaries $x_{\frac{1}{2}}$ (left) and $x_{M+\frac{1}{2}}$ (right). A useful numerical flux $f_{i+\frac{1}{2}}$ is required to satisfy the following properties:

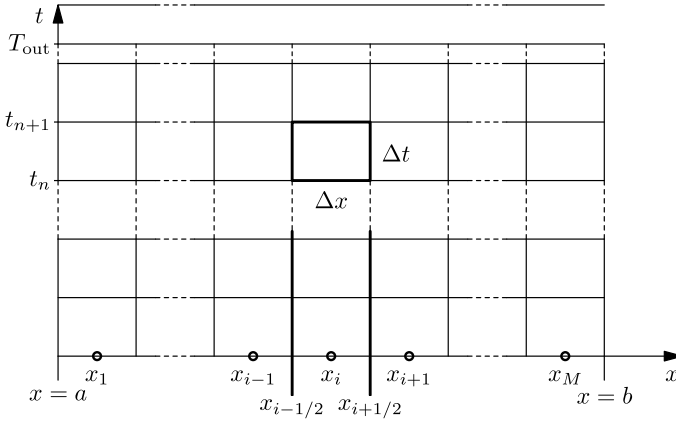


Fig. 12.7 Mesh configuration for a conservative method. The domain of the PDE is discretized into rectangles of dimensions $\Delta x \times \Delta t$, or finite volumes, in the x - t plane

1. **Continuity:** $f_{i+\frac{1}{2}} : \mathbb{R}^{l+r} \rightarrow \mathbb{R}$ is a continuous real-valued function.
2. **Consistency:** when all arguments of $f_{i+\frac{1}{2}}$ in (12.93) are identical we have

$$f_{i+\frac{1}{2}}(\hat{q}, \dots, \hat{q}) = f(\hat{q}), \quad (12.95)$$

that is, the numerical flux coincides with the physical flux.

12.3.2 Finite Difference Methods Written in Conservative Form

Most finite difference methods can be written in conservative form (12.92), by identifying the corresponding numerical flux, as seen below. For some cases we shall assume LAE, with $f(q) = \lambda q$ in (12.91), while for others we shall keep $f(q)$ more general.

1. **The Godunov upwind method** [2]. For LAE the Godunov upwind flux is

$$f_{i+\frac{1}{2}}^{GodU} = \frac{1}{2} [1 + \text{sign}(c)] (\lambda q_i^n) + \frac{1}{2} [1 - \text{sign}(c)] (\lambda q_{i+1}^n). \quad (12.96)$$

This formula is only valid for the linear advection equation; it contains both the $\lambda > 0$ and $\lambda < 0$ cases. For $\lambda > 0$ $f_{i+\frac{1}{2}} = \lambda q_i^n$ and $f_{i-\frac{1}{2}} = \lambda q_{i-1}^n$, with analogous definitions for $\lambda < 0$. Verify that by inserting these fluxes into the conservative formula (12.92), the Godunov upwind method in finite difference form (12.22) is reproduced, for both positive and negative wave speeds.

2. **The Godunov centred method** [22]. For the conservation law in (12.91) with flux function $f(q)$, the numerical flux for the Godunov centred method is

$$f_{i+\frac{1}{2}}^{GodC} = f(q_{i+\frac{1}{2}}^{GodC}) ; \quad q_{i+\frac{1}{2}}^{GodC} = \frac{1}{2}(q_i^n + q_{i+1}^n) - \frac{\Delta t}{\Delta x} [f(q_{i+1}^n) - f(q_i^n)] . \quad (12.97)$$

Here f is the flux function in (12.91) evaluated at various mesh points. This method, also due to Godunov [22], is a centred method, not upwind; it does not depend on the speed λ . Here the scheme is written in a way that is applicable to any conservation law (12.91), or in fact, any system of conservation laws.

3. **The Lax-Friedrichs method (or Lax method)** [23]. For the conservation law in (12.91) with flux function $f(q)$, the numerical flux for the Lax-Friedrichs method is

$$f_{i+\frac{1}{2}}^{LF} = \frac{1}{2} [f(q_i^n) + f(q_{i+1}^n)] - \frac{1}{2} \frac{\Delta x}{\Delta t} (q_{i+1}^n - q_i^n) . \quad (12.98)$$

This is also a centred flux (not upwind) and is easily implemented for solving any conservation law or system of conservation laws.

4. **The Lax-Wendroff method** [24]. For the conservation law in (12.91) with flux function $f(q)$, the numerical flux for the Lax-Wendroff method is

$$f_{i+\frac{1}{2}}^{LW} = f(q_{i+\frac{1}{2}}^{LW}) ; \quad q_{i+\frac{1}{2}}^{LW} = \frac{1}{2}(q_i^n + q_{i+1}^n) - \frac{1}{2} \frac{\Delta t}{\Delta x} [f(q_{i+1}^n) - f(q_i^n)] . \quad (12.99)$$

This is also a centred flux which is easily implemented for solving any conservation law or system of conservation laws, leading to a method of second-order accuracy.

5. **The FORCE method** [8, 9]. For the conservation law in (12.91) with flux function $f(q)$, the numerical flux for the FORCE method is

$$f_{i+\frac{1}{2}}^{FORCE} = \frac{1}{2} [f_{i+\frac{1}{2}}^{LF} + f_{i+\frac{1}{2}}^{LW}] . \quad (12.100)$$

The FORCE flux is precisely an arithmetic average of the Lax-Friedrichs flux and Lax-Wendroff flux. FORCE is also a centred method, which is easily implemented for solving any conservation law or system of conservation laws.

6. **The Warming-Beam method** [25]. For LAE the numerical flux for the Warming-Beam method is

$$f_{i+\frac{1}{2}}^{WB} = \begin{cases} -\frac{1}{2}(1-c)(\lambda q_{i-1}^n) + \frac{1}{2}(3-c)(\lambda q_i^n) & \text{if } \lambda > 0 , \\ \frac{1}{2}(3+c)(\lambda q_{i+1}^n) - \frac{1}{2}(1+c)(\lambda q_{i+2}^n) & \text{if } \lambda < 0 . \end{cases} \quad (12.101)$$

This is an upwind flux, written here specifically for the linear advection equation.

Table 12.1 Properties of seven finite difference methods, written as conservative schemes. Note that not all linearly stable methods have Courant number stability limit $C_{lim} = 1$ and not all first-order schemes are monotone

Scheme	Type	Accuracy	Monotonicity	Stability condition
Godunov upwind	Upwind	1st order	Monotone	$ c \leq 1$
Godunov centred	Centred	1st order	Non-monotone	$ c \leq \frac{1}{2}\sqrt{2}$
Lax-Friedrichs	Centred	1st order	Monotone	$ c \leq 1$
FORCE	Centred	1st order	Monotone	$ c \leq 1$
Lax-Wendroff	Centred	2nd order	Non-monotone	$ c \leq 1$
Warming-Beam	Upwind	2nd order	Non-monotone	$ c \leq 2$
Fromm	Centred	2nd order	Non-monotone	$ c \leq 1$

7. **The Fromm method** [26]. For LAE the numerical flux for the Fromm method is

$$f_{i+\frac{1}{2}}^{FR} = \begin{cases} -\frac{1}{4}(1-c)(\lambda q_{i-1}^n) + (\lambda q_i^n) + \frac{1}{4}(1-c)(\lambda q_{i+1}^n) & \text{if } \lambda > 0, \\ \frac{1}{4}(1+c)(\lambda q_i^n) + (\lambda q_{i+1}^n) - \frac{1}{4}(1+c)(\lambda q_{i+2}^n) & \text{if } \lambda < 0. \end{cases} \quad (12.102)$$

This is a centred method, written here specifically for the linear advection equation.

Table 12.1 summarises the main properties of the listed numerical methods (12.96)–(12.102). The reader is encouraged to verify these properties by using previous definitions given in this chapter. Note that not all linearly stable methods in Table 12.1 have Courant number stability limit $C_{lim} = 1$ and not all first-order schemes are monotone. For the Godunov centred method $C_{lim} = \frac{1}{2}\sqrt{2}$ and for Warming-Beam $C_{lim} = 2$.

Exercise 12.12 Verify that, for both positive and negative speed λ in LAE, the Fromm numerical flux turns out to be an arithmetic average of the Lax-Wendroff numerical flux and the Warming-Beam numerical flux.

Next we address two important properties of conservative numerical methods, namely the telescopic property and monotonicity, this time seen in terms of the numerical flux.

The Telescopic Property of Conservative Methods

It is easily verified that summation applied to formula (12.92), premultiplied by Δx , gives

$$\sum_{i=i_{Left}}^{i=i_{Right}} \Delta x q_i^{n+1} = \sum_{i=i_{Left}}^{i=i_{Right}} \Delta x q_i^n - \Delta t \left(f_{Right+\frac{1}{2}} - f_{Left-\frac{1}{2}} \right). \quad (12.103)$$

Here i_{Left} and i_{Right} are the leftmost and rightmost cells; $f_{Left-\frac{1}{2}}$ and $f_{Right+\frac{1}{2}}$ are leftmost and rightmost intercell boundaries fluxes. Expression (12.103) says that the total amount of q changes only due to the fluxes through the **end boundaries**. The telescopic property also holds for non-linear systems, in one and in multiple space dimensions.

Remark: conservative methods and shocks. For non-linear equations there is a relation between shock waves and the conservation property of numerical methods. In fact, as shown by Lax and Wendroff [24], shock waves are captured with the correct strength and correct propagation speed by conservative methods. On the other hand, non-conservative methods will compute shock waves with the wrong speed, as proved 34 years later by Hou and LeFloch [27].

Monotonicity in Terms of the Numerical Flux

We return to the monotonicity property of numerical methods, this time for three-point conservative methods for non-linear equations.

Theorem 12.13 Necessary conditions for monotonicity. *If a three-point conservative scheme*

$$q_i^{n+1} = q_i^n - \frac{\Delta t}{\Delta x} \left(f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}} \right) \quad (12.104)$$

for a non-linear conservation law in (12.91) is monotone, then

$$\frac{\partial}{\partial q_i^n} f_{i+\frac{1}{2}}(q_i^n, q_{i+1}^n) \geq 0 \quad \text{and} \quad \frac{\partial}{\partial q_{i+1}^n} f_{i+\frac{1}{2}}(q_i^n, q_{i+1}^n) \leq 0. \quad (12.105)$$

Proof We first recall the general form (12.38) of a scheme and then define

$$H(q_{i-1}^n, q_i^n, q_{i+1}^n) \equiv q_i^n - \frac{\Delta t}{\Delta x} \left(f_{i+\frac{1}{2}}(q_i^n, q_{i+1}^n) - f_{i-\frac{1}{2}}(q_{i-1}^n, q_i^n) \right). \quad (12.106)$$

From the definition of monotonicity it is easily proved that

$$\left. \begin{aligned} \frac{\partial}{\partial q_{i-1}^n} H(q_{i-1}^n, q_i^n, q_{i+1}^n) \geq 0 &\rightarrow \frac{\partial}{\partial q_{i-1}^n} f_{i-\frac{1}{2}}(q_{i-1}^n, q_i^n) \geq 0. \\ \frac{\partial}{\partial q_{i+1}^n} H(q_{i-1}^n, q_i^n, q_{i+1}^n) \geq 0 &\rightarrow \frac{\partial}{\partial q_{i+1}^n} f_{i+\frac{1}{2}}(q_i^n, q_{i+1}^n) \leq 0. \end{aligned} \right\} \quad (12.107)$$

Hence, if the scheme is monotone, then the flux is an increasing function of its first argument and a decreasing function of its second argument and the claimed result is thus proved. ■

We have proved **necessary conditions for a conservative scheme to be monotone**. Note that conditions (12.105) are not sufficient for monotonicity.

12.3.3 Time Step and Boundary Conditions

Given an initial-boundary value problem, such as (12.91), one has initial and boundary conditions available. Then, given the conservative method (12.92) to solve it, one needs two items: the numerical flux $f_{i+\frac{1}{2}}$ and the time step Δt . In general, the time step Δt varies from time level to time level, depending on the spacing Δx , the fastest wave speed and the **stability condition** of the specific scheme.

Imposing the CFL Condition to Calculate the Time Step

We first consider the case for the linear advection equation with constant wave speed λ , as solved by the Godunov upwind method. Recalling that the Courant number is

$$c = \frac{\lambda \Delta t}{\Delta x} , \quad (12.108)$$

we use the stability condition (12.68) to select the time step Δt so that the scheme satisfies the stability condition. For both $\lambda > 0$ and $\lambda < 0$ this requires

$$\Delta t \leq \frac{\Delta x}{|\lambda|} . \quad (12.109)$$

In practice, one selects the time step according to

$$\Delta t = C_{cfl} \frac{\Delta x}{|\lambda|} , \quad (12.110)$$

where C_{cfl} , called the **CFL coefficient**, is chosen to satisfy

$$C_{cfl} = SF \times C_{lim} , \quad 0 < SF \leq 1 . \quad (12.111)$$

Here C_{lim} is the linear stability limit of the scheme obtained from a linear stability analysis. For the Godunov upwind method $C_{lim} = 1$, whereas for the Godunov centred scheme $C_{lim} = \frac{1}{2}\sqrt{2}$. The parameter SF is a **safety factor** that may take care of (i) round-off errors and (ii) uncertainties in determining the maximum wave speed emerging from the equations. For the linear advection equation, the speed λ is well defined.

For non-linear conservation laws, such as that in (12.91), the wave speed s analogous to λ , but needs to be estimated. This is usually derived from the characteristic speed $s = df(q)/dq$, noting however, that the characteristic speed is not always identical to a wave speed; see Chap. 8 for relevant definitions. Then, the time step is computed from

$$\Delta t = C_{cfl} \times \frac{\Delta x}{s_{\max}^n} , \quad (12.112)$$

where $s_{\max}^n$ is an **estimate** for the maximum, in absolute value, of the true wave speed s in the full computational domain at the time level n . The estimation of Δt from (12.112) is subject to additional uncertainties, due to uncertainties in estimating $s_{\max}^n$. For this reason the **rule of thumb**

$$C_{\text{eff}} = SF \times C_{\text{lim}}, \quad SF \approx 0.9 \quad (12.113)$$

is applied in practice. As a final remark we note that the subject of estimating wave speeds has not been exhausted; see [28] for recent advances. See also [19]. The case of systems of equations will be dealt with in Chap. 13.

Numerical Boundary Conditions

For conservative methods one must impose boundary conditions at positions that correspond to inter-cell boundaries. Therefore, imposing the boundary conditions requires conditions on the numerical flux at the domain boundary positions.

A practical approach is to add **fictitious cells** to the left and right of the domain $[a, b]$, as depicted in Fig. 12.8. The number of fictitious cells required depends on the accuracy of the scheme. For first-order methods we add one fictitious cell on each side, namely a cell I_0 on the left and a cell I_{M+1} on the right, as depicted in Fig. 12.8. Then, we prescribe corresponding initial values q_0^n and q_{M+1}^n in the fictitious cells so that the corresponding numerical fluxes resulting from the solution of the associated Riemann problem fulfils the desired boundary condition. For the flux $f_{\frac{1}{2}}$ on the left boundary one considers the Riemann problem with initial condition (q_0^n, q_1^n) . For the flux $f_{M+\frac{1}{2}}$ on the right boundary one considers the Riemann problem with initial condition (q_M^n, q_{M+1}^n) . For a second-order method one adds two fictitious cells on either side of the domain.

In general, prescribing suitable boundary conditions is a challenging task and requires a good understanding of the meaning (e.g. physical) of the mathematical problem to be solved. Then, the correct implementation of the boundary conditions is an equally challenging task, for which great care is required. For upwind methods, such as the Godunov method, the implementation of boundary conditions is greatly facilitated by the use of Riemann problem solutions also at the boundaries.

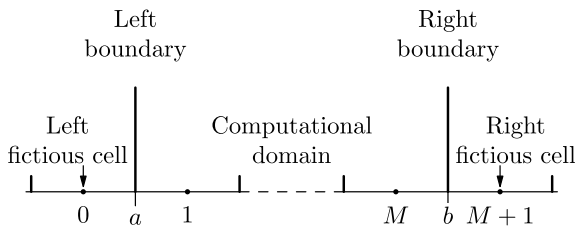


Fig. 12.8 Boundary conditions for a first-order method. Fictitious cells adjacent to the boundaries of the computational domain are added, with corresponding data values so as to fulfil the required boundary condition

As a matter of fact, even when using non-upwind methods in the interior of the domain, one could benefit from keeping the Riemann-problem approach just to implement boundary conditions.

In the context of the simple model equations dealt with in this chapter, the discussion on boundary conditions will be concerned exclusively with the application of just two types of boundary conditions.

Transmissive (or transparent) boundary conditions:

$$\left. \begin{array}{l} \text{1st order: } q_0^n = q_1^n ; \quad q_{M+1}^n = q_M^n , \\ \text{2nd order: } q_{-1}^n = q_2^n , q_0^n = q_1^n ; q_{M+1}^n = q_M^n , q_{M+2}^n = q_{M-1}^n . \end{array} \right\} \quad (12.114)$$

This type of BCs is applied when the presence of the computational boundary should have no effect on the solution. The boundary should behave as if it were transparent to the passage of waves. This approach works reasonably well in one space dimension but is far from perfect.

Periodic boundary conditions:

$$\left. \begin{array}{l} \text{1st order: } q_0^n = q_M^n ; \quad q_{M+1}^n = q_1^n , \\ \text{2nd order: } q_{-1}^n = q_{M-1}^n , q_0^n = q_M^n ; q_{M+1}^n = q_1^n , q_{M+2}^n = q_2^n . \end{array} \right\} \quad (12.115)$$

This type of BCs is applied when one wants to compute the long time evolution of the solution using a short computational domain. For example, a right travelling wave initiated in the computational domain will move right, will pass the right boundary and come back into the computational domain through the left boundary. This approach works very well.

12.4 Computational Results from a Case Study

Here we show computational results for LAE from six methods described in this chapter, as applied to two test problems, one problem with smooth solution and another with discontinuous solution. Numerical results, as compared with the exact solution, are fully discussed.

12.4.1 The Problem, the Methods and Parameters

We solve the following initial-boundary value problem for the linear advection equation

$$\left. \begin{array}{ll} \text{PDE:} & \partial_t q + \lambda \partial_x q = 0, \quad x \in [a, b], \quad t > 0, \\ \text{IC:} & q(x, 0) = h(x), \quad x \in [a, b], \quad t = 0, \\ \text{BCs (periodic):} & q(a, t) = q(b, t), \quad \forall t > 0, \end{array} \right\} \quad (12.116)$$

with $\lambda = 1$, $a = -5$ and $b = 5$. We use 3 first-order methods: Lax-Friedrichs, FORCE and Godunov upwind; and 3 second-order methods: Lax-Wendroff, Warming-Beam and Fromm. We consider two tests problems given by two types of initial conditions $h(x)$:

Test 1: the initial condition is smooth. In this test problem the initial condition $h(x)$ is given by a smooth Gaussian profile

$$q(x, 0) = h(x) = \alpha e^{-\beta x^2}, \quad \alpha = 1, \quad \beta = 2, \quad x \in [-5, 5]. \quad (12.117)$$

Test 2: the initial condition is discontinuous. In this test problem the initial condition $h(x)$ is a (discontinuous) square wave given as

$$q(x, 0) = h(x) = \begin{cases} 0 & \text{if } a = -5 \leq x < -2, \\ 1 & \text{if } -2 \leq x \leq 2, \\ 0 & \text{if } 2 < x \leq b = 5. \end{cases} \quad (12.118)$$

Computational parameters. We use two meshes: $M = 25$ cells and $M = 200$ cells. The time step Δt results from choosing the CFL coefficient set to $C_{cfl} = 0.9$. The output time $T_{out} = 100$ is chosen so as to complete 10 cycles within the domain, at which time the exact solution coincides with the initial condition. In both test problems we impose periodic boundary conditions.

12.4.2 Computational Results and Discussion

Results are displayed in the following four figures. Figure 12.9 shows results for Test 1 for three first-order methods; Fig. 12.10 shows results for Test 1 for three second-order methods; Fig. 12.11 shows results for Test 2 for three first-order methods; Fig. 12.12 shows results for Test 2 for three second-order methods.

Comments on First-Order Methods for Test 1

Results are shown in Fig. 12.9. The exact solution is a very smooth function but has large derivatives and a pronounced peak $q_{max} = 1$ located at $x = 0$ at the initial time

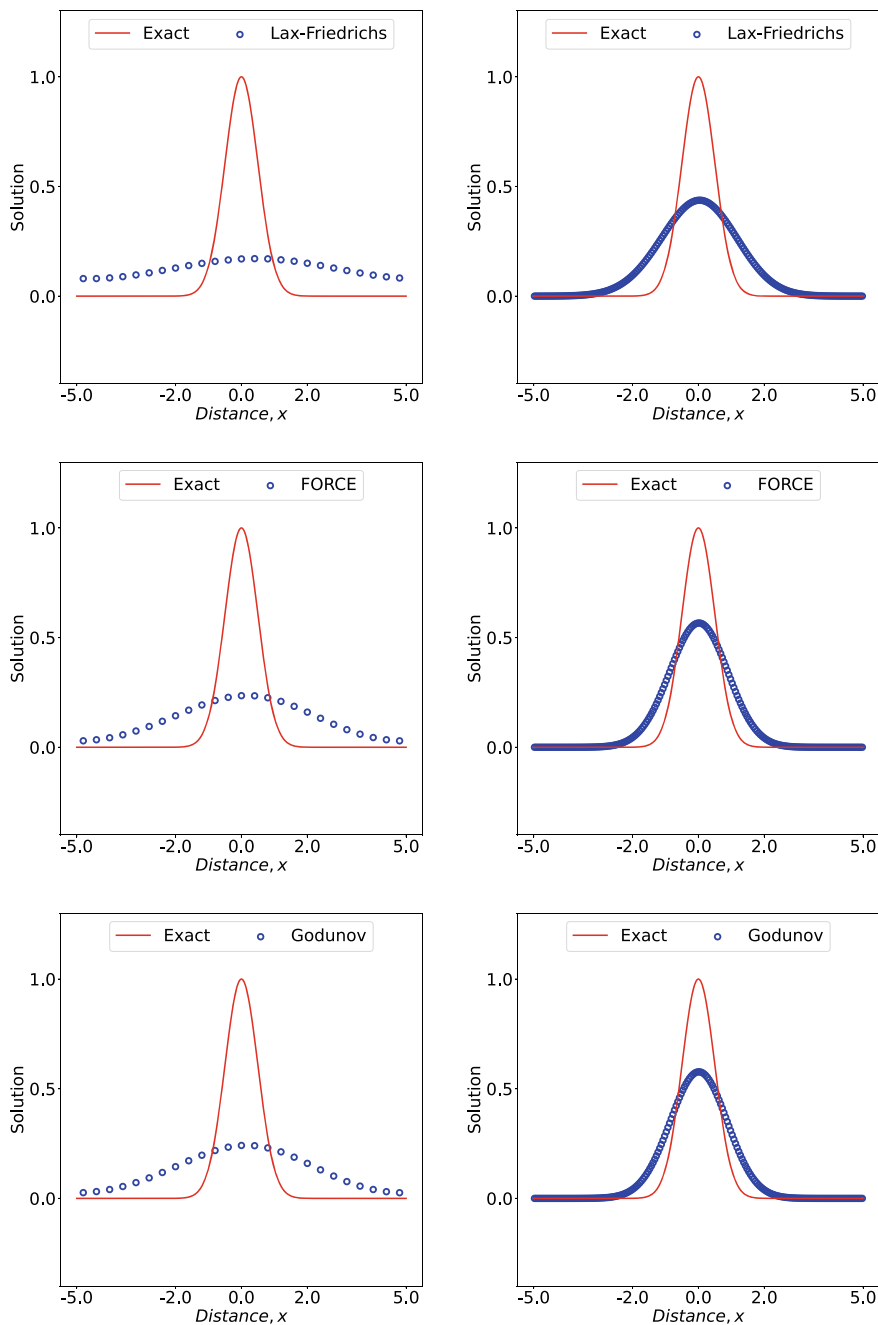


Fig. 12.9 Test 1 solved with three first-order methods: Lax-Friedrichs (top), FORCE (middle) and Godunov upwind (bottom). Results are shown at output time $T_{out} = 100$ and for two meshes: $M = 25$ cells (left column) and $M = 200$ cells (right column); the CFL coefficient is $C_{cfl} = 0.9$. The exact solution is shown by full line and the numerical solution is shown by symbols

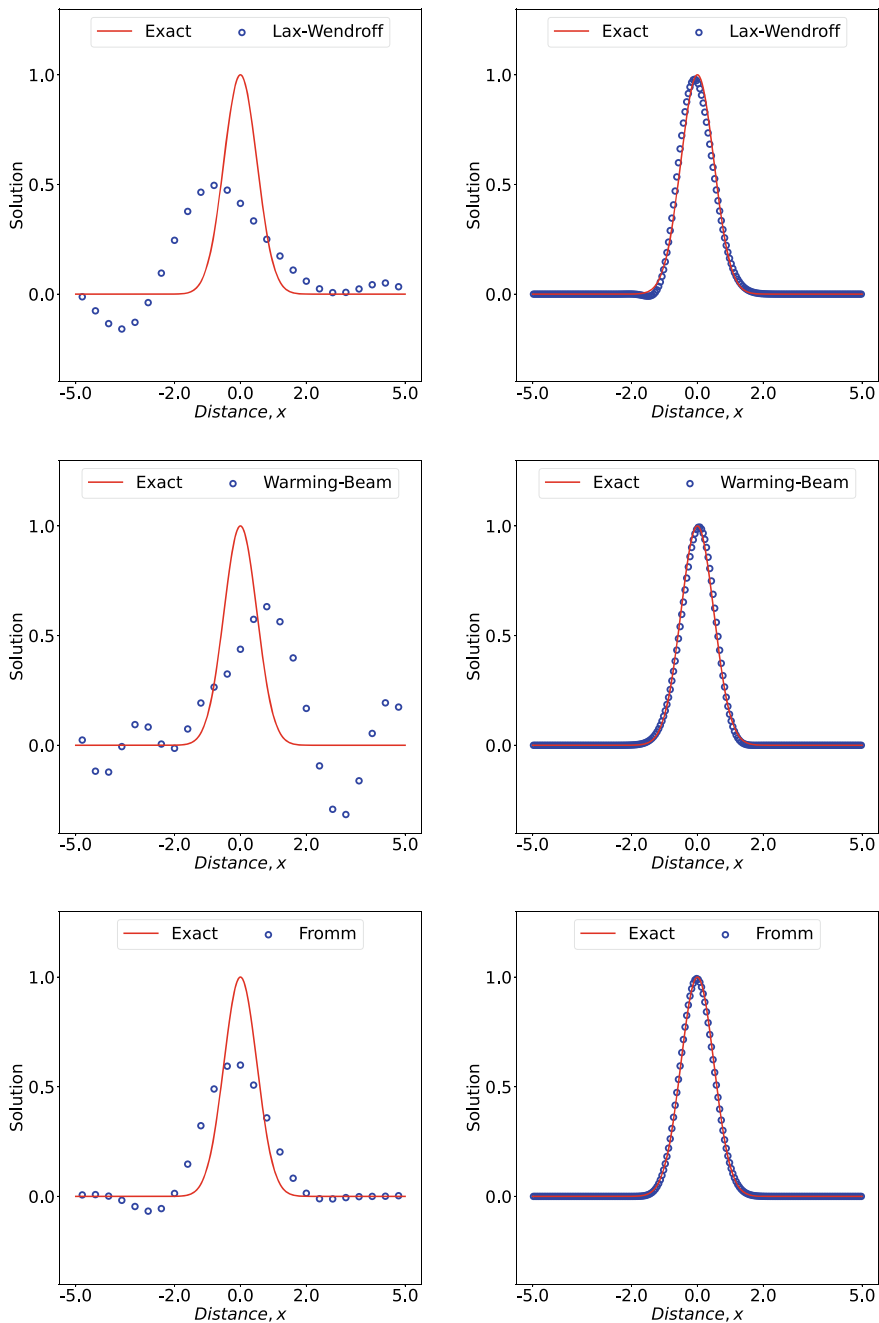


Fig. 12.10 Test 1 solved with three second-order methods: Lax-Wendroff (top), Warming-Beam (middle) and Fromm (bottom). Results are shown at output time $T_{out} = 100$ and for two meshes: $M = 25$ cells (left column) and $M = 200$ cells (right column); the CFL coefficient is $C_{cfl} = 0.9$. The exact solution is shown by full line and the numerical solution is shown by symbols

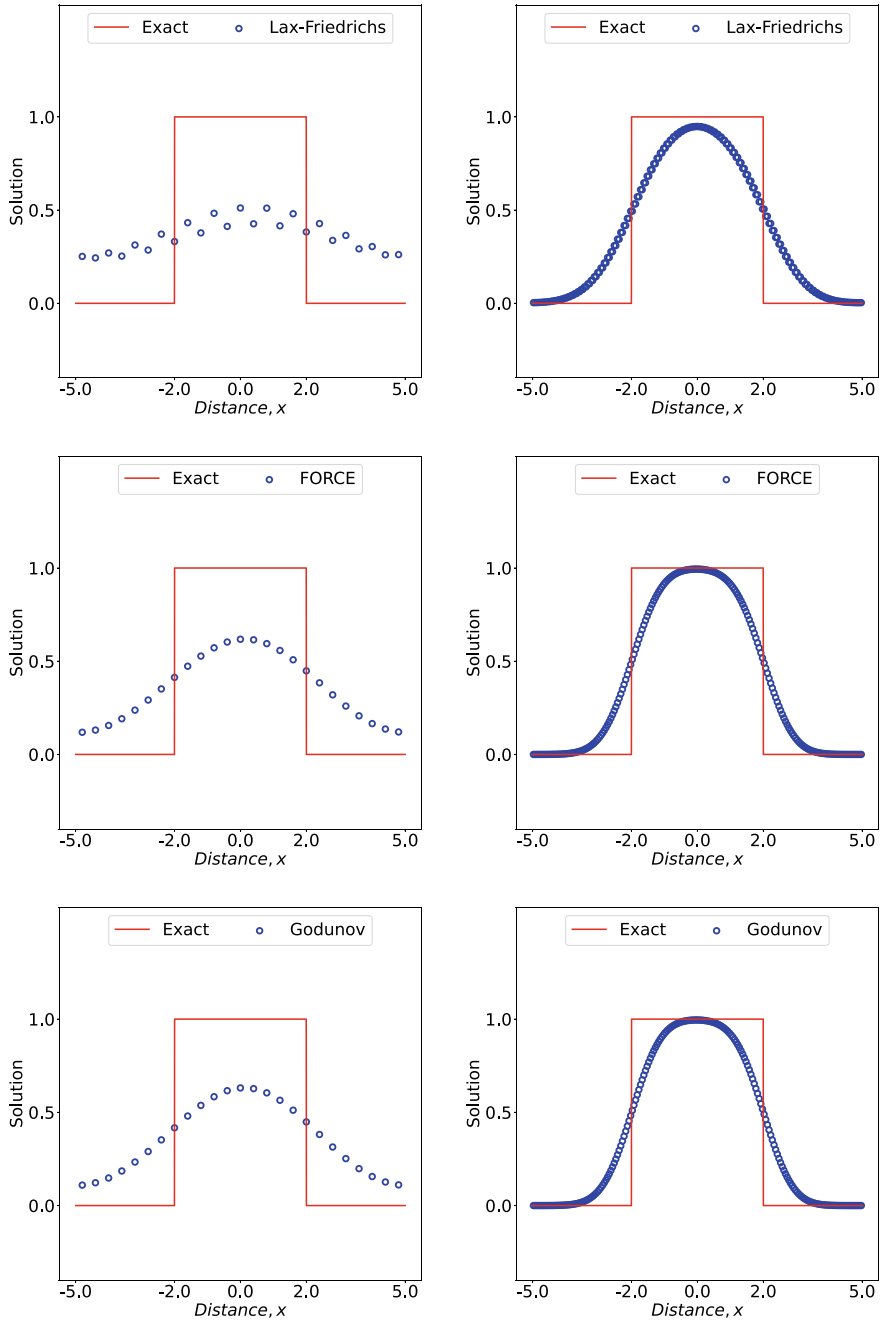


Fig. 12.11 Test 2 solved with three first-order methods: Lax-Friedrichs (top), FORCE (middle) and Godunov upwind (bottom). Results are shown at output time $T_{out} = 100$ and for two meshes: $M = 25$ cells (left column) and $M = 200$ cells (right column); the CFL coefficient is $C_{cfl} = 0.9$. The exact solution is shown by full line and the numerical solution is shown by symbols

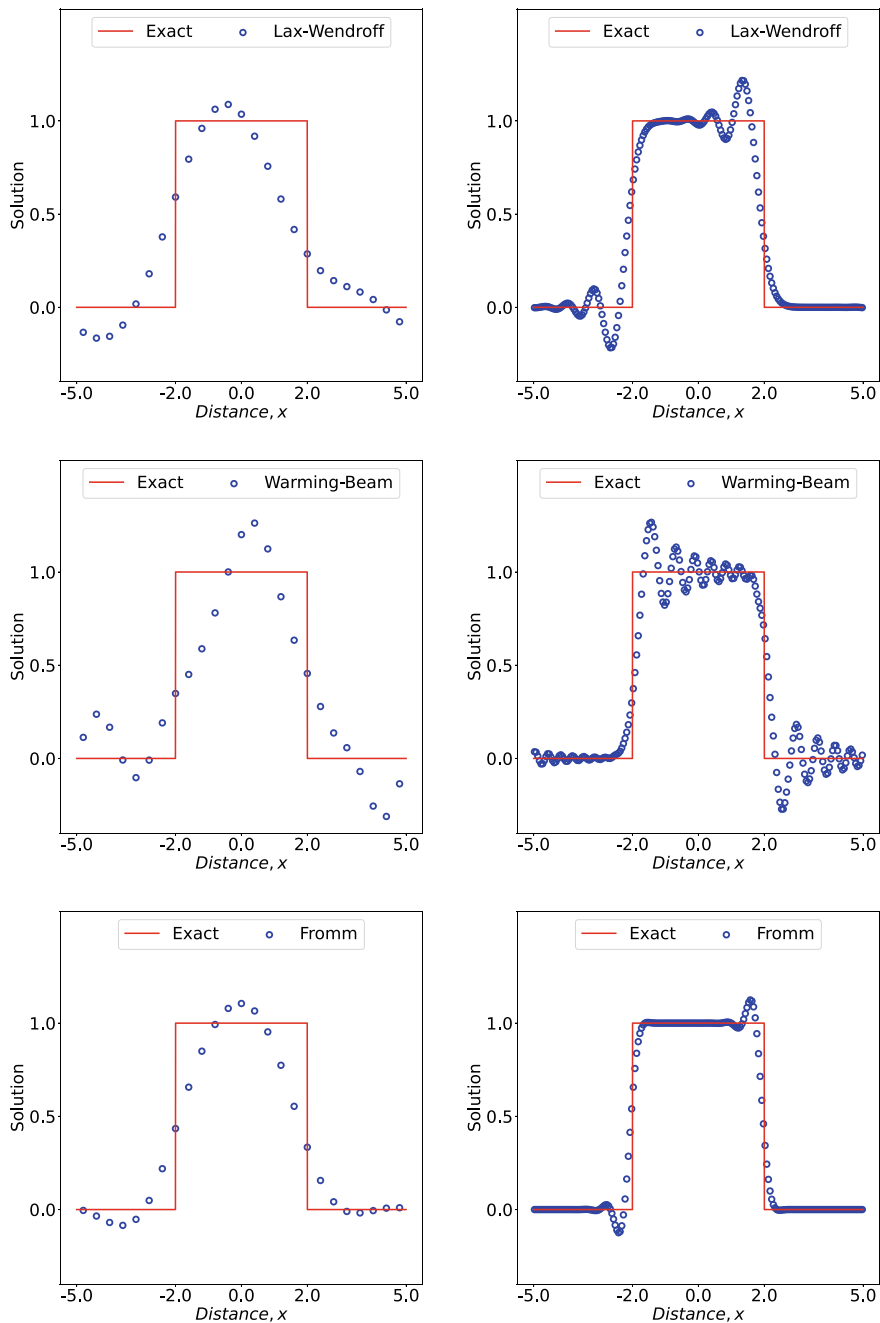


Fig. 12.12 Test 2 solved with three second-order methods: Lax-Wendroff (top), Warming-Beam (middle) and Fromm (bottom). Results are shown at output time $T_{out} = 100$ and for two meshes: $M = 25$ cells (left column) and $M = 200$ cells (right column); the CFL coefficient is $C_{cfl} = 0.9$. The exact solution is shown by full line and the numerical solution is shown by symbols

and at the output time of $T_{out} = 100$ units. Subject to periodic boundary conditions, the initial profile has travelled towards the right, has left the domain through the right boundary and has reentered the domain through the left boundary, completing a total of 10 cycles in 100 time units.

Starting from the Lax-Friedrichs method (top), results are shown for two meshes: 25 cells (left) and 200 cells (right). The coarser mesh result is way too inaccurate; this is more evident at $x = 0$; the numerical error is very large. The finer mesh on the right, has improved the results, but not very appreciably, even after a mesh refinement by a factor 8. In theory, if the mesh continues to be refined the solution should get closer and closer to the exact solution, but clearly convergence will be very slow. The slow convergence is to do with the first-order of accuracy of the scheme.

The results from the FORCE scheme (middle) have improved upon the results of Lax-Friedrichs, but not significantly. Again we see an improvement with the finer mesh but convergence is slow too. The results from Godunov upwind (bottom) are again a little better than the previous two schemes, but not by an appreciable margin.

In summary, these results from three first-order methods are rather disappointing for this problem with smooth solution. Their main feature is **excessive numerical diffusion** that **clips extrema**. This behaviour is understood from the form of the modified equation for first-order methods; the truncation error has a diffusion-like term. Mesh refinement helps but convergence is very slow. This means that getting accurate results from first-order methods may prove very expensive, as **extremely fine meshes** will be required to attain prescribed small errors.

Comments on Second-Order Methods for Test 1

Results are shown in Fig. 12.10. Starting from the Lax-Wendroff method (top), we observe that the coarse mesh result on the left is very inaccurate ($M = 25$ cells). This is disappointing for a second-order method applied to a problem with smooth solution. Note the **undershoots** below zero and **overshoots** above one. These spurious oscillations are typical of (linear) second-order methods and are present even for smooth problems that exhibit large gradients. This is contrary to popular belief that second-order methods (and higher order) are oscillation-free for smooth problems. We also observe a **phase error** in the form of erroneous position of the wave, which in this case trails behind the exact solution. The oscillations and the phase error are linked to the form of the modified equation for the Lax-Wendroff method, which is of **dispersive type**, leading to dispersive-type errors. As the mesh is refined by a factor of 8 the numerical solution improves dramatically (top right). But even so we still see a small undershoot at the tail of the wave.

The results from the Warming-Beam scheme (middle plots) are similar to those of the Lax-Wendroff method. A noticeable difference is that now the large overshoots for the coarse mesh case (middle left) are ahead of the wave. The finer mesh result (middle right) exhibits good agreement with the exact solution. To the eye, the dispersive error has decreased significantly, for the finer mesh computation.

Results from the Fromm scheme are shown in the bottom pictures of Fig. 12.10. This second-order method is distinctly better than the other two second-order methods. The coarser mesh case still shows a visible undershoot behind the wave and the finer-mesh result looks very good. No appreciable dispersive errors are seen, which is typical of the Fromm scheme.

In summary, second-order methods for smooth problems are not as accurate as is sometimes believed. Spurious oscillations are present for smooth problems, even for reasonably fine meshes. These methods are indeed better than first-order methods, in the sense that the errors diminish faster than for first-order methods, as the mesh is refined.

Comments on First-Order Methods for Test 2

Results are shown in Fig. 12.11. The exact solution is a square wave with two discontinuities. Subject to periodic boundary conditions, the initial profile has travelled towards the right, has left the domain through the right boundary and has reentered the domain through the left boundary, completing a total of 10 cycles in 100 time units.

The two top profiles of Fig. 12.11 correspond to the Lax-Friedrichs method. Results are very inaccurate. The numerical method cannot represent the correct shape of the square wave with two discontinuities, even though the finer mesh results look better. A dominant feature of the numerical solution is **numerical diffusion**, linked to the diffusive character of the modified equation, as stated earlier. Each discontinuity is represented by virtually 100 cells (right), instead of none, in the exact solution. A **puzzling** observation is the following: the solution from the Lax-Friedrichs method on the coarser mesh exhibits visible **spurious oscillations**. This is in apparent contradiction with the fact that the method is monotone, strictly speaking. Recall however, the remarks made earlier on this point. In this case the oscillations are **internal** to the interval containing the initial conditions.

The FORCE method improves the results visibly (middle), but in general the quality of the numerical solution is still unsatisfactory, dominated by numerical diffusion. Again convergence seems to be very slow; refining the mesh by a factor of 8 has improved the result but this is still far from satisfactory. The Godunov upwind scheme (bottom pictures) shows again a slight improvement over the FORCE results, and a visibly improvement over the Lax-Friedrichs results. Again, numerical diffusion dominates the numerical solution. In summary, first order monotone methods for discontinuous problems have the unique feature of absence of spurious overshoots and undershoots, unlike second-order linear methods. The best method is the Godunov upwind method and the worst one is the Lax-Friedrichs method.

First-order accuracy and monotonicity. It is important to be aware that the first-order accuracy of a scheme is not a sufficient condition for that scheme to be monotone. The Godunov centred method is the typical example to show this. The scheme is first order accurate but non-monotone. The reader is invited to verify these statements.

Comments on Second-Order Methods for Test 2

Results from second-order methods for this problem with discontinuous solution are shown in Fig. 12.12. One statement that applies to all six results is that second-order (linear) schemes are wholly inadequate for problems with discontinuous solutions. While for the smooth solution case, mesh refinement made the difference, in the present case, mesh refinement does not help visibly. **The spurious oscillations will not vanish under mesh refinement.** Better methods for problems with discontinuities, or even large gradients, are needed, and these, according to Godunov's theorem, must be non-linear methods.

12.5 Errors and Mesh Refinement

The core of this section includes definitions for error, various norms to measure it and various notions linked to the influence of the mesh size on the accuracy of the numerical solution. In particular, we introduce the concept of efficiency of a numerical method, whereby the quality of a numerical method is judged from a plot of **error against CPU cost**.

12.5.1 Types of Errors When Solving PDEs Numerically

When solving PDEs numerically, several kinds of errors are present, excluding involuntary mistakes. First, we have what might be called **modelling errors**; these arise when deriving the PDEs that are meant to represent a phenomenon or process of interest. Such errors come from the assumptions made that might not always hold true, as well as uncertainties from coefficients that may need to be evaluated experimentally, or estimated from partial information available. Then, there are **discretization errors**, due to the conversion of a **continuous** mathematical problem to a **discrete** problem. The local truncation error is just one example. **Round off errors** are always present due to the inability of computers to represent numbers exactly.

Here we are mainly concerned with **discretization errors** that may arise because (a) the domain is represented by a finite set of points or finite elements, or finite volumes; (b) the boundary of the domain is represented approximately; (c) functions are replaced by discrete sets of approximate values; (d) initial and boundary conditions are implemented approximately. Each of these steps introduces an error. Moreover, as computations proceed in time, new errors are introduced and accumulated. A question one may then pose is this: **how sure are we that, as the mesh is refined, the error tends to zero?** This is a very challenging subject and most numerical schemes used to solve practical problems do not have proofs that ensure convergence. But even if they do, convergence may be so slow that in computational

practice it may become impossible to attain nearly converged results. For linear problems we have the **Lax Equivalence Theorem** [29], which states that **a method is convergent if and only if it is consistent and stable**. Therefore, for convergence for linear problems one then needs to ensure (a) consistency and (b) stability.

We now consider errors in finite difference methods. The error will be defined at a chosen time $T_N = N\Delta t$, at which we have, for a purely IVP, the infinite set of solution values

$$q^N = \{q_i^N\}_{i=-\infty}^{i=\infty} . \quad (12.119)$$

At the spatial position x_i and time T_N we can define the **local** error as

$$e_i^N = q_i^N - q(x_i, T_N) , \quad (12.120)$$

where $q(x_i, T_N)$ is the exact solution at the mesh point (x_i, T_N) . The **absolute error** (local) is defined as

$$|e_i^N| = |q_i^N - q(x_i, T_N)| . \quad (12.121)$$

Often, it is also useful to introduce the **relative error** (local), defined as

$$|r_i^N| = \frac{|q_i^N - q(x_i, T_N)|}{|q(x_i, T_N)|} . \quad (12.122)$$

The **global error** E^N at time T_N involves all the **local** errors e_i^N and to measure it we introduce a **norm**. Commonly used norms are the L_p -**norms**, or simply p -norms, where p is a positive integer. A measure of the global error E^N in the L_p -norm is

$$\|E^N\|_p = \left(\Delta x \sum_{i=-\infty}^{i=\infty} |q_i^N - q(x_i, T_N)|^p \right)^{1/p} . \quad (12.123)$$

Special cases used in practice are: $p = 1$, which defines the L_1 -norm; $p = 2$, which defines the L_2 -norm and $p = \infty$, which defines the L_∞ -norm, also called the maximum norm; this takes the special and simple form

$$\|E^N\|_\infty = \max_{-\infty \leq i \leq \infty} |q_i^N - q(x_i, T_N)| . \quad (12.124)$$

Example 12.14 Consider the following IBVP for the linear advection equation

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \lambda \partial_x q = 0 , \quad x \in [a, b] , \quad t > 0 , \\ \text{IC: } q(x, 0) = h(x) = \alpha e^{-\beta x^2} , \quad x \in [a, b] , \\ \text{BCs: } q(a, t) = 0 , \quad q(b, t) = 0 , \quad \forall t > 0 . \end{array} \right\} \quad (12.125)$$

Table 12.2 Global errors from the Godunov scheme in three different norms for three meshes. Compare the size of the errors with those from the Lax-Wendroff method in Table 12.3, for the same mesh size

Mesh M	Δx	L_1 -norm error	L_2 -norm error	L_∞ -norm error
50	0.1960	0.9176	0.5077	0.5359
100	0.0990	0.6462	0.3789	0.4183
200	0.0496	0.4155	0.2558	0.2919

Table 12.3 Global errors from the Lax-Wendroff scheme in three different norms for three meshes. Compare the size of the errors with those from the Godunov method in Table 12.2, for the same mesh size

Mesh M	Δx	L_1 -norm error	L_2 -norm error	L_∞ -norm error
50	0.1960	0.5980	0.3400	0.3205
100	0.0990	0.2157	0.1385	0.1474
200	0.0496	0.0585	0.0396	0.0434

Parameters of the problem are: $\lambda = 1$, $a = -5$, $b = 5$, $\alpha = 1$, $\beta = 2$, $C_{cfl} = 0.9$ and $T_{out} = 50$. Meshes used are $M = 50$, $M = 100$ and $M = 200$. Computed errors are shown in Tables 12.2 and 12.3.

Table 12.2 shows the results for the Godunov method and Table 12.3 shows the results for the Law-Wendroff method. First note that the errors have different values for the three different norms used. The errors of the Lax-Wendroff method are smaller than those of the Godunov method for the same mesh and the same norm. This fact is related to the **order of accuracy of the methods**, the first order for Godunov and the second order for Lax-Wendroff. The main observation is that the error, in all three norms, decreases as the mesh spacing Δx decreases. In addition, we observe that the error of the Lax-Wendroff method decreases faster than that of the Godunov method. In other words, the speed of convergence (or convergence rate) of Lax-Wendroff is larger than that of Godunov. Again, this fact is linked to the order of accuracy of the methods.

12.5.2 Empirical Convergence Rates

For practical problems, determining theoretically the accuracy of a numerical scheme is hard or impossible. However, one can use an empirical convergence rates procedure that is applicable to all problems, provided we have an exact solution $q^e(x, t)$ available. For difficult problems one may resort to the method of **manufactured exact solutions** to obtain $q^e(x, t)$, as will be explained in subsequent chapters. Here, to determined the empirical convergence rate we proceed as follows:

1. Fix an output time $t = T_{out}$.
2. Consider a sequence of K meshes, each one defined by its number of cells M_k , so that the following increasing sequence of integers results

$$M_1 < M_2 < \dots < M_k < M_{k+1} < \dots < M_K . \quad (12.126)$$

A corresponding, decreasing sequence of mesh spacings Δx_k is generated by the K meshes:

$$\Delta x_1 > \Delta x_2 > \dots > \Delta x_k > \Delta x_{k+1} > \dots > \Delta x_K . \quad (12.127)$$

3. At time $t = T_{out}$, evaluate the errors in the L_1 , L_2 and L_∞ norms

$$\left. \begin{aligned} e_1^k(T_{out}, \Delta x_k) &= \Delta x_k \sum_{i=1}^{M_k} |q_i^n - q_i^e| , \\ e_2^k(T_{out}, \Delta x_k) &= \sqrt{\Delta x_k \sum_{i=1}^{M_k} (q_i^n - q_i^e)^2} , \\ e_\infty^k(T_{out}, \Delta x_k) &= \max_{i=1, \dots, M_k} |q_i^n - q_i^e| . \end{aligned} \right\} \quad (12.128)$$

Here q_i^e ($i = 1, \dots, M_k$) is the **exact solution** at $x = x_i$ and time $t = T_{out}$; q_i^n is the numerical solution at $x = x_i$ and $t = T_{out}$.

4. The error e_l^k will generally depend on the space and time discretization

$$e_l^k(T_{out}, \Delta x_k) = C_1 (\Delta x_k)^j + C_2 (\Delta t_k)^m , \quad (12.129)$$

with C_1 and C_2 two positive constants.

5. Using the assumed fixed relationship between Δt and Δx we have

$$e_l^k(T_{out}, \Delta x_k) = C (\Delta x_k)^p , \quad (12.130)$$

where $p = \min(j, m)$ and C is a positive constant.

6. We apply (12.130) to two successive meshes k and $k + 1$, with spacings Δx_k and Δx_{k+1} , respectively. After eliminating the constant C in (12.130), we obtain the **empirical order of accuracy** p as

$$p_{k+1} = \frac{\ln \left(\frac{e_2^{k+1}(T_{out}, \Delta x_{k+1})}{e_2^k(T_{out}, \Delta x_k)} \right)}{\ln \left(\frac{\Delta x_{k+1}}{\Delta x_k} \right)} . \quad (12.131)$$

One expects p_{k+1} to converge asymptotically to the actual accuracy p , for sufficiently large k . A fixed ratio between successive meshes is often used, e.g. $h_k/h_{k+1} = 2$. The ratio $h_k/h_{k+1} = 3$ is specially convenient for testing **mesh convergence** to compute **mesh-independent solutions**, as interpolation between meshes is entirely avoided with this choice.

In measuring errors one needs to discriminate between the various numerical approaches, in order to define the **numerical solution** and the **exact solution**. So far we have given definitions as for finite difference methods, for which solution values are **point values**. Instead, in the finite volume method, the exact solution in the cell $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ is the integral average of the exact solution at time $t = T_{out}$ within I_i and q_i^n is the finite-volume **numerical solution** at time $t = T_{out}$.

Empirical Convergence Rate Tables

Here we display the results from the empirical convergence rates study in the form of tables, as will be explained. In all computations the Courant number coefficient was set to $C_{cfl} = 0.9$ for all methods, except the Godunov centred method, for which we used $C_{cfl} = 0.9 \times C_{lim}$, where $C_{lim} = \sqrt{2}/2$.

Table 12.4 shows results for four first-order methods, namely Lax-Friedrichs, FORCE, Godunov upwind and Godunov centred. The left column shows the sequence of mesh cells M used. Successive columns show pairs (E_l, p_l) for the error and convergence rates in the L_1 , L_2 and L_∞ norms. The final column shows the CPU cost. For the Lax-Friedrichs method, for instance, we see that the error decreases as the mesh is refined and the convergence rate tends to the theoretically expected rate of 1, for all three norms. The results from the FORCE scheme are similar but errors are generally smaller and the sought convergence rate is closer to 1 than that of Lax-Friedrichs, for any given mesh. Results from the Godunov upwind method are similar to those of FORCE. Surprisingly, the results from the Godunov centred method exhibit larger errors than the other schemes, presumably due to its smaller time step. In any case, all schemes display empirical convergence rates that tend to the theoretically expected rate of 1.

Table 12.5 shows empirical convergence rates and errors from three second-order methods, namely Lax-Wendroff, Warming-Beam and Fromm. All methods attain the theoretically expected convergence rate of 2, as the mesh is refined, with Fromm giving the best results. Note that the range of meshes used is different from that of Table 12.4 for first-order methods, illustrating the fact that the expected convergence rate for second-order methods is attained with much coarser meshes, see the Fromm scheme for example.

12.5.3 Mesh-Independent Solution

A mesh refinement exercise is mandatory in practice, in order to assess the level of errors involved, as measured against known solutions, exact, numerical. A common practice is to seek the so-called **mesh-independent solution** in a mesh-refinement exercise. The motivation for this is to identify the mesh, say M_k , such that the finer mesh M_{k+1} will not result in an appreciable gain in accuracy in the computed solution. Recall that in general we do not have an exact solution to compute the error of the numerical solution. A practical approach is to consider a sequence of meshes $\{M_l\}$,

Table 12.4 Empirical convergence rates. Left column shows sequence of successively refined meshes by factor of 3, followed by successive pairs (E_l, p_l) of columns showing error and convergence rate in the L_1 , L_2 and L_∞ norms. The last column shows CPU time. Results are shown for four first-order methods: Lax-Friedrichs, FORCE, Godunov upwind and Godunov centred

M	E_1	p_1	E_2	p_2	E_∞	p_∞	CPU [s]
Lax-Friedrichs							
36	0.7190	0.5772	0.4181	0.3637	0.4415	-0.0679	0.0099
108	0.3479	0.6607	0.2177	0.5941	0.2509	0.5145	0.0854
324	0.1403	0.8268	0.0916	0.7881	0.1093	0.7559	0.7611
972	0.0505	0.9296	0.0336	0.9128	0.0408	0.8974	6.9275
2916	0.0173	0.9747	0.0116	0.9685	0.0142	0.9624	62.7971
FORCE							
36	0.4700	0.7390	0.2845	0.5238	0.3072	0.1029	0.0123
108	0.1997	0.7792	0.1287	0.7216	0.1515	0.6433	0.0984
324	0.0743	0.9003	0.0492	0.8764	0.0594	0.8525	0.8921
972	0.0258	0.9628	0.0172	0.9537	0.0210	0.9447	8.1575
2916	0.0087	0.9871	0.0058	0.9839	0.0072	0.9807	73.7141
Godunov upwind							
36	0.4545	0.6747	0.2755	0.4721	0.2973	0.0520	0.0124
108	0.1908	0.7899	0.1232	0.7323	0.1452	0.6525	0.1045
324	0.0706	0.9052	0.0468	0.8823	0.0565	0.8586	0.9310
972	0.0245	0.9647	0.0164	0.9560	0.0200	0.9474	9.1005
2916	0.0083	0.9878	0.0055	0.9847	0.0068	0.9817	84.8465
Godunov centred							
36	1.1678	0.3244	0.6063	0.1537	0.6153	-0.1778	0.0150
108	0.7139	0.4480	0.4114	0.3530	0.4517	0.2814	0.1264
324	0.3492	0.6509	0.2181	0.5774	0.2515	0.5328	1.1287
972	0.1409	0.8261	0.0920	0.7861	0.1098	0.7547	10.2187
2916	0.0508	0.9295	0.0338	0.9124	0.0410	0.8969	91.8439

as in (12.126), and compute corresponding solutions $\{\mathbf{Q}_l\}$. For $l \geq 2$ compute the change $\Delta \mathbf{Q}_k$ between the current solution $\mathbf{Q}_k$ and the previous solution $\mathbf{Q}_{k-1}$, in a chosen norm, say the L_1 norm. In order to avoid the need for interpolation between meshes, it is convenient to use a refinement factor of 3. In this manner, the cell centre of the coarser mesh will coincide with one cell centre in the finer mesh. When the change $\Delta \mathbf{Q}_k$ in the solution from mesh M_{k-1} to mesh M_k is sufficiently small, the mesh refinement algorithm is stopped. A sufficiently small tolerance TOL is chosen as a stopping criterion. Once the stopping criterion is met, one considers the solution in the finest mesh as a **mesh-independent solution**. Sometimes, the terminology **mesh-converged solution** is used, which is misleading. This is because in general, a so-called mesh-independent solution is not a converged solution and could still exhibit large errors. A **mesh-independent solution** simply means that

Table 12.5 Empirical convergence rates. Left column shows sequence of successively refined meshes by factor of 3, followed by successive pairs (E_l, p_l) of columns showing error in norm and convergence rate in the L_1 , L_2 and L_∞ norms. The last column shows CPU time. Results for three second-order methods: Lax-Wendroff, Warming-Beam and Fromm

M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
Lax-Wendroff							
36	0.2805	1.0215	0.1725	0.8074	0.1720	0.5291	0.0100
108	0.0403	1.7658	0.0272	1.6813	0.0291	1.6167	0.0797
324	0.0046	1.9835	0.0031	1.9752	0.0033	1.9731	0.7062
972	0.0005	1.9990	0.0003	1.9986	0.0004	2.0004	6.4147
Warming-Beam							
36	0.1935	1.3642	0.1309	1.1325	0.1301	0.9109	0.0096
108	0.0236	1.9138	0.0161	1.9082	0.0174	1.8296	0.0787
324	0.0026	1.9938	0.0018	1.9935	0.0019	2.0034	0.6945
972	0.0003	1.9998	0.0002	1.9994	0.0002	2.0003	6.3357
Fromm							
36	0.1013	1.6244	0.0656	1.3680	0.0685	0.8896	0.0097
108	0.0093	2.1774	0.0064	2.1230	0.0069	2.0918	0.0798
324	0.0010	2.0532	0.0007	2.0605	0.0007	2.0680	0.7051
972	0.0001	2.0074	0.0001	2.0085	0.0001	2.0094	6.4016

the particular scheme in use will not benefit from further mesh refinement. This is particularly evident in low-order methods. Figure 12.13 shows a mesh refinement exercise to compute a mesh-independent solution, from the Godunov upwind method (first order) and the Fromm method (second order).

12.5.4 Efficiency: Error Against CPU Cost

Efficiency refers here to the CPU time (cost) required by a method to attain a prescribed, **fixed error** E by suitably adjusting the mesh. This is one of several existing ways for comparing two or more numerical methods.

Figure 12.14 shows computational results from the Godunov upwind method (first order) and the Fromm method (second order). For each method, computations were performed on a sequence of meshes. For each mesh a point on the graph is identified by an open circle. Each point $P_k = (e_k, C_k)$, for $k = 1, \dots, K$, has an error level e_k on the vertical axis and a cost (CPU time) C_k on the horizontal axis. In Fig. 12.14 $K = 8$ for the Godunov method and $K = 5$ for the Fromm method. It is convenient to plot results in the $\log - \log$ plane and then fit a least-square straight line to the resulting points. Two straight lines are obtained, one for each method.

Fig. 12.13 Mesh refinement and mesh-independent solution exercise for the Godunov upwind method (first order) and the Fromm method (second order)

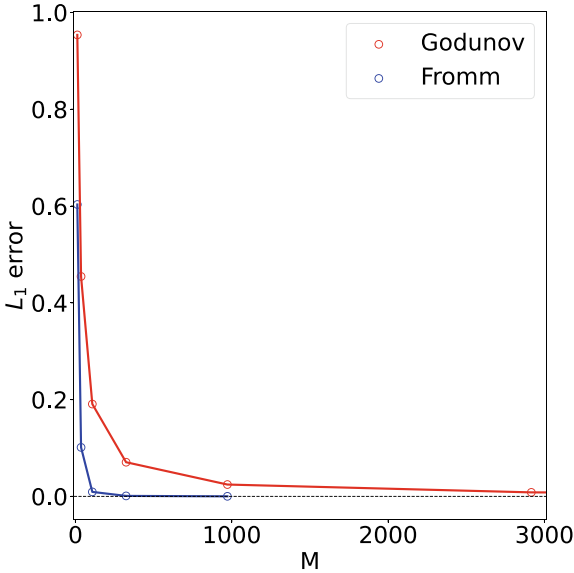
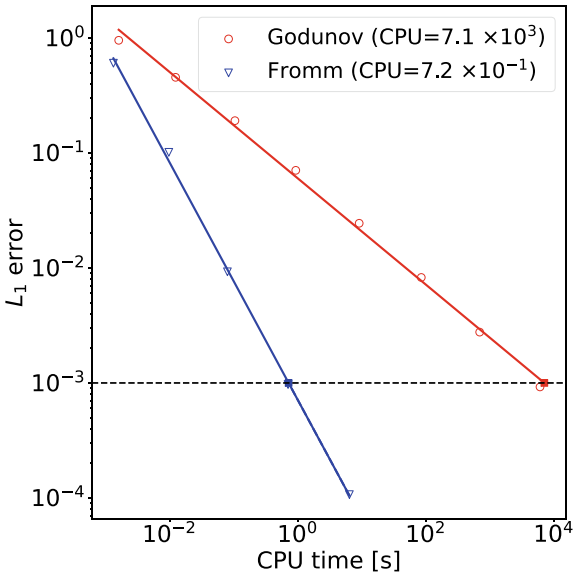


Fig. 12.14 Efficiency plot. Error against CPU cost for the Godunov upwind method (first order) and the Fromm method (second order). For a chosen error $E = 0.001$ the first-order Godunov method is four orders of magnitude more expensive than the second-order Fromm method, to attain the same error



Then the efficiency comparison of the two methods is carried out by tracing the horizontal line corresponding to the fixed error and finding its intersection points with the two straight lines. Each intersection gives a computational cost for the chosen fixed error. For example, for the chosen error $E = 0.001$, the first-order Godunov method is **four orders of magnitude** more expensive than the second-order Fromm method, to attain the same error. These results illustrate the fact that, in general,

high-order methods are much more efficient than low-order methods. That is, they cost less to attain the same error. In other words, refining the mesh to reduce the error by a low-order method is less competitive than increasing the order of accuracy of the scheme to be used in coarse meshes.

12.6 Summary and Further Reading

We have introduced some basic concepts on the subject of numerical approximation of hyperbolic equations, mostly in terms of the model linear advection equation. The most well-known finite difference methods have been introduced as well as the theoretical notions of local truncation error, accuracy, stability and monotonicity. The famous Godunov theorem has been stated and proved. Then, the introduced finite difference methods have been formulated as conservative methods with a corresponding numerical flux. Through a case study for the linear advection equation we have demonstrated the computational performance of all methods studied in this chapter and have discussed their salient features. The chapter is finished with some important notions concerned with the error in the approximations, mesh refinement, empirical convergence rates and efficiency. The numerical notions introduced in this chapter will be further extended in subsequent chapters, concerned with numerical methods for hyperbolic systems, and in particular, for the equations of blood flow.

For further reading on the subject we strongly recommend the following books: Richtmyer and Morton [29]; Smith [30]; Hirsch [10]; Godlewski and Raviart [13]; LeVeque [14]; Toro [6]; Toro [31]; Vázquez-Cendón [32].

12.7 Exercises

12.1 Analysis of schemes. For each of the methods in Table 12.1, perform the following tasks:

1. Finite difference scheme. By writing down the two inter-cell fluxes $f_{i+\frac{1}{2}}$ and $f_{i-\frac{1}{2}}$, verify that by inserting these fluxes in the conservative formula (12.92) one obtains a corresponding finite difference method.
2. Order of accuracy. By use of the definition of local truncation error, find the order of accuracy of the method.
3. Monotonicity. By first writing the finite difference scheme as a linear scheme, analyse the monotonicity of the scheme.
4. Linear stability. Use the von Neumann method to analyse the linear stability of the method.
5. Stability shortcut. For all three-point schemes, apply the Billett shortcut to stability and verify your findings from the previous task.

6. Viscous form. For all three-point schemes, write the methods in viscous form and find the coefficient of viscosity d .
7. Stability shortcut. For all three-point schemes, show that the Billett shortcut to stability leads to the simpler stability condition

$$c^2 \leq d \leq 1 . \quad (12.132)$$

12.2 Implicit Godunov method. Assume the temporal partial derivative to be approximated by the explicit forward formula and the spatial derivative to be approximated by a backward finite difference approximation calculated at the future time level $n + 1$, that is

$$\partial_x q(x_i, t_n) = \frac{q(x_i, t_{n+1}) - q(x_{i-1}, t_{n+1})}{\Delta x} + \mathcal{O}(\Delta x) \quad (12.133)$$

$$L_a(q_i^n) \equiv \frac{q_i^{n+1} - q_{i-1}^n}{\Delta t} + \lambda \left(\frac{q_i^{n+1} - q_{i-1}^{n+1}}{\Delta x} \right) = 0 , \quad (12.134)$$

Solving for q_i^{n+1} we obtain the *implicit* numerical scheme

$$q_i^{n+1} = q_i^n - c (q_i^{n+1} - q_{i-1}^{n+1}) , \quad (12.135)$$

or

$$-\frac{1}{2}c q_{i-1}^{n+1} + q_i^{n+1} + \frac{1}{2}c q_{i+1}^{n+1} = q_i^n , \text{ for } i = 1, \dots, M . \quad (12.136)$$

1. Write the problem as a linear system.
2. Write down the coefficient matrix $\mathbf{A}$, the vector $\mathbf{X}$ of unknowns and the vector $\mathbf{B}$ of known terms.
3. Implement the above implicit Godunov numerical method for the linear advection equation and test its performance against to explicit Godunov upwind method.

12.3 Linear advection with source term. Consider the linear advection equation with a linear source term

$$\partial_t q + \partial_x (\lambda q) = \beta q , \quad \lambda : \text{a constant} , \quad \beta \leq 0 \text{ a constant} . \quad (12.137)$$

For $\lambda > 0$ consider the schemes

$$q_i^{n+1} = q_i^n - c(q_i^n - q_{i-1}^n) + \Delta t (\beta q_i^n) \quad (12.138)$$

and

$$q_i^{n+1} = q_i^n - c(q_i^n - q_{i-1}^n) + \Delta t \left[\frac{1}{2} c \beta q_{i-1}^n + \left(1 - \frac{1}{2} c\right) \beta q_i^n \right]. \quad (12.139)$$

Introduce the dimensionless number $r = \beta \Delta t$, analogous to the Courant number c into the schemes.

1. Show that scheme (12.138) is stable under the condition

$$c + \frac{1}{2} r \leq 1. \quad (12.140)$$

2. Show that the scheme (12.139) is stable under the condition

$$c \leq 1, r \leq 2. \quad (12.141)$$

3. Which of the schemes has the more generous stability condition. Provide an explanation for your answer.

12.4 Monotone scheme in the presence of a source term.

1. Consider the scheme (12.138) above. Write the scheme in linear form and state whether the scheme is monotone, and if so, under what conditions?
2. Consider the scheme (12.139) above. Write the scheme in linear form and state whether the scheme is monotone, and if so, under what conditions?
3. Which of the two schemes is better from the point of view of monotonicity?
4. Is there a relationship between conditions for stability and conditions for monotonicity?

See Sweby [33] for original discussion on this subject.

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Chapter 13

Conservative Methods



Abstract This chapter focuses on conservative numerical methods for solving systems of hyperbolic balance laws in conservation-law form. All methods are characterised by a numerical flux. First, we introduce the finite volume framework, which accommodates both upwind and centred methods. We then present the Godunov method, along with several Riemann solvers for determining the numerical flux. The Riemann solvers explored include the exact solver, a two-rarefaction approximation, the HLL solver, the HLLC solver, the DOT solver, a TV flux vector splitting solver, and the simple Rusanov solver. Next, we discuss classical finite difference methods formulated in conservative form with a numerical flux. These methods include the Lax-Friedrichs method, the FORCE method, the centred Godunov method, and the Lax-Wendroff method. The numerical schemes are applied to the one-dimensional blood flow equations, augmented by an advection equation for a concentration variable. A systematic evaluation of the performance of these methods is conducted through a carefully selected suite of test problems for arteries and veins, with comparisons between the numerical solutions and exact solutions. Finally, suggested exercises are provided at the end of the chapter.

13.1 The Finite Volume Framework

In this section we first introduce the finite volume framework from the integral form of balance laws and derive general finite volume conservative methods. Then we apply the framework to construct the classical Godunov upwind method [1] for a general system of conservation laws. The section is ended with a formulation of the Godunov upwind method, as a finite volume method, for the scalar linear advection equation, with and without source term.

13.1.1 The Finite Volume Method from Integral Equations

Consider the initial-boundary value problem (IBVP) for a generic system of nonlinear hyperbolic balance laws

$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}) , \quad x \in [a, b] , \quad t > 0 , \\ \text{ICs: } \quad \mathbf{Q}(x, 0) = \mathbf{Q}^{(0)}(x) , \quad x \in [a, b] , \\ \text{BCs: } \quad \mathbf{Q}(a, t) = \mathbf{B}_L(t) , \quad \mathbf{Q}(b, t) = \mathbf{B}_R(t) , \quad t \geq 0 . \end{array} \right\} \quad (13.1)$$

$\mathbf{Q}(x, t)$ is the vector of conserved variables; $\mathbf{F}(\mathbf{Q})$ is the flux vector, or **physical flux**; $\mathbf{S}(\mathbf{Q})$ is the source term vector; $\mathbf{Q}^{(0)}(x)$ is the initial condition; $\mathbf{B}_L(t)$ and $\mathbf{B}_R(t)$ are the boundary conditions on the left and right boundaries, respectively, two prescribed functions of time. Assuming that the IBVP (13.1) has a solution, the task at hand is to construct numerical methods to solve the problem approximately.

The finite volume method is based on the integral form of the conservation laws. Exact integration of the PDEs in (13.1) in the **control volume**

$$V_i^n = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [t_n, t_{n+1}] \quad (13.2)$$

gives the following **exact** formula

$$\mathbf{Q}_i^{n+1} = \mathbf{Q}_i^n - \frac{\Delta t}{\Delta x} \left(\mathbf{F}_{i+\frac{1}{2}} - \mathbf{F}_{i-\frac{1}{2}} \right) + \Delta t \mathbf{S}_i , \quad (13.3)$$

with corresponding definitions for each item: $\mathbf{Q}_i^n$, $\mathbf{Q}_i^{n+1}$, $\mathbf{F}_{i+\frac{1}{2}}$, $\mathbf{F}_{i-\frac{1}{2}}$ and $\mathbf{S}_i$. See Chaps. 12 and 8 for background. In the formula (13.3) $\mathbf{Q}_i^n$ is the spatial integral average of the solution $\mathbf{Q}(x, t)$ in the control volume V_i^n at time t_n , namely

$$\mathbf{Q}_i^n = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{Q}(x, t_n) dx . \quad (13.4)$$

Likewise, $\mathbf{Q}_i^{n+1}$ is the spatial integral average of $\mathbf{Q}(x, t)$ at time t_{n+1} . $\mathbf{F}_{i+\frac{1}{2}}$ is the time-integral average of the flux vector $\mathbf{F}(\mathbf{Q}(x, t))$ at the cell interface $x_{i+\frac{1}{2}}$

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}, t)) dt . \quad (13.5)$$

Likewise, $\mathbf{F}_{i-\frac{1}{2}}$ is the time-integral average of the flux vector $\mathbf{F}(\mathbf{Q}(x, t))$ at the cell interface $x_{i-\frac{1}{2}}$. $\mathbf{S}_i$ is the space-time volume integral of the source term

$$\mathbf{S}_i = \frac{1}{\Delta t \Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \int_{t_n}^{t_{n+1}} \mathbf{S}(\mathbf{Q}(x, t)) dt dx . \quad (13.6)$$

Formula (13.3) is exact, with definitions (13.4)–(13.6). From now on we interpret Eq. (13.3) as an **approximate numerical formula** to update $\mathbf{Q}_i^n$ to $\mathbf{Q}_i^{n+1}$ in time. The update formula is completely determined once suitable **approximations** are provided for $\mathbf{F}_{i+\frac{1}{2}}$ in (13.5), called **numerical flux**, and for $\mathbf{S}_i$ in (13.6), called **numerical source**. $\mathbf{Q}_i^n$ in (13.4) requires numerical integration only once, at the initial time $t = 0$, when the integrand $\mathbf{Q}(x, t_n)$ is known from the initial conditions at the initial time. $\mathbf{Q}_i^{n+1}$ does not require calculation, as is given explicitly by formula (13.3) at every time step. The reader is encouraged to revise Chap. 12, where conservative methods were introduced.

13.1.2 The Godunov Numerical Flux from the Riemann Problem

Godunov's method [1] computes the **numerical flux** $\mathbf{F}_{i+\frac{1}{2}}$ from the solution of the local Riemann problem at the interface $x = x_{i+\frac{1}{2}}$ given as

$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0} , \\ \text{ICs: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{Q}_L = \mathbf{Q}_i^n & \text{if } x < 0 , \\ \mathbf{Q}_R = \mathbf{Q}_{i+1}^n & \text{if } x > 0 . \end{cases} \end{array} \right\} \quad (13.7)$$

We remark that for convenience, the IVP (13.7) in global coordinates centred at $(x_{i+\frac{1}{2}}, t_n)$ has been transformed to **local coordinates** $(\bar{x}, \bar{t})$ with origin at $(0, 0)$, by applying the coordinate transformation:

$$\left. \begin{array}{l} \bar{x} = x - x_{i+\frac{1}{2}} , \quad \bar{t} = t - t^n , \\ x \in [x_i, x_{i+1}] , \quad t \in [t^n, t^{n+1}] , \\ \bar{x} \in [-\frac{1}{2}\Delta x, \frac{1}{2}\Delta x] , \quad \bar{t} \in [0, \Delta t] . \end{array} \right\} \quad (13.8)$$

Then in practice we still keep the notation (x, t) to mean the local coordinates $(\bar{x}, \bar{t})$.

The similarity solution of (13.7) is denoted as $\mathbf{Q}_{i+\frac{1}{2}}(x/t)$. To compute the numerical flux one then evaluates the integral (13.5) at $x/t = 0$, that is along the interface (along the t -axis). $\mathbf{Q}_{i+\frac{1}{2}}(0)$ is called the **Godunov state**, which is constant along the interface, see Chap. 9. The Godunov flux is then trivially computed from (13.5) as

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_0^{\Delta t} \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}, t)) dt = \mathbf{F}(\mathbf{Q}_{i+\frac{1}{2}}(0)) . \quad (13.9)$$

Figure 13.1 represents the Godunov method as applied to a homogeneous hyperbolic system, $\mathbf{S} = \mathbf{0}$. Frame (a) represents integral averages in three successive cells at time t_n for a generic component q of vector $\mathbf{Q}$. Frame (b) shows the structure of solutions of two neighbouring Riemann problems at the intercell boundaries $x_{i-\frac{1}{2}}$ and

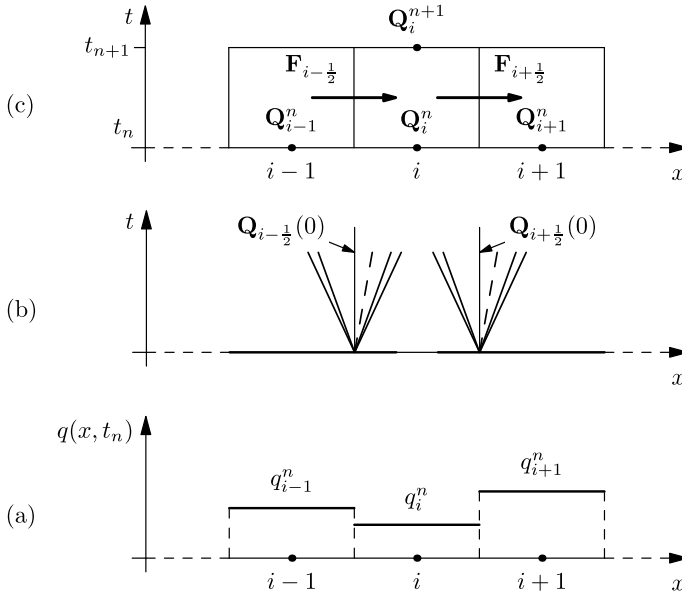


Fig. 13.1 Godunov's method for a homogeneous hyperbolic system. Frame **a**: integral averages give piece-wise constant data at time t_n ; plot represents a generic component q of $\mathbf{Q}$. Frame **b**: structure of solutions of Riemann problems at the intercell boundaries. Frame **c**: representation of finite volume formula (13.3) to update the cell average in cell I_i using intercell numerical fluxes

$x_{i+1/2}$. Frame (c) represents the finite volume formula (13.3) to update cell averages in cell I_i using intercell numerical fluxes $\mathbf{F}_{i-1/2}$ and $\mathbf{F}_{i+1/2}$.

13.1.3 Godunov's Method for the Linear Advection Equation

Here we illustrate in detail the formulation of the Godunov upwind, finite volume method to the linear advection equation (LAE).

Godunov's Method for LAE Without Source Term

We first consider the homogeneous case, $s = 0$, namely

$$\partial_t q(x, t) + \partial_x f(q(x, t)) = 0, \quad \text{flux function: } f(q) = \lambda q. \quad (13.10)$$

Here λ is a constant wave propagation speed. Recall that in Chap. 12 we introduced the Godunov method to solve Eq. (13.10), as a finite difference scheme. The finite

volume version of the Godunov method departs from (13.3), with definitions (13.4)–(13.6), which applied to (13.10) may be written as

$$q_i^{n+1} = q_i^n - \frac{\Delta t}{\Delta x} \left(f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}} \right). \quad (13.11)$$

To compute the numerical flux $f_{i+\frac{1}{2}}$ from (13.5) one solves the local Riemann problem

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \partial_x(\lambda q) = 0, \\ \text{ICs: } q(x, 0) = \begin{cases} q_i^n & \text{if } x < 0, \\ q_{i+1}^n & \text{if } x > 0 \end{cases} \end{array} \right\} \quad (13.12)$$

to find the integrand in (13.5). As seen in Chap. 7 the solution of this problem, in local coordinates, at the cell interface is

$$q(x/t) = \begin{cases} q_i^n & \text{if } \frac{x}{t} < \lambda, \\ q_{i+1}^n & \text{if } \frac{x}{t} > \lambda. \end{cases} \quad (13.13)$$

In order to compute the Godunov flux we need to determine the Godunov state $q_{i+\frac{1}{2}}(0)$. Generally, this requires a process called solution sampling to correctly identify the solution value along the t -axis. From (13.13) we see that for the linear advection equation the sought **Godunov state** $q_{i+\frac{1}{2}}(0)$ is simply given as

$$q_{i+\frac{1}{2}}(0) = \begin{cases} q_i^n & \text{if } \lambda > 0, \\ q_{i+1}^n & \text{if } \lambda < 0. \end{cases} \quad (13.14)$$

The **upwind** character of the Godunov method becomes evident; the sought Godunov state $q_{i+\frac{1}{2}}(0)$ depends on the **direction of the wind**. When $\lambda > 0$ the wind blows from the left and hence the Godunov state is given by the value q_i^n that has moved with the wind from the left. Otherwise the Godunov state is q_{i+1}^n . Therefore, from (13.9), the Godunov numerical flux $f_{i+\frac{1}{2}}$ is

$$f_{i+\frac{1}{2}} = \begin{cases} \lambda q_i^n & \text{if } \lambda > 0, \\ \lambda q_{i+1}^n & \text{if } \lambda < 0. \end{cases} \quad (13.15)$$

Similarly, for the left interface the corresponding numerical flux is

$$f_{i-\frac{1}{2}} = \begin{cases} \lambda q_{i-1}^n & \text{if } \lambda > 0, \\ \lambda q_i^n & \text{if } \lambda < 0. \end{cases} \quad (13.16)$$

After inserting the numerical fluxes (13.15)–(13.16) into the finite volume scheme (13.11) we obtain

$$q_i^{n+1} = \begin{cases} q_i^n - \frac{\Delta t}{\Delta x} (\lambda q_i^n - \lambda q_{i-1}^n) & \text{if } \lambda > 0, \\ q_i^n - \frac{\Delta t}{\Delta x} (\lambda q_{i+1}^n - \lambda q_i^n) & \text{if } \lambda < 0. \end{cases} \quad (13.17)$$

By introducing the Courant number $c = \frac{\lambda \Delta t}{\Delta x}$ in (13.17) we obtain

$$q_i^{n+1} = \begin{cases} q_i^n - c (\lambda q_i^n - \lambda q_{i-1}^n) & \text{if } \lambda > 0, \\ q_i^n - c (\lambda q_{i+1}^n - \lambda q_i^n) & \text{if } \lambda < 0, \end{cases} \quad (13.18)$$

which is identical to the finite difference form of the Godunov method presented in Chap. 12.

Godunov's Method for LAE with a Source Term

The linear advection equation with a linear source term reads

$$\partial_t q + \partial_x f(q) = s(q), \quad f(q) = \lambda q, \quad s(q) = \beta q, \quad (13.19)$$

with λ a constant wave propagation speed and $\beta \leq 0$ a constant reaction rate. Then we apply the finite volume formula (13.3) with the numerical source included. The **numerical source** results from an approximation to the integral average (13.6). The simplest approximation assumes that the integrand is given in terms of the initial data at time t^n , that is $s(q(x, t)) = \beta q_i^n$, a constant. Then, from (13.6) the numerical source is simply

$$s_i = \beta q_i^n. \quad (13.20)$$

If we assume that the numerical flux for the in-homogeneous Eq. (13.19) is the same as that for the homogeneous equation (13.10), then the **extended** Godunov scheme is

$$q_i^{n+1} = \begin{cases} q_i^n - c (\lambda q_i^n - \lambda q_{i-1}^n) + \Delta t \beta q_i^n & \text{if } \lambda > 0, \\ q_i^n - c (\lambda q_{i+1}^n - \lambda q_i^n) + \Delta t \beta q_i^n & \text{if } \lambda < 0. \end{cases} \quad (13.21)$$

Note that the source term is **centred**, it does not depend on the sign of the wave propagation speed λ .

We have used the term **extended** Godunov scheme, because source terms were not included in the original formulation of the Godunov method. The approximation to the numerical source proposed here is simply for illustrative purposes, it is not recommended for practical use. More sophisticated methods to evaluate the numerical source are required in practice.

Upwinding the Source Term

In order to evaluate the source term from an approximation to the integral (13.6) let us assume that the integrand is the solution of the two neighbouring Riemann problems in the control volume $V_i^n = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [t_n, t_{n+1}]$ introduced in (13.2). For $\lambda > 0$ only the left Riemann problem at $x_{i-\frac{1}{2}}$ affects the solution inside the integration volume V_i^n . For $\lambda < 0$ only the right Riemann problem at $x_{i+\frac{1}{2}}$ affects the solution inside the integration volume V_i^n .

Exercise 13.1 *The reader is encouraged to verify that the following steps lead to approximations to the source term that account for the influence of wave propagation in determining the numerical source:*

1. By assuming $|c| \leq 1$ and performing exact integration in space and time in integral (13.6) find the expressions for the numerical source in both cases: $\lambda > 0$ and $\lambda < 0$.
2. Write down the final form of the schemes and note that now the source term, and thus the scheme, responds to the sign of the velocity λ . That is, the source term has been **upwinded**.
3. Instead of exact integration in space and time in integral (13.6), apply the mid-point rule in time and exact integration in space. Verify that the result is identical to that of the previous cases.

Answer: upwinded numerical source

$$s_i = \begin{cases} \frac{1}{2}c\beta q_{i-1}^n + (1 - \frac{1}{2}c)\beta q_i^n & \text{if } \lambda > 0, \\ \frac{1}{2}c\beta q_i^n + (1 - \frac{1}{2}c)\beta q_{i+1}^n & \text{if } \lambda < 0. \end{cases} \quad (13.22)$$

Note that now the numerical source depends on the sign of the wave propagation speed λ . In other words, the source term has been **upwinded**.

Through this very simple exercise we see that solving hyperbolic balance laws in which source terms are present, is more challenging than solving homogeneous conservation laws. Great care is required in dealing with the source term, when present.

13.2 Godunov's Method for Blood Flow

As already explained, in order to apply Godunov's method, one requires a solver for computing the solution of the Riemann problem (13.7). For the equations of blood flow, an exact solver for the solution of the Riemann problem was given in Chap. 9 in full detail, for both arteries and veins.

13.2.1 The Equations and the Riemann Problem

The equations for blood flow in arteries, assuming constant geometrical and mechanical parameters, written in conservation-law form read

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}) , \quad (13.23)$$

where

$$\left. \begin{aligned} \mathbf{Q} &= \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \equiv \begin{bmatrix} A \\ Au \\ A\eta \end{bmatrix} , & \mathbf{F}(\mathbf{Q}) &= \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \equiv \begin{bmatrix} Au \\ Au^2 + \gamma A^{3/2} \\ A\eta u \end{bmatrix} , \\ \mathbf{S}(\mathbf{Q}) &= \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} . \end{aligned} \right\} \quad (13.24)$$

For the Coriolis coefficient we have assumed $\alpha = 1$ and viscosity $\mu = 0$; therefore the source term $s_2 = 0$ the momentum equation. See Chaps. 6, 10 and 9 for details.

To close the system we assume a tube law of the form $p := p(A)$. Specifically, for arterial vessels we adopt

$$p = p_{ext} + \mathbb{P}(A) , \quad (13.25)$$

with

$$\mathbb{P}(A) = K(x) \left[\left(\frac{A(x, t)}{A_0(x)} \right)^{1/2} - 1 \right] , \quad K(x) = \frac{\sqrt{\pi}}{(1 - \nu^2)} \frac{E(x)h_0(x)}{\sqrt{A_0(x)}} . \quad (13.26)$$

Here $\mathbb{P}(A)$ is called the **transmural pressure** and may depend on parameters of the problem. For constant K and Poisson's ratio set to $\nu = 1/2$, the constant coefficient γ in (13.24) is given as

$$\gamma = \frac{K}{3\rho\sqrt{A_0}} = \frac{4\sqrt{\pi}}{9} \frac{Eh_0}{\rho A_0} . \quad (13.27)$$

Recall that the eigenvalues of the 3×3 system (13.23) are

$$\lambda_1 = u - c , \quad \lambda_2 = u , \quad \lambda_3 = u + c , \quad (13.28)$$

where c is the **wave speed**, given as

$$c = \sqrt{\frac{A}{\rho} \frac{\partial p}{\partial A}} = \sqrt{\frac{3}{2} \gamma \sqrt{A}} . \quad (13.29)$$

In order to compute the Godunov numerical flux (13.9) one requires the Godunov state $\mathbf{Q}_{i+\frac{1}{2}}(0)$ along the t -axis from the solution of the Riemann problem. Identifying

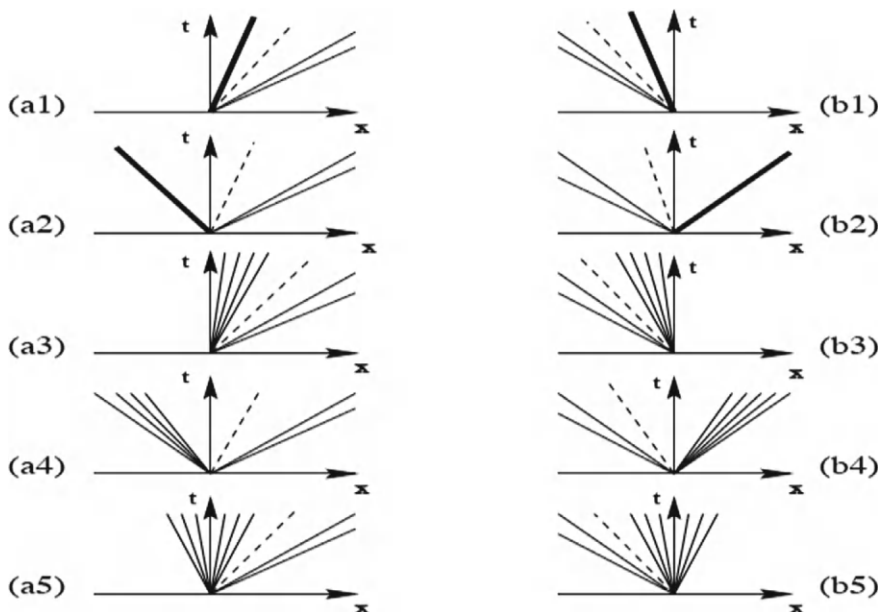


Fig. 13.2 Sampling the solution of the Riemann problem to find the Godunov state along the t -axis. The contact discontinuity $x/t = u_*$, shown by the dashed line, divides the half-plane into two regions. Case a: $u_* \geq 0$; all five possible wave patterns are shown on the left column. Case b: $u_* \leq 0$; all five possible wave patterns are shown on the right column

this particular state in the solution is a relatively complex procedure and depends on the wave pattern emerging from the solution. As shown in Fig. 13.2 there are 10 possibilities. The following two tasks to find $\mathbf{Q}_{i+\frac{1}{2}}(0)$ are required:

1. **The star state $\mathbf{Q}_*$** : computation of the state $\mathbf{Q}_*$ in the **Star Region**.
2. **The Godunov state $\mathbf{Q}_{i+\frac{1}{2}}(0)$** : an algorithm to correctly select the Godunov state $\mathbf{Q}_{i+\frac{1}{2}}(0)$ in the full wave pattern, for flux evaluation.

Only in some cases, not known *a-priori*, the Godunov state coincides with the star state.

13.2.2 Solution for the Star State

Here, in a succinct manner, we discuss methods to find the star state $\mathbf{Q}_*$ in the solution of the Riemann problem. We start from the equations derived in Chap. 9.

Equations for the Star State

From Chap. 9 we know that the star state in the **Star Region**, represented here by A_* and u_* , results from the following non-linear algebraic equation

$$f(A) \equiv f_L(A) + f_R(A) + \Delta u = 0, \quad \Delta u \equiv u_R - u_L. \quad (13.30)$$

The functions $f_L(A)$ and $f_R(A)$ for the left and right waves are respectively given as

$$f_L = \begin{cases} 4(c - c_L) & \text{if } A \leq A_L \text{ (rarefaction)}, \\ \sqrt{\frac{\gamma(A - A_L)(A^{3/2} - A_L^{3/2})}{A_L A}} & \text{if } A > A_L \text{ (shock)} \end{cases} \quad (13.31)$$

and

$$f_R = \begin{cases} 4(c - c_R) & \text{if } A \leq A_R \text{ (rarefaction)}, \\ \sqrt{\frac{\gamma(A - A_R)(A^{3/2} - A_R^{3/2})}{A_R A}} & \text{if } A > A_R \text{ (shock)}. \end{cases} \quad (13.32)$$

From here, there are various possibilities to find the start values A_* and u_* . In what follows we consider two of these.

Approximate Star State: Two-Rarefaction Solution

It is possible to obtain a direct, approximate, solution to the equation $f(A) = 0$ in (13.30) by assuming that both left and right waves are rarefaction waves in (13.31) and (13.32). Then, from (13.30) we obtain

$$4(c - c_L) + 4(c - c_R) + u_R - u_L = 0. \quad (13.33)$$

Direct solution of (13.33) for the wave speed in the Star Region gives

$$c_{TR}^* = \frac{1}{2}(c_L + c_R) - \frac{1}{8}(u_R - u_L). \quad (13.34)$$

From the definition of wave speed (13.29) and Eq. (13.34) we obtain the two-rarefaction approximate solution for the star state in terms of cross sectional area A and velocity u :

$$\left. \begin{aligned} A_{TR}^* &= \left[\frac{2}{3\gamma} \right]^2 (c_{TR}^*)^4, \\ u_{TR}^* &= \frac{1}{2}(u_L + u_R) - 2(c_R - c_L). \end{aligned} \right\} \quad (13.35)$$

Exact Star State: Iterative Solution

Another way to find A_* is to solve the algebraic non-linear Eq. (13.30) iteratively by, for example, the Newton-Raphson method

$$A^{(k+1)} = A^{(k)} - \frac{f(A^{(k)})}{f'(A^{(k)})}, \quad \text{for } k = 0, 1, \dots, K. \quad (13.36)$$

The iterative process (13.36) is stopped whenever the relative change ΔA in A is smaller than a prescribed, small positive tolerance TOL , that is when

$$\Delta A \equiv \frac{|A^{(k+1)} - A^{(k)}|}{(A^{(k+1)} + A^{(k)})/2} < TOL. \quad (13.37)$$

As a guess value $A^{(0)}$ in (13.36) we use the two-rarefaction approximation (13.35). For the purpose of flux evaluation, one typically chooses $TOL = 10^{-6}$ in single-precision calculations. In our experience $TOL = 10^{-3}$ is sufficient in practice. We note that it is advisable to check the number of iterations required for convergence. In our experience, in most of the computational domain just one or two iterations are needed. Larger numbers of iterations may indicate something wrong with the algorithm, or the initial guess $A^{(0)}$.

Once the cross-sectional area $A = A_*$ has been found from (13.36), the velocity u_* is computed as

$$u_* = \frac{1}{2}(u_L + u_R) + \frac{1}{2}[f_R(A_*) - f_L(A_*)]. \quad (13.38)$$

The next stage is to find the Godunov state $\mathbf{Q}_{i+\frac{1}{2}}(0)$ in order to evaluate the Godunov flux as in (13.9). To find this state we must perform a solution sampling procedure to identify the correct wave pattern out of 10 possibilities, as shown in Fig. 13.2. The sampling is analogous to that for the exact Riemann solver presented in Chap. 9.

13.2.3 Godunov Flux from Sampled Solution

So far we know the solution in the star region for A_* and u_* , either approximately or exactly. However, the sought Godunov state $\mathbf{Q}_{i+\frac{1}{2}}(0)$ for the evaluation of the Godunov flux may not coincide with the already known Star State. A solution-sampling procedure is needed to deal with the 10 possibilities depicted in Fig. 13.2.

To perform the sampling it is convenient to identify the Godunov state $\mathbf{Q}_{i+\frac{1}{2}}(0)$ along the t -axis in terms of the **vector of physical variables** denoted as

$$\mathbf{W}(0) \equiv [A(0), u(0)]^T, \quad (13.39)$$

where $A(0)$ is short notation for $A_{i+\frac{1}{2}}(0)$ and $u(0)$ is short notation for $u_{i+\frac{1}{2}}(0)$. The passive scalar η can be dealt with separately and for the moment is excluded from the vector $\mathbf{W}(0)$. There are two main cases, as depicted in Fig. 13.2. **Case a** is identified by the condition $u_* \geq 0$; all five corresponding wave patterns are shown on the left column ($a1, a2, \dots, a5$). **Case b** is identified by the condition $u_* \leq 0$; all five corresponding wave patterns are shown on the right column ($b1, b2, \dots, b5$). In what follows we describe the sampling in algorithmic form.

1. **Case a:** $u_* \geq 0$. The t -axis is to the left of the contact $x/t = u_*$. The solution $\mathbf{W}(0)$ depends on the left wave type. There are two possibilities:

- $A_* > A_L$: the left wave is a shock of speed S_L . There are two possibilities:

$$\mathbf{W}(0) \equiv \begin{cases} \mathbf{W}_L = [A_L, u_L]^T & \text{if } S_L \geq 0 \text{ (a1) ,} \\ \mathbf{W}_L = [A_*, u_*]^T & \text{if } S_L < 0 \text{ (a2) ,} \end{cases} \quad (13.40)$$

where the shock speed S_L is

$$S_L = u_L - \sqrt{\frac{\gamma A_* (A_*^{3/2} - A_L^{3/2})}{A_L (A_* - A_L)}}. \quad (13.41)$$

- $A_* \leq A_L$: the left wave is a rarefaction. There are three possibilities:

$$\mathbf{W}(0) = \begin{cases} \mathbf{W}_L = [A_L, u_L]^T & \text{if } u_L - u_* \geq 0 \quad (a3) , \\ \mathbf{W}_* = [A_*, u_*]^T & \text{if } u_* - c_* \leq 0 \quad (a4) , \\ \mathbf{W}_{Lfan} = [\hat{A}_L, \hat{u}_L]^T & \text{if } u_L - c_L \leq 0 \leq u_* - c_* \text{ (a5) .} \end{cases} \quad (13.42)$$

To compute $\mathbf{W}_{Lfan}$ (left sonic rarefaction) we invoke the exact solution from Chap. 9: for u and c inside the left rarefaction along the t -axis we have

$$\hat{u} = \frac{1}{5}(u_L + 4c_L), \quad \hat{c} = \frac{1}{5}(u_L + 4c_L). \quad (13.43)$$

$\hat{A}_L$ follows from wave speed $\hat{c}$ using definition (13.29).

2. **Case b:** $u_* \leq 0$. The t -axis lies to the right of the contact $x/t = u_*$. The solution $\mathbf{W}(0)$ depends on the right wave type. There are two possibilities:

- $A_* > A_R$: the right wave is a shock of speed S_R . There are two possibilities:

$$\mathbf{W}(0) \equiv \begin{cases} \mathbf{W}_R = [A_R, u_R]^T & \text{if } S_R < 0 \text{ (b1) ,} \\ \mathbf{W}_* = [A_*, u_*]^T & \text{if } S_R \geq 0 \text{ (b2) ,} \end{cases} \quad (13.44)$$

where the shock speed S_R is

$$S_R = u_R + \sqrt{\frac{\gamma A_* (A_*^{3/2} - A_R^{3/2})}{A_R (A_* - A_R)}}. \quad (13.45)$$

- $A_* \leq A_R$: Right wave is a rarefaction. There are three possibilities:

$$\mathbf{W}(0) = \begin{cases} \mathbf{W}_R = [A_R, u_R]^T & \text{if } u_R + c_R \leq 0 & (b3), \\ \mathbf{W}_* = [A_*, u_*]^T & \text{if } u_* + c_* \geq 0 & (b4), \\ \mathbf{W}_{Lfan} = [\hat{A}_R, \hat{u}_R]^T & \text{if } u_* + c_* \leq 0 \leq u_R + c_R & (b5). \end{cases} \quad (13.46)$$

The solutions for u and c inside the right sonic rarefaction along the t -axis are

$$\hat{u} = \frac{1}{5}(u_R - 4c_R), \quad \hat{c} = \frac{1}{5}(-u_R + 4c_R). \quad (13.47)$$

$\hat{A}_R$ follows from wave speed $\hat{c}$ using definition (13.29).

Passive scalar. The vector $\mathbf{W}(0) \equiv [A(0), u(0)]^T$ is known. To complete the solution we need the third component, the passive scalar $\eta(x, t)$. As seen in Chap. 9, this only changes across the contact discontinuity $x/t = u_*$. Therefore

$$\eta(0) = \begin{cases} \eta_L & \text{if } u_* \geq 0, \\ \eta_R & \text{if } u_* < 0. \end{cases} \quad (13.48)$$

Having found the physical variables $A(0)$, $u(0)$ and $\eta(0)$ right at the interface, we now compute the Godunov flux

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}(\mathbf{Q}_{i+\frac{1}{2}}(0)), \quad \mathbf{Q}_{i+\frac{1}{2}}(0) = [A(0), A(0)u(0), A(0)\eta(0)]^T. \quad (13.49)$$

Summarising, the original Godunov method computes the numerical flux by first finding the Godunov state $\mathbf{Q}_{i+\frac{1}{2}}(0)$ and then evaluating the flux function at this state. Before determining $\mathbf{Q}_{i+\frac{1}{2}}(0)$ we need to find the star state $\mathbf{Q}_*$, for which two ways have been presented, namely

1. via the two-rarefaction approximation (13.35), and
2. through the exact iterative solution via (13.36).

In both cases, there follows a sampling procedure to find the Godunov state $\mathbf{Q}_{i+\frac{1}{2}}(0)$ to be used for flux evaluation as in (13.9).

In the next sections we shall present approximate Godunov methods in which the Godunov flux is computed directly, where the computation of $\mathbf{Q}_*$ and $\mathbf{Q}_{i+\frac{1}{2}}(0)$ is bypassed and the solution-sampling process is considerably simplified.

13.3 HLL Two-State Approximate Riemann Solver

In this section we seek a direct approximation to $\mathbf{F}_{i+\frac{1}{2}}$ by considering an approximate solution to the Riemann problem (13.7) following the HLL approach, first proposed by Harten, Lax and van Leer [2].

13.3.1 The HLL State

The aim is to find, **directly**, the intercell numerical flux expressed in the form

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}_0 = \frac{1}{T} \int_0^T \mathbf{F}(\mathbf{Q}(0, t)) dt. \quad (13.50)$$

$T > 0$ is an arbitrary time and $\mathbf{Q}(0, t)$ denotes the solution of the Riemann problem at $x = 0$ along the t -axis. Since $\mathbf{Q}(0, t)$ is constant in time for piece-wise constant data in the Riemann problem, this is effectively the Godunov state. To determine $\mathbf{F}_{i+\frac{1}{2}}$ in (13.50) we first need to determine an HLL state that is analogous to the Godunov state.

We start by establishing some useful relations obtained from the integral form of the conservation laws in appropriately chosen control volumes. Given the initial conditions $\mathbf{Q}_L$ and $\mathbf{Q}_R$ for the Riemann problem (13.7), we assume that S_L and S_R are, respectively, estimates for the speeds of the fastest left and right travelling signals in the solution of the Riemann problem. The **wave speed estimates** S_L and S_R are an essential component of the HLL Riemann solver and are yet to be determined.

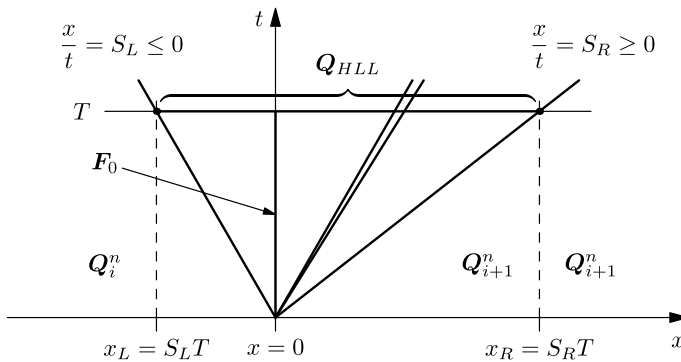


Fig. 13.3 Assumed wave configuration for deriving the HLL flux $\mathbf{F}_0$ for a subcritical, or subsonic, wave pattern: $S_L \leq 0 \leq S_R$. The wave speed estimates S_L and S_R are assumed to have been prescribed

Consider the wave configuration depicted in Fig. 13.3, in which the distances

$$x_L = TS_L, \quad x_R = TS_R \quad (13.51)$$

and time T permit the definition of various, conveniently chosen, control volumes in the x - t half plane. Application of the integral form of the conservation laws in (13.1) in the control volume $[x_L, 0] \times [0, T]$, gives

$$\int_{x_L}^0 \mathbf{Q}(x, T)dx = \int_{x_L}^0 \mathbf{Q}(x, 0)dx + \int_0^T \mathbf{F}(\mathbf{Q}(x_L, t))dt - \int_0^T \mathbf{F}(\mathbf{Q}(0, t))dt. \quad (13.52)$$

Evaluation of the first and second terms on the right hand side yields

$$\int_{x_L}^0 \mathbf{Q}(x, 0)dx = -S_L T \mathbf{Q}_L; \quad \int_0^T \mathbf{F}(\mathbf{Q}(x_L, t))dt = T \mathbf{F}(\mathbf{Q}_L). \quad (13.53)$$

Inserting these into (13.50) and dividing through by T gives

$$\mathbf{F}_0 = \frac{1}{T} \int_0^T \mathbf{F}(\mathbf{Q}(0, t))dt = -S_L \mathbf{Q}_L + \mathbf{F}(\mathbf{Q}_L) - \frac{1}{T} \int_{x_L}^0 \mathbf{Q}(x, T)dx. \quad (13.54)$$

To define $\mathbf{F}_0$ approximately it is sufficient to find an approximation to the integral on the right hand side of (13.54). This is accomplished by finding a suitable state $\mathbf{Q}(x, T)$, as seen below.

By applying the integral form of the conservation laws in the control volume $[x_L, x_R] \times [0, T]$, see Fig. 13.3, we obtain

$$\int_{x_L}^{x_R} \mathbf{Q}(x, T)dx = \int_{x_L}^{x_R} \mathbf{Q}(x, 0)dx + \int_0^T \mathbf{F}(\mathbf{Q}(x_L, t))dt - \int_0^T \mathbf{F}(\mathbf{Q}(x_R, t))dt. \quad (13.55)$$

Evaluation of the first term on the right-hand side gives

$$\int_{x_L}^{x_R} \mathbf{Q}(x, 0)dx = -S_L T \mathbf{Q}_L + S_R T \mathbf{Q}_R. \quad (13.56)$$

Substitution of this expression into (13.55) and evaluating the remaining integrals yield

$$\int_{x_L}^{x_R} \mathbf{Q}(x, T)dx = T[S_R \mathbf{Q}_R - S_L \mathbf{Q}_L + \mathbf{F}(\mathbf{Q}_L) - \mathbf{F}(\mathbf{Q}_R)]. \quad (13.57)$$

On division through by $x_R - x_L = T(S_R - S_L)$ we obtain the averaged state

$$\left. \begin{aligned} \mathbf{Q}^{HLL} &= \frac{1}{(x_R - x_L)} \int_{x_L}^{x_R} \mathbf{Q}(x, T) dx \\ &= \frac{S_R \mathbf{Q}_R - S_L \mathbf{Q}_L + \mathbf{F}(\mathbf{Q}_L) - \mathbf{F}(\mathbf{Q}_R)}{S_R - S_L} . \end{aligned} \right\} \quad (13.58)$$

This is the sought single HLL state.

13.3.2 The HLL Flux

We now use the state $\mathbf{Q}^{HLL}$ given by (13.58), in place of $\mathbf{Q}(x, T)$ to evaluate the integral on the right-hand side in (13.54). The resulting intercell flux is

$$\mathbf{F}_0 = \frac{S_R \mathbf{F}(\mathbf{Q}_L) - S_L \mathbf{F}(\mathbf{Q}_R) + S_L S_R (\mathbf{Q}_R - \mathbf{Q}_L)}{S_R - S_L} . \quad (13.59)$$

This is the sought numerical flux if the wave pattern is sub-critical, or subsonic, see Fig. 13.3. In general, sampling of the wave patterns to correctly identify the sought numerical flux is needed. For the HLL approach this is indeed a very simple process and the resulting HLL flux for the approximate Godunov method is

$$\mathbf{F}_{i+\frac{1}{2}}^{HLL} = \begin{cases} \mathbf{F}_L & \text{if } 0 \leq S_L , \\ \frac{S_R \mathbf{F}(\mathbf{Q}_L) - S_L \mathbf{F}(\mathbf{Q}_R) + S_L S_R (\mathbf{Q}_R - \mathbf{Q}_L)}{S_R - S_L} , & \text{if } S_L \leq 0 \leq S_R , \\ \mathbf{F}_R & \text{if } 0 \geq S_R . \end{cases} \quad (13.60)$$

The HLL flux admits only three possibilities in the sampling procedure. In the exact solver there are ten possibilities. See Fig. 13.2.

Remarks on the HLL Riemann Solver

1. **Wave speed estimates.** To apply the HLL scheme it is necessary to find estimates for S_L and S_R . This is a non-trivial undertaking, as knowledge of the exact solution of the Riemann problem is needed to perform this task correctly. There exist several methods in the literature to estimate S_L and S_R . In Sect. 13.5 we will address this issue in detail.
2. **Missing intermediate waves.** The HLL approach takes an integral average of all the states lying between the slowest and fastest waves in the wave system of the Riemann problem solution. If the system consists of just two waves, then the

Two-Wave Model adopted by HLL is correct. However, if the system has more than two equations, then all intermediate waves are averaged, which results in excessive numerical diffusion for intermediate waves.

3. **Completeness of a Riemann solver.** It is appropriate to recall the definition of **complete Riemann solver**, being that in which the wave model of the solver contains all the characteristic fields of the governing equations. Therefore, HLL is **complete** for a system of two equations, but **incomplete** for a larger system.
4. It is possible to **restore missing intermediate waves**, as first suggested by Toro et al. [3] for the Euler equations, through the HLLC solver. For selected systems HLL can be **completed** by following the **HLLC approach**. See also [4].
5. **HLL strong points.** We first note that HLL is completely general, no matter how many waves the exact Riemann problem has. The specificity of the system comes in when having to estimate the extreme wave speeds S_L and S_R . Also to be noted, HLL is a **non-linear** Riemann solver and is **entropy satisfying**, unlike all linearised Riemann solvers.

Next we study an extension of the HLL scheme.

13.4 HLLC Two-State Approximate Riemann Solver

The HLLC approximate Riemann solver [4, 5] is a modification of the basic HLL scheme to account for the presence of intermediate waves in the structure of the solution of the Riemann problem, for systems of more than two equations in (13.1).

13.4.1 The HLLC Two States

Consider the wave pattern depicted in Fig. 13.4, where an intermediate wave of speed S_* is present. Such speed determines two star states, $\mathbf{Q}_{*L}$ and $\mathbf{Q}_{*R}$. Application of the integral form of the conservation laws in suitably chosen control volumes in Fig. 13.3 contained in $[x_L, 0] \times [0, T]$ and in $[0, x_R] \times [0, T]$ yield

$$\mathbf{F}_{*L} = \mathbf{F}_L + S_L(\mathbf{Q}_{*L} - \mathbf{Q}_L), \quad \mathbf{F}_{*R} = \mathbf{F}_R + S_R(\mathbf{Q}_{*R} - \mathbf{Q}_R). \quad (13.61)$$

Here there are four unknown vectors, namely $\mathbf{Q}_{*L}$, $\mathbf{Q}_{*R}$, $\mathbf{F}_{*L}$ and $\mathbf{F}_{*R}$, that is more unknowns than equations. The problem is to find the correct intercell numerical flux.

To solve this overdetermined problem we impose the following two properties, which are satisfied by the exact solution of the Riemann problem for blood flow, namely

$$A_{*L} = A_{*R} = A_*, \quad u_{*L} = u_{*R} = u_* = S_*. \quad (13.62)$$

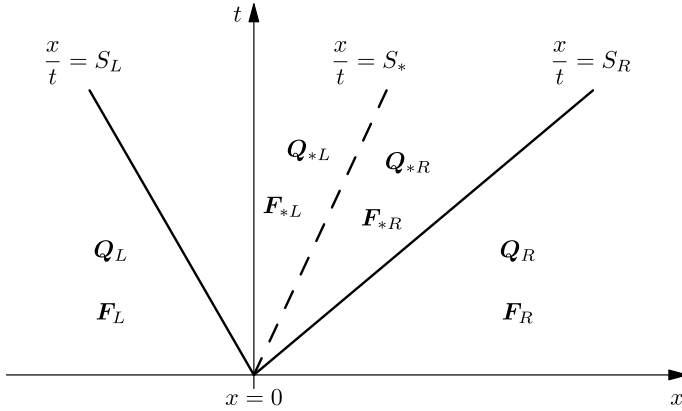


Fig. 13.4 Assumed three-wave wave pattern for the HLLC Riemann solver. Two star states are admitted

For the moment we make no assumption as to the intermediate wave speed u_* . From the first component of the first vector equation in (13.61) we write

$$A_* u_* = A_L u_L + S_L (A_* - A_L) . \quad (13.63)$$

From the first component of the second vector equation in (13.61) we write

$$A_* u_* = A_R u_R + S_R (A_* - A_R) . \quad (13.64)$$

From (13.63) and (13.64) we write

$$A_* = \frac{A_R(u_R - S_R)}{u_* - S_R} = \frac{A_L(u_L - S_L)}{u_* - S_L} . \quad (13.65)$$

From here we obtain the intermediate wave speed S_* purely in terms of S_L and S_R , namely

$$S_* \equiv u_* = \frac{S_L A_R (u_R - S_R) - S_R A_L (u_L - S_L)}{A_R (u_R - S_R) - A_L (u_L - S_L)} . \quad (13.66)$$

Therefore, if S_L and S_R are prescribed, then u_* follows from (13.66) and A_* follows from (13.65). Then, recalling that the passive scalar η is constant across the non-linear waves in the exact problem, the two vectors $\mathbf{Q}_{*L}$ and $\mathbf{Q}_{*R}$ in (13.61) are given as

$$\mathbf{Q}_{*L} = A_* \begin{bmatrix} 1 \\ S_* \\ \psi_L \end{bmatrix} , \quad \mathbf{Q}_{*R} = A_* \begin{bmatrix} 1 \\ S_* \\ \psi_R \end{bmatrix} . \quad (13.67)$$

Next we determine the intercell numerical flux.

13.4.2 The HLLC Flux

Having determined two **Star States** $\mathbf{Q}_{*L}$ and $\mathbf{Q}_{*R}$ we can now compute two **Star Fluxes** $\mathbf{F}_{*L}$ and $\mathbf{F}_{*R}$ from (13.61). Therefore the sought HLLC numerical flux is given as

$$\mathbf{F}_{i+\frac{1}{2}}^{HLLC} = \begin{cases} \mathbf{F}_L & \text{if } 0 \leq S_L, \\ \mathbf{F}_{*L} = \mathbf{F}_L + S_L(\mathbf{Q}_{*L} - \mathbf{Q}_L) & \text{if } S_L \leq 0 \leq S_*, \\ \mathbf{F}_{*R} = \mathbf{F}_R + S_R(\mathbf{Q}_{*R} - \mathbf{Q}_R) & \text{if } S_* \leq 0 \leq S_R, \\ \mathbf{F}_R & \text{if } 0 \geq S_R. \end{cases} \quad (13.68)$$

The HLLC flux admits four possibilities in the sampling procedure, as compared to three in the HLL scheme and 10 in the exact solver. Compare Figs. 13.2 with 13.4 and 13.3.

The use of HLLC instead of HLL for a system including the passive scalar η makes a significant difference to the resolution of the contact wave. This is particularly evident for long-time evolution. In Sect. 13.9 we compare numerical results from HLL, HLLC and the exact Riemann solver.

There is a pending problem, namely to find reliable estimates for the wave speeds S_L and S_R , noting that S_* is already given in (13.66) in terms of S_L and S_R .

13.5 Wave Speed Estimates for HLL and HLLC

The implementation of HLL and HLLC requires wave speed estimates S_L and S_R . Theoretically, it is expected that both S_L and S_R would be bounds, respectively, for the smallest and largest speeds occurring in the Riemann problem solution [2], as suggested in Fig. 13.5. However, this is difficult to guarantee in advance and most proposed estimates in current use do not satisfy this property. This issue is addressed quite comprehensively in a recent work by Toro, Siviglia and Müller [6]. In this section we shall consider two approaches to provide wave speed estimates.

13.5.1 Davis' Wave Speed Estimates

Davis [7] is credited for having provided two very simple wave speed estimates for HLL, as applied to the Euler equations. These estimates rely on evaluation of the eigenvalues $\lambda_1 = u - c$ and $\lambda_3 = u + c$ on data states, for both outer eigenvalues. For the blood flow equations, the first simplest estimates are given as

$$S_L^{Dav_a} = u_L - c_L, \quad S_R^{Dav_a} = u_R + c_R, \quad (13.69)$$

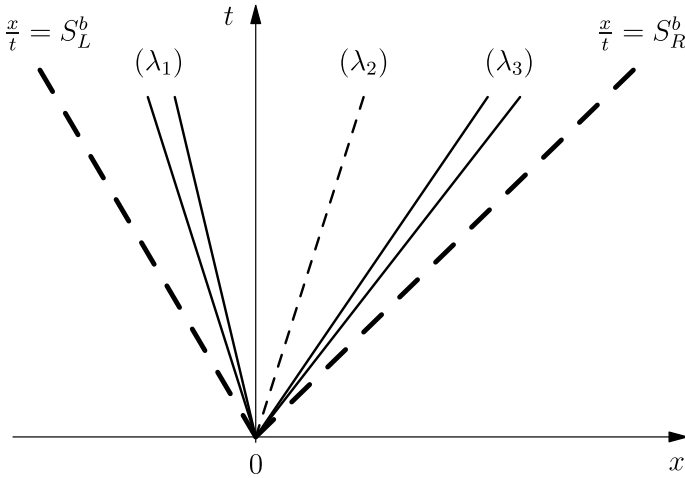


Fig. 13.5 Bounds S_L^b and S_R^b for the minimal and maximal wave speeds emerging from the solution of the Riemann problem (13.7) for general initial conditions $\mathbf{Q}_i^n$ and $\mathbf{Q}_{i+1}^n$

where c is the **wave speed**

$$c = c(A) = \sqrt{\frac{3}{2}\gamma A^{\frac{1}{2}}} . \quad (13.70)$$

We note that in the special case in which both waves are rarefaction waves, these estimates are exact. However, for the general case they are a poor approximation to the true speeds and are far from constituting bounds for the exact wave speeds. See Toro et al. [6] for details. Davis' first estimates (13.69) **are not recommended** for practical use. Davis' second estimates [7] are

$$S_L^{Dav_b} = \min\{u_L - c_L, u_R - c_R\} , \quad S_R^{Dav_b} = \max\{u_L + c_L, u_R + c_R\} . \quad (13.71)$$

Note that the special case of two rarefactions is also recognised by estimates (13.71). In addition, Davis' second estimates also recognise **isolated** shocks. Practical experience suggests that these estimates can be used in practice, but with great care as, again, they fail to bound the true wave speeds in general, as demonstrated in [6].

13.5.2 Toro's Wave Speed Estimates

Wave speed estimates inspired in the exact solution of the Riemann problem were first proposed by Toro and collaborators for the Euler equations in 1992 [8]; see [3, 5] for later developments. For the blood flow equations analogous estimates were

proposed in the HLLC review [4]. In [6] it is shown theoretically that such estimates constitute bounds for the true wave speeds in all circumstances.

From the exact solution of the Riemann problem for blood flow in arteries, see Chap. 9, the exact speed of a **left shock wave** is given as

$$S_L = u_L - c_L q_L, \quad q_L = \sqrt{\frac{2}{3} \frac{(y^{3/2} - 1)y}{(y - 1)}}, \quad y = \frac{A_*}{A_L}. \quad (13.72)$$

For a **right shock wave** we have

$$S_R = u_R + c_R q_R, \quad q_R = \sqrt{\frac{2}{3} \frac{(y^{3/2} - 1)y}{(y - 1)}}, \quad y = \frac{A_*}{A_R}. \quad (13.73)$$

Here A_* is the exact solution in the **Star Region** for the cross-sectional area in the Riemann problem for blood flow in arteries, which is obtained by solving a non-linear algebraic equation; see Chap. 9. If both waves are rarefaction waves then the exact speeds are

$$S_L = u_L - c_L, \quad S_R = u_R + c_R. \quad (13.74)$$

As seen in Sect. 13.2.2, Eqs. (13.33)–(13.35), in the two-rarefaction case we can find the solution in the **Star Region** in closed form, namely

$$A_{*TR} = \left(\frac{2\rho}{\beta} \right)^2 \left[\frac{1}{2}(c_L + c_R) - \frac{1}{8}(u_R - u_L) \right]^4. \quad (13.75)$$

As A_{*TR} approximates A_* from above [6], that is $A_{*TR} \geq A_*$, this turns out to be a desirable property for deriving direct wave speed estimates.

Toro's estimates are then given as follows. For the left wave

$$\left. \begin{aligned} S_L &= u_L - c_L q_L, \\ q_L &= \begin{cases} 1 & \text{if } y \leq 1 \text{ (Rarefaction)}, \\ \sqrt{\frac{2}{3} \frac{(y^{3/2} - 1)y}{(y - 1)}}, \quad y = \frac{A_{*TR}}{A_L} & \text{if } y > 1 \text{ (Shock)}. \end{cases} \end{aligned} \right\} \quad (13.76)$$

For the right wave

$$\left. \begin{aligned} S_R &= u_R + c_R q_R, \\ q_R &= \begin{cases} 1 & \text{if } y \leq 1 \text{ (Rarefaction)}, \\ \sqrt{\frac{2}{3} \frac{(y^{3/2} - 1)y}{(y - 1)}}, \quad y = \frac{A_{*TR}}{A_R} & \text{if } y > 1 \text{ (Shock)}. \end{cases} \end{aligned} \right\} \quad (13.77)$$

Having found estimates S_L and S_R for the left and right waves, the intermediate wave speed S_* follows from (13.66), reproduced here

$$S_* = \frac{S_L A_R(u_R - S_R) - S_R A_L(u_L - S_L)}{A_R(u_R - S_R) - A_L(u_L - S_L)}. \quad (13.78)$$

Therefore, wave speed estimates for both HLL and HLLC are available for blood flow in arteries. Moreover, these estimates constitute bounds for the true wave speeds.

Remarks on wave speeds and the Courant condition.

1. **The Courant stability condition.** As seen in Chap. 12, the time step Δt in formula (13.3) is chosen according to

$$\Delta t = C_{cfl} \frac{\Delta x}{S_{max}^n}, \quad C_{cfl} = SF \times C_{lim}, \quad 0 < SF \leq 1. \quad (13.79)$$

Here C_{lim} is the linear stability limit of the scheme, derived from a linear stability analysis. The parameter SF is a **safety factor** chosen by the user in a rather arbitrary manner, in practice; the closer to unity, the better, the larger the time step, resulting in computing efficiency. However, if the estimate for S_{max}^n in (13.79) is poor and subject to uncertainties, the scheme may become unstable, leading to crashing of the code. In practice one takes $SF = 0.9$, $SF = 0.5$ or even $SF = 0.2$, in desperation!

2. **Wave speed estimates and time step.** The wave speed estimates S_L and S_R in (13.76) and (13.77) can also be used to estimate S_{max}^n in the full domain, reliably. In this case one can use SF very close to 1.0.
3. **Computational expense.** Estimating S_L and S_R according to (13.76)–(13.77) and using these to compute S_{max}^n is more expensive than seeking the maximal eigenvalue evaluated on the initial data at time level n , which is the usual practice, see (13.69) and (13.71). However, by using $SF = 1.0$ instead of $SF = 0.9$ or $SF = 0.2$ in (13.79) one comfortably compensates for the extra expense. More importantly, full reliability of the computations is assured by applying the known theory.

We next summarise, in algorithmic form, the presentation of the HLL and HLLC Riemann solvers.

13.5.3 Algorithm for HLL and HLLC for Blood Flow

To implement the HLL and HLLC solvers to solve the augmented 1D blood flow equations for arteries, the following sequence of steps are performed:

For the HLL flux:

1. Compute A_{*TR} from (13.75).
2. Compute S_L and S_R from (13.76)–(13.77)
3. Compute the HLL flux according to (13.60).

For the HLLC flux:

4. Compute A_{*TR} from (13.75).
5. Compute S_L and S_R from (13.76)–(13.77) and S_* from (13.78).
6. Compute the star states $\mathbf{Q}_{*L}$ and $\mathbf{Q}_{*R}$ as in (13.67)
7. Compute $\mathbf{F}_{*L}$ or $\mathbf{F}_{*R}$ from (13.68), according to the wave pattern; the HLLC flux is thus determined.

The HLL and HLLC numerical fluxes for the augmented equations for blood flow in arteries have been determined.

13.6 The Dumbser-Osher-Toro Riemann Solver: DOT

Here we present a very general approximate Riemann solver that is applicable to any hyperbolic system. The scheme was proposed by Dumbser and Toro [9, 10] and is a modification of the original Osher-Solomon Riemann solver [11]. We first briefly review the original scheme and then present the modified version.

13.6.1 The Osher-Solomon Riemann Solver: Brief Review

The Osher-Solomon approximate Riemann solver [11] has some attractive features, but its complexity has prevented widespread application of the scheme to hyperbolic systems, generally. Here we briefly review the original method for a general $m \times m$ system of hyperbolic conservation laws

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0} , \quad (13.80)$$

with the vectors of conserved variables and fluxes respectively denoted as

$$\mathbf{Q} = [q_1, q_2, \dots, q_m]^T , \quad \mathbf{F} = [f_1, f_2, \dots, f_m]^T . \quad (13.81)$$

The real eigenvalues, written in increasing order, are denoted by $\lambda_i(\mathbf{Q})$ and the corresponding right eigenvectors by $\mathbf{R}_i(\mathbf{Q})$, for $i = 1, 2, \dots, m$. Here we consider Godunov-type schemes of the form (13.3) to solve (13.1), or (13.80). The numerical flux $\mathbf{F}_{i+\frac{1}{2}}$ is found by solving the local Riemann problem for (13.80) with initial condition

$$\mathbf{Q}(x, 0) = \begin{cases} \mathbf{Q}_0 & \text{if } x < 0 , \\ \mathbf{Q}_1 & \text{if } x > 0 . \end{cases} \quad (13.82)$$

Here we have adopted the Osher-Solomon notation for the initial conditions $\mathbf{Q}_0$ and $\mathbf{Q}_1$, instead of $\mathbf{Q}_L$ and $\mathbf{Q}_R$, respectively. Recall that **hyperbolicity** of system (13.80) is equivalent to saying that the Jacobian matrix $\mathbf{A}(\mathbf{Q})$ of the flux $\mathbf{F}(\mathbf{Q})$ is **diagonalizable**, that is

$$\mathbf{A}(\mathbf{Q}) = \mathbf{R}(\mathbf{Q})\Lambda(\mathbf{Q})\mathbf{R}^{-1}(\mathbf{Q}) , \quad (13.83)$$

where $\mathbf{R}(\mathbf{Q})$ is the matrix formed by the right eigenvectors $\mathbf{R}_i(\mathbf{Q})$, $\mathbf{R}^{-1}(\mathbf{Q})$ is its inverse and $\Lambda(\mathbf{Q})$ is the diagonal matrix whose diagonal entries are the eigenvalues $\lambda_i(\mathbf{Q})$. For background see Chap. 8.

We introduce the following definitions

$$\left. \begin{aligned} \lambda_i^+(\mathbf{Q}) &= \max(\lambda_i(\mathbf{Q}), 0) , \\ \lambda_i^-(\mathbf{Q}) &= \min(\lambda_i(\mathbf{Q}), 0) , \\ |\lambda_i(\mathbf{Q})| &= \lambda_i^+(\mathbf{Q}) - \lambda_i^-(\mathbf{Q}) . \end{aligned} \right\} \quad (13.84)$$

Then we consider the associated diagonal matrices $\Lambda^+(\mathbf{Q})$, $\Lambda^-(\mathbf{Q})$ and $|\Lambda^-(\mathbf{Q})|$, whose diagonal entries are respectively $\lambda_i^+(\mathbf{Q})$, $\lambda_i^-(\mathbf{Q})$ and $|\lambda_i(\mathbf{Q})|$. It follows that

$$|\Lambda(\mathbf{Q})| = \Lambda^+(\mathbf{Q}) - \Lambda^-(\mathbf{Q}) . \quad (13.85)$$

Then we introduce the **absolute value matrix** as

$$|\mathbf{A}(\mathbf{Q})| = \mathbf{R}(\mathbf{Q})|\Lambda(\mathbf{Q})|\mathbf{R}^{-1}(\mathbf{Q}) \quad (13.86)$$

to define the Osher-Solomon numerical flux [11] as

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{2} [\mathbf{F}(\mathbf{Q}_0) + \mathbf{F}(\mathbf{Q}_1)] - \frac{1}{2} \int_{\mathbf{Q}_0}^{\mathbf{Q}_1} |\mathbf{A}(\mathbf{Q})| d\mathbf{Q} , \quad (13.87)$$

with $|\mathbf{A}(\mathbf{Q})|$ as in (13.86).

The evaluation of the integral (13.87) in phase space depends on an integration path joining $\mathbf{Q}_0$ to $\mathbf{Q}_1$. Originally, Osher and Solomon proposed two ways of choosing integration paths for the one-dimensional Euler equations for ideal gases. The two choices were proposed **so as to make the actual integration tractable**, and were called the **P-ordering** and the **O-ordering**.

However, the analytical calculations are still too involved to be performed for general hyperbolic systems, a reason for which the Osher-Solomon Riemann solver has remained restricted to just a few examples. Full details on the original Osher-Solomon Riemann solver are found in Chap. 12 of [12].

13.6.2 The DOT Reformulation

Dumbser and Toro [9, 10] proposed an effective modification in order to simplify the Osher-Solomon solver and make it applicable to more general hyperbolic systems. To start with, regarding the computational tractability of the scheme, the choice of **integration path** is no longer a constraint. This is because the challenging integration in the diffusion part of the numerical flux (13.87) in phase space is now performed **numerically**.

A commonly used integration path to evaluate the integral in (13.87) is the **canonical path**

$$\psi(s; \mathbf{Q}_0, \mathbf{Q}_1) = \mathbf{Q}_0 + s(\mathbf{Q}_1 - \mathbf{Q}_0), \quad s \in [0, 1]. \quad (13.88)$$

Other choices may also be considered. Then, under a change of variables (13.87) becomes

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{2} [\mathbf{F}(\mathbf{Q}_0) + \mathbf{F}(\mathbf{Q}_1)] - \frac{1}{2} \left(\int_0^1 |\mathbf{A}(\psi(s; \mathbf{Q}_0, \mathbf{Q}_1))| ds \right) (\mathbf{Q}_1 - \mathbf{Q}_0). \quad (13.89)$$

Now the integral in (13.89) is computed **numerically** along the path ψ using, for instance, a Gauss-Legendre quadrature rule with G points s_j and associated weights ω_j in the unit interval $I = [0, 1]$. We obtain

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{2} [\mathbf{F}(\mathbf{Q}_0) + \mathbf{F}(\mathbf{Q}_1)] - \frac{1}{2} \left(\sum_{j=1}^G \omega_j |\mathbf{A}(\psi(s_j; \mathbf{Q}_0, \mathbf{Q}_1))| \right) (\mathbf{Q}_1 - \mathbf{Q}_0). \quad (13.90)$$

Note that the numerical evaluation of $|\mathbf{A}(\psi(s_j; \mathbf{Q}_0, \mathbf{Q}_1))|$ requires, for each s_j , a matrix characteristic decomposition as in (13.86). To perform this task, the complete eigenstructure of the system is needed, that is eigenvalues and right eigenvectors. This is a straightforward but computationally demanding task. If the eigenstructure of the system is not available in analytical form, then one can compute it numerically, which will obviously add to the computational cost of the scheme but will also contribute to widen its range of applicability.

For the numerical Gaussian-Legendre integration in (13.90), it is found that a three-point quadrature rule is sufficiently accurate and robust, with points s_j and weights ω_j given as

$$s_1 = \frac{1}{2} - \frac{\sqrt{15}}{10}, s_2 = \frac{1}{2}, s_3 = \frac{1}{2} + \frac{\sqrt{15}}{10}, \omega_1 = \frac{5}{18}, \omega_2 = \frac{8}{18}, \omega_3 = \frac{5}{18}. \quad (13.91)$$

See [9] and [10] for details. The resulting DOT scheme is **non-linear**; it is also **complete**, in the sense that contains all the characteristic fields of the exact problem.

Finally, we note that the DOT scheme, as the original Osher-Solomon scheme, is **entropy satisfying**, but unlike the original scheme the reformulated one is applicable to any hyperbolic system (13.80). Moreover, the new Osher-type scheme is also applicable to systems in **non-conservative form**, as presented in Chap. 14.

As seen in (13.90) and (13.86), in order to apply the DOT scheme to the blood flow equations one needs the eigenvalues, the right eigenvectors, the matrix of right eigenvectors and its inverse. The DOT scheme has successfully been implemented in the global closed loop mathematical model for the human circulation in [13] and [14]; see also [15].

In the next section we present some upwind methods belonging to the class of flux vector splitting methods.

13.7 TV-Type Splitting Schemes

In this section we adopt the flux vector splitting approach of Toro and Vázquez [16], the TV splitting, originally proposed for the Euler equations of gas dynamics. For the blood flow equations of interest here we introduce a modification to the original TV scheme, whereby the continuity equation is fully accounted for by the pressure system, as will be explained shortly.

13.7.1 Advection-Pressure Splitting

We depart from the 1D augmented equations for blood flow in arteries, written in conservation-law form (13.23)–(13.24). The TV splitting has two parts.

1. First, we **split** the flux vector as

$$\mathbf{F}(\mathbf{Q}) = \mathcal{A}(\mathbf{Q}) + \mathcal{P}(\mathbf{Q}) , \quad (13.92)$$

with

$$\mathcal{A}(\mathbf{Q}) = \begin{bmatrix} 0 \\ \alpha A u^2 \\ A \eta u \end{bmatrix} , \quad \mathcal{P}(\mathbf{Q}) = \begin{bmatrix} A u \\ \gamma A^{3/2} \\ 0 \end{bmatrix} . \quad (13.93)$$

$\mathcal{A}(\mathbf{Q})$ is called the **advection flux** and $\mathcal{P}(\mathbf{Q})$ is called the **pressure flux**.

2. Second, we make use of the two fluxes $\mathcal{A}(\mathbf{Q})$ and $\mathcal{P}(\mathbf{Q})$ to formulate two distinct systems of PDEs, namely

$$\left. \begin{array}{l} \text{Advection system: } \partial_t \mathbf{Q} + \partial_x \mathcal{A}(\mathbf{Q}) = \mathbf{0} , \\ \text{Pressure system: } \partial_t \mathbf{Q} + \partial_x \mathcal{P}(\mathbf{Q}) = \mathbf{0} . \end{array} \right\} \quad (13.94)$$

The aim of the second step is to exploit each sub-system so as to obtain corresponding numerical fluxes $\mathcal{A}_{i+\frac{1}{2}}$, the **advection numerical flux**, and $\mathcal{P}_{i+\frac{1}{2}}$, the **pressure numerical flux**. The complete numerical flux $\mathbf{F}_{i+\frac{1}{2}}$ for the full system will be given as

$$\mathbf{F}_{i+\frac{1}{2}} = \mathcal{A}_{i+\frac{1}{2}} + \mathcal{P}_{i+\frac{1}{2}}, \quad (13.95)$$

which will then be used in the conservative scheme (13.3) to advance the solution in time, **in one step**.

Remark: We note that the above flux splitting (13.92)–(13.93) differs from the original TV splitting [16]. Here, we assign the continuity equation fully to pressure system in (13.94). The present version is consistent with approximations to the full equations in which the inertial term is neglected for sufficiently small Froude numbers, leading to an approximation that coincides with the pressure system in (13.94). It is also consistent with zero-dimensional approximations to the blood flow equations, in which one effectively replaces the pressure system of PDEs in (13.94) by a system of ordinary differential equations (ODEs) in time, by integrating the system in space, in a control volume; see Chap. 6.

The numerical strategy to determine the advection and pressure numerical fluxes in (13.95) relies on first solving the Riemann problem for the pressure system in (13.94). The solution of this system will fully determine the pressure numerical flux $\mathcal{P}_{i+\frac{1}{2}}$ and will also provide advection information for determining the advection numerical flux $\mathcal{A}_{i+\frac{1}{2}}$ in (13.95). In other words, we only need to solve the Riemann problem for the pressure system in (13.94). This is carried out in the next section.

13.7.2 The Equations for the Pressure System

As the pressure system arising from the TV-type splitting applied to the blood flow equations is new in the literature, we perform here a fairly detailed analysis, starting from some mathematical properties of the equations. The governing equations of the pressure system are

$$\partial_t \mathbf{Q} + \partial_x \mathcal{P}(\mathbf{Q}) = \mathbf{0}, \quad (13.96)$$

with definitions

$$\mathbf{Q} = \begin{bmatrix} A \\ \alpha Au \\ A\eta \end{bmatrix}, \quad \mathcal{P}(\mathbf{Q}) = \begin{bmatrix} Au \\ \gamma A^{3/2} \\ 0 \end{bmatrix}. \quad (13.97)$$

Source terms have been ignored, as they are not relevant to the subject of this section.

Proposition 1 *The eigenvalues of system (13.96)–(13.97) are all real and distinct and given as*

$$\lambda_1 = -c, \quad \lambda_2 = 0, \quad \lambda_3 = c, \quad (13.98)$$

with corresponding linearly independent right eigenvectors

$$\mathbf{R}_1 = \begin{bmatrix} 1 \\ -c \\ 0 \end{bmatrix}, \quad \mathbf{R}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{R}_3 = \begin{bmatrix} 1 \\ c \\ 0 \end{bmatrix}, \quad (13.99)$$

where the scaling factors have been set to unity. The pressure system is thus hyperbolic.

Proof Left to the reader as an exercise.

Note the subsonic (or subcritical) character of system (13.96)–(13.97), that is

$$\lambda_1 = -c < \lambda_2 = 0 < \lambda_3 = c. \quad (13.100)$$

Proposition 2 *The λ_2 -field is linearly degenerate. The λ_1 -field and the λ_3 -field are genuinely non-linear, provided $\frac{\partial c}{\partial A} < 0$ for all states A . For the tube law in use this monotonicity condition is always satisfied.*

Proof Left to the reader as an exercise.

13.7.3 The Riemann Problem for the Pressure System

The Riemann problem at the interface $x = x_{i+\frac{1}{2}}$, in local coordinates is given as follows

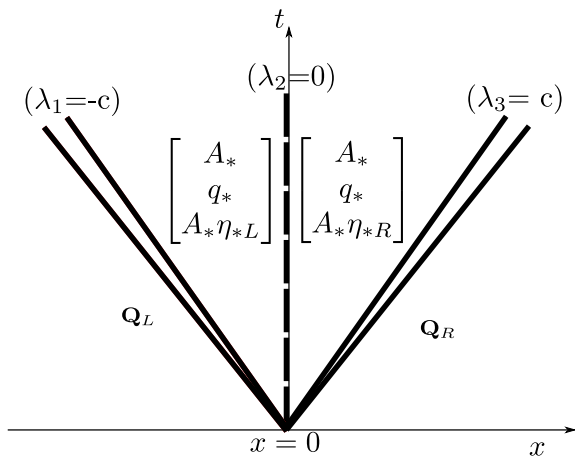
$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathcal{P}(\mathbf{Q}) = \mathbf{0}, \\ \text{ICs: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{Q}_L \equiv \mathbf{Q}_i^n & \text{if } x < 0, \\ \mathbf{Q}_R \equiv \mathbf{Q}_{i+1}^n & \text{if } x > 0. \end{cases} \end{array} \right\} \quad (13.101)$$

Figure 13.6 shows the structure of the solution, where the subsonic character of the system is put in evidence by the two wave families associated to the eigenvalues $\lambda_1 = -c$ and $\lambda_3 = c$. The **start state** between the non-linear waves has two components, namely $\mathbf{Q}_{*L}$ to the left of the contact and $\mathbf{Q}_{*R}$ to the right of the contact. To solve the Riemann problem exactly we proceed to study wave relations for each of the non-linear characteristic fields. We assume two types of non-linear waves, namely rarefaction waves and shock waves. We start from rarefaction waves.

Rarefaction Waves

First, assume an isolated left-facing rarefaction wave associated with the eigenvalue $\lambda_1 = -c$, connecting the data state $\mathbf{Q}_L$ on the left to the unknown state $\mathbf{Q}_{*L}$ on the

Fig. 13.6 Structure of the solution of the Riemann problem (13.101) for the pressure system arising from the TV splitting. The structure of the wave system is symmetric, with a stationary contact discontinuity associated with the eigenvalue $\lambda_2 = 0$ (courtesy of Prof. A. Siviglia, University of Trento, Italy)



right of the wave. The generalised Riemann invariants state that across the wave structure the following relations apply

$$\frac{dA}{1} = \frac{dq}{-c} = \frac{d(A\eta)}{0}. \quad (13.102)$$

There follows

$$q + CA^{\frac{5}{4}} = \text{constant}, \quad A\eta = \text{constant}. \quad (13.103)$$

with

$$C = \frac{4}{5} \sqrt{\frac{3}{2} \gamma}. \quad (13.104)$$

Analogously, the generalised Riemann invariants across a right-facing rarefaction are

$$\frac{dA}{1} = \frac{dq}{c} = \frac{d(A\eta)}{0}, \quad (13.105)$$

which give

$$q - CA^{\frac{5}{4}} = \text{constant}, \quad A\eta = \text{constant}. \quad (13.106)$$

Application of (13.103) to connect $\mathbf{Q}_L$ on the left to $\mathbf{Q}_{*L}$ in the Star Region gives

$$q_* + CA_*^{\frac{5}{4}} = q_L + CA_L^{\frac{5}{4}}. \quad (13.107)$$

Analogously, application of (13.106) to connect $\mathbf{Q}_R$ on the right to $\mathbf{Q}_{*R}$ in the Star Region gives

$$q_* - CA_*^{\frac{5}{4}} = q_R - CA_R^{\frac{5}{4}}. \quad (13.108)$$

In readiness for assembling the equations for the complete solution of the Riemann problem, from (13.107) we write

$$q_* = q_L - f_L(A_*; A_L), \quad f_L = C[A_*^{\frac{5}{4}} - A_L^{\frac{5}{4}}]. \quad (13.109)$$

Analogously, from (13.108) we write

$$q_* = q_R + f_R(A_*; A_R), \quad f_R = C[A_*^{\frac{5}{4}} - A_R^{\frac{5}{4}}]. \quad (13.110)$$

Two-Rarefaction Approximation

Under the *a-priori* assumption that both non-linear waves are rarefaction waves, direct application of (13.109)–(13.110) gives the solution in the Star Region as

$$\left. \begin{aligned} A_* &= \left[\frac{1}{2}(A_L^{\frac{5}{4}} + A_R^{\frac{5}{4}}) - \frac{1}{2C}(q_R - q_L) \right]^{\frac{4}{5}}, \\ q_* &= \frac{1}{2}(q_L + q_R) - \frac{1}{2}C(A_R^{\frac{5}{4}} - A_L^{\frac{5}{4}}), \\ (A\eta)_{*L} &= (A\eta)_L, \quad (A\eta)_{*R} = (A\eta)_R. \end{aligned} \right\} \quad (13.111)$$

If both non-linear waves happen to be rarefaction waves, then solution (13.111) is the exact solution. Otherwise it turns out to be a good approximation to the solution of the Riemann problem, also in the presence of shock waves.

Remark: Exact integration leading (13.103) and (13.106) is possible for the tube law used for arteries. For veins, integration must be performed numerically, and therefore a two-rarefaction approximation for the case of veins is not given in closed form.

Shock Waves

First we assume an isolated left-facing shock wave of speed S_L associated with the eigenvalue $\lambda_1 = -c$ connecting the data state $\mathbf{Q}_L$ on the left to $\mathbf{Q}_{*L}$ on the right, behind the shock. Direct application of the Rankine-Hugoniot conditions gives

$$\left. \begin{aligned} S_L(A_* - A_L) &= q_* - q_L, \\ S_L(q_* - q_L) &= \gamma(A_*^{\frac{3}{2}} - A_L^{\frac{3}{2}}), \\ S_L(A_*\eta_* - A_L\eta_L) &= 0. \end{aligned} \right\} \quad (13.112)$$

Manipulation of the first two equations gives

$$S_L = \pm \left[\frac{\gamma(A_*^{\frac{3}{2}} - A_L^{\frac{3}{2}})}{(A_* - A_L)} \right]^{1/2}. \quad (13.113)$$

The correct choice of sign in (13.113) is yet to be made. From the first equation in (13.112) we may write

$$q_* = q_L + S_L(A_* - A_L) . \quad (13.114)$$

Here we have chosen the negative sign on the grounds of the entropy condition

$$\lambda_1(\mathbf{Q}_L) > S_L > \lambda_1(\mathbf{Q}_{*L}) , \quad (13.115)$$

which implies

$$S_L < 0 \text{ and } A_* > A_L . \quad (13.116)$$

In readiness for assembling the equations for the complete solution of the Riemann problem we write

$$q_* = q_L - f_L(A_*; A_L) , \quad f_L = \left[\gamma(A_* - A_L)(A_*^{\frac{3}{2}} - A_L^{\frac{3}{2}}) \right]^{1/2} . \quad (13.117)$$

Analogous treatment of a right-facing shock wave of speed S_R associated to $\lambda_3 = c$ gives

$$q_* = q_R + f_R(A_*; A_R) , \quad f_R = \left[\gamma(A_* - A_R)(A_*^{\frac{3}{2}} - A_R^{\frac{3}{2}}) \right]^{1/2} . \quad (13.118)$$

Collecting all results we obtain a single, non-linear algebraic equation for A_*

$$f(A) := f_L + f_R + \Delta q = 0 , \quad \text{with } \Delta q = q_R - q_L , \quad (13.119)$$

where

$$f_L(A) = \begin{cases} C[A^{\frac{5}{4}} - A_L^{\frac{5}{4}}] & \text{if } A \leq A_L \text{ (rarefaction)} , \\ \left[\gamma(A - A_L)(A^{\frac{3}{2}} - A_L^{\frac{3}{2}}) \right]^{1/2} & \text{if } A > A_L \text{ (shock)} \end{cases} \quad (13.120)$$

and

$$f_R(A) = \begin{cases} C[A^{\frac{5}{4}} - A_R^{\frac{5}{4}}] & \text{if } A \leq A_R \text{ (rarefaction)} , \\ \left[\gamma(A - A_R)(A^{\frac{3}{2}} - A_R^{\frac{3}{2}}) \right]^{1/2} & \text{if } A > A_R \text{ (shock)} . \end{cases} \quad (13.121)$$

Iterative Solution for A_*

The nonlinear equation (13.119) is conveniently solved with an iterative Newton-Raphson method (13.36), as done for solving (13.30) for the full blood flow equations. As initial guess $A^{(0)}$ we suggest to use the two-rarefaction solution A_* in (13.111).

Numerical Fluxes

We have two reliable choices for computing the numerical fluxes in (13.95). The first choice is the non-iterative two-rarefaction solution (13.111). Numerous tests confirm that this approximation can be used with confidence in practice. Obviously, one can also utilise the exact, iterative solution of (13.119). This has been presented in full detail, essentially, as a reference solution. Then the TV-type numerical fluxes are:

$$\mathcal{P}_{i+\frac{1}{2}} = \begin{bmatrix} q_* \\ \gamma A_{*3} \\ 0 \end{bmatrix}, \quad \mathcal{A}_{i+\frac{1}{2}} = \begin{cases} \begin{bmatrix} 0 \\ q_* u_L \\ q_* \eta_L \end{bmatrix} & \text{if } q_* \geq 0, \\ \begin{bmatrix} 0 \\ q_* u_R \\ q_* \eta_R \end{bmatrix} & \text{if } q_* < 0. \end{cases} \quad (13.122)$$

The sought full numerical flux is then given by (13.95), which can be used in the conservative formula (13.3) to update the full system **in one step**. Recent applications of the TV splitting to the shallow water equations are reported in [17] and to sediment transport in [18]. See also [19].

In the next section we recall some very simple numerical fluxes for systems of conservation laws, some of which were introduced in Chap. 12 for the scalar case.

13.8 Simple Numerical Fluxes

Here we recall some classical, very simple numerical methods. The first one is the Rusanov method, the simplest upwind method. The rest are centred finite difference methods that can be expressed in conservative form (13.3), with corresponding numerical fluxes.

13.8.1 The Rusanov Flux from HLL

The Rusanov scheme [20] is the simplest upwind scheme, requiring just one wave-speed estimate S^+ . The scheme may be derived from the HLL scheme assuming that wave speed estimates S_L and S_R are available, see Sect. 13.5. Then we set

$$S^+ = \max\{|S_L|, |S_R|\}, \quad S_R = S^+, \quad S_L = -S^+. \quad (13.123)$$

Substituting these speeds into the HLL flux (13.59) for the subsonic regime, we obtain the classical Rusanov flux [20]

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{2} [\mathbf{F}(\mathbf{Q}_L) + \mathbf{F}(\mathbf{Q}_R)] - \frac{1}{2} S^+ (\mathbf{Q}_R - \mathbf{Q}_L) . \quad (13.124)$$

It is said that the Rusanov scheme is an **upwind scheme** that adopts a **one-wave model**. The scheme is **incomplete for any system of equations**. Sometimes in the literature, the Rusanov flux is called the **Local Lax-Friedrichs flux**, or simply the **Lax-Friedrichs flux**, which is somehow confusing.

The schemes that follow were introduced in Chap. 12 in the context of the linear advection equations, where appropriate definitions and basic properties were studied. Therefore, the presentation of the schemes for the case of systems will be succinct.

13.8.2 The Lax-Friedrichs Flux from HLL

The (true) **Lax-Friedrichs** flux [21] can be obtained from the Rusanov scheme by choosing the speed as $S^+ = \frac{\Delta x}{\Delta t}$, namely

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{2} [\mathbf{F}(\mathbf{Q}_L) + \mathbf{F}(\mathbf{Q}_R)] - \frac{1}{2} \frac{\Delta x}{\Delta t} (\mathbf{Q}_R - \mathbf{Q}_L) . \quad (13.125)$$

From stability considerations the **mesh speed** $\frac{\Delta x}{\Delta t}$ should always be larger than the largest wave speed in the full computational domain. This is precisely the reason why the Lax-Friedrichs scheme is the most diffusive of all stable schemes, as seen in Chap. 12.

13.8.3 The Godunov Centred Flux

The Godunov centred flux (not the upwind Godunov flux) [22] is a two-step scheme given as

$$\left. \begin{aligned} \mathbf{F}_{i+\frac{1}{2}}^{GodC} &= \mathbf{F}(\mathbf{Q}_{i+\frac{1}{2}}^{GodC}) , \\ \mathbf{Q}_{i+\frac{1}{2}}^{GodC} &= \frac{1}{2} (\mathbf{Q}_i^n + \mathbf{Q}_{i+1}^n) - \frac{\Delta t}{\Delta x} [\mathbf{F}(\mathbf{Q}_{i+1}^n) - \mathbf{F}(\mathbf{Q}_i^n)] . \end{aligned} \right\} \quad (13.126)$$

13.8.4 The Lax-Wendroff Flux

There are three versions of the Lax-Wendroff flux [23] in the literature. The most well-known version is the **two-step version**

$$\left. \begin{aligned} \mathbf{F}_{i+\frac{1}{2}}^{LWa} &= \mathbf{F}(\mathbf{Q}_{i+\frac{1}{2}}^{LWa}), \\ \mathbf{Q}_{i+\frac{1}{2}}^{LWa} &= \frac{1}{2}(\mathbf{Q}_i^n + \mathbf{Q}_{i+1}^n) - \frac{1}{2} \frac{\Delta t}{\Delta x} [\mathbf{F}(\mathbf{Q}_{i+1}^n) - \mathbf{F}(\mathbf{Q}_i^n)]. \end{aligned} \right\} \quad (13.127)$$

Note the similarity between the Godunov centred flux (13.126) and the Lax-Wendroff flux (13.127). The difference is in the coefficient of the diffusion term. Another version of the Lax-Wendroff method has numerical flux

$$\mathbf{F}_{i+\frac{1}{2}}^{LWb} = \frac{1}{2}[\mathbf{F}(\mathbf{Q}_i^n) + \mathbf{F}(\mathbf{Q}_{i+1}^n)] - \frac{1}{2} \frac{\Delta t}{\Delta x} \mathbf{A}_{i+\frac{1}{2}} [\mathbf{F}(\mathbf{Q}_{i+1}^n) - \mathbf{F}(\mathbf{Q}_i^n)]. \quad (13.128)$$

There are two choices for the interface matrix $\mathbf{A}_{i+\frac{1}{2}}$ in (13.128), namely

$$\mathbf{A}_{i+\frac{1}{2}} = \mathbf{A} \left(\frac{1}{2} (\mathbf{Q}_i^n + \mathbf{Q}_{i+1}^n) \right) \text{ and } \mathbf{A}_{i+\frac{1}{2}} = \frac{1}{2} [\mathbf{A}(\mathbf{Q}_i^n) + \mathbf{A}(\mathbf{Q}_{i+1}^n)]. \quad (13.129)$$

Note that flux (13.128) can also be written as

$$\mathbf{F}_{i+\frac{1}{2}}^{LWb} = \frac{1}{2} \left(\mathbf{I} + \frac{\Delta t}{\Delta x} \mathbf{A}_{i+\frac{1}{2}} \right) \mathbf{F}(\mathbf{Q}_i^n) + \frac{1}{2} \left(\mathbf{I} - \frac{\Delta t}{\Delta x} \mathbf{A}_{i+\frac{1}{2}} \right) \mathbf{F}(\mathbf{Q}_{i+1}^n). \quad (13.130)$$

A third version of the Lax-Wendroff flux is

$$\mathbf{F}_{i+\frac{1}{2}}^{LWc} = \frac{1}{2}[\mathbf{F}(\mathbf{Q}_i^n) + \mathbf{F}(\mathbf{Q}_{i+1}^n)] - \frac{1}{2} \frac{\Delta t}{\Delta x} \mathbf{A}_{i+\frac{1}{2}}^2 (\mathbf{Q}_{i+1}^n - \mathbf{Q}_i^n). \quad (13.131)$$

Remark: other versions for the Godunov centred method. Following formulations (13.128) and (13.131) for the Lax-Wendroff method, it is also conceivable to reformulate the Godunov centred flux (13.126) accordingly. For the model linear advection equation reformulations (b) and (c) are identical to the original scheme (a). Verify.

13.8.5 The FORCE Flux

The FORCE scheme [24, 25] was originally [26] derived as a deterministic version of the stochastic Random Choice Method [27]. See also [28–31] and [32]. The

resulting FORCE flux turns out to be the arithmetic mean of the Lax-Friedrichs and Lax-Wendroff fluxes, namely

$$\mathbf{F}_{i+\frac{1}{2}}^{FO} = \frac{1}{2} \left(\mathbf{F}_{i+\frac{1}{2}}^{LF} + \mathbf{F}_{i+\frac{1}{2}}^{LW*} \right). \quad (13.132)$$

Given that the Lax-Wendroff flux has three versions, so does the FORCE flux. It is sufficient to substitute the superscript $LW*$ in (13.132) by $LW a$, $LW b$, $LW c$. For the linear advection equation all three versions are equivalent (verify). Of all these simple fluxes, for practical applications we recommend the Rusanov flux (13.124), which is an **upwind flux**, and the FORCE flux (13.132), which is a **centred flux**. The associated schemes are monotone and hence they can be used as the building block for constructing non-linear (in the sense of Godunov) high-order accurate methods.

Remarks: two and three dimensional versions of FORCE. A multi-dimensional version of the FORCE scheme on unstructured meshes was constructed in [29] for conservative systems and in [30] for non-conservative systems. As a byproduct, corresponding, new multidimensional versions of the Lax-Friedrichs and Lax-Wendroff methods on unstructured meshes resulted. A low-diffusion version of the multidimensional FORCE scheme, called FORCE- α [32], has emerged as an interesting alternative to previous versions.

The last section of this chapter is devoted to assessing the performance of all the numerical methods studied here.

13.9 Numerical Results

In this section we assess the performance of all the numerical methods presented in this chapter. We do so through two sets of carefully chosen test problems for the blood flow equations. The tests have been selected so as to represent the main features of the admissible solutions of the blood flow equations, namely smooth solutions, large spatial gradients and discontinuous solutions in the form of elastic jumps and contact discontinuities. Contact discontinuities occur naturally when extending the blood flow equations to include transport of species.

13.9.1 Schemes and Test Problems

All the methods to be assessed are summarised in Table 13.1, including upwind and centred methods. Initial data for all test problems are given in Table 13.2, while parameter values for such tests are given in Table 13.3 for arteries and in Table 13.4 for veins. The first set of three test problems concerns blood flow in arteries, while the remaining set of test problems concerns blood flow in veins. Both sets of tests share the same initial conditions, as given in Table 13.2.

Table 13.1 Numerical schemes to be assessed by comparing their approximate solutions to the exact solution. The Acronym of column 2 will be used to identify the scheme in the figures displaying numerical results

Scheme	Acronym	Description	Note
1	God+Ex.RS	Godunov method with exact Riem. Solv.	Upwind
2	God+Ap.RS	Godunov method with TR App. Riem. Solv.	Upwind
3	HLL	Godunov method with HLL Riem. Solv.	Upwind
4	HLLC	Godunov method with HLLC Riem. Solv.	Upwind
5	DOT	Godunov with Dumbser-Osher-Toro Riem. Solv.	Upwind
6	TV+Ex.RS	TV Split. with Ex. Riem. Solv. for Press. Syst.	Upwind
7	TV+Ap.RS	TV Split. with App. Riem. Solv. for Press. Syst.	Upwind
8	Rusanov	Rusanov method	Upwind
9	FORCE	FORCE method	Centred
10	Lax-Friedrichs	Lax-Friedrichs method	Centred
11	Lax-Wendroff	Lax-Wendroff method	Centred
12	GodC	Godunov centred method	Centred

Table 13.2 Initial data for five test problems for the augmented blood flow equations in SI units. The 9-th column describes the wave pattern that emerges from the exact solution, where R stands for rarefaction, C stands for contact discontinuity and S stands for shock. The last column identifies the type of blood vessel, artery or vein

Test	A_L	u_L	ψ_L	A_R	u_R	ψ_R	T_{out}	Wave pattern	Blood vessel
1	3.5×10^{-4}	0.0	1.0	3.0×10^{-4}	0.0	0.0	0.05	RCS	Artery
2	10.0×10^{-4}	0.0	1.0	1.0×10^{-4}	0.0	0.0	0.04	R(sonic)CS	Artery
3	3.14×10^{-4}	-0.5	1.0	3.14×10^{-4}	0.5	0.0	0.05	RCR	Artery
4	3.5×10^{-4}	0.0	1.0	3.0×10^{-4}	0.0	0.0	0.05	RCS	Vein
5	3.14×10^{-4}	-0.5	1.0	3.14×10^{-4}	0.5	0.0	0.08	RCR	Vein

Table 13.3 Parameters for tests 1 to 3 in Table 13.2 for arteries. Here x_0 denotes position of initial discontinuity; h_0 is vessel wall thickness; A_0 is equilibrium cross-sectional area; E is Young’s modulus; ν is Poisson ratio and ρ is blood density

Domain length	x_0	h_0	A_0	E	ν	ρ	Blood vessel
0.5	0.25	5.0×10^{-4}	3.14×10^{-4}	3.0×10^5	0.5	1000	Artery

Table 13.4 Parameters for tests 4 to 5 in Table 13.2 for veins. Here x_0 denotes position of initial discontinuity; A_0 is equilibrium cross-sectional area; K is the stiffness coefficient; ρ is blood density; m and n are tube law exponents

Domain length	x_0	A_0	K	ρ	m	n	Blood vessel
0.5	0.25	3.14×10^{-4}	333.0	1000	10	-3/2	Vein

Initial Conditions and Parameters

Initial conditions for all test problems are defined in Table 13.2, given in terms of physical variables A , u and η . Columns 2 to 4 give the initial data to the left of x_0 , while columns 5 to 7 give respective values on the right hand side of x_0 . The output time T_{out} to display results is given in column 7. The last column identifies the kind of blood vessel considered, artery or vein. The initial conditions for tests 4 and 5 for veins are the same as for arteries, tests 1 to 3, but the biomechanical and computational parameters are different, to reflect the fact that arteries are different from veins. Table 13.3 shows the values of parameters used for arteries, while Table 13.4 shows values for veins.

Main Features of the Exact Solution of the Test Problems

Before proceeding with the assessment of the numerical approximations we highlight the salient features of the exact solutions of the test problems, to be compared against, that may challenge the quality of the approximation.

The solution of Test 1 consists of three waves, namely a left-facing rarefaction wave, a middle contact discontinuity, visible for the tracer η , and a right-facing elastic jump, or shock. The left rarefaction wave carries smooth transitions of cross-sectional area A and velocity u , while the right shock carries discontinuous jumps in these quantities. Generally, discontinuities may put in evidence the monotonicity property of the schemes. Also, large spatial gradients, not necessarily discontinuities, may cause the schemes to generate spurious oscillations. Of all three waves, it is usually the contact discontinuity the one that is more challenging, due to the presence of excessive numerical diffusion seen in the form of smearing of the discontinuity.

Test 2 contains the same wave pattern as Test 1, that is left rarefaction, middle contact and right shock. There are however, two important differences. First, the left rarefaction in Test 2 is **transonic**, or **transcritical**; that is the associated left eigenvalue $\lambda_1 = u - c$ goes monotonically from negative values to positive values, passing through a critical point at which $u = c$. Even though the wave is smooth, the sonic point is challenging to all numerical methods. Some schemes will give a jump (shock) instead of a smooth transition across the sonic point; such shock is **entropy violating** and thus unphysical. The second difference between Test 1 and Test 2 is the strength of the waves; this may pose a challenge to the robustness of the methods. It also poses the problem of correctly estimating the maximum wave speed in the domain to reliably define a stable time step Δt from the Courant stability condition.

Test 3 consists of two rarefaction waves, with smooth transitions for A and u , and a middle (stationary) contact discontinuity with a discontinuous jump in tracer η . The main challenge here is to resolve the stationary contact discontinuity with minimal numerical diffusion and without spurious oscillations.

Overall, the main criteria to analyse the numerical results include the following:

1. **Smooth waves.** Generally one expects accurate resolution of this kind of waves (rarefactions).
2. **Position of waves.** From conservative methods one expects these to agree with the exact solution, on average. Shock wave positions are particularly critical.
3. **Discontinuous waves.** For discontinuous waves, such as contacts and shocks, one expects sharp resolution of the discontinuities, with a minimal number of cells across the transition.
4. **Discontinuous waves and large spatial gradients.** For these features of the solution one expects absence of spurious oscillations in their vicinity.

13.9.2 Numerical Results for Arteries and Discussion

Computed results for arteries are compared to exact solutions in Figs. 13.7, 13.8, 13.9, 13.10, 13.11, 13.12, 13.13, 13.14 and 13.15. Each figure displays results from four schemes, for a single test problem, for the cross-sectional area A , velocity u and concentration η .

Results from Test 1 for Arteries

Figure 13.7 shows results for Test 1 for four upwind methods. All results are very similar for the rarefaction wave and the shock. The only obvious difference is in the contact discontinuity, with the HLL scheme exhibiting excessive numerical diffusion, causing spreading of the discontinuity for about 20 mesh points, compared to just 3 to 4 points in the other schemes. This behaviour of HLL is related to the **incompleteness** of the Riemann solver.

Figure 13.8 shows results for Test 1 from another set of four upwind methods. Results, overall, are similar to those of Fig. 13.7. The Rusanov method stands out through excessive numerical diffusion of the contact discontinuity. Recall that the Rusanov scheme is the simplest of all upwind methods, admitting just one wave in the structure of the Riemann problem solution. Again, the point is the **incompleteness** of the Riemann solver.

Figure 13.9 shows results for Test 1 from four centred methods. The Lax-Friedrichs method stands out for its excessive numerical diffusion, especially for the contact wave, and the typical **pairing of points**; no spurious oscillations are seen for this example. The Godunov centred method was applied with a smaller time step than for the other methods. Recall that the stability limit of the Godunov centred scheme is $C_{lim} = \frac{1}{2}\sqrt{2}$. In enforcing the Courant condition we applied the

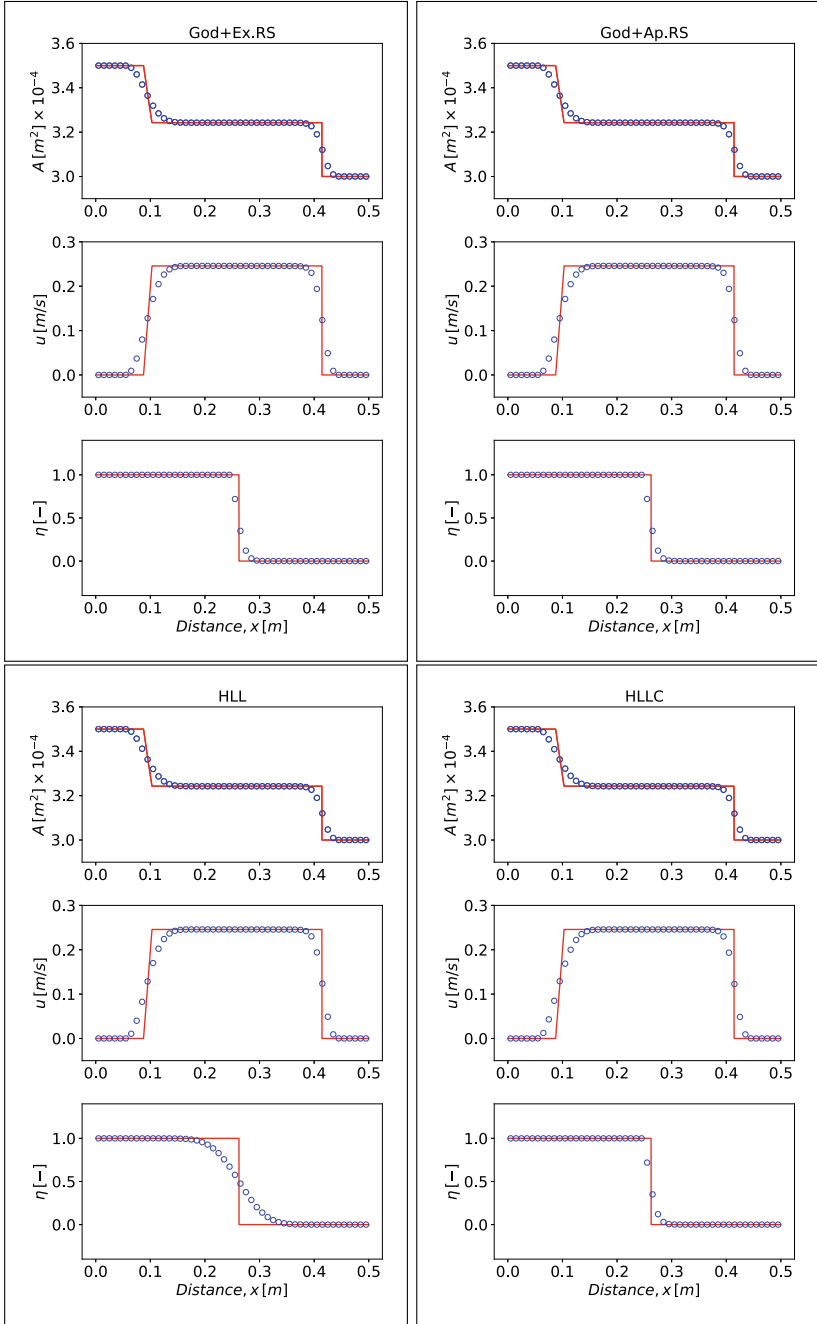


Fig. 13.7 Test 1 (arteries). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$. Initial data is given in Table 13.2 and parameters are given in Table 13.3

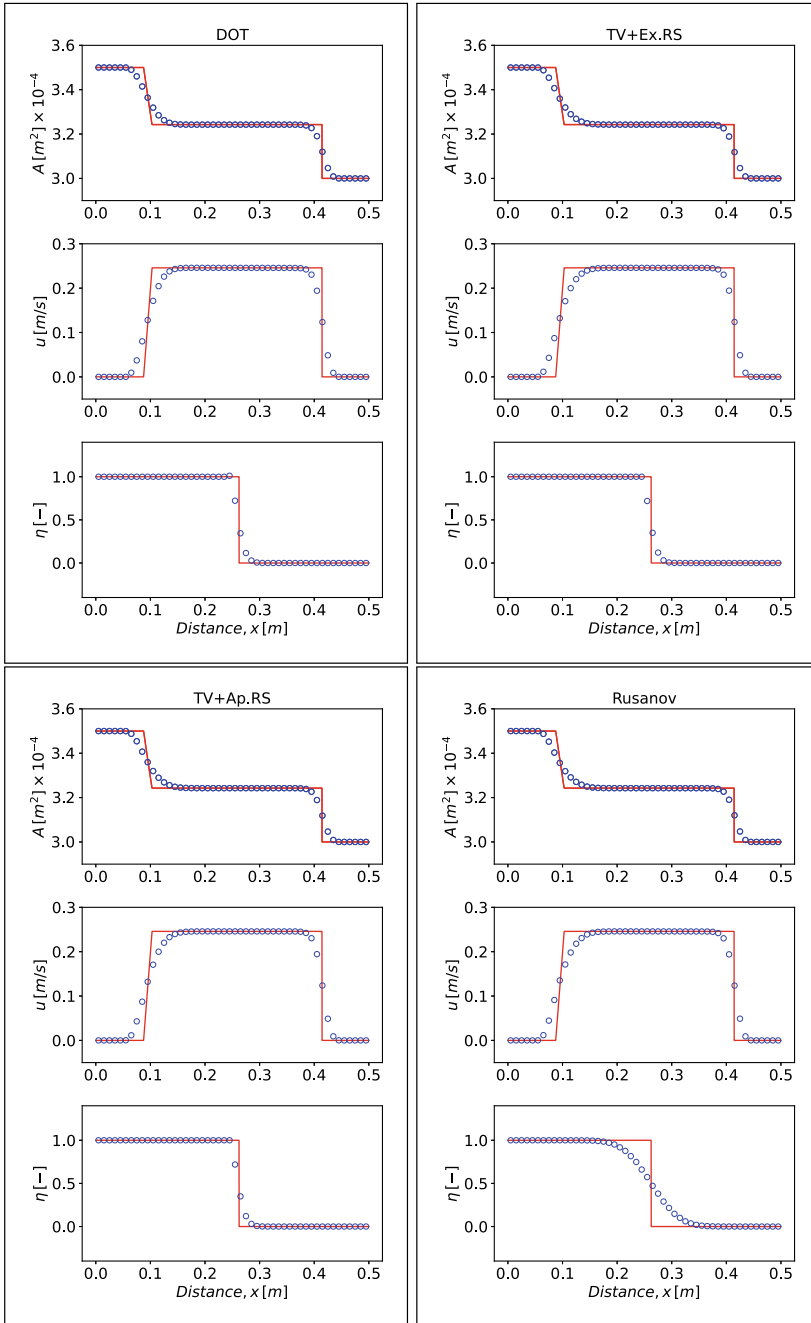


Fig. 13.8 Test 1 (arteries). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$. Initial data is given in Table 13.2 and parameters are given in Table 13.3

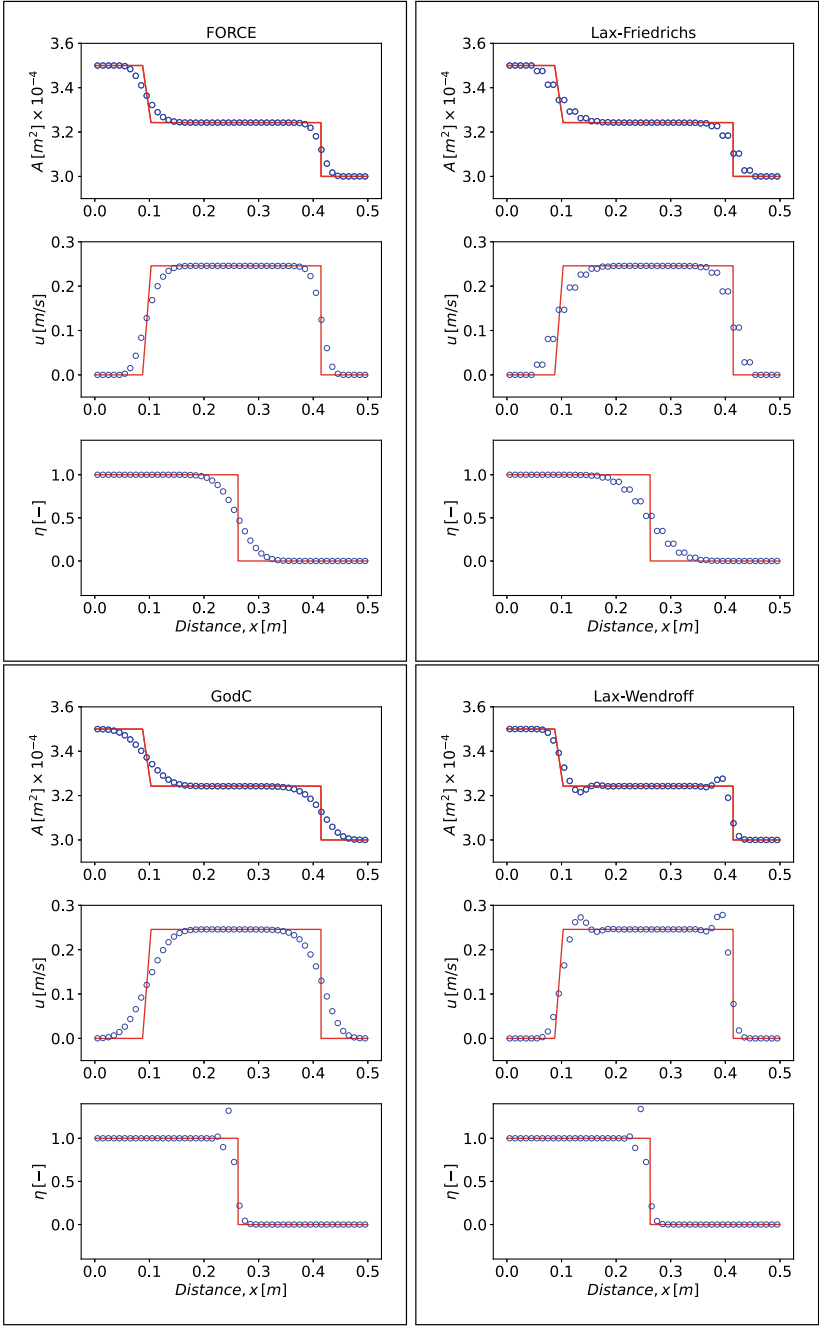


Fig. 13.9 Test 1 (arteries). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$, except for GodC ($C_{cfl} = 0.636$). Initial data is given in Table 13.2 and parameters are given in Table 13.3

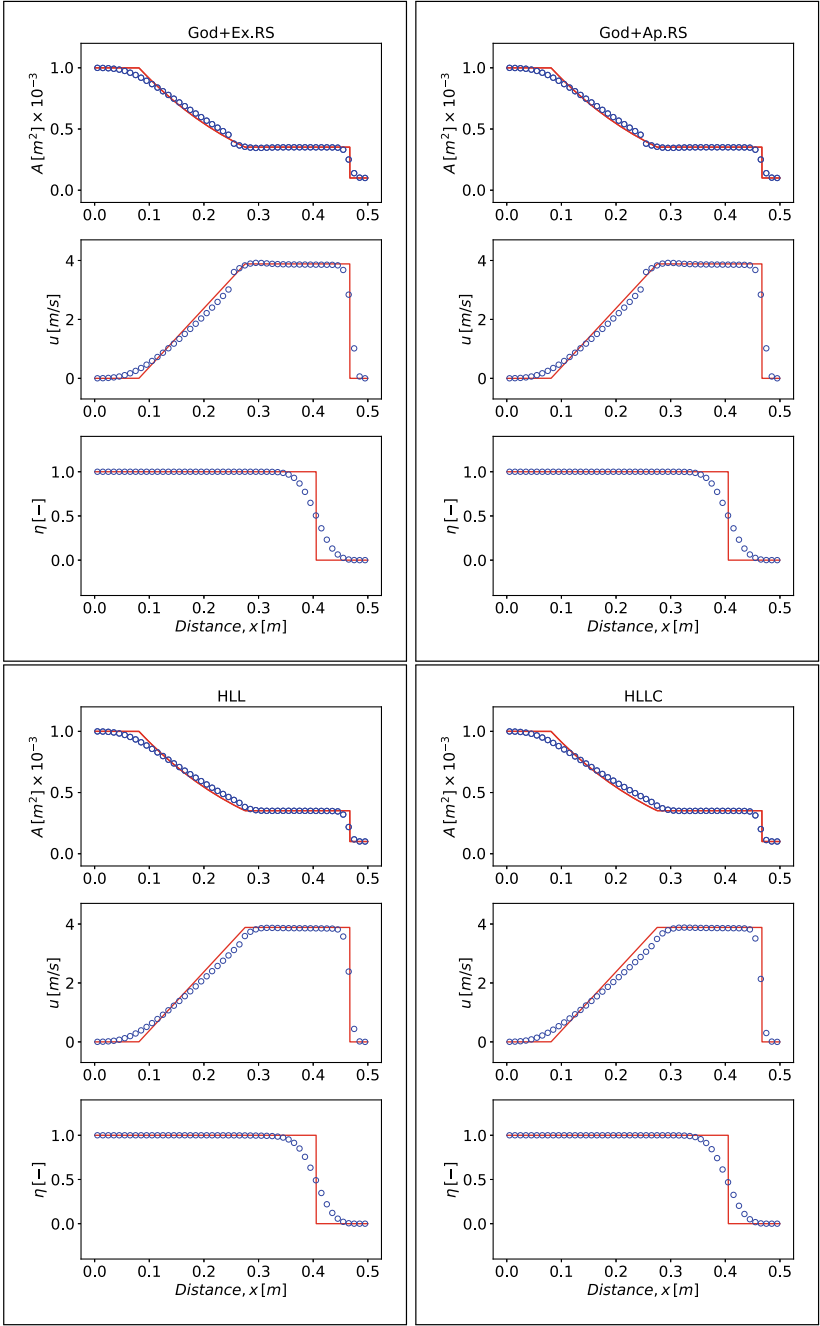


Fig. 13.10 Test 2 (arteries). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$. Initial data is given in Table 13.2 and parameters are given in Table 13.3

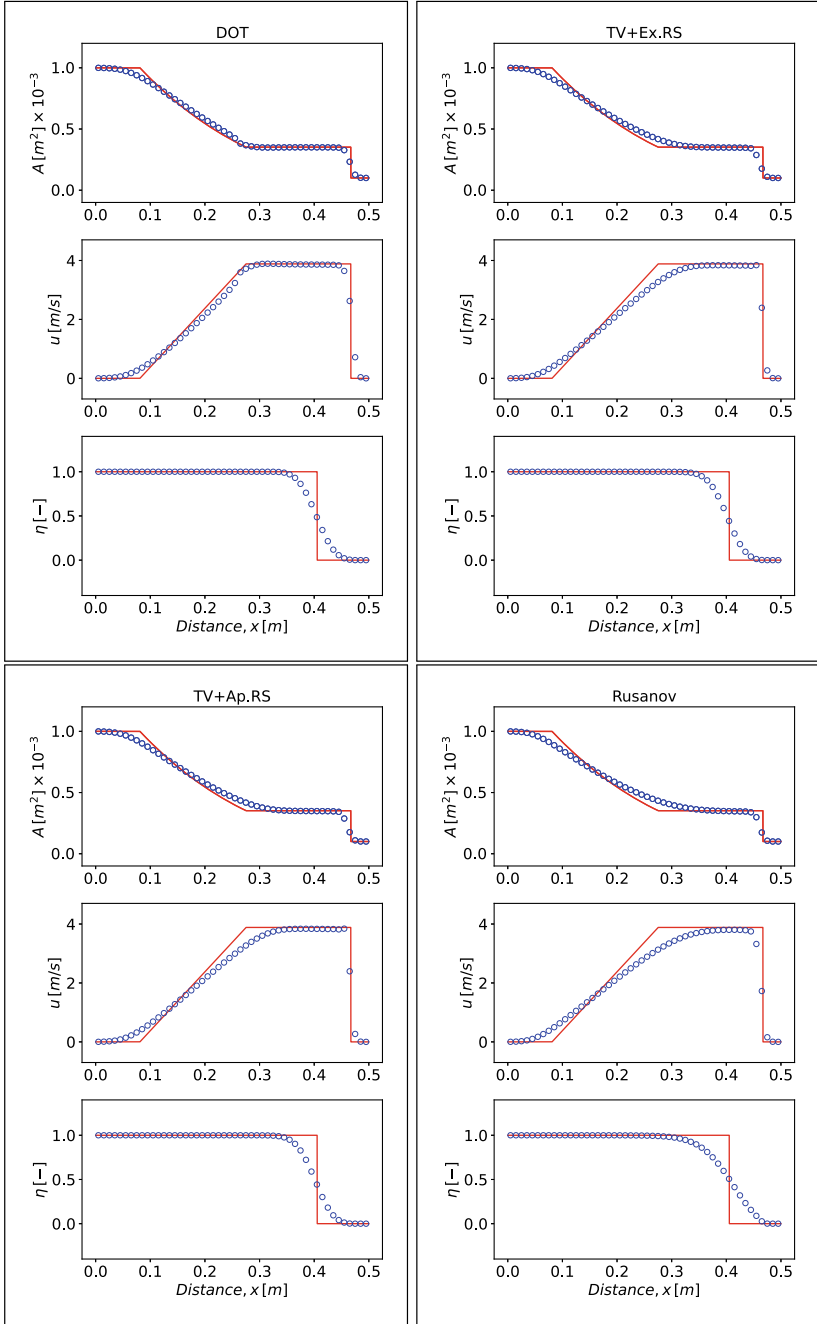


Fig. 13.11 Test 2 (arteries). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$. Initial data is given in Table 13.2 and parameters are given in Table 13.3

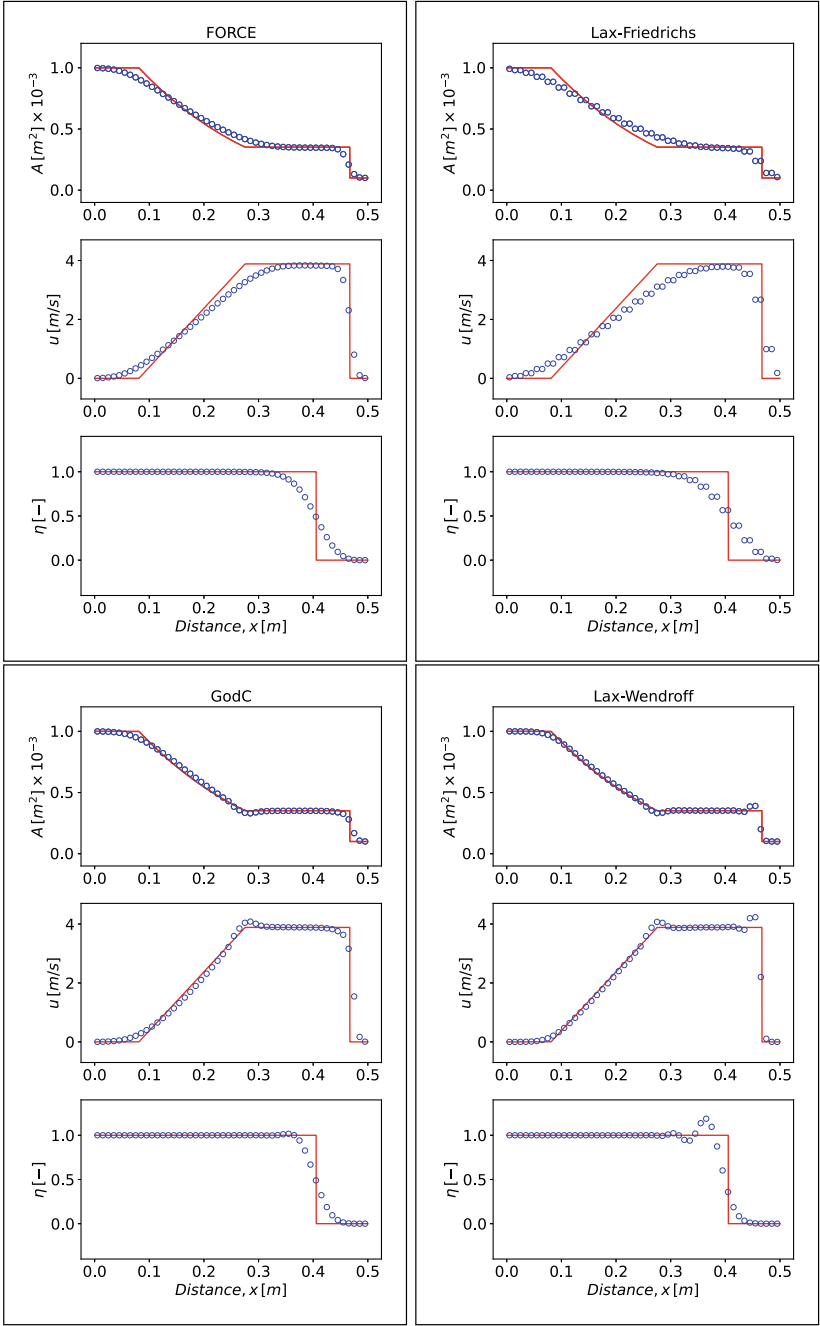


Fig. 13.12 Test 2 (arteries). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$, except for GodC ($C_{cfl} = 0.636$). Initial data is given in Table 13.2 and parameters are given in Table 13.3

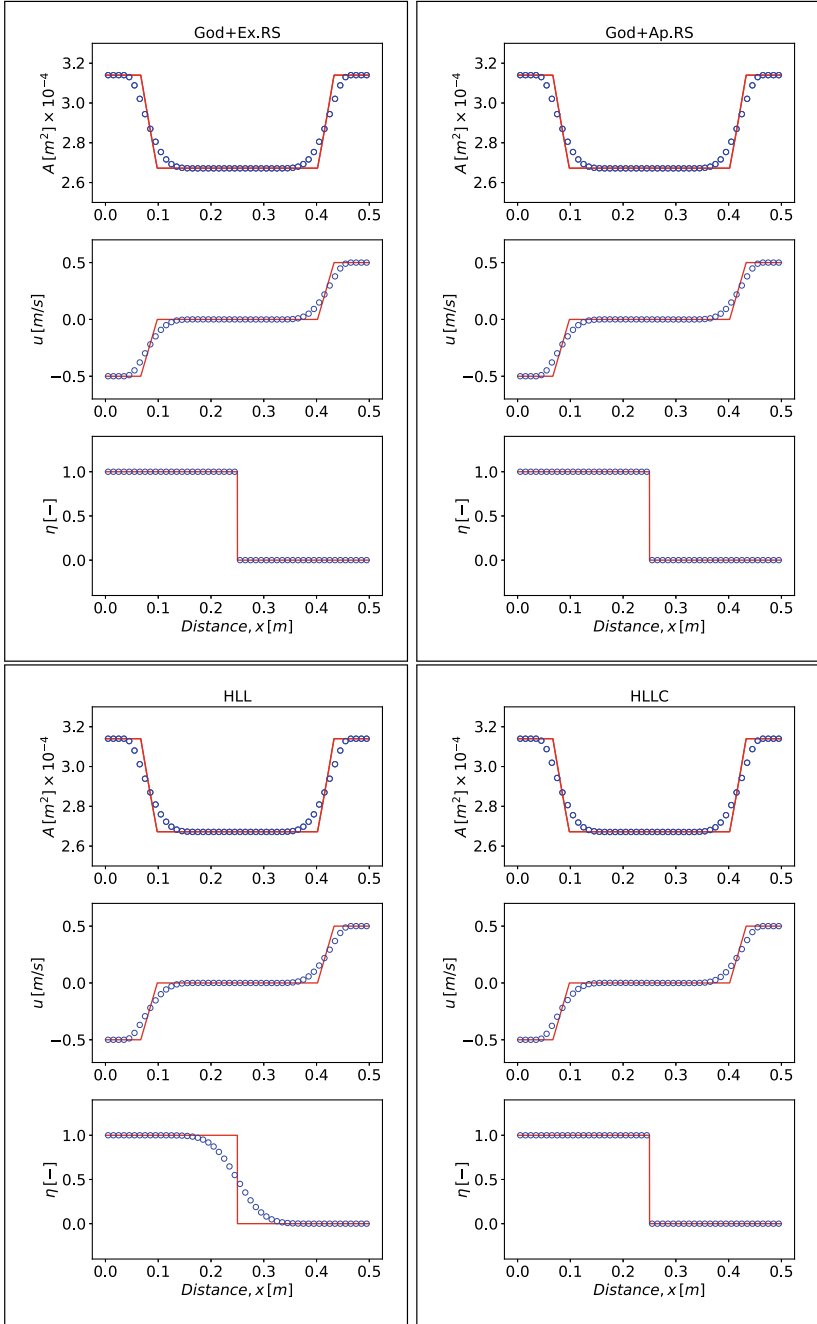


Fig. 13.13 Test 3 (arteries). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$. Initial data is given in Table 13.2 and parameters are given in Table 13.3

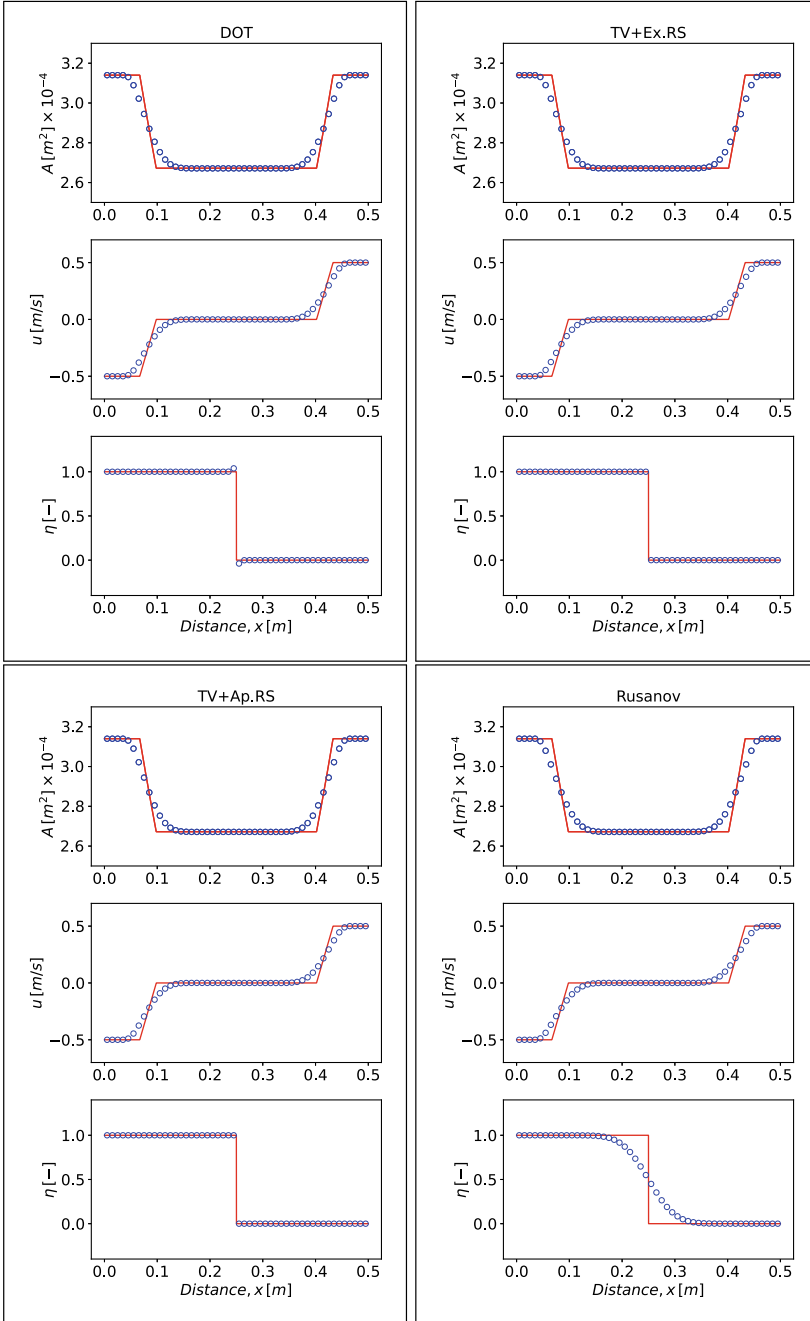


Fig. 13.14 Test 3 (arteries). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$. Initial data is given in Table 13.2 and parameters are given in Table 13.3

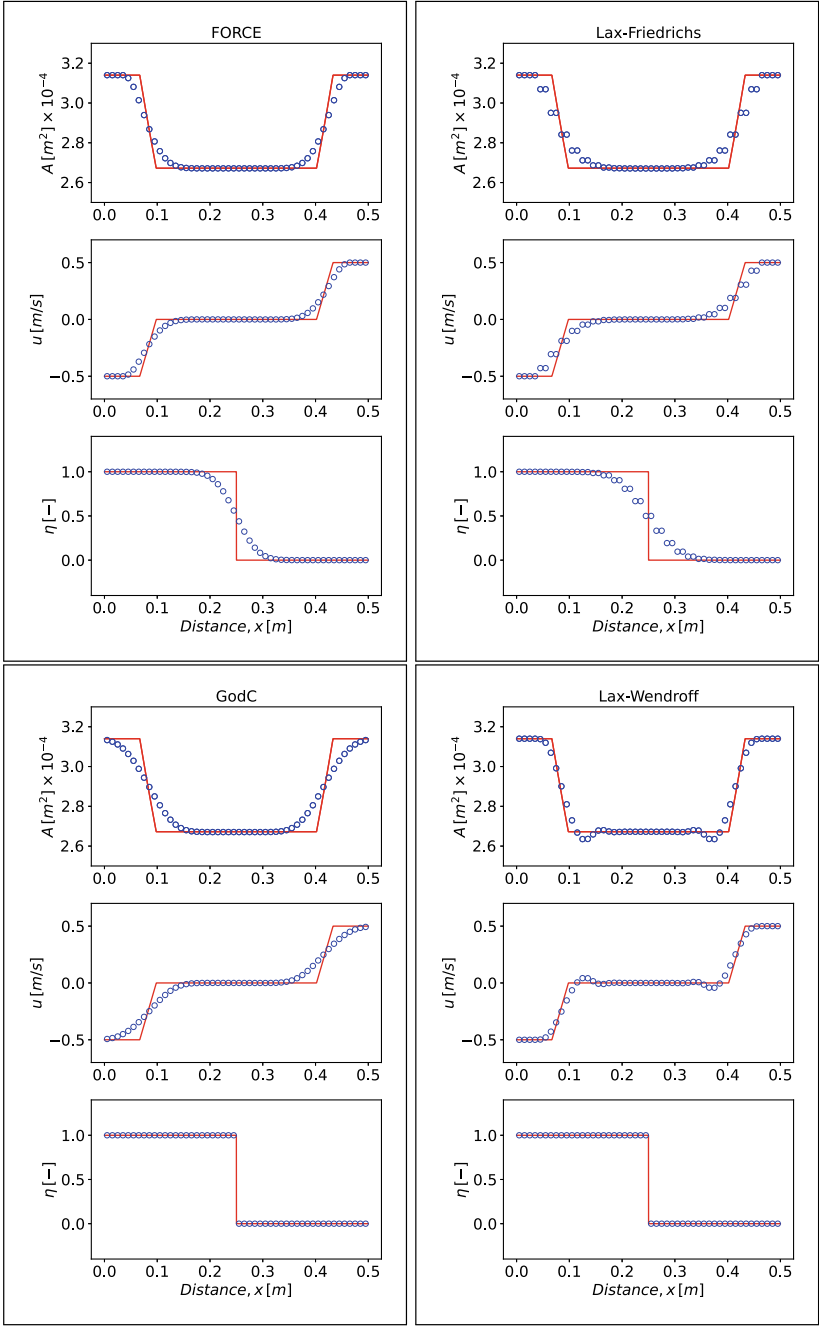


Fig. 13.15 Test 3 (arteries). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$, except for GodC ($C_{cfl} = 0.636$). Initial data is given in Table 13.2 and parameters are given in Table 13.3

same criterion as for the other schemes by taking a C_{cfl} coefficient of 90% of C_{lim} . Moreover, GodC is not monotone in the range $0 < c \leq \frac{1}{2}$. This explains the spurious oscillations seen in the contact discontinuity. As expected, the results from the Lax-Wendroff method display spurious oscillations everywhere, not just in the vicinity of discontinuities, but also around the tail of the rarefaction wave (smooth). This behaviour is due to the fact that **the Lax-Wendroff method is not a monotone method**, as demonstrated for the linear advection equation in Chap. 12.

Results from Test 2 for Arteries

Figure 13.10 shows results for Test 2 from four upwind schemes. As compared to results from Test 1 we observe that all methods are being challenged by the severity of this test, including the presence of a **sonic point**, or **critical point**. The Godunov method with both the exact and an approximate Riemann solver shows the typical **entropy glitch** at the sonic point. It is known that mesh refinement will reduce the size of the entropy-violating jump in these methods. HLL and HLLC cope better with the sonic point than the Godunov method with the other Riemann solvers. This is well known for other hyperbolic systems too. As to the resolution of the contact discontinuity, all four methods give comparable results; this is somehow surprising, as one would have expected the HLL to display the worst performance. This is explained by the large speed of the wave, namely $4m/s$. It is known that numerical diffusion increases with the wave speed, to the point that there is little or no difference between schemes that resolve intermediate waves and those that do not.

Figure 13.11 shows results for Test 2 from another set of four upwind methods. Results are similar to those of Fig. 13.10. DOT shows a small entropy glitch at the sonic point. The two TV splitting schemes, used with two different Riemann solvers for the pressure system, give similar results; both schemes resolve the shock very sharply, but diffuse the tail of the rarefaction, visibly. The Rusanov method gives the most inaccurate approximations, as expected. Figure 13.12 shows results for Test 2 from four centred methods. The numerical approximations are as expected. The Lax-Friedrichs method exhibits large numerical diffusion and pairing of points; it also shows small oscillations, see plot for A. The non-monotone methods, Godunov centred and Lax-Wendroff, exhibit spurious oscillations. For this test problem GodC performs quite satisfactorily, the loss of monotonicity is hardly visible. The first-order monotone FORCE method gives reasonable approximations.

Results from Test 3 for Arteries

The solution of this test problem is smooth everywhere for A and u , but discontinuous for the tracer η , with a stationary contact discontinuity. Figure 13.13 shows results for Test 3 from four upwind methods. All numerical approximations are as expected. All methods, apart from HLL, resolve the stationary contact discontinuity exactly, as a true discontinuity. Figure 13.14 shows results for Test 3 from another set of four upwind methods. Again, all results are as expected. All methods, apart from the Rusanov scheme, resolve the stationary contact discontinuity

exactly, as a true discontinuity; DOT exhibits small oscillations around the contact. Figure 13.15 shows results for Test 3 from four centred methods. The Lax-Friedrichs and Lax-Wendroff schemes give the worst results. Note that the loss of monotonicity of the Lax-Wendroff method is evident also for smooth solutions with large spatial gradients. Surprisingly, GodC gives the best results, with no sign of loss of monotonicity.

13.9.3 Numerical Results for Veins and Discussion

Here we present results for 2 test problems for veins defined in Table 13.2, Tests 4 and 5. The salient features of the exact solution are the same as for arteries, discussed previously. On the whole, the challenges to the numerical schemes to correctly represent these features are similar to those of the arterial case. Test 5 is unique; it represents a nearly collapsed vein; the exact solution is smooth everywhere, but large spatial gradients are present and pose a challenge to non-monotone methods and to poorly designed non-linear methods.

Results from Test 4 for Veins

Figure 13.16 shows results for Test 4 (for veins) from four upwind methods. There is little novelty here, as compared to Test 1 for arteries (same initial conditions). The HLL scheme stands out due to its excessive numerical diffusion at the contact discontinuity. Figure 13.17 shows results for Test 4 (for veins) from another set of four upwind methods. Results from all schemes are consistent with the results from Test 1 for arteries. As expected the least accurate scheme is the Rusanov method. Figure 13.18 shows results for Test 4 (for veins) from four centred methods. Results are as expected, with the non-monotone methods Godunov centred and Lax-Wendroff exhibiting expected spurious oscillations.

Results from Test 5 for Veins

Figure 13.19 shows results for Test 5 (for veins) from four upwind methods. The performance of all the schemes is as expected and consistent with those from previous test problems. Again HLL stands out for the excessive smearing of the contact discontinuity, while all other methods resolve it exactly. Figure 13.20 shows results for Test 5 (for veins) from another set of four upwind methods. Again the approximations are as expected. This time the Rusanov scheme stands out for the excessive smearing of the contact discontinuity. The DOT scheme exhibits some small-amplitude spurious oscillations around the stationary contact discontinuity. Figure 13.21 shows results for Test 5 (for veins) from four centred methods. The Lax-Friedrichs scheme is exceedingly diffusive, as expected and the Lax-Wendroff method shows spurious oscillations for A and u . This is somehow surprising, as the solution for this test problem is continuous, even though it exhibits large spatial gradients near the tail of the rarefactions; these features cause the Lax-Wendroff

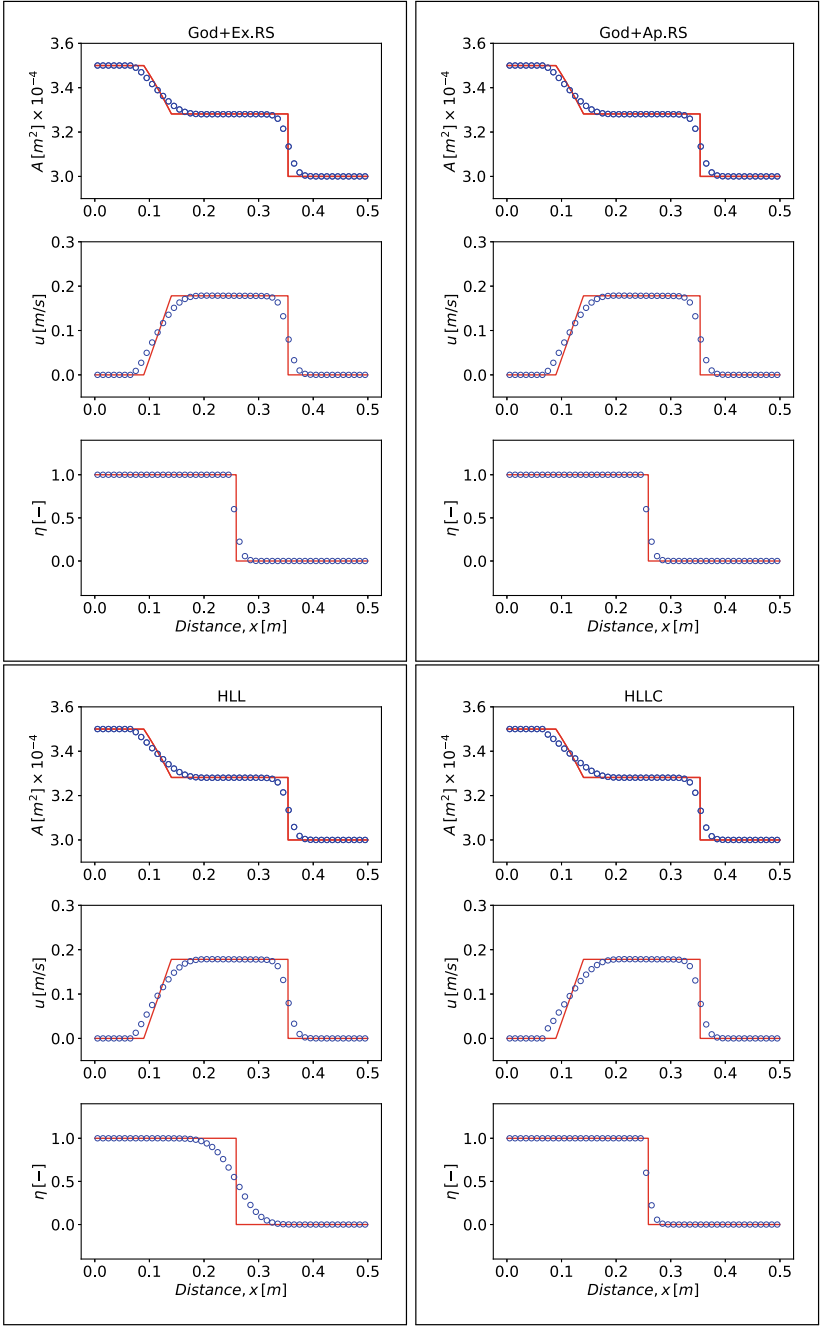


Fig. 13.16 Test 4 (veins). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$. Initial data is given in Table 13.2 and parameters are given in Table 13.4

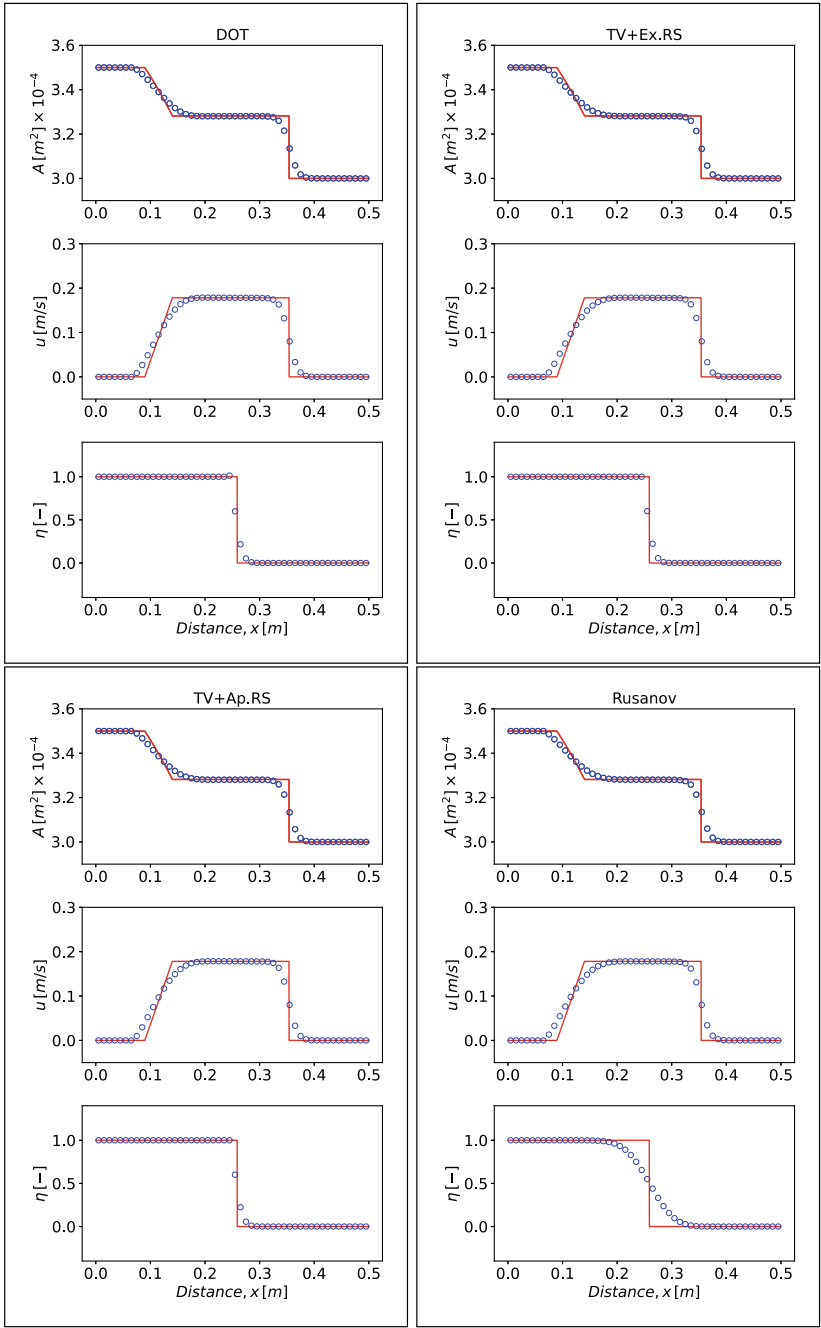


Fig. 13.17 Test 4 (veins). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$. Initial data is given in Table 13.2 and parameters are given in Table 13.4

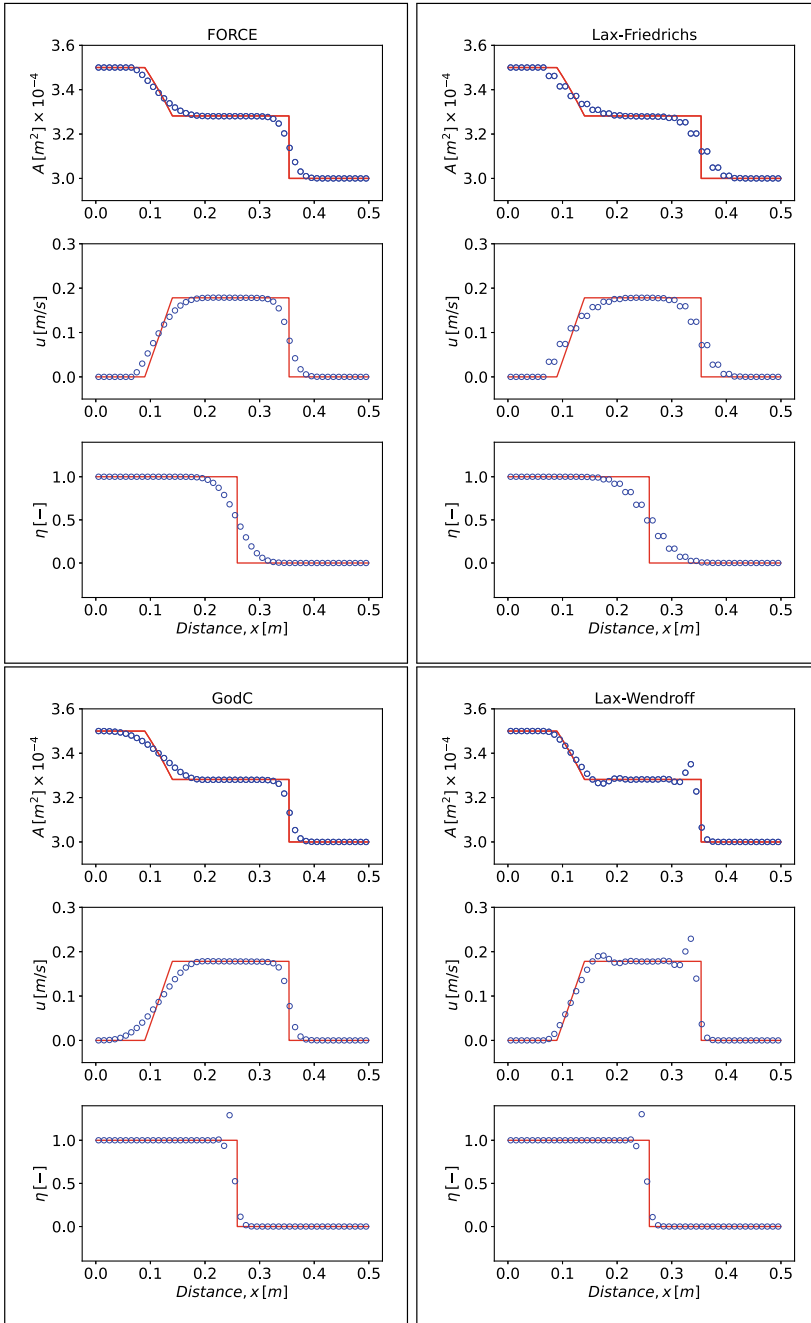


Fig. 13.18 Test 4 (veins). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$, except for GodC ($C_{cfl} = 0.636$). Initial data is given in Table 13.2 and parameters are given in Table 13.4

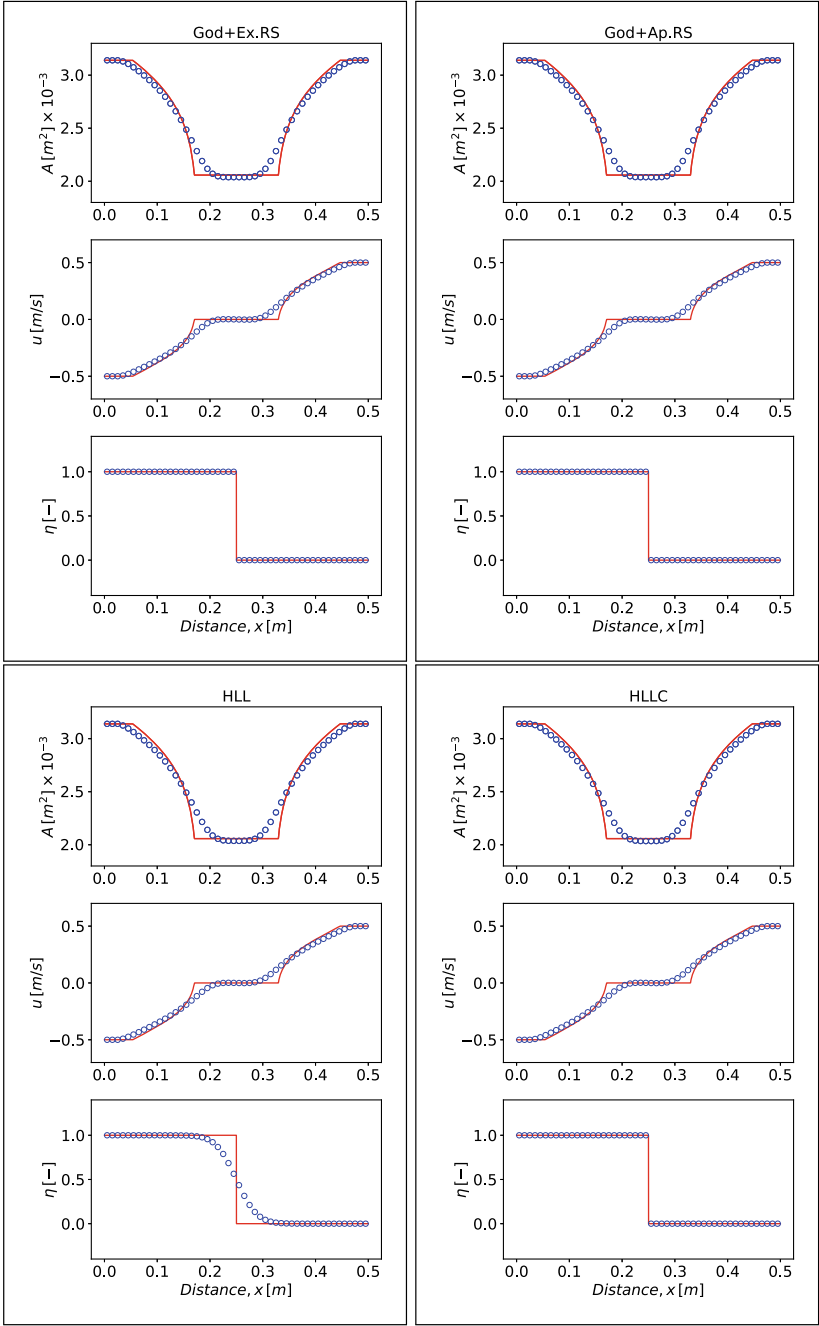


Fig. 13.19 Test 5 (veins). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$. Initial data is given in Table 13.2 and parameters are given in Table 13.4

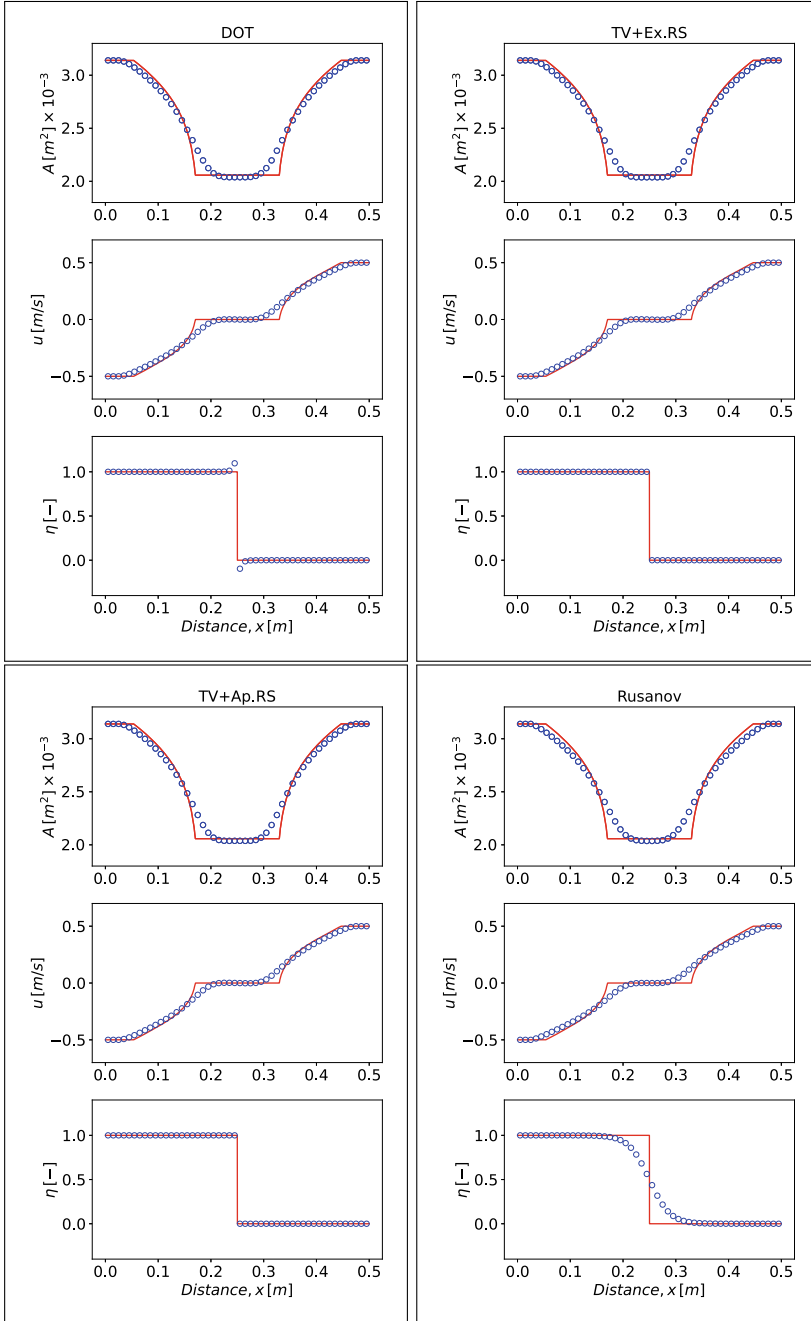


Fig. 13.20 Test 5 (veins). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$. Initial data is given in Table 13.2 and parameters are given in Table 13.4

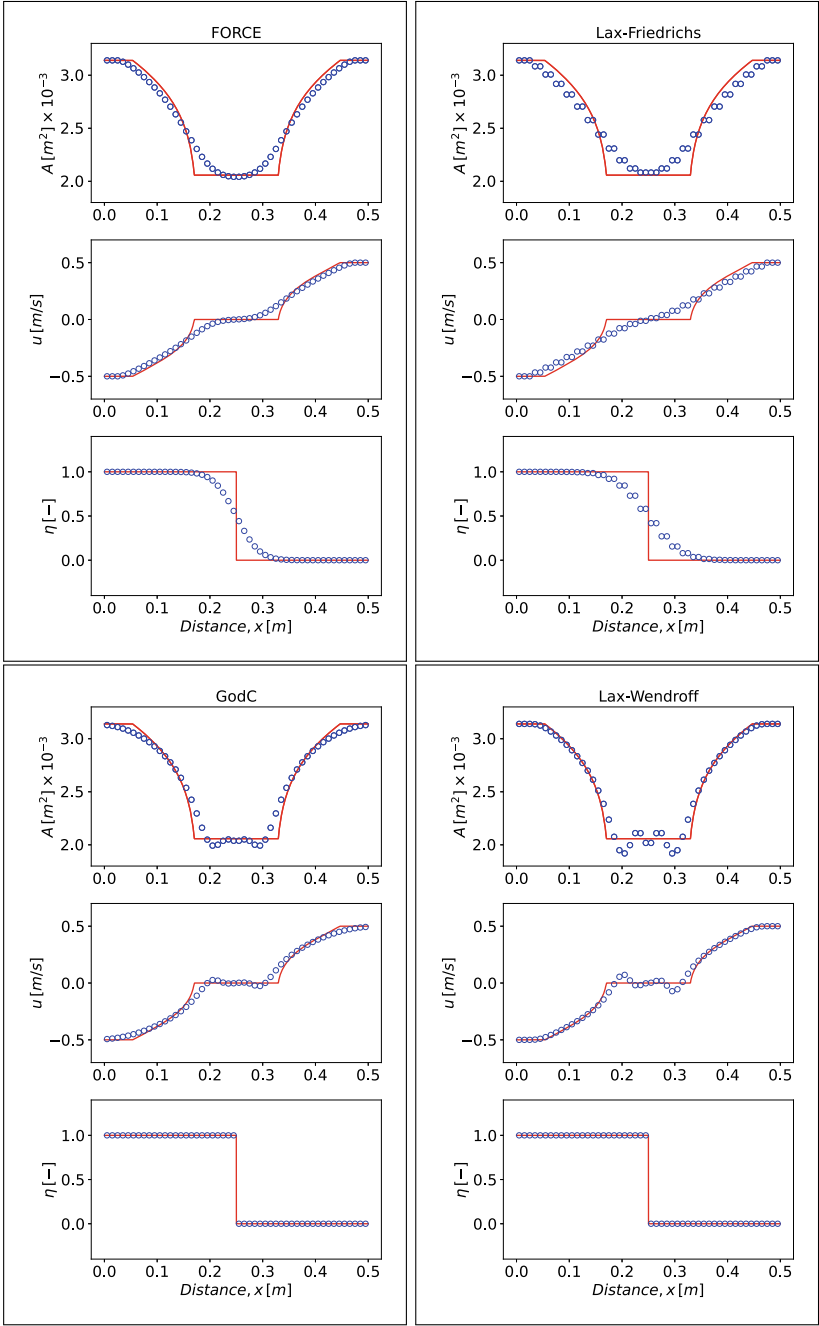


Fig. 13.21 Test 5 (veins). Numerical solution from four methods (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{cfl} = 0.9$, except for GodC ($C_{cfl} = 0.636$). Initial data is given in Table 13.2 and parameters are given in Table 13.4

method to respond as if these features were shocks, causing oscillations. The same remark is applicable to the GodC scheme. Interestingly, however, both GodC and the Lax-Wendroff methods resolve the stationary contact discontinuity perfectly. See exercises at the end of the chapter.

Closing Remarks on Results

We have thoroughly tested all the numerical methods studied in this chapter against exact solutions. Two classes of methods are identified, monotone methods and non-monotone methods, the latter exhibiting spurious oscillations even in smooth zones. A feature that divides all methods into two classes, again, is the contact discontinuity. Some methods are capable of resolving this feature as a true discontinuity. Resolving accurately features associated with advected tracers is very important in extended models that simulate transport of species in the circulation. We have not assessed the methods here regarding speed of convergence under mesh refinement, neither have we evaluated the methods in terms of efficiency, that is error against cost. These issues will be taken up in Chap. 17.

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Chapter 14

Methods for Non-conservative Systems



Abstract This chapter focuses on systems of non-linear hyperbolic equations expressed in non-conservative form. In several areas of scientific and technological interest, mathematical models cannot be represented in conservation-law form, even if they are derived from physical conservation principles. For such systems, the established class of conservative methods does not apply, and alternative approaches are required. The chapter begins with a motivating example, using a non-linear scalar equation, to highlight some of the issues that arise in the presence of discontinuities. The chapter introduces theoretical concepts related to general hyperbolic systems in non-conservative form, including non-conservative products, path functions, and generalised Rankine-Hugoniot conditions. A relevant example is provided by the blood flow equations, incorporating tube laws for arteries and veins, as well as discontinuous mechanical and geometric parameters. A framework for constructing numerical methods for non-conservative systems is introduced, specifically the path-conservative approach. In this methodology, the numerical fluctuation replaces the role of the numerical flux in the conservative case. Several non-conservative methods are examined, including upwind and centred schemes. These methods are applicable to the blood flow equations in arteries and veins, accommodating both constant and discontinuous geometric and biomechanical parameters. The key challenge in these methods is the construction of the numerical fluctuation. A systematic evaluation of the performance of these schemes is conducted using a carefully selected suite of test problems with exact solutions. Numerical results are presented for four of the seven methods studied in this chapter. High-order extensions of these methods are discussed in subsequent chapters. The chapter concludes with a summary, recommendations for further reading, and a list of suggested exercises.

14.1 Forms of the Equations

We begin by highlighting two issues regarding the form in which the partial differential equations are written. First, the desirable form for hyperbolic equations is the conservation-law form, with a **physical flux** function and expressed in terms of the physically conserved variables. This allows the definition of physically mean-

ingful shocks through the classical Rankine-Hugoniot conditions. Additionally, we warn the reader that, mathematically, under the assumption of smoothness, it is sometimes possible to find **alternative conserved variables** that furnish **alternative conservation-law forms** for the equations. However, the alternative variables may not have the meaning of physical conservation. Moreover, the Rankine-Hugoniot conditions will give shocks with different speeds to those in terms of conserved variables.

Then there is a second issue. There are areas of applications in which the governing equations do not have a known conservation-law form, even if the equations have been derived on physical conservation principles and the unknowns are the physically conserved quantities; no physical flux function can be found for these equations. A classical example is furnished by the Baer-Nunziato equations [1] for compressible multiphase flow, commonly adopted in aerospace sciences, for example. In biofluid mechanics there are also various well-known examples, some of which will be presented in this chapter in the form of test problems for the methods studied. In haemodynamics, practical and realistic one-dimensional mathematical models for blood flow in vessels with variable geometric and mechanical properties are non-conservative [2–4]. Cerebrospinal fluid in the spinal column furnishes another example of a model in non-conservative form. This topic is studied in detail in Chap. 11. Hyperbolisation of parabolic equations [5–8] is another source of systems in non-conservative form. The study of methods for non-conservative systems is central to this book. But first, in order to illustrate these issues, in what follows we consider two examples.

14.1.1 Example 1: Burgers' Equation

Consider the inviscid Burgers equation written in conservation-law form as

$$\partial_t q + \partial_x f(q) = 0, \quad f(q) = \frac{1}{2} q^2. \quad (14.1)$$

Here q is the conserved variable and $f(q)$ is the **physical flux function**. Recall from Chap. 8 that application of the chain rule to the term $\partial_x f(q)$ leads to the quasi-linear form of (14.1), namely

$$\partial_t q + \lambda(q) \partial_x q = 0, \quad \lambda(q) = q. \quad (14.2)$$

For smooth solutions both formulations (14.1) and (14.2) are equivalent, the first written in conservation-law form and the latter in non-conservative form, both having the same variable as the unknown. Now, under the assumption of smooth solutions, we define the new variable $v = q^m$, where the exponent m is a positive integer. If $q(x, t)$ is a solution of (14.2) then $v(x, t)$ is a solution of the **conservation law**

$$\partial_t v + \partial_x g(v) = 0, \quad g(v) = \left(\frac{m}{m+1} \right) v^{(\frac{m+1}{m})}. \quad (14.3)$$

Note however that discontinuous solutions of (14.1) are not necessarily solutions of (14.3). Discontinuous solutions of the Burgers equation satisfy the Rankine-Hugoniot condition

$$s(q_R - q_L) = f(q_R) - f(q_L), \quad (14.4)$$

leading to the classical shock speed for the Burgers equation

$$s = \frac{f(q_R) - f(q_L)}{q_R - q_L} = \frac{1}{2}(q_L + q_R). \quad (14.5)$$

The Rankine-Hugoniot condition for the conservation law (14.3) gives the shock speed

$$r = \left(\frac{m}{m+1} \right) \left(\frac{(q_R)^{m+1} - (q_L)^{m+1}}{(q_R)^m - (q_L)^m} \right). \quad (14.6)$$

The reader can verify that the shock speeds s in (14.5) and r in (14.6) are distinct, unless $m = 1$, for which one has the classical Burgers equation (14.1).

14.1.2 Example 2: The Blood Flow Equations

As a second example we consider the homogenous (no source term) one-dimensional blood flow equations for arteries, first written in terms of physically conserved variables and in conservation-law form

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0}, \quad (14.7)$$

where

$$\mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \equiv \begin{bmatrix} A \\ Au \end{bmatrix}, \quad \mathbf{F}(\mathbf{Q}) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \equiv \begin{bmatrix} Au \\ Au^2 + \gamma A^{3/2} \end{bmatrix}. \quad (14.8)$$

Here $\mathbf{Q}$ is the vector of conserved variables; A is the vessel cross-sectional area and u is velocity. $\mathbf{F}(\mathbf{Q})$ is the physical flux vector and γ is a constant. Full details on the equations are found in Chaps. 8 and 9.

Under the assumption of smoothness, these equations can also be written in terms of **new variables**, namely cross-sectional area A and velocity u , also in **conservation-law form** as follows

$$\partial_t \mathbf{V} + \partial_x \mathbf{G}(\mathbf{V}) = \mathbf{0}, \quad (14.9)$$

where

$$\mathbf{V} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \equiv \begin{bmatrix} A \\ u \end{bmatrix}, \quad \mathbf{G}(\mathbf{V}) = \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} \equiv \begin{bmatrix} Au \\ \frac{1}{2}u^2 + 3\gamma A^{1/2} \end{bmatrix}. \quad (14.10)$$

Obviously $v_1 = A$ is a truly conserved variable, but the new variable $v_2 = u$, though a **conserved variable** in system in (14.9)–(14.10) is only so in a **mathematical sense**, not in a **physical sense**. Therefore the new conservative formulation (14.9)–(14.10) is conservative purely in a mathematical sense, not in a physical sense. The purpose of this second example has been to emphasise that much care is required when formulating the governing equations. The much appreciated conservative formulation might not be physically correct.

For smooth solutions all formulations are equivalent, but not so for generalised solutions admitting shock waves (elastic jumps). In fact it can be demonstrated that shock waves from formulation (14.9)–(14.10) are different from those from the physically conservative formulation (14.7)–(14.8); they have different wave strengths and different wave speeds. Great care is needed when adopting formulations of the governing equations for real practical purposes. See exercises at the end of the chapter.

Exercise 14.1 *Verify that under the assumption of smoothness, equations (14.7)–(14.8) can actually be written as equations (14.9)–(14.10). Justify every step.*

There are hyperbolic systems that do not have a conservation-law form at all, for which there is no option but to accept non-conservative forms of the equations and, consequently, consider associated non-conservative numerical methods. This is the subject of the rest of this chapter.

14.2 Hyperbolic Systems in Non-conservative Form

In this chapter we are primarily interested in first-order hyperbolic systems of $m \times m$ equations written in non-conservative form

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{0}. \quad (14.11)$$

Here $\mathbf{Q}$ is the vector of unknowns and $\mathbf{A}(\mathbf{Q})$ is the coefficient matrix, both given as follows

$$\mathbf{Q} = \begin{bmatrix} q_1 \\ q_2 \\ \dots \\ q_m \end{bmatrix}, \quad \mathbf{A}(\mathbf{Q}) = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,m} \\ a_{2,1} & a_{2,2} & \dots & a_{2,m} \\ \dots & \dots & \dots & \dots \\ a_{m,1} & a_{m,2} & \dots & a_{m,m} \end{bmatrix}. \quad (14.12)$$

Depending on the context, $\mathbf{Q}$ may well represent a vector of physically conserved quantities.

14.2.1 Some Theoretical Considerations

Recall that conservative systems

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0} \quad (14.13)$$

can be written in quasilinear form as in (14.11)–(14.12), in which case the coefficient matrix $\mathbf{A}(\mathbf{Q})$ is the Jacobian matrix of a physical flux function $\mathbf{F}(\mathbf{Q})$, that is

$$\mathbf{A}(\mathbf{Q}) = \frac{\partial \mathbf{F}(\mathbf{Q})}{\partial \mathbf{Q}}. \quad (14.14)$$

See Chaps. 8 and 13 for background on conservative hyperbolic systems.

As already pointed out, the conservative and non-conservative formulations are equivalent if solutions are smooth. The difference comes in when discontinuous solutions are present. In this case, the use of the conservative form is mandatory, for various reasons. First, theoretically, the conservative form defines shocks through the classical Rankine-Hugoniot conditions, for which the existence of the flux function is essential. Then, for conservative systems there are many conservative methods available to solve the equations reliably.

The Lax-Wendroff theorem [9] provides some of the theoretical backing for conservative methods. This theorem states that if the conservative method is convergent, then it will converge to a weak solution of the conservation laws. But note that this theorem does not say that a conservative method is convergent; this is another long story. Moreover, it does not say that the converged weak solution is the correct, physically admissible weak solution. Harten [10] later proved that if the conservative method is convergent and **entropy satisfying**, then the converged weak solution is an entropy, physically admissible solution of the conservative system.

Up to this point there is no theoretical statement that says that the only methods that have these properties are conservative methods. But 34 years after the Lax-Wendroff theorem was stated and proved, Hou and LeFloch [11] demonstrated that non-conservative methods, if convergent, will converge to the wrong solution, in the presence of a discontinuity.

In this chapter we are primarily interested in the case in which the coefficient matrix $\mathbf{A}(\mathbf{Q})$ in (14.11) is **not** the Jacobian of some flux function $\mathbf{F}(\mathbf{Q})$. This means that (14.11)–(14.12) cannot be recast in conservation-law form (14.13). The term

$$\mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} \quad (14.15)$$

in (14.11) is called **non-conservative product**. The theory of non-conservative systems like (14.11)–(14.12) is still in the early stages of development. For details see the seminal paper of Dal Maso, LeFloch and Murat [12].

The development of computational methods for non-conservative hyperbolic systems to a status comparable to that of conservative methods remains a challenge for

numerical analysts, although some progress has been made, as we shall see in the rest of this chapter. Nonetheless, a measure of caution is prudent when approaching the subject of numerical methods for non-conservative hyperbolic systems.

14.2.2 The Blood Flow Equations in Non-conservative Form

As a simple example of a system written in non-conservative form, we consider the blood flow equations for arteries (14.7)–(14.8) with constant mechanical and geometric parameters, augmented by an advection equation for a passive scalar $\eta(x, t)$. See Chaps. 8 and 9 for background. In (14.7)–(14.8), the equations are written in conservation-law form. But now, for illustrative purposes, we choose to write the equations in non-conservative form but still formulated in terms of the physically conserved variables, namely

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}) , \quad (14.16)$$

with

$$\mathbf{Q} = [q_1, q_2, q_3]^T := [A, Au, A\eta]^T , \quad \mathbf{S}(\mathbf{Q}) = [s_1, s_2, s_3]^T , \quad (14.17)$$

where $\eta(x, t)$ is a passive scalar and $\mathbf{S}(\mathbf{Q})$ is the source term, assumed to be zero, for the time being. The coefficient matrix $\mathbf{A}$ is

$$\mathbf{A}(\mathbf{Q}) = \begin{bmatrix} 0 & 1 & 0 \\ -u^2 + c^2 & 2u & 0 \\ -u\eta & \eta & u \end{bmatrix} . \quad (14.18)$$

Exercise 14.2 Verify that the coefficient matrix $\mathbf{A}(\mathbf{Q})$ in (14.18) is the Jacobian of a flux function $\mathbf{F}(\mathbf{Q})$ and explicitly state $\mathbf{F}(\mathbf{Q})$.

It is easily shown that the eigenvalues of $\mathbf{A}(\mathbf{Q})$ are

$$\lambda_1 = u - c , \quad \lambda_2 = u , \quad \lambda_3 = u + c , \quad (14.19)$$

with wavespeed given as

$$c = \sqrt{\frac{A}{\rho} \frac{\partial p}{\partial A}} = \sqrt{\frac{3}{2} \gamma \sqrt{A}} \quad (14.20)$$

for the tube law of the form

$$p = p_e + \mathbb{P}(A) . \quad (14.21)$$

Here p_e is the external pressure and $\mathbb{P}(A)$ is the **transmural pressure**; $\mathbb{P}(A)$ is prescribed by a tube law and may also depend on parameters. See Chap. 10.

The right eigenvectors corresponding to the eigenvalues (14.19) are

$$\mathbf{R}_1 = \begin{bmatrix} 1 \\ u - c \\ \eta \end{bmatrix}, \quad \mathbf{R}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{R}_3 = \begin{bmatrix} 1 \\ u + c \\ \eta \end{bmatrix}, \quad (14.22)$$

where the scaling factors have been chosen to be unity.

Remark: non-conservative form. The non-conservative formulation (14.16)–(14.18) is a choice here. However, when extending the blood flow equations to more realistic situations involving, for example, **variable geometric and mechanical parameters**, the non-conservative form arises naturally and there is no conservative counterpart available. The non-conservative system (14.16)–(14.18) is convenient for testing numerical methods intended for solving general systems in non-conservative form. See Sect. 14.7.

14.2.3 Blood Flow in Vessels with Discontinuous Parameters

Here we consider the equations for blood flow in vessels with **variable geometric and mechanical parameters**. Moreover, we consider the case in which variations of parameters are discontinuous. This leads to a formulation of the equations that is **genuinely non-conservative system**, in the sense that a flux function to express the full system in conservation-law form does not exist. Here we study the case in which the Young's modulus of elasticity K of the arterial vessels is discontinuous. Succinctly, the basic equations read

$$\left. \begin{aligned} \partial_t A + \partial_x (Au) &= 0, \\ \partial_t (Au) + \partial_x (\alpha Au^2) + \frac{A}{\rho} \partial_x p &= s_2. \end{aligned} \right\} \quad (14.23)$$

To close system (14.23) we adopt the tube law

$$p = p_e + \mathbb{P}(A; A_0, K), \quad (14.24)$$

with

$$\left. \begin{aligned} \mathbb{P}(A; A_0, K) &= K(x) \phi(A; A_0), \\ K(x) &= \frac{\sqrt{\pi}}{1 - \nu^{\frac{1}{2}}} \frac{E(x) h_0(x)}{\sqrt{A_0}}, \\ \phi(A; A_0) &= \left(\frac{A}{A_0} \right)^m - \left(\frac{A}{A_0} \right)^n. \end{aligned} \right\} \quad (14.25)$$

Definitions for all quantities and parameters are found in Chaps. 6, 8, 9 and 10. Here we choose $m = 1/2$, $n = 0$, as for arteries, and $\nu = 1/2$. We also assume that A_0 and h_0 are constant. Here we adopt the approach proposed by Toro and Siviglia [3] to deal with the discontinuous parameter K , whereby an additional, trivial, PDE is added to the equations. The resulting 3×3 system is

$$\left. \begin{aligned} \partial_t A + \partial_x(Au) &= 0, \\ \partial_t(Au) + \partial_x(\alpha Au^2) + \frac{A}{\rho} \left(\partial_x p_e + \frac{\partial p}{\partial A} \partial_x A + \frac{\partial p}{\partial K} \partial_x K \right) &= s_2, \\ \partial_t K &= 0. \end{aligned} \right\} \quad (14.26)$$

The pressure gradient $\partial_x p$ has been expanded in terms of the tube law (14.24); subscripts are short notation for partial derivatives.

In the next section we address some theoretical issues regarding numerical methodologies to solve systems in non-conservative form.

14.2.4 Non-conservative Products and Path Functions

This section concerns non-conservative products arising in PDEs written in non-conservative form. Following Dal Maso, LeFloch and Murat [12], a rigorous way to deal with non-conservative products is through the introduction of a **family of paths**. A path in phase space is a Lipschitz continuous function

$$\psi : [0, 1] \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n \quad (14.27)$$

that joins two prescribed points $\mathbf{Q}_L$ and $\mathbf{Q}_R$ in a subset of $\mathbb{R}^n$, with

$$\psi(0; \mathbf{Q}_L, \mathbf{Q}_R) = \mathbf{Q}_L, \quad \psi(1; \mathbf{Q}_L, \mathbf{Q}_R) = \mathbf{Q}_R. \quad (14.28)$$

An obvious example of a path is the **canonical path** defined as

$$\psi(s, \mathbf{Q}_L, \mathbf{Q}_R) = \mathbf{Q}_L + s(\mathbf{Q}_R - \mathbf{Q}_L), \quad s \in [0, 1]. \quad (14.29)$$

Admissible path functions are expected to be Lipschitz continuous and to satisfy the following regularity properties:

$$\left. \begin{aligned} \left| \frac{\partial}{\partial s} \psi(s; \mathbf{Q}_L, \mathbf{Q}_R) \right| &\leq C_1 |\mathbf{Q}_R - \mathbf{Q}_L|, \\ \left| \frac{\partial}{\partial s} \psi(s; \mathbf{Q}_L^1, \mathbf{Q}_R^1) - \frac{\partial}{\partial s} \psi(s; \mathbf{Q}_L^2, \mathbf{Q}_R^2) \right| &\leq C_2 (|\mathbf{Q}_L^1 - \mathbf{Q}_L^2| + |\mathbf{Q}_R^1 - \mathbf{Q}_R^2|), \end{aligned} \right\} \quad (14.30)$$

for almost every $s \in [0, 1]$, where C_1 and C_2 are positive constants and the states $\mathbf{Q}_L, \mathbf{Q}_L^1, \mathbf{Q}_L^2, \mathbf{Q}_R, \mathbf{Q}_R^1, \mathbf{Q}_R^2$ lie in a bounded subset of $\mathbb{R}^n$. See [13], for further details

Following the theory of Dal Maso, LeFloch and Murat [12], consider now an isolated shock wave solution of speed S of the non-conservative system (14.11), given as

$$\mathbf{Q}(x, t) = \begin{cases} \mathbf{Q}_L & \text{if } x/t < S, \\ \mathbf{Q}_R & \text{if } x/t > S. \end{cases} \quad (14.31)$$

Across the discontinuity, a weak solution must satisfy the **Generalised Rankine-Hugoniot** condition

$$\int_0^1 S \frac{\partial}{\partial s} \psi(s; \mathbf{Q}_L, \mathbf{Q}_R) ds = \int_0^1 \mathbf{A}(\psi(s; \mathbf{Q}_L, \mathbf{Q}_R)) \frac{\partial}{\partial s} \psi(s; \mathbf{Q}_L, \mathbf{Q}_R) ds. \quad (14.32)$$

That is

$$S(\mathbf{Q}_R - \mathbf{Q}_L) = \int_0^1 \mathbf{A}(\psi(s; \mathbf{Q}_L, \mathbf{Q}_R)) \frac{\partial}{\partial s} \psi(s; \mathbf{Q}_L, \mathbf{Q}_R) ds. \quad (14.33)$$

In the case in which the coefficient matrix $\mathbf{A}(\mathbf{Q})$ in (14.11) is the Jacobian of a flux function $\mathbf{F}(\mathbf{Q})$, that is if

$$\mathbf{A}(\mathbf{Q}) = \frac{\partial \mathbf{F}}{\partial \mathbf{Q}}, \quad (14.34)$$

then the **Generalised Rankine-Hugoniot** condition reduces to the classical Rankine-Hugoniot condition. This follows from a change of variables and the fundamental theorem of calculus. That is

$$S(\mathbf{Q}_R - \mathbf{Q}_L) = \int_0^1 \mathbf{A}(\psi) \frac{\partial \psi}{\partial s} ds = \int_0^1 \frac{\partial \mathbf{F}(\psi)}{\partial \psi} \frac{\partial \psi}{\partial s} ds = \mathbf{F}(\mathbf{Q}_R) - \mathbf{F}(\mathbf{Q}_L). \quad (14.35)$$

Note that if the special case of the canonical path (14.29) is used in (14.33), then the generalised Rankine-Hugoniot condition gives

$$S(\mathbf{Q}_R - \mathbf{Q}_L) = \left(\int_0^1 \mathbf{A}(\mathbf{Q}_L + s(\mathbf{Q}_R - \mathbf{Q}_L)) ds \right) (\mathbf{Q}_R - \mathbf{Q}_L). \quad (14.36)$$

This expression for the Rankine-Hugoniot condition could be expressed as

$$\left. \begin{aligned} S(\mathbf{Q}_R - \mathbf{Q}_L) &= \mathbf{M}(\mathbf{Q}_R - \mathbf{Q}_L) \\ \mathbf{M} &= \int_0^1 \mathbf{A}(\mathbf{Q}_L + s(\mathbf{Q}_R - \mathbf{Q}_L)) ds \end{aligned} \right\}. \quad (14.37)$$

The coefficient matrix $\mathbf{M}$ is a path integral average of the coefficient matrix $\mathbf{A}$ of the system (14.11).

14.2.5 Path-Conservative Methods: The Approach

A new class of numerical methods for solving non-conservative systems has been put forward in recent years [13–15]. These schemes depend on the choice of a path ψ in phase space and have been termed **path-conservative schemes**. In this section we shall present several path-conservative schemes, including upwind and centred approaches. The update formula for a **high-order** path-conservative scheme is

$$\mathbf{Q}_i^{n+1} = \mathbf{Q}_i^n - \frac{\Delta t}{\Delta x} [\mathbf{D}_{i-\frac{1}{2}}^+ + \mathbf{D}_{i+\frac{1}{2}}^-] + \Delta t \mathbf{S}_i - \Delta t \mathbf{H}_i^n. \quad (14.38)$$

Here $\mathbf{D}_{i+\frac{1}{2}}^-$ and $\mathbf{D}_{i+\frac{1}{2}}^+$ are called **fluctuations, or increments** [16] and are related to the numerical flux in the case of a conservative system. $\mathbf{S}_i$ is the usual numerical source. The term $\mathbf{H}_i^n$ is present only in schemes of accuracy greater than one and is defined as

$$\mathbf{H}_i^n = \frac{1}{\Delta t \Delta x} \int_{t_n}^{t_{n+1}} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{A}(\mathbf{P}_i(x, t)) \partial_x \mathbf{P}_i(x, t) dx dt. \quad (14.39)$$

Here $\mathbf{P}_i(x, t)$ is a polynomial representation of the solution in cell I_i , which results from a **spatial reconstruction procedure**, studied in Chap. 15. In the first-order case, of interest in this chapter, the reconstruction reduces to piece-wise constant cell averages; hence $\partial_x \mathbf{P}_i(x, t) = \mathbf{0}$ and therefore $\mathbf{H}_i^n = \mathbf{0}$ in (14.38).

The fluctuations $\mathbf{D}_{i+\frac{1}{2}}^-$ and $\mathbf{D}_{i+\frac{1}{2}}^+$, accounting for the absence of a flux function, are expected to satisfy the **consistency condition**

$$\mathbf{D}_{i+\frac{1}{2}}^-(\mathbf{Q}, \dots, \mathbf{Q}) = \mathbf{0}, \quad \mathbf{D}_{i+\frac{1}{2}}^+(\mathbf{Q}, \dots, \mathbf{Q}) = \mathbf{0} \quad (14.40)$$

and the **compatibility condition**

$$\mathbf{D}_{i+\frac{1}{2}}^- + \mathbf{D}_{i+\frac{1}{2}}^+ = \int_0^1 \mathbf{A}(\psi(s; \mathbf{Q}_i^n, \mathbf{Q}_{i+1}^n)) \frac{\partial}{\partial s} \psi(s; \mathbf{Q}_i^n, \mathbf{Q}_{i+1}^n) ds. \quad (14.41)$$

As already stated, the path function $\psi(s; \mathbf{Q}_i^n, \mathbf{Q}_{i+1}^n)$, with $s \in [0, 1]$, joins $\mathbf{Q}_i^n$ to $\mathbf{Q}_{i+1}^n$ and

$$\psi(0; \mathbf{Q}_i^n, \mathbf{Q}_{i+1}^n) = \mathbf{Q}_i^n, \quad \psi(1; \mathbf{Q}_i^n, \mathbf{Q}_{i+1}^n) = \mathbf{Q}_{i+1}^n. \quad (14.42)$$

While many choices for the path are available, for most of this chapter we assume the **canonical path** (14.29), re-written here as

$$\psi(s; \mathbf{Q}_i^n, \mathbf{Q}_{i+1}^n) = \mathbf{Q}_i^n + s(\mathbf{Q}_{i+1}^n - \mathbf{Q}_i^n) . \quad (14.43)$$

In analogy to conservative methods, which require an intercell numerical flux $\mathbf{F}_{i+\frac{1}{2}}$ to be determined, path-conservative methods (14.38) are defined once the path $\psi(s; \mathbf{Q}_i^n, \mathbf{Q}_{i+1}^n)$ and the fluctuations $\mathbf{D}_{i+\frac{1}{2}}^-, \mathbf{D}_{i+\frac{1}{2}}^+$ are specified.

Several path-conservative methods will be presented. Each scheme is characterised by the way the fluctuations $\mathbf{D}_{i+\frac{1}{2}}^-$ and $\mathbf{D}_{i+\frac{1}{2}}^+$ are determined, with potential variations on the path function too.

14.3 DOT Path-Conservative Scheme

Here we present a path-conservative scheme (14.38) to solve (14.11), which is a variant of the conservative Dumbser-Osher-Toro (DOT) scheme presented for conservative systems in Chap. 13. See [17, 18] for details. We shall study two versions of DOT, namely a conventional non-conservative form and a well-balanced non-conservative version.

14.3.1 Conventional Version: DOT+NC

The path-conservative scheme of this section is analogous to the conservative scheme, for which the reader is advised to revise Chap. 13 before proceeding further. In the DOT scheme for non-conservative systems, the fluctuations are defined as follows

$$\mathbf{D}_{i+\frac{1}{2}}^\pm = \frac{1}{2} \int_0^1 [\mathbf{A}(\psi(s; \mathbf{Q}_0, \mathbf{Q}_1)) \pm |\mathbf{A}(\psi(s; \mathbf{Q}_0, \mathbf{Q}_1))|] \frac{\partial \psi}{\partial s} ds . \quad (14.44)$$

As in Chap. 13, this path-conservative method makes use of the local Riemann problem. Just a note on notation. The original Osher-Solomon [19] notation for the initial conditions for the local Riemann problem was $\mathbf{Q}_0$ and $\mathbf{Q}_1$, instead of $\mathbf{Q}_L$ and $\mathbf{Q}_R$ or $\mathbf{Q}_i^n$ and $\mathbf{Q}_{i+1}^n$.

The path function ψ in (14.44) is completely general. However, if the canonical path (14.43) is chosen, then (14.44) becomes

$$\mathbf{D}_{i+\frac{1}{2}}^\pm = \frac{1}{2} \left(\int_0^1 [\mathbf{A}(\psi(s; \mathbf{Q}_0, \mathbf{Q}_1)) \pm |\mathbf{A}(\psi(s; \mathbf{Q}_0, \mathbf{Q}_1))|] ds \right) (\mathbf{Q}_1 - \mathbf{Q}_0) . \quad (14.45)$$

Now the integrals in (14.45) are computed **numerically** along the path ψ using a Gauss quadrature rule with G points s_j and associated weights ω_j in the unit interval $[0, 1]$. We obtain

$$\mathbf{D}_{i+\frac{1}{2}}^{\pm} = \frac{1}{2} \left(\sum_{j=1}^G \omega_j [\mathbf{A}(\psi(s_j; \mathbf{Q}_0, \mathbf{Q}_1)) \pm |\mathbf{A}(\psi(s_j; \mathbf{Q}_0, \mathbf{Q}_1))|] \right) (\mathbf{Q}_1 - \mathbf{Q}_0) . \quad (14.46)$$

For details on the definition of the **absolute value matrix** $|\mathbf{A}|$ see Chap. 13. More details are given in the original references [17, 18]. Recommended choices for the computational parameters in (14.46) are

$$s_1 = \frac{1}{2} - \frac{\sqrt{15}}{10}, s_2 = \frac{1}{2}, s_3 = \frac{1}{2} + \frac{\sqrt{15}}{10}, \omega_1 = \frac{5}{18}, \omega_2 = \frac{8}{18}, \omega_3 = \frac{5}{18} . \quad (14.47)$$

The first-order upwind, path-conservative method (14.38) with (14.46) for the homogeneous equations, no source term, has been completely defined. The method, supplemented with a way to account for source terms and in higher-order mode has been applied in several areas of study. Here we emphasise its utilization in the global, closed loop mathematical model for the human circulation in [8, 20, 21]. The version of DOT+NC utilised in these cited works is a modified version, the so called **well-balanced** version, as explained below.

14.3.2 Well-Balanced Version: DOT+NC+WB

As presented, the scheme DOT+NC is not well-balanced. The concept of well-balanced schemes was originally introduced for balance laws written in conservation-law form, for which the scheme is expected to preserve the balance between the flux gradient and the source term, when the time derivative of the vector of unknowns vanishes, that is in the steady state. Upwinding the source term is a key ingredient in adopted strategies [22, 23]. See [24] for an up-to-date overview of the subject. For systems written in non-conservative form the definition of well-balanced scheme is suitably modified [25].

Briefly, for the case of blood flow equations the preserved quantities are mass and total pressure, namely

$$q = \text{const} , \quad p + \frac{1}{2} \rho u^2 = \text{const} . \quad (14.48)$$

It has been shown [4, 26] that in this case the canonical integration path (14.43) has to be modified, but only its component corresponding to the cross-sectional area. A . The resulting path reads

$$\Psi(\mathbf{Q}_0, \mathbf{Q}_1, s) = \begin{bmatrix} \bar{A}(s) \\ \bar{q}(s) \\ \bar{K}(s) \end{bmatrix} = \begin{bmatrix} \bar{p}(s)^{-1} \\ q_0 + s(q_1 - q_0) \\ K_0 + s(K_1 - K_0) \end{bmatrix}. \quad (14.49)$$

Here $\bar{p}$ is defined as

$$\bar{p}(s) = p(A_0; K_0) + s(p(A_1; K_1) - p(A_0; K_0)) \quad (14.50)$$

and A_0 denotes the area corresponding to the data state $\mathbf{Q}_0$. Once $\bar{p}(s)$ is known, $\bar{A}(s)$ is computed by finding the unique root of

$$G(\bar{A}(s)) := \bar{p}(s) - (p_e + \mathbb{P}(\bar{A}(s); A_0, \bar{K}(s))) = 0, \quad (14.51)$$

noting that for $n = 0$ in the tube law (14.21)–(14.22) the solution is direct.

The well-balanced version of DOT+NC just described will be denoted as DOT+NC+WB. The scheme can be generalized for an arbitrary number of discontinuous parameters. For the case of 3 varying parameters see [4, 27]. See [28] for background.

14.4 The Parés-Muñoz-Godunov Scheme

14.4.1 Framework of PMG Schemes

Parés and Muñoz [29] proposed a Godunov-type, path-conservative method (14.38) to solve (14.11) with the following definitions for the fluctuations

$$\left. \begin{aligned} \mathbf{D}_{i+\frac{1}{2}}^- &= \int_0^1 \mathbf{A}(\psi(s; \mathbf{Q}_i^n, \mathbf{Q}_{i+\frac{1}{2}}(0))) \frac{\partial}{\partial s} \psi(s, \mathbf{Q}_i^n, \mathbf{Q}_{i+\frac{1}{2}}(0)) ds, \\ \mathbf{D}_{i+\frac{1}{2}}^+ &= \int_0^1 \mathbf{A}(\psi(s; \mathbf{Q}_{i+\frac{1}{2}}(0), \mathbf{Q}_{i+1}^n)) \frac{\partial}{\partial s} \psi(s, \mathbf{Q}_{i+\frac{1}{2}}(0), \mathbf{Q}_{i+1}^n) ds. \end{aligned} \right\} \quad (14.52)$$

Here $\mathbf{Q}_{i+\frac{1}{2}}(x/t)$ denotes the solution of the local Riemann problem

$$\left. \begin{aligned} \partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} &= \mathbf{0}, \\ \mathbf{Q}(x, 0) &= \begin{cases} \mathbf{Q}_i^n & \text{if } x < 0, \\ \mathbf{Q}_{i+1}^n & \text{if } x > 0. \end{cases} \end{aligned} \right\} \quad (14.53)$$

$\mathbf{Q}_{i+\frac{1}{2}}(0)$ is the **Godunov state**, the determination of which requires solving the Riemann problem (14.53) to find $\mathbf{Q}_{i+\frac{1}{2}}(x/t)$ and a solution sampling procedure to find $\mathbf{Q}_{i+\frac{1}{2}}(0)$. See Chap. 13.

A major undertaking here is to find a Riemann solver for (14.53). For the non-conservative case there are less choices available than for the conservative case. The scheme just described will be denoted as the **PMG scheme**, for Parés, Muñoz and Godunov.

14.4.2 Implementing a PMG Scheme

On the assumption that the Godunov state $\mathbf{Q}_{i+\frac{1}{2}}(0)$ is known, the implementation of the scheme proceeds as follows. Assuming the **canonical paths** in (14.52) we have

$$\left. \begin{aligned} \psi(s; \mathbf{Q}_i^n, \mathbf{Q}_{i+\frac{1}{2}}(0)) &= \mathbf{Q}_i^n + s[\mathbf{Q}_{i+\frac{1}{2}}(0) - \mathbf{Q}_i^n], \\ \psi(s; \mathbf{Q}_{i+\frac{1}{2}}(0), \mathbf{Q}_{i+1}^n) &= \mathbf{Q}_{i+\frac{1}{2}}(0) + s[\mathbf{Q}_{i+1}^n - \mathbf{Q}_{i+\frac{1}{2}}(0)]. \end{aligned} \right\} \quad (14.54)$$

Then, from (14.52) and (14.54) we have

$$\mathbf{D}_{i+\frac{1}{2}}^- = \hat{\mathbf{A}}_{i+\frac{1}{2}}^- [\mathbf{Q}_{i+\frac{1}{2}}(0) - \mathbf{Q}_i^n]; \quad \mathbf{D}_{i+\frac{1}{2}}^+ = \hat{\mathbf{A}}_{i+\frac{1}{2}}^+ [\mathbf{Q}_{i+1}^n - \mathbf{Q}_{i+\frac{1}{2}}(0)]. \quad (14.55)$$

Here $\hat{\mathbf{A}}_{i+\frac{1}{2}}^-$ and $\hat{\mathbf{A}}_{i+\frac{1}{2}}^+$ are numerical approximations to the respective integrals in (14.52), that is

$$\left. \begin{aligned} \hat{\mathbf{A}}_{i+\frac{1}{2}}^- &\approx \int_0^1 \mathbf{A}(\psi(s; \mathbf{Q}_i^n, \mathbf{Q}_{i+\frac{1}{2}}(0))) ds, \\ \hat{\mathbf{A}}_{i+\frac{1}{2}}^+ &\approx \int_0^1 \mathbf{A}(\psi(s; \mathbf{Q}_{i+\frac{1}{2}}(0), \mathbf{Q}_{i+1}^n)) ds. \end{aligned} \right\} \quad (14.56)$$

Using a Gaussian-Legendre rule to evaluate the integrals in (14.56) numerically we obtain

$$\left. \begin{aligned} \hat{\mathbf{A}}_{i+\frac{1}{2}}^- &= \sum_{j=1}^G \omega_j \mathbf{A}(\psi(s_j; \mathbf{Q}_i^n, \mathbf{Q}_{i+\frac{1}{2}}(0))), \\ \hat{\mathbf{A}}_{i+\frac{1}{2}}^+ &= \sum_{j=1}^G \omega_j \mathbf{A}(\psi(s_j; \mathbf{Q}_{i+\frac{1}{2}}(0), \mathbf{Q}_{i+1}^n)). \end{aligned} \right\} \quad (14.57)$$

For the integration points and weights in the quadrature formulae (14.57) we use (14.47).

The scheme, as formulated, is very general. The specificity of the problem of interest will enter in resolving the pending task, namely finding a Riemann solver to determine the Godunov state $\mathbf{Q}_{i+\frac{1}{2}}(0)$ to be used in (14.55) and (14.57) to calculate the fluctuations. There appears to be little reported experience on practical implementations of this kind of Godunov-type path-conservative methods. A recent application to the modelling of sediment transport in rivers is found in [30].

In the next section we study a method to provide a Godunov state for use in the PMG scheme just described, as applied to the blood flow equations.

14.5 TV Splitting Schemes

The TV flux vector splitting method, due to Toro and Vázquez [2], was first formulated for partial differential equations written in conservation-law form. This approach was introduced in Chap. 13 of this book for the blood flow equations in conservative form

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}) . \quad (14.58)$$

See Toro et al. [31] for preliminary results for the blood flow equations.

The TV approach for a conservative system splits the physical flux function $\mathbf{F}(\mathbf{Q})$ into an advection flux and a pressure flux, with corresponding advection and pressure systems to compute their respective numerical fluxes that will make up the numerical flux for the full system. For systems written in non-conservative form there is no flux function to split. However, one can still apply the idea to suitably identified advection terms and pressure terms, as explained below.

14.5.1 The TV Framework

Here we tailor the TV approach to solve systems of balance laws written in the mixed form

$$\partial_t \mathbf{Q} + \partial_x \mathcal{A}(\mathbf{Q}) + \mathcal{P}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}) . \quad (14.59)$$

This system has a conservative part $\partial_x \mathcal{A}(\mathbf{Q})$, a non-conservative part $\mathcal{P}(\mathbf{Q}) \partial_x \mathbf{Q}$ and the source term $\mathbf{S}(\mathbf{Q})$. As written, part of the advection term is expressed in conservative form, with flux function $\mathcal{A}(\mathbf{Q})$. Indeed, there is always the option of applying the chain rule to this advection term to obtain a non-conservative form, namely

$$\partial_x \mathcal{A}(\mathbf{Q}) = \frac{\partial \mathcal{A}}{\partial \mathbf{Q}} \partial_x \mathbf{Q} := \mathcal{B}(\mathbf{Q}) \partial_x \mathbf{Q} . \quad (14.60)$$

$\mathcal{B}(\mathbf{Q})$ is the Jacobian matrix of $\mathcal{A}(\mathbf{Q})$. Then, the full system (14.59) could be written in the general non-conservative form, as assumed in previous sections, namely

$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}) , \quad (14.61)$$

with coefficient matrix defined as

$$\mathbf{A}(\mathbf{Q}) := \mathcal{B}(\mathbf{Q}) + \mathcal{P}(\mathbf{Q}) . \quad (14.62)$$

One could apply the previously described non-conservative methods to the full Eq. (14.61). Alternatively, one could apply the TV splitting framework and write two separate non-conservative systems, namely

$$\left. \begin{aligned} \partial_t \mathbf{Q} + \mathcal{B}(\mathbf{Q}) \partial_x \mathbf{Q} &= \mathbf{0} && : \text{Advection non-conservative system} , \\ \partial_t \mathbf{Q} + \mathcal{P}(\mathbf{Q}) \partial_x \mathbf{Q} &= \mathbf{S}(\mathbf{Q}) && : \text{Pressure non-conservative system} . \end{aligned} \right\} \quad (14.63)$$

Application of the TV Approach

In what follows, we shall assume the mixed conservative non-conservative form (14.59). The TV splitting approach will then **split** advection terms from pressure-type terms. The advection term will be kept in conservative form while the pressure term will be kept in non-conservative form. By doing so we use the advection and pressure terms to split the full system of PDEs into two sub-systems of PDEs as follows

$$\left. \begin{aligned} \partial_t \mathbf{Q} + \partial_x \mathcal{A}(\mathbf{Q}) &= \mathbf{0} && : \text{Conservative system for advection} , \\ \partial_t \mathbf{Q} + \mathcal{P}(\mathbf{Q}) \partial_x \mathbf{Q} &= \mathbf{S}(\mathbf{Q}) && : \text{Non-conservative system for pressure} . \end{aligned} \right\} \quad (14.64)$$

In (14.64) we have allocated the source term $\mathbf{S}(\mathbf{Q})$ to the pressure system. However, there may be cases in which it is more convenient to allocate it to the advection system. In fact, this is an issue worth-investigating further.

To solve the mixed-formulation equations (14.59) we propose the mixed conservative/non-conservative simultaneous update formula

$$\mathbf{Q}_i^{n+1} = \mathbf{Q}_i^n - \frac{\Delta t}{\Delta x} [\mathcal{A}_{i+\frac{1}{2}} - \mathcal{A}_{i-\frac{1}{2}}] - \frac{\Delta t}{\Delta x} [\mathbf{D}_{i-\frac{1}{2}}^+ + \mathbf{D}_{i+\frac{1}{2}}^-] + \Delta t \mathbf{S}_i - \Delta t \mathbf{H}_i^n , \quad (14.65)$$

where

1. $\mathcal{A}_{i+\frac{1}{2}}$ is the **numerical flux** for the advection part in conservative form in (14.64).
2. $\mathbf{D}_{i+\frac{1}{2}}^\pm$ are the **fluctuations** in the path-conservative scheme for the non-conservative part in (14.64).
3. $\mathbf{S}_i$ is the numerical source.
4. $\mathbf{H}_i^n$ arises in schemes of accuracy greater than one. As this chapter deals exclusively with first-order schemes, the term $\mathbf{H}_i^n$ will be assumed to be zero.

The whole point of the TV splitting approach is its ability to provide simplified and effective schemes for computing the needed items in formula (14.65), that is the fluctuations $\mathbf{D}_{i+\frac{1}{2}}^\pm$ and the numerical flux $\mathcal{A}_{i+\frac{1}{2}}$, as we shall see. There is also evidence, even if partial, that the TV splitting approach is advantageous in dealing with source terms, notably, of the discontinuous type.

Next we consider a prototype of a genuinely non-conservative system written in the mixed form (14.59), for which the application of the TV splitting will be illustrated in full detail.

14.5.2 TV Schemes for a Non-conservative System

We consider a model for blood flow in vessels with discontinuous stiffness coefficient K . This gives rise to the non-conservative system (14.26), which can be written in the mixed form (14.59), namely

$$\partial_t \mathbf{Q} + \partial_x \mathcal{A}(\mathbf{Q}) + \mathcal{P}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}) , \quad (14.66)$$

with

$$\left. \begin{aligned} \mathbf{Q} &= \begin{bmatrix} A \\ Au \\ K \end{bmatrix} , \quad \mathcal{A}(\mathbf{Q}) = \begin{bmatrix} 0 \\ \alpha Au^2 \\ 0 \end{bmatrix} , \\ \mathcal{P}(\mathbf{Q}) &= \begin{bmatrix} 0 & 1 & 0 \\ \frac{A}{\rho} \frac{\partial p}{\partial A} & 0 & \frac{A}{\rho} \frac{\partial p}{\partial K} \\ 0 & 0 & 0 \end{bmatrix} , \quad \mathbf{S}(\mathbf{Q}) = \begin{bmatrix} 0 \\ s_2 - \frac{A}{\rho} \partial_x p_e \\ 0 \end{bmatrix} . \end{aligned} \right\} \quad (14.67)$$

In order to solve system (14.66)–(14.67) numerically we shall use the scheme (14.65). This requires a numerical flux for advection part and the fluctuations for the non-conservative pressure term. We do not address here, the task of devising general numerical schemes for the source terms, by exploiting the present splitting schemes. It turns out that the solution of the Riemann problem for the pressure system in (14.64) provides all the required items to compute the fluctuations for the pressure term and the numerical flux for the advection term.

Note that in this model system there is no passive scalar equation, but an equation for the **parameter** K following the approach proposed by Toro and Siviglia [3].

14.5.3 The Pressure System from the TV Splitting

The pressure system determines the fluctuations $\mathbf{D}_{i+\frac{1}{2}}^\pm$, but will also provide sufficient information to compute the numerical flux $\mathcal{A}_{i+\frac{1}{2}}$ for the advection term. We therefore study the pressure system in some detail. Consider the homogeneous version of the pressure system (14.66), namely

$$\partial_t \mathbf{Q} + \mathcal{M}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{0} , \quad (14.68)$$

with definitions (14.67). The eigenvalues of the matrix $\mathcal{M}(\mathbf{Q})$ are

$$\lambda_1 = -c , \quad \lambda_2 = 0 , \quad \lambda_3 = c , \quad (14.69)$$

where c is the wave speed

$$c = \sqrt{\frac{A}{\rho} \frac{\partial p}{\partial A}}. \quad (14.70)$$

The corresponding right eigenvectors are

$$\mathbf{R}_1 = \begin{bmatrix} 1 \\ -c \\ 0 \end{bmatrix}, \quad \mathbf{R}_2 = \begin{bmatrix} D \\ 0 \\ -c^2 \end{bmatrix}, \quad \mathbf{R}_3 = \begin{bmatrix} 1 \\ c \\ 0 \end{bmatrix}, \quad (14.71)$$

with

$$D = \frac{A}{\rho} \frac{\partial p}{\partial K}. \quad (14.72)$$

The scaling factors in the eigenvectors (14.71) have been assumed to be unity. Note the subsonic character of the pressure system, that is

$$\lambda_1 = -c < \lambda_2 = 0 < \lambda_3 = c. \quad (14.73)$$

The Riemann problem for the pressure system is

$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \mathcal{M}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{0}, \quad -\infty < x < \infty, \quad t > 0, \\ \text{ICs: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{Q}_L & \text{if } x < 0, \\ \mathbf{Q}_R & \text{if } x > 0. \end{cases} \end{array} \right\} \quad (14.74)$$

The initial data are the constant states $\mathbf{Q}_L$ and $\mathbf{Q}_R$. The structure of the solution is depicted in Fig. 14.1. There are three wave families corresponding to the three eigenvalues (14.69). The sought Godunov state is precisely the **Star State**, and

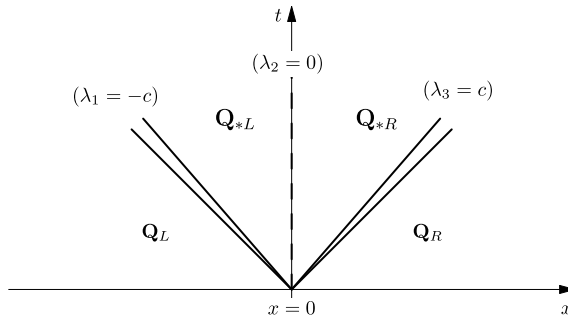


Fig. 14.1 Structure of the solution of the Riemann problem for the pressure system (14.74) arising from the TV splitting, for the case of one discontinuous parameter. The structure of the wave system is symmetric, with a stationary contact discontinuity associated with the eigenvalue $\lambda_2 = 0$

hence no sampling is required to determine the sought information to compute the fluctuations. Note that the star state includes $\mathbf{Q}_{*L}$ and $\mathbf{Q}_{*R}$.

The presentation of the exact Riemann solver for the pressure system (14.74) is omitted here. A detail to be noted. Although the equations are written in non-conservative form due to the jump in the coefficient of elasticity K , across the non-linear outer waves K is constant; therefore the Rankine-Hugoniot conditions are still applicable for the case of elastic jumps. See [32] for general background on solving Riemann problems. See Chap. 9.

14.5.4 Approximate Riemann Solver

In what follows we seek an approximate solution based on the assumption that the two outer waves associated with $\lambda_1 = -c$ and $\lambda_3 = c$ are rarefaction waves. For algebraic convenience, in the tube law (14.24)–(14.25) we set

$$\sqrt{\frac{A}{A_0}} = \left(\frac{A}{A_0} \right)^m \quad \text{with } m = 1/2.$$

Across the λ_1 and λ_3 fields, the generalised Riemann invariants yield

$$\left. \begin{aligned} q_* + T_L A_{*L}^{\frac{m+2}{2}} &= q_L + T_L A_L^{\frac{m+2}{2}}, & K_{*L} &= K_L, \\ q_* - T_R A_{*R}^{\frac{m+2}{2}} &= q_R - T_R A_R^{\frac{m+2}{2}}, & K_{*R} &= K_R, \end{aligned} \right\} \quad (14.75)$$

with

$$T_L = \frac{2}{m+2} \sqrt{\frac{mK_L}{\rho A_0^m}}, \quad T_R = \frac{2}{m+2} \sqrt{\frac{mK_R}{\rho A_0^m}}. \quad (14.76)$$

Across the λ_2 -field we have

$$dq = 0, \quad DdK + c^2 dA = 0 \rightarrow dp(A; K) = 0. \quad (14.77)$$

Therefore the flow q and the pressure $p(A; K)$ are constant in the star region, that is

$$q_* : \text{constant}, \quad p_*(A; K) : \text{constant}. \quad (14.78)$$

Algebraic manipulations yield

$$\left. \begin{aligned} q_{*L} &= q_{*R} = q_* : \text{constant} , \\ A_{*L} &= A_0 \left\{ \frac{K_R}{K_L} \left[\left(\frac{A_{*R}}{A_0} \right)^m - 1 \right] + 1 \right\}^{1/m} . \end{aligned} \right\} \quad (14.79)$$

There are three unknowns to be determined, namely q_* , A_{*L} and A_{*R} , which satisfy the following three equations:

$$\left. \begin{aligned} q_* + T_L A_{*L}^{\frac{m+2}{2}} &= q_L + T_L A_L^{\frac{m+2}{2}} , \quad K_{*L} = K_L , \\ q_* - T_R A_{*R}^{\frac{m+2}{2}} &= q_R - T_R A_R^{\frac{m+2}{2}} , \quad K_{*R} = K_R , \\ A_{*L} &= A_0 \left\{ \frac{K_R}{K_L} \left[\left(\frac{A_{*R}}{A_0} \right)^m - 1 \right] + 1 \right\}^{1/m} . \end{aligned} \right\} \quad (14.80)$$

From the first and second equations in (14.80) we obtain

$$A_{*L} = \left[-\frac{T_R}{T_L} A_{*R}^{\frac{m+2}{2}} + \frac{1}{T_L} \left(q_L - q_R + T_L A_L^{\frac{m+2}{2}} + T_R A_R^{\frac{m+2}{2}} \right) \right]^{\frac{2}{m+2}} . \quad (14.81)$$

Combining (14.81) with the third equation in (14.80) we obtain a single algebraic equation for A_{*R} , namely

$$\left. \begin{aligned} f(A_{*R}) &:= A_0 \left\{ \frac{K_R}{K_L} \left[\left(\frac{A_{*R}}{A_0} \right)^m - 1 \right] + 1 \right\}^{1/m} - \left[-\frac{T_R}{T_L} A_{*R}^{\frac{m+2}{2}} + Z \right]^{\frac{2}{m+2}} = 0 , \\ Z &= \frac{1}{T_L} \left(q_L - q_R + T_L A_L^{\frac{m+2}{2}} + T_R A_R^{\frac{m+2}{2}} \right) . \end{aligned} \right\} \quad (14.82)$$

The non-linear algebraic equation (14.82), of the form $f(A) = 0$, may be solved iteratively via a Newton-Raphson method

$$A^{(k+1)} = A^{(k)} - \frac{f(A^{(k)})}{f'(A^{(k)})} , \quad \text{for } k = 0, 1, \dots, K , \quad (14.83)$$

with A replacing A_{*R} , for notational convenience. The iterative process (14.83) is stopped whenever the relative change ΔA in A is smaller than a prescribed, small positive tolerance TOL , that is

$$\Delta A \equiv \frac{|A^{(k+1)} - A^{(k)}|}{(A^{(k+1)} + A^{(k)})/2} < TOL . \quad (14.84)$$

Noting that the equation $f(A) = 0$ is itself an approximation, we suggest to use $TOL = 10^{-4}$ in practice. As a guess value $A^{(0)}$ one could use a mean value of the initial data for the cross-sectional area, but with great care. The Newton-Raphson method is very sensitive to the initial guess.

Having found A_{*R} from the iterative process (14.83) we compute A_{*L} from (14.81), while q_* follows from (14.80), from which there are two one-sided choices. Another choice results from the mean value

$$q_* = \frac{1}{2} \left(q_L + q_R + T_R A_{*R}^{\frac{m+2}{2}} - T_L A_{*L}^{\frac{m+2}{2}} + T_L A_L^{\frac{m+2}{2}} - T_R A_R^{\frac{m+2}{2}} \right). \quad (14.85)$$

The full solution in the star region is known, resulting in two star vectors, namely

$$\mathbf{Q}_{*L} = [A_{*L}, q_*, K_L]^T, \quad \mathbf{Q}_{*R} = [A_{*R}, q_*, K_R]^T. \quad (14.86)$$

14.5.5 Fluctuations and Advection Numerical Flux

In order to implement the update formula (14.65) we need to determine the fluctuations for the non-conservative part (the pressure term) and the numerical flux for the conservative part (the advection part). To compute the fluctuations we adopt the Parés-Muñoz-Godunov scheme (PMG) of Sect. 14.4. By defining the two Godunov states

$$\mathbf{Q}_{i+\frac{1}{2}}^-(0) = \mathbf{Q}_{*L}, \quad \mathbf{Q}_{i+\frac{1}{2}}^+(0) = \mathbf{Q}_{*R} \quad (14.87)$$

the PMG scheme computes the fluctuations as

$$\mathbf{D}_{i+\frac{1}{2}}^- = \hat{\mathcal{M}}_{i+\frac{1}{2}}^- [\mathbf{Q}_{i+\frac{1}{2}}^-(0) - \mathbf{Q}_i^n]; \quad \mathbf{D}_{i+\frac{1}{2}}^+ = \hat{\mathcal{M}}_{i+\frac{1}{2}}^+ [\mathbf{Q}_{i+1}^n - \mathbf{Q}_{i+\frac{1}{2}}^+(0)], \quad (14.88)$$

with

$$\left. \begin{aligned} \hat{\mathcal{M}}_{i+\frac{1}{2}}^- &= \sum_{j=1}^G \omega_j \mathcal{M} \left(\psi(s_j; \mathbf{Q}_i^n, \mathbf{Q}_{i+\frac{1}{2}}^-(0)) \right), \\ \hat{\mathcal{M}}_{i+\frac{1}{2}}^+ &= \sum_{j=1}^G \omega_j \mathcal{M} \left(\psi(s_j; \mathbf{Q}_{i+\frac{1}{2}}^+(0), \mathbf{Q}_{i+1}^n) \right). \end{aligned} \right\} \quad (14.89)$$

In computing the advection flux we shall directly utilise the available value q_* from the pressure system, so that

$$\mathcal{A}_{i+\frac{1}{2}} = \begin{bmatrix} 0 \\ q_* u_K \\ 0 \end{bmatrix}, \quad u_K = \begin{cases} u_L & \text{if } q_* \geq 0, \\ u_R & \text{if } q_* < 0. \end{cases} \quad (14.90)$$

Therefore, all the required items for the TV scheme (14.65) to solve system (14.66) are determined, namely the numerical flux $\mathcal{A}_{i+\frac{1}{2}}$ for the advection part in conservative form, and the fluctuations $\mathbf{D}_{i+\frac{1}{2}}^-$ and $\mathbf{D}_{i+\frac{1}{2}}^+$ for the non-conservative part.

Two classes of TV schemes will be tested. The first one is denoted as TV+PMG+Ex.RS. This means TV splitting with PMG and exact Riemann solver for the pressure system to find the Godunov state, fluctuations and the advection numerical flux. The second TV scheme is denoted as TV+PMG+Ap.RS, in which the exact Riemann solver is replaced by the approximate Riemann solver just described.

Closing remark: The presentation of the exact Riemann solver for the pressure system (14.74) has been omitted. The reader is assumed sufficiently well equipped to derive it in full and implement it. See Chap. 13 for the exact solution of the pressure system for the case of constant parameters.

14.6 FORCE Path-Conservative Scheme

Here we present a centred path-conservative scheme based on the FORCE approach, first proposed for conservative systems in [33]. For further developments on FORCE for conservative systems see [34–37].

14.6.1 Fluctuations for FORCE Schemes

For general non-conservative systems, a FORCE-type scheme was presented by Cenestrelli et al. [38] and then by Dumbser et al. [39].

Here we propose a generalisation of the FORCE approach following the recent FORCE- α variant [37] to solve non-conservative systems, such as the blood flow equations for arteries in non-conservative form (14.16)–(14.18) for continuous parameters or (14.66)–(14.67) for discontinuous parameters.

The FORCE- α fluctuations are given as

$$\mathbf{D}_{i+\frac{1}{2}}^- = \mathbf{A}_{i+\frac{1}{2}}^- (\mathbf{Q}_{i+1}^n - \mathbf{Q}_i^n), \quad \mathbf{D}_{i+\frac{1}{2}}^+ = \mathbf{A}_{i+\frac{1}{2}}^+ (\mathbf{Q}_{i+1}^n - \mathbf{Q}_i^n), \quad (14.91)$$

with

$$\left. \begin{aligned} \mathbf{A}_{i+\frac{1}{2}}^- &= \frac{1}{2} \hat{\mathbf{A}}_{i+\frac{1}{2}} - \frac{1}{4} \frac{\alpha \Delta t}{\Delta x} \left[\hat{\mathbf{A}}_{i+\frac{1}{2}}^2 + \left(\frac{\Delta x}{\alpha \Delta t} \right)^2 \mathbf{I} \right], \\ \mathbf{A}_{i+\frac{1}{2}}^+ &= \frac{1}{2} \hat{\mathbf{A}}_{i+\frac{1}{2}} + \frac{1}{4} \frac{\alpha \Delta t}{\Delta x} \left[\hat{\mathbf{A}}_{i+\frac{1}{2}}^2 + \left(\frac{\Delta x}{\alpha \Delta t} \right)^2 \mathbf{I} \right]. \end{aligned} \right\} \quad (14.92)$$

In (14.92) $\mathbf{I}$ is the identity matrix and $\hat{\mathbf{A}}_{i+\frac{1}{2}}$ is a path integral of the coefficient matrix $\mathbf{A}$ in the governing equations, which is approximated numerically as

$$\hat{\mathbf{A}}_{i+\frac{1}{2}} = \sum_{j=1}^G \omega_j \mathbf{A}(\psi(s_j; \mathbf{Q}_i^n, \mathbf{Q}_{i+1}^n)) \approx \int_0^1 \mathbf{A}(\psi(s; \mathbf{Q}_i^n, \mathbf{Q}_{i+1}^n)) ds . \quad (14.93)$$

See (14.57). It is suggested to use a Gauss-Legendre rule with three integration points as in (17.57).

14.6.2 Stability and Viscosity for FORCE- α Schemes

The parameter α in (14.92) must satisfy $\alpha \geq 1$. For $\alpha = 1$ one obtains the scheme of Canestrelli et al. [38]. More sophisticated options admit $\alpha > 1$ [37]. Each value of α gives a scheme with different properties regarding stability and numerical viscosity. These schemes enhance accuracy, which is particularly useful for slowly-moving intermediate waves. We recall that these are the most challenging features for centred methods and for incomplete Riemann solvers.

The penalty of the enhanced accuracy of the FORCE- α scheme is the reduction of the stability range, as α increases. The stability condition is

$$C_{max} \leq \frac{1}{\alpha} \sqrt{2\alpha - 1} . \quad (14.94)$$

It turns out that (14.94) is also the condition for monotonicity. Figure 14.2 shows two curves as functions of α , assuming $\alpha \geq 1$; the blue line above shows the restriction for stability and monotonicity and the red line below shows the variation of the numerical viscosity function d_0 for stationary waves. As α increases d_0 decreases but at the cost of decreasing the Courant number C_{lim} . However d_0 goes to zero as $\frac{1}{\alpha}$, while C_{lim} goes to zero as $\frac{1}{\sqrt{\alpha}}$. That is, the numerical viscosity for stationary waves decreases faster than the Courant number, which means the cost is less than the benefit.

Table 14.1 shows some numerical values from Fig. 14.2 for selected values of α . The value $\alpha = 1$ simply reproduces the classical FORCE scheme in one space dimension, while the values $\alpha = 2$ and $\alpha = 3$ give the multidimensional FORCE interpreted as a one-dimensional flux. Note the remarkable decrease of d_0 from $d_0 = 1$ for the Lax-Friedrichs flux to $d_0 = 0.1667$ for $\alpha = 3$; this is the value corresponding to the multidimensional FORCE flux in three space dimensions [36]. Figure 14.2 shows that the strongest rate of decrease for C_{lim} takes place for values of α between $\alpha = 1$ and $\alpha = 25$. For $\alpha = 20$ a reduction of d_0 relative to the Lax-Friedrichs scheme is by a factor of 40, while the corresponding reduction of C_{lim} is by a factor of 3.2. For $\alpha = 25$ these numbers are respectively 50 and 3.57. For large α , for example

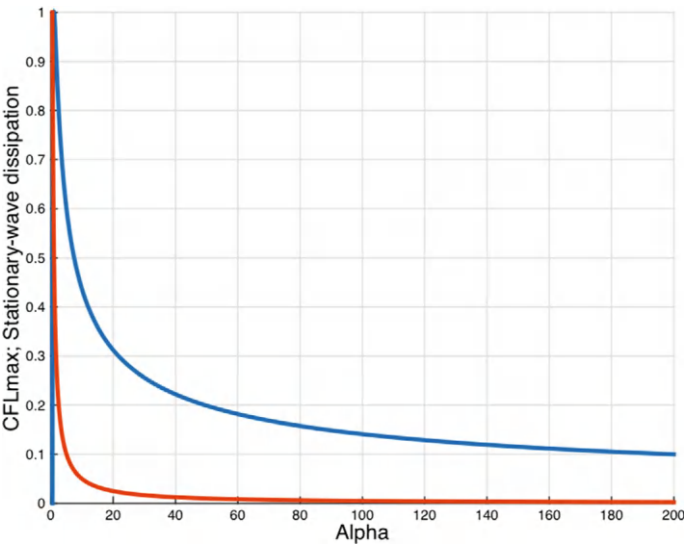


Fig. 14.2 Courant number C_{lim} for stability/monotonicity (blue) and coefficient of numerical viscosity (red) for stationary wave, both as functions of α

Table 14.1 Variation of the Courant number C_{lim} and numerical viscosity function d_0 for stationary waves as functions of selected increasing values of the parameter α . See Fig. 14.4

α	Courant number C_{lim}	Viscosity for stationary waves d_0
1	1.0000	0.5000
2	0.8660	0.2500
3	0.7454	0.1667
4	0.6614	0.1250
5	0.6000	0.1000
10	0.4359	0.0500
20	0.3122	0.0250
25	0.2800	0.0200
30	0.2560	0.0167
40	0.2222	0.0125
50	0.1990	0.0100
60	0.1818	0.0083
199.4987	0.1000	0.0025

for $\alpha = 199.4987$, the reduction of C_{lim} is by a factor of 10, to $C_{lim} = 0.1$, and the corresponding reduction of d_0 is by a factor of 400.

These schemes are particularly well suited for the Discontinuous Galerkin Finite Element Methods (DG), which actually have a more stringent stability condition than the FORCE- α schemes. Furthermore, given the order of a DG scheme, there is a precise value of α that matches the stability range. Full details on implementation issues regarding the choice of α are given in [37].

We have presented several first-order numerical methods of the path-conservative type to solve systems of hyperbolic equations written in non-conservative form. Table 14.2 shows a summary of all schemes studied. In the next section we assess the performance of some of these methods, via a suite of test problems with exact solution.

14.7 Test Problems and Numerical Results

In this section we assess the performance of four out of the seven numerical methods presented in this chapter, namely the first four schemes in Table 14.2.

14.7.1 Test Problems and Parameters

We have selected a suite of three test problems with exact solution. Table 14.3 gives the initial conditions for the three primitive variables A , u and η . The 8th column shows the output time, while the last column describes the wave pattern in the exact

Table 14.2 Path-conservative schemes for solving non-conservative systems

Scheme	Name	Description	Note
1	PMG+Ex.RS	Páres-Muñoz-Godunov and Ex. Riem. Solv.	Upwind
2	DOT+NC	Dumbser-Osher-Toro non-conservative	Upwind
3	DOT+NC+WB	DOT non-conservative and well-balanced	Upwind
4	TV+PMG+Ex.RS	TV Split. +PMG+Ex. RS. for Press. Syst.	Upwind
5	PMG+Ap.RS	PMG with approximate Riemann solver	Upwind
6	TV+PMG+Ap.RS	TV Split. +PMG+App. RS. for Press. Syst.	Upwind
7	FORCE+NC	FORCE non-conservative	Centred

Table 14.3 Three test problems for the augmented equations for blood flow in vessels with discontinuous parameters. The last column describes the wave pattern that emerges from the exact solution of the Riemann problem, where R: rarefaction; C: contact and S: shock

Test	A_L	u_L	η_L	A_R	u_R	η_R	T_{out}	Wave pattern
1	$A(p = 80\text{mmHg})$	0.0	1.0	$A(p = 80\text{ mmHg})$	0.0	0.5	0.007	–
2	$1.6A_{0L}$	1.0	1.0	$1.05A_{0R}$	0.0	0.5	0.007	SCS
3	$0.9A_{0L}$	0.0	1.0	$1.6A_{0R}$	0.0	0.5	0.025	SCR

Table 14.4 Parameters for test problems in Table 14.3

Parameter	Units	Test 1	Test 2	Test 3
$A_{0,\text{ref}}$	m^2	3.1353×10^{-4}	3.1353×10^{-4}	1.0000×10^{-4}
K_{ref}	Pa	58725	58725	58725
K_L	Pa	K_{ref}	K_{ref}	K_{ref}
A_{0L}	m^2	$2 A_{0,\text{ref}}$	$\frac{1}{2} A_{0,\text{ref}}$	$1.1 A_{0,\text{ref}}$
p_{eL}	Pa	9999.15	3999.66	1333.22
K_R	Pa	$10 K_{\text{ref}}$	$10 K_{\text{ref}}$	$10 K_{\text{ref}}$
A_{0R}	m^2	$A_{0,\text{ref}}$	$A_{0,\text{ref}}$	$1.3 A_{0,\text{ref}}$
p_{eR}	Pa	0	0	666.61
m	–	$\frac{1}{2}$	$\frac{1}{2}$	10
n	–	0	0	$-\frac{3}{2}$
l	m	0.2	0.2	0.2
x_0	m	0.1	0.06	0.06
ρ	$\frac{kg}{m^3}$	1000	1000	1000

solution. Table 14.4 gives the values of parameters for each test problem. We note that in the selected test problems the geometric and biomechanical parameters are allowed to change discontinuously across x_0 , which is also the position of the initial discontinuity in the flow variables.

For the test problems chosen here we need a generalisation of the model equations studied in detail in Sect. 14.5.2 for one discontinuous parameter, namely K . See Chap. 10.

14.7.2 Discussion of Results

The test problems chosen here are extremely demanding. First, not only flow variables have discontinuous initial conditions (Riemann problems) but also several geometric and biomechanical parameters are variable; moreover, they change discontinuously

across the position x_0 . Even sophisticated non-conservative methods are known to fail for this kind of problems.

Figure 14.3 shows results for Test 1 from the four methods tested, namely methods 1–4 in Table 14.2. The results from PMG+Ex.RS (Páres-Muñoz-Godunov and exact Riemann solver) are perfect. The contact discontinuity is preserved as a true discontinuity with no spurious oscillations. The results from DOT+NC (Dumbser-Osher-Toro, non-conservative) are not satisfactory. The resolution of the contact discontinuity is good for the cross-sectional area and the passive scalar (though smeared by two points), but the result for velocity is unacceptable. The results from the well-balanced version of this scheme, DOT+NC+WB, is perfect. The last results are also perfect and correspond to TV+PMG+Ex.RS (TV splitting with PMG and the exact Riemann solver for the pressure system).

Figure 14.4 shows the results for Test 2 from methods 1–4 of Table 14.2. These results are consistent with those for Test 1. All methods perform very satisfactorily, apart from DOT+NC. The three methods that perform well give virtually identical results. Of course the outer (left and right) discontinuities are smeared very visibly, but no spurious oscillations are seen around the discontinuities.

Figure 14.5 shows the results for Test 3 from methods 1–4 of Table 14.2. For this test problem, DOT+NC performs better than for Tests 1 and 2, though visible undershoots are seen near the tail of the rarefaction. Again, the other three methods perform very satisfactorily and give, overall, comparable results.

Recall that all the methods of this chapter are first-order accurate. For higher-order extensions see Chaps. 16 and 17.

14.8 Summary and Further Reading

We have first succinctly reviewed the field of hyperbolic systems in non-conservative form and the class of path-conservative methods to solve such systems numerically. Then, we presented several first-order path-conservative methods for solving hyperbolic systems in non-conservative form, including upwind and centred methods. We have shown results for three test problems from four out of the seven methods studied. Three of the four methods tested perform very satisfactorily and are recommended for practical applications. Non-linear extension of these methods to higher order of accuracy will be carried out in Chap. 17. Suggested exercises are given at the end of the chapter.

For further reading we suggest: Parés [13]; Parés and Muñoz-Ruiz [29]; Dumbser and Toro [17]; Müller and Toro [26]; Toro and Vázquez [2]; Toro, Saggiorato, Tokareva and Hidalgo [37].

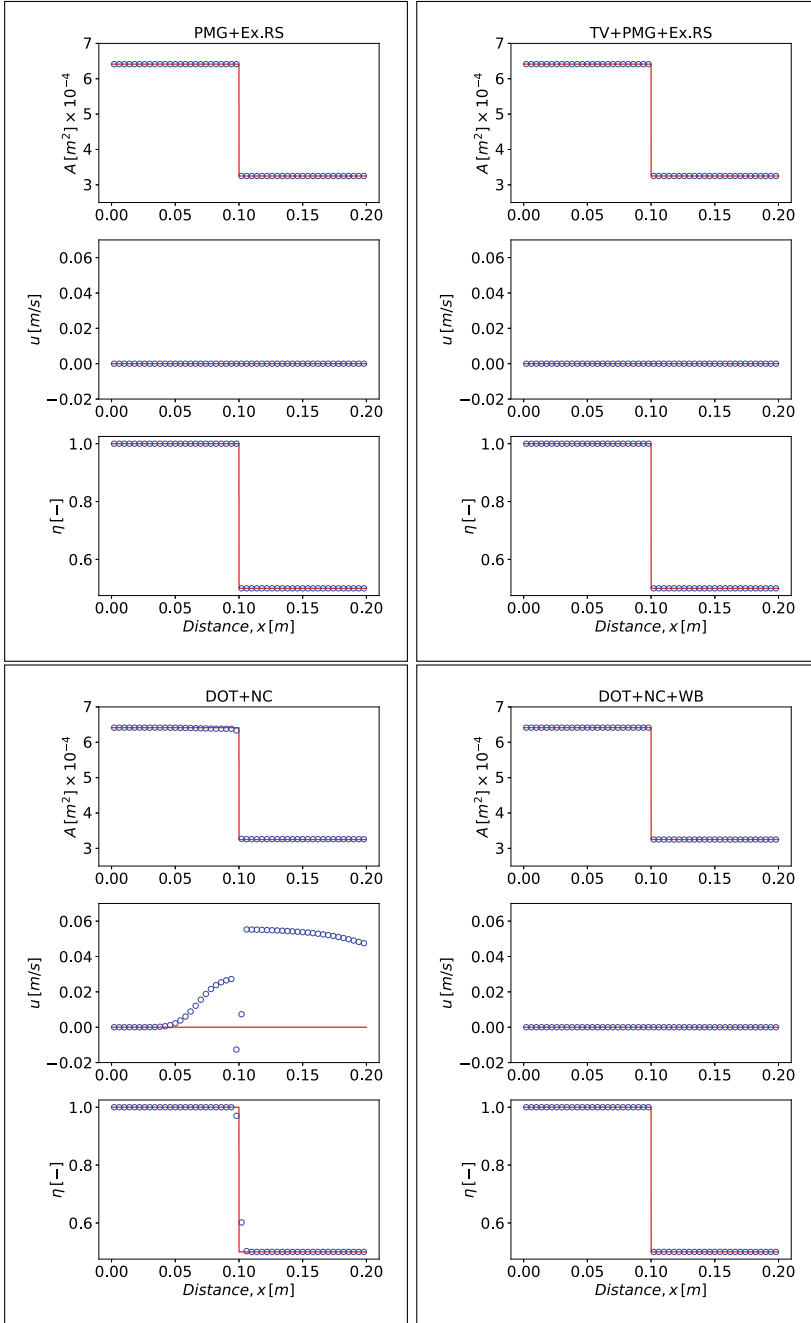


Fig. 14.3 Test 1 (arteries). Numerical solutions from the first four methods of Table 14.2 (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{efl} = 0.9$. Initial data is given in Table 14.3 and parameters are given in Table 14.4

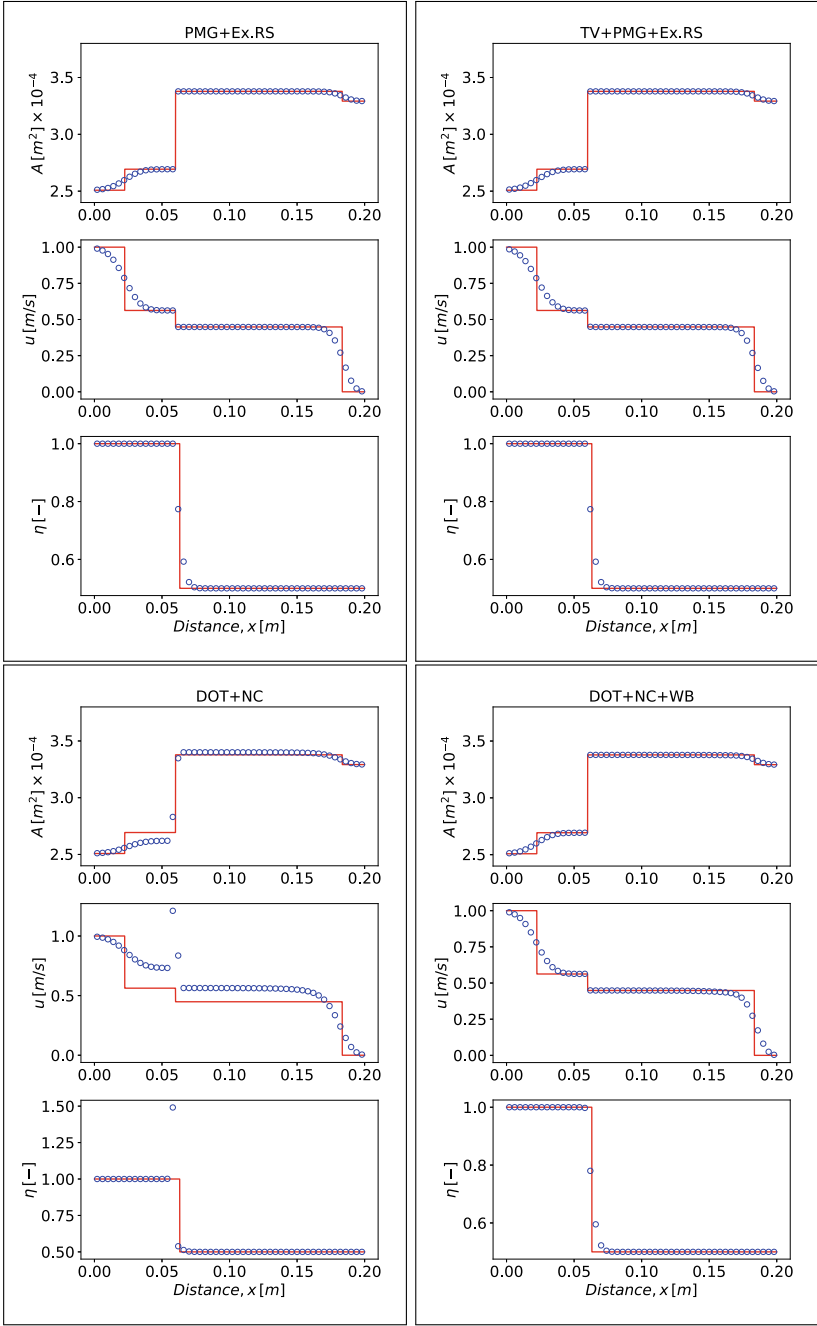


Fig. 14.4 Test 2 (arteries). Numerical solutions from the first four methods of Table 14.2 (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{efl} = 0.9$. Initial data is given in Table 14.3 and parameters are given in Table 14.4

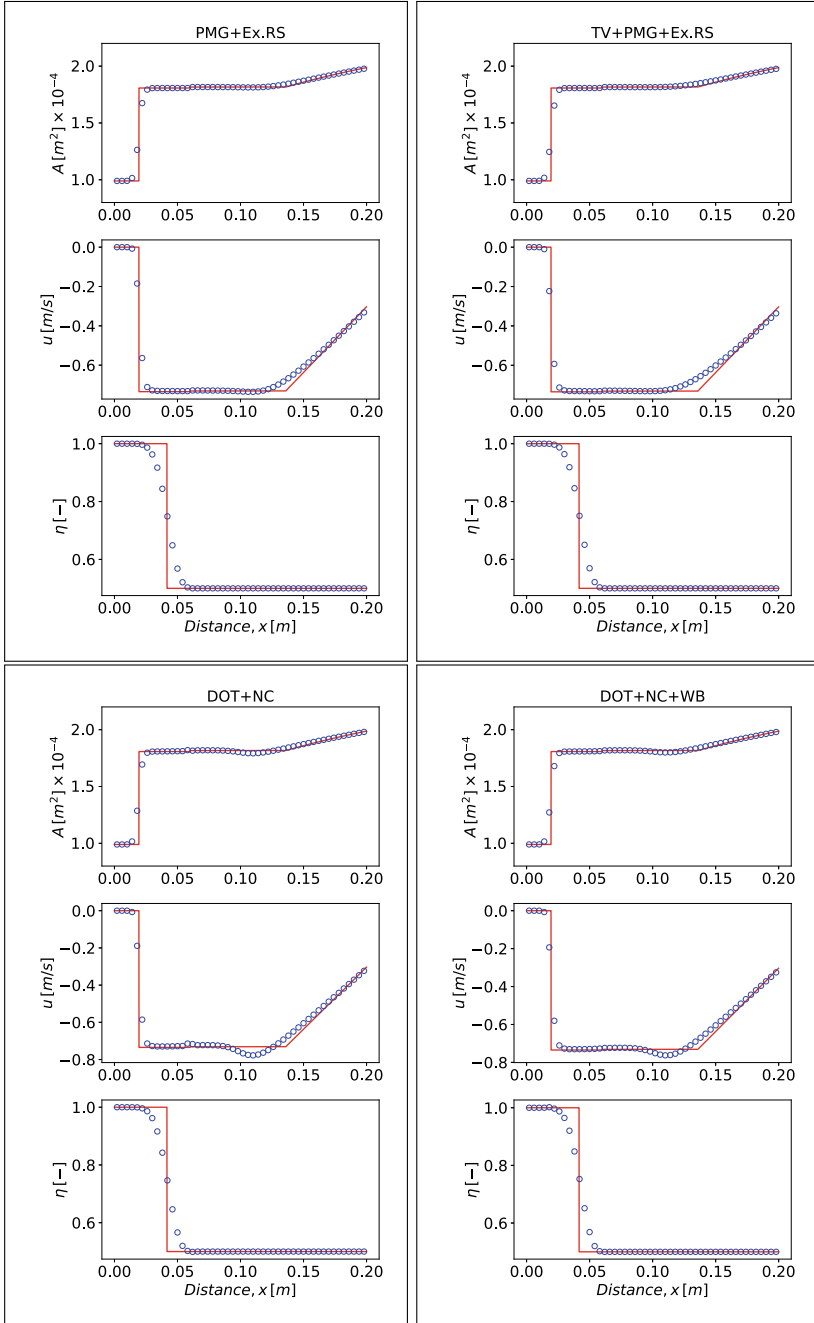


Fig. 14.5 Test 3 (veins). Numerical solutions from the first four methods of Table 14.2 (symbol) compared to the exact solution (line). $M = 50$ cells; $C_{efl} = 0.9$. Initial data is given in Table 14.3 and parameters are given in Table 14.4

14.9 Exercises

14.1 Path-conservative method for LAE. Consider the linear advection equation with constant wave propagation speed λ .

1. Construct a path-conservative scheme to solve it.
2. Show that the scheme reproduces the Godunov's upwind scheme, for both positive and negative wave speeds λ .

14.2 Path-conservative method for Burgers' equation. Consider the Burgers equation

$$\partial_t q + \partial_x f(q) = 0, \quad f(q) = \frac{1}{2} q^2. \quad (14.95)$$

1. Construct a path-conservative scheme to solve it.
2. Implement the above method numerically and test it for problems involving shock waves.

14.3 Rankine-Hugoniot conditions. Consider the scalar non-linear conservation law

$$\partial_t q + \partial_x f(q) = 0, \quad (14.96)$$

with physical flux

$$f(q) = b_0 + b_1 q + b_2 q^2, \quad \text{with constant coefficients } b_0, b_1, b_2. \quad (14.97)$$

1. Apply the classical Rankine-Hugoniot condition and verify that the speed S of a shock connecting the constant states q_L and q_R is given as

$$S = b_1 + b_2(q_L + q_R). \quad (14.98)$$

2. Apply the generalised Rankine-Hugoniot condition

$$\int_0^1 S \frac{\partial}{\partial s} \psi(s; q_L, q_R) ds = \int_0^1 \lambda(\psi(s; q_L, q_R)) \frac{\partial}{\partial s} \psi(s; q_L, q_R) ds, \quad (14.99)$$

where $\lambda(q)$ is the characteristic speed of (14.96). Using the canonical path

$$\psi(s; q_L, q_R) = q_L + s(q_R - q_L) \quad (14.100)$$

in (14.99) show that the shock speed S coincides with (14.98).

14.4 Consider an isolated shock wave for Eqs. (14.7)–(14.8).

1. Derive the shock speed S_c using the classical Rankine-Hugoniot conditions.
2. Repeat the same exercise for Eqs. (14.9)–(14.10) to obtain a shock speed S_{nc} . Show that $S_{nc} \neq S_c$.

Hint: see [41], pp. 107–110 and [42].

14.5 Solve numerically a Riemann problem that contains a shocks wave using system (14.7)–(14.8) and system (14.9)–(14.10). Compare your numerical results against the exact solution for the physically conservative system (14.7)–(14.8) and comment on your results. Pay special attention to the shock position. Hint: see [43].

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Chapter 15

Spatial Reconstruction Methods



Abstract This chapter addresses the need to design non-linear methods that circumvent Godunov's theorem for solving hyperbolic partial differential equations. After providing some background on classical polynomial interpolation theory, we examine modern, non-linear polynomial interpolation methods. These methods determine polynomial coefficients based on the behaviour of the data. Such polynomials are designed to handle not only smooth data but also rapidly changing data, even discontinuities. We begin with the standard ENO (Essentially Non-Oscillatory) method, along with two modifications that produce conservative ENO schemes. One modification is classical and based on the primitive function, while the other is more direct and conceptually simpler, relying on enforcing conservation on a preselected ENO stencil. Next, we introduce the classical WENO (Weighted Essentially Non-Oscillatory) methodology, along with a more recent variation. These non-linear spatial reconstruction methods are introduced for a single function. We then explore extensions of these methods to vector functions. The simplest extension is the component-by-component approach, in which the scalar reconstruction scheme is applied to each component of the vector of unknowns, typically the conserved variables. However, collective experience suggests that this simple approach is inadequate for problems involving discontinuities or large spatial gradients, particularly for schemes of accuracy higher than two. A more successful extension involves the use of characteristic variables. Although the primary application of these methods is in the design of non-linear, high-order numerical schemes that attempt to circumvent Godunov's theorem to solve hyperbolic partial differential equations, in this chapter we evaluate their performance purely as interpolation techniques. The chapter concludes with a summary, a brief review of methods not covered here, and suggestions for further study. A list of illustrative exercises is also provided.

15.1 Polynomial Interpolation: Basic Theory

This section first gives a brief presentation of the basic theory of polynomial interpolation, as can be found in any classical book of numerical analysis [1–3]; see also [4] for more advanced material. These theoretical concepts are then used in the

following sections that are more directly relevant to the main themes of this book. We consider two ways of constructing polynomial functions. The first one deals with the case of point values, as in the classical literature. The second way of constructing the polynomial makes use of integral cell averages, rather than point values. This part is more directly relevant to the subject of numerical schemes for solving hyperbolic partial differential equations.

15.1.1 Introductory Concepts

Consider a real-valued function $q(x)$, with $x \in I = [a, b] \subset \mathbb{R}$, along with a set of distinct **nodes** $x_i \in I$ and the set of $m + 1$ function values $q_i = q(x_i)$, for $i = 0, 1, 2, \dots, m$. Let us also consider the set of corresponding $m + 1$ **knots**

$$P_0 = (x_0, q_0), P_1 = (x_1, q_1), \dots, P_m = (x_m, q_m), \quad (15.1)$$

as conveniently displayed in Table 15.1.

The nodes $x_i \in I$, **not necessarily displayed in their natural order**, range from a minimum x_{min} to a maximum x_{max} , with

$$\left. \begin{aligned} x_{min} &= \min \{x_0, x_1, x_2, \dots, x_m\}, \\ x_{max} &= \max \{x_0, x_1, x_2, \dots, x_m\}. \end{aligned} \right\} \quad (15.2)$$

The task at hand is to find a polynomial function $p(x)$ of **degree** $\leq m$. Written in **power form**, or **standard form**, the polynomial function $p(x)$ reads

$$p(x) = c_0 + c_1x + c_2x^2 + \dots + c_mx^m. \quad (15.3)$$

The coefficients $c_0, c_1, \dots, c_m$ are found by requiring that $p(x)$ **coincides** with $q(x)$ at all nodes, that is the following **polynomial equations** are satisfied

$$p(x_0) = q_0, \quad p(x_1) = q_1, \quad p(x_2) = q_2, \quad \dots, \quad p(x_m) = q_m. \quad (15.4)$$

The sought polynomial $p(x)$ is expected to pass through all the knots on the graph of $q(x)$ in the x - $q(x)$ plane. The sought polynomial function $p(x)$ is called the **interpolating polynomial**, defined in the **interpolation interval** given by $[x_{min}, x_{max}] \subset I$. The polynomial $p(x)$ in (15.3) is known once the coefficients $c_0, c_1, \dots, c_m$ have been determined.

Table 15.1 Tabulated $m + 1$ knots $P_i = (x_i, q_i)$ for nodes x_i and function values $q_i = q(x_i)$ of a real-valued function $q(x)$, $x \in I \subset \mathbb{R}$

Nodes:	x_i	x_0	x_1	x_2	$\dots$	x_{m-1}	x_m
Function values:	q_i	q_0	q_1	q_2	$\dots$	q_{m-1}	q_m

In addition to the standard form (15.3), there are other forms of expressing the polynomial function. One variation is the **shifted power form**, which gives the Taylor expansion for $p(x)$ around a chosen **centre** $x = \hat{x}$, usually taken in the interpolation sub-interval $[x_{min}, x_{max}]$ of interest. The shifted power form is also called the **Taylor form** of the polynomial and reads

$$p(x) = c_0 + c_1(x - \hat{x}) + c_2(x - \hat{x})^2 + \dots + c_m(x - \hat{x})^m. \quad (15.5)$$

Remarks:

1. For any point $\hat{x} \in [x_{min}, x_{max}]$ that coincides with a node, the polynomial $p(x)$ coincides with the function $q(x)$, that is $p(\hat{x}) = q(\hat{x})$. If $\hat{x}$ is not a node then $p(\hat{x})$ is an approximation to $q(\hat{x})$, that is $p(\hat{x}) \approx q(\hat{x})$, with a corresponding error.
2. Often, the nodes and knots do not derive from a function $q(x)$, that may or may not exist. The nodes and knots may come from measured experimental data, for example. In this case $p(x)$ is a function that passes through the given data points and defines values for every point $\hat{x}$ in the interval $[x_{min}, x_{max}]$, not necessarily part of the given data set.
3. The interpolating polynomial may be useful to estimate not only values away from the nodes but also to estimate derivatives. The interpolating polynomial may also be useful for estimating integrals and for finding approximate solutions to ordinary differential equations.
4. The interpolation problem of interest here is distinct from the **discrete-curve fitting problem**, in which the graph of the constructed function $p(x)$ **passes near**, though not necessarily through, the given knots. This problem, called **data-smoothing problem**, is not studied here.

An important theoretical question is whether the sought interpolation polynomial actually exists, and if so, whether it is unique. This and related questions will be addressed later in this section. Regarding the practical issue of finding the interpolating polynomial, there are several methods available in the literature. Next we give one example of such methods.

15.1.2 The Method of Undetermined Coefficients

As already indicated, the polynomial is determined once the coefficients are known. Here we first consider the simple case arising from the tabulated values in Table 15.2.

As $m = 2$ we look for a second-degree interpolating polynomial $p(x)$ written in the standard form

$$p(x) = c_0 + c_1x + c_2x^2 \quad (15.6)$$

Table 15.2 Example: knots $P_i = (x_i, q_i)$ for nodes x_i and function values $q_i = q(x_i)$, for $i = 0, 1, 2$. Here $m = 2$, leading to a second-degree polynomial

Nodes:	x_i	-1	0	2
Function values:	q_i	2	1	11

passing through the three chosen knots in Table 15.2. The condition that $p(x)$ passes through all three knots gives the polynomial equations

$$p(x_i) = c_0 + c_1x_i + c_2x_i^2 = q_i, \quad \text{for } i = 0, 1, 2. \quad (15.7)$$

Equations (15.7) constitute a 3×3 linear system for the three unknown coefficients c_0, c_1, c_2 , namely

$$\mathbf{AC} = \mathbf{B}, \quad (15.8)$$

with

$$\mathbf{C} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 2 \\ 1 \\ 11 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 2 & 4 \end{bmatrix}. \quad (15.9)$$

Here $\mathbf{C}$ is the vector of unknown coefficients, $\mathbf{B}$ is the vector of known function values and $\mathbf{A}$ is the coefficient matrix. It is easily verified that the solution of system (15.8)–(15.9) is $c_0 = 1, c_1 = 1, c_2 = 2$. Therefore the sought quadratic polynomial is

$$p(x) = 1 + x + 2x^2. \quad (15.10)$$

The method of undetermined coefficients is quite obvious and requires the solution of a linear system of algebraic equations of the type (15.8), for the coefficients of the sought polynomial. For low-degree polynomials this method can be of practical use, but for high-degree polynomials the system becomes very large, rendering the method unattractive in practice.

In general, for the $m + 1$ knots as given in Table 15.1, the interpolation equations read

$$p(x_i) = c_0 + c_1x_i + c_2x_i^2 + \dots + c_mx_i^m = q_i, \quad \text{for } i = 0, 1, \dots, m. \quad (15.11)$$

The corresponding $(m + 1) \times (m + 1)$ linear algebraic system has the form (15.8), where the vector of unknowns $\mathbf{C}$, the vector of prescribed function values $\mathbf{B}$ and the coefficient matrix $\mathbf{A}$ are given as follows:

$$\mathbf{C} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} q(x_0) \\ q(x_1) \\ q(x_2) \\ \vdots \\ q(x_m) \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^m \\ 1 & x_1 & x_1^2 & \dots & x_1^m \\ 1 & x_2 & x_2^2 & \dots & x_2^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \dots & x_m^m \end{bmatrix}. \quad (15.12)$$

The reader is encouraged to verify the form of the above system of algebraic equations.

Remark: Vandermonde matrix. The matrix $\mathbf{A}$ in (15.12) is called the **Vandermonde matrix**, whose determinant, called the Vandermonde determinant, is non-zero if all x_i are distinct, in which case the system has a unique solution. But note that the Vandermonde matrix can be ill-conditioned.

System (15.8), with definitions (15.12), can be solved by existing very reliable software, publicly available, for which it is sufficient to supply the known entries of the matrix $\mathbf{A}$ and the known components of the right-hand side vector $\mathbf{B}$ in (15.12). In the rest of this chapter we shall assume the reader to be familiar with existing software for solving linear algebraic systems, such as (15.8), (15.12).

There are more methods for constructing the interpolating polynomial, with corresponding forms for the polynomial. The **Lagrangian form** is one of such forms, not studied here. **Newton's form** of the interpolation polynomial gives an attractive method, in which the sought polynomial coefficients are recursively obtained from **divided differences**. This is the subject of the next subsection.

15.1.3 Newton's Form of the Interpolation Polynomial

Here we adopt the Newton form for constructing the interpolation polynomial and introduce its essential component, namely, divided differences. We first illustrate the approach through a couple of examples and then deal with the general case.

Newton's Form: Illustrative Examples

Example 15.1 Let us begin by considering the quadratic polynomial that interpolates Table 15.2, which by using the method of undetermined coefficients yielded the second-degree polynomial (15.10). Newton's method constructs the sought polynomial in a sequential manner by expressing it as the sum of a sequence of low-degree polynomials, as illustrated below. The sought second-degree polynomial is first written thus

$$p(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)(x - x_1). \quad (15.13)$$

This polynomial is made up of

- A constant polynomial $p_0(x) = c_0$;
- A first-degree polynomial $p_1(x) = p_0(x) + c_1(x - x_0)$, and
- The full polynomial $p(x) := p_2(x) = p_1(x) + c_2(x - x_0)(x - x_1)$.

The problem at hand is to find the coefficients c_0, c_1, c_2 in (15.13). To this end one considers each polynomial $p_0(x), p_1(x), p_2(x)$ and their corresponding polynomial equations, as follows:

1. Set

$$p_0(x) = q_0 = c_0 = 2.$$

2. Define $p_1(x) = p_0(x) + c_1(x - x_0)$ that adds node x_0 explicitly and apply the polynomial equation at node $x = x_1$, that is

$$p_1(x_1) = p_0(x_1) + c_1(x_1 - x_0) = q_1 = 1.$$

This gives the coefficient $c_1 = -1$ and hence

$$p_1(x) = 1 - x.$$

3. Define $p_2(x) = p_1(x) + c_2(x - x_0)(x - x_1)$ that adds node x_1 explicitly and apply the polynomial equation at node $x = x_2$

$$p_2(x_2) = q_2 = 11.$$

This gives the coefficient $c_2 = 2$ and hence

$$p(x) := p_2(x) = 1 - x + 2(x + 1)x,$$

which is identical to the quadratic polynomial (15.10).

Remark: Note that each new, non-constant, term in (15.13) adds a new node.

Example 15.2 Consider now the general quadratic polynomial that interpolates Table 15.3.

We proceed as in the previous example by considering (15.13).

Table 15.3 Data for the general quadratic polynomial. Knots $P_i = (x_i, q_i)$ for nodes x_i and function values $q_i = q(x_i)$, for $i = 0, 1, 2$

Nodes:	x_i	x_0	x_1	x_2
Function values:	q_i	q_0	q_1	q_2

1. Set the constant polynomial

$$p_0(x) = q_0 = c_0 .$$

2. Define $p_1(x) = p_0(x) + c_1(x - x_0)$ and apply the polynomial equation at $x = x_1$, that is

$$p_1(x_1) = p_0(x) + c_1(x_1 - x_0) = q_1 .$$

This gives the coefficient c_1 in terms of a **divided difference**, namely

$$c_1 = \frac{q_1 - q_0}{x_1 - x_0} := q[x_0, x_1]$$

and therefore

$$p_1(x) = q_0 + q[x_0, x_1](x - x_0) .$$

3. Define $p_2(x) = p_1(x) + c_2(x - x_0)(x - x_1)$ and apply the polynomial equation at $x = x_2$, that is

$$p_2(x_2) = q_2 .$$

This gives the coefficient c_2 as

$$c_2 := q[x_0, x_1, x_2] ,$$

with notation for $q[x_0, x_1]$ and $q[x_0, x_1, x_2]$ to be established shortly. Therefore, the general quadratic polynomial for Table 15.3 can be written in Newton's form as

$$p(x) = q_0 + q[x_0, x_1](x - x_0) + q[x_0, x_1, x_2](x - x_0)(x - x_1) . \quad (15.14)$$

Divided Differences

In the definitions for c_1 and c_2 above, we effectively introduced the definition for **divided differences**, recursively, as follows

$$\left. \begin{aligned} q_k &= f(x_k) = q[x_k] , \\ q[x_0, x_1] &= \frac{q_1 - q_0}{x_1 - x_0} = \frac{q[x_1] - q[x_0]}{x_1 - x_0} , \\ q[x_0, x_1, x_2] &= \frac{q[x_1, x_2] - q[x_0, x_1]}{x_2 - x_0} . \end{aligned} \right\} \quad (15.15)$$

Divided differences will be used to define the general polynomial in Newton's form.

Newton's Form: the General Case

In general, the Newton's form of the interpolation polynomial reads

$$\left. \begin{aligned} p(x) = & c_0 + c_1(x - x_0) + c_2(x - x_0)(x - x_1) + \dots \\ & \dots + c_m(x - x_0)(x - x_1) \dots (x - x_{m-2})(x - x_{m-1}) . \end{aligned} \right\} \quad (15.16)$$

As noted earlier, each new term adds a new node. The final form explicitly includes $x_0, x_1, \dots, x_{m-1}$. In more succinct form, (15.16) may be written as

$$p(x) = \sum_{i=0}^m c_i \prod_{j=0}^{i-1} (x - x_j) , \quad (15.17)$$

where by convention $\prod_{j=0}^{-1} (x - x_j) = 1$. The coefficients c_i are found by generalising the procedure employed in the quadratic polynomial examples above, through evaluation of $p(x)$ in (15.16) at the nodes $x_0, x_1, x_2, \dots, x_m$ and imposing the polynomial equations, namely

$$\left. \begin{aligned} p(x_0) &= c_0 = q_0 , \\ p(x_1) &= c_0 + c_1(x_1 - x_0) = q_1 , \\ p(x_2) &= c_0 + c_1(x_2 - x_0) + c_2(x_2 - x_0)(x_2 - x_1) = q_2 , \\ &\dots \\ p(x_m) &= c_0 + \dots + c_m(x_m - x_0)(x_m - x_1) \dots (x_m - x_{m-1}) = q_m . \end{aligned} \right\} \quad (15.18)$$

From equations (15.18), it turns out that the coefficients c_i in (15.16) can be found in terms of divided differences, that is

$$c_i = q[x_0, x_1, x_2, x_3, \dots, x_i] . \quad (15.19)$$

This is called the i th order **divided difference (DD)**. Then, from (15.16) and (15.18) the Newton's form of the interpolation polynomial reads

$$p(x) = \sum_{i=0}^m q[x_0, x_1, x_2, x_3, \dots, x_i] \prod_{j=0}^{i-1} (x - x_j) . \quad (15.20)$$

The divided differences can be calculated recursively, as suggested in (15.15) for a simple example. For the general case the following theorem holds:

Theorem 15.3 *For the set of knots (15.1) of Table 15.1, starting from the 0th order divided difference*

$$q[x_0] = q_0 \quad (15.21)$$

the k th order divided difference, for $k = 1, 2, \dots$, is defined recursively as

$$q[x_0, x_1, \dots, x_k] = \frac{q[x_1, x_2, \dots, x_k] - q[x_0, x_1, \dots, x_{k-1}]}{x_k - x_0}. \quad (15.22)$$

Proof Omitted. For details see any book of elementary numerical analysis, e.g. [1, 2]. ■

Remark: an advantage of the Newton form is that, thanks to the divided differences, no matrix inversion is needed to build up the polynomial.

Example 15.4 For the knots of Table 15.2 we have the following DDs

$$\left. \begin{aligned} 0\text{-th DD : } q[x_0] &= q_0 = 2, \\ 0\text{-th DD : } q[x_1] &= q_1 = 1, \\ 0\text{-th DD : } q[x_2] &= q_2 = 11, \\ 1\text{st DD : } q[x_0, x_1] &= \frac{q[x_1] - q[x_0]}{x_1 - x_0} = -1, \\ 1\text{st DD : } q[x_1, x_2] &= \frac{q[x_2] - q[x_1]}{x_2 - x_1} = 5, \\ 2\text{nd DD : } q[x_0, x_1, x_2] &= \frac{q[x_1, x_2] - q[x_0, x_1]}{x_2 - x_0} = 2. \end{aligned} \right\} \quad (15.23)$$

Example 15.5 Consider the function $q(x) = \sqrt{x}$ and the four knots given in Table 15.4. Find the third-degree interpolating polynomial in Newton's form.

To solve this problem all we need is to find the coefficients, which are given in terms of divided differences, see example (15.14)–(15.15) and the general case (15.21)–(15.22). Table 15.5 shows the required divided differences in tabular form, corresponding to Table 15.4. This way of displaying the divided differences allows the clear identification of the sought polynomial coefficients.

The top diagonal of Table 15.5 in bold face gives the sought solution:

$$q[x_0] = 0, \quad q[x_0, x_1] = 1, \quad q[x_0, x_1, x_2] = -1/6, \quad q[x_0, x_1, x_2, x_3] = 1/60. \quad (15.24)$$

Table 15.4 Tabulated knots $P_i = (x_i, q_i)$, with nodes x_i and function values q_i , for the function $q(x) = \sqrt{x}$, with $i = 0, 1, 2, 3$

Nodes:	x_i	0	1	4	9
Function values:	q_i	0	1	2	3

Table 15.5 Table of divided differences for function values of $q(x) = \sqrt{x}$, corresponding to data of Table 15.4. The top diagonal (in bold face) forms the coefficients of the sought polynomial in Newton form

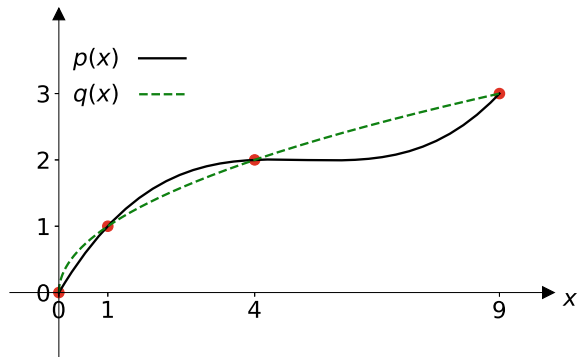
x	$q(x) = \sqrt{x}$	$q[x_i, x_{i+1}]$	$q[x_i, x_{i+1}, x_{i+2}]$	$q[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$
$x_0 = 0$	$q[x_0] = \mathbf{0}$			
		$q[x_0, x_1] = \mathbf{1}$		
$x_1 = 1$	$q[x_1] = 1$		$q[x_0, x_1, x_2] = -\mathbf{1/6}$	
		$q[x_1, x_2] = 1/3$		$q[x_0, x_1, x_2, x_3] = \mathbf{1/60}$
$x_2 = 4$	$q[x_2] = 2$		$q[x_1, x_2, x_3] = -1/60$	
		$q[x_2, x_3] = 1/5$		
$x_3 = 9$	$q[x_3] = 3$			

Therefore, the interpolating polynomial in Newton's form is

$$p(x) = 0 + x - \frac{1}{6}x(x - 1) + \frac{1}{60}x(x - 1)(x - 4) . \quad (15.25)$$

The polynomial $p(x)$ in (15.25) approximates the function $q(x) = \sqrt{x}$. Figure 15.1 shows both $q(x) = \sqrt{x}$ (dotted line) and its polynomial approximation $p(x)$ (full line). Clearly, the polynomial passes exactly through the knots, but notice that the error of the polynomial approximation is quite large, especially in the interval $[4, 9]$.

Fig. 15.1 Polynomial approximation (15.25) (full line) to the square root function $\sqrt{x}$ (dotted line) using data of Table 15.4. Note the large error in the interval $[4, 9]$



15.1.4 Taylor Form of the Interpolation Polynomial

Consider the Taylor form of the interpolating polynomial centred at $\hat{x}$

$$p(x) = c_0 + c_1(x - \hat{x}) + c_2(x - \hat{x})^2 + \dots + c_m(x - \hat{x})^m . \quad (15.26)$$

The Taylor form, in comparison with other forms, is attractive for performing operations on the polynomial, such differentiation and integration. The coefficients $c_0, c_1, \dots, c_m$ are found by requiring that $p(x)$ **coincide** with $q(x)$ at all nodes, that is

$$p(x_0) = q_0 , \quad p(x_1) = q_1 , \quad p(x_2) = q_2 , \quad \dots , \quad p(x_m) = q_m . \quad (15.27)$$

The interpolation equations are

$$p(x_i) = c_0 + c_1(x_i - \hat{x}) + \dots + c_m(x_i - \hat{x})^m = q_i , \quad \text{for } i = 0, 1, \dots, m , \quad (15.28)$$

which constitute a corresponding $(m + 1) \times (m + 1)$ linear algebraic system of the form (15.8), $\mathbf{AC} = \mathbf{B}$. The vector of unknowns $\mathbf{C}$, the vector of prescribed function values $\mathbf{B}$ and the coefficient matrix $\mathbf{A}$ are given as follows:

$$\mathbf{C} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \dots \\ c_m \end{bmatrix} , \quad \mathbf{B} = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ \dots \\ q_m \end{bmatrix} , \quad \mathbf{A} = \begin{bmatrix} 1 & (x_0 - \hat{x}) & \dots & (x_0 - \hat{x})^m \\ 1 & (x_1 - \hat{x}) & \dots & (x_1 - \hat{x})^m \\ 1 & (x_2 - \hat{x}) & \dots & (x_2 - \hat{x})^m \\ \dots & \dots & \dots & \dots \\ 1 & (x_m - \hat{x}) & \dots & (x_m - \hat{x})^m \end{bmatrix} . \quad (15.29)$$

System (15.8), with definitions (15.29), can be solved by existing very reliable software, publicly available, for which it is sufficient to supply the known entries of the matrix $\mathbf{A}$ and of the right-hand side vector $\mathbf{B}$ in (15.29).

Alternatively, following Laney [4], the coefficients $c_0, c_1, \dots, c_m$ in (15.26) can be computed as follows

$$c_j = \sum_{k=0}^{m-j} d_{k,j} q[x_0, \dots, x_{k+j}] , \quad j = 0, \dots, m , \quad (15.30)$$

where $q[x_0, \dots, x_{k+j}]$ are divided differences and $d_{k,j}$ can be interpreted as (some) entries of the $(m + 1) \times (m + 1)$ matrix $\mathbf{D}$

$$\mathbf{D} = \begin{bmatrix} 1 & 1 & \dots & 1 & 1 & \dots & 1 \\ (\hat{x} - x_0) & d_{1,1} & \dots & d_{1,j-1} & d_{1,j} & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ \dots & \dots & \dots & d_{k-1,j-1} & d_{k-1,j} & \dots & 0 \\ (\hat{x} - x_0) \dots (\hat{x} - x_{k-1}) & d_{k,1} & \dots & d_{k,j-1} & d_{k,j} & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ (\hat{x} - x_0) \dots (\hat{x} - x_{m-1}) & 0 & \dots & 0 & \dots & \dots & 0 \end{bmatrix}. \quad (15.31)$$

The entries of $\mathbf{D}$ are defined recursively as

$$\left. \begin{aligned} d_{0,j} &= 1, \quad j = 0, 1, \dots, m, \\ d_{k,0} &= (\hat{x} - x_{k-1})d_{k-1,0}, \quad k = 0, 1, \dots, m, \\ d_{k,j} &= d_{k,j-1} + (\hat{x} - x_{k+j-1})d_{k-1,j}, \quad k, j = 1, \dots, m. \end{aligned} \right\} \quad (15.32)$$

Having computed the coefficients c_j , the polynomial in Taylor form (15.26) is completely determined.

Example 15.6 Quadratic polynomial in Taylor form. Here $m = 2$ and the polynomial reads

$$p(x) = c_0 + c_1(x - \hat{x}) + c_2(x - \hat{x})^2. \quad (15.33)$$

To compute the coefficients c_j we first use (15.30) to obtain

$$\left. \begin{aligned} c_0 &= \sum_{k=0}^2 d_{k,0}q[x_0, \dots, x_k] = d_{0,0}q[x_0] + d_{1,0}q[x_0, x_1] + d_{2,0}q[x_0, x_1, x_2], \\ c_1 &= \sum_{k=0}^1 d_{k,1}q[x_0, \dots, x_{k+1}] = d_{0,1}q[x_0, x_1] + d_{1,1}q[x_0, x_1, x_2], \\ c_2 &= \sum_{k=0}^0 d_{k,2}q[x_0, \dots, x_{k+2}] = d_{0,2}q[x_0, x_1, x_2]. \end{aligned} \right\} \quad (15.34)$$

After calculating the coefficients $d_{i,j}$ from (15.32) and substitution into (15.34) gives

$$\left. \begin{aligned} c_0 &= q[x_0] + (\hat{x} - x_0)q[x_0, x_1] + (\hat{x} - x_0)(\hat{x} - x_1)q[x_0, x_1, x_2], \\ c_1 &= q[x_0, x_1] + (\hat{x} - x_0 + \hat{x} - x_1)q[x_0, x_1, x_2], \\ c_2 &= q[x_0, x_1, x_2]. \end{aligned} \right\} \quad (15.35)$$

Example 15.7 Consider again the quadratic polynomial in Taylor form of the previous example.

1. Write down the corresponding linear system emerging from applying the polynomial equations.

2. Find by hand analytical expressions for the coefficients of the polynomial c_0, c_1, c_2 .
3. Compare your results with those given by (15.35) above.

We next address some theoretical issues regarding the interpolating polynomial.

15.1.5 Properties of the Interpolation Polynomial

We conclude this section on basic theory of polynomial interpolation by stating some useful, classical results concerning existence, uniqueness, properties of divided differences, which are the coefficients of the interpolating polynomial in Newton's form, and the error of the interpolation. So far, we have assumed that the interpolation polynomial exists and is unique, but in fact this can be proved.

Theorem 15.8 (Existence and uniqueness) *Given the knots of Table 15.1, in which the nodes $x_0, x_1, \dots, x_m$ are distinct and the function values $q_0, q_1, \dots, q_m$ are real and arbitrary, there is a unique polynomial $p(x)$ of degree at most m such that the graph of $p(x)$ passes through the knots, that is $q_i = p(x_i)$ for $i = 0, 1, \dots, m$.*

Proof Sketch of proof. The proof of existence is constructive in nature. The easiest way to see this is to write the polynomial in Lagrangian form [1], not studied here. Also, the Newton form allows a constructive way to prove existence, as already seen through some examples and the statement of the general methodology. As to uniqueness, this is proved by contradiction, assuming the existence of a second polynomial $r(x)$ and arguing that the difference polynomial $p(x) - r(x)$ must be identically zero, from which one concludes that $p(x)$ is unique. See [1, 2] for further details. ■

Next we address an important property of divided differences, a crucial component in the Newton form of the interpolation polynomial. The property concerns the order of the arguments in divided differences. For example, one can easily verify that $q[x_0, x_1] = q[x_1, x_0]$. But in fact this result is more general, as stated below.

Theorem 15.9 Invariance property of divided differences. *The divided difference $q[x_0, x_1, \dots, x_k]$ is invariant under all permutations of its arguments, that is the nodes $x_0, x_1, \dots, x_k$. Formally:*

$$q[x_0, x_1, \dots, x_k] = q[x_{i_0}, x_{i_1}, \dots, x_{i_k}], \quad (15.36)$$

where $x_{i_0}, x_{i_1}, \dots, x_{i_k}$ is an arbitrary permutation of nodes $x_0, x_1, \dots, x_k$.

Proof Omitted. For details see any book of elementary numerical analysis, e.g. [1], [2]. ■

Next we address some results on the error $p(x) - q(x)$ of the interpolation polynomial. Following [1] we shall state three results.

Theorem 15.10 The error of the interpolation polynomial I. *If $p(x)$ is the interpolation polynomial of a function $q(x)$ at $m + 1$ distinct nodes $x_0, x_1, \dots, x_m$ in $[a, b] \subset \mathbb{R}$ and if $q(x)$ has continuous derivative $q^{(m+1)}(x)$, then there exists a point $\eta \in (a, b)$ such that*

$$p(x) - q(x) = \frac{1}{(m+1)!} q^{(m+1)}(\eta) \prod_{j=0}^m (x - x_j) . \quad (15.37)$$

Proof Omitted. See [1, 2]. ■

Needless to say that the error in (15.37) will be zero for any x that coincides with the nodes. The next result gives a bound on the error.

Theorem 15.11 The error of the interpolation polynomial II. *If $p(x)$ is the interpolation polynomial of degree at most m of a function $q(x)$ at $m + 1$ distinct and equally spaced nodes $x_0, x_1, \dots, x_m$ in $[a, b] \subset \mathbb{R}$ and if $q(x)$ has continuous derivative $q^{(m+1)}(x)$ in $[a, b]$, with $|q^{(m+1)}(x)| \leq C$, then*

$$|p(x) - q(x)| \leq \frac{1}{4(m+1)} C \left(\frac{b-a}{m} \right)^{m+1} , \quad \forall x \in [a, b] . \quad (15.38)$$

Proof Omitted. ■

The next result gives a precise expression for the error in terms of divided differences, namely

Theorem 15.12 The error of the interpolation polynomial III. *If $p(x)$ is the interpolation polynomial of degree m of a function $q(x)$ at nodes $x_0, x_1, \dots, x_m$, then for any $x \neq x_k$ we have*

$$p(x) - q(x) = q[x_0, x_1, \dots, x_m, x] \prod_{j=0}^m (x - x_j) . \quad (15.39)$$

Proof Omitted. See [1, 2]. ■

The last theoretical result stated here concerns the fact that divided differences actually represent approximations to derivatives of functions, when the samples q_k are point-function values $q_k = q(x_k)$ for a given function $q(x)$. As a matter of fact, for regularly-spaced three nodes x_0, x_1, x_2 , with spacing Δx , one can easily show that

$$q[x_0, x_1, x_2] = \frac{1}{2} \left(\frac{q_2 - 2q_1 + q_0}{\Delta x^2} \right) \approx \frac{1}{2} q^{(2)}(x) . \quad (15.40)$$

See Chap. 12 for finite difference approximation to derivatives. In fact this result is more general, as stated below.

Theorem 15.13 Divided differences as approximate derivatives. *If the derivative $q^{(m)}$ of a function $q(x)$ in $[a, b] \subset \mathbb{R}$ is continuous and if $x_0, x_1, \dots, x_m$ in $[a, b]$ are any $m + 1$ distinct nodes, then*

$$q[x_0, x_1, \dots, x_m] = \frac{1}{m!} q^{(m)}(\eta) \quad (15.41)$$

for some value η in the open interval (a, b) .

Proof Omitted. See [1, 2]. ■

In this section on the basic theory of polynomial interpolation we always assumed point values q_k corresponding to nodes x_k , in which $q_k = q(x_k)$ for some, sufficiently smooth, function $q(x)$. The q_k values could also derive from measurements, but still regarded as point values. In the next section, as well as the succeeding ones, we shall do away with these assumptions and accept discontinuous functions. Then, the q_k values will represent integral averages in appropriate sub-intervals. This viewpoint goes hand in hand with solutions of hyperbolic conservation laws, see Chaps. 7 and 8, and associated numerical methods, see Chaps. 12 and 13.

15.1.6 Interpolation from Cell Averages

In this section we study polynomial interpolation in which function values q_k , or measurements, at points x_k are understood as **cell averages**, still denoted as q_k , or as $\bar{q}_k$. In the framework of finite volume methods introduced in Chap. 13, we assumed the computational domain $[a, b] \subset \mathbb{R}$ to be discretized into a set of M cells I_i , with definitions

$$I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}], \quad \Delta x = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}, \quad x_i = \frac{1}{2}(x_{i-\frac{1}{2}} + x_{i+\frac{1}{2}}), \quad i = 1, 2, \dots, M. \quad (15.42)$$

Figure 15.2 illustrates a set of M averages in the full computational domain. For notational convenience we omit the time superscript that appears in the context of finite volume methods. Here x_i is the **cell centre** and Δx is the **cell width**, assumed constant, for simplicity. Now, in the knots $P_i = (x_i, q_i)$, q_i is the I_i -cell integral average. More generally, for any given function $q(x)$ we define cell averages as

$$q_i = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} q(x) dx. \quad (15.43)$$

Here the function $q(x)$ is not necessarily smooth; it may even contain discontinuities. Figure 15.2 illustrates a set of M cell averages that may arise from (15.43).

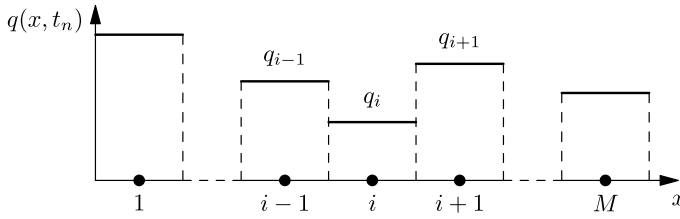


Fig. 15.2 Cell averages q_k arising in the finite volume method, in which the full computational domain has been discretized into M computational cells I_k of width Δx and cell centre x_k . The time superscript is omitted. Cell averages may also arise as integral averages (15.43) of any function $q(x)$ in cells I_k

Remark on notation. In what follows we shall consider subsets of the full finite volume domain $\{I_1, I_2, \dots, I_M\}$. Consistent with the notation of the previous section, such subsets will contain $m+1$ cells and will be re-labelled as $\{I_0, I_1, \dots, I_m\}$, with corresponding knots $P_i = (x_i, q_i)$, for $i = 0, 1, \dots, m$.

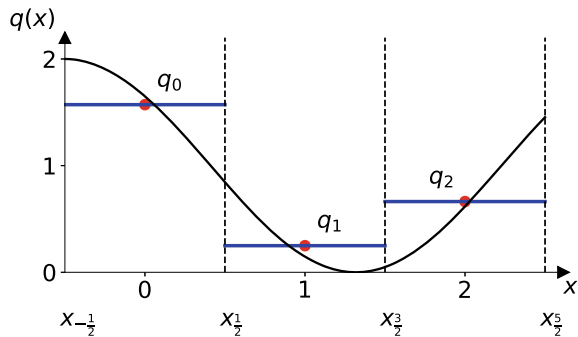
Quadratic Polynomial

First consider a function $q(x)$ defined in the subdomain formed by the following three successive cells: $I_0 = [x_{-\frac{1}{2}}, x_{\frac{1}{2}}]$, $I_1 = [x_{\frac{1}{2}}, x_{\frac{3}{2}}]$, $I_2 = [x_{\frac{3}{2}}, x_{\frac{5}{2}}]$, as shown in Fig. 15.3. Then consider cell integral averages of $q(x)$ in each cell, namely

$$q_0 = \frac{1}{\Delta x} \int_{x_{-\frac{1}{2}}}^{x_{\frac{1}{2}}} q(x) dx, \quad q_1 = \frac{1}{\Delta x} \int_{x_{\frac{1}{2}}}^{x_{\frac{3}{2}}} q(x) dx, \quad q_2 = \frac{1}{\Delta x} \int_{x_{\frac{3}{2}}}^{x_{\frac{5}{2}}} q(x) dx. \quad (15.44)$$

The three corresponding knots $P_0 = (x_0, q_0)$, $P_1 = (x_1, q_1)$, $P_2 = (x_2, q_2)$ are conveniently displayed as in Table 15.3.

Fig. 15.3 Cell integral averages $\{q_0, q_1, q_2\}$ of function $q(x)$ in cells $\{I_0, I_1, I_2\}$. Here $q(x) = 1 + \sin(x)$, for $-0.5\pi \leq x \leq 2.15\pi$. Note that $q(x)$ does not in general pass through the mid point of each cell



Now we tackle the problem of approximating a function $q(x)$ through a quadratic polynomial $p(x)$ of the form

$$p(x) = c_0 + c_1x + c_2x^2, \quad (15.45)$$

in which the unknown coefficients c_k in (15.45) are determined by imposing the following **integral-average interpolation conditions**

$$\left. \begin{aligned} \frac{1}{\Delta x} \int_{x_{-\frac{1}{2}}}^{x_{\frac{1}{2}}} p(x) dx &= \frac{1}{\Delta x} \int_{x_{-\frac{1}{2}}}^{x_{\frac{1}{2}}} q(x) dx = q_0, \\ \frac{1}{\Delta x} \int_{x_{\frac{1}{2}}}^{x_{\frac{3}{2}}} p(x) dx &= \frac{1}{\Delta x} \int_{x_{\frac{1}{2}}}^{x_{\frac{3}{2}}} q(x) dx = q_1, \\ \frac{1}{\Delta x} \int_{x_{\frac{3}{2}}}^{x_{\frac{5}{2}}} p(x) dx &= \frac{1}{\Delta x} \int_{x_{\frac{3}{2}}}^{x_{\frac{5}{2}}} q(x) dx = q_2. \end{aligned} \right\} \quad (15.46)$$

That is, the sought interpolation polynomial $p(x)$ is required to satisfy the above integral conditions (15.46). In other words, for each cell, the integral of the sought polynomial $p(x)$ reproduces the given cell-integral averages of the function $q(x)$. This is called the **conservation condition**. Recall that for point values q_i , studied in previous sections, the interpolation equations required $q_i = q(x_i)$.

Application of the first expression in (15.46) in cell I_0 gives

$$\frac{1}{\Delta x} \int_{x_{-\frac{1}{2}}}^{x_{\frac{1}{2}}} p(x) dx = \frac{1}{\Delta x} \int_{x_{-\frac{1}{2}}}^{x_{\frac{1}{2}}} (c_0 + c_1x + c_2x^2) dx = q_0, \quad (15.47)$$

which in turn yields

$$c_0 + a_{0,1}c_1 + a_{0,2}c_2 = q_0, \quad (15.48)$$

with

$$a_{0,1} = \frac{1}{\Delta x} \int_{x_{0-\frac{1}{2}}}^{x_{0+\frac{1}{2}}} x dx, \quad a_{0,2} = \frac{1}{\Delta x} \int_{x_{0-\frac{1}{2}}}^{x_{0+\frac{1}{2}}} x^2 dx. \quad (15.49)$$

Application of the second expression in (15.46) in cell I_1 yields

$$c_0 + a_{1,1}c_1 + a_{1,2}c_2 = q_1, \quad (15.50)$$

while application of the third expression in (15.46) in cell I_2 yields

$$c_0 + a_{2,1}c_1 + a_{2,2}c_2 = q_2. \quad (15.51)$$

Equations (15.48), (15.50) and (15.51) constitute a linear algebraic system for the sought coefficients c_k in (15.45), which may be written in the form

$$\mathbf{AC} = \mathbf{B} , \quad (15.52)$$

with

$$\mathbf{C} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} , \quad \mathbf{B} = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \end{bmatrix} , \quad \mathbf{A} = \begin{bmatrix} 1 & a_{0,1} & a_{0,2} \\ 1 & a_{1,1} & a_{1,2} \\ 1 & a_{2,1} & a_{2,2} \end{bmatrix} . \quad (15.53)$$

The entries $a_{i,j}$ of the matrix $\mathbf{A}$ are given as

$$a_{i,j} = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} x^j dx = \frac{1}{(j+1)\Delta x} [(x_{i+\frac{1}{2}})^{j+1} - (x_{i-\frac{1}{2}})^{j+1}] , \quad (15.54)$$

with i denoting the cell number and j denoting the power of x in the polynomial (15.45).

The General Case

For a general cell-average interpolation polynomial (15.3) defined in cells $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$, with cell centres x_i and cell averages q_i , with $i = 0, 1, 2, \dots, m$ the linear system (15.52) has

$$\mathbf{C} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \dots \\ c_m \end{bmatrix} , \quad \mathbf{B} = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ \dots \\ q_m \end{bmatrix} , \quad \mathbf{A} = \begin{bmatrix} 1 & a_{0,1} & a_{0,2} & \dots & a_{0,m} \\ 1 & a_{1,1} & a_{1,2} & \dots & a_{1,m} \\ 1 & a_{2,1} & a_{2,2} & \dots & a_{2,m} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & a_{m,1} & a_{m,2} & \dots & a_{m,m} \end{bmatrix} , \quad (15.55)$$

with definition (15.54) for the entries $a_{i,j}$ of the coefficient matrix $\mathbf{A}$. System (15.52) with definitions (15.55) can be solved by existing very reliable software, publicly available, for which it is sufficient to supply the known entries of the matrix $\mathbf{A}$ and the components of the right-hand side vector $\mathbf{B}$ in (15.55). Once the sought coefficients c_i are found, the full polynomial (15.3) is determined.

Remarks:

1. **Local coordinates.** In practice one may choose appropriate, local, x -coordinates so as to obtain constant coefficient matrices $\mathbf{A}$ in system (15.52), (15.55).
2. **Cell averages versus cell-centred point values.** It is important to realise that the integral-average interpolation condition giving $\bar{q}_i$, is in general different from the cell-centre point value $q_i = q(x_i)$ condition, that is

$$\frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} p(x) dx = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} q(x) dx = \bar{q}_i \neq q(x_i) . \quad (15.56)$$

In other words, the set of knots $\{P_i = (x_i, \bar{q}_i)\}$ gives rise to a polynomial that is different from that obtained from the set of knots $\{P_i = (x_i, q(x_i))\}$. The first-degree polynomial is an exception (verify).

3. **Methods.** The methods studied, such as the method of undetermined coefficients and Newton's method, are applicable to both sets of knots, namely

- Knots $\{P_i = (x_i, q_i)\}$ in which $q_i = q(x_i)$ is a point value.
- Knots $\{P_i = (x_i, \bar{q}_i)\}$ in which $\bar{q}_i$ is a cell integral average.

In all cases considered the problem is reduced to solving a linear system of algebraic equations to find the coefficients of the sought polynomial. Methods to solve linear systems are described in classical books on numerical analysis; see [1], for example.

15.1.7 Closing Remarks on Polynomial Interpolation

We have reviewed, very succinctly, some basic concepts in the classical theory of polynomial interpolation. Here we have put the emphasis on the Newton form and the Taylor form of the interpolating polynomial, the latter being very convenient for functional operations, such as integration and differentiation. But of course there are other forms, such as the Lagrange form and the Neville form, each of these having their own merits.

A shortcoming of interpolation polynomials, especially of very high degree, is their **oscillatory behaviour**. This is manifested in the form of the Runge phenomenon and the Gibbs oscillations; see [4] for details. To overcome the oscillatory behaviour of polynomials there are at least two possibilities. First, instead of using very high-degree (very oscillatory) global polynomials one can use piece-wise, lower-degree local polynomials, even for individual cells, to make up composite polynomial representations for the full domain of interest.

The second route results from a **non-linear polynomial interpolation technique** first proposed by Harten and collaborators [5] and presented in the next section. Here, the local polynomial is chosen according to the behaviour of the data, for which the Newton form of the interpolating polynomial turns out to be most convenient. For a local cell I_i , one builds up the polynomial by adding points to the left or to the right according to a measure of smoothness of the data. The resulting **non-linear** polynomial interpolation technique turns out to be very effective in the construction of non-linear numerical methods of higher order of accuracy, which is the ultimate aim of this chapter.

15.2 Essentially Non-Oscillatory (ENO) Reconstruction

The motivation for this section emanates from Godunov's theorem [6], which concerns **spurious oscillations** produced by **linear numerical methods**, see Chap. 12. Such oscillations are especially evident when we attempt to construct high-order methods, even second-order methods. See results at the end of Chap. 12. One way to, at least partly, circumvent Godunov's theorem comes from a novel polynomial interpolation technique, called ENO (Essentially Non-Oscillatory), proposed by Harten and collaborators [5, 7]. ENO facilitates the construction of **non-linear methods**, even when applied to solve linear equations.

The construction of ENO polynomials departs from a given set of knots $P_i = (x_i, \bar{q}_i)$ in a corresponding stencil, where $\bar{q}_i$ denotes the integral cell average in cell I_i . A distinguishing feature of ENO is that the stencil is **adaptive**, rather than fixed. The discussion also applies to point values $q_i = q(x_i)$, but for the rest of this chapter we shall assume cell averages, unless otherwise stated. ENO polynomials have two basic properties:

1. **Locality.** Each knot $P_i = (x_i, \bar{q}_i)$ will have its own associated polynomial. No global polynomial is used; as already pointed out, global, high-degree polynomials are in general highly oscillatory.
2. **Adaptivity (non-linearity).** The polynomial coefficients depend on the choice of the **stencil**, which in turn depends on the behaviour of the cell-average data $\{\bar{q}_i\}$ in various potential stencils. This procedure makes polynomial interpolation a non-linear procedure.

The local and adaptive nature of the ENO interpolation method results in a reconstruction polynomial that can be used to construct **non-linear** numerical methods to solve PDEs. Such methods are **Essentially Non-Oscillatory**. ENO numerical schemes may still exhibit spurious oscillations in the vicinity of large gradients, but their amplitudes tend to vanish as the mesh is refined. This is not the case for **linear higher order methods**, for which mesh refinement will not eliminate the oscillations, see Chap. 12.

When applying ENO interpolation polynomials to design non-linear numerical methods to solve PDEs, a polynomial of degree m based on $m + 1$ knots may give rise to a numerical method of order of accuracy $m + 1$. This will depend on the particular numerical methodology. For example, second-order numerical methods to solve PDEs are based on first-degree polynomial reconstructions.

15.2.1 ENO Interpolation from Cell Averages

ENO is a piece-wise polynomial interpolation method. Instead of constructing a global very high-degree polynomial for the full domain $[a, b]$, ENO constructs **local**

polynomials for each cell I_i of $[a, b]$. Then, the polynomial representation in the full domain $[a, b]$ is made up of the union of local polynomials. We next discuss some simple cases.

Piece-Wise Constant Polynomial

The simplest example of piece-wise polynomial is

$$p(x) = \cup_{i=1}^M p_i(x) \quad \text{with} \quad p_i(x) = \begin{cases} \bar{q}_i & \text{if } x \in I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \\ 0 & \text{if } x \notin I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \end{cases} \quad (15.57)$$

Trivially, each local polynomial consists of a single (constant) $\bar{q}_i$. Figure 15.2 depicts an example of piece-wise constant polynomial.

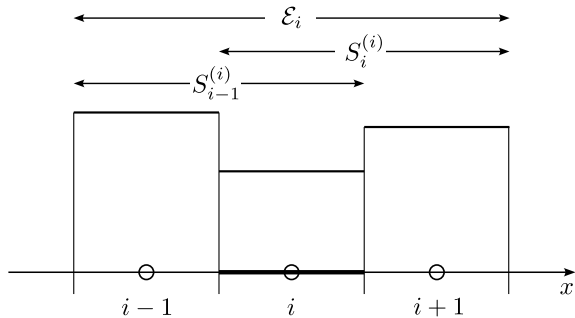
Piece-Wise 1st-Degree ENO Polynomial

The simplest non-trivial piece-wise polynomial representation in $[a, b]$ is based on polynomials of degree $m = 1$, that is, straight lines. As illustrated in Fig. 15.4 for the 1st-degree polynomial $p_i(x)$ in the generic cell I_i , in addition to cell I_i one needs one more cell, either left or right. In terms of knots, see Fig. 15.4, there are two candidates: $P_{i-1} = (x_{i-1}, \bar{q}_{i-1})$ on the left and $P_{i+1} = (x_{i+1}, \bar{q}_{i+1})$ on the right. Therefore, there are two possible **stencils** that would allow the construction of a candidate ENO polynomial $p_i(x)$, namely

$$\begin{aligned} \text{Left stencil: } S_{i-1}^{(i)} &= \{I_{i-1}, I_i\} , \\ \text{Right stencil: } S_i^{(i)} &= \{I_i, I_{i+1}\} . \end{aligned} \quad (15.58)$$

The task at hand is to find the stencil with the smoothest data. In general, the search for the ENO stencil and thus the ENO polynomial of degree m will determine an **extended stencil** $\mathcal{E}_i$ of $2m + 1$ cells. For the 1st-degree case depicted in Fig. 15.4 the extended stencil $\mathcal{E}_i$ has $2m + 1 = 3$ cells and contains the two candidate stencils

Fig. 15.4 Two candidate stencils $S_{i-1}^{(i)}$ and $S_i^{(i)}$ for constructing a 1st-degree ENO polynomial $p_i(x)$ for cell I_i . The extended stencil $\mathcal{E}_i$ of $2m + 1 = 3$ cells is also shown



$S_{i-1}^{(i)} = \{I_{i-1}, I_i\}$ and $S_i^{(i)} = \{I_i, I_{i+1}\}$. Each stencil in (15.58) generates a 1st degree polynomial, as illustrated in Fig. 15.5. By expressing each polynomial in Newton form we may write

$$\begin{aligned} \text{Left polynomial: } p_{i-1}^{(i)}(x) &= \bar{q}_i + q[x_{i-1}, x_i](x - x_i) , \\ \text{Right polynomial: } p_i^{(i)}(x) &= \bar{q}_i + q[x_i, x_{i+1}](x - x_i) . \end{aligned} \quad (15.59)$$

That is

$$p_i(x) = \bar{q}_i + \left\{ \begin{array}{l} q[x_{i-1}, x_i] \\ q[x_i, x_{i+1}] \end{array} \right\} (x - x_i) , \quad (15.60)$$

with two possibilities for choosing the coefficient of $(x - x_i)$ given in terms of divided differences, either $q[x_{i-1}, x_i]$ or $q[x_i, x_{i+1}]$.

Figure 15.5 shows the two candidate 1st degree polynomials: $p_{i-1}^{(i)}(x)$ and $p_i^{(i)}(x)$. Out of the two available candidates, ENO chooses the polynomial such that the coefficient of $(x - x_i)$ in (15.60) is the divided difference with the **smallest absolute value**. Thus, in algorithmic form we write

$$p_i(x) = \bar{q}_i + q[x_j, x_{j+1}](x - x_i) , \quad (15.61)$$

with

$$j = \begin{cases} i-1 & \text{if } |q[x_{i-1}, x_i]| < |q[x_i, x_{i+1}]| , \\ i & \text{if } |q[x_{i-1}, x_i]| \geq |q[x_i, x_{i+1}]| . \end{cases} \quad (15.62)$$

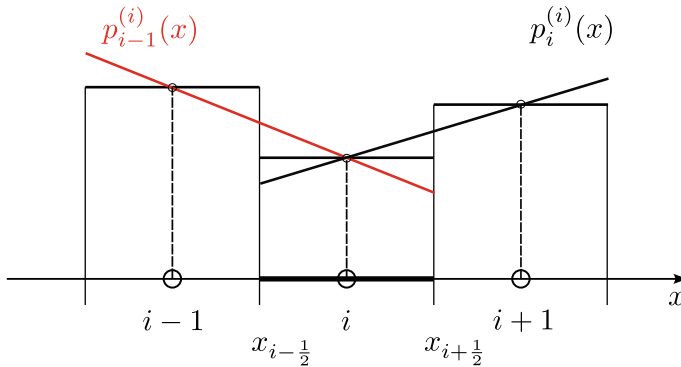


Fig. 15.5 Two candidate ENO polynomials: $p_{i-1}^{(i)}(x)$ in stencil $S_{i-1}^{(i)}$ (left) and $p_i^{(i)}(x)$ in stencil $S_i^{(i)}$ (right). Note that the 1st-degree polynomials pass exactly through the knots to uphold conservation; this property does not hold in general

Then, a composite, piece-wise first degree polynomial $p(x)$ may be defined from (15.61)–(15.62) as

$$p(x) = \begin{cases} p_i(x) & \text{if } x \in I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] , \\ 0 & \text{if } x \notin I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] . \end{cases} \quad (15.63)$$

In the composite polynomial, for each cell I_i one takes the **restriction** of $p_i(x)$, even if $p_i(x)$ is also defined outside cell I_i .

We note that the adaptivity of the ENO stencil search cannot be applied directly to the first cell I_1 on the left and the last cell I_M on the right; see Fig. 15.2. This difficulty is resolved in practice by imposing the desired boundary conditions, which usually results in an enlarged domain from adding a suitable number of fictitious cells to those in the actual computational domain.

Remark: slopes and TVD methods. We can also write polynomials (15.61)–(15.62) as

$$p_i(x) = \bar{q}_i + \Delta_i(x - x_i) , \quad (15.64)$$

where Δ_i is the **slope**, a divided difference, for which there are two choices. Large divided differences are associated with large derivatives and possible oscillations; see Section 15.1.5. Thus, effectively, the ENO method just described selects the polynomial with the smallest slope in absolute value, namely

$$\Delta_i^{ENO} = \begin{cases} \Delta_{i-\frac{1}{2}} & \text{if } |\Delta_{i-\frac{1}{2}}| < |\Delta_{i+\frac{1}{2}}| , \\ \Delta_{i+\frac{1}{2}} & \text{if } |\Delta_{i-\frac{1}{2}}| \geq |\Delta_{i+\frac{1}{2}}| , \end{cases} \quad (15.65)$$

with appropriate definitions for the slopes $\Delta_{i-\frac{1}{2}}$ and $\Delta_{i+\frac{1}{2}}$. Total Variation Diminishing methods, or TVD methods, find suitable slopes Δ_i so as to guarantee an oscillation-free method. For details on TVD methods see for example [8, 9], and references therein.

Piece-Wise Quadratic Polynomial

The aim is to construct a polynomial $p_i(x)$ of degree $m = 2$ for a cell I_i , associated to knot $P_i = (x_i, \bar{q}_i)$, requiring $m + 1 = 3$ successive cells, including cell I_i . As illustrated in Fig. 15.6, the ENO procedure determines an enlarged stencil $\mathcal{E}_i$ of $2m + 1 = 5$ consecutive cells around the target cell I_i . There are $m + 1 = 3$ possible stencils of $m + 1 = 3$ cells each, namely

$$\left. \begin{array}{l} \text{Left stencil: } S_{i-2}^{(i)} = \{I_{i-2}, I_{i-1}, I_i\} , \\ \text{Middle stencil: } S_{i-1}^{(i)} = \{I_{i-1}, I_i, I_{i+1}\} , \\ \text{Right stencil: } S_i^{(i)} = \{I_i, I_{i+1}, I_{i+2}\} . \end{array} \right\} \quad (15.66)$$

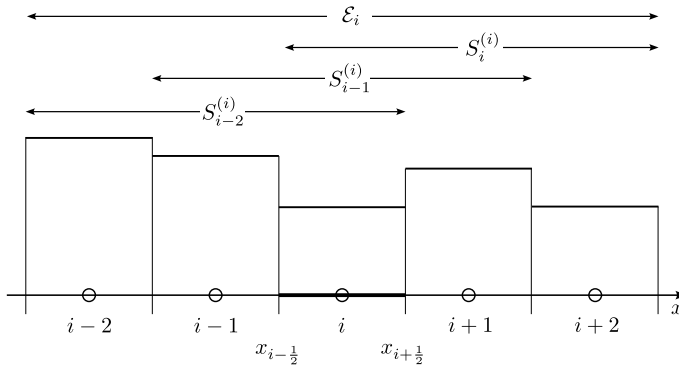


Fig. 15.6 Three candidate stencils $S_{i-2}^{(i)}$ (left), $S_{i-1}^{(i)}$ (middle) and $S_i^{(i)}$ (right) for constructing a 2nd degree ENO polynomial for cell I_i . The extended stencil $\mathcal{E}_i$ of $2m + 1 = 5$ cells is also shown

The sought 2nd-degree ENO polynomial can be adaptively constructed from

$$p_i(x) = \bar{q}_i + \left\{ \begin{array}{l} q[x_{i-1}, x_i] \\ q[x_i, x_{i+1}] \end{array} \right\} (x - x_i) + \left\{ \begin{array}{l} \left\{ \begin{array}{l} q[x_{i-2}, x_{i-1}, x_i] \\ q[x_{i-1}, x_i, x_{i+1}] \end{array} \right\} (x - x_i)(x - x_{i-1}) , \\ \left\{ \begin{array}{l} q[x_{i-1}, x_i, x_{i+1}] \\ q[x_i, x_{i+1}, x_{i+2}] \end{array} \right\} (x - x_i)(x - x_{i+1}) . \end{array} \right. \quad (15.67)$$

On the right hand side of (15.67), in addition to the constant first term $\bar{q}_i$, there is a first-degree term and a second-degree term, for which there are various choices for the coefficients given in terms of divided differences. The final choice for the ENO polynomial will depend on the relative size of divided differences in absolute value, as illustrated in Fig. 15.7 and explained below.

1. The 1st-degree part of the ENO polynomial (15.67) is just like the 1st-degree polynomial (15.61)–(15.62); the choice of slope will depend on the relative size of the 1st divided differences in absolute value.
2. The 2nd-degree part in (15.67) will depend on the choice of the 1st divided difference, as detailed below.
3. If $|q[x_{i-1}, x_i]| < |q[x_i, x_{i+1}]|$, then the 2nd-degree part will choose the coefficient from the top quadratics in (15.67).

This in turn has two choices, which are determined from comparing the divided differences $q[x_{i-2}, x_{i-1}, x_i]$ and $q[x_{i-1}, x_i, x_{i+1}]$ in absolute value.

4. If the 1st-degree polynomial (15.67) chooses the bottom divided difference, then the 2nd-degree part will choose the coefficient from the bottom quadratics in (15.67), for which again there are two choices, whichever has the smallest 2nd divided difference.

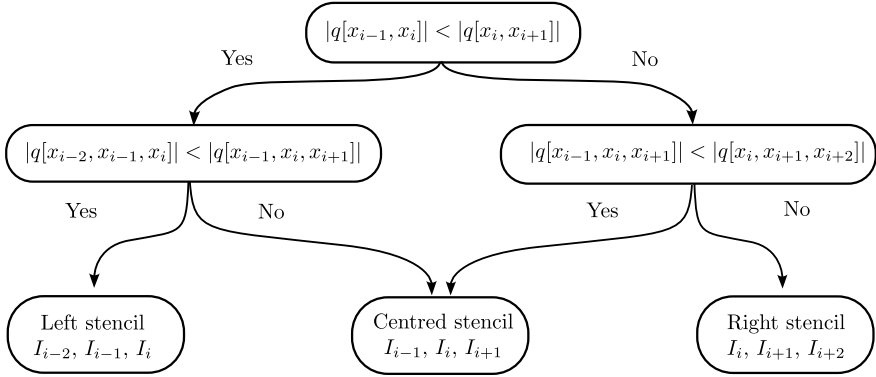


Fig. 15.7 Algorithm for selecting the ENO stencil for constructing the quadratic ENO polynomial in cell I_i . Potentially, there are three candidates: $S_{i-2}^{(i)}$ (left), $S_{i-1}^{(i)}$ (centre) and $S_i^{(i)}$ (right) (adapted from Laney [4])

The final expression for the chosen ENO polynomial from (15.67) may succinctly be written as

$$p_i(x) = q_i + q[x_j, x_{j+1}](x - x_i) + q[x_k, x_{k+1}, x_{k+2}](x - x_j)(x - x_{j+1}), \quad (15.68)$$

with j chosen as in (15.62) and k chosen as

$$k = \begin{cases} j-1 & \text{if } |q[x_{j-1}, x_j, x_{j+1}]| < |q[x_j, x_{j+1}, x_{j+2}]|, \\ j & \text{if } |q[x_{j-1}, x_j, x_{j+1}]| \geq |q[x_j, x_{j+1}, x_{j+2}]|. \end{cases} \quad (15.69)$$

The above procedure for constructing the 1st degree and 2nd degree ENO polynomials can be generalised to any degree m , as explained below.

15.2.2 The General ENO Polynomial

The aim is to construct an ENO polynomial $p_i(x)$ of arbitrary degree m for a cell I_i in the computational domain $[a, b]$. As illustrated in Figure 15.8, the search for the ENO stencil takes place within an **extended stencil** $\mathcal{E}_i$ of $2m+1$ consecutive cells around the target cell I_i , with m cells to the left of I_i and m cells to the right of I_i . The extended stencil $\mathcal{E}_i$ contains $m+1$ stencils of $m+1$ cells each, which are candidates to become the sought ENO stencil on which to construct the ENO polynomial. We denote a generic, potential ENO stencil as $S_s^{(i)}$, with

$$S_s^{(i)} = \{I_s, I_{s+1}, I_{s+2}, \dots, I_{s+m}\}, \quad i-m \leq s \leq i. \quad (15.70)$$

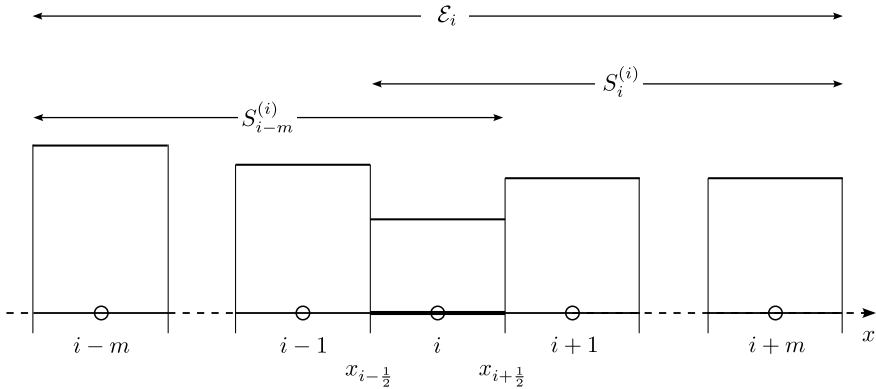


Fig. 15.8 Extended stencil $\mathcal{E}_i$ centred at cell I_i and containing $2m + 1$ contiguous cells. $\mathcal{E}_i$ includes $m + 1$ local ENO-candidate stencils $S_s^{(i)}$ of $m + 1$ cells each. Each local stencil will be identified by their left-most cell I_s , with $i - m \leq s \leq i$. The figure also shows the two extreme stencils: $S_{i-m}^{(i)}$ on the left and $S_i^{(i)}$ on the right. All stencils contain cell I_i

Here, the positive integer s represents the left displacement with respect to the index i , called the **shift**. I_s is the left-most cell of stencil $S_s^{(i)}$. The left-most possible sub-stencil is $S_{i-m}^{(i)}$, while the right-most possible sub-stencil is $S_i^{(i)}$. See Fig. 15.8.

Next we describe the search procedure to identify the ENO stencil, by identifying its left-most cell I_s , that is by identifying the **shift** s , with $i - m \leq s \leq i$.

ENO Stencil Search Procedure

The general search procedure for the **shift** s is as follows:

- Set $s_0 = i$
- For $k = 0, \dots, m - 1$ set

$$s_{k+1} = \begin{cases} s_k - 1 & \text{if } |q[x_{s_k-1}, \dots, x_{s_k+k}]| < |q[x_{s_k}, \dots, x_{s_k+k+1}]|, \\ s_k & \text{otherwise.} \end{cases} \quad (15.71)$$

- At the end of the search, when $k = m - 1$, the left-most subindex that identifies the ENO stencil for cell I_i is $s := s_m$, and we have

$$\begin{cases} \text{Stencil:} & S_s^{(i)} = \{I_s, I_{s+1}, \dots, I_{s+m}\}, \\ \text{Nodes:} & [x_s, x_{s+1}, \dots, x_{s+m-1}, x_{s+m}], \\ \text{Cell averages:} & [\bar{q}_s, \bar{q}_{s+1}, \dots, \bar{q}_{s+m-1}, \bar{q}_{s+m}]. \end{cases} \quad (15.72)$$

Having found the ENO stencil, we express the m th degree ENO reconstruction polynomial in Newton form as

$$p_i(x) = \sum_{j=0}^m q[x_s, \dots, x_{s+j}] \prod_{k=0}^{j-1} (x - x_{s+k}) . \quad (15.73)$$

Compare (15.73) with equation (15.20).

Remarks on the ENO polynomial.

1. The ENO reconstruction procedure just described applies to knots $P_i = (x_i, \bar{q}_i)$, where $\bar{q}_i$ is a cell integral average, and also the case $P_i = (x_i, q(x_i))$.
2. The ENO polynomial $p_i(x)$ (15.73) does not, in general, satisfy the conservation property (15.46).
3. For a conservative ENO reconstruction one requires a modification of the ENO reconstruction procedure described above. There are two well-established ways of doing this. One is called **Reconstruction via Deconvolution** [5], for which the mesh spacing Δx is assumed to be constant. The second one does away with this limitation and is called the **Primitive Function** reconstruction [10]. In Sect. 15.3.1 we give a succinct presentation of the primitive function reconstruction procedure.
4. In Sect. 15.3.2 we shall present an alternative, more direct way of obtaining the conservative ENO polynomial that avoids the primitive function approach. In such method one simply applies the linear interpolation methods described in Sect. 15.1.6 on the identified ENO stencil (15.72) found through the ENO search procedure.

15.3 Conservative ENO Reconstruction Methods

As already pointed out, the previously described ENO reconstruction method does not produce a conservative reconstruction, even if the samples are cell-integral averages. In this section we remedy this shortcoming by studying two ways of constructing conservative ENO polynomials $p_i(x)$ for cell I_i , in the sense that the integral average of $p_i(x)$ in cell I_i reproduces the given cell average $\bar{q}_i$.

15.3.1 ENO Reconstruction via the Primitive Function

The **primitive-function** approach [4, 11, 12] to construct a conservative ENO polynomial considers, in addition to cell-integral averages $\bar{q}_j$, samples $Q_{i+1/2}$ at the

intercell boundaries $x_{i+1/2}$, as we shall explain shortly. Recall that a function $Q(x)$ is the primitive function of $q(x)$, or antiderivative, if

$$q(x) = \frac{d}{dx} Q(x) , \quad \text{or} \quad Q(x) = \int_{x_{-1/2}}^x q(y) dy , \quad (15.74)$$

where $x_{-1/2}$ is the left most interface in the computational domain made out of cells I_j , with $j = 1, \dots, M$. Then, **point-samples** $Q_{i+1/2}$ of the function $Q(x)$ are defined at cell interfaces as follows

$$Q_{i+1/2} := Q(x_{i+1/2}) = \sum_{j=0}^i \int_{x_{j-1/2}}^{x_{j+1/2}} q(y) dy = \sum_{j=0}^i \Delta x_j \bar{q}_j . \quad (15.75)$$

That is, the partial sums of cell averages $\bar{q}_j$ are the samples of the primitive function $Q(x)$. Given the knots $P_{j+1/2} = (x_{j+1/2}, Q_{j+1/2})$, the task at hand is to construct the interpolating polynomial $P_i(x)$ of degree $m + 1$ passing through $(x_{j+1/2}, Q_{j+1/2})$; that is the primitive function polynomial is of **one degree higher** than the sought reconstruction polynomial $p_i(x)$. Figure 15.9 illustrates the samples for the construction of a general primitive function ENO polynomial.

The sought ENO reconstruction polynomial $p_i(x)$ is obtained from differentiation, as

$$p_i(x) = \frac{d}{dx} P_i(x) . \quad (15.76)$$

The sought reconstruction polynomial $P_i(x)$ will be expressed in Newton form and will be of the ENO type, for which we can apply directly what has been presented in Section 15.2. However, we note the following differences:

1. The sample values are point values $Q_{i+1/2} := Q(x_{i+1/2})$.
2. The nodes are positioned at interfaces $x_{i+1/2}$.
3. There is one extra point to the stencil, resulting in $m + 2$ knots that will support a polynomial of degree $m + 1$.

Figure 15.9 depicts the general case for the primitive function formulation. Compare with the general case of the non-conservative ENO reconstruction, see Fig. 15.8. In the primitive function case the stencils are based on interface point values $Q_{i+1/2}$, noting that for a polynomial of degree $m + 1$ there are $m + 2$ sample knots. Before proceeding any further, we revisit Newton's divided differences for some simple cases, in this new context of the primitive function. Consider the generic cell $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ with intercell boundaries $x_{i-\frac{1}{2}}$ (left) and $x_{i+\frac{1}{2}}$ (right). Recall that from (15.75) we have

$$Q_{i+1/2} = \sum_{j=0}^i \Delta x_j \bar{q}_j . \quad (15.77)$$

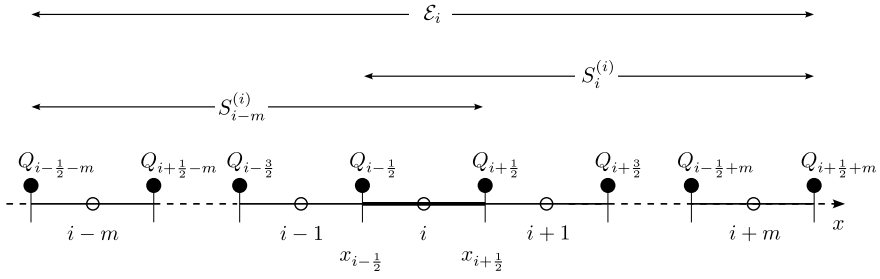


Fig. 15.9 Stencils and samples for ENO polynomial $P_i(x)$ for the primitive function. The algorithm starts from two knots placed at the boundaries of the target cell I_i , identified by filled circles. A new node is added either left or right, depending on divided differences in absolute value, in the usual ENO manner described in Sect. 15.2

Then, the following examples of divided differences result:

$$\left. \begin{aligned} Q[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] &= \frac{Q_{i+\frac{1}{2}} - Q_{i-\frac{1}{2}}}{x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}} = \bar{q}_i, \\ Q[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}, x_{i+\frac{3}{2}}] &= \frac{Q[x_{i+\frac{1}{2}}, x_{i+\frac{3}{2}}] - Q[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]}{x_{i+\frac{3}{2}} - x_{i-\frac{1}{2}}} = \frac{\bar{q}_{i+1} - \bar{q}_i}{2\Delta x}, \\ Q[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}, x_{i+\frac{3}{2}}, x_{i+\frac{5}{2}}] &= \frac{\bar{q}_{i+2} - 2\bar{q}_{i+1} + \bar{q}_i}{6\Delta x^2}. \end{aligned} \right\} \quad (15.78)$$

The construction of the polynomial $P_i(x)$ in Newton form is started from the two knots $P_{i-\frac{1}{2}} = (x_{i-\frac{1}{2}}, Q_{i-\frac{1}{2}})$ and $P_{i+\frac{1}{2}} = (x_{i+\frac{1}{2}}, Q_{i+\frac{1}{2}})$, see Fig. 15.9. To this end we apply the ENO-stencil selection algorithm of Sect. 15.2.2, which was started from just one knot. The sought stencil is built by adding one point at the time. The new point is added either to the left or to the right, depending on the relative size of divided differences in absolute value. Each time one adds one degree to the sought polynomial of degree $m + 1$.

ENO Polynomial Selection Procedure

Consider the general case as depicted in Fig. 15.9. The selection procedure for the shift s is as follows:

- Set $s_0 = i + \frac{1}{2}$ and $s_1 = i - \frac{1}{2}$.
- For $k = 1, \dots, m$

$$s_{k+1} = \begin{cases} s_k - 1 & \text{if } |Q[x_{s_k-1}, \dots, x_{s_k+k}]| < |Q[x_{s_k}, \dots, x_{s_k+k+1}]|, \\ s_k & \text{otherwise.} \end{cases} \quad (15.79)$$

At the end of the search, when $k = m$, the left-most subindex that identifies the ENO stencil for cell I_i is $s := s_{m+1}$. We have

$$\text{Nodes for ENO stencil: } [x_s, x_{s+1}, \dots, x_{s+m-1}, x_{s+m+1}] . \quad (15.80)$$

Having found the stencil, we express the $(m + 1)$ th degree ENO reconstruction polynomial in Newton form as

$$P_i(x) = \sum_{j=0}^{m+1} Q[x_s, \dots, x_{s+j}] \prod_{k=0}^{j-1} (x - x_{s+k}) . \quad (15.81)$$

This is not the sought ENO polynomial from the primitive function formulation. Compare with Eq. (15.20). This is yet to be determined.

Primitive Function Polynomial in Taylor Form

As the sought ENO polynomial $p_i(x)$ is the derivative of the ENO primitive polynomial $P_i(x)$, see Equation (15.76), it is convenient to express $P_i(x)$ in Taylor form as follows

$$P_i(x) = \sum_{j=0}^{m+1} c_j (x - x_i)^j . \quad (15.82)$$

The coefficients $c_0, c_1, \dots, c_{m+1}$ depend on the stencil of the polynomial. The task at hand is to transform the coefficients from (15.81) to (15.82). Following [4] we have

$$c_j = \sum_{k=0}^{m+1-j} d_{k,j} Q[x_s, \dots, x_{s+j+k}] , \quad j = 0, \dots, m+1 , \quad (15.83)$$

where

$$\left. \begin{aligned} d_{0,j} &= 1 , \\ d_{k,0} &= (x_i - x_{s+k-1}) d_{k-1,0} , \\ d_{k,j} &= d_{k,j-1} + (x_i - x_{s+k+j-1}) d_{k-1,j} . \end{aligned} \right\} \quad (15.84)$$

Finally, the sought conservative polynomial is obtained from differentiating $P_i(x)$ in (15.82), to obtain

$$p_i(x) = \sum_{j=1}^{m+1} j c_j (x - x_i)^{j-1} , \quad (15.85)$$

with coefficients c_j given in (15.83)–(15.84).

15.3.2 Direct ENO (dENO) Conservative Reconstruction

There is an alternative, more direct, way to construct the conservative ENO polynomial of degree m , without having to construct a polynomial of one degree higher and then differentiate it. To this end we assume that the ENO stencil $S_s^{(i)}$ is available from the usual **ENO search** described in Sect. 15.2.2, and given in (15.72), namely

$$S_s^{(i)} = \{I_s, I_{s+1}, \dots, I_{s+m}\} . \quad (15.86)$$

Recall that s is the shift, which results from direct application of the ENO search algorithm (15.71). Next one chooses the form of the sought reconstruction polynomial $p_i(x)$ for cell $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$, centred at $x = x_i$. For convenience we assume the Taylor form

$$p_i(x) = \sum_{k=0}^m c_k (x - x_i)^k . \quad (15.87)$$

The coefficients $c_0, c_1, \dots, c_m$ depend on the stencil of the polynomial and are obtained by enforcing the conservation property in each cell of the stencil (15.86). That is

$$\frac{1}{\Delta x} \int_{I_j} p_i(x) dx = \bar{q}_j , \text{ for } j = s, s+1, \dots, s+m . \quad (15.88)$$

This is entirely analogous to the linear case in Sect. 15.1.6. The non-linearity of the scheme is built in via the determination of the ENO stencil (15.86), which is a non-linear procedure.

By enforcing conservation (15.88) on polynomial (15.87), in each cell I_j , with $j = s, s+1, \dots, s+m$, we obtain the following linear system

$$\mathbf{A}\mathbf{C} = \mathbf{B} \quad (15.89)$$

with

$$\mathbf{C} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \bar{q}_0 \\ \bar{q}_1 \\ \bar{q}_2 \\ \vdots \\ \bar{q}_m \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & a_{0,1} & a_{0,2} & \dots & a_{0,m} \\ 1 & a_{1,1} & a_{1,2} & \dots & a_{1,m} \\ 1 & a_{2,1} & a_{2,2} & \dots & a_{2,m} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & a_{m,1} & a_{m,2} & \dots & a_{m,m} \end{bmatrix} . \quad (15.90)$$

The entries $a_{j,k}$ of the matrix $\mathbf{A}$ are given as

$$a_{j,k} = \frac{1}{\Delta x} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} (x - x_i)^k dx , \quad a_{j,0} = 1 , \quad (15.91)$$

for $j = s, s + 1, \dots, s + m$ and $k = 1, 2, \dots, m$. Here i denotes the number of the cell of interest, j denotes the cell in the stencil in which conservation is enforced and k is the power of $(x - x_i)$ in the polynomial (15.87) expressed in Taylor form. Once the sought coefficients c_k are found by solving the linear system (15.88)–(15.89), the full conservative ENO polynomial (15.87) is determined.

In the following we make some remarks regarding the three available forms of ENO reconstruction polynomials.

Remarks.

1. The ENO reconstruction polynomial in Newton form described in Sect. 15.2.2 is not conservative.
2. The primitive function version of the ENO reconstruction polynomial described in Sect. 15.3.1 is conservative, but requires (i) extension of the stencil of $m + 1$ cells to $m + 2$ cells and (ii) construction of a non-conservative ENO polynomial $P_i(x)$ of degree $m + 1$. Then the sought conservative ENO polynomial $p_i(x)$ results from differentiation of $P_i(x)$.
3. The direct ENO polynomial of this section is conservative and uses a stencil of $m + 1$ cells. The coefficients of the sought polynomial are obtained from enforcing conservation, as in the linear case. The non-linear part of dENO is in the identification of the stencil (15.86) via the usual ENO search. Compare with Sect. 15.1.6.
4. The resulting conservative ENO polynomials based on primitive function and the direct form just described are entirely equivalent, as demonstrated in [13]. The dENO methodology is conceptually simpler; it is straightforward to implement in practice and may prove useful for constructing variations of ENO; see [12, 14, 15].

15.4 Weighted Essentially Non-oscillatory Reconstruction

The Weighted Essentially Non-Oscillatory (WENO) reconstruction method, or weighted ENO, was originally introduced by Liu, Osher and Chan [16] and was further developed by Jiang and Shu [17], amongst others. WENO departs from the collection of all $m + 1$ potential ENO polynomials of degree m defined in the $m + 1$ ENO stencils with $m + 1$ cells each. Recall that the union of all ENO stencils form the **extended** stencil $\mathcal{E}_i$ of $2m + 1$ cells. See Fig. 15.8 and Sect. 15.2.2 for notation. The WENO polynomial results from taking a **linear combination** of all ENO-candidate polynomials in the $m + 1$ stencils contained in $\mathcal{E}_i$.

The motivation for introducing WENO was to overcome some shortcomings of ENO, such as the high sensitivity of the stencil selection procedure to small perturbations, even round-off perturbations, in the vicinity of zeros of the solution and

its derivatives [17]. This difficulty is evident when computing steady state solutions through time marching [14]. Another observed shortcoming of ENO is that for smooth functions exhibiting large gradients, the scheme may fail to attain the theoretically expected order of accuracy when performing empirical convergence rate studies [12, 13, 18].

In this section we first present the original WENO reconstruction scheme [16, 17] and then present one of its most recent variations due to Dumbser and Käser [19]. The comprehensive review of Shu [22] is highly recommended.

15.4.1 The Original WENO Scheme

As in Sect. 15.2, the aim here is to construct a local non-linear polynomial for a generic cell $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$, based on integral cell averages and defined in the entire cell. For convenience we recall the direct ENO scheme of Sect. 15.3.2. There is a total number of $m + 1$ potential ENO stencils $S_s^{(i)}$ defined by (15.72) of $m + 1$ cells each. Each $S_s^{(i)}$ supports a potential ENO polynomial $p_s^{(i)}(x)$ of degree m . The ENO polynomial representation in cell I_i will consist of a single polynomial out of these $m + 1$ candidates.

Figure 15.8 depicts potential ENO stencils contained in the extended stencil $\mathcal{E}_i$, while Table 15.6 establishes the notation for the potential ENO stencils and corresponding polynomials $p_s^{(i)}(x)$ of degree m . The construction of the WENO polynomial for the cell $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ relies on two items, namely

- The entire set of $m + 1$ potential ENO polynomials $p_s^{(i)}$ of degree m .
- The extended polynomial $\mathcal{P}_i^E(x)$ of degree $2m$ defined in the extended stencil $\mathcal{E}_i$ of $2m + 1$ cells that is the union of all $m + 1$ potential ENO stencils.

Figure 15.8 illustrates the $m + 1$ potential ENO stencils and the extended stencil $\mathcal{E}_i$ of $2m + 1$ cells for the extended polynomial $\mathcal{P}_i^E(x)$ of degree $2m$.

Table 15.6 Potential ENO stencils $S_s^{(i)}$ of $m + 1$ cells and corresponding polynomials $p_s^{(i)}(x)$ of degree m , with $s = i - m, i - m + 1, \dots, i - 1, i$. The subindex in $S_s^{(i)}$ denotes the left-most cell in the stencil. Figure 15.8 depicts the local stencils $S_s^{(i)}$ and the extended stencil $\mathcal{E}_i$

Sub-index: s	Stencil: $S_s^{(i)}$	Polynomial: $p_s^{(i)}(x)$
$i - m$	$\{I_{i-m}, I_{i-m+1}, \dots, I_i\}$	$p_{i-m}^{(i)}(x)$
$i - m + 1$	$\{I_{i-m+1}, I_{i-m+2}, \dots, I_{i+1}\}$	$p_{i-m+1}^{(i)}(x)$
$\dots$	$\dots$	$\dots$
$i - 1$	$\{I_{i-1}, I_i, \dots, I_{i+m-1}\}$	$p_{i-1}^{(i)}(x)$
i	$\{I_i, I_{i+2}, \dots, I_{i+m}\}$	$p_i^{(i)}(x)$

Note that each of the local polynomials $p_s^{(i)}$ of degree m (potentially an ENO polynomial) and the extended polynomial $\mathcal{P}_i^E(x)$ of degree $2m$ are **linear**, and are constructed by imposing conservation, using any of the methodologies studied in previous sections.

The sought WENO polynomial $P_i^W(x)$ for cell I_i is constructed as follows:

1. **Linear combination.** Express the sought WENO, linear polynomial $P_i^W(x)$ as the linear combination of all lower-degree linear polynomials $p_s^{(i)}(x)$, namely

$$P_i^W(x) = \sum_{s=i-m}^i \lambda_s p_s^{(i)}(x) . \quad (15.92)$$

2. **Linear weights.** The still unknown coefficients λ_s in (15.92), called **linear weights**, are obtained by enforcing the condition

$$P_i^W(x_{i+\frac{1}{2}}) = \mathcal{P}_i^E(x_{i+\frac{1}{2}}) . \quad (15.93)$$

Condition (15.93) requires that the polynomial (15.92) formed by the linear combination coincides with the higher-degree, extended polynomial $\mathcal{P}_i^E(x)$ at the fixed intercell position $x = x_{i+\frac{1}{2}}$. For details on the derivation of the linear weights λ_s from (15.93) for a simple case see Sect. 15.4.2. For the general case see Section 15.4.3. See also Shu [22], Qiu and Shu in [23] and Montecinos and Toro [15]. As defined so far, the resulting WENO polynomial is **linear**, that is the weights do not depend on the data $\{\bar{q}_s\}$. The resulting numerical method will be linear and therefore oscillatory.

3. **Smoothness indicators.** To construct a non-linear version of WENO one modifies the linear weights as follows:

- First, compute the smoothness indicators

$$\beta_s = \sum_{l=1}^m \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \Delta x^{2l-1} \left[\frac{d^l}{dx^l} p_s^{(i)}(x) \right]^2 dx , \text{ for } i-m \leq s \leq i . \quad (15.94)$$

- Then, define non-linear weights as

$$\alpha_s = \frac{\lambda_s}{(\epsilon + \beta_s)^r} , \text{ for } i-m \leq s \leq i . \quad (15.95)$$

Here r and ϵ are two parameters of the WENO method, to be specified by the user. The role of ϵ is to avoid division by zero.

- Define re-scaled non-linear weights as

$$w_s = \frac{\alpha_s}{\sum_{k=i-m}^i \alpha_k} , \text{ for } i-m \leq s \leq i . \quad (15.96)$$

4. **The non-linear WENO polynomial.** Finally, the sought non-linear WENO polynomial, defined in the entire cell $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$, is given thus

$$P_i^W(x) = \sum_{s=i-m}^i w_s p_s^{(i)}(x) \quad \forall x \in I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] . \quad (15.97)$$

This completes the description of the WENO method, though some additional remarks are in order.

Remarks on the WENO scheme:

1. The WENO polynomial $P_i^W(x)$, as described, is defined in the entire cell I_i . This will allow the computation of limiting values left and right of the cell interface to define initial data for Riemann problems, possibly generalised, at the interface to compute the numerical flux.
2. The fact that $P_i^W(x)$ is defined in the interior of cell I_i makes it possible to treat source terms to the required high-order of accuracy, consistent with that of the numerical flux. Source terms are represented by volume integrals within the entire control volume $V_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [t^n, t^{n+1}]$.
3. The extended stencil $\mathcal{E}_i$ is the union of all potential ENO sub-stencils, ranging from cell I_{i-m} on the left to cell I_{i+m} on the right, with a total of $2m + 1$ cells.
4. The WENO parameters r and ϵ in (15.95) must be specified by the user. The choice is largely empirical. Recommended values for the WENO method just described are $r = 2$ and $\epsilon = 10^{-6}$. For theoretical considerations on the choice of these parameters see the works of Arándiga et al. [24] and Baeza et al. [25, 26]. For more details on the WENO reconstruction methodology we recommend Shu's review [22].
5. There are several variations of WENO in the literature. See [19, 27, 28], for example.

Next we deal with the pending issue of constructing the linear weights λ_s in (15.92) and (15.95). We start with an example.

15.4.2 WENO Linear Weights: Example

Here, via an example, we illustrate in full detail the derivation of the linear weights for the WENO polynomial of first degree. Consider the following polynomials in their respective stencils

$$\left. \begin{aligned} p_{i-1}^{(i)}(x) &= c_0 + c_1(x - x_i) , \quad S_{i-1}^i : \{I_{i-1}, I_i\} , \\ p_i^{(i)}(x) &= c_0 + c_1(x - x_i) , \quad S_i^i : \{I_i, I_{i+1}\} , \\ \mathcal{P}_i^E(x) &= c_0 + c_1(x - x_i) + c_2(x - x_i)^2 , \quad \mathcal{E}_i : \{I_{i-1}, I_i, I_{i+1}\} . \end{aligned} \right\} \quad (15.98)$$

Here $p_{i-1}^{(i)}(x)$ and $p_i^{(i)}(x)$ are two first-degree polynomials considered to be ENO candidate polynomials; $\mathcal{P}_i^E(x)$ is the extended second degree polynomial in the enlarged stencil. In what follows we shall construct conservative versions of these polynomials, whereby the unknown coefficients c_k are obtained by enforcing conservation, in the usual manner.

Potential ENO polynomials. First, evaluation of the integral

$$\frac{1}{\Delta x} \int_{I_{i-1}} p_{i-1}^{(i)}(x) dx = \frac{1}{\Delta x} \int_{I_{i-1}} [c_0 + c_1(x - x_i)] dx = \bar{q}_{i-1} \quad (15.99)$$

leads to the algebraic equation

$$c_0 - c_1 \Delta x = \bar{q}_{i-1} . \quad (15.100)$$

Analogously, integration of $p_i^{(i)}(x)$ in cell I_i

$$\frac{1}{\Delta x} \int_{I_i} p_i^{(i)}(x) dx = \frac{1}{\Delta x} \int_{I_i} [c_0 + c_1(x - x_i)] dx = \bar{q}_i \quad (15.101)$$

leads to

$$c_0 = \bar{q}_i . \quad (15.102)$$

Solution of the algebraic system formed by (15.100) and (15.102) gives

$$c_0 = \bar{q}_i , \quad c_1 = \frac{\bar{q}_i - \bar{q}_{i-1}}{\Delta x} . \quad (15.103)$$

Therefore

$$p_{i-1}^{(i)}(x) = \bar{q}_i + \frac{\bar{q}_i - \bar{q}_{i-1}}{\Delta x} (x - x_i) . \quad (15.104)$$

Analogously, by enforcing conservation on $p_i^{(i)}(x)$ in cells I_i and I_{i+1} gives

$$c_0 = \bar{q}_i , \quad c_1 = \frac{\bar{q}_{i+1} - \bar{q}_i}{\Delta x} . \quad (15.105)$$

Therefore

$$p_i^{(i)}(x) = \bar{q}_i + \frac{\bar{q}_{i+1} - \bar{q}_i}{\Delta x} (x - x_i) . \quad (15.106)$$

So far, we have expressed the polynomials $p_{i-1}^{(i)}(x)$ and $p_i^{(i)}(x)$ in (15.98) as resulting from enforcing conservation in each of the cells of their respective stencils.

Extended higher-degree polynomial. Next, the same approach is applied to construct the second-degree polynomial $p_i^E(x)$ in (15.98) in the extended stencil $\mathcal{E}_i$. Integration of $p_i^E(x)$ in the three cells I_{i-1} , I_i , I_{i+1} leads to the following linear algebraic system of three equations for the three unknown coefficients c_0 , c_1 and c_2 , namely

$$\left. \begin{aligned} c_0 - c_1 \Delta x + \frac{13}{12} c_2 \Delta x^2 &= \bar{q}_{i-1} , \\ c_0 + \frac{1}{12} c_2 \Delta x^2 &= \bar{q}_i , \\ c_0 + c_1 \Delta x + \frac{13}{12} c_2 \Delta x^2 &= \bar{q}_{i+1} . \end{aligned} \right\} \quad (15.107)$$

The solution of (15.107) is

$$\left. \begin{aligned} c_0 &= \frac{-\bar{q}_{i-1} + 26\bar{q}_i - \bar{q}_{i+1}}{24} , \\ c_1 &= \frac{\bar{q}_{i+1} - \bar{q}_{i-1}}{2\Delta x} , \\ c_2 &= \frac{\bar{q}_{i-1} - 2\bar{q}_i + \bar{q}_{i+1}}{2\Delta x^2} . \end{aligned} \right\} \quad (15.108)$$

Determination of the linear weights. Next, following (15.93), we evaluate all relevant polynomials at the fixed intercell position $x = x_{i+\frac{1}{2}}$, in which case the polynomials can be expressed as linear combinations of the set of cells averages $\bar{q}_s$. The two (potential) ENO polynomials are

$$\left. \begin{aligned} p_{i-1}(x_{i+\frac{1}{2}}) &= \gamma_{i-1}^{(i-1)} \bar{q}_{i-1} + \gamma_i^{(i-1)} \bar{q}_i , \\ p_i(x_{i+\frac{1}{2}}) &= \gamma_i^{(i)} \bar{q}_i + \gamma_{i+1}^{(i)} \bar{q}_{i+1} . \end{aligned} \right\} \quad (15.109)$$

The reader can easily verify that the coefficients above are given as

$$\gamma_{i-1}^{(i-1)} = -\frac{1}{2} , \quad \gamma_i^{(i-1)} = \frac{3}{2} , \quad \gamma_i^{(i)} = \frac{1}{2} , \quad \gamma_{i+1}^{(i)} = \frac{1}{2} . \quad (15.110)$$

Analogously, the extended polynomial evaluated at $x = x_{i+\frac{1}{2}}$ is

$$\mathcal{P}_i^E(x_{i+\frac{1}{2}}) = \beta_{i-1} \bar{q}_{i-1} + \beta_i \bar{q}_i + \beta_{i+1} \bar{q}_{i+1} , \quad (15.111)$$

with

$$\beta_{i-1} = -\frac{1}{6} , \quad \beta_i = \frac{5}{6} , \quad \beta_{i+1} = \frac{1}{3} . \quad (15.112)$$

Then we form a linear combination of polynomials $p_{i-1}(x_{i+\frac{1}{2}})$ and $p_i(x_{i+\frac{1}{2}})$ involving two **linear weights** λ_{i-1} and λ_i , namely

$$P_i^W(x) = \lambda_{i-1}p_{i-1}(x) + \lambda_i p_i(x) . \quad (15.113)$$

As stated earlier, the linear weights are determined from the condition

$$P_i^W(x_{i+\frac{1}{2}}) = \mathcal{P}_i^E(x_{i+\frac{1}{2}}) , \quad (15.114)$$

which after collecting coefficients give the following, overdetermined, system of linear algebraic equations for the linear weights λ_{i-1} and λ_i

$$\left. \begin{aligned} \beta_{i-1} - \lambda_{i-1}\gamma_{i-1}^{(i-1)} &= 0 && \text{Eq. } (i-1) , \\ \beta_i - (\lambda_{i-1}\gamma_i^{(i-1)} + \lambda_i\gamma_i^{(i)}) &= 0 && \text{Eq. } (i) , \\ \beta_{i+1} - \lambda_i\gamma_{i+1}^{(i)} &= 0 && \text{Eq. } (i+1) . \end{aligned} \right\} \quad (15.115)$$

Using Eqs. $(i-1)$ and (i) in (15.115) gives the sought solution for the linear weights

$$\lambda_{i-1} = \frac{1}{3} , \quad \lambda_i = \frac{2}{3} . \quad (15.116)$$

Verify that by using Eqs. (i) and $(i+1)$ in (15.115) gives the same solution.

Next, we deal with the general case.

15.4.3 WENO Linear Weights: The General Case

For the general case of ENO polynomials of degree $m+1$ we need to determine the $m+1$ linear weights λ_s , for $s = i-m, \dots, i$. As seen in the example of Sect. 15.4.2 the polynomials in (15.93) can be written as a linear combination of cell averages $\bar{q}_k$ with constant coefficients that depend only on the fixed value of $x = x_{i+\frac{1}{2}}$, at which the polynomials are evaluated. The $m+1$ ENO polynomials may be written as

$$p_k^{(i)}(x_{i+\frac{1}{2}}) = \sum_{j=k}^{k+m} \gamma_j^{(k)} \bar{q}_j \quad \text{for } k = i-m, \dots, i , \quad (15.117)$$

while the extended higher-degree polynomial $\mathcal{P}_i^E(x)$ reads

$$\mathcal{P}_i^E(x_{i+\frac{1}{2}}) = \sum_{j=i-m}^{i+m} \beta_j \bar{q}_j . \quad (15.118)$$

The linear combination of ENO plynomials reads

$$P_i^W(x_{i+\frac{1}{2}}) = \sum_{k=i-m}^i \lambda_k p_k^{(i)}(x_{i+\frac{1}{2}}) , \quad (15.119)$$

which from (15.117) becomes

$$P_i^W(x_{i+\frac{1}{2}}) = \sum_{k=i-m}^i \lambda_k \left(\sum_{j=k}^{k+m} \gamma_j^{(k)} \bar{q}_j \right) . \quad (15.120)$$

Then, by imposing condition (15.93), rewritten below

$$P_i^W(x_{i+\frac{1}{2}}) = \mathcal{P}_i^E(x_{i+\frac{1}{2}}) , \quad (15.121)$$

we equate (15.118) with (15.120), leading to

$$\sum_{j=i-m}^i \bar{q}_j \left(\beta_j - \sum_{k=i-m}^j \lambda_k \gamma_j^{(k)} \right) + \sum_{j=i+1}^{i+m} \bar{q}_j \left(\beta_j - \sum_{k=j-m}^j \lambda_k \gamma_j^{(k)} \right) = 0 . \quad (15.122)$$

Since the above expression must hold for any set of cell averages $\{\bar{q}_j\}$, it follows that

$$\beta_j - \sum_{k=i-m}^j \lambda_k \gamma_j^{(k)} = 0 , \quad j = i - m, \dots, i . \quad (15.123)$$

and

$$\beta_j - \sum_{k=j-m}^j \lambda_k \gamma_j^{(k)} = 0 , \quad j = i + 1, \dots, i + m . \quad (15.124)$$

From system (15.123) we can compute the linear weights λ_k , $k = i - m, \dots, i$ recursively as follows

$$\left. \begin{aligned} \lambda_{i-m} &= \frac{\beta_{i-m}}{\gamma_{i-m}^{(i-m)}} , \\ \lambda_j &= \frac{1}{\gamma_j^{(j)}} \left(\beta_j - \sum_{k=i-m}^{j-1} \gamma_j^{(k)} \lambda_k \right) , \quad j = i - m + 1, \dots, i . \end{aligned} \right\} \quad (15.125)$$

Table 15.7 shows linear weights for WENO plynomials of degree m , for $m = 1, \dots, 9$. High-order finite volume methods can thus be constructed using semi-discrete or fully discrete approaches. In particular, with the available WENO weights in Table 15.7, fully discrete ADER schemes of orders 2nd to 10th in space and time can be implemented. See Chaps. 16 and 17.

Table 15.7 WENO linear weights for polynomials of degree m shown in first column, for $m = 1, 2, \dots, 9$. Non-linear ADER schemes of orders 2nd to 10th in both space and time can thus be constructed

m	λ_{i-9}	λ_{i-8}	λ_{i-7}	λ_{i-6}	λ_{i-5}	λ_{i-4}	λ_{i-3}	λ_{i-2}	λ_{i-1}	λ_i
1	—	—	—	—	—	—	—	—	$\frac{1}{3}$	$\frac{2}{3}$
2	—	—	—	—	—	—	—	$\frac{1}{10}$	$\frac{3}{5}$	$\frac{3}{10}$
3	—	—	—	—	—	—	$\frac{1}{35}$	$\frac{12}{35}$	$\frac{18}{35}$	$\frac{4}{35}$
4	—	—	—	—	—	$\frac{1}{126}$	$\frac{10}{63}$	$\frac{10}{21}$	$\frac{20}{63}$	$\frac{5}{126}$
5	—	—	—	—	$\frac{1}{462}$	$\frac{5}{77}$	$\frac{25}{77}$	$\frac{100}{231}$	$\frac{25}{154}$	$\frac{1}{77}$
6	—	—	—	$\frac{1}{1716}$	$\frac{7}{286}$	$\frac{105}{572}$	$\frac{175}{429}$	$\frac{175}{572}$	$\frac{21}{286}$	$\frac{7}{1716}$
7	—	—	$\frac{1}{6435}$	$\frac{56}{6435}$	$\frac{196}{2145}$	$\frac{392}{1287}$	$\frac{490}{1287}$	$\frac{392}{2145}$	$\frac{196}{6435}$	$\frac{8}{6435}$
8	—	$\frac{1}{24310}$	$\frac{36}{12155}$	$\frac{504}{12155}$	$\frac{2352}{12155}$	$\frac{882}{2431}$	$\frac{3528}{12155}$	$\frac{1176}{12155}$	$\frac{144}{12155}$	$\frac{9}{24310}$
9	$\frac{1}{92378}$	$\frac{45}{46189}$	$\frac{810}{46189}$	$\frac{5040}{46189}$	$\frac{13230}{46189}$	$\frac{15876}{46189}$	$\frac{8820}{46189}$	$\frac{2160}{46189}$	$\frac{405}{92378}$	$\frac{5}{46189}$

Remarks on the WENO Linear Weights

1. **Computation of the linear weights.** The linear weights λ_k in the linear combination that defines the WENO polynomial

$$P_i^W(x) = \sum_{s=i-m}^i \lambda_s p_s^{(i)}(x) \quad (15.126)$$

can be computed by solving the linear system

$$\sum_{k=i-m}^j \lambda_k \gamma_j^{(k)} = \beta_j, \quad j = i-m, \dots, i, \quad (15.127)$$

or

$$\sum_{k=j-m}^j \lambda_k \gamma_j^{(k)} = \beta_j, \quad j = i+1, \dots, i+m. \quad (15.128)$$

See (15.123) and (15.124).

2. **Consistency property of the WENO polynomial.** A kind of **consistency condition** for the WENO polynomial emerges from the case in which each of the polynomials $p_s^{(i)}(x)$ in (15.126) are equal to a constant (non zero) in cell I_i and the linear weights add up to unity. Formally, if

$$p_s^{(i)}(\hat{x}) = C : \text{constant} \neq 0 \quad \forall s = i-m, \dots, i \quad (15.129)$$

and

$$\sum_{k=i-m}^i \lambda_k = 1, \quad (15.130)$$

then $P_i^W(\hat{x}) = C$. This follows trivially from (15.126), (15.129) and (15.130). That is

$$P_i^W(\hat{x}) = \sum_{s=i-m}^i \lambda_s p_s^{(i)}(\hat{x}) = \sum_{s=i-m}^i \lambda_s C = C \sum_{s=i-m}^i \lambda_s = C.$$

3. Conservative property of the WENO polynomial.

More relevantly, if each of the polynomials $p_s^{(i)}(x)$ in (15.126) satisfies the conservation property and the linear weights satisfy (15.130), then the WENO polynomial satisfies the conservation property. In particular

$$\left. \begin{aligned} \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} P_i^W(x) dx &= \sum_{s=i-m}^i \lambda_s \left(\frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} p_s^{(i)}(x) dx \right) \\ &= \bar{q}_i \sum_{s=i-m}^i \lambda_s = \bar{q}_i. \end{aligned} \right\} \quad (15.131)$$

Property (15.130) of the linear weights is necessary for both the consistency condition and the conservation property of the WENO polynomial.

Next, we study a variation of the classical WENO method.

15.4.4 The Dumbser-Käser WENO Variant

There is a variation of the classical WENO reconstruction method just presented, due to Dumbser and Käser [19], called hereafter WENO-DK. This scheme starts from the extended stencil $\mathcal{E}_i$ of $2m + 1$ cells associated with all $m + 1$ ENO candidate polynomials of degree m , studied in Sect. 15.2. See also Sect. 15.4.1 on the classical WENO, which uses the $m + 1$ sub-stencils of $\mathcal{E}_i$ to form the combination (15.92) for the WENO polynomial.

The Three Stencils in the WENO-DK Method

Out of the collection of $m + 1$ sub-stencils in WENO, the WENO-DK version considers just three such sub-stencils. Two of these are the extreme sub-stencils left and right of the extended stencil $\mathcal{E}_i$, namely

$$\left. \begin{aligned} S_L^{(i)} = S_{i-m}^{(i)} &= \{I_{i-m}, I_{i-m+1}, \dots, I_{i-1}, I_i\} , \\ S_R^{(i)} = S_i^{(i)} &= \{I_i, I_{i+1}, \dots, I_{i+m-1}, I_{i+m}\} . \end{aligned} \right\} \quad (15.132)$$

For the third, middle stencil $S_M^{(i)}$ it is desirable that this be perfectly centred around cell I_i . However this depends on whether m is odd or even. When m is even the definition of $S_M^{(i)}$ is unambiguous and given by

$$S_M^{(i)} = \{I_{i-m/2}, I_{i-m/2+1}, \dots, I_{i+m/2-1}, I_{i+m/2}\} . \quad (15.133)$$

Figure 15.10 depicts the even case of $m = 2$. The perfectly centred stencil

$$S_M^{(i)} = \{I_{i-1}, I_i, I_{i+1}\} \quad (15.134)$$

has $m/2 = 1$ cell to the left and $m/2 = 1$ cell to the right of the central cell I_i .

When m is odd there is no perfectly centred stencil $S_M^{(i)}$, as illustrated in Fig. 15.11 for the case $m = 3$. There are two candidates that are the closest approximations to the perfectly centred stencil, namely

$$S_M^{(i)} = \begin{cases} S_{ML}^{(i)} = \{I_{i-2}, I_{i-1}, I_i, I_{i+1}\} & \text{left biased} , \\ S_{MR}^{(i)} = \{I_{i-1}, I_i, I_{i+1}, I_{i+2}\} & \text{right biased} . \end{cases} \quad (15.135)$$

In general, for m odd the two closest candidate stencils are

$$S_M^{(i)} = \begin{cases} S_{ML}^{(i)} = \{I_{i-(c+1)}, I_{i-c}, \dots, I_{i+c-1}, I_{i+c}\} & \text{left biased} , \\ S_{MR}^{(i)} = \{I_{i-c}, I_{i-c+1}, \dots, I_{i+c}, I_{i+c+1}\} & \text{right biased} . \end{cases} \quad (15.136)$$

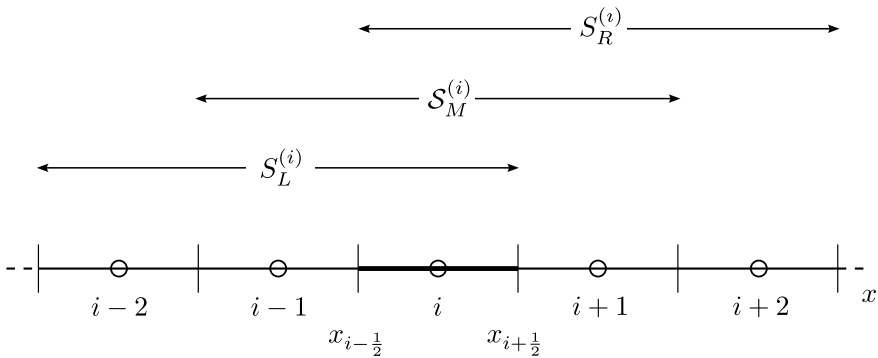


Fig. 15.10 The three stencils of WENO-DK for the even case $m = 2$. The middle stencil is perfectly centred, with one cell to the left and one cell to the right of the central cell I_i

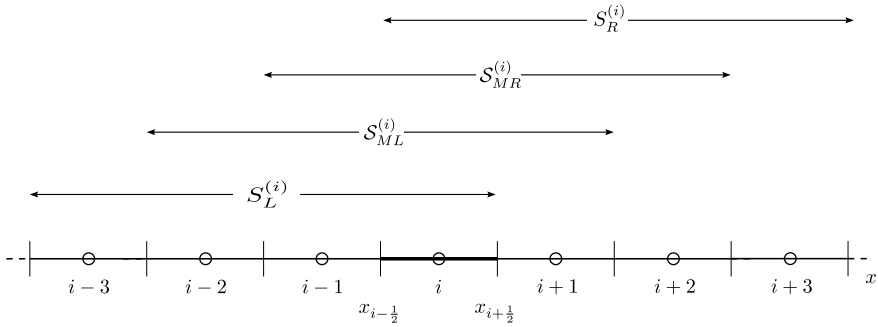


Fig. 15.11 Stencils for WENO-DK for the odd case $m = 3$. There are two non-perfectly centred plausible choices for the middle stencil, namely $S_{ML}^{(i)}$ and $S_{MR}^{(i)}$

Here $c = \text{floor}(m/2)$, where the floor function gives the greatest integer less or equal to x . Note that the bias in the stencils in (15.136) is just by one cell and, as the degree m of the polynomials considered increases, such bias becomes negligible, in some sense. This special case will be discussed shortly.

The WENO-DK Polynomial

Now assume that three sub-stencils have been identified and that for each of them one has constructed corresponding polynomials $p_i^L(x)$, $p_i^M(x)$, $p_i^R(x)$ by imposing conservation, as usual. Then the WENO-DK polynomial starts from the combination

$$P_i^{dk}(x) = \lambda_L p_i^L(x) + \lambda_M p_i^M(x) + \lambda_R p_i^R(x) . \quad (15.137)$$

The linear weights λ_L , λ_M and λ_R in the WENO-DK reconstruction scheme are prescribed a priori. The coefficients λ_K in (15.137) have nothing to do with the conventional WENO linear weights constructed in Sect. 15.4.3. Following [19] these weights are set as

$$\lambda_L = 1 , \quad \lambda_M = 10^5 , \quad \lambda_R = 1 . \quad (15.138)$$

The weights λ_K are then modified according to the oscillatory character of the corresponding polynomials, to obtain non-linear weights ω_K , for $K \in \{L, M, R\}$. To this end one first computes oscillation indicators β_K for each component polynomial in (15.137), as routinely done in the WENO approach described in Sect. 15.4.1, namely

$$\beta_K = \sum_{l=1}^m \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \Delta x^{2l-1} \left[\frac{d^l}{dx^l} p_i^K(x) \right]^2 dx , \text{ for } K \in \{L, M, R\} . \quad (15.139)$$

Then, the non-linear weights are given as

$$\omega_K = \frac{\hat{\omega}_K}{\hat{\omega}_L + \hat{\omega}_M + \hat{\omega}_R}, \quad \hat{\omega}_K = \frac{\lambda_K}{(\epsilon + \beta_K)^r}. \quad (15.140)$$

The sought WENO-DK polynomial is

$$P_i^{dk}(x) = \omega_L p_i^L(x) + \omega_M p_i^M(x) + \omega_R p_i^R(x). \quad (15.141)$$

See Sect. 15.4.1 regarding the choice of the parameters ϵ and r . For their method Dumbser and Käser [19] suggest $r = 4$ and $\epsilon = 10^{-12}$.

Next we discuss the special case in which degree m is odd.

The Middle Polynomial When the Degree m Is Odd

As already pointed out, when m is odd there is no perfectly centred stencil, but two potential approximations, instead. Dumbser et al. [19, 21] addressed this issue and proposed to consider the enlarged middle stencil formed by the union of the two potential candidates (15.136), namely

$$S_M^{(i)} = \{I_{i-(c+1)}, I_{i-c}, \dots, I_{i+c-1}, I_{i+(c+1)}\}. \quad (15.142)$$

This enlarged stencil is perfectly centred and has $2(c+1) + 1$ cells, where as before $c = \text{floor}(m/2)$. Figure 15.11 depicts the case for $m = 3$, $c = \text{floor}(3/2) = 1$, with $2(c+1) + 1 = 5$ cells. The difficulty is that $S_M^{(i)}$ has one more cell than desired, namely $m + 2$, for which a constrained least squares method to find the middle polynomial of degree m is proposed. Dumbser and Käser [19] proposed to find a polynomial $p_i^M(x)$ of degree m in the *enlarged* stencil (15.142) of $m + 2$ cells with corresponding $m + 2$ cell averages $\bar{q}_j$.

Here we assume the Taylor form of the sought polynomial for cell I_i as follows

$$p_i^M(x) = \sum_{k=0}^m c_k (x - x_i)^k. \quad (15.143)$$

To find the coefficients $c_0, c_1, \dots, c_{m-1}, c_m$ of such polynomial we follow [19] by formulating a least squares minimization problem. We look for the minimum of a function $F(c_0, c_1, \dots, c_{m-1}, c_m)$ subject to a constraint. That is

$$\min_C F(C), \quad (15.144)$$

with

$$C = [c_0, c_1, \dots, c_{m-1}, c_m] \quad (15.145)$$

and

$$F(C) = \sum_{j=i-(c+1)}^{i+(c+1)} \left(\frac{1}{\Delta x} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} p_i^M(x) dx - \bar{q}_j \right)^2 . \quad (15.146)$$

But

$$\frac{1}{\Delta x} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} p_i^M(x) dx = \sum_{k=0}^m a_{j,k} c_k , \quad (15.147)$$

with

$$a_{j,k} = \frac{1}{\Delta x} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} (x - x_i)^k dx . \quad (15.148)$$

Therefore the function $F(C)$ to minimise becomes

$$F(C) = \sum_{j=i-(c+1)}^{i+(c+1)} \left(\frac{1}{\Delta x} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} p_i^M(x) dx - \bar{q}_j \right)^2 = \sum_{j=i-(c+1)}^{i+(c+1)} \left(\sum_{k=0}^m a_{j,k} c_k - \bar{q}_j \right)^2 , \quad (15.149)$$

subject to the constraint

$$\frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} p_i^M(x) dx = \sum_{k=0}^m a_{i,k} c_k = \bar{q}_i . \quad (15.150)$$

On application of the method of Lagrange multipliers, problem (15.144), (15.145), (15.149), 15.150 becomes a minimization problem in which

$$G(C, \lambda) = F(C) + \lambda \left(\sum_{k=0}^m a_{i,k} c_k - \bar{q}_i \right) \quad (15.151)$$

is minimized with respect to both variables C and the Lagrange multiplier λ . This requires the following conditions to be satisfied

$$\left. \begin{aligned} \frac{\partial G}{\partial c_l} &= 0 , \text{ for } l = 0, 1, \dots, m , \\ \frac{\partial G}{\partial \lambda} &= 0 . \end{aligned} \right\} \quad (15.152)$$

That is

$$\left. \begin{aligned} \frac{\partial G}{\partial c_l} &= \sum_{k=0}^m \left[\sum_{j=i-(c+1)}^{i+(c+1)} (2a_{j,k} a_{j,l} - 2a_{j,l} \bar{q}_j) \right] = 0 , \\ \frac{\partial G}{\partial \lambda} &= \sum_{k=0}^m a_{i,k} c_k - \bar{q}_i = 0 . \end{aligned} \right\} \quad (15.153)$$

Conditions (15.153) lead to the following linear system for the unknowns (C, λ)

$$\mathbf{AC} = \mathbf{B} , \quad (15.154)$$

with

$$\mathbf{C} = \begin{bmatrix} c_0 \\ \vdots \\ c_m \\ \lambda \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_0 \\ \vdots \\ b_m \\ \bar{q}_i \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} A_{0,0} & \cdots & A_{0,m} & a_{i,0} \\ \vdots & \cdots & \vdots & \vdots \\ A_{m,0} & \cdots & A_{m,m} & a_{i,m} \\ a_{i,0} & \cdots & a_{i,m} & 0 \end{bmatrix} \quad (15.155)$$

where

$$A_{k,l} = 2 \sum_{j=i-c}^{i+c} a_{j,k} a_{j,l}, \quad b_k = 2 \sum_{j=i-c}^{i+c} \bar{q}_j a_{j,k}. \quad (15.156)$$

Numerical solution of the linear system (15.154)–(15.155) gives the coefficients of the sought polynomial $p_i^M(x)$ in (15.143) for the case in which m is odd. This completes the description of the WENO-DK reconstruction procedure.

So far, we have described non-linear reconstruction methods assuming the scalar case, that is a single variable $q(x)$. In the next section we address the extension of the methodology to be applied to systems of partial differential equations.

15.5 Reconstruction Methods for Systems

We have presented two major families of methods for building the reconstruction polynomial for the scalar case, namely ENO and WENO methods. In the scalar case there is just one unknown in the partial differential equation, denoted as $q(x, t)$. When extending the methodologies to systems of equations, in which there are more than one dependent variable, we must make a choice as to the variables to be used. One obvious choice is the set of conserved variables. In this case one would apply the scalar theory, separately, to each component $q_k(x, t)$ of the vector of conserved variables in the equations.

A more elaborate choice is the set of characteristic variables, which tends to be preferred in practice. As a matter of fact, for methods of order of accuracy greater than two, collective experience suggests that the use of characteristic variables is mandatory [23]. In this section we review both methods.

15.5.1 Component by Component Reconstruction

Consider the generic system of N hyperbolic balance laws

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}), \quad (15.157)$$

where

$$\mathbf{Q}(x, t) = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_N \end{bmatrix}, \quad \mathbf{F}(\mathbf{Q}) = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_N \end{bmatrix}, \quad \mathbf{S}(\mathbf{Q}) = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_N \end{bmatrix}. \quad (15.158)$$

The reconstruction procedure for a generic cell $I_i = [x_{i-1/2}, x_{i+1/2}]$ studied for the single unknown $q(x, t)$ in the scalar case is now applied component wise. That is, for each component $q_k(x, t)$ of system (15.157) we find a reconstruction polynomial $p_k(x)$, as explained in previous sections. In this manner we end up with a vector of reconstruction polynomials

$$\mathbf{P}_i(x) = [p_1(x), p_2(x), \dots, p_N(x)]^T. \quad (15.159)$$

The component-by-component reconstruction procedure is indeed very simple. It uses, component-by-component, the procedures studied for the scalar case. Experience suggests that for second-order methods, that is 1st-degree polynomials, the performance of the component-by-component reconstruction procedure gives acceptable results. As already pointed out, for schemes of accuracy greater than two, that is for $m > 1$, this method is not recommended in practice, as it produces obvious spurious oscillations in the vicinity of large spatial gradients.

15.5.2 Reconstruction in Characteristic Variables

An alternative to the component-by-component approach is to perform the reconstruction procedure in terms of *characteristic variables*. In this manner, the theory developed for the scalar case is closer to that of the system case, in some sense. For each cell $I_i = [x_{i-1/2}, x_{i+1/2}]$ we construct an m th degree polynomial $\mathbf{P}_i^c(x)$ of *characteristic variables* by first transforming the equations from conserved variables to characteristic variables, in the usual way; see Chap. 7.

This transformation is based on the *frozen* Jacobian matrix $\mathbf{A}_i^n = \mathbf{A}(\mathbf{Q}_i^n)$, which is used to produce a set of characteristic-variable cell averages $\mathbf{C}_j^n$ from conserved-variable cell averages $\mathbf{Q}_j^n$. Note that for each cell I_i the transformation is carried

out not just for cell I_i but for a set of neighbouring cells I_j around I_i that form the reconstruction stencil $S_s^{(i)}$, with the same matrix $\mathbf{A}_i^n$, namely

$$\mathbf{C}_j^n = (\mathbf{R}(\mathbf{Q}_i^n))^{-1} \mathbf{Q}_j^n, \quad I_j \in S_s^{(i)}. \quad (15.160)$$

In full

$$\mathbf{C}_j^n = [c_{1,j}, c_{2,j}, \dots, c_{N,j}]^T, \quad I_j \in S_s^{(i)}. \quad (15.161)$$

$\mathbf{R}(\mathbf{Q}_i^n)$ is the matrix whose columns are the right eigenvectors of $\mathbf{A}_i^n$, namely

$$\mathbf{R}(\mathbf{Q}_i^n) = [\mathbf{R}_1^n; \mathbf{R}_2^n; \dots; \mathbf{R}_N^n], \quad (15.162)$$

with $\mathbf{R}_k^n$ the right eigenvector of $\mathbf{A}_i^n$ corresponding to the eigenvalue $\lambda_k(\mathbf{Q}_i^n)$, for $k = 1, \dots, N$. $(\mathbf{R}(\mathbf{Q}_i^n))^{-1}$ is the inverse of matrix $\mathbf{R}(\mathbf{Q}_i^n)$ and $S_s^{(i)}$ is the set of cells that form the stencil for the non-linear reconstruction polynomial $\mathbf{P}_i(x)$. For example, for $m = 2$, the ENO method will have three potential stencils, namely $S_{i-2}^{(i)} = \{I_{i-2}, I_{i-1}, I_i\}$, $S_{i-1}^{(i)} = \{I_{i-1}, I_i, I_{i+1}\}$, $S_i^{(i)} = \{I_i, I_{i+1}, I_{i+2}\}$. For each characteristic variable $c_{k,i}$ in (15.161) we build up a reconstructed polynomial in cell $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ in a component-by-component fashion, as done for the scalar case, using ENO or WENO.

Example: Second-Order ENO Scheme

For second-order schemes we need reconstruction polynomials of degree $m = 1$. For each component $c_{k,i}$ we write the first degree polynomial

$$p_{k,i}(x) = c_{k,i} + (x - x_i) \Delta_{k,i}. \quad (15.163)$$

In choosing the slopes $\Delta_{k,i}$ one applies the ENO criterion studied for the scalar case in the usual way, for each component k of the vector of characteristic variables. The slopes are chosen as follows

$$\Delta_{k,i} = \begin{cases} \frac{c_{k,i} - c_{k,i-1}}{\Delta x} & \text{if } |c_{k,i} - c_{k,i-1}| \leq |c_{k,i+1} - c_{k,i}|, \\ \frac{c_{k,i+1} - c_{k,i}}{\Delta x} & \text{if } |c_{k,i} - c_{k,i-1}| > |c_{k,i+1} - c_{k,i}|. \end{cases} \quad (15.164)$$

The method just described gives the complete polynomial $\mathbf{P}_i^c(x)$ in cell I_i for the characteristic variables, namely

$$\mathbf{P}_i^c(x) = [p_{1,i}(x), p_{2,i}(x), \dots, p_{N,i}(x)]^T. \quad (15.165)$$

But we now need the reconstructed polynomial $\mathbf{P}_i(x)$ for the conserved variables. These are simply obtained from (15.160) by transforming back as follows

$$\mathbf{P}_i(x) = \mathbf{R}(\mathbf{Q}_i^n) \mathbf{P}_i^c(x) . \quad (15.166)$$

In particular, one may compute intercell boundary values, or *boundary extrapolated values*, as follows

$$\left. \begin{aligned} \mathbf{C}_i^L &= \mathbf{P}_i^c(x_{i-\frac{1}{2}}) , & \mathbf{C}_i^R &= \mathbf{P}_i^c(x_{i+\frac{1}{2}}) , \\ \mathbf{Q}_i^L &= \mathbf{P}_i(x_{i-\frac{1}{2}}) = \mathbf{R}(\mathbf{Q}_i^n) \mathbf{C}_i^L , & \mathbf{Q}_i^R &= \mathbf{P}_i(x_{i+\frac{1}{2}}) = \mathbf{R}(\mathbf{Q}_i^n) \mathbf{C}_i^R . \end{aligned} \right\} \quad (15.167)$$

In the next section we carry out a partial assessment of the non-linear spatial reconstruction methods studied in this chapter.

15.6 Assessment of Reconstruction Polynomials

Non-linear reconstruction polynomials of the ENO and WENO type were originally developed for designing high-order non-linear (in the sense of Godunov) numerical methods for solving partial differential equations. As pointed out by Shu [22], these polynomials may also be used as interpolation polynomials, as in approximation theory.

In this section we briefly assess the performance of the reconstruction polynomials studied in this chapter in interpolating discrete data in the form of integral cell averages of a known function. To this end we consider two functions, the Gaussian profile

$$q(x) = \alpha e^{-\beta x^2} , \quad \alpha = 1 , \quad \beta = 2 , \quad x \in [a, b] \equiv [-5, 5] \quad (15.168)$$

and the discontinuous function

$$q(x) = \begin{cases} \sin(x + \frac{\pi}{2}) & \text{if } x < 0 , \\ \sin(x - \frac{\pi}{2}) & \text{if } x > 0 , \end{cases} \quad (15.169)$$

with $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. The discontinuity is placed in the middle of the cell I_i of interest. The non-linear interpolation exercise is carried out in a **single cell** I_i positioned in the middle of a suitably chosen spatial domain. Recall that such exercise requires data left and right of cell I_i in order to fully exploit the behaviour of the complete data set $\{\tilde{q}_k\}$. As a matter of fact, this is exactly what is done in the numerical solution of PDEs using non-linear reconstruction polynomials.

We shall assess the interpolation errors in cell I_i of the reconstruction polynomial $p_i(x)$, with respect to the prescribed function $q(x)$ restricted to cell I_i , via the error

$$\mathcal{E} = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} |p_i(x) - q(x)| dx . \quad (15.170)$$

The evaluation of the integral in (15.170) is performed numerically, with an appropriate quadrature rule. We use $\mathcal{E}^E$ for the error of the extended polynomial $\mathcal{P}_i^E(x)$ and $\mathcal{E}$ for the error of the non-linear interpolation polynomial of interest $p_i^{zz}(x)$, where zz may represent dENO, WENO or WENO-DK. Results are shown in Figs. 15.12, 15.13, 15.14 and 15.15.

Figure 15.12 shows a comparison of the reconstruction polynomial (WENO-DK) $p_i^{dk}(x)$ and the extended (linear) higher-degree polynomial $\mathcal{P}_i^E(x)$ against the exact smooth function $q(x)$ of (15.168), within a single cell I_i in the centre. The full domain $[a, b]$ contains 25 cells. Errors from both polynomials within cell I_i are shown at the top of the figure and they are seen to coincide, to the number of decimal places shown. Plots for the exact function $q(x)$ (full line) and the polynomial $\mathcal{P}_i^E(x)$ (dotted line) are shown for the full domain, not just for the cell I_i . Although very accurate in the centre, the linear polynomial $\mathcal{P}_i^E(x)$ is highly oscillatory away from the centre, as is to be expected from very-high degree linear polynomials.

Figure 15.13 shows a zoom of Fig. 15.12 to display results for the smooth function (15.168), from three reconstruction methods, namely dENO (left column), WENO (middle column) and WENO-DK (right column), for polynomials of degrees $m = 2$ (top row), $m = 4$ (middle row), and $m = 9$ (bottom row). The results from dENO, from top to bottom, show a marked increase in accuracy as m increases from 2 to 4 and then to 9. The gains in accuracy are seen through the plots and the errors shown

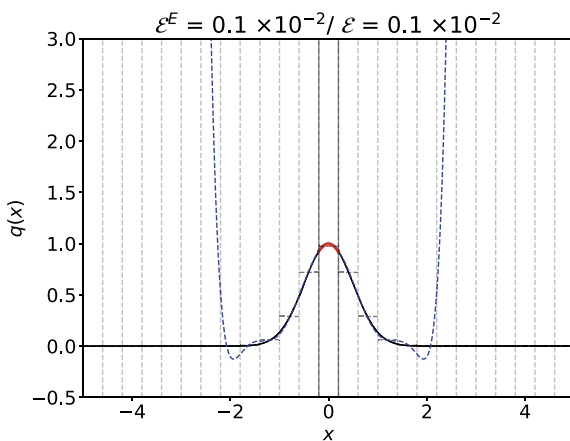


Fig. 15.12 Comparison of non-linear reconstruction polynomial $p_i^{dk}(x)$ (WENO-DK), shown by red curve, and the extended polynomial $\mathcal{P}_i^E(x)$ against the exact smooth function $q(x)$ given by (15.168) within a single cell I_i in the middle. Errors from both polynomials within cell I_i are shown at the top. $\mathcal{P}_i^E(x)$ (dotted line) and $q(x)$ (full line) are plotted in the full domain. Note the highly oscillatory behaviour of very-high degree ($m = 9$) linear polynomials, displaying very large amplitude errors near the boundaries

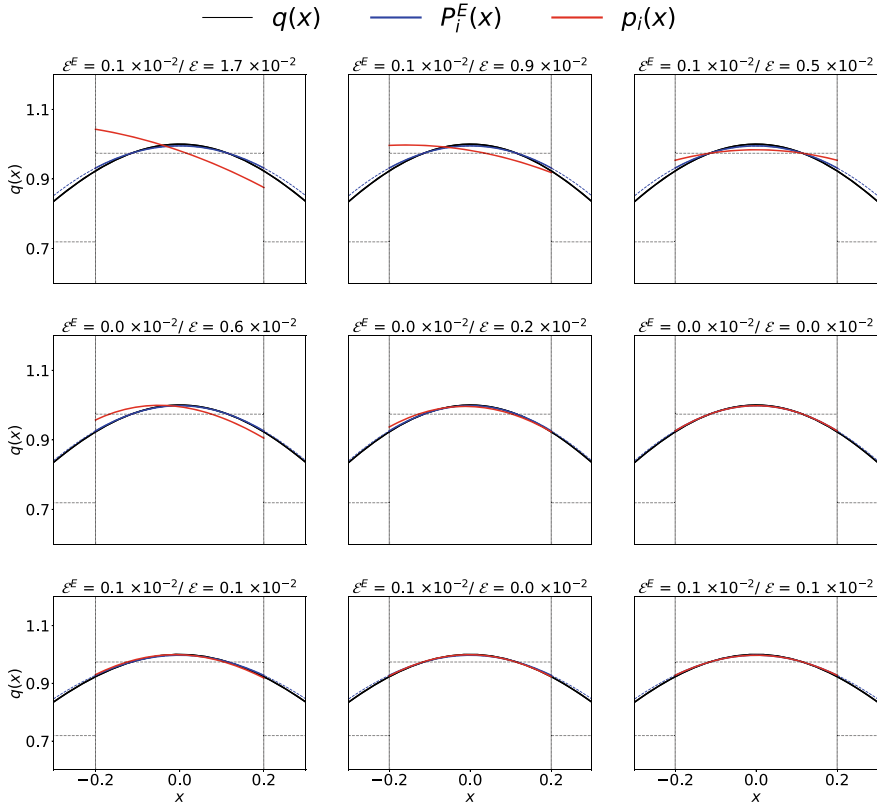


Fig. 15.13 Comparison of the three reconstruction polynomials dENO (left column), WENO (middle column) and WENO-DK (right column), against the Gaussian function $q(x)$ in (15.168) and the extended linear polynomial $\mathcal{P}_i^E(x)$. The comparison is performed for a single cell immersed in a mesh of $M = 25$ cells (not shown), for polynomials of degree $m = 2$ (top row), $m = 4$ (middle row), $m = 9$ (bottom row). Corresponding errors are found on top of each frame

on top of each frame. The first error $\mathcal{E}^E$ corresponds to the extended higher-degree polynomial $\mathcal{P}_i^E(x)$, while the second error $\mathcal{E}$ corresponds to the interpolation polynomial of interest. Results on the second column corresponding to WENO exhibit a similar tendency as dENO, as the degree m is increased from top to bottom. However, comparing horizontally between dENO and WENO for a fixed degree m , we see that WENO gives superior results to dENO; this is not surprising for this test for a very smooth function. Results from the right-hand side column correspond to WENO-DK. Again the tendency to improve accuracy is observed, as the degree m increases from top to bottom. Comparing horizontally for fixed m we observe that WENO-DK gives the best results for this test, which is more clearly seen for the lower degree polynomials. Compare results for $m = 4$.

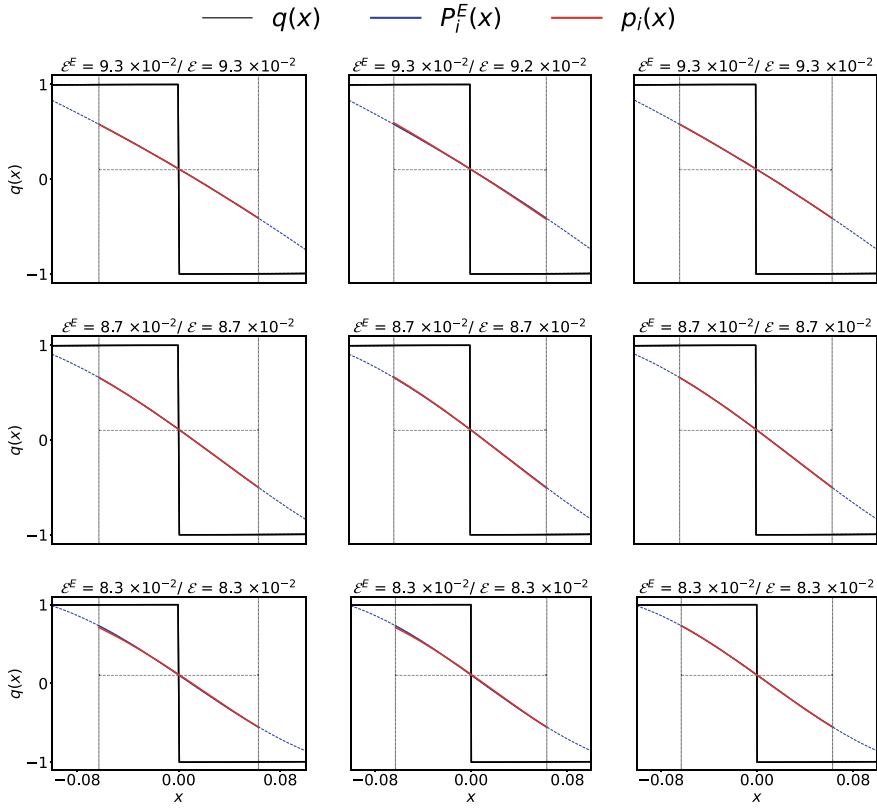


Fig. 15.14 Comparison of the three reconstruction polynomials dENO (left column), WENO (middle column) and WENO-DK (right column), against the discontinuous function $q(x)$ in Equation (15.169) and the extended linear polynomial $\mathcal{P}_i^E(x)$. The comparison is performed for a single cell immersed in a mesh of $M = 25$ cells (not shown), for polynomials of degree $m = 2$ (top row), $m = 4$ (middle row), $m = 9$ (bottom row). Corresponding errors are found on top of each frame

Figure 15.14 shows a zoom of results for the discontinuous function $q(x)$ in (15.169), from three reconstruction methods, namely dENO (left column), WENO (middle column) and WENO-DK (right column), for polynomials of degrees $m = 2$ (top row), $m = 4$ (middle row), and $m = 9$ (bottom row). The comparison is performed for a single cell immersed in a mesh of $M = 25$ cells (not shown), for polynomials of degree $m = 2$ (top row), $m = 4$ (middle row), $m = 9$ (bottom row). Corresponding errors are found on top of each frame.

Figure 15.15 shows error bars for all three methods dENO, WENO and WENO-DK. The top row shows results for the smooth Gaussian function (15.168), while the bottom row shows results for the discontinuous function (15.169). The assessment is performed for three meshes ($M = 15, 25, 51$) and various polynomial degrees ($m = 1, \dots, 6$). The left column shows results for the case of 15 cells for the domain

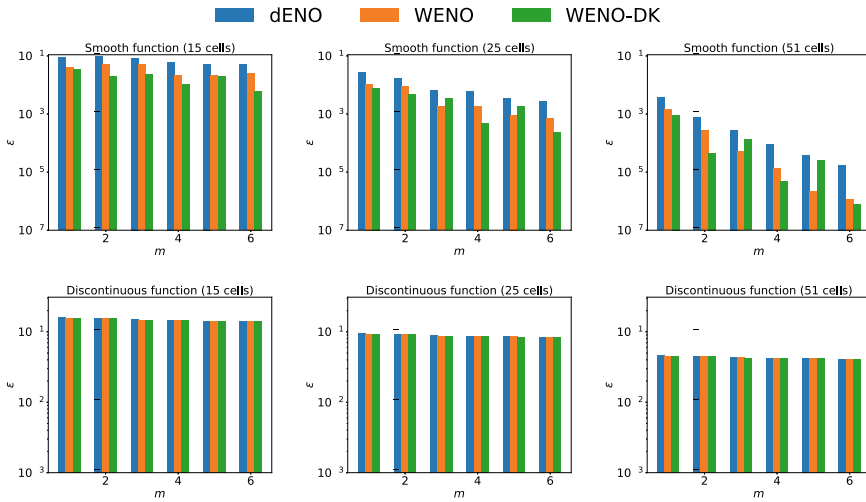


Fig. 15.15 Error plots for all three nonlinear reconstruction methods: dENO, WENO and WENO-DK. The top row shows results for the smooth function (15.168), while the bottom row shows results for the discontinuous function (15.169). The left column shows results for mesh of $M = 15$ cells; the middle column shows results for mesh of $M = 25$ cells and the right column shows results for mesh of $M = 51$ cells. In all cases the degree of the polynomials varies in the range $m = 2, \dots, 6$

$[a, b]$. The middle column shows results for 25 cells and the right column shows results for 51 cells. The top three frames refer to the smooth function (15.168) and overall, the error decreases as the degree of the polynomials increases. For the coarse mesh of 15 cells, dENO has the largest errors, followed by WENO. For this case ENO-DK has the smallest errors. For the case of 25 cells dENO remains the least accurate method, but now WENO has the smallest errors, rather than WENO-DK. For the finest mesh, errors from all three methods are comparable. Overall, dENO is the least accurate interpolation polynomial for the smooth profile. The bottom row of results corresponds to errors from dENO, WENO and WENO-DK, for three different meshes, as applied to the discontinuous function (15.169). Overall, errors decrease as the number of mesh cells is increased. Errors also decrease, slightly, as the degree of the polynomials increases.

In this section we have partially assessed the performance of three non-linear reconstruction methods, purely as interpolation tools, for two functions, a smooth Gaussian profile and a discontinuous profile. Indeed, this is not the main intended use of non-linear reconstruction polynomials. Instead, their main intended use is in conjunction with high-order finite volume numerical methods to solve partial differential equations. This will be done, first in Chap. 16 for the scalar case and then in Chap. 17 for non-linear hyperbolic systems.

15.7 Closing Remarks and Further Reading

In this chapter we have presented non-linear spatial reconstruction techniques, which constitute the bases for constructing high-order accurate non-linear finite volume numerical methods for solving hyperbolic balance laws. We started from a succinct presentation of the basic theory of polynomial interpolation; this is essential background to study spatial reconstruction methods. Then we studied the ENO approach, which historically was the first method that appeared in the open literature. We presented two ways of constructing conservative ENO polynomials, one through the use of the primitive function and another through direct application of conservation to a pre-selected ENO stencil. We then presented the WENO method, starting from the classical version, to then present a variation called here WENO-DK. The bases for these polynomial interpolation methods were first presented for the scalar case and then extended to the vector case for solving systems of partial differential equations. A partial assessment was carried out of the performance of three non-linear reconstruction methods, purely as interpolation tools.

There are several spatial reconstruction approaches that emanate from ENO and WENO that have not been considered here. Shu [18] proposed a modification to the ENO search algorithm, called mENO, so as to bias it towards the central stencil. Fu and collaborators [29] proposed the so called targeted ENO scheme, or TENO. Toro and collaborators [12] proposed a weighted averaged ENO scheme, or AENO, whereby the interpolation polynomial is a weighted average of the ENO polynomial and its immediate neighbour in the central direction. Montecinos and Toro [13, 15] proposed a variation of AENO, called ENO-ET, whereby the reconstruction polynomial results from extending the ENO stencil by one extra cell, constructing a higher degree polynomial and then truncating the highest-degree term. Regarding the original WENO method of Jiang and Shu, there are several modifications in the literature. See for example Dumbser et al. [19], Shu [22], Levy et al. [27] and Dumbser et al. [28]. In this chapter we presented the modification due to Dumbser and Käser [19].

This chapter has focused exclusively on non-linear polynomials for constructing non-linear methods. For a wider outlook of the field, see Iske [30] for the use of radial functions, for example, as well as various issues related to approximation theory.

For further study we suggest the following references: Harten, Engquist, Osher and Chakravarthy [5]; Jiang and Shu [17]; Dumbser and Käser [19]; Shu [22]; Laney [4].

The chapter is concluded with a list of exercises for the reader to solve.

15.8 Exercises

15.1 Consider a cubic polynomial in Taylor form ($m = 3$)

$$p(x) = c_0 + c_1(x - \hat{x}) + c_2(x - \hat{x})^2 + c_3(x - \hat{x})^3, \quad (15.171)$$

with knots $(x_0, q(x_0)), (x_1, q(x_1)), (x_2, q(x_2)), (x_3, q(x_3))$.

1. Write down the corresponding linear system emerging from application of the polynomial equations.
2. Either by hand or by using an algebraic manipulator find analytical expressions for the coefficients of the polynomial c_0, c_1, c_2, c_3 .

15.2 Consider the following polynomials in Taylor form

$$\left. \begin{aligned} p_{i-2}^{(i)}(x) &= c_0 + c_1(x - x_i) + c_2(x - x_i)^2, & S_{i-2}^{(i)} &: \{I_{i-2}, I_{i-1}, I_i\}, \\ p_{i-1}^{(i)}(x) &= c_0 + c_1(x - x_i) + c_2(x - x_i)^2, & S_{i-1}^{(i)} &: \{I_{i-1}, I_i, I_{i+1}\}, \\ p_i^{(i)}(x) &= c_0 + c_1(x - x_i) + c_2(x - x_i)^2, & S_i^{(i)} &: \{I_i, I_{i+1}, I_{i+2}\}. \end{aligned} \right\} \quad (15.172)$$

1. Determine the coefficients c_0, c_1, c_2 of $p_{i-2}^{(i)}(x)$ by enforcing the conservation property in each cell in the stencil $\{I_{i-2}, I_{i-1}, I_i\}$.
2. Determine the coefficients c_0, c_1, c_2 of $p_{i-1}^{(i)}(x)$ by enforcing the conservation property in each cell in the stencil $\{I_{i-1}, I_i, I_{i+1}\}$.
3. Determine the coefficients c_0, c_1, c_2 of $p_i^{(i)}(x)$ by enforcing the conservation property in each cell in the stencil $\{I_i, I_{i+1}, I_{i+2}\}$.

15.3 Consider the **extended** third-degree polynomial in Taylor form in the **extended** stencil $\mathcal{E}_i$

$$\left. \begin{aligned} \mathcal{P}_i^E(x) &= c_0 + c_1(x - x_i) + c_2(x - x_i)^2 + c_3(x - x_i)^3, \\ \mathcal{E}_i &: \{I_{i-2}, I_{i-1}, I_i, I_{i+1}, I_{i+2}\}. \end{aligned} \right\} \quad (15.173)$$

1. Determine the coefficients c_0, c_1, c_2, c_3 of $\mathcal{P}_i^E(x)$ by enforcing the conservation property in each cell in the stencil $\mathcal{E}_i$.
2. Evaluate $\mathcal{P}_i^E(x_{i+\frac{1}{2}})$ and express it a linear combination of the cell averages in $\mathcal{E}_i$

$$\mathcal{P}_i^E(x_{i+\frac{1}{2}}) = \sum_{k=i-2}^{i+2} \beta_k \bar{q}_k, \quad (15.174)$$

stating the values of all coefficients β_k .

15.4 Evaluate each of the polynomials in (15.172) at the interface $x = x_{i+\frac{1}{2}}$ and express the result as a linear combination of the cell averages $\bar{q}_k$ in their respective stencils

$$p_s^{(i)}(x_{i+\frac{1}{2}}) = \sum_{k=s}^{s+2} \gamma_k^{(s)} \bar{q}_k, \text{ for } s = i-2, i-1, i. \quad (15.175)$$

Determine all coefficients $\gamma_k^{(s)}$.

15.5 WENO linear combination.

1. Consider the WENO linear combination of all lower-degree ($m = 2$) polynomials in (15.172)

$$P_i^W(x) = \sum_{k=i-2}^i \lambda_k p_k^{(i)}(x). \quad (15.176)$$

2. Evaluate $P_i^W(x_{i+\frac{1}{2}})$ and express it as a linear combination of the cell averages in $\mathcal{E}_i$

$$P_i^W(x_{i+\frac{1}{2}}) = \sum_{k=i-2}^i \lambda_k p_k^{(i)}(x_{i+\frac{1}{2}}), \quad (15.177)$$

with $p_k^{(i)}(x_{i+\frac{1}{2}})$ as in (15.175) and corresponding coefficients $\gamma_k^{(s)}$.

3. Evaluate $\mathcal{P}_i^E(x_{i+\frac{1}{2}})$ and express it as a linear combination of the cell averages in $\mathcal{E}_i$

$$\mathcal{P}_i^E(x_{i+\frac{1}{2}}) = \sum_{k=i-2}^{i+2} \beta_k \bar{q}_k \quad (15.178)$$

15.6 WENO linear weights.

1. Enforce the condition

$$P_i^W(x_{i+\frac{1}{2}}) = \mathcal{P}_i^E(x_{i+\frac{1}{2}}) \quad (15.179)$$

and write down the system of linear equations for the linear weights λ_k in terms of the known coefficients β_k and $\gamma_k^{(s)}$.

2. Determine the linear weights λ_k in (15.176) and compare your results with those of Table 15.7.
3. Verify that all the weights λ_k satisfy the condition

$$0 \leq \lambda_k \leq 1. \quad (15.180)$$

4. Verify that all the weights λ_k satisfy the condition

$$\sum_{k=i-2}^i \lambda_k = 1. \quad (15.181)$$

HINT: for problems (15.2)–(15.6) see Sects. 15.4.2 and 15.4.3.

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Chapter 16

High-Order Methods for Scalar Equations



Abstract This chapter focuses on high-order numerical methods for scalar equations in both the finite volume and discontinuous Galerkin FE frameworks. We present two classes of methods for approximating hyperbolic equations with high-order accuracy: fully-discrete methods and semi-discrete methods. In the finite volume setting both classes rely on the non-linear spatial reconstruction methodology, in view of Godunov's theorem. Fully-discrete methods are represented here by the ADER methodology. These methods, within the finite volume framework, require a non-linear reconstruction procedure using polynomials of degree m , along with solving the generalised Riemann problem GRP_m to compute a high-order numerical flux. The numerical source is calculated from a volume integral, evaluated on the time-evolved reconstruction polynomial within the volume. Semi-discrete methods is represented here by ENO/WENO spatial reconstructions combined with TVD Runge-Kutta schemes for solving the associated ordinary differential equations. The performance of the numerical methods is thoroughly assessed through a carefully selected suite of test problems, which include isolated smooth profiles, isolated discontinuous profiles, and mixed smooth/discontinuous multi-wave profiles. Results are presented for schemes with accuracy up to the 10th order in both space and time. The assessment includes convergence rate studies for smooth profiles, resolution of discontinuous profiles, and efficiency (i.e., error versus cost). Two key issues are worth noting. First, the higher-order range (from 5th to 10th order) reveals limitations of existing spatial reconstruction schemes in handling discontinuous solutions. Second, higher-order methods are orders of magnitude more efficient than low-order methods. Consequently, high-order methods are essential in the active field of modelling and simulation in science and engineering. A list of suggested exercises is provided at the end of the chapter.

16.1 Fully-Discrete ADER Schemes

We consider the initial-boundary value problem (IBVP) for a generic, single non-linear balance law written in conservation-law form

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \partial_x f(q) = s(q) , \quad x \in [a, b] , \quad t > 0 , \\ \text{ICs: } q(x, 0) = q^{(0)}(x) , \quad x \in [a, b] , \\ \text{BCs: } q(a, t) = b_L(t) , \quad q(b, t) = b_R(t) , \quad t \geq 0 . \end{array} \right\} \quad (16.1)$$

Here $q(x, t)$ is the conserved variable; $f(q)$ is the **physical flux**; $s(q)$ is the source term; $q^{(0)}(x)$ is the initial condition; $b_L(t)$ and $b_R(t)$ are the boundary conditions on the left and right boundaries, respectively, two prescribed functions of time. Assuming that the IBVP (16.1) has a solution, the task at hand is to construct numerical methods to compute the solution approximately.

The reader is encouraged to review Chaps. 7 and 8 concerned with the theoretical properties of the partial differential equations, prior to engaging in their numerical approximation.

16.1.1 ADER in the Finite Volume Framework

In this section we study the ADER approach in the finite volume framework. We assume the spatial domain $[a, b]$ in (16.1) is discretised into M cells $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ and the time domain $[0, T_{out}]$ is partitioned by time levels t_n , with $t_n = n\Delta t$, for $n = 0, 1, \dots, N_{out}$. Here T_{out} is a chosen output time and t_0 is the initial time. Provisionally, we assume the time step Δt is constant. The PDE in (16.1) is solved numerically by a finite volume method, with space-time volumes defined as

$$V_i^n = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [t_n, t_{n+1}] . \quad (16.2)$$

Exact integration of (16.1) on V_i^n in (16.2) gives the exact formula

$$q_i^{n+1} = q_i^n - \frac{\Delta t}{\Delta x} [f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}}] + \Delta t s_i , \quad (16.3)$$

with definitions

$$\left. \begin{array}{l} q_i^n = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} q(x, t_n) dx , \\ f_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} f(q(x_{i+\frac{1}{2}}, t)) dt , \\ s_i = \frac{1}{\Delta t} \frac{1}{\Delta x} \int_{t_n}^{t_{n+1}} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} s(q(x, t)) dx dt . \end{array} \right\} \quad (16.4)$$

The computational parameters are:

$$\Delta x = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}, \quad \Delta t = t_{n+1} - t_n, \quad x_i = \frac{1}{2}(x_{i-\frac{1}{2}} + x_{i+\frac{1}{2}}). \quad (16.5)$$

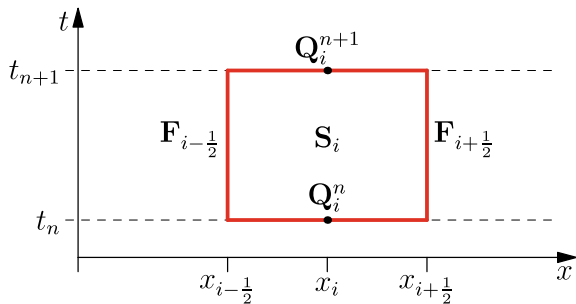
Δx is the mesh spacing, Δt is the time step, $x_{i-\frac{1}{2}}$ and $x_{i+\frac{1}{2}}$ are the left and right cell interfaces, while x_i is the cell centre. Figure 16.1 illustrates the finite volume update formula (16.3), as for a system of hyperbolic balance laws.

We construct one-step, fully discrete high-order finite volume methods, based on approximations to formulae (16.3) and (16.4), for solving scalar balance laws. The building blocks of the scheme are a **non-linear spatial reconstruction** procedure and the **generalised Riemann problem (GRP)**. The **GRP** is a two-fold generalisation of the conventional Riemann problem, that is (i) the initial condition is piece-wise smooth, instead of piece-wise constant, and (ii) the partial differential equations include source terms.

The ADER method, first put forward in [1–3], departs from an approximation to the exact finite volume formula (16.3) resulting from suitable approximations to the integrals in (16.4). The one-step formula (16.3) updates the cell average q_i^n to q_i^{n+1} , for which definitions for the **numerical flux** $f_{i+\frac{1}{2}}$ and the **numerical source** s_i are required. The resulting approximate one-step formula (16.3) represents a **fully-discrete method**, in which the discretisation in both space and time are performed simultaneously, in a coupled manner. The space-time coupling results from the solution of the generalised Riemann problem.

For most of the presentation of ADER in this chapter, we will assume that (16.1) represents the linear advection-reaction equation, in which $f(q) = \lambda q$ and $s(q) = \beta q$, with λ a constant wave propagation speed and β constant reaction rate satisfying the condition $\beta \leq 0$.

Fig. 16.1 Illustration of the update formula for a finite volume method as applied to a system of hyperbolic balance laws. The cell average at time level t_n is updated to a cell average at time level t_{n+1} by virtue of the intercell numerical fluxes and the numerical source



16.1.2 The Generalised Riemann Problem GRP_m

To determine the numerical flux $f_{i+\frac{1}{2}}$ in (16.3) we compute a high-order accurate approximation to the flux integral in (16.4), which in turn requires a high-order approximation to $q(x_{i+\frac{1}{2}}, t)$ in the integrand. This is achieved through two main steps, namely a reconstruction procedure, as presented in Chap. 15, and the solution of a generalised Riemann problem, as will be explained below.

Spatial Reconstruction

Recall that the spatial reconstruction procedure studied in Chap. 15 is a non-linear procedure aimed at circumventing the Godunov theorem, and thus designing high-order methods without, or much reduced, spurious oscillations in the vicinity of large spatial gradients. Given cell-integral averaged data $\{q_i^n\}$ at time level n in the full computational domain, one performs spatial piece-wise polynomial reconstruction, as described in Chap. 15, where two main methods were studied, ENO (Essentially Non-Oscillatory) and WENO (Weighted ENO). Such reconstruction procedures convert the constant cell average q_i^n into a polynomial $p_i(x)$, for each cell I_i in the domain. Figure 16.2 depicts reconstruction polynomials in three successive cells.

The Generalised Riemann Problem

The generalised Riemann problem GRP_m at the cell interface $x = x_{i+\frac{1}{2}}$ for the linear advection-reaction equation is the following initial-value problem (or Cauchy problem)

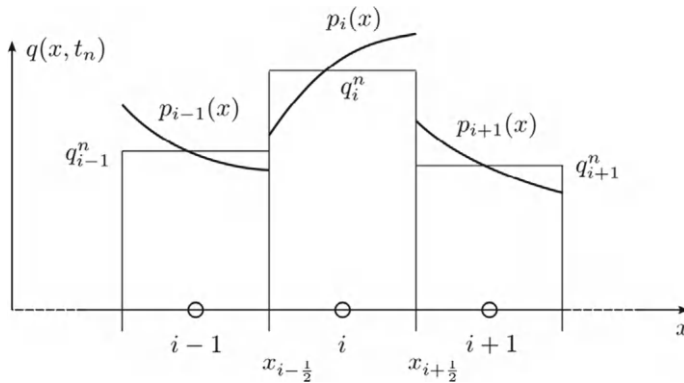


Fig. 16.2 Reconstruction polynomials $p_{i-1}(x)$, $p_i(x)$ and $p_{i+1}(x)$ in cells I_{i-1} , I_i and I_{i+1} replace cell-integral averages q_{i-1}^n , q_i^n and q_{i+1}^n . The non-linear reconstruction method could be ENO, WENO or variations of these

$$GRP_m : \left\{ \begin{array}{l} \text{PDE: } \partial_t q + \lambda \partial_x q = \beta q, \quad x \in \mathcal{R}, \quad t > 0, \\ \text{IC: } q(x, 0) = \begin{cases} p_i(x) & \text{if } x < x_{i+\frac{1}{2}}, \\ p_{i+1}(x) & \text{if } x > x_{i+\frac{1}{2}}. \end{cases} \end{array} \right\} \quad (16.6)$$

The initial condition is given by two polynomials $p_i(x)$ and $p_{i+1}(x)$, which are assumed to be **reconstruction** polynomials of arbitrary degree m . Moreover, and as anticipated, the GRP_m includes the source term in the equation. We next present a non-linear example of a generalised Riemann problem.

Example of GRP

A scalar, non-linear inhomogeneous example of a generalised Riemann problem is given as follows:

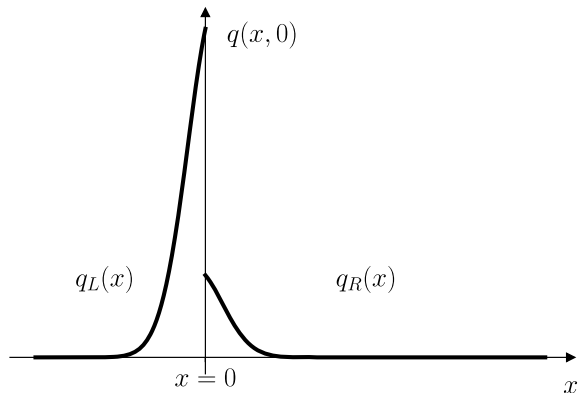
$$\left\{ \begin{array}{l} \text{PDE: } \partial_t q + \partial_x (\frac{1}{2} q^2) = e^{-q}, \quad x \in \mathcal{R}, \quad t > 0, \\ \text{IC: } q(x, 0) = \begin{cases} q_L(x) = e^{-2(x-\frac{1}{3})^2} & \text{if } x < 0, \\ q_R(x) = \frac{1}{4} e^{-2(x+\frac{1}{3})^2} & \text{if } x > 0. \end{cases} \end{array} \right\} \quad (16.7)$$

In this example the equation is non-linear and includes a non-linear source term. The initial condition given by $q_L(x)$ and $q_R(x)$ is piece-wise smooth, with a strong discontinuity at the interface $x = 0$. Figure 16.3 shows the initial condition and Fig. 16.4 shows the solution $q(x, t)$ in the x - t plane computed on a very fine mesh. Note that wave paths are curved lines, unlike the conventional piece-wise constant, homogeneous Riemann problem GRP_0 used in the first-order Godunov method. See Dematté et al. [4] for further details.

Solvers for the GRP_m

There are by now several solvers for the GRP_m available in the literature. Here we shall apply the first practical method published and used in the ADER framework,

Fig. 16.3 Generalised Riemann problem for a non-linear equation defined by (16.7). Piece-wise smooth initial condition with a strong discontinuity at the interface $x = 0$. See solution in Fig. 16.4



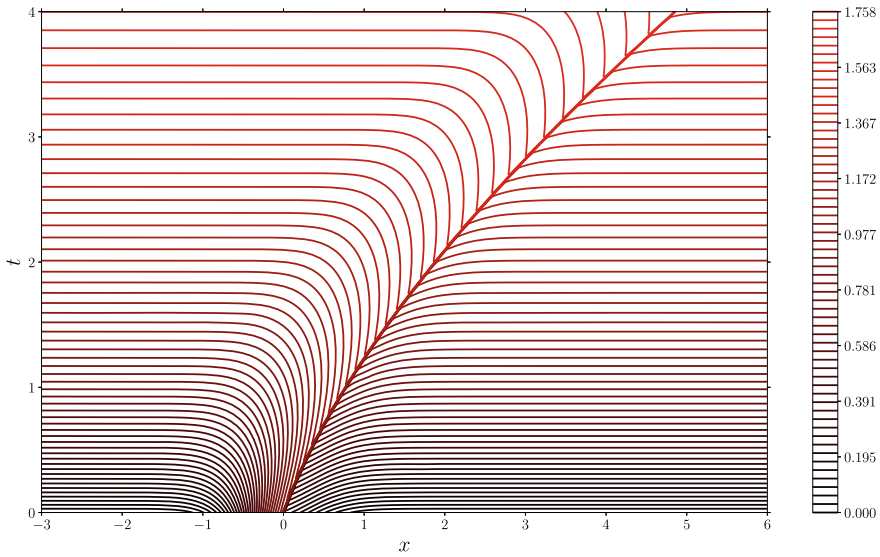


Fig. 16.4 Generalised Riemann problem for a non-linear equation defined by (16.7), with initial condition depicted in Fig. 16.3. Emerging solution wave structure in the x - t plane, computed on a very fine mesh. Note that characteristic paths are curved lines (courtesy of Siddharth Yajaman, University of Toronto Institute for Aerospace Studies, Toronto, Canada)

due to Toro and Titarev (TT) [2]. In the TT solver, the solution is sought directly at the cell interface, and is reduced to a sequence of $m + 1$ conventional Riemann problems for spatial derivatives, with piece-wise constant data. The TT method was inspired in the second-order method of Ben-Artzi and Falcoviz [5], known in the literature as the **GRP method**. But note that the Ben-Artzi/Falcoviz method is only second-order accurate and in our notation corresponds to GRP_1 . Another class of GRP_m solvers emerges from a re-interpretation of the high-order method of Harten, Enquist, Osher and Chakravarthy [16], called here HEOC [7]. Additional solvers for the GRP_m are found in [7], [4, 8, 10–12]. See the recent review by Toro et al. [13]. More details are given in Chap. 17 in the context of non-linear systems.

Next we describe one GRP solver in full detail, for the case of the linear advection-reaction equation.

16.1.3 The Toro-Titarev (TT) Solver for the GRP_m

In this section we present the Toro-Titarev method [2, 3, 14] to solve the GRP_m (16.6). The starting point of the TT solver is the LeFloch-Raviart [15] time expansion for the approximate solution at the cell interface, namely

$$\left. \begin{aligned} q_{i+\frac{1}{2}}(\tau) &= q(x_{i+\frac{1}{2}}, 0_+) + \sum_{k=1}^m \left[\partial_t^{(k)} q(x_{i+\frac{1}{2}}, 0_+) \right] \frac{\tau^k}{k!}, \\ q(x_{i+\frac{1}{2}}, 0_+) &= \lim_{t \rightarrow 0^+} q(x_{i+\frac{1}{2}}, t). \end{aligned} \right\} \quad (16.8)$$

Here the global time t_n corresponds to the local time $t = 0$. To fully determine the expansion (16.8) we need to deal with (i) the leading term and (ii) the high-order terms, as explained below.

The Leading Term of the Expansion

The leading term $q(x_{i+\frac{1}{2}}, 0_+)$ in (16.8) results from solving the **conventional** piece-wise-constant data Riemann problem

$$\left. \begin{aligned} \text{PDE: } \partial_t q + \lambda \partial_x q &= 0, \\ \text{IC: } q(x, 0) &= \begin{cases} p_i(x_{i+\frac{1}{2}}) & \text{if } x < x_{i+\frac{1}{2}}, \\ p_{i+1}(x_{i+\frac{1}{2}}) & \text{if } x > x_{i+\frac{1}{2}}. \end{cases} \end{aligned} \right\} \quad (16.9)$$

The piece-wise constant initial conditions consist of the intercell-boundary evaluated polynomials $p_i(x_{i+\frac{1}{2}})$ on the left and $p_{i+1}(x_{i+\frac{1}{2}})$ on the right. The solution $q(x_{i+\frac{1}{2}}, 0_+)$ of (16.9) represents the **first-instant interaction of the initial data** at the interface $x_{i+\frac{1}{2}}$, that is at $x = 0$ in local coordinates. The solution of (16.9) is

$$q(x_{i+\frac{1}{2}}, 0_+) = \begin{cases} p_i(x_{i+\frac{1}{2}}) & \text{if } \lambda > 0, \\ p_{i+1}(x_{i+\frac{1}{2}}) & \text{if } \lambda < 0. \end{cases} \quad (16.10)$$

See Chap. 7 for background on the conventional Riemann problem for linear equations.

The Higher-Order Terms of the Expansion

To compute the higher-order terms in (16.8) one requires the determination of the time partial derivatives $\partial_t^{(k)} q(x_{i+\frac{1}{2}}, 0_+)$. In the first version of ADER [1] it was proposed to apply the Cauchy-Kovalevskaya (CK) procedure to the governing equation in (16.6). In order to illustrate the procedure, let us just for a moment assume the homogenous case in (16.6), with $\beta = 0$. Then, it is easily verified that the CK procedure gives the following time derivatives in terms of spatial derivatives, as follows

$$\partial_t q + \lambda \partial_x q = 0, \quad \partial_t q = -\lambda \partial_x q, \quad \partial_t^{(2)} q = \lambda^2 \partial_x^{(2)} q. \quad (16.11)$$

In fact, it is easily shown by induction that for an arbitrary positive integer k

$$\partial_t^{(k)} q = (-1)^k \lambda^k \partial_x^{(k)} q. \quad (16.12)$$

Analogously, for the in-homogeneous case with non-zero source term in (16.6), it can be shown that the time derivatives can be expressed as

$$\partial_t^{(k)} q(x, t) = \sum_{l=0}^k \left[\frac{k!}{(k-l)!l!} \beta^l (-\lambda)^{k-l} \right] \partial_x^{(k-l)} q(x, t) . \quad (16.13)$$

Having expressed time derivatives in terms of spatial derivatives, now the pending problem is to determine the spatial derivatives $\partial_x^{(l)} q(x_{i+\frac{1}{2}}, 0_+)$ in (16.12) and (16.13), evaluated at the leading term.

Evolution Equations for Spatial Derivatives

A first, useful step is the construction of **evolution equations for the spatial derivatives**. For the homogenous case it is easily shown that $\partial_x^{(l)} q(x, t)$ obeys the same evolution equation as $q(x, t)$, that is

$$\partial_t(\partial_x^{(l)} q(x, t)) + \lambda \partial_x(\partial_x^{(l)} q(x, t)) = 0 , \text{ for } l = 1, 2, \dots, m . \quad (16.14)$$

The next problem is to find initial conditions for (16.14), in order to define corresponding initial-value problems. Assuming that for each computational cell we have spatial polynomial representations $\{p_i(x)\}$ for $q(x, t_n)$, then it is possible to pose **conventional Riemann problems** for spatial derivatives as follows:

$$\begin{aligned} \text{PDE: } & \partial_t(\partial_x^{(l)} q(x, t)) + \lambda \partial_x(\partial_x^{(l)} q(x, t)) = 0 , \\ \text{IC: } & \partial_x^{(l)} q(x, 0) = \left\{ \begin{array}{ll} \frac{d^l}{dx^l} p_i(x_{i+\frac{1}{2}}) & := p_i^{(l)}(x_{i+\frac{1}{2}}) \text{ if } x < x_{i+\frac{1}{2}} , \\ \frac{d^l}{dx^l} p_{i+1}(x_{i+\frac{1}{2}}) & := p_{i+1}^{(l)}(x_{i+\frac{1}{2}}) \text{ if } x > x_{i+\frac{1}{2}} . \end{array} \right\} \end{aligned} \quad (16.15)$$

The initial conditions for these conventional Riemann problems (piece-wise constant data) consists of intercell-boundary evaluated spatial derivatives of the reconstruction polynomials, $p_i(x)$ on the left and $p_{i+1}(x)$ on the right. The solution of the conventional Riemann problems (16.15) will give the spatial derivatives $\partial_x^{(l)} q(x_{i+\frac{1}{2}}, 0_+)$ needed in (16.8) as

$$\partial_x^{(l)} q(x_{i+\frac{1}{2}}, 0_+) = \begin{cases} p_i^{(l)}(x_{i+\frac{1}{2}}) & \text{if } \lambda > x_{i+\frac{1}{2}} , \\ p_{i+1}^{(l)}(x_{i+\frac{1}{2}}) & \text{if } \lambda < x_{i+\frac{1}{2}} . \end{cases} \quad (16.16)$$

We have therefore found the complete solution of homogeneous version of the GRP_m (16.6), given as

$$q_{i+\frac{1}{2}}(\tau) = \begin{cases} p_i(x_{i+\frac{1}{2}}) + \sum_{k=1}^m \left((-1)^k \lambda^k p_i^{(l)}(x_{i+\frac{1}{2}}) \right) \frac{\tau^k}{(k+1)!} & \text{if } \lambda > 0 , \\ p_{i+1}(x_{i+\frac{1}{2}}) + \sum_{k=1}^m \left((-1)^k \lambda^k p_{i+1}^{(l)}(x_{i+\frac{1}{2}}) \right) \frac{\tau^k}{(k+1)!} & \text{if } \lambda < 0 . \end{cases} \quad (16.17)$$

The solution for the in-homogeneous case is similar. Details are omitted.

The ADER Numerical Flux

The numerical flux results from the flux integral in (16.4), with (16.17) used in the integrand

$$f_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_0^{\Delta t} f(q_{i+\frac{1}{2}}(\tau)) d\tau . \quad (16.18)$$

Exact evaluation of (16.18) gives the numerical flux as

$$f_{i+\frac{1}{2}} = \begin{cases} \lambda \left[p_i(x_{i+\frac{1}{2}}) + \sum_{k=1}^m \left((-1)^k \lambda^k p_i^{(l)}(x_{i+\frac{1}{2}}) \right) \frac{(\Delta t)^k}{(k+1)!} \right] & \text{if } \lambda > 0 , \\ \lambda \left[p_{i+1}(x_{i+\frac{1}{2}}) + \sum_{k=1}^m \left((-1)^k \lambda^k p_{i+1}^{(l)}(x_{i+\frac{1}{2}}) \right) \frac{(\Delta t)^k}{(k+1)!} \right] & \text{if } \lambda < 0 . \end{cases} \quad (16.19)$$

In general, the evaluation of the numerical flux (16.18) will require numerical integration of the appropriate order. For the simple, linear case, the integration can be performed exactly.

Inserting the numerical flux (16.19) into the conservative finite volume formula (16.3) will give an ADER numerical approximation of order $m + 1$ in both space and time. In the presence of source terms in the governing equations, one must include the source terms when performing the CK procedure, as these enter in the calculation of the numerical flux. In addition, one must compute the numerical source to the appropriate order of accuracy. This requires an approximation of the source integral in (16.4) in the entire control volume V_i^n in (16.2), for which a high-order approximation to $q(x, t)$ within V_i^n is needed. Such approximation is found in a manner similar to that of the generalised Riemann problem, as will be explained.

The described ADER approach based on ENO or WENO reconstruction polynomials of degree m results in non-linear numerical schemes of order of accuracy $m + 1$ in both space and time. Here $m \geq 1$ is an arbitrary positive integer, meaning that there are no accuracy barriers for the ADER approach just introduced. Next we illustrate the scheme in full detail for the second-order case.

16.2 Example: ADER in Second-Order Mode ($m = 1$)

To provide further insight into the ADER methodology, here we present the ADER approach in full detail for the second-order case ($m=1$), as applied to the linear advection equation with a source term. First we summarise the main steps:

Step A: Reconstruction. The simplest case is that of a first-degree polynomial reconstruction ($m = 1$), with reconstructed polynomials left and right of the interface given as

$$p_i(x) = q_i^n + \Delta_i(x - x_i) , \quad p_{i+1}(x) = q_{i+1}^n + \Delta_{i+1}(x - x_{i+1}) . \quad (16.20)$$

To determine the polynomials that result in a non-linear method, the slopes Δ_i and Δ_{i+1} will be chosen here according to the ENO method, see Chap. 15.

Step B: Generalised Riemann problem GRP_1 . For the linear advection equation with linear source term, the GRP_1 is

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \lambda \partial_x q = \beta q , \\ \text{IC: } q(x, 0) = \begin{cases} p_i(x) = q_i^n + \Delta_i(x - x_i) & \text{if } x < x_{i+\frac{1}{2}} , \\ p_{i+1}(x) = q_{i+1}^n + \Delta_{i+1}(x - x_{i+1}) & \text{if } x > x_{i+\frac{1}{2}} , \end{cases} \end{array} \right\} \quad (16.21)$$

whose solution is used to find the numerical flux $f_{i+\frac{1}{2}}$ by evaluation of the flux integral in (16.18).

Step C: Numerical source. One requires an approximation $\hat{s}_i$, which is obtained from an appropriate evaluation of the source integral in (16.4).

Next we give full details of each step, starting from Step B, the generalised Riemann problem.

16.2.1 The Generalised Riemann Problem GRP_1 and the Flux

To start with, we express the solution $q_{i+\frac{1}{2}}(\tau)$ of (16.21) in terms of the truncated LeFloch-Raviart expansion (16.8) with $m = 1$

$$q_{i+\frac{1}{2}}(\tau) = q(x_{i+\frac{1}{2}}, 0_+) + \tau \partial_t q(x_{i+\frac{1}{2}}, 0_+) . \quad (16.22)$$

Substep B1: The leading term $q(x_{i+\frac{1}{2}}, 0_+)$ is found by solving the conventional Riemann problem

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \lambda \partial_x q = 0 , \\ \text{IC: } q(x, 0) = \begin{cases} p_i(x_{i+\frac{1}{2}}) = q_i^n + \frac{1}{2} \Delta x \Delta_i & \text{if } x < x_{i+\frac{1}{2}} , \\ p_{i+1}(x_{i+\frac{1}{2}}) = q_{i+1}^n - \frac{1}{2} \Delta x \Delta_{i+1} & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{array} \right\} \quad (16.23)$$

The similarity solution $d_{i+\frac{1}{2}}^{(0)}(x/t)$ of (16.23) in local coordinates is

$$d_{i+\frac{1}{2}}^{(0)}(x/t) = \begin{cases} q_i^n + \frac{1}{2}\Delta x \Delta_i & \text{if } x/t < \lambda, \\ q_{i+1}^n - \frac{1}{2}\Delta x \Delta_{i+1} & \text{if } x/t > \lambda. \end{cases} \quad (16.24)$$

The leading term in (16.22) is called **the Godunov state**; it is the solution (16.24) evaluated along the interface $x/t = 0$ (in local coordinates), that is

$$q(x_{i+\frac{1}{2}}, 0_+) = d_{i+\frac{1}{2}}^{(0)}(0) = \begin{cases} q_i^n + \frac{1}{2}\Delta x \Delta_i & \text{if } \lambda > 0, \\ q_{i+1}^n - \frac{1}{2}\Delta x \Delta_{i+1} & \text{if } \lambda < 0. \end{cases} \quad (16.25)$$

Substep B2: The higher order term. To compute the second term in (16.22) the following steps are performed:

- *Time derivative in terms of the spatial derivative.* We apply the Cauchy-Kovalevskaya procedure whereby the PDE in (16.21) is used to express the time derivative in terms of the space derivative and the source term, that is

$$\partial_t q(x, t) = -\lambda \partial_x q(x, t) + \beta q(x, t). \quad (16.26)$$

The sought term $\partial_t q(x, t)$ will be known once the source function $\beta q(x, t)$ and the spatial gradient $\partial_x q(x, t)$ in (16.26) are known. These items are evaluated at the leading term $q(x_{i+\frac{1}{2}}, 0_+)$, so that the expansion (16.22) reads

$$q_{i+\frac{1}{2}}(\tau) = q(x_{i+\frac{1}{2}}, 0_+) + [-\lambda \partial_x q(x_{i+\frac{1}{2}}, 0_+) + \beta q(x_{i+\frac{1}{2}}, 0_+)]\tau. \quad (16.27)$$

The gradient $\partial_x q(x_{i+\frac{1}{2}}, 0_+)$ is still unknown.

- *Evolution equation for the spatial derivative.*

$$\partial_t(\partial_x q(x, t)) + \lambda \partial_x(\partial_x q(x, t)) = \beta \partial_x q(x, t). \quad (16.28)$$

We simplify (16.28) by neglecting the source term $\beta \partial_x q(x, t)$ for the gradient. For the model equation this can be justified rigorously.

- *Riemann problem for the spatial derivative.* We pose the conventional homogeneous Riemann problem for the spatial derivative, namely

$$\left. \begin{aligned} \text{PDE: } & \partial_t(\partial_x q(x, t)) + \lambda \partial_x(\partial_x q(x, t)) = 0, \\ \text{IC: } & \partial_x q(x, 0) = \begin{cases} p'_i(x_{i+\frac{1}{2}}) = \Delta_i & \text{if } x < x_{i+\frac{1}{2}}, \\ p'_{i+1}(x_{i+\frac{1}{2}}) = \Delta_{i+1} & \text{if } x > x_{i+\frac{1}{2}}, \end{cases} \end{aligned} \right\} \quad (16.29)$$

the solution of which, in local coordinates, is

$$d_{i+\frac{1}{2}}^{(1)}(x/t) = \begin{cases} \Delta_i & \text{if } x/t < \lambda, \\ \Delta_{i+1} & \text{if } x/t > \lambda. \end{cases} \quad (16.30)$$

We need the Godunov state at $x/t = 0$ at the interface, which is

$$\partial_x q(x_{i+\frac{1}{2}}, 0_+) = d_{i+\frac{1}{2}}^{(1)}(0) = \begin{cases} \Delta_i & \text{if } \lambda > 0, \\ \Delta_{i+1} & \text{if } \lambda < 0. \end{cases} \quad (16.31)$$

- Therefore the sought solution (16.22) is

$$q_{i+\frac{1}{2}}(\tau) = \begin{cases} q_i^n + \frac{1}{2}\Delta x \Delta_i + \tau [-\lambda \Delta_i + \beta(q_i^n + \frac{1}{2}\Delta x \Delta_i)] & \text{if } \lambda > 0, \\ q_{i+1}^n - \frac{1}{2}\Delta x \Delta_{i+1} + \tau [-\lambda \Delta_{i+1} + \beta(q_{i+1}^n - \frac{1}{2}\Delta x \Delta_{i+1})] & \text{if } \lambda < 0. \end{cases} \quad (16.32)$$

- The numerical flux

$$f_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_0^{\Delta t} f(q_{i+\frac{1}{2}}(\tau)) d\tau \quad (16.33)$$

becomes

$$f_{i+\frac{1}{2}} = \begin{cases} \lambda (q_i^n + \frac{1}{2}(1-c)\Delta x \Delta_i + \frac{1}{2}r(q_i^n + \frac{1}{2}\Delta x \Delta_i)) & \text{if } \lambda > 0, \\ \lambda (q_{i+1}^n - \frac{1}{2}(1+c)\Delta x \Delta_{i+1} + \frac{1}{2}r(q_{i+1}^n - \frac{1}{2}\Delta x \Delta_{i+1})) & \text{if } \lambda < 0, \end{cases} \quad (16.34)$$

where $c = \lambda \Delta t / \Delta x$ is the Courant number and $r = \Delta t \beta$ is a dimensionless *reaction* number.

Remarks: The ADER numerical flux (16.34) includes a contribution due to the source term. In other words, the flux knows of the source term, as one would have expected. But note that there is still another contribution to the scheme resulting from the *numerical source*, yet to be specified.

16.2.2 The Numerical Source

The **Step C** concerns the determination of the numerical source s_i in scheme (16.3), which is obtained according to the source integral in (16.4), namely

$$s_i = \frac{1}{\Delta t} \frac{1}{\Delta x} \int_0^{\Delta t} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} s(q_i(x, t)) dx dt, \quad (16.35)$$

where $q_i(x, t)$ is an approximation to the solution in the full control volume V_i^n . To second-order, the mid-point integration rule suffices. One therefore requires $q_i(x_i, \frac{1}{2}\Delta t)$ for the integrand. The following tasks are carried out

- Taylor expanding in conjunction with the Cauchy-Kovalevskaya procedure give

$$\left. \begin{aligned} q_i(x_i, \tau) &= q_i(x_i, 0_+) + \tau \partial_t q_i(x_i, 0_+) \\ &= q_i(x_i, 0_+) + \tau (-\lambda \partial_x q_i(x_i, 0_+) + \beta q_i(x_i, 0_+)) \end{aligned} \right\} \quad (16.36)$$

- In the case of interest we have

$$\tau = \frac{1}{2}\Delta t, \quad q_i(x_i, 0_+) = q_i^n = p_i(x_i), \quad \partial_x q_i(x_i, 0_+) = \Delta_i.$$

- From (16.35) the numerical source becomes

$$s_i = \beta \left(q_i^n + \frac{1}{2}\Delta t (-\lambda \Delta_i + \beta q_i^n) \right). \quad (16.37)$$

Hence the numerical source to second-order has been obtained.

The Final Second-Order ADER Scheme

The final form of the second-order ADER scheme for the linear advection equation with a linear source term is

$$q_i^{n+1} = q_i^n - \frac{\Delta t}{\Delta x} [f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}}] + \Delta t s_i, \quad (16.38)$$

with numerical flux

$$f_{i+\frac{1}{2}} = \begin{cases} \lambda (q_i^n + \frac{1}{2}(1-c)\Delta x \Delta_i + \frac{1}{2}r(q_i^n + \frac{1}{2}\Delta x \Delta_i)) & \text{if } \lambda > 0, \\ \lambda (q_{i+1}^n - \frac{1}{2}(1+c)\Delta x \Delta_{i+1} + \frac{1}{2}r(q_{i+1}^n - \frac{1}{2}\Delta x \Delta_{i+1})) & \text{if } \lambda < 0 \end{cases} \quad (16.39)$$

and numerical source

$$s_i = \beta \left(q_i^n + \frac{1}{2}\Delta t (-\lambda \Delta_i + \beta q_i^n) \right). \quad (16.40)$$

In full, for the case $\lambda > 0$, the scheme (16.38) reads

$$\left. \begin{aligned} q_i^{n+1} &= q_i^n - c \left[(q_i^n - q_{i-1}^n) + \frac{1}{2}(1-c)\Delta x (\Delta_i - \Delta_{i-1}) \right] \\ &\quad - cr \left[\frac{1}{4}(q_i^n - q_{i-1}^n) + \frac{1}{2}\Delta x (\Delta_i - \Delta_{i-1}) \right] \\ &\quad + r \left[(1 + \frac{1}{2}r)q_i^n - \frac{1}{2}c\Delta x \Delta_i \right] \end{aligned} \right\} \quad (16.41)$$

The reader is encouraged to verify the final scheme (16.41).

16.2.3 Resulting Second-Order Methods

We discuss two classes of second-order methods resulting from (16.38)–(16.40).

Linear Second-Order Methods

Classical second-order methods for the homogeneous linear advection equation ($\beta = 0$) result from (16.41), by choosing appropriate slopes in the reconstruction polynomials. In particular, the following choices reproduce three well-known methods for the homogeneous case, namely

$$\Delta_i = \begin{cases} \Delta_{i-\frac{1}{2}} = \frac{q_i^n - q_{i-1}^n}{\Delta x} & \text{(Warming-Beam) } \lambda > 0, \\ \Delta_{i+\frac{1}{2}} = \frac{q_{i+1}^n - q_i^n}{\Delta x} & \text{(Lax-Wendroff) } \lambda > 0, \\ \Delta_c = \frac{q_{i+1}^n - q_{i-1}^n}{2\Delta x} & \text{(Fromm).} \end{cases} \quad (16.42)$$

See Chap. 12 for background on these schemes.

Exercise 16.1 Verify that (16.38)–(16.40) with slopes (16.42) reproduce these well-known second order schemes.

The schemes resulting from (16.42) are linear, of second-order accuracy and produce spurious oscillations with new extrema. In order to avoid, or at least control, the spurious oscillations one must select the slopes in a non-linear manner, as discussed below.

Non-linear Second-Order ADER Method

Here we construct a non-linear version of the second-order ADER method by adopting ENO first-degree reconstruction polynomials $p_i(x)$ for each cell i , of the form

$$p_i(x) = q_i^n + (x - x_i)\Delta_i. \quad (16.43)$$

In the ENO reconstruction method the choice of the slope Δ_i is made by analysing two potential candidate polynomials $p_L(x)$ and $p_R(x)$, as depicted in Figure 16.5. There are two possible choices for the slope in (16.43)

$$\Delta_i = \frac{q_i^n - q_{i-1}^n}{\Delta x} := \Delta_{i-\frac{1}{2}} \quad \text{and} \quad \Delta_i = \frac{q_{i+1}^n - q_i^n}{\Delta x} := \Delta_{i+\frac{1}{2}}. \quad (16.44)$$

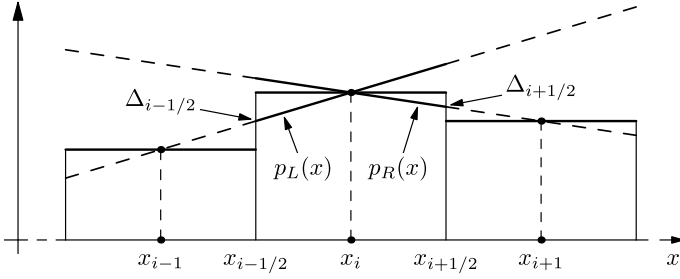


Fig. 16.5 Candidate first-degree polynomials $p_L(x)$ and $p_R(x)$ reconstructed from cell-integral averages $q_{i-1}^n, q_i^n, q_{i+1}^n$. The resulting ENO polynomial $p_i(x)$ is the one with the smallest slope in absolute value

The resulting two possible choices for the ENO reconstruction polynomial are

$$p_i^{ENO}(x) = \begin{cases} p_L(x) = q_i^n + (x - x_i) \left(\frac{q_i^n - q_{i-1}^n}{\Delta x} \right) & \text{if } |\Delta_{i-1/2}| \leq |\Delta_{i+1/2}|, \\ p_R(x) = q_i^n + (x - x_i) \left(\frac{q_{i+1}^n - q_i^n}{\Delta x} \right) & \text{if } |\Delta_{i-1/2}| > |\Delta_{i+1/2}|. \end{cases} \quad (16.45)$$

The ENO method chooses the polynomial $p_i(x)$ with the smallest slope in absolute value, for cell I_i .

Remarks. Full details on non-linear spatial reconstruction methods are found in Chap. 15, where ENO and WENO (Weighted ENO) methods for polynomials of any degree m are presented. See the classical works on ENO and WENO [6], [16–18]. It is noted that ENO reconstructions admit spurious oscillations. However, on mesh refinement the amplitude of the spurious oscillations tends to zero as the mesh width Δx tends to zero [6, 16]. This property is not satisfied by linear schemes. See [19, 20] on recent advances on reconstruction procedures in one and multiple space dimensions.

TVD Variants

A Total Variation Diminishing (TVD) version of the second-order ADER scheme results from choosing the slopes based on TVD constraints. See [21] for background on TVD methods. The following choices for TVD slopes are available.

1. **The minmod slope.** This has the form

$$\Delta_i = \text{minmod}(\Delta_{i-1/2}, \Delta_{i+1/2}), \quad (16.46)$$

where the minmod function is defined as

$$\minmod(x, y) = \begin{cases} x & \text{if } |x| \leq |y| \text{ and } xy > 0, \\ y & \text{if } |x| > |y| \text{ and } xy > 0, \\ 0 & \text{if } xy < 0. \end{cases} \quad (16.47)$$

2. **The superbee slope.** This has the form

$$\Delta_i = \maxmod(\Delta_i^{(1)}, \Delta_i^{(2)}), \quad (16.48)$$

where

$$\Delta_i^{(1)} = \minmod(\Delta_{i+\frac{1}{2}}, 2\Delta_{i-\frac{1}{2}}), \quad \Delta_i^{(2)} = \minmod(2\Delta_{i+\frac{1}{2}}, \Delta_{i-\frac{1}{2}}) \quad (16.49)$$

and

$$\maxmod(x, y) = \begin{cases} x & \text{if } |x| \geq |y| \text{ and } xy > 0, \\ y & \text{if } |x| \leq |y| \text{ and } xy > 0, \\ 0 & \text{if } xy < 0. \end{cases} \quad (16.50)$$

3. **The MC slope.** This has the following form

$$\Delta_i = \minmod(\Delta_i^{(c)}, 2\Delta_{i-\frac{1}{2}}, 2\Delta_{i+\frac{1}{2}}), \quad (16.51)$$

where

$$\Delta_i^{(c)} = \frac{1}{2}(\Delta_{i-\frac{1}{2}} + \Delta_{i+\frac{1}{2}}) \quad (16.52)$$

is called the *centred* slope.

Special First-Order Scheme

Note that if the slopes in scheme (16.41) are all set to zero and we also set to zero the contribution of the source to the numerical flux, then we produce the following first-order numerical scheme for the linear advection-reaction equation, namely

$$q_i^{n+1} = q_i^n - c(q_i^n - q_{i-1}^n) + r \left[\frac{1}{2}cq_{i-1}^n + (1 - \frac{1}{2}c)q_i^n \right]. \quad (16.53)$$

This scheme consists of the first-order upwind method of Godunov for the advection part and a special upwind discretisation of the source term. This corresponds to one of the schemes analysed by Roe [22] and by Bermúdez and Vázquez [23], in which the concept of *upwinding the source term* was employed.

We have completed the presentation of the fully-discrete ADER scheme as applied to the linear advection-reaction equation, with full details given for the second-order case. The algorithm may seem rather complex to the reader, even for the second-order case. However, for the higher-order versions of the method, any order in fact, the steps

remain the same. The fully-discrete ADER methodology is a general framework that includes all orders of accuracy, starting from the first-order Godunov method. Next we present the family of semi-discrete methods.

16.3 Semi-Discrete Schemes

Semi-discrete methods, as applied to an evolutionary partial differential equation (PDE), separate the discretisation in time from that in space. To advance the solution from time t_n to the next time t_{n+1} , the method relies on the solution of an initial-value problem for a system of Ordinary Differential Equations (ODEs). Classical Runge-Kutta methods are potential candidates for this task. However, in view of the Godunov theorem that requires the construction of non-linear methods [24], special ODE solvers have been devised, namely TVD (Total Variation Diminishing) ODE solvers [25], also called strong stability preserving methods [26]. Here we present semi-discrete methods of up to third order of accuracy, as applied to a single balance law.

16.3.1 The Bases and First-Order Methods

Let us consider the general hyperbolic balance law

$$\partial_t q(x, t) + \partial_x f(q(x, t)) = s(q(x, t)) , \quad x \in [a, b] \subset \mathcal{R} , \quad t > 0 \quad (16.54)$$

solved in computational cells $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$, with prescribed initial data at time t_n . Integrating (16.54) in I_i with respect to x gives an integral form of (16.54), namely

$$\frac{d}{dt} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} q(x, t) dx = f(q(x_{i-\frac{1}{2}}, t)) - f(q(x_{i+\frac{1}{2}}, t)) + \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} s(q(x, t)) dx . \quad (16.55)$$

See Chap. 8 for background on integral forms of hyperbolic balance laws. We now adopt the following definition for the cell-integral average in a generic cell I_i at time t

$$\bar{q}_i(t) = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} q(x, t) dx , \quad (16.56)$$

with $\Delta x = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}$, as usual. We also define the spatial operator L as

$$L(\bar{q}_i(t)) = -\frac{1}{\Delta x} [f(q(x_{i+\frac{1}{2}}, t)) - f(q(x_{i-\frac{1}{2}}, t))] \quad (16.57)$$

and the cell-averaged source

$$\bar{s}_i(t) = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} s(q(x, t)) dx . \quad (16.58)$$

For convenience we also adopt the notation $q_i^n = \bar{q}_i(t_n)$. Substituting (16.56), (16.57) and (16.58) into (16.55) yields the following ordinary differential equation (ODE) in time t

$$\frac{d}{dt} \bar{q}_i(t) = L(\bar{q}_i(t)) + \bar{s}_i(t) . \quad (16.59)$$

To fully define (16.59) we need to specify a **numerical flux** corresponding to the operator $L(\bar{q}_i(t))$ in (16.57) and a **numerical source** corresponding to $\bar{s}_i(t)$ in (16.58).

The Numerical Flux and the Numerical Source

To define the numerical flux at the interface position $x_{i+\frac{1}{2}}$ one relates the set of cell averages $\{q_i^n\}$ at time $t = t_n$ to two limiting intercell boundary values, namely $q_{i+\frac{1}{2}}^-$ (left of the interface) and $q_{i+\frac{1}{2}}^+$ (right of the interface). Such limiting intercell values, or boundary extrapolated values, will arise from spatial reconstruction of cell averages $\{q_i^n\}$. In the special case of no reconstruction, the case associated to first-order methods, such limiting values are simply q_i^n and q_{i+1}^n respectively. It is customary to define the numerical flux as

$$f_{i+\frac{1}{2}} = f_{i+\frac{1}{2}}(q_{i+\frac{1}{2}}^-, q_{i+\frac{1}{2}}^+) \approx f(q(x_{i+\frac{1}{2}}, t)) , \quad (16.60)$$

an approximation to $f(q(x_{i+\frac{1}{2}}, t))$ right at the interface. Recall that a useful numerical flux function $f_{i+\frac{1}{2}}(v_L, v_R)$ is required to satisfy the following properties

1. **Monotonicity:** recall from Chap. 12 for the scalar case, that **necessary conditions** for $f_{i+\frac{1}{2}}$ to be monotone are (i) $f_{i+\frac{1}{2}}(v_L, v_R)$ is a monotone increasing function of its first argument v_L and (ii) $f_{i+\frac{1}{2}}(v_L, v_R)$ is a monotone decreasing function of its second argument v_R . We emphasise that these are only necessary conditions.
2. **Consistency:**

$$f_{i+\frac{1}{2}}(v, v) = f(v) . \quad (16.61)$$

3. **Lipschitz continuity**, or still better, continuity.

Based on the above considerations the spatial operator (16.57) is replaced by

$$L(q^n) = -\frac{1}{\Delta x} [f_{i+\frac{1}{2}}(q_{i+\frac{1}{2}}^-, q_{i+\frac{1}{2}}^+) - f_{i-\frac{1}{2}}(q_{i-\frac{1}{2}}^-, q_{i-\frac{1}{2}}^+)] . \quad (16.62)$$

The notation q^n for the argument of L conveys the fact that L may depend on several neighbouring values in the set of cell averages $\{q_i^n\}$. In the first-order case $L(q^n) = L(q_{i-1}^n, q_i^n, q_{i+1}^n)$.

Regarding the **numerical source**, this is given by an approximation to the integral in (16.58), for which several options are available.

We are now in a position to state the numerical scheme for the balance law (16.54) in terms of the solution of the following initial-value problem involving an ODE, namely

$$\left. \begin{aligned} \frac{d}{dt} \bar{q}_i(t) &= L(q^n) + \bar{s}_i, \\ \bar{q}_i(t_n) &= q_i^n. \end{aligned} \right\} \quad (16.63)$$

Up to this point, Eq. (16.54) has only been discretised in space, remaining fully continuous in time in the form of the initial-value problem for an ODE (16.63).

To complete the solution procedure one must consider solvers for ordinary differential equations. In particular, we may apply the first-order explicit Euler method with time step Δt to the ODE in (16.63) to obtain q_i^{n+1} ; this requires the first-order approximation to the time derivative as

$$\frac{d}{dt} \bar{q}_i(t) \approx \frac{q_i^{n+1} - q_i^n}{\Delta t}. \quad (16.64)$$

Such approximation leads to the explicit first-order Euler scheme

$$q_i^{n+1} = q_i^n + \Delta t L(q^n) + \Delta t \bar{s}_i. \quad (16.65)$$

In the absence of the source term, $\bar{s}_i = 0$, the resulting method for (16.54) is the familiar first-order conservative scheme.

First-Order Godunov Method

Consider the linear advection-reaction equation

$$\partial_t q + \lambda \partial_x q = \beta q, \quad (16.66)$$

with $\lambda > 0$ constant and $\beta \leq 0$ constant. Assume the numerical flux to be the Godunov flux $f_{i+\frac{1}{2}}(q_i^n, q_{i+1}^n) = \lambda q_i^n$ and the numerical source to be approximated as $s_i = \beta q_i^n$. Then, substitution of the numerical flux and the numerical source into (16.65) gives

$$q_i^{n+1} = q_i^n - c(q_i^n - q_i^n) + r q_i^n. \quad (16.67)$$

As usual, $c = \frac{\Delta t \lambda}{\Delta x}$ is the Courant number and $r = \Delta t \beta$ is the reaction-rate number. In the absence of the source term, scheme (16.67) is, identically, the first-order Godunov upwind method; see Chap. 12. We note that, even at the first-order level, there are other options to deal with the source term.

In the next section we construct higher-order semi-discrete methods by means of high-order solvers for ODEs.

16.3.2 TVD Runge-Kutta Schemes

Here we consider the initial-value problem of the form (16.63) for a first-order ODE, where the unknown, for simplicity denoted as $q(t)$, is a real-valued function of time t and, for convenience, the source term is neglected for now. In what follows we introduce two ODE solvers of the Runge-Kutta type that satisfy the TVD property. For details on their construction and properties see [25, 26].

Second-Order Scheme (Heun's Method)

A two-stage, second-order TVD Runge-Kutta scheme to solve (16.63) is the following two-stage method

$$\left. \begin{array}{l} \text{Stage 1: } q^{(1)} = q^n + \Delta t L(q^n) , \\ \text{Stage 2: } q^{(2)} = q^{(1)} + \Delta t L(q^{(1)}) , \\ \text{Solution: } q^{n+1} = \frac{1}{2}(q^n + q^{(2)}) . \end{array} \right\} \quad (16.68)$$

Note that at each stage one applies the first-order explicit Euler scheme (16.65). The numerical fluxes for stage 1 are obtained in the following steps:

1. **Reconstruction.** Perform a non-linear spatial reconstruction to produce first-degree polynomials

$$p_i(x) = q_i^n + (x - x_i) \Delta_i , \quad (16.69)$$

where Δ_i is the slope in cell $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$. Figure 16.5 depicts candidate polynomials $p_L(x)$ and $p_R(x)$ in cell I_i . Here we assume the slope has been obtained from a non-linear procedure, such as the ENO slopes (16.44). See Chap. 15 for the general case.

2. **Interface Riemann problem.** Consider the local Riemann problem stated in global coordinates

$$\left. \begin{array}{l} \partial_t q + \partial_x f(q) = 0 , \\ q(x, 0) = \begin{cases} q_{i+\frac{1}{2}}^- = p_i(x_{i+\frac{1}{2}}) = q_i^n + \Delta x \Delta_i & \text{if } x < x_{i+\frac{1}{2}} , \\ q_{i+\frac{1}{2}}^+ = p_{i+1}(x_{i+\frac{1}{2}}) = q_{i+1}^n - \Delta x \Delta_{i+1} & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{array} \right\} \quad (16.70)$$

Here $q_{i+\frac{1}{2}}^-$ denotes the boundary extrapolated value of the **left** reconstructed polynomial $p_i(x)$ on the **right boundary** of cell I_i . Likewise, $q_{i+\frac{1}{2}}^+$ is the boundary extrapolated value of the **right** reconstructed polynomial $p_{i+1}(x)$ on the **left boundary** of cell I_{i+1} .

3. **Numerical flux.** Denote the solution of (16.70) in local coordinates by $g_{i+\frac{1}{2}}(x/t)$. Then the Godunov flux is

$$f_{i+\frac{1}{2}} = f(g_{i+\frac{1}{2}}(0)) . \quad (16.71)$$

There are several suitable numerical fluxes than can be used here, including centred fluxes, such as the FORCE flux. See Chap. 13 for more choices.

Having computed the intercell numerical fluxes $f_{i+\frac{1}{2}}$ for all interfaces, the operator L at time t for each cell I_i is defined as

$$L = -\frac{1}{\Delta x}(f_{i+\frac{1}{2}} - f_{i-\frac{1}{2}}). \quad (16.72)$$

Stage 1 in (16.68) has been performed to obtain $\{q_i^{(1)}\}$. Stage 2 in (16.68) is performed in an entirely analogous manner by applying (16.65) to obtain $\{q_i^{(2)}\}$. The final solution $\{q_i^{n+1}\}$ results from averaging in (16.68).

The second-order scheme (16.68) resulting from the semi-discrete approach looks exceedingly simple, as compared to the corresponding second-order fully discrete ADER method presented in Sect. 16.2. However, it is important to recall that, in the context of PDEs, the operator $L(q)$ is a spatial operator that involves the difference of two intercell fluxes. Therefore, each of the two stages requires the solution of a sequence of Riemann problems, exact or approximate, to define numerical fluxes and thus $L(q)$. Moreover, each stage requires a spatial reconstruction procedure. In other words, the second-order semi-discrete scheme requires, in addition to the ODE solver, two sequences of spatial reconstructions and two sequences of Riemann problem solutions.

Third-Order TVD Runge-Kutta Scheme

A three-stage, third-order accurate TVD Runge-Kutta method is the following

$$\left. \begin{array}{l} \text{Stage 1: } q^{(1)} = q^n + \Delta t L(q^n), \\ \text{Stage 2: } \hat{q}^{(2)} = q^{(1)} + \Delta t L(q^{(1)}), \\ \text{Averaging: } q^{(2)} = \frac{3}{4}q^n + \frac{1}{4}\hat{q}^{(2)}, \\ \text{Stage 3: } q^{(3)} = q^{(2)} + \Delta t L(q^{(2)}), \\ \text{Solution: } q^{n+1} = \frac{1}{3}q^n + \frac{2}{3}q^{(3)}. \end{array} \right\} \quad (16.73)$$

At each stage, one applies the explicit first-order Euler operator (16.65). Now, each evaluation of the operator $L(q)$ implies a sequence of Riemann problems and a reconstruction procedure. Therefore this method requires three reconstruction procedures and three rows of Riemann solvers per time step. To achieve third-order accuracy each reconstruction stage requires reconstructed polynomials of degree two; see Chap. 15 for details.

The TVD Property

The TVD (total variation diminishing) property is a scalar property and relates to the definition of total variation (TV) for a given, scalar, mesh function $q^n = \{q_i^n\}$, namely

$$TV(q^n) = \sum_{-\infty}^{\infty} |q_{i+1}^n - q_i^n|, \quad (16.74)$$

where $q_i^n \rightarrow 0$ as $|i| \rightarrow \infty$. For relevant background see Chap. 13 of [21], for example. In the context of semi-discrete methods we now prove the following result:

Proposition 16.2 *If the spatial operator $L(q^n)$ is such that the Euler first-order step*

$$q^{n+1} = q^n + \Delta t L(q^n) \quad (16.75)$$

is TVD, then the second-order two-stage scheme (16.68) is TVD.

Proof From hypothesis, steps 1 and 2 give respectively

$$TV(q^{(1)}) \leq TV(q^n) \quad \text{and} \quad TV(q^{(2)}) \leq TV(q^{(1)}). \quad (16.76)$$

Then, the total variation of the solution from (16.68) leads to

$$TV(q^{n+1}) = TV\left[\frac{1}{2}(q^n + q^{(2)})\right] \leq \frac{1}{2}[TV(q^n) + TV(q^{(2)})], \quad (16.77)$$

in which the inequality follows from the triangular inequality applied to definition (16.74). Then, from (16.76) and (16.77) we deduce that $TV(q^{(2)}) \leq TV(q^n)$ and therefore the sought result follows, that is $TV(q^{n+1}) \leq TV(q^n)$ and the proof is completed. ■

Proposition 16.3 *The third-order TVD Runge-Kutta scheme (16.73) is TVD, if the spatial operator $L(q^n)$ in the Euler first-order step (16.75) is TVD.*

Proof Left to the reader. ■

We have presented the class of semi-discrete methods to solve scalar first-order equations to up to third-order in space and time. In the next section we study the class of discontinuous Galerkin finite element methods, which can be formulated both as semi discrete methods, as well as fully discrete methods.

16.4 Discontinuous Galerkin Finite Element Methods

In this section we introduce the class of discontinuous Galerkin finite element methods (DG) for solving the initial-boundary value problem (IBVP) for a generic, single non-linear balance law written in conservation-law form

$$\left. \begin{array}{l} \text{PDE: } \partial_t q + \partial_x f(q) = s(q) , \quad x \in [a, b] , \quad t > 0 , \\ \text{ICs: } q(x, 0) = q^{(0)}(x) , \quad x \in [a, b] , \\ \text{BCs: } q(a, t) = b_L(t) , \quad q(b, t) = b_R(t) , \quad t \geq 0 . \end{array} \right\} \quad (16.78)$$

See Eq. 16.1 and subsequent definitions.

The origins of DG are traced back to the work of Reed and Hills [27]. DG methods were further developed by Cockburn, Shu and others; see for example [28–38]. As for finite volume methods, for DG methods there are also two versions: semi-discrete and fully-discrete formulations. Most of the publications so far are on the semi-discrete DG formulations, such as in Sect. 16.3.

The fully-discrete formulation for DG is more recent and was pioneered by Dumbser and collaborators [39, 40]. The fully discrete DG formulation connects to the ADER approach, presented in Sect. 16.1 in the finite volume framework. The fully discrete DG formulation allows the construction of very high order DG schemes, in both space and time; see also [41, 42].

There are several commonalities between finite volume methods, studied so far, and DG finite element methods, to be introduced now. These include integral form of the conservation laws, numerical fluxes, the Riemann problem and non-linear procedures, in view of the Godunov theorem.

16.4.1 Semi-Discrete Formulation

Here we introduce the discontinuous Galerkin finite element method in the context of the IBVP (16.78), for the homogeneous scalar conservation law. Consider the finite element space

$$\mathbf{V}^m = \{\psi : \psi|_{I_i} \in P^m(I_i) , \quad i = 1, 2, \dots, M\} , \quad (16.79)$$

where $P^m(I_i)$ is the set of polynomials of degree $m \geq d$, and a set of test functions $\psi_k(x) \in \mathbf{V}^m$, with $k = 0, 1, \dots, m$. Now, multiply the PDE in (16.78) by a fixed test function $\psi_k(x)$ and integrate in space within cell I_i , that is

$$\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} [\psi_k(x) \partial_t q(x, t) + \psi_k(x) \partial_x f(q(x, t))] dx = 0 . \quad (16.80)$$

Integration by parts in (16.80) gives

$$\left. \begin{aligned} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \psi_k \partial_t q dx &= -[\psi_k(x_{i+\frac{1}{2}}) f(q(x_{i+\frac{1}{2}}, t)) - \psi_k(x_{i-\frac{1}{2}}) f(q(x_{i-\frac{1}{2}}, t))] \\ &\quad + \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \partial_x \psi_k f(q) dx . \end{aligned} \right\} \quad (16.81)$$

Now consider an approximation $q_i(x, t)$ to the solution $q(x, t)$ in element I_i , expressed as

$$q_i(x, t) = \sum_{l=0}^m \bar{q}_l(t) \psi_l(x) . \quad (16.82)$$

The space-dependent test functions $\psi_l(x)$ are assumed to be known. The unknowns are the time-dependent coefficients $\bar{q}_l(t)$, for $l = 0, \dots, m$, called **degrees of freedom**. Inserting $q_i(x, t)$ from (16.82) in place of $q(x, t)$ into (16.81) gives

$$\sum_{l=0}^m \left[\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \psi_k(x) \psi_l(x) dx \right] \frac{d}{dt} \bar{q}_l(t) = \left. \begin{aligned} & -[\psi_{k,i+\frac{1}{2}}^- f_{i+\frac{1}{2}} - \psi_{k,i-\frac{1}{2}}^+ f_{i-\frac{1}{2}}] \\ & + \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \frac{d}{dx} \psi_k(x) f(q_i(x, t)) dx . \end{aligned} \right\} \quad (16.83)$$

Here $\psi_{k,i-\frac{1}{2}}^+$ and $\psi_{k,i+\frac{1}{2}}^-$ denote, respectively, the limiting values of $\psi_k(x)$ from the right side at interface $x_{i-\frac{1}{2}}$ and from the left side at interface $x_{i+\frac{1}{2}}$. The functions $f_{i-\frac{1}{2}}$ and $f_{i+\frac{1}{2}}$ denote **numerical fluxes** at interfaces $x_{i-\frac{1}{2}}$ and $x_{i+\frac{1}{2}}$, respectively, yet to be defined. The intercell numerical flux is a common ingredient to finite volume and DG methods.

16.4.2 ODE System, Numerical Fluxes and Volumetric Integrals

Evaluation of (16.83) for each $k = 0, 1, \dots, m$ yields the following system of ODEs

$$\frac{d}{dt} \bar{\mathbf{Q}}(t) = \mathbf{A}^{-1}(-\mathbf{F} + \mathbf{S}) \equiv \mathbf{L}(\bar{\mathbf{Q}}(t)) , \quad (16.84)$$

with definitions

$$\left. \begin{aligned} \bar{\mathbf{Q}} &= \begin{bmatrix} \bar{q}_0(t) \\ \bar{q}_1(t) \\ \vdots \\ \bar{q}_m(t) \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a_{00} & a_{01} & \cdots & a_{0m} \\ a_{10} & a_{11} & & a_{1m} \\ \vdots & & \ddots & \vdots \\ a_{m0} & a_{m1} & \cdots & a_{mm} \end{bmatrix}, \\ \Delta \mathbf{F} &= \begin{bmatrix} \Delta f_0 \\ \Delta f_1 \\ \vdots \\ \Delta f_m \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} s_0 \\ s_1 \\ \vdots \\ s_m \end{bmatrix} . \end{aligned} \right\} \quad (16.85)$$

The components of vectors $\Delta \mathbf{F}$ and $\mathbf{S}$ and the matrix $\mathbf{A}$ are defined as follows

$$\left. \begin{aligned} a_{kl} &= \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \psi_k(x) \psi_l(x) dx , \\ \Delta f_k &= \psi_{k,i+\frac{1}{2}}^- f_{i+\frac{1}{2}} - \psi_{k,i-\frac{1}{2}}^+ f_{i-\frac{1}{2}} , \\ s_k &= \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \frac{d}{dx} \psi_k(x) f(q_i(x, t)) dx . \end{aligned} \right\} \quad (16.86)$$

Solving system (16.84) gives all the degrees of freedom $\bar{q}_0(t), \bar{q}_1(t), \dots, \bar{q}_m(t)$ and hence the solution (16.82) in cell I_i is determined. Pending items to completely determine the DG scheme include: the calculation of the entries a_{kl} of matrix $\mathbf{A}$; numerical fluxes $f_{i+\frac{1}{2}}$; the volumetric integrals for s_k ; the initial conditions in (16.84) and the choice of test functions $\psi_l(x)$.

Numerical Flux

In order to fully define the right hand side operator $\mathbf{L}(\bar{\mathbf{Q}}(t))$ (16.84) it is necessary to determine the numerical fluxes $f_{i+\frac{1}{2}}$ in (16.86). Here $f_{i+\frac{1}{2}}$ denotes the numerical flux at the interface $x_{i+\frac{1}{2}}$. As for finite volume methods, the numerical flux emerges naturally from the conventional Riemann problem with piece-wise constant initial data, namely

$$\left. \begin{aligned} \text{PDE: } \partial_t q + \partial_x f(q) &= 0 , \\ \text{IC: } q(x, 0) &= \begin{cases} q_{i+\frac{1}{2}}^- = \lim_{x \rightarrow x_{i+\frac{1}{2}}} q_i(x, t) & \text{if } x < x_{i+\frac{1}{2}} , \\ q_{i+\frac{1}{2}}^+ = \lim_{x \rightarrow x_{i+\frac{1}{2}}} q_{i+1}(x, t) & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{aligned} \right\} \quad (16.87)$$

By denoting the similarity solution of (16.87), in local coordinates, as $g_{i+\frac{1}{2}}(x/t)$, the numerical flux is

$$f_{i+\frac{1}{2}} = f_{i+\frac{1}{2}}(q_{i+\frac{1}{2}}^-, q_{i+\frac{1}{2}}^+) = f(g_{i+\frac{1}{2}}(0)) . \quad (16.88)$$

As for all methods studied in this chapter, see for example Sect. 16.3, the flux is required to be monotone (for the scalar case), consistent and Lipschitz continuous. For a range of choices for upwind and centred-type fluxes, see Chaps. 12 and 13.

Volumetric Integral

In determining the right-hand side operator $\mathbf{L}(\bar{\mathbf{Q}}(t))$ (16.84) it is necessary to compute the volume integrals s_k in (16.86) within the element I_i , for $k = 0, 1, \dots, m$. For given space-dependent test functions $\psi_l(x)$, these spatial integrals can be evaluated with an appropriate numerical quadrature rule.

16.4.3 Solution of ODE Systems: TVD Runge-Kutta Schemes

In the semi-discrete formulation of the DG method, for each cell, we solve the following initial-value problem

$$\left. \begin{aligned} \frac{d}{dt} \bar{\mathbf{Q}}(t) &= \mathbf{A}^{-1}(-\mathbf{F} + \mathbf{S}) \equiv \mathbf{L}(\bar{\mathbf{Q}}(t)) , \\ \bar{\mathbf{Q}}(t_n) &= \mathbf{Q}^n , \end{aligned} \right\} \quad (16.89)$$

with definitions (16.85)-(16.86) for the right-hand side operator $\mathbf{L}(\bar{\mathbf{Q}}(t))$ in the ODE system.

Next we define the initial conditions at time $t = 0$, the ODE solvers to be used and the stability condition to be imposed to select the time step Δt .

Initial Conditions

The initial condition $q(x, 0) = h(x)$ at time $t = 0$ in (16.78) is projected onto the basis functions in order to obtain the initial vector

$$\bar{\mathbf{Q}}(0) = [\bar{q}_0(0), \bar{q}_1(0), \dots, \bar{q}_m(0)]^T ,$$

namely

$$\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \psi_k(x) \sum_{l=0}^m \bar{q}_l(0) \psi_l(x) dx = \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \psi_k(x) h(x) dx = h_k . \quad (16.90)$$

That is

$$\sum_{l=0}^m \left(\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \psi_k(x) \psi_l(x) dx \right) \bar{q}_l(0) = h_k . \quad (16.91)$$

Applying (16.91) for each $\psi_k(x)$, for $k = 0, 1, \dots, m$ we obtain the system

$$\mathbf{A} \bar{\mathbf{Q}}(0) = \mathbf{H} = [h_0, h_1, \dots, h_m]^T , \quad (16.92)$$

where the coefficient matrix $\mathbf{A}$ is given in (16.85). Therefore the initial condition in element I_i is given by

$$\bar{\mathbf{Q}}(0) = \mathbf{A}^{-1} \mathbf{H} . \quad (16.93)$$

In the semi-discrete formulation of the DG method one makes use of the TVD Runge-Kutta schemes presented in Sect. 16.3. Second and third-order accurate methods are available.

CFL Stability Condition and Time Step

The Courant condition for choosing a stable time step Δt is particularly restrictive for the DG method, as compared with the finite volume methods presented in previous sections. The Courant number c must obey the following condition

$$c \leq C_{lim} = \frac{1}{2m+1}, \quad (16.94)$$

where m is the degree of the polynomials and $m+1$ is the accuracy of the method. For second order accuracy $m=1$ and thus $C_{lim}=1/3$. For third-order accuracy $m=2$ and hence $C_{lim}=1/5$. These restrictions are to be compared with $C_{lim}=1$ for the ADER fully-discrete approach, for any order of accuracy.

The stability restriction (16.94) is a major drawback of the discontinuous Galerkin finite element method, as compared to finite volume methods. Another drawback of the DG method is its performance for discontinuous solutions, although progress has been made in devising the equivalent of non-linear schemes in finite volume methods, in order to account for Godunov's theorem. A distinctive feature of DG is its spatial resolution of the solution within each element, as compared to the finite volume method.

So far we have presented the semi-discrete version of the DG method, in which the formal accuracy of the method does not exceed 3, if the TVD Runge-Kutta methods presented in Sect. 16.3 are to be used.

16.4.4 Fully-Discrete ADER-Type DG Schemes

Finite volume methods have a fully discrete, one-step high-order formulation via the ADER framework [1], as seen in Sect. 16.1, as well as a semi-discrete, multi-step formulation, as seen in Sect. 16.3. Discontinuous Galerkin finite element methods are most well known for their semi-discrete, multi-step version, as presented in this section. However DG methods also have a fully-discrete, one-step formulation via the ADER framework, originally due to Dumbser [39, 40]. The resulting ADER-DG schemes have arbitrary accuracy in both space and time, that is the ADER-DG schemes have no theoretical accuracy barrier. Further developments of the fully-discrete DG schemes are reported, for example, in [42, 43]. Additional refinements to the numerical techniques and very ambitious application of the resulting fully-discrete ADER discontinuous Galerkin finite element schemes are reported in [44, 45].

In the next section we assess the performance of most of the numerical methods studied in this chapter.

16.5 Numerical Results

In this section we carry out a comprehensive computational assessment of the performance of some of the numerical methods presented in this chapter. The assessment is performed in three ways, namely (i) convergence rates study for two kinds of smooth profiles, (ii) comparison of numerical solutions against exact solutions at specified output times, for three kinds of profiles, and (iii) efficiency, that is error against cost, for orders of accuracy 3, 5 and 10. From all the methods studied in this chapter we assess those listed in Table 16.1.

16.5.1 Test Problems

We solve the following initial-boundary value problem for the linear advection equation

$$\left. \begin{array}{l} \text{PDE:} \quad \partial_t q + \lambda \partial_x q = 0, \quad \lambda = 1, \quad x \in [a, b], \quad t > 0, \\ \text{IC:} \quad q(x, 0) = h(x), \quad x \in [a, b], \quad t = 0, \\ \text{BCs (periodic):} \quad q(a, t) = q(b, t), \quad t > 0. \end{array} \right\} \quad (16.95)$$

We consider four test problems, with appropriate choice for the initial condition $h(x)$, the computational domain $[a, b]$ and the output time.

Test 1: sinusoidal profile. The initial condition $h(x)$ for (16.95) in the spatial domain $[a, b] = [0, 10]$ is given as

$$q(x, 0) = h(x) = \sin(2\pi x). \quad (16.96)$$

This smooth solution test problem is used for convergence rates studies.

Table 16.1 High-order ADER schemes, along with two solvers for the generalised Riemann problem GRP_m and three spatial reconstruction methods. The GRP_m solvers are the Toro-Titarev (TT) solver and the Dumbser-Enaux-Toro (DET) solver. The reconstruction methods are WENO, WENO-DK and dENO (direct ENO). Orders of accuracy in space and time: from 2 to 10

Scheme	Description	Order
ADER+TT+WENO	ADER+TT solver for GRP_m and WENO	2–10
ADER+TT+WENO-DK	ADER+TT solver for GRP_m and WENO-DK	2–10
ADER+TT+dENO	ADER+TT solver for GRP_m and dENO	2–10
ADER+DET+WENO	ADER+DET solver for GRP_m and WENO	2–10
ADER+DET+WENO-DK	ADER+DET solver for GRP_m and WENO-DK	2–10
ADER+DET+dENO	ADER+DET solver for GRP_m and dENO	2–10

Test 2: Gaussian profile. The initial condition $h(x)$ for (16.95) is smooth in the spatial domain $[a, b] = [-5, 5]$ and is given by

$$q(x, 0) = h(x) = \alpha e^{-\beta x^2}, \quad \alpha = 1, \quad \beta = 2. \quad (16.97)$$

This test problem is used for comparing computed solution profiles against the exact solution and for convergence rates studies.

Test 3: square wave. The initial condition $h(x)$ for (16.95) is discontinuous in the spatial domain $[a, b] = [-5, 5]$ and given by the square wave profile

$$q(x, 0) = h(x) = \begin{cases} 0 & \text{if } a = -5 \leq x < -2, \\ 1 & \text{if } -2 \leq x \leq 2, \\ 0 & \text{if } 2 < x \leq b = 5. \end{cases} \quad (16.98)$$

This test problem is used for comparing computed solution profiles against the exact solution.

Test 4: multi-wave profile. The initial condition $h(x)$ for (16.95) consists of a multi-wave profile [18] in the spatial domain $[a, b] = [-1, 1]$ and is given as

$$q(x, 0) = h(x) = \begin{cases} \exp(-\log(2)(x + 0.7)^2/0.0009) & -0.8 \leq x \leq -0.6, \\ 1 & -0.4 \leq x \leq -0.2, \\ 1 - |10x - 1| & 0 \leq x \leq 0.2, \\ \sqrt{1 - 100(x - 0.5)^2} & 0.4 \leq x \leq 0.6, \\ 0 & \text{otherwise.} \end{cases} \quad (16.99)$$

This test problem is used for comparing computed solution profiles against the exact solution.

In the next section we assess empirically convergence rates, in order to verify whether the high-order non-linear methods attain the theoretically expected convergence rates for smooth solutions.

16.5.2 Convergence Rates Study

In this section we assess a key aspect of high-order numerical methods, namely convergence rates. The study, though empirical, provides a reliable indication of whether the non-linear schemes of interest attain the theoretically expected rate of convergence for smooth solutions. This is a most important issue when developing algorithms of expected high-order of accuracy. The study is carried for all schemes of Table 16.1 and for two initial boundary value problems (16.95). Results are shown in Tables 16.2 to 16.7.

Results from Test 1

Results from Test 1 for IBVP(16.95) with initial condition (16.96) are shown in Tables 16.2 to 16.4, respectively for three reconstruction methods for ADER with the Toro-Titarev (TT) solver for the generalised Riemann problem GRP_m , at output time $T_{out} = 0.5$. Errors L_k and convergence rates p_k are computed in norms L_k , with $k = 1, 2, \infty$. The mesh M is given in the first column and CPU time in seconds is given in the last column.

Table 16.2 shows results for Test 1 for the WENO reconstruction method. It is seen that WENO attains, comfortably, all the expected convergence rates, in all three norms, up to 10th order of accuracy in space and time. In fact from the 6th order upwards, there are signs of super convergence.

Table 16.3 shows results for the WENO-DK reconstruction method. It is seen that WENO-DK also attains the expected convergence rates, in all three norms, up to the 10th order of accuracy in space and time.

Table 16.4 shows results for the dENO reconstruction method. It is seen that dENO attains the theoretically expected convergence rates for the odd orders, while for the even orders convergence is sub-optimal. Convergence difficulties are more obvious for the L_∞ norm.

The reader is invited to exam the errors and CPU times for all three reconstruction schemes. For example, compare the last row of Table 16.2 with the corresponding rows from WENO-DK and dENO in Tables 16.3 and 16.4, respectively. The L_2 error of WENO is 2.8×10^{-12} and CPU is 2.010 s. The corresponding results from WENO-DK are 2.0×10^{-9} and CPU is 0.886 s. WENO-DK is cheaper by a factor of about 2, but the WENO error is about 3 orders of magnitude smaller. The results from dENO are 5.6×10^{-9} and 0.706 s.

Results from Test 2

Results from Test 2 for IBVP(16.95) with initial condition (16.97) are shown in Tables 16.5 to 16.7 for three reconstruction methods for ADER with Toro-Titarev (TT) solver for the generalised Riemann problem GRP_m , at output time $T_{out} = 50$. Errors L_k and convergence rates p_k are computed in norms L_k , with $k = 1, 2, \infty$. The mesh M is given in the first column and CPU time in seconds is given in the last column.

Table 16.5 shows results from WENO. As for Test 1, WENO reaches the expected convergence rates, for all tree norms. However, this test problem is more demanding than Test 1. Still, for orders 9th and 10th, again there are signs of super convergence.

Table 16.6 shows results from WENO-DK. The expected convergence rates are attained in all three norms from 3rd order to 6th order. However, for the higher order cases 9th and 10th, convergence rates tend to be sub-optimal. Again comparing errors and CPU times for WENO and WENO-DK we see that WENO gives much smaller errors, but the CPU cost is higher.

Table 16.7 shows results from dENO (direct ENO). For the 3rd order case, convergence rates tend towards the expected order of accuracy, but sub-optimally. However, for all remaining orders, dENO does not attain the expected orders at all, in none of

Table 16.2 Convergence rates for ADER with TT solver for the generalised Riemann problem GRP_m and WENO reconstruction applied to IBVP (16.95), with initial condition (16.96) at output time $T_{out} = 0.5$, with $C_{cf1} = 0.9$. Errors L_k and convergence rates p_k are computed in norms L_k , with $k = 1, 2, \infty$. The mesh M is given in the first column and CPU time in seconds is given in the last column

M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
ADER-TT+WENO Order 3							
26	3.3e+00	-0.2	1.1e+00	-0.1	5.1e-01	-0.1	0.001
52	3.2e-01	3.4	1.3e-01	3.1	8.2e-02	2.6	0.004
104	3.0e-02	3.4	1.2e-02	3.5	7.3e-03	3.5	0.017
208	1.4e-03	4.4	4.8e-04	4.6	3.0e-04	4.6	0.066
ADER-TT+WENO Order 4							
26	3.2e+00	-0.1	1.1e+00	-0.0	5.3e-01	-0.2	0.002
52	1.4e-01	4.5	5.3e-02	4.4	3.1e-02	4.1	0.006
104	5.8e-03	4.6	2.1e-03	4.7	1.1e-03	4.8	0.024
208	1.2e-04	5.5	4.5e-05	5.5	2.6e-05	5.4	0.097
ADER-TT+WENO Order 5							
26	3.7e+00	-0.3	1.3e+00	-0.2	6.8e-01	-0.5	0.005
52	9.6e-02	5.3	3.6e-02	5.1	2.0e-02	5.0	0.013
104	3.4e-03	4.8	1.2e-03	4.9	7.0e-04	4.9	0.053
208	1.1e-04	5.0	3.8e-05	5.1	2.0e-05	5.1	0.207
ADER-TT+WENO Order 6							
26	2.6e+00	0.2	9.0e-01	0.3	4.3e-01	0.2	0.005
52	2.9e-02	6.5	1.1e-02	6.4	6.2e-03	6.1	0.013
104	7.5e-05	8.6	2.7e-05	8.6	1.6e-05	8.6	0.051
208	3.7e-07	7.7	1.3e-07	7.7	7.0e-08	7.8	0.205
ADER-TT+WENO Order 9							
26	1.9e+00	0.7	6.7e-01	0.7	3.3e-01	0.6	0.016
52	2.9e-03	9.4	1.1e-03	9.2	6.4e-04	9.0	0.039
104	4.5e-08	16.0	1.7e-08	16.0	1.0e-08	16.0	0.140
208	1.1e-10	8.6	4.0e-11	8.7	1.8e-11	9.1	0.533
ADER-TT+WENO Order 10							
26	1.7e+00	0.9	5.9e-01	0.9	2.8e-01	0.8	0.042
52	1.4e-03	10.3	5.4e-04	10.1	3.2e-04	9.8	0.127
104	4.1e-09	18.3	1.5e-09	18.4	8.1e-10	18.6	0.502
208	7.8e-12	9.0	2.8e-12	9.1	1.2e-12	9.4	2.010

Table 16.3 Convergence rates for ADER with TT solver for the generalised Riemann problem GRP_m and WENO-DK reconstruction applied to IBVP (16.95), with initial condition (16.96) at output time $T_{out} = 0.5$, with $C_{cfl} = 0.9$. Errors L_k and convergence rates p_k are computed in norms L_k , with $k = 1, 2, \infty$. The mesh M is given in the first column and CPU time in seconds is given in the last column

M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
ADER-TT+WENO-DK Order 3							
26	3.1e+00	-0.1	1.1e+00	-0.0	5.0e-01	-0.1	0.002
52	3.4e-01	3.2	1.2e-01	3.2	5.2e-02	3.3	0.004
104	4.7e-02	2.8	1.7e-02	2.8	7.4e-03	2.8	0.017
208	5.7e-03	3.0	2.0e-03	3.0	9.0e-04	3.0	0.069
ADER-TT+WENO-DK Order 4							
26	2.2e+00	0.5	7.7e-01	0.5	3.4e-01	0.5	0.002
52	1.2e-01	4.2	4.3e-02	4.2	1.9e-02	4.2	0.006
104	7.0e-03	4.1	2.4e-03	4.1	1.1e-03	4.1	0.023
208	3.7e-04	4.2	1.3e-04	4.2	5.8e-05	4.2	0.092
ADER-TT+WENO-DK Order 5							
26	2.7e+00	0.2	9.7e-01	0.2	5.4e-01	-0.2	0.003
52	9.0e-02	4.9	3.2e-02	4.9	1.4e-02	5.3	0.007
104	3.4e-03	4.7	1.2e-03	4.7	5.4e-04	4.7	0.029
208	1.1e-04	5.0	3.7e-05	5.0	1.7e-05	5.0	0.114
ADER-TT+WENO-DK Order 6							
26	1.8e+00	0.7	6.5e-01	0.7	2.9e-01	0.7	0.003
52	4.0e-02	5.5	1.4e-02	5.5	6.2e-03	5.5	0.010
104	6.7e-04	5.9	2.4e-04	5.9	1.1e-04	5.9	0.037
208	9.5e-06	6.1	3.4e-06	6.1	1.5e-06	6.1	0.146
ADER-TT+WENO-DK Order 9							
26	1.4e+00	1.2	4.8e-01	1.2	2.1e-01	1.2	0.010
52	7.4e-03	7.5	2.6e-03	7.5	1.2e-03	7.5	0.021
104	2.1e-05	8.5	7.4e-06	8.5	3.3e-06	8.5	0.070
208	4.2e-08	9.0	1.5e-08	9.0	6.6e-09	9.0	0.258
ADER-TT+WENO-DK Order 10							
26	1.3e+00	1.2	4.6e-01	1.2	2.1e-01	1.2	0.019
52	4.2e-03	8.3	1.5e-03	8.3	6.6e-04	8.3	0.056
104	5.9e-06	9.5	2.1e-06	9.5	9.2e-07	9.5	0.222
208	5.6e-09	10.0	2.0e-09	10.0	8.9e-10	10.0	0.886

Table 16.4 Convergence rates for ADER with TT solver for the generalised Riemann problem GRP_m and dENO reconstruction applied to IBVP (16.95), with initial condition (16.96) at output time $T_{out} = 0.5$, with $C_{cfI} = 0.9$. Errors L_k and convergence rates p_k are computed in norms L_k , with $k = 1, 2, \infty$. The mesh M is given in the first column and CPU time in seconds is given in the last column

M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
ADER-TT+dENO Order 3							
26	4.2e+00	-0.5	1.5e+00	-0.5	7.3e-01	-0.7	0.003
52	3.6e-01	3.6	1.4e-01	3.4	8.0e-02	3.2	0.007
104	4.7e-02	2.9	1.7e-02	3.0	9.8e-03	3.0	0.029
208	5.7e-03	3.0	2.0e-03	3.1	1.1e-03	3.1	0.114
ADER-TT+dENO Order 4							
26	4.1e+00	-0.4	1.4e+00	-0.4	7.0e-01	-0.6	0.003
52	1.9e-01	4.4	8.4e-02	4.1	4.8e-02	3.9	0.010
104	2.3e-02	3.1	8.9e-03	3.2	5.6e-03	3.1	0.039
208	1.7e-03	3.8	7.3e-04	3.6	6.4e-04	3.1	0.157
ADER-TT+dENO Order 5							
26	3.7e+00	-0.3	1.3e+00	-0.2	6.8e-01	-0.5	0.005
52	9.6e-02	5.3	3.6e-02	5.1	2.0e-02	5.0	0.013
104	3.4e-03	4.8	1.2e-03	4.9	7.0e-04	4.9	0.053
208	1.1e-04	5.0	3.8e-05	5.1	2.0e-05	5.1	0.207
ADER-TT+dENO Order 6							
26	3.1e+00	-0.0	1.1e+00	0.0	5.6e-01	-0.2	0.006
52	5.3e-02	5.9	2.3e-02	5.5	1.3e-02	5.4	0.017
104	1.7e-03	5.0	6.7e-04	5.1	4.2e-04	5.0	0.067
208	3.1e-05	5.8	1.4e-05	5.6	1.3e-05	5.1	0.268
ADER-TT+dENO Order 9							
26	2.2e+00	0.5	7.5e-01	0.5	3.8e-01	0.4	0.016
52	7.8e-03	8.1	3.2e-03	7.9	2.0e-03	7.6	0.040
104	2.1e-05	8.5	7.6e-06	8.7	4.2e-06	8.9	0.140
208	4.2e-08	9.0	1.5e-08	9.0	8.2e-09	9.0	0.539
ADER-TT+dENO Order 10							
26	1.9e+00	0.7	6.6e-01	0.7	3.2e-01	0.6	0.015
52	4.4e-03	8.8	1.9e-03	8.4	1.1e-03	8.2	0.044
104	1.1e-05	8.7	4.2e-06	8.8	2.7e-06	8.7	0.178
208	1.2e-08	9.8	5.6e-09	9.6	5.2e-09	9.0	0.706

Table 16.5 Convergence rates for ADER with TT solver for the generalised Riemann problem GRP_m and WENO reconstruction applied to IBVP (16.95), with initial condition (16.97) at output time $T_{out} = 50$, with $C_{cfl} = 0.9$. Errors L_k and convergence rates p_k are computed in norms L_k , with $k = 1, 2, \infty$. The mesh M is given in the first column and CPU time in seconds is given in the last column

M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
ADER-TT+WENO Order 3							
26	3.7e-01	0.9	2.3e-01	1.0	2.5e-01	0.9	0.107
52	6.3e-02	2.6	4.3e-02	2.4	4.7e-02	2.4	0.414
104	6.5e-03	3.3	4.2e-03	3.3	5.1e-03	3.2	1.649
208	7.9e-04	3.0	5.3e-04	3.0	6.8e-04	2.9	6.631
ADER-TT+WENO Order 4							
26	1.7e-01	1.8	1.1e-01	1.7	1.2e-01	1.6	0.159
52	1.2e-02	3.9	8.3e-03	3.8	1.1e-02	3.5	0.621
104	4.9e-04	4.6	3.7e-04	4.5	5.4e-04	4.4	2.460
208	3.2e-05	3.9	2.2e-05	4.0	2.7e-05	4.3	9.772
ADER-TT+WENO Order 5							
26	1.1e-01	2.3	7.3e-02	2.2	7.9e-02	2.1	0.236
52	2.5e-03	5.5	1.7e-03	5.4	2.0e-03	5.3	0.931
104	4.2e-05	5.9	3.0e-05	5.9	4.4e-05	5.5	3.703
208	1.6e-06	4.7	1.1e-06	4.8	1.4e-06	5.0	14.821
ADER-TT+WENO Order 6							
26	8.5e-02	2.5	5.4e-02	2.5	5.3e-02	2.6	0.334
52	7.0e-04	6.9	4.8e-04	6.8	5.5e-04	6.6	1.321
104	4.9e-06	7.2	3.5e-06	7.1	4.8e-06	6.8	5.252
208	8.4e-08	5.9	5.8e-08	5.9	6.8e-08	6.2	20.940
ADER-TT+WENO Order 9							
26	5.1e-02	3.3	3.1e-02	3.1	2.6e-02	3.3	0.845
52	4.2e-05	10.2	2.7e-05	10.1	3.0e-05	9.8	3.371
104	9.3e-09	12.1	6.6e-09	12.0	8.0e-09	11.9	13.515
208	2.1e-11	8.8	1.4e-11	8.8	1.7e-11	8.8	53.928
ADER-TT+WENO Order 10							
26	4.3e-02	3.5	2.6e-02	3.3	2.3e-02	3.4	2.968
52	2.0e-05	11.1	1.3e-05	11.0	1.6e-05	10.4	11.890
104	1.3e-09	14.0	9.0e-10	13.8	1.0e-09	13.9	47.334
208	1.9e-12	9.4	1.4e-12	9.3	2.1e-12	9.0	189.476

Table 16.6 Convergence rates for ADER with TT solver for the generalised Riemann problem GRP_m and WENO-DK reconstruction applied to IBVP (16.95), with initial condition (16.97) at output time $T_{out} = 50$, with $C_{cfl} = 0.9$. Errors L_k and convergence rates p_k are computed in norms L_k , with $k = 1, 2, \infty$. The mesh M is given in the first column and CPU time in seconds is given in the last column

M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
ADER-TT+WENO-DK Order 3							
26	3.6e-01	1.1	2.4e-01	1.0	2.4e-01	1.0	0.110
52	9.8e-02	1.9	7.1e-02	1.7	8.5e-02	1.5	0.434
104	2.0e-02	2.3	1.4e-02	2.4	1.7e-02	2.4	1.730
208	2.7e-03	2.9	1.8e-03	2.9	2.3e-03	2.9	6.861
ADER-TT+WENO-DK Order 4							
26	1.7e-01	1.7	1.2e-01	1.6	1.3e-01	1.5	0.153
52	3.3e-02	2.4	2.2e-02	2.4	2.4e-02	2.4	0.599
104	2.3e-03	3.8	1.6e-03	3.8	1.8e-03	3.7	2.396
208	1.4e-04	4.0	9.9e-05	4.0	1.1e-04	4.0	9.395
ADER-TT+WENO-DK Order 5							
26	1.4e-01	2.0	9.3e-02	1.9	9.6e-02	1.9	0.189
52	2.0e-02	2.8	1.3e-02	2.8	1.4e-02	2.7	0.743
104	7.7e-04	4.7	5.3e-04	4.6	6.5e-04	4.5	2.988
208	2.5e-05	4.9	1.7e-05	4.9	2.2e-05	4.9	11.809
ADER-TT+WENO-DK Order 6							
26	9.5e-02	2.3	6.3e-02	2.2	6.9e-02	2.1	0.247
52	7.7e-03	3.6	5.0e-03	3.7	5.5e-03	3.6	0.974
104	1.4e-04	5.8	9.8e-05	5.7	1.1e-04	5.6	3.874
208	2.3e-06	6.0	1.6e-06	6.0	1.8e-06	5.9	15.431
ADER-TT+WENO-DK Order 9							
26	8.8e-02	3.3	6.2e-02	2.7	6.4e-02	2.3	0.441
52	1.0e-03	6.5	6.6e-04	6.6	7.6e-04	6.4	1.744
104	2.9e-06	8.4	2.0e-06	8.4	2.5e-06	8.2	6.937
208	6.3e-09	8.9	4.4e-09	8.9	5.5e-09	8.8	27.708
ADER-TT+WENO-DK Order 10							
26	4.7e-02	3.4	2.7e-02	3.1	2.6e-02	2.9	1.320
52	6.4e-04	6.2	4.1e-04	6.1	4.6e-04	5.8	5.241
104	1.0e-06	9.3	6.8e-07	9.2	8.1e-07	9.1	20.886
208	1.1e-09	9.8	7.6e-10	9.8	9.1e-10	9.8	83.415

Table 16.7 Convergence rates for ADER with TT solver for the generalised Riemann problem GRP_m and dENO reconstruction applied to IBVP (16.95), with initial condition (16.97) at output time $T_{out} = 50$, with $C_{cfl} = 0.9$. Errors L_k and convergence rates p_k are computed in norms L_k , with $k = 1, 2, \infty$. The mesh M is given in the first column and CPU time in seconds is given in the last column

M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
ADER-TT+dENO Order 3							
26	4.0e-01	0.9	2.5e-01	0.9	2.6e-01	0.9	0.188
52	1.0e-01	2.0	6.7e-02	1.9	8.4e-02	1.6	0.735
104	1.6e-02	2.6	1.2e-02	2.5	1.7e-02	2.3	2.910
208	2.7e-03	2.6	1.8e-03	2.7	2.3e-03	2.8	11.462
ADER-TT+dENO Order 4							
26	2.8e-01	1.2	1.8e-01	1.2	1.9e-01	1.2	0.248
52	4.0e-02	2.8	2.8e-02	2.7	4.0e-02	2.2	0.972
104	3.5e-03	3.5	3.0e-03	3.2	5.7e-03	2.8	3.884
208	3.9e-02	-3.5	3.8e-02	-3.7	6.2e-02	-3.4	15.536
ADER-TT+dENO Order 5							
26	1.9e-01	1.7	1.2e-01	1.6	1.3e-01	1.6	0.331
52	1.3e-02	3.9	9.2e-03	3.8	1.3e-02	3.3	1.303
104	6.9e-04	4.2	4.6e-04	4.3	6.7e-04	4.3	5.181
208	1.4e-02	-4.4	1.6e-02	-5.1	3.7e-02	-5.8	20.690
ADER-TT+dENO Order 6							
26	1.4e-01	2.0	9.3e-02	2.0	9.6e-02	1.9	0.426
52	6.0e-03	4.5	4.5e-03	4.4	6.6e-03	3.9	1.677
104	2.1e-01	-5.1	1.6e-01	-5.2	1.9e-01	-4.8	6.698
208	4.9e-03	5.4	5.1e-03	5.0	8.2e-03	4.5	26.804
ADER-TT+dENO Order 9							
26	6.5e-02	2.9	4.3e-02	2.9	4.5e-02	2.8	0.861
52	6.4e-04	6.7	4.6e-04	6.6	7.3e-04	6.0	3.404
104	8.3e-03	-3.7	8.1e-03	-4.1	1.7e-02	-4.5	13.625
208	7.1e-03	0.2	8.4e-03	-0.0	1.7e-02	0.0	54.014
ADER-TT+dENO Order 10							
26	5.3e-02	3.1	3.5e-02	3.1	3.7e-02	3.0	1.039
52	4.4e-04	6.9	2.9e-04	6.9	3.6e-04	6.7	4.118
104	1.9e-02	-5.4	2.0e-02	-6.1	3.4e-02	-6.6	16.548
208	7.7e-04	4.6	6.1e-04	5.0	9.5e-04	5.2	65.328

the norms, in fact. This is not a surprising result. There are plenty of studies in the literature that confirm this kind of results for ENO, though for orders below 5th in space and time, for smooth data, but with large spatial gradients. See [46, 47], for example.

For sufficiently smooth solutions, such as for Test 1, all methods assessed attain the theoretically expected convergence rates. This is a very positive result, because it shows that the non-linear character (adaptive stencil) of the designed high-order methods does not spoil the theoretically expected accuracy for sufficiently smooth solutions. Moreover, as we shall see in the next set of numerical experiments, and contrary to widespread belief, smooth solutions with sufficiently large gradients need non-linear methods to avoid spurious oscillations.

16.5.3 Results for Solution Profiles

Here we compute solutions to IBVP (16.95) with three initial conditions $h(x)$ and compare results with the exact solution at specified output times T_{out} . We use two meshes: $M = 25$ cells and $M = 50$ cells. The time step Δt results from choosing the CFL coefficient as $C_{cfl} = 0.9$. The output time is set to $T_{out} = 100$, so that the initial profiles complete 10 cycles within the domain (periodic boundary conditions), at which time the exact solution coincides with the initial condition at time $t = 0$.

Gaussian Profile

Test 2 results for the Gaussian profile, IBVP (16.95), (16.97), are shown in Figs. 16.6 to 16.11 from the high-order ADER method, along with two solvers for the GRP_m , three reconstruction methods and two meshes. Selected orders of accuracy in space and time shown are: 3rd and 4th (top row), 5th and 6th (middle row), and 9th and 10th (bottom row).

Figures 16.6 and 16.7 display results for ADER+TT+WENO for meshes $M = 25$ and $M = 50$ respectively. For the coarser mesh of $M = 25$ cells it is seen that the method in the low-order range is unable to capture the profile correctly; there are visible errors in amplitude. Only the higher-order methods of orders 9th and 10th capture the profile accurately. For the finer mesh of $M = 50$ cells the results of Fig. 16.7 show that the 3rd order method is still unable to capture the wave correctly. The 4th order method improves upon the 3rd order scheme, as expected. For all the higher orders the schemes capture the profile very accurately.

Figures 16.8 and 16.9 display results for ADER+TT+WENO-DK for meshes $M = 25$ and $M = 50$ respectively. The only difference with the previous two results is the reconstruction method, namely WENO-DK instead of WENO. Again, for the coarser mesh of $M = 25$ cells it is seen that the method in the low-order range produces visible errors in amplitude. The higher-order methods show improvements in capturing the peak value, but for orders 9th and 10th the schemes show some slight overshoots and undershoots. Results from the finer mesh of $M = 50$ cells of Fig. 16.9

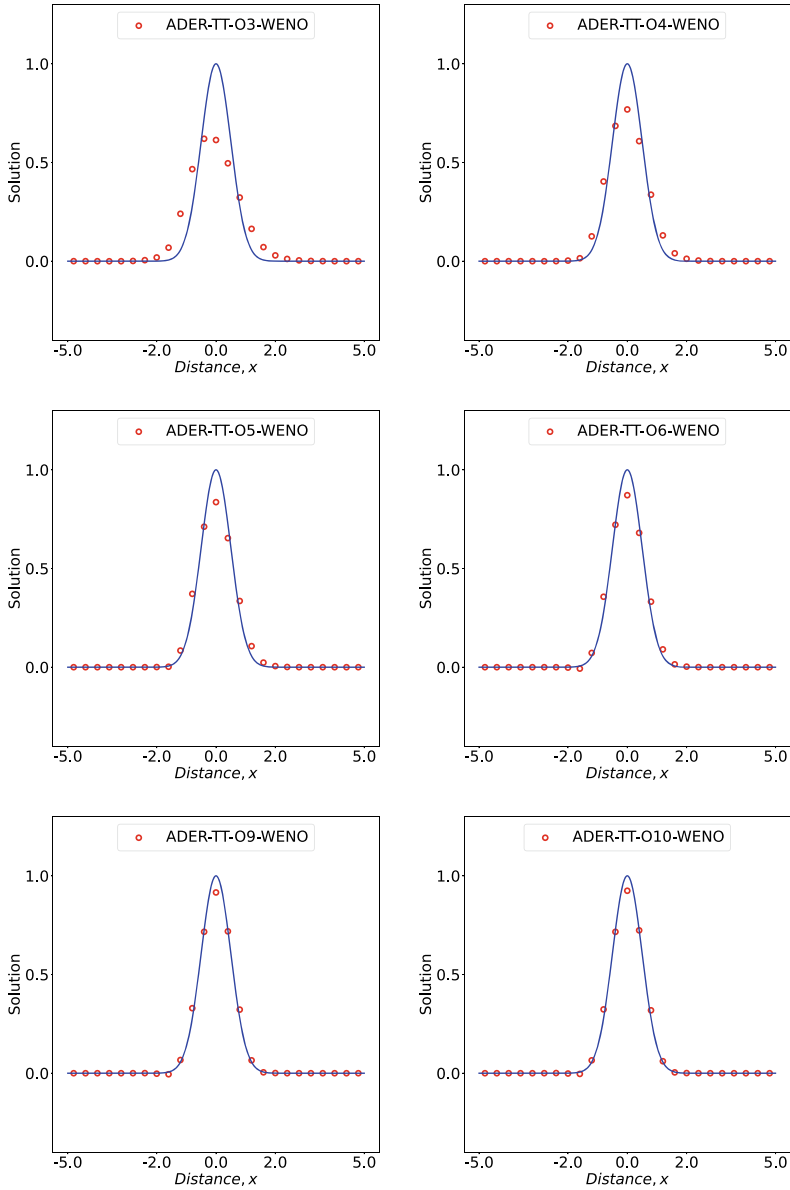


Fig. 16.6 Comparison (for $M = 25$ cells) between the numerical (symbols) and the exact solutions (line) for the smooth initial condition (16.97), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

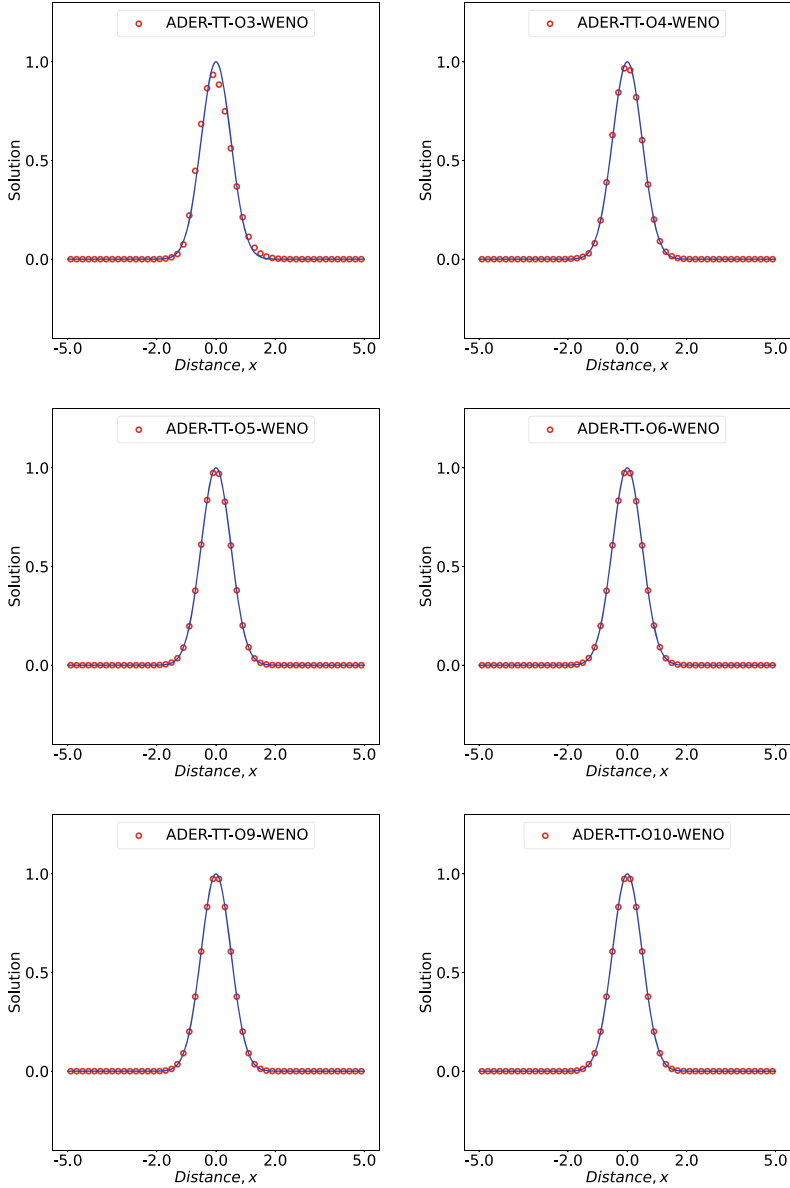


Fig. 16.7 Comparison (for $M = 50$ cells) between the numerical (symbols) and the exact solutions (line) for the smooth initial condition (16.97), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

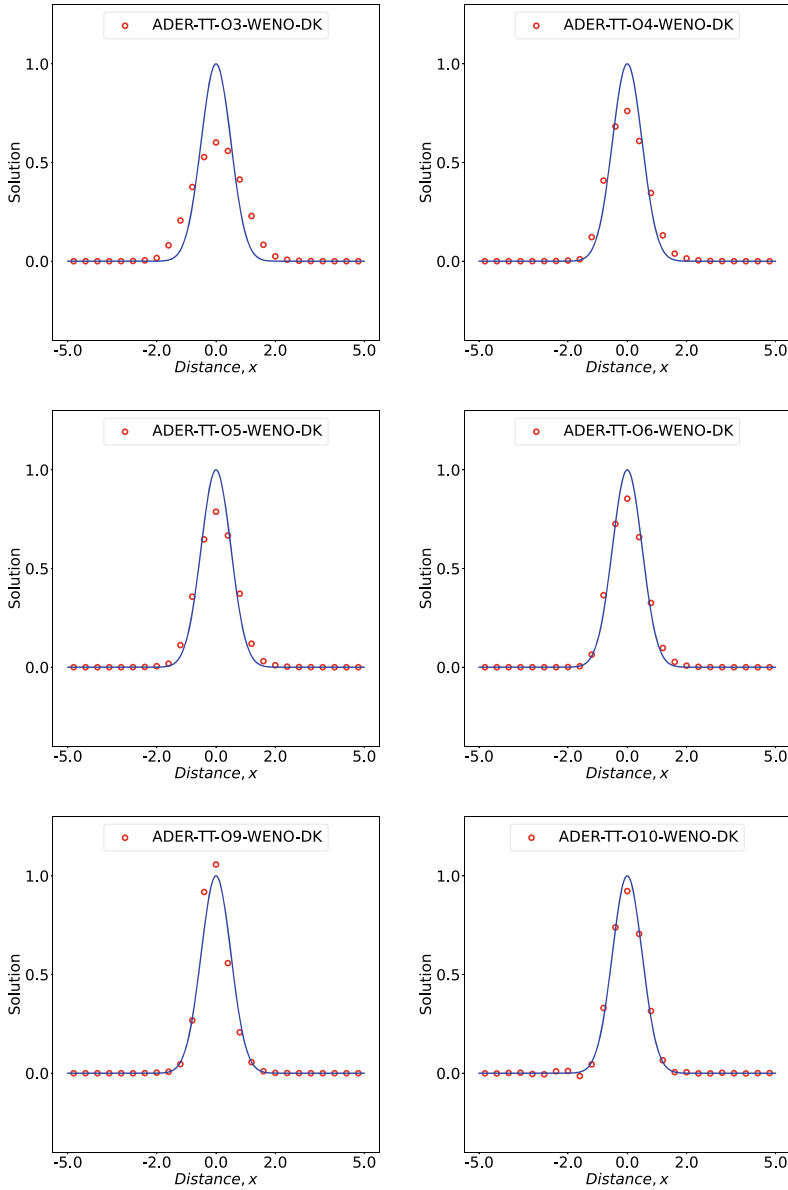


Fig. 16.8 Comparison (for $M = 25$ cells) between the numerical (symbols) and the exact solutions (line) for the smooth initial condition (16.97), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO-DK spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

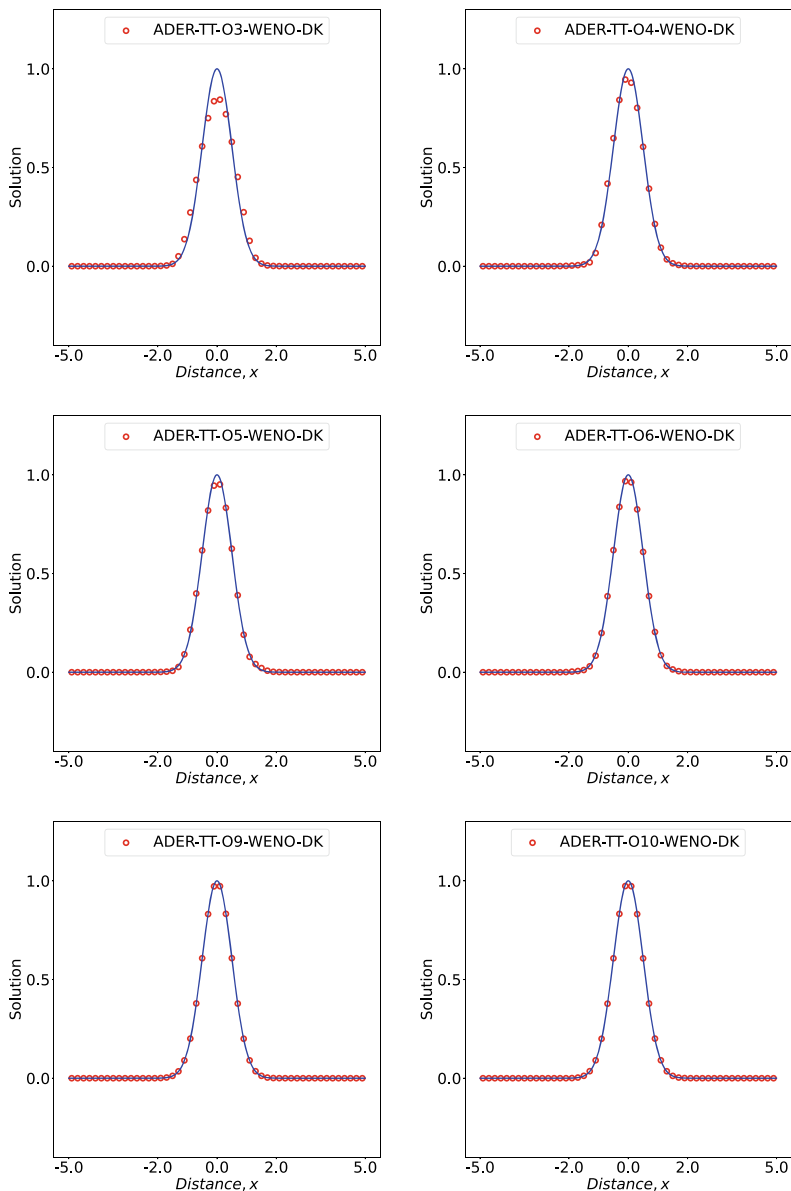


Fig. 16.9 Comparison (for $M = 50$ cells) between the numerical (symbols) and the exact solutions (line) for the smooth initial condition (16.97), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO-DK spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

show a visible improvement. From the 6th order upwards, the method shows very accurate results. The over and undershoots shown for the mesh of $M = 25$ cells have disappeared.

Figures 16.10 and 16.11 show results for ADER+TT+dENO for meshes $M = 25$ and $M = 50$ respectively. For the coarser mesh case of Figure 16.10, amplitude errors are visible, even for the 10th order scheme. No over and undershoots are seen. The finer-mesh results of Fig. 16.11 are very accurate from the 6th order upwards. Amplitude errors are still seen for the lower-order schemes, namely 3rd to 5th.

Summary. Three sets of results have been shown for the Gaussian profile, given by three reconstruction methods: WENO, WENO-DK and dENO. Overall, for all schemes we observe accurate wave propagation speed but visible amplitude errors for the lower-order of accuracy range, say from 3rd to 5th. The results from WENO-DK show some over and undershoots for the coarser mesh case. Overall, the results from dENO (direct ENO) are less accurate than those from the WENO-type methods, though no over and undershoots are visible. The above observations may be modified for shorter and longer output times.

For sufficiently short output times, most methods will display more accurate results. The challenge to methods emerges for long evolution times, for which only the higher-order methods may capture the profiles more accurately. Results from the second set of methods in Table 16.1 using DET instead of TT are virtually identical to those shown above, and are therefore omitted. A general conclusion is that even for very smooth solutions, one requires the use of non-linear methods (in the sense of Godunov). See results from linear methods shown in Chap. 12. A related observation is that not all non-linear methods are perfect for smooth solutions, see Fig. 16.8.

Square Wave Profiles

Test 3 results for the square wave profile, IBVP (16.95), (16.98), with two discontinuities, are shown in Figs. 16.12 to 16.17, from three reconstruction methods (WENO, WENO-DK, dENO) and two meshes. The challenge for numerical methods is to propagate the wave with the correct speed, so that on average discontinuities are placed correctly, and without spurious oscillations in the vicinity of the discontinuities.

The WENO results are shown in Figs. 16.12 and 16.13 for meshes $M = 25$ and $M = 50$ respectively. The position of the two discontinuities is, on average, correct for all orders. However, spurious oscillations in the form of over and undershoots are seen for the higher-order range, namely from 5th to 10th. The severity of the oscillations increases as the order of accuracy increases. This feature of the numerical results for the higher-order range may be unexpected to the reader and appears to be particular to **very high accuracy in both space and time**. Note that the resolution (sharpness) of the discontinuities does benefit from higher order of accuracy, but upto a limit. Beyond the threshold of about 5th order, isolated discontinuities do not benefit from very high order of accuracy, as one may have expected.

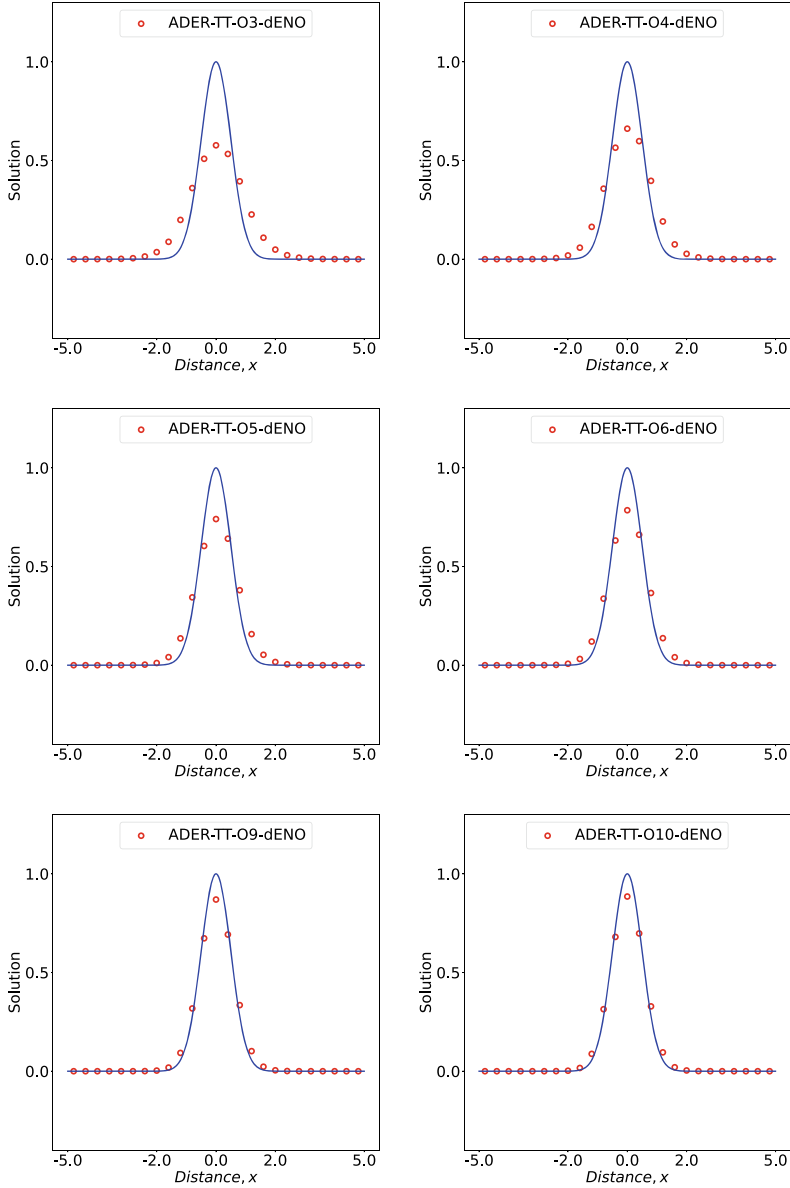


Fig. 16.10 Comparison (for $M = 25$ cells) between the numerical (symbols) and the exact solutions (line) for the smooth initial condition (16.97), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with dENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

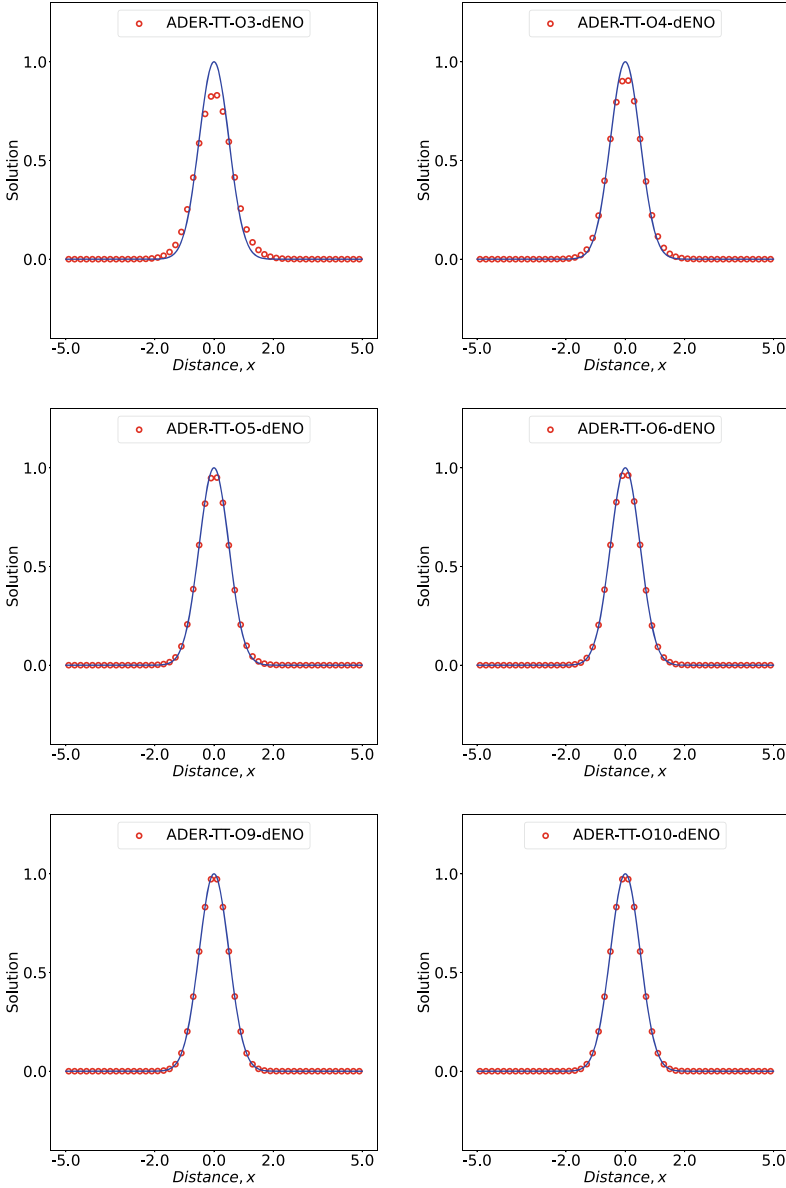


Fig. 16.11 Comparison (for $M = 50$ cells) between the numerical (symbols) and the exact solutions (line) for the smooth initial condition (16.97), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with dENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

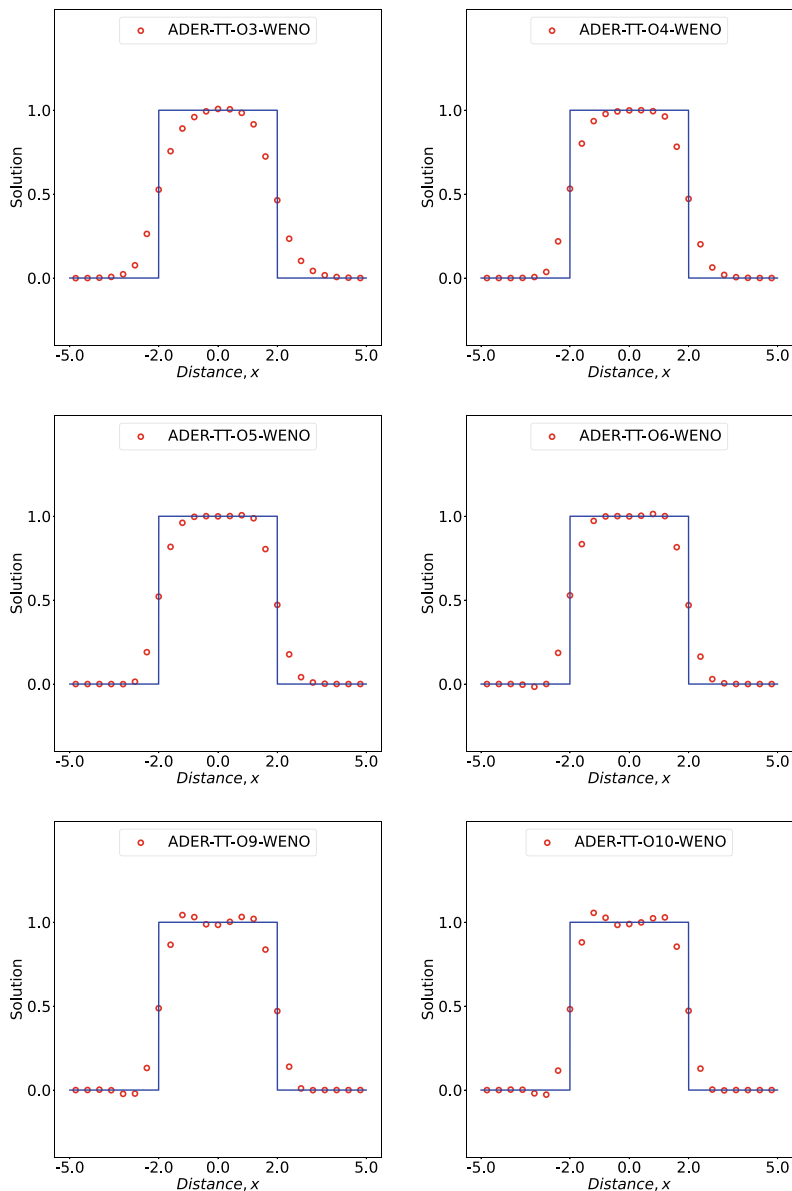


Fig. 16.12 Comparison (for $M = 25$ cells) between the numerical (symbols) and the exact solutions (line) for the square wave initial condition (16.98), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

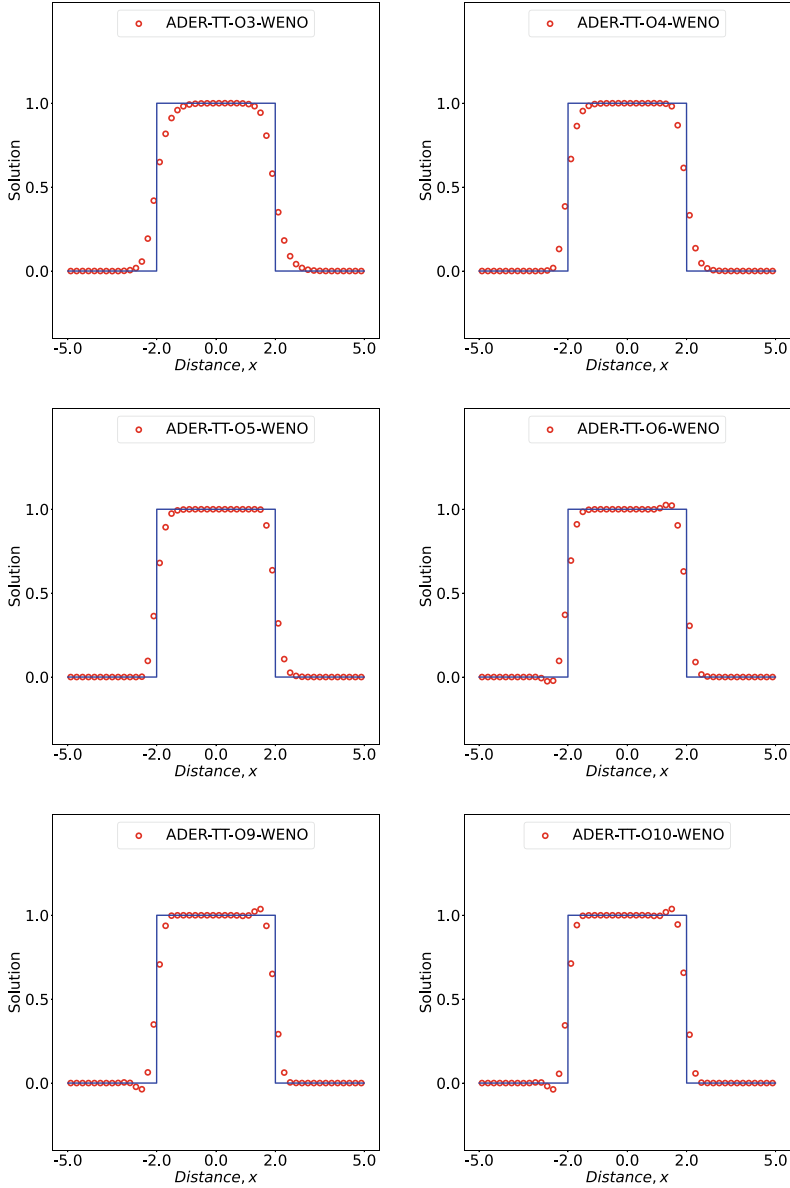


Fig. 16.13 Comparison (for $M = 50$ cells) between the numerical (symbols) and the exact solutions (line) for the square wave initial condition (16.98), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

Results from the WENO-DK reconstruction method are shown in Figs. 16.14 and 16.15 for meshes $M = 25$ and $M = 50$ respectively. The results are similar to those of WENO of Figs. 16.12 and 16.13, but with a difference regarding the spurious oscillations for the higher-order of accuracy range. The amplitudes of the over and undershoots in the vicinity of the discontinuities are larger and increase with mesh refinement, apart from the 6th order case.

Results from the dENO (direct ENO) reconstruction method are shown in Figs. 16.16 and 16.17 for meshes $M = 25$ and $M = 50$ respectively. As compared to the previous two sets of results, these are very satisfactory. Wave positions are correct and there are no visible over and undershoots in the vicinity of the discontinuities. But of course, as already pointed out for the previous results, the resolution of the discontinuities does not improve with the order accuracy beyond the threshold of about 5th order.

Summary. The expected feature from all the numerical results is increased resolution of the discontinuities, as the accuracy of the methods increases. This is true up to the threshold of about 5th order, beyond which no benefits are seen from methods in the higher-order range. An unexpected feature is that the WENO-type reconstruction schemes begin to show over and undershoots in the vicinity of discontinuities, starting from the 5th order method. The spurious oscillations worsen as the order of accuracy increases, the worse case being the 10th order result. Mesh refinement tends to favour the suppression of the spurious oscillations for the 5th and 6th order cases but not for the 9 and 10th order schemes. In this respect the classical WENO reconstruction tends to produce better results than the more recent WENO-DK scheme. The classical ENO reconstruction scheme (dENO) produces the best results for this example, for both meshes. No visible spurious oscillations are seen for any of the orders of accuracy, including the 9th and 10th order, for both $M = 25$ and $M = 50$ cells.

Multi-wave Profiles

Test 4 results for the multi-wave profiles, IBVP (16.95), (16.99), are shown in Figs. 16.18 to 16.23, from three reconstruction methods (WENO, WENO-DK, ENO) and the two meshes $M = 100$ and $M = 200$. In the two previous test problems for solution profiles we considered one isolated very smooth wave (Test 2) and one isolated square wave with two discontinuities (Test 3). The multi-wave test problem [18] considered here includes an array of four distinct wave profiles, smooth and discontinuous, each contributing to a rich set of numerical challenges.

Figures 16.18 and 16.19 show results from WENO for meshes $M = 100$ and $M = 200$, respectively. Overall, the results are rather poor, plagued with large amplitude spurious oscillations, with large over and undershoots. Results tend to improve as the order of accuracy increases, and also as the mesh is refined.

Figures 16.20 and 16.21 show results from WENO-DK for meshes $M = 100$ and $M = 200$, respectively. As for the previous set of results from WENO, the WENO-DK results are also very oscillatory. For the lower-order range the WENO-DK are

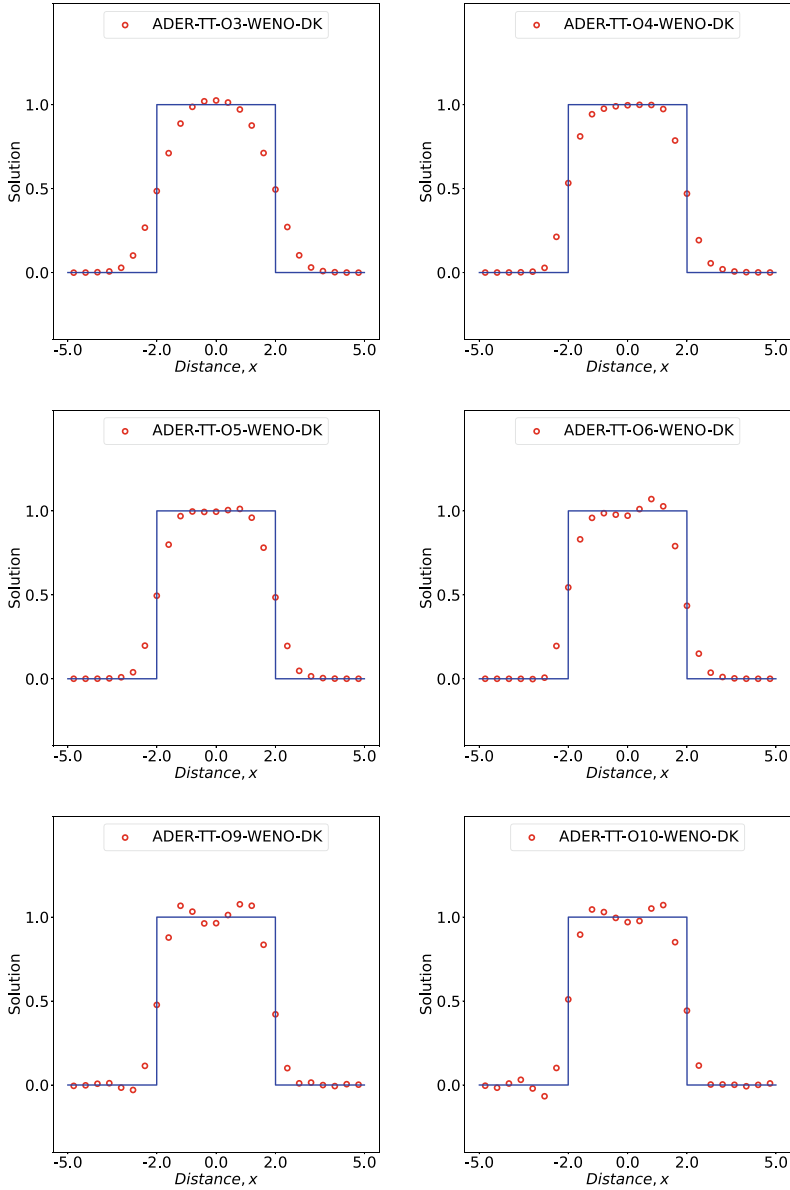


Fig. 16.14 Comparison (for $M = 25$ cells) between the numerical (symbols) and the exact solutions (line) for the square wave initial condition (16.98), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO-DK spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

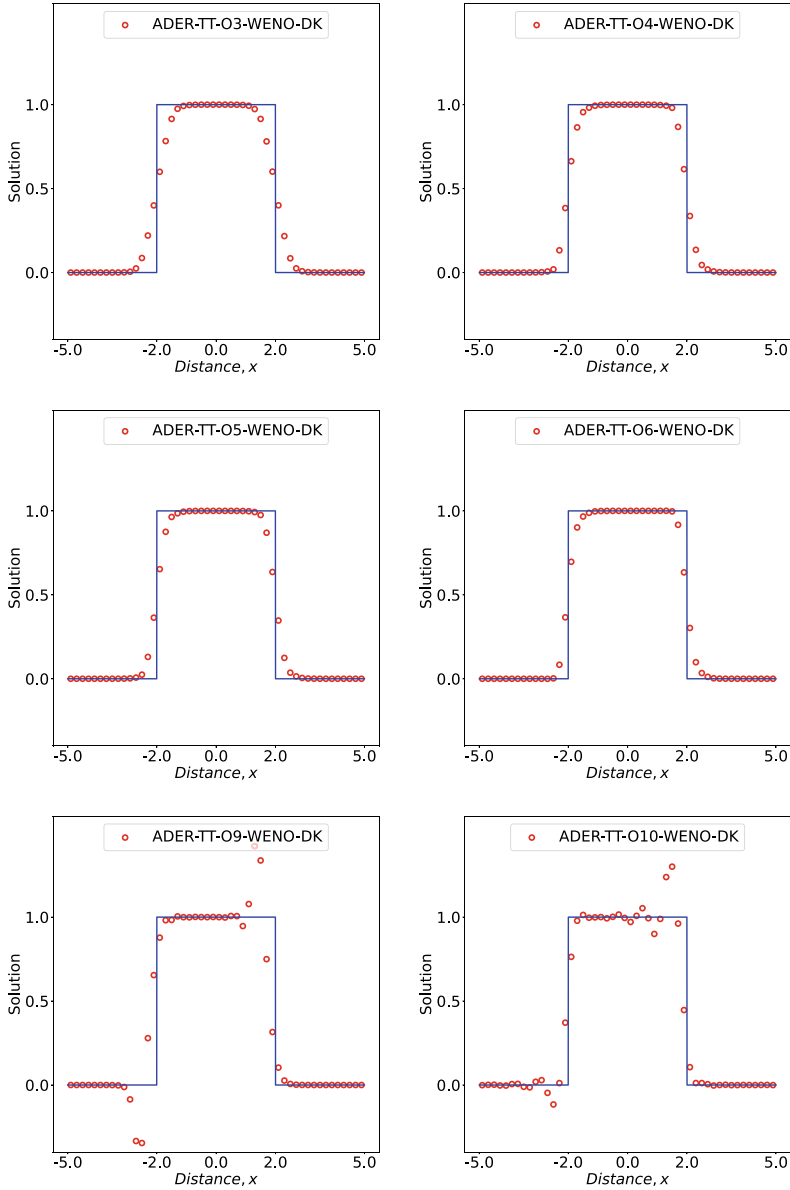


Fig. 16.15 Comparison (for $M = 50$ cells) between the numerical (symbols) and the exact solutions (line) for the square wave initial condition (16.98), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO-DK spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

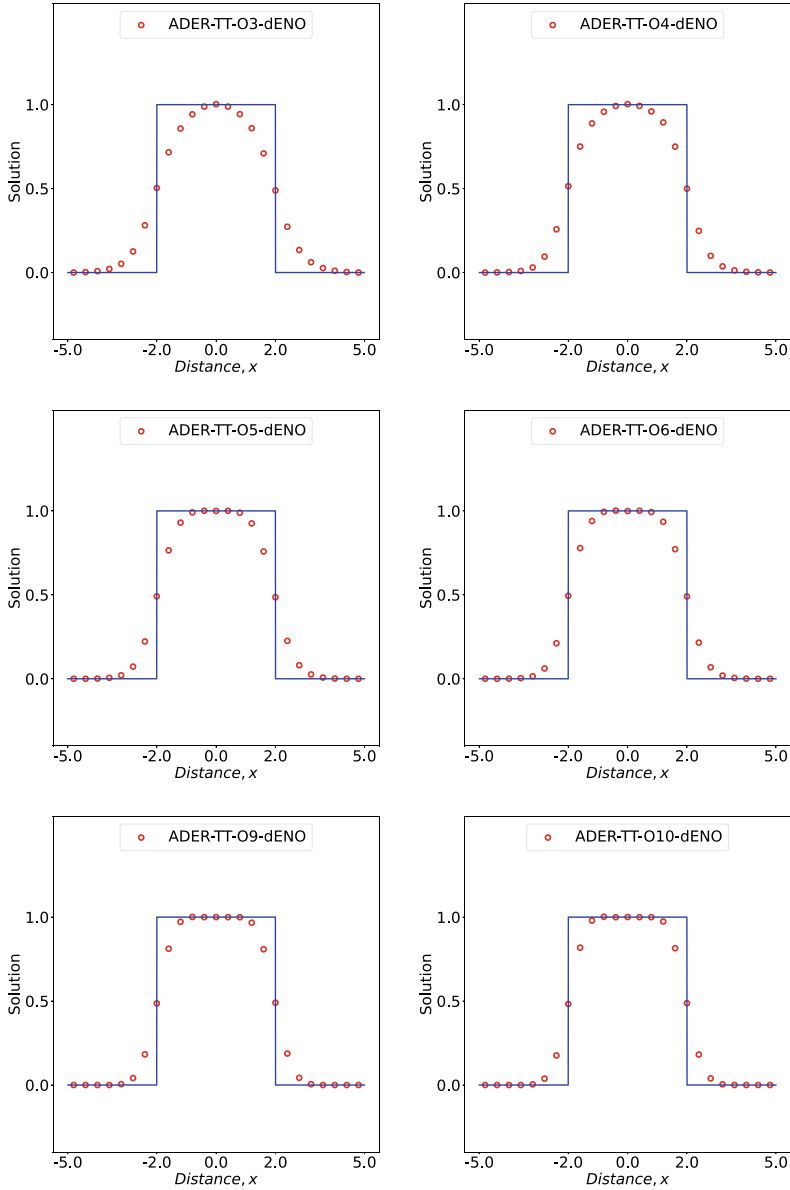


Fig. 16.16 Comparison (for $M = 25$ cells) between the numerical (symbols) and the exact solutions (line) for the square wave initial condition (16.98), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with dENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

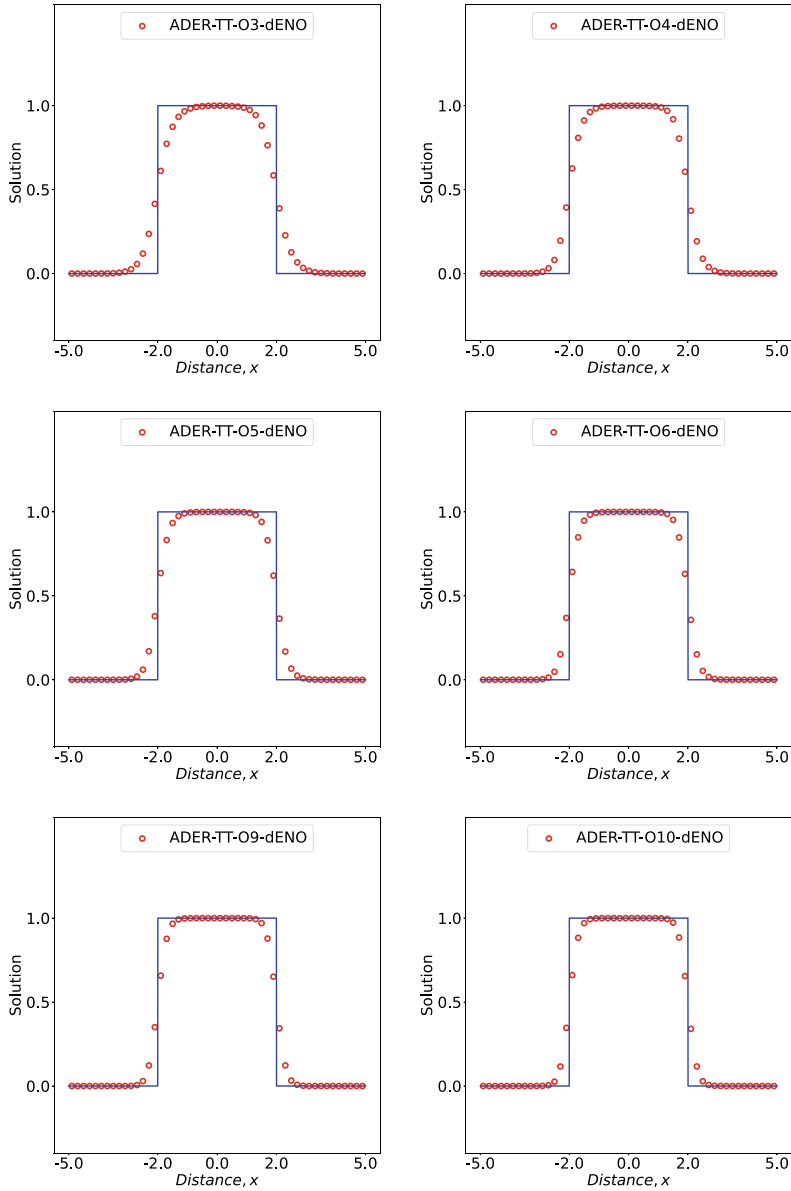


Fig. 16.17 Comparison (for $M = 50$ cells) between the numerical (symbols) and the exact solutions (line) for the square wave initial condition (16.98), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with dENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

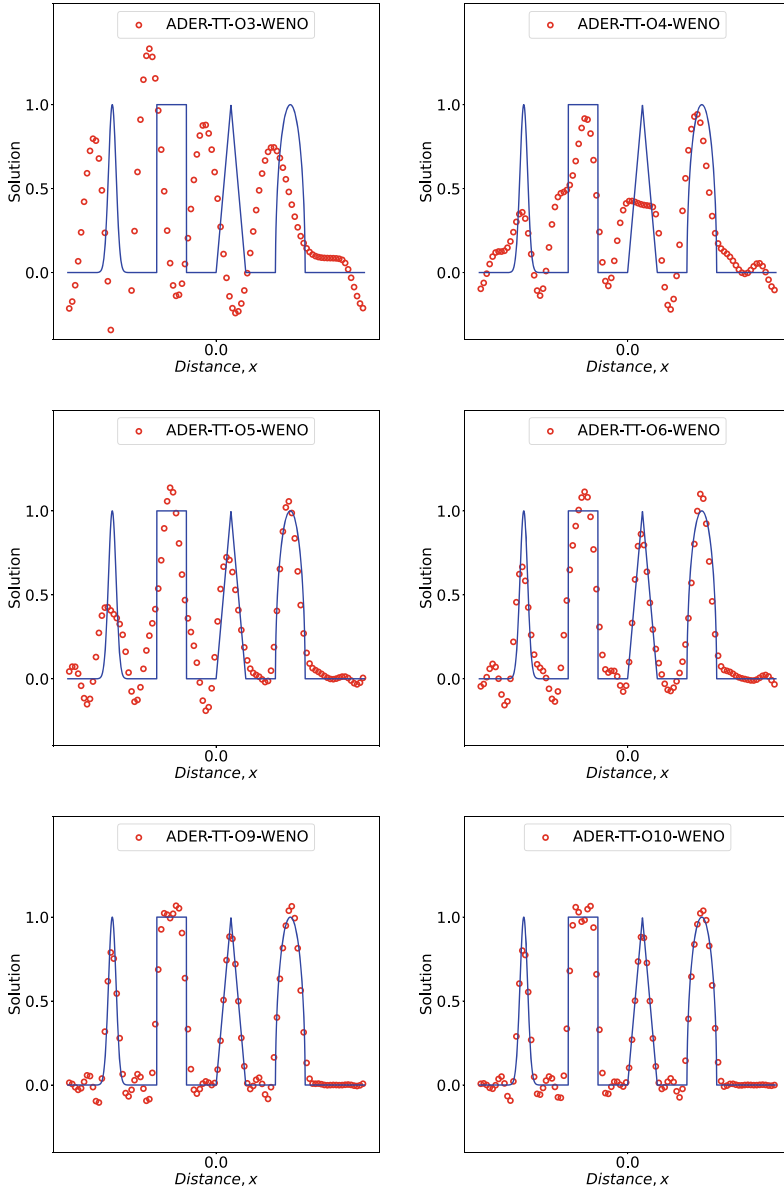


Fig. 16.18 Comparison (for $M = 100$ cells) between the numerical (symbols) and the exact solutions (line) for the multi-wave initial condition (16.99), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

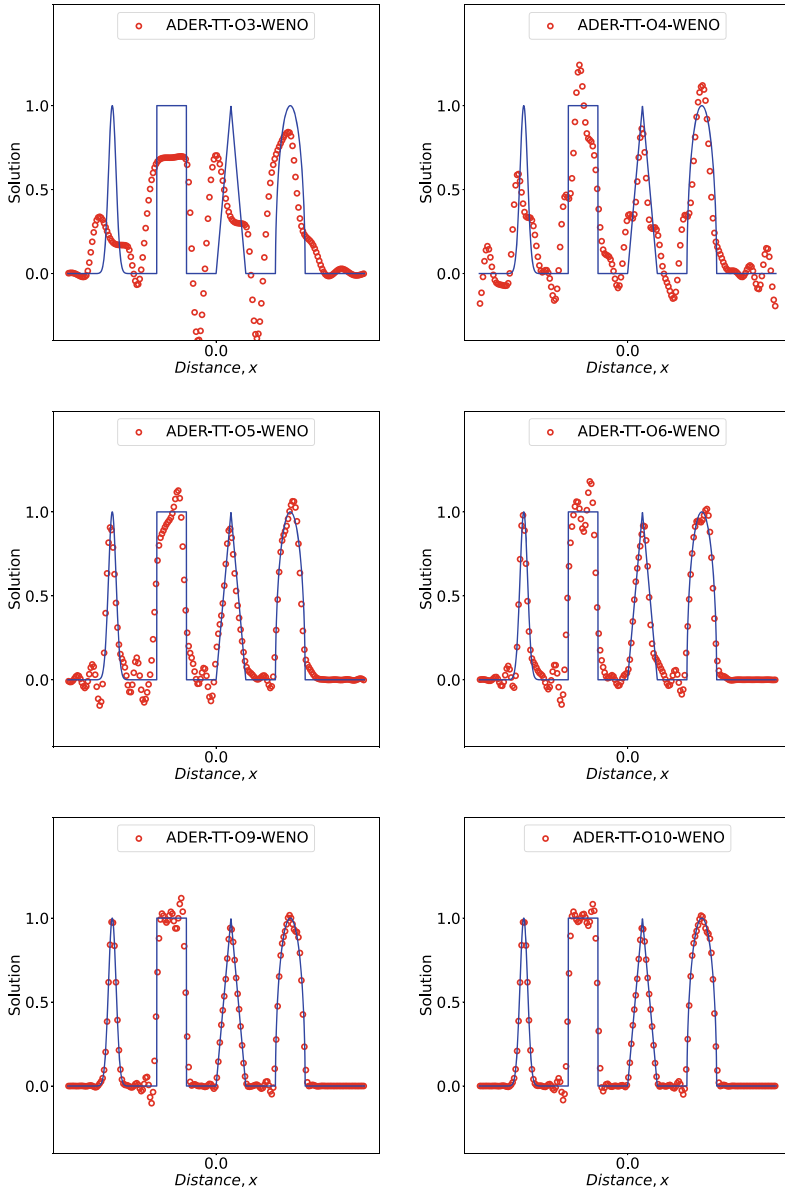


Fig. 16.19 Comparison (for $M = 200$ cells) between the numerical (symbols) and the exact solutions (line) for the multi-wave initial condition (16.99), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

somehow better than those from WENO, while for the higher-order range the WENO and WENO-DK results tend to be similar, with WENO slightly better than WENO-DK.

Results from ENO (dENO) are shown in Figs. 16.22 and 16.23 for meshes $M = 100$ and $M = 200$, respectively. As compared with the previous sets of results from WENO and WENO-DK, the ENO schemes perform better as regards over and undershoots, completely absent in the ENO results, for all orders of accuracy. Still, errors for the lower-order cases, such as 3rd and 4th, seem to be larger than for WENO and WENO-DK. As expected, the ENO results improve as the mesh is refined, but only the methods of accuracy above 5th give satisfactory results.

Summary for multi-wave profiles. Test 3, with the multi-wave profiles, combines both smooth and discontinuous waves. Moreover, all four waves are close to each other, posing a challenge to all spatial reconstruction methods that examine the data left and right of each cell. This test emulates more closely, realistic situations found in computational practice. Surprisingly, for this test problem ENO appears to give better results, in the high-order range, than WENO and WENO-DK. See [47] for analogous observations.

Overall summary. Non-linear ADER schemes from 3rd to 10th orders of accuracy in both space and time have been presented for three test problems defined by wave profiles embodying different features and numerical challenges. Three variations of the schemes result from the application of three classical spatial reconstruction methods: WENO, WENO-DK and dENO (direct ENO).

The results presented confirm those available in the literature for the lower-order range. For the higher-order range, say from 6th to 10th, there are some surprises. In particular, WENO-type reconstruction methods, as implemented in the ADER approach, are highly oscillatory for problems containing discontinuities. The classical ENO method tends to perform much better than the WENO-type schemes for discontinuous solutions and for solutions containing both smooth parts and discontinuities. Results, generally will depend on various factors, computational parameters and evolution time, for example. Informal discussions with collaborators Dr Gino Montecinos (UFRO, Chile) and Dr Riccardo Dematté (University of Cambridge, UK) suggest that the oscillatory character of the WENO-type reconstructions for **very high order of accuracy** depends quite crucially on the choice of computational parameters.

Very short times, with output profiles close to the prescribed initial conditions will not reveal some of the numerical difficulties under discussion. Very long evolution times will test the accuracy of the methods, usually requiring very high accuracy in order to capture main wave features. Time accuracy is another issue. Semi-discrete methods will couple high-order spatial reconstruction polynomials with ODE solvers of at most third-order of accuracy, in most publications. This combination will favour the smoothing of spurious oscillations. The results shown in this chapter refer to fully discrete schemes in which the time accuracy is matched to the spatial accuracy, without resorting to ad-hoc adjustments of the time step. Examples shown include

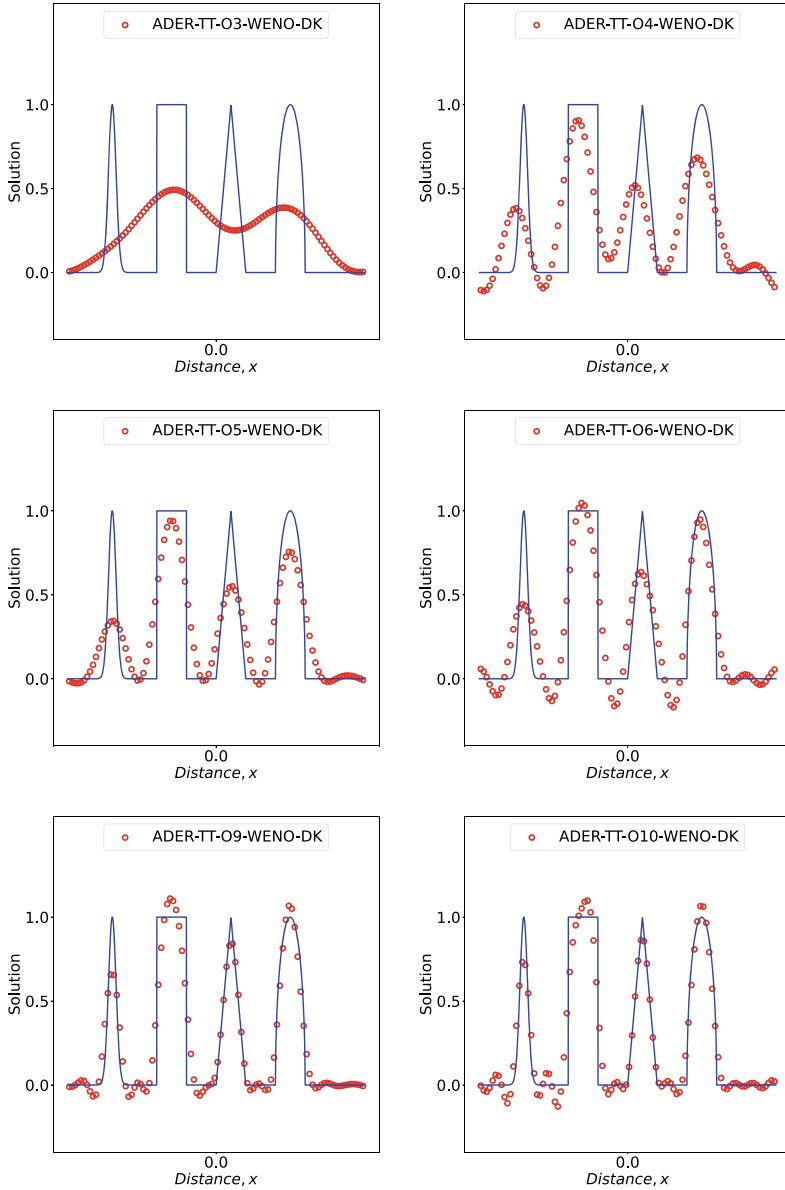


Fig. 16.20 Comparison (for $M = 100$ cells) between the numerical (symbols) and the exact solutions (line) for the multi-wave initial condition (16.99), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO-DK spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

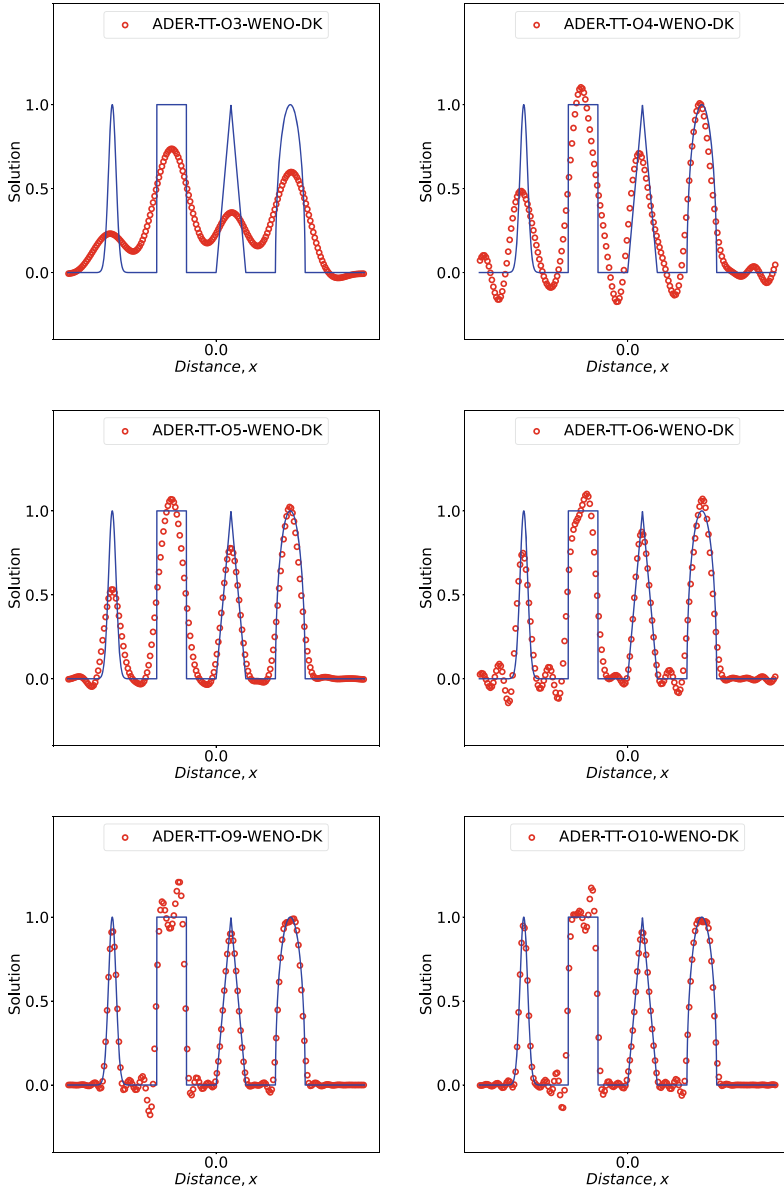


Fig. 16.21 Comparison (for $M = 200$ cells) between the numerical (symbols) and the exact solutions (line) for the multi-wave initial condition (16.99), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with WENO-DK spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

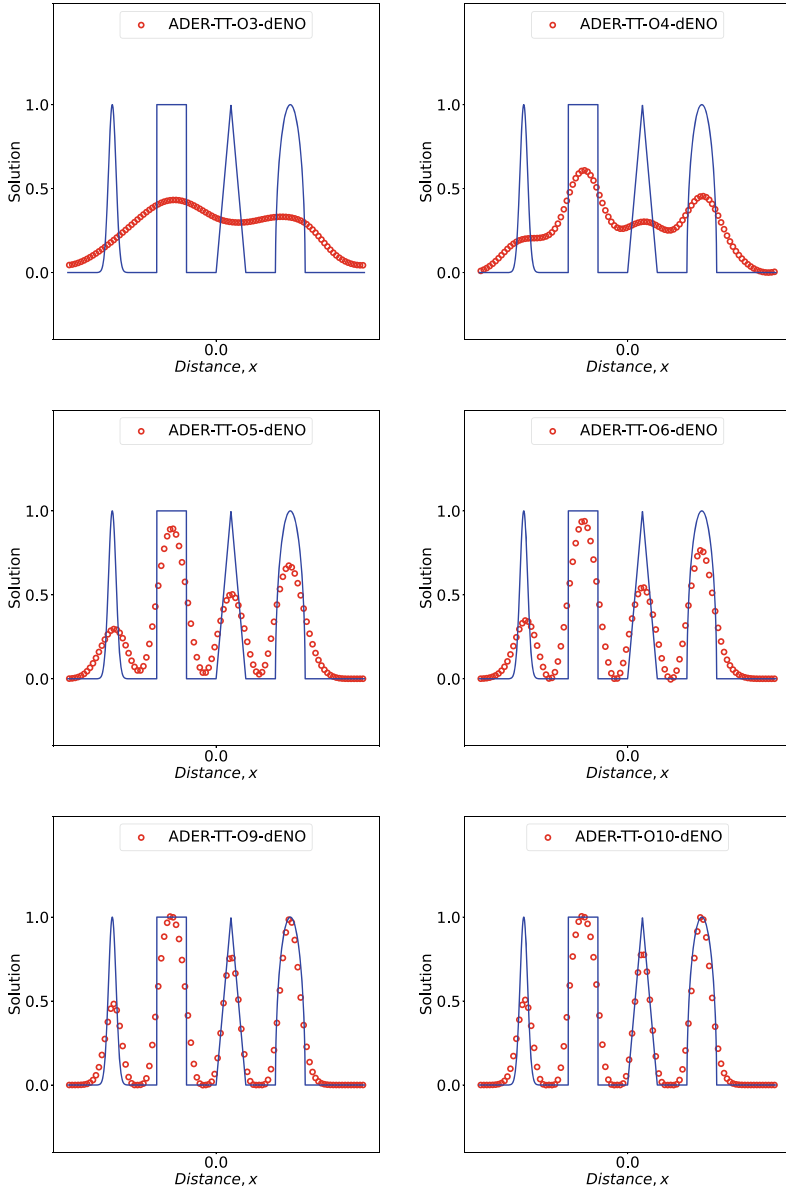


Fig. 16.22 Comparison (for $M = 100$ cells) between the numerical (symbols) and the exact solutions (line) for the multi-wave initial condition (16.99), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with dENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

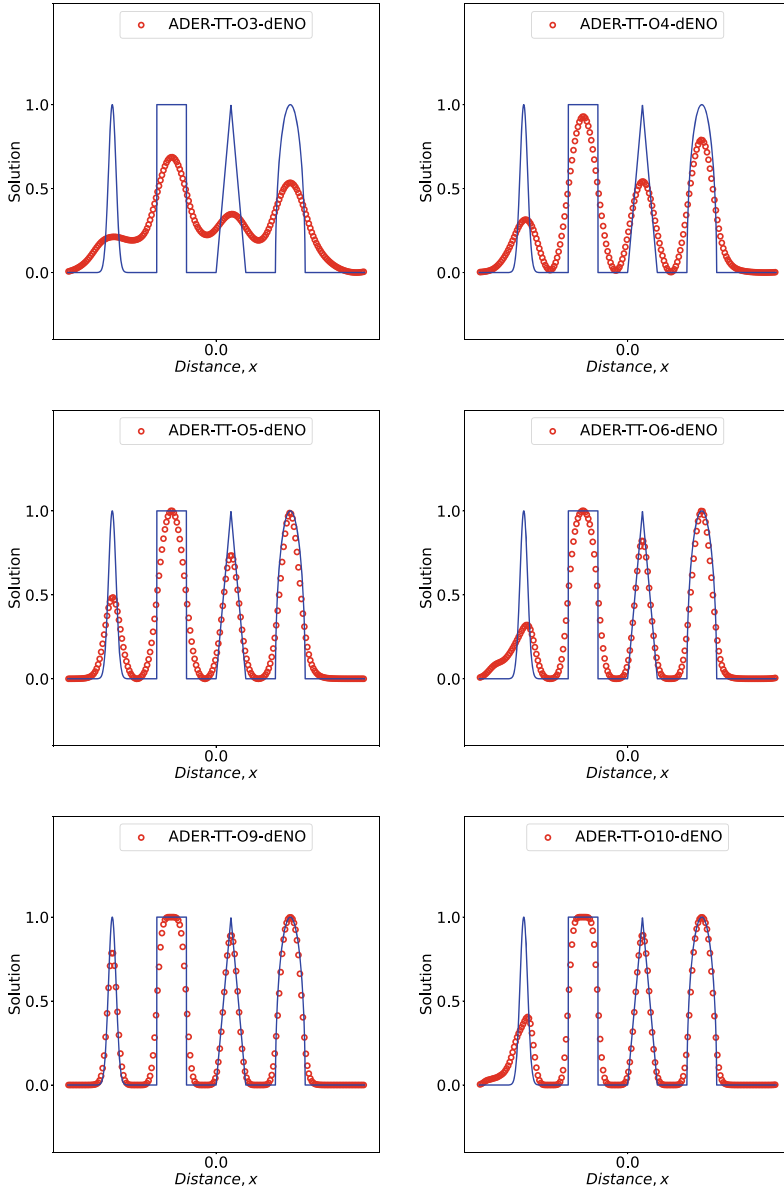


Fig. 16.23 Comparison (for $M = 200$ cells) between the numerical (symbols) and the exact solutions (line) for the multi-wave initial condition (16.99), at the output time $T_{out} = 100$ and CFL coefficient $C_{cfl} = 0.9$. Methods used: ADER with the TT solver for the GRP_m , along with dENO spatial reconstruction. Orders of accuracy in space and time: 3, 4 (top row); 5, 6 (middle row) and 9, 10 (bottom row)

schemes of 10th order of accuracy in both space and time. It seems as if it is this combination of very high space and time accuracy that reveals new issues regarding the approximation of hyperbolic equations.

16.5.4 Efficiency: Error Against CPU Cost

Efficiency is a unique feature of very high-order methods, such as the ones presented here. That is, given a problem with a reference solution and a **prescribed small error** E to be attained, the question that we pose is this: are low-order methods on fine meshes cheaper than high-order methods on coarse meshes for attaining the same error E ? The answer is no. In fact, methods of very high accuracy, such as the ones presented here, on coarse meshes are orders-of-magnitude cheaper than lower-order methods on fine meshes for attaining a prescribed, small error E . Here we illustrate this fact through Test problem 1.

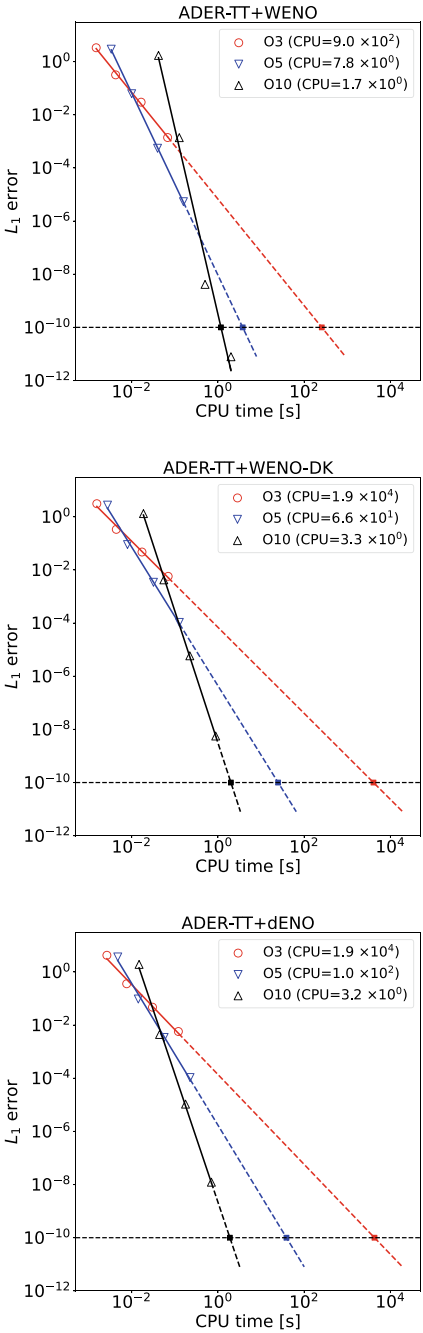
Figure 16.24 shows efficiency plots in $\log/\log$ scale (L_1 error against CPU time) for ADER with the TT solver for the generalised Riemann problem GRP_m for orders 3rd, 5th and 10th, along with WENO, WENO-DK and dENO reconstructions, applied to IBVP (16.95), with initial condition (16.96), at $T_{out} = 0.5$ and Courant number coefficient $C_{cfl} = 0.9$.

For each order of accuracy, each of the methods is executed for a sequence of meshes. The calculated errors for each mesh and each order are shown by symbols in the figure. The procedure produces, for each order, a sequence of points (CPU_k, E_k) . A corresponding straight line in the $\log/\log$ plane is obtained by a least-squares fit to the points (CPU_k, E_k) . Therefore, to each of the three chosen orders of accuracy for the exercise (3rd, 5th, 10th) there corresponds a straight line. To assess the efficiency of the methods we show the intersections of these lines with the horizontal line for the fixed, **prescribed error** $E = 10^{-10}$ in the L_1 norm. We look for a single number for the CPU cost, for each method.

On the top right-hand corner of each frame we show the results. For WENO (top frame) we see CPU times for orders 3rd, 5th and 10th. We see that going from 3rd order to 10th order there is a gain in CPU time of about three orders of magnitude. In other words, the 10th order method on a coarse mesh is much more efficient than the 3rd order method on a fine mesh for attaining the fixed error $E = 10^{-10}$. High-order methods are much more efficient than low-order methods to compute accurate solutions (small errors). Similar gains are seen from the other reconstruction methods WENO-DK (middle frame) and ENO (bottom frame). In all three cases the computational cost decreases with increasing order of accuracy. That is, for small errors, higher-order methods are more efficient than lower order methods, by orders of magnitude.

New reconstruction method: AENO. Fig. 16.25 displays results obtained from a new spatial non-linear reconstruction method called AENO-C. The AENO (Averaged ENO) approach, first put forward in [47], is based on averaging the ENO polynomial

Fig. 16.24 Efficiency plots (error against CPU time) for ADER with TT solver for GRP_m for orders 3, 5 and 10, with WENO, WENO-DK and dENO reconstructions applied to IBVP (16.95), with initial condition (16.96) at $T_{out} = 0.5$, with $C_{cfl} = 0.9$



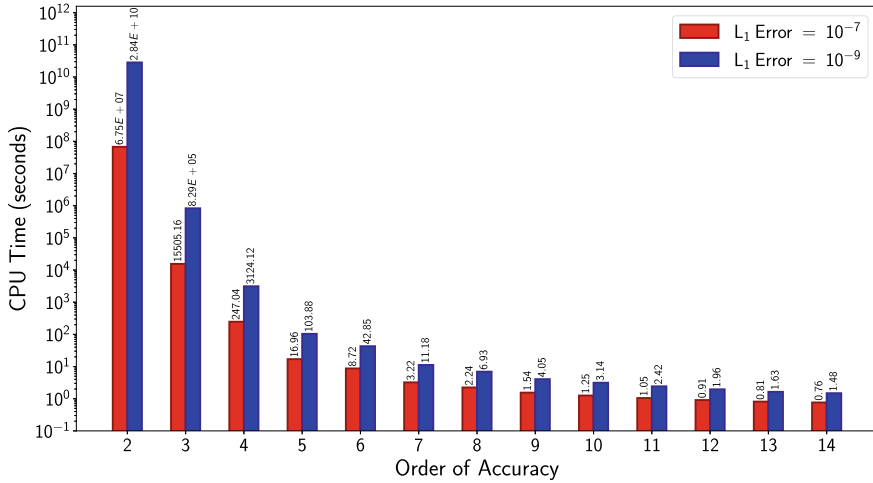


Fig. 16.25 Efficiency plot: error versus order of accuracy in space and time for two prescribed errors in L_1 -norm: $E = 10^{-7}$ and $E = 10^{-9}$. Equation solved is the linear advection equation with $\lambda = 1$, domain is $x \in [-1, 1]$, initial condition is $q(x, 0) = \sin^4(\pi x)$, CFL coefficient is $C_{cfl} = 0.9$, output time is $T_{out} = 100$. Method used is ADER with AENO-C spatial reconstruction [51]

with a neighbouring polynomial. The issues are: which neighbour and what averaging. Further developments are reported in [48–50]. Current developments on the AENO approach look promising for overcoming the difficulties seen with current reconstruction methods in the higher-order range, namely orders above 5. The results shown in Fig. 16.25 were obtained from the version AENO-C, see [51] for details on the scheme. The figure shows an efficiency plot, that is error versus order of accuracy in space and time, for two prescribed errors, for ADER schemes from order 2 to 14. The main message of the figure is that high-order methods (in both space and time) are much more efficient than low order methods to attain a prescribed (small) error, as has already been pointed out.

16.6 Summary and Further Reading

In this chapter we have presented two classes of high-order methods as applied to the linear advection-reaction equation, namely fully discrete ADER methods and semi-discrete methods with TVD Runge-Kutta time stepping. Both classes of schemes make use of ENO or WENO spatial reconstructions. The methods have been assessed on four test problems. Tests 1 and 2 are for smooth solutions and serve to assess the empirical convergence rates and verify that the schemes attain their theoretically expected orders of accuracy.

Test 3 has initial conditions consisting of an array of composite profiles including both smooth regions and discontinuities. Test 4 compares all methods in terms of efficiency displayed in a plot of error against CPU time. The contents of this chapter have been designed as an introduction to high-order numerical methods for solving hyperbolic balance laws, with the intention of introducing, in the simple setting of a scalar equation, important concepts such as order of accuracy, convergence rates, semi-discrete schemes, fully discrete schemes and efficiency.

In Chap. 17, most of these methods will be presented in the more realistic setting of non-linear systems of hyperbolic balance laws, with emphasis on equations that govern the dynamics of biological flows, such as arterial blood, venous blood and cerebrospinal fluid.

As further reading material we suggest: Toro, Millington and Nejad [1]; Toro and Titarev [2]; Titarev and Toro [3]; Jiang and Shu [18]; Dumbser and Käser [19] and Toro [21], Chaps. 19 and 20.

Suggested exercises are found at the end of the chapter.

16.7 Exercises

16.1 Show that for the homogeneous linear advection Eq. (16.6), the Cauchy-Kovalevskaya procedure gives the time derivative of order k as

$$\partial_t^{(k)} q = (-1)^k \lambda^k \partial_x^{(k)} q . \quad (16.100)$$

16.2 Show that for the inhomogeneous linear advection-reaction Eq. (16.6), the Cauchy-Kovalevskaya procedure gives the time derivative of order k as

$$\partial_t^{(k)} q(x, t) = \sum_{l=0}^k \left[\frac{k!}{(k-l)! l!} \beta^l (-\lambda)^{k-l} \right] \partial_x^{(k-l)} q(x, t) . \quad (16.101)$$

16.3 For the homogeneous linear advection Eq. (16.6) show that the spatial derivative $\partial_x^{(l)} q(x, t)$ obeys the same evolution equation as $q(x, t)$, that is

$$\partial_t (\partial_x^{(l)} q(x, t)) + \lambda \partial_x (\partial_x^{(l)} q(x, t)) = 0 , \text{ for } l = 1, 2, \dots, m . \quad (16.102)$$

16.4 For the inhomogeneous linear advection Eq. (16.6) derive an evolution equation for the spatial derivative $\partial_x^{(l)} q(x, t)$. Compare your result with that for the homogeneous case.

16.5 The Kolgan scheme (1972). Kolgan [52] is recognised for having been the first to propose non-linear schemes. Kolgan's scheme may be reproduced from the ADER scheme in *second-order* mode by relying entirely on Step B1 and neglecting Step B2 to compute the flux. That is, one takes the Riemann problem solution (16.25), the data of which are the boundary extrapolated values of the reconstruction. In other words, only the leading term of the expansion (16.22) is used. Perform the following tasks:

1. Verify that for the homogeneous linear advection equation the numerical flux for the Kolgan scheme is

$$f_{i+\frac{1}{2}} = \begin{cases} \lambda(q_i^n + \frac{1}{2}\Delta x \Delta_i) & \text{if } \lambda > 0, \\ \lambda(q_{i+1}^n - \frac{1}{2}\Delta x \Delta_{i+1}) & \text{if } \lambda < 0. \end{cases} \quad (16.103)$$

2. For $\lambda > 0$ consider the *linear* version of Kolgan's scheme by taking linear (fixed) slopes

$$\Delta_i = \Delta_{i-\frac{1}{2}} = \frac{q_i^n - q_{i-1}^n}{\Delta x} \quad \forall i$$

and write the resulting three-point scheme as

$$q_i^{n+1} = b_{-1}q_{i-1}^n + b_0q_i^n + b_1q_{i+1}^n, \quad (16.104)$$

explicitly identifying the coefficients b_{-1} , b_0 and b_1 .

3. Derive the local truncation error and state the order of accuracy of the scheme.
4. Analyse the stability of the scheme by using the (a) the von Neumann method, (b) Billett's method and (c) the viscosity form of the scheme. See Chap. 12.
5. Repeat the above with slopes chosen as

$$\Delta_i = \Delta_{i+\frac{1}{2}} = \frac{q_{i+1}^n - q_i^n}{\Delta x} \quad \forall i. \quad (16.105)$$

Closing remark on the Kolgan scheme. As seen above the linear version of Kolgan's scheme is **unconditionally unstable**. However the non-linear version proposed originally by Kolgan uses a non-linear reconstruction, whereby the choice of the slopes is non-linear. In particular Kolgan proposed the use of *minimum slopes*, which is equivalent to choosing the MINMOD slope limiter from TVD methods. See [52, 53] and Chap. 13 of [21].

16.6 For semi-discrete schemes, show that if the spatial operator $L(q^n)$ is such that the Euler first-order step

$$q^{n+1} = q^n + \Delta t L(q^n) \quad (16.106)$$

is TVD, then the third-order three-stage scheme (16.73) is TVD.

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Chapter 17

High-Order Methods for Non-Linear Systems



Abstract This chapter presents high-order, non-linear numerical methods for solving non-linear systems of hyperbolic equations with source terms, written in both conservative and non-conservative forms. Two major classes of methods are discussed: fully-discrete and semi-discrete methods. From the fully-discrete class, we study ADER schemes, which are based on non-linear ENO/WENO spatial reconstructions using polynomials of degree m . Their space-time coupling is achieved through solving the generalised Riemann problem GRP_m for the numerical flux and a similar procedure for the integral of the source term. The resulting schemes achieve accuracy of $m + 1$ in both space and time, where m is arbitrary. Semi-discrete methods discretise the PDEs separately in space and time. These methods use ENO/WENO spatial reconstructions and, typically, TVD Runge-Kutta time integrators for the associated ordinary differential equations in time. As examples, we provide detailed descriptions of various fully-discrete and semi-discrete methods of second-order accuracy for both conservative and non-conservative systems. We also examine fully-discrete ADER methods of arbitrary accuracy in both space and time for both conservative and non-conservative systems. Additionally, semi-discrete schemes of second- and third-order accuracy are presented for conservative systems of balance laws. Computational results for selected test problems are provided for four of the methods studied. First, a convergence rate study is conducted to verify that the theoretically expected orders of accuracy are attained. Results for schemes of second to fifth order are presented. The same schemes are also evaluated through Riemann problems with exact solutions, involving rarefactions, shocks, and contact discontinuities. The chapter concludes with a summary, conclusions, and suggestions for further reading.

17.1 Second-Order ADER with TT Solver for GRP_1

In this section we study a fully-discrete finite volume ADER scheme [1–6] in second-order mode, in both space and time, based on first-degree non-linear spatial reconstruction polynomials and the Toro-Titarev (TT) solver for the generalised Riemann problem GRP_1 [3, 4]. The resulting scheme is denoted as ADER2-TT.

17.1.1 The Finite Volume Framework

Consider the initial-boundary value problem (IBVP) for a generic system of non-linear hyperbolic balance laws written in conservation-law form

$$\left. \begin{aligned} \text{PDEs: } & \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}) , \quad x \in [a, b] , \quad t > 0 , \\ \text{ICs: } & \mathbf{Q}(x, 0) = \mathbf{Q}^{(0)}(x) , \quad x \in [a, b] , \\ \text{BCs: } & \mathbf{Q}(a, t) = \mathbf{B}_L(t) , \quad \mathbf{Q}(b, t) = \mathbf{B}_R(t) , \quad t \geq 0 . \end{aligned} \right\} \quad (17.1)$$

$\mathbf{Q}(x, t)$ is the vector of conserved variables; $\mathbf{F}(\mathbf{Q})$ is the flux vector, or **physical flux** vector; $\mathbf{S}(\mathbf{Q})$ is the **source term** vector; $\mathbf{Q}^{(0)}(x)$ is the initial condition; $\mathbf{B}_L(t)$ and $\mathbf{B}_R(t)$ are the boundary conditions on the left and right boundaries, respectively, two prescribed functions of time. Assuming that the IBVP (17.1) has a solution, the task at hand is to construct numerical methods to compute the solution approximately.

We first recall the finite volume framework for a general system of one-dimensional hyperbolic balance laws written in conservation-law form as in (17.1), assuming the space-time domain to be discretised by finite volumes of the form

$$V_i^n = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \times [t_n, t_{n+1}] . \quad (17.2)$$

Then, exact integration of the PDEs in (17.1) in the control volume V_i^n gives the exact formula

$$\mathbf{Q}_i^{n+1} = \mathbf{Q}_i^n - \frac{\Delta t}{\Delta x} (\mathbf{F}_{i+\frac{1}{2}} - \mathbf{F}_{i-\frac{1}{2}}) + \Delta t \mathbf{S}_i , \quad (17.3)$$

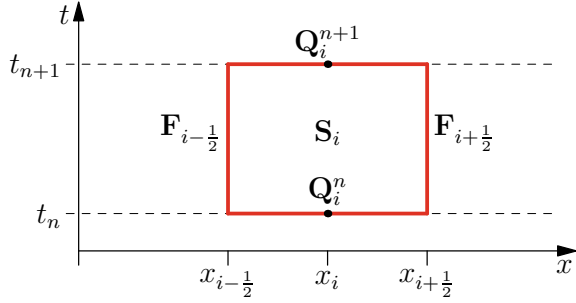
with definitions

$$\left. \begin{aligned} \mathbf{Q}_i^n &= \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{Q}(x, t_n) dx , \\ \mathbf{F}_{i+\frac{1}{2}} &= \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}, t)) dt , \\ \mathbf{S}_i &= \frac{1}{\Delta t \Delta x} \int_{t_n}^{t_{n+1}} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{S}(\mathbf{Q}(x, t)) dx dt . \end{aligned} \right\} \quad (17.4)$$

Here $\Delta x = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}$ and $\Delta t = t_{n+1} - t_n$ are the mesh spacing and the time step, respectively. The domain $[a, b]$ is assumed to be discretised into M cells $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$.

Finite volume numerical methods to solve (17.1) are constructed from formula (17.3) by suitably approximating the integrals in (17.4) to obtain the **numerical flux** $\mathbf{F}_{i+\frac{1}{2}}$ and the **numerical source** $\mathbf{S}_i$ in (17.3). Figure 17.1 illustrates the finite volume update formula (17.3) for a system of balance laws as in (17.1). See Chap. 13 for details on the finite volume method and Chap. 16 for high-order finite volume methods for scalar equations.

Fig. 17.1 Illustration of the finite volume update formula (17.3) for a system of balance laws expressed in conservation-law form, such as (17.1)



Remark: initial conditions. The first integral in (17.4) gives the initial conditions $\mathbf{Q}_i^0$ in each computational cell $I_i \in [a, b]$ at the start of the computations, that is at $t = t_n = 0$. This integral average must be computed numerically to the required order of accuracy, in order to correctly provide the initial condition $\mathbf{Q}(x, 0) = \mathbf{Q}^{(0)}(x)$ in (17.1). For $n > 0$, the integral averages are computed automatically from the scheme (17.3), which in effect is a formula for updating integral averages. The correct evaluation of the initial conditions is mandatory for high-order schemes. This is most obvious when carrying out convergence rate studies.

17.1.2 Ingredients of the Numerical Flux

The determination of the numerical flux first requires a non-linear reconstruction procedure, such as ENO or WENO; see Chap. 15 for background. This step provides reconstruction polynomials $\mathbf{P}_i(x)$, based on the set of cell averages $\{\mathbf{Q}_i^n\}_{i=1}^M$. Figure 17.2 illustrates the setting of reconstruction polynomials for a single component q of $\mathbf{Q}$, in three successive cells.

To compute the numerical flux $\mathbf{F}_{i+\frac{1}{2}}$ from the flux integral in (17.4) we need to define the integrand under the integral, for which we solve the **generalised Riemann problem** GRP_m :

$$GRP_m : \left\{ \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}) \text{ , } x \in \mathcal{R} \text{ , } t > 0 \text{ ,} \\ \text{IC: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{P}_i(x) & \text{if } x < x_{i+\frac{1}{2}} \text{ ,} \\ \mathbf{P}_{i+1}(x) & \text{if } x > x_{i+\frac{1}{2}} \text{ .} \end{cases} \end{array} \right\} \quad (17.5)$$

In general, $\mathbf{P}_i(x)$ and $\mathbf{P}_{i+1}(x)$ are assumed to be reconstructed polynomial vectors of degree m in order to obtain an ADER method (17.3) of order $m + 1$. Figure 17.3 depicts the typical wave structure of the solution of a local GRP_m (17.5) at $x_{i+\frac{1}{2}}$ (locally at $x = 0$) for a 3×3 non-linear system, in the x - t plane. For numer-

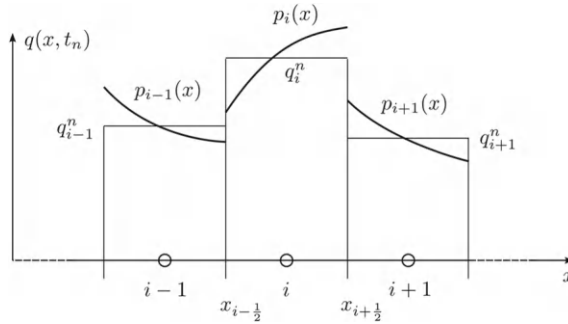


Fig. 17.2 Reconstruction polynomials for single component $p(x)$ of $\mathbf{P}_k(x)$ in cells I_{i-1} , I_i and I_{i+1} . Such polynomial functions constitute the initial conditions for a generalised Riemann problem at the interface between the two neighbouring cells

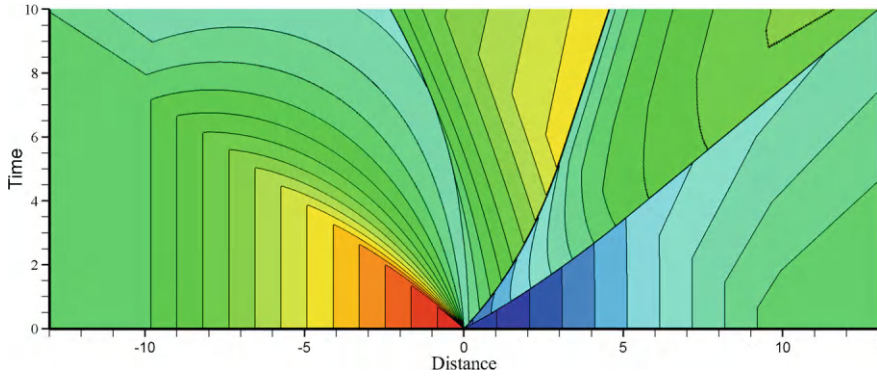


Fig. 17.3 Typical wave structure of the solution of the generalised Riemann problem in the x - t plane. To compute the numerical flux one seeks an approximate solution right at the interface, locally at $x = 0$, along the t -axis (courtesy of Dr. V A Titarev, Russia)

ical purposes we shall partially utilise solutions of this type. As in the first-order Godunov method, see Chap. 13, we only use the solution along the interface, that is along the t -axis.

In this section we study second-order numerical methods, for which the required first-degree polynomials ($m = 1$) $\mathbf{P}_i(x)$ in cell I_i take the form

$$\mathbf{P}_i(x) = \mathbf{Q}_i^n + (x - x_i)\mathbf{\Delta}_i, \quad i = 1, 2, \dots, M, \quad (17.6)$$

where $\mathbf{\Delta}_i$ is the slope of $\mathbf{P}_i(x)$ (a vector) in cell $I_i = [x_{i-1/2}, x_{i+1/2}]$. Here, for simplicity, we have assumed a component-by-component reconstruction procedure. A better choice is reconstruction in terms of characteristic variables. See Chap. 15 for details.

17.1.3 Solution of GRP_1 and Numerical Flux

We now seek a time-dependent approximate solution $\mathbf{J}_{i+\frac{1}{2}}(\tau)$ of the generalised Riemann problem (17.5) for the case $m = 1$, right at the interface, along the t -axis. The sought approximate solution is expressed in terms of the truncated LeFloch-Raviart expansion [7], given as

$$\mathbf{J}_{i+\frac{1}{2}}(\tau) = \mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) + \tau \partial_t \mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) . \quad (17.7)$$

The leading term is defined as the time limit from above

$$\mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) = \lim_{t \rightarrow 0+} \mathbf{Q}(x_{i+\frac{1}{2}}, t) \quad (17.8)$$

and $\mathbf{Q}(x, t)$ denotes the general solution of the PDEs in (17.5). The task at hand is to compute the leading term $\mathbf{Q}(x_{i+\frac{1}{2}}, 0_+)$ and the higher-order term $\tau \partial_t \mathbf{Q}(x_{i+\frac{1}{2}}, 0_+)$ in (17.7).

The Leading Term of the Expansion

To compute the leading term $\mathbf{Q}(x_{i+\frac{1}{2}}, 0_+)$ in (17.7) one solves the conventional Riemann problem (piece-wise constant data and no source term)

$$GRP_0 : \left\{ \begin{array}{l} \text{PDE: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0} , \quad x \in \mathcal{R} , t > 0 , \\ \text{IC: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{P}_i(x_{i+\frac{1}{2}}) & \text{if } x < x_{i+\frac{1}{2}} , \\ \mathbf{P}_{i+1}(x_{i+\frac{1}{2}}) & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{array} \right\} \quad (17.9)$$

Denoting the similarity solution of (17.9), in local coordinates, by $\mathbf{G}_{i+\frac{1}{2}}^{(0)}(x/t)$ the leading term in (17.7) is given by the Godunov state $\mathbf{G}_{i+\frac{1}{2}}^{(0)}(0)$ right at the interface, that is

$$\mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) = \mathbf{G}_{i+\frac{1}{2}}^{(0)}(0) . \quad (17.10)$$

In this step one can use any state solver for the conventional Riemann problem (17.9), exact or approximate. In effect, this step is entirely analogous to the first-order Godunov scheme; see Chap. 13 on the Godunov method for systems.

The Higher-Order Term of the Expansion

The computation of the higher-order term $\tau \partial_t \mathbf{Q}(x_{i+\frac{1}{2}}, 0_+)$ in (17.7) involves the following steps:

1. The **Cauchy-Kovalevskaya (CK) procedure** is applied to the governing equations in (17.5), to express the time derivative in terms of the remaining terms, namely

$$\partial_t \mathbf{Q}(x, t) = -\partial_x \mathbf{F}(\mathbf{Q}) + \mathbf{S}(\mathbf{Q}) = -\mathbf{A}(\mathbf{Q})\partial_x \mathbf{Q} + \mathbf{S}(\mathbf{Q}) , \quad (17.11)$$

where the chain rule has been used to obtain the Jacobian matrix $\mathbf{A}(\mathbf{Q})$.

2. **Evolution equation for the space derivative.** Now, one unknown in (17.11) is the spatial gradient $\partial_x \mathbf{Q}$, right at the interface. To determine it, we first write an evolution equation for $\partial_x \mathbf{Q}$, namely

$$\partial_t (\partial_x \mathbf{Q}) + \mathbf{A}(\mathbf{Q})\partial_x (\partial_x \mathbf{Q}) = \mathbf{K} . \quad (17.12)$$

3. **Simplifications to the evolution equation.** We next make two simplifications to system (17.12), namely the system is linearised and the source term is neglected. That is

$$\mathbf{A}(\mathbf{Q}) \approx \mathbf{A}_{i+\frac{1}{2}}^{(0)} = \mathbf{A}(\mathbf{G}_{i+\frac{1}{2}}^{(0)}(0)) , \quad \mathbf{K} = \mathbf{0} . \quad (17.13)$$

The above simplifications can be justified rigorously for some simple cases [8], while for the general case numerical evidence suggests that for smooth solutions the sought order of accuracy is attained. See recent review [9] and references therein.

It is remarked that the leading term (17.10) has been used to evaluate the coefficient matrix in the linearised system (17.13); it will also be used to evaluate the source term contribution to the flux in (17.11).

4. **Linearised Riemann problem for spatial derivatives.** The resulting linearised conventional Riemann problem from (17.12) for the spatial derivative reads

$$\left. \begin{array}{l} \text{PDEs: } \partial_t (\partial_x \mathbf{Q}) + \mathbf{A}_{i+\frac{1}{2}}^{(0)} \partial_x (\partial_x \mathbf{Q}) = \mathbf{0} , \\ \text{IC: } \partial_x \mathbf{Q}(x, 0) = \left\{ \begin{array}{ll} \Delta_i \equiv \frac{d}{dx} \mathbf{P}_i(x) & \text{if } x < x_{i+\frac{1}{2}} , \\ \Delta_{i+1} \equiv \frac{d}{dx} \mathbf{P}_{i+1}(x) & \text{if } x > x_{i+\frac{1}{2}} . \end{array} \right. \end{array} \right\} \quad (17.14)$$

We solve (17.14) in local coordinates to obtain the solution $\mathbf{G}_{i+\frac{1}{2}}^{(1)}(x/t)$. To find the Godunov state right at the interface, this solution is sampled, giving $\mathbf{G}_{i+\frac{1}{2}}^{(1)}(0)$ and therefore

$$\partial_x \mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) = \mathbf{G}_{i+\frac{1}{2}}^{(1)}(0) . \quad (17.15)$$

5. **The complete solution and the flux.** By collecting results from (17.7), (17.10) and (17.15), the complete sought solution becomes

$$\mathbf{J}_{i+\frac{1}{2}}(\tau) = \mathbf{G}_{i+\frac{1}{2}}^{(0)}(0) + \tau [-\mathbf{A}_{i+\frac{1}{2}}^{(0)} \mathbf{G}_{i+\frac{1}{2}}^{(1)}(0) + \mathbf{S}(\mathbf{G}_{i+\frac{1}{2}}^{(0)}(0))] . \quad (17.16)$$

From the flux integral in (17.4) the numerical flux can be determined by inserting the approximate solution $\mathbf{J}_{i+\frac{1}{2}}(\tau)$ under the integral, namely

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_0^{\Delta t} \mathbf{F}(\mathbf{J}_{i+\frac{1}{2}}(\tau)) d\tau . \quad (17.17)$$

Integration in (17.17) gives the numerical flux as

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F} \left(\mathbf{G}_{i+\frac{1}{2}}^{(0)}(0) + \frac{1}{2} \Delta t [-\mathbf{A}_{i+\frac{1}{2}}^{(0)} \mathbf{G}_{i+\frac{1}{2}}^{(1)}(0) + \mathbf{S}(\mathbf{G}_{i+\frac{1}{2}}^{(0)}(0))] \right) , \quad (17.18)$$

where the leading term $\mathbf{G}_{i+\frac{1}{2}}^{(0)}(0)$ has been used to compute the source term contribution to the numerical flux.

Note that for higher-order ADER methods ($m \geq 2$), one generally performs numerical integration in (17.17) to the required accuracy, in order to determine the numerical flux.

17.1.4 The Numerical Source

To compute the numerical source $\mathbf{S}_i$ in (17.3) from the volume integral in (17.4) we need to know the integrand, that is the function $\mathbf{Q}(x, t)$ in the space-time control volume V_i^n in (17.2). All we know in V_i^n is the initial condition at the current time $t = t_n$, locally at $t = 0$. As a matter of fact the initial condition is represented by the reconstruction polynomial $\mathbf{P}_i(x)$ at time $t = t_n$. Given the second-order accuracy of the scheme, a mid-point rule evaluation of the volume integral in (17.4) gives the sought second-order approximation.

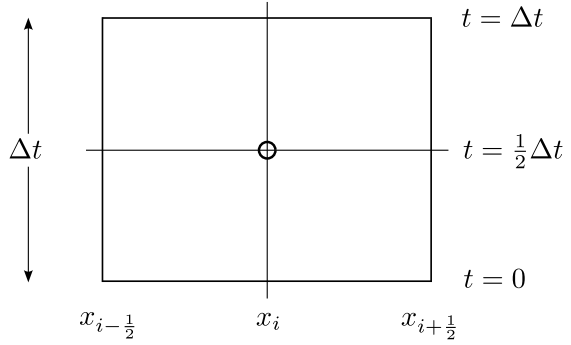
To achieve this we need an approximate solution at the mid point $(x_i, \frac{1}{2} \Delta t)$, as depicted in Fig. 17.4. For any smooth function $\mathbf{Q}(x, t)$, at local time zero, with $x = x_k \in I_i$, we may write the truncated time Taylor series expansion

$$\mathbf{Q}(x_k, \tau) = \mathbf{Q}(x_k, 0) + \tau \partial_t \mathbf{Q}(x_k, 0) . \quad (17.19)$$

In particular, at $x = x_i$, after applying the Cauchy-Kovalevskaya procedure we obtain

$$\mathbf{Q}(x_i, \frac{1}{2} \Delta t) = \mathbf{Q}_i^n + \frac{1}{2} \Delta t \partial_t \mathbf{Q}(x_i, \frac{1}{2} \Delta t) = \mathbf{Q}_i^n + \frac{1}{2} \Delta t [-\mathbf{A}(\mathbf{Q}_i^n) \Delta_i + \mathbf{S}(\mathbf{Q}_i^n)] , \quad (17.20)$$

Fig. 17.4 The numerical source is obtained by approximating the volume integral in (17.4) in the control volume V_i^n . To obtain second-order accuracy, it is enough to find an approximation to the integrand at the mid point $(x_i, \Delta t/2)$



where Δ_i is as in (17.6) and (17.14). From the space-time integral (17.4), approximated by the mid-point rule in space and time, we obtain the numerical source as

$$\mathbf{S}_i = \mathbf{S} \left(\mathbf{Q}_i^n + \frac{1}{2} \Delta t [-\mathbf{A}(\mathbf{Q}_i^n) \Delta_i + \mathbf{S}(\mathbf{Q}_i^n)] \right). \quad (17.21)$$

The complete second-order ADER2-TT scheme is given by the update formula (17.3) with numerical flux (17.18) and numerical source (17.21). See Fig. 17.1.

The numerical flux for the ADER2-TT scheme just described, in addition to the non-linear spatial reconstruction, requires the solution of one non-linear conventional Riemann problem for the leading term of the GRP_1 solution at the interface, and the solution of one linear Riemann problem for the gradients to find the higher order term.

For the homogeneous case (no source term), this scheme is a **modified version** of the second-order GRP scheme of Ben-Artzi and Falcovitz [10] to solve hyperbolic systems.

The ADER2-TT, as presented, is a two-fold extension of the Ben-Artzi and Falcovitz scheme:

1. ADER2-TT applies to hyperbolic systems with source terms, computed to second-order accuracy in space and time, and
2. ADER2-TT extends to **arbitrary** order of accuracy $m + 1$ in space and time for hyperbolic systems with source terms, with reconstruction polynomials of degree m , as we shall see in Sect. 17.4.

In the next section we construct a second-order ADER scheme resulting from a different approximation to the solution of the generalised Riemann problem GRP_1 .

17.2 Second-Order ADER with HEOC Solver for GRP_1

Here we present a second-order accurate ADER scheme ($m = 1$) based on a solver for GRP_m derived from a reinterpretation of the high-order method of Harten, Engquist, Osher and Chakravarthy (HEOC) [11]. Details of this reinterpretation are found in [12]. The resulting scheme is denoted as ADER2-HEOC. The general case ($m \geq 2$) is dealt with in Sect. 17.4.

17.2.1 HEOC Solver for GRP_1 and Numerical Flux

To compute the numerical flux $\mathbf{F}_{i+\frac{1}{2}}$ in (17.3) from the flux integral in (17.4) we solve the generalised Riemann problem (17.5) to find the approximate solution $\mathbf{J}_{i+\frac{1}{2}}(\tau)$, where $\mathbf{P}_i(x)$ and $\mathbf{P}_{i+1}(x)$ are reconstructed polynomial vectors of degree $m = 1$ in cells I_i and I_{i+1} , respectively; see (17.6). The HEOC solver has three distinct steps to find the numerical flux $\mathbf{F}_{i+\frac{1}{2}}(\tau)$: (i) data-evolution and (ii) evolved-data interaction via a conventional Riemann problem, and (iii) computation of the numerical flux; see Fig. 17.5. The numerical source $\mathbf{S}_i$ is computed as in the previous section; see Eq. (17.21) and Fig. 17.4.

Data Evolution in the HEOC Solver

The GRP_m defined in (17.5) has initial conditions consisting of two polynomials $\mathbf{P}_i$ and $\mathbf{P}_{i+1}$ in cells I_i and I_{i+1} respectively. Here we deal with the case of degree $m = 1$; see Eq. (17.6). At the interface $x_{i+\frac{1}{2}}$, the left and right limiting values of the reconstruction polynomials, often called **boundary extrapolated values**, are defined as

$$\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) = \mathbf{P}_i(x_{i+\frac{1}{2}}) , \quad \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) = \mathbf{P}_{i+1}(x_{i+\frac{1}{2}}) . \quad (17.22)$$

See Fig. 17.5. In the HEOC solver, these boundary values are evolved in time using a time Taylor series expansion of the type (17.7) as follows

$$\left. \begin{aligned} \mathbf{J}_{i+\frac{1}{2}}^-(\tau) &= \mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) + \tau \partial_t \mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) , \\ \mathbf{J}_{i+\frac{1}{2}}^+(\tau) &= \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) + \tau \partial_t \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) . \end{aligned} \right\} \quad (17.23)$$

Figure 17.5 illustrates, via the vertical curved arrows, the time evolution up to time $t = \frac{1}{2} \Delta t$, of the limiting values $\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0)$ from the left cell I_i and $\mathbf{Q}(x_{i+\frac{1}{2}}^+, 0)$ from the right cell I_{i+1} . After using the Cauchy-Kovalevskaya procedure, the time derivatives in (17.23) can be expressed as

$$\left. \begin{aligned} \partial_t \mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) &= -\partial_x \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0)) + \tau \mathbf{S}(\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0)) , \\ \partial_t \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) &= -\partial_x \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}^+, 0)) + \tau \mathbf{S}(\mathbf{Q}(x_{i+\frac{1}{2}}^+, 0)) . \end{aligned} \right\} \quad (17.24)$$

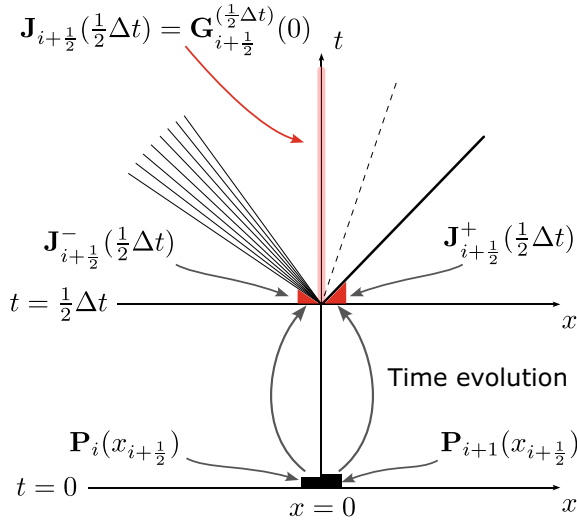


Fig. 17.5 Solution of GRP_1 (17.5) ($m = 1$) by the HEOC method. The left and right limiting polynomial values at the interface are evolved in time as shown by the vertical curved arrows either side of the interface. The time-evolved values are the initial conditions for a conventional Riemann problem, whose solution structure is represented by straight characteristics and straight wave paths. The Godunov state is then the sought solution of GRP_1 at the interface (courtesy of Dr Cristóbal Castro, Chile)

The flux gradients in (17.24), to second-order, are approximated as follows

$$\left. \begin{aligned} \partial_x \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0)) &= \frac{\mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0)) - \mathbf{F}(\mathbf{Q}(x_{i-\frac{1}{2}}^+, 0))}{\Delta x}, \\ \partial_x \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}^+, 0)) &= \frac{\mathbf{F}(\mathbf{Q}(x_{i+\frac{3}{2}}^-, 0)) - \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}^+, 0))}{\Delta x}. \end{aligned} \right\} \quad (17.25)$$

Then, the evolved boundary values left and right of the interface, at time $\tau = \frac{1}{2} \Delta t$, after using (17.23), (17.24) and (17.25) in 17.23), become

$$\left. \begin{aligned} \mathbf{J}_{i+\frac{1}{2}}^-(\tfrac{1}{2} \Delta t) &= \mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) - \frac{1}{2} \frac{\Delta t}{\Delta x} \left[\mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0)) - \mathbf{F}(\mathbf{Q}(x_{i-\frac{1}{2}}^+, 0)) \right] \\ &\quad + \tfrac{1}{2} \Delta t \mathbf{S}(\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0)) \end{aligned} \right\} \quad (17.26)$$

and

$$\left. \begin{aligned} \mathbf{J}_{i+\frac{1}{2}}^+(\tfrac{1}{2} \Delta t) &= \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) - \frac{1}{2} \frac{\Delta t}{\Delta x} \left[\mathbf{F}(\mathbf{Q}(x_{i+\frac{3}{2}}^-, 0)) - \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}^+, 0)) \right] \\ &\quad + \tfrac{1}{2} \Delta t \mathbf{S}(\mathbf{Q}(x_{i+\frac{1}{2}}^+, 0)). \end{aligned} \right\} \quad (17.27)$$

Next resolve the interaction of the two evolved data states (17.26)–(17.27) right at the interface at time $\tau = \frac{1}{2}\Delta t$, see Fig. 17.5.

Evolved-Data Interaction

In order to find the numerical flux right at the interface $x_{i+\frac{1}{2}}$ at time $\tau = \frac{1}{2}\Delta t$, the next stage is to resolve the interaction of the evolved left data $\mathbf{J}_{i+\frac{1}{2}}^-(\frac{1}{2}\Delta t)$ and right data $\mathbf{J}_{i+\frac{1}{2}}^+(\frac{1}{2}\Delta t)$ in (17.26) and (17.27) respectively, via the solution of the conventional (piece-wise-constant data, homogeneous) Riemann problem

$$GRP_0 : \left\{ \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0} , \\ \text{ICs: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{J}_{i+\frac{1}{2}}^-(\frac{1}{2}\Delta t) & \text{if } x < x_{i+\frac{1}{2}} , \\ \mathbf{J}_{i+\frac{1}{2}}^+(\frac{1}{2}\Delta t) & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{array} \right\} \quad (17.28)$$

The structure of the solution of this conventional Riemann problem at the interface, at the half time $\tau = \frac{1}{2}\Delta t$ in Fig. 17.5 is represented by the wave configuration made up of straight characteristics and wave paths.

The Numerical Flux

Let us denote the similarity solution of (17.28) in local coordinates as $\mathbf{G}_{i+\frac{1}{2}}^{\frac{1}{2}\Delta t}(x/t)$.

The corresponding sought Godunov state right at the interface is $\mathbf{G}_{i+\frac{1}{2}}^{\frac{1}{2}\Delta t}(0)$, which is obtained after sampling the solution. The numerical flux follows directly from the solution of (17.28) and becomes

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_0^{\Delta t} \mathbf{F}(\mathbf{G}_{i+\frac{1}{2}}^{\frac{1}{2}\Delta t}(0)) dt = \mathbf{F}(\mathbf{G}_{i+\frac{1}{2}}^{\frac{1}{2}\Delta t}(0)) . \quad (17.29)$$

Note that stages (17.28) and (17.29) are just like in the conventional first-order Godunov method. See Chaps. 9 and 13.

The Numerical Source

The numerical source for this version of the ADER scheme is identical to that for the ADER2-TT scheme of the previous section, namely

$$\mathbf{S}_i = \mathbf{S} \left(\mathbf{Q}_i^n + \frac{1}{2} \Delta t [-\mathbf{A}(\mathbf{Q}_i^n) \mathbf{A}_i + \mathbf{S}(\mathbf{Q}_i^n)] \right) . \quad (17.30)$$

Therefore the complete numerical scheme ADER2-HEOC via the update formula (17.3) is defined by the numerical flux (17.29) and the numerical source (17.30).

17.2.2 ADER2-TT Versus ADER2-HEOC

We have constructed two ADER-type schemes of second order of accuracy to solve hyperbolic balance laws, source terms included. The difference between these two schemes lies in the way the solution of the generalised Riemann problem is approximated. To be noted, in the original scheme of Harten, Engquist, Osher and Chakravarthy [11], no mention was made of the concept of generalised Riemann problem and their scheme was constructed for the homogeneous case only, no source terms.

Regarding the computational effort involved in the two second-order schemes just constructed we note the following:

1. In the ADER2-TT, to solve the GRP_1 , there is one power series expansion to evolve the solution in time, right at the interface, followed by the Cauchy-Kovalevskaya procedure, applied once.
2. Instead, in the ADER2-HEOC scheme there are two power series expansions to evolve the data either side of the interface, followed by the Cauchy-Kovalevskaya procedure applied twice.
3. In the ADER2-TT solver for the GRP_1 there are two conventional Riemann problems to find the solution, one non-linear for the leading term and one linear for the gradients.
4. Instead, in the ADER2-HEOC scheme there is one non-linear Riemann problem for the evolved-data interaction step.
5. The ADER2-HEOC scheme for the homogenous case (no source term) reproduces identically the MUSCL-Hancock scheme, first communicated by van Leer in [13], and attributed to Steve Hancock. However, the ADER2-HEOC scheme is more general. In fact it is a two-fold extension of the MUSCL-Hancock scheme: (i) ADER2-HEOC is applicable to inhomogeneous hyperbolic systems, and (ii) it extends to arbitrary order of accuracy.
6. For schemes of very higher-order of accuracy the ADER-HEOC variant is more expensive than the ADER-TT scheme, see [14].

17.2.3 FORCE Variant of the ADER2-HEOC Scheme

The ADER2-HEOC scheme just described, may be simplified by replacing the Godunov scheme in the data-interaction stage (17.28) by the centred FORCE flux [15]. See Chap. 13. The resulting scheme is a **centred** variant of the ADER2-HEOC scheme. This centred second-order scheme was termed **SLIC** in [15]. For the implementation of the approach let us first define the initial data of (17.28) as

$$\mathbf{V}_L = \mathbf{J}_{i+\frac{1}{2}}^-(\frac{1}{2}\Delta t), \quad \mathbf{V}_R = \mathbf{J}_{i+\frac{1}{2}}^+(\frac{1}{2}\Delta t). \quad (17.31)$$

Then, the FORCE flux is computed as the arithmetic average of the Lax-Friedrichs and the Lax-Wendroff fluxes

$$\mathbf{F}_{i+\frac{1}{2}}(\mathbf{V}_L, \mathbf{V}_R) = \frac{1}{2} \left[\mathbf{F}_{i+\frac{1}{2}}^{LF}(\mathbf{V}_L, \mathbf{V}_R) + \mathbf{F}_{i+\frac{1}{2}}^{LW}(\mathbf{V}_L, \mathbf{V}_R) \right], \quad (17.32)$$

where the Lax-Friedrichs flux is given as

$$\mathbf{F}_{i+\frac{1}{2}}^{LF}(\mathbf{V}_L, \mathbf{V}_R) = \frac{1}{2} [\mathbf{F}(\mathbf{V}_L) + \mathbf{F}(\mathbf{V}_R)] - \frac{1}{2} \frac{\Delta x}{\Delta t} (\mathbf{V}_R - \mathbf{V}_L) \quad (17.33)$$

and the Lax-Wendroff flux is given as

$$\left. \begin{aligned} \mathbf{F}_{i+\frac{1}{2}}^{LW}(\mathbf{V}_L, \mathbf{V}_R) &= \mathbf{F}(\mathbf{V}_{i+\frac{1}{2}}^{LW}), \\ \mathbf{V}_{i+\frac{1}{2}}^{LW} &= \frac{1}{2} (\mathbf{V}_L + \mathbf{V}_R) - \frac{1}{2} \frac{\Delta t}{\Delta x} [\mathbf{F}(\mathbf{V}_R) - \mathbf{F}(\mathbf{V}_L)] \end{aligned} \right\} \quad (17.34)$$

The upwind flux (17.29) is replaced by the centred flux (17.32). All the remaining components remain the same, including the numerical source. A high-order, locally implicit generalisation of this **centred** approach was recently communicated by Montecinos [16], in which a low-dissipation variant of the FORCE scheme, called FORCE- α [17], was employed.

In the next section we introduce second-order methods for hyperbolic systems in non-conservative form.

17.3 ADER2-HEOC for Non-Conservative Systems

As seen Chap. 14, there are systems of balance laws that cannot be expressed in conservation-law form, for which we adopted the path-conservative approach to design first-order numerical methods. In such schemes, instead of the numerical flux for the update formula, one employs the so-called fluctuations. The reader is advised to review that chapter before proceeding with the rest of this chapter. Here we first briefly recall first-order path conservative schemes and then extend the approach to second-order of accuracy in space and time following the fully-discrete ADER approach, in conjunction with the HEOC solver for the generalised Riemann problem GRP_1 .

17.3.1 First-Order Path-Conservative Scheme

Consider the initial-boundary value problem (IBVP) for a generic system of non-linear hyperbolic balance laws written in non-conservative form

$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}) , \quad x \in [a, b] , \quad t > 0 , \\ \text{ICs: } \quad \mathbf{Q}(x, 0) = \mathbf{Q}^{(0)}(x) , \quad x \in [a, b] , \\ \text{BCs: } \quad \mathbf{Q}(a, t) = \mathbf{B}_L(t) , \quad \mathbf{Q}(b, t) = \mathbf{B}_R(t) , \quad t \geq 0 . \end{array} \right\} \quad (17.35)$$

$\mathbf{Q}(x, t)$ is the vector of unknowns; $\mathbf{A}(\mathbf{Q})$ is the coefficient matrix; $\mathbf{S}(\mathbf{Q})$ is the source term vector; $\mathbf{Q}^{(0)}(x)$ is the initial condition; $\mathbf{B}_L(t)$ and $\mathbf{B}_R(t)$ are the boundary conditions on the left and right boundaries, respectively, two prescribed functions of time. The coefficient matrix $\mathbf{A}(\mathbf{Q})$ is not necessarily the Jacobian of a flux vector $\mathbf{F}(\mathbf{Q})$, which may or may not exist. Assuming that the IBVP (17.35) has a solution, the task at hand is to construct numerical methods to solve the problem approximately.

Here we first briefly recall first-order path conservative schemes for solving non-conservative problems (17.35). As seen Chap. 14 the update path-conservative numerical formula to solve (17.35) reads

$$\mathbf{Q}_i^{n+1} = \mathbf{Q}_i^n - \frac{\Delta t}{\Delta x} (\mathbf{D}_{i+\frac{1}{2}}^- + \mathbf{D}_{i-\frac{1}{2}}^+) + \Delta t \mathbf{S}_i - \Delta t \mathbf{H}_i . \quad (17.36)$$

This scheme will be defined once expressions for the fluctuations $\mathbf{D}_{i+\frac{1}{2}}^-$, $\mathbf{D}_{i+\frac{1}{2}}^+$ are specified. The terms $\mathbf{S}_i$ and $\mathbf{H}_i$ will be discussed later, in the context of second-order methods.

To define the fluctuations we adopt the Parés-Muñoz-Godunov (PMG) approach [18], which yields a Godunov-type scheme of first-order accuracy to solve non-conservative systems. As for conservative schemes, here we also make use of the local Riemann problem to define the fluctuations for a first-order scheme. We therefore introduce the Riemann problem at the interface $x = x_{i+\frac{1}{2}}$ for a non-conservative system

$$GRP_0 : \left\{ \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}) , \\ \text{ICs: } \quad \mathbf{Q}(x, 0) = \begin{cases} \mathbf{Q}_L := \mathbf{Q}_i^n & \text{if } x < x_{i+\frac{1}{2}} , \\ \mathbf{Q}_R := \mathbf{Q}_{i+1}^n & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{array} \right\} \quad (17.37)$$

We assume that (17.37) can be solved to find the similarity solution $\mathbf{G}_{i+\frac{1}{2}}(x/t)$, in local coordinates. Then, the Godunov state $\mathbf{G}_{i+\frac{1}{2}}(0)$ follows after sampling the solution appropriately. As seen in Chap. 14 path functions are used in order to complete the definition of the fluctuations. As required by the PMG approach, two path functions must be specified; here we assume two canonical paths

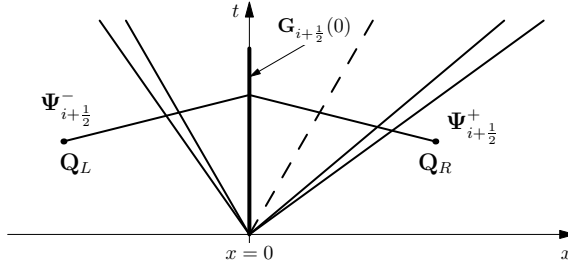


Fig. 17.6 The Parés-Muñoz-Godunov (PMG) scheme for non-conservative systems. Illustration of two path functions joining the Godunov state of the solution of the conventional Riemann problem GRP_0 on the t -axis to the initial data states $\mathbf{Q}_L$ and $\mathbf{Q}_R$

$$\left. \begin{aligned} \psi_{i+\frac{1}{2}}^-(s; \mathbf{Q}_L, \mathbf{G}_{i+\frac{1}{2}}(0)) &= \mathbf{Q}_L + s(\mathbf{G}_{i+\frac{1}{2}}(0) - \mathbf{Q}_L), \quad s \in [0, 1], \\ \psi_{i+\frac{1}{2}}^+(s; \mathbf{G}_{i+\frac{1}{2}}(0), \mathbf{Q}_R) &= \mathbf{G}_{i+\frac{1}{2}}(0) + s(\mathbf{Q}_R - \mathbf{G}_{i+\frac{1}{2}}(0)), \quad s \in [0, 1]. \end{aligned} \right\} \quad (17.38)$$

The path function $\psi_{i+\frac{1}{2}}^-(s; \mathbf{Q}_L, \mathbf{G}_{i+\frac{1}{2}}(0))$ joints the data state $\mathbf{Q}_L$ to the Godunov state $\mathbf{G}_{i+\frac{1}{2}}(0)$. Similarly, the path function $\psi_{i+\frac{1}{2}}^+(s; \mathbf{G}_{i+\frac{1}{2}}(0), \mathbf{Q}_R)$ joints Godunov state $\mathbf{G}_{i+\frac{1}{2}}(0)$ to the data state $\mathbf{Q}_R$. Figure 17.6 depicts the situation. The path function $\psi_{i+\frac{1}{2}}^-$ accounts for the left running waves in the Riemann problem solution, while $\psi_{i+\frac{1}{2}}^+$ accounts for the right running waves in the Riemann problem solution.

The Fluctuations

The numerical fluctuations in the scheme (17.36), for general path functions are defined as

$$\left. \begin{aligned} \mathbf{D}_{i+\frac{1}{2}}^- &= \frac{1}{\Delta t} \int_0^{\Delta t} \int_0^1 \mathbf{A}(\psi_{i+\frac{1}{2}}^-(s; \mathbf{Q}_L, \mathbf{G}_{i+\frac{1}{2}}(0))) \partial_s \psi_{i+\frac{1}{2}}^- ds dt, \\ \mathbf{D}_{i+\frac{1}{2}}^+ &= \frac{1}{\Delta t} \int_0^{\Delta t} \int_0^1 \mathbf{A}(\psi_{i+\frac{1}{2}}^+(s; \mathbf{G}_{i+\frac{1}{2}}(0), \mathbf{Q}_R)) \partial_s \psi_{i+\frac{1}{2}}^+ ds dt. \end{aligned} \right\} \quad (17.39)$$

For the canonical path functions (17.38) the integrals in (17.39) simplify significantly, as the derivative of the path under the integral becomes a constant difference of states and hence comes out of the integral, and the fluctuations become

$$\left. \begin{aligned} \mathbf{D}_{i+\frac{1}{2}}^- &= \left(\int_0^1 \mathbf{A}(\psi_{i+\frac{1}{2}}^-(s; \mathbf{Q}_L, \mathbf{G}_{i+\frac{1}{2}}(0))) ds \right) (\mathbf{G}_{i+\frac{1}{2}}(0) - \mathbf{Q}_L), \\ \mathbf{D}_{i+\frac{1}{2}}^+ &= \left(\int_0^1 \mathbf{A}(\psi_{i+\frac{1}{2}}^+(s; \mathbf{G}_{i+\frac{1}{2}}(0), \mathbf{Q}_R)) ds \right) (\mathbf{Q}_R - \mathbf{G}_{i+\frac{1}{2}}(0)). \end{aligned} \right\} \quad (17.40)$$

The integrals in (17.40) are computed **numerically** along the path functions $\psi_{i+\frac{1}{2}}^-$ and $\psi_{i+\frac{1}{2}}^+$ using a Gauss quadrature rule with G points s_j and associated weights ω_j in the unit interval $[0, 1]$. We obtain

$$\left. \begin{aligned} \mathbf{D}_{i+\frac{1}{2}}^- &= \sum_{j=1}^G \omega_j \left[\mathbf{A}(\psi_{i+\frac{1}{2}}^-(s_j; \mathbf{Q}_L, \mathbf{G}_{i+\frac{1}{2}}(0))) \right] \left(\mathbf{G}_{i+\frac{1}{2}}(0) - \mathbf{Q}_L \right), \\ \mathbf{D}_{i+\frac{1}{2}}^+ &= \sum_{j=1}^G \omega_j \left[\mathbf{A}(\psi_{i+\frac{1}{2}}^+(s_j; \mathbf{G}_{i+\frac{1}{2}}(0), \mathbf{Q}_R)) \right] \left(\mathbf{Q}_R - \mathbf{G}_{i+\frac{1}{2}}(0) \right). \end{aligned} \right\} \quad (17.41)$$

Recommended choices for the computational parameters in (17.41) are

$$s_1 = \frac{1}{2} - \frac{\sqrt{15}}{10}, s_2 = \frac{1}{2}, s_3 = \frac{1}{2} + \frac{\sqrt{15}}{10}, \omega_1 = \frac{5}{18}, \omega_2 = \frac{8}{18}, \omega_3 = \frac{5}{18}. \quad (17.42)$$

The Godunov State via the TV Splitting

A pending problem is to compute the fluctuations in (17.41), for which the Godunov state $\mathbf{G}_{i+\frac{1}{2}}(0)$ corresponding to the non-conservative Riemann problem (17.37) is needed. For this task there are less options available than for the conservative case. One choice is the Toro-Vázquez (TV) splitting method [19, 20] as applied to non-conservative systems and fully explained in Chap. 14. Details are omitted here.

An alternative to the Godunov framework is the **centred approach** offered by the FORCE scheme [15], in non-conservative mode [17], to find the solution of the conventional Riemann problem (17.37). See Chap. 14 and [17] for details.

17.3.2 Godunov-Type ADER2-HEOC Scheme

Here we extend the first-order Parés-Muñoz-Godunov approach [18] to compute the numerical fluctuations in the scheme (17.36), to then construct an ADER scheme of second-order accuracy to solve non-conservative systems. Full details on the PMG approach are given in Chap. 14. The generalised Riemann problem GRP_1 at the interface $x = x_{i+\frac{1}{2}}$ is solved with the HEOC method, in conjunction with the TV splitting. The GRP_1 at $x = x_{i+\frac{1}{2}}$ for a system of balance laws in non-conservative form reads

$$GRP_1 : \left\{ \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}), \\ \text{ICs: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{P}_i(x) & \text{if } x < x_{i+\frac{1}{2}}, \\ \mathbf{P}_{i+1}(x) & \text{if } x > x_{i+\frac{1}{2}}. \end{cases} \end{array} \right\} \quad (17.43)$$

$\mathbf{P}_i(x)$ and $\mathbf{P}_{i+1}(x)$ are first-degree reconstruction polynomials in cells I_i and I_{i+1} respectively. Here we consider the case of polynomials of degree $m = 1$, as given in

(17.6). As for the conservative case, the HEOC solution procedure to solve (17.43) contains a data evolution step and an evolved-data interaction step.

Data Evolution

With notation (17.22) the left and right polynomial limiting values are evolved in time to a generic time $t = \tau$ according to (17.23), that is

$$\left. \begin{aligned} \mathbf{J}_{i+\frac{1}{2}}^-(\tau) &= \mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) + \tau \partial_t \mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) , \\ \mathbf{J}_{i+\frac{1}{2}}^+(\tau) &= \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) + \tau \partial_t \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) . \end{aligned} \right\} \quad (17.44)$$

After applying the Cauchy-Kovalevskaya procedure to the time derivatives in (17.44) for the non-conservative form of the equations in (17.43), the time evolved boundary extrapolated values at time $\tau = \frac{1}{2}\Delta t$ are

$$\left. \begin{aligned} \mathbf{J}_{i+\frac{1}{2}}^-(\tfrac{1}{2}\Delta t) &= \mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) - \tfrac{1}{2}\Delta t \mathbf{A}(\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0)) \mathbf{A}_i + \tfrac{1}{2}\Delta t \mathbf{S}(\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0)) , \\ \mathbf{J}_{i+\frac{1}{2}}^+(\tfrac{1}{2}\Delta t) &= \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) - \tfrac{1}{2}\Delta t \mathbf{A}(\mathbf{Q}(x_{i+\frac{1}{2}}^+, 0)) \mathbf{A}_i + \tfrac{1}{2}\Delta t \mathbf{S}(\mathbf{Q}(x_{i+\frac{1}{2}}^+, 0)) . \end{aligned} \right\} \quad (17.45)$$

Figure 17.5 illustrates the evolution step of the boundary extrapolated data.

Evolved-Data Interaction via the Riemann Problem

Now, the required solution of the generalised Riemann problem (17.43) at the interface $x_{i+\frac{1}{2}}$ is completed by solving the following conventional (piece-wise-constant data, homogeneous) Riemann problem for a non-conservative system

$$GRP_0 : \left\{ \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{0} , \\ \text{ICs: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{Q}_L := \mathbf{J}_{i+\frac{1}{2}}^-(\tfrac{1}{2}\Delta t) & \text{if } x < x_{i+\frac{1}{2}} , \\ \mathbf{Q}_R := \mathbf{J}_{i+\frac{1}{2}}^+(\tfrac{1}{2}\Delta t) & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{array} \right\} \quad (17.46)$$

We assume that (17.46) can be solved to find the similarity solution $\mathbf{G}_{i+\frac{1}{2}}(x/t)$, in local coordinates. Then, after sampling the solution appropriately we obtain the Godunov state $\mathbf{G}_{i+\frac{1}{2}}^{\frac{1}{2}\Delta t}(0)$, which is the sought approximate solution of GRP_1 (17.43) at time $t = \frac{1}{2}\Delta t$.

Godunov State, TV Splitting and Fluctuations

To complete the computation of the fluctuations for the second-order ADER2-HEOC scheme, the following tasks are carried out

1. **Godunov state.** Evaluation of $\mathbf{G}_{i+\frac{1}{2}}^{\frac{1}{2}\Delta t}(0)$ in (17.46) using the TV splitting [19, 20] method described in Chap. 14.

2. **Path functions.** Use $\mathbf{G}_{i+\frac{1}{2}}^{\frac{1}{2}\Delta t}(0)$, instead of $\mathbf{G}_{i+\frac{1}{2}}(0)$, to define the path functions in (17.38).
3. **Fluctuations.** Compute the fluctuations as in (17.41) with path functions evaluated on $\mathbf{G}_{i+\frac{1}{2}}^{\frac{1}{2}\Delta t}(0)$.

The Source Term and the High-Order Term

The numerical source $\mathbf{S}_i$ in the scheme (17.36) is calculated as in (17.21), namely

$$\mathbf{S}_i = \mathbf{S} \left(\mathbf{Q}_i^n + \frac{1}{2} \Delta t [-\mathbf{A}(\mathbf{Q}_i^n) \mathbf{A}_i + \mathbf{S}(\mathbf{Q}_i^n)] \right). \quad (17.47)$$

The high-order term $\mathbf{H}_i$ in the scheme (17.36) is present only in schemes of order greater or equal to 2. For the second-order ADER2-HEOC scheme we compute

$$\mathbf{H}_i = \frac{1}{\Delta t} \int_0^{\Delta t} \int_0^1 \mathbf{A}(\mathbf{Q}(x_i, \frac{1}{2} \Delta t)) \partial_x \mathbf{Q}(x_i, \frac{1}{2} \Delta t) dx dt, \quad (17.48)$$

where $\mathbf{Q}(x_i, \frac{1}{2} \Delta t)$ is given in Eq. (17.20). The spatial derivative in cell I_i to second-order is

$$\partial_x \mathbf{Q}(x_i, \frac{1}{2} \Delta t) = \frac{\mathbf{J}_{i+\frac{1}{2}}^-(\frac{1}{2} \Delta t) - \mathbf{J}_{i-\frac{1}{2}}^+(\frac{1}{2} \Delta t)}{\Delta x}. \quad (17.49)$$

See (17.45). Substitution of (17.49) into (17.48), followed by integration, gives

$$\mathbf{H}_i = \mathbf{A}(\mathbf{Q}(x_i, \frac{1}{2} \Delta t)) \left[\frac{\mathbf{J}_{i+\frac{1}{2}}^-(\frac{1}{2} \Delta t) - \mathbf{J}_{i-\frac{1}{2}}^+(\frac{1}{2} \Delta t)}{\Delta x} \right]. \quad (17.50)$$

Therefore the second-order ADER2-HEOC path conservative scheme (17.36) is completely determined by the fluctuations (17.41), the numerical source (17.47) and the high-order term in (17.50).

17.3.3 Osher-Type ADER2-HEOC Scheme

Another upwind-based second-order ADER scheme for non-conservative systems may be constructed by employing the Dumbser-Osher-Toro (DOT) first-order scheme for non-conservative systems [21], coupled to the HEOC solution of the GRP_1 given in (17.43). Background on the DOT scheme is found in Chap. 14. In the HEOC method the data evolution step is as described in Equations (17.44) and (17.45), ending up with the conventional Riemann problem GRP_0 defined in (17.46). For

convenience we reproduce here the relevant GRP_0 , with the notation used in the DOT approach for the initial data, namely

$$GRP_0 : \left\{ \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \partial_x \mathbf{Q} = \mathbf{0} , \\ \text{ICs: } \mathbf{Q}(x, 0) = \left\{ \begin{array}{ll} \mathbf{Q}_0 := \mathbf{J}_{i+\frac{1}{2}}^- (\frac{1}{2} \Delta t) & \text{if } x < x_{i+\frac{1}{2}} , \\ \mathbf{Q}_1 := \mathbf{J}_{i+\frac{1}{2}}^+ (\frac{1}{2} \Delta t) & \text{if } x > x_{i+\frac{1}{2}} . \end{array} \right. \end{array} \right\} \quad (17.51)$$

Next we define the fluctuations for scheme (17.36) following the DOT approach.

The Fluctuations in the DOT Scheme

For the path conservative scheme (17.36), the DOT scheme for the non-conservative GRP_0 (17.51) the fluctuations are defined as follows

$$\mathbf{D}_{i+\frac{1}{2}}^\pm = \frac{1}{2} \int_0^1 [\mathbf{A}(\psi(s; \mathbf{Q}_0, \mathbf{Q}_1)) \pm |\mathbf{A}(\psi(s; \mathbf{Q}_0, \mathbf{Q}_1))|] \frac{\partial \psi}{\partial s} ds , \quad (17.52)$$

where the path function ψ is completely general. For the particular choice of the canonical path

$$\psi(s; \mathbf{Q}_0, \mathbf{Q}_1) = \mathbf{Q}_0 + s(\mathbf{Q}_1 - \mathbf{Q}_0) \quad (17.53)$$

the fluctuations (17.52) simplify to

$$\mathbf{D}_{i+\frac{1}{2}}^\pm = \frac{1}{2} \left(\int_0^1 [\mathbf{A}(\psi(s; \mathbf{Q}_0, \mathbf{Q}_1)) \pm |\mathbf{A}(\psi(s; \mathbf{Q}_0, \mathbf{Q}_1))|] ds \right) (\mathbf{Q}_1 - \mathbf{Q}_0) . \quad (17.54)$$

The integrals in (17.54) are computed **numerically** along the path ψ using a Gauss quadrature rule with G points s_j and associated weights ω_j in the unit interval $[0, 1]$. We obtain

$$\mathbf{D}_{i+\frac{1}{2}}^\pm = \frac{1}{2} \left(\sum_{j=1}^G \omega_j [\mathbf{A}(\psi(s_j; \mathbf{Q}_0, \mathbf{Q}_1)) \pm |\mathbf{A}(\psi(s_j; \mathbf{Q}_0, \mathbf{Q}_1))|] \right) (\mathbf{Q}_1 - \mathbf{Q}_0) . \quad (17.55)$$

For details on the definition of the **absolute value matrix** $|\mathbf{A}|$ see Chap. 13. More details are given in the original references [21, 22]. Recommended choices for the computational parameters in (17.54) are given in (17.42).

The upwind, path-conservative method (17.36) with (17.55) has been applied in the global, closed loop model for the human circulation in [23–25]. The version of DOT+NC utilised in these cited works is a modified version, so called **well-balanced** version, for which a different path ψ must be used. More details on the well-balanced version of DOT are given in Chap. 13.

Expressions for the numerical source $\mathbf{S}_i$ and the high-order term $\mathbf{H}_i$ in (17.36) remain the same as those given in (17.47) and (17.50), respectively.

17.3.4 *FORCE Variant of ADER2-HEOC*

A centred path conservative approach for non-conservative systems using the FORCE approach [15] is a feasible alternative to the upwind-based methods presented so far. See Toro et al. [17], where low dissipation, first-order FORCE- α schemes in non-conservative mode were presented and applied to the compressible two-phase flow equations of Baer and Nunziato [26]. For a second-order extension the FORCE- α scheme would be applied to the conventional Riemann problem GRP_0 (17.51) arising from the generalised Riemann problem GRP_1 in the HEOC solution method. Expressions for the numerical source $\mathbf{S}_i$ and the high-order term $\mathbf{H}_i$ in (17.36) remain the same as those given in (17.47) and (17.50), respectively. See Sect. 17.2.3 for the analogous conservative case.

We anticipate that in the work of Montecinos [16], an advanced version of this approach was communicated, where the LeFlock-Raviart expansion is carried out implicitly for the equations in non-conservative form, allowing the treatment of stiff source terms. Moreover, in the HEOC approach to solve the GRP_m , the FORCE- α scheme was applied to the conventional Riemann problem GRP_0 at each time-integration point; a code is provided by the author [16] for the Cauchy-Kovalevskaya procedure

So far we have presented fully-discrete ADER methods to solve hyperbolic balance laws to second order of accuracy. In the next section we extend the approach to construct schemes of arbitrary (any) order of accuracy in both space and time.

17.4 Arbitrary-Order ADER for Non-Linear Systems

This section is concerned with the fully-discrete ADER methodology for solving non-linear systems of balance laws to arbitrary order of accuracy in space and time. Useful background is provided by the scalar case in Chap. 16 and the second-order schemes for systems presented in previous sections of this chapter.

The presentation here will be succinct. As already pointed out, the ADER schemes rest upon two major building blocks, non-linear reconstruction polynomials of degree m and the solution of the generalised Riemann problem GRP_m , to obtain ADER schemes of order $m + 1$ in space and time.

The aim is to solve non-linear systems of balance laws in conservation-law form (17.1) via the finite volume method (17.3) illustrated in Fig. 17.1, in which the numerical flux and the numerical source derive from approximations to the respective inte-

grals in (17.4). The ADER numerical flux is found from the solution of the following generalised Riemann problem

$$GRP_m : \left\{ \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}) , \quad x \in \mathcal{R} , t > 0 , \\ \text{IC: } \quad \mathbf{Q}(x, 0) = \begin{cases} \mathbf{P}_i(x) & \text{if } x < x_{i+\frac{1}{2}} , \\ \mathbf{P}_{i+1}(x) & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{array} \right. \quad (17.56)$$

In general, $\mathbf{P}_i(x)$ and $\mathbf{P}_{i+1}(x)$ are assumed to be reconstructed polynomial vectors of degree m . In this section we present the ADER approach in conjunction with four solvers for the generalised Riemann problem GRP_m (17.56), namely the Toro-Titarev (TT) solver [3], the Harten-Engquist-Osher-Chakravarthy (HEOC) solver [11, 12], the Montecinos-Toro (MT) solver [14] and the Dumbser-Enaux-Toro (DET) solver [28].

17.4.1 Toro-Titarev Solver for GRP_m

To compute the numerical flux in (17.3) we solve the generalised Riemann problem (17.56) using the Toro-Titarev (TT) method [3, 4]. The initial conditions for the GRP_m (17.56) are reconstruction polynomials of arbitrary degree m and the resulting ADER method will be of $(m + 1)$ th order of accuracy in space and time. The TT method seeks an approximate time-dependent solution $\mathbf{J}_{i+\frac{1}{2}}(\tau)$ of (17.56) at the interface, through suitable evaluation of the LeFloch-Raviart expansion [7]

$$\mathbf{J}_{i+\frac{1}{2}}(\tau) = \mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) + \sum_{k=1}^m \left[\partial_t^{(k)} \mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) \right] \frac{\tau^k}{k!} . \quad (17.57)$$

The task at hand consists of finding the leading term $\mathbf{Q}(x_{i+\frac{1}{2}}, 0_+)$ and the remaining m higher-order terms in (17.57).

The Leading Term

The leading term of the expansion (17.57) is defined as

$$\mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) = \lim_{t \rightarrow 0_+} \mathbf{Q}(x_{i+\frac{1}{2}}, t) , \quad (17.58)$$

which is computed by solving the conventional Riemann problem

$$GRP_0 : \left\{ \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0} , \quad t > 0 , \\ \text{ICs: } \quad \mathbf{Q}(x, 0) = \begin{cases} \mathbf{P}_i(x_{i+\frac{1}{2}}) & \text{if } x < x_{i+\frac{1}{2}} , \\ \mathbf{P}_{i+1}(x_{i+\frac{1}{2}}) & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{array} \right. \quad (17.59)$$

The initial conditions are **piecewise constant**, given by the limiting values of $\mathbf{P}_i(x)$ and $\mathbf{P}_{i+1}(x)$ from left and right respectively. Denoting by $\mathbf{G}_{i+\frac{1}{2}}^{(0)}(\hat{x}/\tau)$, in local coordinates, the similarity solution of (17.59) we set

$$\mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) = \mathbf{G}_{i+\frac{1}{2}}^{(0)}(0) . \quad (17.60)$$

Exact or approximate solvers can be used to find $\mathbf{G}_{i+\frac{1}{2}}^{(0)}(\hat{x}/\tau)$, the sampling of which gives the **Godunov state** $\mathbf{G}_{i+\frac{1}{2}}^{(0)}(0)$.

Higher Order Terms

The higher-order terms in (17.57) involve the following steps:

1. The Cauchy-Kovalevskaya procedure to express time derivatives as functionals of spatial derivatives:

$$\partial_t^{(l)} \mathbf{Q}(x, t) = \mathbf{T}^{(l)}(\partial_x^{(0)} \mathbf{Q}, \partial_x^{(1)} \mathbf{Q}, \dots, \partial_x^{(l)} \mathbf{Q}) . \quad (17.61)$$

$\mathbf{T}^{(l)}$ is specific to the PDE system of interest. The pending problem is to determine its arguments, which are spatial derivatives.

2. Evolution equations for spatial derivatives of order l , for $l = 1, \dots, m$:

$$\partial_t(\partial_x^{(l)} \mathbf{Q}(x, t)) + \mathbf{A}(\mathbf{Q}) \partial_x(\partial_x^{(l)} \mathbf{Q}(x, t)) = \mathbf{K}^{(l)} . \quad (17.62)$$

Here $\mathbf{A}(\mathbf{Q})$ is the Jacobian matrix of the PDEs in (17.56) and the source term $\mathbf{K}^{(l)}$ arises from the CK procedure.

3. Simplifications to (17.62) involve a linearisation of the PDEs with a constant Jacobian matrix and neglecting source terms, namely

$$\mathbf{A}(\mathbf{Q}) \approx \mathbf{A}(\mathbf{G}_{i+\frac{1}{2}}^{(0)}(0)) = \mathbf{A}_{i+\frac{1}{2}}^{(0)} \quad \text{and} \quad \mathbf{K}^{(l)} = \mathbf{0} . \quad (17.63)$$

The above simplifications can be justified rigorously for some special cases [8]. For the general case, empirical evidence suggests that, for smooth solutions, such simplifications do not sacrifice the theoretically aimed order of accuracy $m + 1$.

4. Conventional, linearised Riemann problems for spatial derivatives

$$GRP_0 : \left\{ \begin{array}{l} \partial_t(\partial_x^{(l)} \mathbf{Q}(x, t)) + \mathbf{A}_{i+\frac{1}{2}}^{(0)} \partial_x(\partial_x^{(l)} \mathbf{Q}(x, t)) = \mathbf{0} , \\ \partial_x^{(l)} \mathbf{Q}(x, 0) = \begin{cases} \mathbf{P}_i^{(l)}(x_{i+\frac{1}{2}}) & \text{if } x < x_{i+\frac{1}{2}} , \\ \mathbf{P}_{i+1}^{(l)}(x_{i+\frac{1}{2}}) & \text{if } x > x_{i+\frac{1}{2}} , \end{cases} \end{array} \right\} \quad (17.64)$$

where $\mathbf{P}_i^{(l)}(x) = \partial_x^{(l)} \mathbf{Q}(x, t)$; likewise $\mathbf{P}_{i+1}^{(l)}(x)$. The similarity solution of (17.64) in local coordinates is denoted as $\mathbf{G}_{i+\frac{1}{2}}^{(l)}(\hat{x}/\tau)$, with Godunov states denoted as $\mathbf{G}_{i+\frac{1}{2}}^{(l)}(0)$, for $l = 1, \dots, m$.

5. Complete solution. Having determined the leading term and all the arguments of $\mathbf{T}^{(l)}$ in (17.61), the final solution reads

$$\mathbf{J}_{i+\frac{1}{2}}(\tau) = \mathbf{G}_{i+\frac{1}{2}}^{(0)}(0) + \sum_{l=1}^m \left[\mathbf{T}^{(l)} \left(\mathbf{G}_{i+\frac{1}{2}}^{(0)}(0), \dots, \mathbf{G}_{i+\frac{1}{2}}^{(l)}(0) \right) \right] \frac{\tau^l}{l!}. \quad (17.65)$$

The Numerical Flux

The approximate solution $\mathbf{J}_{i+\frac{1}{2}}(\tau)$ to the GRP_m (17.65) is inserted into the flux integral in (17.4) and computed numerically via a suitable quadrature rule

$$\mathbf{F}_{i+\frac{1}{2}}^{Ader} = \frac{1}{\Delta t} \int_0^{\Delta t} \mathbf{F}(\mathbf{J}_{i+\frac{1}{2}}(\tau)) d\tau \approx \sum_{\alpha=0}^{\hat{\alpha}} \omega_{\alpha} \mathbf{F}(\mathbf{J}_{i+\frac{1}{2}}(\gamma_{\alpha} \Delta t)). \quad (17.66)$$

The ADER flux $\mathbf{F}_{i+\frac{1}{2}}^{Ader}$ is used in the one-step conservative formula (17.3) that gives a method of order $m+1$ in space and time. Further details are found in [3, 4] and in Chaps. 19 and 20 of [29].

Remark: flux expansion. In the solution of the GRP_m in (17.56), instead of the state expansion (17.57), one can perform a flux expansion, as demonstrated by Toro and Titarev in [30]. The advantage of the flux expansion over the state expansion lies in the choices available for the leading term. There are many choices for first-order fluxes to be used in the flux expansion, including upwind and centred [29].

The Numerical Source

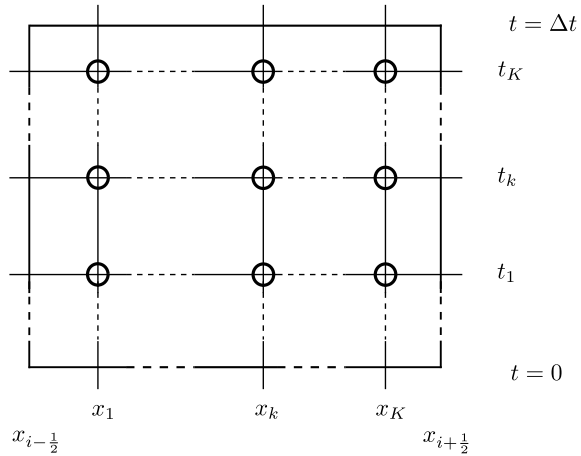
The numerical source in (17.3) follows from the volume integral in (17.4) as

$$\mathbf{S}_i = \frac{1}{\Delta t \Delta x} \int_0^{\Delta t} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{S}(\mathbf{Q}_i(x, t)) dx dt, \quad (17.67)$$

where $\mathbf{Q}_i(x, t)$ is an approximate solution in the control volume V_i^n (17.2). The approximation $\mathbf{Q}_i(x, t)$ is obtained as a generalisation of the data evolution procedure presented in Sect. 17.1.4 for the second-order case, see Eq. (17.19).

Recall that for second-order, the mid-point integration rule in space and time required the solution value at a single point $(x_i, \frac{1}{2} \Delta t)$ in the volume V_i^n ; see Fig. 17.4. For orders of accuracy higher than 2, one needs an appropriate number of spatial positions inside the cell I_i and time levels within the time interval $[0, \Delta t]$, as illustrated in Fig. 17.7.

Fig. 17.7 Numerical source. Integration points within the control volume V_i^n to compute the source term to high-order of accuracy



For a generic, fixed, spatial integration point x_α within the cell I_i , as depicted in Fig. 17.7, we write a time power series expansion of the form

$$\mathbf{Q}(x_\alpha, \tau) = \mathbf{Q}(x_\alpha, 0) + \sum_{k=1}^m \partial_t^{(k)} \mathbf{Q}(x_\alpha, 0) \frac{\tau^k}{k!}. \quad (17.68)$$

This expansion is entirely analogous to the LeFloch-Raviart expansion (17.57) used for the solution of the GRP_m at the interface between two cells. Here, however, the expansion is performed in the interior of the relevant cell I_i and the leading term is simply the reconstruction polynomial evaluated at the position x_α , at which point the function is continuous. Recall that at the local time $t = 0$, the solution coincides with the spatial reconstruction polynomial, that is $\mathbf{Q}_i(x, 0) = \mathbf{P}_i(x)$, a continuous function. Therefore, there is not need to solve a conventional Riemann problem to find the leading term, as

$$\mathbf{Q}(x_\alpha, 0) = \mathbf{P}_i(x_\alpha). \quad (17.69)$$

The higher-order terms in (17.68) are computed via the Cauchy-Kovalevskaya procedure, as for the GRP_m , whereby time derivatives in (17.68) are expressed as functionals of space derivatives and the source terms $\mathbf{S}(\mathbf{Q})$, namely

$$\partial_t^{(l)} \mathbf{Q}(x, t) = \mathbf{T}^{(l)} \left(\partial_x^{(0)} \mathbf{Q}, \partial_x^{(1)} \mathbf{Q}, \dots, \partial_x^{(l)} \mathbf{Q} \right). \quad (17.70)$$

The arguments of the functionals (17.70) are well defined in the entire cell I_i , as the solution at the local time $t = 0$ is represented by the smooth reconstructed polynomial $\mathbf{P}_i(x)$, with well-defined spatial derivatives

$$\mathbf{P}_i^{(l)}(x) \equiv \frac{d^l}{dx^l} \mathbf{P}_i(x), \quad x \in [x_{i-1/2}, x_{i+1/2}]. \quad (17.71)$$

Therefore

$$\mathbf{Q}(x_\alpha, \tau) = \mathbf{P}_i(x_\alpha) + \sum_{k=1}^m \mathbf{T}^{(k)} \left(\mathbf{P}_i^{(0)}(x_\alpha), \mathbf{P}_i^{(1)}(x_\alpha), \dots, \mathbf{P}_i^{(k)}(x_\alpha) \right) \frac{\tau^k}{k!}. \quad (17.72)$$

Expression (17.72) is capable of reproducing all necessary space-time grid points within the control volume V_i^n , illustrated in Fig. 17.7.

There are several integration rules available for computing the multidimensional integral (17.67); see [31], for example. Using a Gaussian quadrature rule we obtain

$$\mathbf{S}_i = \sum_{\alpha=1}^{m_\alpha} \sum_{\beta=1}^{m_\beta} \omega_\alpha \omega_\beta \mathbf{S}(\mathbf{Q}(x_\alpha, \gamma_\beta \Delta t)). \quad (17.73)$$

Here $x_\alpha = x_{i-\frac{1}{2}} + \gamma_\alpha \Delta x$, where γ and ω are properly scaled nodes and weights of the rule.

An ADER-TT scheme, for systems of hyperbolic balance laws in conservation law form, of arbitrary order of accuracy in space and time has been presented. The one-step scheme (17.3) uses the numerical flux (17.66) and the numerical source (17.73).

In the next section we study arbitrary high-order ADER methods in conjunction with the Harten, Engquist, Osher and Chakravarthy (HEOC) solver for the GRP_m [11].

17.4.2 HEOC Solver for GRP_m

The HEOC solver for the GRP_m defined in (17.5) emerges from a reinterpretation by Castro and Toro [12] of the high-order method of Harten et al. [11] as an ADER method. By doing so a corresponding solver for GRP_m is derived, called the HEOC solver here, for Harten, Engquist, Osher and Chakravarthy. The HEOC solver is used to compute the numerical flux in (17.3) and consists of a **time evolution** step and an evolved-data **interaction step**, as seen below.

The Time Evolution Step

This step evolves the limiting values of the data left and right of the interface via two conventional Taylor expansions

$$\left. \begin{aligned} \mathbf{J}_{i+\frac{1}{2}}^-(\tau) &= \mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) + \sum_{l=1}^m \left[\partial_t^{(l)} \mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) \right] \frac{\tau^l}{l!}, \\ \mathbf{J}_{i+\frac{1}{2}}^+(\tau) &= \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) + \sum_{l=1}^m \left[\partial_t^{(l)} \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) \right] \frac{\tau^l}{l!}, \end{aligned} \right\} \quad (17.74)$$

with

$$\mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) = \mathbf{P}_i(x_{i+\frac{1}{2}}) , \quad \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) = \mathbf{P}_{i+1}(x_{i+\frac{1}{2}}) . \quad (17.75)$$

By means of the Cauchy-Kovalevskaya procedure we express all time derivatives in (17.74) as functionals of spatial derivatives of left and right reconstruction polynomials

$$\left. \begin{aligned} \partial_t^{(l)} \mathbf{Q}(x_{i+\frac{1}{2}}^-, 0) &= \mathbf{T}^{(l)} \left(\mathbf{P}_i^{(0)}(x_{i+\frac{1}{2}}), \mathbf{P}_i^{(1)}(x_{i+\frac{1}{2}}), \dots, \mathbf{P}_i^{(l)}(x_{i+\frac{1}{2}}) \right) , \\ \partial_t^{(l)} \mathbf{Q}(x_{i+\frac{1}{2}}^+, 0) &= \mathbf{T}^{(l)} \left(\mathbf{P}_{i+1}^{(0)}(x_{i+\frac{1}{2}}), \mathbf{P}_{i+1}^{(1)}(x_{i+\frac{1}{2}}), \dots, \mathbf{P}_{i+1}^{(l)}(x_{i+\frac{1}{2}}) \right) , \end{aligned} \right\} \quad (17.76)$$

with

$$\mathbf{P}_i^{(l)}(x_{i+\frac{1}{2}}) = \frac{d^l}{dx^l} \mathbf{P}_i(x_{i+\frac{1}{2}}) , \quad \mathbf{P}_{i+1}^{(l)}(x_{i+\frac{1}{2}}) = \frac{d^l}{dx^l} \mathbf{P}_{i+1}(x_{i+\frac{1}{2}}) . \quad (17.77)$$

Expressions (17.76) are substituted into (17.74).

Data Interaction Step

In this step one solves the classical Riemann problem

$$GRP_0 : \left\{ \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0} , \\ \text{ICs: } \mathbf{Q}(x, 0) = \begin{cases} \mathbf{J}_{i+\frac{1}{2}}^-(\tau) & \text{if } x < x_{i+\frac{1}{2}} , \\ \mathbf{J}_{i+\frac{1}{2}}^+(\tau) & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{array} \right\} \quad (17.78)$$

Denoting the similarity solution of (17.78) in local coordinates as $\mathbf{G}_{i+\frac{1}{2}}^{(\tau)}(x/t)$ we obtain an approximate solution of the GRP_m (17.5) at the interface $x = x_{i+\frac{1}{2}}$, at an arbitrary time $t = \tau$, with $0 < \tau \leq \Delta t$, as

$$\mathbf{J}_{i+\frac{1}{2}}(\tau) = \mathbf{G}_{i+\frac{1}{2}}^{(\tau)}(0) . \quad (17.79)$$

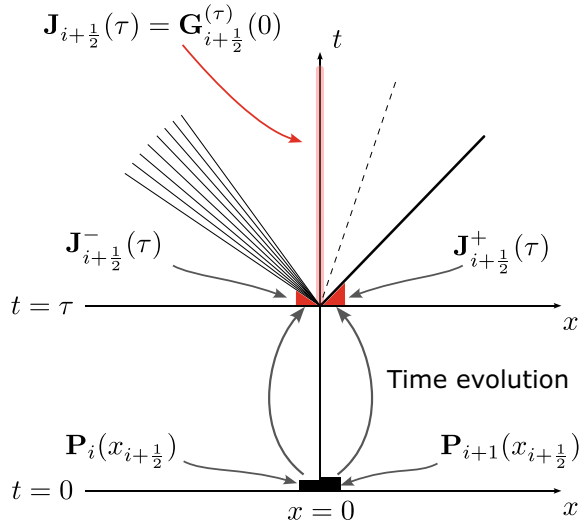
Figure 17.8 illustrates the HEOC solver, with its data evolution step in time via two Taylor expansions, followed by the solution of a conventional Riemann problem, whose initial conditions are the time evolved left and right data.

The Numerical Flux

The ADER-HEOC numerical flux follows as in (17.66). The flux integral in (17.4) is computed numerically via a Gaussian integration rule as follows

$$\mathbf{F}_{i+\frac{1}{2}} = \sum_{j=1}^G \omega_j \mathbf{F}(\mathbf{G}_{i+\frac{1}{2}}^{(\tau_j)}) , \quad (17.80)$$

Fig. 17.8 HEOC solver for GRP_m . Bottom: limiting data values from left and right reconstruction polynomials are evolved in time via expansions (17.74) to arbitrary time τ . Top: the HEOC solution of GRP_m at time $t = \tau$ is found by computing the similarity solution of the conventional Riemann problem (17.78) with piecewise constant data



with an appropriate sequence of times $\tau_1, \tau_2, \dots, \tau_G$ in (17.79). For each integration time τ_j one needs to solve the conventional Riemann problem (17.78) to obtain the Godunov state $\mathbf{G}_{i+1/2}^{(\tau_j)}$.

The Numerical Source $\mathbf{S}_i$

To complete the high-order ADER scheme one computes the numerical source $\mathbf{S}_i$ in identically the same way as in Sect. 17.4.1. See Eq. (17.73).

Variations of the HEOC Solver

Castro and Toro [12] also proposed a variation of the HEOC solver for the GRP_m just presented, whereby the solution is obtained from a sequence of conventional Riemann problems for **time derivatives**, directly. Such solver, called the CT solver for the GRP_m , directly based on time derivatives, inspired the **time reconstruction (TR)** solver for the GRP_m , which eliminates the need for the Cauchy-Kovalevskaya procedure altogether. Preliminary results obtained by Dematté and collaborators [33] are very encouraging. Another variation of the ADER-HEOC scheme is offered by the centred approach based on the FORCE scheme. See Sect. 17.2.3 and [16].

A complete description of the ADER scheme to solve systems of hyperbolic balance laws (17.1) to arbitrary order of accuracy $m + 1$, in conjunction with the HEOC solver for the GRP_m , has been given. As usual the update formula (17.3) makes use of the numerical flux (17.80) and the numerical source (17.73).

In the next section we present ADER schemes of arbitrary order of accuracy, in conjunction with a solver for the GRP_m that is applicable to balance laws with stiff source terms.

17.4.3 Montecinos-Toro Implicit Solver for GRP_m

This solver for the GRP_m [14, 34] is distinct from those studied so far in two respects. Firstly, this method is (locally) **implicit**, which makes it suitable for solving systems of hyperbolic balance laws with stiff source terms, to arbitrarily high order of accuracy. The resulting ADER method, still explicit, reconciles stiffness with high-order of accuracy, for smooth solutions. Secondly, the scheme is an implicit version of both the Toro-Titarev solver and of the HEOC solver.

Implicit Time Expansion

The key step of the method is the following result:

Lemma 17.1 *Let $\mathbf{Q}(x, \tau)$ be an analytic function of τ , then*

$$\mathbf{Q}(x, \tau) = \mathbf{Q}(x, 0_+) - \sum_{k=1}^{\infty} \frac{(-\tau)^k}{k!} \partial_t^{(k)} \mathbf{Q}(x, \tau) . \quad (17.81)$$

Proof Omitted. See [14, 35, 36] for details. ■

The time expansion (17.81) is the **implicit** counterpart of the explicit LeFloch-Raviart expansion (17.57). When utilising the implicit expansion (17.81), in truncated form, for solving approximately the generalised Riemann problem GRP_m , there are at least two possible frameworks that can be adopted, namely the Toro-Titarev [3] of Sect. 17.4.1 and the HEOC approaches [11, 12] of Sect. 17.4.2. The former applies the explicit LeFloch-Raviart expansion right at the interface, while the latter applies it left and right of the interface in the data-evolution step. In [14] both the TT and HEOC approaches were extended on the basis of the truncated implicit expansion (17.81).

The Toro-Titarev framework as applied to the truncated expansion (17.81) right at the interface, placed locally at $x = 0$, reads

$$\mathbf{Q}(x_{i+\frac{1}{2}}, \tau) = \mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) - \sum_{k=1}^m \frac{(-\tau)^k}{k!} \partial_t^{(k)} \mathbf{Q}(x_{i+\frac{1}{2}}, \tau) . \quad (17.82)$$

$\mathbf{Q}(x_{i+\frac{1}{2}}, \tau)$ denotes the sought solution at the interface as a function of time τ and $\mathbf{Q}(x_{i+\frac{1}{2}}, 0_+)$ is the leading term of the expansion

$$\mathbf{Q}(x_{i+\frac{1}{2}}, 0_+) = \lim_{t \rightarrow 0_+} \mathbf{Q}(x_{i+\frac{1}{2}}, t) . \quad (17.83)$$

Now, the **implicit** Cauchy-Kovalevskaya procedure leads to the following implicit expressions for the time derivatives

$$\partial_t^{(k)} \mathbf{Q}(x_{i+\frac{1}{2}}, \tau) = \mathbf{T}^{(k)}(\partial_x^{(0)} \mathbf{Q}(x_{i+\frac{1}{2}}, \tau), \partial_x^{(1)} \mathbf{Q}(x_{i+\frac{1}{2}}, \tau), \dots, \partial_x^{(k)} \mathbf{Q}(x_{i+\frac{1}{2}}, \tau)) . \quad (17.84)$$

Then the expansion becomes

$$\mathbf{Q}(0, \tau) = \mathbf{Q}(0, 0_+) - \sum_{k=1}^m \frac{(-\tau)^k}{k!} \mathbf{T}^{(k)}(\partial_x^{(0)} \mathbf{Q}(x_{i+\frac{1}{2}}, \tau), \dots, \partial_x^{(k)} \mathbf{Q}(x_{i+\frac{1}{2}}, \tau)) . \quad (17.85)$$

A key step is to determine the arguments of the functionals $\mathbf{T}^{(k)}$ in (17.85), right at the interface. For details see [14].

Second-Order Scheme for the Linear Scalar Case

As a simple example consider the scalar balance law

$$\partial_t q(x, t) + \partial_x f(q(x, t)) = s(q(x, t)) , \quad (17.86)$$

with linear physical flux and source functions given as

$$f(q) = \lambda q , \quad s(q) = \beta q . \quad (17.87)$$

As usual, λ is a constant wave propagation speed and β a constant reaction rate, with $\beta \leq 0$. In the case of stiff source term the reaction rate may be very large in absolute value, that is $|\beta| \gg 0$. In this case, the methods studied so far are unsuitable, as the stability requirement on the choice of the time step Δt would be prohibitively restrictive.

For a second-order scheme, application of the implicit expansion (17.82) reads

$$q(x_{i+\frac{1}{2}}, \tau) = q(x_{i+\frac{1}{2}}, 0_+) + \tau \partial_t q(x_{i+\frac{1}{2}}, \tau) . \quad (17.88)$$

Application of the Cauchy-Kovalevskaya procedure to the governing equations (17.86)–(17.87) gives

$$\partial_t q(x_{i+\frac{1}{2}}, \tau) = -\lambda \partial_x q(x_{i+\frac{1}{2}}, \tau) + \beta q(x_{i+\frac{1}{2}}, \tau) . \quad (17.89)$$

Application of the implicit expansion (17.82) to $\partial_x q(x_{i+\frac{1}{2}}, \tau)$ gives

$$\partial_x q(x_{i+\frac{1}{2}}, \tau) = \partial_x q(x_{i+\frac{1}{2}}, 0_+) + \tau \left[-\lambda \partial_x^{(2)} q(x_{i+\frac{1}{2}}, \tau) + \beta \partial_x q(x_{i+\frac{1}{2}}, \tau) \right] . \quad (17.90)$$

As the underlying reconstruction polynomial is of degree $m = 1$, the second-order derivative is neglected, leading to

$$\partial_x q(x_{i+\frac{1}{2}}, \tau) = \frac{1}{(1 - \beta \tau)} \partial_x q(x_{i+\frac{1}{2}}, 0_+) . \quad (17.91)$$

Further manipulations lead to

$$q(x_{i+\frac{1}{2}}, \tau) = \frac{1}{(1 - \beta\tau)} \left[q(x_{i+\frac{1}{2}}, 0_+) - \frac{\tau\lambda}{(1 - \beta\tau)} \partial_x q(x_{i+\frac{1}{2}}, 0_+) \right]. \quad (17.92)$$

Evaluation of the flux integral in (17.4) with (17.92) in the integrand will give the numerical flux for use in (17.3).

For the numerical source in (17.86), one first applies the implicit time expansion (17.82) at the mid-point x_i followed by the Cauchy-Kovalevskaya procedure. For comparison, see Eq. (17.20) for the explicit case. Then the resulting function $q(x_i, \frac{1}{2}\Delta t)$ is used in the integrand to evaluate the source integral in (17.4). The reader is encouraged to complete the details.

The resulting scheme (17.3) is second-order accurate and locally implicit. The method is suitable for balance laws with stiff source terms. This locally implicit methodology can also be applied in the framework of the HEOC solver for the GRP_m . See [14] for full details and applications of higher order versions of the method in both the TT and the HEOC frameworks for solving the GRP_m .

Recently, Montecinos and Balsara [37] have developed a simplified Cauchy-Kovalevskaya procedure for these locally implicit schemes. See [16] for further developments.

17.4.4 Remarks on the Cauchy-Kovalevskaya Procedure

Here we briefly review the explicit version of the Cauchy-Kovalevskaya procedure, which is applied to the ADER method in conjunction with the Toro-Titarev solver and the HEOC solver for the GRP_m . The Cauchy-Kovalevskaya procedure [29] expresses time derivatives in terms of spatial derivatives. For second-order explicit GRP_m solvers, as seen in Sect. 17.1, the procedure is trivial, but not so for locally implicit solvers, as seen in the previous section. The real challenge is brought about by higher-order schemes and by locally implicit schemes. To gain some insight into the complexity of the procedure, even for the locally explicit approach, let us consider the single non-linear wave equation with source term

$$\partial_t q(x, t) + \partial_x f(q(x, t)) = s(q(x, t)). \quad (17.93)$$

Assuming $q(x, t)$ to be continuous and differentiable up to order m , from (17.93) it follows that the first-order time derivative is given as

$$\partial_t q(x, t) = -a(q) \partial_x q(x, t) + s(q) \equiv T^{(1)}(\partial_x^{(0)} q, \partial_x^{(1)} q), \quad (17.94)$$

where

$$a(q) = \frac{d}{dq} f(q) = f'(q) \quad (17.95)$$

is the wave velocity and the functional $T^{(1)}$ is

$$T^{(1)}(\partial_x^{(0)} q, \partial_x^{(1)} q) = -f'(\partial_x^{(0)} q) \times \partial_x^{(1)} q + s(\partial_x^{(0)} q) , \quad \partial_x^{(0)} q = q . \quad (17.96)$$

Analogously, the second-order time derivative is

$$\begin{aligned} \partial_t^{(2)} q(x, t) &= (f'(q))^2 \partial_x^{(2)} q(x, t) + 2f'(q)f''(q) (\partial_x^{(1)} q(x, t))^2 \\ &\quad - \partial_x^{(1)} q(x, t) (f''(q)s(q) + 2f''(q)s'(q)) + s'(q)s(q) \\ &\equiv T^{(2)}(\partial_x^{(0)} q, \partial_x^{(1)} q, \partial_x^{(2)} q) . \end{aligned} \quad (17.97)$$

In this case the functional $T^{(2)}$ is more complicated; its arguments are spatial derivatives of $q(x, t)$. In general, the k th order time derivative is written as a functional depending on the space derivatives up to order k , namely

$$\partial_t^{(k)} q(x, t) = T^{(k)}(\partial_x^{(0)} q, \partial_x^{(1)} q, \dots, \partial_x^{(k-1)} q, \partial_x^{(k)} q) , \quad (17.98)$$

for $k = 1 \dots m$.

For general non-linear systems, the Cauchy-Kovalevskaya procedure becomes a very complex analytical procedure to be performed by hand, as the order of accuracy increases. In practice one utilises symbolic manipulators to perform the algebra to compute the functionals $T^{(k)}$ in (17.98) and to convert it directly into code. We highly recommend the paper by Dumbser et al. [38], who applied ADER schemes to the three-dimensional Euler equations to very high order of accuracy, which fully involved the application of the Cauchy-Kovalevskaya procedure. At the end of that paper the reader will find a FORTRAN code to apply the Cauchy-Kovalevskaya procedure via a computer. See also [16].

Nowadays there are alternative solvers for the GRP_m that avoid the direct use of the Cauchy-Kovalevskaya procedure [28, 33]. In the next section we study one of such methods.

17.4.5 Dumbser-Enaux-Toro (DET) Solver for GRP_m

Here we present the method proposed by Dumbser, Enaux and Toro [28] to solve the generalised Riemann problem (17.5) to find the numerical flux $\mathbf{F}_{i+\frac{1}{2}}$ in (17.4) to be used in the conservative scheme (17.3) to solve the system of balance laws (17.1). The DET solver has two distinctive features: it bypasses the Cauchy-Kovalevskaya procedure and can deal with stiff source terms in (17.1). In analogy with HEOC, the DET solver for the GRP_m has the following main steps:

- **Evolution of the initial data.** The initial conditions represented by reconstruction polynomials in adjacent control volumes to the left and right of the interface $x_{i+\frac{1}{2}}$ are evolved in time in the entire control volumes, **numerically**. In the literature, this part is called the prediction step.
- **Interaction of the evolved data.** This is accounted for, at any specified time $t_k \in (t_n, t_{n+1}] := (0, \Delta t]$, by solving a conventional Riemann problem, just as done for the ADER-HEOC method of Sect. 17.4.2.
- **Numerical flux.** The sought solution of the generalised Riemann problem (17.5) at the interface is given by the solutions of a sequence of conventional Riemann problems in the interaction step, which are then utilised to determine the numerical flux $\mathbf{F}_{i+\frac{1}{2}}$ by evaluating the flux integral in (17.4), just as done for the ADER-HEOC method of Sect. 17.4.2.
- **Numerical source.** If the governing equations contain source terms then the numerical source $\mathbf{S}_i$ in the scheme (17.3) is computed by evaluating the volume integral (17.4) for $\mathbf{S}(\mathbf{Q})$ in the control volume V_i^n (17.2) utilising the implicitly evolved data. See Fig. 17.7.

The DET method is akin to the HEOC method [11, 12], see Sect. 17.4.2. The chief difference between these two methods lies in the evolution step. In DET, the initial conditions represented by reconstruction polynomials left and right of the interface, are **numerically** evolved in time in the interval $(t_n, t_{n+1}]$. In HEOC the evolution is done **analytically** via Taylor expansions followed by application of the Cauchy-Kovalevskaya procedure. In the presence of source terms, both methods are able to provide a space-time distribution in each cell so that the source integral in (17.4) may be evaluated to the required accuracy. Furthermore, use of an implicit numerical method for the data evolution in DET results in high-order ADER methods for balance laws with stiff source terms. This aspect is akin to the Montecinos-Toro solver for the GRP_m described in Sect. 17.4.3, which is also able to deal with stiff source terms. In what follows we describe the two steps of the original DET method. The presentation is succinct and follows [14, 28].

Evolution Step in the DET Solver

The data-evolution step is performed in the two space-time control volumes adjacent to the interface $x_{i+\frac{1}{2}}$, locally at $x = 0$, namely

$$\left. \begin{aligned} V_L &= [-\Delta x, 0] \times [0, \Delta t] \quad \text{to the left of the interface} , \\ V_R &= [0, \Delta x] \times [0, \Delta t] \quad \text{to the right of the interface} . \end{aligned} \right\} \quad (17.99)$$

In each of these domains one defines the following local initial-value problems (Cauchy problems)

$$\left. \begin{aligned} \text{PDE: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) &= \mathbf{S}(\mathbf{Q}) , \\ \text{IC: } \mathbf{Q}(x, 0) &= \mathbf{P}_k(x) , \end{aligned} \right\} \quad (17.100)$$

where $k = i$ for V_L and $k = i + 1$ for V_R . Here $\mathbf{P}_k(x)$ is a reconstruction polynomial of degree m . For convenience one transforms V_L and V_R into the reference space-time domain $[0, 1] \times [0, 1]$ through

$$\left. \begin{aligned} x(\xi) &= (\xi - 1)\Delta x, \quad t(\tau) = \tau \Delta t, \quad \text{for } (x, t) \in V_L, \\ x(\xi) &= \xi \Delta x, \quad t(\tau) = \tau \Delta t, \quad \text{for } (x, t) \in V_R. \end{aligned} \right\} \quad (17.101)$$

In $\xi - \tau$ coordinates the Cauchy problem (17.100) becomes

$$\left. \begin{aligned} \text{PDE: } \partial_\tau r(\xi, \tau) + \partial_\xi g(r(\xi, \tau)) &= z(r(\xi, \tau)), \\ \text{IC: } r(\xi, 0) &= p_k(x(\xi)), \end{aligned} \right\} \quad (17.102)$$

where

$$r(\xi, \tau) = \mathbf{Q}(x(\xi), t(\tau)), \quad g(r) = \frac{\Delta t}{\Delta x} \mathbf{F}(r), \quad z(r) = \Delta t \mathbf{S}(r). \quad (17.103)$$

Then, the time evolution of the data in domains (17.99) is performed by solving numerically the IVP (17.102). In the original paper [28], an implicit, space-time discontinuous Galerkin numerical method was used to performed this task, as explained below.

Consider the space $\mathcal{P}$ of nodal space-time polynomials $\theta_p(\xi, \tau)$ defined in $[0, 1] \times [0, 1]$, whose base is denoted by $\{\theta_1, \dots, \theta_s\}$. Here $s = (m + 1)^2$, with m the degree of the reconstruction polynomials in (17.100). There are $m + 1$ degrees of freedom, which coincides with the order of accuracy of the resulting ADER numerical scheme. Multiplying the PDE system in (17.102) by a test function $\theta_l \in \mathcal{P}$ and integrating in $[0, 1] \times [0, 1]$, we have

$$\int_0^1 \int_0^1 [(\partial_\tau r)\theta_l + (\partial_\xi g(r))\theta_l] d\xi d\tau = \int_0^1 \int_0^1 z(r)\theta_l d\xi d\tau. \quad (17.104)$$

Integrating by parts in time τ , the first term on the left hand side yields

$$\left. \begin{aligned} \int_0^1 \int_0^1 \partial_\tau r(\xi, \tau)\theta_l(\xi, \tau) d\xi d\tau &= \int_0^1 r(\xi, 1)\theta_l(\xi, 1) d\xi \\ &- \int_0^1 p_k(x(\xi))\theta_l(\xi, 0) d\xi - \int_0^1 \int_0^1 r(\xi, \tau)\partial_\tau \theta_l(\xi, \tau) d\xi d\tau. \end{aligned} \right\} \quad (17.105)$$

Substituting (17.105) into (17.104) gives

$$\left. \begin{aligned} & \int_0^1 r(\xi, 1)\theta_l(\xi, 1)s - \int_0^1 \int_0^1 r(\xi, \tau)\partial_\tau\theta_l(\xi, \tau)d\xi d\tau \\ & + \int_0^1 \int_0^1 \partial_\xi g(r(\xi, \tau))\theta_l(\xi, \tau)d\xi d\tau \\ & = \int_0^1 \int_0^1 z(r(\xi, \tau))\theta_l(\xi, \tau)d\xi d\tau + \int_0^1 p_k(x(\xi))\theta_l(\xi, 0)d\xi . \end{aligned} \right\} \quad (17.106)$$

Now, for any two functions $\phi(\xi, \tau)$ and $\psi(\xi, \tau)$ we introduce the following operators

$$\left. \begin{aligned} [\phi, \psi]_\tau &= \int_0^1 \phi(\xi, \tau)\psi(\xi, \tau)d\xi , \\ \langle \phi, \psi \rangle &= \int_0^1 \int_0^1 \phi(\xi, \tau)\psi(\xi, \tau)d\xi d\tau . \end{aligned} \right\} \quad (17.107)$$

Introducing definitions (17.107) into (17.106) yields

$$[r, \theta_l]_1 - \langle r, \partial_\tau\theta_l \rangle + \langle \partial_\xi g(r), \theta_l \rangle = \langle z(r), \theta_l \rangle + [p_k, \theta_l]_0 , \quad (17.108)$$

with

$$[p_k, \theta_l]_0 = \int_0^1 p_k(x(\xi))\theta_l(\xi, 0)d\xi . \quad (17.109)$$

We seek solutions of the form

$$r(\xi, \tau) = \sum_{p=1}^s \theta_p(\xi, \tau)\hat{r}_p , \quad (17.110)$$

with coefficients $\hat{r}_k$ yet to be determined. As we are considering a nodal base we then have

$$g(r) = g\left(\sum_{p=1}^s \theta_p(\xi, \tau)\hat{r}_p\right) \approx \sum_{p=1}^s \theta_p(\xi, \tau)g(\hat{r}_p) , \quad (17.111)$$

which is the projection of $g(r)$ on $\mathbf{V}$, see (17.99). Then

$$z(r) \approx \sum_{p=1}^s \theta_p(\xi, \tau)z(\hat{r}_p) . \quad (17.112)$$

We now define the matrices

$$\left. \begin{aligned} \mathbf{K}_{k,l}^1 &= [\theta_k, \theta_l]_1 - \langle \theta_k, \partial_\tau \theta_l \rangle, \\ \mathbf{K}_{k,l}^\xi &= \langle \partial_\xi \theta_k, \theta_l \rangle, \\ \mathbf{M}_{k,l} &= \langle \theta_k, \theta_l \rangle, \\ \mathbf{W}_l &= [p_k, \theta_l]_0 \end{aligned} \right\} \quad (17.113)$$

and vectors

$$\mathcal{R} = \begin{bmatrix} \hat{r}_1 \\ \vdots \\ \hat{r}_s \end{bmatrix}, \quad \mathcal{G}(\mathcal{R}) = \begin{bmatrix} g(\hat{r}_1) \\ \vdots \\ g(\hat{r}_s) \end{bmatrix}, \quad \mathcal{Z}(\mathcal{R}) = \begin{bmatrix} z(\hat{r}_1) \\ \vdots \\ z(\hat{r}_s) \end{bmatrix}. \quad (17.114)$$

Then, (17.108) can be written as

$$\mathbf{K}^1 \mathcal{R} + \mathbf{K}^\xi \mathcal{G}(\mathcal{R}) - \mathbf{M} \mathcal{Z}(\mathcal{R}) = \mathbf{W}. \quad (17.115)$$

This is a system of non-linear algebraic equations for $\mathcal{R}$. Standard Newton methods can be used. Here we suggest the nested Newton method proposed by Casulli and Zanolli [39], namely

$$\mathbf{K}^1 \mathcal{R}^{l+1} + \mathbf{K}^\xi \mathcal{G}(\mathcal{R}^l) - \mathbf{M} \mathcal{Z}(\mathcal{R}^{l+1}) = \mathbf{W}, \quad (17.116)$$

where l stands for the Newton iteration step. Once $\mathcal{R}$ is known, the sought coefficients are known and the polynomial representations of the form (17.110) for the evolved data on both sides of the interface are available, which are denoted by r_i and r_{i+1} respectively.

Interaction Step for Flux Evaluation in the DET Solver

The computation of the numerical flux is identical to that for the HEOC solver, see Sect. 17.4.2. One needs to define a function $\mathbf{Q}_{i+\frac{1}{2}}(\tau)$ of time along the interface $x = 0$ between V_L and V_R . To evaluate the flux integral in time one requires values at time-integration points τ_k . To find the solution at $\mathbf{Q}_{i+\frac{1}{2}}(\tau_k)$ one solves the following **conventional Riemann problem**, with piece-wise constant initial conditions $r_i(1, \tau_k)$ on the left and $r_{i+1}(0, \tau_k)$ on the right, namely

$$GRP_0 : \left\{ \begin{array}{l} \text{PDE: } \partial_t \mathbf{Q}(x, t) + \partial_x \mathbf{F}(\mathbf{Q}(x, t)) = \mathbf{0}, \\ \text{IC: } \mathbf{Q}(x, 0) = \begin{cases} r_i(1, \tau_k) & \text{if } x < 0, \\ r_{i+1}(0, \tau_k) & \text{if } x > 0. \end{cases} \end{array} \right\} \quad (17.117)$$

Denote by $\mathbf{G}_{i+\frac{1}{2}}^k(\xi/\tau)$ the self-similar solution of (17.117) in local coordinates, then the sought Godunov state is

$$\mathbf{Q}_{i+\frac{1}{2}}(\tau_k) = \mathbf{G}_{i+\frac{1}{2}}^k(0). \quad (17.118)$$

A wide range of possible solvers for (17.117) to obtain (17.118) are available, approximate, exact, upwind, and even centred solvers; see [29] and references there in.

Finally, the explicit finite volume update formula (17.3) is completed by computing the numerical flux as in (17.80). For the numerical source $\mathbf{S}_i$ see Sect. 17.4.1 and Eq. (17.73).

17.4.6 Closing Remarks on GRP_m Solvers

There are by now several solvers for the GRP_m available in the literature. Here we have presented the first practical method published and used in the ADER framework, due to Toro and Titarev (TT) [3, 4]. In the TT solver, the solution is sought directly at the cell interface, and is reduced to a sequence of $m + 1$ conventional Riemann solvers with piece-wise constant data. The TT method was inspired in the second-order method of Ben-Artzi and Falcoviz [10], known in the literature as the **GRP method**, but note that the Ben-Artzi/Falcoviz method is only second-order accurate. The modified version of the Ben-Artzi/Falcoviz solver was first communicated in the first edition of [40] in 1997, p. 399. A generalisation of this version gave rise to ADER schemes of arbitrary order of accuracy [1].

As already seen, another class of GRP_m solvers emerges from a re-interpretation [12] of the high-order method of Harten, Enquist, Osher and Chakravarthy [11], called here HEOC. Interestingly, in the original HEOC method, no mention of the generalised Riemann problem concept was made. In the HEOC approach the initial data left and right of the interface is evolved in time via Taylor expansions and use of the Cauchy-Kovalevskaya procedure is made to convert time derivatives to functionals of space derivatives; then the sought solution at a time τ results from interaction of evolved data at the interface via a conventional Riemann problem. The re-interpretation of HEOC as a GRP_m solver to be used in ADER was put forward by Castro and Toro [12]. The ADER method in conjunction with the HEOC solver for the GRP_m in second-order mode reproduces the MUSCL-Hancock second-order method [41].

The GRP_m solvers referred to above are **explicit** and suitable for hyperbolic balance laws with **non-stiff source terms**. There are by now GRP_m solvers for ADER that are able to deal with hyperbolic balance laws with stiff source terms. Such solvers are implicit, leading to explicit ADER methods, with locally implicit solvers to determine the numerical flux and the numerical source.

The first implicit solver is due to Dumbser, Enaux and Toro [28], the DET solver; this scheme is a numerical extension of the HEOC method, whereby the data left and

right of the interface is evolved numerically via an implicit space-time discontinuous Galerkin approach; see also [12]. DET avoids the Taylor expansion and the Cauchy-Kovalevskaya procedure. As in the HEOC solver, the actual DET solution at the interface is found by interacting the evolved data at selected time points and solving conventional Riemann problems. The implicit DET method can deal with stiff source terms and the corresponding ADER methods are then able to reconcile two conflicting requirements: stiffness and high accuracy.

A more recent implicit GRP_m solver was proposed by Toro and Montecinos [14], which still relies on the Cauchy-Kovalevskaya procedure. Even more recently, Montecinos and Balsara [37] proposed a simplification of the Montecinos-Toro solver by modifying the Cauchy-Kovalevskaya procedure. See also [16]. Other recent solvers for the generalised Riemann problem are due to Goetz and Iske [8]; Goetz and Dumbser [42], Goetz et al. [43] and Dematté et al. [33].

We have studied fully-discrete, high-order methods for systems of hyperbolic balance laws. In the next section we study fully-discrete high-order methods for systems. This class of methods was introduced in Chap. 16 for the scalar case, which the reader is encouraged to revise before proceeding further.

17.5 Semi-Discrete High Order Methods

Semi-discrete methods, applied to system of evolutionary partial differential equations (PDEs), separate the discretization in time from that in space. To advance the solution from time t_n to the next time t_{n+1} , for a given time discretization, the method relies on the solution of an initial-value problem for a system of Ordinary Differential Equations (ODEs) in time. Classical Runge-Kutta methods are potential candidates for this task. However, in view of the Godunov theorem that requires the construction of non-linear methods [44], one must devise special ODE solvers. TVD (Total Variation Diminishing) ODE solvers [45], later called strong stability preserving methods [46], are a prominent class of special ODE solvers. Here we present semi-discrete methods of up to third order of accuracy for solving non-linear systems of balance laws. Before proceeding with this section, the reader is encouraged to revise Chap. 16, which presents semi-discrete methods for the scalar case.

17.5.1 The Bases and First-Order Methods

As usual, let us start from the IBVP for a general system of hyperbolic balance laws

$$\left. \begin{array}{l} \text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}) , \quad x \in [a, b] , \quad t > 0 , \\ \text{ICs: } \mathbf{Q}(x, 0) = \mathbf{Q}^{(0)}(x) , \quad x \in [a, b] , \\ \text{BCs: } \mathbf{Q}(a, t) = \mathbf{B}_L(t) , \quad \mathbf{Q}(b, t) = \mathbf{B}_R(t) , \quad t \geq 0 , \end{array} \right\} \quad (17.119)$$

with $x \in [a, b]$ and computational cells $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$, with prescribed initial data at time t_n . Integrating the PDEs in (17.119) in I_i with respect to x gives an integral form of the PDEs, namely

$$\frac{d}{dt} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{Q}(x, t) dx = \mathbf{F}(\mathbf{Q}(x_{i-\frac{1}{2}}, t)) - \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}, t)) + \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{S}(\mathbf{Q}(x, t)) dx . \quad (17.120)$$

See Chap. 8 for background on integral forms of hyperbolic balance laws. We now adopt the following definition for the cell-integral average in a generic cell I_i at time t

$$\bar{\mathbf{Q}}_i(t) = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{Q}(x, t) dx , \quad (17.121)$$

with $\Delta x = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}$. We also define the spatial operator

$$\mathbf{L}(\bar{\mathbf{Q}}_i(t)) = -\frac{1}{\Delta x} [\mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}, t)) - \mathbf{F}(\mathbf{Q}(x_{i-\frac{1}{2}}, t))] \quad (17.122)$$

and the cell-averaged source

$$\bar{\mathbf{S}}_i(t) = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \mathbf{S}(\mathbf{Q}(x, t)) dx . \quad (17.123)$$

For convenience we also adopt the notation $\mathbf{Q}_i^n = \bar{\mathbf{Q}}_i(t_n)$. Substituting (17.121), (17.122) and (17.123) into (17.120) yields the following system of ordinary differential equations in time

$$\frac{d}{dt} \bar{\mathbf{Q}}_i(t) = \mathbf{L}(\bar{\mathbf{Q}}_i(t)) + \bar{\mathbf{S}}_i(t) . \quad (17.124)$$

To fully determine the ODE system (17.124) we need to define a **numerical flux** corresponding to the operator $\mathbf{L}(\bar{\mathbf{Q}}_i(t))$ in (17.122) and a **numerical source** corresponding to $\bar{\mathbf{S}}_i(t)$ in (17.123).

The Numerical Flux and the Numerical Source

To define the numerical flux at the interface position $x_{i+\frac{1}{2}}$ one relates the set of cell averages $\{\mathbf{Q}_i^n\}$ at time $t = t_n$ to two limiting intercell boundary values $\mathbf{Q}_{i+\frac{1}{2}}^-$ (left of the interface) and $\mathbf{Q}_{i+\frac{1}{2}}^+$ (right of the interface). Such limiting intercell values, or boundary extrapolated values, will arise from spatial reconstruction polynomials based on cell averages $\{\mathbf{Q}_i^n\}$. In the special case of no reconstruction, the case associated to first-order methods, such limiting values are simply $\mathbf{Q}_i^n$ and $\mathbf{Q}_{i+1}^n$ respectively. It is customary to define the numerical flux as

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}_{i+\frac{1}{2}}(\mathbf{Q}_{i+\frac{1}{2}}^-, \mathbf{Q}_{i+\frac{1}{2}}^+) \approx \mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}, t)) , \quad (17.125)$$

an approximation to $\mathbf{F}(\mathbf{Q}(x_{i+\frac{1}{2}}, t))$ right at the interface. Recall that a useful numerical flux function $\mathbf{F}_{i+\frac{1}{2}}(\mathbf{V}_L, \mathbf{V}_R)$ is required to satisfy the following properties

1. **Monotonicity:** in the scalar case $\mathbf{F}_{i+\frac{1}{2}}$ is required to be a monotone increasing function of its first argument $\mathbf{V}_L$ and monotone decreasing in its second argument $\mathbf{V}_R$. From Chap. 12 recall that this is only a necessary condition.

2. **Consistency:**

$$\mathbf{F}_{i+\frac{1}{2}}(\mathbf{V}, \mathbf{V}) = \mathbf{F}(\mathbf{V}) . \quad (17.126)$$

3. **Lipschitz continuity**, or still better, continuity.

Based on the above considerations the spatial operator (17.122) is replaced by

$$\mathbf{L}(\mathbf{Q}^n) = -\frac{1}{\Delta x} [\mathbf{F}_{i+\frac{1}{2}}(\mathbf{Q}_{i+\frac{1}{2}}^-, \mathbf{Q}_{i+\frac{1}{2}}^+) - \mathbf{F}_{i-\frac{1}{2}}(\mathbf{Q}_{i-\frac{1}{2}}^-, \mathbf{Q}_{i-\frac{1}{2}}^+)] . \quad (17.127)$$

The notation $\mathbf{Q}^n$ for the argument of $\mathbf{L}$ conveys the fact that $\mathbf{L}$ may depend on several neighbouring values in the set of cell averages $\{\mathbf{Q}_i^n\}$. In the first-order case $\mathbf{L}(\mathbf{Q}^n) = \mathbf{L}(\mathbf{Q}_{i-1}^n, \mathbf{Q}_i^n, \mathbf{Q}_{i+1}^n)$. As to the **numerical source**, this is simply given by an approximation to the integral in (17.123).

We are now in a position to state the numerical scheme for the system of hyperbolic balance laws in (17.119) in terms of the solution of the following initial-value problem involving a system of ODEs in time, namely

$$\left. \begin{array}{l} \text{ODEs: } \frac{d}{dt} \bar{\mathbf{Q}}_i(t) = \mathbf{L}(\mathbf{Q}^n) + \bar{\mathbf{S}}_i , \\ \text{ICs: } \bar{\mathbf{Q}}_i(t_n) = \mathbf{Q}_i^n . \end{array} \right\} \quad (17.128)$$

Consider the first-order explicit Euler method for solving the ODE system in (17.128). A first-order approximation to the time derivative in (17.128) gives

$$\frac{d}{dt} \bar{\mathbf{Q}}_i(t) \approx \frac{\mathbf{Q}_i^{n+1} - \mathbf{Q}_i^n}{\Delta t} . \quad (17.129)$$

Such approximation gives the explicit first-order Euler scheme

$$\mathbf{Q}_i^{n+1} = \mathbf{Q}_i^n + \Delta t \mathbf{L}(\mathbf{Q}^n) + \Delta t \bar{\mathbf{S}}_i \quad (17.130)$$

to update $\mathbf{Q}_i^n$ to $\mathbf{Q}_i^{n+1}$ by a time step Δt . In the absence of the source term, $\bar{\mathbf{S}}_i = 0$, the resulting method for the PDE system in (17.119) is the familiar first-order conservative scheme.

In the next section we construct higher-order semi-discrete methods by means of high-order solvers for ODEs.

17.5.2 TVD Runge-Kutta Schemes

Here we consider the initial-value problem of the form (17.128) for a first-order ODE system, where the unknown, denoted as $\mathbf{Q}(t)$, is a vector-valued function of time t and, for convenience, the source term has been provisionally neglected. In what follows we introduce two ODE solvers of the Runge-Kutta type that satisfy the TVD property. For details on their construction and properties see [45, 46].

Second-Order Scheme (Heun's Method)

A two-stage, second-order TVD Runge-Kutta scheme to solve (17.128) is the following

$$\left. \begin{aligned} \text{Stage 1: } \mathbf{Q}^{(1)} &= \mathbf{Q}^n + \Delta t \mathbf{L}(\mathbf{Q}^n) , \\ \text{Stage 2: } \mathbf{Q}^{(2)} &= \mathbf{Q}^{(1)} + \Delta t \mathbf{L}(\mathbf{Q}^{(1)}) , \\ \text{Solution: } \mathbf{Q}^{n+1} &= \frac{1}{2}(\mathbf{Q}^n + \mathbf{Q}^{(2)}) . \end{aligned} \right\} \quad (17.131)$$

Note that at each stage one applies the first-order explicit Euler scheme (17.130).

Numerical Flux

For stage 1 the numerical flux is obtained as follows:

1. **Reconstruction.** Perform a non-linear spatial reconstruction to produce first-degree polynomials

$$\mathbf{P}_i(x) = \mathbf{Q}_i^n + (x - x_i) \mathbf{\Delta}_i , \quad (17.132)$$

where $\mathbf{\Delta}_i$ is the slope vector in cell $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$. Here we assume the slope vector has been obtained from a non-linear reconstruction procedure, such as ENO, WENO or even from a TVD method. See Chap. 15.

2. **Interface Riemann problem.** Consider the local Riemann problem

$$\left. \begin{aligned} \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) &= \mathbf{0} , \\ \mathbf{Q}(x, 0) &= \begin{cases} \mathbf{P}_i(x_{i+\frac{1}{2}}^+) = \mathbf{Q}_{i+\frac{1}{2}}^- = \mathbf{Q}_i^n + \Delta x_i & \text{if } x < x_{i+\frac{1}{2}} , \\ \mathbf{P}_{i+1}(x_{i+\frac{1}{2}}^-) = \mathbf{Q}_{i+\frac{1}{2}}^+ = \mathbf{Q}_{i+1}^n - \Delta x_{i+1} & \text{if } x > x_{i+\frac{1}{2}} . \end{cases} \end{aligned} \right\} \quad (17.133)$$

3. **Numerical flux.** Denote the solution of (17.133) by $\mathbf{G}_{i+\frac{1}{2}}(x/t)$ in local coordinates. Then the Godunov flux is

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}(\mathbf{G}_{i+\frac{1}{2}}(0)) . \quad (17.134)$$

There are several suitable numerical fluxes than can be used here, including centred fluxes, such as the FORCE flux. See Chap. 13 for more choices.

Having computed the intercell numerical fluxes $\mathbf{F}_{i+\frac{1}{2}}$ for all interfaces, the operator $\mathbf{L}$ at time t for each cell I_i is defined as

$$\mathbf{L} = -\frac{1}{\Delta x}(\mathbf{F}_{i+\frac{1}{2}} - \mathbf{F}_{i-\frac{1}{2}}). \quad (17.135)$$

Now, stage 1 in (17.131) can be performed to obtain $\{\mathbf{Q}_i^{(1)}\}$. Stage 2 in (17.131) is performed in an entirely analogous manner by applying (17.130) to obtain $\{\mathbf{Q}_i^{(2)}\}$. The final solution $\{\mathbf{Q}_i^{n+1}\}$ results from averaging in (17.131).

Scheme (17.131) looks exceeding simple. Note however, that each stage requires a non-linear reconstruction step and solution of a row of Riemann problems for computing the numerical fluxes. Hence, the second-order semi-discrete method (17.131) requires two reconstruction procedures and two rows of (conventional) Riemann problems for the numerical fluxes.

Third-Order TVD Runge-Kutta Scheme

A three-stage, third order TVD Runge-Kutta method is the following

$$\left. \begin{array}{ll} \text{Stage 1:} & \mathbf{Q}^{(1)} = \mathbf{Q}^n + \Delta t \mathbf{L}(\mathbf{Q}^n), \\ \text{Stage 2:} & \hat{\mathbf{Q}}^{(2)} = \mathbf{Q}^{(1)} + \Delta t \mathbf{L}(\mathbf{Q}^{(1)}), \\ \text{Averaging:} & \mathbf{Q}^{(2)} = \frac{3}{4}\mathbf{Q}^n + \frac{1}{4}\hat{\mathbf{Q}}^{(2)}, \\ \text{Stage 3:} & \mathbf{Q}^{(3)} = \mathbf{Q}^{(2)} + \Delta t \mathbf{L}(\mathbf{Q}^{(2)}), \\ \text{Solution:} & \mathbf{Q}^{n+1} = \frac{1}{3}\mathbf{Q}^n + \frac{2}{3}\mathbf{Q}^{(3)}. \end{array} \right\} \quad (17.136)$$

At each stage, one applies the explicit first-order Euler operator (17.130). Now, each evaluation of the operator $\mathbf{L}(\mathbf{Q})$ implies a sequence of Riemann problems and a reconstruction procedure involving polynomials of degree $m = 2$. Therefore this method requires three reconstruction procedures and three rows of (conventional) Riemann problems per time step.

The TVD property: In Chap. 16 we discussed the TVD property in the context of ODE solvers and presented the proof for the second-order scheme (17.131). The proof for the third-order scheme (17.136) was left to the reader as an exercise.

In the next section we present illustrative numerical results from some of the methods studied.

17.6 Test Problems and Numerical Results

In this section we illustrate the performance of some of the high-order numerical methods presented in this chapter for solving non-linear systems. Table 17.1 shows the four selected schemes to be tested. The methods will be assessed in two ways.

Table 17.1 Selected high-order numerical schemes tested for the system of non-linear blood flow equations. Schemes 1 to 3 use the fully discrete ADER approach along with the DET method for solving the generalised Riemann problem GRP_m . These schemes also require a solver for the classical Riemann problem GRP_0 , for which two solvers are used, HLLC in schemes 1 and 2, and DOT for scheme 3. Scheme 4 is the fully discrete ADER approach in conjunction with the TT solver for the generalised Riemann problem GRP_m , along with the two-rarefaction Riemann solver (TRRS) for the leading term in the LeFloch-Raviart expansion obtained by solving the classical Riemann problem GRP_0 . Schemes 2 and 3 are able to solve problems with stiff source terms. All four schemes use the WENO-DK spatial reconstruction method presented in Chap. 15

Scheme	Name	Source term capability	Orders
1	ADER-DET+HLLC	Non-Stiff	2 to 5
2	ADER-DET+HLLC	Stiff	2 to 5
3	ADER-DET+DOT	Stiff	2 to 5
4	ADER-TT+TRRS	Non-Stiff	2 to 5

First, a test problem with smooth, whose exact solution is used in order to compute errors and to calculate convergence rates in order to verify that the theoretically expected orders of accuracy are attained. This solution is also used to asses the efficiency of the methods, that is their performance in the error versus CPU cost graph, for a prescribed fixed error. The exact, smooth solution is constructed via the method of manufactured solutions.

Another way of assessing the performance of the high-order numerical methods is through test problems with discontinuous solutions, such as elastic jumps and contact discontinuities. Numerical solution profiles from these test problems will be compared with exact solution profiles.

The four methods to be tested are described in Table 17.1. They are all in the class of fully discrete ADER methods, with two solvers for the generalised Riemann problem GRP_m , DET and TT, and three schemes for solving the classical Riemann problem GRP_0 , namely HLLC, DOT and the two-rarefaction Riemann solver TRRS. Schemes 2 and 3 are locally implicit and thus suitable for solving problems with stiff source terms. All four methods are assessed for orders of accuracy from 2nd to 5th in both space and time.

17.6.1 Convergence Rates and Efficiency

The main purpose of this section is to first verify empirically that the high-order methods presented in this chapter attain the theoretically expected convergence rates for smooth solutions. We also study efficiency and show that high-order methods are orders-of-magnitude more efficient than low-order methods, for a fixed error. To assess convergence rates one needs an exact solution, in order to define errors. But exact solutions to equations of interest in realistic applications are rare. In what follows we introduce a method to obtain exact solutions.

Manufactured Exact Solutions

A common practice to resolve the difficulty of defining errors is to adopt the *method of manufactured solutions*, as follows. One first assumes the original PDE system to be of the form

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}) , \quad (17.137)$$

where $\mathbf{S}(\mathbf{Q})$ is the source term, which could be admitted to be zero. Then one *invents a smooth function* $\tilde{\mathbf{Q}}(x, t)$ and inserts it in (17.137). As $\tilde{\mathbf{Q}}(x, t)$ does not necessarily satisfy (17.137), one would obtain a new balance law with a new source term, namely

$$\partial_t \tilde{\mathbf{Q}} + \partial_x \mathbf{F}(\tilde{\mathbf{Q}}) = \mathbf{S}(\tilde{\mathbf{Q}}) + \mathbf{S}_{new}(\tilde{\mathbf{Q}}) := \tilde{\mathbf{S}}(\tilde{\mathbf{Q}}) . \quad (17.138)$$

Then the function $\tilde{\mathbf{Q}}(x, t)$ is the exact solution of the new balance law

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \tilde{\mathbf{S}}(\mathbf{Q}, x, t) . \quad (17.139)$$

Note the explicit dependence of the new source term on the independent variables x and t .

The methods are tested by solving system (17.139) numerically to compute errors with respect to $\tilde{\mathbf{Q}}(x, t)$ and hence convergence rates. It is expected that the method to solve (17.139) being tested is able to deal with the source term $\tilde{\mathbf{S}}(\mathbf{Q})$ in a way that the accuracy for the full PDE system is preserved. The method of manufactured solutions will not be useful for assessing convergence rates if source terms are not treated to the expected high order of accuracy.

As an example, let us consider the scalar homogenous conservation law

$$\partial_t q + \partial_x f(q) = 0 , \quad f(q) = \lambda q , \quad \lambda : \text{constant} . \quad (17.140)$$

Assume the manufactured solution $\tilde{q}(x, t) = \sin(xt)$. It is easy to verify that the new balance law for which $\tilde{q}(x, t) = \sin(xt)$ is the exact solution, is given by

$$\partial_t q + \partial_x f(q) = (x + \lambda t) \cos(xt) = \tilde{s}(q, x, t) . \quad (17.141)$$

Another example is furnished by the scalar balance law

$$\partial_t q + \partial_x f(q) = \beta q , \quad f(q) = \lambda q , \quad \beta : \text{constant} . \quad (17.142)$$

Assuming again the manufactured solution to be $\tilde{q}(x, t) = \sin(xt)$, the new balance law for which $\tilde{q}(x, t) = \sin(xt)$ is the exact solution is given by

$$\partial_t q + \partial_x f(q) = \beta q + (x + \lambda t) \cos(xt) = \tilde{s}(q, x, t) . \quad (17.143)$$

We note that the choice of $\tilde{q}(x, t)$ is completely arbitrary, though some common sense is advisable. What matters is smoothness, for the purpose of computing convergence rates.

Smooth-Solution Test Problem for Blood Flow

We now turn to the specific problem at hand concerning the augmented blood flow equations written in conservation-law form

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{0} , \quad (17.144)$$

where

$$\mathbf{Q} = \begin{bmatrix} A \\ Au \\ A\eta \end{bmatrix} , \quad \mathbf{F}(\mathbf{Q}) = \begin{bmatrix} Au \\ Au^2 + \gamma A^{3/2} \\ Au\eta \end{bmatrix} , \quad (17.145)$$

with

$$\gamma = \frac{4\sqrt{\pi} E h_0}{9\rho A_0} \quad (17.146)$$

and tube law

$$p = K \left[\left(\frac{A}{A_0} \right)^{\frac{1}{2}} - 1 \right] , \quad \text{with } K = \frac{4}{3} \frac{E h_0}{R_0} . \quad (17.147)$$

We use the method of manufactured solutions to obtain a reference exact solution to a modified, inhomogeneous system, as described previously. We first prescribe the vector-valued function $\tilde{\mathbf{Q}}$ in terms of the following variables:

$$\left. \begin{aligned} \tilde{A} &= A_0 + b_0 A_0 \sin \left(2\pi \frac{x}{L} \right) \cos \left(2\pi \frac{t}{T} \right) , \\ \tilde{q} &= q_0 - b_0 A_0 \frac{L}{T} \cos \left(2\pi \frac{x}{L} \right) \sin \left(2\pi \frac{t}{T} \right) , \\ \tilde{\eta} &= \frac{1}{2} + \frac{1}{10} \sin \left(2\pi \frac{x}{L} \right) \cos \left(2\pi \frac{t}{T} \right) . \end{aligned} \right\} \quad (17.148)$$

Here $q = Au$ is flow, as usual. By inserting $\tilde{\mathbf{Q}}$ into the homogeneous 3×3 system (17.144)–(17.147) representing the blood flow equations, one verifies that $\tilde{\mathbf{Q}}$ satisfies exactly an inhomogeneous system, with algebraic source term $\tilde{S}(\mathbf{Q}, x, t)$. For the case of interest here, the source term $\tilde{S}(\mathbf{Q}, x, t)$ is calculated using an algebraic manipulator; its final expression is rather involved and is therefore omitted. The interested reader is encouraged to calculate $\tilde{S}(\mathbf{Q}, x, t)$ either by hand or via a symbolic manipulator.

Then, in order to calculate convergence rates, one solves numerically the inhomogeneous system

$$\partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \tilde{S}(\mathbf{Q}, x, t), \quad (17.149)$$

for which $\tilde{\mathbf{Q}}$ is the exact solution and hence errors between the numerically computed solution and the exact solution can be defined. Note that in the present case $\tilde{\mathbf{Q}}$ is periodic in the spatial interval $[0, L]$. The parameters of the test problem are given as follows:

1. Reference radius: $R_0 = 0.01$ m ($A_0 = \pi R_0^2$)
2. Vessel length: $L = 1$ m
3. Young modulus: $E = 1.2 \times 10^6$ Pa
4. Wall thickness: $h_0 = 0.5 \times 10^{-3}$ m
5. Baseline flow: $q_0 = 100 \times 10^{-6}$ m³/s
6. Perturbation constant: $b_0 = 0.1$
7. Characteristic time: $T = 1$ s
8. Output time: $T_{\text{out}} = 0.5$ s

Computed Convergence Rates

For background on empirical convergence rates see Chaps. 12 and 16.

Here, Tables 17.5 to 17.12 show empirical convergence rates for Schemes 1 and 2 of Table 17.1. For each table, column 1 shows the unknown for which convergence rates are computed. Column 2 shows the sequence of meshes M . Then, each of three pairs (L_k, p_k) shows the error in the L_k norm and the corresponding convergence rate p_k . CPU times are shown in the last column.

Tables 17.5, 17.10, 17.11 and 17.8 show convergence rates for Scheme 1 in Table 17.1, for orders 2 to 5. The theoretically expected convergence rates are attained in all cases, for all three norms and for all three unknowns.

Tables 17.9, 17.10, 17.11 and 17.12 show convergence rates for Scheme 2 in Table 17.1, for orders 2 to 5. The theoretically expected convergence rates are attained satisfactorily for all orders.

Efficiency: Error Against Cost

Efficiency graphs displaying errors against CPU cost for Schemes 1 and 2 of Table 17.1, for a prescribed fixed error of 10^{-13} , are shown in Fig. 17.9. Results from Scheme 1 (non-stiff) are shown on the top frame and those from Scheme 2 (stiff) are shown in the bottom frame. For each scheme, results are shown for three orders of accuracy, namely 2nd, 3rd and 5th. The corresponding CPU costs are shown on the top right corner of each frame. It is seen that higher-order methods are orders-of-magnitude more efficient than the lower-order schemes.

17.6.2 Test Problems with Discontinuous Solutions

Here we assess the performance of some of the high-order methods of this chapter, for test problems with exact solution including both smooth and discontinuous features. Next we define the test problems.

Riemann Problem Tests

Initial conditions for five test problems are defined in Table 17.2, given in terms of physical variables A , u and η . Columns 2 to 4 give the initial data to the left of the position x_0 of the initial discontinuity, while columns 5 to 7 give respective values on the right hand side of x_0 . The output time T_{out} at which results are displayed is given in column 8, while column 9 identifies the type of wave pattern in the solution. The last column identifies the kind of blood vessel considered, artery or vein. The initial conditions for Tests 4 and 5 for veins are the same as for arteries, Tests 1 and 3 respectively. But the biomechanical and computational parameters are different, in order to account for the fact that arteries are different from veins. Table 17.3 shows the values of parameters used for arteries, while Table 17.4 shows values for veins. The features of the solution for each test problem are fully described in Chap. 13, which the reader is encouraged to consult.

Table 17.2 Initial data for five test problems for the augmented blood flow equations. SI units are used. The 9th column describes the wave pattern that emerges from the exact solution, where R stands for rarefaction, C stands for contact discontinuity and S stands for shock. The last column identifies the type of blood vessel, artery or vein

Test	A_L	u_L	ψ_L	A_R	u_R	ψ_R	T_{out}	Wave pattern	Blood vessel
1	3.5×10^{-4}	0.0	1.0	3.0×10^{-4}	0.0	0.0	0.05	RCS	Artery
2	10.0×10^{-4}	0.0	1.0	1.0×10^{-4}	0.0	0.0	0.04	R(sonic)CS	Artery
3	3.14×10^{-4}	-0.5	1.0	3.14×10^{-4}	0.5	0.0	0.05	RCR	Artery
4	3.5×10^{-4}	0.0	1.0	3.0×10^{-4}	0.0	0.0	0.05	RCS	Vein
5	3.14×10^{-4}	-0.5	1.0	3.14×10^{-4}	0.5	0.0	0.08	RCR	Vein

Table 17.3 Parameters for tests 1 to 3 in Table 17.2 for arteries. SI units are used. Here x_0 denotes position of initial discontinuity; α is the Coriolis coefficient; h_0 is vessel wall thickness; A_0 is equilibrium cross-sectional area; E is Young's modulus; ν is Poisson ratio and ρ is blood density

Domain length	x_0	α	h_0	A_0	E	ν	ρ	Blood vessel
0.5	0.25	1.0	5.0×10^{-4}	3.14×10^{-4}	3.0×10^5	0.5	1000	Artery

Table 17.4 Parameters for tests 4 to 5 in Table 17.2 for veins. Here x_0 denotes position of initial discontinuity; α is the Coriolis coefficient; A_0 is equilibrium cross-sectional area; K is the stiffness coefficient; ρ is blood density; m and n are tube law exponents

Domain length	x_0	α	A_0	K	ρ	m	n	Blood vessel
0.5	0.25	1.0	3.14×10^{-4}	333.0	1000	10	$-3/2$	Vein

Table 17.5 Convergence rates for Scheme 1 of Table 17.1, that is ADER with DET solver for GRP_m and HLLC solver for classical GRP_0 . Non-stiff case; theoretical order: 2

Var.	M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
$A [m^2]$	16	1.9e-07	2.0	2.2e-07	2.1	3.2e-07	2.3	1.3
	32	4.2e-08	2.2	4.9e-08	2.2	8.9e-08	1.9	4.7
	64	1.1e-08	1.9	1.2e-08	2.0	2.1e-08	2.1	18.0
	128	2.8e-09	2.0	3.1e-09	2.0	5.4e-09	2.0	70.0
$q [m^3/s]$	16	4.9e-07	2.7	8.0e-07	2.5	2.0e-06	1.7	1.3
	32	2.1e-07	1.2	2.6e-07	1.6	7.2e-07	1.4	4.7
	64	5.3e-08	2.0	6.3e-08	2.1	1.9e-07	1.9	18.0
	128	1.2e-08	2.2	1.5e-08	2.1	5.9e-08	1.7	70.0
$A\eta [m^2]$	16	3.9e-07	2.1	5.1e-07	2.1	1.3e-06	1.8	1.3
	32	9.4e-08	2.1	1.2e-07	2.1	3.2e-07	2.0	4.7
	64	2.4e-08	1.9	3.0e-08	2.0	7.4e-08	2.1	18.0
	128	6.0e-09	2.0	7.3e-09	2.0	1.9e-08	2.0	70.0

Table 17.6 Convergence rates for Scheme 1 of Table 17.1, that is ADER with DET solver for GRP_m and HLLC solver for classical GRP_0 . Non-stiff case; theoretical order: 3

Var.	M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
$A [m^2]$	16	2.1e-08	2.8	2.3e-08	2.8	3.3e-08	2.8	2.9
	32	2.7e-09	3.0	3.0e-09	3.0	4.2e-09	3.0	10.2
	64	3.3e-10	3.0	3.7e-10	3.0	5.3e-10	3.0	39.6
	128	4.2e-11	3.0	4.6e-11	3.0	6.6e-11	3.0	148.5
$q [m^3/s]$	16	1.7e-07	2.9	1.9e-07	2.9	2.7e-07	2.8	2.9
	32	2.2e-08	3.0	2.4e-08	3.0	3.4e-08	3.0	10.2
	64	2.7e-09	3.0	3.0e-09	3.0	4.3e-09	3.0	39.6
	128	3.4e-10	3.0	3.8e-10	3.0	5.4e-10	3.0	148.5
$A\eta [m^2]$	16	4.0e-08	2.6	4.5e-08	2.6	8.0e-08	2.6	2.9
	32	5.2e-09	2.9	5.9e-09	2.9	1.0e-08	3.0	10.2
	64	6.6e-10	3.0	7.5e-10	3.0	1.3e-09	3.0	39.6
	128	8.3e-11	3.0	9.4e-11	3.0	1.6e-10	3.0	148.5

Table 17.7 Convergence rates for Scheme 1 of Table 17.1, that is ADER with DET solver for GRP_m and HLLC solver for classical GRP_0 . Non-stiff case; theoretical order: 4

Var.	M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
$A [m^2]$	16	2.5e-09	4.2	2.8e-09	4.2	4.0e-09	4.1	5.2
	32	1.5e-10	4.1	1.6e-10	4.1	2.4e-10	4.1	18.3
	64	8.9e-12	4.0	9.8e-12	4.0	1.5e-11	4.0	78.0
	128	5.3e-13	4.1	5.8e-13	4.1	8.9e-13	4.0	264.2
$q [m^3/s]$	16	7.4e-09	4.2	8.2e-09	4.2	1.2e-08	4.1	5.2
	32	5.1e-10	3.9	5.7e-10	3.8	8.1e-10	3.8	18.3
	64	3.4e-11	3.9	3.8e-11	3.9	5.5e-11	3.9	78.0
	128	2.1e-12	4.0	2.4e-12	4.0	3.5e-12	4.0	264.2
$A\eta [m^2]$	16	8.2e-09	3.6	9.1e-09	3.7	1.4e-08	3.9	5.2
	32	5.1e-10	4.0	5.7e-10	4.0	8.9e-10	4.0	18.3
	64	3.2e-11	4.0	3.6e-11	4.0	5.6e-11	4.0	78.0
	128	2.0e-12	4.0	2.2e-12	4.0	3.5e-12	4.0	264.2

Table 17.8 Convergence rates for Scheme 1 of Table 17.1, that is ADER with DET solver for GRP_m and HLLC solver for classical GRP_0 . Non-stiff case; theoretical order: 5

Var.	M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
$A [m^2]$	16	6.8e-10	4.8	7.5e-10	4.9	1.1e-09	4.9	8.6
	32	2.1e-11	5.0	2.4e-11	5.0	3.4e-11	5.0	65.3
	64	6.8e-13	5.0	7.6e-13	5.0	1.2e-12	4.8	115.1
	128	2.4e-14	4.9	2.9e-14	4.7	6.9e-14	4.1	421.5
$q [m^3/s]$	16	5.2e-09	4.9	5.9e-09	4.9	8.3e-09	4.8	8.6
	32	1.7e-10	4.9	1.9e-10	5.0	2.7e-10	5.0	65.3
	64	5.6e-12	4.9	6.2e-12	4.9	9.3e-12	4.9	115.1
	128	1.6e-13	5.1	1.9e-13	5.1	4.3e-13	4.4	421.5
$A\eta [m^2]$	16	3.6e-09	4.6	4.0e-09	4.5	6.0e-09	4.2	8.6
	32	1.2e-10	4.9	1.4e-10	4.9	2.0e-10	4.9	65.3
	64	3.9e-12	5.0	4.3e-12	5.0	6.4e-12	5.0	115.1
	128	1.3e-13	4.9	1.4e-13	5.0	2.2e-13	4.9	421.5

Numerical Results

Figures 17.10 to 17.24 show computed results compared to exact solutions for all test problems, for orders of accuracy 2nd to 5th. These computational results are to be compared with those from first-order methods shown at the end of Chap. 13.

Figures 17.10, 17.11, 17.12, 17.13 and 17.14 show results for Scheme 2 of Table 17.1 (ADER-DET+HLLC, Stiff), for all five test problems of Table 17.2 and for orders of accuracy 2nd to 5th.

Figure 17.10 shows results from Scheme 1 of Table 17.1 for Test 1 (arteries) for orders of accuracy 2nd to 5th. Overall, results are very satisfactory, with sharp

Table 17.9 Convergence rates for Scheme 2 of Table 17.1, that is ADER with DET solver for GRP_m and HLLC solver for classical GRP_0 . Stiff case; theoretical order: 2

Var.	M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
$A [m^2]$	16	1.9e-07	2.0	2.2e-07	2.1	3.2e-07	2.3	3.3
	32	4.2e-08	2.2	4.9e-08	2.2	8.9e-08	1.9	9.2
	64	1.1e-08	1.9	1.2e-08	2.0	2.1e-08	2.1	26.2
	128	2.8e-09	2.0	3.1e-09	2.0	5.4e-09	2.0	108.8
$q [m^3/s]$	16	4.9e-07	2.7	8.0e-07	2.5	2.0e-06	1.7	3.3
	32	2.1e-07	1.2	2.6e-07	1.6	7.2e-07	1.4	9.2
	64	5.3e-08	2.0	6.3e-08	2.1	1.9e-07	1.9	26.2
	128	1.2e-08	2.2	1.5e-08	2.1	5.9e-08	1.7	108.8
$A\eta [m^2]$	16	3.9e-07	2.1	5.1e-07	2.1	1.3e-06	1.8	3.3
	32	9.4e-08	2.1	1.2e-07	2.1	3.2e-07	2.0	9.2
	64	2.4e-08	1.9	3.0e-08	2.0	7.4e-08	2.1	26.2
	128	6.0e-09	2.0	7.3e-09	2.0	1.9e-08	2.0	108.8

Table 17.10 Convergence rates for Scheme 2 of Table 17.1, that is ADER with DET solver for GRP_m and HLLC solver for classical GRP_0 . Stiff case; theoretical order: 3

Var.	M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
$A [m^2]$	16	2.1e-08	2.8	2.3e-08	2.8	3.3e-08	2.8	11.8
	32	2.7e-09	3.0	3.0e-09	3.0	4.2e-09	3.0	50.2
	64	3.3e-10	3.0	3.7e-10	3.0	5.3e-10	3.0	129.3
	128	4.2e-11	3.0	4.6e-11	3.0	6.6e-11	3.0	306.0
$q [m^3/s]$	16	1.7e-07	2.9	1.9e-07	2.9	2.7e-07	2.8	11.8
	32	2.2e-08	3.0	2.4e-08	3.0	3.4e-08	3.0	50.2
	64	2.7e-09	3.0	3.0e-09	3.0	4.3e-09	3.0	129.3
	128	3.4e-10	3.0	3.8e-10	3.0	5.4e-10	3.0	306.0
$A\eta [m^2]$	16	4.0e-08	2.6	4.5e-08	2.6	8.0e-08	2.6	11.8
	32	5.2e-09	2.9	5.9e-09	2.9	1.0e-08	3.0	50.2
	64	6.6e-10	3.0	7.5e-10	3.0	1.3e-09	3.0	129.3
	128	8.3e-11	3.0	9.4e-11	3.0	1.6e-10	3.0	306.0

resolution of discontinuities, without spurious oscillations in the vicinity of large spatial gradients. The resolution of the rarefaction wave (smooth) improves as the order of accuracy increases. Note however, that for this test problem dominated by discontinuous features, the gains from the higher-order methods with respect to the lower-order ones are not striking.

Figure 17.11 shows results from Scheme 2 for Test 2 for arteries, and note that the scheme in fourth-order mode fails for this test (no plot). The results shown deserve similar comments as for Test 1. The discontinuities are the dominant features and

Table 17.11 Convergence rates for Scheme 2 of Table 17.1, that is ADER with DET solver for GRP_m and HLLC solver for classical GRP_0 . Stiff case; theoretical order: 4

Var.	M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
$A [m^2]$	16	2.5e-09	4.2	2.8e-09	4.2	4.0e-09	4.1	16.9
	32	1.5e-10	4.1	1.6e-10	4.1	2.4e-10	4.1	62.5
	64	8.9e-12	4.0	9.8e-12	4.0	1.5e-11	4.0	194.9
	128	5.3e-13	4.1	5.8e-13	4.1	9.0e-13	4.0	746.1
$q [m^3/s]$	16	7.4e-09	4.2	8.2e-09	4.2	1.2e-08	4.1	16.9
	32	5.1e-10	3.9	5.7e-10	3.8	8.1e-10	3.8	62.5
	64	3.4e-11	3.9	3.8e-11	3.9	5.5e-11	3.9	194.9
	128	2.1e-12	4.0	2.4e-12	4.0	3.5e-12	4.0	746.1
$A\eta [m^2]$	16	8.2e-09	3.6	9.1e-09	3.7	1.4e-08	3.9	16.9
	32	5.1e-10	4.0	5.7e-10	4.0	8.9e-10	4.0	62.5
	64	3.2e-11	4.0	3.6e-11	4.0	5.6e-11	4.0	194.9
	128	2.0e-12	4.0	2.2e-12	4.0	3.5e-12	4.0	746.1

Table 17.12 Convergence rates for Scheme 2 of Table 17.1, that is ADER with DET solver for GRP_m and HLLC solver for classical GRP_0 . Stiff case; theoretical order: 5

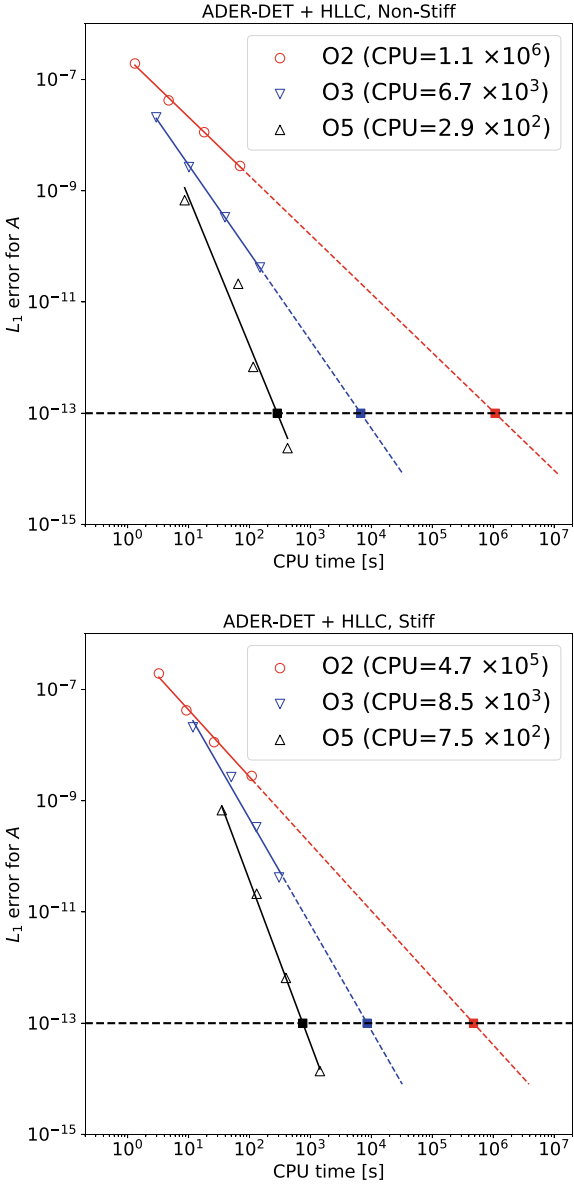
Var.	M	L_1	p_1	L_2	p_2	L_∞	p_∞	CPU [s]
$A [m^2]$	16	6.8e-10	4.8	7.5e-10	4.9	1.1e-09	4.9	35.1
	32	2.1e-11	5.0	2.4e-11	5.0	3.4e-11	5.0	131.1
	64	6.5e-13	5.0	7.2e-13	5.0	1.1e-12	5.0	396.2
	128	1.4e-14	5.6	1.5e-14	5.6	2.7e-14	5.3	1430.9
$q [m^3/s]$	16	5.2e-09	4.9	5.9e-09	4.9	8.3e-09	4.8	35.1
	32	1.7e-10	4.9	1.9e-10	5.0	2.7e-10	5.0	131.1
	64	5.3e-12	5.0	5.9e-12	5.0	8.3e-12	5.0	396.2
	128	1.3e-13	5.4	1.4e-13	5.3	2.5e-13	5.1	1430.9
$A\eta [m^2]$	16	3.6e-09	4.6	4.0e-09	4.5	6.0e-09	4.2	35.1
	32	1.2e-10	4.9	1.4e-10	4.9	2.0e-10	4.9	131.1
	64	3.9e-12	5.0	4.3e-12	5.0	6.5e-12	5.0	396.2
	128	1.2e-13	5.0	1.3e-13	5.0	2.0e-13	5.0	1430.9

increasing the order of accuracy does not visibly improve the computational results, as compared with the exact solution.

Figure 17.12 shows results from Scheme 2 for Test 3 for arteries. Note the perfect resolution of the stationary contact discontinuity, for all orders of accuracy.

Figures 17.13 and 17.14 show results from Scheme 2 for Tests 4 and 5 for veins for orders of accuracy 2nd to 5th. The performance of the methods is similar to that for previous tests. Note again the perfect resolution of the stationary contact discontinuity for Test 5.

Fig. 17.9 Efficiency. Error against cost for Schemes 1 (top) and 2 (bottom) of Table 17.1 for a prescribed fixed error of 10^{-13} . For each scheme, results are shown for three orders of accuracy, namely 2nd, 3rd and 5th. The corresponding CPU costs are shown on the top right corner. It is seen that higher-order methods are orders of magnitude more efficient than low-order schemes



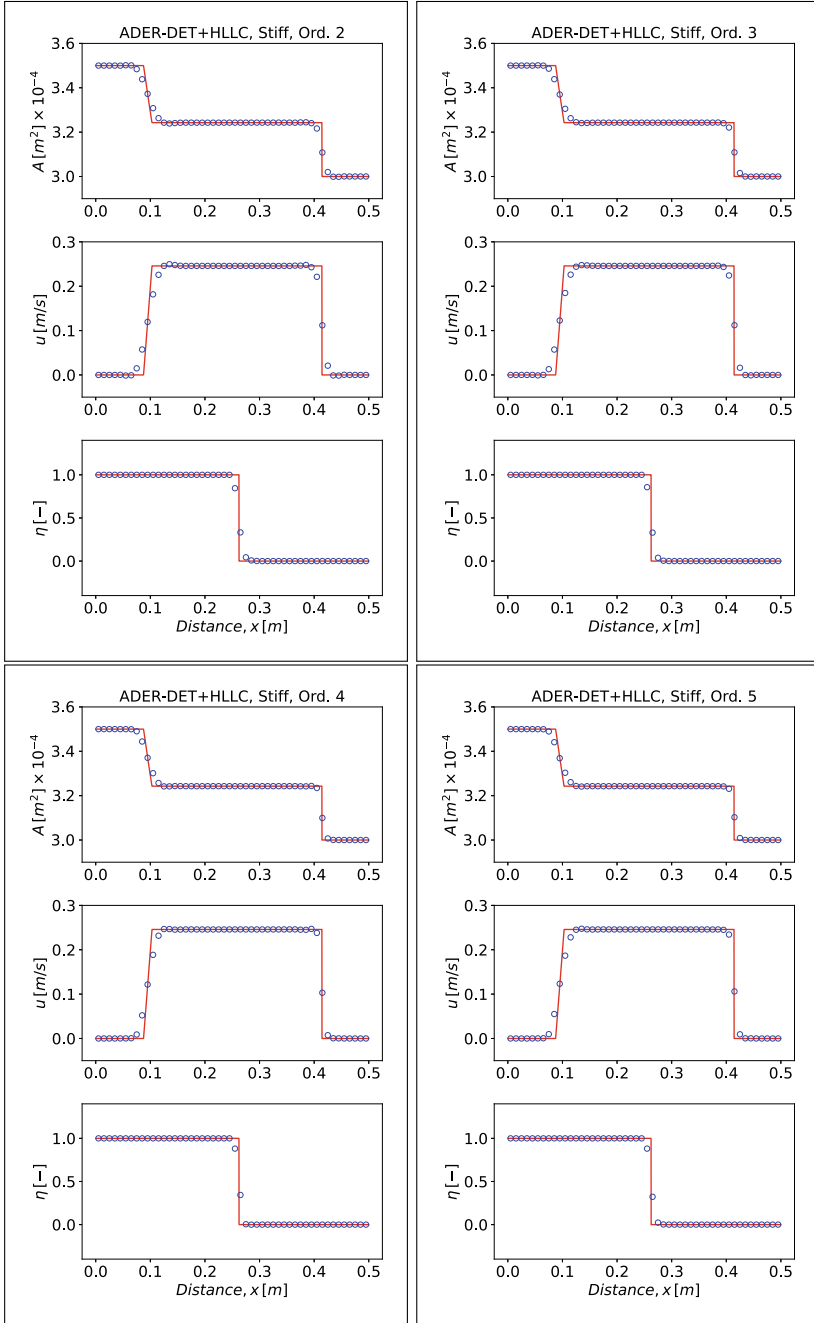


Fig. 17.10 Test 1 (arteries). Numerical solution from Scheme 2 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cf} = 0.9$

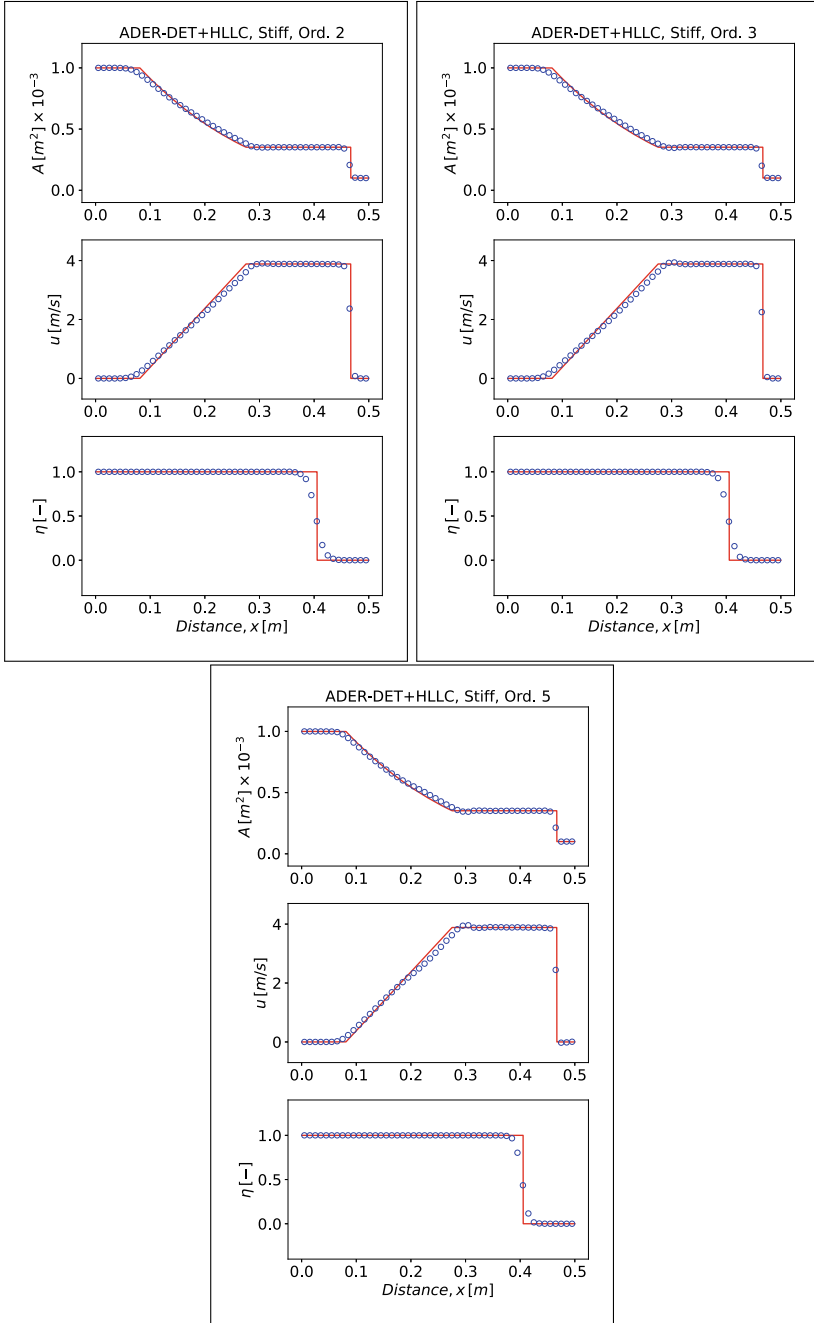


Fig. 17.11 Test 2 (arteries). Numerical solution from Scheme 2 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

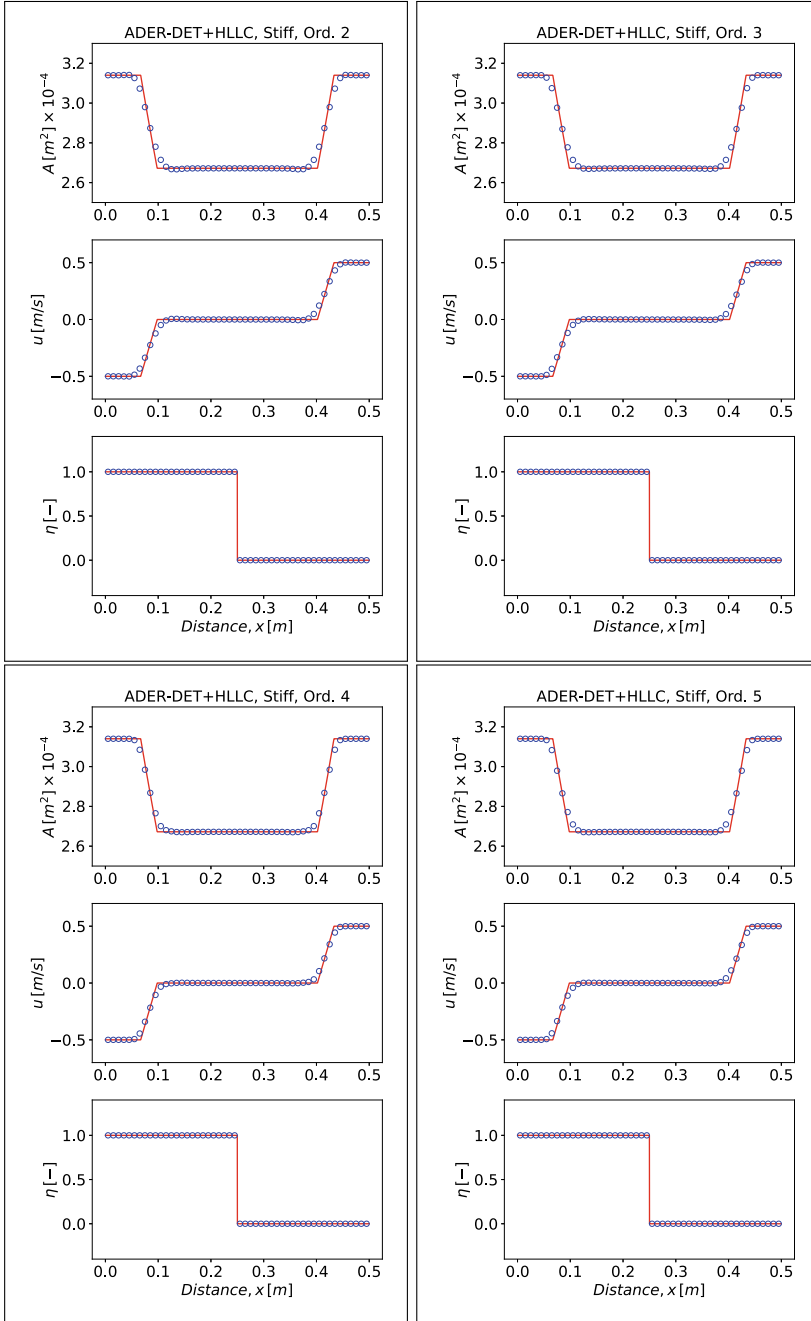


Fig. 17.12 Test 3 (arteries). Numerical solution from Scheme 2 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

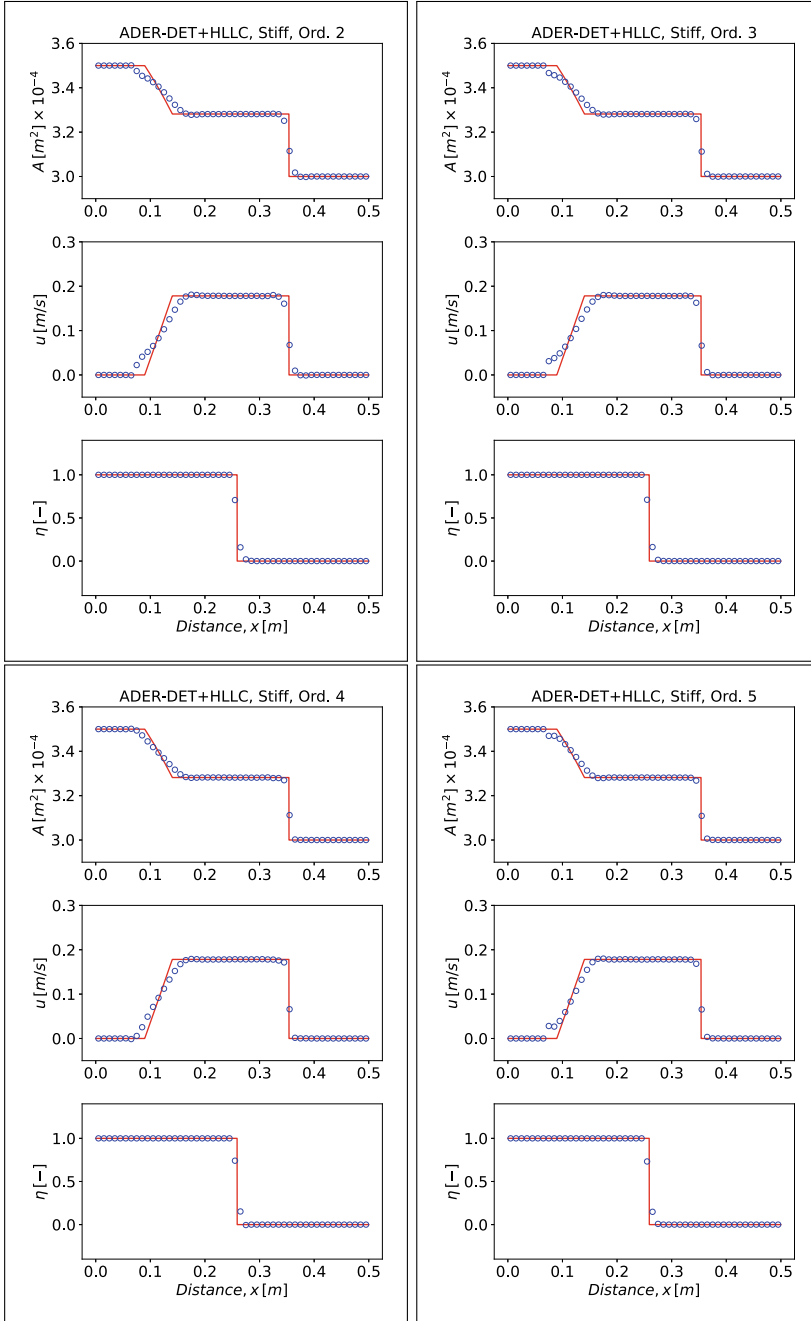


Fig. 17.13 Test 4 (veins). Numerical solution from Scheme 2 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.4. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

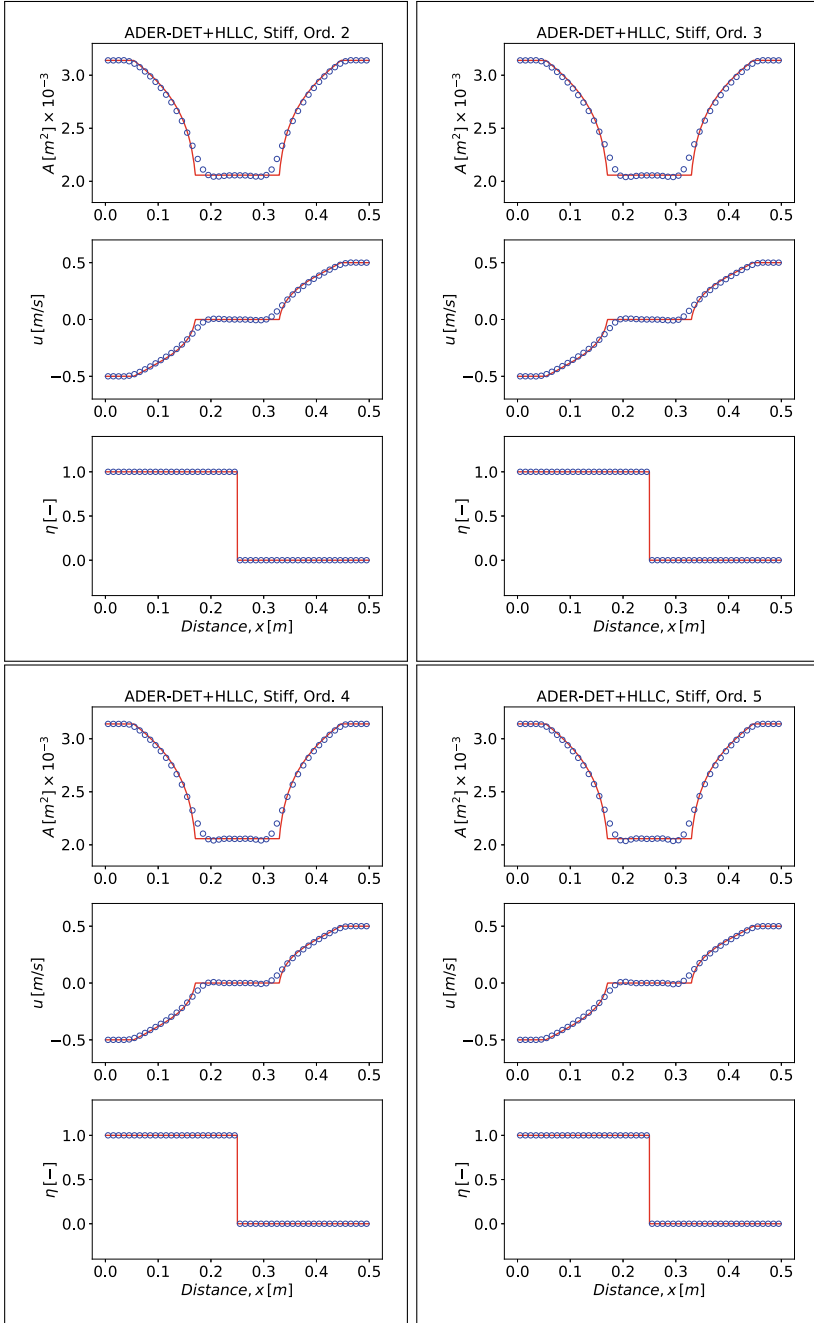


Fig. 17.14 Test 5 (veins). Numerical solution from Scheme 2 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.4. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

Figures 17.15, 17.16, 17.17, 17.18 and 17.19 show results from Scheme 3 of Table 17.1, for all five test problems. These are similar to those from Scheme 2. For Test 2, Scheme 3 in fourth order mode fails (no plot), as did Scheme 2 for the same test.

Figures 17.20, 17.21, 17.22, 17.23 and 17.24 show results from Scheme 4 of Table 17.1, for all five test problems. These are similar to those from Schemes 2 and 3. Note however, that in fourth order mode Scheme 4, though oscillatory for the unknowns A and u , does not crush, as was the case for Schemes 2 and 3 for the same test.

As already pointed out, for these discontinuity-dominated test problems, the gains in accuracy from higher order schemes, relative to the 2nd order method, are not striking. A clear advantage of the higher-order methods would be seen for problems involving solutions with highly varying but smooth regions, separated by discontinuities. Overall, schemes 2, 3 and 4 give similar results, with scheme 4, visually displaying a slight advantage. Scheme 4 runs for all test problems, for all orders of accuracy, and displays a slight advantage for the stationary contact discontinuity.

17.7 Concluding Remarks and Further Reading

In this chapter we have presented two classes of high-order, non-linear numerical methods to compute approximate solutions to hyperbolic systems of balance laws. The common ingredient for these schemes is the polynomial representation of cell integral averages of the approximate solution, via a non-linear spatial reconstruction scheme. First, the class of **fully-discrete**, one-step ADER methods was presented, along with several solvers for the generalised Riemann problem GPR_m , including explicit and implicit solvers. The implicit solvers are needed for approximating balance laws with stiff sources terms. The second-class of high-order methods presented are **semi-discrete**, multi-step schemes that rely on solvers for systems of Ordinary Differential Equations of the TVD Runge-Kutta type. Numerical examples to illustrate the performance of some of the high-order methods have also been presented.

For further reading we recommend the following selected references: Toro, Millington and Nejad [1]; Toro and Titarev [3]; Titarev and Toro [4]; Gottlieb, Shu and Tadmor [46]; Dumbser, Enaux and Toro [28]; Dumbser, Balsara, Toro and Munz [47]; Shu [45, 48, 49].

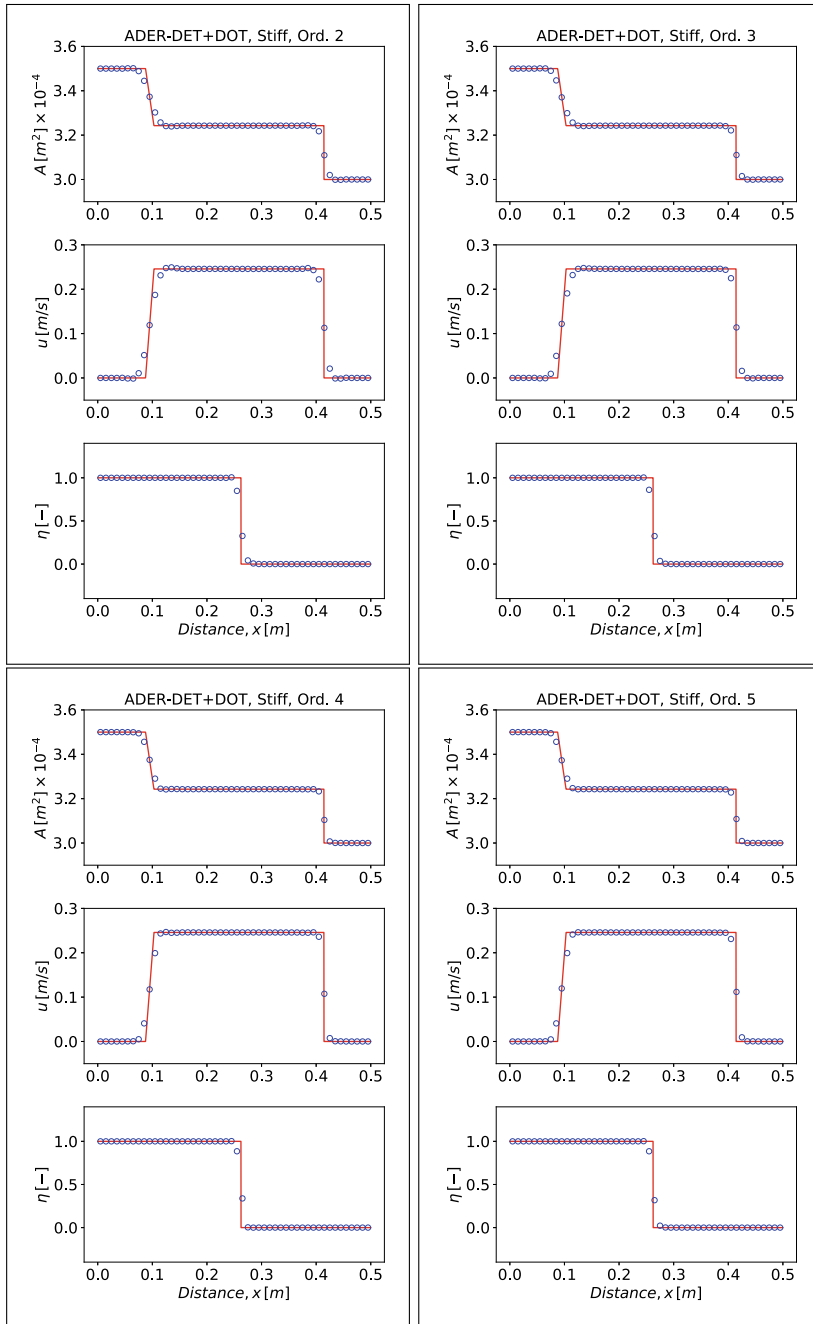


Fig. 17.15 Test 1 (arteries). Numerical solution from Scheme 3 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

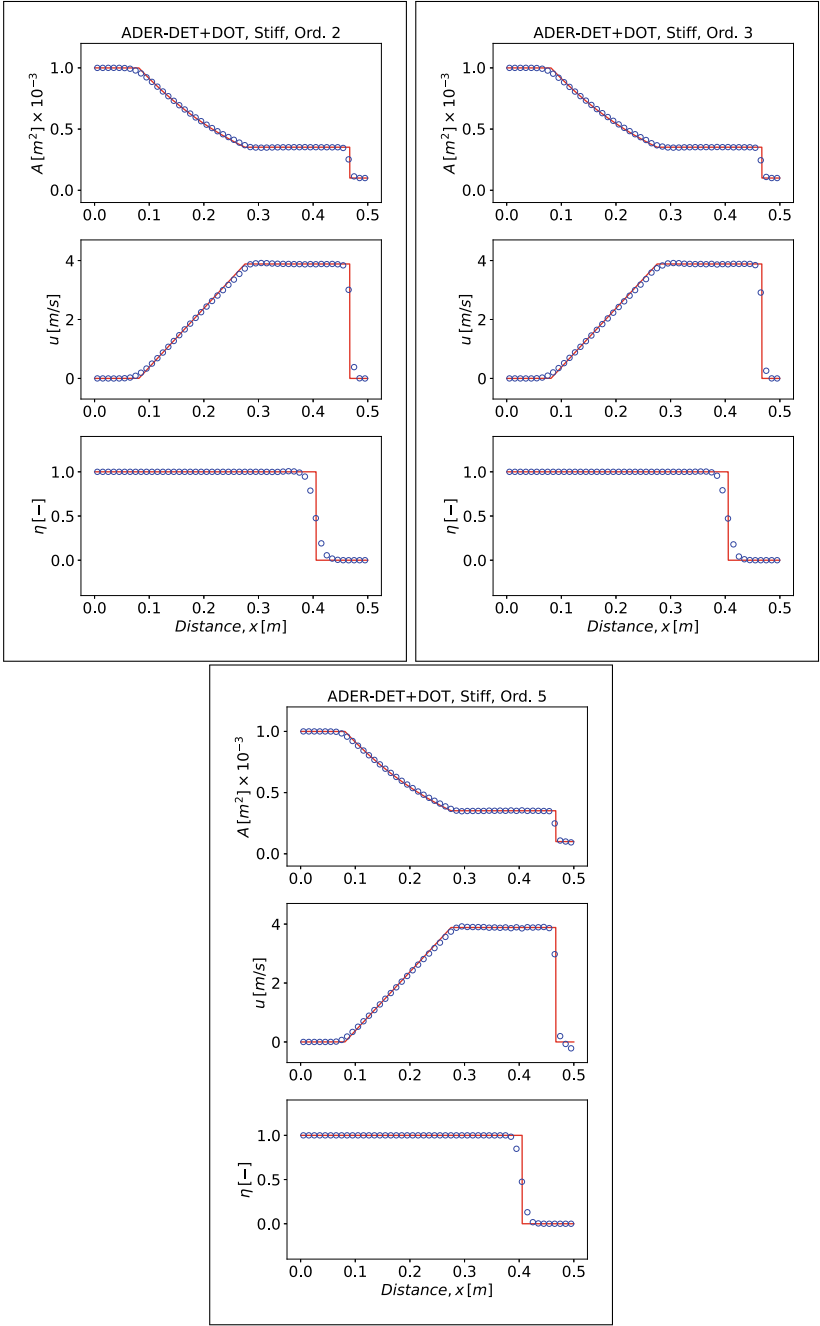


Fig. 17.16 Test 2 (arteries). Numerical solution from Scheme 3 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

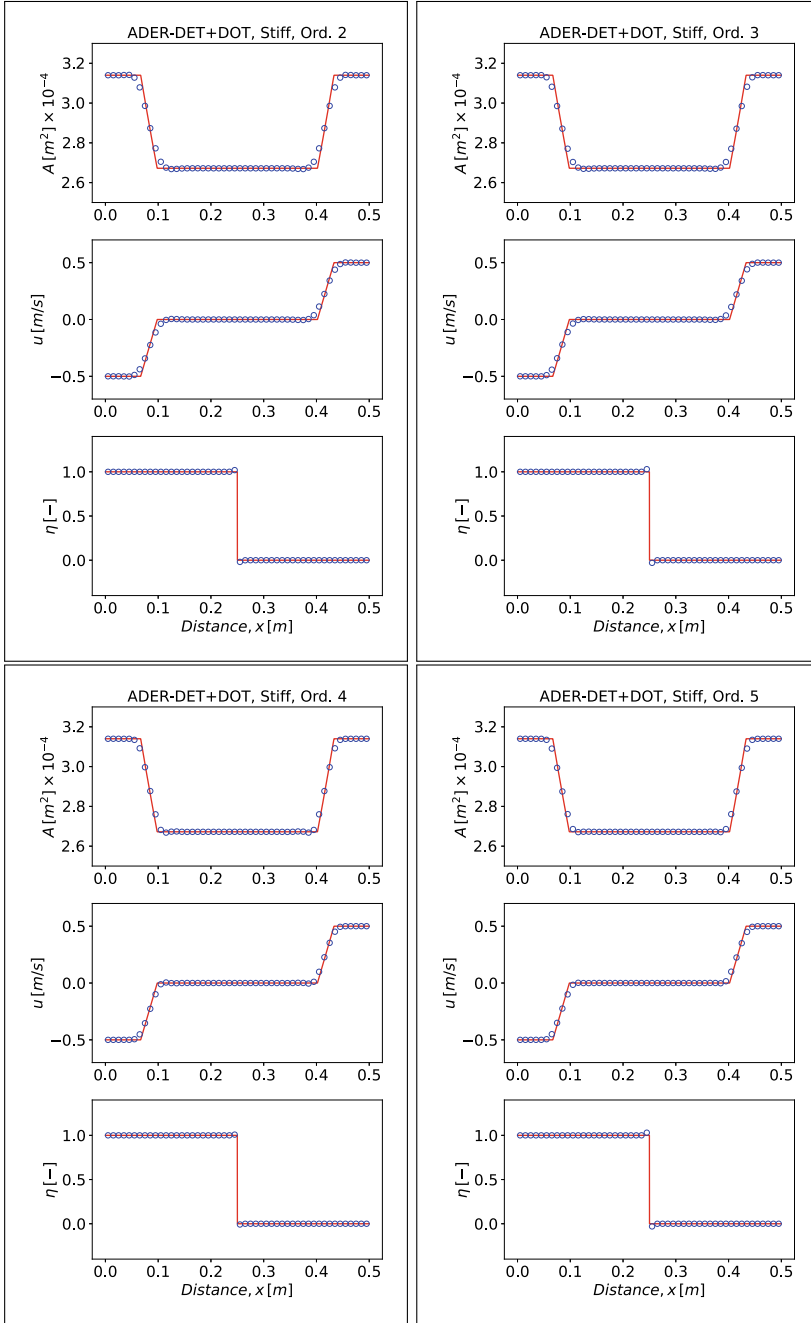


Fig. 17.17 Test 3 (arteries). Numerical solution from Scheme 3 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

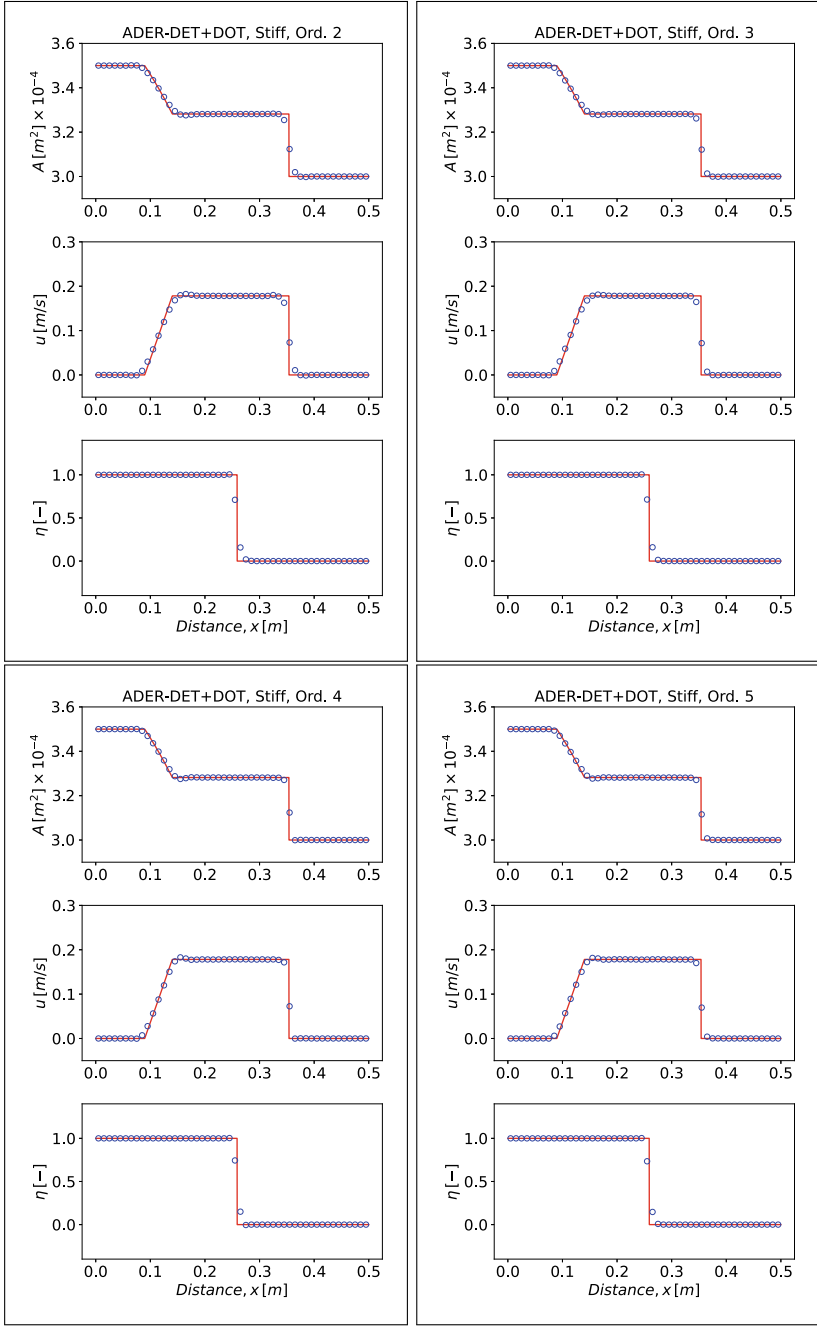


Fig. 17.18 Test 4 (veins). Numerical solution from Scheme 3 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

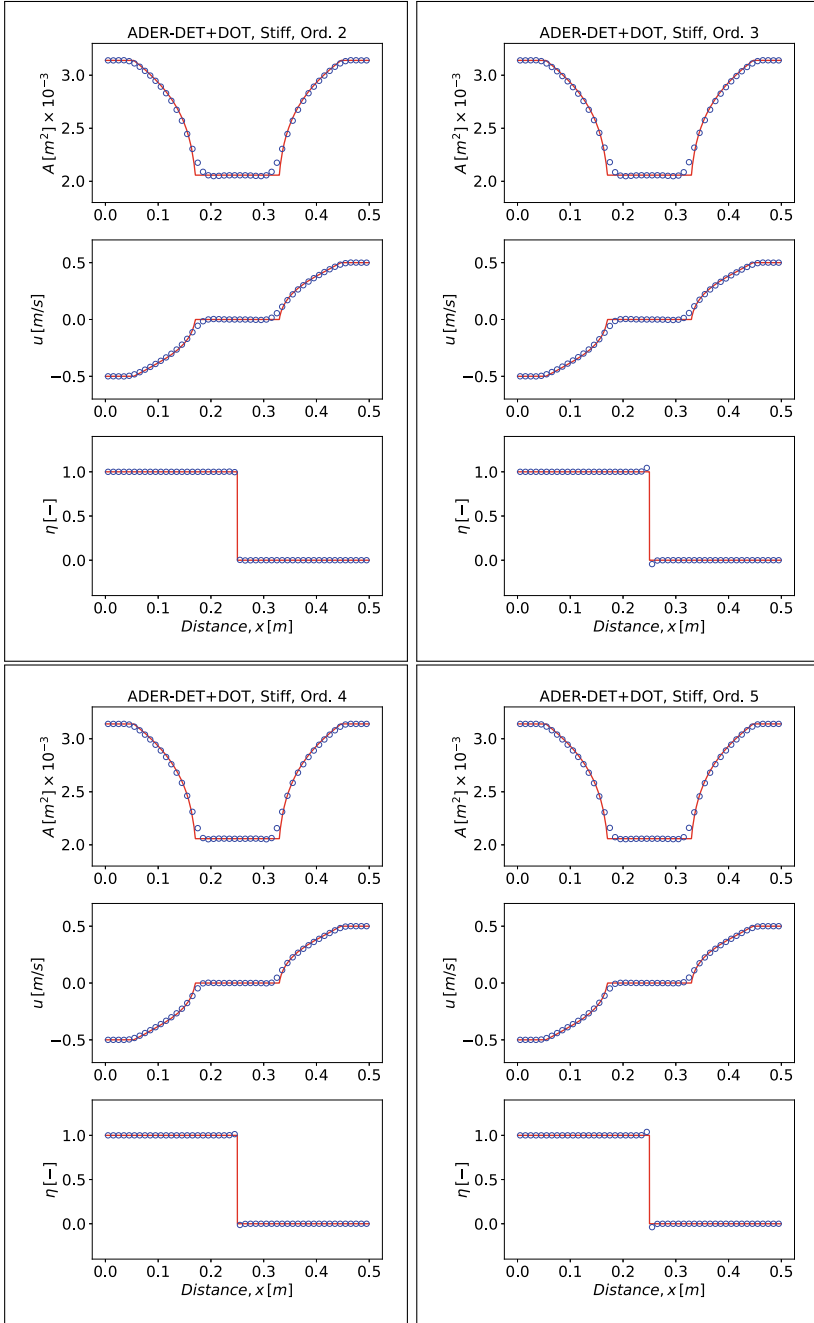


Fig. 17.19 Test 5 (veins). Numerical solution from Scheme 3 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

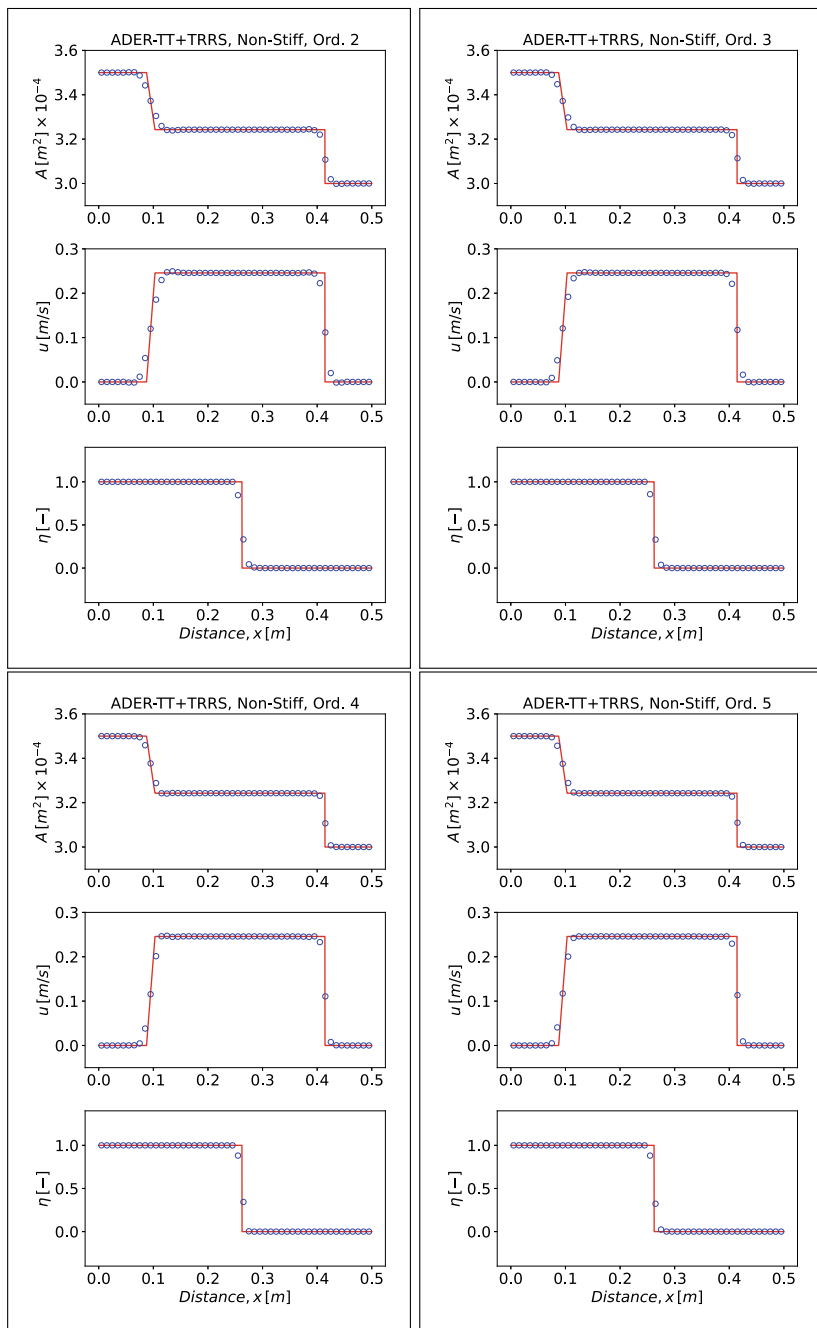


Fig. 17.20 Test 1 (arteries). Numerical solution from Scheme 4 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

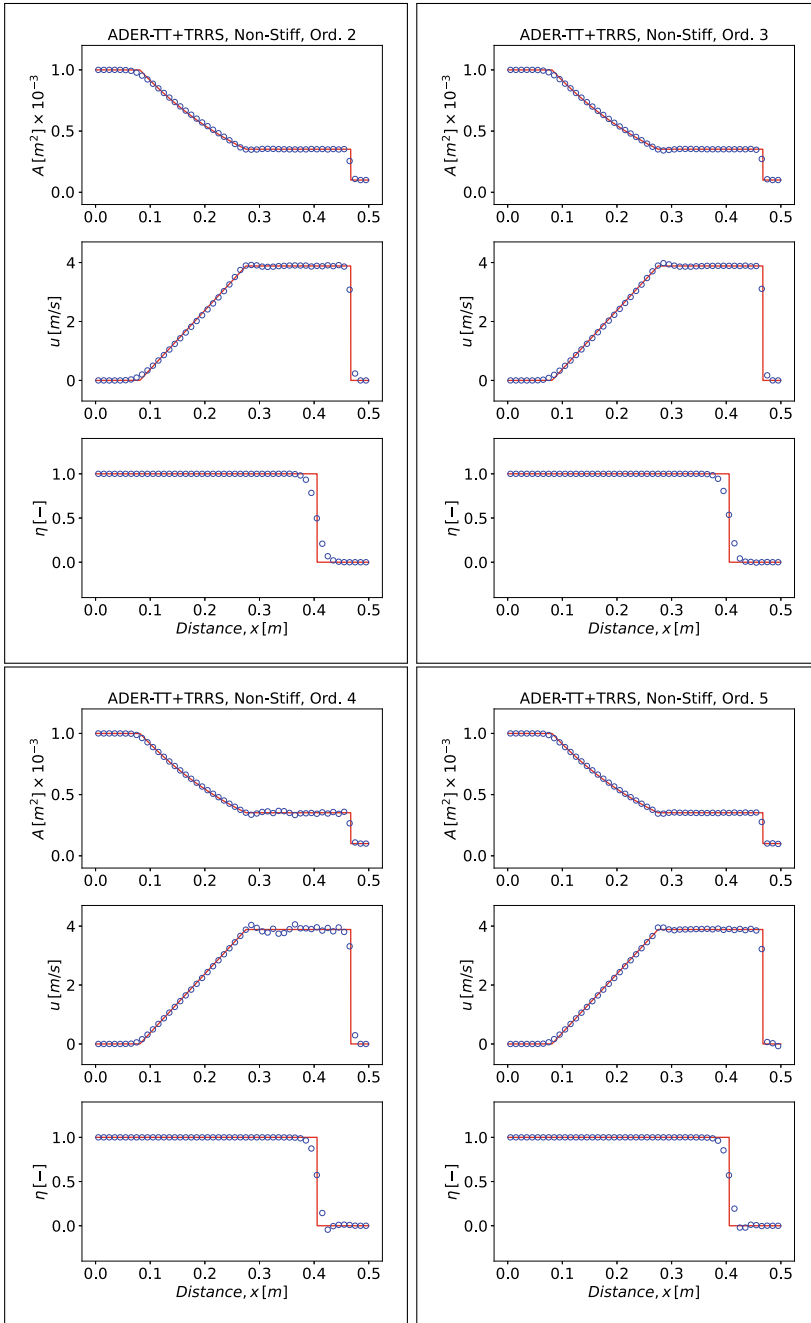


Fig. 17.21 Test 2 (arteries). Numerical solution from Scheme 4 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

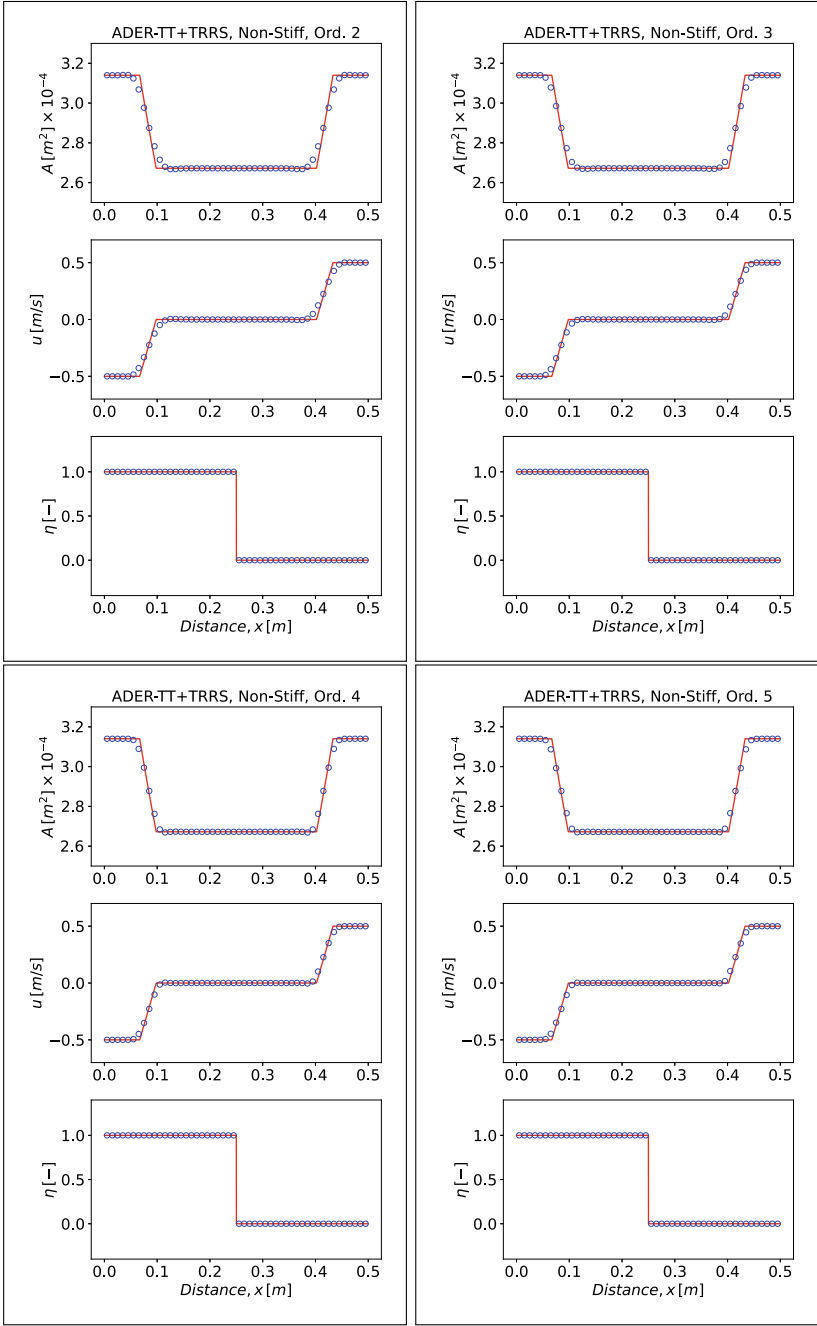


Fig. 17.22 Test 3 (arteries). Numerical solution from Scheme 4 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

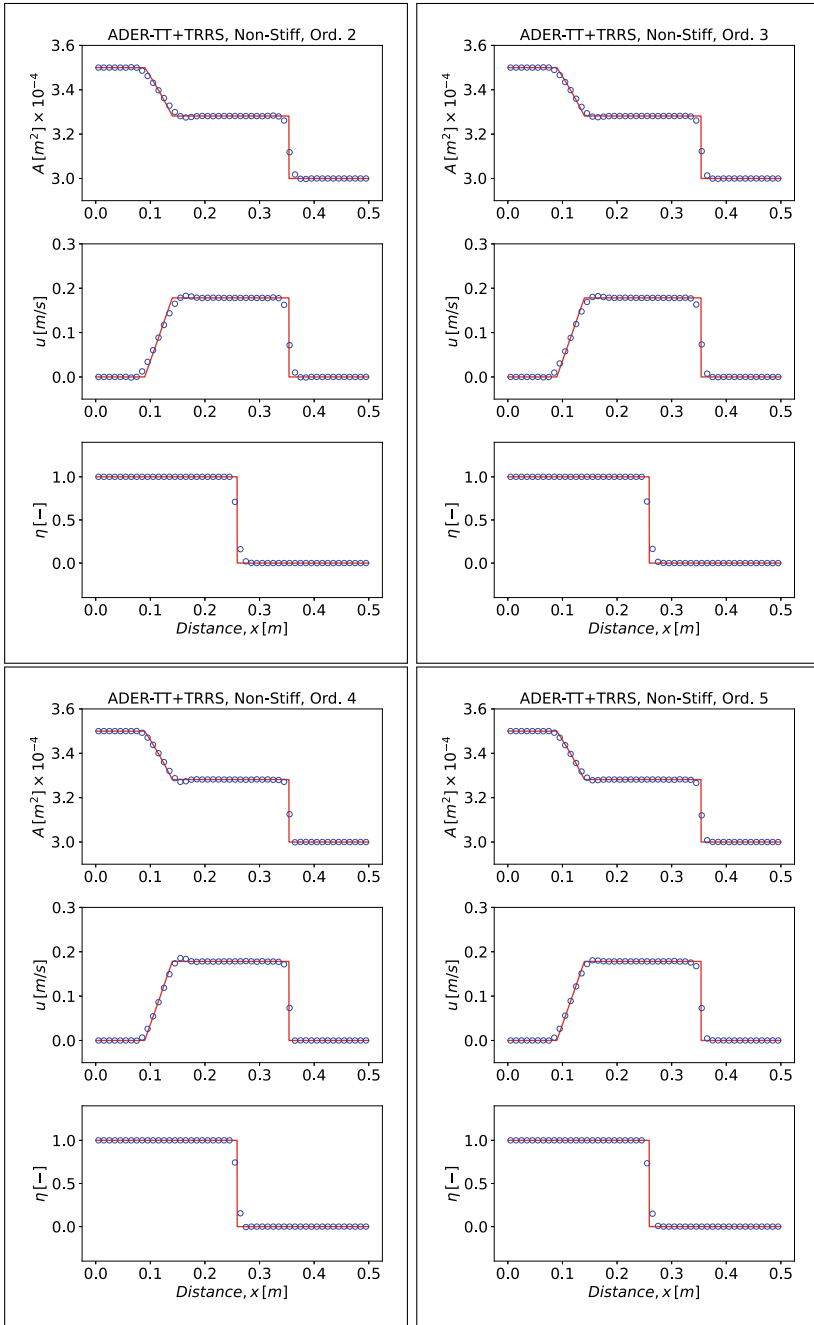


Fig. 17.23 Test 4 (veins). Numerical solution from Scheme 4 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

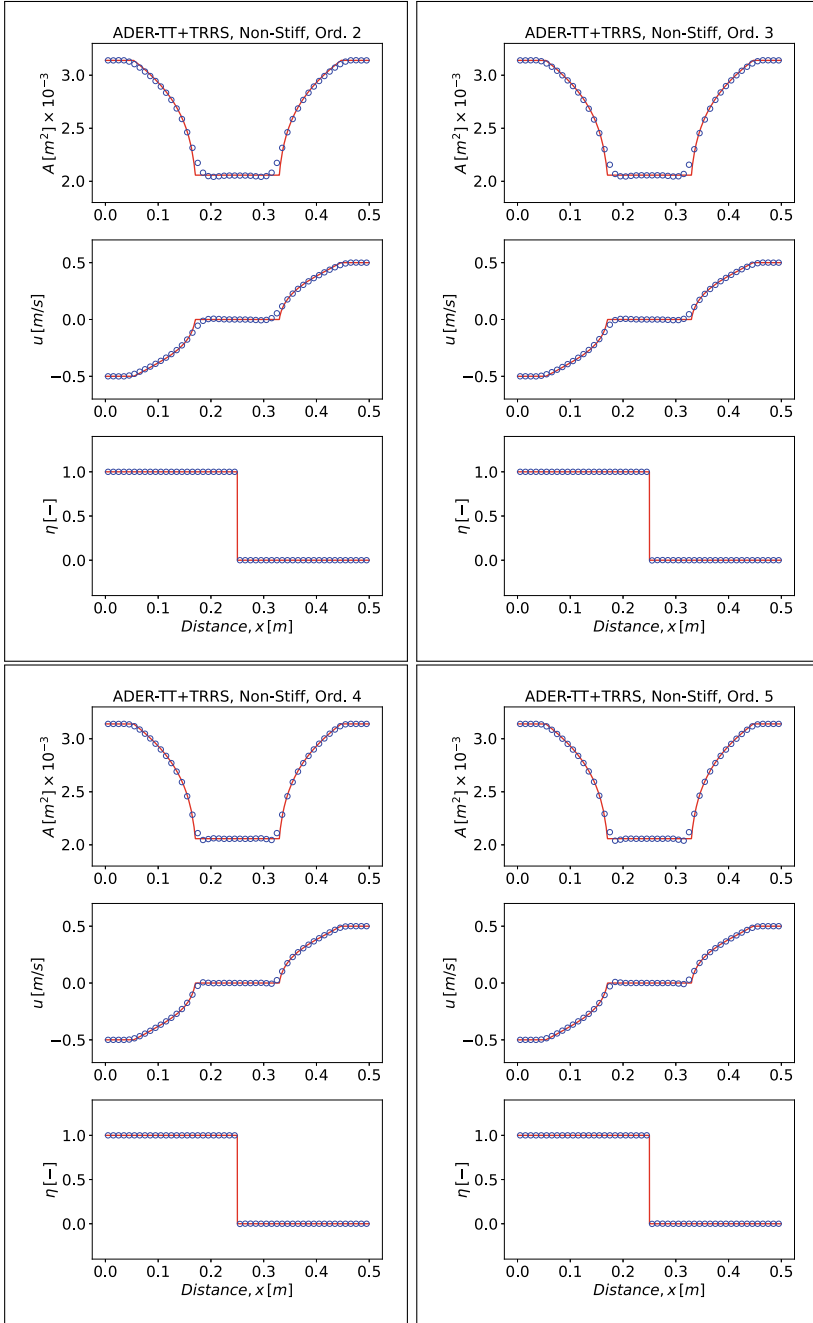


Fig. 17.24 Test 5 (veins). Numerical solution from Scheme 4 of Table 17.1 (symbol) compared with the exact solution (line), for orders of accuracy 2 to 5. Initial data is given in Table 17.2 and parameters are given in Table 17.3. Mesh $M = 50$ cells and CFL coefficient $C_{cfl} = 0.9$

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Part IV
Major Cardiovascular Diseases

Chapter 18

Neurological Disorders and Neurofluids



Abstract This chapter reviews a broad class of neurological disorders linked to disturbed fluid dynamics, specifically the interactions of neurofluids, including arterial and venous blood, cerebrospinal fluid, interstitial fluid, and lymph. Cerebral venous outflow strictures, both intracranial and extracranial, are identified as major triggering mechanisms for CNS pathological states. The head and neck venous system is examined, building on the related anatomy and fluid physiology covered in previous chapters. Jugular and extra-jugular venous districts, as well as physiological cerebral venous outflow patterns, are explored. Multiple sclerosis is reviewed in light of the currently dominant autoimmune theory and the re-emerging vascular theory, which historically assigned a prominent role to the cerebral venous system. Also examined are retinal abnormalities, transient monocular blindness, transient global amnesia, Ménière's disease, idiopathic intracranial hypertension, idiopathic Parkinson's disease, and Alzheimer's disease. Two methods for restoring normal, or nearly normal, cerebral venous outflow are discussed: Percutaneous Transluminal Angioplasty (PTA) and dedicated venous stents. Studies on the application of these methods to patients with anomalous cerebral venous outflow have shown that restoration of venous flow, close to physiological conditions, has a net positive effect on selected patient groups. However, due to re-stenosis, PTA must be excluded as a viable therapy. Dedicated venous stents appear promising but still need to prove their long-term effectiveness. Much research is still required to elucidate the underlying mechanisms and design effective therapeutic strategies. We advocate for a comprehensive, multi-disciplinary, and holistic approach, one that involves the interactive dynamics of all neurofluids, in which mathematical modeling and simulation can play a significant role.

18.1 Cerebral Venous Outflow

Useful background material for this chapter is found in previous chapters: Chap. 1 on the cardiovascular system; Chap. 2 on basic notions on fluid mechanics; Chap. 3 on interstitial fluid and the lymphatic system; Chap. 4 on cerebrospinal fluid; Chap. 5

on fluid systems in the inner ear and the eye; and Chap. 11 on the fluid dynamics of CSF in the spinal canal. More details on each of these topics are found in the references given in each chapter.

Recent studies in craniospinal fluid dynamics have established an increasingly strong association between cerebral venous outflow anomalies, such as stenoses, and a range of diseases with no cure and unknown cause, affecting the central nervous system (CNS). These findings have generated considerable controversy with established medical theories, but have also drawn increasing attention to the venous system, a largely neglected district of the human circulation, even after William Harvey's breakthrough four hundred years ago.

Before proceeding to the next section we encourage the reader to revise Chap. 4, with particular emphasis on craniospinal fluid dynamics and the dynamic time-dependent interaction of all craniospinal compartments in one cardiac cycle.

Here we focus specifically on the venous system. For general background see Pang [1, 2]. For specific background on the head and neck venous system see Valdueza et al. [3], Schaller [4], Schaller and Graf [5], and Doepp et al. [6]. As is well known, the brain receives a privileged input of arterial blood that carries oxygen and nutrients to the brain tissues through the capillary bed. Blood flows to venules and veins, being then transported back to the right atrium of the heart.

The forces responsible for the venous return include the residual arterial pressure, intrathoracic and abdominal pressures and modes of respiration, with strong dependence on postural changes. Veins are responsible for collecting and eliminating waste products of tissue metabolism. The role of the venous system in conjunction with the glymphatic system is currently a major topic of research; see Chap. 3 for more details, and references therein. This metabolic-waste clearing process performed by the venous system parallels that of the arterial system that delivers oxygen and nutrients to tissues. Noteworthy, most of the literature focus exclusively on the oxygen-provision function performed by the arterial system.

18.1.1 Main Veins of the Head and Neck

Venous vessels, also called **capacitance vessels**, hold about 2/3 of the total intravascular volume, noting that some authors quote the figure of 80% to 85% [7]. Given the large proportion of venous intravascular volume and the high compliance of venous vessels, the venous system is a major player in the regulation of fluid volumes and translocation of fluids in the craniospinal cavity.

Figure 18.1 depicts a simplified sketch of the main veins of the head and neck. Two draining pathways returning venous blood from the brain back to the heart intersect at the confluence of sinuses, namely the superior sagittal sinus and the straight sinus; the latter collects venous blood from the inferior sagittal sinus and the vein of Galen. From the confluence of sinuses the flow is bifurcated to the left and right into the left transverse sinus and the right transverse sinus, each connecting to the left sigmoid sinus and the right sigmoid sinus, respectively. Both sigmoid sinuses

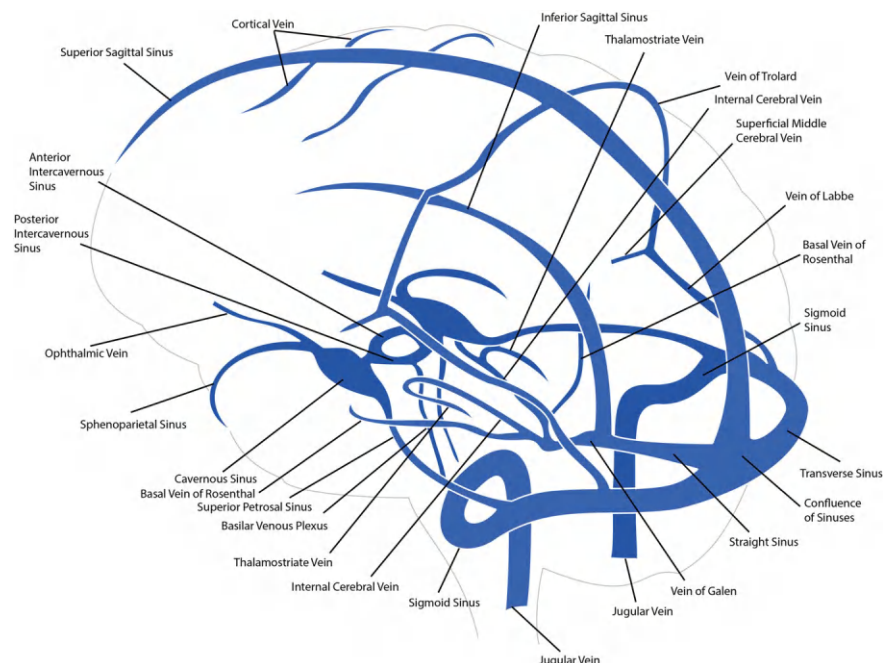


Fig. 18.1 Simplified illustration of the main veins of the head and neck in humans (edited by Violeta Da Rold. Data from [8])

then drain directly into their respective left internal jugular vein and right internal jugular vein, which then transport the brain venous blood to the superior vena cava and then to the right atrium. Drainage from deep regions of the brain, left and right, is carried out by the left and right basal veins of Rosenthal and the internal cerebral veins; these meet just before joining the single, great vein of Galen, which after its confluence with the inferior sagittal sinus conducts cerebral venous blood to the straight sinus, that ends up at the confluence of sinuses. Four more pathways empty their venous blood content into the left and right transverse sinuses, namely the left and right veins of Labbé (inferior anastomotic vein) and the left and right superior petrosal sinuses. Each vein of Labbé drains blood from the vein of Trolard (superior anastomotic vein), which is connected to the superior sagittal sinus.

Each superior petrosal sinus, left and right, collects blood from the cavernous sinuses, into which venous blood from the sphenoparietal sinus and the ophthalmic vein is drained. The two inferior petrosal sinuses, not shown in Fig. 18.1 also collect venous blood from the cavernous sinuses and drain it into their respective sigmoid sinuses. Note that the left and right cavernous sinuses are connected to each other through the anterior intercavernous sinus and the posterior intercavernous sinus. The above simplified description refers to the *standard human*. The venous system, as compared with the arterial system, tends to vary significantly from individual to individual, also exhibiting strong asymmetries.

18.1.2 *Jugular and Extra-Jugular Venous Systems*

Following Doepp et al. [6], two modes of cerebral venous return can be identified, namely (a) the jugular venous system and (b) the extra-jugular venous system. The former returns venous blood from the brain through the left and right internal jugular veins. It is currently accepted that the internal jugular veins are the main route for cerebral venous return in most healthy subjects, in the supine position. The extra-jugular venous system can be further divided into the *intra-spinal venous compartment* and the *extra-spinal venous compartment*. Each of these extends along the entire vertebral column. The longitudinal vertebral veins, deep neck (cervical) veins and external jugular veins are potential collectors of these compartments. The extra-jugular venous system has a huge volume capacity, of up to 1000 ml, which, according to Doepp et al. [6], it is sufficiently large to take over the complete brain venous drainage.

The vertebral veins are formed by the junction of the retromandibular vein with the posterior auricular vein. The vertebral veins collect venous blood from the sub-occipital sinus, deep cervical veins and emissary veins, which connect the veins outside the cranium to the venous system inside the cranium [6]. See also Valdueza et al. [3]. The external jugular veins, in the main, drain the exterior part of the cranium and the deep parts of the face.

The main routes for venous return from the central nervous system to the heart are therefore the internal jugular veins, the vertebral veins, and the azygos vein for the spinal district [9]. At this stage we note that jugular veins possess valves which, under physiological conditions, ensure unidirectional flow towards the heart [10]. Valves are also found in the vertebral veins [6, 11], but are more common in the internal and external jugular veins.

18.1.3 *Cerebral Venous Return Patterns*

There are at present various studies that have used different methodologies for quantifying the venous blood flow from the central nervous system. Doepp and collaborators [6] analysed the types and prevalence of human cerebral venous outflow patterns in healthy individuals by means of ultrasound and MRI applied to 50 healthy volunteers, in the **supine position**. The authors quantified arterial blood flow volume into the brain as the sum of the measured flow volumes in both internal carotid and vertebral arteries. To quantify the venous blood volume flow they measured blood volume flow in both internal jugular and vertebral veins. Arterial cerebral blood flow was measured as $CBF = 752 \pm 133$ ml/min and a total venous blood volume flow (vBVF) in the IJVs and VVs as $vBVF = 669 \pm 240$ ml/min, noting that for internal jugular veins $vBVF = 634 \pm 246$ ml/min and for vertebral veins $vBVF = 36 \pm 34$ ml/min. The authors defined $\overline{vBVF}$ as venous blood volume flow through IJVs, normalised by

Table 18.1 Cerebral venous return patterns in healthy individuals in supine position, according to Doepp et al. [6], give rise to three types of cerebral venous drainers shown in column 1. Here $\overline{vBV\overline{F}}$ in column 2 is venous blood flow volume through IJVs, normalised by the global arterial blood flow. Column 3 shows percentages of subjects corresponding to each of the three types of drainers. Note that only 72% of healthy subjects are jugular drainers in the supine position

Type	$\overline{vBV\overline{F}}$	Percentage
1 (Jugular)	$\overline{vBV\overline{F}} \geq 2/3$	72 %
2 (Jugular/Extra-jugular)	$1/3 \leq \overline{vBV\overline{F}} \leq 2/3$	22 %
3 (Extra-jugular)	$\overline{vBV\overline{F}} \leq 1/3$	6 %

the global arterial blood flow. They then identified $\overline{vBV\overline{F}}$ as a key quantity to define three types of venous drainage. See Table 18.1.

Doepp and collaborators [6] found that all subjects in their study could be classified into three groups, according to the dominant mode of venous return from the brain. The first column of Table 18.1 shows the types, the second column quantifies jugular venous blood outflow (relative to the global arterial blood inflow), while the third column shows the percentage of the population in each of the three categories. The results shown in Table 18.1 say that in 72% of the healthy subjects studied, in the supine position, the internal jugular veins are the main drainage vessels. A sizeable 22% are defined as **neck drainers**, while a 6% of subjects are predominantly non-jugular drainers, defined as **spinal drainers**. We emphasise that in nearly one third of the healthy population studied, the main draining vessels were not the internal jugular veins.

We remark once more that the above results refer to healthy subjects in the **supine body position**, as it is known that in the upright position, blood is redirected through the vertebral veins, which become the predominant venous outflow pathway, while internal jugular veins tend to collapse. Menegatti and Zamboni [12] have presented a more complete picture of cerebral venous return, that includes changes of posture.

Next we briefly review multiple sclerosis, a neurological disorder that has been associated with impairment of the venous system, in recent years.

18.2 Multiple Sclerosis and the Venous System

Here we succinctly review the main aspects of Multiple Sclerosis (MS), starting from a simplified description of the disease. More comprehensive accounts of MS are found in [13–16], and in the many relevant references therein. The presentation is largely based on [8].

18.2.1 *Definition, History and Theories*

Multiple Sclerosis is a disease of the Central Nervous System, characterised by damage to the myelin (demyelination), the protective layer of axons in the CNS, and to axons themselves. Axons are responsible for the transmission of signals in the body and **demyelination** leads to disturbances to the speed and quality of inter-neuronal signalling, which may result in disability. MS is the most common neurological disease in young adults and affects predominantly subjects aged 20 to 50 years and in most cases involves: increasing degrees of disability; cognitive impairment; muscle spasticity; mobility, coordination, balance and speech disorders; visual disturbances; fatigue; acute or chronic pain; and bladder and bowel complications.

The body of literature on MS is huge and goes from massive medical textbooks, research journal articles, historical medical books, to anecdotal communications. A comprehensive up-to-date source of information on MS on a global scale, published by the World Health Organisation (WHO), is found in [17]. Globally, the median **prevalence** of MS (proportion of a population found to have MS) is 30 (range 5 to 80) in 100000 people, making MS one of the most common neurological disorders that cause disability in young adults. Regionally, Europe has the largest prevalence (80) in the world. The global median **incidence** (probability of occurrence) is 2.5 (range 1.1–4) per 100000, with Europe having the largest value of 2.8, regionally. Interestingly, the largest median prevalence is found in countries with the largest income.

Europe has currently an estimated number of 400 thousand MS patients and the world MS population is estimated at 2.5 million. One of the many unexplained curiosities of MS is that people in the northern and southern hemispheres, living furthest from the equator, have an increased risk of MS, the so-called **latitude gradient of MS**. Another peculiarity of MS, is that women are more likely to have it, by an estimated factor of three. The cause of MS is unknown and there is no cure for it.

Historically, Jean-Martin Charcot (1860) is credited with the first comprehensive account of MS up to his time. Other prominent names associated with the early study of MS include Georg Eduard Rindfleisch (1863), Pierre Marie (1884) and James Dawson (1916). Interestingly, they all confirmed the presence of **inflamed veins, blood stains or clots around blood vessels inside the MS lesion**, signalling a strong vascular (venous) component to the MS aetiology.

However, they had different ideas regarding cause and effect of this vascular fact. To start with, the presence of blood was regarded as part of an infectious disease. Later, in the period 1931 to 1936, the neurologist Tracy Putnam strongly pursued the vascular connection of MS. His experimental research with dogs managed to create brain lesions very similar to MS lesions, by simply **occluding the main veins draining blood from the brain** for a certain period of time.

Putnam suggested the use of blood thinners and vasodilators as a way of **curing** the disease. These ideas were followed by others, suggesting that easing blood flow may have something to do with the disease, shaping what could be described as a **vascular theory of MS**. For more precise information on MS see [13–15], for example.

From the 1950s onwards, the bases for the currently dominant **autoimmune theory of MS** began to be created, whereby the immune system is at the centre. This is apparent through currently accepted definitions of Multiple Sclerosis, as seen below:

- Laupacis et al. [18] state: *Multiple Sclerosis is a chronic demyelinating and degenerative disease of the central nervous system. The exact cause remains unknown, but most evidence favours an autoimmune mechanism.*
- Talbert [19] states: *Multiple sclerosis is characterized by multiple plaques of demyelination within the brain and/or spinal cord, currently attributed to an autoimmune process, following some as yet unidentified event.*
- Dake [15] states: *Multiple sclerosis is a chronic inflammatory disorder of the central nervous system (CNS) most commonly characterized by focal areas of myelin destruction, inflammation and axonal transection. The multicentric inflammation and demyelination of the brain and spinal cord are associated with variable neurologic symptoms ranging from mild dysfunction to debilitating. Typically, these symptoms are marked by episodes of clinical worsening followed by improvement. The cause of this disease remains unclear currently, but the underlying aetiology is generally considered to be immunologically based. Other factors, including genetic, environmental and infectious influences have been implicated, as well.*

As already stated, the dominant theory of MS is currently the **autoimmune theory** and therefore most health systems around the world are shaped around this view of MS [17], for which immune-system targeted pharmaceuticals are provided in what is known as **Disease Modifying Treatment (DMT)**, which has been reported to be effective in reducing inflammation.

However, these treatments are unable to stop the progression of the disease. Radiologists diagnose MS through the use of magnetic resonance imaging, from which they identify patches in the brain, brain lesions, which may be indicative of inflammation, demyelination or scars from older lesions. Not all visible lesions may be active, while brain inflammatory and demyelinating lesions cause a corresponding temporary loss of function, called a **relapse**. There usually follows a **remission** phase, characterised by a remyelination activity and associated regain of function, though mostly partial. The relapsing-remitting pattern of each patient allows specialists to classify MS patients into four categories.

The most common category is Relapsing-Remitting Multiple Sclerosis (RRMS). Usually RRMS is converted to Secondary Progressive MS (SPMS). Primary Progressive MS (PPMS) is another category of MS patients in which progressive loss of function is accumulated, with virtually absent relapses. The very first manifestation of what could become MS is termed **Clinically Isolated Syndrome (CIS)**, which will show neurological symptoms caused by inflammation and demyelination in the CNS. CIS may or may not become MS.

During the 1980s, some researchers continued to pursue the vascular theory of MS. In their cadaver studies Adams and collaborators [20, 21] showed beyond a shadow of doubt that plaques were formed around veins and venules in the brain and that periventricular lesions and plaques were often accompanied by **lymphocytic**

infiltration. This last point is important as it brings in the role of the vessel wall and the Blood-Brain Barrier. In [21] Adams states: *the involvement of the cerebral venous system in multiple sclerosis has been recognised for many years. In particular, plaques are centred on small veins or venules and extend outwards therefrom along branch veins as Dawson's fingers.*

In his work Adams identified three important issues that persisted in later developments of the vascular theory, namely (a) cerebral vein wall damage and increased permeability, (b) perivenular lymphocytic infiltration and (c) iron deposition.

More recent works advocating the vascular route include those of Schelling [22] and Talbert [19]. The work of Talbert [19] addresses more specifically the condition associated with **Dural Arteriovenous Fistulas (DAVFs)**. He put forward the hypothesis that DAVFs may raise venous pressure, cause vessel wall filtration of colloids, alteration of the osmotic pressure, disruption of the internal transport system of axons, potentially triggering immune response and thus MS. Note that this proposed chain of events includes **venous hypertension**. This is indeed very credible, as DAVFs involve direct connections between the arterial system and the venous system, bypassing the microvasculature. Thus the transmission of arterial high pressure and velocity to the venous system is direct, an unquestionable event.

What appears to be a commonly accepted fact, both in the vascular and the immune theory, is the transport of **T cells** across the vessel wall, through the Blood-Brain Barrier, briefly described below.

18.2.2 The Blood-Brain Barrier

Blood vessels have an inner layer of adjacent endothelial cells, the endothelium, which forms a selective membrane that controls the transport of molecules across the vessel wall and thus the exchange of nutrients and waste products between tissue and blood stream. This is a very complex process, for which useful background and references are found in Chaps. 1, 2, 3 and 4.

In the brain, the endothelium is further specialised so that its selectivity function becomes even more stringent, contributing to keep **neurons separate from blood stream components** in the circulation. This is called the **Blood-Brain Barrier (BBB)** and is formed by specialised endothelial cell junctions that constitute an unbroken seal between the cells [23–25]. In general, while being highly permeable to lipophilic solutes, brain capillaries are exceptionally impermeable to lipophobic solutes, such as L-glucose, catecholamines and plasma proteins.

Following [24], we see that the BBB has a two-way filtering function consisting of:

1. protecting the delicate neuronal circuits from interference by circulating stimulants and solutes such as catecholamines, which are also neurotransmitters;
2. maintaining a tight control of Ph and K^+ concentration in the cerebrospinal fluid and interstitial fluid;

3. preventing the washout, by the blood stream, of synaptic neurotransmitters from the brain parenchyma.

Two further points about the BBB are worth mentioning. First, there are specific endothelial carriers to aid the transport of lipophobic solutes into the brain parenchyma, and second, there are regions of the CNS in which the BBB is absent or is defective, as for example in the circumventricular region and the area postrema.

Breakdown of the BBB is associated with severe acute hypertension, cerebral haemorrhage, ischaemia, hypoxia (high altitude exposure), and inflammation, such as in meningitis and MS [24]. Blood components include red and white blood cells, the latter in charge of the immune function for the full body. T cells are one of three types of white blood cells (or lymphocytes). Breakdown of the BBB allows the passage of T cells into the brain tissue. See Chap. 1 for background on blood cellular elements.

Talbert [19] argues that T cells then target myelin as foreign tissue, initiating an attack by the immune system, thereby starting the inflammation process.

In the autoimmune theory of MS the immune system, for an as yet unknown reason, attacks self cells, following some triggering mechanisms that may be of genetic, environmental or infectious type. In this manner, as observed for example, by Grammas and collaborators [26, 27], most current research on neurodegenerative diseases is highly **neuron centred**. The question being asked is: *why does the immune system attack the myelin?* Instead, it would seem more natural to ask: *why do T cells cross the BBB?* It would therefore seem reasonable to take a step back and conduct research into the blood vessel wall integrity, as a necessary step to begin to understand the phenomenon.

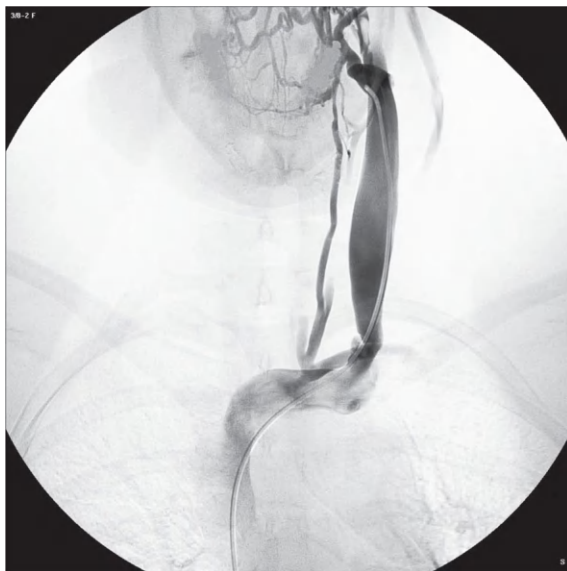
In the context of Alzheimer's disease and its cardiovascular risk factors, Grammas and collaborators [26, 27] advocate the development of pharmaceuticals specifically targeted to the vessel wall. In the context of MS, an even bolder proposition would be to take a further step back and investigate the complex control mechanisms and the transient mechanical forces present in the interaction of the flowing blood stream in the vessel lumen and the vessel wall, such as wall shear stress on the endothelium and Starling forces; see Chap. 2. An approach of this kind would necessarily involve the vascular system more comprehensively, as part of the puzzle.

18.2.3 Extracranial Venous Outflow Strictures

In this section we succinctly review some relevant aspects of anomalous cerebral venous haemodynamics resulting from cerebral venous outflow strictures located extracranially. For background material we recommend [4–6, 28, 29] and references therein.

As seen earlier, the main routes for venous return from the central nervous system to the heart are the internal jugular veins, the vertebral veins and the azygos vein. Zamboni and collaborators [30] examined the extracranial venous outflow routes in

Fig. 18.2 Obstructed internal jugular vein (courtesy of Professor Paolo Zamboni, from Zamboni's archives, University of Ferrara, Italy)



patients with clinically defined MS and found that such venous outflow routes are often affected by a range of obstructions. These include membranous obstructions at junctions, twisting, septum and atresia, inverted valves, intraluminal stenoses and external muscle or bone compression stenoses.

Figure 18.2 shows the case of an obstructed Internal Jugular Vein [31]. Purely from a fluid mechanics point of view, it is very reasonable to expect that such strictures in the venous outflow routes would disturb venous drainage from the central nervous system. To partially verify this expectation, Zamboni and collaborators [30] quantified the haemodynamical impact of such anomalies by measuring two key physical quantities: pressure and flow direction. Regarding pressure measurements, given the invasive nature of their method, they restricted the study to the vicinity of extracranial venous strictures.

Remarks on Pressure

To our knowledge, no direct pressure measurements in intracranial districts, in connection with extracranial venous outflow strictures, are yet available. But see [32] for results from indirect methods. Pressure in the head and neck veins is a key physical quantity at play, but its determination remains illusive. MRI cannot provide values for pressure, unfortunately. Nonetheless, many publications assume for a fact, even if reasonable, that intracranial venous hypertension follows from cranial venous outflow strictures. Global closed-loop mathematical models can nowadays compute the pressure throughout the cardiovascular system [33, 34]. Modelling results confirm that cerebral venous outflow strictures cause intracranial venous hypertension and elevated intracranial pressure (ICP). See Chap. 20.

Disturbed Cerebral Venous Outflow

Concerning flow direction, one expects unidirectional flow of venous blood, from the brain to the heart. Thus, reflux or stagnant flow, is a strong indication of disturbed venous haemodynamics. Zamboni et al. [30] found, as expected, that stenoses produced a pressure jump, though of modest size, under the conditions in which it was measured.

Regarding reflux, they found this to occur, even in deep cerebral veins (internal cerebral vein, basal vein of Rosenthal and great vein of Galen). Their methodology combined EcoColour-Doppler (ECD) with Transcranial Color-Coded Duplex Sonography (TCCS). In order to distinguish normal subjects from individuals with anomalous extracranial venous configurations, the authors identified the following five criteria:

1. reflux (constantly) in the Internal Jugular Veins and/or Vertebral Veins in sitting and supine posture;
2. reflux in the Deep Cerebral Veins;
3. high-resolution B-mode evidence of Internal Jugular Vein stenoses;
4. flow not Doppler-detectable in the Internal Jugular Veins and/or Vertebral Veins;
5. reverted postural control of the main cerebral venous outflow pathways.

Subjects meeting two or more of the above criteria were defined as suffering from **Chronic Cerebro Spinal Venous Insufficiency (CCSVI)**. Zamboni et al. [30] found that all MS patients in their study had CCSVI, while none of the healthy controls met **more than one** of the above criteria. Further studies [35] included the dynamics of the cerebrospinal fluid and the role of collaterals. Collaterals, under the action of venous strictures, fulfil the functions of redirecting the venous blood flow and reducing presumed CNS venous pressure.

The proposed CCSVI condition has generated intensive world-wide interest, as well as controversy, confronting the CCSVI-related findings against the currently dominant autoimmune theory of MS. Many studies on the CCSVI and its relation to MS have been carried out after Zamboni's early publications, with visible differences in the results, with some of them supporting Zamboni's findings, others denying them. See for example [32, 36, 37]. Laupacis et al. [18] conducted a meta-analysis of studies up to 2011 that reported the frequency of CCSVI among patients with and without MS, excluding the works published by Zamboni and collaborators. They identified eight eligible studies and found overall positive association between CCSVI and multiple sclerosis. In the more recent meta-analysis of Zwischenberger et al. [38] it was concluded: *The meta-analysis demonstrated a correlation between CCSVI and MS. However, there was no evidence that CCSVI has a causative role in MS.*

Methodologies and Discrepancies

It seems as if the discrepancies, to some extent, can be explained by the differences in methodologies used and by the precise manner in which the five criteria defining CCSVI according to Zamboni et al. [30], is applied.

Haacke and collaborators [39] have adopted a different methodology to that of Zamboni, by introducing a well-defined protocol using Magnetic Resonance Imaging. It is said that MRI has the advantage of being less operator-dependent and therefore more objective than sonography. In [39] the authors set out to study, using MRI, extracranial venous structural and flow characteristics in patients with multiple sclerosis, using 200 MS patients from two sites. Their approach was to consider two types of abnormalities: structural and fluid dynamical. They found that 68% of MS patients showed structural abnormalities (e.g. unilateral or bilateral stenoses at different levels in the neck and atresia), which allowed them to split the complete sample of patients into **stenotic** and **non-stenotic** MS patients.

Figure 18.3 shows examples of structural venous abnormalities in MS patients. Concerning flow, they found that the stenotic group carried significantly less flow than the non-stenotic group. To quantify flow they normalised total Internal Jugular Vein flow to the total arterial inflow. The stenotic group had normalised flow $56\% \pm 22\%$, while the non-stenotic group had normalised flow of $77\% \pm 14\%$. See also [40, 41].

Mancini and collaborators [42], using contrast-enhanced ultra sonography, carried out a study aimed at measuring **cerebral circulation time**, involving 103 MS

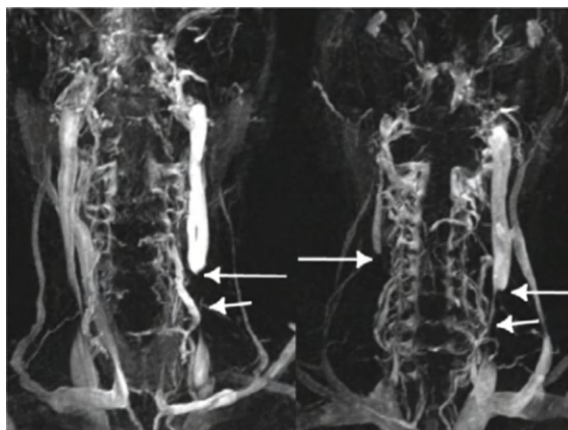


Fig. 18.3 Structural venous anomalies. The left hand side shows that the left IJV is truncated at the midneck level (long arrow) with a thin connection between the midneck level and the inferior jugular bulb near the confluence with the subclavian vein (short arrow). The right hand side shows that both IJVs display continuous enhancement from the sigmoid sinuses through the upper neck level, but are truncated near the midneck level (long arrows). A thin connection is seen in the left IJV between the truncated end at the midneck level and the inferior jugular bulb (short arrow). The right IJV shows no connection between the midneck level and the subclavian vein (taken from Fig. 1 of [39], frames (c) and (d), by permission)

patients and 42 healthy controls. First, they found that the prevalence of CCSVI was higher in patients with MS (77%) than in controls (28%). Cerebral circulation time was significantly longer in MS patients (6.47 seconds) than in controls (5.54 seconds). According to the authors, the longer cerebral circulation time in MS patients suggests that MS may be associated with vascular impairment. Analogous results, using dynamic susceptibility contrast MRI, had been reported earlier by Ge and collaborators [43].

Zivadinov and collaborators [44], using susceptibility-weighted imaging venography, reported reduced brain venous vasculature in MS patients with the CCSVI condition, as compared to healthy controls. They state: *the pathogenesis of these findings has to be further investigated, but they suggest that reduced metabolism and morphological changes of venous vasculature may be taking place in patients with MS.*

18.2.4 Resurfacing a Vascular Theory of MS

So far, the studies regarding extracranial venous anomalies, tell us that the association of these to MS patients is not as strong as that initially reported by Zamboni and collaborators [30], though it seems as if, independently of the methodology used, a conservative figure of about 60% of MS patients exhibit extracranial venous anomalies.

From both structural and fluid mechanical considerations, the association between extracranial anomalies and MS seems to be sufficiently strong to warrant further investigation [45]. Interestingly, a proportion of healthy controls have also been found to be affected by the CCSVI pathology, though percentages vary from study to study. It would be informative to monitor the time evolution of these healthy subjects diagnosed with the CCSVI condition. Longitudinal studies are essential.

The start of a new era for the vascular theory of MS was signalled by the works of Adams et al. [20, 21], Schelling [22] and Talbert [19], amongst others. Zamboni and collaborators [30] drew renewed attention to the vascular theory through their reported association between MS and anomalous **extra-cranial venous outflow strictures**.

The **intracranial** venous association to MS had been known for a long time, e.g. [20, 21]. But what Zamboni and collaborators reported was different, not alluded to in previous works on the pathophysiology of MS. Their findings related MS to **extracranial** morphological and functional venous anomalies in the form of head and neck veins strictures resulting in anomalous cerebral venous outflow from the brain. They proposed that such extracranial venous outflow strictures could contribute to explain [46] the intracranial venous anomalies observed in MS patients by Adams [20, 21] and others.

These advances have put the vascular theory on firmer bases, by identifying a potential **triggering mechanism**, namely extracranial morphological and functional venous anomalies. The findings also bring fluid dynamics more decisively into play. A recent, comprehensive review of these issues is found in the paper by Haacke and collaborators [47]; see also the recent work of Laganá et al. [48]. Having established that a significant percentage of MS patients have extracranial venous outflow anomalies, what is now required is an explanation of the chain of events between CCSVI and MS.

Singh and Zamboni [46] hypothesized a CCSVI-MS link. Their argument goes like this: altered venous fluid dynamics caused by extracranial venous outflow strictures may lead to a range of anomalies including reflux, stagnant flow, oscillatory flow and thus oscillatory wall shear stresses, altered transport and anomalous deposition of iron, venous hypertension, disruption of the BBB and penetration of auto-aggressive immune cells across vessel walls and into the CNS, with the known consequences of axon demyelination and inflammation.

It is worth mentioning related venous anomalies and their connection to MS reported independently by other groups of researchers. An analogous hypothesis to that of Singh and Zamboni [46] was put forward by Talbert [19], though this is specifically related to *Dural Arteriovenous Fistulas* (DAVs), see Borden et al. [49]. In this case, the triggering fluid-dynamics disturbance is built into the very nature of DAVF, a direct anatomical connection between an arterial vessel and a venous vessel. Talbert suggested that DAVFs may raise venous pressure, cause vessel wall filtration of colloids, altering the osmotic pressure in the interstitial space, disrupting the internal transport system of axons and potentially triggering immune response and thus MS. See the earlier work of Wuerfel et al. [50] and their proposed chain of events for MS.

Another vascular condition, called **Thoracic Outlet Syndrome (TOS)**, has also been linked to MS, and could be of potential interest in the current context, see Sanders et al. [51] and Panther et al. [52]. TOS is characterized by compression of vessels by bones, nerves, arteries, resulting in inadequate passageways through an area between the base of the neck and the armpit. TOS is therefore analogous to CCSVI, even though they have separate histories. In TOS the culprit is also identified as extracranial venous outflow strictures, with haemodynamical consequences entirely analogous to those of CCSVI.

In the meantime, the very concept of CCSVI has evolved, as noted by Simka [53], who states: *The research is currently suggesting that CCSVI is not a single entity, but rather a group of different anatomic and functional venous abnormalities.* See [40, 41] for related discussions. See also the recent review paper by Haacke et al. [47] for a re-evaluation of the vascular theory of MS as shaped around the CCSVI or similar conditions.

Mathematical modelling, coupled to MRI and other advanced technologies, might be able to help in advancing our understanding of MS and other neurological disorders. Preliminary results from a global, closed-loop mathematical model for the entire circulation, coupled to other neurofluids in the craniospinal cavity have been

reported in [33, 34]. There might also be room for modelling the BBB under anomalous flow conditions, by extending the mathematical modelling approach proposed in [54, 55].

The vascular aspect of MS discussed above also applies to other CNS pathologies, some of which are discussed in the next section.

18.3 More Venous-Related Neurological Disorders

Neurological disorders are known to run well into the hundreds, with many of them exhibiting a vascular association. The main purpose of this section is to identify a commonality, namely *venous haemodynamics*, occurring across a wide range of neuropathologies. The neurological disorders included here have been studied by medical researchers and their results clearly show a strong association with disturbed cerebral venous haemodynamics. The research reported so far is valuable but incipient and whether this association is merely correlative or whether there is causation between these pathologies and the venous system remains to be demonstrated.

18.3.1 *Retinal Abnormalities, MS and CCSVI*

Adamczyk-Ludyga and collaborators [56] set out to study the relationship between retinal abnormalities and CCSVI in MS patients, distinguishing between MS patients with the CCSVI condition and MS patients without it. For background on the eye anatomy and physiology see Chap. 5 and references therein. The authors [56] first re-interpreted the CCSVI concept as: **vascular pathology comprising venous malformations in the main veins draining the central nervous system**.

In this work the re-interpreted CCSVI condition was ascertained through the presence, or otherwise, of haemodynamical or structural venous anomalies, using catheter venography and following a procedure described in one of their previously published works. They define a venographic flow pattern as abnormal if any of the following four criteria applies, with increasing levels of severity:

1. Grade 1: venous outflow slowed down, no reflux detected.
2. Grade 2: venous outflow slowed down, mild reflux and/or pre-stenotic dilation of the vein.
3. Grade 3: venous outflow slowed down, with reflux and outflow through collaterals.
4. Grade 4: no outflow through the vein, huge outflow through collaterals.

They examined 239 MS patients, of which 220 individuals had CCSVI (92.1%) as defined in the paper. Then the authors in [56] made use of clearly defined, measurable quantities to ascertain retinal abnormalities.

To ascertain eye manifestations they used the non-invasive technique called **Optical Coherence Tomography (OCT)** to examine the retina and the optic nerve. They measured four OCT parameters:

1. average ganglion cell complex thickness,
2. global loss volume,
3. focal loss volume and
4. average retinal nerve fibre layer thickness.

Their main result was that about 60% of MS patients showed abnormal OCT parameters. A result described by the authors as intriguing was that MS patients with a **unilateral stenosis** of internal jugular veins had a higher prevalence of eye abnormalities. They found that the most severe flow abnormalities (grades 3 and 4) were associated with increased prevalence of pathological OCT values, as compared with the patients with mild-degree lesions. Patients with no CCSVI or other kinds of abnormalities did not show anomalous OCT parameters. In summary, severity of OCT abnormalities were correlated to severity of haemodynamical abnormalities.

18.3.2 Transient Monocular Blindness (TMB)

Transient Monocular Blindness is an ophthalmic condition characterised by loss of vision, including blindness, dimming, fogging and blurring, usually for brief periods of time, from seconds to minutes. The general literature on TMB proposes a broad range of potential causes, including carotid stenoses; emboli reaching the retinal circulation from the carotid vessels or from the heart; migraine; intracranial hypertension (elevated ICP) and malignant hypertension.

Chung and collaborators [57] performed a case-control study involving 31 TMB patients and 31 healthy controls. They noted that the frequency of jugular venous reflux is higher in patients with transient monocular blindness. They measured retinal venule diameter and found that TMB patients had a wider retinal venule diameter, especially TMB patients with jugular venous reflux. They concluded that jugular venous reflux disturbed ocular venous return and that the resulting disturbed ocular haemodynamics could be a cause of TMB. See also the more recent related work [58].

18.3.3 Transient Global Amnesia (TGA)

Transient Global Amnesia is a syndrome that affects memory, transiently. In one TGA manifestation, called **anterograde amnesia**, the subject is unable to form new memories. In the TGA manifestation called **retrograde amnesia** the subject has difficulty in recalling recent memories [59, 60]. TGA has an incidence of 5 per 100000 population per year and it usually affects patients between the ages of 40 and 80 years, at an average age of 61 years. TGA appears to show no male or female

preponderance. According to Owen and collaborators [60], there is currently no agreement on the aetiology of TGA, although several theories have been suggested. These authors also state that most of the patients do not have significant vascular risk factors and that recurrence is rare.

Chung and collaborators [59], however, have suggested that TGA is related to abnormal venous flow in the brain, more specifically intracranial venous reflux. They found that occlusion of the left brachiocephalic vein by compression from the sternum and aorta during regular breathing occurred consistently amongst TGA patients with abnormal venous signals and that such occlusion and associated venous reflux was found only in patients with transient global amnesia. They concluded by stating that TGA patient's vulnerability to a TGA attack is increased by an underlying venous flow impairment. An updated review of the subject is found in [61], where the authors confirm the association of TGA with venous outflow anomalies. Related research is found in [51].

18.3.4 Ménière's Disease (MD)

Ménière's disease is a pathology of the inner ear first described by the French physician Prosper Ménière in 1861. MD is characterized by intermittent vertigo episodes, tinnitus, aural pressure and temporal sensorineural hearing loss.

The fluid dynamics of the inner ear is central to the understanding of various mechanisms associated with several hearing disorders. Potentially therefore, such disorders involve the dynamics of the arterial and venous blood systems, the cerebrospinal fluid system, as well as the perilymph and endolymph fluid compartments. A description of MD was given in Chap. 5 of this book, along with relevant references. A strong association of MD with cerebral venous outflow anomalies has been reported in the literature. See for example the animal experiments of Friis and Qvortrup [62] and their discussion on the potential role of venous occlusions in various pathologies, including inner-ear related diseases, such as MD.

See the clinical studies of Alpini et al. [63, 64] and Bruno et al. [65, 66]. See also the computational study of Toro et al. [67]. We refer the reader to Chap. 5 for more details.

18.3.5 Idiopathic Intracranial Hypertension (IIH)

Idiopathic intracranial hypertension (IIH) is a neurological condition of currently unknown aetiology [28, 68–71] that is strongly associated to altered craniospinal fluid dynamics, typically elevated intracranial pressure (ICP), that is, elevated CSF pressure.

IIH tends to be more prevalent in obese females of childbearing age. IIH requires prompt diagnosis, which if left untreated could result in rapidly progressive visual

loss [69]. The diagnosis of IIH is usually performed by invasive measurement of CSF pressure in the spinal canal (lumbar puncture, LP), coupled to neuroimaging findings. Conventional medical therapy consists of carbonic anhydrase inhibitors or topiramate, but little success is reported [28].

Interestingly, IIH is one of those neurological conditions in which the biophysical aspects of altered craniospinal fluid dynamics is more evident and more readily accepted by the medical community. In particular, regarding the role of the venous system, there is evidence that about 93% of patients with IIH will harbour some degree of dural sinus stenosis resulting in **intracranial venous outflow strictures** [28]. The precise mechanisms of intracranial fluid dynamics remain unclear.

Intracranial venous outflow strictures, on the one hand, may cause a **tumoural** effect of increased intracranial venous volume. Such increased volume, in the frame of the Monroe-Kellie hypothesis, will increase the intracranial venous pressure, resulting in anomalous CSF reabsorption and thus increased CSF volume and pressure.

On the other hand, increased CSF pressure, acting as external pressure to the intracranial venous system, may generate external venous compression affecting venous outflow. Widely accepted physically-motivated therapeutic interventions for IIH include the periodic removal of CSF through LP, use of CSF diversion procedures with the aim of reducing ICP by reducing craniospinal CSF volume.

Stenting has also been used to restore venous outflow, hence favouring CSF reabsorption and reduction of intracranial volume and pressure [28, 72].

18.3.6 Idiopathic Parkinson's Disease (IPD)

Parkinson's disease (PD), first described in detail by James Parkinson in 1817, is a progressive, neurodegenerative disorder of the central nervous system and affects mainly people older than 50 years of age. Symptoms include shaking; rigidity; difficulty with walking; depression; speech, cognitive and emotional problems. In most diagnosed PD patients, the origin of the disease is unknown, thus referred to as Idiopathic Parkinson's Disease (IPD), for which there is not cure, though medication and other forms of treatment may help alleviate symptoms.

Many clinically diagnosed cases of IPD are associated with cerebrovascular disease and white matter hyperintensities (WMH). Liu and collaborators have researched the venous connection to IPD [73]. The authors investigated the presence of transverse sinus and extracranial venous abnormalities in IPD patients, and their relationship with brain WMH. In their study they considered 23 IPD patients and 23 age-matched healthy controls. Their results show that venous abnormalities were seen in 57% of IPD patients, with **anatomical and flow abnormalities on the left side of the brain and neck**. Moreover, abnormalities seem to correlate with WMH volume.

This is another example in which cerebral venous haemodynamics and venous malformations appear to be playing an important role, to be investigated further.

18.3.7 *Alzheimer's Disease (AD)*

Alzheimer's Disease is a brain disorder, considered to be the most common cause of dementia. Dementia refers to a group of symptoms including gradual loss of cognitive functioning (thinking, remembering and reasoning), eventually rendering the patient unable to carry out the simplest daily tasks [74].

Currently, it is estimated that 57 million people worldwide are affected by dementia, of which, about 36 million are AD patients [26, 27]. The AD brain is characterised by shrinking, abnormal build-ups of proteins and neuron degeneration. Age is an important risk factor in AD. Moreover, the incidence of AD increases with age, doubling every five to ten years. For example, for persons aged between 65 and 69 years, the incidence is estimated at 0.6%, reaching 8.4% for persons above 85 years [75].

There is currently no cure for AD, as the causes of the disease remain illusive. There are, however, some interrelated vascular-based events that are helpful in attempting to unravel the pathophysiology of AD.

Cerebral Amyloid Angiopathy (CAA) is described as a cerebrovascular disorder involving the aggregation and deposition of $A\beta$ peptides in the wall of cerebral blood vessels [76]. It is currently widely accepted that the pathophysiology of AD is strongly associated with CAA, which impairs drainage pathways from the brain and contributes to accumulation of $A\beta$ and cerebrovascular dysfunction in AD. For decades, fundamental advances in our understanding of AD has focused on the connection of CAA with $A\beta$ accumulation in the arterial system [77–79].

Recently, however, there has been considerable interest in the potentially additional role of the venous system in the pathophysiology of AD, for example, in the accumulation of $A\beta$ in cerebral veins and venules [80–84].

White matter hyperintensities (WMHs), also known as **leukoariosis**, are another piece of the AD puzzle. WMHs, detected by MRI and computed tomography (CT) scans, are brain white matter lesions common in individuals older than 65 years, and are prevalent in the elderly and in AD patients. WMH is a confirmed increased risk of cognitive decline, dementia, stroke, and death. Despite their prevalence, the aetiology and neuropathology of WMH remain elusive [82], with most theories of the pathogenesis of WMH focussing on vascular aetiologies.

Venous collagenosis (VC) was first described as collagenous thickening of periventricular venules and veins associated with WMHs and aging [74]. Moody et al. [85], as early as 1995, identified venous collagenosis of the small deep vessels in autopsy cases with WMH. They found VC in 65% of subjects aged 65 years or over, characterised by excessive accumulation of collagen in the venous vessel wall, resulting in thicker walls. See also Black et al. [80]. Later, Keith and collaborators [82] performed a study of 24 pathologically confirmed AD from archived autopsy brains and 18 age-matched non-AD patients. They found: (i) collagenosis of veins less than 150 mm in diameter; (ii) significant stenosis of veins greater than 200 mm in diameter; (iii) venous collagenosis correlated positively with WMH scores. They stated that their description of histologically confirmed stenosis of the large calibre

deep medullary veins in the setting of WMH was novel. Yan et al. [86], using SWI, had previously observed a strong relationship between WMH volumes and *visibility* of the large deep medullary veins. This, according to [82], was a *visual manifestation of venous collagenosis*.

Houck and collaborators [87] performed T2-weighted FLAIR and Susceptibility Weighted Imaging (SWI) in 682 subjects mean aged 73.9 years, without dementia. They found increased diameters of the internal cerebral veins and the basal veins of Rosenthal. Moreover, this venous vascular finding was associated with greater WMH volume. More recently, Morrone et al. [74] have added new evidence on the potential role of the venous system in the pathogenesis of Alzheimer's Disease. AD has been closely associated with solute transport in perivascular spaces and dilated cerebral veins have been associated with the size of perivenous spaces [87]. It is therefore reasonable to consider the AD-venous connection in the frame of the glymphatic theory [88–91]. See Chap. 4 for background.

It seems appropriate to remark that the increasing recognition of a role for the venous system in AD is not surprising. This is in light of the established role of the venous system in the overall clearance function of wastes from tissues and the strong association of AD with the clearance function of the perivascular (periarterial and perivenous) transport system. This is more so when considering the recent work of Hu et al. [92] regarding the enhanced reabsorption function of the brain-parenchyma venous system from perivenous spaces.

Though research on the role of the venous system in the AD pathophysiology is still incipient, the evidence is sufficiently strong so as to advocate a holistic approach involving all neurofluids in AD research, as for other neurological disorders reviewed in this section.

Cerebral venous outflow strictures and disturbed venous outflow are relevant features of several neurological disorders, some of which have been discussed above. The next section deals with one method that attempts to restore physiological venous outflow.

18.4 Angioplasty for Restoring Cerebral Venous Outflow

This section is concerned with the application of **Percutaneous Transluminal Angioplasty (PTA)**, aiming at the restoration of cerebral venous outflow. The technique is known as **balloon angioplasty**, or just **angioplasty**.

PTA is described as a minimally invasive technique, whereby, through a catheter, a balloon is inserted inside the vessel stricture and inflated to a given maximum pressure, typically 6 atm. Inflation is maintained for a short period of time and repeated several times. Figure 18.4 shows three stages of PTA, as first applied by Zamboni et al. [35] to MS patients with CCSVI. The expectation of the PTA application was twofold:

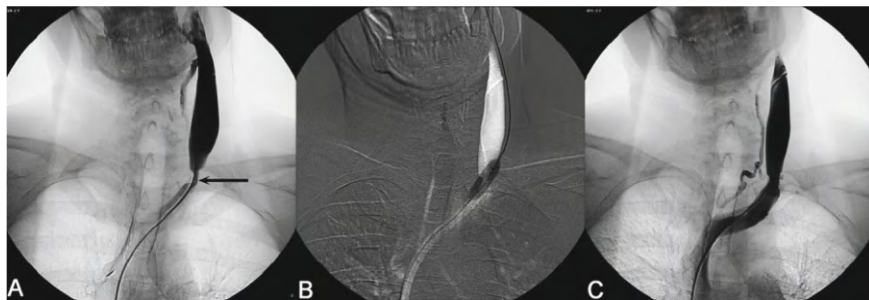


Fig. 18.4 Application of PTA. A: selective venography shows a closed annular stenosis of the proximal left internal jugular vein (arrow). B: percutaneous transluminal angioplasty was performed by road map facility. C: selective venography shows the postoperative result, where restoration of blood flow is visible (taken from [35] by permission)

1. to demonstrate that it is a feasible and safe procedure, and crucially
2. to demonstrate that by restoring physiologically correct venous return from the brain has a *positive medical impact* on patients, as described in the original papers referred to below.

18.4.1 Early Studies on PTA for MS/CCSVI Patients

Zamboni et al. [35] reported the first application of PTA to a group of 65 patients affected by clinically defined MS. The authors assessed the effect of the treatment by evaluating (i) vascular outcomes and (ii) neurological outcomes. Regarding the former, they confirmed that venous pressure was reduced. This was confirmed by, invasively, taking measurements of pressure in various **extracranial** locations. As to the neurological outcomes. These were regarded as positive, the measures being relapse rate, MRI measures and quality of life. A disappointing result of the study was **restenosing**, particularly in the internal jugular veins.

In the study, the authors isolated the effect of PTA from that of DMT (Disease Modifying Treatment) by considering two distinct groups: the ITG group was made out of patients who had *immediate* PTA in addition to DMT, and the DTG group was made out of patients on DMT who had *delayed* PTA by 6 months. In the study the authors confirmed the feasibility and safety of PTA and of re-stenosing, as a negative outcome. They also observed that PTA led to significant improvements in functional score. Annualised relapse rate was 0.12% in the ITG group, compared to 0.66% in the DTG group. Magnetic resonance imaging blindly demonstrated a trend for fewer T2 lesions in ITG, corresponding to a 10 % decrease in the ITG compared with a 23% increase in the DTG over the first 6 months of the study.

A relevant, subsequent study by Magnano and collaborators [93] showed that CSF flow dynamics is altered in patients with MS and that more severe clinical and

MR imaging outcomes in Relapsing-Remitting MS (RRMS) and clinically isolated syndrome (CIS) relate to altered CSF flow and velocity measures. In particular, it was noted that increased CSF flow and decreased CSF velocity are indicative of improved brain parenchyma venous drainage.

Zivadinov et al. [94] then carried out another study of the same group of subjects considered in [95]. They found evidence that PTA treatment for CCSVI may play a role in changing CSF dynamics of the brain parenchyma. In particular, they found that PTA changed CSF flow and velocity measures at 6 months in the ITG versus the DTG MS-treatment groups. They also found that the PTA treatment of CCSVI was related to partial improvement in the clinical and MRI outcomes of patients with MS, and there was a trend for lower accumulation of T2 lesion volume in the ITG group versus the DTG group, over the period 0–6 months. Moreover, they also found that in the DTG treatment group, patients who did not exhibit a change in CSF flow outcomes showed a greater number of relapses and accumulation of T2 lesion and whole brain T2 lesion volume.

The early studies on the effect of PTA indicate that restoration of venous outflow has a positive effect on patients. But the drawback of PTA is restenosing. This means that PTA has been useful in proving the concept but cannot be proposed as a long-term *cure*, but rather as a diagnostic tool. More research is needed. Even though the results of this first study were encouraging, as the authors themselves pointed out, the study suffered from a number of shortcomings, later corrected in a second study [95], not reported here. For more precise details the reader is referred to the original works.

18.4.2 *More Recent Studies on PTA*

A major Italian multicentre research project lead by the University of Ferrara, called the **The Brave Dreams Trial**, took place in the period 2012 to 2016. This was a randomized, parallel group, double-blind, sham-controlled trial aimed at assessing the efficacy of jugular PTA in MS patients with the CCSVI condition (ClinicalTrials.gov, identifier NCT01371760) [96]. The assessment was through two primary outcomes:

1. Combined measurement of five functional indexes: walking, balance, manual dexterity, bladder control and visual acuity.
2. Accumulation of new cerebral lesions, as detected by magnetic resonance imaging.

From the results of the study based on 115 relapsing-remitting MS patients (RRMS), it was reported that (i) PTA did not increase the proportion of patients in the RRMS class who improved functionally; (ii) PTA did not reduce the mean number of new combined brain lesions on MRI at 12 months, but (iii) there was a tendency for more patients to become free of new lesions after venous PTA, due in the main, to a reduction in new lesions appearing 6 to 12 months after randomization.

It was concluded that *Venous PTA cannot be recommended for patients with relapsing-remitting multiple sclerosis*. However, it was suggested the possibility that PTA may be beneficial for a particular subgroup of patients. This statement was based on *the observed delayed effect of venous PTA 6 months after the procedure on the magnetic resonance biomarker*.

There are three points from the paper that draw our attention.

- First, it was noted that the procedure was *was largely ineffective in restoring blood flow in nearly half the patients in the PTA group*. This means that PTA may not be *the right cure* for restoring physiological venous outflow in subjects with the CCSVI condition, due to recurrent re-stenosing.
- There is a second aspect of PTA, not discussed in the study, that is **lumen patency** after PTA. How effective is PTA in *recovering and maintaining normal venous outflow*? This question is relevant for any procedure aimed at reversing CCSVI-like conditions.
- The third point of our interest is contained in the author's statement *The delayed effect of venous PTA 6 months after the procedure on the magnetic resonance biomarker suggests a possibility that PTA may produce benefit for a subgroup of patients with MS*. This is precisely the subject of a follow up study [97], which benefited from an investigation reported in [98].

Giaquinta and collaborators [98] carried out a detailed study on the anatomic factors and patient characteristics that may influence the efficacy of PTA, as applied to the internal jugular veins, aimed at improving IJV outflow. Their study made use of data from 797 patients, who underwent catheter venography and PTA. They documented, before and after PTA, morphologic and haemodynamic anomalies of the IJVs. In this manner they could somehow predict the efficacy of PTA in the treatment of IJV anomalies. They concluded: *PTA resulted in an increased outflow through the IJVs in most patients. However, younger individuals with transverse endoluminal defects and higher pre-PTA flows are more likely to respond well to PTA, compared with those who exhibit hypoplasia, stenosis, or longitudinal endoluminal defects*.

In referring to the work of Giaquinta et al. [98], Moneta [99] recognised that the work contained a positive aspect, in spite of being rather critical of both the CCSVI-MS theory and procedural aspects of the associated research activity. He states: *The paper does address a potentially important issue of how to select the best anatomic candidates for angioplasty of veins draining the cerebral circulation and suggests a way to quantitate the haemodynamic response to angioplasty of those veins*.

Building upon the work of Giaquinta et al. [98], Zamboni and collaborators [97] set out to verify if jugular vein flow restoration, by PTA again, in the presence of CCSVI in MS patients, has or not a positive effect on **cerebral lesions**, identified through MRI. With respect to the earlier work [96] two new aspects stand out, namely (i) by making use of venograms, they discriminated between **favourable** (38) and **unfavourable** (30) patients; and (ii) they included results also from the secondary progressive MS group (SPMS).

Analysis of their results showed that *the likelihood of being free of new cerebral lesions at 1 year was significantly higher after venoplasty compared to the sham*

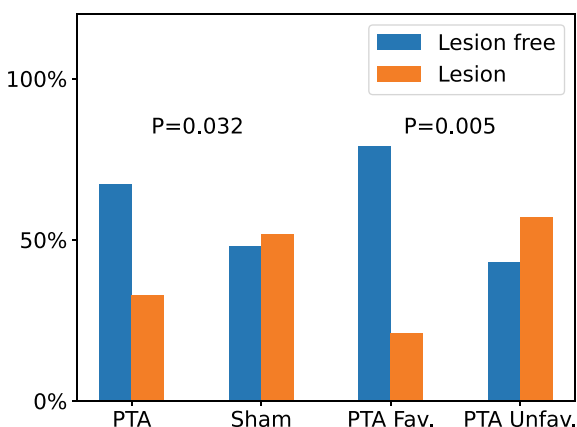


Fig. 18.5 Comparison of MS patients with CCSVI with respect to measures of new cerebral lesions on MRI: lesion-free patients (blue) and patients with new lesions (red). The left two pairs of bars compare patients who underwent PTA with those who did not (Sham). The right two pairs of bars compare PTA patients from the favourable group (PTA favourable) with PTA patients from the unfavourable group (PTA unfavourable) (Data from Fig. 3 of [97])

group. Moreover, they noted that *patients with favourable venograms had a significantly higher probability of being free of new cerebral lesions than patients with unfavourable venograms, or patients in the sham arm.*

From the results of Fig. 18.5 it is clear that PTA treated patients in the favourable group have a significantly higher probability of a lesion-free status than PTA treated patients in the unfavourable group, as well as the non-PTA group (sham). For further details on the study, the reader is referred to the original works [96–98].

The first conclusion that may be drawn from these studies is that restoration of physiological venous outflow via PTA has a positive effect on MS patients. However, PTA will only be effective on CCSVI-affected patients with PTA-favourable anatomical characteristics. That is, not all venous outflow strictures may be corrected via PTA. Moreover, PTA suffers from re-stenosing and therefore it is not a durable, long-term solution. In addition, there remains an open question regarding the effectiveness of PTA for achieving lumen patency, even before re-stenosing.

Next we address the possibility of stenting to restore physiological venous outflow, as an alternative to PTA.

18.5 Stents for Restoring Cerebral Venous Outflow

A stent is a medical device, typically an expandable tubular mesh, which is positioned in the diseased vessel segment to support the vessel wall and restore the patency of physiological fluid flow [100]. Stents can be applied in various affected human body conduits, including arteries, the urinary system and more recently, veins. Alternative

methods to maintain fluid flow include bypasses and PTA. Stents appear to be more successful in resolving the problem of recurrent restenosing observed with PTA applied to veins.

18.5.1 Stent Design Considerations

A major challenge is the design of the stent, targeted to the particular type of narrowed conduits of interest. This involves, for example, the assessment of the mechanical properties of both the conduits and the stent, the conduit function, as well the associated fluid dynamics conditions under physiological and pathological conditions [101].

The design and application of stents for arteries has a long and successful history. For veins, however, the situation is different and many new challenges need to be addressed. As seen in Chap. 1, veins differ significantly from arteries in terms of mechanical properties, function and fluid dynamics conditions. Dural sinuses are, to some extent an exception; mechanically, they are similar to arteries and the use of artery-derived stents to treat intracranial venous sinus strictures is quite widespread; see for example, [28, 72, 102, 103], and references therein. Potential complications of stenting include intra-stent thrombosis, intracranial haemorrhage and, notoriously, stent migration. For a discussion on the desirable properties of stents for veins, see the papers by Shamimi-Noori et al. [104] and by Massi and Zamboni [105].

The choice of the material properties of the stent are important at the design stage. These will respond to deformation and interaction with the mechanics of fluid flow. One would expect fluid flow conditions after stenting, to be close to physiological conditions. In practice, the diseased vessel segment with the stent in place will necessarily present mechanical properties that are different to those of a healthy vessel. Such differences in material properties will affect the fluid dynamics and wave propagation properties of flow waves and pressure waves. This can be seen at the level of the governing equations introduced in Chap. 6; see also Chap. 10.

The material properties enter explicitly in the tube law. The interfaces between the stented segment and the non-stented segment will present abrupt changes in material properties, such as the Young's modulus of elasticity. As such, these locations will be the source of a standing wave pattern that will interfere with the fluid dynamics and wave propagation. Li et al. [106] have reported a computational biofluid mechanics study addressing some of these issues. More work is needed in this direction in order to assist in the development of better stents. See also [100].

18.5.2 Stents Specifically Designed for Veins

Stents designed specifically for veins, or dedicated venous stents, are a recent development. As set out in [104, 105], stents for veins demand very specific properties.

These include compliance to easily adapt to the shape of the vein. The vein stent is also expected to be crush resistant, corrosion resistant, fatigue resistant and radiopaque, for follow-up via magnetic resonance imaging and computed tomography. The scaffold should avoid occlusion of collaterals.

Stent migration is an important issue. In order to avoid such occurrence, it is necessary to correctly chose, for example, the dimensions of the stent, as well as the force exerted radially by the stent structure; these parameters of the problem will ultimately, affect the patency of the thin-walled venous vessel.

Currently available dedicated venous stents include:

1. **The Cook Zilver Vena venous stent.** This devise is produced by Cook Medical. O'Sullivan and collaborators [107] carried out a prospective case series of 20 patients, using the Zilver Vena stent for malignant or nonmalignant iliofemoral obstruction. The results showed a 30-day patency rate of 85%. The manufacturers claim (October 2021) a 3-year rate of patency of $90.3\% \pm 2.2\%$. See also Hofmann et al. [108].
2. **The Optimed Sinus-Venous stent.** This stent is developed by Optimed. In a study carried out by de Wolf and collaborators [109] 75 patients were treated with this stent. The authors report excellent 12 month patency rates, of over 90%. They also show significant symptomatic relief, with low morbidity and no mortality. Variations of this stent include: the Optimed sinus-XL stent, the Optimed sinus-XL 6F and the Optimed sinus-Obliquus stent. See Stuck et al. [110] for a retrospective study evaluating 24 patients treated with the sinus-Obliquus stent.
3. **The Abre Venous Self-expanding stent.** This devise is produced by Medtronic. In their website the manufacturers claim a 36 month overall patency of 81.6%. See the recent study by Murphy et al. [111].
4. **Venovo Venous Stent system.** This stent is produced by Boston Scientific. An assessment of safety and patency of the Venovo venous stent for the treatment of iliofemoral vein obstruction was carried by Dake and collaborators [112]. This was a three-year study for the treatment of iliac and femoral vein obstructions.
5. **The Petalo stent.** This is a recent addition to venous dedicated stents due to Veroux and collaborators [113, 114]. The design of the stent followed an extensive prior study in a porcine model, reported in [115]. A clinical trial is underway, to assess *Safety Profile of PETALO CVS in the Treatment of Non-thrombotic Internal Jugular Vein Stenosis and Chronic Headache (PETALO)*; the trail is registered at <https://classic.clinicaltrials.gov/ct2/show/NCT04352218>.

We recommend the systematic review and meta-analysis by Razavi et al. [116] on the safety and stent placement for iliofemoral venous outflow obstructions.

18.5.3 Recent Treatments of Venous Strictures by Stenting

There has been a resurgence of interest in the once controversial subject of cerebral venous outflow strictures, presumed venous hypertension and neurological manifestations. Here we mention two examples in which alternatives to PTA were applied in order to restore internal jugular venous outflow.

Zhou et al. [72] successfully applied venous stenting to fifteen patients presumably affected by Idiopathic Intracranial Hypertension (IIH) and confirmed with isolated internal jugular vein stenoses at the J3 level. Patients presented with headache, tinnitus, papilledema, visual impairment and intracranial hypertension. Elevated ICP was confirmed by lumbar puncture. The stent used by the authors was the Sinus Super Flex 635, with a diameter of 6–10 mm and length of 40 mm, made by Optimed, Ettlingen, Germany. The authors reported success, as evidenced by resolution of headache in 4 out of 15 patients; 10 out of 15 patients recovered from visual impairments; tinnitus was resolved in 10 out of 11 patients; and in 12 out of 15 cases, the papilledema grade scores declined.

Interestingly, *the issues of IJV stenosis were soon corrected and the abnormal collateral veins disappeared or shrank immediately following stent implantation*. They concluded that internal jugular outflow strictures caused by stenosis may underlie the aetiology of IIH, for which stenting appears as a promising alternative to resolve the consequences, particularly after the failure of traditional medical treatments. This study is also interesting because it deals with presumed IIH patients with **extracranial** venous anomalies, rather than **intracranial** dural sinus anomalies, which has been the dominant IIH variant studied in the past.

Primiani et al. [117] have recently presented a case study in which they applied right C1 tuberculectomy to restore jugular venous outflow in a 30 year old man with new cognitive fatigue and brain fog in the setting of increased venous pressure. Cognitive dysfunction is just one of many medical manifestations associated with strictures in the veins of the head and neck. The authors recall that cognitive decline is irreversible in many neurological conditions. However, they recognise that in the cases related to venous outflow strictures, cognitive decline can be reversed if diagnosed and treated promptly.

Recent works include Higgins et al. [118, 119] and Bai et al. [120]. In [120] the authors address a relevant issue, that is the long-term safety and efficacy of stenting in correcting (intracranial) cerebral venous sinus stenosis and (extracranial) internal jugular venous stenosis. They conclude that using stents to treat intracranial hypertension and related neurological issues has shown to be a safe and effective approach for both internal jugular veins and cerebral venous sinus stenoses.

18.5.4 Pending Issues with Venous Stents

The effort directed at the designing venous dedicated stents is very recent, and although progress has been made, there are still a number of pending issues to be addressed, as noted in the recent study of Ma et al. [121]. The authors raised various matters regarding the performance of venous-dedicated stents already approved by the Food and Drug Administration (FDA) for treatment of iliofemoral vein obstructions. One major issue identified in the study was stent migration. The reported problem caused the recalling of two venous stent systems.

Stent durability is a major characteristic of stents. For veins, stents would generally require longer durability than those for arteries. This is because patients with chronic venous disease are relatively younger than those with atherosclerotic disease. Venous stents may be expected to last 50 years or more [104]. For this, and from additional considerations, the scaffolding of the ideal stent should disappear over time. This brings into the discussion the bioresorbable vascular scaffold technology, first developed for the coronary arterial system. So far, such technology has not proved successful, but the experience gained may be applied to the venous system, according to Shamimi-Noori et al. [104], with potentially better results. Again, stent patency comes up when discussing stent durability. As seen in [107, 109, 111, 112], stent patency is reported to be between 80 and 90%, as monitored in short time periods, from months to no more than 3 years. One wonders what happens to stent patency in a period of 50 years. Most of the reported studies on venous dedicated stents have focused on iliofemoral obstructions. While stenting dural sinuses in the head has been around for sometime, reported systematic experience on stenting neck veins is so far lacking.

It is reasonable to expect that in the future, computational modelling might help in the design of stents for veins, being aware however, of the potential challenges that the strong non-linearities of veins may pose.

18.6 Summary and Concluding Remarks

In this chapter we have reviewed neurological disorders, a large class of human diseases, and their potential link to the fluid dynamic interaction of major extracellular fluid compartments in a holistic framework, a view advocated throughout this book.

There is evidence that anomalous cerebral venous outflow has a detrimental impact on individuals health and that restoration of flow, close to physiological conditions, has a net positive effect on patients. Most of the reported experience so far relates to MS patients, though there is an increasing number of studies concerned with Ménière's Disease, with similar results. A pending question is the means by which the flow restoration may be achieved best, ensuring safety, durability and long-term vessel-lumen patency.

PTA has simply served the purpose of **making the point**, by at least partially contributing to demonstrate health benefits in an appropriately chosen subset of patients. But, from the reported experience, it seems as if PTA cannot be proposed as a long-term method for restoring venous outflow close to physiological conditions. Dedicated vein stents, subject to further technological developments in their design, hold promise as a treatment for cerebral venous outflow strictures, though experience so far is very limited, especially for neck veins.

The empirical observations so far call for the formulation of a coherent theory capable of describing a clear chain of events that goes from cerebral venous outflow strictures, to venous hypertension, vessel-wall damage, extravasation of plasma contents, reduced perfusion, reduced reabsorption, inflammation, and ultimately, to the hallmark of the disease, **demyelination**, in the case of MS.

The early works of Schelling [22], Talbert [19], and Singh and Zamboni [46] identified interesting elements for an acceptable theory applicable to subjects afflicted by various types of cerebral venous outflow strictures. On the whole, such early attempts are consistent with the reported experience so far. From here, it may be conjectured that the triggering mechanism for a subsequent chain of events is the anatomical, possibly congenital, vessel stricture. Of course this framework is only applicable to cases in which the downstream outflow stricture is properly identified and characterised.

Subsequent to the outflow stricture is the presumed venous hypertension upstream, the assumption of which is physically reasonable. However, the actual quantitative verification of cerebral venous hypertension is scarce, at least for the case of extracranial vein strictures. For intracranial strictures associated with IIH, cerebral venous pressure and ICP are measured invasively, especially in the intensive care unit, where this information is of vital importance.

For both the intracranial and extracranial cases, it has been shown theoretically, through a global closed-loop mathematical model, that anatomical venous outflow strictures cause **elevated venous pressure** and **elevated intracranial pressure**, as has been hypothesised. The computational results show disturbed craniospinal fluid dynamics, including reversed venous flow and activation of collaterals [33, 34, 122–124]. These theoretical results are consistent with some early, direct pressure measurements [35], as well as intracranial and extracranial **flow measurements** through MRI [39–41].

More challenging is the case of patients that do not exhibit obvious anatomical malformations resulting in cerebral venous outflow strictures. This is the case for some of the neurological disorders discussed in this chapter, such as Alzheimer's disease, in which the most apparent physical feature is vessel wall damage, such as collagenosis.

Haacke and collaborators [47] have recently communicated an updated, comprehensive review on **vascular effects on multiple sclerosis**, though much of the contents are applicable to other neurological disorders. The review focused specifically on: **changes in the venous vasculature and inflammation, which are likely to be the earliest processes in the development of the disease**. From an initial data base of about 900 papers, the authors selected about 200 publications to be

reviewed in detail, addressing 17 specific subtopics, including: collagenosis and vessel narrowing; ischemia and abnormal perfusion.

The authors set out to understand the chain of events in the pathophysiology of MS related to venous vascular effects. Admitting that the source of progression of MS is currently not well understood, from their extensive review they deduce that, potentially, **locally disrupted flow in turn leads to remodeling of the medullary veins, followed by endothelial damage, with the subsequent escape of glial cells, cytokines, etc.** Nonetheless, Haacke et al. [47] admit the classical **Chicken or Egg** causality paradox. That is, **does venous-wall damage precede inflammatory response or does inflammation lead to vasculitis, followed by an autoimmune response and demyelination?**

We conclude that the task of elucidating the mechanisms at play in neurological disorders requires a comprehensive, multidisciplinary and holistic approach, involving the full circulation and the interactive dynamics of all neurofluids. In this context, mathematical modelling, as advocated in this book, may be a useful tool to be used in conjunction with experimental methods, advanced modalities of imaging methods [125, 126] and the latest technological advances.

Much of the material of this chapter was gathered from classical textbooks and from many research papers. The literature on the subject is vast, for which it is challenging to make recommendations for further reading. Still we recommend the following works: Lassmann, Bruck and Lucchinetti [13]; Adams, Abdulla, Torres and Poston [20]; Talbert [19]; Haacke, Ge, Sethi, Buch and Zamboni [47]; Morrone, Bishay, McLaurin [74].

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Chapter 19

Arterial Hypertension and Bodily Fluids



Abstract Arterial hypertension affects one-quarter of the global population. It is the leading single contributor to global mortality and a major risk factor for a broad range of cardiovascular diseases, including stroke, myocardial infarction, heart failure, chronic kidney disease, peripheral vascular disease, cognitive decline, and premature death, to name just a few. The contents of this chapter are framed in the holistic fluid dynamics spirit of this book, with a focus on mathematical modelling and computation. The basics of hypertension, as found in most classical books or reviews, are briefly summarised, including definitions, prevalence, aetiology, and risk factors. We then examine associated cardiovascular changes, initially focusing on classical descriptions of events in the arterial side of the circulation, such as vascular remodelling and increased resistance. Next, we discuss the role of the venous system in arterial hypertension, starting with a description of the relevant properties of the venous district—such as its large proportion of total blood volume (between 70 and 85%) and its high compliance, particularly in the splanchnic region, which stores most of the blood. Reduced compliance of the venous system in arterial hypertension is then examined, beginning with the seminal works by London and Safar in the 1970s and 80s. Reduced venous compliance leads to the displacement of blood volumes from the peripheral venous system to the central circulation, causing disturbances in bodily fluid balance and an increase in blood pressure. The role of the sympathetic nervous system in regulating venous compliance is reviewed, including both clinical studies and recent experimental animal research. Finally, we briefly discuss some current theories on the aetiology of hypertension and highlight issues that warrant further investigation. The chapter concludes with a summary of key aspects in hypertension and recommendations for further reading.

19.1 Arterial Hypertension

The aetiology of primary (or essential) hypertension and its causes are currently unknown, but are likely to be multifactorial in character. Secondary hypertension, on the other hand, is associated with an identified organic cause and constitutes about 5% of hypertension cases [1], a proportion regarded as very conservative by

others [2]. Here we discuss the very basics of arterial hypertension following classical, mainstream sources, such as the textbooks of [1, 3–7]. Advanced material is found, for example, in the seminal works of [8–10].

19.1.1 Definitions of Hypertension and Classification

Arterial hypertension is commonly understood as a chronic disease characterised by **raised arterial blood pressure**. However, precise quantification of *raised* is a non-trivial task, for which a clinical criterion is adopted in practice. That is, evidence of substantial, treatable harm caused by the condition; this leads to a more precise definition of hypertension. Following Levick [1]: *clinical hypertension is a rise in arterial blood pressure sufficient to raise the incidence of strokes, myocardial infarction, heart failure and renal failure*. Thus, the definition of hypertension is referred to blood pressure readings, at which the benefits of treatment clearly outweigh the risks of no treatment.

Based on Levick's definition, hypertension is established [1] from repeated measurements of brachial artery systolic pressure p , but distinguishing two groups of patients according to age. Thus, patients under 50 years of age are defined as hypertensive if systolic blood pressure (SBP) is greater than 140 mmHg and diastolic blood pressure (DBP) is greater than 90 mmHg. Conventionally, this is expressed as $p > 140/90$ mmHg [1]. Older patients are defined as hypertensive if $p > 160/95$ mmHg. The adopted thresholds for defining hypertension vary across the relevant literature. For example, Ramzy [11] argues that, as documented by clinical trials, the threshold $p > 140/90$ mmHg applies universally to all subjects regarded as hypertensives.

Various terminologies are used for arterial hypertension. **Essential hypertension**, or **primary hypertension**, is understood as elevated blood pressure without an obvious organic cause. The terminology **essential** has been used from the times when it was thought that arterial pressure elevation was essential to preserve organ perfusion. Currently, the term **essential** is meant to imply that the precise mechanisms that lead to elevated arterial blood pressure are unknown.

Idiopathic hypertension is also used for primary hypertension, conveying again the meaning that the precise aetiology of the disease is unknown. That essential hypertension is idiopathic is disputed by Malpas [10], who argues that many of the contributing factors to hypertension are known already, the problem being that of unravelling the complex interconnections of these in a multifactorial scenario; see also Rossi et al. [2]. Other terms include **benign hypertension** and **malignant hypertension**, with obvious meaning for each case.

As already stated, **secondary hypertension** refers to the case in which an identifiable cause exists. One example is renal artery stenosis, a narrowing of one or both renal arteries [1, 12]. Analogous vascular conditions have recently been identified by Warnert et al. [13]; they suggest that congenital vascular variants of the

posterior cerebral circulation and cerebral hypoperfusion could partially explain the pathogenesis of essential hypertension; see Sect. 19.5.1.

It is important to note that, in establishing arterial hypertension, the pressure readings, as well as the various classes of patients to take into account, are not set in stone and thresholds and classes of patients may vary across the relevant literature [11]. According to guidelines set out by the European Society of Hypertension [14], hypertension is diagnosed by repeated office Systolic Blood Pressure (SBP) ≥ 140 mmHg and/or Diastolic Blood Pressure (DBP) ≥ 90 mmHg. See Fig. 19.1. These thresholds relate to clear evidence of arterial-pressure-lowering treatment benefit. It is also noted in [14] that much lower arterial pressure values, office SBP ≥ 115 mmHg and DBP ≥ 75 mmHg, are already related to cardiovascular or renal morbid or fatal events. Following [14] hypertension is classified in terms of grades or stages. Grade relates to blood pressure levels, as defined in Fig. 19.1.

The stage classification type is based on the impact of the disease on the patient, and is defined as follows:

- 1. **Stage 1:** uncomplicated hypertension; for example, without hypertension-mediated organ damage or established cardiovascular disease.
- 2. **Stage 2:** presence of hypertension-mediated organ damage or intermediate chronic kidney disease or diabetes.
- 3. **Stage 3:** established cardiovascular disease or advanced-stage chronic kidney disease.

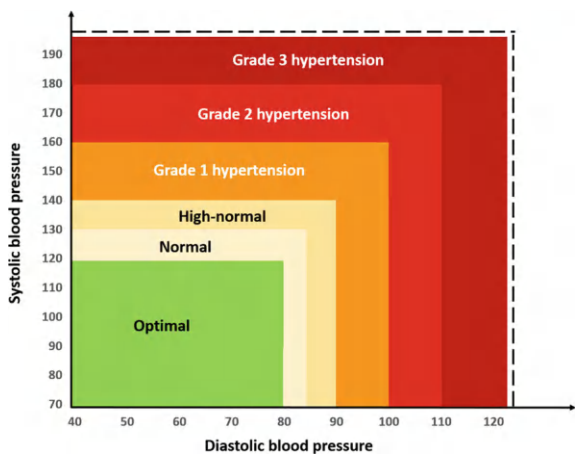


Fig. 19.1 Hypertension stages: optimal, normal, high-normal, grade 1 hypertension, grade 2 hypertension and grade 3 hypertension. Two additional cases are **a** isolated systolic hypertension: $p_{sys} \geq 140$ mmHg and $p_{dia} < 90$ mmHg, and **b** isolated diastolic hypertension: $p_{sys} < 140$ mmHg and $p_{dia} \geq 90$ mmHg (courtesy of Dr. Brigitte Toro. Data from the 2023 European Society of Hypertension guidelines for the management of arterial hypertension [14])

19.1.2 *Prevalence of Hypertension*

The prevalence of a disease is the proportion of a population affected by it, usually reported as a percentage, or alternatively, as the number of cases per ten thousand or per one hundred thousand individuals. Arterial hypertension is a disease that affects more than 25% of the world population. Naturally, the prevalence, usually estimated, depends strongly on several aspects, such as the very definition of hypertension adopted, the age brackets considered, the population under study, amongst others. Regarding the sampled population, the prevalence of hypertension depends strongly on the countries or regions of the world considered, as well as on the socio-economic groups of populations inside a country or region. For example, Levick [1] reports that in England in 1999, at age 16–24 years, the prevalence of pressures above 140/90 mmHg (hypertension) was 16% in males and 4% in females; but at age 75 years, these figures increased to 74% and 78%, for males and females respectively.

The comprehensive study of Zhou et al. [15] estimated the prevalence of hypertension, defined as pressures above 140/90 mmHg, or taking medication for hypertension, using data from 1990 to 2019 on people aged 30 to 79 years, for 200 countries and territories. They estimated that in the year 2019, approximately 1.28 billion adults aged 30 to 79 were affected by arterial hypertension, with a global age-standardized prevalence of 34% in men and 32% in women. They also found that the number of people with hypertension, in the age bracket considered, doubled from the year 1990 to 2019. In men the number of hypertensives increased from 317 million in the year 1990 to 652 million in the year 2019. In females the increase was from 331 million in the year 1990 to 626 million in the year 2019.

Interestingly, the prevalence does not seem to depend too obviously on the average income level of countries. For example, they found that *in 2019, age-standardised hypertension prevalence was lowest in Canada and Perú for both men and women; in Taiwan, South Korea, Japan, and some countries in western Europe, including Switzerland, Spain, and the UK for women; and in several low-income and middle-income countries such as Eritrea, Bangladesh, Ethiopia, and Solomon Islands for men*. However, the prevalence of hypertension varies quite substantially by regions within a country. For example, as reported by LaCruz et al. [16], the eastern parts of Germany reveal a higher prevalence of hypertension than the western parts.

Hypertension-related complications account for 7.7–10.4 million deaths worldwide every year [15]. Moreover, the prevalence of hypertension and its complications are increasing and there is a trend of decreasing control of hypertension [17].

19.1.3 *Aetiology, Diagnosis and Treatment*

Hypertension is analogous to a tsunami. It is a silent, virtually undetected, fast-moving wave in the deep sea that only becomes visible when it reaches the shoreline—by then, the disaster is imminent.

Aetiology and Factors Contributing to Hypertension

The precise aetiology of essential hypertension remains largely unknown. There are, however, many changes associated with the disease, with current understanding pointing to a very complex combination of many factors as the cause [10, 14]. These factors can be grouped into:

1. **Environmental factors.** These include geographic factors, such as sunlight exposure, altitude, temperature and environmental and noise pollution; air pollution; circadian rhythm; green spaces, such as areas with vegetation, parks, and forests; infections; and socio-economic deprivation [17–19].
2. **Lifestyle factors.** These include diet, stress, sleep quality, alcohol consumption, smoking, general adiposity, abdominal adiposity and physical activity (or lack of) [14]. Importantly, lifestyle factors and environmental determinants are interconnected, as small changes in one factor might affect others [17]. This highlights the relevance of lifestyle factors, which being in principle modifiable, have the potential for reducing and controlling blood pressure [17, 20].
3. **Genetic factors.** For decades the association between elevated blood pressure and genetic factors has been intensively studied [21–23]. These studies have highlighted the association of elevated blood pressure with well over 1000 genetic factors [24], and that genetic predisposition affects disease progression.
4. **Regulatory factors.** The disruption of regulatory mechanisms linked to hypertension is also well known. Classically, the main regulatory mechanisms involved in the pathogenesis of essential hypertension include sympathetic nervous system activation, vascular smooth muscle dysfunction, endothelial dysregulation, increased oxidative stress and disrupted renin-angiotensin-aldosterone regulatory activity [25, 26].

Clearly, all of these factors are linked to one another and the precise role played by each of them and by all their possible combinations remain poorly understood [10].

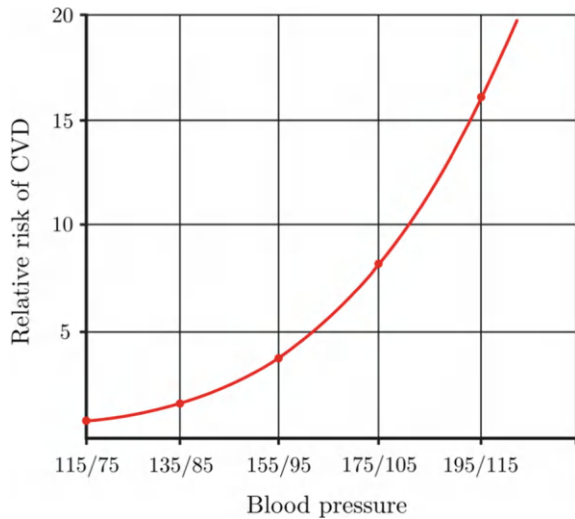
Hypertension: a Risk Factor for Cardiovascular Diseases

It is well established that hypertension is a risk factor for many cardiovascular diseases (CVD). Moreover, as shown in Fig. 19.2, the relative risk factor increases non-linearly with blood pressure elevation. For each 20/10 *mmHg* increase in blood pressure, the risk of developing cardiovascular disease is doubled [7].

Lim et al. [27] reported that in the year 2010 elevated blood pressure was the leading risk factor for the overall global burden of disease. In particular, Rapsomaniki et al. [28], reported a broad range of 12 identified cardiovascular diseases, namely:

- stable and unstable angina,
- myocardial infarction,
- unheralded coronary heart disease death,
- heart failure,
- cardiac arrest/sudden cardiac death,
- transient ischaemic attack,

Fig. 19.2 Cardiovascular disease (CVD) risk as a consequence of hypertension. For each 20/10 mmHg increase in blood pressure, the risk of developing cardiovascular disease doubles (courtesy of Violeta Da Rold. Data from [7])



- ischaemic stroke,
- subarachnoid haemorrhage,
- intracerebral haemorrhage,
- peripheral arterial disease and
- abdominal aortic aneurysm.

Interestingly, the reported associations were different for systolic and diastolic blood pressure, and they were also heterogeneous amongst different pathologies [28]. Moreover, hypertension may affect other organs not alluded to above. Examples include:

- The ocular system. Hypertension has profound effects on various parts of the eye. Elevated blood pressure results in retinal microvascular changes, called hypertensive retinopathy; these comprise generalised and focal retinal arteriolar narrowing, arteriovenous nicking (compression of venules by high pressure arteries), retinal haemorrhages, microaneurysms and, in severe cases, optic disc and macular oedema [29, 30].
- The hearing system. The link between hypertension and the development of hearing loss has recently been examined by Toyama and Mogi [31]. High blood pressure is a significant risk factor for the development of hearing impairment [32].

Diagnosing Hypertension

Like the tsunami wave, a major problem with hypertension is detection. But detection is only one problem, as not all detected hypertensive subjects are treated, and then, of those treated not all respond to the treatment, for a wide variety of reasons.

An idea of the problem is given by Fig. 19.3, which refers to world figures for the year 2019 given by Zhou et al. [15]. The total estimated number of hypertensives

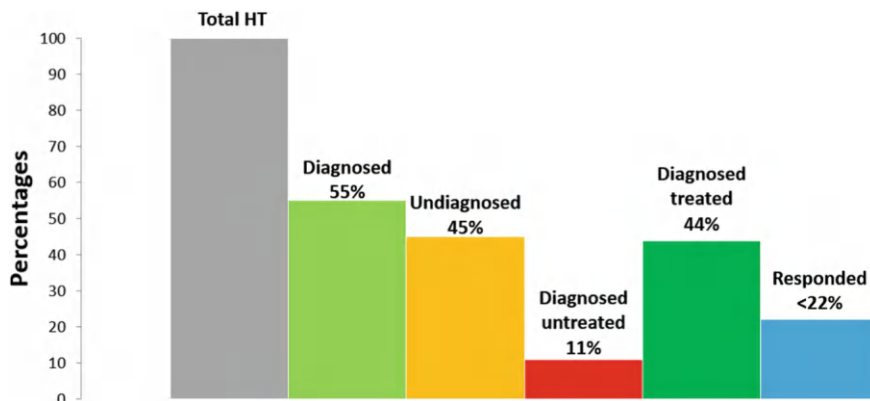


Fig. 19.3 Hypertensives worldwide in the year 2019 were 1.28 billion (Total HT). Relative percentages for sub-classes are given: Diagnosed, Undiagnosed, Diagnosed untreated, Diagnosed treated, and Responded (courtesy of Dr. Brigitte Toro. Data from [15])

worldwide was 1.28 billion, shown as 100% in the figure. Of these HT subjects only 55% were diagnosed, leaving 45% undiagnosed. Of the diagnosed, 11% of the total were not treated, leaving 44% of HT diagnosed and treated. Finally of these, less than one half responded well to the treatment (see bar Responded). That is, less than 22% of the total HTs worldwide, that is less than 0.28 billion, received treatment and the therapy was effective.

The numbers illustrated in Fig. 19.3 point to two main issues, namely

- **Detection.** Diagnosing hypertensive patients relies heavily on the health care systems around the world. Availability of care depends on socio-economical conditions of each country or region. It is known that there is a large disparity amongst countries and also in various regions within a single country. But the geographical criterion is not enough to account for the heterogeneity, as in a single town or village there will be socio-economic and cultural differences that will show when it comes to detection, and detection and treatment.
- **Therapy.** This points to the current lack of effectiveness in treating hypertensive patients. It is reasonable to suggest that this is due to a poor understanding of the aetiology of the disease.

Hypertension Treatment Options

Conventional treatment options fall into two main categories, roughly aligned with the four types of factors identified above that are thought to contribute to the aetiology of hypertension, namely

- **Life style.** These respond to environmental and lifestyle factors identified above. Examples include exercise and physical activity, diet, reduction of body weight, discontinuing smoking, and many more [14, 33].

- **Pharmacological treatments.** These respond to genetic and regulatory factors identified above. See Mancina et al. [14] for a comprehensive list of antihypertensive drugs. One class of such drugs are blockers of the renin-angiotensin system; this hormonal system regulates fluid and electrolytes, which in turn are strong determinants of blood pressure. Another class of drugs are of the diuretic type, which can be used to control body fluid volume and in turn modify blood pressure. Other drugs concentrate in altering muscle contractility and, consequently, vessel stiffness and/or heart contractility. Calcium channel blockers reduce vessel stiffness. Beta-blockers act to slow down the heart rate and reduce the force at which blood is pumped in the cardiovascular system. For a comprehensive review see [34].

19.1.4 Obesity and Hypertension

Obesity is strongly associated with numerous co-morbidities, with arterial hypertension being among the most prevalent. Furthermore, the global prevalence of obesity has been steadily increasing. Obesity is usually measured by the body mass index (BMI), even though, BMI does not reflect body fat distribution, which is an important factor. There is evidence that the prevalence of hypertension increases with body mass index BMI [35, 36], and therefore obesity deserves special mention. The body mass index BMI is defined as follows:

$$BMI = \frac{W}{H^2}, \quad (19.1)$$

where W is the person's weight, in kg , and H is the person's height, in m . The World Health Organization (WHO) has suggested the following classification of the population, according to BMI

$$\text{Classes according to } BMI : \left\{ \begin{array}{ll} 18.5 \leq BMI < 25 & : \text{Normal ,} \\ 25 \leq BMI < 30 & : \text{Overweight ,} \\ 30 \leq BMI < 35 & : \text{Obese class I ,} \\ 35 \leq BMI < 40 & : \text{Obese class II ,} \\ 40 \leq BMI & : \text{Obese class III .} \end{array} \right. \quad (19.2)$$

We note that in the original definitions in [36], there were some small gaps. Definitions (19.2) are slightly modified versions, by the author of this book.

Traditionally, only the first three classes had been considered. In what follows, as a practical example, we consider a recent study aimed at establishing an association between body mass index and arterial hypertension. Landi et al. [36] considered a sample of 7907 Italian participants (males: 3427; females: 4480) of average age 55.4 ± 15.0 . On application of criterion (19.2), participants were classed as follows:

$$\left\{ \begin{array}{l} 57\% \quad : \text{Normal weight} , \\ 38\% \quad : \text{Overweight} , \\ 11.8\% : \text{Obese class I} , \\ 2.5\% \quad : \text{Obese class II} , \\ 0.7\% \quad : \text{Obese class III} . \end{array} \right. \tag{19.3}$$

It is interesting to note that in this population sample, only 57% had normal weight according to the adopted *BMI* criterion (19.2).

Figure 19.4 shows relative percentages of normotensives and hypertensives for each of the five classes (19.3) defined according to the *BMI* criterion (19.2). Two main conclusions emerge from Fig. 19.4:

- 45% of the normal-weight subjects are hypertensive, and
- the hypertensive proportion increases (non-linearly) as the obesity-class *BMI* increases.

The strong association of obesity to hypertension points to the **life-style treatment** option as potentially promising. As a matter of fact it is a well established option in practice. It rests basically upon an adequate diet and systematic physical activity. But, *easier said than done*, in practice.

An alternative way to deal with obesity is bariatric, or metabolic, surgery. Moriconi et al. [37] state that bariatric surgery is the most effective strategy for inducing significant and sustained weight loss in patients with obesity. Bariatric surgery is a radical measure, with potential risks and side effects, and is applied only in cases in which other strategies have proved ineffective. See [37, 38], and references therein.

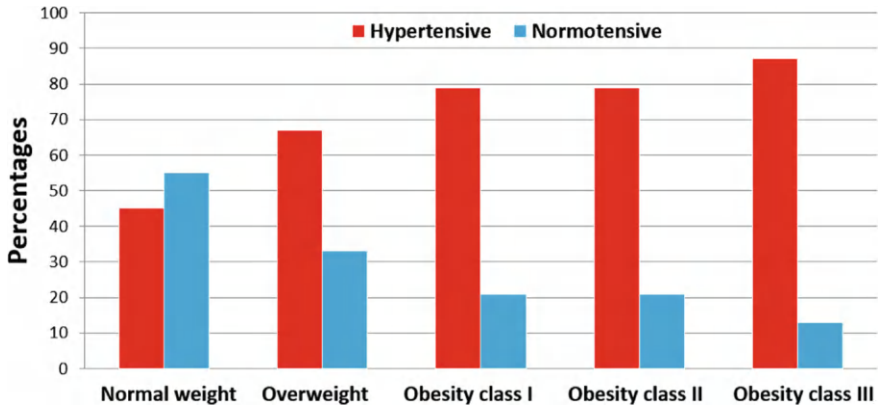


Fig. 19.4 Landi’s study of 7907 Italian individuals (males: 3427; females: 4480) of average age 55.4 ± 15.0 . Relative percentages of normotensives and hypertensives for each of the five classes defined according to the *BMI* criterion (19.2). Note that 45% of the normal-weight subjects are hypertensive and the proportion of hypertensives increases (non-linearly) as the obesity-class *BMI* increases (courtesy of Dr. Brigitte Toro. Data from Landi et al. [36])

19.1.5 Other Forms of Hypertension

So far we have spoken of essential (or primary, or idiopathic) hypertension to mean elevated arterial blood pressure without an obvious organic cause to explain it. But there are other forms of hypertension. These include elevated arterial pressure, but also elevated pressure in some other bodily fluid compartment, as briefly explained below.

Isolated systolic hypertension is defined as the condition determined by the following pressure bounds $p_{sys} \geq 140$ mmHg and $p_{dia} < 90$ mmHg; see Fig. 19.1.

Isolated diastolic hypertension is defined by the pressure-value conditions $p_{sys} < 140$ mmHg and $p_{dia} \geq 90$ mmHg; see Fig. 19.1.

Secondary hypertension refers to the case in which an identifiable cause exists for elevated arterial blood pressures, consistent with definitions for primary hypertension. It is much less common than essential hypertension (no cause known), about 5% of hypertension in general [1]. See also Rossi et al. [2], who argue that the proportion of secondary hypertension patients is much higher.

Stenosis, the narrowing of the blood vessel lumen, accounts for some of the cases of secondary hypertension. **Renal stenosis**, narrowing of arteries that carry blood to the kidney, is a prominent example [12]. A very interesting case that may be regarded as secondary hypertension has recently been studied by Warnert et al. [13]; they suggest that congenital vascular variants of the posterior cerebral circulation and cerebral hypoperfusion could partially explain the pathogenesis of hypertension; see Sect. 19.5.1 for details.

Pulmonary arterial hypertension (PAH) is a disease manifested through sustained elevated resting mean pulmonary arterial pressures $p \geq 20$ mmHg and characterised by structural remodelling of the pulmonary arteries [39–41]. PAH is associated with right ventricle failure and premature death.

Cerebral hypertension refers to elevated pressure of the cerebrospinal fluid immersed and surrounding the brain, spinal cord and the optic nerve; this pressure is often referred to as Intracranial Pressure (ICP); for background see Chaps. 3, 4, 5, 11 and 18. Though not often recognised, intracranial hypertension is accompanied by elevated venous pressure in the craniospinal cavity and disturbed cerebral haemodynamics [42, 43]. Normal values of ICP are about 11 mmHg.

Ocular hypertension is loosely defined as elevated pressure inside the eye; this pressure is termed Intraocular Pressure (IOP). By pressure inside the eye is meant the pressure of the ocular fluid called **aqueous humour**, both in the anterior and posterior chambers; a role of the vitreous chamber housing the vitreous humour is

also recognised [3]. See Chap. 5 for background and precise definitions. The range of IOP considered to be normal is 12–20 mmHg, with an average IOP of about 15 mmHg.

Inner-ear fluids hypertension refers to elevated pressures, or pressure differences, in the inner-ear fluid compartments endolymph and perilymph; see Chap. 5 for background and definitions. Ménière’s disease is an inner-ear illness characterised by **endolymphatic hydrops**, which features elevated endolymphatic pressure in the membranous labyrinth [44]. The pressures in these inner-ear compartments are related to pressures in the cerebrospinal fluid, via the cochlear aqueduct and endolymphatic sac.

Most importantly, the hypertension cases mentioned above refer to elevated (above normal) pressure in a specific fluid compartment, not just in the arterial blood. The definition of hypertension extends to other fluid compartments, and although it may not be widely recognised, it certainly involves the disrupted dynamics and interactions between adjacent fluid compartments, including the circulation

19.2 Arterial Changes in Hypertension

For a long time it has been known that arterial hypertension modifies the circulatory architecture in the heart and blood vessels. Classical accounts of hypertension recognise that the heart is affected by hypertrophy, as are the large and medium-sized blood vessels [45]. It is also accepted that changes in the large elastic arteries lead to increased wall stiffness and to increased systolic pressure; and that in small arterial vessels there is wall thickening and lumen narrowing, contributing to increased vascular resistance [1]. In this section we succinctly examine some of these issues, but only regarding the arterial side of the circulation. Other aspects will be reviewed separately.

19.2.1 Arterial Remodelling

Figure 19.5 motivates the contents of this section; it depicts the various districts of the systemic circulation and their functions. In this section we briefly review blood vessel pathological changes associated with arterial hypertension. Specifically, we review changes in the wall of (i) small arterial resistance vessels and (ii) large elastic vessels [1]. See Chap. 1 for a description of the vascular system, and blood vessels in particular.

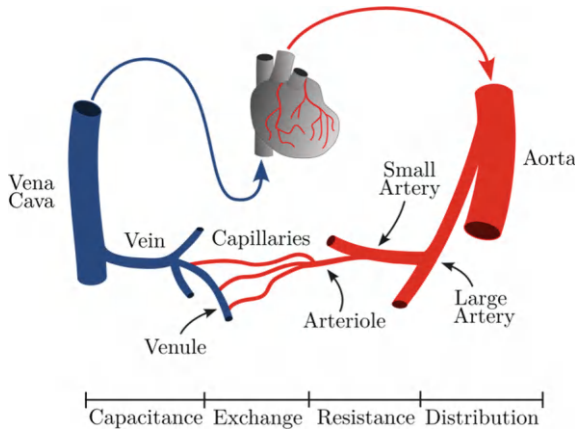


Fig. 19.5 Sketch of various districts of the systemic circulation and their main functions. In arterial hypertension large arteries and resistance vessels undergo changes. Capacitance vessels of the venous system hold most of the blood volume, have much larger compliance than their arterial counterpart, and under reduced compliance may redistribute blood volume to the central circulation (courtesy of Violeta Da Rold. Data from [46])

Small Arteries and Resistance

Vascular resistance, as seen from Poiseuille's law in Chap. 2, is given as

$$R = \frac{128}{\pi} \frac{L\mu}{d^4}. \quad (19.4)$$

See [3]. Thus, vascular resistance R is directly proportional to vessel length L and viscosity μ ; and is inversely proportional to the fourth power of the vessel diameter d . Here we are specifically interested in the following hypertension-related changes in small arterial vessels characterised by small lumen diameter d :

1. **Rarefaction**,
2. **Narrowing of vessel lumen** (reduction of d) and
3. **Thickening of the vessel wall** h_0 .

Rarefaction refers to a reduction in the number of vessels per unit tissue volume, which by itself contributes to increased resistance to flow. According to [1], in established hypertension, rarefaction in the skin results in a reduction of the capillary density by about 21%.

Figure 19.6 illustrates changes in resistance vessels in hypertension, namely vessel wall thickening and vessel lumen reduction. Figure 19.7 illustrates the relative changes in vascular resistance (left) and mean arterial pressure (right), resulting from changes in resistance vessels, as illustrated in Fig. 19.6.

Fig. 19.6 Illustration of changes in resistance vessel (right) with respect to the normal original vessel (left); these changes are characterised by narrowing of the lumen and thickening of the vessel wall (courtesy of Dr. Brigitte Toro. Data from [47])

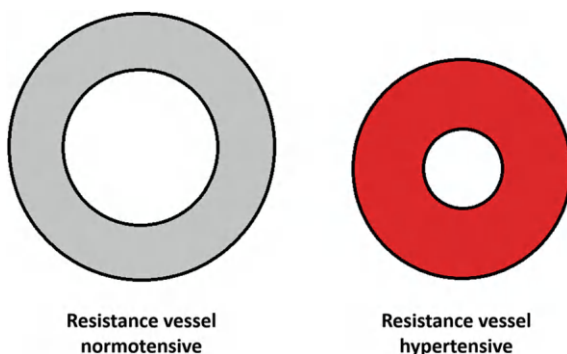
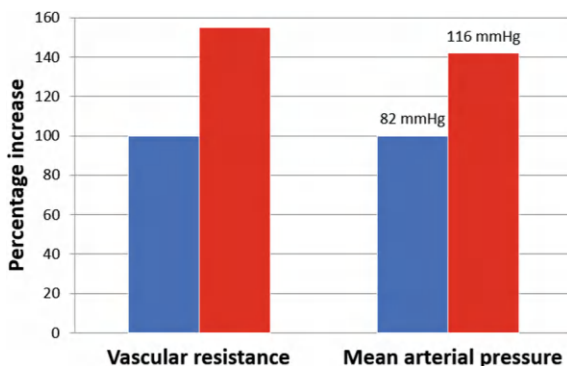


Fig. 19.7 Relative changes in vascular resistance (left) and mean arterial pressure (right) resulting from changes in resistance vessels, as illustrated in Fig. 19.6. Normotensive shown in blue and hypertensive in red (courtesy of Dr. Brigitte Toro. Data from [47])



From Darcy's law of flow, see Chap. 2, mean arterial pressure MAP is related to cardiac output CO and vascular resistance R via the equation

$$CO = MAP/R \rightarrow MAP = CO \times R. \quad (19.5)$$

Generally, in the early stages of hypertension the cardiac output CO is raised and the resistance R increases, even if only marginally, above normal [1]. Hence, from (19.5) mean arterial pressure MAP increases, as expected; the rise in pressure is due to increases in both cardiac output and resistance. It is also accepted that changes in cardiac output in established hypertension are insignificant, that is CO remains essentially constant; see [45] for relevant discussion on CO in hypertension. Therefore, in established hypertension, the pressure elevation is caused essentially by increased resistance alone, in every organ of the body, as expressed in equation (19.5). The vessel wall changes, in addition to increased resistance, also affect the sensitivity of the **baroreflex**, even if this remains active in hypertension.

Large Artery Stiffening

We now consider the role of elastic arteries in arterial hypertension, see Fig. 19.5. These are the largest arteries in the body and include pulmonary arteries, the aorta and its branches, that is, brachiocephalic artery, common carotid arteries, subclavian arteries and common iliac arteries; they have the largest diameter d amongst arteries, but their vessel walls thickness h_0 are relatively thin compared with the overall size of the vessel [3]; see Chap. 1 for background on the cardiovascular system. In hypertension, the structure of these arteries deteriorates through elastin fragmentation, dilatation and increased wall stiffness, which results in reduced compliance. The effects of the changes manifest themselves in magnifying the increases of (i) systolic hypertension and (ii) pulse pressure. It is relevant to recall that morbidity and mortality relate more closely to systolic hypertension than to diastolic hypertension [1].

The increase in systolic pressure is realised through the following two mechanisms:

1. **Increased arterial stiffness.** Recall that stiffness is elastance (E) and compliance C is the reciprocal of elastance, that is $C = 1/E$. Also recall that pulse pressure (PP) is the pressure wave generated by left ventricular contraction at each heart-beat, that travels through the blood vessels. PP is the difference between systolic and diastolic pressures and depends on the ratio of stroke volume (SV) to compliance. Therefore a rise in elastance (a reduction in compliance) results in an increased pulse pressure, leading to a larger increase in systolic pressure than in diastolic pressure.
2. **Increased pulse wave velocity.** Pulse wave velocity is the speed of propagation of the pulse wave [1, 7, 48]. Increased wall stiffness causes increases in pulse wave velocity. Therefore, the times for reflected waves to arrive back in the ascending aorta are shorter. Furthermore, peripheral resistance also contributes to enhance wave reflection. These phenomena cause **systolic augmentation**, whereby the reflected waves augment the incident systolic pressure wave. See Fig. 19.8.

We have examined how arterial wall remodelling results in increased resistance and increased arterial pressure. A general discussion on how ageing affects the cardiovascular system is found in the recent review by Climie et al. [48]. We now discuss how blood itself can also contribute to increased resistance.

19.2.2 Haemorheology and Resistance

As seen from Poiseuille's law (19.4), resistance depends directly on viscosity. The influence of blood rheology and blood viscosity μ on resistance is a largely neglected aspect of haemodynamics in hypertension. A motivation for introducing blood rheology into the discussion emanates from two accepted facts in hypertensive

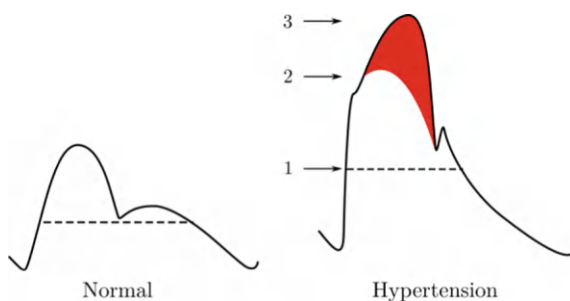


Fig. 19.8 Pressure waves in normal (left frame) and hypertensive (right frame) subjects. Dashed horizontal lines depict mean arterial pressures. In hypertensive case (right) level 1 dashed-line shows the increase in mean arterial pressure due to increased vascular resistance from small resistance vessels. Level 2 represents the effect of reduced aortic compliance in large elastic vessels, generating an increase in the systolic incident wave; compare to original incident wave. Level 3 shows the systolic augmentation in pressure caused by the return of large reflected wave depicted by the area portion in red (courtesy of Violeta Da Rold. Data from [1])

subjects, namely (i) normal, or quasi-normal, total blood volume and (ii) reduced plasma volume [49]. Then, it naturally follows that haematocrit will be increased in hypertensive subjects. Haematocrit is the volume fraction occupied by red blood cells, see Chap. 1. As is well known, the volume of red blood cells accounts for almost the entire volume of formed elements in blood. Elevated haematocrit would significantly impact blood viscosity, thereby increasing resistance. Surprisingly, little attention has been paid to this issue in the main stream hypertension literature. As stated by Jae et al. [50], little is known on the relationship between hematocrit and the risk of hypertension.

Blood Rheology, Haematocrit and Viscosity

Following Chap. 2, see also [51], the apparent viscosity of blood depends on the existing shear forces (i.e., blood behaves as a non-Newtonian fluid) and is determined by (i) the viscosity of plasma, (ii) the rheological behaviour of formed elements and (iii) the haematocrit (He), defined as the proportion of blood volume occupied by red blood cells. Figure 19.9 shows the distribution of relative viscosity of normal blood, as a function of haematocrit, compared to the viscosity of blood plasma and water. Recall that blood plasma is composed of 91.5% water and 7% dissolved substances. Clearly, He is a major determinant of whole blood viscosity [51, 52].

In humans, given normal values for haematocrit in the literature vary somehow. Hall [6], page 165, gives average haematocrit in females as 0.38 and in men as 0.42. But on page 287 these figures are 0.36 and 0.40, respectively. On the other hand Levick [1] gives 0.40 and 0.45 for women and men respectively. As to normal ranges $He \in [a, b]$ in humans, there are again considerable variations in the literature. Figures published depend on specific populations studied. For example Jae et al. [50], in their study of more than 5000 men, quote the range $0.39 \leq He \leq 0.50$ as normal.

Fig. 19.9 Relative viscosity of normal blood as function of haematocrit He , compared to viscosity of blood plasma and viscosity of water. At the reference value of $He = 0.45$ blood is about 4 to 5 times more viscous than water (Data from [6]). See Chap. 2

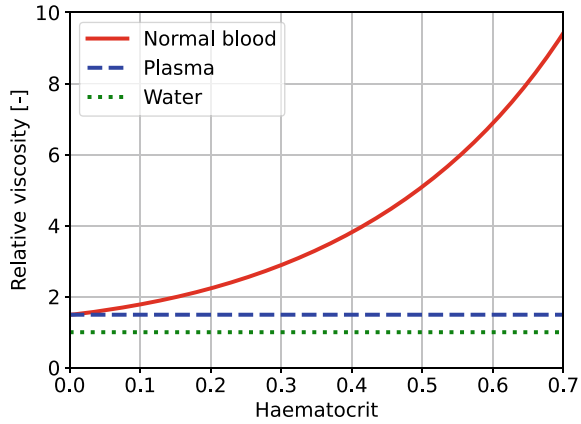
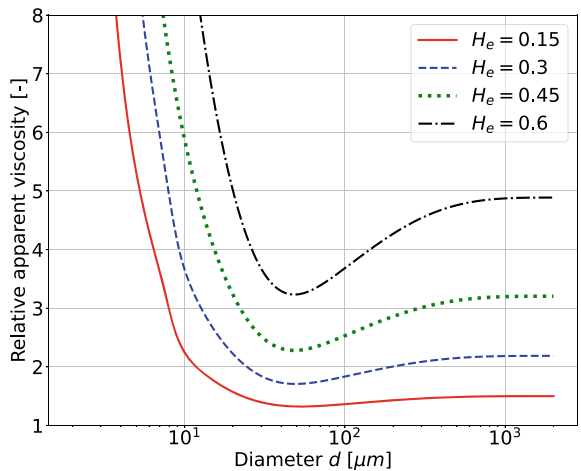


Fig. 19.10 In vivo relative viscosity of blood as a function of vessel diameter for four fixed values of haematocrit He : 0.15, 0.3, 0.45, 0.6. Curves were obtained by fitting experimental data (data from Pries, Neuhaus and Gaehgins [54]). See Chap. 2



Paul and collaborators [53], from their study of more than ten thousand subjects, give the normal range as $0.38 \leq He \leq 0.43$ for females and $0.42 \leq He \leq 0.47$ for males.

In Figure 19.9 it is seen that at the reference value $He = 0.45$, normal blood is about 4 to 5 times more viscous than water. Figure 19.10 shows in vivo relative viscosity of blood as a function of vessel diameter for four fixed values of haematocrit, $He : 0.15, 0.3, 0.45, 0.6$. Curves were obtained by fitting experimental data [54]. Note the very sharp increase in viscosity with reduction of vessel diameter, in the lower range of vessel diameter for all values of haematocrit. At a fixed vessel radius of 10 *micron*, which is typical of capillaries, viscosity increases sharply with haematocrit.

Haematocrit as a Risk Marker of Hypertension

The literature on the potential association of haematocrit and hypertension is not abundant, but there are some important studies, some of which are briefly reviewed here. The comprehensive study of Paul et al. [53], the Glasgow Blood Pressure Clinic Study, establishes that haematocrit is able to predict long-term mortality in hypertensive adult subjects. The investigation is based on 10951 hypertensive patients, who had haematocrit measured at their initial clinic visit and followed for up to 35 years. Higher baseline haematocrit was associated with higher on-treatment blood pressure during follow-up. They concluded that *hypertensive patients with hematocrit levels outside of the sex-specific reference ranges identified in this study should be targeted for more aggressive blood pressure and cardiovascular risk reduction treatment.*

In the study of Jae et al. [50] it is established that higher blood haematocrit predicts hypertension in men. They considered a cohort of 3620 men of mean age 48 ± 6 years, free from hypertension, with He in the range $[0.39, 0.50]$ at baseline. During the 17856 person-years of follow-up they observed that every 1% increment in haematocrit was associated with 7% higher incidence of hypertension after adjusting for risk factors. They concluded that *higher haematocrit levels, even within the normal range, were associated with the incidence of hypertension independent of other risk factors in men.*

Piesanen et al. [55] have recently communicated results from a study involving a 50-year-old cohort, including 307 hypertensives and 579 normotensives. They considered two subgroups of subjects according to measured He at the age of 35. Group 1 had $He < 0.45$ ($n = 581$) and Group 2 had $He \geq 0.45$ ($n = 305$). Hypertensives and normotensives were followed retrospectively at the age of 35 years, and prospectively up to the age of 65. The aim was to determine whether an early haematocrit measurement would predict later hypertension, or cardiovascular complications. Group 2 ($He \geq 0.45$) was associated with hypertension and coronary artery disease by the age of 60. When the subjects were followed up to the age of 65, Group 2 was associated with premature cardiovascular death, and death by any cause.

In conclusion, the role of haematocrit in hypertension is well worth studying in more detail, noting that He is routinely measured in any blood sample test. Assessing the role of He in hypertension would contribute to a more refined apportioning of the contributing roles of the various determinants of overall cardiovascular peripheral resistance in health and disease. See [56, 57] for further details.

19.3 The Venous System and Arterial Hypertension

There is growing consensus in the cardiovascular research community that arterial hypertension involves the full circulation and not just the arterial system; see Fig. 19.5. Classical accounts of arterial hypertension make no mention of the potential role of the venous system, or other bodily fluid compartments, in the aetiology of

the disease [1, 3, 9]. There is evidence however, that essential hypertension is linked to other fluid compartments, such as the venous system [10, 49]. In this section we examine this aspect of hypertension, for which useful background is found in Chaps. 1, 2, 3, 4, 5, 11 of this book.

19.3.1 The Venous System and Its Functions

Relevant, authoritative references on the venous system are Pang [58, 59] and Schaller [60]; see also Tansey et al. [61]. Chap. 15 of [6] is particularly recommended. Classically, veins and their functions in the circulatory system are not given the prominence they deserve. They are very often considered to be passive passageways for blood to be transported to the heart. However, evidence is mounting in recognising that, physiologically, veins perform additional functions that are necessary for the full circulation system to operate. Moreover, the pathophysiological role of the venous system is becoming essential for the understanding of a range of diseases, such as neurological diseases, as described in Chap. 18.

Venous Volumes

We first recall two basic concepts from haemodynamics, namely vascular distensibility and vascular compliance, or just **distensibility** and **compliance**, respectively. Distensibility is defined as

$$\text{Distensibility} := D = \frac{\text{Volume increase}}{\text{Pressure increase}} \times \frac{1}{V_{\text{orig}}} = \frac{\Delta V}{\Delta p} \times \frac{1}{V_{\text{orig}}} . \quad (19.6)$$

That is, distensibility is the ratio of volume change ΔV to pressure change Δp , normalised (divided) by the original volume V_{orig} . Veins are much more distensible than arteries, by a factor of about 8 in the systemic circulation, and by a factor of about 6 in the pulmonary circulation [6].

More generally used is the related concept of **vascular compliance** (C), also called **vascular capacitance**. This is defined as

$$\text{Compliance} := C = \frac{\text{Volume increase}}{\text{Pressure increase}} = \frac{\Delta V}{\Delta p} . \quad (19.7)$$

Therefore

$$C = D \times V_{\text{orig}} . \quad (19.8)$$

The conservative estimate of venous volume of 70% of total blood volume gives a venous compliance estimate of 24 times that of arteries. Fink [62] gives a factor of 30. The precise figures for D and C of course depend on more precise estimates for venous volumes.

The venous system contains about 70%–85% of the total blood volume [58, 59, 63]. The precise figures of venous volume percentage vary significantly in the literature, but clearly the venous compartment stores most of the blood. Therefore, alterations in venous capacity, by its sheer volume, will affect the function of all the components of the cardiac and circulatory systems [59]. In addition to holding 70%–85% of the total blood volume, veins are capable of constricting and enlarging, converting the venous system into a storage for blood, which when required, is made available to the full circulation. Contrary to what is often believed, the venous system is innervated; of the major regulatory systems that participate in vasoconstrictor tone under various physiological and pathophysiological conditions, the sympathetic nervous system is by far the most significant vasopressor system. Local venous blood reservoirs include: the veins in the spleen and the liver; the large abdominal veins; the venous plexus beneath the skin; and also the veins in the heart and the lungs.

All veins in the systemic circulation carry deoxygenated blood, which transports carbon dioxide, waste products, and heat from the tissues. This blood is directed into the superior vena cava, inferior vena cava, or coronary sinus, eventually reaching the right atrium. From there, it flows into the right ventricle and is pumped into the pulmonary circulation

Venous Pressures

In the context of arterial hypertension, a discussion of blood pressures involved is necessary. To this end it is first necessary to recall the accepted convention to quantify pressure. Gravitational pressure, or hydrostatic pressure, occurs in the vascular system of humans due to the weight of the blood in the vessels. Therefore pressure values are stated with reference to the gravitational level of the heart, which is at or near the level of the tricuspid valve [6].

The flow of blood from the venous system into the right atrium is a major determinant of the right atrial pressure, also called the **central venous pressure (CVP)**, of major physiological significance. CVP is normally about 0 mmHg, which corresponds to the atmospheric pressure around the body. CVP is in great part regulated by the ability of the right atrium and ventricle to pump blood into the pulmonary circulation, peripheral venous tone and the tendency of venous blood to flow from the peripheral veins into the right atrium. Therefore, variations of CVP reveal the state of the heart and of the **venous return**, which can increase through the following:

- increased blood volume;
- increased tone of large vessels in the full body, resulting in increased peripheral venous pressures;
- dilation of the arterioles, decreasing peripheral resistance, thus allowing fast flow of arterial blood into the veins.

Blood flow from the peripheral veins into the right atrium is also enhanced by the **venous pump** or **muscular pump** [6], which contributes to regulate the cardiac output. Venous pump refers to the blood propelling effect resulting from moving the

legs and tightening of the muscles; this causes compression of adjacent veins and squeezing of the blood out of the veins, a mechanism that works in conjunction with venous valves that prevent back flow. See Hall [6], p. 173 for further details.

CVP affects peripheral venous pressure. If CVP rises above its normal values of 0 mmHg, blood volume in large veins begins to build up, causing enlargement of these veins. Further increases of CVP cause corresponding increases in peripheral venous pressure in the limbs and elsewhere.

Venous Resistance

It is widely acknowledged that distended large veins offer virtually no resistance to flow. However this is not so when these veins are compressed, as happens with most of the large veins that enter the thorax, for example. These are compressed by surrounded tissue, obstructing venous blood flow at the compressed locations. For example, veins from the arms are compressed by their sharp angulations over the first rib.

Veins are compressed at many points by the surrounding tissues, so that blood flow is hampered at these points. Veins in the abdomen are compressed by intra-abdominal pressure, causing them to partially collapse. Intra-abdominal pressure can increase to values of the order of 15 mmHg to 30 mmHg in cases such as pregnancy, large tumours, abdominal obesity, or excessive fluid in the abdominal cavity.

Compression of abdominal veins by different organs has the net result of (i) increasing venous resistance to flow and (ii) causing pressure increase in the small peripheral veins. These chronic pressure increases are of the order of +4 mmHg to +6 mmHg with respect to CVP. Transient pressure increases can be much larger. Compression of intra-abdominal veins by organs and elevated intra-abdominal pressure due to **abdominal obesity** is relevant to the discussion of arterial hypertension and its relationship to obesity [64].

Venous Functions

As already stated, the primary function, classically ascribed to the venous system, is the return of oxygen-poor blood, transporting carbon dioxide and other wastes and heat from the tissues to the right atrium, and thus to the circulation.

Here we add a few more functions and characteristics of the venous system. The venous system collects all neurofluids, that is interstitial fluid, cerebrospinal fluid, aqueous humor, vitreous humour, endolymph and perilymph. It collects lymph at lymph nodes (about 700 in the human body) and at neck veins, that is, at the thoracic duct (left) and at the right lymphatic duct (right); see Chap. 3.

In recent years, researchers have proposed the **glymphatic theory**, involving a brain-wide transport system of waste clearing from the central nervous system [65]. The venous system in the central nervous system is claimed to play an important role in this theory. Hu et al. [66] suggest that there is enhanced transport from the **perivenous space** into the vein lumen in the cranial cavity.

From the alluded fact that the venous system is strongly and actively connected to the circulation and other fluid compartments, in addition to its responsiveness to

the sympathetic nervous system, one may infer that the venous system has an effect in regulating the dynamics of bodily fluid volumes and pressures.

As pointed out previously, e.g. Chap. 1, veins exhibit highly variable geometrical and mechanical properties; they are highly deformable (non-linear), even collapsible. These features of veins make the construction of mechanical and mathematical models a challenging exercise. See Chaps. 6 and 10.

19.3.2 Reduced Venous Compliance

Changes in haemodynamic parameters in hypertensive animals were reported in the 60s and 70s, particularly concerning decreased venous distensibility [67]. In humans, similar conclusions were arrived at, but from studies restricted to the skin [68]. The seminal works of London, Safar and collaborators in the 70s and 80s resulted in significant advances in establishing a relationship between arterial hypertension in humans and reduced vascular compliance [49, 63, 69–71].

In their earliest work, London et al. [49] studied 32 men, 18 sustained essential hypertensives and 14 normotensives. They measured several parameters, including total effective compliance, plasma volume, interstitial fluid volume, cardiopulmonary blood volume and total blood volume. They calculated effective compliance from the changes in central venous pressure, which was recorded simultaneously with changes in blood volume obtained after a rapid Dextran infusion. Table 19.1 shows measured parameters, from which the following main conclusions emerge:

- Hypertensive and normotensives have similar values for: cardiac index, extracellular fluid volume, cardiopulmonary blood volume and the ratio $CPBV/TBV$;

Table 19.1 Measured parameters in normotensive and hypertensive humans. Abbreviations: CI = cardiac index; CVP = central venous pressure; EFV = extracellular fluid volume; TBV = total blood volume; PV = plasma volume; $CPBV$ = cardiopulmonary blood volume; $ISFV$ = interstitial fluid volume; C = effective vascular compliance (Data from [49])

Parameters	Normotensive	Hypertensives	P values
CI ($mL/min/m^2$)	3668 ± 130	3413 ± 105	—
CVP ($mmHg$)	4.5 ± 0.6	5.5 ± 0.6	—
EFV (mL/kg)	158 ± 5	150 ± 6	—
TBV (mL)	5454 ± 237	5197 ± 174	—
TBV_* (mL/kg)	81 ± 2	68 ± 1	0.001
PV (mL/kg)	41 ± 1	35 ± 1	0.001
$CPBV$ (mL/kg)	18 ± 1	17 ± 1	—
$PV/ISFV$	37 ± 1	31 ± 2	0.05
$CPBV/TBV$	23 ± 1	25 ± 1	—
C ($mL/mmHg/kg$)	2.08 ± 0.09	1.49 ± 0.06	0.001

- Hypertensives, compared to normotensives, have a significantly decreased total blood and plasma volumes and ratio $PV/ISFV$.
- In hypertensives, compared to normotensives, total effective compliance C is significantly reduced.

Concerning correlations between parameters, they noted differences between normotensives and hypertensives, from which we highlight the following: for normotensives they found a strong **positive correlation** between total effective compliance and plasma volume; and between compliance and the extracellular fluid volume. For hypertensives, they found no significant correlation between compliance and fluid volumes; moreover, compliance was **negatively correlated** with cardiac index. Overall, from their results, the following observations emerge:

1. Given that the venous system holds 70%–85% of the total blood volume, the contribution of the arterial system to the measured total compliance is negligible, and therefore the reduction in compliance means reduction in venous compliance.
2. Based on their results and previously reported animal experiments they speculated that: ... *reduction in compliance could cause an elevation of the filling pressure with activation of the mechano-receptors and neuro-humoral mechanisms of volume control*. In particular, this observation connects reduced compliance to the sympathetic nervous system; see the very early clinical work of Hoobler et al. [72].
3. The partition of extracellular fluid volume is disturbed in hypertensives; there is a displacement of blood plasma from the intravascular to the extravascular space.

Reduced systemic venous distensibility associated with hypertension, in animals, had been reported previously, as noted by London et al. [49]. Reduced compliance alters the entire circulation in hypertensives. In particular, a redistribution of blood volumes takes place, with a shift of blood to the central circulation.

19.3.3 Disturbed Bodily Fluid Dynamics

From the evidence reported above, it is clear that arterial hypertension takes place in a scenario of disturbed bodily fluid dynamics, involving fluids redistribution and regulatory mechanisms that include baroreflex, renal, hormonal, and neural counter regulatory responses. The role of the sympathetic nervous system is important [49, 62, 72–74]; see Sect. 19.4. The study of arterial hypertension therefore, requires an integrative, holistic approach, as advocated in the pioneering work of Guyton and collaborators [62, 75], and also in this book. See [56, 57, 76, 77].

Redistribution of Fluids

The role of the venous system is fundamental to understand arterial hypertension, in a large measure due to the large blood volume fraction it hosts. Moreover, venous compliance is estimated to be 30 times greater than arterial compliance in humans, which is a consequence of vein structure and function [62]. Compared to arteries,

vein walls have much less smooth muscle, but sufficient to actively modulate venous tone [78, 79]. Consequently, even small changes in venous tone can significantly affect the blood volume stored in the venous system, the redistribution of blood in the circulation, blood pressure and thus global haemodynamics.

Concerning venous blood volumes and compliance, a distinction is made between the central venous district and the peripheral venous district. The latter is more compliant and accounts for the larger proportion of blood, with most of it located in the splanchnic region, storing about 60% of blood in the circulation. As will be discussed in Sect. 19.4, the venous system in the splanchnic region is densely innervated and is highly responsive to sympathetic nerve activity, resulting in reduced compliance and hence venous blood redistribution.

The redistribution of fluids is an important aspect of the haemodynamics of hypertension. As Fink puts it: *My overall conclusions are that venous smooth muscle tone and vascular capacitance contribute to long-term blood pressure regulation and that this is one important mechanism by which elevated sympathetic nervous system activity can lead to hypertension.* On this point, Wolff and collaborators say: *this concept of a volume displacement, without any change in blood flow, is unfamiliar to most people but is an essential feature of the regulatory process* [73]. In fact, Wolff and collaborators [73] have recently proposed an improved model for the circulation; their theory is consistent with earlier findings and assigns a major role to the venous system in the arterial hypertension scenario.

The fact that the splanchnic region contains a large proportion of venous blood and that venous compliance is larger there than in the central venous district is also interesting from the point of view of resistance. As already discussed in Sect. 19.3.1 veins are easily compressed by surrounding tissue, all within physiological scenarios. Moreover, according to Tortora and Derrickson [4], the extra adipose tissue in obese subjects increases vessel length L , and that for each kilogram of fatty tissue, an estimated 650 km of vasculature are developed.

Fluid volume redistribution also affects blood plasma volume, as seen in the results from London et al. [49], who reported a reduction in plasma volume, see Table 19.1. Potentially, this may cause an increase in haematocrit, which may also impact blood viscosity, as discussed in Sect. 19.2.2. The redistribution of fluid volumes affects pressure and velocity throughout the circulation and associated fluid compartments. In other words, fluid volumes redistribution configures a highly dynamic scenario in which, for example, wall shear stresses are affected, as are pulse pressure and pulse pressure velocity. Importantly, disturbed fluid dynamics may affect Starling forces governing filtration and reabsorption across the vessel wall [1]. See also Chap. 2 and [56, 57].

Interstitial Fluid, Lymph and Cerebrospinal Fluid

The participation of other fluid compartments, such as interstitial fluid (ISF), lymph and cerebrospinal fluid (CSF), in the dynamics of hypertension cannot be discarded. As a matter of fact, in the early works of Safar and London [70, 71], see Table 19.1, the ratio $G = PV/ISFV$ is seen to be meaningful, where PV is intravascular blood

volume and $ISFV$ is interstitial fluid volume. From their measurements in hypertensive and normotensive humans they observed that G is decreased in hypertensive humans.

As already observed in Sect. 19.3.2, hypertension results in a disturbed partition of extracellular fluids with a larger proportion of interstitial fluid than in normotensives. Though this finding does not seem to have been pursued in recent years, in the framework of multifluids as advocated in this book, it is an appealing concept, worth of speculation on its potential significance. Two lines of enquiry would emerge.

The first regards the ISF reabsorption function of post-capillary venules. Under physiological conditions, most ISF is reabsorbed into the circulation at the level of post-capillary venules, a process governed by Starling forces [1, 56, 57]; see also Chap. 3. Thus, impaired venule reabsorption of ISF is one possibility, which may well be coupled to **venous hypertension**, even if slight [56, 57, 80]. As discussed in [56, 57] this line of enquiry does not seem to have been pursued in the literature, in the frame of arterial hypertension.

The second possibility concerns the role of the lymphatic system in hypertension. As is well known, the excess of ISF not reabsorbed into the circulation by post-capillary venules is absorbed by the lymphatics and transported back into the venous system, partly at the level of lymph nodes and the rest into the neck veins via the thoracic duct and the right lymphatic duct. See Chap. 3. In any case, the increased interstitial fluid volume $ISFV$ may result from either anomalous reabsorption by the post-capillary venules, as discussed above, or from an imbalance between lymph absorption in the interstitium and lymph drainage into the venous system. In any case, as already established, hypertension is strongly linked to changes in extracellular fluid volumes and the role of the lymphatics in the buffering of these changes is well recognised.

In the last two decades or so there has been interest in the role of the lymphatics in hypertension, but only concerning the lymphatic vessels in the skin. In the recent review by Mattson [81], the work of Thowsen et al. [82] is described, in which the potential role of dermal lymphatics in the development of salt-sensitive hypertension is studied. There seems to be an intriguing link between skin sodium, lymphatics, and blood pressure. However, this link remains unexplained [81]. See also the paper by Mekarski [83].

In the context of hypertension, the role of cerebrospinal fluid dynamics in the craniospinal cavity warrants consideration. This is particularly the case regarding the role of cerebral haemodynamics in hypertension, from the perspective of the selfish-brain hypothesis [13]. CSF dynamics could also be important in light of the disturbed partition of extracellular fluids in hypertensives, as observed by Safar and London [70, 71]. See also the recent work of Phillips and Schwartz [84], who assign an important role to intracranial hypertension, elevated CSF pressure, in the genesis of arterial hypertension. See Sect. 19.5.1.

In conclusion, the study of arterial hypertension warrants a multi-fluid compartment approach.

19.4 The Sympathetic Nervous System in Hypertension

During decades of research on hypertension, two organs have been hypothesised to be major controllers of blood pressure, with increased total peripheral resistance being an indisputable component, namely the brain and the kidney [8–10]. Renal denervation is a catheter-based procedure that ablates the renal sympathetic nerves; it is seen as an alternative treatment for hypertensives who are resistant to other therapies, such as lifestyle modifications and antihypertensive medications [85, 86].

In this section, however, we focus on the activity of the sympathetic nervous system in the splanchnic vascular region, specifically including the mesenteric venous blood vessel network, and its effect on compliance and blood volume redistribution in hypertension.

19.4.1 Sympathetic Activity and Splanchnic Veins

It is established that the sympathetic nervous system (SNS) plays a major role in controlling vascular resistance, tone and hence blood pressure. SNS acts through both neural and hormonal mechanisms. Recall that **neural** regulation of blood pressure occurs via baroreceptor reflexes and chemoreceptor reflexes. **Hormonal** regulation includes, for example, the hormone epinephrine and norepinephrine. These are released from the adrenal medulla in response to sympathetic stimulation; they cause **vasoconstriction of arterioles and veins in the skin and abdominal organs**.

In general, sympathetic stimulation of most veins causes constriction that moves blood out of venous blood reservoirs into the central circulation, thus increasing blood pressure; see Tortora and Derrickson [4]. For general background on the **autonomic nervous system** (ANS) see the review article of Malpas [10] and the recent book by Biaggioni et al. [87]. From the available evidence, it seems amply justified to focus on activity of the **sympathetic nervous system** (SNS) in the splanchnic vascular region, the venous system in particular.

First, the splanchnic venous bed is densely innervated by the sympathetic nervous system and small veins and venules have been shown to be more sensitive to sympathetic activation than arterioles [10]. Birch et al. [88] studied the morphology of sympathetic innervation in mesenteric blood vessels in humans and examined the relationship of nerves to smooth muscle cells. They found that there was a significantly greater nerve density in veins than in arteries, by a factor larger than 5. They concluded thus: *It is suggested that in humans with an upright stance, the mesenteric venous system plays a particularly important role in controlling mesenteric capacitance, which is reflected by their dense innervation.* See [89] for a follow up to this work.

Second, capacitance of the venous system in the splanchnic region represents the most important active capacitance bed in the whole body [90]. As stated by Fink [62], small veins and venules in the splanchnic district make up the bulk of

the peripheral venous compartment; in addition, they express the highest degree of active venoconstriction.

Quantitatively, it is sympathetic venoconstrictor activity the most important determinant of venomotor tone in these veins [91]. This is also the case for venomotor tone in the full circulation [59]. Following [62] ..., *veins respond to much lower frequencies of sympathetic nerve activity than do arteries. Therefore, it seems reasonable to hypothesize that moderately increased sympathetic nervous system activity could contribute to the development of hypertension by reducing vascular capacitance.* In the context of arterial hypertension, Julius [92] identifies the role of neural and hormonal activity in the decreased compliance of the peripheral capacitance space, again with a shifting of blood volume from the peripheral to the central portion of the circulation.

The neural and hormonal venous constriction of veins, especially in the splanchnic region, has motivated researchers to pursue this line of investigation in the context of arterial hypertension [62, 72, 74, 93].

19.4.2 Clinical and Experimental Studies

Sympathectomy (surgical procedure in which the splanchnic nerves are dissected) was one of the treatment options for essential hypertension in the early 1900s. However, the treatment was discontinued due to its high complication rate [94]. Hoobler et al. [72] was one of the pioneers of that period in arterial hypertension clinical research, in the framework of the link between the splanchnic vascular system and the sympathetic nervous system.

Hoobler and collaborators studied the effect of splanchnicectomy on the arterial blood pressure of 294 hypertensive patients, followed for 10 to 18 months after surgery. They then compared their results with the effects of non-specific medical management in a control group of 79 patients. *The operation consisted of the bilateral removal at one stage of the greater and lesser splanchnic nerves and sympathetic ganglions D-12 and above.* They concluded that 29 per cent of the hypertensive patients had reductions in arterial blood pressure.

Peripheral chemoreceptors, such as carotid body, have been implicated in neurogenic hypertension in animals and humans. Smith and Pacchia [95] discuss mechanisms by which chemoreflex dysfunction may contribute to chronically elevated sympathetic tone and ultimately hypertension. They state that chronically elevated sympathetic activity may explain the causal, well documented, relationship between sleep apnoea and hypertension.

Moreover, Tan et al. [96] tested the hypothesis that enhanced peripheral chemoreceptor sensitivity, which increases sympathetic nerve activity, is a possible mechanism for initiating an increase in sympathetic nerve activity **before the onset of hypertension**. They concluded: *Our results establish a novel molecular basis for increased chemotransduction that contributes to excessive sympathetic activity before the onset of hypertension.*

In line with these works, denervation of carotid chemoreceptors has been carried out. For example, Abdala et al. [97] denervated carotid chemoreceptors in both young and adult spontaneously hypertensive rats; their results showed that in the young animals the operation prevented arterial pressure from reaching hypertensive levels; in the adult animals arterial pressure fell by about 20 mmHg. All these works, and many more, have demonstrated that carotid body denervation does have an effect in controlling the rise of arterial pressure, or even reduction of this. What apparently has not been done is to study the effect of carotid body denervation on the venous system, that is on pressures, flows and compliances.

Very recently, Emans and collaborators [74] presented the first-ever functional demonstration of impaired mesenteric venous capacitance in arterial hypertension. They conducted combined measurements of arterial and mesenteric venous pressures in conscious male Wistar and spontaneously hypertensive (SH) rats. Their findings revealed that, in hypertensive rats, excessive sympathetic activity induced venoconstriction, which in turn reduced venous capacitance. This impaired the venous system's natural volume-buffering function and increased the mean circulatory filling pressure. By monitoring the mesenteric response, they observed that a reduction in sympathetic activity improved the mesenteric vascular bed's ability to accommodate excess volume.

In their words: *Increased venous tone causing reduced buffering capacity of volume within the mesenteric veins might contribute to hypertension in the hypertensive rat. These findings support the idea that venodilatory capacity is reduced in individuals with increased venous tone and suggest targeting the venous mesenteric circulation to enhance venous capacitance as a novel therapeutic opportunity to ameliorate hypertension.*

19.5 Hypotheses and Discussion

We close this chapter by succinctly touching upon some ideas and some controversial suggestions that may well be worth exploring in future research on arterial hypertension.

19.5.1 *The Selfish Brain Hypothesis*

The Cushing's mechanism is rooted in an original observation by Cushing in 1901, from his studies in anaesthetized dogs. He related brainstem compression to the resulting blood pressure elevation, and stated *any increase in intracranial pressure causes a rise of blood pressure...* See Paton et al. [98]. The interpretation is that reduction in cerebral blood flow and associated brain hypoperfusion caused by brainstem compression, triggers hypertension as a self-defence action, in the form of an emergency response to restore adequate blood supply.

For this reason Cushing's mechanism [98] has also been termed the selfish brain hypothesis [99]. Dickinson and Thomson [100] hypothesised that *restriction of flow in the vertebral arteries, and to a lesser extent in the internal carotid arteries, actually causes hypertension*. They went on to state that “*Essential*” hypertension, on this basis, might be a necessary adaptation; and generalised narrowing of terminal arteries in the body would also have a beneficial effect, because it would allow the autonomic nervous system to function efficiently in postural readjustments of blood-pressure. Based on their observations they hypothesised the following chain of events: (1) narrowing of the vertebral arteries, (2) brain stem hypoperfusion and (3) hypertension, as a consequence.

Dickinson [101] showed that the vertebral arteries in hypertensive patients were narrower than those observed in normotensive individuals, and later still, in animal experiments [102]. In sum, the **selfish brain hypothesis** refers to the situation in which *decreased cerebral perfusion triggers elevated sympathetic nerve activity and increased blood pressure in order to maintain the brain perfusion requirements*.

Warnert and collaborators [13] investigated the potential involvement of Cushing's mechanism in the development of hypertension in humans. To this end, they conducted a series of retrospective and mechanistic case-control studies involving 259 participants with varying levels of blood pressure and classifications of hypertension. A distinctive feature of their study was the examination of the architecture of the posterior arterial cerebral vasculature. They found that hypertensive individuals had a higher prevalence of congenital cerebrovascular variants, including vertebral artery hypoplasia and an incomplete posterior circle of Willis. These vascular abnormalities were associated with increased cerebral vascular resistance, reduced cerebral blood flow, and a higher incidence of lacunar-type infarcts.

Focusing specifically on the posterior cerebral circulation, Warnert and collaborators [13] evaluated the temporal relationship between changes in cerebral arterial vascular structure, associated vascular resistance, cerebral blood flow, and sympathetic nerve activity. They found these factors to be closely linked to the development of hypertension.

They concluded thus: *Our data indicate that congenital cerebrovascular variants in the posterior circulation and the associated cerebral hypoperfusion may be a factor in triggering hypertension. Therefore, lowering blood pressure may worsen cerebral perfusion in susceptible individuals*.

In line with the selfish-brain hypothesis, this work constitutes a confirmation, in conscious humans, that anomalous cerebral vasculature and the resulting cerebral hypoperfusion might be important factors for the development of hypertension. For a follow-up, see Manghat et al. [103].

Finally, we note that a different, perhaps not totally unrelated, definition of selfish brain theory has been put forward by Peters et al. [104]. In this theory *the brain occupies a special hierarchical position in human energy metabolism*.

19.5.2 *Lymphatic Obstruction and Elevated ICP*

Very recently, Phillips and Schwartz [84] have published a paper entitled: *Nasal turbinate lymphatic obstruction: a proposed new paradigm in the aetiology of essential hypertension*. They propose, as a primary cause for hypertension, the regulation of cerebral perfusion, as in the selfish-brain hypothesis, in fact. Phillips and Schwartz involve the dynamics of cerebrospinal fluid and lymph. The theory relies on the existence of a CSF passageway connecting the cranial subarachnoid space containing CSF to cervical lymph nodes, exiting the cranial cavity through the foramina of the cribriform plate, alongside the olfactory nerves, reaching the nasal lymphatics, and finally, the cervical lymph nodes; see Chae et al. [105].

The Phillips-Schwartz hypothesis is based on their observations that in hypertensive subjects there is a significant **increase in nasal turbinate vasodilation with blood pooling**. They propose that such blood pooling causes obstruction of CFS outflow through the nasal turbinate lymphatics. Such obstruction would then cause elevation of the intracranial pressure, that is elevation of the cerebrospinal fluid in the craniospinal cavity. They then hypothesise that such increased intracranial pressure would lead to *compensatory arterial hypertension via Cushing's mechanism*. They claim that their hypothesis proposes a *novel mechanism and a new paradigm for the aetiology of essential hypertension related to nasal turbinate obstruction*.

One obvious question that arises is this: do hypertensives, as we have defined them so far, suffer from intracranial hypertension, that is elevated CSF pressure? For related recent work see [106].

19.5.3 *Venous Pressure in Arterial Hypertension*

The suggestion of increased venous pressure in arterial hypertension may appear far fetched, at first. However, there is some evidence, even if partial, that venous pressure is elevated in arterial hypertensive subjects.

Elevated Venous Pressure

The work of Caliva et al. [68] is one of the earliest on the haemodynamics of hypertensive subjects, in which measurements of venous pressure were conducted. These authors employed rheoplethysmography to study the haemodynamics of the finger in the hypertensive state, confirming that the finger tip reveals some of the haemodynamic characteristics of hypertension. Their results showed that the mean finger venous pressure in the normotensive subjects was 8.5 ± 3.1 mmHg, as compared with 10.9 ± 3.1 mmHg in the hypertensive patients ($p < 0.01$). Moreover, they observed that 90% of the hypertensive subjects had a venous pressure between 8.9 and 13.8 mmHg, while 90% of the normotensive ones fell between 5.9 and 10.4 mmHg. They concluded thus: *there is a small but definite increase in digital venous pressure in the hypertensive state*.

More recently, Novo et al. [107] studied the behaviour of venous pressure in two groups of hypertensive patients and one group of normotensive controls. The venous pressure was estimated at the fold of the elbow on the basilic vein, using a Doppler ultrasound instrument. Venous pressure was found to be significantly higher in two groups of hypertensive patients, in comparison with the healthy controls. Venous pressure was 14.76 ± 1.90 mmHg in one hypertensive group, versus 12.53 ± 2.39 mmHg in the other hypertensive group, versus 8.7 ± 2.02 mmHg in the control group ($p < 0.0001$). From their data the authors concluded: *that hypertension is a disease in which both the arterial and the venous vascular beds are involved with increased pressure in both circulatory beds.*

From the experimental study of London et al. [63], reported measured central venous pressure values are 4.5 ± 0.6 mmHg in normotensives and 5.5 ± 0.6 mmHg in hypertensives, with the increase regarded as non-statistically significant. Similar results are found in [63], their Table 2; they reported measured central venous pressure as 4.2 ± 0.8 mmHg in normotensives and 5.7 ± 0.4 mmHg in hypertensives; again, the increase was described as non-statistically significant by the authors.

The computational study conducted by Toro and Müller [56, 57] predicts a rise in venous pressure of about 25%, in the hypertensive state. See also the earlier work of Segers et al. [108]. Still, the evidence for increased venous pressure in arterial hypertension is currently not sufficiently strong, warranting further studies.

Potential Consequences

The occurrence of venous hypertension in conjunction with arterial hypertension is controversial. However, if venous pressure were elevated in arterial hypertension, one could speculate that the increased post-capillary pressure might disturb interstitial fluid reabsorption into the circulation at the end-capillary venule. This possibility is supported by the Starling equation of fluid exchange, introduced in Chap. 2. If so, this would contribute to explain, at least partly, the reduction in plasma volume in hypertensive humans, first reported by London et al. [49]. See Table 19.1.

Additionally, elevated venous pressure may contribute to reduce perfusion pressure, defined as $P_{perf} = MAP - \max(ICP, JVP)$, where MAP is mean arterial pressure, ICP is intracranial pressure and JVP is jugular venous pressure [109]. Therefore, in view of the selfish-brain hypothesis, increased venous pressure may potentially enhance the Cushin response.

Moreover, arterial hypertension is bound to enhance extravasation of fluids and solutes from the capillary lumen in the hypertensive state with elevated capillary pressure, again from the Starling equation of fluid exchange. This is indeed well understood. More intriguing are the results communicated by Yablonski et al. [80]. They conducted a study to elucidate the physiological mechanisms responsible for the intestinal secretion produced by venous pressure elevation in dogs. They measured rate and composition of small intestinal secretion, rate of flow and composition of intestinal lymph, plasma composition, and mucosal water content, all in response to

elevations of intestinal venous pressure. They found that above a threshold value in the range 22–26 mmHg, there were large increases in the rates of lymph flow and net sustained transcapillary filtration.

The findings in [80] refer to increases in local intestinal venous pressure. However, their connection to the observed modest elevation of venous pressure in arterial hypertension is unclear. Nonetheless, the fact that elevated venous pressure could explain transcapillary filtration is relevant to the broader discussion of additional phenomena involved in arterial hypertension. Based on the above discussion, we can infer that arterial hypertension creates a potentially complex biophysical scenario, involving the dynamics of arterial and venous blood, lymph, interstitial fluid, and other local fluid compartments, such as cerebrospinal fluid [110].

19.6 Concluding Remarks and Further Reading

In this chapter, we first briefly reviewed the fundamentals of arterial (or essential) hypertension, based on material that can be found in textbooks. Most classical accounts of arterial hypertension focus exclusively on the arterial side of the circulation. Instead, we take the view that arterial hypertension is a systemic disease, involving the entire circulation as well as associated local bodily fluid compartments.

In line with this perspective, we examined the role of the venous system, which is well documented in the pioneering works of London and Safar in the 1970s and 1980s. These works demonstrated, in humans, that the combination of the large volume held by the venous system (between 70% and 85% of the total blood volume), its naturally high compliance, and its severe reduction in the hypertensive state leads to the displacement of blood from the peripheral to the central circulation, contributing to the rise in arterial pressure.

The role of the sympathetic nervous system in regulating venous compliance has also been reviewed, including both pioneering clinical studies and recent experimental animal research.

Finally, we briefly discussed some current theories of hypertension, such as the selfish-brain hypothesis. We also indulged in speculation and highlighted some intriguing issues that warrant further investigation. One such issue concerns the possibility of elevated venous pressure in the context of arterial hypertension. Some potential consequences include disturbances in fluid and solute transport across the capillary and end-capillary venule walls, which could involve other fluid compartments, thereby broadening the framework in which arterial hypertension occurs.

Not covered in this chapter is the role of mathematical modelling and other modern technologies in the study of arterial hypertension. This will be the subject of Chap. 20.

Regarding recommended reading for further study, the classical textbooks by Levick [1, 5], and Tortora and Derrickson [3, 4] are essential, particularly for beginners,

but also for advanced readers. The review by Malpas [10], the specialised book by Korner [9], and the recent review by Wolff et al. [73] are highly recommended. As for key research publications, the following are suggested: the pioneering works of London et al. [49, 70], and the recent works of Warnert et al. [13] Emans et al. [74] and Toro et al. [56, 57].

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Chapter 20

Simulating Intracranial and Arterial Hypertension



Abstract In this chapter we bring together most of the material presented in previous chapters of this book, starting from cardiovascular fluid physiology, followed by mathematical models for the dynamics of bodily fluids and computational algorithms of high order of accuracy in space and time to solve the partial differential equations of blood flow. The end product is a global, closed-loop mathematical model for the entire human circulation, including the arterial system, the venous system, the heart, the microvasculature and the cerebrospinal fluid. We illustrate the practical applicability of all the previous theoretical content and the future potential of the overall subject of computational bodily fluid dynamics.

In Sect. 20.1 we review all major components of the global, closed-loop mathematical model originally developed by Müller and Toro [1, 2]. Appropriate references are given for the components not studied in this book.

In Sect. 20.2 we apply the mathematical model to simulate the bodily fluid dynamics in the craniospinal cavity to demonstrate that strictures to venous outflow, either intracranial or extracranial, disturb the entire craniospinal dynamics involving the interaction of arterial blood, venous blood, brain tissue and cerebrospinal fluid. One of the consequences is intracranial hypertension, that is elevated intracranial pressure and associated pathologies. Background on the pathophysiology of neurological disorders is given in Chap. 18.

In Sect. 20.3 we apply the global, closed-loop mathematical model to simulate arterial (essential, or primary) hypertension. Background on the pathophysiology is given in Chap. 19. By means of the mathematical model it is shown that the venous system is the second most important determinant of arterial hypertension, after arterial-based cardiovascular resistance; moreover, our results suggest that the hypertensive state cannot be attained without the contribution from the venous system.

The chapter ends with Sect. 20.4, with a summary and conclusions.

20.1 Global Mathematical Model for the Circulation

In this section we review the global, closed-loop mathematical model for the human circulation developed by Müller and Toro [1, 2], restricting attention to its main components.

20.1.1 Overview

The development of global, closed-loop mathematical models for the entire human circulation has seen a resurgence in recent years. Examples include the works of Liang et al. [3]; Mueller and Toro [1, 4]; Mynard and Smolich [5]; Caiazzo et al. [6, 7]; Strocchi et al. [8]; Toro et al. [2] and Mueller et al. [9].

These advances have relied on progress in several related scientific disciplines, starting from the works of Harvey on the human circulation [10], see also the early work of Mackenzie [11]. The mechanics of bodily fluids, in which the works of Pedley [12], Fung [13, 14] and Parker [15] must be mentioned. Mathematical modelling of fluid flow brings in the historic contributions of Euler [16, 17]. For mathematical modelling and computation, see the pioneering works of Guyton and collaborators [18]; Quarteroni and collaborators [19, 20]; Sherwin and collaborators [21]; Alastruey and collaborators [22, 23]. Such contributions rely on advances on computational methods for differential equations, see Godlewski and Raviart [24], Cockburn, Karniadakis and Shu [25], LeVeque [26] and Toro [27]. Advances in magnetic resonance, computed tomography (CT) and ultrasound imaging techniques are of fundamental importance in these developments, see Haacke and collaborators [28]. And last but not least, advances in clinical studies and cardiovascular system physiology; see Zamboni and collaborators [29], [30], Carare et al. [31], Weller et al. [32, 33], for example.

For this chapter we shall essentially consider two papers on the Müller-Toro model for the human circulation, the first communication by Müller and Toro in 2014 [1], and an updated version of 2022 by Toro et al. [2]. See also Müller et al. [9] for more recent developments. The reader is referred to these publications for full details.

Figure 20.1 depicts a schematic representation of the global, closed-loop mathematical model for the entire human circulation [1, 2]. Major anatomic components include: the heart with its four chambers; the arterial system; the venous system; the microvasculature and the cerebrospinal fluid in the craniospinal cavity. Figure 20.2 shows more details of cerebrospinal fluid compartments in the cerebrospinal cavity.

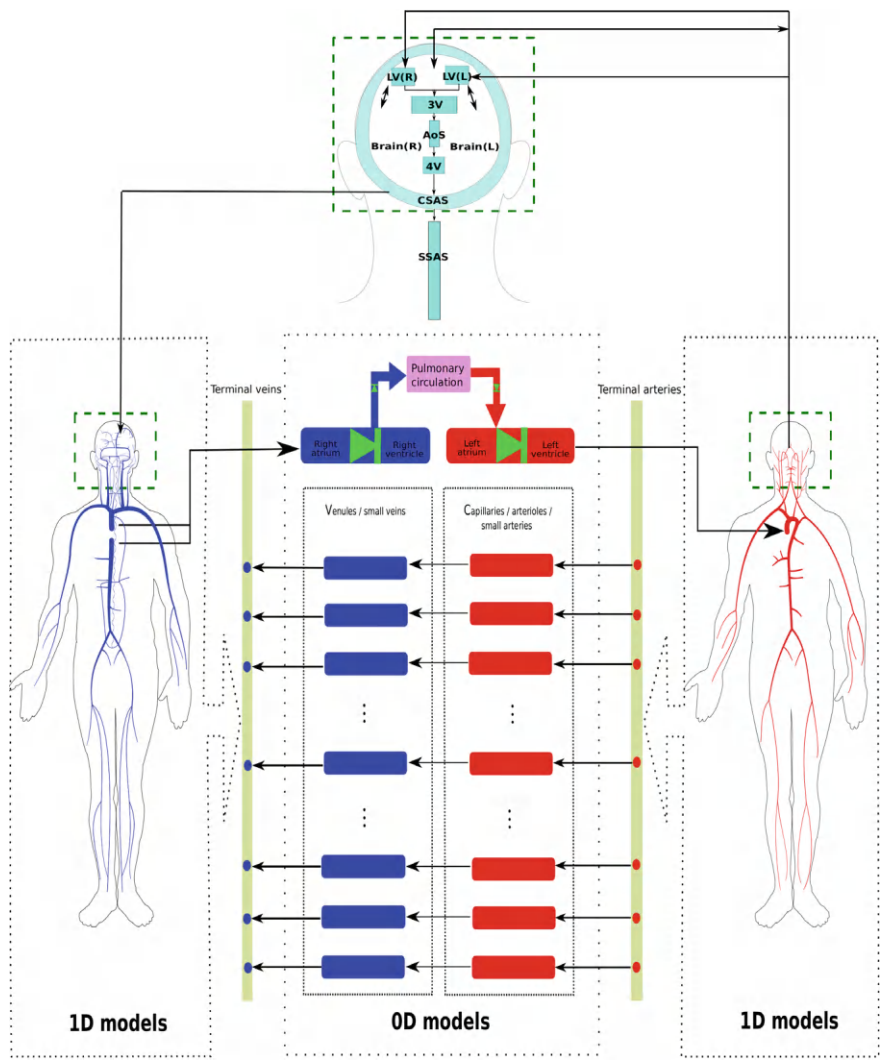
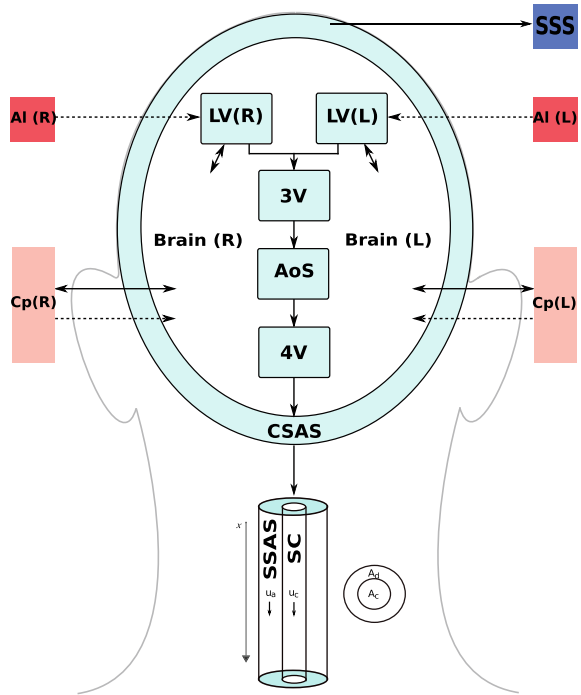


Fig. 20.1 Schematic representation of the global model for the full human circulatory system coupled to cerebrospinal fluids and brain parenchyma. The arterial one-dimensional network (1D) is represented in the right hand side rectangle while the venous 1D network is displayed in the left hand side rectangle. Terminal arteries of the arterial network are connected to draining veins of the venous circulation through 0D models representing arterioles, capillaries, small arteries (red boxes) and venules, small veins (blue boxes). The dotted arrows indicate the connection between 1D networks and terminal vessels, depicted as dots in the green/yellow vertical bars. Cardiac chambers are connected to the pulmonary circulation, comprising arteries, capillaries and veins, represented by the pink box. Cerebrospinal fluid compartments are represented by cyan boxes; see Fig. 20.2 for more details. The arrows between the cardiovascular system and the CSF circulation represent the fluid exchange between these systems through production and absorption, while green dashed boxes indicate the mechanical interaction between all components in the cranial cavity through the Monro-Kellie hypothesis (taken from [2] by permission)

Fig. 20.2 Cerebrospinal fluid (CSF) compartments in global, closed-loop model [1, 2]: Lateral ventricles LV(L) and LV(R); Third Ventricle (3V); Fourth Ventricle (4V); Aqueduct of Sylvius (AoS); Cranial Subarachnoid Space (CSAS); Spinal Subarachnoid Space (SSAS). Also depicted are: Superior Saggital Sinus (SSS); fluid part of brain parenchyma (Brain); cerebral arterioles (Al) and cerebral capillaries (Cp) (taken from [2] by permission)



20.1.2 Equations and Methods

Here we briefly refer to one-dimensional models for major vessels, zero-dimensional systems for compartmental models and junctions. For submodels, parameters and other model components, readers are referred to [1, 2] and references therein.

One-Dimensional Models

Governing equations for all major vessels, arteries and veins, were presented in Chaps. 6 and 10. Mathematical bases for the analysis of these equations are found in Chaps. 7 and 8. See Toro and Siviglia [34], [35] for early works on the governing partial differential equations (PDEs). For arterial and venous blood flow in all major vessels the governing PDEs constitute a system of time-dependent, nonlinear hyperbolic partial differential equations with stiff source terms. The stiff source terms arise from a hyperbolisation procedure applied to the parabolic equations associated with viscoelastic tube laws. Through friction, purely elastic tube laws could also generate stiff source terms, as shown in Müller et al. [36]. See Chap. 10 for more details.

To approximate numerically the one-dimensional equations we use the ADER approach, first communicated in [37] and subsequently developed in collaboration with many researchers. Examples include [38–44] and the recent review [45]. ADER is a one-step, explicit fully discrete method, based on non-linear spatial reconstruction

polynomials $\mathbf{P}_i(x)$ of degree m in each cell I_i and the solution of the generalised Riemann problem GRP_m at the cell interface $i + \frac{1}{2}$ to determine the numerical flux $\mathbf{F}_{i+\frac{1}{2}}$. The numerical source $\mathbf{S}_i$ is computed through evaluation of a volume integral using time-evolved reconstructed polynomials $\mathbf{P}_i(x)$. The resulting method is of (arbitrary) accuracy $m + 1$. See Chaps. 16 and 17 for full details about ADER and the various methods for solving GRP_m .

Current versions of the global model for the human circulation use the DET (Dumbser-Enaux-Toro) solver for GRP_m [43], which is locally implicit and thus able to handle stiff source terms, if present. DET requires the solution of a sequence of conventional Riemann problems GRP_0 , for which the DOT (Dumbser-Osher-Toro) solver [46, 47] is used, along with a modification to make the scheme well balanced [48]. See Chap. 17 for details. An enhancement to the numerical methodology for the present model of the human circulatory system is the Local Time Stepping (LTS) technique [49, 50].

Zero-D Compartments

Compartmental models were introduced in Chap. 6. These consist of systems of Ordinary Differential Equations (ODEs). The key simplification of these models lies on the assumption of no spatial variations in the considered compartment. Only variations in time are retained. To solve these equations there are many mature methods available in the literature, including classical books on numerical analysis. See also Chaps. 16 and 17, where special ODE solvers were presented in the context of semidiscrete methods for solving PDEs.

Junctions

Networks of blood vessels, and other biofluid conduits, are characterised by branching. Arteries display diverging branching while veins exhibit converging branching. At each branching location, junction or node, one may assume the intersection of one-dimensional vessels (or zero-dimensional compartments). There are established methods to account for the flow at junctions. Briefly, the most commonly adopted coupling strategy enforces mass conservation and total pressure continuity among vessels sharing a node. Part of the methodology includes the application of generalised Riemann invariants, see Chaps. 8 and 9. See [1, 50–55]. For a review of the subject see [20].

Most existing methods for junctions are used for subjects in the supine position. Challenges arise for subjects under postural changes, for which local flow conditions may become critical or even supercritical. More research is needed in this area; see [55–58].

20.1.3 Sample Computations and MRI Data

The resulting global, closed-loop mathematical model [1, 2] has been applied to simulate the entire human circulatory system, involving the systemic and pulmonary circulations, coupled to the dynamics of cerebrospinal fluid and brain dynamics. Multiple applications have been carried out to validate the model, comparing computational results against literature data and bespoke MRI measurements.

Just to give the reader an idea of the capabilities of the resulting model, here we present a couple of results. Figure 20.3 shows computations for cerebral arterial inflow and cerebrospinal fluid flow in the spinal canal. Importantly, in the global model, computations are conducted for the full body. Predicted computational results are compared against MRI data. Note that flow quantities have been normalised by their maximal absolute values.

The left frame of Fig. 20.3 shows cerebral arterial inflow against time and the right hand side frame shows cerebrospinal fluid flow in the spinal canal, as a function of time, at the C2/C3 level. Figure 20.4 shows computed flow and velocity in the aqueduct of Sylvius, as functions of time. The left frame shows computed CSF flow over a cardiac cycle, compared with MRI data, and the right frame shows computed CSF velocity over a cardiac cycle, compared with MRI data from [59]. These sample results are quite satisfactory, considering that simulations have been performed for the full body.

The exhibited results of Figs. 20.3 and 20.4 are meant to illustrate the potential of a global closed-loop mathematical model to simulate bodily fluid dynamics in realistic scenarios. In the next two sections we show results from application of the global model to simulate flow conditions related to intracranial hypertension and arterial (essential, or primary) hypertension.

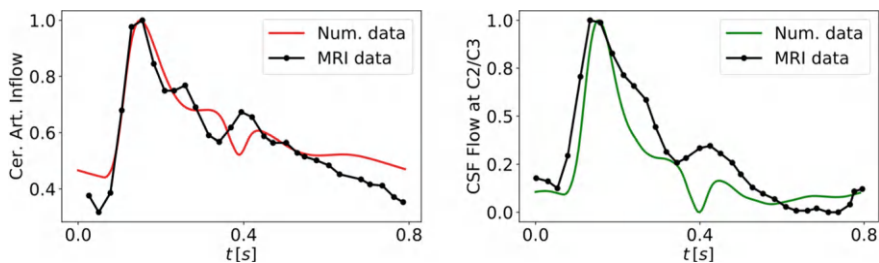


Fig. 20.3 Cerebral arterial inflow and CSF flow as functions of time. Computed results are compared against MRI measured data [60]. Left frame: computed cerebral arterial inflow (internal carotid arteries and vertebral arteries) over a cardiac cycle. Right frame: computed CSF flow at the C2/C3 level over a cardiac cycle. Flow variables have been normalised by their maximal absolute values and thus lie between 0 and 1 (taken from [2] by permission)

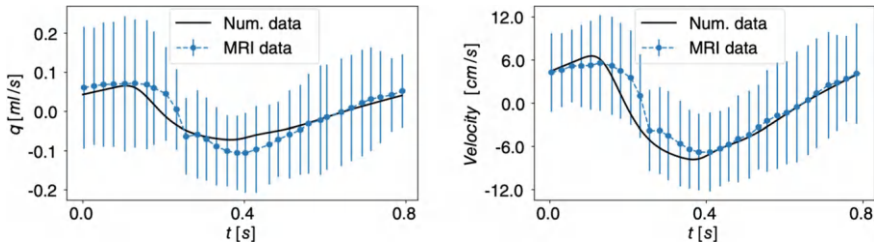


Fig. 20.4 Computed flow and velocity in the aqueduct of Sylvius compared with MRI data [59]. Left frame: computed cerebrospinal fluid flow through the aqueduct of Sylvius over a cardiac cycle. Right frame: computed CSF velocity in the aqueduct of Sylvius over a cardiac cycle (taken from [2] by permission)

20.2 Intracranial Hypertension

This section is primarily concerned with intracranial pressure (ICP). We show results from the computational study performed in [2] using a global, closed-loop mathematical model for the entire circulation, coupled to CSF dynamics in the craniospinal cavity; see Sect. 20.1. The focus is on elevated ICP resulting from morphological anomalies in the venous system. We consider two cases: intracranial and extracranial venous vessel stenoses, both situations leading to hampered venous outflow and pressure elevation in the venous and CSF systems.

20.2.1 Venous Strictures and Neurological Disorders

It is by now well documented that venous flow strictures in head and neck veins are strongly associated with neurological disorders. For background see Chap. 18.

Venous Strictures and Associated Compartments

Strictures in venous vessels result in reduction of the calibre of these blood conduits. These anatomical anomalies include membranous obstructions at junctions, twisting, septum and atresia, intraluminal stenoses and external muscle or bone-compression stenoses. See Chap. 18. For the case of arterial strictures and resulting arterial hypertension, see Warnert et al. [61].

Intracranial and extracranial venous vessels are shown in Fig. 20.5. Intracranial vessel of interest include: Superior Sagittal Sinus (SSS); Inferior Sagittal Sinus (ISS); Basal Veins of Rosenthal (BVR); Straight Sinus (SS); Transverse Sinuses (TS); Superior Petrosal Sinus (SPS); Inferior Petrosal Sinus (IPS) and Sigmoid Sinuses. Extracranial venous vessels of interest include internal jugular veins (left and right), external jugular veins (left and right) and the azygos vein. Strictures to venous flow

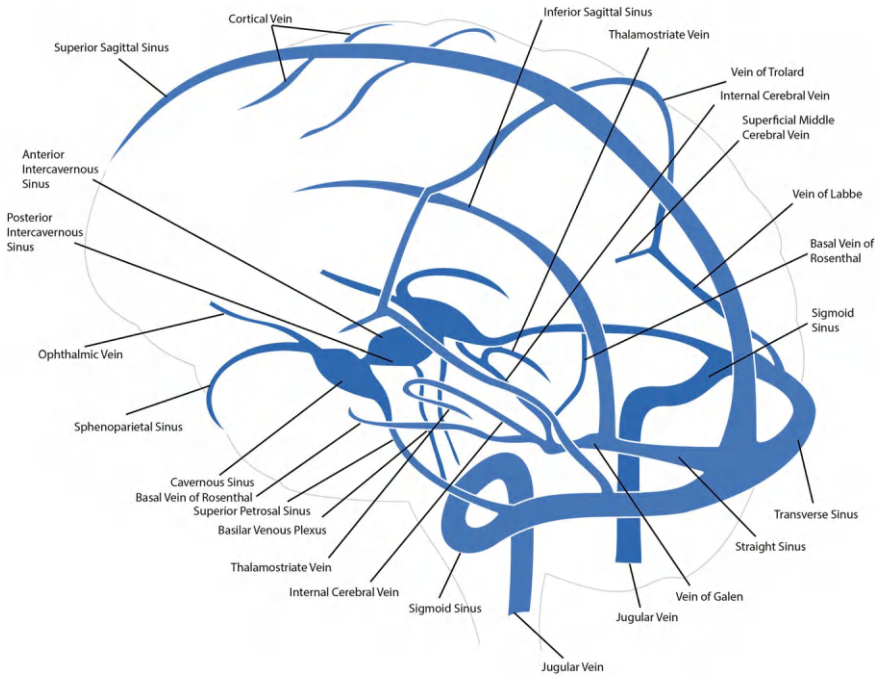


Fig. 20.5 Simplified illustration of the main veins of the head and neck in humans. Azygos vein not shown (courtesy of Violeta Da Rold. Data from [68])

in these vessels include anomalous venous valves. For studies on the anatomy, prevalence and competence of neck vein valves see [62–66]. For a theoretical study of intracranial flow disturbances caused by anomalous valves see [67].

Cerebrospinal fluid compartments of specific interest here are depicted in Fig. 20.2. These include Brain (brain tissue and brain parenchyma fluid phase); Cranial Subarachnoid Space (CSAS); Lateral Ventricles (left and right), Third Ventricle (3V); Aqueduct of Sylvius (AoS); Fourth Ventricle (4V) and Spinal Subarachnoid Space (SSAS).

Neurological Disorders

As reported in Chap. 18, there are many neurological disorders that have been linked to disturbed craniospinal fluid dynamics resulting from morphological anomalies in the venous system. Examples include:

- Idiopathic Intracranial Hypertension (IIH) [69], [70–74]. The aetiology of IIH is currently regarded as unknown, and thus the term *idiopathic*, despite its strong association with venous outflow strictures, both intracranial and also extracranial [60, 75];
- Meniere’s Disease [76–80];

- Retinal Abnormalities [81];
- Transient Monocular Blindness [82];
- Transient Global Amnesia [83–85];
- Cognitive dysfunctions and *brain fog*. See recent work of Primiani and collaborators [86].

Elevated intracranial pressure, idiopathic or secondary, is an overarching condition common to several neuropathologies associated with disturbed craniospinal fluid dynamics. Idiopathic Intracranial Hypertension is a prominent example of a neurological condition strongly associated with both intracranial and extracranial venous outflow strictures. IIH requires prompt diagnosis, as if left untreated could result in rapidly progressive visual loss [70]. A large proportion of IIH patients harbour some degree of dural sinus stenosis, which hampers craniospinal venous outflow, thus altering the entire craniospinal fluid dynamics. IIH has also been associated with extracranial venous outflow strictures by Alperin and collaborators [75]. The reader is encouraged to review Chap. 4, in which a case study involving an IIH patient (pre- and post therapy) and a healthy control were presented. Detailed MRI measurements of flow of arterial, venous and CSF were reported. See also [72].

As is well known, measuring ICP is complicated. Reliable measurements usually involve invasive procedures, such as lumbar puncture [72], though methods to estimate ICP indirectly exist. See [87] for a review on non-invasive methods to monitor ICP. In the intensive care unit, given the emergency situation of death or life, very invasive procedures are applied to monitor ICP constantly.

In the remainder of this section, we will examine two types of venous anomalies and their impact on fluid dynamics in the craniospinal district: intracranial and extracranial venous strictures. The first group focuses on the effects of dural sinus stenosis, while the second group explores the consequences of stenosis in the internal jugular veins and the azygos vein. We shall focus on pressure in head and neck veins and in CSF compartments. The computational results were obtained from applying the Müller-Toro model, briefly described in Sect. 20.1. For full details see [1, 2].

20.2.2 Transverse Sinus Stenoses

As already stated, TS stenoses cause narrowing of the calibre of these intracranial venous vessels, which drain cerebral venous blood to the sigmoid sinuses, then to the internal jugular veins and eventually to the right atrium of the heart. TS narrowing may also be caused by external compression. See Fig. 20.5.

Following [2] we shall assume bilateral transverse sinus stenoses, that is stenoses in both L.TS (left transverse sinus) and R.TS (right transverse sinus); see Fig. 20.5. Computed results for venous pressure in intracranial vessels are shown in Fig. 20.6. Computed results for CSF pressure in CSF compartments depicted in Fig. 20.2, are shown in Fig. 20.7.

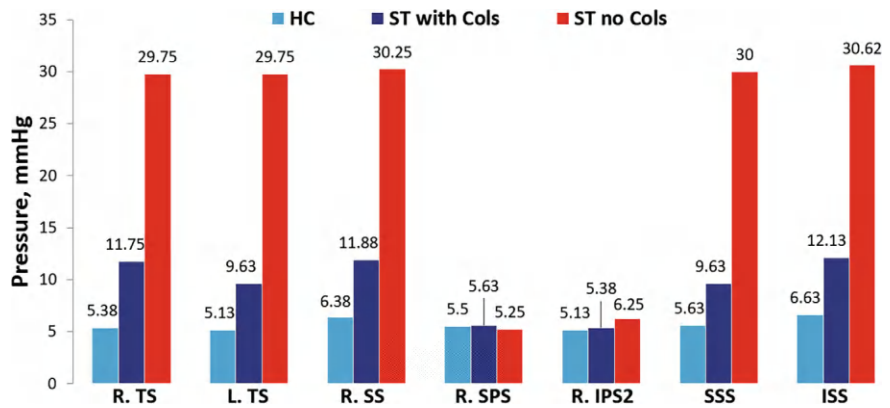


Fig. 20.6 Venous pressure resulting from bilateral transverse sinus stenoses. Pressure in healthy control (HC) is compared against pressures from two cases of transverse sinus stenotic subjects: stenotic with fully active collaterals (ST with Cols) and stenotic without active collaterals (ST no Cols). Venous pressure results are shown for the following compartments: Right Transverse Sinus (R.TS); Left Transverse Sinus (L.TS); Right Sigmoid Sinus (R.SS); the Right Superior Petrosal Sinus (R.SPS); Right Inferior Petrosal Sinus (R.IPS2); Superior Sagittal Sinus (SSS) and Inferior Sagittal Sinus (ISS). For location of these venous vessels see Fig. 20.5 (courtesy of Dr. Brigitte Toro. Data from [2])

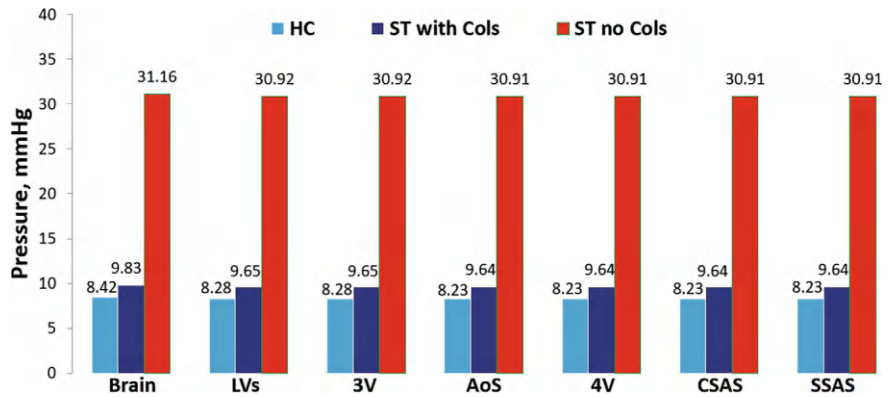


Fig. 20.7 Cerebrospinal fluid pressure resulting from bilateral transverse sinuses stenoses. Cardiac-cycle averaged pressures are shown for a healthy control (HC) and compared against pressures from two cases of transverse sinus stenotic subjects: stenotic with fully active collaterals (ST with Cols) and stenotic without active collaterals (ST no Cols). Results are shown for fluid content of brain parenchyma (Brain); Lateral Ventricles (LVs); Third Ventricle (3V); Aqueduct of Sylvius (AoS); Fourth Ventricle (4V); Cranial Subarachnoid Space (CSAS) and Spinal Subarachnoid Space (SSAS). See Figs. 20.5 and 20.2 for location of relevant compartments (courtesy of Dr. Brigitte Toro. Data from [2])

Intracranial Venous Pressure

Figure 20.6 shows computed venous pressure in seven intracranial venous vessel locations. Pressure values correspond to cardiac-cycle averaged pressure obtained from integrating time-dependent computed pressure distributions. See Fig. 20.5 for location of these compartments.

For each compartment in Fig. 20.6 we compare the healthy control **HC** against two stenotic cases. One stenotic case is denoted **ST with Cols** and refers to bilateral transverse sinus stenoses in which all collaterals are functional. The second stenotic case is denoted **ST no Cols** and refers to bilateral transverse sinus stenoses in which the entire collateral network is inactive, so that cerebral venous outflow is forced to pass through stenotic transverse sinuses.

In the predictions of Fig. 20.6 there are virtually no pressure changes in the superior and inferior petrosal sinuses, whereas in all other vessels the changes are significant, with and without active collaterals. Changes are large, even in the best-case scenario of fully active collaterals, with pressure increases of the order of 100%. In the worse-case scenario, no active collaterals, increases are huge, by a factor of about four, reaching venous pressures of about 30 mmHg. Venous hypertension is predicted.

A real subject with TS stenoses will have a collateral network the function of which will lie between the two extremes shown, namely **ST with Cols** and **ST no Cols**, that is, fully active collaterals and no active collaterals at all. The pressure changes will depend on the level of collateralisation of the specific subject. As already noted, the best case scenario of fully active collaterals, still shows large increases in venous pressure.

We next examine cerebrospinal fluid pressures, that is ICP, resulting from bilateral TS stenoses.

Cerebrospinal Fluid Pressure

Computed pressures in the CSF compartments of Fig. 20.2 are shown in Fig. 20.7. Results are shown for the model in [2], in which each of these compartments is represented by a zero-dimensional model. Recall that in 0D models variables are time dependent but constant in space. In the updated version of [88] the spinal canal is represented by the four-equation, one-dimensional model presented in Chap. 11.

In line with the results for venous pressure shown in Fig. 20.6, a large difference in CSF pressure is also seen between the worst-case scenario of no active collaterals, and the other two cases. The CSF pressure elevation from the healthy control to the stenotic patient with fully active collaterals is not large. For **Brain** for example, the predicted CSF pressure increase is of the order of 17%. In contrast, the venous pressure changes of Fig. 20.6, in the stenotic case with fully active collaterals, are much larger. These venous changes are transmitted to the CSF compartments, but the effect in terms of CFS pressure is relatively modest. However, for the case of no collaterals, the pressure increases in the CSF compartments are very large, by a factor of about four, relative to the healthy control; see Fig. 20.7.

At first sight, these results may look unrealistic. Interestingly however, Alperin and collaborators [89] measured intracranial pressure in four patients with chronically elevated ICP. They found that in three of the four patients, ICP measurements were in the range 20.5–34.0 mmHg.

20.2.3 Extracranial Venous Stenoses

Here we study the case of venous strictures located outside the cranial cavity. Details on geometry and parameters are found in [2]. Results are shown in Figs. 20.8, 20.9 and 20.10. Following [2] we consider the case of stenotic internal jugular veins and the azygos vein. See Fig. 20.5. This anomalous situation is of more recent interest in the biomedical community. Alperin and collaborators are amongst the pioneers in associating IIH also with extracranial venous outflow strictures [75].

More recently Zamboni and collaborators have associated multiple sclerosis with extracranial venous outflow strictures, notably internal jugular veins, vertebral vein and the azygos vein [30], [90–92], [93–97]. Such condition was termed Chronic Cerebrospinal Venous Insufficiency (CCSVI). See Chap. 18 for full details.

Venous Pressure

Figure 20.8 shows computed pressure in various compartments for the case of stenosed left and right internal jugular veins and the azygos vein. Results from a

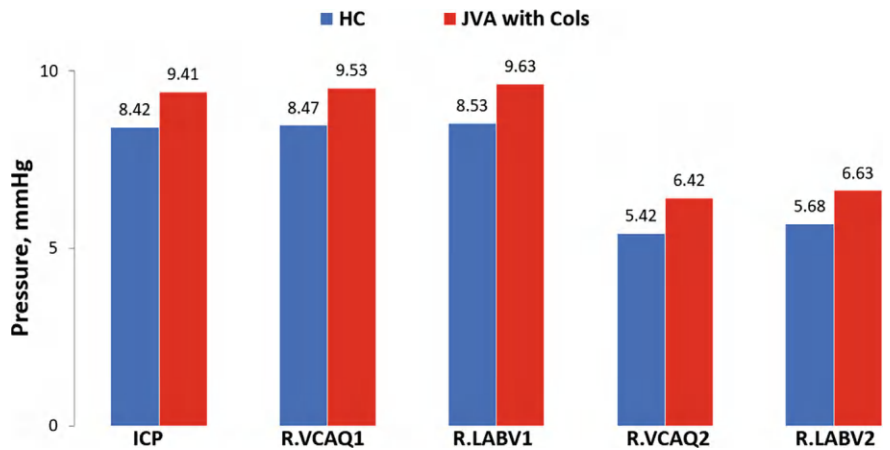


Fig. 20.8 Venous pressure in various compartments resulting from stenosed Left and Right Internal Jugular Veins and Azygos Vein. Cardiac-cycle averaged pressures for healthy control (HC) are compared against venous stenotic subject with fully active collaterals (JVA with Cols). Compartments include: ICP (CSF pressure as a reference value), and venous pressures in 2 positions of the Right Vein of the Cochlear Aqueduct (R.VCAQ1), (R.VCAQ2), and the Labyrinthine Vein (R.LABV1), (R.LABV2) (courtesy of Dr. Brigitte Toro. Data from [2])

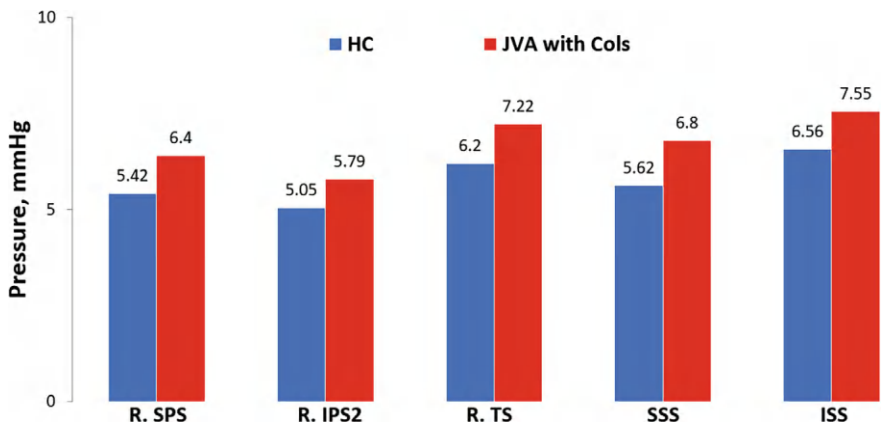


Fig. 20.9 Venous pressure resulting from stenosed left and right internal jugular veins and azygos vein. Cardiac-cycle averaged pressure values for healthy control (HC) are compared against venous stenotic subject with fully active collaterals (JVA with Cols). Computed venous pressure in several intracranial venous compartments: Right Superior Petrosal Sinus (R.SPS); Right Inferior Petrosal Sinus (R.IPS2); Right Transverse Sinus (R.TS); Superior Sagittal Sinus (SSS) and Inferior Sagittal Sinus (ISS). See Figures 20.5 (courtesy of Dr. Brigitte Toro. Data from [2])

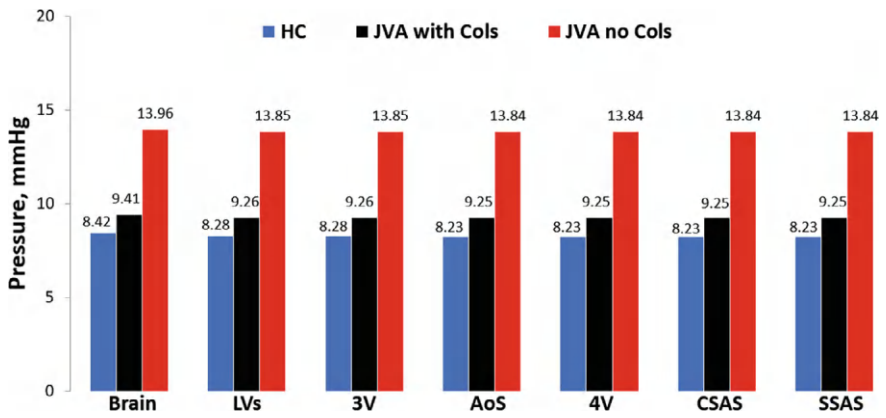


Fig. 20.10 Cerebrospinal fluid pressure in various CSF compartments resulting from stenosed Left and Right Internal Jugular Veins and Azygos Vein. Cardiac-cycle averaged pressure values for a healthy control (HC) are compared against venous stenotic subject with fully active collaterals (JVA with Cols) and subject with no active collaterals (JVA no Cols). CSF compartments are: Brain (brain parenchyma fluid phase); Lateral Ventricles (LVs), Third Ventricle (3V), Aqueduct of Sylvius (AoS), Fourth Ventricle (4V); Cranial Subarachnoid Space (CSAS) and Spinal Subarachnoid Space (SSAS). See Fig. 20.2 for location of relevant compartments (courtesy of Dr. Brigitte Toro. Data from [2])

healthy control (HC) are compared to those from a subject with venous stenoses and fully active collaterals (JVA with Cols) in various compartments. As a reference, we also include ICP, that is CSF pressure in the brain parenchyma. The other venous compartments are the vein of the cochlear aqueduct (VCAQ) and the labyrinthine vein (LABV) [80]. These two veins are relevant to inner-ear haemodynamics and play an important role in Ménière's disease (MD). Importantly, these intracranial changes in the inner ear circulation are caused by venous flow strictures located extracranially. See Chap. 18 for more details on MD.

Figure 20.9 shows results for pressure in various intracranial venous vessels. See Fig. 20.5 for location of these compartments. Overall, venous pressure changes for the case of fully active collaterals are relatively modest but chronic, bearing in mind that these results are for subjects assumed in the supine position. Again, venous hypertension is observed.

Cerebrospinal Fluid Pressure

Figure 20.10 shows computed cerebrospinal fluid pressure in various CSF compartments, resulting from stenosed left and right internal jugular veins and azygos vein, for a subject with fully active collaterals (JVA with Cols) and a subject with no active collaterals (JVA no Cols). CSF compartments include: Brain (brain parenchyma fluid phase); lateral ventricles (LVs), third ventricle (3V), aqueduct of Sylvius (AoS), fourth ventricle (4V); cranial subarachnoid space (CSAS) and spinal subarachnoid space (SSAS).

Closing Remarks

In this section, we have demonstrated the capabilities of a global, closed-loop mathematical model for simulating the circulation in the entire human body, dynamically interacting with cerebrospinal fluid in the craniospinal compartment. We have omitted the computational details regarding the domain, geometric and mechanical aspects of the circulation network, and regulatory mechanisms; these can be found in [1, 2], and associated publications.

Instead, we have focused on the computational results, particularly for pressure, a variable that is difficult to quantify using current methodologies. Importantly, the global model can also compute flow rates, velocities, and wall shear stresses throughout the computational network. Intracranial hypertension and venous hypertension have been theoretically demonstrated in various scenarios. These results are clinically significant, as elevated intracranial pressure and elevated venous pressure in the craniospinal cavity may trigger complex chains of events in the setting of various neurological disorders. For background, see Chap. 18 and references therein.

20.3 Arterial Hypertension

This section is concerned with mathematical modelling studies of arterial hypertension. Arterial hypertension affects one quarter of the global population and remains enigmatic in about 90% of hypertensive subjects. Hypertension is the leading single contributor to global mortality and a major risk factor for a wide range of cardiovascular diseases, including stroke, myocardial infarction, heart failure, chronic kidney disease, peripheral vascular disease, cognitive decline, and premature death, to name just a few. Chapter 19 provides a comprehensive review of both classical and non-classical issues related to hypertension, from a biofluid dynamics perspective. Notable non-classical issues include the role of the venous system, the sympathetic nervous system, and various theories of hypertension.

20.3.1 Modelling Studies of Hypertension

There is consensus in the mainstream relevant literature that the underlying mechanisms of primary (or essential) hypertension are currently poorly understood, thus preventing the design of effective therapeutical strategies. In this scenario, mathematical modelling of bodily fluid dynamics can assist in the unravelling of the complex interconnections of many factors at play and in delineating the relative roles of various vascular compartments that are likely to play a part, as evidenced from available data.

A major enterprise consists in **quantitatively** identifying the array of potential determinants of hypertension along with explanations that are consistent with available data. As Liang et al. put it: *model-based numerical experiments enable quantitative assessment of the inherent relationships between haemodynamic variables and cardiovascular properties, without suffering from the limitations of traditional clinical studies, which are often restricted by available in vivo measurements and complicated by inter-patient differences* [98].

Historically, the pioneering work of Guyton and Coleman in 1969 [18] is widely regarded as the first significant contribution to the application of mathematical modelling in the study of hypertension. In this context, we provide a brief overview of contemporary research on mathematical models used to investigate arterial hypertension.

Segers and collaborators [99] applied a mathematical model to study the interaction between the left ventricle (LV) and the arterial system under conditions of chronic pressure overload, reminiscent of hypertensive patients with compensated concentric hypertrophy. Their theoretical framework consisted of a LV time-varying elastance model coupled to a lumped-parameter model for the arterial district.

Out of the three scenarios they considered in their simulations, the one that included a parameter P_v , called **venous filling pressure**, was regarded as the most successful in terms of reproducing patient data. P_v was explicitly included in the

model and served to normalise end-diastolic wall stress. In their words *When we allowed venous filling pressure to rise to normalize end-diastolic wall stress while still normalizing systolic wall stress, the hemodynamic data better matched the in vivo observations, with a marked elevation of systolic and diastolic pressure.*

Interestingly, the authors pointed out the relationship of P_v (which the authors also called preload and LV filling pressure) to sympathetic nervous system (SNS) activity on the compliance of the venous system. They also refer to the early works of London and Safar in the 70s and 80s on reduced venous compliance in hypertensive individuals [100]. See Chap. 19. The SNS connection to reduced venous compliance has been pursued in recent years. According to Malpas [101], SNS activity provides a critical aspect in the control of arterial pressure. See also [102–104], Chap. 19 of this book, and Sect. 20.3.3 of this chapter.

Liang, Guan and Alastruey [98] conducted a computational study aimed at identifying determinants for the haemodynamics in the hypertensive state. They deployed an advanced global, closed-loop mathematical model that explicitly incorporated both the arterial and venous components of the human circulatory system. In their multiscale approach, major arterial vessels were represented through one-dimensional systems of partial differential equations, while distal arteries and arterioles were modelled through a structured-tree approach. Lumped-parameter models were used for the left heart, the pulmonary circulation, the right heart, capillaries and the venous system. After the model proposed by Guyton and collaborators [18], theirs appears to be the first to have integrated most of the main cardiovascular components in order to study arterial hypertension.

From their results the authors identified three major determinants for blood pressure and flow pulsatility indices in both large arteries and the microcirculation. These were: heart period (or cardiac cycle), central arterial stiffness and arteriolar radius. The venous component was not considered as a potential determinant of hypertension, even though the model explicitly included the venous system. From the ability of their model to account for many cardiovascular factors and their complex interaction the authors state: *These model-based findings highlight the importance of comprehensively considering patient-specific cardiovascular conditions when selecting anti-hypertensive drugs or exploring the mechanisms underlying hemodynamic changes associated with anti-hypertensive treatment.*

More recently, Celant et al. [105] adopted the global closed-loop multi-scale mathematical model of Müller and Toro [1, 2] for the entire human circulation, to reproduce a hypertensive scenario; see Sect. 20.1 of this chapter. The original model was suitably modified to reproduce alterations in the cardiovascular system that are cause and/or consequence of the hypertensive state. They considered cardiovascular adaptations to hypertension from the following five systems: large systemic arteries, the heart, the microcirculation, the pulmonary circulation and, most importantly, the venous system. Model results for the hypertensive scenario were validated through comparison against current knowledge on the impact of hypertension on the cardiovascular system; see [106].

20.3.2 Normotensive and Hypertensive States

Recently, a stochastic computational study aimed at identifying major determinants of essential hypertension in humans was conducted by Toro and Müller [103, 104]. They built upon extensions [105, 106] of the global, closed-loop multiscale model reported in [1, 2], starting from a parametrisation of the model in order to reproduce a **normotensive state** and the **hypertensive state** of a hypertensive middle-aged subject (50–65 years old).

Five candidate cardiovascular compartments were first considered in order to define a hypertensive state, deterministically, following the literature [105]. Full details of parameters are found in [103, 104]. A succinct review is given below.

Cardiovascular Compartments of Interest

1. **The heart.** As is well documented, elevated systemic load induced by hypertension results in left ventricular hypertrophy. The enlargement and thickening of the left ventricular heart muscle help maintain normal systolic wall stress, despite the increased load. Active and passive elastances in the heart model were increased by 30%, in order to normalise left ventricle wall stress to a normotensive state.
2. **Systemic arteries.** Hypertension-induced arterial stiffening due to remodelling of vessel walls is also well documented; see Chap. 19. Distinction was made, regarding some parameters, between different types of arteries, as follows:
 - Elastic arteries. The stiffness was raised so as to obtain a wave propagation velocity increase of 40%. Diameter enlargement of elastic arteries due to ageing was accounted for by introducing a 10% increment.
 - Muscular arteries. The stiffness was raised in order to obtain a wave propagation velocity increment of 20%.

Concerning viscous properties of the arterial vessel walls. These were increased in line with the arterial stiffness parameter, which is proportional to changes in wall thickness; see [103, 104] for details.

3. **Pulmonary circulation.** In hypertensives, changes in arterial pressure and resistance in the pulmonary circulation are similar to those in the systemic circulation. In line with the literature [100, 107], no changes to pulmonary compliance were incorporated in the model parametrisation for the hypertensive state.
4. **Systemic peripheral resistance.** It is established that the main site for structural resistance elevation is the proximal part of the microcirculation.
 - Proximal arterial resistance was increased by 40%, in line with a decrease of vessel radii of 8% and an increase in blood viscosity.
 - Capillary resistance was increased by 10% to account for the reduction of capillary density that characterises the ageing process [108, 109].
5. **Veins.** As discussed in Chap. 19, the venous system compliance dominates by far the total vascular compliance [110]. Furthermore, human and animal

studies have shown that total vascular compliance is reduced in hypertension [107, 110–114]. Accordingly, systemic venous compliance was reduced by 25% in the hypertensive model.

Further details are found in [103, 104]. For background on arterial hypertension see Chap. 19 of this book.

20.3.3 Hypertension Determinants via Stochastic Modelling

A major enterprise is to identify hypertension determinants. Here, we succinctly describe the stochastic modelling approach of Toro and Müller [103, 104] to assess the potential role of the five cardiovascular compartments described above.

Stochastic Approach

Adaptations of the cardiovascular system to ageing and hypertension may be simulated via representative parameters for each of the vascular compartments defined in Sect. 20.3.2. In the work of Celant et al. [105], such parameters were initially treated as deterministic quantities. See also [103, 104] for minor variations to the parametrisation of [105].

Instead, in [103, 104], these quantities were treated as random variables, and an assessment was made of how uncertainty in the chosen parametrisation affected the relevance of the adaptations considered. Random parameters were defined as follows:

$$q_{ht} = q_{nt} \Delta_q \quad \text{with } q \in \mathcal{P} . \quad (20.1)$$

Here $\mathcal{P}$ is the set of all parameters q in the above five cardiovascular compartments, subject to potential changes in the hypertensive state; q_{nt} and q_{ht} are values of the variable q in the normotensive and hypertensive states respectively. Δ_q is a uniformly distributed random variable defined as

$$\Delta_q = (1 + \delta_q \gamma_c) . \quad (20.2)$$

Here δ_q is a constant perturbation, such that $(1 + \delta_q)$ is the modified, deterministic variable used in [103–105]; $\gamma_c \in U(0, 2)$, so that Δ_q is a uniformly distributed random variable with average $(1 + \delta_q)$; values of Δ_q vary between the normotensive case and a hypertensive case with a perturbation twice as large as that used in [103–105]. Values for δ_q and γ_c are reported in Table 1 of [103, 104]. Polynomial chaos with stochastic collocation, as described by Xiu [115], is used to discretise the problem and to analyse the impact of parameter variability on the results; see also [116].

Cardiovascular Indexes

Results for a set of 29 cardiovascular indexes were reported in Table 2 of [103, 104]. Here, in Fig. 20.11 we reproduce three of those 29 indexes. Results are shown for brachial systolic blood pressure (top frame), brachial diastolic blood pressure (middle frame) and mean central venous pressure (bottom frame). In each case there are four quantities (bars) corresponding to:

1. a deterministic normotensive (NTdeter);
2. a deterministic hypertensive (HTdeter), in which the parameter perturbations are set to $q_{ht} = q_{nt} (1 + \delta_q)$;
3. a stochastic hypertensive (HTstoch), in which random parameter perturbations are as defined in Eqs. (20.1) and (20.2). HTstoch corresponds to the mean, or expected value; note the standard deviation;
4. a reference hypertensive case (HTrefer) from the literature; note standard deviation.

Regarding standard deviations for HTstoch and HTrefer in Fig. 20.11, a clarification is in order. The standard deviation in HTstoch originates from the stochastic model, which reflects the impact of uncertain model input parameters on computed indexes. Instead, the standard deviation in HTrefer indicates the variability of indexes in the population for which the indexes were computed. In other words, the standard deviations in HTstoch and HTrefer are unrelated.

Wave Forms in normotensive and hypertensives

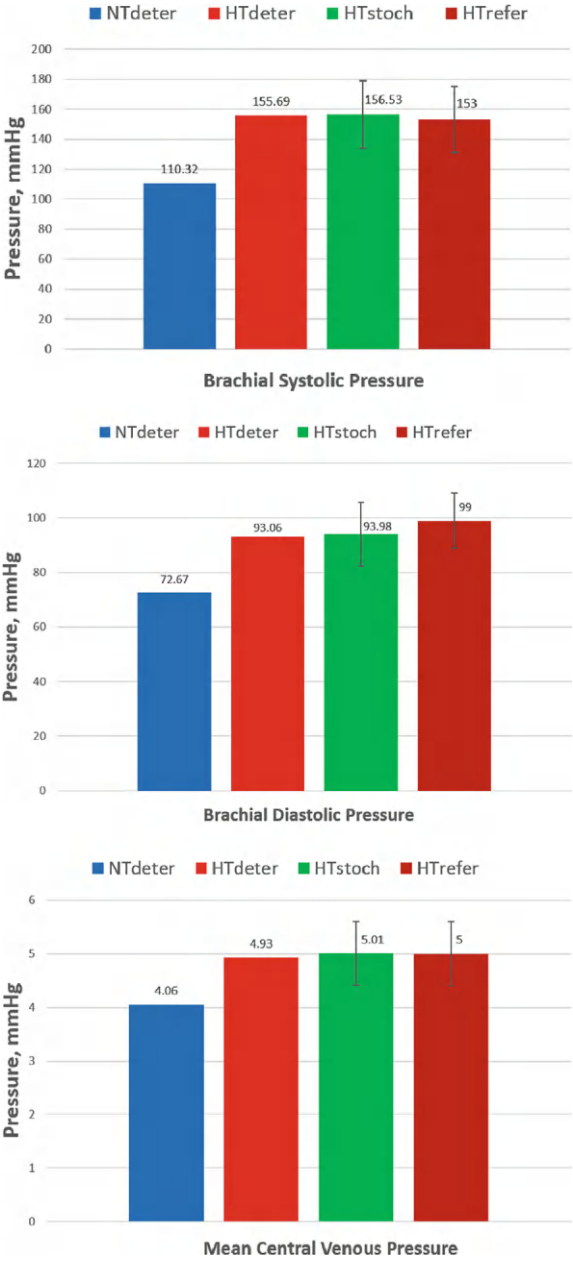
Comparison of wave forms from a normotensive and a hypertensive in various arterial and venous compartments are shown in Fig. 20.12. Generally, it is seen that in the (arterial) hypertensive state, venous pressure is also elevated, though the arterial pressure increase is more prominent than the venous pressure increment. To quantify the changes let us consider cardiac-cycle averaged pressures p_{nt} and p_{ht} in Fig. 20.12. In the ascending aorta, top left frame in Fig. 20.12, the average arterial pressure increase is 36.8%. In the left internal jugular vein, top right frame in Fig. 20.12, the venous pressure increase is 20%. In the right femoral artery, bottom left frame in Fig. 20.12, the arterial pressure increase is 36.6%. In the right femoral vein, bottom right frame in Fig. 20.12, the venous pressure increase is 16.3%.

In summary, from the model results we can state that in arterial hypertension, the venous system is also pressurised, though the arterial pressure increase is more prominent than the venous pressure increase. This finding is currently controversial but is consistent with supportive evidence reported elsewhere; see Chap. 19 for discussion on elevated venous pressure in arterial hypertension.

Total Sensitivity Indexes

The sensitivity of predicted outputs to uncertain parameters is represented in Fig. 20.13, which shows total sensitivity indexes for: mean (MBP), systolic (SBP) and diastolic (DBP) brachial pressures. The total sensitivity index represents the

Fig. 20.11 Cardiovascular indexes for brachial systolic pressure (top), brachial diastolic pressure (middle) and mean central venous pressure (bottom). NTdeter: deterministic normotensive; HTdeter: deterministic hypertensive; HTstoch: stochastic hypertensive and HTrefer: reference hypertensive (courtesy of Dr. Brigitte Toro. Data from Table 2 of [103, 104])



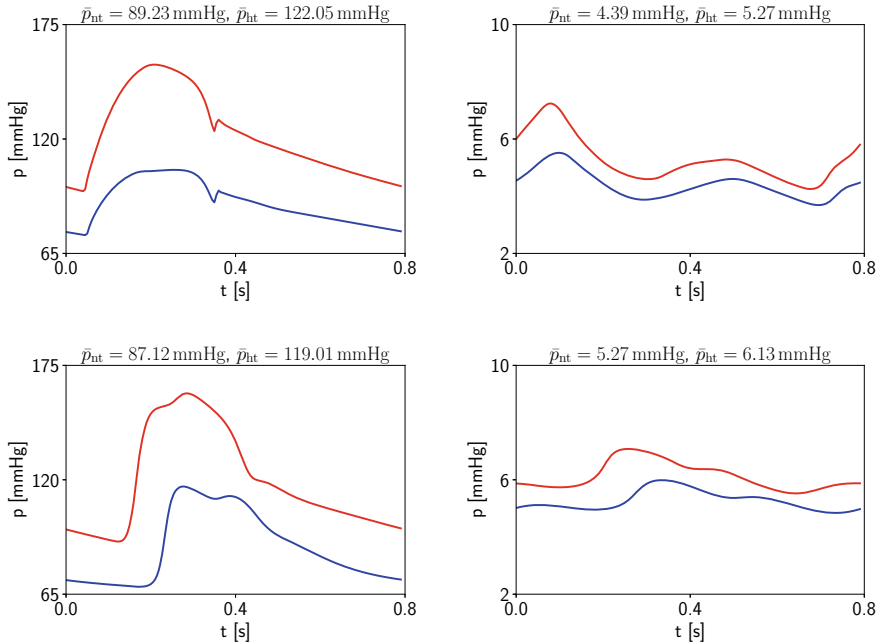


Fig. 20.12 Pressure wave forms for normotensive (blue) and hypertensive (red) subjects. Top left: ascending aorta. Top right: left internal jugular vein. Bottom left: right femoral artery. Bottom right: right femoral vein. Cardiac-cycle averaged values are $\bar{p}_{nt}$ (normotensive) and $\bar{p}_{ht}$ (hypertensive). More details are found in [103, 104]

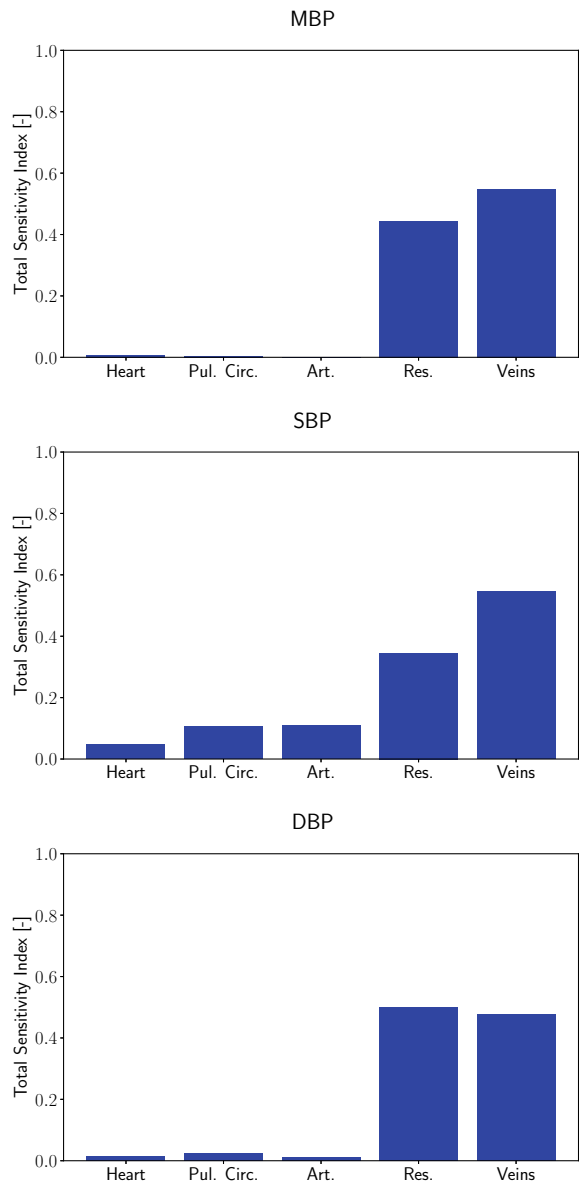
impact of the variability of a random parameter plus the effects from its interactions, of any order, with other random parameters.

The results of Fig. 20.13 show that the increment of peripheral resistance (Res) and the reduction of venous compliance (Veins) are the two most important adaptations in determining the variability of mean blood pressure (top), systolic blood pressure (middle) and diastolic blood pressure (bottom). Uncertainty in the other three compartments considered in the study, has little impact on the final uncertainty shown by variables of interest.

Impact of Parameters on Systemic Arterial Pressure

Figure 20.14 illustrates absolute changes, with respect to normotensive values, in left brachial arterial pressure: mean (MBP), systolic (SBP), diastolic (DBP) and pulse pressure (PP). These were obtained from stochastic simulations by considering the adaptation to hypertension of each of the five cardiovascular system compartments defined above in Sect. 20.3.2, in a *one-at-a-time* fashion, while maintaining normotensive parameter values in the remaining compartments. Dashed horizontal lines denote normotensive model results, square symbols denote mean values and error bars denote standard deviation. Again, the results show that the largest impact comes

Fig. 20.13 Total sensitivity indexes for brachial pressures: mean (MBP), systolic (SBP) and diastolic (DBP). Increment of peripheral resistance (Res) and the reduction of venous compliance (Veins) emerge as the two most important factors. For details see [103, 104]



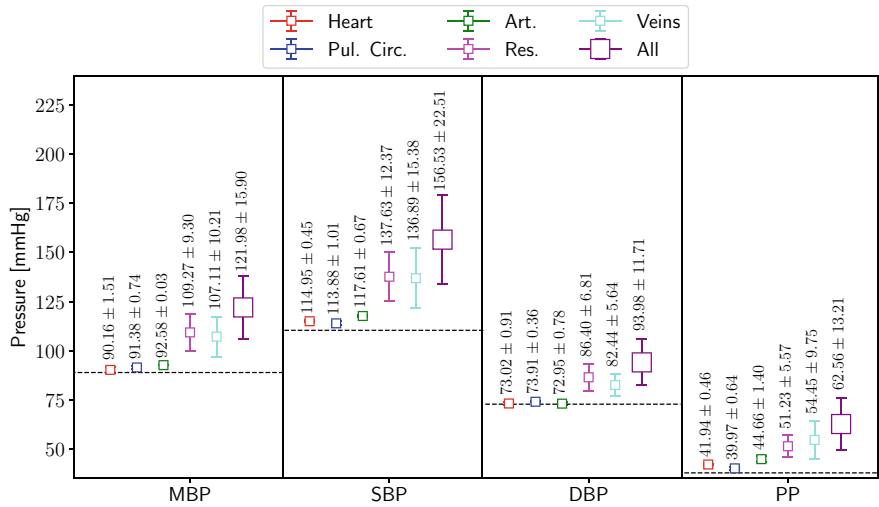


Fig. 20.14 Impact on pressure in left brachial artery, from parameter variations in five individual compartments defined in Sect. 20.3.2 and in all compartments (All). Dashed horizontal lines represent normotensive model results for MBP: mean blood pressure; SBP: systolic blood pressure; DBP: diastolic blood pressure; PP: pulse pressure. Square symbols are mean values and error bars denote the standard deviation. More details are found in [103, 104]

from modifications in peripheral resistance and venous compliance. This finding is also supported by brachial artery pressure waveforms, as seen below.

Brachial Artery Pressure Waveforms

Figure 20.15 displays the brachial artery pressure waveforms for six cases. Five cases are obtained from the stochastic hypertensive model with a *one-at-a-time parameter modification*; these correspond to each of the cardiovascular compartments considered in Sect. 20.3.2, shown as Heart, Pul. Circ., Art., Res. and Veins. The bottom right frame (All) shows the case in which all parameters were modified to represent the hypertensive state.

These results are useful for appreciating the temporal distribution of uncertainty propagation. It is seen that adaptation of different compartments manifest as uncertainty in brachial pressure at different times within the cardiac cycle. Overall, the results from [103, 104] suggest two main **determinants for arterial hypertension**, namely

- increased peripheral vascular resistance and
- reduced venous compliance.

The prominent role of increased peripheral vascular resistance is well established in the scientific community, even though the precise mechanisms and contributing factors are still the subject of investigations. Admittedly, the role of the venous

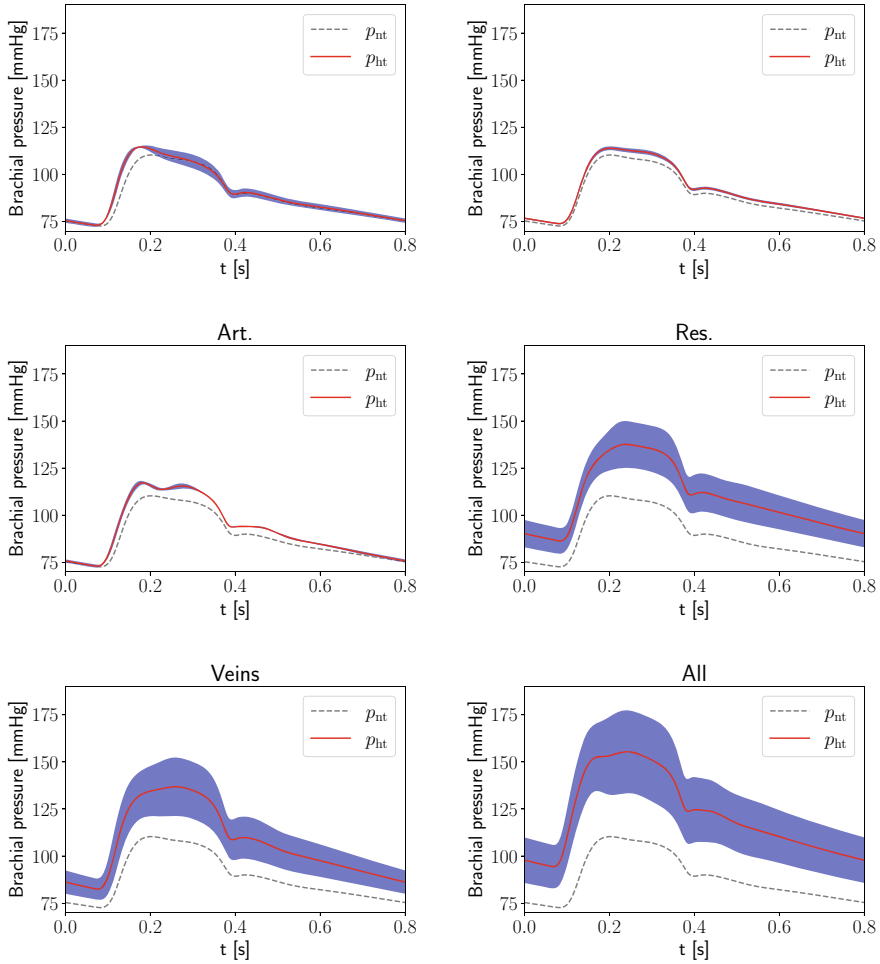


Fig. 20.15 Brachial artery pressure waveforms. Deterministic normotensive pressure p_{nt} shown by dashed grey line and stochastic p_{ht} is shown by continuous red line. Shaded areas represent standard deviations. Results are shown for one-at-a-time compartment parameter changes and for the case in which all (All) compartments are affected. For details see [103, 104]

system as a prominent determinant of arterial hypertension, is still controversial. See [103, 104] for further details and discussion on the role of the venous system in arterial hypertension.

Discussion of Results

The mathematical model results [103, 104] have shown that reduced venous vascular compliance significantly contributes to augmentation of arterial pressure. As a matter of fact, reduced venous compliance is the second most important factor

in defining the hypertensive state, after increased peripheral resistance. Moreover, if alterations in the venous circulation are neglected, the mathematical model does not reach the hypertensive state. Future developments should involve a more refined model of the venous system to account, for example, for regional differences. See Chap. 19. Refined description of current model components and addition of new ones, as indicated by new evidence or intuition, would help elucidate therapeutical strategies that address specific cardiovascular components, generally. Moreover, refined modelling may even be helpful in tackling patient-specific requirements, particularly for patients in the class of secondary hypertension.

20.4 Summary and Concluding Remarks

This chapter builds upon the content of Parts 1–3 of this book, particularly Chaps. 18 and 19. Our focus has been to illustrate the practical applicability of the book's final product through computational simulations of two major cardiovascular pathologies affecting the human population today: neurological disorders and arterial hypertension. In order to attain this goal we have omitted computational details regarding domains, geometric and mechanical aspects of the circulatory network, as well as parameters and regulatory mechanisms. These details can be found in [1, 2, 103, 104], and related publications.

Several neurological disorders, as discussed in Chap. 18, are associated with intra- and extracranial morphological anomalies of the venous system. Using the global, closed-loop mathematical model described in Sect. 20.1, we have simulated disturbed craniospinal fluid dynamics in two scenarios: venous sinus stenosis (intracranial) and venous stenosis in the internal jugular veins and the azygos vein (extracranial). Our primary focus has been on computed pressure, a variable that is difficult to quantify with other methodologies. However, it is important to note that the global model can also compute flow rates, velocities, and wall shear stresses throughout the computational network.

Both intracranial hypertension (elevated ICP) and venous hypertension in the head and neck have been theoretically demonstrated in the two scenarios considered. These findings are clinically significant, as elevated intracranial pressure and venous pressure in the craniospinal cavity are associated with events in various neurological disorders, some of which were discussed in Chap. 18.

Arterial (essential or primary) hypertension was also studied in this chapter using the global, closed-loop model described in Sect. 20.1, with the aim of identifying the determinants of hypertension. The computational results suggest that the venous system is the second most important determinant of arterial hypertension, after the well-established arterial-based cardiovascular resistance. Moreover, the results indicate that the hypertensive state cannot be attained without the contribution of the venous system.

The venous system contributes to arterial hypertension through reduced venous compliance. This reduction causes a displacement of blood volumes from the peripheral venous system to the central circulation, leading to disturbances in bodily fluid balance and an increase in blood pressure. As discussed in Chap. 19, reduced venous compliance is secondary to sympathetic nervous system (SNS) activity. SNS activity has been shown to play a role in regulating venous compliance, both in clinical studies and recent experimental animal research.

We have focused on essential (or primary) hypertension. However, as discussed in Chap. 19, there are also various cases of secondary hypertension, each associated with specific, well-identified mechanisms. Most of the literature suggests that the prevalence of secondary hypertension is small compared to primary hypertension, with estimates ranging from 5% according to some studies, to 10% according to others. However, these figures are disputed. Rossi et al. [117] argue that the proportion of secondary hypertension cases is much higher than is commonly accepted in the literature. Potentially, the methods presented in this book could be extended or adapted to study specific cases of secondary hypertension.

Our study supports the view that both intracranial hypertension and arterial hypertension should be considered systemic pathologies. Therefore, their theoretical study must be conducted using mathematical models that account for the global nature of bodily fluid dynamics.

Interestingly, intracranial hypertension and arterial hypertension are traditionally studied by different scientific communities, yet their theoretical simulation tends to unify them. Moreover, as noted in Chap. 19, some researchers have suggested that these two conditions may, in fact, be linked. See the recent works of Phillips and Schwartz [118], and da Costa et al. [119].

Through mathematical modelling of human bodily fluid dynamics, we have explored research issues related to complex pathologies that are currently the focus of extensive global study. Additionally, we have presented novel results from our recent theoretical investigations. While these findings appear promising, it is evident that further research is needed to validate their scientific accuracy and assess their practical applicability in developing effective therapeutic strategies.

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Chapter 21

Summary and Perspectives



Abstract Here we briefly summarise the main contents of this book and touch upon challenging topics for future developments.

Summary

This book has addressed a broad multi/inter/transdisciplinary subject that may be encapsulated in the following extended title:

Human anatomy and fluid physiology; mechanics of fluid flow and governing equations; mathematical analysis of the equations; computational algorithms to solve the equations; components of a global, closed loop mathematical model for the entire human circulation and associated local fluid systems; application of methodologies for simulating fluid flow phenomena in two major families of diseases: neurological disorders and arterial hypertension.

The contents of the entire book are divided into four parts, each having a distinct identity and a specific role within the overall book's framework.

Part I focused on major extracellular bodily fluid systems in humans, their anatomy and physiology, basic concepts in fluid mechanics, and introductory notions of mathematical modelling. This part consists of five chapters and is intended for scientists wishing to engage in the field of computational bodily fluid dynamics, including engineers, physicists, applied mathematicians, numerical analysts, and computer scientists.

Part II includes six chapters that delve into the governing partial differential equations of bodily fluid mechanics, their mathematical analysis, and the construction of exact solutions for specially simple initial conditions.

Part III also contains six chapters, concentrating on computational algorithms for solving evolutionary partial differential equations, with bodily fluid flow equations serving as a key example.

Finally, **Part IV** contains three chapters; two of these are dedicated to comprehensive reviews of two major fields of application: neurological disorders and arterial hypertension. The third chapter contains selected simulations in these areas, using a global, closed-loop mathematical model that integrates much of the content presented throughout the book.

The entire book is framed from an integrative, holistic perspective of human anatomy and fluid physiology [1], based on the belief that this approach offers a productive way to understand the underlying mechanisms of pathological events. It is hoped that this understanding will contribute to the development of effective therapeutic strategies.

Pending Issues and Perspectives

Inevitably, in a subject as broad as the one undertaken here, there are many aspects that have not been included, or have not been dealt with in sufficient depth. In what follows we mention just a few of these, with the intention of helping the reader to pursue the subjects further.

1. **The lymphatic system.** The lymphatic system, also called the third circulation, was introduced in Chap. 3 of this book, emphasising its role in preserving body-fluid balance via its connection with interstitial fluid and the venous system, its nutritional function through its connection to the digestive system and its immune function through its connection to the immune system.

Recent research on the lymphatic system and its multiple connections and functions has yielded important advances, particularly regarding the cerebral circulation. The discovery of a meningeal lymphatic system was announced in 2015 by two independent groups [2, 3].

Research on the lymphatic system is very active and the literature is vast. Examples of research publications include Swartz [4]; Gashev [5]; Margaris and Black [6]; Roose and Swartz [7]; Kunert et al. [8]; Moore and Bertram [9]; Contarino and Toro [10]; Jafarnejad et al. [11]; Shou et al. [12], to name but a few. In spite of advances in the mathematical modelling of bodily fluids, particularly on the arterial side of the circulatory system, the lymphatic system—an integral part of the human fluid system—remains behind in this regard.

In this book we have not included models for the lymphatic system. In principle, some of the mathematical models proposed for blood flow could be adapted to the flow of lymph; obviously the numerical methods of this book could also be applied quite directly to a network of lymphatic vessels.

A major challenge is the inclusion of nodes, reported to be between 500 and 700 in the human body. Lymph is absorbed into the venous system in each node, which attests to the complexity of modelling the connection between ISF in the interstitium, the lymphatic system and the venous system. Mathematical modelling of the lymphatic system in a realistic physiological setting remains a major challenge.

2. **Local fluid compartments and systems.** We have not studied some important local fluid compartments, such as the urinary and respiratory systems, for which some of the methods in this book may be applicable. This may also extend to renal, ocular, and inner ear circulations. Major systems not included are the digestive, immune, central nervous, and endocrine systems. Additionally, key processes such as heat transfer, transport, and vasomotion have been excluded. For more details, see Part I of this book.

3. **The cell and intracellular fluids.** Recall from Chap. 3 that two thirds of the total body fluid volume is intracellular. In this book we have limited ourselves to the modelling of extracellular fluids. We have not addressed intracellular fluid physiology and transport processes inside the cell and between the cell and the interstitium, across the cell membrane [13, 14].
4. **The microcirculation.** The microcirculation connects the two main extracellular fluid compartments examined in this book: the arterial and venous systems. However, the anatomical and physiological aspects of the microcirculation have not been discussed in sufficient detail. Mathematical modeling of the microcirculation, within the context of global circulation models, remains a challenging topic [15–20].
5. **Refined characterisation of fluid conduits.** A more detailed anatomical and mechanical characterisation of the vasculature in current mathematical models deserves attention. Advances have been made on the arterial side of the circulation, while refinements to the venous counterpart are a pending task [21]. All current one-dimensional descriptions of blood vessels assume straight conduits. A step forward would be to include curvature [22]. Modelling vessel junctions remains a challenging exercise. Current methods work satisfactorily for subjects in the supine position, but such methods become unsuitable for individuals subject to postural changes. Potentially successful approaches are found in [23–26].
6. **Perivascular spaces.** Periarterial and perivenous spaces are relevant in the new lymphatic theory [27–29] and constitute a major area for mathematical modelling research.
The four-equation model for CSF and the spinal cord presented in Chap. 11 could be adapted to describe blood flow in vessels enclosed by the perivascular space containing cerebrospinal fluid.
The mechanical role of tissues and structures surrounding fluid conduits needs to be recognised in the framework of global models for the circulation [30, 31].
7. **Postural changes and gravitational forces.** Over two thirds of an individual's life time is spent in constant movement. Most of the haemodynamics research as well as global mathematical models for blood flow assume the subject in the supine position. Sport physiology [32], high speed flight and space exploration pose a host of challenges to the mathematical modelling of bodily fluids, both at the level of mathematical descriptions as well as for developing suitable computational algorithms [33–36].
8. **Parameters and uncertainty.** Complex mathematical descriptions of bodily fluid dynamics contain many parameters. Choosing specific parameter values is subject to considerable uncertainty and is perhaps the most challenging task in modelling physiological flows, particularly in the holistic setting as advocated in this book. Parameter determination and uncertainty quantification, not just in bodily fluid dynamics, is a relevant and dynamic subject of current research. The literature on this subject is vast, see [37–44], to name just a few.

9. **Three-dimensional fluid/structure interaction models.** The most sophisticated description of flow in blood vessels, or other compliant conduits, are three-dimensional fluid/structure interaction models. Over the years, these models have reached a high degree of maturity. For a comprehensive presentation of these methods we recommend the book edited by Formaggia, Quarteroni and Veneziani [22], and the many references therein. For 3D models see also [45–49]. These methods form the advanced, sophisticated component of global multiscale models, designed to resolve local mechanical features with high fidelity. Algebraic, zero-dimensional, one-dimensional, and three-dimensional FSI models can be effectively integrated into a single multi-module simulation tool.

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